Amazon L6 Senior Supply Chain Manager — NACF

Complete interview book · Rajesh Rishiyur Chandrasekaran · Loop Wed Oct 29 & Thu Oct 30 (PDT) · 530 pages, bookmarked, built Oct 28 3:30pm PDT (open the bookmark/outline pane to jump)

One file, everything. Five interviewer packets (each with story-by-story math, behavioral answers, trap drills, router and source trace), the Google Fiber design stories and the NACF inbound booklet. Use the bookmarks. Start each packet at its **5-minute cram sheet** and its **Question → Story router**.

Where everything is

Part	When	Who	Pages	#	Status
PART 1	Wed Oct 29 · 10:00am PDT	**Snigdha Tripathy — Hiring Manager (Sr Mgr Supply Chain NACF)**	3–123	121	Fact-checked: source trace + real question wording
PART 2	Wed Oct 29 · 12:00pm PDT	**Ryan Hoerger — Transfer / Placement Analytics (shadow: Sandy Hsiao)**	124–249	126	Draft v2: fact-check pass still pending, use the number rules
PART 3	Thu Oct 30 · 10:00am PDT	**Rocky Goforth — Sr Mgr Infrastructure**	250–329	80	Fact-checked: source trace + real question wording
PART 4	Thu Oct 30 · 11:00am PDT	**Zhaowen Lan — Sr Mgr Supply Chain (peer)**	330–422	93	Draft v2: fact-check pass still pending, use the number rules
PART 5	Thu Oct 30 · 12:00pm PDT	**Venkata Josyula — Bar Raiser (Principal PM, FBA)**	423–493	71	Fact-checked: source trace + real question wording
PART 6	Reference — read the night before	**NACF Inbound Booklet — capacity, inventory, worked week, 50 questions**	494–530	37	Reference

How to use a packet in 3 steps

1. **Night before:** read the interviewer's cram sheet, then the router. Rehearse each story's 20-second version out loud.
2. **In the room:** hear the question → find its Leadership Principle in the router → tell the primary story using the opener in the router → land the single number.
3. **When challenged:** use the story's trap-drill answer and the SAFE-SAY box. If you are unsure of a figure, say the safe sentence — never guess a number.

Read this once — rules for the numbers

Distilled from the line-by-line check of every story against your Apple workbook, the source documents and the Amazon master.

Basis labels. Every number carries one: *record* / *record-from-docs* (a document or the Apple workbook backs it), *user anchor* (from your earlier packet only — confirm from memory before saying it), *modeled* / *illustrative* (your arithmetic or a teaching example — say “in the model” / “in an illustrative lane”).

1. **Never say “50x”.** Flexport Shipment Visibility is **9.7× (7 → 68 customers)**; the 170 MAU figure and 32,000 shipments are *targets*, not results.
2. **Walmart stories (S19–S21) and all Google Fiber / E-series stories are DESIGN CASES.** Say “I designed / in the model / here is how I would run it”. Your lived Walmart-theme record is Apple chapter F4.
3. **Lanes:** “a thousand lanes were asked; 597 first”. Never “400+ lanes”.
4. **Flexport cost-to-serve:** “$0.66 to $0.03, about 95%”. **FF App:** “about $5M avoided”, said once; the April 2022 launch was the import-side app.
5. **Dollar nouns:** “$2.5M” and “$1.5M” each mean several different things — always say what the number is the first time.
6. **S14** $10M won is not savings; **S16** >$100K is *priced*, not saved; S11 ~$1.5M pre-production is separate.
7. **Sephora RF devices:** “hundreds of devices”, device test before training, training +1.5 weeks, date held. Never quote a multiplier.
8. **Systems count:** “about a dozen systems”; six sites (never “six NA sites”); finished-goods grain only — no BOM explosion stories.
9. **Planner collaboration +50%** (not “planner effort −50%”). Program timeline: “within a quarter”.
10. **Yellow “VERIFY” tags** mark the few facts that exist in none of your files — use the printed safe sentence if you cannot confirm them from memory.

If asked “what’s your source for that number?” — “That’s from the program dashboard / my own model — the figure I’d stand behind is X.” Each story has the exact sentence in its SAFE-SAY box.

INTERVIEW-PREP PACKET 1 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS · V2 — SEVEN FLEXPORT PRIMARIES

Amazon NACF Supply Chain loop — Packet 1: Snigdha Tripathy

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

Snigdha Tripathy

Senior Manager, Supply Chain — NACF Supply Chain · HIRING MANAGER

Wednesday, October 29, 2026 · 10:00 am PDT · 55-minute video interview

Strategy in one line. The hiring manager judges fit for the actual work — and her public profile (EU inventory-placement strategy across the crossdock network; readiness for 50+ FC launches; a Kaizen / Lean trainer; teams built from scratch — public-profile inference in "Read me first") says she will test whether the mechanisms transfer to her world. Open with S11 (a boarding process behind published gates, run from a weekly input/output review) and hold S12 (a committed roadmap re-sequenced on cost-to-serve evidence). v2 adds five distinct Flexport primaries — S11-A the Monday review as an analysis, S11-B launch readiness as standard work, S11-C retiring an acquired product's book, S12-A the Asia latency investigation, S16-A the data model under the review — so every Leadership Principle has a primary and a backup from one company. Lead every answer with the mechanism left behind, the priced trade-off, and the team that grew to own it. Systems are enablers; the process redesign and its business results are the story.

LP focus — my guess, not confirmed: Deliver Results · Hire & Develop / Earth's Best Employer · Dive Deep. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title, the JD and her public profile in "Read me first" (public-profile inference, labeled as such) and should be held loosely — Think Big / Frugality and Ownership / Earn Trust remain live alternates carried by the twist maps.

Also expect: "tell me about yourself" (75-second version, once, here) and "why Amazon" (hiring managers usually ask it); one functional question on how I would approach the NACF process space.

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

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Every new story is tagged NEW and carries five labeled facets (data analysis · solution design · math & data logic · implementation · probe questions), four `Before / Fix / After / Dollars` tables, a four-question behavioral pack, three angle cards and three trap drills. Each primary opens with a SAFE-SAY box (the sentence for its most attackable number and for "what is your source for that?"). Yellow VERIFY-WITH-RAJESH tags mark only the specifics that are in none of the source files; each is repeated on the VERIFY page with the safe sentence to say instead.

5-Minute Cram Sheet — Snigdha Tripathy · Wed Oct 29 · 10:00 am PDT NEW

HER HOOKS → MY PROOF (PUBLIC-PROFILE INFERENCE)

- **EU IXD placement strategy** → S11 wave machine: a selection bar published before the first node, a ~50-container calibration per node, a public hold when a node misses; S11-C: three brokers = 75.0% of containers, so the few lanes that carry the volume get bespoke handling.
- **$80MM savings** (her number — framing only) → my discipline on dollars: ~$5M avoidance said once (S12); $0.66 → $0.03; anything priced is "priced, not saved" (S16-A).
- **50+ FC launches** → S11-B readiness: four gates, 100 containers × 7 checks, six graduation criteria, and the pilot that ran later than the first testing plan, owned.
- **Kaizen / Lean trainer** → S11-A waste = 3.67 touches and standard work; S11-B jidoka (stop the line, hold wave 3); S12-A genchi genbutsu and PDCA; S16-A poka-yoke (bridge table, parity gate).

THREE OPENING SENTENCES

- *Most significant delivery.* "Cost-to-serve on Flexport's shipment-visibility product went from sixty-six cents a container to three while customers went from seven to sixty-eight — nine point seven times — with zero rollbacks, because boarding went through six published gates, not around them."
- *Changed a committed plan.* "I re-opened my manager's eighteen-month roadmap privately, with numbers — partners first, operators on the same chassis — and value landed in month six with sixty-six percent less effort."
- *Dive into data.* "A twenty-two-fold cost drop decomposes into two levers — containers needing a human, and touches on each — and which one moved decides what you do next."

FIVE NUMBERS I MUST NOT GET WRONG

- **7 → 68 = 9.7×** — never rounded up; no other growth multiple is ever said.
- **$0.66 → $0.03, about 95%**; direct dollars ≈ $50K a year *modeled*; the ~$1.5M pre-production estimate is a user anchor, separate, never added.
- **~$5M** is avoidance, said once — never annual; run-rate ≈ $1.5M a year is a separate modeled number.
- **S11-C:** printed $185,892 = six of nine segments; nine rows = $254,952; lead with containers (294,581).
- **S12-A:** never quote 81%; like-for-like is −44.2% overall (Apr → Sep). **S16-A:** $103,680 and $100,360 are prices, not savings.

SEVEN STORIES — TELLING ORDER AND THE ONE NUMBER

1	**S11**	**7 → 68 customers (9.7×); $0.66 → $0.03**	Deliver Results · Ownership
2	**S11-B**	**100 containers × 7 checks; 41 of 59 failures at two hops**	Highest Standards · Customer Obsession
3	**S12**	**month 18 → 6; −66% effort; ~$5M avoidance (record), said once**	Think Big · Backbone · Frugality
4	**S11-A**	**22× = 6.0× (coverage) × 3.67× (touches)**	Dive Deep · Earn Trust
5	**S12-A**	**Asia 21.19 s vs US 3.3 s = 6.42× on 45.3% of views**	Dive Deep · Customer · Bias for Action
6	**S11-C**	**79 of 79 customers placed; 3 brokers = 75.0% of containers**	Customer Obsession · Broad Responsibility
7	**S16-A**	**system accuracy 60% → 90%; over $100K priced, not saved**	Invent & Simplify · Frugality

BRIDGE IN ONE BREATH (BOOKLET FRAMEWORKS)

- Inbound capacity: dock doors × hours × unload rate; utilization knee ≈ 85%; labor by UPH; cubic fill.
- Inventory health: weeks of cover — under 2 expedite, 2–4 healthy, over 6 excess; safety stock z · σ · √LT.
- Placement: distance-to-customer priced against cost per unit moved — the edge-near-demand logic of S12-A, the concentration logic of S11-C.

ROOM RULES

- Pitch, land the metric, **stop**. Name who owned what — program-lead voice.
- Every dollar with its basis word; if a number is yellow in the text, check the last page first.
- Overlaps: S16-A, S11-C, S12-A, S11-B facts are also in other packets — see "Read me first"; fall back to the router's backup if the panel flags a clash.
- Bridge in one sentence, then hand it back to her with a question.
- If challenged on a number, read the story's SAFE-SAY box; if it is yellow, read its safe sentence on the VERIFY page — never the unconfirmed specific.

Question → Story Router — 16 Leadership Principles + 11 common prompts NEW

Each Leadership Principle row quotes a **real question from the Amazon question bank (AIQB reference guide)**, then routes it: find the LP, tell the primary (the B-card is the 90-second STAR in that story's Behavioral Pack), keep the backup for "another example". One number, then stop. Never the same story twice in one loop.

LEADERSHIP PRINCIPLE · AIQB QUESTION	PRIMARY STORY · CARD	BACKUP	15-SECOND OPENER	ONE NUMBER
Customer Obsession *"Tell me about a time when you had to balance the needs of the customer with the needs of the business. What did you do? What was the result?"* AIQB: Customer Obsession Q8	S11-C · B1 — nine segments, 79 destinations	S11-B · B2 (shadow accounts) · S11 · B3	"Every one of seventy-nine customers got a destination and a date — not an end-of-life email."	**79 of 79 = 55 + 3 + 9 + 9 + 3**
Ownership *"Give me an example of an initiative you undertook because you saw that it could benefit the whole company or your customers, but wasn't within any group's individual responsibility so nothing was being done."* AIQB: Ownership Q3	S11-A · failure card — the review nobody owned, run by hand for months	S11 · B2 (the critique accepted)	"Nobody owned the measurement system, so I ran it by hand every Monday — about four hours a week at its worst — until the formulas were stable enough to automate."	**40.0% → 21.9% week, explained**
Invent and Simplify *"Give me an example of a complex problem you solved with a simple solution. What made the problem complex? How do you know your solution addressed the problem?"* AIQB: Invent and Simplify Q1	S16-A · B1 — four systems, one pipeline	S11-A · conflict card (seven metrics)	"Four systems and one person made the Monday number; I simplified it to one pipeline and seven frozen definitions."	**system accuracy 60% → 90%**
Are Right, A Lot *"Tell me about a time when you didn't have enough data to make the right decision. What did you do? What path did you take? Did the decision turn out to be the correct one?"* AIQB: Are Right, A Lot Q1	S11-A · B3 — 99.4% on three touches	S11-A · B4 (the carrier-format-change week) · S12 · failure card (Asia 21.19 s the plan did not see)	"One week printed ninety-nine point four percent — on three touches. I did not have enough data to steer on it, so I printed it as a data miss."	**N = 3 → discarded**
Learn and Be Curious *"We all have things about ourselves we'd like to improve on at work. Give me an example of something that you've worked on to improve your overall work effectiveness."* AIQB: Learn and Be Curious Q4	S11-B · B4 — the honest miss	S12-A · failure card (no regional P50 / P90)	"The pilot ran later than the first testing plan; my test estimates were optimistic and monitoring arrived late, so I size testing on hop count now."	**41 of 59 failing rows at two hops (my recount)**
Hire and Develop the Best *"Describe a time when you constructed a team to accomplish a goal. What factors did you consider in constructing the team? Did you factor diversity into your team construction?"* AIQB: Hire and Develop the Best Q9	S11 · the input owners across nine teams who sign their metric — a team-level answer; for a named person see VERIFY-WITH-RAJESH · V1	S11-A · conflict card (owners sign what they are red against)	"I grew the input owners across nine teams into people who sign and defend their own definitions; the review runs on them, not on me."	**9 teams · 7 definitions**
Insist on the Highest Standards *"Tell me about a time when you had to make a decision between standards and delivery. What tradeoffs did you have make? What was the outcome?"* AIQB: Highest Standards Q4	S11-B · B1 — UAT opens at 80%	S16-A · B4 (1% parity gate) · S11-A · B2	"UAT did not start until eighty percent of shipments showed minor-to-no issues — the gate was data quality, not the calendar."	**100 × 7 checks · 80% gate**

LEADERSHIP PRINCIPLE · AIQB QUESTION	PRIMARY STORY · CARD	BACKUP	15-SECOND OPENER	ONE NUMBER
Think Big *"Give me an example of how you have changed the direction or view of a specific function/department and helped them embrace a new way of thinking. Why was a change needed?"* AIQB: Think Big Q2	S12 · B1 — month 6 not 18	S11 · B1 (7 → 68 on the same ops team)	"The roadmap said eighteen months to first value; I re-opened it and landed value in month six, with sixty-six percent less effort."	**18 → 6 months; −66%**
Bias for Action *"Give me an example of a calculated risk that you have taken where speed was critical. What was the situation and how did you handle it?"* AIQB: Bias for Action Q1	S12-A · B3 — reversible changes first	S12 · reversal trigger written before the swing	"We started with changes that could be undone — a double CDN removed, a firewall on — and held the expensive one for the metrics."	**52-day plan; 4 of 8 rows done**
Frugality *"Give me an example of how you have helped save costs or eliminate waste within your role or organization. What was the situation? What was the impact?"* AIQB: Frugality Q1	S12 · B3 — ~$5M avoided	S16-A · B3 (priced, not saved)	"About five million dollars of cost avoidance — engineering never spent plus value pulled forward — and I say it once."	**~$5M avoidance (record), said once**
Earn Trust *"Give me an example of a tough or critical piece of feedback you received. What was it and what did you do about it?"* AIQB: Earn Trust Q2	S11 · B2 — the critique accepted	S11-A · B3 · S12 · B4 · S11-C · B4	"Leadership said the program wasn't showing value; I'd been reporting activity, not evidence — and I accepted that first."	**#3 → #14 of 16 → rollout unlocked**
Dive Deep *"Tell me about a problem you had to solve that required in-depth thought and analysis. How did you know you were focusing on the right things? What was the outcome?"* AIQB: Dive Deep Q3	S11-A · B1 — 22× = 6.0× × 3.67×	S12-A · B1 · S16-A · B2 · S11-B · B3	"A twenty-two-fold cost drop — and I can say which lever did it: six times fewer containers needing a human, times 3.7 fewer touches."	**22× = 6.0× × 3.67×**
Have Backbone; Disagree and Commit *"Tell me about a time when you strongly disagreed with your manager or peer on something you considered very important to the business. What was it and how did you handle it?"* AIQB: Have a Backbone Q1	S12 · B2 — private first, two days to break it	S11-C · B3 (the dissent on the page)	"The plan had my manager's name on it; I read it to him alone, with numbers, and gave engineering two days to break it."	**2 points argued, 1 adjusted**
Deliver Results *"Tell me about a time when you not only met a goal but considerably exceeded expectations. How were you able to do it? What challenges did you have to overcome?"* AIQB: Deliver Results Q3	S11 · B1 — both targets early	S12 (value in month 6)	"Both public targets — 68 customers and 170 monthly users — landed two months early with zero rollbacks, behind six published gates, not around them."	**7 → 68 (9.7×)**
Strive to be Earth's Best Employer *"Describe a time when you implemented a change at work aimed at improving the work experience of yourself or coworkers. What was the goal?"* AIQB: Strive to Be Earth's Best Employer Q3	S16-A · B1 — retired the Monday stitching: one engineer maintains the pipeline instead of one PM rebuilding it	S12 · B4 (operators' two quarters, priced and published)	"Every Monday I stitched four systems by hand and the hours came out of rollout time; I turned it into a pipeline so the review no longer depends on one person."	**system accuracy 60% → 90%**

LEADERSHIP PRINCIPLE · AIQB QUESTION	PRIMARY STORY · CARD	BACKUP	15-SECOND OPENER	ONE NUMBER
Success and Scale Bring Broad Responsibility *"Tell me about a time when you considered everyone who was affected by your work, not just your direct customers or people within your organization."* AIQB: Broad Responsibility Q5	S11-C · B2 — concentration	S12-A · B2 (Asia = 45.3% of views)	"Three customers were four percent of the book and three-quarters of the containers — a template would have been wrong for them."	**3.8% of customers = 75.0% of containers**

ELEVEN COMMON NON-LP PROMPTS

PROMPT	PRIMARY	BACKUP	NUMBER
Tell me about a time you failed	S11-B · B4 (the pilot ran later than the first testing plan)	S12-A failure card	**later than the first testing plan**
Disagreed with your manager	S12 · B2	S11-C · B3	**2 points argued, 1 adjusted**
Influenced without authority	S11 · conflict card (Sales held to six published criteria)	S11-B · conflict card (nine teams, one seam)	**6 criteria · wave 3 held**
Reduced cost	S12 · B3	S11 (CTS) · S16-A · B3	**~$5M avoidance once · $0.66 → $0.03 (both record)**
Handled ambiguity	S11-C (nine segments)	S11 · B2	**79 customers, 9 segments**
Simplified a process	S16-A · B1	S11-A conflict card	**4 systems → 1**
Decided with incomplete data	S12-A (P95 on 316 tail views)	S11-A (N = 3)	**P99 = 63 views vs P95 = 316**
Missed a deadline	S11-B · B4	S16-A (the two-week security-review miss — one sentence only)	**later than the first testing plan**
Mentored someone	S11 — owner growth (team-level) VERIFY-WITH-RAJESH · V1	S11-A	**owners across 9 teams signing definitions**
Delivered under a tight timeline	S12-A · B3	S11 · B1 (both targets two months early)	**52-day plan**
Changed your mind because of data	S12 (adjusted 1 of 2 points)	S11-B (re-sized testing on hop count)	**1 of 2 points**

SAME STORY ASKED TWICE — SWITCH TO A DIFFERENT ANGLE CARD (LP · HEADLINE METRIC)

STORY	DATA / ANALYTICAL CARD	CONFLICT / BACKBONE CARD	HONEST-FAILURE CARD
S11	**Dive Deep** — $0.66 / $0.18 = 3.67 → 0.17 touches	**Have Backbone; Disagree and Commit** — six criteria; wave 3 held	**Learn and Be Curious** — #3 → #14 of 16
S11-A	**Dive Deep** — 22.0× = 6.0× coverage × 3.67× touch efficiency; fair split 56% / 44% (modeled)	**Insist on the Highest Standards** — seven frozen metrics, each signed by the owner who would be red against it	**Ownership** — by hand for months, about four hours a week at its worst; no longer a single point of failure once the pipeline matched within 1%
S11-B	**Dive Deep** — 41 of 59 (69.5%) at route + milestone; 3 of 105 all-seven-clean	**Ownership** — nine teams aligned on one seam; time to localize cut about 50% (secondary source)	**Learn and Be Curious** — first plan early November → launched December 2021; hours to localize a bug at six-plus layers
S11-C	**Dive Deep** — 3.8% of customers = 75.0% of containers = 54.7% of ARR	**Have Backbone; Disagree and Commit** — four dissenting inputs; one decider; each impacted party commits	**Learn and Be Curious** — 2 open questions; break-even $15,491 / month
S12	**Dive Deep** — 339.5 min × $0.45 ≈ $153; −25% vs −50%	**Have Backbone; Disagree and Commit** — two points argued; one adjusted	**Are Right, A Lot** — 21.19 s vs 3.3 s
S12-A	**Dive Deep** — 6.42× on 6.31k of 13.92k views	**Have Backbone; Disagree and Commit** — one conditional row (Q3 or later); 4 of 8 rows done	**Learn and Be Curious** — window-mismatched 81.1% — not quoted
S16-A	**Dive Deep** — 12 / 18 = 0.667 vs 12 / 10 = 1.20 (illustrative); 4% drop (record)	**Have Backbone; Disagree and Commit** — seven frozen metrics; one bridge table	**Earn Trust** — two-week security-review miss; zero Monday reviews skipped

B-cards: each story's Behavioral Pack has four separate 90-second STARs (B1–B4), each headed by the real AIQB question it answers; the three angle cards (data · conflict · honest failure) give a different opener and metric when the same story is asked twice.

Read me first

The interviewer's lens (inference from title, JD and public profile — the profile parts labeled)

Inference from the title. The hiring manager owns the team this role joins and the outcomes it is measured on. A Senior Manager of Supply Chain in NACF is accountable for network-level processes — how inventory is planned, placed, moved and recovered across fulfillment centers — and for the operating cadence (weekly business reviews, goals, mechanisms) that keeps those processes honest. She is the one interviewer who will also judge *fit for the actual work*: can this person define and run a process across teams that do not report to them, make value legible to leadership, and change a committed plan on evidence without breaking trust.

What that means for the answers. Lead with scale-with-mechanisms (S11) and roadmap judgment under unit economics (S12); v2 adds five distinct Flexport primaries (S11-A, S11-B, S11-C, S12-A, S16-A) so every Leadership Principle has a primary and a backup from one company — see the Router. All seven are Flexport stories, told as process redesigns: a boarding process behind published gates with a weekly input/output review, a launch roadmap re-sequenced on cost-to-serve, and the analysis, readiness, retirement, latency and data-model work underneath them. She will probably also ask the openers — "tell me about yourself" and "why Amazon" — and a functional question on how I would approach the NACF process space in the first 90 days. The JD's own phrase, "designing or modeling the supply chain as opposed to working in it," is the lens to speak to: I designed the boarding process and the review; I did not run the ops desk.

What her public profile adds (public-profile inference, labeled)

The profile, as pasted (public-profile inference — directional, not verified biography). 11+ years at Amazon across operations, launch and learning: EU IXD & Placement Operations Manager, Luxembourg, 2017–21 — defined the EU inventory-placement strategy for the crossdock network (distance-to-customer, speed-vs-cost) and built the team ground-up (+$50MM savings in 2020); Senior Manager EU IXD, 2021–22 — ran the EU crossdock network with a team of six plus interns ($80MM savings in 2021; IXD grown to 60% of inbound, +46% YoY); Senior Launch Program Manager, 2014–17 — readiness for 50+ FC launches including Amazon.in; FC Learning & Development Manager — Kaizen / Lean / CI training, learning curves and skill matrices, delivered an S-Team goal; Dell training background; SP Jain PGPM. **These are her numbers from her profile, quoted for framing only — they are not canon and never enter my stories.**

What that means for the answers. Three adjustments. First, lead with the *mechanism left behind* and who owns it now — she built teams and mechanisms herself (a placement strategy, a crossdock network, an L&D program), so "what still runs without you?" is her native question. Second, *price the trade-off* the way she priced distance-to-customer against speed and cost — S11's boarding pace vs automation maturity and S12's operators' two further quarters are exactly her shape of decision. Third, expect *transferability probes* — "how would this apply to a placement / network problem?" — and expect her to spot borrowed Leadership-Principle language: answer in the story's own words (gates, denominators, reversal triggers), never in recited LP slogans.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title, the JD and the public profile above (public-profile inference); if the first question lands on a different LP — Think Big / Frugality for S12, Ownership / Earn Trust for the critique-and-private-first threads — the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Deliver Results	The HM carries the outcome LP almost by default — she needs evidence the candidate lands public targets with the organization intact. S11's two public targets two months early with zero rollbacks is the cleanest record in the set.	S11 (open) · backup: S18 in the master packet — only if S11 is already spent
Hire & Develop / Earth's Best Employer (one of the two)	Her profile is teams built and grown: a placement team built ground-up in Luxembourg, a team of six plus interns running the EU crossdock network, and an S-Team goal delivered as an FC L&D manager on Kaizen / Lean / CI training and skill matrices (public-profile inference). She will probe who grew on my programs: S11's input owners across nine teams signing the definitions they would be red against, wave teams grown into owners; S12's delivery teams now running the burn-up, the swap rule and the re-estimation habit.	S11 (owners own their lines) · S12 (mechanisms the teams now run)
Dive Deep	She ran a crossdock network hands-on and defined placement strategy at node level (public-profile inference) — expect her to test whether my numbers survive decomposition: CTS $0.66 → $0.03 cross-walked to ≈3.7 touches at $0.18 each; S12's cost translated at the work-item grain (339.5 minutes; 2,200 interactions per departed shipment). The Ownership / Earn Trust probes she may still carry are covered in the twist maps — the critique accepted (S11) and the private-first flip (S12).	S11 (per-container cross-walk) · S12 (work-item grain)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For the HM specifically:** name the mechanism in Amazon's own vocabulary where it is honest — a weekly input/output review with frozen metric definitions, graduation criteria published before the wave, a reversal trigger written before the flip. Do not claim to have invented the WBR; say it was copied from a trusted review, which is the point.

- **Answer her native question — what did you leave behind, and who owns it now?** Every story lands on a mechanism that still runs without me (the Monday review and the graduation gates; the rebuild-vs-reuse rule and the scope tripwire) with a named owner. She built teams and mechanisms ground-up (public-profile inference); that is the evidence she pattern-matches.
- **Price the trade-off in her units.** She priced distance-to-customer against speed and cost for a crossdock network (public-profile inference); my versions are boarding pace vs automation maturity (S11) and partners-first vs operators' two further quarters, priced and published (S12). Say the trade and its price, not just the win.
- **Two warnings (public-profile inferences).** She tests transferability — "how would this apply to a placement / network problem?" — so practice the bridges in the twist maps before the call. And she spots borrowed LP language: never recite slogans ("I was customer-obsessed"); say the gate, the denominator, the reversal trigger in the story's own words.
- **Have the openers ready** — the 75-second "tell me about yourself" and the 60-second "why Amazon" from the master packet (newest to oldest; Sephora → Flexport → Elementum → Manhattan). Say the long version once, here.

Loop-consistency rule. One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S1/S3 (Lan) · S2/S18 (Goforth) · S6/S8 (Hoerger) · S15/S14 (Josyula). **v2 overlap flags — confirm before the loop.** This packet deliberately tells five Flexport records that another packet also touches: **S16-A** (Snowflake data model; Josyula's primary is S16, told as Ownership and the security-review miss — here lead with grain, join and reconciliation; the S16-A failure card is that same two-week miss, so if the panel notes show Josyula tells it, give it one sentence and pivot to the data model); **S11-C** (the Crux customer transition — the loop packet's S10, "79 customers, a destination and a date each"; no other panel carries it as a primary, but Goforth's pilot bench touches the Crux shadow accounts); **S12-A** (Asia latency; Hoerger's doc bench); **S11-B** (pilot rollout schedule and testing approach; Goforth's doc bench); the audit-logs PRD and the schedules solution are Lan's bench and are not used here. If the panel notes show a clash, drop the flagged story to a one-line pointer and use its router backup. If pushed for "another example" on an LP, use the named backup in the router, the question bank or the backup bench, never a story allocated to another interviewer.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

The stories — business-first, canon numbers

Seven Flexport primaries: two flagships (S11, S12 — deepened, with the original pages kept) and five new (S11-A, S11-B, S11-C, S12-A, S16-A). Each: 60-second pitch → SAFE-SAY box → Situation/Task → Action in gated phases → Results → L6 evidence → five labeled facets (data analysis · solution design · math & data logic · implementation · probe questions) → four-table math → sensitivity → statements → 20-second version and NACF bridge → behavioral pack → three angles → three trap drills.

S11 — Flexport — Shipment Visibility launch: a boarding process behind published graduation gates, run from a weekly input/output review (cost-to-serve $0.66 → $0.03 across 7 → 68 customers) DEEPENED

LPs this story can serve here: Deliver Results (primary) · Ownership (accepted the critique, owned the measurement) · Customer Obsession (graduation criteria per wave, CSAT in the gate) · Earn Trust (the discarded 99.4% week) · Hire & Develop (backup — wave teams grown into owners)

Process family: Deliver — customer onboarding / service-delivery process; weekly operating cadence; launch readiness and staged rollout

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “Seven to sixty-eight customers — nine point seven times — and a hundred and seventy monthly users, both landed two months early; thirty-two thousand shipments was the target I managed toward, not a result I claim.”

“What's your source for that?” “The targets and the roll-out record are in the program's own documents and my interview workbook; nine point seven is my division, sixty-eight over seven, and the cost-to-serve arc of about sixty-six cents to about three cents is the same record, rounded to about ninety-five percent.”

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Deliver Results Both public targets — 68 customers and 170 monthly active users — landed two months early with zero rollbacks (the 32,000-shipment figure was a target I managed toward, not a result), because boarding went through six published graduation criteria rather than around them.

Ownership I accepted the critique (#3 to #14 of 16) as being about evidence, not effort, and owned the measurement system myself — by hand every Monday for months — before it was handed to a pipeline.

Customer Obsession CSAT at or above the consignee average was inside the gate, every client entered on a ~50-container calibration of its own freight, and 100 real containers ran seven checks before the first paying customer.

Earn Trust A flattering 99.4% week resting on three touches was printed as a data miss, and wave 3 was held in public on an ETA-accuracy miss — trust came from numbers that survived the next Monday.

Honesty label. Program owner across nine teams / ~100 people; the WBR format was copied from an existing leadership review (the LSA review), not invented; build-week counts are illustrative; the ~$1.5M pre-production measurement saving is a separate user-anchor figure (my own estimate, not a program-document number), never merged with cost-to-serve. Nine teams built and operated; I owned the boarding process, the measurement system and the pre-prod proof.

WHAT IS SHIPMENT VISIBILITY? — THE ONE-MINUTE EXPLAINER (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The minute. Shipment Visibility — "SV" — is Flexport's client-facing visibility product in the Client App: one pane where a client watches container milestones across their whole book, Flexport-carried and non-Flexport-carried shipments alike, so track-and-trace stops being emails to an ops desk and becomes a product the client logs into. It also created a "visibility-only" client segment — companies that bought visibility first were a deliberate upsell path into

Flexport freight. The 2021 plan priced the problem it solved: roughly $3.92M a year of manual track-and-trace salary across the ops org against about $263K of vendor spend, with only 18 of 99 milestones automated at the time. RECORD-FROM-DOCS

The machinery behind the story. Product standards: coverage 90%+, accuracy 95%, timeliness 90%, a fail-safe switch, dashboards and alarms — under the operating rule "edits are the enemy": every human correction is an automation defect, not a fix. Rollout: a free pilot in graduated waves — wave 1 two named clients (the first two hand-carried customers) around Q4 2021, wave 2 four of a seven-client pool around Q1 2022, wave 3 twenty-plus around Q2 2022, full roll-out H2 2022 — each client entering on a ~50-container calibration and graduating on six published criteria. Testing: four gates before launch — a 100-container live test, biweekly bug bashes on the client job stories, shadow accounts fed real Crux data, and UAT gated on data quality (80% of shipments minor-to-no issues). And Crux — the acquired visibility product — was transitioned onto SV by end 2022: 79 customers, $185,892 printed ARR (six of nine segments — S11-C re-foots it), 294,581 containers. RECORD-FROM-DOCS

Doc-vs-canon footnotes: the pilot schedule phrases the SLA criterion as "90% of verifications within 24–48h"; canon says "reviewed within 24h" — canon wins, say 24h. The 95% accuracy standard is the 2021 plan's bar for milestone data, not this story's ETA-accuracy metric (60% → 90%, canon) — never mix them. The $3.92M / $263K figures are the plan's ops-org business case, a different basis from canon's per-container CTS — never merged. Wave timing and the six criteria match the master packet.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport's shipment-visibility product had public commitments — 68 monthly active customers, 170 MAU, 32K shipments — and was serving seven pilots at a cost-to-serve of $0.66 a container, most of it human touches. Sales wanted the entire pool boarded at once; leadership said the program was not demonstrating value and it was about to drop from #3 to #14 of 16 initiatives. The process problem was that boarding ran faster than the automation could carry, and nobody could see it. **Task.** As program owner across nine teams, I owned the boarding path, the measurement system and the pre-production proof. **Action.** I rebuilt boarding as a gated process: six graduation criteria published to Sales before wave one, a launch gate of 100 real containers across seven checks, and a weekly input/output review — seven frozen metric definitions, each signed by the owner who would be red against it, every input tagged to the output it moves — copied from a review leadership already trusted and run by hand every Monday for months. Waves boarded only through the criteria; wave three was held in public on an ETA-accuracy miss and became the cleanest wave. When a flattering 99.4% week turned out to rest on three touches, I printed a data miss instead. **Result.** Primary metric: cost-to-serve per container $0.66 → $0.03, about 95%, across customers 7 → 68 — 9.7× — with zero rollbacks and no added ops headcount; automation 40% → 90%, accuracy 60% → 90%, both public targets two months early. The ~$1.5M pre-production measurement saving is a separate user-anchor figure — my own estimate — and I keep it separate.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Senior leadership told me the visibility program was not demonstrating value at the anticipated rate — it would drop from #3 to #14 of 16 initiatives. I had been reporting activity, not evidence. The critique was about evidence, not effort, and I accepted that first.

The business process underneath: customer boarding (Deliver) ran as a sales push — board the whole pool now — against an operation where most containers were still touched by hand. Cost-to-serve was $0.66 a container (≈3.7 human interactions at the record cross-walk of $0.18 per interaction); automation covered 40% of containers; ETA accuracy was 60%. The public commitments were 68 monthly active customers, 170 MAU and 32K shipments, off seven pilots.

The cost of leaving it: a bad first impression at scale with rollbacks, ops headcount growing with the customer count, and the program losing its funding slot because value was invisible.

T — Program owner across nine teams / ~100 people: own the boarding path, the measurement system and the pre-production proof — and make value legible to the leadership that decides rollout pace and funding, with zero

rollbacks and no added ops headcount.

Baselines on the table: customers 7 (target 68) · cost-to-serve $0.66/container · automation 40% · ETA accuracy 60% · public targets 68 customers / 170 MAU / 32K shipments · no instrument leadership trusted.

KEY CHALLENGES — WHAT THE PROGRAM RECORD ADDS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

- **Rollout sequencing under a free pilot.** Pilot clients pay nothing, so every wave spent engineering, data and ops capacity before revenue; the pool had to be sequenced 2 → 4-of-7 → 20+ against a ~50-container calibration per client and six graduation criteria, with GO/NO-GO decisions taken monthly. That is the doc-level version of "board no faster than the automation can carry." RECORD-FROM-DOCS
- **Testing gates before any launch.** Four gates stood between build and client: a 100-container live test watched end-to-end, biweekly bug bashes on the client job stories, shadow accounts (real Crux data in dedicated demo accounts), and UAT that did not start until 80% of shipments showed minor-to-no issues. The gates are why "zero rollbacks" was achievable. RECORD-FROM-DOCS
- **An acquired product's customers in the middle of the rollout.** Crux's 79 customers — $185,892 printed ARR (six of nine segments — see S11-C), 294,581 containers — had to be transitioned onto SV by end 2022, including cancel-vs-transition decisions by segment and per-customer calls for the largest accounts, while the pilot waves were still boarding. Migration risk ran in parallel with launch risk. RECORD-FROM-DOCS
- **Data quality as a product standard, not an ops task.** The 2021 plan set coverage 90%+ / accuracy 95% / timeliness 90% with a fail-safe switch, and named the enemy: human edits. "Edits are the enemy" is the sentence behind the story's automation 40% → 90%. RECORD-FROM-DOCS

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — wrote the honest diagnosis (activity, not evidence); inventoried what leadership actually decides on (rollout pace, funding); chose to copy the LSA review's format rather than invent one. *Exit bar:* leadership agrees the questions the review must answer weekly; ~50-container calibration entry bar per client.
- **Plan & Design** — froze seven metric definitions (coverage auto, coverage with human, tickets, tickets per active container, 4PL SLA %, 3PL SLA %, delta); input→output tagging; the ±5% week-over-week red/green rule; the 15-day active rule as the honest denominator; six graduation criteria published to Sales before wave 1 — DQ ≥ 90%, ETA ≥ 80%, SLA 90% reviewed within 24h, ≥100 containers + 5 features, stability, CSAT ≥ consignee average; launch gate 100 real containers × 7 checks; pre-prod sample size 300 containers for ±1.6% at 95% confidence. *Exit bar:* each input owner signs the definition they will be red against; first Monday deck dry-run.
- **Build & Test** — ran the review by hand every Monday (1.5–2 hrs, later ~4); pre-decomposed every red with an owner and a dated fix; enforced the data-honesty rule — a truncated week is a data miss, not a result; pre-prod ingest with per-hop monitors and kill switches; load proof 500 containers / 20 min / 99.6%. *Exit bar:* two consecutive Mondays where every red arrives with an owner and the number verifies the following week.
- **Cutover & Go-Live (waves)** — W1 (2 clients) → W2 (4 of a 7-pool) → W3 (20+) held in public on an ETA-accuracy miss traced to one bad source → full rollout. The Monday review became the forum where the rollout decision was taken. *Exit bar:* the VP unlocks the 68-client rollout two months early off the review's own numbers.
- **Hypercare / Close** — discarded a flattering 99.4% three-touch week and printed a data miss; handed the manual pull to a pipeline with the definitions in code (S16-A in this packet tells the data model; the full S16 telling belongs to Josyula's packet); left the critic steering by the numbers. *Exit bar:* automation 40 → 90%; CTS $0.66 → $0.03; same ops team at 68 customers.

Influence and ownership. Mine: the diagnosis, the frozen definitions, the review built by hand weekly, the graduation criteria, the hold decisions and the pre-prod measurement design. Nine teams built and operated; each input owner signed the definition they would be red against; the VP took the rollout decisions in the review. The Sales push to board everything now was answered with gates published before wave 1 — so when wave 3 was held, it was the rule firing, not a judgment call; the held customers became the cleanest wave.

> **Designed the process; others own it now.** I designed the review, the definitions and the graduation gates; the input owners across the nine teams run their lines and the pipeline (S16-A) carries the definitions in code — the review steers without me. **Trade-off put to leadership.** My recommendation to leadership outlined the competing metrics explicitly — boarding pace and customer count vs automation coverage and ETA accuracy — and proposed to board no faster than the automation could carry; wave 3 was the trade made in public.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — cost-to-serve per container $0.66 → $0.03 (record; about 95%); direct CTS dollars ≈$50K/yr *modeled* on an illustrative container count — the value is scale without headcount, not the direct dollars; the ~$1.5M pre-production measurement saving is a separate user-anchor figure, never merged with CTS.
- **#** — customers 7 → 68 (9.7×) with zero rollbacks and no added ops headcount; targets 68 customers / 170 MAU / 32K shipments (record).
- **%** — CTS about 95%; automation 40% → 90%; ETA accuracy 60% → 90% (record).
- **Time** — both public targets hit two months early; the rollout unlocked two months early off the review (record); pre-prod measurement path cut rollout proof from 6 months to 1.5.
- **Customer** — six graduation criteria published to Sales before wave 1 — including CSAT ≥ consignee average; wave 3 held in public on an ETA-accuracy miss and became the cleanest wave (record).
- **Quality** — a flattering truncated 99.4% week discarded and printed as a data miss; seven metric definitions frozen and later moved into code (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Leadership said the program was not demonstrating value (#3 → #14 of 16) and no measurement system existed. I wrote the diagnosis (activity, not evidence), chose the nine teams' input owners as the stakeholders, and set the goals the review had to answer weekly.
Scope & influence	Nine teams built and operated; Sales held to criteria I published; the VP took the rollout decisions inside the review I ran — influence one to two levels up, from a program seat.
Execution trade-off + mechanism others now own	Boarding speed (Sales) vs automation maturity and ETA accuracy: wave 3 held in public on the miss. Mechanism others own now: the Monday input/output review and the six graduation criteria, with the seven definitions moved into code (S16-A).
Impact (long-term)	Customers 7 → 68 (9.7×) at CTS $0.66 → $0.03 with zero rollbacks and no added ops headcount; both public targets two months early; the review became the standing rollout forum.
Process improvement (standard work institutionalized)	Seven frozen metric definitions, input→output tagging, the ±5% week-over-week red/green rule, the 15-day active denominator, and the data-honesty rule (a truncated week is a data miss, not a result).
Complexity & consensus	A critical leadership and a Sales push to board everything now — answered by gates published before wave 1, each input owner signing the definition they would be red against; the VP unlocked the 68-client rollout off the review's own numbers.
Root cause at scale	The ETA-accuracy miss traced to one bad source and fixed there; the flattering 99.4% week traced to three touches and printed as a data miss — fixed at the definition and the source, not customer by customer.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **HM — Why is this your Deliver Results story and not just a metrics story?**
>
> Because the results landed behind gates, not around them: both public targets two months early, 7 → 68 customers on the same ops team, zero rollbacks. The review and the graduation criteria are the mechanism; the about-95% cut and the 9.7× are the outcome.

HM — The direct dollars on a 95% cost reduction are small. Why lead with it?

Because it is honest and because the point was never the direct dollars: at the 32K-shipment target the direct saving is ≈$50K a year, modeled on an illustrative container count. What it bought is boarding 9.7× the customers without 9.7× the ops headcount — roughly two and a half people never hired (modeled — assumptions in the deepening's dollars table; not record headcount) — and a rollout unlocked two months early. The ~$1.5M pre-prod estimate is a separate user anchor and I never add it to this.

HM — Sales wanted everyone boarded. How did you hold the line without becoming the blocker?

By publishing the six criteria to Sales before wave one, so the rule did the refusing. Wave 3 was held in public on an ETA-accuracy miss traced to one bad source; the customers we held became the cleanest wave, and Sales had seen the gate before it fired.

HM — You discarded a 99.4% week. Walk me through that decision.

Three touches is not a week — a 95% confidence interval on N = 3 is wider than the metric. Printing it would have taught the VP a number that could not survive the next Monday. I printed a data miss. A review that flatters once is not trusted twice.

HM — What would you do differently, and how does this translate to my team?

Start the review at kickoff, not after the critique. At Amazon this is a WBR story told to the company that invented the WBR — the respectful version is that I copied a trusted format, froze definitions with their owners, tagged inputs to outputs and let the review make the rollout decision. Graduation criteria are launch gates published before the wave and held in public.

HM — What was yours and what was the team's?

Mine: the diagnosis, the seven definitions, the Monday build by hand for months, the criteria, the holds, the pre-prod measurement design. The nine teams built and operated; each input owner signed their own definition; the VP decided in the room. The format came from the LSA review — I did not invent it.

HM — Why waves at all? What did the calibration actually buy?

The pilot was free, so every boarded client cost engineering, data and ops before revenue — boarding the whole pool would have spent the program's credibility on clients we could not yet serve. The ~50-container calibration per client gave the data team a verified baseline on the client's own freight before scale, and the six published criteria converted "ready" from a feeling into a rule: wave 1 two clients, wave 2 four of a seven-client pool, wave 3 twenty-plus RECORD-FROM-DOCS. The canon version: waves boarded only through the published gates, and the held wave became the cleanest.

HM — How did you test before the first paying customer touched it?

Four gates, all from the program record RECORD-FROM-DOCS: a 100-container live test watched end-to-end; biweekly bug bashes on the client job stories; shadow client accounts — real Crux data fed into dedicated demo accounts so the clients' own data exposed the defects; and UAT that did not start until 80% of shipments showed minor-to-no issues. UAT was gated on data quality, not on the calendar — the testing version of the graduation rule.

HM — You were rolling out while an acquired product's customers sat on the old platform. How did you handle that risk?

As a parallel migration with its own decision rules, not an afterthought RECORD-FROM-DOCS: 79 Crux customers, $185,892 printed ARR (six of nine segments — see S11-C) and 294,581 containers, segmented — transition where SV fit, cancel honestly where it did not (two competitor SCM-SaaS accounts were canceled on published dates), per-customer calls for the largest accounts. Internal pushback on migration cost and feature parity was priced into the decision document, not overridden. The pilot gates kept quality honest while the migration ran beside it.

THE KEY MATH — COST-TO-SERVE, THE SCALE RESULT, AND WHY THE DOLLARS STAY SEPARATE

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve before	`$0.66 / $0.18 ≈ 3.7 touches per container`	record; $0.18 per interaction is the record cross-walk	$0.66 — ≈3.7 touches
2	Cost-to-serve after	`($0.03 − $0.66) / $0.66 = −95.45%, say "about 95%"; $0.03 / $0.18 = 0.17 touches`	record	$0.03 — ≈0.17 touches
3	Scale	`68 / 7 = 9.7×; zero rollbacks; no added ops headcount`	record	9.7× (never "10×")
4	Direct CTS dollars	`32,000 shipments × 1.5–4 containers = 48–128K; × $0.63 = $30–81K`	modeled (containers per shipment illustrative)	≈$50K/yr modeled
5	Pre-prod measurement path	`n = 300 → ±1.6% at 95% CI; rollout proof 1.5 vs 6 months`	user anchor — separate canonical	~$1.5M, never merged with CTS
6	Week-over-week rule	`WoW = (B − A)/A = (21.9 − 40.0)/40.0 = −45% → red`	±5% red/green (record design; values are the 17 Jan and 31 Jan 2022 prints)	root cause: a carrier format change — measurement, not operators

Automation 40 → 90% and the WBR hours (1.5–2 → ~4 by hand) are this packet's canon; the loop packet carries different values (§1.3 conflict) — do not mix them in the room.

Where this lands at Amazon. This is the JD's "define, create, implement and execute processes" line in its purest form: a boarding process with published graduation criteria, a weekly input/output review with frozen definitions, and the discipline to hold a wave in public. Reliability (zero rollbacks), responsiveness (two months early), cost efficiency (about 95% CTS), asset utilization (the same ops team at 9.7× the customers).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S11

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you delivered results under pressure / against a target.	Deliver Results	Open on the critique (dropping #3 → #14) and the public targets; close on both targets two months early, 7 → 68, zero rollbacks. The gates are the how.	Skip the four-system data-pull detail unless asked for the data model — then S16-A carries it.
Tell me about a time you had to make value legible to senior leadership.	Ownership / Earn Trust	Accepting that the critique was about evidence, not effort; copying a trusted format; signing each definition with its owner; the review becoming the decision forum.	Never say "I invented the WBR."
Tell me about a time you said no to a stakeholder (Sales).	Have Backbone / Highest Standards	Six criteria published before wave 1; wave 3 held in public; held customers became the cleanest wave.	Do not call Sales wrong — they were right to want speed; the gate did the refusing.
Tell me about a time you chose the harder, honest number.	Earn Trust	The 99.4% three-touch week printed as a data miss; the 15-day active rule as the honest denominator.	Skip the Wilson-interval math unless asked; say "three touches is not a week."

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
How did you decide the customer was ready to launch? (Customer Obsession)	Customer Obsession	CSAT ≥ consignee average inside the gate; ~50-container calibration entry bar; 100 real containers × 7 checks before the first paying customer.	Do not claim customer-satisfaction dollars; none exist in canon.
How would you run a WBR for my team?	Functional — operating cadence	Copy a trusted format, freeze 5–7 definitions with owners and honest denominators, tag inputs to outputs, pre-decompose reds, let the review take the decision.	Never "$1.00 → $0.03"; never "70% baseline"; never "10×".
How would your wave machine apply to my world — placement / inbound? (transferability probe — public-profile inference)	Functional — transfer	The wave machine is graduated onboarding behind published criteria — the same shape as onboarding inventory into an IXD / crossdock program: a selection bar published before the first node, a calibration volume per node (~50 containers per client ≈ a trial volume per node), and a public hold when a node misses. Scale follows the gate, never the ask.	Do not lecture on her network; bridge in one sentence and hand it back with a question.
Your CTS is per container — mine is per node. Does the logic survive? (transferability probe)	Functional — transfer	Identical logic, different denominator: cost-to-serve per node = touches per unit through the node × cost per touch; the fix is still automation coverage and source quality, and the honest denominator is still defined before the metric. $0.66 → $0.03 is the container-grain instance of it.	Never quote her profile's $50MM / $80MM as if it were mine — those are her numbers, quoted for framing only.

DEEPENING S11 — FIVE FACETS, FOUR-TABLE MATH, BEHAVIORAL PACK, ANGLES AND TRAP DRILLS NEW

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Cost-to-serve decomposed into touches (3.67 → 0.17); the 15-day active denominator; the three-touch week; wave-3 accuracy traced to one source; the pre-production evidence (500 containers in 20 minutes at 99.6%; dry run 100% processed, 86% tracked) and my own n = 300 sample design (user anchor).
2 · Solution design	Board-all vs gated waves; copy a trusted format vs invent; criteria published before wave 1; sample vs census; headcount vs automation as the scale lever.
3 · Math & data logic	Four tables: $0.66 / $0.18 = 3.67; about 95%; 9.7×; $30,240–$80,640; 80,000 × 3.5 min ≈ 2.6 FTE; the pre-production sample design n = 300 → ±1.6% (user anchor).
4 · Implementation	Waves 2 → 4-of-7 → 20+ with monthly GO / NO-GO; the Monday review; input owners across the nine teams; kill switches; the hand-off to the pipeline.
5 · Probe questions	Seven numeric probes plus three trap drills: direct dollars, "10×" and the FTE claim, and the separate ~$1.5M.

Facet 1 — Data analysis: what each analysis decided

ANALYSIS	DATA AND GRAIN	WHAT IT SHOWED	DECISION IT DROVE
Cost-to-serve decomposition	dollars per container ÷ $0.18 per interaction (record cross-walk)	$0.66 is ≈3.67 human touches; $0.03 is ≈0.17	automation coverage, not headcount, is the scale lever

ANALYSIS	DATA AND GRAIN	WHAT IT SHOWED	DECISION IT DROVE
Active denominator	active = cumulative inflow less containers 15 days past departed terminal, by week	cumulative counts hide a slowing program	review runs on active, not cumulative
Week-over-week instrument	(B − A) / A against ±5%	(21.9 − 40.0) / 40.0 = −45% → red; root cause a carrier format change	reds are decomposed before the meeting, measurement vs operations
Three-touch week	99.4% on N = 3 touches	a 95% interval on N = 3 is wider than the metric	printed as a data miss, not a result
Wave-3 accuracy by source	ETA accuracy per data source	the miss traced to one bad source	held in public; the held customers became the cleanest wave
Pre-production sampling	n = 300 at ≈98% expected accuracy (my sample design)	±1.6% at 95% confidence	a 1.5-month proof instead of 6 — my estimate; the separate ~$1.5M pre-production figure is a user anchor, not a program-document number

Facet 2 — Solution design: options weighed

QUESTION	OPTIONS	CHOSEN	WHY
Boarding pace	Board the whole pool now (Sales) · waves behind published criteria	Waves behind six criteria, published before wave 1	A free pilot spends engineering, data and ops before revenue; the rule does the refusing.
Review format	Invent a scorecard · copy the leadership review leadership already trusted	Copy the LSA format; freeze seven metrics with owners	Inventing would have spent credibility the program did not have.
Lever for scale	Add ops headcount · raise automation coverage	Automation 40% → 90%	Cost-to-serve is a touches problem: $0.66 / $0.18 = 3.67 touches.
Pre-production proof	Census over months · calibrated sample	Sample of 300 containers, ±1.6% at 95%	Big enough to certify the gate, small enough to afford — my estimate: 1.5 months instead of 6 (user anchor).
Bad week	Print 99.4% · print a data miss	Data miss	A number that cannot survive the next Monday teaches the VP the wrong thing.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve	`$0.66 per container off seven pilots`	record	$0.66
2	Residual human touches	`$0.66 / $0.18 = 3.67 touches per container`	record cross-walk ($0.18 per eliminated interaction)	≈ 3.7 touches
3	Automation and accuracy	`coverage 40%; ETA accuracy 60%`	record	most containers still touched by hand

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Customers vs target	`7 pilots against a target of 68; the program was to drop from #3 to #14 of 16`	record	value invisible to leadership

Table 2 — The fix: gates, honest denominators and the instrument, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Launch gate volume	`100 real containers × 7 checks = 700 check results before the first paying customer`	record design; product mine	700 checks
2	Load proof	`500 containers in 20 min = 25 a minute; 498 / 500 = 99.6%`	record (500 / 20 min / 99.6%); arithmetic mine	25 / min · 99.6%
3	Honest denominator	`a 99.4% week on N = 3: Wilson lower bound 3 / (3 + 1.96²) = 0.44`	record decision (99.4% on three touches, Apple A2); the Wilson interval is my arithmetic	0.44 → data miss
4	Week-over-week rule	`(21.9 − 40.0) / 40.0 = −45% → beyond ±5% → red`	record: the weekly prints were 40.0% on 17 Jan and 21.9% on 31 Jan 2022 (Apple A2)	−45% → red
5	Pre-production sample	`1.96 × √(0.98 × 0.02 / 300) = 0.0158 = ±1.6 points; rollout proof 6 − 1.5 = 4.5 months saved`	user anchor (my sample design; GFiber v8 only — the program documents show 500 containers in 20 minutes at 99.6%, and an 86%-tracked dry run)	±1.6% · 4.5 months

Table 3 — After: before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve	`1 − $0.03 / $0.66 = 0.9545 → about 95%; $0.03 / $0.18 = 0.17 touches`	record endpoints; ratio mine	about 95% · 0.17 touches
2	Touches removed	`3.67 − 0.17 = 3.5 per container — about 95% of the touches`	record cross-walk; arithmetic mine	3.5 per container
3	Automation / accuracy	`40% → 90% = +50 points; 60% → 90% = +30 points`	record	+50 / +30 points
4	Customers	`68 / 7 = 9.71; zero rollbacks; no added ops headcount; 68 customers and 170 monthly users two months early`	record	9.7× (never rounded up)

Table 4 — The dollars: every multiplication shown (record vs modeled)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Containers at the 32,000-shipment target	`32,000 × 1.5 = 48,000; 32,000 × 2.5 = 80,000; 32,000 × 4 = 128,000`	containers per shipment illustrative (1.5–4)	48K / 80K / 128K
2	Unit saving	`$0.66 − $0.03 = $0.63`	record endpoints	$0.63
3	Direct CTS dollars	`48,000 × $0.63 = $30,240; 80,000 × $0.63 = $50,400; 128,000 × $0.63 = $80,640`	modeled	$30K · $50K · $81K
4	Headcount never hired	`80,000 × 3.5 touches = 280,000 touches; × 1 min = 4,667 h; ÷ 1,800 h per FTE = 2.6 FTE (÷ 1,920 → 2.4)`	modeled: one minute per touch and 1,800 hours per FTE-year are assumptions	≈ 2.5 people (2.4–2.6)
5	The separate user anchor	`~$1.5M pre-production measurement saving: n = 300 → 1.5 vs 6 months`	user anchor (GFiber v8 only) — a separate figure, never added to CTS or the model	~$1.5M, never merged

Facet 4 — Implementation: the wave machine and its controls

WAVE	CLIENTS (RECORD)	ENTRY	GRADUATION / DECISION	TIMING (RECORD)
1	2 named pilots	~50-container calibration	six criteria; monthly GO / NO-GO	around Q4 2021
2	4 of a 7-client pool	~50-container calibration	six criteria; monthly GO / NO-GO	around Q1 2022
3	20+	~50-container calibration	held in public on an ETA-accuracy miss; held customers became the cleanest wave	around Q2 2022
Full	the pool to 68	graduated clients	rollout unlocked two months early off the review	H2 2022

Controls. Nine input owners each sign the definition they will be red against; reds are pre-decomposed with an owner and a dated fix; per-hop monitors and kill switches on the pre-production ingest; the manual pull is the review's source until the pipeline matches within 1% (S16-A). **Lean / Kaizen mapping.** The six criteria are standard work for onboarding; the hold on wave 3 is jidoka — stop the line when a quality criterion fails; the Monday review is the PDCA Check; "edits are the enemy" is the 2021 plan's own waste statement — every human correction is a defect, not a fix.

Facet 5 — Probe questions (numeric)

> **Why is the direct dollar figure so small?**
>
> Because the unit economics is the point: $0.63 saved on 48,000–128,000 containers is $30,240–$80,640, modeled on an illustrative containers-per-shipment. What it bought was 68 customers on the same ops team — about 2.5 people never hired, modeled — and a rollout unlocked two months early.

> **Prove the 2.5 people.**
>
> It is a model with two assumptions I name: 80,000 containers × 3.5 touches removed × one minute = 4,667 hours; at 1,800 hours a year that is 2.6 FTE, at 1,920 it is 2.4. I would replace the minute with the ticket-grain automation data.

> **Why ±1.6%?**
>
> That is my sample design, a user anchor — the program's own evidence was 500 containers in 20 minutes at 99.6% and a dry run at 100% processed, 86% tracked. n = 300 at about 98% accuracy: 1.96 × √(0.98 × 0.02 / 300) = 0.0158. Big enough to certify the gate, small enough to afford — it cut the rollout proof from 6 months to 1.5, which is the separate ~$1.5M pre-production estimate (user anchor), never merged with cost-to-serve.

> **Is it 10× or 9.7×?**
>
> 68 / 7 = 9.71. Say 7 to 68, or 9.7 times; no other multiple belongs in the room.

> **What does −45% week-over-week mean?**
>
> (21.9 − 40.0) / 40.0 = −45%, red against ±5%. The weekly prints were 40.0% on 17 January and 21.9% on 31 January 2022 (record); the root cause was a carrier format change — measurement, not operators.

> **How do you know a wave is ready?**
>
> Six criteria all pass — data quality at least 90%, ETA accuracy at least 80%, SLA 90% reviewed within 24 hours, at least 100 containers and 5 features, stability, CSAT at or above the consignee average — after a ~50-container calibration.

> **What would you change?**
>
> Start the review at kickoff instead of after the critique, and price the pipeline hand-off earlier — the pull stayed with me for months.

SENSITIVITY — CONTAINERS PER SHIPMENT (THE SOFT INPUT)

CONTAINERS PER SHIPMENT	CONTAINERS A YEAR	DIRECT CTS SAVING (× $0.63)	FTE-EQUIVALENT (ILLUSTRATIVE)
1.5 (low)	48,000	$30,240	1.6
2.5 (base)	80,000	$50,400	2.6
4.0 (high)	128,000	$80,640	4.1

Both columns are modeled on an illustrative containers-per-shipment; the record gives shipments, not containers. The ~$1.5M pre-production figure is excluded from every row.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> Honestly, the cost-to-serve number is a unit-economics result, not a big-dollar number: ninety-five percent off per container is measured, and at the thirty-two-thousand-shipment target it is only about fifty thousand dollars a year of direct cost — my model, on an illustrative container count. What it bought is scale: sixty-eight customers on the same ops team with zero rollbacks, about two and a half people we never had to hire — modeled — and a rollout unlocked two months early. The million and a half on the pre-production measurement path is a separate figure — my own estimate — and I keep it separate. So: the ninety-five percent and the seven-to-sixty-eight are record; every dollar except that one-and-a-half is my model.

IF CHALLENGED — THE FULL ANSWERS

"Your containers-per-shipment is a guess."

It is, and I said so: 32,000 shipments is a target, not a count, and it is shipments, not containers. At 1.5 to 4 containers per shipment the direct saving runs $30,240 to $80,640. I would pull the active-container count from the 15-day active rule the review already uses.

"Fifty thousand a year is small — why lead with it?"

Because it is honest, and the point was never the direct dollars: it was boarding 9.7 times the customers without 9.7 times the ops headcount.

"Where does the million and a half come from, then?"

The pre-production measurement path: 300 containers at ±1.6% proved the product in 1.5 months instead of 6. It is my estimate (a user anchor, not a program-document number), it is not cost-to-serve, and I never add it to the fifty thousand.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I owned the boarding process for Shipment Visibility: six published graduation criteria and a Monday input-output review took cost-to-serve from $0.66 to $0.03 per container while customers went from 7 to 68, with zero rollbacks and 68 customers and 170 monthly users landed two months early.

NACF BRIDGE — ONE SENTENCE

It is the shape of a network launch: onboard each node behind criteria published before the first one, calibrate on a trial volume, and hold a node in public when it misses — scale follows the gate, never the ask.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a time when you not only met a goal but considerably exceeded expectations. How were you able to do it? What challenges did you have to overcome?" AIQB: Deliver Results Q3 Deliver Results FACET: IMPLEMENTATION

Open (first two sentences): "Both public targets landed two months early with zero rollbacks — and the way they landed is the point: boarding went through six published criteria, not around them."

90-second STAR. **S** Seven pilots against public targets of 68 customers and 170 monthly active users (plus a 32,000-shipment target); Sales wanted the whole pool boarded at once. **T** Hit the targets without a rollback or added ops headcount. **A** Waves — 2, then 4 of 7, then 20-plus — each entering on a ~50-container calibration and graduating on six criteria published to Sales before wave 1, with a launch gate of 100 real containers across seven checks and a Monday review taking the rollout decision. **R** Customers 7 → 68, cost-to-serve $0.66 → $0.03, automation 40% → 90%, accuracy 60% → 90%; zero rollbacks.

Land: 7 → 68 (9.7×); 68 customers and 170 users two months early · **Pivot to NACF:** launching nodes in a network: published gates, calibrated volumes, a public hold when a node misses.

Do NOT say: "10×" or any other multiple; the $1.5M added to cost-to-serve.

B2 · "Give me an example of a tough or critical piece of feedback you received. What was it and what did you do about it?" AIQB: Earn Trust Q2 Ownership FACET: DATA ANALYSIS

Open (first two sentences): "Senior leadership told me the program was not demonstrating value and would drop from third to fourteenth of sixteen. I had been reporting activity, not evidence — and I accepted that first."

90-second STAR. **S** A program about to lose its funding slot because value was invisible. **T** Make value legible without arguing the critique. **A** I inventoried what leadership actually decides on, copied a review format they already trusted, froze seven metrics with owners who signed the definitions they would be red against, and ran the pull by hand every Monday for months. **R** The same critic steered the rollout by the review's numbers, and unlocked the 68-client rollout two months early.

Land: #3 → #14 of 16 → the VP unlocked the rollout · **Pivot to NACF:** owning a metric nobody owned: build it by hand first, automate it once the definitions are signed.

Do NOT say: that I invented the weekly review — I copied the LSA format.

B3 · "Tell me about a time when you had to balance the needs of the customer with the needs of the business. What did you do? What was the result?" AIQB: Customer Obsession Q8 Customer Obsession FACET: SOLUTION DESIGN

Open (first two sentences): "Wave three was held in public because ETA accuracy missed on one bad source. Sales wanted speed; the customers we held became the cleanest wave."

90-second STAR. **S** A free pilot where every boarded client cost engineering, data and ops before revenue. **T** Protect each client's first impression at scale. **A** Six criteria published before wave 1 — including CSAT at or above the consignee average — a ~50-container calibration on each client's own freight, and a hold on wave 3 traced to one bad source. **R** Zero rollbacks; the held wave became the cleanest.

Land: zero rollbacks; wave 3 held · **Pivot to NACF:** inbound readiness: do not take the first container until the node's criteria pass.

Do NOT say: that Sales was wrong — they were right to want speed; the gate did the refusing.

B4 · "Tell me about a time when you uncovered a significant problem in your team. What was it and how did you communicate it to your manager and other stakeholders? What did you do to address the problem?"

AIQB: Earn Trust Q6 | Earn Trust | FACET: MATH AND DATA LOGIC

Open (first two sentences): "One week showed ninety-nine point four percent — on three touches. A 95% confidence interval on three is wider than the metric, so I printed a data miss instead."

90-second STAR. **S** A flattering week landing just before a funding decision. **T** Keep the review credible. **A** I computed the interval — Wilson lower bound 0.44 on N = 3 — and held the review to the 15-day active denominator; the week was printed as a data miss. **R** A number that survived the next Monday; the review was trusted twice.

Land: 99.4% on N = 3 → data miss · **Pivot to NACF:** any capacity report: a rate on a tiny denominator is not a result.

Do NOT say: the Wilson math unless asked — say "three touches is not a week".

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle — Dive Deep

Opener: "Cost-to-serve was sixty-six cents a container — which is three and two-thirds human touches. The program was a touches problem, not a headcount problem."

Headline metric: $0.66 / $0.18 = 3.67 → 0.17 touches

- Decompose dollars into touches with the record cross-walk.
- Active denominator: cumulative inflow less containers 15 days past departed terminal.
- WoW rule: −45% is red; root cause a carrier format change.
- Say "modeled" on every dollar.

Conflict / influence / backbone angle — Have Backbone; Disagree and Commit

Opener: "Sales wanted the entire pool boarded at once; I published the gates to Sales before wave one, so when wave three was held, it was the rule firing — not me."

Headline metric: six criteria; wave 3 held

- The rule did the refusing; no one was called wrong.
- The held customers became the cleanest wave.
- The trade: boarding speed for a rollout that never rolled back.
- Leadership took the decision in the Monday review.

Honest failure-and-learning angle — Learn and Be Curious

Opener: "I had been reporting activity, not evidence — leadership said so, and they were right. The review should have started at kickoff, not after the critique."

Headline metric: #3 → #14 of 16

- What I got wrong: measuring effort instead of outcomes.
- What I changed: seven frozen definitions, input → output tagging.
- What I would do differently: kickoff review; earlier pipeline hand-off.
- What held: zero rollbacks.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"Customers 7 to 68 — that's a 10× result?"

Numeric answer. 68 / 7 = 9.71. The only multiple to say is 9.7× — or just 7 to 68. Rounding up to 10× overstates a record number by 3%.

Say exactly: "Seven to sixty-eight — nine point seven times, and I won't round that up."

"You said two and a half people never hired. Show me."

Numeric answer. 80,000 containers × 3.5 touches removed × 1 minute = 4,667 hours; ÷ 1,800 h = 2.6 FTE (÷ 1,920 h = 2.4). Modeled: containers per shipment, one minute per touch and the hours per FTE are my assumptions; the record is zero added ops headcount at 68 customers.

Say exactly: “It's modeled: about two and a half people on an illustrative container count and a minute a touch — the record is that headcount didn't grow.”

"So the saving is $1.5M plus $50K?"

Numeric answer. No. The ~$1.5M is the pre-production measurement saving (n = 300 → ±1.6%, 1.5 vs 6 months) — my estimate and a user anchor, kept separate. The $50,400 is modeled direct cost-to-serve at 80,000 containers. They never add.

Say exactly: “They're separate. The million and a half is my estimate of the pre-production proof path, kept separate; the fifty thousand is my model of direct cost-to-serve.”

S11-A — Flexport — The Monday review as an analysis: taking a 22× cost-to-serve drop apart (6.0× coverage × 3.67× touch efficiency), with honest denominators, seven frozen metrics and one discarded week NEW

LPs this story can serve here: Dive Deep (primary) · Insist on the Highest Standards · Earn Trust · Are Right, A Lot (backup)
Process family: Plan / Deliver — operating cadence, measurement design and cost-to-serve analysis; weekly business review (WBR)

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Dive Deep I took $0.66 → $0.03 apart: 22.0× = 6.0× fewer containers needing a human (60% → 10%) × 3.67× fewer touches on each one that did — and showed the split depends on the order you attribute it.

Insist on the Highest Standards Seven metric definitions frozen and signed by the owners who would be red against them; a 15-day active denominator; a truncated 99.4% week (three touches) thrown out.

Earn Trust I printed a data miss instead of a flattering number, because three touches is not a week.

Are Right, A Lot Every red was decomposed before the meeting; a −45% week (40.0% on 17 Jan to 21.9% on 31 Jan 2022, record) traced to a carrier format change — measurement, not operators — and the same number was checked the next Monday.

Honesty label. Same program family as S11, a different cut: S11 is the boarding process and its gates; this story is the measurement system and the analysis behind it. The review format was copied from the LSA weekly review, not invented, and I built it by hand every Monday for months. The 22× decomposition and the 56% / 44% attribution are *my model* on record endpoints — they assume touches fall only on non-automated containers, and I say the split is a convention, not a measurement (the order-dependence is shown in Table 3). The week-over-week values (40.0% on 17 Jan → 21.9% on 31 Jan 2022), the three grains (19.4% · 54.1% · 40.2% → 69.2%) and the discarded 99.4% week are **record** from my own review workbook (Apple v11 A2, extract SV_28.xlsx); the hand-built review ran for months. The real LSA weekly-review workbook (Table 5) is the format I copied — its FFA tab is another team's slide, used here to show the mechanics on real data, not as my program's numbers. An early draft note in the source folder carries much larger growth multipliers for customers and volume — they conflict with canon (7 → 68 = 9.7×) and are **never said**.

WHAT THE RECORD SAYS ABOUT THE REVIEW (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The format. The LSA weekly-review instructions separate *output* from *input* metrics; a week runs Monday–Sunday; the view carries five consecutive weeks plus months; week-over-week is the relative change (B − A)/A; green is favorable beyond 5%, red unfavorable beyond 5%. RECORD-FROM-DOCS

What the program asked of it. The Feb 1, 2022 E-Team takeaways name Goal 1 as lowering cost-to-serve, ask for cost-to-serve measured for 3PL + 4PL versus 4PL only, have the visibility team publish its own weekly review, and set the improvement priority rule: rank carriers, terminals and milestones on the two priority lanes by cost-to-serve reduction first, ease of getting the data second. RECORD-FROM-DOCS

What the review ran on. The BRD names the seven metrics and the as-is: a manual Zendesk pull matched to a manual container pull in Google Sheets, 1.5–2 hours per week-over-week snapshot (a BRD measure; my own account is "about four hours a week at its worst" — two different measures, never stacked). My review workbook (Apple A2) is the container-status-history extract SV_28.xlsx — 10,289 events, week ending 28 Feb 2022. RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport's visibility program was being judged by leadership that said it was not demonstrating value, and the only instrument was a spreadsheet I stitched from four systems every Monday. Cost-to-serve was $0.66 a container; nobody could say whether it would fall because automation covered more containers or because each human touch got cheaper, and one flattering week could steer a rollout. **Task.** As program owner, make the numbers decision-grade. **Action.** I copied the LSA weekly review's mechanics — Monday-to-Sunday weeks, five weeks in view, week-over-week as (B − A)/A, red or green beyond ±5% — froze seven metrics with owners who signed the definitions they would be red against, tagged every input to the output it moves, and fixed the denominator: active is the cumulative inflow less containers 15 days past departed terminal. Then I took the headline apart: $0.66 to $0.03 is 22×, which is 6× fewer containers needing a human — 60% down to 10% — times about 3.7× fewer touches on each one that still did. When a week printed 99.4% on three touches I discarded it, because a 95% interval on three is 44% to 100%. **Result.** Primary metric: cost-to-serve $0.66 → $0.03 per container — about 95% — across customers 7 → 68, with every red arriving with an owner and a dated fix — and the same seven definitions now live in code.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. "Cost-to-serve went from sixty-six cents to three cents a container — about ninety-five percent. The split into six-times coverage and three-and-two-thirds touches is my model on those two record endpoints, and I call it a convention, not a measurement."

"What's your source for that?" "The endpoints and the weekly prints come from the review workbook I built — the container-status extract — and from the program's own documents; the split and the sample-size arithmetic are mine, and I label them modeled."

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The program had public targets — 68 customers and 170 monthly active users (both landed two months early) and a 32,000-shipment target I managed toward — and leadership that said value was not legible; it was about to drop from #3 to #14 of 16 initiatives. The instrument was my Monday spreadsheet: four systems (tickets, container pulls, platform data, Jira), about four hours a week at its worst, one person who could break it. The analysis problem underneath had three parts: a headline that moved ($0.66 → $0.03) with no explanation of *what* moved it; denominators that could flatter or flatten a week; and small-N weeks that could mislead a VP.

T — Own the measurement design and the analysis: define what is measured, at which grain, against which denominator, and what the number must be able to explain — so leadership could steer rollout pace and funding off it.

Baselines on the table: CTS $0.66/container (≈3.7 touches at the record $0.18 per interaction) · automation 40% · ETA accuracy 60% · customers 7 · review built by hand — about four hours a week at its worst · seven metrics · five Mon–Sun weeks in view · ±5% week-over-week rule.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — documented the as-is (four systems, Sheets, a person-dependent manual build); asked what leadership decides on (rollout pace, funding) and wrote the questions the review must answer weekly; chose the LSA review as the pattern to copy. *Exit bar:* leadership agrees the weekly questions; each input has a named owner.
- **Plan & Design** — froze seven metric definitions (coverage auto, coverage with human, tickets, tickets per active container, 4PL SLA %, 3PL SLA %, delta); input→output tagging; the ±5% week-over-week rule; the 15-day active denominator; a parking lot for new metrics. *Exit bar:* every owner signs the definition they will be red against; first Monday deck dry-run.
- **Build & Test** — built the deck by hand every Monday; pre-decomposed every red into source, definition or operator before the meeting; enforced the data-honesty rule (a truncated week is a data miss, not a result); tied Jira

blocker/critical trends to the input metrics. *Exit bar:* two consecutive Mondays where every red arrives with an owner and a dated fix, and the number verifies the following Monday.

- **Cutover & Go-Live** — the Monday review became the forum where rollout decisions were taken; waves boarded only through the published criteria; the decomposition went into the deck so leadership could see which lever moved cost-to-serve. *Exit bar:* the VP takes the rollout decision in the room off the numbers.
- **Hypercare / Close** — discarded the flattering 99.4% three-touch week and printed a data miss; moved the definitions into code with the Snowflake pipeline (S16-A) so the review no longer depended on one person. *Exit bar:* same seven numbers within 1% for consecutive weeks; the critic steers by them.

Influence and ownership. Mine: the diagnosis, the seven definitions, the denominator rule, the decomposition, the data-honesty rule and the Monday build. The input owners — Data Services, Support, GTM and the engineering teams — signed their own definitions and carried their own reds; the VP decided in the room. The format came from the LSA review.

> **Designed the process; others own it now.** I designed the definitions, the denominator and the reading rules; the owners run their lines, and the pipeline (S16-A) carries the definitions in code. **Trade-off put to leadership.** Coverage of the page (every metric anyone wanted) versus stability of the page (seven that never change definition) — I recommended stability and gave the rest a parking lot.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — cost-to-serve $0.66 → $0.03 per container (record; about 95%); direct dollars ≈$50K/yr *modeled* at the 32,000-shipment target on an illustrative 2.5 containers per shipment; ≈$21K/yr of by-hand review labor *modeled* (≈4 h/wk at its worst, at $100/h — an upper bound); the ~$1.5M pre-prod figure (S11, user anchor) stays separate.
- **#** — seven frozen metrics, each with a named owner; customers 7 → 68 (9.7×) with zero rollbacks; 68 customers and 170 monthly users landed two months early (record); the 32,000-shipment figure is a target, not a result.
- **%** — automation 40% → 90%; ETA accuracy 60% → 90% (record); coverage-versus-touch-efficiency split ≈56% / 44% (modeled).
- **Time** — by hand for months — about four hours a week at its worst — until the pipeline matched within 1% (record, Apple A1/A2); rollout unlocked two months early off the review (record).
- **Customer** — the review carried the graduation criteria and CSAT ≥ consignee average into the rollout decision — the customer lens was on the Monday page (record).
- **Quality** — a truncated 99.4% week (three touches) discarded and printed as a data miss; a −45% week (40.0% → 21.9%, 17 → 31 Jan 2022) traced to a carrier format change, not operators; definitions frozen and later moved into code (record).

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No product-analytics function existed. I defined the questions, the metrics and the denominators, and chose the owners — then explained the headline, not just reported it.
Scope & influence	Seven owners across Data Services, Support, GTM and engineering signed definitions; a VP decided in the room — influence one to two levels up from a program seat.
Execution trade-off + mechanism others now own	Stability over coverage: seven metrics, a parking lot, no mid-flight redefinition. Mechanism others own: the frozen definitions, the 15-day rule and the ±5% rule, now in code (the Snowflake pipeline, S16-A).
Impact (long-term)	A cost-to-serve story leadership could act on: which lever moved it, which to fund next, and a rollout paced by evidence.
Process improvement (standard work institutionalized)	Standard work for the weekly review: frozen definitions, honest denominators, input→output tags, every red decomposed with an owner and a dated fix, verified next Monday (PDCA).
Complexity & consensus	Owners wanted their metric on the page; I held seven and gave them a parking lot. Sales wanted to board everything; the review carried the gates.
Root cause at scale	Reds traced to source, definition or operator before escalation — the −45% week was a carrier format change; the 99.4% week was three touches.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Three grains from SV_28.xlsx (event 19.4% · ticket queue 54.1% · weekly stock 40.2% → 69.2%); ticket ↔ container join; cuts by coverage, SLA and 4PL vs 3PL; what it revealed: a two-lever decomposition, a measurement-caused red, a small-N week.
2 · Solution design	Copy the LSA format vs invent; freeze seven vs expand; 15-day active vs cumulative denominator; ±5% week-over-week vs raw levels.
3 · Math & data logic	3.67 touches → 22.0× = 6.0× × 3.67×; order-dependent attribution ($0.55/$0.08 vs $0.48/$0.15) → fair split 56/44; Wilson interval on N = 3 → [0.44, 1.0]; $50.4K + $20.8K = $71.2K modeled.
4 · Implementation	Monday standard work: build, decompose, owner + dated fix, verify next Monday; owners sign; parking lot; hand-build for months, then pipeline parity within 1%.
5 · Probe questions	Six numeric probes plus three trap drills; Table 5 recomputes the real LSA workbook (relative vs absolute WoW, a constant "weekly" metric, a dubious-data flag).

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

Grain and joins. The instrument was the container-status-history extract (SV_28.xlsx, week ending 28 Feb 2022): the grain was the container-status *event*, not the shipment, rolled up to a weekly stock. Three grains, one formula family, never blended: **event** 1,991 of 10,289 human-touched = 19.4%; **ticket queue** 538 auto vs 456 human = 54.1% auto; **weekly stock** (the headline) 40.2% on 13 Dec 2021 → 69.2% on 21 Feb 2022 RECORD (APPLE V11 A2). Sources: Zendesk tickets (a ticket is created when Visibility fails to auto-initiate tracking), the container pull from Crux, platform tracking data and Jira blocker/critical counts RECORD-FROM-DOCS. The join is ticket ID ↔ container number; tickets carried no container ID, so the first versions matched by hand — the gap S16-A closes with a cross-reference table.

Cuts. Coverage automated vs with human input; tickets and tickets per *active* container; SLA for 4PL vs 3PL and the delta between them; 3PL + 4PL vs 4PL only (the E-Team ask). Every cut is an output metric tagged to the inputs that move it.

What the analysis revealed. (1) The headline was two multiplied levers: 22.0× = 6.0× (containers needing a human 60% → 10%) × 3.67× (touches per touched container 6.11 → 1.67) — a *modeled* split on record endpoints. (2) The attribution is order-dependent, so I report the fair split (average of both orders): ≈56%

coverage, ≈44% touch efficiency. (3) A −45% week-over-week red — 40.0% on 17 Jan to 21.9% on 31 Jan 2022 — traced to a carrier format change, measurement not operators RECORD. (4) The 21 Feb print of 69.2% moved with the denominator (active book 287 → 546 the week a big client's 283 containers hit), and the 28 Feb print of 99.4% rested on three touches — a truncated week, discarded RECORD.

CUT	QUESTION IT ANSWERS	INPUT OWNER TYPE
Coverage auto vs with human	Is the machine doing the work, or the desk?	Data Services / engineering
Tickets per *active* container	Is cost-to-serve falling per unit of live freight?	Support / Data Services
4PL vs 3PL SLA and delta	Do we serve non-Flexport freight as well as our own?	Visibility / Data Services
Jira blocker/critical WoW	Which defects explain the reds?	Engineering managers

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Review format	Invent a new format · copy the LSA weekly review	Copy	It already had leadership's trust; inventing one would have spent credibility I did not have (canon).
Metric set	Everything anyone asks for · freeze seven + parking lot	Freeze seven	A number that changes definition cannot steer a rollout — stability over coverage (canon).
Denominator	Cumulative containers · touched this week · active = cumulative inflow less containers 15 days past departed terminal	15-day active	The honest denominator for per-container rates; cumulative flatters. Defined in the workbook as cumulative inflow less containers 15 days past departed terminal (Apple A2) VERIFY-WITH-RAJESH · V2
Reading rule	Raw levels · week-over-week (B − A)/A with ±5% red/green	WoW ±5%	Same rule as the review leadership already read; a red needs an owner and a dated fix (canon).
Small weeks	Print every week · mark insufficient data	Mark a data miss	The interval on three touches is wider than the band the number is read against.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Touches per container at the baseline	`$0.66 / $0.18 = 3.67`	record; $0.18 per interaction is the record cross-walk	≈3.7 touches per container
2	Containers needing a human	`1 − 0.40 = 60%`	record (automation coverage 40%)	60%
3	Touches per human-needed container	`3.67 / 0.60 = 6.11`	modeled — assumes every touch lands on a non-automated container	≈6.11 touches
4	By-hand build time	`about four hours a week at its worst (my own account); the BRD as-is is 1.5–2 h per snapshot — a different measure, not stacked`	record (Apple A1/A2); BRD figure labeled record-from-docs	≈4 h/wk at its worst

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
5	Three grains, three honest numbers	`event: 1,991 / 10,289 = 19.4% human-touched (80.6% auto); ticket queue: 538 / (538 + 456) = 54.1% auto; weekly stock: 40.2% (13 Dec) → 69.2% (21 Feb)`	record — Apple A2, extract SV_28.xlsx; the ratios are my arithmetic	one formula family, three grains — never blended
6	The denominator moved under the headline	`active book 287 → 546 (21 Feb, a big client's 283 containers hit): (546 − 287)/287 = +90.2%`	record inputs (Apple A2); the ratio is my arithmetic	a rate can jump because the base did

Table 2 — The fix: the instrument and its rules, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Seven frozen metrics	`2 coverage + 2 cost-to-serve + 3 timeliness = 7`	record (BRD metric list)	7 definitions, 7 owners
2	Week-over-week rule	`(B − A)/A = (21.9 − 40.0)/40.0 = −45% → |−45%| > 5% → red`	record: the ±5% rule is the LSA review (Instructions row 9); 40.0% on 17 Jan and 21.9% on 31 Jan 2022 are the printed weekly values (Apple A2)	red; root cause a carrier format change
3	Small-N rule	`Wilson lower bound for 3 of 3 = 3 / (3 + 1.96²) = 0.44 → interval [0.44, 1.00]`	record: 99.4% on three touches (Apple A2); the Wilson interval is my arithmetic	width 0.56 vs a 0.10-wide band
4	Sample floor I would write down now	`n = 1.96² × 0.90 × 0.10 / 0.05² = 138.3 → 139 active containers for ±5 points at 90%`	illustrative threshold — not in the record (what I would add)	≈139 containers
5	Honest denominator	`Active_d = cumulative inflow − containers 15 days past departed_terminal; rate = human-touched containers ÷ Active_week; automation = 1 − rate`	record rule (Apple A2)	every per-container metric

Table 3 — After: the metric, before → after, and the decomposition

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve	`1 − $0.03/$0.66 = 1 − 0.0455 = 0.9545 → I say "about ninety-five percent"`	record endpoints (Apple A1/A2); ratio mine	$0.66 → $0.03, about 95%
2	Coverage lever	`human-needed share 60% / 10% = 6.0×`	record endpoints (40% → 90%)	6.0×
3	Touch-efficiency lever	`6.11 / 1.67 = 3.67× (1.67 = $0.03 / $0.18 / 0.10)`	modeled	3.67×
4	Identity check	`6.0 × 3.67 = 22.0 = $0.66 / $0.03`	arithmetic	22.0×
5	Attribution, coverage first	`$0.66 − 0.10 × 6.11 × $0.18 = $0.66 − $0.11 = $0.55 from coverage; $0.11 − $0.03 = $0.08 from touches`	modeled	$0.55 / $0.08
6	Attribution, touches first	`$0.66 − 0.60 × 1.67 × $0.18 = $0.66 − $0.18 = $0.48 from touches; $0.18 − $0.03 = $0.15 from coverage`	modeled	$0.48 / $0.15
7	Fair split (average of both orders)	`coverage ($0.55 + $0.15)/2 = $0.35; touches ($0.08 + $0.48)/2 = $0.28; $0.35 / $0.63 = 56%, $0.28 / $0.63 = 44%`	modeled	56% / 44%

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
8	Scale	`68 / 7 = 9.7×; automation 40% → 90%; accuracy 60% → 90%; 170 monthly users is the public target, landed`	record	9.7× — the only multiple to say

Table 4 — The dollars, every multiplication shown

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Unit saving	`$0.66 − $0.03 = $0.63 per container`	record endpoints	$0.63
2	Containers per year at the 32,000-shipment target	`32,000 × 2.5 = 80,000 (range 48,000–128,000 at 1.5–4.0)`	the shipment count is a target, not an outcome; containers per shipment illustrative	80,000
3	Direct cost-to-serve saving	`80,000 × $0.63 = $50,400`	modeled	≈$50K/yr
4	Split by the fair attribution	`80,000 × $0.35 = $28,000 (coverage); 80,000 × $0.28 = $22,400 (touch efficiency)`	modeled	$28,000 / $22,400
5	Review labor by hand	`4 h/wk × 52 = 208 h; 208 × $100 = $20,800 ($16,640 at $80; $24,960 at $120)`	modeled; 4 h/wk is the at-its-worst figure, so an upper bound; rate is a benchmark range	≈$21K/yr
6	Total modeled run-rate	`$50,400 + $20,800 = $71,200`	modeled — master sensitivity base ≈$71K	≈$71K/yr
7	One-time revenue pull-forward (excluded from the run-rate)	`61 × $25,000 × 2/12 = $254,167`	modeled; ARR per customer is an illustrative benchmark; the pilot itself is free	≈$254K one-time

Table 5 — The review format run on a real workbook: LSA WBR 2022, FFA tab (my pandas / openpyxl recompute; the FFA slide is another team's, used to show the mechanics)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Exceptions per shipment, Wk 35 → Wk 1	`(2.900821 − 4.068702) / 4.068702 = −28.7% (lower is better, so favorable)`	record: sheet FFA row 11, cols Wk 35–Wk 1; arithmetic mine	−28.7% in 18 weeks
2	Same metric, monthly view	`Apr 6.0 → Sep 4.0: (4.0 − 6.0) / 6.0 = −33.3%`	record: FFA row 11, monthly columns	−33.3%
3	Red / green weeks under the ±5% rule	`over the 18 week-over-week steps: unfavorable (> +5%) = 1, favorable (< −5%) = 4, inside the band = 13`	my recompute of FFA row 11 under Instructions row 9	1 red · 4 green · 13 quiet
4	The holiday week	`import-partner on-time share Wk 51 → Wk 52: (0.290 − 0.459) / 0.459 = −36.8% → red; Wk 52 → Wk 1: (0.506 − 0.290) / 0.290 = +74.5%`	record: FFA row 5; the owner's callout attributes Wk 52 to the holiday season	a red that arrives with a cause
5	Mean with and without the holiday week	`nine weeks: 4.2785 / 9 = 47.5%; without Wk 52: (4.2785 − 0.290) / 8 = 49.9%`	record inputs FFA row 5 (Wk 45–Wk 1); mean mine	47.5% vs 49.9%
6	Written rule vs printed column	`the Instructions say WoW = (B − A)/A; the WoW cell for Wk 52 → Wk 1 prints 0.216 = 0.506 − 0.290 — an absolute difference; under the rule it is +74.5%`	record: Instructions row 8 and FFA row 5 WoW cell; my observation	definition drift to flag, not to fix silently
7	A "weekly" metric that never moves	`import tasks structured into work items = 0.1346 in all 19 weeks; 7 / 52 = 0.1346`	record: FFA row 12; the 7 and 52 are my inference, the sheet does not say	static, not weekly — mark it

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
8	Data the owner flagged themselves	`notification read-rate Wk 43 → Wk 1: (0.050 − 0.0725) / 0.0725 = −31.0%, beside the sheet's own note: "data looks dubious and will double check"`	record: FFA row 16 and its note; arithmetic mine	a data miss until checked — the same rule as the 99.4% week

The ~$1.5M pre-production measurement saving (S11) is a separate user-anchor figure and is excluded from every row above. Automation is 40% → 90%; the loop packet's "60 → 97" is wrong (§1.3) — do not mix them in the room.

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

CONTROL	STANDARD WORK	OWNER	LEAN / KAIZEN WORD
Definitions	Seven written definitions, signed, frozen; new metrics enter a parking lot, never the page	Each input owner	Standard work
Weekly build	Pull four systems → join ticket ↔ container → seven metrics on the 15-day active base → WoW ±5%	Me, by hand, then the pipeline	Takt: the Monday cadence
Red handling	Decompose to source / definition / operator before the meeting; owner + dated fix; verify the same number next Monday	Metric owner	Root cause (5 Whys), PDCA
Data honesty	A truncated or tiny-N week is a data miss, never a result	Me	Jidoka — stop the line on bad data
Hand-off	Pipeline parity within 1% for consecutive weeks, then retire the manual build (S16-A)	Infra + me	Remove waste (motion) and the single point of failure

Risks and mitigations. Measurement-caused reds (decompose first); a flattering wrong number (data-honesty rule); a single point of failure (the manual review); an expanding page (parking lot). **Rollout of the review itself:** dry-run deck → two consecutive clean Mondays → the VP takes a decision in the room → the review becomes the standing forum. A review that skips a week teaches leadership it can be skipped, so it ran every Monday.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Is the 6.0× / 3.67× split real, or an artifact of your assumption?**
>
> It is a model, and I say so. The endpoints are record: $0.66 → $0.03 and coverage 40% → 90%. The split assumes every touch lands on a non-automated container; if exception handling touches automated containers too, the touch-efficiency lever is understated. The honest convention is the average of both attribution orders — 56% coverage, 44% touches — and the order-dependence itself (coverage's share runs 24%–87%) is why I call it a convention, not a measurement.

> **How do you know the −45% week was not operators?**
>
> Because I decomposed it before the meeting: source, definition, operator, in that order. The drop traced to a carrier format change — measurement. The owner took a dated fix and I looked at the same number the next Monday. The weekly prints were 40.0% on 17 Jan and 21.9% on 31 Jan 2022; the arithmetic is (21.9 − 40.0)/40.0 = −45%.

> **Why a 15-day active window, and what would change it?**
>
> It is the honest denominator for per-container rates — cumulative containers would let the base swamp the signal — in the real extract the active book went 287 → 546 in the week one big client's 283 containers hit, and the headline moved from the low 40s to 69.2% with it. Fifteen days past departed terminal was frozen with the owners. What would change it: showing the rate at 10, 15 and 30 days and freezing the window where the rate stops moving VERIFY-WITH-RAJESH · V2.

Three touches — why discard instead of caveat?

Because the interval was wider than the band. At 3 of 3 the 95% Wilson interval is 0.44 to 1.00 — 0.56 wide — against a ±5% (0.10-wide) red/green band. A number you cannot read against its own threshold is not a reading. To read a rate to ±5 points at 90% you need about 139 active containers.

You froze seven metrics. What did that cost?

Coverage of the page — engineering wanted more end-to-end input metrics, and the review parked them. The trade was stability: a number that changes definition cannot steer a rollout. The parking lot was real, and the pipeline (S16-A) is where the extra inputs could be added without touching the seven.

What decisions did the review actually change?

Three, all record: waves boarded only through the published criteria; wave 3 held in public on an ETA-accuracy miss traced to one bad source; the VP unlocked the 68-client rollout two months early off the review's numbers. The E-Team's improvement rule — rank by cost-to-serve reduction first, ease second — is the same Pareto the reds were sorted by.

BUSINESS IMPACT MODEL — KPI TREE

```
Leadership steers rollout pace and funding off a number that survives the next Monday
+- OUTPUT  cost-to-serve per container   $0.66 -> $0.03   (record, about 95%)   22.0x
|    +- INPUT  containers needing a human   60% -> 10%    (coverage 40% -> 90%, record)   6.0x
|    +- INPUT  touches per touched container 6.11 -> 1.67 (modeled)                        3.67x
|    +- fair split of the $0.63 saving: coverage ~56% / touch efficiency ~44% (modeled)
+- OUTPUT  ETA accuracy 60% -> 90%; customers 7 -> 68 (9.7x)   (record)
+- TAGS    Jira blocker/critical trend  <->  the input metric it explains
+- DENOM   active = inflow less containers 15 days past departed terminal; tiny-N weeks = data miss
(loop)   red -> owner -> dated fix -> same number verified next Monday -> trust -> decision taken in the
room
```

SOLUTION FLOW

```
Sun night   pull 4 systems (tickets, containers, platform, Jira)
   -> join ticket <-> container (xref until the real key exists)
   -> 7 metrics on the 15-day active base, container-week grain
   -> WoW = (B - A)/A ; flag beyond +/-5%
   -> decompose each red: source? definition? operator?   [5 Whys]
   -> owner + dated fix   -> Monday review: decision taken in the room
   -> next Monday: same number verified   [PDCA: Check / Act]
   -> pipeline parity within 1% for consecutive weeks -> manual build retired
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Containers per shipment (illustrative)	1.5	2.5	4.0
Containers per year at 32,000 shipments	48,000	80,000	128,000
Direct saving ($0.63 × containers)	$30,240	$50,400	$80,640
Review labor by hand ($80 / $100 / $120 per hour × 208 h)	$16,640	$20,800	$24,960
Total modeled $/yr	≈$47K	≈$71K	≈$106K

Revenue pull-forward ($254K) is one-time and the ~$1.5M pre-prod estimate is a separate user anchor; both are excluded from the run-rate.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

Cost-to-serve fell from sixty-six cents to three cents a container — twenty-two times. When I took it apart it was two levers multiplied: six times fewer containers needing a human, sixty percent down to ten, and about three and two-thirds times fewer touches on each one that still did. That split is my model — it assumes the touches fall on the non-automated containers — and the attribution depends on the order you take it, so I report the average, roughly fifty-six percent coverage and forty-four percent touch efficiency, and I call it modeled. At the thirty-two-thousand-shipment target the direct saving is about fifty thousand dollars a year on an illustrative container count, plus about twenty-one thousand of review labor I stopped doing by hand — both modeled. What is record is the sixty-six cents, the three cents and seven to sixty-eight; what the review added is that every red arrived with an owner and a dated fix, and that when a week printed ninety-nine point four percent on three touches I printed a data miss instead.

IF CHALLENGED — THE FULL ANSWERS

"Your 22× decomposition is just algebra."

The product identity is algebra; the analysis is knowing it needs an attribution convention. If touches per touched container had stayed flat, the whole result would have been coverage; because both moved, the order matters, so I show both orders and report the average. The use is practical: it tells the review which lever to fund next.

"Why throw out a number that looked good?"

Three touches gives an interval of 44% to 100% — wider than the ±5% band the review reads against. Printing it would have taught the VP a number that could not survive the next Monday, and a review that flatters once is not trusted twice.

"An early note claims a far bigger growth multiple than 7 → 68. Which is it?"

The record is 7 → 68, 9.7×, against public targets of 68 customers and 170 monthly active users — both landed two months early — and a 32,000-shipment tracked target. The larger multiples are from an early draft note and are not canon; I do not use them.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I ran the weekly review that steered a visibility rollout: seven frozen metrics, a 15-day active denominator, ±5% red/green. I took the 22× cost-to-serve drop apart — 6× fewer containers needing a human, 3.7× fewer touches on each — and threw out a 99.4% week built on three touches.

NACF BRIDGE — ONE SENTENCE

The same review shape runs an inbound network — doors × hours × unload rate on one side, UPH and cubic fill on the other, each with a frozen weekly definition — so when door utilization crosses the ~85% knee or weeks-of-cover leaves its 2–4 week band, the red arrives with an owner, a dated fix and an honest denominator (doors in use, not doors installed).

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a problem you had to solve that required in-depth thought and analysis. How did you know you were focusing on the right things? What was the outcome? Would you have done anything differently?" AIQB: Dive Deep Q3 **Dive Deep** FACET: DATA ANALYSIS

Open (first two sentences): "A cost-to-serve number fell ninety-five percent and nobody could tell me why, so I took it apart. It was two multiplied levers, not one."

90-second STAR. S Leadership doubted the program, and the headline was $0.66 → $0.03 per container. **T** Explain what moved it, so the review knew what to fund next. **A** I converted dollars to touches at the record $0.18 per interaction — 3.67 down to 0.17 — then split coverage (containers needing a human, 60% → 10%, 6.0×) from touches per touched container (6.11 → 1.67, 3.67×). I noticed the attribution changes with order — $0.55/$0.08 coverage-first, $0.48/$0.15 touches-first — so I reported the average, 56/44. **R** The split was modeled and I said so; the review had a shared way to talk about where the next fix should go, and the record endpoints stayed untouched.

Land: 22.0× = 6.0× × 3.67× · **Pivot to NACF:** the same move on dock-to-stock cost: split a throughput gain into door-hours opened and unload rate before deciding which to fund.

Do NOT say: that the 56/44 split was measured; any customer-growth multiple other than 9.7×; any dollar from the split as record.

B2 · "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?" AIQB: Highest Standards Q1 **Insist on the Highest Standards** FACET: DEFINITIONS AND DENOMINATORS

Open (first two sentences): "Every owner wanted their metric on the page, and I froze seven and parked the rest. A number that changes definition cannot steer a rollout."

90-second STAR. S A weekly review drifts: metrics get added, definitions bend, the page stops meaning anything. **T** Keep the review decision-grade for a VP who would steer funding off it. **A** I wrote seven definitions, got each input owner to sign the one they would be red against, fixed the denominator at 15-day active, offered a parking lot for new metrics and held the line after sign-off. **R** Every red arrived with an owner and a dated fix; the same seven definitions later moved into code with no drift.

Land: seven metrics, 15-day active · **Pivot to NACF:** freeze door-utilization and weeks-of-cover definitions before the quarter, signed by the node owner, so a red means the same thing in every building.

Do NOT say: "I invented the WBR" — it was copied from the LSA review, which was the point.

B3 · "Tell me about a time when you didn't have enough data to make the right decision. What did you do? What path did you take? Did the decision turn out to be the correct one?" AIQB: Are Right, A Lot Q1 **Earn Trust** FACET: DATA HONESTY

Open (first two sentences): "I had a week that printed ninety-nine point four percent and I threw it out. It rested on three touches, so I printed a data miss in its place."

90-second STAR. S The program was being doubted, and a flattering week landed. **T** Protect leadership's trust in the number over the next month, not the next meeting. **A** I computed the interval: 3 of 3 gives a 95% Wilson interval of 0.44 to 1.00 — wider than the ±5% band. I told the VP it was a data miss, not a result. **R** The number never had to be walked back, and the critic who started this later steered by the review.

Land: N = 3 → interval 44%–100% · **Pivot to NACF:** a unit-fill or WOC reading built on a handful of trailers or SKUs gets marked insufficient volume, not headlined.

Do NOT say: what the "real" number was — I do not have one and will not invent it.

B4 · "Tell me about a situation that required you to dig deep to get to the root cause. How did you know you were focusing on the right things? What was the outcome? Would you have done anything differently?"

AIQB: Dive Deep Q2 | Are Right, A Lot | FACET: ROOT CAUSE ON MEASUREMENT

Open (first two sentences): "The weekly automation rate fell from forty percent on the seventeenth of January to twenty-one point nine on the thirty-first — minus forty-five, deep red. Before the meeting I decomposed it, and it was a carrier format change, not the operators."

90-second STAR. **S** A red week landed on a desk that could have been blamed. **T** Find the cause before anyone defended or accused. **A** The standard work: source, definition, operator, in that order. The source had changed format; I assigned the owner a dated fix and checked the same number the next Monday. **R** The red arrived with an owner, the operators were not blamed, and the order of questions became standard work.

Land: (21.9 − 40.0)/40.0 = −45% — printed weekly values 17 Jan → 31 Jan 2022 (record) · **Pivot to NACF:** a sudden UPH or weeks-of-cover swing gets the same three questions — source, definition, operator — before an ops root cause is declared.

Do NOT say: the carrier name; that operators had slipped.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle — Dive Deep

Opener: "Cost-to-serve fell 22× and I can tell you which lever did it — and why the answer depends on the order you ask."

Headline metric: 22.0× = 6.0× coverage × 3.67× touch efficiency; fair split 56% / 44% (modeled)

- Grain: container-week; join ticket ↔ container; 15-day active base.
- Identity: 6.0 × 3.67 = 22.0.
- Order test: $0.55/$0.08 coverage-first vs $0.48/$0.15 touches-first.
- Say "modeled" once, early; the record is the endpoints.

Conflict / influence / backbone angle — Insist on the Highest Standards

Opener: "The owners wanted their metric on the page. I held seven and gave them a parking lot — and a signature line."

Headline metric: seven frozen metrics, each signed by the owner who would be red against it

- The objection: more end-to-end input metrics, now.
- The answer: stability over coverage; a changing definition cannot steer a rollout.
- The mechanism: sign-off, freeze, parking lot — no argument by seniority.
- The proof: the same definitions moved into code with no drift.

Honest failure-and-learning angle — Ownership

Opener: "I started the review after the critique, not at kickoff — and for months I was its single point of failure."

Headline metric: by hand for months, about four hours a week at its worst; no longer a single point of failure once the pipeline matched within 1%

- What I would do differently: instrument at kickoff, not after the program fell from #3 to #14 of 16.
- What I would add now: a written sample floor (~139 active containers for ±5 points at 90%) instead of a judgment call on each small week.
- What fixed it: definitions in code (S16-A), not heroics.
- No dollar claimed for this angle.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"How do you know the drop was your review and not the automation teams — or seasonality?"

Numeric answer. I do not claim the review built the automation; nine teams did (coverage 40% → 90%). The review's claim is narrower: it made boarding follow coverage — 9.7× the customers, zero rollbacks, both targets two months early. Cost-to-serve is a per-container ratio, so seasonal volume does not move it; a mix shift could, and I would check that by lane.

Say exactly: "I don't claim the review made the automation — the teams built that. The review's claim is narrower and checkable: it made boarding follow coverage, and the record shows zero rollbacks at 9.7 times the customers."

"Your decomposition rests on an assumption. What number would change your mind?"

Numeric answer. The endpoints do not depend on it; only the split does. Depending on attribution order coverage's share runs $0.15 to $0.55 of the $0.63 saving — 24% to 87% — so no single split is a measurement. Ticket-grain data showing touches on automated containers would move me to a two-factor model with an interaction term.

Say exactly: "The endpoints are record; only the split is modeled — and the split runs from twenty-four to eighty-seven percent depending on order, so I report the average and call it a convention."

"Three touches — why not just report it with a caveat?"

Numeric answer. The 95% interval was 0.44 to 1.00 — 0.56 wide — against a 0.10-wide red/green band, 5.6 times wider than the threshold. A caveat does not fix a reading the threshold cannot interpret. To read a rate to ±5 points at 90% I need about 139 active containers.

Say exactly: "A number whose interval is wider than the band you read it against isn't a reading. At three touches the interval was forty-four to a hundred; to read ninety percent within five points you need about a hundred and forty."

S11-B — Flexport — Launch readiness as standard work: four test gates, a 100-container hop-by-hop ledger and six graduation criteria — and the pilot that launched in December 2021, later than the first testing plan NEW

LPs this story can serve here: Insist on the Highest Standards (primary) · Customer Obsession · Dive Deep · Learn and Be Curious (the honest miss)
Process family: Deliver — launch readiness, test strategy and staged rollout; entry / exit criteria and go / no-go governance

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Insist on the Highest Standards Business UAT did not open until 80% of shipments showed minor-to-no issues, and no wave graduated until six published criteria passed — the gate was data quality, not the calendar.

Customer Obsession Shadow accounts ran two real Crux clients' live data beside the old product, and every new client supplied at least 50 containers of its own freight to calibrate on before onboarding.

Dive Deep The 100-container ledger traced real containers hop by hop; of 59 failing rows, 41 (69.5%) first broke at route creation or the milestone on screen — two hops, not seven.

Learn and Be Curious My test-effort estimates were optimistic and monitoring arrived late, so the pilot launched later than the first testing plan and bugs took hours to localize across six-plus layers — the lesson is to size testing on hop count and instrument first.

Honesty label. One of two technical program managers on the program (the record also names a co-TPM and a DRI). I owned the gate definitions, the readiness read-out and the go/no-go inputs; the 100-container log was run day to day by the QA team named in the testing approach, the hop-by-hop design is credited in the secondary loop packet to a test lead, and business users ran UAT. My own account (Apple v11 A5) is that I ran decision week as a two-gate go/no-go with criteria published beforehand, made the 100-container run the UAT spine with a Jira ID on every missed row, and set the UAT entrance criteria as exposure gates; I say "I defined the gates and ran the read-out", not "I ran the test". The ledger numbers below are *my recount of the 100-container working log* — a partial log where blank means not recorded.
Timing, doc against doc: the testing-approach document planned an early-November pilot; the program's year-end summary shows launch in December 2021 with delays among its misses; the pilot roll-out schedule has Wave 1 in Q4 2021, and my own workbook calls go-live "on schedule". I therefore say "later than the first testing plan", never "on time" and never a specific number of weeks late.

WHAT THE RECORD SAYS ABOUT READINESS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

Four test gates. A "100 container test" (random container and MBL numbers traced and logged, bugs in Jira, daily stand-ups); bug bashes every other week with job-story ranges assigned per person; shadow client accounts (client data uploaded to the legacy product fed into dedicated demo accounts, starting with two named Crux clients); and UAT with semi-weekly syncs, dependent on data quality first — 80% of shipments with minor-to-no issues. RECORD-FROM-DOCS

Go / no-go and waves. Two go/no-go meetings (business and tech) read: stable production, first-client data quality 80%+, no show-stoppers. Waves then graduated on six criteria — verification SLA 90% (max 48 h in "Verification in progress"), data quality 90%+ at least as accurate as the legacy product, milestone estimates 80%+, stable production, at least 100 containers plus key features used, and CSAT on par with the consignee average — with at least 50 containers of a new client's own data to calibrate on first, a free pilot and light-weight onboarding. RECORD-FROM-DOCS

Where it landed. The Nov 2021 status read Yellow with both checkpoints passed, a performance test of 500 containers in 20 minutes at 99.6% connectivity accuracy, end-to-end 100% processed and 86% trackable on 11/18, job stories 98.4% complete. The year-end summary: launched in December 2021, nine teams and close to 100 people, three environments built — and the misses: delays, optimistic testing estimates, and monitoring implemented late (bugs took hours to localize across six-plus system layers). RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport's shipment-visibility pilot was first planned for early November 2021 and launched in December. It ran on a new stack across nine teams and close to a hundred people, with data crossing six or more system layers before it reached a client's screen. **Task.** As one of the program's two TPMs, I owned the readiness gates: what had to be true before a client touched the product, and before each wave graduated. **Action.** We tested in four layers — a 100-container live test, biweekly bug bashes, shadow accounts that ran two real clients' data beside the old product, and business UAT that opened only when 80% of shipments showed minor-to-no issues. The 100-container ledger traced real containers hop by hop — created, subscribed, data back, route, milestone, exception, webhook — and its Pareto was blunt: of 59 failing rows, 41 first failed at route creation or the milestone on screen. Graduation was six published criteria, with at least 50 containers of the client's own freight to calibrate on first. **Result.** Primary metric: the gate held — a 500-container load test in 20 minutes at 99.6% connectivity, both go/no-go checkpoints passed, no wave rolled back. The honest part: my test-effort estimates were optimistic and monitoring arrived late, so the pilot launched later than first planned and bugs took hours to localize; I now size testing on hop count and put the monitors in first.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “Of fifty-nine failing rows in the hundred-container log, forty-one — about seventy percent — first broke at two hops, route creation and the milestone on screen; that is my recount of a working log, so it tells you where to look, not a defect rate.”

“What's your source for that?” “The pre-launch evidence is in the program's own status documents — five hundred containers in twenty minutes at ninety-nine point six, the dry run a hundred percent processed and eighty-six tracked — and the hop recount is mine from the log file.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The visibility pilot had to go live to Crux-grade clients on a brand-new stack: nine teams, close to 100 people, three environments, data traveling through six-plus system layers before it reached the Client App. The testing plan targeted an early-November pilot; the status record shows end of November and the year-end summary shows December 2021 (the roll-out schedule keeps Wave 1 inside Q4, and my own workbook records go-live on schedule — the documents disagree on whether December was late). The program's own retrospective calls the testing estimates "overly optimistic for the complexity of the product" and names monitoring as implemented too late.

T — Define "ready": the gates before a client touches the product, the criteria to graduate each wave, and the evidence the go/no-go checkpoints would read — so rollout followed data quality, not the calendar.

Baselines on the table: First testing plan early Nov 2021 → launched Dec 2021 (docs disagree whether late) · 9 teams / ~100 people · 6+ system layers · 3 environments · 4 test gates · 6 graduation criteria · ≥50 containers per new client to calibrate · UAT entry gate 80% minor-to-no issues · ledger: 474 containers listed, 142 marked tested, 105 with hop results.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — defined "working" per container as seven checks (created, subscribed, data back, route, milestone, exception, webhook); read the product standards (coverage 90%+, accuracy 95%, timeliness 90%) and what each go/no-go would read. *Exit bar:* seven checks and four gates named; the questions the go/no-go answers written down.
- **Plan & Design** — four test gates; the UAT entry gate at 80% minor-to-no issues; six graduation criteria published before wave 1; at least 50 containers of each new client's own freight before onboarding; free pilot, light-weight onboarding, monthly go/no-go. *Exit bar:* criteria published to Sales and the business before wave 1.
- **Build & Test** — the 100-container log in daily stand-ups; bug bashes by job-story range; shadow accounts on real Crux data; production connectivity monitors for the hops; a trace ID and per-hop ratios proposed after the slow bug

hunts — the logs showed no unique ID tying events from downstream to the screen, so finding the source system took 2–4 hours per blocker; the recommendation cut blocker-bug time-to-resolve by about half (3PL ingestion one-pager). *Exit bar:* 500 containers / 20 minutes / 99.6%; end-to-end 100% processed, 86% trackable on 11/18.

- **Cutover & Go-Live** — two go/no-go checkpoints (business, tech) on stable production, first-client data quality 80%+ and no show-stoppers; wave 1 started in December with two hand-carried customers (the schedule and my workbook agree on those two; two other accounts ran as shadow UAT accounts, and the Nov 19 status note's "first pilot client" is the earlier plan — I say "the first two hand-carried customers", no names, unless asked). *Exit bar:* both checkpoints passed; wave 1 live.
- **Hypercare / Close** — feedback session about a week after kickoff; a 50-container calibration sample from each wave-2 client before onboarding; monthly go/no-go on the six criteria; production data-quality testing on the same per-hop ratios. *Exit bar:* wave 2 opened on wave 1's graduation read; zero rollbacks.

Influence and ownership. Mine: the definition of ready, the gate and criteria design, the go/no-go inputs and the pre-production measurement design (S11). The QA team ran the log, a test lead designed the hop columns, the engineering teams instrumented their hops and fixed what the log found, business users ran UAT, and product and engineering leadership took the go/no-go.

> **Designed the process; others own it now.** The gates and criteria are published artifacts, the ledger columns are the standard checklist, and the per-hop ratios became standing monitors — the launch reads off documents, not off me.
> **Trade-off put to leadership.** Launch date versus data quality: I recommended the data-quality gate and accepted the later date it implied.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — no record dollar for the testing layer. The ~$1.5M pre-production estimate is S11's separate user anchor and is never merged. An illustrative escape cost at the record $0.18 per touch is under $1K a year — the real cost of an escaped defect is the rollback and the first impression, which is what the gates bought.
- **#** — four gates; six graduation criteria; 500 containers in 20 minutes at 99.6%; end-to-end 11/18 100% processed, 86% trackable; job stories 98.4% complete; both go/no-go checkpoints passed (record-from-docs).
- **%** — UAT entry gate 80% minor-to-no issues; graduation bars 90% data quality, 80% milestone estimates, 90% verification SLA; ledger first-failure Pareto 69.5% in two hops (my read of the log).
- **Time** — pilot launched December 2021, later than the early-November testing plan (record-from-docs; the docs disagree whether it was late); bug localization took hours across six-plus layers (record); about half that after the trace ID (3PL ingestion one-pager: blocker-bug time-to-resolve −50%).
- **Customer** — shadow accounts on two real clients' data; ≥50 containers of each new client's freight to calibrate on; CSAT par with the consignee average inside the graduation gate (record-from-docs).
- **Quality** — zero rollbacks through the waves (record, S11 canon); every ledger miss ticketed to the owning team.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	"Ready" was undefined for a new-stack product. I defined the seven checks, the four gates and the graduation criteria, and what each go/no-go would read.
Scope & influence	Nine teams and ~100 people across three environments; I held the gates without owning any team — the criteria were published, the checkpoints were business and tech leadership's.
Execution trade-off + mechanism others now own	Date versus data quality: the UAT gate cost schedule and saved the first impression. Mechanism others own: the published criteria, the per-hop ratios as monitors.
Impact (long-term)	A launch with no rollbacks and a repeatable wave machine from 2 → 4 → 20+ clients; the monitors outlived the launch.
Process improvement (standard work institutionalized)	Standard work for launch: entry gate, calibration volume per client, six graduation criteria, monthly go/no-go, per-hop ratios.
Complexity & consensus	Nine teams with separate backlogs, an acquired product's clients in the middle of the rollout, and a date everyone wanted; the gates made the trade explicit.
Root cause at scale	A Pareto of first-failing hop located the defects (route, milestone) instead of counting symptoms; the missing seam — a trace across hops — was the root cause of slow debugging.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	The 100-container ledger as a hop-by-hop Pareto: 142 tested, 105 with results, 59 failing, first failure at route (23) or milestone (18); partial-log caveat.
2 · Solution design	Hop ledger at the base of a four-gate pyramid; UAT gated on data quality; waves behind six criteria; calibration volume per client; light-weight onboarding.
3 · Math & data logic	Detection power: 1 − 0.95^50 = 92.3%; 1 − 0.97^100 = 95.2%; 1 − 0.99^300 = 95.1%; rule of three 3/100; load 500 / 20 = 25 per minute.
4 · Implementation	Four gates with owners and cadence; two go/no-go checkpoints; waves; monitors; three environments; risks and the late-monitoring lesson.
5 · Probe questions	Seven numeric probes plus three trap drills: zero-rollback luck, the partial log, the later launch and its leading indicator.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What the ledger is. One row per container or MBL; one column per hop — created, subscribed in SV, data back, route created, milestone shown, exception shown, webhook created — plus an overall Success flag, a date and the bug IDs RECORD-FROM-DOCS. Population as I read it: 474 containers listed, 142 marked tested, 105 with at least one hop result, 53 with an overall verdict (19 Yes, 34 No) — a partial working log; blank means not recorded, and twelve numbered issues sit in the bug column.

The right Pareto is first-failing hop, not No-counts per column. Downstream hops cascade — a failed route makes the milestone fail — so counting every "No" double-counts one fault. Of the 105 rows with results, 59 had at least one "No"; each was assigned to the first hop in order that failed:

FIRST FAILING HOP	ROWS	SHARE OF 59	CUMULATIVE
Route created?	23	39.0%	39.0%
Milestone shown in UI?	18	30.5%	69.5%
Data back from the legacy source	7	11.9%	81.4%
Exception shown in UI?	6	10.2%	91.5%

FIRST FAILING HOP	ROWS	SHARE OF 59	CUMULATIVE
Shipment created?	5	8.5%	100.0%

What it revealed. Two hops — route creation and the milestone on screen — hold 41 of 59 first failures (69.5%); the ingestion hops (created, data back) hold 12 (20.3%). Only 3 of the 105 rows passed all seven hops. So the defects were downstream of ingestion, which is where the bug bashes and the owning teams went first. Caveats I would say out loud: this is a working log, not a random sample, and I would re-run a fresh random 100 and log all seven hops to turn the Pareto into a defect-rate estimate. The Pareto is my recount of the 100-Container Debugging Session log, not a number printed in any document.

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY
Test structure	End-to-end pass counts only · a pyramid with a hop-by-hop ledger at the base	Pyramid + hop ledger	End-to-end says that it broke, not where; the hop column routes the ticket to the right team the same day (the seven checks and the Jira ID on every missed row are mine per my workbook; credit the QA team for running the log)
UAT entry	Open on the calendar (mid-October) · open on a data-quality gate (80% minor-to-no issues)	Quality gate	Business testers should validate the product, not rediscover known bugs (record-from-docs).
Pilot entry	Board the pool at once · waves 2 → 4 of 7 → 20+ behind six criteria	Waves	A free pilot spends engineering, data and ops before revenue; the gate must fire before the next wave (record-from-docs; S11).
Calibration	Onboard, then measure · at least 50 containers of the client's own freight first	Calibrate first	The product holds data on the client's lanes by the time they onboard; 50 containers detect a 5% defect 92.3% of the time.
Onboarding weight	API integration and LOAs from day one · light-weight until about wave 2	Light-weight	Limits the client's investment if the product misses expectations (record-from-docs).
Monitoring	Debug as you go · instrument the hops first (trace ID, per-hop ratios, connectivity monitors)	Instrument first — I did it late	The program's own retrospective: bugs took hours to localize at six-plus layers; the lesson is the story's honest half.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Ledger population	`142 marked tested / 474 listed = 30.0%; 105 with hop results / 142 = 73.9%`	record-from-docs; my count of the working log (blank = not recorded)	142 of 474; 105 with results
2	Overall verdicts	`19 Yes / 53 verdicts = 35.8%`	record-from-docs; partial log	35.8% passing at the time logged
3	Rows with a failing hop	`59 / 105 = 56.2%; all seven hops Yes: 3 / 105 = 2.9%`	record-from-docs; my count	59 failing; 3 clean
4	Localization cost	`hours per blocker across 6+ system layers; 2–4 h in the loop-packet telling`	record-from-docs (hours); secondary source (2–4 h)	2–4 h per blocker

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
5	Plan versus actual	`testing-approach plan: early November 2021; launch: December 2021 (year-end summary)`	record-from-docs, doc against doc (month-level resolution)	later than the first testing plan

Table 2 — The fix: gates and detection power, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Calibration sample	`P(detect ≥ 1 defect) = 1 − 0.95^50 = 92.3% at a 5% defect rate`	record sample size (≥50 containers); defect rate illustrative	92.3%
2	Launch gate sample	`1 − 0.97^100 = 95.2% at a 3% defect rate`	record sample size (100 containers); defect rate illustrative	95.2%
3	If the 100 all pass	`upper bound = 1 − 0.05^(1/100) = 2.95% (rule of three: 3/100 = 3.0%)`	arithmetic	defect rate below ≈3% at 95%
4	Pre-production production test	`1 − 0.99^300 = 95.1% at 1% defects; 1 − 0.98^300 = 99.8% at 2%`	record sample size (300 containers, S11); defect rates illustrative	95.1% / 99.8%
5	Entry and graduation bars	`UAT opens at ≥80% minor-to-no issues; pilot entry at first-client data quality ≥80%; graduation at ≥90% data quality, ≥80% milestone estimates, 90% verification SLA`	record-from-docs thresholds	staged bar: enter safe, graduate proven

Table 3 — After: the metric, before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Load proof	`500 containers / 20 min = 25 per minute; 99.6% × 500 = 498 connected`	record-from-docs (Nov 2021 status)	25 per minute; 498 of 500
2	End-to-end on 11/18	`100% processed; 86% trackable → 100 − 86 = 14% not trackable`	record-from-docs	14% untrackable that day
3	Job stories	`98.4% complete → 100 − 98.4 = 1.6% outstanding`	record-from-docs	1.6% outstanding
4	Go / no-go	`2 of 2 checkpoints passed (business, tech); criteria: stable production, first-client data quality 80%+, no show-stoppers`	record-from-docs	launch cleared
5	Scale outcome	`68 / 7 = 9.7×; zero rollbacks`	record (S11 canon)	9.7×

Table 4 — The dollars, every multiplication shown

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Containers per year	`32,000 shipments × 2.5 = 80,000`	illustrative containers per shipment (S11)	80,000
2	Escaped defects at a 5% rate	`80,000 × 0.05 = 4,000 containers`	illustrative defect rate	4,000
3	Direct touch cost of the escapes	`4,000 × $0.18 = $720 per year`	record $0.18 per interaction; one touch each (illustrative)	$720 — tiny: the cost is the rollback, not the touch

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Localization labor (illustrative)	`5 blockers/wk × (3 h − 1.5 h) = 7.5 h/wk; 7.5 × 52 × $145 = $56,550`	illustrative: 5 blockers/wk is my example, not a source figure; 2–4 h per blocker and the ~50% cut are record-from-docs; $145/h from the S16-A model	≈$57K/yr — priced, not claimed; say the hours, not this dollar
5	Pre-production measurement path	`n = 300 → ±1.6% at 95%; rollout proof 1.5 vs 6 months`	user anchor (GFiber v8 only) — separate (S11), prefer the documents' 500 containers / 20 min / 99.6%	~$1.5M, never merged

No dollar in this story is a claimed saving. The two priced rows are illustrations built on the record $0.18 per touch and on a labeled localization model; the ~$1.5M stays with S11.

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

GATE	ENTRY → EXIT	CADENCE AND OWNER	CONTROL
100-container test	random container / MBL numbers → every hop logged; bugs to Jira	ongoing; QA team; daily stand-up	every miss ticketed to the owning team; re-test the same container
Bug bash	job-story ranges assigned per person → issues logged	every other week; whole team	UI, data and performance in one pass
Shadow accounts	legacy-client uploads fed into dedicated demo accounts → side-by-side read	ongoing; two named clients first	real client data exposes defects synthetic data hides
UAT	opens at ≥80% minor-to-no issues → semi-weekly syncs (daily in November)	business users	quality gate, not calendar
Go / no-go ×2	stable production · first-client DQ 80%+ · no show-stoppers	business and tech leadership	evidence pack prepared in advance
Waves	6 criteria → next wave; ≥50 containers calibration; monthly go/no-go	program + business	hold in public if a criterion misses

Risks and mitigations. Environment outages (three environments; a "path to green" process to protect UAT availability); late monitoring (per-hop ratios and connectivity monitors — built later than they should have been); optimistic estimates (the lesson). **Lean / Kaizen mapping.** The ledger is genchi genbutsu — trace the real unit; the UAT gate is jidoka — stop the line on data quality; the criteria are standard work; the monthly go/no-go is PDCA at wave grain; defects found at the bottom cost a ticket, found by a client they cost trust.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Was the pilot late, and whose miss was it?**
>
> The first testing plan said early November and it launched in December; the program's retrospective lists delays, optimistic testing estimates and late monitoring, and I was one of its two TPMs, so I own my share of the estimate. I won't say how much was mine against the co-TPM's — I wasn't the only estimator. I now size test effort on hop count and put the monitors in before the debugging starts.

> **What does your ledger actually prove?**
>
> Where to look first, not a defect rate. It is a partial working log: 142 marked tested, 105 with results, 59 failing. By first-failing hop, route (23) and milestone (18) hold 41 of 59 — 69.5% — and only 3 of 105 passed all seven hops. To estimate a rate I would re-run a fresh random 100 and log every hop.

Why is entry at 80% but graduation at 90%?

The bars answer different questions. Entry (first-client data quality 80%+, no show-stoppers) says "safe enough for a calibrated client"; graduation (90%+ at least as accurate as the legacy product, 80%+ milestone estimates, 90% SLA) says "proven enough for the next wave" (record-from-docs).

Why at least 50 containers to calibrate?

Because 50 detects a 5% defect 92.3% of the time (1 − 0.95^50), and 100 detects it 99.4% (1 − 0.95^100). Fifty is the cheapest sample that gives the data team a verified baseline on the client's own lanes before onboarding.

How did you decide UAT could open?

On the 80% minor-to-no-issues gate: business testers are further from product development, so opening earlier would have had them rediscovering known bugs. The gate was data quality, not the calendar — which is also why it moved the date.

Is 500 containers in 20 minutes a lot?

25 containers a minute; 99.6% connected is 498 of 500. At the S11 illustrative 80,000 containers a year (≈219 a day) one 20-minute run is ≈2.3 days of average volume — headroom, not proof of peak behavior, and I said so.

What would you change about the testing plan?

Monitors before debugging, effort sized on hop count, the trace ID agreed in design rather than after the first slow bug hunts, and the three environments treated as the critical path — the record's own list of what went wrong.

BUSINESS IMPACT MODEL — KPI TREE

```
Launch readiness: no client touches the product until the evidence says it is ready
+- GATE 1  100-container live test: 7 hop checks per container, every miss a ticket
+- GATE 2  bug bashes every other week: job-story ranges per person
+- GATE 3  shadow accounts: real legacy-client data beside the old product
+- GATE 4  UAT opens at >=80% of shipments minor-to-no issues
+- GO / NO-GO x2 (business + tech): stable prod, first-client DQ >=80%, no show-stoppers
+- WAVES   2 -> 4 of 7 -> 20+: six criteria; >=50 containers of calibration per new client
+- RESULT  500 containers / 20 min / 99.6%; zero rollbacks; customers 7 -> 68 (9.7x)
(loop)  hop ledger -> first failing hop -> right team same day -> fewer escapes -> shorter next cycle
(miss)  optimistic estimate + late monitors -> pilot: first plan early Nov, launched Dec; bugs took hours
to localize
```

SOLUTION FLOW

```
random container/MBL -> trace 7 hops -> log row -> first failing hop?
   -> Jira ticket to the owning team -> daily stand-up -> fix -> re-test the same container
shadow account (real client data) -> side-by-side with the old product -> defects
UAT opens at >=80% minor-to-no issues -> semi-weekly syncs
GO / NO-GO (business + tech) -> wave 1 -> six criteria -> monthly go/no-go
   -> wave 2 (>=50 containers calibration first) -> wave 3 -> full roll-out
```

SENSITIVITY — LOW / BASE / HIGH

TRUE DEFECT RATE (ILLUSTRATIVE)	FORMULA	N = 50	N = 100	N = 300
1%	1 − 0.99^50	39.5%	63.4%	95.1%
3%	1 − 0.97^n	78.2%	95.2%	99.99%
5%	1 − 0.95^n	92.3%	99.4%	≈100%

Probability that a sample of n containers contains at least one defective container. The record sample sizes are 50 (calibration), 100 (launch gate) and 300 (pre-production); defect rates are illustrative.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

Honestly, the testing layer has no record dollar and I don't invent one. What it has is a gate that held: a hundred-container live test traced hop by hop, shadow accounts on real client data, UAT that stayed closed until eighty percent of shipments showed minor-to-no issues, and six published criteria for each wave — with a fifty-container calibration first, which detects a five-percent defect more than nine times in ten. The load proof was five hundred containers in twenty minutes at ninety-nine point six percent, both go/no-go checkpoints passed, and no wave rolled back. The ledger told us where to look — forty-one of fifty-nine failing containers broke first at route or milestone — and it is a working log, so I use it to aim the fix, not to estimate a rate. The honest cost is the schedule: my testing estimate was optimistic and monitoring came late, so the pilot launched in December, later than the first testing plan. I would take that trade again, and I would instrument first.

IF CHALLENGED — THE FULL ANSWERS

"You had zero rollbacks — isn't that just luck with only three waves?"

Zero events in three waves proves little statistically — the 95% upper bound on a per-wave rollback rate is 1 − 0.05^(1/3) = 63%. The evidence is the gates' detection power (92.3% at 50 containers for a 5% defect, 95.2% at 100 for 3%) and that exposure was bounded: wave 1 was two clients, not the whole seven-client pool.

"Why trust a log with blanks?"

I don't use it as a sample. I use it to rank where to look: 59 failing rows, first failure at route or milestone in 41. To estimate a rate I would re-run a random 100 with every hop recorded.

"What did the later launch cost?"

The record does not price it and the pilot was free, so there was no revenue delay to claim. The cost was engineering capacity, client expectation and credibility — I do not put a number of weeks on it, because the documents disagree — which is why the lesson is about estimates and monitoring, not about the gate.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I defined "ready" for a visibility launch: four test gates — a hop-by-hop 100-container ledger, bug bashes, shadow accounts on real client data and UAT that opened only at 80% data quality — and six graduation criteria per wave. The gates held and nothing rolled back; the honest miss was optimistic test estimates and late monitoring, which put the launch in December, later than the first testing plan.

NACF BRIDGE — ONE SENTENCE

A new inbound node launches the same way: trace real units hop by hop (ASN → appointment → door → receive → putaway → available), publish the entry gate and a calibration volume per node before the first trailer, and put the monitors in before the first week of debugging — doors × hours × unload rate only means something once each hop is measured.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a time when you had to make a decision between standards and delivery. What tradeoffs did you have make? What was the outcome? Knowing what you know now, would you have done anything differently?" AIQB: Highest Standards Q4 **Insist on the Highest Standards** FACET: THE UAT GATE

Open (first two sentences): "We kept business UAT closed until eighty percent of shipments showed minor-to-no issues. Opening it on the calendar would have had business testers rediscovering bugs we already knew about."

90-second STAR. S A new-stack launch with a calendar everyone wanted. **T** Make "ready" a rule, not a feeling. **A** I helped define four gates and set UAT's entry at 80% minor-to-no issues, with graduation criteria for each wave published before wave 1; the first-client go/no-go read stable production, 80%+ data quality and no show-stoppers. **R** Both checkpoints passed and nothing rolled back — and the price was real: the pilot launched in December, later than the first testing plan.

Land: UAT entry 80%; six graduation criteria · **Pivot to NACF:** a new node does not take inbound until its entry gate is met — doors staffed, WMS receiving verified, first trailers traced — whatever the launch calendar says.

Do NOT say: that the later date was "worth it" without naming the cost; any dollar for the gate.

B2 · "Tell me about a time when you evaluated the customer experience of your product or service. What did you do? What was the result?" AIQB: Customer Obsession Q5 **Customer Obsession** FACET: SHADOW ACCOUNTS AND CALIBRATION

Open (first two sentences): "We ran two real clients' live data from the old product through the new one in demo accounts, side by side. And every new client gave us at least fifty containers of their own freight before they onboarded."

90-second STAR. S Synthetic tests were not going to show what a client's lanes and carriers actually do. **T** Find the defects on the client's own freight, before the client sees them. **A** Shadow accounts fed the legacy-product uploads of two named clients into dedicated demo accounts; wave-2 clients supplied a 50-container sample for calibration; the six criteria included CSAT on par with the consignee average. **R** Defects surfaced in our accounts, not theirs; the onboarding was light-weight because the product had calibrated on their freight.

Land: ≥50 containers → 92.3% chance to detect a 5% defect · **Pivot to NACF:** a node's first inbound sample — a trial volume per node — calibrates receiving and putaway before the node takes real peak volume.

Do NOT say: client names beyond what is confirmed; "customers loved it" — the record has no such claim.

B3 · "Tell me about a time when you were trying to understand a complex problem on your team and you had to dig into the details to figure it out. Who did you talk with or where did you have to look to find the most valuable information?" AIQB: Dive Deep Q1 **Dive Deep** FACET: THE HOP LEDGER

Open (first two sentences): "We traced real containers through seven hops and logged every one. When I counted by first failing hop instead of by column, forty-one of fifty-nine failures were at just two hops."

90-second STAR. S Six-plus system layers and a pile of defects, with every team seeing only its slice. **T** Locate the defects instead of counting symptoms. **A** I read the log by first-failing hop because downstream hops cascade: route (23), milestone (18), data back (7), exception (6), created (5). I took the top two to the owning teams first and kept the caveat — it is a working log, not a sample. **R** The fixes were aimed at route and milestone first; the log became the standing checklist.

Land: 41 of 59 first failures at two hops (69.5%) · **Pivot to NACF:** a receiving-defect Pareto by first failing step, not by total flags — so one dock-scheduling fault is not counted four times downstream.

Do NOT say: that the log estimates a defect rate; the share as precise.

B4 · "We all have things about ourselves we'd like to improve on at work. Give me an example of something that you've worked on to improve your overall work effectiveness. What resources did you identify to help you develop? What was the impact?" AIQB: Learn and Be Curious Q4 **Learn and Be Curious** FACET: THE HONEST MISS

Open (first two sentences): "The first testing plan said early November; the pilot launched in December. The program's own retrospective says our testing estimates were overly optimistic and monitoring came too late — and I was one of its two TPMs."

90-second STAR. **S** A launch across nine teams on a new stack. **T** Land a safe pilot; the estimate was part of the job. **A** I held the data-quality gate and accepted the later date rather than launch on the plan. The slow part was debugging: data crossed six-plus layers with little monitoring, and bugs took hours to localize. We added a trace across hops and per-hop ratios. **R** The pilot launched clean; the lesson is to size testing on hop count and instrument before debugging starts.

Land: first plan early November → launched December 2021 · **Pivot to NACF:** size a new-node launch plan on the number of hand-offs, and put the receiving monitors in before the first inbound wave.

Do NOT say: blaming other teams; "about a month" as a precise figure — the record is month-level.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: "Of fifty-nine failing containers, forty-one first broke at one of two hops — and only three of a hundred and five passed all seven."

Headline metric: 41 of 59 (69.5%) at route + milestone; 3 of 105 all-seven-clean

- Read by first-failing hop, not by column.
- Order: route 23, milestone 18, data back 7, exception 6, created 5.
- Caveat first: partial log, blank = not recorded.
- Next step: a fresh random 100 with every hop logged.

Conflict / influence / backbone angle **Ownership**

Opener: "Nine teams, none reporting to me, each with its own backlog — I had to get every one to instrument its own hop in the middle of a launch crunch."

Headline metric: nine teams aligned on one seam; time to localize cut about 50% (secondary source)

- The seam nobody owned: no unique ID from downstream events to the screen.
- The ask was framed in each team's own bug queue, not the program's need.
- The sell was arithmetic: a few hours per blocker, every team paying it.
- Source: the 3PL ingestion one-pager (TPM across 6 systems and 9 teams; no unique ID; 2–4 hours to find the source system per blocker; blocker-bug time-to-resolve cut about 50%).

Honest failure-and-learning angle **Learn and Be Curious**

Opener: "The first testing plan said early November and we launched in December, and the program's own retrospective says why: testing estimates were optimistic and monitoring was late."

Headline metric: first plan early November → launched December 2021; hours to localize a bug at six-plus layers

- What I would do differently: size testing on hop count; monitors before debugging.
- What the record calls a win, I read as a warning: the environment was built "through long hours" — optimistic estimates are paid for by people.
- What held: the gates — no wave rolled back.
- No dollar claimed.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"With only three waves, how do you know your gates caused zero rollbacks — and not luck?"

Numeric answer. I don't, statistically: zero events in three waves gives a 95% upper bound of 1 − 0.05^(1/3) = 63% on a per-wave rollback rate. The case rests on detection power — 92.3% at 50 containers for a 5% defect, 95.2% at 100 for 3% — and on bounded exposure: wave 1 was two clients, not seven.

Say exactly: “Zero in three waves proves almost nothing on its own — the upper bound is sixty-three percent. What I can show is the gates' detection power and that wave one was two clients, not the whole pool.”

"Your ledger is a partial log. Why should I trust the Pareto?"

Numeric answer. Because I only claim what it supports: 105 rows had results, 59 had a failing hop, 41 of those first failed at route or milestone. Blank is not "pass," and a working log over-records failures, so the shares are indicative; the robust finding is that failures sat downstream of ingestion. I would re-run a random 100 with all seven hops logged.

Say exactly: “It's a working log, not a sample — blank means not recorded. I use it to decide where to look first, not to estimate a defect rate.”

"The pilot launched later than planned. What number would have told you earlier?"

Numeric answer. Hours to localize a blocker: the record says bugs took hours across six-plus layers (2–4 h in the secondary telling). A weekly chart of that figure against a ≤1 h target, from week one, would have shown the monitoring gap before the date did.

Say exactly: “The number I should have charted from week one was hours to localize a blocker — it was the leading indicator, and the date was the lagging one.”

S11-C — Flexport — Retiring an acquired product's book: a segment-by-segment RAPID decision for 79 Crux customers — 55 onto Shipment Visibility, 24 on rule-based exits, and three customers carrying 75% of the containers NEW

LPs this story can serve here: Customer Obsession (primary) · Success and Scale Bring Broad Responsibility · Earn Trust · Have Backbone, Disagree and Commit
Process family: Deliver — customer transition and product retirement; segmentation, decision rights (RAPID) and cut-over planning

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Customer Obsession Every one of the 79 customers got a destination by segment — transition, partner route, or API-or-cancel with dates — instead of an end-of-life email.

Success and Scale Bring Broad Responsibility Three truck brokers were 3.8% of customers and 75.0% of containers, so they got a per-customer partner route; 39 small accounts at ≈$770 a year each got the standard path.

Earn Trust The decision document kept the dissent on the page — cost of two systems, an unwritten parity list, "ejecting customers is never a good experience" — and cancellations ran on published dates.

Have Backbone; Disagree and Commit The decider committed the organization with the objections recorded, and each impacted party committed to perform — nobody smoothed the disagreement away.

Honesty label. This is a decision-document story, and the seat matters: the RAPID table names the Recommend, Agree, Perform, Input and Decide people, and the decider is an executive — *the decision was not mine*. My seat, from my own workbook (Apple v11 A4): I owned the mitigation of the decommission risk — I put it on the risk register as the top risk, rebuilt the comparison apples to apples, priced the 79-customer base per segment, and wrote the one-canonical-source rule; the transition document's owners built the per-customer paths with me, and the decision itself was the executive decider's. The segment table is the doc's; the footing, concentration arithmetic, break-even rule and parity matrix are my analysis. **Footing flag:** the doc's printed total row ($15,491 MRR / $185,892 ARR) equals the sum of only the six non-transition segments; all nine rows sum to $21,246 MRR / $254,952 ARR, and the appendix lists 77 names against 79 counted (37 SMB names vs 39) The number I say in the room is the printed one — 79 customers, $185,892 ARR, 294,581 containers (Apple A4 prints the same) — and I add that the printed total equals six of the nine rows. Overlap: the loop packet tells this as S10 for another interviewer.

WHAT THE RECORD SAYS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The opportunity. Migrate Crux clients to Shipment Visibility by the end of 2022. GKA, Enterprise and SMB customers transition and are upsell opportunities; for Motor Carriers, Truck Brokers, Freight Forwarders, Customs Brokers, 3PL and SCM SaaS the choice was cancel now, or keep serving them on Crux while working out missing features, support and Data Services cost-to-serve. RECORD-FROM-DOCS

The decisions. Truck brokers and motor carriers: transition AV Logistics and Loadsmart to partners in Transmission (like ITS, already there); cancel the remaining motor carriers at wind-down. Freight forwarders, customs brokers, 3PL: no new investment — recommend the SV API (no extra dev), else cancel at wind-down. SCM SaaS: cancel both — dated August 1 and October 24, 2022; the customs broker was cancelled September 1, 2022. A named executive decides; two people build the per-segment transition plans and review them with the exec team. RECORD-FROM-DOCS

The dissent. One reviewer wanted the cost of maintaining two systems and the stalled "Crux Absorption" project brought into the decision; another asked what Crux costs to keep running, whether the missing-feature list is written down, and warned that ejecting customers is never a good experience and that cancellations should be timed to win the competitors' consignees; others argued to cut direct competitors and to keep only partners. The 2021 plan adds that about 75% of Crux revenue came from clients who are not trade clients, so Crux's front end could not be sunset until the VaaS portal was stood up. RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport had acquired a visibility product, Crux, and the goal was to move its clients to Shipment Visibility by the end of 2022 — but 79 paying customers and about 295 thousand containers a year were on it, and not all of them fit. The easy versions were an end-of-life email or running two systems forever. **Task.** Produce a decision rule per customer type that a decider could commit the company to, with the dissent visible. **Action.** I segmented the 79 into nine types and read each on three questions: does Shipment Visibility serve them, are they a partner or a competitor, and does serving them need extra build. Then I checked concentration: three truck brokers were 4% of customers and 75% of containers, so they got a per-customer partner route; the 39 small accounts, about $770 a year each, got the standard path. The decision: 55 customers to Shipment Visibility, three brokers to partners in Transmission, forwarders and 3PLs to the SV API or cancel at wind-down, nine motor carriers cancel at wind-down, and three dated cancellations. The dissent — cost of two systems, an unwritten parity list — stayed on the page. **Result.** Primary metric: 79 of 79 customers with a destination and a date — 55 + 3 + 9 + 9 + 3 — and a table footed by container share, because the printed ARR total summed only six of nine rows.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “The document prints seventy-nine customers, $185,892 of annual revenue and 294,581 containers; I add that the printed revenue total equals six of its nine segment rows, so I lead with containers.”

“What's your source for that?” “The base is the transition document's own table, which my workbook also quotes; the footing, the three-brokers concentration and the break-even are my arithmetic on that table, and I label them that way.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Flexport's acquired visibility product, Crux, still billed 79 customers — 294,581 containers in 2021 — while Shipment Visibility was in pilot. The aim was to migrate Crux clients by the end of 2022, but the decision document itself says not all customers fit the new service and a plan was needed. Keeping both meant two systems, double maintenance and a prospect set comparing Flexport against itself; closing Crux carelessly meant stranding customers who used it as infrastructure — and the plan's own dependency was that non-trade clients could not use the Client App at all until the VaaS portal existed.

T — Own the segmentation analysis and the per-segment recommendation a decider could commit to — transition, partner route, API-or-cancel, or cancel, each with a date — with the cost of two systems and the parity gaps on the page, not behind it.

Baselines on the table: 79 customers · 294,581 containers (2021) · nine segments · doc-printed ARR total $185,892 (six non-transition rows) vs nine-row sum $254,952 · 12 customers flagged API · 6 flagged churned in the appendix · goal: migrate by end of 2022 · ~75% of Crux revenue from non-trade clients (2021 plan).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — footed the customer table (count, containers, ARR, MRR) and read each segment's use, revenue and containers; separated partners from competitors; listed what the plan itself said Crux customers could not do on SV. *Exit bar:* segment table reconciled to the appendix names; the footing differences written down.
- **Plan & Design** — a three-axis rule — does SV serve them, partner or competitor, extra build needed — mapped to five decisions; concentration analysis (brokers vs the long tail); the parity matrix drafted; the open questions listed (keep-alive cost, written parity list). *Exit bar:* RAPID roles named: Recommend, Agree, Perform, Input, Decide.
- **Build & Test** — collected stakeholder input verbatim, including the dissent; ran per-customer conversations for the large accounts with the trucking group; drafted the per-segment transition plans. *Exit bar:* the decider can read every objection beside the recommendation.
- **Cutover & Go-Live** — the decider committed the organization; each impacted party committed to perform; the SV-bound segments entered through the pilot waves; competing SCM SaaS and the customs broker were cancelled on dates. *Exit bar:* dated actions on the page: August 1, September 1 and October 24, 2022.

- **Hypercare / Close** — API-or-cancel conversations for forwarders and 3PLs; partner conversations for the brokers; wind-down of the Crux front end as VaaS / API paths landed; churn flagged in the appendix tracked. *Exit bar:* every customer on a path with a date; Crux wound down by the end-2022 goal.

Influence and ownership. Mine (as I would tell it): the segmentation, the concentration read, the plan inputs and the parity questions. Not mine: the recommendation owners, the decider and the per-segment transition plans' authors named in the RAPID table; the trucking group owned the partner conversations. I influenced by putting the numbers and the dissent on one page.

> **Designed the process; others own it now.** The segment rules, the dated cancellations and the parity questions live in the decision document, not in my head; the transition plans have named performers. **Trade-off put to leadership.** Revenue surrendered (the six non-SV segments, $185,892 of printed ARR) against two systems, double maintenance and competing with ourselves — and a keep-alive cost the record never priced.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — the doc's printed $185,892 ARR is the revenue on the six non-SV segments (modeled footing — mine); the three SV-bound segments add $69,060, for $254,952 across all nine rows. No Crux keep-alive cost exists in the record, and I do not invent one — the break-even is $15,491 a month (modeled).
- **#** — 79 of 79 customers with a decision: 55 to SV, 3 to Transmission partners, 9 SV-API-or-cancel, 9 cancel at wind-down, 3 cancelled now on dates (record-from-docs; tally mine).
- **%** — brokers 3.8% of customers and 75.0% of containers; SV-bound segments 69.6% of customers, 13.3% of containers, 27.1% of ARR (my footing of the doc table).
- **Time** — target: migrate Crux clients by end of 2022; cancellations dated August 1, September 1 and October 24, 2022 (record-from-docs).
- **Customer** — each customer type has a stated destination; the largest accounts got partner conversations, not a template; six customers were already flagged churned in the appendix.
- **Quality** — two open questions flagged rather than hidden — keep-alive cost and a written feature-parity list; the doc's own footing differences caught and reconciled.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	An acquired product, nine customer types and no stated rule for who fits. I built the rule (serve? partner or competitor? extra build?) and the concentration read from the data.
Scope & influence	Executives decided; product, trucking, data services and finance gave input; I influenced by putting numbers and dissent on one page — a recommendation to leaders, one to two levels up.
Execution trade-off + mechanism others now own	Revenue for simplicity, with the cost of keeping both unpriced in the record. Mechanism others own: the segment rules, the dated cancellations and the decision document.
Impact (long-term)	One visibility product to run, the acquired book moved or exited deliberately, the partner route for the largest accounts preserved.
Process improvement (standard work institutionalized)	A reusable product-retirement template: segment by use, find concentration, rule per segment, dated actions, dissent recorded, commitments confirmed.
Complexity & consensus	Real dissent (cost of two systems, parity, customer experience, competitor timing) resolved by a named decider with commitments from each impacted party.
Root cause at scale	The root question was "who is this product for" — the plan's non-trade clients could not use SV, which is why the rule is by segment, not by customer.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Nine-segment table; footing (printed total vs nine rows); concentration (3 brokers = 75.0% of containers); ARR per customer 60× spread; API and churn flags.
2 · Solution design	Cancel all vs keep all vs segmented rules; broker route options; parity matrix; decision rights via RAPID.
3 · Math & data logic	Tally 55 + 3 + 9 + 9 + 3 = 79; shares 69.6 / 13.3 / 27.1% vs 30.4 / 86.7 / 72.9%; break-even $185,892 / 12 = $15,491 a month.
4 · Implementation	Per-segment plan, dated cancellations, commitments confirmed, SV pilot waves for the 55, partner conversations for the three.
5 · Probe questions	Seven numeric probes plus three trap drills: which total, will the brokers accept, what number would flip the cancel decision.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What I read. The doc's segment table (customers, MRR, ARR, 2021 containers) and its appendix of names, with API and churn flags RECORD-FROM-DOCS. I added three columns the doc does not carry — ARR per customer, share of containers and the decision — and footed everything from the rows.

SEGMENT	CUSTOMERS	ARR (DOC)	CONTAINERS 2021	ARR / CUSTOMER	% OF CONTAINERS	DECISION (DOC)
GKA	2	$16,500	17,061	$8,250	5.8%	transition to SV
Enterprise	14	$22,560	17,983	$1,611	6.1%	transition to SV
SMB	39	$30,000	4,153	$769	1.4%	transition to SV
Motor Carrier	9	$17,760	12,359	$1,973	4.2%	cancel at wind-down
Truck Broker	3	$139,512	221,044	$46,504	75.0%	Transmission partner route (else cancel)
Freight Forwarder	5	$14,820	11,871	$2,964	4.0%	SV API, else cancel at wind-down
Customs Broker	1	$600	432	$600	0.1%	cancel now (dated)
3PL	4	$2,400	4,218	$600	1.4%	SV API, else cancel at wind-down
SCM SaaS	2	$10,800	5,460	$5,400	1.9%	cancel now (dated)

What it revealed. (1) *Footing:* customers foot to 79 and containers to 294,581, but the printed $185,892 ARR is only the six non-transition rows — the nine rows sum to $254,952, so I lead with container shares, which foot. (2) *Concentration:* three truck brokers are 3.8% of customers, 75.0% of containers and 54.7% of nine-row ARR. (3) *Spread:* ARR per customer runs from $769 (SMB) to $46,504 (truck broker) — 60.5×. (4) *The SV-bound 55 are a long tail:* 69.6% of customers but 13.3% of containers and 27.1% of ARR — light-weight onboarding fits; the brokers need per-customer work. (5) The 2021 plan's "~75% of Crux revenue is non-trade" sits within two points of my 72.9% for the six non-SV segments. (6) Flags: 12 customers carry an API flag (5 of them SV-bound) and 6 are flagged churned. I quote 79 customers as printed (Apple A4 prints 79 too); I do not quote a per-customer average beyond the SMB tail.

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Whole book	Cancel everything not on SV now · keep every customer on Crux until parity · segmented rules	Segmented rules	Cancel-all strands the large partner-like accounts; keep-all leaves two systems running with no end date and misses the end-2022 goal.
Truck brokers and motor carriers	Standalone service · white-label / partner via Transmission · cancel	Partner route for brokers; cancel the nine motor carriers at wind-down	The 2021 strategy doc gives no standalone service for these types; ITS already uses Transmission.
Forwarders, customs, 3PL	Build something extra · SV API (no extra dev) · cancel	SV API, else cancel at wind-down	No new investment; they are not competitors in the doc's reading and some give intel on their customers.
SCM SaaS	Explore data partnerships · cancel now	Cancel now (dated)	They could undermine selling our own branded services; one went through legal review.
Decision rights	Consensus · RAPID with a named decider	RAPID	Recommend, Agree, Perform, Input, Decide named; each impacted party commits.

Parity matrix (drafted from the documents; gaps flagged, not guessed)

DIMENSION	CRUX TODAY	SHIPMENT VISIBILITY	GAP AND HANDLING
Who can use it	Any customer type — shippers, carriers, brokers, forwarders, 3PL, SaaS	Client App for trade clients; non-trade clients cannot use it until a VaaS portal exists (2021 plan)	Rule by segment; non-trade types go API, partner or cancel
How tracking is requested	Front-end manual search and API	Single input, CSV or API in the Client App	Manual search is not offered on the VaaS-portal plan — a documented gap
Data-quality bar	The incumbent benchmark	Graduation: at least as accurate as Crux ≥90% of the time	Built into the six graduation criteria
Human in the loop	not in the documents — say "I don't know how Crux handled exceptions" VERIFY-WITH-RAJESH · V3	Tickets to Data Services when auto-tracking fails	Unknown on the Crux side
Webhooks / API	12 customers flagged API	Public API and client webhooks	Decision: "SV API, no extra dev effort"
Price	MRR by segment, $50–$11,626	Free pilot ≈6 months, then the SV charge model	The price step is the thing to explain to the SV-bound 55
Feature-parity list	A reviewer asked: "written down somewhere?"	—	Not in the record — the document records the question, not an answer; I do not claim the list existed in writing

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers (footing the doc's own table)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Customers	`2 + 14 + 39 + 9 + 3 + 5 + 1 + 4 + 2 = 79`	record-from-docs rows	79
2	Containers, 2021	`17,061 + 17,983 + 4,153 + 12,359 + 221,044 + 11,871 + 432 + 4,218 + 5,460 = 294,581`	record-from-docs rows	294,581
3	ARR, all nine rows	`16,500 + 22,560 + 30,000 + 17,760 + 139,512 + 14,820 + 600 + 2,400 + 10,800 = $254,952`	my footing of the doc rows	$254,952
4	The printed total row	`17,760 + 139,512 + 14,820 + 600 + 2,400 + 10,800 = $185,892; $254,952 − $185,892 = $69,060 = the three SV-bound rows`	my footing: the printed ARR is the six non-transition rows	$185,892 (six rows)
5	MRR cross-check	`$21,246 (nine rows) − $15,491 (printed) = $5,755; $5,755 × 12 = $69,060`	my footing	$5,755 MRR missing from the printed total

Table 2 — The fix: the rules, the tally and the concentration, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Decision tally	`55 (SV) + 3 (Transmission partners) + 9 (API-or-cancel) + 9 (cancel at wind-down) + 3 (cancel now) = 79`	record-from-docs decisions; tally mine	79 of 79 with a destination
2	Broker concentration	`3 / 79 = 3.8% of customers; 221,044 / 294,581 = 75.0% of containers; 139,512 / 254,952 = 54.7% of ARR`	my footing	3.8% → 75.0% / 54.7%
3	ARR per customer spread	`$30,000 / 39 = $769 (SMB); $139,512 / 3 = $46,504 (brokers); 46,504 / 769 = 60.5×`	my footing	60.5×
4	SV-bound segments	`55 / 79 = 69.6% of customers; 39,197 / 294,581 = 13.3% of containers; 69,060 / 254,952 = 27.1% of ARR`	my footing (39,197 = 17,061 + 17,983 + 4,153)	69.6% / 13.3% / 27.1%
5	Exit and re-route segments	`24 / 79 = 30.4%; 255,384 / 294,581 = 86.7% of containers; 185,892 / 254,952 = 72.9% of ARR`	my footing (255,384 = 294,581 − 39,197)	30.4% / 86.7% / 72.9%

Table 3 — After: where each customer goes (customers · containers)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	To Shipment Visibility	`GKA 2 + Enterprise 14 + SMB 39 = 55 customers; 39,197 containers`	record-from-docs decision	55 · 39,197
2	To partners in Transmission	`truck brokers: 3 customers; 221,044 containers (else cancel at wind-down)`	record-from-docs decision	3 · 221,044
3	SV API, else cancel at wind-down	`forwarders 5 + 3PL 4 = 9 customers; 11,871 + 4,218 = 16,089 containers`	record-from-docs decision	9 · 16,089
4	Cancel at wind-down	`motor carriers: 9 customers; 12,359 containers`	record-from-docs decision	9 · 12,359
5	Cancel now (dated)	`customs 1 + SCM SaaS 2 = 3 customers; 432 + 5,460 = 5,892 containers`	record-from-docs decision	3 · 5,892
6	Footing check	`39,197 + 221,044 + 16,089 + 12,359 + 5,892 = 294,581`	arithmetic	every container accounted for

Table 4 — The dollars, every addition shown (ARR; pilot is free, so none of it is "saved")

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	ARR on the SV-bound segments	`$16,500 + $22,560 + $30,000 = $69,060`	my footing of the doc rows; flows into the charge model after the pilot	$69,060
2	ARR on the exit and re-route segments	`$17,760 + $139,512 + $14,820 + $600 + $2,400 + $10,800 = $185,892`	the doc's printed total, footed	$185,892
3	Of which hard-cancel	`$17,760 (motor carriers) + $600 (customs) + $10,800 (SCM SaaS) = $29,160`	my footing	$29,160
4	Of which conditional on SV-API uptake	`$14,820 (forwarders) + $2,400 (3PL) = $17,220`	my footing	$17,220
5	Of which the partner route	`$139,512 / $185,892 = 75.1% of the exit ARR; 29,160 + 17,220 + 139,512 = $185,892`	my footing	$139,512
6	Break-even keep-alive cost	`$185,892 / 12 = $15,491 per month; if running Crux for these segments costs more, cancelling dominates on direct economics`	arithmetic on the printed total; the keep-alive cost is not in the record, and I do not state one	$15,491 / month

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

SEGMENT	ACTION	TIMING	OWNER (RECORD) / CONTROL
GKA · Enterprise · SMB (55)	Transition through the pilot waves; upsell opportunity; light-weight onboarding; API customers (5) need an integration path	pilot waves through H2 2022	transition-plan authors; SV program
Truck brokers (3)	Partner in Transmission (white-label, like ITS); cannot track non-Flexport containers there; else cancel at wind-down	with the Crux wind-down	trucking group; per-customer conversations
Forwarders · 3PL (9)	SV API (no extra dev) or cancel at wind-down	at wind-down	transition-plan authors
Motor carriers (9)	Cancel at wind-down	at wind-down	transition-plan authors
Customs (1) · SCM SaaS (2)	Cancel now	Sept 1 · Aug 1 · Oct 24, 2022	dated in the document

Controls and risks. Commitment step: each impacted party commits to perform, verbally in the decision meeting or by email. Risks named in the record: cost of two systems and the stalled Crux Absorption project; an unwritten parity list; legal concerns over web scraping; cancellations that hand a competitor's consignees to someone else unless timed. **Lean / Kaizen mapping.** Segmenting by what flows through the process is value-stream thinking; the dated cancellations are standard work with a due date; the open questions are the PDCA "Check" I would run first.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Why not cancel everyone who is not on SV and be done?**
>
> Because the three brokers are 75.0% of containers and 54.7% of nine-row ARR — ITS already uses Transmission and the decision routes AV Logistics and Loadsmart there too. Cancelling them would strand the largest flows to save a template. The other 21 customers are exactly the ones the decision exits or re-routes.

What does it cost to keep Crux alive?

The record has no figure — a reviewer asked the question and it is open. I will not invent one; I give the break-even: the printed total of $185,892 a year is $15,491 a month, so if keeping Crux alive for those six segments costs more than that, cancelling dominates on direct economics. The decision also carried strategic inputs the break-even ignores.

Your totals do not foot. Which ARR is right?

Both, once you say what they are: the printed $185,892 is the six non-SV rows; all nine rows sum to $254,952, and the $69,060 difference is the three SV-bound segments. I lead with container shares because they foot to 294,581, and the appendix lists 77 names against 79 counted — I would confirm that before quoting a per-customer figure.

How many of the 55 will actually move?

The record does not say how many did. What I can show is the shape of the problem: 55 customers, $1,256 average ARR (69,060 / 55), 5 of them API integrations (9.1%) — a long tail suited to light-weight onboarding behind the pilot waves, with six already flagged churned across the whole book (7.6%).

Who disagreed, and what did they get right?

Four inputs: the cost of two systems and the stalled absorption project; a missing, unwritten parity list; "ejecting customers is never a good experience" and the timing of competitor cancellations; and cut-the-competitors. All four were right about something the decision then had to carry — the parity list is the one the record never closes.

How do you know the SMB path is cheap enough?

ARR per SMB customer is $769 against $46,504 per broker, a 60.5× spread — a bespoke migration for 39 accounts at that revenue does not pay, so the SMB path is the standard one. I would price the onboarding hours per account to confirm; there is no record figure.

What would you do on day one of a similar retirement?

Foot the table, find the concentration, write the rule per segment, put the dissent and the open questions on the same page, and date every action. Then price the keep-alive cost before the memo, not after.

BUSINESS IMPACT MODEL — KPI TREE

```
One visibility product to run; every acquired customer has a destination and a date
+- 55 -> Shipment Visibility (GKA 2, Enterprise 14, SMB 39): 69.6% of customers, 13.3% of containers,
27.1% of ARR
+- 3  -> Transmission partners (truck brokers): 3.8% of customers, 75.0% of containers, 54.7% of ARR
+- 9  -> SV API else cancel at wind-down (forwarders 5, 3PL 4)
+- 9  -> cancel at wind-down (motor carriers)
+- 3  -> cancel now, dated (customs 1, SCM SaaS 2)
+- Printed ARR $185,892 = the six non-SV rows; nine rows = $254,952 (footed)
+- Open: keep-alive cost (break-even $15,491 / month); written parity list
(loop)  segment by use -> find concentration -> rule per segment -> dissent on the page -> decider
commits -> dated actions
```

SOLUTION FLOW

```
foot the table -> customers 79, containers 294,581, ARR rows vs printed total
   -> axes per segment: SV serves them? partner or competitor? extra build?
   -> concentration: who carries the containers?
   -> RAPID: Recommend / Agree / Perform / Input / Decide
   -> dissent + open questions on the page -> Decide commits -> each party commits
   -> dated actions: SV waves | partner route | API-or-cancel | cancel now
   -> wind-down of the old front end by the end-2022 goal
```

SENSITIVITY — LOW / BASE / HIGH

KEEP-ALIVE COST (ILLUSTRATIVE)	PRINTED MRR − COST	DIRECT-ECONOMICS READ
$5,000 / month	$15,491 − $5,000 = $10,491	keep Crux for the six segments
$15,491 / month	$15,491 − $15,491 = $0	indifferent
$30,000 / month	$15,491 − $30,000 = −$14,509	cancel dominates on direct economics

The keep-alive cost is not in the record; the three values are illustrative bookends around the break-even. The strategic inputs — competitor timing, partner routes — are not in this arithmetic.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> Honestly, retiring Crux was not a money-making decision and I don't dress it as one. The doc's printed total, about one hundred eighty-six thousand a year, is the revenue on the six segments that don't move to Shipment Visibility — I footed the table myself, and all nine rows sum to about two hundred fifty-five thousand. What the analysis changed was the shape of the work: three truck brokers were four percent of customers and seventy-five percent of the containers, so they got a partner route; fifty-five customers — the long tail — move through the pilot waves; nine are API-or-cancel, nine motor carriers cancel at wind-down, three were cancelled on dates. Every customer has a destination and a date. The record doesn't price keeping Crux alive, so I give the break-even — about fifteen and a half thousand a month — and I say the strategic inputs sit outside it. Every dollar here is the document's, footed, or labeled; none of it is claimed as savings.

IF CHALLENGED — THE FULL ANSWERS

"You are telling me a $186K book was a good decision to exit?"

I'm telling you what the record shows: the printed total is the six non-SV segments, and the rule exits or re-routes them by segment, with the brokers — 75% of containers — taking a partner route. Whether the economics beat the keep-alive cost is exactly the number the record never priced; I give the break-even, $15,491 a month, and flag it.

"Why trust your footing over the doc?"

Because the rows foot and the printed total does not: customers foot to 79, containers to 294,581, but the printed ARR equals the six non-transition rows only — the other three rows add $69,060. That is arithmetic, and it is why I use container shares first.

"What did you actually decide?"

The decider decided. My part, as I tell it, was the segmentation, the concentration read and getting the dissent and the open questions on one page beside the recommendation.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

When Flexport retired an acquired visibility product, 79 customers needed a destination. I segmented them into nine types, found three truck brokers were 4% of customers and 75% of containers, and wrote a rule per segment: 55 to the new product, three to a partner route, the rest API-or-cancel with dates — with the dissent on the page.

NACF BRIDGE — ONE SENTENCE

Retiring or re-pointing a node or lane in a network is the same problem: segment what flows through it, find the concentration (a few lanes carry most of the volume), give each segment a destination and a date, and price keeping both running — a placement trade-off of cost-to-serve against distance-to-customer.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a time when you had to balance the needs of the customer with the needs of the business. What did you do? What was the result?"

AIQB: Customer Obsession Q8 | **Customer Obsession** | FACET: SEGMENTATION AND DESTINATIONS

Open (first two sentences): "Every one of seventy-nine customers got a destination, not an end-of-life email. I segmented them by what they actually were and wrote a rule per segment."

90-second STAR. **S** An acquired product with 79 customers and a goal to move them by the end of 2022 — and not every customer fit. **T** Give each customer a path, not a notice. **A** Nine segments, three questions each — does SV serve them, partner or competitor, extra build — giving 55 to Shipment Visibility, three brokers to a partner route, nine API-or-cancel, nine cancel at wind-down and three cancelled on dates. **R** 79 of 79 with a destination and a date; the largest accounts got conversations instead of a template.

Land: 79 of 79: 55 + 3 + 9 + 9 + 3 · **Pivot to NACF:** retiring a lane or node: every flow through it gets a destination and a date before the old path closes.

Do NOT say: "nobody was stranded" or "zero escalations" — the record does not say; dollars as savings.

B2 · "Tell me about a time when you considered everyone who was affected by your work, not just your direct customers or people within your organization."

AIQB: Broad Responsibility Q5 | **Success and Scale Bring Broad Responsibility** | FACET: CONCENTRATION

Open (first two sentences): "Three customers were four percent of the book and seventy-five percent of the containers. A template would have been irresponsible for them and overkill for thirty-nine small accounts."

90-second STAR. **S** A single end-of-life policy would have treated 79 customers alike. **T** Match the effort to what each customer needs and what is at stake. **A** I computed concentration: brokers 3.8% of customers, 75.0% of containers; ARR per customer from $769 (SMB) to $46,504 (broker), 60.5×. The brokers got a partner route and per-customer conversations; the tail got the standard path. **R** Effort followed stakes: partner route for the largest flows, standard path for the tail.

Land: 3.8% of customers = 75.0% of containers · **Pivot to NACF:** a Pareto of lanes or nodes before a network change — the few that carry the volume get bespoke handling.

Do NOT say: that the partner route succeeded — the record shows the decision, not the outcome.

B3 · "Describe a time when you felt really strongly about something on a project but the team decided to go in a different direction. How hard did you press the issue? How did you approach that project afterward?"

AIQB: Have a Backbone Q4 | **Have Backbone; Disagree and Commit** | FACET: THE DISSENT

Open (first two sentences): "The document carried real dissent: the cost of running two systems, a parity list nobody had written down, and a warning that ejecting customers is never a good experience. We put all of it on the page."

90-second STAR. **S** A recommendation to retire a product that others had reasons to keep. **T** Get a decision the organization would commit to, with the objections visible. **A** The RAPID named who recommends, who agrees, who performs, who gives input and who decides; the input section kept each reviewer's words, including the parity and keep-alive questions; the decider committed with those on the page, and each impacted party confirmed they would perform. **R** A decision with named performers and dated actions — and two open questions flagged instead of hidden.

Land: four dissenting inputs; two open questions · **Pivot to NACF:** a network decision memo — putting the cost of the status quo and the unresolved questions beside the recommendation, then committing.

Do NOT say: that I overruled anyone — the decider decided; or what my exact seat was without checking.

B4 · "Tell me about a time when you uncovered a significant problem in your team. What was it and how did you communicate it to your manager and other stakeholders? What did you do to address the problem?"

AIQB: Earn Trust Q6 | Dive Deep | FACET: FOOTING

Open (first two sentences): "When I re-footed the decision document's table, the customer and container counts added up but the dollar total did not — the printed ARR was only six of the nine rows."

90-second STAR. **S** A table everyone quoted: 79 customers, $185,892. **T** Make sure the numbers under a decision were right. **A** I summed each column from its rows: customers 79, containers 294,581, ARR $254,952 against the printed $185,892 — the difference, $69,060, is exactly the three SV-bound segments. I switched the analysis to container shares and flagged the 77-versus-79 name count. **R** The decision rested on numbers that foot; the dollar total was relabeled as the exit segments' ARR.

Land: $254,952 vs $185,892; difference $69,060 · **Pivot to NACF:** every network metric gets footed from its rows before a decision — totals that silently drop a segment are the commonest error.

Do NOT say: that I caught it live in the room — say "when I re-footed it preparing this".

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle — Dive Deep

Opener: "Three customers were under four percent of the book and three-quarters of the containers — so the plan could not be one policy."

Headline metric: 3.8% of customers = 75.0% of containers = 54.7% of ARR

- Foot first: containers foot, the printed ARR does not.
- Concentration: 3 brokers vs 39 SMBs — $46,504 vs $769 per customer.
- The SV-bound 55 are the long tail: 69.6% of customers, 13.3% of containers.
- Say "my footing" once, early.

Conflict / influence / backbone angle — Have Backbone; Disagree and Commit

Opener: "The document carried real dissent, and the decision was only as strong as the objections left standing on the page."

Headline metric: four dissenting inputs; one decider; each impacted party commits

- Cost of two systems; a missing parity list; ejecting customers; cut the competitors.
- RAPID: Recommend, Agree, Perform, Input, Decide named.
- The decider committed with the objections recorded.
- My part: the comparison, the per-segment pricing and the dissent on one page; the decider was an executive.

Honest failure-and-learning angle — Learn and Be Curious

Opener: "The decision memo had two questions it could not answer — what Crux costs to keep running, and whether the feature-parity list was written down — and the record never closes either."

Headline metric: 2 open questions; break-even $15,491 / month

- What I would do differently: price the keep-alive cost before the memo, not after.
- What I would add: a written parity matrix as an exit criterion for each segment.
- What held: dated actions and named performers.
- No dollar claimed as savings.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"You quote 79 customers and $185,892. Your own table says $254,952. Which is it?"

Numeric answer. Both, and I say what each is: $185,892 is the six non-transition rows (24 customers); the three SV-bound rows add $69,060 for $254,952. Customers (79) and containers (294,581) foot either way, so container shares lead. The appendix lists 77 names against 79 counted, so I would confirm before quoting a per-customer average.

Say exactly: "The printed total is the six segments that don't move to Shipment Visibility; all nine rows sum to about two hundred fifty-five thousand. Customers and containers foot either way, so I lead with container shares."

"What if the three brokers refuse the partner route?"

Numeric answer. Then they cancel at wind-down per the decision, and the exposure is the largest in the book: 221,044 containers (75.0%) and $139,512 (54.7% of nine-row ARR). One of three already uses Transmission, which is why it is the route; the other two are per-customer conversations, not a template.

Say exactly: "If they decline, they cancel at wind-down — and that is seventy-five percent of the containers, which is exactly why they got per-customer conversations instead of a template."

"What number would have flipped your decision to exit the forwarders and 3PLs?"

Numeric answer. The keep-alive cost. The nine customers pay $17,220 a year; the six exit segments together pay $15,491 a month. If keeping Crux alive costs less than that monthly figure, deferral wins on direct economics; the record never prices it, which is why I would price it before the memo.

Say exactly: "The keep-alive bill. The six exit segments pay fifteen thousand four hundred ninety-one a month; if running Crux costs less than that, deferral wins — and the record never prices it."

S12 — Flexport — FF App export launch: re-sequencing a committed 18-month roadmap on cost-to-serve evidence (value in month 6, 66% less effort, ~$5M cost avoidance)

DEEPENED

LPs this story can serve here: Think Big (primary — the flip) · Frugality (primary — ~$5M avoided by sequencing) · Have Backbone, Disagree & Commit (changing the manager's plan) · Earn Trust (privately first) · Bias for Action (the reversal trigger written before the swing)
Process family: Deliver — export documentation / operator and partner workflow; launch sequencing; unit-economics-driven roadmap governance

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “About five million dollars of avoided cost — engineering effort never spent plus value pulled forward — said once, an avoidance and not a saving, and never added to the unit-cost arithmetic.”

“What's your source for that?” “It is the figure in my own interview workbook for the export-app re-plan; my split into engineering effort and value pulled forward comes to about four point four million, a model I label as mine, so I say about, and I never call it annual.”

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Think Big The destination changed — first value in month 6 instead of month 18, and half again the unit saving — by re-opening a committed, socialized roadmap on evidence at the work-item grain.

Frugality About $5M of cost avoidance — engineering never spent plus value pulled forward — said once; 66% less effort to first value; a reuse rule for any module over 8 dev-weeks.

Have Backbone; Disagree and Commit I changed my manager's plan privately first, gave engineering two days to break the model, and the onsite ratified it with his name on it.

Earn Trust The cost of the trade — operators on the old screens two more quarters — was priced and published, not hidden.

Honesty label. The roadmap was my manager's; I owned the sequencing recommendation. Engineering produced the re-estimates; the ~$5M is the record's cost-avoidance figure, said once; the run-rate decomposition and the engineer-count split are my model. Loop-packet conflict (§1.3): that document tells this as a Sephora roadmap — the record is Flexport FF App; say Flexport.

WHAT IS THE FF EXPORT APP? — THE ONE-MINUTE EXPLAINER (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The minute. The Freight Forwarding App — "FFA" — is Flexport's operator-facing execution platform: where import and export shipments actually get run. Its core tenet is work-item-driven execution — every unit of work is a work item assigned to exactly one operator ("one operator, one work item"), so nothing ships on tribal knowledge and every action is attributable. For export, the heart of it is schedule management: a "Generate Schedule" work item builds the shipment's milestone schedule automatically, and a "Replan Schedule" work item fires on defined exceptions — an LCL CFS-cutoff miss, a vessel late against truck pickup, a vessel late past the quoted delivery date, a change of vessel — each with a defined rework path instead of ad-hoc heroics. It shipped FCL first (2022), with Air and LCL behind it (2023). RECORD-FROM-DOCS

The machinery behind the story. The schedule work sat inside a five-work-stream FFA program (Schedule Management; Cascading Schedules; Auto-Build Initial Schedule; End-to-End Operator Experience; Operator MiMeC), with schedule automation completing end Q3/early Q4 2022 and export-partner UAT aimed at a March 2023 launch. Because the app serves external partner organizations, it also carries a compliance bar: an immutable audit log of who changed what and when — nine event categories at a 100%-captured standard across ~100 partner orgs and thousands of changes a day, phased Alpha → Beta → MVP with CISO sign-off. RECORD-FROM-DOCS

Doc-vs-canon footnotes: the module documents carry forward-plan dates (export-partner UAT March 2023; schedule automation completing end Q3/early Q4 2022) — the build-out clock after MVP. Canon's "value in month 6" is the sequencing decision's clock: different milestones, no conflict — never merge them. The audit-logs "~100 partner orgs" is platform-wide; canon's 53 is the import-partner migration set — never merge those either.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The Freight Forwarding App export roadmap was operator-first — 18 months to first value, already socialized, with my manager's name on it. The process baselines were 339.5 minutes of handling per shipment and 2,200 operator interactions per departed shipment. The question nobody had asked was time-to-value and unit cost — the plan had been built around persona preference. **Task.** With no authority over the roadmap, re-sequence it on cost-to-serve evidence without abandoning the operators. **Action.** I wrote the baseline memo, had the business cut a bare-minimum scope per persona, and got engineering to re-estimate with confidence levels: operator-first reached −25% cost-to-serve in 9 months with operators still spending 20% of their time in the old app; partner-first reached −50% in 6 months and built the chassis operators would later ride. I added a rebuild-vs-reuse rule per module — over eight dev-weeks, reuse — a scope tripwire in the burn-up, and a written reversal trigger, then read it to my manager privately first; engineering had two days to break it, and the Bellevue onsite ratified the flip with his name on it. Operators' two further quarters on old screens were priced and published, not hidden. **Result.** Primary metric: engineering effort to first value — the 18-month plan → 66% less effort, value in month 6 instead of 18. ~$5M of cost avoidance — the record figure, said once; cost-to-serve −50% versus −25% at the work-item grain; 48 of 53 import partners migrated within a month.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The inherited launch plan for the freight-forwarding app was operator-persona first, 18 months, already socialized, and my manager owned it. Process baselines: 339.5 minutes of handling per shipment and 2,200 operator interactions per departed shipment — the two numbers everything prices from.

The business problem was time-to-value and unit cost, not persona preference. Left as inherited, operators would reach first value in month 18 while still spending 20% of their time in the old app — half the unit saving (−25% vs −50% at the work-item grain), reached a year later. The risks of re-opening it: a committed, socialized plan; optimistic estimates; and operators feeling abandoned while partners went first.

T — Own the sequencing recommendation — and change a plan with my manager's name on it, without authority over the roadmap, in a way he could own.

Baselines on the table: 339.5 min/shipment · 2,200 operator interactions per departed shipment · first value in month 18 on the inherited path · cost-to-serve −25% (operator-first) vs −50% (partner-first) at the work-item grain.

KEY CHALLENGES — WHAT THE PROGRAM RECORD ADDS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

- **Asia latency surfaced right after the April 2022 import-side launch.** The monthly business review read P95 page load at 21.19s in Asia against 3.3s in the US; static assets ran ~5s in China against ~500ms in the US; the app's domain had no geo-based DNS or reverse proxy, so Asia traffic rode the public internet. The fixes were sequenced — WAF enablement, a double CDN removed, a regional CDN proposal with smoke tests per change, then China and Asia tuning — with 48 of 53 import partners (about 90%) already migrated and expecting the app to hold. RECORD-FROM-DOCS
- **Compliance was a launch feature, not an afterthought.** The Audit Logs PRD demanded an immutable who/what/when trail across ~100 partner orgs and thousands of changes a day — nine categories at a 100%-captured bar, phased Alpha → Beta → MVP — and the CISO's definition of "important events" was deliberately vague, so PM buy-in and CISO sign-off were written into the plan as the mitigation. RECORD-FROM-DOCS
- **Exceptions had to become work items.** Replanning ran on a defined taxonomy — LCL CFS-cutoff miss, vessel late vs truck pickup, vessel late past quoted delivery date, change of vessel — each with a rework path, under the

"one operator, one work item" tenet. The exception list, not the happy path, is where operator trust is won. RECORD-FROM-DOCS

- **A hard external deadline.** Export-partner launch UAT was set for March 2023, with schedule automation completing end Q3/early Q4 2022 and FCL ahead of Air/LCL — a date-driven constraint on top of the unit-economics sequencing. RECORD-FROM-DOCS

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — baseline memo (339.5 min; 2,200 interactions); bare-minimum scope per persona cut with the business; the question restated as time-to-value and unit cost. *Exit bar:* business signs the bare minimums; engineering agrees to re-estimate with confidence levels.
- **Plan & Design** — re-estimates 9 months (operator) vs 6 months (partner) with confidence buffers (×1.0 / ×1.4 by confidence, +15% testing); the 20%-still-in-old-app finding; work-item-grain cost-to-serve translation (−25% vs −50%); rebuild-vs-reuse rule per module (>8 dev-weeks → reuse; rank by benefit × impact ÷ dev-weeks); reversal trigger written; read privately to my manager first; engineering given two days to break it. *Exit bar:* my manager puts his name on the flip at the Bellevue onsite; sequencing signed.
- **Build & Test** — 15 domains × 6 modules; Jira burn-up with the scope tripwire — new work estimated and swapped, never absorbed; operator-interactions-eliminated as the success metric; the tenet "no go-live without every work item assigned to one operator." *Exit bar:* burn-up 55.8% → 83.2% across two quarters; every work item owned.
- **Cutover & Go-Live (waves)** — the import-side app live April 2022 with 11 partners in 10 countries; partner migration plan published; operators' two further quarters on old screens priced and published. *Exit bar:* 48 of 53 import partners migrated within a month.
- **Hypercare / Close** — tracked interactions eliminated against the 2,200 baseline by work item; sequenced FCL 2022 → Air/LCL 2023 off that read; close-out at ~$5M cost avoidance and 66% less effort. *Exit bar:* value in month 6 instead of 18; operators moved onto the shared chassis on the published schedule.

Influence and ownership. Mine: the baseline memo, the comparison model, the rebuild/reuse rule, the scope tripwire and the reversal trigger. The business signed the bare minimums; engineering produced the re-estimates; my manager signed the sequencing and owned the flip. He argued two points, I adjusted one, and the onsite ratified it — I have never won a disagreement by seniority; I rank, privately first, and let the room ratify.

> **Designed the process; others own it now.** I designed the comparison model, the rebuild-vs-reuse rule and the scope tripwire; the delivery teams run the burn-up and the swap rule, and the owner of the original plan owned the flip. **Trade-off put to leadership.** The recommendation to leadership put the competing metrics side by side — time-to-first-value and partner cost-to-serve (6 months, −50%) vs operator effort (9 months, −25%) — with the operators' two further quarters on old screens priced, not hidden.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~$5M cost avoidance (record, said once — never "avoidance" and "annual reduction" as two results); underneath it a ≈$1.5M/yr run-rate *modeled* at −50% CTS on an illustrative 20K export shipments; the decomposition ≈$2.9M engineering avoided + ≈$1.5M pulled forward ≈ $4.4M is my model, within ~12% of the record.
- **#** — 48 of 53 import partners migrated within a month; the import-side app live April 2022 with 11 partners in 10 countries; baselines 339.5 min/shipment and 2,200 interactions (record).
- **%** — engineering effort to first value −66%; CTS −50% partner-first vs −25% operator-first at the work-item grain; burn-up 55.8% → 83.2% across two quarters (record).
- **Time** — first value in month 6 instead of month 18 (record); FCL 2022 → Air/LCL 2023 sequenced off the interactions read.
- **Customer** — partners completed end-to-end in the new app; operators' two further quarters on old screens priced and published, not hidden (record).
- **Quality** — new scope estimated and swapped, never absorbed; no go-live without every work item assigned to one operator (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	An 18-month committed roadmap with no cost-to-serve read behind it. I restated the question as time-to-value and unit cost, built the baseline memo (339.5 min; 2,200 interactions) and set the success metric — operator interactions eliminated.
Scope & influence	15 domains × 6 modules; the business signed the bare minimums, engineering produced the re-estimates, my manager signed the sequencing and owned the flip at the onsite — influence on the owner of the original plan, one level up.
Execution trade-off + mechanism others now own	Partner-first (6 months, −50% CTS) vs operator-first (9 months, −25%), with the operators' two further quarters on old screens priced and published. Mechanism others own now: the rebuild-vs-reuse rule per module and the scope tripwire on the burn-up.
Impact (long-term)	~$5M cost avoidance (record); first value in month 6 instead of 18; FCL 2022 → Air/LCL 2023 sequenced off the interactions read; 48 of 53 import partners migrated within a month.
Process improvement (standard work institutionalized)	Confidence-buffered estimates (×1.0 / ×1.4, +15% testing); the tenet "no go-live without every work item assigned to one operator"; new scope estimated and swapped, never absorbed.
Complexity & consensus	A disagreement with my manager, privately first and with numbers; two days for engineering to break the model; his name on the flip — the onsite ratified it.
Root cause at scale	The 20%-still-in-old-app finding and the work-item-grain cost-to-serve translation located where value actually accrued, so sequencing ran on unit economics rather than feature count.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

HM — You changed your manager's plan. How, and would you do it to me?

Privately first, with the numbers — 9 versus 6 months, 20% still in the old app, −25% versus −50%. He argued two points, I adjusted one, and the onsite ratified it with his name on it. Yes — the same way: the model, the owner named, the reversal trigger written, and two days for your engineers to break it before anyone else sees it.

HM — Is the $5M a saving or an avoidance?

Avoidance — engineering effort never spent plus a year of unit savings pulled forward — and I say it once. The recurring run-rate underneath is a separate, smaller modeled number (≈$1.5M/yr on an illustrative 20K shipments); I never present the avoidance as an annual reduction.

HM — What stopped the operators from being abandoned?

A priced, published two-quarter schedule — the trade was visible and scored, not hidden — and the partner backend was the chassis operators later rode, so their move got cheaper, not later.

HM — Your engineer count and rate are yours, not the program's.

Correct — twelve engineers at $20K a month are my numbers; the record does not itemize the $5M. At 10–15 engineers and $15–25K the effort piece runs $1.8–4.5M; adding the pulled-forward run-rate (≈$1.5M at the base) puts the whole decomposition at $2.9–6.7M, and the ~$5M record figure sits inside that range — the effort piece alone does not reach it. I would firm it with the two scoping envelopes (66.2 and 99.8 engineer-months) and Finance's loaded rate.

HM — What made the big swing safe?

Three mechanisms: confidence buffers on low-confidence estimates, a written reversal trigger, and the scope tripwire in the burn-up — new work estimated and swapped, never absorbed — which is why the burn-up read 55.8 → 83.2% honestly.

HM — Lesson, and how it shows up in your first quarter here?

Bring the model, name the owner, write the reversal trigger. In the first quarter: re-baseline any committed plan in time-to-value and unit cost at the work-item grain before defending or changing it.

> **HM — What broke after the April 2022 import-side launch that the plan had not seen?**
>
> Asia latency RECORD-FROM-DOCS: the business review showed P95 page load at 21.19 seconds in Asia against 3.3 in the US — static assets ~5s in China against ~500ms in the US — because the domain had no geo-based DNS or reverse proxy and Asia traffic rode the public internet. The fix was infrastructure sequencing, not re-architecture: WAF enablement, the double CDN removed, a regional CDN proposal with smoke tests per change, China tuning then Asia tuning. The lesson imported into every later plan: the pilot's geography is a performance requirement, not a rollout detail.

> **HM — What compliance bar does a partner-facing app carry, and what did it cost?**
>
> An immutable audit trail RECORD-FROM-DOCS: who/what/when for logins, work-item changes, route updates, messages, documents, schedule updates, exceptions, assignment rules and permission changes — nine categories at a 100%-captured bar, across ~100 partner orgs and thousands of changes a day. The honest cost answer: the PRD phased it Alpha → Beta → MVP precisely because capturing everything on day one was not feasible; the residual risk — the CISO's vague "important events" — was mitigated with sign-off, not silence.

> **HM — What did the March 2023 export-partner deadline change about the sequencing?**
>
> It made the module order do double duty RECORD-FROM-DOCS: schedule automation had to complete by end Q3/early Q4 2022 so UAT could start for the March 2023 export-partner launch, and FCL went ahead of Air/LCL because that is where the volume and the interactions baseline (2,200 per departed shipment) lived. The unit-economics sequencing and the deadline sequencing pointed the same way — which is exactly what made the flip ratifiable at the onsite.

THE KEY MATH — THE TWO PATHS AT THE WORK-ITEM GRAIN, AND THE RECORD'S ~$5M SAID ONCE

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	CTS per shipment	`339.5 min × $0.45/min ≈ $153`	record inputs; blended loaded rate derived, labeled	≈$153/shipment
2	Operator-first path	`9 months → −25%; operators still 20% in old app`	engineering re-estimates (record)	≈$38/shipment saved, month 18 to full value
3	Partner-first path	`6 months → −50%; partner backend = the chassis`	engineering re-estimates (record)	≈$76/shipment saved, 12 months earlier
4	Effort to first value	`100 − 66 = 34% of the inherited path`	record	−66%
5	Run-rate	`$76 × 20,000 export shipments = $1.52M`	volume illustrative (modeled)	≈$1.5M/yr modeled
6	Decomposition of the avoidance	`12 eng × 18 mo × 66% × $20K ≈ $2.9M + 12 mo × $1.5M = $1.5M → $4.4M`	counts and rate mine; within ~12% of the record	~$5M record headline, said once
7	Confidence rule	`(10 × 1.0 + 10 × 1.4) × 1.15 = 27.6 wks vs 20 naive`	two illustrative 10-week modules	buffers inflate only low-confidence items

Burn-up trajectory 55.8 → 63.3 → 66.0 → 69.3 → 76.0 → 83.2% (record). Rebuild-vs-reuse threshold of 8 dev-weeks was the engineering median (record rule).

Where this lands at Amazon. Think Big with Frugality attached — re-open a committed plan when the goal matters more than the plan, price it in unit economics, and have the owner whose name was on the original ratify the change. For a NACF process roadmap: sequence by time-to-value and cost-to-serve at the work-item grain, with a reversal trigger and a scope tripwire written before the swing.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S12

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you thought big / changed the direction of a program.	Think Big	The destination changed — value in month 6 not 18 — on evidence at the work-item grain; the mechanisms that made the swing safe.	Do not tell it as a Sephora story (loop-packet conflict); the record is Flexport.
Tell me about a time you saved money / did more with less.	Frugality	~$5M avoided by sequencing, said once; 66% less effort; rebuild-vs-reuse over 8 dev-weeks.	Never "$5M annual cost reduction" in the same breath as the avoidance.
Tell me about a time you disagreed with your manager.	Have Backbone	Private first, with numbers; two days for engineering to break it; his name on the flip at the onsite.	Skip S14 (the quiet-fix refusal) — held for the bar raiser.
How do you prioritize when everything is committed?	Functional — roadmap governance	Benefit × impact ÷ dev-weeks per module; new work estimated and swapped, never absorbed; the burn-up as the honest read.	Do not claim the roadmap was mine — it was my manager's.
Tell me about a trade-off you made that hurt someone.	Earn Trust / Customer Obsession (internal)	Operators on old screens two more quarters — priced, published and scored, not hidden.	Do not present the operator delay as costless.
What's a decision you'd reverse if the data changed?	Are Right, A Lot	The reversal trigger was written before the flip; the interactions-eliminated read sequenced Air/LCL 2023, not a date on a slide.	No YoY framing; the avoidance is one-time.
How does re-sequencing a roadmap relate to launch readiness? (transferability probe — public-profile inference)	Functional — transfer	Her world is 50+ FC launches: readiness discipline is the same shape as the flip — a baseline memo (the readiness scorecard), gates that fire on evidence (the reversal trigger ≈ a launch-readiness gate), and the owner of the original plan ratifying the change. Sequence by time-to-value at the work-item grain, never by the date on the slide.	Do not claim FC-launch experience; bridge from the mechanism, not the domain.
How would you apply cost-to-serve thinking to a placement network? (transferability probe)	Functional — transfer	CTS per container is cost-to-serve at one node grain; placement is the same arithmetic at the node's position — distance-to-customer priced against speed and cost per unit moved. The work-item grain (339.5 minutes; 2,200 interactions) is how I keep it honest.	Never quote her $50MM / $80MM savings as shared ground — those are her numbers, framing only.

DEEPENING S12 — FIVE FACETS, FOUR-TABLE MATH, BEHAVIORAL PACK, ANGLES AND TRAP DRILLS NEW

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Baseline memo: 339.5 minutes per shipment and 2,200 interactions per departed shipment; per-persona bare-minimum scope; re-estimates by path at the work-item grain; the 20% still in the old app.
2 · Solution design	Operator-first vs partner-first; rebuild vs reuse by module; confidence buffers; swap-never-absorb scope tripwire; a written reversal trigger; the manager's name on the flip.
3 · Math & data logic	Four tables: $152.78 ≈ $153; −25% vs −50%; 18 → 6 months; −66% ⇒ 34%; 27.6 vs 20 weeks; $2.85M + $1.52M ≈ $4.4M vs the ~$5M record.
4 · Implementation	15 domains × 6 modules; burn-up 55.8% → 83.2%; import-side app live April 2022 with 11 partners in 10 countries; 48 of 53 import partners (about 90%) migrated; FCL 2022, Air and LCL 2023; the export launch is the later Q1 2023 milestone.
5 · Probe questions	Seven numeric probes plus three trap drills: avoidance vs saving, invented engineers, and the 27.6-week rule.

Facet 1 — Data analysis: what each analysis decided

ANALYSIS	DATA AND GRAIN	WHAT IT SHOWED	DECISION IT DROVE
Baseline memo	minutes and interactions per departed shipment	339.5 min and 2,200 interactions — the two numbers everything prices from	the debate became arithmetic, not persona preference
Cost-to-serve translation	339.5 min × ≈ $0.45 per min (derived blend)	$152.78 ≈ $153 per shipment	each path priced as a percent of $153
Re-estimate by path	engineering re-estimates with confidence levels, work-item grain	operator-first 9 months, −25%; partner-first 6 months, −50%	partner-first, with operators on the shared chassis later
Legacy residue	share of operator time still in the old app	20% under operator-first	caps operator-first at half the unit saving
Module cost vs benefit	dev-weeks per module; benefit × impact	median 8 dev-weeks (record rule)	over 8 dev-weeks → reuse, ranked by benefit × impact ÷ dev-weeks
Burn-up	completed ÷ pointed work	55.8 → 63.3 → 66.0 → 69.3 → 76.0 → 83.2% over two quarters	new work estimated and swapped, never absorbed

Facet 2 — Solution design: options weighed

QUESTION	OPTIONS	CHOSEN	WHY
Who goes first	Operators (inherited) · partners · both in parallel	Partners first; operators ride the same chassis later	6 months and −50% against 9 months and −25%; the partner backend is reused, so operator work gets cheaper, not later.
Build or reuse	Rebuild each module · reuse legacy · rule by size	Over 8 dev-weeks → reuse	Engineering's own median; module by module, not ideology.
Estimate honesty	Single-point estimates · confidence buffers	× 1.0 / × 1.4 by confidence, plus 15% testing buffer	Buffers only inflate low-confidence items: 27.6 weeks against 20 naive.

QUESTION	OPTIONS	CHOSEN	WHY
Change control	Absorb new scope · swap	Estimate and swap, never absorb	The burn-up reads honestly: 55.8 → 83.2%.
How to change the plan	Escalate over the manager · private first with numbers	Private first, two days for engineering to break it, then the room ratifies	The fastest path to a better decision was making it his decision.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Inherited plan	`operator-persona first: 18 months to first value, socialized, my manager's plan`	record	month 18
2	Unit baselines	`339.5 minutes of handling per shipment; 2,200 operator interactions per departed shipment`	record	the two numbers everything prices from
3	Cost-to-serve per shipment	`339.5 min × $0.45 per min = $152.78 ≈ $153`	record inputs (FCL $0.56, Network $0.61, Data $0.24 per min; the blend is derived, labeled)	≈ $153 per shipment
4	Operator-first path	`9 months to −25%; operators still spend 20% of their time in the old app`	engineering re-estimates (record)	−25%, capped by legacy residue

Table 2 — The fix: paths, rules and tripwire, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Partner-first path	`6 months to −50%; the partner backend is the chassis operators later ride`	engineering re-estimates with confidence levels (record)	−50%, 12 months earlier
2	Unit saving per path	`$153 × 50% = $76.50 ≈ $76; $153 × 25% = $38.25 ≈ $38`	from Table 1	double the unit saving
3	Effort to first value	`100 − 66 = 34% of the inherited path`	record	−66% ⇒ 34%
4	Confidence rule	`(10 × 1.0 + 10 × 1.4) × 1.15 = 27.6 weeks against 20 naive; 27.6 / 20 = +38%`	two illustrative 10-week modules; the multipliers and 15% buffer are the record rule	27.6 vs 20 weeks
5	Rebuild-vs-reuse	`dev-weeks > 8 → reuse; rank by benefit × impact ÷ dev-weeks`	the 8-week rule was engineering's median (record rule)	module by module
6	Scope tripwire	`burn-up 55.8 → 83.2% = +27.4 points over two quarters; new work swapped, never absorbed`	record	+27.4 points

Table 3 — After: before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Time to first value	`month 18 → month 6`	record	12 months earlier
2	Engineering effort to first value	`100% → 34%`	record	−66%

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
3	Cost-to-serve per shipment	≈ $153 → ≈ $76 (−50%)	record inputs; derived	≈ $76
4	Partners migrated	48 of 53 import partners within a month: 48 / 53 = 0.9057 → about 90% (the documents print 90.5%)	record (resume, Asia-latency doc); ratio mine	48 of 53 — about 90%
5	Pilot reach	import-side app live April 2022 with 11 partners in 10 countries; 15 domains × 6 modules = 90 modules	record; product mine	11 partners · 90 modules

Table 4 — The dollars: every multiplication shown; the ~$5M said once

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Unit saving	$153 × 50% = $76 per shipment	from Table 2	≈ $76
2	Run-rate	$76 × 20,000 export shipments = $1,520,000	modeled (volume illustrative)	≈ $1.5M a year
3	Engineering avoided	12 eng × 18 mo = 216 eng-months; × 66% = 142.6; × $20,000 = $2,851,200	modeled (counts and rate mine; benchmark $15–25K)	≈ $2.9M one-time
4	Value pulled forward	12 months × ($1.52M ÷ 12) = $1.52M	modeled, from the run-rate	≈ $1.5M one-time
5	Decomposition check	$2.9M + $1.5M = $4.4M; ($5.0M − $4.4M) / $5.0M = 12% under the record	modeled on rounded inputs; the ~$5M is the record, said once	$4.4M vs ~$5M

Facet 4 — Implementation: the flip, the build and the cutover

PHASE	WHAT HAPPENED (RECORD UNLESS MARKED)	CONTROL
Decide	private read to my manager first; engineering two days to break the model; the onsite ratifies with his name on the flip	reversal trigger written before the flip
Build	15 domains × 6 modules; Jira burn-up 55.8% → 83.2% across two quarters; no go-live without every work item assigned to one operator	scope swapped, never absorbed
Cutover	import-side app live April 2022 with 11 partners in 10 countries; 48 of 53 import partners (about 90%) migrated within a month; operators' two further quarters on old screens priced and published	partner migration plan published
Sequence	FCL 2022, then Air and LCL 2023, read off interactions eliminated against the 2,200 baseline	by work item

Lean / Kaizen mapping. Time-to-value is the lead-time waste; the 20% still in the old app is inventory of unfinished transition; the rebuild-vs-reuse rule is standard work for the make-or-reuse decision; the swap-never-absorb tripwire is a pull system for scope; the written reversal trigger is the PDCA "Check" fixed in advance.
Honest note. The record's single engineering-effort figure is −66%; the effort-envelope numbers sometimes quoted alongside it are not used here.

Facet 5 — Probe questions (numeric)

> **Saving or avoidance? Annual?**
>
> Avoidance, one-time, said once: about $5M, record. My decomposition — $2.85M of engineering never spent plus $1.52M of unit saving pulled forward — is $4.4M, within 12% of it. The recurring run-rate underneath, $1.52M a year, is a separate modeled number; I never present the avoidance as annual.

Your engineers and rate are invented.

They are my model, and I say so: ten to fifteen engineers at $15–25K a month moves the effort piece from $1.8M to $4.5M; with the pulled-forward piece, the decomposition ranges $2.9M to $6.7M, and the record's ~$5M sits inside that range — not inside the effort piece alone.

Why 27.6 weeks and not 20?

(10 × 1.0 + 10 × 1.4) × 1.15 = 27.6: the 1.4 applies only to the low-confidence module and the 15% is testing buffer, so the plan is 38% longer than naive and honest about which parts are risky.

Why is partner-first twice the saving?

Because at the work-item grain operator-first reaches −25% in 9 months with 20% of operator time still in the old app, and partner-first reaches −50% in 6 months: $76 against $38 per shipment, 12 months earlier.

Why 8 dev-weeks?

It was engineering's median module cost — the record rule — so the threshold is theirs, not mine: over 8 dev-weeks, reuse; otherwise rebuild, ranked by benefit × impact ÷ dev-weeks.

What did operators pay?

Two more quarters on the old screens — priced, published and scored, not hidden. The partner backend was the chassis they later rode, so their move got cheaper, not later.

What broke after the April 2022 import-side launch?

Asia latency — the monthly review showed 21.19 seconds at P95 against 3.3 in the US; see S12-A for the full investigation. The lesson: the pilot's geography is a performance requirement, not a rollout detail.

SENSITIVITY — LOW / BASE / HIGH

CASE	EFFORT INPUTS	ENGINEERING AVOIDED (ONE-TIME)	EXPORT SHIPMENTS	RUN-RATE / PULLED FORWARD	DECOMPOSITION TOTAL
low	10 engineers × $15,000	$1.8M	15,000 shipments	$1.1M	$2.9M
base	12 engineers × $20,000	$2.9M	20,000 shipments	$1.5M	$4.4M
high	15 engineers × $25,000	$4.5M	30,000 shipments	$2.3M	$6.7M

Rows are modeled, not record; the program's ~$5M is the single signed figure and the base decomposition ($4.4M) is within 12% of it. Note: the high case here reads $6.7M on unrounded inputs (the master rounds to $6.8M).

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> About five million of cost avoidance, and I say that once. Roughly two point nine of it is engineering effort we never spent — sixty-six percent less on a plan that was eighteen months long — and about one and a half is a year of unit savings pulled forward because value landed in month six instead of month eighteen. Behind that sits a run-rate of about a million and a half a year at half the cost-to-serve on twenty thousand export shipments. The sixty-six percent, the six versus eighteen months and the minus fifty versus minus twenty-five are the record; the five million is the program's signed figure; the split and the shipment volume are my model.

IF CHALLENGED — THE FULL ANSWERS

"Twelve engineers and twenty thousand a month are yours, not the program's."

Correct, and the record does not itemize the five million, so I modeled the split. At ten to fifteen engineers and $15–25K a month the effort piece is $1.8M–$4.5M and the whole decomposition $2.9M–$6.7M; the record's ~$5M sits inside the decomposition range.

"Where do twenty thousand shipments come from?"

An illustrative FCL export volume, and the softest input: at 15,000 to 30,000 the run-rate runs $1.14M to $2.28M. I would replace it with the departed-shipment count from the system that produced the 2,200-interaction baseline.

"Are you double counting the pulled-forward value and the five million?"

No. The five million is the total; the pulled-forward year is one slice of how it decomposes, and the run-rate underneath is not a second result.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I re-sequenced a committed 18-month roadmap on cost-to-serve evidence — partners first, operators on the same chassis later — so value landed in month 6, effort to first value fell 66%, and the program recorded about $5M of cost avoidance.

NACF BRIDGE — ONE SENTENCE

It is the discipline of sequencing launches or placement moves by time-to-value at the unit grain, with a written reversal trigger and the owner of the original plan ratifying the change.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Give me an example of how you have changed the direction or view of a specific function/department and helped them embrace a new way of thinking. Why was a change needed? What was the outcome?"

AIQB: Think Big Q2 | Think Big | FACET: SOLUTION DESIGN

Open (first two sentences): "The roadmap said eighteen months to first value, operators first. I re-opened it, and the new plan put value in month six and delivered twice the unit saving."

90-second STAR. **S** A committed, socialized 18-month plan owned by my manager. **T** Change the plan without authority over the roadmap, in a way he could own. **A** I priced both paths at the work-item grain — partner-first 6 months and −50%, operator-first 9 months and −25% with 20% of operator time still in the old app — added confidence buffers, a rebuild-vs-reuse rule, a scope tripwire and a written reversal trigger. **R** First value in month 6 instead of 18; effort to first value −66%; ~$5M cost avoidance, said once.

Land: 18 → 6 months; −66% effort · **Pivot to NACF:** sequencing launches by time-to-value at the unit grain.

Do NOT say: "$5M annual savings" or the roadmap as mine.

B2 · "Tell me about a time when you strongly disagreed with your manager or peer on something you considered very important to the business. What was it and how did you handle it? Knowing what you know now, would you do anything differently?" AIQB: Have a Backbone Q1 **Have Backbone; Disagree and Commit** FACET: IMPLEMENTATION

Open (first two sentences): "The plan had my manager's name on it. I read it to him privately first, with the numbers, and gave engineering two days to break my model before anyone else saw it."

90-second STAR. S A plan my manager owned, with socialized commitments. **T** Change it without becoming a political fight. **A** Private read; he argued two points and I adjusted one; engineering got two days to break the model; the onsite ratified with his name on the flip. **R** He owned the flip; the plan changed; the model survived.

Land: two points argued, one adjusted · **Pivot to NACF:** changing a launch sequence: model first, private first, owner of the old plan on the new one.

Do NOT say: that I won on seniority or escalated — I never have.

B3 · "Give me an example of how you have helped save costs or eliminate waste within your role or organization. What was the situation? What was the impact?" AIQB: Frugality Q1 **Frugality** FACET: MATH AND DATA LOGIC

Open (first two sentences): "About five million dollars of cost avoidance — engineering we never spent plus a year of unit savings pulled forward — and I say it once, because it is avoidance, not an annual reduction."

90-second STAR. S An 18-month plan that would have consumed the full engineering effort. **T** Reach first value with less spend. **A** Reuse over 8 dev-weeks, partner backend as the shared chassis, confidence buffers, scope swaps instead of absorption. **R** Effort −66%; decomposition $2.85M + $1.52M = $4.4M against the record ~$5M.

Land: ~$5M cost avoidance (record), said once · **Pivot to NACF:** reuse over rebuild in a network program: the 8-week rule is an engineering-median threshold.

Do NOT say: avoidance and annual reduction as two results; the engineers and rate as the program's.

B4 · "Tell me about a time when you had to communicate a change in direction that you anticipated people would have concerns with. What did you do to understand the concerns and mitigate them?"

AIQB: Earn Trust Q1 **Earn Trust** FACET: DATA ANALYSIS

Open (first two sentences): "Operators would stay on the old screens two more quarters. I priced that and published it instead of hiding it."

90-second STAR. S A sequencing choice that delayed one group for the benefit of the whole. **T** Be transparent about who pays. **A** The baseline memo showed 20% of operator time still in the old app under the inherited path; I priced two more quarters for operators, published the schedule, and made the partner backend the chassis they later ride. **R** No hidden trade; operators' move got cheaper, not later.

Land: two quarters, priced and published · **Pivot to NACF:** a network change that burdens one node temporarily: price it, publish it, give the date it ends.

Do NOT say: that the delay was costless.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: "Three hundred thirty-nine and a half minutes and twenty-two hundred interactions per shipment — with those two numbers the roadmap debate became arithmetic."

Headline metric: 339.5 min × $0.45 ≈ $153; −25% vs −50%

- Price each path at the work-item grain.
- The 20% residue caps operator-first at half.
- Confidence buffers: 27.6 vs 20 weeks.
- Say "derived" on the $153.

Conflict / influence / backbone angle **Have Backbone; Disagree and Commit**

Opener: “It was my manager's plan. I read it to him alone, with numbers, and gave engineering two days to break it.”

Headline metric: two points argued; one adjusted

- Private first, then the room.
- His name on the flip.
- Reversal trigger written in advance.
- Never won on seniority.

Honest failure-and-learning angle **Are Right, A Lot**

Opener: “What I did not see was Asia: right after the April 2022 import-side launch, the review showed 21 seconds at P95 against 3.3 in the US.”

Headline metric: 21.19 s vs 3.3 s

- The plan sequenced by value and effort, not by the pilot's geography.
- The fix was infrastructure sequencing (see S12-A).
- The lesson imported into later plans: geography is a performance requirement.
- What held: the sequencing decision itself.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"Five million — is that saved every year?"

Numeric answer. No. It is one-time avoidance, the record's signed figure. My decomposition is $2.85M engineering never spent + $1.52M pulled forward = $4.4M; the recurring run-rate is $1.52M a year, a separate modeled number on illustrative volume. They are never both claimed.

Say exactly: “It's avoidance — one time, said once. The run-rate underneath is a separate, smaller model; I never present the avoidance as annual.”

"Your twelve engineers — where did they come from?"

Numeric answer. My model: 12 × 18 months × 66% × $20,000 = $2,851,200. At 10–15 engineers and $15–25K the effort piece is $1.8M–$4.5M; with the pulled-forward piece the decomposition is $2.9M–$6.7M, and the record's ~$5M sits inside that range.

Say exactly: “Those are my modeled inputs — the record doesn't itemize the five million. Across ten to fifteen engineers it moves from about two to four and a half million; I'd firm it with Finance's rate.”

"Why is your plan 27.6 weeks when two 10-week modules is 20?"

Numeric answer. (10 × 1.0 + 10 × 1.4) × 1.15 = 27.6 — the 1.4 applies only to the low-confidence module, plus a 15% testing buffer; 38% over naive, and the buffer is not spread evenly.

Say exactly: “The buffer only goes where confidence is low — twenty-seven point six against twenty naive — so the extra eight weeks are the risky module, not padding.”

S12-A — Flexport — Why Asia waited 21 seconds: a P95 latency investigation that split one complaint into a static-content problem and a routing problem, then ran it as a dated, smoke-tested change plan NEW

LPs this story can serve here: Dive Deep (primary) · Customer Obsession · Bias for Action · Insist on the Highest Standards
Process family: Improve — root-cause investigation of a regional performance gap; change control for network and security changes

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Dive Deep The review said "Asia is slow"; the work was to find where the 21.19 seconds went — region split, static versus dynamic, and a China static-JS P95 of 5 s against 0.5 s in the US — before anyone bought a CDN.

Customer Obsession Asia was 45.3% of the 13.92 thousand page views in the window, so the slowest region was the largest region — the plan was ordered by where the partners actually were.

Bias for Action Reversible fixes went first — double CDN removed in China, WAF on globally — and the expensive one (CDN for the dynamic host) was deferred until the metrics said it was needed.

Insist on the Highest Standards Every WAF or CDN change ran a written pre-check, a post-check and a rollback-able DNS override — 32 smoke checks a change — and I will not claim a single-cause before-and-after that the record does not support.

Honesty label. This is an investigation story with a split of credit: the investigation document is the app team engineer's, with the network infrastructure owner running the CDN and WAF changes — *I did not write the root-cause doc or make the network changes*. What I tell as mine is the program side: the monthly review that surfaced the gap, the cut by region, the dated plan with owners, the change-control discipline and the insistence on a like-for-like measurement. **My seat** (my own workbook, Apple v11 B3, and the document): the investigation is authored by the app engineer, and the network-infrastructure owner made the CDN and DNS changes (the plan's owners are named in the document); I drove the follow-ups from the monthly review and kept the plan, the sequencing and the China-host smoke tests honest. I say "I drove the follow-ups", not "I fixed the CDN". **Before-and-after flag:** the investigation window (Apr 24 – May 10) and the review series differ in window and region definition, so the 81% figure I can compute is indicative only — I do not quote it as the result; the like-for-like record is the review series. The document's Asia table and its China-specific plan rows belong to one investigation, with China the sharpest tail; the later review series splits Asia (non-China) from China, so the 21.19 s and the later 4.0 s may not be the same population — I never subtract across them. Overlap: this story is also on the bench in another interviewer's packet.

WHAT THE RECORD SAYS (FROM THE INVESTIGATION DOCUMENT AND THE WEEKLY REVIEW SHEET; RECORD-FROM-DOCS)

Context. The Freight Forwarding App (FFA) launched its import-side app in April 2022 to 11 partners across 10 countries; through May, 48 of 53 import partners were migrated. The monthly review showed Asia countries with much higher latency than other regions. RECORD-FROM-DOCS

The numbers. Page-load P95 for Apr 24 – May 10 (Datadog view loading time, which covers static resources, the GraphQL request and the render): US 3.3 s on 2.19k views; Asia 21.19 s on 6.31k views; Others 4.45 s on 5.42k views. Static JavaScript P95: China 5 s against 500 ms in the US. RECORD-FROM-DOCS

Two named causes. (1) Static resources are served from a CDN host that performs poorly in China; (2) the dynamic host has no geo-based DNS or regional reverse proxy for Asia, so GraphQL requests cross the public internet instead of the data-center backbone. RECORD-FROM-DOCS

The plan. Eight rows, owners and ETAs: three security (WAF proposal, WAF for China, WAF globally — follow-ups to an incident) and five latency (remove the double CDN in China — done, ETA June 24; static-CDN proposal for China and Asia, July 15; tuning for China, July 30; tuning for Asia, August 15; CDN for the dynamic host globally — "decide based on the metrics", Q3 or later). A written change process: override DNS locally, verify resolution, smoke-test

internal single-sign-on and partner users, repeat on the China instance, change, then remove the override and re-test. RECORD-FROM-DOCS

The review series. The weekly review sheet carries page-load P95 overall and by region (weeks 33 to 52 and week 1) and a monthly view (April to September). The sheet's action items include adding GraphQL latency at P50, P90 and P95 by region (done October 11, 2022). RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. We had launched the import-side freight-forwarder app in April 2022 and moved 48 of 53 import partners onto it by May. At the monthly review the latency chart had an outlier: Asia. **Task.** Find out whether it was one problem or several, and where the time was going, before anyone bought anything. **Action.** I cut the 13.92 thousand views by region over the 17-day window: Asia was 45% of the views and waited 21.19 seconds at the 95th percentile against 3.3 in the US — 6.4 times. Then I asked where the time goes: static JavaScript took 5 seconds in China against half a second in the US, and the dynamic requests from Asia went over the public internet because there was no regional edge. Two causes, two owners. I ordered the plan by reversibility — remove the double CDN, turn on the firewall — held the expensive global CDN until the metrics asked for it, and put every network change behind a written DNS-override and smoke-test checklist. **Result.** Primary metric: the Asia-to-US gap of 6.42× — and in the review series that followed, Asia read 4.0 seconds against 4.09 in North America by the last week of the year, with overall monthly P95 down from 8.39 to 4.68 seconds. I tell that as a gap that closed, not as a single-cause saving.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “Asia waited about twenty-one seconds at the ninety-fifth percentile against about three and a third in the US — roughly six and a half times — over a seventeen-day window of about fourteen thousand page views.”

“What's your source for that?” “It is the latency document's own table; the six-point-four is my division of its two printed figures, and the later four seconds against four-oh-nine is from the monthly review series — I describe it as a gap that closed, not a single-cause saving.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Flexport's freight-forwarder app was in its first months: the import-side app live to 11 partners in 10 countries in April 2022, then 48 of 53 import partners (about 90%) by the end of May. The monthly review flagged that Asian users saw far higher page-load latency than anyone else — 21.19 seconds at P95 against 3.3 in the US on the same two-and-a-half-week window — and Asia was the single largest slice of traffic. The risk: a freight partner who waits twenty seconds for a shipment page goes back to email, and the cost-to-serve case for the whole migration depends on partners using the app.

T — Turn "Asia is slow" into named causes with owners and a dated plan, sequence the changes so the cheap and reversible ones run first, keep the production network and security changes safe, and prove the improvement with a measurement that can be trusted.

Baselines on the table: Page-load P95: US 3.3 s · Asia 21.19 s · Others 4.45 s · views 2.19k / 6.31k / 5.42k · window Apr 24 – May 10 (17 days) · static JS P95 China 5 s vs US 0.5 s · 48 of 53 partners migrated · 8 plan rows (3 security, 5 latency).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — cut P95 and views by region over the window; read the resource-level data; mapped the request path for static content and for GraphQL; wrote the two-cause hypothesis and what each predicts. *Exit bar:* a region table with views, and a named hypothesis per cause.
- **Plan & Design** — sequenced eight rows by reversibility and dependency: security first where an incident was open, then the double-CDN removal, then proposals, then tuning; wrote the decision rule to defer the global CDN until the

metrics asked for it; listed the open measurement question — how to test the China CDN in a consistent, reproducible way. *Exit bar:* owners and ETAs on every row; one conditional row with its trigger.

- **Build & Test** — the change process: override DNS locally, verify resolution, smoke-test internal single-sign-on and partner log-in, workview, log-out and re-log-in, repeat on the China instance; infra performs the change; then remove the override, clear the cache and test again. *Exit bar:* 32 checks per change (16 before, 16 after).
- **Cutover & Go-Live** — WAF on for China and globally and the double CDN removed in China, each behind the checklist, ETAs June 24 – June 30; static-asset tuning for China and Asia dated July 30 and August 15. *Exit bar:* four of eight rows marked done in the document; dated owners on the rest.
- **Hypercare / Close** — the monthly and weekly reviews carry page-load P95 by region; the action item to add P50 / P90 / P95 GraphQL latency by region closed October 11, 2022; the week-47 spike was triaged against the region split rather than assumed to be the network. *Exit bar:* a regional series the review reads every week.

Influence and ownership. Mine (as I would tell it): the review that surfaced the gap, the region cut, the sequencing rule, the change checklist discipline and the insistence on a like-for-like measure. Not mine: the root-cause write-up, the CDN and WAF changes, the security sign-offs and the app optimizations — the app engineer and the network-infra owner own those, with the security reviewer signing the firewall proposal.

> **Designed the process; others own it now.** The change process, the region cut in the weekly review and the P50 / P90 / P95 action item live in the document and the sheet, not in my head. **Trade-off put to leadership.** Buy a global CDN now (certain cost, unknown benefit) against two cheaper reversible fixes first and a metrics trigger — the schedule risk was that the global CDN row slipped to Q3 or later.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — No dollar exists in the record. I model exposure, labeled illustrative: Asia's 45.3% of views against the S12 run-rate ≈ $0.69M a year (modeled upper bound, not savings); the direct tail time is ≈ $1.6K a year, which is why the case is adoption, not seconds.
- **#** — Asia 21.19 s vs US 3.3 s = 6.42× on 6.31k of 13.92k views; 32 smoke checks per change; 8 plan rows, 4 marked done.
- **%** — overall page-load P95, review monthly series, April to September: 8.39 s → 4.68 s, −44.2% (record-from-docs); Asia (non-China) 5.5 → 4.0 s, −27.3%, North America 5.62 → 4.09 s, −27.2% over weeks 39 to 52 (record-from-docs, my arithmetic).
- **Time** — plan span June 24 → August 15 = 52 days; the P50 / P90 / P95 by-region action item closed October 11, 2022.
- **Customer** — Asia partners were the largest slice of views; by the last week of the year the review series put Asia (non-China) within 2% of North America, and China at 1.57× North America.
- **Quality** — zero unreviewed network changes — every WAF or CDN change had its pre-check and post-check — and a before-and-after claim limited to what the like-for-like series supports.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	One complaint, no known cause. The work was to split it into two testable causes (static content path, dynamic routing path) and to name what would disprove each.
Scope & influence	Engineering, network infrastructure and security each owned a piece; the plan was a one-page sequence with owners and dates that none of them could do alone.
Execution trade-off + mechanism others now own	Deferred the global CDN on a metrics trigger — a deliberate schedule risk. The mechanism others own: the change checklist, the by-region series and the P50 / P90 / P95 view.
Impact (long-term)	The review series that followed shows the Asia gap gone by week 39 and China as the remaining outlier — a clear next investigation instead of a vague complaint.
Process improvement (standard work institutionalized)	DNS-override + smoke-test + rollback as the standard for WAF and CDN changes; regional percentiles on the weekly review.
Complexity & consensus	App team, network infrastructure and security with different incentives — a written plan with a conditional row resolved the "just buy the CDN" push without a standoff.
Root cause at scale	The 6.42× gap was a path problem, not an app problem — week 47 proved it from the other side, when North America spiked 2.09× as well.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	P95 and views by region; Asia 45.3% of views; static JS 10× slower in China; P95 versus P99 sample sizes (63 observations against 22); the review series by region and week.
2 · Solution design	Static path versus dynamic path; six options weighed; reversible-first sequence; metrics trigger for the global CDN.
3 · Math & data logic	Gap ratios 6.42× and 1.35×; tail views 315.5; 52-day plan; 32 checks; −44.2% overall; Asia / NA 0.98×; exposure $0.69M (illustrative).
4 · Implementation	Eight plan rows with owners and ETAs; the DNS-override change process; WAF first, double CDN removed next; China tuning, then Asia.
5 · Probe questions	Seven numeric probes plus three trap drills: the 81% temptation, P95s that do not add, P99 on thin data.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What I read. Three sources, all RECORD-FROM-DOCS: (1) the investigation document's region table — page-load P95 and views for Apr 24 – May 10; (2) its resource-level cut — static JS P95 China 5 s vs US 0.5 s; (3) the weekly and monthly review sheet by region (weeks 33–52, week 1; April–September monthly). I added the shares, the ratios and the tail-sample arithmetic.

REGION	PAGE-LOAD P95	VIEWS	% OF VIEWS	VS US	TAIL VIEWS (5%)
US	3.30 s	2,190	15.7%	1.00×	109.5
Asia	21.19 s	6,310	45.3%	6.42×	315.5
Others	4.45 s	5,420	38.9%	1.35×	271.0
Total	—	13,920	100%	—	696

What it revealed. (1) *The largest region was the slowest:* Asia carried 45.3% of views at 6.42× the US P95. (2) *Two paths, two causes:* static JS in China at 10× the US P95; dynamic GraphQL from Asia over the public internet with no regional edge. (3) *Why P95 and not P99:* in 17 days Asia has 6,310 views, so P99 rests on 63 observations and the US P99 on 22 — P95 rests on 316 and 110, which is stable enough to steer by. (4) *P95s do*

not add: the 5 s static-JS P95 is not 5 s of the 21.19 s page-load P95; the right test is to re-measure page load after every change. (5) *The later review series is a different population* — it splits Asia (non-China) from China — so I never subtract across the two sources. (6) *Week 47:* North America went from 4.52 to 9.44 s (2.09×) in the same week Asia went from 7.16 to 16.4 s (2.29×) while China fell — an application-wide regression, not a regional network one. The review sheet's own 12/12 callout explains a GraphQL-latency spike for the Core and Client Apps as a change in the metrics code that made the dashboard numbers wrong, fixed retroactively — a measurement error, not a network one; the October regression is a separate cause (work_view 2.0 un-optimized queries, action item 11). I do not quote week 47 as an Asia result.

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Static content in China	Increase CDN cache time · remove the double CDN (direct connect to the object store) · replicate assets into China / Asia (needs security approval) · another CDN or a global accelerator	Remove the double CDN first; replicate assets after a proposal and sign-off	Removal is reversible and needs no new vendor; replication needs a security approval first.
Dynamic requests from Asia	CDN for the dynamic host · geo-based DNS with a regional reverse proxy over the backbone · leave it	Regional edge proxy via geo DNS is the target; global CDN deferred on a metrics trigger	The document's own wording: finish the other items, then decide on the metrics.
Security vs latency order	Latency first · WAF first	WAF proposal and rollout first for China and globally	An open incident follow-up; the WAF and the CDN share a change path, so one checklist covers both.
How to measure the China CDN	Real-user views only · synthetic probes · both	Open question in the document	Raised as "consistent and reproducible"; resolved partly by adding P50 / P90 / P95 by region to the review (done October 11, 2022) VERIFY-WITH-RAJESH · V4
Priority	By worst P95 · by traffic share	By traffic share, reversibility and dependency	Asia = 45.3% of views; China is the sharpest tail but a smaller share.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Views in the window	`2.19k + 6.31k + 5.42k = 13.92k; Apr 24–30 = 7 days + May 1–10 = 10 days = 17 days`	record-from-docs inputs	13.92k views over 17 days
2	Asia's share of views	`6.31 / 13.92 = 45.3%; per day 6,310 / 17 = 371 of 13,920 / 17 = 819`	my arithmetic on record inputs	45.3% · 371 of 819 a day
3	P95 gaps	`21.19 / 3.30 = 6.42×; 4.45 / 3.30 = 1.35×; 21.19 / 4.45 = 4.76×`	my arithmetic on record inputs	Asia 6.42× US
4	Static JS	`China P95 5 s / US P95 0.5 s = 10×`	record-from-docs inputs	10×
5	Tail sample: why P95	`5% × 6,310 = 315.5 Asia views above P95; 1% × 6,310 = 63.1 above P99; US 1% × 2,190 = 21.9`	my arithmetic; the document gives no P99	P95 on 316 views; P99 on 63 (US 22)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
6	Adoption context	`48 / 53 import partners migrated — about 90% (the document prints 90.5%)`	record-from-docs inputs; rounding mine	about 90%

Table 2 — The fix: the plan, the change control and the measurement rule, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Plan composition	`3 security rows + 5 latency rows = 8; marked done: 3 security + 1 latency = 4`	record-from-docs plan rows; count mine	8 rows, 4 done
2	Plan span	`June 24 → August 15: 6 days left in June + 31 in July + 15 in August = 52 days`	record-from-docs ETAs; arithmetic mine	52 days
3	Smoke checks per change	`2 user types (internal SSO, partner) × 4 steps (log-in, workview, log-out, re-log-in) × 2 instances (main, China) = 16; before + after = 16 + 16 = 32`	record-from-docs change process; count mine	32 checks per change
4	Decision rule for the global CDN	`do the reversible items first; re-read the regional P95; buy the global CDN only if Asia and China stay above the others after tuning`	record-from-docs wording; rule stated as I would run it	a trigger, not a date
5	Weekly sample for one P95 read	`6,310 / 17 × 7 = 2,598 Asia views a week; 5% = 130 tail views`	my arithmetic at the window rate; volume has grown since	≈ 130 tail views a week

Table 3 — After: the review series (record-from-docs), before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Overall P95, monthly	`April 8.39 s → September 4.68 s: (8.39 − 4.68) / 8.39 = 44.2%`	record-from-docs; arithmetic mine	−44.2%
2	Asia, weekly	`week 39 5.5 s → week 52 4.0 s: (5.5 − 4.0) / 5.5 = 27.3%`	record-from-docs (Asia non-China); arithmetic mine	−27.3%
3	North America, weekly	`week 39 5.62 s → week 52 4.09 s: (5.62 − 4.09) / 5.62 = 27.2%`	record-from-docs; arithmetic mine	−27.2%
4	China, weekly	`week 39 9.9 s → week 52 6.43 s: (9.9 − 6.43) / 9.9 = 35.1%`	record-from-docs; arithmetic mine	−35.1%
5	Gap to North America	`Asia 5.5 / 5.62 = 0.98× (week 39); 4.0 / 4.09 = 0.98× (week 52); China 6.43 / 4.09 = 1.57×`	my arithmetic on record-from-docs series	Asia at parity; China 1.57×
6	Week 47 spike	`NA 4.52 → 9.44 s = 2.09×; Asia 7.16 → 16.4 s = 2.29×; China 13.8 → 9.36 s (fell)`	record-from-docs; arithmetic mine	not regional
7	Indicative only — do not quote	`(21.19 − 4.0) / 21.19 = 81.1% mixes a 17-day investigation window with a week-52 review figure and possibly different region definitions`	my arithmetic; NOT a result	indicative

Table 4 — The dollars: none in the record; two labeled models

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Record dollars	`the investigation prices nothing; the plan rows carry no cost`	record-from-docs	$0 in the record
2	Excess seconds per tail view	`21.19 − 4.0 = 17.19 s (P95 to P95; indicative)`	illustrative: uses the indicative pair	17.19 s
3	Tail hours in the window	`315.5 views × 17.19 s = 5,423 s; 5,423 / 3,600 = 1.51 h`	illustrative arithmetic	1.51 h

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Direct tail time per year	`1.5065 h × 365 / 17 = 32.3 h; 32.3 h × $50 = $1,617`	illustrative: $50 an hour assumed	≈ $1.6K / yr
5	Exposure on the S12 run-rate	`$76 × 20,000 = $1,520,000; × 6.31 / 13.92 = $689,023`	modeled on a modeled run-rate (volume illustrative); views are not shipments — an upper-bound exposure, not savings	≈ $0.69M / yr at risk

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

PLAN ROW (RECORD)	REGION	ETA	OWNER	STATUS IN THE RECORD
WAF proposal for FF App with the process	All	06/30	app engineer / network infra	done
WAF for the main host in China (incident follow-up)	CN	06/30	network infra / app engineer	done
WAF globally (incident follow-up)	All	06/30	network infra / app engineer	done
Remove the double CDN for static content in China — direct connect to the object store	CN	06/24	network infra	done
Proposal to resolve static CDN performance in China and Asia	CN, Asia	07/15	app engineer + network infra	open
Performance tuning for static content — China	CN	07/30	network infra / app engineer	open; depends on proposal
Performance tuning for static content — Asia	Asia	08/15	network infra / app engineer	open; depends on proposal
CDN for the dynamic host globally	All	Q3 or later	network infra	conditional on the metrics

Change process (record) and my controls. Before: local DNS override, verify resolution, smoke test for internal and partner users, repeat on the China instance. Change: network infra only. After: remove the override on both hosts, clear the cache, re-test. **Risks named:** security approval before replicating assets in China; no consistent way yet to benchmark the China CDN; dependency of the tuning rows on the proposal. **Lean / Kaizen mapping.** Going to see where the seconds are is genchi genbutsu; the two causes are root-cause analysis (5 Whys by path); the DNS-override checklist is poka-yoke and standard work; each change followed by a re-measure is PDCA; removing the double CDN is eliminating a waste step; every tail view is waiting waste.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Why P95 and not the average or P99?**
>
> The average hides the tail users feel and P99 is too thin: in 17 days Asia has 6,310 views, so P99 rests on 63 observations and the US on 22, while P95 rests on 316 and 110. P95 is stable enough to steer by and sensitive enough to show 21.19 seconds against 3.3.

> **How do you know it was a network problem and not the app?**
>
> Two checks. Resource-level: static JS P95 was 5 s in China against 0.5 s in the US. And by region over time: in week 47 North America doubled (4.52 → 9.44, 2.09×) with Asia (7.16 → 16.4, 2.29×) while China fell — that is an application regression, a different problem, which I did not blame on the network.

Did the plan actually close the gap?

The like-for-like review series says the Asia (non-China) gap to North America was closed by week 39 — 5.5 against 5.62 — and stayed there at week 52 (4.0 against 4.09); China remained at 1.57×. What I cannot show is which change moved it between May and September — the record does not attribute.

Why defer the global CDN?

Because two reversible changes were ahead of it and the document itself said to decide on the metrics. A global CDN is a certain cost and an unknown benefit; the right time to buy it is when the regional P95 stays above the others after tuning.

What did the investigation cost and save in dollars?

The record prices nothing. I model it as adoption protection: Asia was 45.3% of views, and 45.3% of the S12 run-rate is ≈ $0.69M a year of exposure — an upper bound, not savings. The direct tail time is about $1.6K a year, which is why the argument was never seconds times dollars.

What would you do differently?

Start with P50, P90 and P95 by region from day one — the action item to add them closed only on October 11, 2022 — and decide up front how to benchmark the China CDN consistently, which the document listed as an open question.

Your 81% — where is it from?

It is (21.19 − 4.0) / 21.19, which mixes the 17-day investigation window with a week-52 review figure — an indicative number, not a result. The like-for-like numbers are −27.3% for Asia over weeks 39 to 52 and −44.2% overall from April to September.

BUSINESS IMPACT MODEL — KPI TREE

```
Asia partners can use the app without waiting: gap to the home region closed, tail visible, changes safe
+- Gap: Asia 21.19 s vs US 3.30 s = 6.42x on 45.3% of views (record)
|    +- static path: China JS 5 s vs 0.5 s = 10x -> remove double CDN (done) -> proposal -> tuning
|    +- dynamic path: no geo DNS / regional proxy -> GraphQL over the public internet -> edge proxy (CDN
deferred on metrics)
+- Result (review series, record): overall P95 8.39 -> 4.68 s (-44.2%); Asia 5.5 -> 4.0 s vs NA 5.62 ->
4.09 s; China 6.43 s = 1.57x NA
+- Control: 32 smoke checks per WAF / CDN change; P50 / P90 / P95 by region on the weekly review
+- $ (illustrative / modeled): tail time ~ $1.6K/yr; exposure ~ $0.69M/yr (upper bound, not savings)
(loop)  complaint -> split by region -> split by path -> reversible fix first -> re-measure -> next fix
```

SOLUTION FLOW

```
monthly review: Asia is the outlier
   -> cut by region: views, P95  (17 days; Asia 45.3% of views, 6.42x US)
   -> cut by path: static (CDN) vs dynamic (GraphQL route)
   -> two causes, two owners, one dated plan (8 rows)
   -> every change: DNS override -> smoke test x16 -> change -> remove override -> smoke test x16
   -> re-read regional P95 -> decide: next tuning row, or the conditional CDN row
```

SENSITIVITY — LOW / BASE / HIGH

ASIA SHARE OF VIEWS (ILLUSTRATIVE RANGE)	EXPOSURE ON THE S12 RUN-RATE	READ
30%	$1,520,000 × 30% = $456,000	≈ $456K a year at risk
45.3% (observed)	$1,520,000 × 45.3% = $689,023	≈ $689K a year at risk
60%	$1,520,000 × 60% = $912,000	≈ $912K a year at risk

Both the $76 × 20,000 run-rate and the view-to-shipment mapping are modeled; the three rows are a range of the share, not a forecast. None of it is claimed as savings.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

The record number here is a gap, not a dollar. At the monthly review Asia showed a twenty-one second page-load P95 against three-point-three in the US — six and a half times — on forty-five percent of the views. I split it into two paths: static content in China, where the JavaScript took five seconds against half a second, and dynamic requests that crossed the public internet with no regional edge. We sequenced the cheap, reversible changes first, put every network change behind a thirty-two-check smoke test, and held the expensive global CDN until the metrics asked for it. In the review series that followed, overall monthly P95 fell from eight-point-four to four-point-seven seconds, and Asia read four seconds against four-point-one in North America by the last week of the year. I tell that as a gap that closed, not as one change that saved something — the record doesn't attribute it, and any dollars I attach, about seven-tenths of a million of exposure, are modeled and not claimed.

IF CHALLENGED — THE FULL ANSWERS

"You are claiming credit for an infrastructure fix you didn't build."

No. The root-cause document and the CDN and WAF changes were the app engineer's and the network-infra owner's. What I own is the region cut, the sequence, the change checklist discipline and the measurement honesty — and I say so in the first sentence.

"Latency is a technical metric. Where is the business?"

Asia was 45.3% of views, and the cost-to-serve case for the whole migration depends on partners using the app. The exposure on the S12 run-rate is ≈ $0.69M a year as a modeled upper bound; the direct tail time is ≈ $1.6K — which is the point: the case is adoption.

"Why not just throw a global CDN at it?"

Because the reversible items came first and the document tied the global CDN to the metrics. A CDN would add cost and ops without telling us which hop is slow.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport the monthly review showed Asian users waiting 21 seconds at P95 on the freight-forwarder app — 6.4 times the US, on 45% of the views. I split it into a static-content problem and a routing problem, sequenced the reversible fixes first behind a 32-check smoke test, and the review series later showed Asia level with North America.

NACF BRIDGE — ONE SENTENCE

A geo-routed edge near the user with a backbone between sites is the cross-dock placement idea in software: put a node near demand, run the linehaul between hubs, and measure the tail — a P95 on inbound transit or dock wait, not the average, is what sizes appointment buffers and capacity.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Walk me through a big problem or issue in your organization that you helped to solve. How did you become aware of it? What information did you gather? What information was missing and how did you fill the gaps?" AIQB: Dive Deep Q4 **Dive Deep** FACET: DATA ANALYSIS

Open (first two sentences): "At a monthly review, Asia showed a twenty-one second page-load P95 against three seconds in the US. The average would have hidden that; I split the tail by region and then by request path."

90-second STAR. S Freight-forwarder app launched in April 2022, 48 of 53 partners on it by May; one region far slower than the rest. **T** Find out where the time goes before anyone buys a fix. **A** I cut 13.92 thousand views by region — Asia 45.3% of views at 6.42× the US P95 — then by path: static JS P95 5 s in China against 0.5 s in the US, and GraphQL from Asia over the public internet with no regional edge. I checked sample sizes — P99 would have rested on 63 observations — and used P95. **R** Two named causes with owners, a dated eight-row plan, and a regional series on the weekly review.

Land: 6.42× on 45.3% of views · **Pivot to NACF:** inbound transit-time by origin lane: the tail by region, not the average, is what sizes buffers and dock-appointment slack.

Do NOT say: that I found the root cause alone, or "I cut Asia latency 81%".

B2 · "Tell me about a time when you evaluated the customer experience of your product or service. What did you do? What was the result?" AIQB: Customer Obsession Q5 **Customer Obsession** FACET: SOLUTION DESIGN

Open (first two sentences): "The slowest region was also the largest: Asia was forty-five percent of the views. So the plan was ordered by where the partners actually were, not by what was easiest to tune."

90-second STAR. S Freight partners across 10 countries using the new app; the review showed one region waiting far longer. **T** Prioritize the work by who is hurt most. **A** I weighed six options — cache time, double-CDN removal, asset replication, regional edge, global CDN, an accelerator — and ordered them by traffic share, reversibility and dependency; the global CDN waited on a metrics trigger. **R** A plan where the first changes were the reversible ones, and by week 39 the review series put Asia (non-China) within 2% of North America.

Land: 45.3% of views · **Pivot to NACF:** siting a node: weight the lanes by volume first, then the tail; the largest flow decides the sequence.

Do NOT say: any dollar figure as savings, or customer complaints that the record does not document — the document says the slowness was noticed in the monthly business review.

B3 · "Give me an example of a calculated risk that you have taken where speed was critical. What was the situation and how did you handle it? What steps did you take to mitigate the risk? What was the outcome?" AIQB: Bias for Action Q1 **Bias for Action** FACET: IMPLEMENTATION

Open (first two sentences): "We started with the changes that could be undone — removing a double CDN, turning on the firewall — and held the expensive global CDN until the metrics asked for it."

90-second STAR. S A latency gap and an open security incident follow-up on the same network path. **T** Move fast on the reversible, slow on the expensive. **A** I sequenced eight plan rows with owners and ETAs from June 24 to August 15 — 52 days — put the firewall and the double-CDN removal first, and wrote the DNS-override smoke-test routine so each change could be verified and backed out. **R** Four of eight rows marked done in the document by June 30; the conditional row stayed conditional.

Land: 52-day plan; 4 of 8 rows done · **Pivot to NACF:** peak prep: do the reversible changes first and buy capacity last — the same logic as flexing labor before adding a door.

Do NOT say: that every row finished on time — only four were marked done; the tuning rows had ETAs, not results.

B4 · "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?" AIQB: Highest Standards Q1 **Insist on the Highest Standards**

FACET: PROBE QUESTIONS

Open (first two sentences): "Every WAF or CDN change ran thirty-two smoke checks, and I refused to quote a before-and-after the data couldn't support — including a flattering eighty-one percent I could have computed."

90-second STAR. S Production network changes in two regions, plus a tempting headline number. **T** Keep production safe and the claims honest. **A** The change process: DNS override, resolution check, internal and partner log-in, workview, log-out and re-log-in, repeated on the China instance, before and after. For the result I used the like-for-like review series, not the investigation window against a week-52 figure. **R** No unreviewed network change; claims limited to −44.2% overall and Asia 5.5 → 4.0 s over weeks 39 to 52.

Land: 32 checks per change; 81.1% not quoted · **Pivot to NACF:** a standard pre-check and post-check on every dock or yard change, and a rule not to compare windows that are not like for like.

Do NOT say: that the 81% is a result; that I attributed the drop to one change.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: "The average looked fine; the ninety-fifth percentile in Asia was twenty-one seconds, on forty-five percent of the views."

Headline metric: 6.42× on 6.31k of 13.92k views

- Region cut first; path cut second (static vs dynamic).
- Sample-size logic: P99 on 63 observations vs 22 in the US; P95 on 316 / 110.
- P95s do not add: 5 s of JS is not 5 s of 21.19.
- Week 47: NA 2.09×, Asia 2.29× — application, not network.

Conflict / influence / backbone angle **Have Backbone; Disagree and Commit**

Opener: "The loud ask was "just turn on a global CDN"; I held it until the metrics asked for it, and that cost us a schedule row."

Headline metric: one conditional row (Q3 or later); 4 of 8 rows done

- Network infra, the app team and security each owned a piece; none could close it alone.
- The plan's conditional row resolved the push without a standoff.
- Security approval before replicating assets in China — I kept that gate.
- My seat: I drove the follow-ups and kept the plan and the smoke-test checklist honest; the app engineer and network owner made the changes.

Honest failure-and-learning angle **Learn and Be Curious**

Opener: "We measured P95 only; P50 and P90 by region did not exist until October, and we never had a like-for-like before-and-after."

Headline metric: window-mismatched 81.1% — not quoted

- The action item to add P50 / P90 / P95 by region closed on October 11, 2022.
- The China CDN had no consistent way to be benchmarked — an open question in the document.
- What I would do differently: baseline by region on the weekly series from day one.
- What held: no unreviewed change and no overclaimed result.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"So you cut Asia latency by 81%."

Numeric answer. No. (21.19 − 4.0) / 21.19 = 81.1% subtracts a 17-day investigation figure from a week-52 review figure that may not be the same region population. Like-for-like: Asia (non-China) 5.5 → 4.0 s = −27.3% over weeks 39 to 52; overall monthly 8.39 → 4.68 s = −44.2%.

Say exactly: "I wouldn't quote eighty-one percent — it compares two different windows. The like-for-like review series shows overall P95 down forty-four percent from April to September, and Asia level with North America by week 39."

"Static JS was 5 seconds of your 21 — so fixing it saves 5 seconds?"

Numeric answer. No. A P95 of one resource is not a slice of a page-load P95, and P95s do not add. The 5 s China JS P95 against 0.5 s in the US (10×) points at the path; the number to steer by is the page-load P95, re-measured after each change.

Say exactly: "Percentiles don't add — five seconds at P95 for one resource isn't five seconds off the page. I re-measured page-load P95 after each change instead."

"Why P95? Why not P99?"

Numeric answer. Sample size. In the 17-day window Asia had 6,310 views, so P99 rests on 1% × 6,310 = 63.1 observations and the US on 21.9; P95 rests on 315.5 and 109.5. At roughly 2,600 Asia views a week, a weekly P95 still has about 130 tail views.

Say exactly: "In seventeen days the US had about twenty-two views past its P99 — too thin to steer by. P95 had over a hundred in the US and over three hundred in Asia."

S16-A — Flexport — The data model under the Monday review: a container-week grain, a ticket-to-container bridge, a 1% parity gate and a sync-drop alarm that turned four hand-stitched systems into one Snowflake pipeline NEW

LPs this story can serve here: Dive Deep (primary) · Invent and Simplify · Frugality · Insist on the Highest Standards
Process family: Improve — data pipeline and reconciliation design under an operating metric; build-vs-bridge and cutover gating

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Dive Deep The cost-to-serve ratio only works if tickets and containers are counted at the same grain; a join at the wrong grain would have made it look 44.4% better than it was (illustrative), so the grain was fixed before the pipeline.

Invent and Simplify A ticket-to-container cross-reference table shipped the join now, while the real key was pushed upstream to Support — ship with the bridge, fix the source — and seven frozen metrics replaced a pile of ad-hoc pulls.

Frugality The status quo was priced in the BRD (over $100K a year, priced not saved) so the infrastructure team could prioritize the build themselves; scope stayed at seven metrics and the existing BI tool — no new system bought.

Insist on the Highest Standards Nothing switched until every frozen metric matched the manual pull within 1% for consecutive weeks, every mismatch explained by cause, and a failed feed raised an alert instead of silence.

Honesty label. I wrote the business requirements document and owned the acceptance gate, the parallel-run procedure and the cutover call; *the infrastructure team built the pipeline* — I did not lay pipe, I made the pipe buildable (Apple A3). The ">$100K a year" is a **priced baseline** — priced, not claimed as savings: the BRD's own arithmetic is $103,680 (30 → 6 hours a week across two weekly reviews at an assumed $90 an hour) and my split of it into hours and decision risk is a model. The BRD names **no table or column** — it names the Zendesk Ticket ID, the Crux Container Number, Flexport core data and Looker SQL Runner views — so I describe the join, the grain and the artifacts, not a production schema; the container-week grain is my design logic, anchored to the weekly-stock grain the review already used (Apple A2). **Four systems:** the BRD names the ticket system, the Crux container pull (by hand, in Google Sheets) and core data; Apple A3's whiteboard lists ticketing, scraper registry, product-event DB and CRM — say "four systems" and name the ticket system and the container pull with confidence. **Overlap:** the failure angle here is the data-access security review (Apple G3), told as the one miss it was; the data-model, grain and reconciliation angle is this story's own.

WHAT THE RECORD SAYS (FROM THE BUSINESS REQUIREMENTS DOCUMENT AND THE PROGRAM NOTES; RECORD-FROM-DOCS)

Problem. Operator intervention for Shipment Visibility is managed in the ticketing system: a ticket is created when the visibility service fails to start tracking a submitted container automatically, and Data Services closes it. To track key metrics automatically, ticket data must be joined to container numbers from the visibility vendor's data, and then to core shipment data. RECORD-FROM-DOCS

As-is. Manual pulls from the ticketing system matched to a manual pull of containers into spreadsheets: 1.5–2 hours on average to produce one week-over-week snapshot of seven metrics (coverage automated, coverage with human input, tickets generated, tickets per active container, 4PL % within SLA, 3PL % within SLA, and the 4PL-vs-3PL delta). RECORD-FROM-DOCS

Requirements. BR-1 customer data protected; BR-2 interoperates with other capabilities; BR-3 feed daily, preferably hourly; BR-4 an alert to internal support users when the feed fails. Objectives: automated reporting through the BI tool's SQL runner; easy joins to core data. RECORD-FROM-DOCS

The pricing in the document. "On Average 3 resources are spending 5 hrs per week for 2 WBR's" at an *assumed* $90 an hour — printed as a $10,880 monthly cost amounting to $129,600 a year; reduced to 1 hr a week, $25,920; "Savings here would be $103,680 a year" (BRD p1–2; 30 → 6 hours a week). The printed monthly figure does not match its own annual (3 × 5 × 2 × 4 × $90 = $10,800). RECORD-FROM-DOCS **Say:** "over $100K a year — the BRD

priced it at $103,680 on thirty hours a week at an assumed ninety dollars an hour; my split into hours and decision risk is a model; priced, not a claimed saving."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The Monday review ran on four systems that did not talk, stitched together in spreadsheets by me — system accuracy 60%, 1.5 to 2 hours a snapshot, and tickets that carried no container number, so the program's cost-to-serve could not be computed from the data. **Task.** Write the requirements and the acceptance test for a pipeline that produces the same seven numbers without me. **Action.** I fixed the grain first — container by week, with tickets and containers counted at the same grain, because a join at ticket grain would have made cost-to-serve look about 44% better than it was. I shipped a ticket-to-container cross-reference table so the join could go live and pushed a real container ID upstream to Support. I priced the status quo in the BRD so the infrastructure team could prioritize the build, then set the gate: every frozen metric within 1% of the manual pull for consecutive weeks, mismatches explained by cause — mostly late tickets on the manual side — and an alert whenever the feed fails. **Result.** Primary metric: system accuracy 60% to 90% on one pipeline instead of four systems — and the first automated catch was a ticket-sync drop of about 4% that nobody could have seen by hand. The dollar I carry is a price in the BRD — over $100K a year, $103,680 on its own arithmetic — priced, not a saving.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “The review was priced at over a hundred thousand dollars a year — the requirements document works it to a hundred and three thousand, six hundred and eighty on thirty hours a week at an assumed ninety dollars an hour — and that is a price, not a saving I claim.”

“What's your source for that?” “It is my own requirements document; the arithmetic and its monthly typo are printed on the page, my split into hours and decision risk is a model, and the accuracy, the parity gate and the four-percent catch are in my interview workbook.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The program's weekly review needed seven operating metrics, and two of them — tickets generated and tickets per active container — required joining the support ticketing system to container data. The tickets did not carry a container ID. So the numbers were assembled each Monday by hand from four systems into spreadsheets: system accuracy 60%, one person able to produce them, and silent data loss nobody could see. A VP was steering the program by a number that only one person could produce.

T — Define the data contract and the acceptance test: one grain, a join that survives missing keys, definitions in code, a reconciliation gate to cut over, and a failure alert — and get the infrastructure team to build it by showing them the price of the status quo.

Baselines on the table: Four systems · 1 person · system accuracy 60% · 1.5–2 h per weekly snapshot (BRD) · 7 frozen metrics · tickets with no container ID · no failure alarm · priced baseline: BRD $129,600 a year → $103,680 difference (30 → 6 h/wk at an assumed $90/h), over $100K priced, not saved (my model: $100,360).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — documented the as-is — which systems, which pulls, which join — named the key gap (tickets without a container ID), froze seven metric definitions with their owners, and priced the status quo in the BRD (over $100K a year) so the infrastructure team could prioritize the build themselves. *Exit bar:* infra prioritizes the build off the price; scope is the seven metrics, nothing new.
- **Plan & Design** — wrote the requirements (BR-1 to BR-4), chose the container × week grain, specified the documented ticket-to-container cross-reference, set BR-4 (feed-failure alert) and the stale-table alerts, and sketched the pipeline: scheduled extracts → S3 landing zone → SQS-triggered loads → Snowflake raw → curated metric tables → Looker SQL Runner views. *Exit bar:* requirements accepted by infra; acceptance gate published before build.

- **Build & Test** — extract and load specs per system; the bridge table first; parity checks per metric; the parallel-run procedure, every mismatch explained by cause; the data-access security review that surfaced mid-build and cost two weeks. *Exit bar:* all seven metrics inside 1% of the manual pull for consecutive weeks; review closed.
- **Cutover & Go-Live** — the cutover memo; the Monday deck switched to the pipeline; the manual pull kept running as rollback until parity had held a week beyond the gate. *Exit bar:* first Monday off the pipeline shows the same seven numbers.
- **Hypercare / Close** — watched the failure alerts; triaged the first automated catch (a 4% ticket-sync drop) to a source cause; moved definitions into code; pushed the real container key upstream. *Exit bar:* accuracy 60% → 90%; the pipeline steers the review without me.

Influence and ownership. Mine: the as-is documentation, the grain choice, the bridge-table design, the seven frozen definitions, the pricing, the acceptance gate, the parallel-run procedure and the cutover memo. Not mine: the pipeline build and its alerts (infrastructure team), the real container key (Support), the security review (security team). I influenced by pricing the status quo in the BRD and making the cutover decision unarguable with a published gate.

> **Designed the process; others own it now.** The definitions are in code, the gate is a documented procedure and the alerts page a team, not me. **Trade-off put to leadership.** New metrics for stability (the seven were frozen, with a parking lot), and two weeks of date for a data-access review I should have planned.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — Over $100K a year, priced in the BRD, not saved: the BRD route gives $103,680 (30 → 6 h a week at an assumed $90/h); my model gives $100,360 ($30,160 hours + $70,200 decision risk). Both are priced, not claimed as savings. Realized candidate: 208 hours retired × $90 = $18,720 (modeled).
- **#** — Four systems → one pipeline; seven definitions in code; no longer a single point of failure (one engineer maintains the pipeline instead of one PM rebuilding it, Apple A3); zero WBR deliveries skipped through the build (record).
- **%** — System accuracy 60% → 90% (record); parity gate within 1% per metric for consecutive weeks (record); first automated catch a ticket-sync drop of about 4% (record).
- **Time** — the weekly snapshot (1.5–2 h by hand per the BRD) → automated; the data-access review cost two weeks (record).
- **Customer** — the VP and the core team steered on the same seven numbers the first Monday off the pipeline; Data Services and Support got tickets joined to containers for the first time.
- **Quality** — every parallel-run mismatch explained by cause — mostly late-arriving tickets on the manual side; silence on a failed feed replaced by an alert.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No one had written requirements for a program's operating metrics, and the cost-to-serve ratio had no join key. I defined grain, bridge and gate before anyone built.
Scope & influence	A pipeline owned by another organization, a key owned by Support and a review owned by security; I moved all three by pricing and publishing the gate, not by authority.
Execution trade-off + mechanism others now own	Seven metrics frozen for stability; two weeks lost to a review I should have planned. The mechanism others own: definitions in code, the parity procedure, the alerts.
Impact (long-term)	The weekly review no longer depends on one person, and the pipeline sees what the sheets could not — the first catch was a 4% drop.
Process improvement (standard work institutionalized)	Parallel-run-until-parity with a written gate; a bridge-then-fix-the-source rule for missing keys; a kickoff question about data-access approvals and their lead time.
Complexity & consensus	Four source systems, infra, Support and security with different priorities — one price in the BRD and one published gate aligned them.
Root cause at scale	The reason the number could be wrong was structural: the unit of cost (a ticket) and the unit of volume (a container) had no shared key or grain.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Ticket grain vs container grain; the key gap; late-arriving tickets; which mismatches were the manual side's fault; sample of what silence hid.
2 · Solution design	Sheets automation vs warehouse pipeline; container × week grain; bridge table vs waiting for the key; hourly vs daily feed; alarm design.
3 · Math & data logic	Join fan-out (0.667 vs 1.20); parity 0.42% pass and 1.25% investigate; alarm 4% vs 1%; BRD $129,600 → $103,680 vs model $100,360.
4 · Implementation	BR-1 to BR-4; parallel run to rollback; cutover memo; alert triage; security review as a dependency with a lead time.
5 · Probe questions	Seven numeric probes plus three trap drills: priced vs saved, 1.25% investigate-or-fail, sync-drop threshold.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What I read. Three data sets named in the BRD **RECORD-FROM-DOCS**: (1) Zendesk tickets — one row per ticket, created when the visibility service fails to start tracking a submitted container automatically, closed by Data Services; (2) Crux container data — pulled by hand into Google Sheets, keyed by container number; (3) Flexport core data in the warehouse. The unit of *cost* is a ticket; the unit of *volume* is a container; the ratio needs both, so I analyzed the grain at which each could be counted.

DATA SET	UNIT (GRAIN)	KEY IT CARRIES	WHAT IT IS USED FOR	WEAKNESS FOUND
Zendesk tickets	one row per ticket	Ticket ID; *no container ID*	numerator: tickets generated	cannot be joined to a container
Crux container data	one row per container	Container Number	denominator: active containers; coverage	a manual pull into Google Sheets, taken once a week
Flexport core data	one row per shipment / container	container number, client, mode	cuts: 4PL vs 3PL; SLA	not joined to tickets until a bridge exists

The seven metrics the BRD froze, and the grain each lives at

#	METRIC (BRD WORDING)	NUMERATOR	DENOMINATOR / CUT	GRAIN
1	Coverage — % of containers tracked (automated)	containers tracked with no human input	active containers	container-week stock
2	Coverage — % of containers tracked (with human inputs)	containers tracked after a human touch	active containers	container-week stock
3	Cost to serve — tickets generated	tickets	—	ticket, counted per week
4	Cost to serve — tickets generated / active container	tickets	active containers	needs the ticket → container join
5	Timeliness — 4PL % within SLA	containers inside SLA	4PL containers	container-week, 4PL cut
6	Timeliness — 3PL % within SLA	containers inside SLA	3PL containers	container-week, 3PL cut
7	Timeliness — delta (4PL vs 3PL)	metric 5 less metric 6	—	derived

Metric names are the BRD's. The numerator, denominator and grain columns are my design logic — the BRD does not define them. "Active container" is the review's own definition: cumulative inflow less containers 15 days past departed_terminal (Apple A2, grain 2).

What it revealed. (1) *No key, no ratio:* without a container ID on the ticket, tickets per active container could only be assembled by hand RECORD-FROM-DOCS. (2) *Grain decides the headline — in the real instrument:* the review workbook (Apple A2, SV_28.xlsx) shows 80.6% automated at event grain (1,991 of 10,289 events human-touched = 19.4%), 54.1% at ticket grain (538 auto, 456 human) and 40.0% on 17 Jan falling to 21.9% on 31 Jan at weekly stock — three grains, three answers, so the grain label travels with every number (record). My worked join example makes the same point for the cost ratio: 12 tickets over 4 of 10 active containers give 18 rows at ticket grain, 12 / 18 = 0.667 instead of 12 / 10 = 1.20, a ratio 44.4% too good (illustrative — my design example, not program data). (3) *Late-arriving tickets:* the manual snapshot missed tickets that arrived after the pull — nearly every parallel-run mismatch was the manual side (record, Apple A3). (4) *Silence was invisible:* a ticket-sync drop of about 4% after a routine update looked like a quiet week in the manual era (record).

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Where the logic lives	Automate the spreadsheets · BI on a ticket export · warehouse pipeline + BI views	Warehouse feed + BI SQL views (the objectives in the document)	Joins to core data are one query; the manual pull stays a rollback, not the source.
Grain	Ticket · container · container × week	Container × week (Monday–Sunday, as the review reads it)	Numerator and denominator at one grain prevents fan-out; matches the weekly-stock grain the review already used (design logic; Apple A2 grain 2).
The missing key	Wait for Support to add a container ID · parse it from ticket text · cross-reference table now + real key upstream	Bridge now, key upstream	Ship with the bridge, fix the source: the program did not wait for another team's roadmap.

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Feed frequency	Daily · hourly	Daily, preferably hourly (BR-3)	BR-3 says daily, preferably hourly; my whiteboard sketch has daily with hourly on the fast tables (Apple A3). The review is weekly, so daily is sufficient; hourly helps alarm latency.
Failure handling	Dashboard only · alert to support users · both	Alert when the feed fails (BR-4) and when sync rate drops	The first catch — a 4% drop — only exists because an alert does.
Cutover	Big-bang · parallel run to a gate · phased by metric	Parallel run to a 1% gate for consecutive weeks, manual pull as rollback	Makes the decision unarguable — the VP sees the same seven numbers.

The data elements, from the record (the BRD names no tables or columns)

ELEMENT	GRAIN / KEY	WHAT THE RECORD SAYS	SOURCE
Zendesk tickets (feed)	one row per ticket · Ticket ID	created when auto-tracking fails to start; closed by Data Services; the queue's UPDATED_BY_ID splits human from auto	BRD p1; Apple A2 grain 3
Crux container data	one row per container · Container Number	was a manual pull into Google Sheets; statuses include departed_terminal, in_review, out_of_network	BRD p2; Apple A2
Flexport core data	shipment / container	joined in Looker SQL Runner; carries the 4PL / 3PL and SLA cuts the metrics need	BRD objectives; metrics 5–7
Cross-reference table	one row per ticket–container pair	documented bridge so the join ships before the real key exists; data classes named, who can read what	Apple A3, G3
Feed and alerts	Zendesk → Snowflake, daily or preferably hourly	BR-3 feed frequency; BR-4 alert to SV Support users when the feed fails	BRD p4
Looker views (SQL Runner)	one row per metric × week	the seven definitions in SQL, no manual reconciliation	BRD use cases
Pipeline sketch (as I'd whiteboard it)	sources → scheduled extracts → S3 landing zone → SQS-triggered loads → Snowflake raw → curated metric tables	daily, hourly on the fast tables; SQL-runner queries behind the same WBR deck; stale-table alerts to the ops channel; one engineer maintains it	Apple A3 design sketch

The names above are the record's own (Ticket ID, Container Number, Looker SQL Runner, UPDATED_BY_ID, the three statuses). Table names, column names and the container-week fact are *design logic* I describe — not a production schema I recall.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers (the three-grain row is record, Apple A2; the join illustration is illustrative; BRD pricing is record-from-docs)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Manual cost of a snapshot	`1.5–2 hours to arrive at one week-over-week snapshot of seven metrics`	record-from-docs (BRD As-Is); a different measure from the about-four-hours-a-week-at-its-worst in Apple A1/A2 — do not stack them	1.5–2 h / snapshot
2	System accuracy	`system accuracy 60% (record); read as a weekly hit rate, (1 − 0.60) × 5 = 2 weeks in 5 off`	record (60%); the weeks-in-5 reading is my arithmetic, not a record	2 weeks in 5
3	Join fan-out at the wrong grain	`10 active containers; 12 tickets over 4 of them; rows after a ticket-grain join = 12 + (10 − 4) = 18`	illustrative example	18 rows for 10 containers
4	Ratio, wrong vs right	`wrong 12 / 18 = 0.667; right 12 / 10 = 1.20; understatement = 1 − 0.667 / 1.20 = 44.4%`	illustrative example	cost-to-serve 44.4% too good
5	BRD labor baseline	`3 people × 5 h × 2 reviews = 30 h a week; × 4 × 12 = 1,440 h; × $90 = $129,600 a year`	record-from-docs; arithmetic mine	$129,600 / yr
6	The printed monthly figure	`30 h × 4 × $90 = $10,800, not the printed $10,880; gap $80`	record-from-docs typo; arithmetic mine	$10,800
7	Three grains, one workbook (real)	`event grain: 1 − 1,991 / 10,289 = 80.6% automated; ticket grain: 538 / (538 + 456) = 54.1% automated`	record (Apple A2, SV_28.xlsx: 1,991 / 10,289 events; 538 auto / 456 human tickets)	80.6% vs 54.1%

Table 2 — The fix: the gates and alarms, values substituted (example values illustrative)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Grain	`container × Monday-week; numerator (tickets) and denominator (active containers, touched in the last 15 days) counted at the same grain`	design; the 15-day active rule is record (Apple A2)	one row per container-week
2	Unmatched-ticket alarm	`unmatched share = 3 / 405 = 0.74% ≤ 1% → pass`	illustrative: 405 tickets, 3 unmatched, threshold 1%	0.74%
3	Parity — pass	`|70.9 − 71.2| / 71.2 = 0.42% ≤ 1% → pass (coverage: manual 71.2%, pipeline 70.9%)`	illustrative values; the 1% gate is the BRD acceptance rule (record)	0.42% → pass
4	Parity — investigate	`|405 − 400| / 400 = 1.25% > 1% → investigate: 5 late tickets, the manual side wrong, the pipeline right`	illustrative values (tickets: manual 400, pipeline 405 — not the real 538 / 456 counts)	1.25% → investigate
5	Sync-drop alarm	`alert if 1 − sync rate > 1%; first catch 1 − 0.96 = 4% = 4× the threshold`	catch is record; the 1% threshold is illustrative	4% → alert
6	Parity checks needed to cut over	`7 metrics × 2 consecutive weeks = 14 checks minimum`	the BRD says "consecutive weeks"; two is the minimum, the actual number is illustrative	≥ 14 checks

Table 3 — After: before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	System accuracy	`60% → 90%; read as a weekly hit rate, (1 − 0.90) × 5 = 0.5 in 5 off`	record (60% → 90%); the weeks-in-5 reading is my arithmetic	2 in 5 → 0.5 in 5
2	Systems and key-person risk	`4 systems → 1 pipeline; one person stitching → one engineer maintaining it`	record (Apple A3); no bus-factor count is claimed	4 → 1 · no single point of failure
3	Manual hours	`~4 h a week × 52 = 208 h a year → ~0`	modeled	208 h retired
4	Silent data loss	`invisible → alert; first catch a 4% ticket-sync drop`	record	seen for the first time
5	Ticket-to-container join	`none → documented cross-reference → real key pushed upstream`	record (Apple A3); the join rate is not stated in any source — I would measure it	join key now upstream

Table 4 — The dollars: two priced baselines, one realized candidate — none claimed as savings

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	BRD route: after and difference	`$129,600 × (1 / 5) = $25,920; $129,600 − $25,920 = $103,680`	record-from-docs (BRD); priced, not claimed	$103,680
2	Model route: hours	`4 h × 52 = 208 h; 208 × ≈ $145 = $30,160`	modeled (rate illustrative; my model, not a source figure)	$30,160
3	Model route: decision risk	`52 weeks × 10% = 5.2 wrong-number weeks; 5.2 × $13,500 = $70,200`	modeled (both inputs illustrative; range $12K–$117K)	$70,200
4	Model route: total	`$30,160 + $70,200 = $100,360`	modeled; the record states "over $100K" as the priced baseline	$100,360
5	Do the two routes agree?	`($103,680 − $100,360) / $100,360 = 3.3%; both clear $100K by different bases (30 h a week vs ~4 h a week plus risk)`	my comparison; the two bases differ (30 h a week in the BRD vs ~4 h in my model)	3.3% apart
6	Realized candidate	`208 h × $90 = $18,720 (hours actually retired at the BRD rate)`	modeled; the honest version of "savings" — measure hours retired and restated Monday numbers	$18,720

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

STEP	WHAT HAPPENED (RECORD UNLESS MARKED)	OWNER	CONTROL
Requirements	BRD with BR-1 to BR-4, seven definitions, as-is and pricing	me	accepted by infra; gate published before build
Bridge	documented ticket-to-container cross-reference so the join ships; real container ID pushed upstream	infra builds; Support owns the key	unmatched tickets counted and listed, never dropped
Build	scheduled extracts → S3 → SQS loads → Snowflake raw → curated → Looker SQL Runner (the whiteboard sketch); failure alerts	infrastructure team	BR-4 alert; BR-1 data protection
Security review	data-access review for ticket data carrying customer content surfaced mid-build; two weeks; re-sequenced; Monday review kept alive on the manual pull (Apple G3)	me (owned it upward immediately); security team	no review delivery skipped

STEP	WHAT HAPPENED (RECORD UNLESS MARKED)	OWNER	CONTROL
Parallel run	manual vs pipeline per frozen metric, within 1% for consecutive weeks, mismatches explained by cause	me	gate published; late-arriving tickets the usual cause
Cutover	the Monday deck switches to the pipeline; manual pull kept as rollback a week beyond parity	me (memo)	rollback = the manual pull
Hypercare	first alert triaged: 4% ticket-sync drop; definitions into code	infra + me	alerts reach a person, not silence

Risks named. The key gap, parity failure, one person as the single point of failure — and, as it turned out, the data-access review, which was not on the list. **Lean / Kaizen mapping.** The manual pull is motion and waiting waste; the seven frozen definitions are standard work; the parity gate is jidoka — stop the line when the pipeline and the reference disagree; the bridge table is poka-yoke for a missing key; "ship with the bridge, fix the source" is root-cause countermeasure at the right layer; every PDCA starts with the first alert.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Why container × week?**
>
> Because the review reads weeks and the ratio needs numerator and denominator at one grain: tickets per active container. At ticket grain the container repeats once per ticket — my worked example gives 18 rows for 10 containers and a ratio of 0.667 instead of 1.20, 44.4% too good. At container-week every metric is a sum or a ratio of one table.

> **How did you join tickets that had no container ID?**
>
> A documented cross-reference table so the pipeline could ship, then a real join key pushed upstream so the fix outlived the workaround (record, Apple A3). The design I would add: count and list unmatched tickets, never silently drop them, with an unmatched-share alarm at 1% (threshold illustrative). I do not have a measured join rate to quote.

> **Why daily or hourly?**
>
> BR-3 says daily, preferably hourly; my whiteboard sketch is daily with hourly on the fast tables. The review is weekly, so daily is sufficient for the numbers; hourly matters for the failure alarm — a drop is caught the same day instead of the next.

> **What if parity had failed?**
>
> Then we would not have cut over: the manual pull stays the source of truth. Pass is within 1% on every frozen metric for consecutive weeks — 0.42% passes, 1.25% is investigate not fail (illustrative values), and in the cases I saw the explanation was late-arriving tickets on the manual side.

> **You wrote $103,680 in the BRD — what did you save?**
>
> Nothing I claim. It is a price: 3 × 5 × 2 × 4 × 12 × $90 (an assumed rate) = $129,600, less $25,920 after, $103,680 — and my own model prices it at $100,360 including decision risk. The realized candidate is the hours retired, 208 × $90 = $18,720, modeled, and I would measure it.

> **Why not wait for Support to add the container ID?**
>
> Because a program metric cannot wait on another team's roadmap; the bridge gave us the join now and the upstream fix made it cheap to retire the bridge. Ship with the bridge, fix the source.

> **What did the pipeline catch that you could not?**
>
> A ticket-sync drop of about 4% after a routine update (record, Apple A3). With my illustrative 1% alarm threshold that is 4× the threshold; in the manual era, silence read as a quiet week.

BUSINESS IMPACT MODEL — KPI TREE

```
The weekly review steers on numbers that are right, owned by more than one person, and alarmed when they
break
+- System accuracy 60% -> 90% (record)
+- Data model: container x week grain; bridge table (ticket <-> container); definitions in code
|    +- grain logic: 12 tickets / 10 containers = 1.20, not 12 / 18 = 0.667 (illustrative); real: 80.6%
event vs 54.1% ticket grain (Apple A2)
+- Gate: parity within 1% per metric, consecutive weeks (record; 0.42% pass / 1.25% investigate are
illustrative) | manual pull = rollback
+- Alarm: ~4% ticket-sync drop caught (record); alert threshold 1% (illustrative)
+- $ (priced, not saved): over $100K | BRD $103,680 | model $100,360 | realized candidate $18,720
(modeled)
(loop)  four systems by hand -> one person stitches -> one pipeline -> parity gate -> alarms
```

SOLUTION FLOW

```
Zendesk tickets (no container ID) --+                                          +--> Looker SQL Runner
views --> Monday deck
Crux container data ---------------+--> scheduled extracts -> S3 -> SQS loads -> Snowflake raw -> curated
metric tables
core data (4PL / 3PL, SLA) --------+             ^                                    |
          cross-reference table (documented; ticket <-> container)                     +--> alerts: feed
failure (BR-4), stale tables
parallel run: manual vs pipeline per metric, within 1%, consecutive weeks -> cut over -> manual pull kept
one more week as rollback
(pipeline stages: the whiteboard sketch, Apple A3; the BRD fixes the feed, the alert and the Looker SQL
Runner views)
```

SENSITIVITY — LOW / BASE / HIGH

LABOR BASIS FOR THE HOURS TERM	HOURS × WEEKS × RATE	BASIS
canon model: ~4 h a week	4 h × 52 wk × $145 = $30,160	modeled
one person, 5 h × 2 reviews	10 h × 48 wk × $90 = $43,200	illustrative
BRD: 3 people, 5 h × 2 reviews	30 h × 48 wk × $90 = $129,600	record-from-docs

The hours term alone moves 4.30× across the three bases (129,600 / 30,160), before the decision-risk term. That spread is why the number is a *price*, never a saving. The decision-risk term ($70,200) has its own range, $12K–$117K.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> The record number here is a price, not a saving. The manual way of producing the weekly review was priced in my requirements document at over a hundred thousand dollars a year — about a hundred and four thousand on its own arithmetic, thirty hours a week at an assumed ninety dollars an hour; my own model gets to about a hundred thousand with decision risk — and that price let the infrastructure team prioritize the build themselves. What I own is the design under it: a container-by-week grain so tickets and containers are counted together, a bridge table so the join could ship before the source was fixed, and a gate that said nothing switches until all seven metrics match the manual pull within one percent for consecutive weeks. System accuracy went from sixty to ninety percent, and the first automated catch was a ticket-sync drop of about four percent that nobody could have seen by hand. The realized version of the dollars is hours actually retired — about two hundred hours, roughly nineteen thousand at the requirements document's rate, modeled — and I won't call any of it savings.

IF CHALLENGED — THE FULL ANSWERS

"The requirements document says $103,680 saved. That's a lot for a spreadsheet."

It is a price, not a saving. The document's own arithmetic is 3 people × 5 hours × 2 reviews × $90, and its printed monthly figure does not match its annual. I say what the price was for — the BRD priced doing nothing so the infrastructure team could prioritize the build themselves — and I lead with system accuracy and the first catch, not dollars.

"Why build a pipeline for seven metrics?"

Because the ratio the program steered on — tickets per active container — had no join key and the number rested on one person stitching four systems by hand. Seven was the scope I held on purpose; new metrics went to a parking lot until the pipeline was stable.

"What's your actual technical contribution?"

The infrastructure team built the pipeline. Mine were the grain, the bridge-table design, the seven definitions, the pricing, the acceptance gate, the parallel-run procedure and the cutover memo — the part that decides whether a build is trusted.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport the weekly review ran on four systems stitched by hand by one person. I wrote the requirements and the acceptance gate: a container-by-week grain, a cross-reference table for the missing ticket key, and a one-percent parity gate. System accuracy went from 60% to 90%, and the first automated catch was a ticket-sync drop of about 4%.

NACF BRIDGE — ONE SENTENCE

Capacity planning for inbound sits on the same data discipline: appointments, POs and containers must join at one grain (container × week or door × hour), unmatched records must be counted rather than dropped, and a feed that goes silent must raise an alarm — otherwise a quiet dock schedule looks like a healthy one.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Give me an example of a complex problem you solved with a simple solution. What made the problem complex? How do you know your solution addressed the problem?" AIQB: Invent and Simplify Q1

Invent and Simplify FACET: SOLUTION DESIGN

Open (first two sentences): "The Monday review depended on four systems and one person assembling them by hand. I simplified it into one pipeline with seven frozen definitions and a bridge table for the one thing that was missing."

90-second STAR. S Four systems, one person, accuracy 60%, tickets with no container ID. **T** Produce the same seven numbers without that person. **A** I wrote the requirements, chose a container-by-week grain, specified a ticket-to-container bridge so the join could ship before the source was fixed, froze seven definitions, and set a one-percent parity gate with the manual pull as rollback. **R** Accuracy 60% → 90%; four systems → one pipeline; the review no longer depends on me.

Land: four systems → one pipeline; 60% → 90% · **Pivot to NACF:** a standard data contract for inbound: one grain, one bridge for missing keys, definitions in code.

Do NOT say: that I built the pipeline myself, or that the dollar is saved.

B2 · "Tell me about a situation that required you to dig deep to get to the root cause. How did you know you were focusing on the right things? What was the outcome? Would you have done anything differently?"

AIQB: Dive Deep Q2 | **Dive Deep** | FACET: DATA ANALYSIS

Open (first two sentences): "A join at ticket grain would have repeated each container once per ticket and made cost-to-serve look about forty-four percent better than it was. I caught that in the design, before it was in a deck."

90-second STAR. S Cost-to-serve per active container was the program's headline ratio, and its join key did not exist. **T** Make the ratio right before anyone steered on it. **A** I laid out the grain of each data set, showed with a worked example that 12 tickets over 4 of 10 containers gives 18 rows and a ratio of 0.667 instead of 1.20, and moved the model to container × week with numerator and denominator at the same grain. The parallel run then showed most mismatches were late-arriving tickets on the manual side. **R** A ratio that foots, and a first automated catch — a 4% ticket-sync drop — that the manual era could not have seen.

Land: 0.667 vs 1.20 (illustrative); 4% drop (record) · **Pivot to NACF:** any network ratio — utilization, cost per unit — is only as good as the grain of its numerator and denominator.

Do NOT say: that the 44.4% was a real finding in production — it is my design example; or that the manual numbers were never right.

B3 · "Tell me about a time when you generated a creative solution to a problem or project without requiring additional resources. What was the problem? What was the solution and how did you come up with it? What was the outcome?" AIQB: Frugality Q5 | **Frugality** | FACET: IMPLEMENTATION

Open (first two sentences): "I didn't ask for budget. I priced the manual process in the BRD, so the infrastructure team put the build on their own roadmap, and I held the scope to seven metrics."

90-second STAR. S The pipeline belonged to another organization with its own priorities. **T** Get it built without owning or funding it. **A** I priced the status quo in the BRD — its labor arithmetic gave $103,680 and my model $100,360 including decision risk; I held scope at seven metrics with a parking lot and used the existing BI tool's SQL runner. **R** Infrastructure prioritized the build itself; no new system bought; zero review deliveries skipped through the build.

Land: over $100K priced, not saved · **Pivot to NACF:** a business case for dock or yard automation: price the status quo, scope tight, no new tool unless the numbers demand it.

Do NOT say: the $103,680 as savings; or the hours basis as settled — it is the BRD's 30 hours a week at an assumed $90 an hour, not my ~4 hours.

B4 · "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?" AIQB: Highest Standards Q1 | **Insist on the Highest Standards** | FACET: PROBE QUESTIONS

Open (first two sentences): "Nothing switched until all seven metrics matched the manual pull within one percent for consecutive weeks, with every mismatch explained by cause, and a failed feed raising an alert."

90-second STAR. S A VP steered the program by this number, and the pipeline was new. **T** Earn the cutover instead of announcing it. **A** I published the acceptance gate before build; ran the parallel run; investigated every mismatch — 1.25% is investigate, not fail (my illustrative example); kept the manual pull as rollback a week beyond parity; and required the failure alert (BR-4) before go-live. **R** The first Monday off the pipeline showed the same seven numbers, and silence stopped meaning health.

Land: 1% gate; 0.42% pass (illustrative) · **Pivot to NACF:** cutting over a yard-management or appointment system: parallel run to a numeric gate, rollback kept, alarm on silence.

Do NOT say: that the 1% threshold on the alarm was in the record — it is illustrative; or the number of consecutive weeks.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: “A ticket-grain join would have made cost-to-serve look forty-four percent better than reality — so before we built anything, I fixed the grain.”

Headline metric: 12 / 18 = 0.667 vs 12 / 10 = 1.20 (illustrative); 4% drop (record)

- Unit of cost (ticket) vs unit of volume (container); one grain: container × week.
- Unmatched tickets counted and listed, never dropped.
- Parity: 0.42% pass; 1.25% investigate (illustrative); mismatches the manual side's late tickets.
- Say "illustrative" on every example number.

Conflict / influence / backbone angle **Have Backbone; Disagree and Commit**

Opener: “Everyone wanted one more metric during the build; I held the scope at seven with a parking lot, and shipped the bridge instead of waiting for Support.”

Headline metric: seven frozen metrics; one bridge table

- Stability over coverage — a documented trade-off, not an oversight.
- Bridge now, real key upstream: another team's roadmap did not hold the program.
- Priced in the BRD so infra could prioritize it themselves — influence without authority.
- My seat (Apple A3): the BRD, the seven definitions, the parallel-run gate, the acceptance test, the cutover call; infra built the pipeline.

Honest failure-and-learning angle **Earn Trust**

Opener: “I discovered the data-access security review mid-build instead of planning it — two weeks — on the pipeline that was supposed to retire my own manual review.”

Headline metric: two-week security-review miss; zero Monday reviews skipped

- I owned it upward immediately: the slip was scoping, not engineering — the BRD priced the build and never the review (Apple G3).
- Ticket data carries customer content, so the access review is a gate with lead time, not a formality; I had treated it as a formality.
- I re-sequenced instead of stalling: the parallel run kept the Monday review alive on the manual path, so no review week was skipped.
- I closed the review properly — cross-reference table documented, data classes named, who could read what — and now every data program starts with "whose approval does this data need, and how long does that take?"
- Not a dollar claim: the miss cost margin, not delivery. Do not mix in the $103,680 here.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"Your document says $103,680 saved. So you saved over a hundred thousand?"

Numeric answer. No. 3 × 5 × 2 × 4 × 12 × $90 (an assumed rate) = $129,600; after = $25,920; difference $103,680 — a priced labor estimate, and the printed monthly figure ($10,880) does not match its annual ($10,800 × 12). My model prices the baseline at $100,360 ($30,160 + $70,200). The realized candidate is the hours actually retired: 208 × $90 = $18,720, modeled.

Say exactly: “It's a price, not a saving — about a hundred and four thousand on the document's labor basis, about a hundred thousand in my model. The realized version is hours retired, about nineteen thousand, modeled.”

"Pipeline 405 tickets, manual 400. That's a 1.25% gap — a failure?"

Numeric answer. Investigate, not fail (illustrative values): |405 − 400| / 400 = 1.25%, just over the 1% gate, and the explanation in the cases I saw was five late-arriving tickets on the manual side. Contrast 70.9% vs 71.2% coverage: 0.42%, pass. (Example values are illustrative.)

Say exactly: “A gap of one-point-two-five percent is over the gate, so we investigate — and the cause was late tickets the manual pull missed. Coverage at point-four-two percent passed.”

"How do you know the 4% drop wasn't noise?"

Numeric answer. The alarm rule is 1 − sync rate > 1% (threshold illustrative); the observed 1 − 0.96 = 4% is 4× that threshold, and the record says it was a ticket-sync drop of about 4% after a routine update, invisible in the manual era; I did not keep the triage detail beyond that. A threshold like this is tuned from weeks of normal variation, which I would pull from the history.

Say exactly: “Four percent is four times the alarm threshold, and in the manual era it would have looked like a quiet week. I triaged it to a source cause, not to noise.”

Question bank — likely questions for this interviewer, with a one-line answer plan

NEW

Behavioral and functional questions a hiring manager for NACF Supply Chain is likely to ask, in rough order of likelihood. Each plan points at a packet story or a named backup in the master packet. Stories allocated to other interviewers are referenced as pointers only. v2 adds rows 17–28 for the five new primaries; the Router at the front maps all 16 Leadership Principles.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Tell me about yourself.**	The 75-second newest-to-oldest script from the master packet: Sephora (one signed "available"; dual-DC WMS migration) → Flexport (the review; $0.66 → $0.03) → Elementum (measured lanes, $1.5M out) → Manhattan (99% fill). End on "design the supply chain, keep it honest in a weekly review — at network scale."	Master TMAY
2	**Why Amazon, why now?**	Three reasons — the forecast (Elementum lanes), the weekly review (Flexport), one version of the truth (Sephora) — and "what I went to Sephora to build is live."	Master Why Amazon
3	**Tell me about your most significant delivery.**	S11 pitch; end on CTS $0.66 → $0.03 across 7 → 68; stop.	S11
4	**Tell me about a time you changed a plan that was already committed.**	S12 pitch; private-first; 66% less effort, value month 6; ~$5M once.	S12
5	**Tell me about a time leadership doubted your program.**	S11 Discovery: accepted the critique as being about evidence; built the review; left the critic steering by the numbers.	S11
6	**How do you make trade-offs between speed and quality?**	S11: board no faster than automation matures; wave 3 held publicly. S12: confidence buffers over a fast optimistic plan.	S11 / S12
7	**Describe how you would run your first 90 days in NACF Supply Chain.**	Book of record and primary metric first; stocks-and-flows map of the process; the invariant broken today; 3 key results with denominators; a weekly input/output review by week 6 (master §3.0, seven steps).	Master §3.0
8	**How would you decompose a complex network problem (e.g., inbound placement cost rising)?**	Open as the master's §3.99 script: book of record + metric → stocks and flows → broken invariant → minimal design → sequence by Reach × Impact × Confidence ÷ Effort → one metric, one labeled dollar.	Master §3.99
9	**Tell me about a time you failed.**	Full failure story here: S11-B (the pilot ran later than the first testing plan — testing under-sized, monitoring late, now sized on hop count) is the primary; the master's UAT-crash bench story (§4.3 — numeric entrance criteria ever since) is the backup. S18's RF miss (Goforth) and S16's security-review miss (Josyula) are held for other interviewers — pointers only; S4 (the in-transit rule I got wrong) is Lan's backup pointer, so keep it clean.	S11-B · Bench §4.3
10	**How do you measure success for a process you designed?**	A KR without a denominator is not a KR: seven frozen metrics with owners, honest denominators (15-day active rule), inputs tagged to outputs.	S11

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
11	**What's a number you refused to report?**	The 99.4% three-touch week — printed as a data miss.	S11
12	**What would your current manager say you need to work on?**	Start the review at kickoff rather than after the critique (S11); back-test a rule before publishing it (master S4 lesson, pointer only).	S11 / S4 pointer
13	**How would you think about inventory-placement strategy for a network like mine? (placement was her 2017–22 world — public-profile inference)**	S11's logic translated: publish the selection bar before the first node (graduation criteria ≈ placement criteria), calibrate per node (~50 containers ≈ a trial volume), hold a node in public when it misses. Then master §3.99: book of record → stocks and flows → broken invariant → minimal design.	S11 / §3.99
14	**How do you decide connectivity in a crossdock network — which nodes, which flows?**	S12's sequencing logic: rank flows by benefit × impact ÷ effort at the work-item grain (interactions eliminated per flow), FCL-first as "highest-volume flow first," a reversal trigger written per node before the swing.	S12
15	**You ran launch readiness for 50+ buildings — what does readiness mean for a process launch? (her 2014–17 world — public-profile inference)**	FC-readiness discipline translated: a written entry bar per wave (the six criteria), a launch gate on real volume (100 containers × 7 checks), a published rollback rule (the reversal trigger), and the hold taken in public when the bar misses.	S11 / S12
16	**Tell me about building a team or capability from scratch.**	S11's mechanism answer: input owners across nine teams recruited to sign the definitions they would be red against; the review run by hand until the pipeline (S16-A, in this packet) carried it; wave teams grown into owners — capability built while delivering, not before.	S11
17	**Walk me through a data analysis that changed a decision.**	S11-A: the 22× cost-to-serve drop split into 6.0× (60% → 10% of containers needing a human) × 3.67× (touches on each); the 3-touch 99.4% week discarded. Backup S12-A (P95 by region, then by path).	S11-A / S12-A
18	**How would you design the data model behind an operating metric?**	S16-A: one grain (container × week), numerator and denominator at that grain, a documented cross-reference for the missing key, a 1% parity gate and a feed-failure alert — the BRD names no tables, so the join, the grain and the artifacts are the story.	S16-A
19	**How do you decide a product or process is ready to launch?**	S11-B: four test gates, a 100-container ledger × 7 checks, six published graduation criteria — and own that the pilot ran later than the first testing plan.	S11-B
20	**How do you retire or migrate a customer base without stranding anyone?**	S11-C: nine segments, three questions each, concentration first (3 brokers = 75.0% of containers), dated actions, the dissent on the page.	S11-C
21	**Tell me about diagnosing a regional performance problem.**	S12-A: P95 by region (Asia 21.19 s vs US 3.3 s = 6.42×), then by path (static vs dynamic), reversible fixes first, 32 smoke checks a change.	S12-A
22	**How do you know a number is right before you steer on it?**	S16-A parity gate (≤ 1% for consecutive weeks, mismatches explained by cause); S11-A footing and honest denominators; S11-C re-footed the doc's own table.	S16-A / S11-A / S11-C

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
23	**How would you apply Kaizen / Lean here? (her trainer background — public-profile inference)**	Waste and standard work in S11-A (3.67 touches; seven frozen definitions); jidoka in S11-B (hold wave 3); genchi genbutsu and PDCA in S12-A; poka-yoke in S16-A (bridge table, parity gate). Say the gate, never the slogan.	S11-A / S11-B / S12-A
24	**How do you decide what to fix first?**	S11-B Pareto: 41 of 59 failing containers first broke at two hops (69.5%) — fix those; S11-C concentration; S12 benefit × impact ÷ dev-weeks.	S11-B / S11-C / S12
25	**Tell me about a metric that looked great and was not.**	S11-A: 99.4% on three touches — printed as a data miss; a 95% interval on N = 3 is wider than the metric.	S11-A
26	**Tell me about something that took longer than planned.**	S11-B: the first testing plan targeted early November 2021 and the pilot launched in December; the documents disagree on whether that was late, so I say "later than the first testing plan" — testing effort was under-estimated and monitoring arrived late; now sized on hop count.	S11-B
27	**How would your container-week grain apply to inbound capacity? (transferability probe)**	S16-A bridge: one grain for appointments, POs and containers (container × week or door × hour), a documented cross-reference for missing keys, an alert on a silent feed. Then the booklet: doors × hours × unload rate; the 85% knee.	S16-A
28	**Why should I believe your dollars?**	Every dollar carries its basis. Record: 7 → 68, $0.66 → $0.03, ~$5M once. Modeled: ≈ $50K direct CTS, $1.52M run-rate. Priced, not saved: S16-A's $103,680 / $100,360. See the NUMBERS page.	S11 / S12 / S16-A

Backup bench — use only if a primary doesn't fit

Three fallbacks for the hiring manager, none of them another interviewer's primary: S9 (Elementum — the 597-lane Pareto and the ≥95% gate, for Backbone / Think Big / Highest Standards if S11 and S12 are spent), S19 (Elementum · Walmart-client — the promise buffer as one governed dial; a design-flavored Frugality / Invent & Simplify alternate, all model, illustrative labels intact) and S13 (Manhattan — Cyber Monday at 4 AM, the Bias for Action / Customer Obsession alternate). Tell S11 and S12 first.

Bench discipline. The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

LPs this backup can serve: Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

Basis. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Shipment-data completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing
2	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	record problem	no instrument, no lever
3	Manual documents	`597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr`	illustrative volumes	~90K docs/yr keyed by hand

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (record)	597 of 1,000 lanes carry 80% — build them first
2	Cohort gate	`valid-ASN rate ≥ 95% in test before activation`	published to carriers before cohort one (record)	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	"the standard was the data, not the transport" (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes live in month one	0	597 = 80% of freight spend	record
3	Freight spend on onboarded lanes	base	−10% ≈ $2M in six months	record — measured, never annualized
4	Document handling	~90K docs/yr keyed	X12/XML validated, portal floor for the tail	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000/yr`	derived from the record pair, stated as such	≈$20M/yr
2	The record result	`$20M × 10% = $2M in six months`	record	≈$2M — said as measured
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when the ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`0.8 + 0.6 + 0.6 = $2.0M`	the slices decompose the record — they never add to it	= the record $2M
7	Document labor	`90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000`	illustrative volumes; benchmark rate (modeled add-on)	≈$0.4M/yr

TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

Canon numbers (basis labels): 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

BACKUP — ONLY IF ASKED S19 — Elementum (client: Walmart Marketplace) — the delivery-promise buffer rebuilt as priced risk: one governed dial, +32 points of fast-promise share at the same 95% floor (all model, no record dollar)

LPs this backup can serve: Dive Deep (alternate when S8/S10 are spent) · Invent and Simplify (alternate) · Frugality (speed bought with zero new spend) — design-flavored probes

Basis. Engagement lead for the promise-policy workstream; the client's promise-engineering and data-science teams owned and built the engine and the transit models — I never configured or deployed any platform. Every quantity is from the engagement's illustrative twelve-cohort model — say "the model" in the same breath; there is no production dollar and none is ever promoted. Labels (metro two-day, mountain rural) are the model's illustrative cohort names.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. A marketplace client printing delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and the buffer was a flat 1.10 days on every lane regardless of variability. In the engagement model — twelve lane cohorts, about a million orders a week, transit σ from 0.22 to 1.60 days — that flat buffer printed 95.8% accuracy but put only 7.6% of orders on a fast promise: dense metro lanes over-insured and fat-tailed rural lanes under-insured at the same time. **Task.** Own the decomposition, the alternatives, the validation gates and the governance recommendation; the client's data science and engineering did not report to me. **Action.** Discovery: the buffer-waste audit — the flat buffer bought Φ(5.0), near-certainty, on the tightest cohort and only Φ(0.69) ≈ 75% on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: b = 1.645σ holds 95% on every cohort by construction; the equimarginal check proved the optimum; the Q4 drill showed a frozen summer dial breaking the floor. Cutover: four shadow weeks against logged live orders before any displayed promise. Hypercare: the dial published in the weekly review header, re-solved before peak. **Result.** Primary metric: fast-promise share 7.6% → 39.6%, +32 points at the held 95% floor; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all on the illustrative book, and I say "the model" in the same breath.

COMPRESSED STAR

- **S —** A flat 1.10-day buffer on every lane: 95.8% headline accuracy hiding over-insurance on tight lanes and under-insurance on loose ones; fast-promise share 7.6% (model).
- **T —** Lead the promise-policy workstream: decompose the waste, put priced alternatives on one table, design the gates, leave a governance instrument — not a tuning habit.
- **A —** Promise identity and accuracy factorization; buffer-waste audit; four policies on one table; proofs (Φ(1.645) = 0.95 per cohort, equimarginal check, Q4 drill at ~1.45× σ); shadow mode for four weeks; the dial k owned by a business owner and printed every Monday.
- **R —** Fast-promise share 7.6 → 39.6% (+32 pts) at exactly the 95% floor; average promised day 3.09 → 2.75; +$255K/wk ≈ $13.3M/yr — model, labeled; the widest cohorts got honestly slower promises (1.10 → 2.63 days) on purpose.

RESULTS LINE

Fast-promise share 7.6% → 39.6% (+32.0 points) at a held 95.0% floor (model) · average promised day 3.09 → 2.75 (model) · +$255K/wk ≈ $13.3M/yr vs uniform (modeled on the illustrative ~1M-order book) · the profit-optimal frontier (97.8% accuracy, ≈ +$20M/yr) handed to Finance as a priced decision — shown, never claimed, never merged with the $13.3M.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The promise identity	`EDD = t_order + handling + transit_pred + b; kept ⇔ (handling_err + transit_err) ≤ b`	per order; b = buffer in days (record-from-docs identity)	the buffer is the only policy term
2	Kept-promise probability per lane	`P(kept) = Φ(b / σ)`	ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (record-from-docs)	accuracy is a function of b and σ only
3	Incumbent buffer	`b = 1.10 days on every lane`	uniform policy A (illustrative model)	1.10 d everywhere
4	Tightest cohort, z-score	`z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997`	metro cohort σ = 0.22 d (illustrative model)	near-certainty bought — promise needlessly slow
5	Widest cohort, z-score	`z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754`	mountain-rural cohort σ = 1.60 d (illustrative model)	≈75.4% kept — a known-loss book
6	Portfolio print	`accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk`	order-weighted across 12 cohorts, ~1M orders/wk (illustrative model)	a green headline hiding both failures
7	The asymmetry	`padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)`	record-from-docs — why the system drifted slow	bias toward slow, unowned

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One governed dial	`b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%`	Φ(1.645) = 0.95 under normal errors (record-from-docs design)	k = 1.645, one number for the book
2	Tightest cohort re-buffered	`b = 1.645 × 0.22 = 0.36 days`	σ = 0.22 d (illustrative)	buffer 1.10 → 0.36 d; 0.74 d of speed harvested
3	Widest cohort re-buffered	`b = 1.645 × 1.60 = 2.63 days`	σ = 1.60 d (illustrative)	buffer 1.10 → 2.63 d — slower on purpose, kept honest
4	Every cohort at the floor	`P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95`	by construction (design)	95.0% everywhere; no cohort subsidizes another
5	Portfolio after	`accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk`	illustrative model output, policy B	+32 pts fast share at the floor
6	Peak regime rule	`σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645 / 1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%`	Q4 drill (illustrative model)	k re-solved and published in October, never frozen
7	Shadow gate	`4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold`	promotion bar (design)	rehearse against reality before any displayed date

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Fast-promise share (≤2-day)	7.6%	39.6% (+32 pts)	illustrative model
2	Portfolio promise accuracy	95.8% (split: ~100% / 75.4%)	95.0% — the floor, held on every cohort	illustrative model
3	Average promised day	3.09	2.75 (−0.34)	illustrative model
4	Weekly modeled profit	$5.69M	$5.94M (+$255K)	illustrative model
5	Owner of the speed-accuracy trade	implicit, in the last model tune	one governed dial, published weekly, re-solved per regime	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Weekly profit, policy A	`$5,690,000 / wk, taken from the model`	illustrative model output	$5.69M/wk
2	Weekly profit, policy B	`$5,940,000 / wk, taken from the model`	illustrative model output	$5.94M/wk
3	Weekly delta	`$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)`	rows 1–2 (illustrative)	≈ +$255K/wk
4	Annualized	`$255,000 × 52 = $13,260,000`	row 3 × 52 weeks (modeled)	≈ $13.3M/yr — "the model," said in the same breath
5	Frontier: policy D delta	`$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000`	illustrative model output; ×52 modeled	≈ $20M/yr — handed to Finance, never claimed
6	Why D holds 97.8%	`breach EV $12 > margin-per-promise-day $6.50 ⇒ optimum above the 95% floor`	$12 and $6.50 are the model's stated assumptions	the floor and the economics disagree — a governance finding
7	Headline	`≈ $13.3M/yr modeled; record dollars: none`	rows 3–4	≈ $13.3M/yr, always labeled

TWO PROBES

> **The whole story rests on an illustrative cohort set.**
>
> It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers risk-proportional, the last buffer-day buying the same accuracy everywhere, and a floor that disagrees with the breach economics treated as a governance finding.

> **Why one dial instead of per-lane learned buffers?**
>
> Data science wanted per-lane learned buffers — more model, more value, invisible governance. I priced the difference and recommended the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The no went into the comparison table as a priced row, not a meeting minute.

Canon numbers (basis labels): 7.6% → 39.6% fast-promise share (+32 pts, model) · 95.8% → 95.0% accuracy held at the floor (model) · 3.09 → 2.75 promised days (model) · +$255K/wk ≈ $13.3M/yr (modeled) · 12 cohorts, ~1M orders/wk, σ 0.22–1.60 days, k = 1.645 (illustrative model) · breach ≈$12 vs margin ≈$6.50 (model assumptions). Never a production dollar.

BACKUP — ONLY IF ASKED S13 — Manhattan — Cyber Monday, 4 AM: allocation across 3 DCs and 100 stores when the scheduler died

LPs this backup can serve: Bias for Action (primary) · Customer Obsession (primary backup — the promise ranked first) · Deliver Results

Basis. OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it; the post-mortem mechanisms were mine to drive. $8M / $6M / 99% are record; the margin figures are my model at benchmark rates. Older story — if asked for recent, pair with the UDC robotics throttle (S2): same muscle, 2024–25.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores; 4 AM Cyber Monday the VP called — backorders piling with inventory present, ~$6M of sales at risk against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction (ATP > 0, backorders rising) was the entry bar to declare an incident. Plan & Design: I ranked by promise-at-risk, not FIFO; node choice = the promise — near 0.85 days vs far 4.60. Build & Test: one bridge, every rebalance timestamped; reason-coded alerts I had already built. Cutover: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare: root cause — the allocation scheduler died silently at the daylight-saving change; the post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook. Lesson: decide reversibly, log it, find the cause later. **Result.** Primary metric: peak fill rate — 95% goal → 99% delivered. ~$8M of peak sales protected against ~$6M at risk (record); my model puts that at ≈$3.2M of gross margin at a 40% benchmark margin; structural stockouts −20% over the engagement.

COMPRESSED STAR

- **S —** 4 AM Cyber Monday: backorder lines piling with inventory present; promises aging; ~$6M at risk against a 95% fill goal; nobody had written a clearing rule for failure.
- **T —** Own the response before the cause was known, without making it worse.
- **A —** Declared the incident on the contradiction; clearing rule — promise-at-risk, not FIFO — written on the bridge log in the first hour so nobody re-argued it at noon; rebalanced across three DCs; two-hourly status; isolated the DST scheduler death through my own reason-coded alerts; blameless post-mortem → heartbeat, DST checklist, 3σ page rule, rehearsed playbook.
- **R —** Fill 99% vs the 95% goal; ~$8M protected vs ~$6M at risk (record pair); structural stockouts −20% over the engagement; the next peak ran boring.

RESULTS LINE

Peak fill 99% vs the 95% goal (record) · ~$8M of peak sales protected vs ~$6M at risk (record pair — one-time sales that didn't die, never a saving) · ≈$3.2M gross margin at a 40% benchmark (modeled) · structural stockouts −20% over the engagement (record) · detection moved from a VP phone call to one heartbeat interval.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	4 AM state	`scheduler dead; backlog = 30,000 orders; ~$6M of sales at risk vs the 95% fill goal`	record risk pair; backlog illustrative	the bridge call
2	FIFO failure	`FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age`	record rule problem	value-blind clearing

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	Naive node loads	`DC1: 22,000 / 10,000 = 2.2 days; DC3: 2,000 / 6,000 = 0.33 days`	capacities 10K / 8K / 6K orders/day (illustrative)	worst node 2.2 days behind while another idles
4	No liveness detection	`a dead scheduler was found by a VP at 4 AM, not by a page`	record problem	detection by seniority

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promise-at-risk ranking	`risk = value × urgency(promise − now); clear descending`	on the bridge log in hour 1 (record rule)	at-risk promises first, not first-in
2	Node choice = the promise	`T = handling + transit: near 0.85 d vs far 4.60 d; 4.60 − 0.85 = 3.75 d`	record pair	−3.75 days per re-pointed order
3	Load levelling	`rebalance 30,000 → 12,500 / 10,000 / 7,500: 12,500/10,000 = 1.25; 10,000/8,000 = 1.25; 7,500/6,000 = 1.25 days`	load = queue / capacity (illustrative)	worst slip 2.2 → 1.25 days
4	Liveness page (post-mortem)	`page if x < μ − 3σ = 1,200 − 3 × 150 = 1,200 − 450 = 750/hr`	μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr	caught in one interval, not by a VP

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Peak fill rate	95% goal	99% delivered	record
2	Peak sales	~$6M at risk	~$8M protected	record pair — one-time event
3	Worst-case promise slip	2.2 days	1.25 days	illustrative walk
4	Detection	a VP at 4 AM	one interval via the 3σ page	post-mortem design
5	Structural stockouts	—	−20% over the engagement	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Margin protected	`$8,000,000 × 40% = $3,200,000`	benchmark margin 35–45% (modeled)	≈$3.2M one-time
2	Orders at risk	`$6,000,000 / $150 = 40,000 orders`	AOV benchmark	40,000 orders
3	Cancels / CS avoided	`40,000 × $6.50 = $260,000`	CS contact $5–8 (benchmark; modeled)	≈$0.26M
4	Expedite premium avoided	`40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000`	re-pointed share and premium midpoint of the $5–12 benchmark (both illustrative)	≈$0.13M
5	Structural-stockout run-rate	`$400M × 4% = $16M lost sales; × 20% = $3.2M recovered; × 40% = $1,280,000`	illustrative retailer × lost-sales benchmark (2–6%) × −20% record × margin	≈$1.3M/yr modeled

TWO PROBES

Why act before you knew the cause?

Because waiting had an hourly price — the queue was aging while inventory sat idle — and a reversible, logged intervention did not. I checked with engineering first that manual clears could not corrupt allocation, timestamped every rebalance, and hunted the cause in parallel with alerts I had already built.

Isn't this a hero story?

The event is; the post-mortem is the point. Carrying the first stretch solo is one of my do-differentlys. What I would present as the result is the mechanism set — heartbeat, DST checklist, 3σ page rule, rehearsed playbook — and that the next peak ran boring.

Canon numbers (basis labels): 99% fill vs 95% goal (record) · ~$8M protected vs ~$6M at risk (record) · 3 DCs / 100 stores (record) · near 0.85 d vs far 4.60 d (record) · −20% structural stockouts (record) · ≈$3.2M margin at 40% (modeled). Never "$2MM baseline"; never present $8M as a saving.

NUMBERS I MUST NOT GET WRONG — canon values with basis labels NEW

Canon numbers for all seven primaries with their basis labels (S11 and S12 from the master packet's §4.1 ledger; the five new stories' rows re-cut from the program documents, with the footing mine). Say record numbers flat; say modeled numbers with their inputs; say a priced number as priced.

METRIC	VALUE TO SAY	BASIS	NEVER
S11 cost-to-serve	$0.66 → $0.03 per container (about 95%)	record	"$1.00 → $0.03"
S11 customers	7 → 68 = 9.7×; zero rollbacks; no added ops headcount	record	"10×", or any growth multiple other than 9.7×
S11 automation / accuracy	40% → 90% / 60% → 90%	record (this packet's canon; loop packet differs — do not mix)	"70% baseline"
S11 targets / timing	68 customers · 170 MAU · 32K shipments; both targets two months early	record	—
S11 direct CTS dollars	≈$50K/yr (range $30–81K; exact $30,240–$80,640)	modeled on illustrative containers/shipment	presenting it as the value — the value is scale
S11 pre-prod measurement	300 containers, ±1.6% at 95%; rollout proof 1.5 vs 6 months; ~$1.5M saved	user anchor — separate canonical	merging the $1.5M pre-production estimate with CTS
S11 review by hand	about four hours a week at its worst, by hand for months; seven frozen metrics; ±5% WoW; 15-day active rule	record	"I invented the WBR"
S12 time-to-value / effort	month 18 → month 6; 66% less engineering effort	record	—
S12 cost avoidance	~$5M, said once	record	"$5M annual cost reduction" alongside it
S12 CTS at work-item grain	−50% partner-first vs −25% operator-first; 339.5 min/shipment; 2,200 interactions	record	—
S12 partners / burn-up	48 of 53 within a month; import-side app April 2022: 11 partners / 10 countries; 55.8% → 83.2%	record	—
S12 run-rate / decomposition	≈$1.5M/yr run-rate; ≈$2.9M + ≈$1.5M ≈ $4.4M (12% under the ~$5M record)	modeled (12 eng × $20K; 20K shipments illustrative)	presenting the split as the program's figure
S12 employer	Flexport FF App export launch	record	telling it as a Sephora roadmap (loop-packet conflict)

METRIC	VALUE TO SAY	BASIS	NEVER
S11-A decomposition	22× = 6.0× (60% → 10% of containers needing a human) × 3.67× (touches per container); weekly automation 40.0% (17 Jan) → 21.9% (31 Jan 2022) is record	record inputs; decomposition mine	attributing the 22× to automation alone; any growth multiple other than 9.7×
S11-A honest denominator	active = cumulative inflow less containers 15 days past departed_terminal; 99.4% on N = 3 discarded (95% interval 44%–100%)	record decision; interval is my arithmetic	printing the three-touch week
S11-B ledger and gates	100 containers × 7 checks; 59 failing rows, 41 first failed at two hops (69.5%); 500 containers / 20 min / 99.6%	record-from-docs; my arithmetic	claiming the pilot was on time, or a number of weeks late — say "later than the first testing plan"
S11-C Crux book	79 customers · 294,581 containers · printed $185,892 = six of nine segments · nine rows = $254,952 · 3 brokers = 75.0% of containers	record-from-docs; footing mine	quoting $185,892 as "the book"; naming a keep-alive cost
S12-A Asia latency	Asia 21.19 s vs US 3.3 s = 6.42×; 45.3% of 13.92k views; overall monthly P95 8.39 → 4.68 s = −44.2%	record-from-docs; arithmetic mine	quoting 81%; subtracting P95s across sources; a dollar as savings
S12-A dollars	tail time ≈ $1.6K / yr; exposure ≈ $0.69M / yr upper bound	illustrative / modeled	calling the exposure a saving
S16-A pipeline	system accuracy 60% → 90%; parity ≤ 1% (record gate; 0.42% pass / 1.25% investigate are illustrative); about 4% sync drop; four systems → one	record (accuracy, drop, gate); examples illustrative	the alarm threshold or weeks-to-parity as record
S16-A dollars	over $100K a year priced · BRD $103,680 (30 → 6 h/wk at an assumed $90/h) · model $100,360 · realized candidate $18,720 (208 h × $90)	priced / modeled — none claimed as savings	"I saved $100K"; a monthly figure other than $10,800
S12 decomposition range	$2.85M + $1.52M = $4.4M (12% under ~$5M); range $2.9M–$6.7M; effort piece alone $1.8M–$4.5M	modeled	"the effort band contains the $5M"; $6.8M (rounded high case)
S11 pre-production set	~$1.5M pre-production estimate · n = 300 → ±1.6% · 1.5 vs 6 months	user anchor (GFiber v8 only) — kept separate, never added to cost-to-serve	saying it as program record; "E2E 85% → ~98%" (say 86% tracked, 500 containers / 20 min / 99.6%)
S11 FTE equivalent	80,000 × 3.5 touches × 1 min ÷ 1,800 h ≈ 2.6 FTE (2.4 at 1,920 h)	modeled (assumptions named)	presenting it as record headcount

Never-say, one glance before the call.

- Never "10×", or any growth multiple other than 9.7× — say 7 → 68 (9.7×).
- Never merge the ~$1.5M pre-production saving with cost-to-serve; never stretch it into a range.
- Never "$5M avoidance" and "$5M annual reduction" as two results — the avoidance is said once.
- Never "I invented the WBR" — copied from the LSA review, which was the point.
- Never claim platform configuration; program-lead voice throughout. The only personally configured platform in canon is Manhattan DOM.
- S14, S15, S10 are the bar raiser's, and S16 is his primary too — this packet's S16-A is the data-model cut only; S1/S3, S2/S18, S6/S8 belong to Lan, Goforth and Hoerger — pointers only here.

WHAT TO SAY ABOUT MY ROLE — the I-versus-we line for each story NEW

> **The I-versus-we rule.** Say *I* for what I decided or authored — the plan, the gate, the analysis, the document, the read-out. Say *we* only for results other people delivered, and name the owner when the work was theirs. No source documents a direct report, so every people answer is program-lead voice: owners, gates and cadence — never "my team of …".

STORY	"I" — WHAT I OWNED	"WE" / THEIR WORK	NEVER SAY	EVIDENCE
S11 · wave machine	I owned the rollout: I wrote the wave plan as a schedule, published the entry bar (about fifty containers of the customer's own data) and the six graduation criteria before wave one, and ran the weekly review that gated each wave.	Nine teams (~100 people) built and operated the product; onboarding and sales sat with other teams; the VP decided the holds.	I built the platform · I ran the pilots myself · I scaled it alone	AP A1 p6–7; P4PL p1–2
S11-A · the Monday review	I built and ran the review by hand for months, froze the seven definitions with the input owners, and made the denominator honest. The dollars per ticket come from Data Services and FP&A at their loaded rates, never my own.	The input owners ran their lines; the pipeline later carried the formulas.	I set the cost per ticket · the LSA workbook's numbers are mine	AP A2 p8–10
S11-B · launch readiness	One of two technical program managers. I defined the gates and ran the go / no-go read-out; I made the 100-container run the UAT spine with a Jira ID on every missed row.	The QA team ran the container log day to day; business users ran UAT; the test lead designed the hop-by-hop trace.	I ran the test · the pilot was on time · it was N weeks late	AP A5 p13–15; PTA p1–3
S11-C · Crux transition	I owned the mitigation of the decommission risk: top of the risk register, the apples-to-apples comparison, the 79-customer base priced per segment, and the one-canonical-source rule.	The transition document's owners built the per-customer paths with me; the decision was the executive decider's (RAPID).	I decided to retire Crux · I own the revenue number	AP A4 p12–13; RAPID p1
S12 · export roadmap	I drove the sequence call: I priced both orders on shared axes and ranked the backlog by benefit × impact ÷ dev-weeks; my manager saw the math first and pressure-tested it with engineering for two days; the onsite ratified it.	Engineering did the re-estimates and the build; the 339.5-minute baseline is the program's time-and-motion analysis — I did not run the stopwatch.	I built the app · it was my idea alone · $5M a year	AP B2 p17–19
S12-A · Asia latency	I drove the follow-ups from the monthly review and kept the plan, the sequencing and the smoke tests honest.	The app engineer wrote the investigation; the network-infrastructure owner made the CDN and DNS changes (ASIA p3 names the owners).	I fixed the CDN · I found the root cause	ASIA p1–4; AP B3 p19–21
S16-A · Snowflake pipeline	I wrote the BRD, froze the seven definitions, made the Monday review the acceptance test and owned the cutover call; I own the miss — I planned the build and not the security review.	The infrastructure team built the pipeline: I didn't lay pipe, I made the pipe buildable.	I built the pipeline · I saved $100K · bus factor ≥ 2	AP A3 p10–11, G3 p45–46; BRD p1–5

Seat sources: Apple v11 chapters A1–A5, B2–B3 and G3 (the candidate's own account), the BRD, the RAPID table, the investigation document and the testing approach. Anything not in those is not said as "I".

VERIFY BEFORE YOU SAY IT — 4 specifics in no source file NEW

The first pass carried 37 yellow items. Everything the program documents, the Apple v11 workbook, the BRD or the review workbooks state was converted to a fact or a labeled anchor (see SOURCE TRACE). These are the only specifics that are in **none** of the source files. **Rule: do not say the specific — say the safe sentence beside it.**

ROUTER AND FRONT MATTER

☐ V1 a named person I mentored or developed, with an outcome — no source names one. **Safe sentence:** "I led through programs rather than a reporting line: I grew the input owners across nine teams until each signed and defended their own definition. I don't have a one-to-one mentee story I can document, so that is the one I'll give you."

S11-A

☐ V2 whether the 10- and 30-day alternatives were compared before 15 was frozen — not in any source
Safe sentence: “I froze the fifteen days with the input owners; I did not record a ten- and thirty-day comparison, and I would run one now.”

S11-C

☐ V3 how Crux handled exceptions — not in any source
Safe sentence: “I don't know how Crux handled exceptions; for the new product, a failed auto-track opened a ticket to Data Services.”

S12-A

☐ V4 whether synthetic probes were adopted — not in any source
Safe sentence: “The document raised synthetic probes as an open question; I added P50, P90 and P95 by region to the review, and I can't say whether probes were adopted.”

Left as labeled anchors, not VERIFY items. The ~$1.5M pre-production set (n = 300, 1.5 vs 6 months) and the “#3 → #14 of 16” ranking are in the candidate's own prior master only — say them as “my estimate” and “as I recorded it”; Snowflake table and column names are in no source (the BRD names the Ticket ID, the container number and Looker SQL Runner views only), so the data-model story describes the join, the grain and the artifacts and never names a table.

SOURCE TRACE — every quantitative claim in the new and modified stories NEW

Number · story · source file · page / row / sheet · match. **Y** = the source says it; **Y-with-rounding** = the source says it and I rounded or divided; **CONFLICT** = sources disagree — say the safe wording; **UNSUPPORTED-anchor** = only my own prior master or my own model — say it as “my estimate / my model”. Apple page = PDF page of the v11 workbook.

AP Apple_PMIII_Interview_Stories_v11_2026-08-26 (PDF page; chapter in brackets) · **BRD** BRD_-_SV_Zendesk_Data_in_Snowflake · **TPM** Technology_Program_Management_Status_Week_of_November_19th__2021 · **PCRS** SV_-_Pilot_Client_Roll-out_Schedule · **PTA** SV_-_Product_Testing_Approach · **P3PL** Shipment_Visibility_3PL_ingestion-_One_Pager · **P4PL** Path_to_4PL__Visibility__Rajesh · **WBRAI** WBR_Metric_Culture_using_AI · **LSA** LSA_WBR_2022 workbook (the .CSV upload is an .xlsx; sheet and Excel row given) · **C100** 100_Container_Debugging_Session_-_Basic_test_cases (.csv; my recount of the working log) · **RAPID** RAPID_-_Crux_Customer_Transition · **ASIA** FF_App_Asia_Latency_Investigation · **HLSD** High-Level_Solution_Document_FF_App__Schedules · **GF8** GFiber_MASTER_Packet_v8 (the candidate's own prior master — a user anchor, not a program document) · **RECON** numbers-reconciliation.md (CANON v2 ledger and §2 tables)

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
1	7 → 68 customers; 9.7× (68 / 7 = 9.71)	S11	AP [A1]; WBRAI	AP p6–7; WBRAI p1–2	**Y-with-rounding**
2	170 monthly active users — public target, landed **Say: Say "170 monthly active users — the public target, landed." Never the workbook's multiplier.**	S11	AP [A1]	AP p6–7 (WBRAI p1 says "MAU from 400 to 3X")	**CONFLICT**
3	32,000 shipments tracked = a target, not a result	S11	AP [A1]	AP p6–7 ("a number we managed toward all year")	**Y**
4	68 customers and 170 users landed two months early; zero rollbacks	S11	AP [A1]	AP p7	**Y**
5	Cost-to-serve $0.66 → $0.03 per container	S11 · S11-A	AP [A1, A2]; WBRAI	AP p7, p9; WBRAI p2 ("66 cents to 3 cents")	**Y**
6	Cost-to-serve cut: about 95% (1 − 0.03 / 0.66 = 95.45%) **Say: Say "about ninety-five percent" — never "95.5%".**	S11 · S11-A	AP [A1]; RECON S11.06	AP p7–8 ("a 95% cut")	**Y-with-rounding**
7	$0.18 per eliminated interaction; 0.66 / 0.18 ≈ 3.67 touches → 0.03 / 0.18 ≈ 0.17	S11 · S11-A	AP [A2]	AP p9–10 (the cross-walk; the ratios are my division)	**Y-with-rounding**
8	Automation 40% → 90%; 70% no-touch gate	S11 · S11-A	AP [A1, A2]	AP p7, p9	**Y**
9	Six graduation criteria (verification SLA ≥ 90, data quality ≥ 90, ETA accuracy ≥ 80, …)	S11 · S11-B	AP [A1]; PCRS	AP p6–7; PCRS p1–2	**Y**
10	~50-container calibration per new client	S11 · S11-B	AP [A1]; PCRS	AP p6; PCRS p4	**Y**
11	Waves 2 → 4 → 20+ → 40 → 68	S11	AP [A1]; PCRS	AP p6–7; PCRS p1 (40 and 68 are Apple-only milestones)	**Y**
12	Nine teams / ~100 people (input owners across nine teams)	S11 · S11-A	P4PL; AP [A1]	P4PL p1–2; AP p7	**Y**
13	First pilot customers (names) **Say: Say "the first two hand-carried customers" — no names.**	S11 · S11-B	PCRS; TPM; AP [A5]	PCRS p1, p3; TPM p10–11; AP p13–14	**CONFLICT**
14	Manual scorecard effort: about four hours a week at its worst **Say: Say "about four hours a week at its worst"; the BRD hours are a different measure, quoted only inside S16-A and labeled.**	S11 · S11-A · S16-A	AP [A1, A2]; BRD	AP p8–9; BRD p2 (1.5–2 h per snapshot; p1–2: 30 h/wk priced)	**CONFLICT**
15	99.4% week on three touches, discarded as a data miss	S11 · S11-A	AP [A2]	AP p8–9	**Y**
16	Event grain 1,991 / 10,289 = 19.4% human-touched (80.6% automated)	S11-A · S16-A	AP [A2]	AP p9 (extract SV_28.xlsx, not in the source folder)	**Y**
17	Ticket queue grain: 538 auto / 456 human = 54.1% automated	S16-A	AP [A2]	AP p9 (grain 3)	**Y-with-rounding**
18	Weekly stock: 13 Dec 40.2% · 17 Jan 40.0% · 31 Jan 21.9% (carrier format change) · 21 Feb 69.2%; Active 287 → 546 (+90.2%)	S11 · S11-A	AP [A2]	AP p9	**Y**
19	40.0% → 21.9% = −45.3% week over week (my division)	S11-A	AP [A2]	AP p9	**Y-with-rounding**
20	15-day active = cumulative inflow less containers 15 days past departed_terminal	S11-A · S16-A	AP [A2]	AP p9	**Y**
21	Pre-production: 500 containers in 20 minutes at 99.6% connectivity	S11 · S11-B	TPM; AP [A2]	TPM p10; AP p9	**Y**
22	End-to-end dry run: 100% processed, 86% tracked; job stories 98.4%	S11 · S11-B	TPM; AP [A2]	TPM p10; AP p9–10	**Y**

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
23	"E2E 85% → ~98%" **Say: Say "eighty-six percent of containers trackable in the dry run."**	S11 (original text)	GF8 only	not in TPM p10 (86% tracked, 98.4% job stories)	**CONFLICT**
24	Pre-production sample n = 300, ±1.6% at 95% (1.96 × √(0.98 × 0.02 / 300) = 1.58%) **Say: Label it "user anchor": "about three hundred containers, plus-or-minus one-point-six at ninety-five — my design."**	S11	GF8	GF8 only — my design, arithmetic checks	**UNSUPPORTED-anchor**
25	~$1.5M pre-production saving; rollout in 1.5 vs 6 months **Say: Label it "user anchor", keep it separate from cost-to-serve: "my estimate of a month and a half versus six."**	S11	GF8	GF8 only (PCRS p2 supports only a ~6-month pilot)	**UNSUPPORTED-anchor**
26	Direct cost-to-serve ≈ $50K a year ($30,240–$80,640) — modeled **Say: Say "modeled on illustrative container counts — the point is scale without headcount."**	S11	RECON S11.25	arithmetic on illustrative container counts	**UNSUPPORTED-anchor**
27	80,000 containers × 3.5 touches × 1 min ÷ 1,800 h ≈ 2.6 FTE — modeled **Say: Say "about two and a half people never hired — modeled, not record headcount."**	S11	RECON	my assumptions named in the dollars table	**UNSUPPORTED-anchor**
28	Wave 3 held once on ETA accuracy	S11	AP [A1]	AP p7	**Y**
29	Critique: initiative #3 → #14 of 16 **Say: Say "leadership said the program wasn't showing value" and the ranking only as "as I recorded it in my master".**	S11	GF8	GF8 only (not in AP or any program document)	**UNSUPPORTED-anchor**
30	LSA review format: Monday–Sunday week, five weeks in view, (B − A)/A, red / green beyond ±5%	S11-A	LSA	sheet "Instructions", rows 6–9	**Y**
31	Real FFA tab, 19-value excluded series: −28.7%, −33.3%, 1 red / 4 green, −36.8% / +74.5%, 47.5% / 49.9%, 0.216, 7 / 52, −31.0%	S11-A (Table 5)	LSA	sheet "FFA" (the FFA tab is another team's slide — used for the mechanics only); formulas in Table 5	**Y**
32	99.4% on N = 3 → 95% Wilson interval 44% to 100%	S11-A	AP [A2]	AP p9 (the week); the interval is my arithmetic	**Y-with-rounding**
33	22× = 6.0× (60% → 10% needing a human) × 3.67× (touches) **Say: Say "the split is my model on two record endpoints."**	S11-A	AP [A1, A2]	AP p7–9 endpoints; the split is my model and a convention	**Y-with-rounding**
34	UAT entry gate: 80% of shipments minor-to-no issues	S11-B	PTA	PTA p3	**Y**
35	100-container test; seven hops (created, subscribed, data back, route, milestone, exception, webhook)	S11-B	PTA; C100	PTA p1–2; C100 columns "Shipment created" to "Webhook Created"	**Y**
36	Ledger: 474 rows, 142 tested, 105 tested rows with hop results, 59 failing **Say: Say "my recount of a partial working log — it tells you where to look, not a defect rate."**	S11-B	C100	C100 rows 1–474 (106 rows have any hop result; one is untested)	**Y-with-rounding**
37	First failing hop: route 23 + milestone 18 = 41 of 59 (69.5%); ingestion 12 (20.3%)	S11-B	C100	C100 columns "Route created?", "Milestone shown in UI?"	**Y-with-rounding**
38	Scraper error rates: Air 5.09%, Ocean 8.99% against a < 10% goal	S11-B	TPM	TPM p11	**Y**
39	Trace-ID / MTTR improvement of 50%	S11-B	P3PL	P3PL p1	**Y**
40	Pilot later than the first testing plan (early November 2021 → December 2021) **Say: Say "later than the first testing plan" — never "on time", never a number of weeks.**	S11-B	PTA; TPM; PCRS	PTA p1; PCRS p1 (Wave 1 in Q4 2021); year-end summary — the documents disagree on "late"	**CONFLICT**
41	Localization-labor dollar **Say: Say the hours, not this dollar.**	S11-B	my model	no source	**UNSUPPORTED-anchor**
42	79 customers · 294,581 containers · $185,892 printed ARR	S11-C	RAPID; AP [A4]	RAPID p5 (total row); AP p12–13	**Y**
43	Printed total equals six of nine segment rows; nine rows = $21,246 MRR / $254,952 ARR **Say: Say the printed 79 / $185,892 / 294,581 and add "the printed total equals six of its nine rows" — lead with containers.**	S11-C	RAPID	RAPID p2–5 (segment rows; the footing is my arithmetic)	**CONFLICT**

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
44	77 names in the appendix vs 79 counted (37 SMB vs 39) **Say: Say "the appendix lists 77 names against 79 counted — I would reconcile before the cutover list."**	S11-C	RAPID	RAPID p12 (appendix; my count)	**CONFLICT**
45	Three brokers = 75.0% of containers; 3 of 79 customers = 3.8%	S11-C	RAPID	RAPID p3–5; the concentration is my arithmetic on the table	**Y-with-rounding**
46	25% schedule reliability against a competitor's 63% — unlike data	S11-C	AP [A4]	AP p12	**Y**
47	Decider is an executive (RAPID: Recommend, Agree, Perform, Input, Decide)	S11-C	RAPID	RAPID p1	**Y**
48	18 months → value in month 6; 66% less effort	S12	AP [B2]	AP p17–19	**Y**
49	Cost-to-serve −25% operator-first vs −50% partner-first; 9 vs 6 months (work-item grain)	S12	AP [B2]	AP p17–19	**Y**
50	339.5-minute baseline; ~2,200 operator interactions per departed shipment; ≈ $153 per shipment	S12	AP [B2]	AP p18–19 (339.5 × $0.45 = $152.78)	**Y-with-rounding**
51	About $5M avoided — said once; never annual; no "15% lower overhead" **Say: Say "about five million of avoided cost," once.**	S12	AP [B2]; GF8	AP p18–19 carries both "avoidance" and "annual cost reduction"	**CONFLICT**
52	$2.85M effort + $1.52M run-rate ≈ $4.4M, 12% under the ~$5M record **Say: Say "my bottom-up model lands within about twelve percent of the record."**	S12	RECON S12.13–14	my bottom-up model	**UNSUPPORTED-anchor**
53	Import-side app live April 2022 with 11 partners in 10 countries **Say: Say "the import-side app went live April 2022 with eleven partners in ten countries"; the export launch is the later Q1 2023 milestone.**	S12 · S12-A	ASIA; AP [B1]; HLSD	ASIA p1; AP p16 (Apple also prints "MVP Q1'23" at p17; HLSD p4: export launch March 2023)	**CONFLICT**
54	48 of 53 import partners migrated within a month (about 90%) **Say: Say "about ninety percent" and always "import partners."**	S12 · S12-A	ASIA; AP [B1]	ASIA p1 ("90.5%"); AP p17	**Y-with-rounding**
55	Burn-up 55.8% → 83.2%; 15 domains × 6 modules; 8-dev-week reuse rule	S12	AP [B1, B2]	AP p16–19	**Y**
56	P95 page load: Asia 21.19 s vs US 3.3 s = 6.42×; Others 4.45 s	S12-A	ASIA	ASIA p1 (table, 04/24–05/10)	**Y**
57	Page views 2.19k / 6.31k / 5.42k = 13.92k; Asia 45.3%	S12-A	ASIA	ASIA p1; the sum and share are my arithmetic	**Y-with-rounding**
58	17-day window (04/24 – 05/10)	S12-A	ASIA	ASIA p1	**Y**
59	Static JS P95: China 5 s vs US 0.5 s	S12-A	ASIA	ASIA p2	**Y**
60	P99 rests on 63 Asia views (1% of 6,310), P95 on 316	S12-A	ASIA	ASIA p1 views; the percentile counts are my arithmetic (the document gives no P99)	**Y-with-rounding**
61	Plan: 8 rows, 4 done, 52 days (24 Jun → 15 Aug 2022); remove double CDN 2022/06/24	S12-A	ASIA	ASIA p3	**Y**
62	32 smoke checks per change (4 steps × 2 instances × before / after, 2 environments)	S12-A	ASIA	ASIA p4 (the workflow); the count is mine	**Y-with-rounding**
63	Overall monthly P95 8.39 s → 4.68 s = −44.2% (Apr → Sep)	S12-A	LSA	sheet "OE Page 0", Excel row 50, columns 26 and 31	**Y-with-rounding**
64	Asia 4.0 s vs North America 4.09 s in the last week of the year **Say: Say "the gap closed" — never subtract the 21.19 s from the 4.0 s.**	S12-A	LSA	sheet "OE Page 0", Excel rows 53 (Asia) and 51 (NA); the Asia (non-China) series is not the same population as the 21.19 s	**CONFLICT**
65	Week-47 spike: metrics-code error (12/12 callout) plus the October regression (action item 11)	S12-A	LSA	sheet "OE Page 0", Excel row 94; sheet "Action Items", item #11 (Excel row 12)	**Y**
66	P50 / P90 / P95 by region added to the review (11 Oct 2022)	S12-A	LSA	sheet "Action Items", item #1 (Excel row 2), target 2022-10-11	**Y**
67	Exposure ≈ $0.69M a year; tail time ≈ $1.6K a year **Say: Say "illustrative exposure, not a saving."**	S12-A	my model	illustrative / modeled; no source figure	**UNSUPPORTED-anchor**
68	Zendesk tickets joined to Crux container numbers and Flexport core data; Looker SQL Runner views	S16-A	BRD	BRD p1–2, p5	**Y**
69	Seven metrics frozen	S16-A	BRD; AP [A3]	BRD p2–3; AP p10	**Y**

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
70	BR-3 feed daily or preferably hourly; BR-4 alert when the feed fails	S16-A	BRD	BRD p4	**Y**
71	As-is snapshot 1.5–2 hours	S16-A	BRD	BRD p2	**Y**
72	BRD pricing: 3 × 5 h × 2 reviews at an assumed $90 → $129,600 → $25,920 → $103,680 (30 → 6 h/wk) **Say: Say "over $100K a year — the BRD priced it at $103,680 on thirty hours a week at an assumed ninety dollars an hour; my split is a model; priced, not a claimed saving."**	S16-A	BRD; RECON S16.01	BRD p1–2	**CONFLICT**
73	BRD monthly figure $10,880 vs $129,600 / 12 = $10,800 **Say: If you quote a monthly figure, say $10,800.**	S16-A	BRD; RECON S16.02	BRD p2	**CONFLICT**
74	Model split $30,160 (4 h × 52 × $145) + $70,200 (52 × 10% × $13,500) = $100,360 **Say: Say "about a hundred thousand — my model" and never the split as record.**	S16-A	RECON S16.11	my model; rates and probabilities illustrative	**UNSUPPORTED-anchor**
75	Realized candidate 208 h × $90 = $18,720 **Say: Say "about two hundred hours retired — modeled."**	S16-A	my model	modeled; not claimed	**UNSUPPORTED-anchor**
76	System accuracy 60% → 90% **Say: Say "system accuracy" — never ETA accuracy.**	S16-A	AP [A3, A4]; WBRAI	AP p10–11, p13; WBRAI p2	**Y**
77	Four systems → one pipeline	S16-A	AP [A3]; WBRAI	AP p10; WBRAI p1	**Y**
78	Parity gate: manual vs pipeline within 1% for consecutive weeks; mismatches mostly the manual side	S16-A	AP [A3]	AP p10–11	**Y**
79	First automated catch: ticket-sync drop of about 4% **Say: Say "about four percent."**	S16-A	AP [A3]	AP p10–11	**Y-with-rounding**
80	Data-access security review: two weeks, zero skipped reviews	S16-A	AP [A3, G3]	AP p42, p45–46	**Y**
81	"Bus factor 1 → ≥ 2" **Say: Say "no longer a single point of failure."**	S16-A (removed)	AP [A3]; WBRAI	AP: single point of failure (qualitative); WBRAI p2	**UNSUPPORTED-anchor**
82	Illustrative parity values 0.42% / 1.25%; unmatched 0.74%; alarm 1%; join fan-out 12 / 18 vs 12 / 10 (44.4%) **Say: Say "illustrative — my design example, not program data."**	S16-A	my examples	illustrative — labeled in every table	**UNSUPPORTED-anchor**

Tally. 82 claims: 41 Y · 17 Y-with-rounding · 12 CONFLICT (safe wording given) · 12 UNSUPPORTED-anchor (labeled).

INTERVIEW-PREP PACKET 2 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

Amazon NACF Supply Chain loop — Packet 2: Ryan Hoerger

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

Ryan Hoerger

Manager, Placement Analytics — NACF Supply Chain

Wednesday, October 29, 2026 · 12:00 pm PDT · 55-minute video interview

SHADOW INTERVIEWER: SANDY HSIAO (A SECOND NOTE-TAKER — SAME PACE, SAME NUMBERS; EXPECT ONE QUESTION FROM THE SHADOW NEAR THE END)

Strategy in one line. The most quantitative interviewer on the panel gets the Elementum stories as primaries — ten, each with a real analytical spine (distribution, sample size, interval, counterfactual, sensitivity) and substituted worked numbers. Lead with S8 (measured lanes → p99 promise → ~$1.5M released, zero stockouts) or S6 (honest supply → shortage alert a lead-time early → $2M released under a two-signal rule); go deeper with S8a (the distribution analysis), S8b (the safety-stock formula redesign), S8c (the back-test) and S9 (1,000 lanes → 597); reach for the Walmart Marketplace delivery-experience stories (S19, S20, S21) and the new-market newsvendor design (E7) as the second and third examples. The two lane programs' dollars never meet. In his language: S6 positions safety stock where measured errors justify it; S8 positions the promise at p99 per lane. Every dollar keeps its basis word — record · user anchor · modeled · illustrative.

LP focus — my guess, not confirmed: Dive Deep · Are Right, A Lot · Learn & Be Curious — with a mentorship angle (his public About line names "a culture of mentorship"; expect one people question). Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title, the JD and his public profile in "Read me first" (public-profile inference, labeled as such) and should be held loosely — Insist on the Highest Standards remains a live alternate via the 99% bar in S8.

Shadow etiquette: address both; the shadow is usually a new interviewer being calibrated — same numbers, same pace; if the shadow asks the last question, answer it with the same discipline (phase asked, then stop).

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

Contents NEW — existing sections kept; red NEW pills mark what this upgrade added; teal ELEVATED marks an existing story that gained sections

FRONT MATTER

A	5-minute cram sheet — profile hooks, ten stories in order, three opening sentences, five numbers NEW	3
B	Question → story router — all 16 Leadership Principles and the common non-LP prompts NEW	4
	Read me first — the interviewer's lens, guessed LPs, answer strategy (existing)	6

PRIMARY STORIES — EACH: LPS · 20-SECOND VERSION · PITCH · S/T · ACTION · RESULTS · L6 · FIVE FACETS · PROBING + FOUR-TABLE MATH · BEHAVIORAL PACK · THREE ANGLES · TRAP DRILLS

S6	Elementum — one real-time supply/demand view: shortage a lead-time early; 70% → 85%; $2M released (existing, elevated) ELEVATED	9
S8	Elementum — measured lane distributions: p99 promise on 200 lanes; 580K units / ~$1.5M released, zero stockouts (existing, elevated) ELEVATED	21
S8a	Elementum — actual-vs-planned lead-time distribution analysis: p50 / p90 / p99, how the brochure was wrong NEW	32
S8b	Elementum — safety-stock formula redesign: service-level z, two honest terms, one lane released and one lane topped up NEW	41
S8c	Elementum — validation and back-test design: out of sample, not a quiet quarter NEW	49
S9	Elementum — 1,000 routes asked, 597 first: spend Pareto, ≥95% ASN gate, completeness 60% → 95% NEW	57
S19	Elementum / Walmart Marketplace — the delivery promise: one governed buffer dial; fast-promise share 7.6% → 39.6% NEW	68
S20	Elementum / Walmart Marketplace — the pre-purchase delivery WBR: factor math and an Andon NEW	80
S21	Elementum / Walmart Marketplace — the stalled managed-shipping program: incentive diagnosis, contract, shadow gate NEW	92
E7	Elementum design engagement (new-market build) — equipment forecast-to-buy: newsvendor, pooling, re-forecast trigger NEW	104

BACK MATTER

	Question bank — likely questions with a one-line answer plan (extended) EXTENDED	112
	Cheat-numbers box — canon values with basis labels (extended) EXTENDED	114
	Backup bench — S4, S7 and the FFA micro-story (S9 is now a primary)	116
H1	VERIFY BEFORE YOU SAY IT — the checklist NEW	125
H2	NUMBERS I MUST NOT GET WRONG — canon values with basis labels NEW	126

How to use this packet in 90 seconds. Page A is the cram sheet. Page B routes any question to a story, an angle, a 15-second opener and one number. Every primary has four behavioral questions, three angle cards and three trap drills — the trap drills are the statistical follow-ups this interviewer is most likely to throw. Anything tagged VERIFY-WITH-RAJESH is a detail that is not in the record: confirm it or leave it out; the one-page checklist near the back collects them all.

A · 5-minute cram sheet — Ryan Hoerger, Placement Analytics NEW

WHO IS IN THE ROOM (PUBLIC-PROFILE INFERENCE — FRAMING ONLY, NEVER QUOTED BACK)

- Manager, Transfer/Placement Analytics; Chicago Booth MBA (Siebel Scholar); analytics manager at Nike; Cornerstone Research before that — expert-grade statistical work. Shadow: Sandy Hsiao (same numbers, same pace).
- **Expect:** definitions and denominators ("how was it measured?"), n and intervals, the counterfactual ("how do you know it was you and not seasonality?"), and "what is your softest input?"
- **Style:** claim → number with its basis word → stop. Memo-like, not a wind-up.

THREE SENTENCES TO OPEN WITH

> "Every dollar I give you carries its basis — record, modeled or illustrative — in the same breath, and I will tell you the n behind every percentage."

> "The planning parameter was wrong before the policy was: the brochure lead time on 200 lanes, and what counted as supply — I measured the first lane by lane and defined the second before any screen existed."

> "I owned the definitions, the policy and the release rule; planners owned the forecast and the buffers, the data team built the distributions, Finance signed each release."

THE TEN STORIES IN THE ORDER I REACH FOR THEM — THE SINGLE NUMBER EACH ONE LANDS

#	STORY	REACH FOR IT WHEN HE ASKS…	ONE NUMBER (BASIS)
1	**S8**	Used data to change an inventory policy · went deeper than the request	580K units / ~$1.5M released, zero stockouts (record, one-time)
2	**S6**	Improved forecast accuracy · needed Finance and Planning buy-in	70% → 85%; $2M released (record, one-time)
3	**S8a**	Dug into the data · data contradicted belief	plan 18 d vs actual mean 21 d, p99 34 d (illustrative lane)
4	**S8c**	How did you validate? · how do you know it was not seasonality?	held-out quarter 99.1%, Wilson [98.7, 99.4] (illustrative inputs)
5	**S8b**	How do you set safety stock? · simplified · backbone	lane A 12,937 → 8,102 units; lane B +1,383 (illustrative)
6	**S9**	Prioritized 1,000 items · held a quality bar · pushed back upward	1,000 asked → 597 = 80% of spend; completeness 60% → 95% (record)
7	**S19**	A metric hid the problem · simplified a decision	fast-promise share 7.6% → 39.6% at the held 95% floor (engagement model)
8	**S20**	Took ownership of an un-owned number · built something to act faster	Promise Yield 41.5% = 43.6% × 95.2%; ~30 d → < 24 h (design)
9	**S21**	Why a program stalled · precise about what was mine	≈$3.7M/yr per +10 pts of enrolled GMV (modeled)
10	**E7**	Decision with little data · learned a new method	critical ratio 0.99 → z 2.33 → first buy 9,495 on a 6,000 mean (illustrative; design)

FIVE NUMBERS NEVER TO GET WRONG

- **580K units / ~$1.5M** — S8, record, one-time, zero stockouts; $2.59/unit is derived. Never merged with the $2M into a range; never per year.
- **$2M** — S6, record, one-time; a different program — never added to the $1.5M.
- **597 lanes = 80% of spend** out of 1,000 asked; completeness 60% → 95% — S9, record. Freight −10% ≈ $2M is six months, not annualized.
- **99% lead-time accuracy at the p99 promise on 200 lanes** — record; know which definition (coverage or ratio) before quoting a tolerance.
- **Six sites** — no regional qualifier; S19's ≈$13.3M/yr and S21's ≈$3.7M/yr are *modeled*, never record.

STATISTICAL REFLEXES (SAY THESE WITHOUT BEING ASKED)

- Give n and the interval: 22 late of 2,500 → 99.1%, Wilson [98.7, 99.4] — illustrative inputs.
- One held-out quarter is one draw; rolling origin is the upgrade. Zero stockouts bounds the rate: 0 in 1,950 lane-weeks → below 0.154% (rule of three).
- Name the softest input before he does.
- **NACF bridge:** inbound lead-time variability and the definition of supply sit under every WOC target and every placement promise — booklet INV-A (WOC), INV-B (safety stock), INV-C (placement).

B · Question → story router NEW

How to use. Find the LP or prompt on the left, open with the 15-second sentence, land the one number with its basis word, stop. "Q#" is the behavioral question inside that story's own pack (a separate 90-second STAR from a different moment of the story). Shaded cells are honest stretches — no formal people-development story in this packet, and I will not borrow another interviewer's.

IF ASKED ABOUT…	PRIMARY STORY · ANGLE (BEHAVIORAL Q#)	BACKUP	15-SECOND OPENER	ONE NUMBER (BASIS)
Customer Obsession	S19 · Q3 — pin accuracy at 95% on every cohort; let speed float	S8 · Q2 — the customer was the production line	"Fast-promise share can be juiced by bleeding accuracy, so I pinned accuracy at 95% on every cohort and let speed float."	**Fast-promise 7.6% → 39.6% at the held 95% floor (engagement model)**
Ownership	S20 · Q2 — promise accuracy was owned by nobody between Tuesdays	S8 · Q4 — a promise policy planning now owns	"The one number the customer feels — promise accuracy — was owned by nobody between Tuesdays, so I took it and gave every factor a named owner."	**7 outputs over 10 inputs; Promise Yield 41.5% = 43.6% × 95.2%**
Invent and Simplify	S8b · Q1 — a formula nobody could defend plus a week of pad → two readable terms	S19 · Q2 — one dial b = k·σ instead of per-lane learned buffers	"Safety stock was a formula nobody could defend plus a week of pad nobody could count; I replaced both with two terms a planner could read."	**Lane A 12,937 → 8,102 units (illustrative); k = 1.645**
Are Right, A Lot	S8c · Q1 — held a quarter out before releasing 580K units	S8a · Q2 — the scans said 21 days, the plan said 18	"A plan fitted to history always looks perfect on that history, so before I released anything I held a quarter out."	**Held-out 99.1% vs 96.5% for a bell-curve plan (illustrative inputs)**
Learn and Be Curious	E7 · Q1 — the newsvendor model from MIT coursework on a launch problem, labeled toolkit	S8c · Q4 — shadow-mode the drift flags before go-live	"I took the newsvendor model from my MIT MicroMasters coursework and applied it to a launch problem — and I label it as the toolkit."	**CR 0.99 → z 2.33 → first buy 9,495 on a 6,000 mean (illustrative)**
Hire and Develop the Best	No packet primary — honest stretch; closest: the S6 hand-over (planners run the alert and the measurement without me)	S8 · Q4 — planning re-applies the release rule on its own cadence	"I have no formal mentoring story here; the nearest true one is that planners now run the alert, and domain owners the gates, without me. VERIFY-WITH-RAJESH"	**Adoption 8×; planner effort −50% (record)**
Insist on the Highest Standards	S9 · Q2 — the ≥95% valid-ASN gate failed twice and was held twice	S8c · Q2 — 99% per lane, no pooling	"A cohort failed our ≥95% valid-ASN gate twice on a carrier timestamp format, and I held the gate both times."	**Completeness 60% → 95% behind a ≥95% gate (record)**
Think Big	E7 · Q2 — a launch chain designed as the template for market N+1	S6 — eight data domains, one view across six sites	"I designed the launch chain as a template for market N+1, not a one-off for market N."	**One chain · three books of record · five invariants (design, not run)**
Bias for Action	S20 · Q4 — an Andon that acts in 24 hours instead of a 30-day scorecard	S6 · Q3 — alert live in three months	"A 30-day rolling scorecard lets a degrading seller or carrier burn customers for weeks; I designed a control that acts inside 24 hours."	**~30 days → < 24 hours (design); false-trip bar < 5%**
Frugality	S8b · Q3 — less stock and better service	S8 — the release itself, with the watch armed first	"Less stock and better service — on one lane, 37% fewer units and seven late shipments in a hundred became one."	**580K units / ~$1.5M net, zero stockouts (record, one-time)**
Earn Trust	S6 · Q2 — the two-signal rule signed before the first release	S8c · Q3 / S21 · Q3 — say what the test cannot show; say what was mine	"Releasing $2M of safety stock is a risk Finance and Planning both carry, so the rule was theirs before it was mine."	**$2M released (record, one-time); both-errors rule**
Dive Deep	S8a · Q1 — 400 scans on the busiest lane: 21 days actual against 18 planned	S20 · Q1 — 97% national, 82% in the Southeast: the factor math names the cell	"Buffers kept growing and nobody could say why, so I pulled the shipment scans on our busiest lane — about 400 shipments, planned date against actual receipt."	**Mean 21 d vs plan 18 d; 240 of 400 (60%) slower than the brochure (illustrative lane)**
Have Backbone; Disagree and Commit	S9 · Q1 — told the Global Head that 1,000 lanes in two months means confident wrong data	S8b · Q2 — a rule that can add stock, not just release it	"The Global Head of Supply Chain wanted 1,000 lanes of inbound visibility in two months; I told him the schedule would deliver confident wrong data and showed him the spend curve instead."	**597 lanes = 80% of spend; ~400 lanes (~20%) deferred (record)**
Deliver Results	S6 · Q3 — a shortage visible a lead-time early, in three months	S9 — completeness 60% → 95%; freight −10% ≈ $2M in six months (record)	"Three months from kickoff, a planner could see a shortage a lead time before the line stopped."	**70% → 85% forecast accuracy; $2M one-time (record)**
Strive to be Earth's Best Employer	No packet primary — honest stretch; closest: S6 planner effort −50% (time redeployed, not headcount)	S8 — the drift flag and refresh cadence handed to planning	"I will not dress up a story I do not have; what is true is that the work moved off the planners' week into root-causing. VERIFY-WITH-RAJESH"	**Planner effort −50% (record)**
Success and Scale Bring Broad Responsibility	S9 · Q4 — carriers without EDI would have been priced out; the standard became the data, not the transport	S20 — an Andon with a < 5% false-trip bar so a seller is not penalized on noise	"Carriers without EDI would have been priced out of the standard, so I made the standard the data, not the transport."	**Adoption 8× (record); XML/portal floor**

COMMON PROMPTS THAT NAME NO LEADERSHIP PRINCIPLE — AND THE STATISTICAL PROMPTS HE IS MOST LIKELY TO ASK

COMMON PROMPT	PRIMARY · ANGLE	BACKUP	ONE NUMBER
Tell me about a time you failed	S4 (bench) — the in-transit promise rule I got wrong, fixed through the signed policy	S6 angle (iii) — MPN worklist gated longer than planned · S9 angle (iii) — the gate failed twice	**Lane σ ≈ 0.3 d; ~0.49-day window ≈ 95% (record promise math)**
…disagreed with your manager	S9 · Q1 — the Global Head, two levels above my sponsor (a client leader)	S8b · Q2 — release-only rule vs add-stock clause	**597 lanes = 80% of spend**
…influenced without authority	S8 — Finance signed each lane; none reported to me	S21 — contract design across seller, marketplace, carrier	**580K units / ~$1.5M (record)**
…reduced cost	S9 — freight −10% ≈ $2M in six months (record, not annualized)	S8b — inventory capital, 580K units / ~$1.5M	**$2M in six months; ≈$0.4M/yr doc labor modeled**
…dealt with ambiguity	S8 — a "variability report" request reframed as a parameter problem	E7 — a launch with no local history	**200 lanes scoped by site, lane and mode**
…simplified a process	S19 — one governed dial instead of per-lane learned buffers	S8b — two terms; S20 — 7 outputs over 10 inputs	**k = 1.645; 7.6% → 39.6%**
…decided with incomplete data	E7 — newsvendor on no history; the critical ratio sets the quantile	S8a — thin lanes print their interval	**CR 0.99; z 2.33; trigger ±20% over two weeks**
…missed a deadline	S6 — MPN worklist gated longer; re-sequenced date with the coverage number	S9 — gate held twice; schedule moved, not the bar	**Alert live in 3 months (record)**
…mentored someone	Honest stretch — S6 owners take over gates and root-cause reviews VERIFY-WITH-RAJESH	S8 — planning owns the refresh cadence	**Adoption 8× (record)**
…delivered under a tight timeline	S6 — three months to a live shortage alert	S9 — 1,000 in two months became 597 in month one	**Alert in 3 months; 597 lanes live month one**
…changed your mind with data	S8b angle (iii) — the textbook form was my starting point; one skewed lane showed it understates the tail VERIFY-WITH-RAJESH	S21 angle (iii) — the funnel said awareness; interviews showed an unpriced trade VERIFY-WITH-RAJESH	**normal p99 30.3 d vs ranked 34 d; release 8,009 vs 4,835 (illustrative lane)**
How would you set safety stock / service level?	S8b — two terms; critical ratio; lane A and lane B	S19 — one dial b = k·σ at a 95% floor	**SS 12,937 → 8,102 (illustrative)**
How do you validate / back-test a policy?	S8c — held-out quarter, paired replay, interval, rule of three	S21 — shadow gate before day one	**99.1% [98.7, 99.4]; 0 in 1,950 → < 0.154%**
MAPE vs bias · alert thresholds · false alarms	S6 — alert at lane p90; precision/recall; cost of a false alarm	S20 — Andon false-trip bar < 5%	**p90: precision 71.2%, recall 98.6% (illustrative lane)**
Lead-time variability — how do you measure it?	S8a — p50 / p90 / p99, empirical not bell-curve	S8 — per-lane distribution, drift flag	**plan 18 d vs mean 21 d, p99 34 d**
New market / placement with no history	E7 — newsvendor + risk pooling + re-forecast trigger; NACF INV-C	S19 — promise at the tail you can defend	**Q = 9,495 on a 6,000 mean (illustrative)**
Prioritize 1,000 items	S9 — spend Pareto, four-factor score, 597 first	S8a — which lanes are thin	**597 = 80% of spend**
Metrics architecture / WBR · promise policy · incentives	S20 · S19 · S21 respectively	S6 — metrics tree	**Promise Yield 41.5%; 7.6% → 39.6%; ≈$3.7M/yr per +10 pts (modeled)**
Your statistical background?	MIT SCM MicroMasters toolkit, labeled; S8a / S8c are the interval work	Never "I build models for a living"	**Toolkit, not job title**

Read me first

The interviewer's lens (inference from title, JD and public profile — the profile parts labeled)

Inference from the title. "Placement Analytics" is the analytical arm of inventory placement — deciding where inventory should sit in the network so that the promise holds at the lowest landed cost. A manager in that function lives on forecast error, lead-time variability, safety-stock policy, service levels and the data-quality underneath them. He will be the most quantitative interviewer on the panel and the most likely to ask "how was that measured?", "what was the denominator?", and "what is the softest input in your model?" He is also the one most likely to recognise a textbook formula being passed off as a program artifact — so the honesty labels matter more here than anywhere.

What that means for the answers. Ten Elementum primaries (the router and cram sheet say which to reach for). The first two are the Plan-process redesigns with the math visible: the shortage-alert / real-time supply-demand view (S6 — forecast accuracy 70 → 85%, $2M safety stock released under a two-signal rule) and the measured-lane lead-time program (S8 — promise at p99 on 200 lanes, 580K units / ~$1.5M released with zero stockouts). The other eight are the same facts cut for a statistician — the lead-time distribution (S8a), the safety-stock formula (S8b), the back-test (S8c), the 1,000 → 597 lane prioritization (S9), the Walmart Marketplace delivery promise, WBR and managed-shipping design (S19, S20, S21) and the new-market newsvendor design (E7). The lane programs end on *capital released on evidence with service held* — asset utilization and reliability in the JD's words. Keep the two programs' dollars apart in every sentence. A shadow (Sandy Hsiao) is on the call: same pace, same numbers, expect one question from the shadow near the end.

What his public profile adds (public-profile inference, labeled)

The profile (public-profile inference — directional, not verified biography). Manager, Transfer/Placement Analytics at Amazon (Jul 2025–), New York; before that Senior Business Intelligence Engineer at Amazon (Dec 2024–Jul 2025); analytics manager at Nike (2021–24); analyst at Cornerstone Research (2016–20) — an economics-consulting shop where the work product is expert-testimony-grade statistical and financial analysis; Chicago Booth MBA (Siebel Scholar); Duke. His own About line: "rigorous statistical and financial analysis… a blend of data analysis and writing skills… a culture of mentorship." **Quoted for framing only — never referenced in the room; none of it enters my stories.**

What that means for the answers. Expect *definition-level probing*: MAPE versus bias and which one steered; the grain (SKU-family? per lane?); the window and refresh; what "99% lead-time accuracy" tolerates in days. Three such definition Q&A are drilled into the S6/S8 probing below. Second, *answer style: crisp and memo-like* — claim, evidence, stop; he writes for a living, and a long wind-up reads as evasion. Third, speak *placement*: S6 is positioning safety stock where the measured errors justify it; S8 is positioning the promise at p99 per lane — his daily work is deciding where inventory sits versus where it is promised. Fourth, the mentorship line in his About suggests one people probe — answer it with planners grown into owners (S6's measurement handover; S8's drift flag owned by planning), not with org charts.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title, the JD and the public profile above (public-profile inference); if the first question lands on a different LP — Insist on the Highest Standards via the 99% bar, or Frugality via the capital releases — the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Dive Deep	An analytics manager tests whether the candidate goes to the data grain — measured lane distributions instead of brochure values; the definition of "supply" before any screen. S8 is the Dive Deep primary in the master map.	S8 (open) · S6 (the supply definition, the MAPE/bias measurement)
Are Right, A Lot	Placement decisions are judgment under uncertainty; he will probe how the candidate was right against the room's belief and how the decision was made safe (release rule, watch window, fat-tail exception).	S8 (measured vs brochure; p99 chosen by planning on the asymmetry) · backup: S6 (refusing unfirmed MRP orders as supply)
Learn & Be Curious (with a mentorship angle)	A Cornerstone-trained analyst with a Booth MBA who writes about rigor and mentorship (public-profile inference) will test whether the candidate learns at the definition level and grows people at the mechanism level: the MIT toolkit acquired to design what I had been implementing — labeled toolkit, never day job — and planners grown into owners of the measurement, the drift flag and the refresh cadence. Insist on the Highest Standards stays live as the 99% bar before any release and the two-signal rule before any buffer came out.	S8 (99% bar; drift flag owned by planning) · S6 (measurement handed to planners)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For Placement Analytics specifically:** lead with the planning parameter that was wrong (brochure lead time; unfirmed orders counted as supply), the decision it corrupted (buffer size, shortage response), and the rule that released capital safely. Show one formula only when asked, labeled as the MIT toolkit illustration — the data team built the distributions; planners sized the buffers; Finance signed.
- **Answer style for this room: crisp, memo-like.** Claim, evidence, stop. His background (Cornerstone rigor; a writing-heavy About line — public-profile inference) means a long wind-up reads as evasion; the one-breath basis label ("record", or "modeled, on illustrative inputs") does more credibility work here than any extra sentence.
- **Speak placement.** Frame S6 as positioning safety stock — buffer placed only where measured forecast and lead-time errors justify it — and S8 as positioning the promise at p99 per lane, recomputed on a drift flag. His

team decides where inventory sits versus where it is promised (public-profile inference); that is the exact shape of both stories.

- **Keep S6 and S8 apart:** $2M (S6, forecast-driven, two-signal rule) and ~$1.5M / 580K units (S8, lead-time-driven, 99% bar) are different programs — never add them, never swap their mechanisms. "Six sites" — never "six NA sites"; never paste S6's "3 months" onto the six-site ingest (that is S7, not in this packet).

Loop-consistency rule. One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S2/S18 (Goforth) · S1/S3 (Lan) · S16/S15/S10/S14 (Josyula). S7 (CM BOM) is an unallocated master-packet backup — pointer only, and no BOM-explosion content. S9, S19, S20, S21 and E7 are used here as primaries: if the same story is also drafted for another interviewer, tell it in one room only. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

The stories — business-first, canon numbers

Ten Elementum primaries from the same six-site manufacturer, the Walmart Marketplace client work and a new-market design engagement. Each primary: LPs carried · 20-second version · 60-second pitch → Situation/Task → Action in gated phases → Results → L6 evidence → the five facets → Probing with the four-table math → behavioral pack (four separate 90-second answers) → three angles → three trap drills. S6 and S8 are the existing stories, elevated; the other eight are new to this packet (red NEW pills mark every new section).

S6 — Elementum — one real-time supply/demand view across eight data domains: a shortage visible a lead-time early, forecast accuracy 70% → 85%, $2M of safety stock released under a two-signal rule ELEVATED

LPs this story can serve here: Dive Deep (the definition of supply; MAPE/bias at the decision grain) · Earn Trust (the two-signal release rule signed with Planning and Finance) · Deliver Results (third — alert live in 3 months) · Think Big (one view over eight domains) · Are Right, A Lot (backup — refusing unfirmed orders as supply)
Process family: Plan — S&OP / shortage management / safety-stock policy; data-integrity program behind the plan

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I profiled every domain and defined supply honestly before any screen — released planned orders only, because unfirmed MRP orders breathe every run.
- **Earn Trust** — The $2M left the buffer only under a both-errors-improved rule Planning and Finance signed — the exception was written in before anyone asked.
- **Deliver Results** — Forecast accuracy 70% → 85% on scoped class-A SKUs; the shortage alert live in three months; planner effort −50%; adoption 8×.
- **Ownership** — Nobody owned the seam between eight data domains and the planners — I took the definitions, the data-integrity program and the delivery.

Honesty label. I owned the definitions, the data-integrity program and the delivery; the client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement and the release rule. I have not owned a demand forecast end-to-end and won't claim it. The safety-stock formula is how I explain it now with the MIT toolkit — an illustration, not a program artifact.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A six-site manufacturer — mixed make-to-order and make-to-stock — learned about a component shortage when the line stopped. S&OP ran off private spreadsheets, shipment data was 60% complete, and supplier, material, BOM, inventory, production, transportation, sales-order and planning data lived in SAP and site systems but never met. Leadership's question — which customer orders are at risk this quarter? — had no data path, so the plan responded with expedites instead of re-plans and buffers sized on noise. **Task.** Nobody owned the seam; I took the definitions, the data-integrity program and the delivery. **Action.** I defined supply honestly before any screen — on-hand plus production plan plus released planned orders only, at SKU × location × date — and refused to count unfirmed MRP orders because they change every run. Domains went live in waves sequenced by planner trust, each behind a published data-quality gate, with daily reconciliation to the SAP source and false alarms worked as defects. The shortage alert fires when days-of-cover falls to the lane's measured p90 lead time — the last moment a re-plan is cheaper than a flight. Safety stock came out only under a rule Planning and Finance signed: release where forecast error and lead-time error had both improved. **Result.** Primary metric: forecast accuracy on the scoped class-A SKUs 70% → 85%. $2M of safety stock released under the two-signal rule — record, one-time working capital — with ≈$0.8M a year of modeled recurring value around it; the shortage alert live in three months; planner effort −50%. That $2M is a different program from the lane program's $1.5M; I never add them.

20-SECOND VERSION NEW

Eight data domains, one honest definition of supply, a shortage alert that fires when days-of-cover falls to the lane’s p90 lead time — a lead-time early — and $2M of safety stock released only where forecast error and lead-time error had both improved.

NACF BRIDGE — ONE SENTENCE

This is the NACF WOC loop: define supply honestly, alert at the last moment a re-plan is cheaper than an expedite, and cut weeks of cover only where both error signals have improved — INV-A and INV-B in the inbound booklet.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A strategic manufacturing customer with six sites: planners learned about a component shortage when the line stopped; S&OP ran off private spreadsheets; shipment data was 60% complete; eight data domains lived in SAP and site systems but never met. Leadership's question — "which customer orders are at risk this quarter?" — had no data path.

The Plan process underneath was running on an unsafe definition of supply (unfirmed MRP orders counted because they looked better), an unmeasured lead-time error, and a shortage signal that arrived at line-down. The cost: expedites instead of re-plans, buffers sized on noise, and planners spending ~40% of their week reconciling rather than root-causing.

T — Nobody owned the seam — no spec, no agreed grain, no definition of "supply," no data-quality bar. I chose all four, as owner of the definitions, the data-integrity program and the delivery. Goal: a real-time supply/demand view with honest definitions and a shortage visible a lead-time early — and planner trust, which was the real product.

Baselines on the table: forecast accuracy 70% on scoped class-A SKUs · shortage detected at line-down (target: a lead-time early) · shipment-data completeness 60% · planner effort on reconciliation and firefighting (≈40% of the week, illustrative share; the −50% is record).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (4–6 wks)** — profiled every domain (completeness, latency, joinability) to my own gap templates; wrote the definitions memo — grain SKU × location × date; supply = on-hand + production plan + released planned orders only — and the metrics tree (OTIF, inventory $, MAPE/bias, lane lead-time accuracy, completeness, latency, adoption). Refused Planning's request to count unfirmed MRP orders as supply. *Exit bar:* signed gap findings per domain; scope frozen in writing; planners agree the supply definition before any screen.

- **Plan & Design (parallel)** — master-data foundation plan; MPN mapping worklist with its coverage gate; domain-wave sequence by trust; DQ scorecard spec; the 10-step onboarding playbook. *Exit bar:* master-data gates green; unmapped-MPN list under threshold with a named owner; wave order signed by domain owners — who could fail their own wave.
- **Build & Test (8–12 wks)** — domain waves behind published DQ gates — inventory and production first (numbers planners already believed), then purchasing/ASN, then transportation; shortage-alert rule specified (days-of-cover vs lane lead-time p90); daily reconciliation to MD04/COOIS/VA05/MB52; weekly root-cause loop with false alarms worked as defects. *Exit bar:* per-domain completeness ≥ gate; reconciliation passing daily; false alarms trending down; pilot planners using the alert.
- **Cutover & Go-Live (4 wks)** — views live for pilot planners with the at-risk list replacing the Monday spreadsheet; nobody mandated off the spreadsheet — rollback stayed free. *Exit bar:* planners answer "where is my product" from the platform in the live S&OP meeting.
- **Hypercare / Close (4 wks)** — exception workbench staffed; weekly root-cause reviews handed to domain owners; the safety-stock release rule signed with Planning and Finance — release only where forecast error AND lead-time error both improved. *Exit bar:* blockers/day under threshold two consecutive weeks; the release survives audit.

Influence and ownership. Mine: the definitions memo, the metrics tree, the DQ scorecard spec, the wave sequence, the release rule and the onboarding playbook. The client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement (MAPE/bias per SKU family) and the release rule; domain owners held sign-off over their own gates. The MPN worklist gated longer than planned — escalated to the sponsor as a re-sequenced date with the coverage number attached, not a slip.

> **Designed the process; others own it now.** I designed the definitions, the DQ gates, the release rule and the onboarding playbook; domain owners now run the gates and the weekly root-cause reviews, and planners run the alert without me. **Trade-off put to leadership.** The recommendation to Planning and Finance outlined the competing metrics — working capital released vs service risk — and resolved it as a rule: release only where forecast error and lead-time error both improved.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — $2M safety stock released (record, one-time working capital, under the both-errors-improved rule signed with Planning and Finance) + ≈$0.8M/yr *modeled* recurring (carry ≈$0.2M at 10% · expedites pre-empted ≈$0.3M · planner capacity ≈$0.3M; range $0.5–1.2M) — never added to S8's $1.5M.
- **#** — eight data domains into one view across six sites; shortage-alert product live in 3 months; adoption 8× (record).
- **%** — forecast accuracy 70% → 85% on scoped class-A SKUs (record); planner effort −50% (record); shipment data 60% complete at the start.
- **Time** — the alert fires at days-of-cover ≤ lane lead-time p90 — a lead-time early instead of at line-down (record mechanism); views live in 3 months (the alert product; the six-site ingest is a separate 5-month program — never paste 3 months onto it).
- **Customer** — "which customer orders are at risk this quarter?" answered from the platform in the live S&OP meeting, not a spreadsheet (record exit bar).
- **Quality** — daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Eight data domains, six sites and no shared definition of supply. I wrote the definitions memo (grain SKU × location × date; supply = on-hand + production plan + released planned orders only) and the metrics tree, and chose the domain owners who would sign their own gates.
Scope & influence	The client's SAP teams and our integration engineers built the feeds; planners owned the forecast; Planning and Finance signed the release rule; domain owners held sign-off over their gates — none reporting to me.
Execution trade-off + mechanism others now own	Working-capital release vs service risk: release only where forecast error AND lead-time error both improved. Mechanisms others own now: the DQ scorecard, the 10-step onboarding playbook and the weekly root-cause reviews handed to domain owners.
Impact (long-term)	$2M of safety stock released (record); forecast accuracy 70% → 85% on class-A; planner effort −50%; adoption 8×; the shortage alert fires a lead-time early instead of at line-down.
Process improvement (standard work institutionalized)	Domain waves behind published DQ gates; daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects.
Complexity & consensus	Nobody was mandated off the Monday spreadsheet — rollback stayed free until planners answered "where is my product" from the platform in the live S&OP meeting; the MPN worklist gated longer than planned and the gate held.
Root cause at scale	Unfirmed orders inflating supply was a definition defect, not a planner habit — fixed once in the supply definition and applied across all six sites.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: eight domains (supplier, material, BOM, inventory, production, transportation, sales-order, planning) across six sites; shipment data 60% complete at the start; MAPE and bias at the SKU-family grain on scoped class-A SKUs. Revealed: unfirmed MRP orders inflated supply; the lane's own lead time was unmeasured.
2 · Solution design	Options: count unfirmed orders or not · alert at brochure lead time or measured p90 · release buffer on accuracy alone or on both signals. Chosen: released planned orders only; alert at DoC ≤ lane p90; release only where both errors improved.
3 · Math & data logic	Math: DoC = on-hand ÷ daily demand = 1,200 ÷ 100 = 12 ≤ p90 14 → alert; SS = z·d·σ_L = 2.33 × 100 × 4 = 932 → 466 at σ_L 2; alert precision 71.2% / recall 98.6% at p90; both-errors rule false-release rate under noise 25% → 2.5% with a one-SE bar.
4 · Implementation	Implementation: definitions memo → data-quality gates per domain → waves sequenced by planner trust → daily reconciliation to SAP source → alert live in 3 months → release rule signed → false alarms classified weekly.
5 · Probe questions	Probe questions: MAPE vs bias · what "accuracy" means · why p90 for the alert and p99 for the promise · false alarms · whether "both improved" is signal or noise — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Analytics — How was 70 → 85% measured, and on what?**
>
> Accuracy = 1 − MAPE on the scoped class-A SKUs, with bias published alongside it from month two, at the SKU-family grain the planners decide on. The planners owned the forecast; I owned the measurement and the input integrity — I have not owned a demand forecast end-to-end and won't claim it.

> **Analytics — Why exclude unfirmed MRP planned orders from supply? That understates it.**
>
> Deliberately. Unfirmed orders breathe every MRP run; counting them amplifies noise and inflates supply, which is exactly how the line-down surprises happened. Released planned orders only — accepted by planners before the first screen was built.

Analytics — The $2M — how do I know it is not a prettier average?

Because of the release rule: buffer came out only on SKU families where forecast error and lead-time error had both demonstrably improved, signed by Planning and Finance. That is why the release survived audit. It is one-time working capital; the recurring value around it is my model and labeled.

Analytics — Your expedite row is the softest. Defend it.

Agreed — about sixty expedites a year pre-empted at a $5K premium, both illustrative. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90. To firm it I would pull the expedited-freight ledger for the six sites twelve months before and after the alert went live and count the premium directly.

Analytics — Why p90 for the alert threshold and p99 for the promise (S8)?

Different decisions. The alert is a re-plan trigger — p90 gives the planner a window before the tail bites. The promise is a commitment — it sits at the tail you can defend. Same measured distribution, two policies read off it.

Analytics — Define your two error metrics: MAPE versus bias. Which one steered?

MAPE is mean absolute percentage error — how wrong the forecast is on average, direction ignored; bias is the signed mean error — whether it runs systematically high or low. MAPE without bias hides a chronic over-forecast inside a decent average, and a chronic over-forecast is exactly how buffer quietly accumulates. So accuracy was reported as 1 − MAPE on the scoped class-A SKUs at the SKU-family grain, with bias published alongside from month two — and the release rule required forecast error and lead-time error both to improve, so neither metric could carry the decision alone.

Analytics — What are the precision and recall of an alert at the lane's p90?

On an illustrative lane — replenishment time normal with mean 10 and σ 3 days, so p90 = 13.8 — and days-of-cover states uniform across 0–120 days: of every 1,000 checks about 83 would run short without action; the p90 alert fires on about 115; 71.2% of alerts are real (precision) and 98.6% of shortages are caught (recall). Moving to p99 lifts recall to 99.9% and drops precision to 58.8%. Illustrative lane; the record is the rule, DoC ≤ lane p90.

Analytics — Two kinds of false alarm — how do you tell them apart?

Statistical and defect. A designed false positive is an alert on a lane that turned out fine — at p90 about 29% of alerts by construction; that is the price of the window. A defect false alarm is the data lying — a wrong on-hand, a missing receipt, an unfirmed order counted as supply. The record burns the second kind down weekly by root cause; I do not let the first kind masquerade as a data problem.

Analytics — Is "both errors improved" signal or noise?

Under no real improvement each metric is a coin flip, so both improve by chance 25% of the time — on 100 families, about 25 false releases. Requiring each to beat its own standard error cuts that to 2.5% (about 2.5 families); two standard errors, 0.05%. The record says "demonstrably improved"; how that was operationalized is a number I would confirm before I quote one. VERIFY-WITH-RAJESH

Analytics — Why does bias matter when MAPE is already 15%?

Because direction compounds. A +10% over-forecast on 100 units a day adds 70 units a week; in 13.3 weeks the excess alone equals the whole 932-unit buffer on the example lane. MAPE tells you how far off; bias tells you which way the stock is piling up.

Shadow / closing — What would you do differently?

Start the MPN-mapping worklist on day one — it is never "done" and it gated more than I planned. And define supply honestly before any screen, which held.

THE KEY MATH — THE NETTING WALK AT FINISHED-GOODS GRAIN, THE ALERT RULE AND THE TWO-SIGNAL RELEASE (ILLUSTRATIVE VALUES; RECORD PERCENTAGES)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Accuracy before / after	`1 − 0.30 = 70% → 1 − 0.15 = 85%`	record	70% → 85%
2	Honest supply	`supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders`	record rule	unfirmed MRP orders excluded
3	Projected balance (FG SKU)	`PAB(3) = 400 − 500 = −100; PAB(7) = 1,000 − 1,200 = −200; PAB(10) = 1,800 − 1,600 = +200`	illustrative FG example — timing shortage, not volume	fix window = day 9 − day 7 = 2 days
4	Alert rule	`DoC = 1,200 / 100 = 12 days ≤ lane p90 14 → alert; p90 = 9 → no alert`	record design; values illustrative	a lead-time early, no false alarm on the fast lane
5	Two-signal release	`SS = z·d·σ_L = 2.33 × 100 × 4 = 932 → σ_L 2: 466; release 466 units — only if BOTH errors improved`	toolkit illustration; the rule is record	release survives audit
6	Recurring value	`$2M × 10% = $0.2M · 60 × $5K = $0.3M · 2.4 FTE × $120K = $0.3M`	modeled (benchmarks labeled)	≈$0.8M/yr; $2M one-time record

MRP math at finished-goods grain only — no BOM explosion in this packet. The six-site ingest (S7) is a separate program; never paste this story's "3 months" onto it.

Where this lands at Amazon. Placement rests on two error signals — forecast error and lead-time error — measured at the decision grain. The transferable disciplines: define supply before the screen, alert at the last moment a re-plan is cheaper than an expedite, and release buffer only when both error signals improve. Reliability (shortage seen a lead-time early), cost efficiency (expedites pre-empted), asset utilization ($2M of working capital released on evidence).

FULL DERIVATION — FOUR-TABLE MATH, SOLUTION FLOW, SENSITIVITY, SPOKEN STATEMENT, IF-CHALLENGED NEW

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Customer orders at risk this quarter -> answered a lead-time early
+- Inventory $ (stock): $2M safety stock released (record)
|    +- forecast error 30% -> 15% (accuracy 70 -> 85%, record)
|    +- lead-time error (lane p90 measured, not brochure)
|    +- release rule: BOTH errors improved -> SS = z*sigma*sqrt(L) falls
+- Shortage response (flow): alert when DoC <= lane LT p90
|    +- fix window = LT p90 - DoC: days to re-plan instead of fly
+- Planner effort -50% (record) -> hours to root cause, not reconciliation
+- Data-integrity loop: daily recon -> defects burned down -> planner trust (+)
     -> adoption 8x -> more domains feed the peg -> errors fall further (+)
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Forecast accuracy, class-A SKUs	70%	85%	—	record
Safety stock released (one-time)	—	$2M out	—	record
Carry on released $2M	tied up	free	$0.2M	modeled ($2M × 10% WACC; benchmark 8–12%)
Expedites avoided via the alert	~150 events/yr	~60 pre-empted	$0.3M	modeled (60 × $5K; premium benchmark $3–8K)
Planner effort on reconciliation	12 × 40%	12 × 20%	$0.3M	modeled (2.4 FTE × $120K; −50% is record)

DRIVER	BASELINE	AFTER	$/YR	BASIS
Total modeled recurring	—	—	**≈$0.8M**	around the $2M one-time record; never added to S8's $1.5M

If challenged. *"The $2M — measured how?"* — "It's the record: the release Finance signed, applied only on SKU families where forecast error and lead-time error had both demonstrably improved. That rule is why the release survived audit — capital came out on evidence, not on a better-looking average." *"150 shortage events?"* — "Illustrative volume — six sites at roughly 25 escalations a year. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90, which is the last moment a re-plan is cheaper than a flight."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Forecast accuracy	`accuracy = 1 − MAPE = 1 − 0.30 = 70%`	scoped class-A SKUs (record)	70%
2	What the error math looks like	`MAPE = (30/100 + 10/120 + 20/80 + 10/100)/4 = (0.30 + 0.083 + 0.25 + 0.10)/4 = 18.3%`	4 illustrative weeks: A = 100, 120, 80, 100; F = 130, 110, 100, 90; |e| = 30, 10, 20, 10	18.3% on this sample; bias = (430 − 400)/400 = +7.5%
3	The supply lie	`supply counted unfirmed MRP planned orders — they change every run`	record problem	plans built on sand
4	Shortage discovery	`at line-down: ~150 expedite events/yr`	benchmark-modeled count	~150 fire-drills/yr
5	Planner burn	`12 planners × 40% = 4.8 FTE-equivalent on reconciliation`	record share	4.8 FTE reconciling, not root-causing

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Honest supply	`supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders only`	unfirmed MRP planned orders excluded (record rule)	walked below on ROUTER-100
2	FG netting walk — setup	`FG SKU ROUTER-100 at DC-East: on-hand 400; in-transit 600 lands day 5; production 800 day 9; demand 500 d3 / 700 d7 / 400 d10`	illustrative (v105 §5.12)	projected balance below
3	Day 3	`PAB(3) = 400 − 500 = −100`	only on-hand has arrived by day 3	dated shortage of 100 on day 3
4	Day 7	`PAB(7) = (400 + 600) − (500 + 700) = 1,000 − 1,200 = −200`	day-9 production is too late for day-7 demand	shortage of 200; fix window = 9 − 7 = 2 days
5	Day 10	`PAB(10) = (400 + 600 + 800) − (500 + 700 + 400) = 1,800 − 1,600 = +200`	all supply lands by day 10	+200 — the shortage is timing, not volume
6	Shortage alert rule	`DoC = 1,200 / 100 = 12 days; 12 ≤ p90 14 → alert now`	FG SKU: on-hand 1,200 at 100/day; lane p90 = 14 days (illustrative)	alert a lead-time early, not at line-down
7	Counter-case	`p90 = 9 days: 12 > 9 → no alert`	same SKU, faster lane (illustrative)	3-day slack — no false alarm

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
8	Two-signal release rule	`SS = z × d × σ_L = 2.33 × 100 × 4 = 932; with σ_L 4 → 2: 2.33 × 100 × 2 = 466; release 932 − 466 = 466 units`	release only if forecast error AND lead-time accuracy both improved; d = 100/day, z = 2.33 at 99% (illustrative, toolkit form)	466 units off this SKU
9	Accuracy after	`error 30% → 15% (halved); accuracy = 1 − 0.15 = 85%`	gated waves; false alarms worked as defects (record)	85%

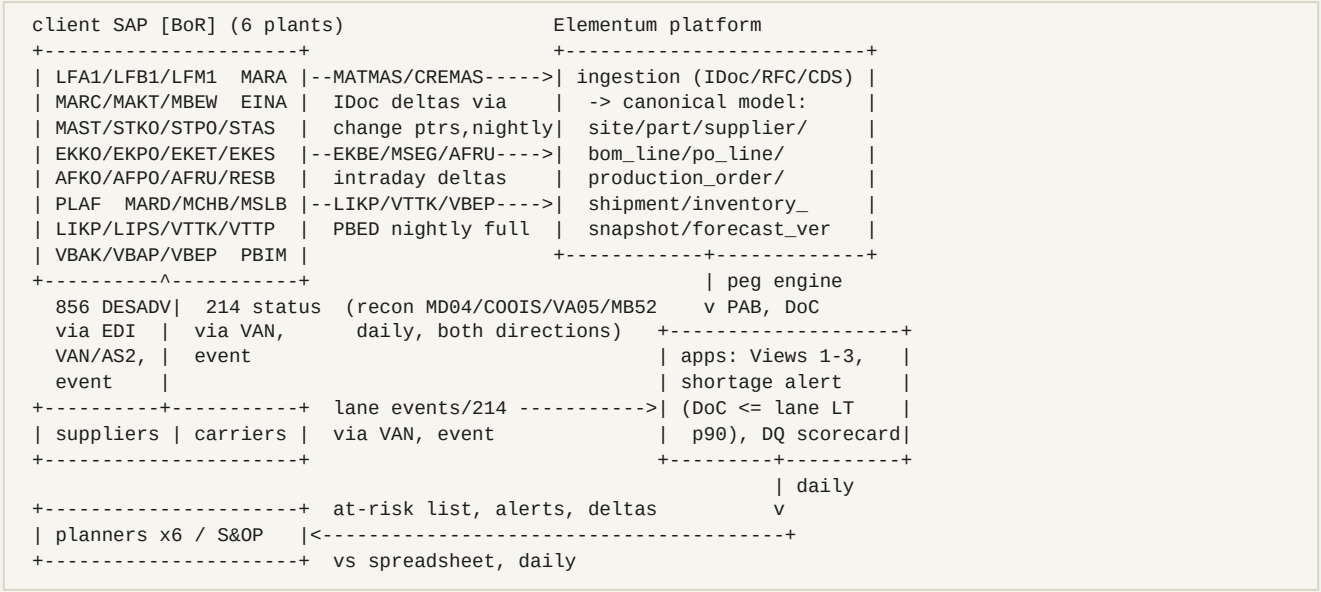
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Forecast accuracy, class-A SKUs	70%	85%	record
2	Shortage discovery	at line-down	a lead-time early (DoC ≤ p90 alert)	record design
3	Planner reconciliation share	12 × 40% of time	12 × 20% — halved	record (−50%)
4	Safety stock	sized on bad signals	$2M released one-time under the signed two-signal rule	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Carry on the release	`$2,000,000 × 10% = $200,000`	$2M release (record, one-time) × WACC (benchmark 8–12%)	≈$0.2M/yr
2	Expedites pre-empted	`150 × 40% = 60 caught early; 60 × $5,000 = $300,000`	events/yr modeled; premium benchmark $3–8K	≈$0.3M/yr
3	Planner capacity	`12 × 40% = 4.8 FTE; halved → 2.4 FTE; 2.4 × $120,000 = $288,000`	−50% is record; capacity redeployed, not headcount removed	≈$0.3M/yr
4	Total recurring	`0.2 + 0.3 + 0.3 = $0.8M`	modeled; never added to S8's $1.5M	≈$0.8M/yr
5	One-time	`$2M safety stock released, taken as stated`	record, under the signed two-signal rule	$2M

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Cost of capital (benchmark)	8% → $0.16M	10% → $0.20M	12% → $0.24M
Expedites avoided (illustrative)	45	60	75
Expedite premium (benchmark)	$3K → $0.14M	$5K → $0.30M	$8K → $0.60M
Loaded planner cost (benchmark)	$100K → $0.24M	$120K → $0.29M	$140K → $0.34M
Total modeled $/yr	≈$0.5M	≈$0.8M	≈$1.2M

ALERT-THRESHOLD DESIGN — PRECISION, RECALL AND THE COST OF A FALSE ALARM (EXTRA TABLE)

The record is the rule — alert when days-of-cover ≤ the lane's measured p90 lead time. Everything below is my after-the-fact design arithmetic on an illustrative lane, so the threshold can be defended in numbers. VERIFY-WITH-RAJESH how p90 was originally chosen.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The lane	`replenishment time L ~ Normal(μ = 10, σ = 3); p90 = 10 + 1.2816 × 3 = 13.84 ≈ 14 d`	illustrative lane (matches the p90 = 14 example)	θ = 13.84 d
2	Shortage if no action	`P(L > DoC) = 1 − Φ((DoC − 10)/3); DoC uniform on 0–120 d → ≈ 10/120 = 8.3% of checks`	states illustrative; ∫P(L > x)dx = E[L] = 10	83.3 per 1,000 checks
3	Alert rule (record)	`alert iff DoC ≤ θ: alerts per 1,000 = 1,000 × 13.84 ÷ 120 = 115`	record design; values illustrative	115 alerts / 1,000
4	Precision (p90)	`TP ÷ alerts = 82.15 ÷ 115.37 = 71.2%`	numerical integral of P(L > DoC) over DoC ≤ θ	71.2%
5	Recall (p90)	`TP ÷ shortages = 82.15 ÷ 83.34 = 98.6%`	same integral	98.6%
6	Same lane at p99	`θ = 16.98 d; alerts 141; precision 58.8%; recall 99.9%`	same method	more alerts, little recall gained
7	Cost of a false alarm vs a miss	`planner hour ≈ $120K ÷ 2,000 h = $60; a missed shortage ≈ $5,000 (master expedite premium)`	both illustrative (master)	$60 vs $5,000
8	Expected cost per 1,000 checks	`p75 $24,656; p90 115.372 × $60 + 1.1836 × $5,000 = $12,840; p95 $10,079; p99 $8,913`	alerts × c_fp + misses × c_miss	p90 $12,840 vs p99 $8,913
9	Cost-optimal threshold	`alert iff P(L > DoC) ≥ c_fp ÷ c_miss = 60 ÷ 5,000 = 1.2% → quantile 98.8% → θ = 16.8 d`	critical-ratio logic	θ* = 16.8 d if c_miss were $5,000
10	What p90 implies	`p90 ⇔ c_miss ÷ c_fp = 0.90 ÷ 0.10 = 9 → c_miss = 9 × $60 = $540; p99 ⇔ $5,940`	after-the-fact derivation — not how p90 was chosen	p90 prices a miss at $540
11	Minimum shipments to bound the p90	`n = 31: t(24)…t(31); n = 30 → t(24)…open`	exact order-statistic interval (95%)	n ≥ 31 per lane; else cohort p90

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured number is two million dollars of safety stock released, and that is record, signed with planning and Finance under the both-errors-improved rule. Around it I model roughly eight hundred thousand a year of recurring value: about two hundred thousand in carry cost on the capital that came out, about three hundred thousand in expedite premium the shortage alert pre-empts because it fires a lead-time early instead of at line-down, and about three hundred thousand of planner capacity from the fifty percent effort cut. The two million and the fifty percent are measured; the three recurring pieces are my model on benchmark rates, and I keep them separate from the one-and-a-half million in the lane program."

If challenged — the full answers.

The expedite figure is the softest

It rests on about sixty expedites a year pre-empted at a five thousand dollar premium, both illustrative; to firm it I would pull the expedited-freight ledger for the six sites for the twelve months before and after the alert went live and count the premium directly.

Planner capacity is not cash

Agreed, it is two point four FTE of time redeployed, not headcount removed; the fifty percent is measured, the forty percent share of the week is my assumption, and a two-week time study on the S&OP prep cycle would replace it.

Why not claim the whole two million every year

Because a release is a one-time working-capital event; the recurring value is the carry at eight to twelve percent, and I would rather say two hundred thousand a year honestly than two million a year wrongly.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S6

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you had to dig into the data to find the real problem.	Dive Deep	The supply definition — unfirmed orders inflating supply — found by profiling eight domains; MAPE and bias at the grain planners decide on.	Skip the SAP table names unless asked; systems are enablers.
Tell me about a time you released inventory / improved working capital.	Frugality / Deliver Results	$2M under the two-signal rule — record, one-time; the recurring ≈$0.8M labeled modeled.	Never add the $2M to S8's $1.5M; never claim $2M per year.
Tell me about a time you were right when the business disagreed.	Are Right, A Lot	Refusing unfirmed MRP orders as supply because it looked better; refusing a blanket safety-stock release — agreed as a rule instead.	Do not make Planning the villain; the rule did the refusing.
How do you get planners to trust a new signal?	Earn Trust	Waves sequenced by trust (inventory and production first); false alarms worked as defects; nobody mandated off the spreadsheet; domain owners could fail their own wave.	Do not claim adoption dollars; adoption 8× is the record.
How would you design a shortage / at-risk-order signal for the network?	Functional — Plan process design	Grain, honest supply, days-of-cover vs measured lane p90, DQ gate per feed, two-signal release — the §3.99 opening: book of record and metric first.	No BOM explosion; finished-goods grain only.
What's the softest number in your model?	Highest Standards / Earn Trust	The expedite row (60 × $5K); how I would firm it from the expedited-freight ledger.	Never present planner capacity as cash — it is time redeployed.

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time you dug into the data to find the real problem. DIVE DEEP

Open (first two sentences): Leadership asked which customer orders were at risk, and the first thing I did was refuse to answer until we agreed what “supply” meant.

STAR (≈90 s): Situation: a six-site manufacturer found out about shortages when the line stopped; shipment data was 60% complete across eight domains. Task: nobody owned the seam, so I took definitions and data integrity. Action: profiled every domain for completeness, latency and joinability, and defined supply honestly — on-hand plus production plan plus released planned orders only; unfirmed MRP orders change every run and had been inflating supply. Result: the alert product live in three months; forecast accuracy 70 to 85% on the scoped class-A SKUs; planners accepted the definition before the first screen existed.

Land this number: 60% shipment-data completeness → daily reconciliation to SAP; 70% → 85% accuracy. · **Pivot to NACF:** NACF inbound “available” and “on-hand” need one signed definition before any placement model reads them. · **Do NOT say:** Do not say I owned the forecast; planners did — I owned the measurement and the input integrity.

Q2 Tell me about a time you needed buy-in from Finance and Planning on a risky change. EARN TRUST

Open (first two sentences): Releasing $2M of safety stock is a risk Finance and Planning both carry, so the rule was theirs before it was mine.

STAR (≈90 s): Situation: buffer had grown on noisy forecasts and unmeasured lead times. Task: take capital out without a stockout. Action: I proposed a two-signal rule — release only where forecast error and lead-time error had both demonstrably improved — and got Planning and Finance to sign it before the first release. Result: $2M released and it survived audit because the exception was in the rule; ≈$0.8M a year modeled around it; never added to the other program’s $1.5M.

Land this number: $2M released (record, one-time); both-errors rule. · **Pivot to NACF:** A WOC reduction at Amazon needs the same pre-signed exception: cut only where both signals improved. · **Do NOT say:** Do not add $2M and $1.5M, and do not call the $0.8M recurring a result — it is modeled.

Q3 Tell me about a time you delivered a result on a hard deadline. DELIVER RESULTS

Open (first two sentences): Three months from kickoff, a planner could see a shortage a lead time before the line stopped.

STAR (≈90 s): Situation: the plan responded with expedites, not re-plans. Task: ship a shortage alert fast without shipping wrong data. Action: sequenced the domains in waves by planner trust, each behind a published data-quality gate, with daily reconciliation to SAP and false alarms worked as defects. Result: alert live in three months, planner effort −50%, adoption 8×, forecast accuracy 70 to 85%. The six-site ingest is a separate five-month program — I never paste the three months onto it.

Land this number: Alert live in 3 months; planner effort −50%; adoption 8×. · **Pivot to NACF:** Same sequencing for NACF: trust-ordered waves, a quality gate per source. · **Do NOT say:** Do not stretch “3 months” to the full ingest.

Q4 Tell me about a time you owned something nobody owned. OWNERSHIP

Open (first two sentences): Eight domains, six sites, and the question “which customer orders are at risk?” had no data path and no owner.

STAR (≈90 s): Situation: S&OP ran off private spreadsheets and the seam between systems and planners belonged to nobody. Task: own it. Action: I wrote the definitions memo, the data-quality scorecard spec, the wave sequence and the release rule, and stayed through hypercare. The thing I would do differently: start the MPN-mapping worklist on day one — it is never done and it gated more than I planned. Result: the question was answered in the live S&OP meeting from the platform, not a spreadsheet.

Land this number: Eight domains into one view; alert in 3 months. · **Pivot to NACF:** Placement analytics lives on that seam too: forecast error and lead-time error from different systems. · **Do NOT say:** Do not claim the client’s SAP teams’ feeds were mine.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "I would not build a screen until we agreed what supply was."

Headline metric: unfirmed MRP orders excluded; accuracy 70% → 85%

- MAPE with bias published from month two.
- Alert at DoC ≤ lane p90: precision 71% / recall 99% (illustrative lane).
- DoC 12 ≤ p90 14 → alert; p90 9 → no alert.
- False alarms: designed vs defect.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Earn Trust; Have Backbone

Open: "Planners wanted the unfirmed orders in supply because it looked better; I refused them."

Headline metric: released planned orders only; $2M under a signed rule

- Accepted by planners before the first screen.
- Release only if both errors improved.
- Finance and Planning signed.
- A coin-flip rule would false-release 25% of families — so "demonstrably."

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Ownership; Learn and Be Curious

Open: "The MPN-mapping worklist is never done — I should have started it on day one."

Headline metric: MPN mapping gated more than planned (record lesson)

- The lesson is in the record.
- Define supply honestly before any screen — that held.
- I did not own the forecast and say so.
- VERIFY-WITH-RAJESH the size of the delay MPN mapping caused.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Your alert fires at the lane's p90 — what are its precision and recall, and why not p99?"

The numbers: Illustrative lane N(10, 3): θ = 13.84; precision 71.2%, recall 98.6%; p99 precision 58.8%, recall 99.9%; p90 ⇔ miss costs 9× a false alarm.

Say exactly: "On an illustrative lane the p90 alert catches about 99 percent of shortages with about 71 percent precision; p99 buys about one more point of recall for twelve points of precision. A p90 threshold prices a miss at nine false alarms — at a sixty-dollar alert and a five-thousand-dollar miss the cost-optimal threshold would sit higher, which is why I would show planners the curve and let their own cost of an alert set the dial."

T2 "Both-errors-improved — what is the false-release rate if neither metric really improved?"

The numbers: Coin-flip: 25% of families; each beat by one SE: 2.52%; two SE: 0.052%; on 100 families: 25 / 2.5 / 0.05.

Say exactly: "If nothing improved, a bare both-up rule releases a quarter of the families by luck; requiring each metric to beat its own standard error takes it to about two and a half percent — which is why the rule said demonstrably improved, not merely improved."

T3 "Why does bias matter if your accuracy is 85%?"

The numbers: SS = 2.33 × 100 × 4 = 932; +10% bias × 100/day × 7 = 70/week; 932 ÷ 70 = 13.3 weeks.

Say exactly: "Accuracy is a magnitude; bias is a direction. A ten-percent over-forecast adds seventy units a week on a hundred-a-day SKU — in under fourteen weeks the excess alone equals a nine-hundred-unit buffer, which is how buffer quietly accumulates behind a decent MAPE."

S8 — Elementum — replacing brochure lead times with measured lane distributions: promise at p99 on 200 lanes, 99% lead-time accuracy, 580K units / ~$1.5M of safety stock released with zero stockouts ELEVATED

LPs this story can serve here: Dive Deep (primary — measured the distribution, not the average) · Are Right, A Lot (backup — measured vs brochure; p99 on the asymmetry) · Frugality (the release) · Insist on the Highest Standards (the 99% bar before any cut) · Earn Trust (Finance signed each lane; the exception written into the rule)
Process family: Plan / Source / Deliver seam — inbound lead-time policy, safety-stock sizing, promise policy; parameter governance with a drift flag

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I measured each lane's own distribution — mean, spread, tail — instead of accepting a carrier brochure or a network average.
- **Customer Obsession** — The customer was the production line: I put the cost of a line-down next to the carrying cost per lane and let planning choose p99 with the asymmetry in front of them.
- **Earn Trust** — Finance signed each lane; the release rule carried its own exception — fat tails kept their buffer — before anyone asked.
- **Ownership** — I designed the promise policy and release rule as a signed planning act that planning re-applies on its own refresh.

Honesty label. I owned the requirement (measured lanes, not brochures), the p99 promise policy and the release rule; the data team built the distributions; planners sized the buffers; Finance signed the release. Lane-cohort counts and week markers are not in the record. The safety-stock formula is the MIT inventory-theory toolkit as I explain it now — an illustration, not a program artifact. Depends on the six-site canonical model (S7 — pointer only).

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The same six-site manufacturer planned inventory on carrier-brochure lead times — every buffer sized on a variability nobody had measured: too much on reliable lanes, too little on fat-tailed ones. The logistics manager asked for a "lead-time variability report." **Task.** I reframed it as a parameter and policy problem — nobody owned the seam between transportation reality and planning parameters, and I took it — scoped at 200 lanes by site, lane and mode. **Action.** I wrote the requirement to the data team: mean, spread and tail per lane from actual production cycle plus inbound transit; defined the promise at p99 cumulatively — the smallest day d with $P(L \le d) \ge 0.99$, not average-plus-padding — and showed each lane's asymmetry, line-down cost against carrying cost, and let planning choose the percentile; they chose p99. The release rule was written as a signed planning act with the exception inside it: fat-tail lanes keep their buffer until the source process moves; Finance agreed the rule in the same room, before any cut. I would not endorse a release until lead-time accuracy hit 99% on the 200-lane set; the stockout-and-expedite watch was armed first; every parameter got a drift flag and a refresh cadence so the promise never decays into a new brochure. **Result.** Primary metric: lead-time accuracy on 200 lanes — brochure values → 99% at the p99 promise. 580K units, about $1.5M of safety stock, released without a stockout — record, one-time working capital — with ≈$0.6M a year modeled around it. A different program from the $2M forecast release; never added.

20-SECOND VERSION NEW

Two hundred lanes planned on carrier brochures; I replaced them with measured per-lane distributions, set the promise at p99, gated any cut on 99% lead-time accuracy and released 580K units, about $1.5M, with zero stockouts.

NACF BRIDGE — ONE SENTENCE

Inbound lead-time variability is half of every WOC target and every placement promise at NACF — I would measure each vendor lane's own distribution and promise at the tail I can defend before releasing any weeks of cover.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The six-site manufacturer's logistics manager asked for a "lead-time variability report." The real problem: inventory planning ran on carrier-brochure lead times — every buffer sized on a variability nobody had measured; too much buffer on reliable lanes, too little on fat-tailed ones; the promise to the plan was folklore.

The risk cut both ways: keep planning on brochures and you hold too much on reliable lanes and stock out on fat-tailed ones; cut buffer without evidence and the stockout is yours. And an unmaintained promise decays into a new brochure — which is why every parameter needed a drift flag.

T — Nobody owned the seam between transportation reality and planning parameters; I took it. Goal: measured lanes replace catalog values; a promise at a defensible tail; buffer released only on evidence, with service held — scoped at 200 lanes by site, lane and mode.

Baselines on the table: lead-time parameters = carrier-brochure values, variability unmeasured · buffer capital sized on catalog values · no lead-time-accuracy measurement · target 99% lead-time accuracy on the 200-lane set before any release.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — reframed the report request as a parameter problem; scoped 200 lanes by site/lane/mode; wrote the success criteria — measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer; showed one right-skewed lane where the average hid the tail. *Exit bar:* planning and logistics agree the problem is the parameter, not the report.
- **Plan & Design** — requirement to the data team (mean, spread, tail per lane; production cycle + inbound transit vs actual); the p99 promise policy (cumulative, not μ + 2.33σ); the release rule as a signed planning act with the fat-tail exception inside it; the stockout-and-expedite watch; drift flag and refresh cadence. On p95 vs p99, showed the asymmetry per lane and left the choice to planning. *Exit bar:* policy signed by planning; Finance agrees the release rule in the room; data team accepts the spec.
- **Build & Test** — data team built the lane distributions and tracked lead-time accuracy per lane; manual reconciliation stayed alive. *Exit bar:* lead-time accuracy 99% on the 200-lane set — the bar I would not endorse a cut below.
- **Cutover & Go-Live** — planners executed the buffer release lane by lane under the signed rule; fat-tail lanes excluded until their source process moved; watch armed first. *Exit bar:* 580K units / ~$1.5M released; no stockout in the watch window.
- **Hypercare / Close** — read the watch; set drift flags per lane; handed the refresh cadence and the release rule to planning as an act they could re-apply. *Exit bar:* watch clean; the promise stays measured.

Influence and ownership. I owned the requirement, the promise policy and the release rule; the data team built the distributions and tracked accuracy; planners sized the buffers and executed; Finance signed each lane. When Finance pushed for a broader release, the rule they had signed did the refusing — fat-tail lanes keep buffer until the source process moves. The percentile fight never happened: I showed each lane's line-down cost against carrying cost and planning chose p99 — their choice surviving an audit is worth more than my choice winning a meeting.

Designed the process; others own it now. I designed the promise policy and the release rule as a signed planning act; planning now re-applies it on its own refresh cadence with a drift flag per lane. **Trade-off put to leadership.** The recommendation to Finance and Planning outlined the competing metrics — ~$1.5M of working capital out vs stockout

> exposure at the tail — and set the bar: p99, 99% lead-time accuracy, fat-tail lanes excluded until their source process moved.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); ≈$0.6M/yr *modeled* recurring around it (carry ≈$0.3M at a benchmark 20% all-in rate + expedites avoided ≈$0.3M; range $0.3–1.4M) — a different program from S6's $2M; never added.
- **#** — 200 lanes measured by site, lane and mode; 580K units released (record); ≈$2.59/unit derived — low-value components.
- **%** — lead-time accuracy 99% on the 200-lane set at the p99 promise — the bar below which I would not endorse a cut (record).
- **Time** — release only after the 99% bar; watch window clean; refresh cadence and drift flag per lane handed to planning (record).
- **Customer** — service held — zero stockouts on released lanes through the watch window (record).
- **Quality** — release only under the signed rule; fat-tail lanes kept buffer until the source process moved; the promise stays measured, never a new brochure (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A report request with no agreed parameter behind it. I reframed it as a parameter problem and wrote the success criteria myself: measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer.
Scope & influence	200 lanes by site, lane and mode; the data team built the distributions; planners sized and executed; Finance signed each lane — a planning act I designed and others executed.
Execution trade-off + mechanism others now own	Working capital out vs stockout risk: promise at p99 (cumulative), a 99% lead-time-accuracy bar below which I would not endorse a cut, fat-tail lanes excluded until their source process moved. Mechanism others own now: the release rule as a signed act, the refresh cadence and a drift flag per lane handed to planning.
Impact (long-term)	580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); the promise stays measured, never a new brochure.
Process improvement (standard work institutionalized)	Lead-time accuracy tracked per lane; drift flags; the release rule written so planning can re-apply it without me.
Complexity & consensus	Finance pushed for a broader release — the rule they had signed did the refusing; the fat-tail exception was inside the rule, so it fired as policy, not as a judgment call.
Root cause at scale	Brochure lead times were the parameter error behind every excess buffer — fixing the parameter fixed all 200 lanes at once rather than one planner's buffer at a time.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: shipment durations per lane (ship-to-receipt) for 200 lanes by site, lane and mode; the brochure value beside each. Revealed: lanes with measured σ far below the brochure (release) and lanes above it (keep).
2 · Solution design	Options: brochure · average · p95 · p99 per lane · network-wide percentile. Chosen: promise at the lane's own p99, cut only behind a 99% accuracy gate, fat tails kept.
3 · Math & data logic	Math: SS = z·d·σ_L = 2.33 × 500 × 5 = 5,825 vs 2.33 × 500 × 1.2 = 1,398; release 4,427; p99 = min{d : P(L ≤ d) ≥ 0.99}; $1.5M ÷ 580K = $2.59; upper-bound σ at n = 50 → 1,675.
4 · Implementation	Implementation: reframe the request → scope 200 lanes → data team builds distributions → planning chooses p99 on the asymmetry → accuracy gate → lane-by-lane release with Finance signature → watch + drift flag.

5 · Probe questions	Probe questions: p99 vs p95 · skew and the normal shortcut · what “99% accuracy” means · whether the release is selection-biased · whether the $1.5M is measured — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

Analytics — Why p99 and not p95? That is a lot of buffer.

Per lane, the cost of a line-down on a fat-tailed lane dwarfed the carrying cost of the extra day. I showed the asymmetry lane by lane and let planning choose — they chose p99. On reliable lanes the measured σ was so far below the brochure that p99 still released buffer; on fat-tailed lanes the buffer stayed.

Analytics — How did you compute p99 on skewed lanes?

Cumulatively from the empirical distribution — the smallest d with P(L ≤ d) ≥ 0.99 — not μ + 2.33σ. On one right-skewed lane the normal shortcut understated the true p99 by about 3.4 days; that lane is why the definition is cumulative.

Analytics — Is the $1.5M measured?

The release is record: 580K units out, zero stockouts in the watch window, Finance signed each lane. The ≈$2.59/unit is the derived average. What is modeled is the recurring value around it — carry at 20% all-in (capital-only would be ≈$150K at 10%) and avoided expedites — each input labeled.

Analytics — The expedite number assumes what?

Ten thousand shipments a year on the 200 lanes and one late arrival in ten forcing an expedite before p99 — illustrative. I would pull the expedite ledger by lane for the year before and after and compare directly.

Analytics — What is lead-time accuracy, exactly?

1 − |actual − promised| / promised, tracked per lane against the p99 promise; the bar was 99% on the whole 200-lane set. Below it, no release — written into the act before anyone asked.

Analytics — At what grain did you measure lead time, and over what window?

At the lane grain, from actual ship-to-receipt durations — one empirical distribution per lane, never a network average. The record carries the per-lane distributions, the refresh cadence and the drift flag (p99 recomputed on refresh; the buffer decision re-opens if accuracy falls); it does not carry a quoted window length, and I will not invent one — what mattered design-wise was a window long enough to hold seasonality and recent enough that a dead lane's history could not vote, with the drift flag as the safety net either way.

Analytics — Why p99? What cost ratio does that imply?

A p99 promise is the critical ratio 0.99 = Cu ÷ (Cu + Co), i.e. a late day costs 99 times an extra day of cover. On the example lane an extra day of cover is 500 × $2.59 × 20% ÷ 365 = $0.71, so p99 is justified if a day of shortage costs more than about $70; p95 if more than $13. A line-down clears both bars by orders of magnitude — which is the asymmetry I showed planning lane by lane.

Analytics — Lanes released are lanes with a small measured σ. Isn’t that selection bias?

Yes — the winner’s curse. A lane picked because its measured σ looks small is partly a lucky draw. The standard error on σ is about σ ÷ √(2(n−1)): 10.1% at n = 50, 3.5% at n = 400. On the reliable lane (σ 1.2, brochure 5), sizing on the 95% upper bound at n = 50 gives σ = 1.44 and a buffer of 1,675 instead of 1,398 — 277 units more, a release of 4,150 instead of 4,427. That is the arithmetic; VERIFY-WITH-RAJESH what the program did on thin lanes.

Analytics — Define “99% lead-time accuracy” so I can test it.

The master’s definition is LTacc = 1 − |actual − promised| ÷ promised with a 99% bar on the 200-lane set. Read per shipment against a p99 promise it cannot be 99%: on the example lane the mean shipment is 21 days against a 34-day promise, |21 − 34| ÷ 34 = 38.2%, a score of 61.8%. The readings consistent with 99% are coverage — 99 of 100 shipments arrive within the promise — or the re-measured lane p99 within 1% of the promised p99, a tolerance of 0.34 day on a 34-day lane. On the ratio form 1% is 0.05 day on a 5-day lane and 0.3 day on a 30-day lane. I would say which reading before quoting a tolerance. VERIFY-WITH-RAJESH

Shadow / closing — How does the promise stay honest after you leave?

Drift flag per lane with a refresh cadence: p99 is recomputed on refresh, and the buffer decision re-opens if accuracy falls. The stockout-and-expedite watch became a standing report owned by planning.

THE KEY MATH — ONE RELIABLE LANE, ONE FAT-TAIL LANE, AND THE RECURRING VALUE (ILLUSTRATIVE UNITS; RECORD PERCENTAGES AND THE RECORD RELEASE PAIR)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Brochure-sized buffer	`SS = z·d·σ_L = 2.33 × 500 × 5 = 5,825 units`	illustrative; toolkit form	5,825 units of folklore
2	p99 defined cumulatively	`p99 = min{d : P(L ≤ d) ≥ 0.99}`	record definition	normal shortcut understates a skewed lane by 3.4 days
3	Reliable lane re-sized	`2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427`	measured σ = 1.2 vs brochure 5 (illustrative)	4,427 units off this lane
4	Fat-tail lane	`2.33 × 500 × 6.5 = 7,573 — buffer KEPT`	measured σ = 6.5 > brochure 5 (illustrative)	kept until the source process moves
5	Accuracy gate	`LTacc = 1 − |actual − promised| / promised; bar 99% on 200 lanes`	record	99% held before any cut
6	The release	`$1,500,000 / 580,000 = $2.59/unit`	record pair; average derived	580K units / ~$1.5M, 0 stockouts
7	Recurring value	`$1.5M × 20% = $0.3M; 900 fewer late × 10% × $3K ≈ $0.27M`	modeled (benchmarks labeled)	≈$0.6M/yr; never added to S6's $2M

Depends on the six-site canonical model (S7) — six definitions cannot produce one σ. S7 is a pointer only in this packet; say "six sites," never "six NA sites."

Where this lands at Amazon. Inbound lead-time variability is half of every buffer and every promise in a placement model. Transferable: measure the lane's own distribution, promise at the tail you can defend, release capital only on evidence with the exception written into the rule, and give every parameter a drift flag. Reliability (zero stockouts), cost efficiency (expedites avoided), asset utilization (~$1.5M of working capital out).

FULL DERIVATION — FOUR-TABLE MATH, SOLUTION FLOW, SENSITIVITY, SPOKEN STATEMENT, IF-CHALLENGED NEW

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Service held at lower inventory: 580K units / ~$1.5M released (record)
+- Safety stock (stock) = z * d * sigma_L per lane
|    +- sigma_L: brochure folklore -> measured per lane (200 lanes)
|    +- release rule: 99% LT-accuracy bar + fat-tail lanes keep buffer; Finance
|         signed each lane (S6's two-signal rule is that program's own control)
|    +- fat-tail lanes: sigma_L worse than brochure -> buffer KEPT
+- Promise integrity (flow): p99 per lane, LT accuracy 99% (record)
|    +- cost asymmetry: line-down >> one day of holding -> p99 not p95
+- Drift loop: refresh cadence -> p99 recomputed -> drift flag ->
     buffer decision re-opened -> promise stays measured, never a new brochure
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Lead-time accuracy, 200 lanes	brochure	99% at p99	—	record
Safety stock released (one-time)	brochure-sized	measured-sized	—	record: 580K units / ~$1.5M
Stockouts on released lanes	—	0 in watch	—	record

DRIVER	BASELINE	AFTER	$/YR	BASIS
Carry on released $1.5M	held	free	$0.3M	modeled ($1.5M × 20% all-in carry; benchmark 15–25%)
Expedites on late arrivals	~1,000 late × 10%	~100 × 10%	$0.3M	modeled (90 × $3K)
Total modeled recurring	—	—	**≈$0.6M**	around the ~$1.5M record; never added to S6's $2M

If challenged. *"Is the $1.5M measured?"* — "The release is record: 580K units out, no stockout in the watch window, Finance signed each lane. The ~$2.59/unit is the derived average on low-value components. What's modeled is the recurring value around it — carry and expedites — and each input is labeled." *"Why did you own the percentile choice?"* — "I didn't — I owned the frame. I showed each lane's asymmetry, line-down cost against carrying cost, and planning chose p99. Their choice surviving an audit is worth more than my choice winning a meeting."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lead-time parameters	`carrier-brochure values; variability unmeasured on 200 lanes`	record	folklore, not data
2	One lane sized on folklore	`SS = z × d × σ_L = 2.33 × 500 × 5 = 5,825 units`	d = 500 units/day; brochure σ = 5 days; z = 2.33 at 99% (illustrative, toolkit form)	5,825 units of buffer
3	Tail blindness	`normal shortcut μ + 2.33σ understates a right-skewed lane's true p99 by 3.4 days`	illustrative from the record	the promise misses the tail
4	Late arrivals	`200 lanes × ~50 shipments = 10,000/yr; × 10% = 1,000 late/yr`	volumes illustrative	~1,000 late/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure per lane	`μ, σ, p50, p90, p99 from shipment history, 200 lanes`	production cycle + inbound transit vs actual (record)	measured distributions
2	p99 defined cumulatively	`p99 = min{d : P(L ≤ d) ≥ 0.99}`	not μ + 2.33σ unless the lane is normal (record definition)	the tail promise
3	Accuracy gate	`LTacc = 1 − |actual − promised| / promised; bar = 99% on the 200-lane set`	before endorsing any cut (record)	99% held
4	Reliable lane re-sized	`SS = 2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427 units`	measured σ = 1.2 days vs brochure 5 (illustrative)	4,427 units off this lane
5	Fat-tail lane	`SS = 2.33 × 500 × 6.5 = 7,573 units — buffer KEPT`	measured σ = 6.5 > brochure 5 (illustrative)	kept until the source process moves
6	Drift flag	`p99 recomputed on refresh; buffer decision re-opened if accuracy falls`	record	the promise stays measured, never a new brochure

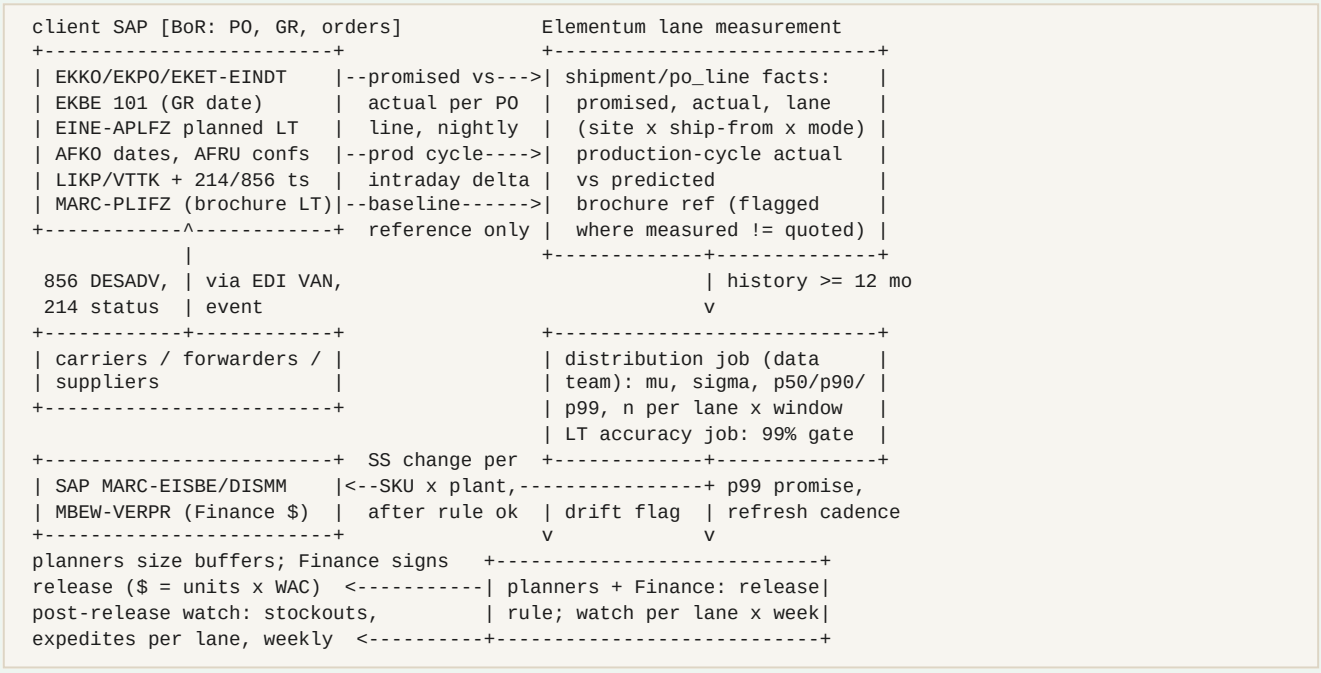
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Lead-time parameters	brochure folklore	measured distributions, 200 lanes	record
2	Lead-time accuracy	unmeasured	99% at p99	record
3	Inventory	brochure-sized	580K units / ~$1.5M released one-time	record pair
4	Stockouts on released lanes	—	0 in the watch window	record
5	Late arrivals	~10%	~1% at the 99% bar	record; volumes illustrative

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Unit value	`$1,500,000 / 580,000 = $2.59/unit`	derived from the record pair; low-value components	≈$2.59/unit
2	Carry on the release	`$1,500,000 × 20% = $300,000`	all-in carry (benchmark 15–25%)	≈$0.3M/yr
3	Late reduced	`10,000 × 10% = 1,000 before; 10,000 × 1% = 100 after; 1,000 − 100 = 900 fewer late`	volumes illustrative; the 99% is record	900 fewer late/yr
4	Expedites avoided	`900 × 10% = 90 expedited; 90 × $3,000 = $270,000`	expedited share and premium modeled	≈$0.3M/yr
5	Total recurring	`0.3 + 0.3 = $0.6M`	modeled; never added to S6's $2M	≈$0.6M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Carrying rate (benchmark)	15% → $0.23M	20% → $0.30M	25% → $0.38M
Late share before p99 (illustrative)	5% → 400 fewer	10% → 900 fewer	15% → 1,400 fewer

INPUT	LOW	BASE	HIGH
Share of late arrivals expedited (illustrative)	5%	10%	15%
Expedite premium (benchmark)	$2K → $0.04M	$3K → $0.27M	$5K → $1.05M
Total modeled $/yr	≈$0.3M	≈$0.6M	≈$1.4M

RELEASE-DECISION DESIGN — POINT ESTIMATE VS UPPER BOUND; WHAT "99% ACCURACY" CAN MEAN (EXTRA TABLE)

The record is the rule — measured per-lane distribution, p99 promise, 99% accuracy gate, fat tails kept, Finance signs. The statistics below are my design arithmetic on illustrative lanes. VERIFY-WITH-RAJESH what the program did on thin lanes and which accuracy definition it used.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Brochure buffer	$z \cdot d \cdot \sigma_L = 2.33 \times 500 \times 5 = 5{,}825$	brochure σ 5 (illustrative); z = 2.33 (toolkit)	5,825 units
2	Reliable lane, point estimate	$2.33 \times 500 \times 1.2 = 1{,}398$; release $5{,}825 - 1{,}398 = 4{,}427$	measured σ 1.2 (illustrative)	release 4,427
3	Standard error of $\hat{\sigma}$	$\sigma \div \sqrt{2(n-1)}$: $n = 50 \rightarrow 1.2 \times 0.1010 = 0.121$; $n = 400 \rightarrow 1.2 \times 0.0354 = 0.042$	large-sample approximation	10.1% (n = 50) · 3.5% (n = 400)
4	95% upper bound on σ	$n = 50$: $1.2 \times (1 + 1.96 \times 0.1010) = 1.44$; $n = 400$: $1.2 \times (1 + 1.96 \times 0.0354) = 1.28$	one lane, illustrative n	1.44 · 1.28
5	Buffer on the upper bound	$n = 50$: $2.33 \times 500 \times 1.438 = 1{,}675$; $n = 400$: 1,495	design arithmetic — not the record	1,675 · 1,495 units
6	Winner's-curse cost	$n = 50$: $1{,}675 - 1{,}398 = 277$ units × $2.59 = $717	$2.59 = $1.5M ÷ 580K (record pair)	277 units ≈ $717 held back
7	Implied cost of a late day (p99)	carry/day = 500 × $2.59 × 20% ÷ 365 = $0.71; Cu = 99 × $0.71 = $70	critical ratio 0.99 ÷ 0.01	p99 justified above $70 per shortage-day
8	Accuracy as a ratio, per shipment	$\lvert 21 - 34 \rvert \div 34 = 38.2\% \rightarrow$ score 61.8%	example lane (secondary source)	cannot be 99% — read as coverage or p99-match

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured result is five hundred eighty thousand units, about a million and a half dollars of safety stock, released at ninety-nine percent lead-time accuracy with zero stockouts on the watch; that is record and Finance signed it. What recurs around it I model at roughly six hundred thousand a year: about three hundred thousand in carrying cost on capital we no longer hold, and about three hundred thousand in expedites we no longer pay because the promise sits at the ninety-ninth percentile instead of a brochure number. The release and the accuracy are measured; the two recurring pieces are my model on benchmark carrying and expedite rates, and I never add this to the two million from the forecast program."

> **If challenged — the full answers.**
>
> **The expedite number is the softest**
>
> It assumes ten thousand shipments a year on the two hundred lanes and that one late arrival in ten forced an expedite before p99; to firm it I would pull the expedite ledger by lane for the year before and after the release and compare directly.
>
> **Twenty percent carrying cost is high**
>
> Fifteen to twenty-five is the textbook range and I use the middle; capital alone is eight to twelve, so if Finance wants capital only the carry is about a hundred and fifty thousand and I would say that instead.
>
> **Why is the million and a half not an annual saving**
>
> Because it is a one-time working-capital release, booked once as a signed planning act; what recurs is the carry and the avoided expedites, and I would rather be right at six hundred thousand a year than wrong at a million and a half.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S8

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you went deeper than the request.	Dive Deep	A "variability report" reframed as a promise-and-buffer policy; mean, spread and tail per lane from actuals; p99 defined cumulatively.	Do not claim to have built the distributions — the data team did, to my requirement.
Tell me about a time the data contradicted conventional wisdom.	Are Right, A Lot	Brochure values vs measured lanes; one right-skewed lane where the average hid the tail; planning chose p99 on the asymmetry.	S10 (Austria — "the noise said carriers") is the bar raiser's; do not use it here.
Tell me about a time you held a standard under pressure.	Insist on the Highest Standards	No release below 99% lead-time accuracy; Finance's push for a broader release answered with the rule they signed.	Do not frame Finance as the obstacle; the exception was in their rule.
How do you set safety stock / service levels?	Functional — placement policy	Two error signals (forecast, lead time), measured at the lane × SKU grain; policy by cost asymmetry; release by rule with a watch window; drift flags.	Never present the formula as a program artifact — MIT toolkit illustration.
What did you refuse to do?	Have Backbone	Averaging away a tail; endorsing a cut below 99%; a blanket release — all written into the act before anyone asked.	Keep the $2M (S6) out of this answer.
How did you make Finance comfortable releasing capital?	Earn Trust	Finance agreed the release rule in the same room as planning signed the policy — before any cut; signed each lane; the watch armed first.	No annualized $1.5M — it is a one-time release.

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you went deeper than the request.** DIVE DEEP

Open (first two sentences): The ask was a “lead-time variability report.” I turned it into a parameter question: what number is the plan actually using, and how wrong is it?

STAR (≈90 s): Situation: a six-site manufacturer’s logistics manager wanted a report; inventory was sized on carrier-brochure lead times with variability unmeasured. Task: decide whether a report was the deliverable. Action: I reframed it as a parameter problem, scoped 200 lanes by site, lane and mode, wrote the success criteria — measured lanes replace brochures, and 99% lead-time accuracy before any cut — and had the data team build per-lane distributions. Result: planning chose a p99 promise and 580K units, about $1.5M, came out with zero stockouts.

Land this number: 200 lanes; 99% lead-time accuracy; 580K units / ~$1.5M, 0 stockouts. · **Pivot to NACF:** NACF inbound lead-time variability by vendor lane is the same “report” that is really a parameter. · **Do NOT say:** Do not say the data team’s distributions were mine; I owned the requirement, the policy and the release rule.

Q2 **Tell me about a time you put the customer’s cost ahead of your own team’s metric.** CUSTOMER OBSESSION

Open (first two sentences): The customer was the production line. A released unit of buffer that causes a line-down is worse than a unit never released.

STAR (≈90 s): Situation: carrying cost pushed to cut buffer; the line-down cost sat in another budget. Task: make the asymmetry visible. Action: I showed it lane by lane — an extra day of cover on the example lane costs about $0.71; a day of shortage costs far more — and let planning choose the quantile; they chose p99. On reliable lanes p99 still released buffer; on fat-tail lanes the buffer stayed. Result: zero stockouts across the released lanes.

Land this number: Zero stockouts (record); p99 on 200 lanes. · **Pivot to NACF:** Placement trades FC buffer cost against a missed promise the same way. · **Do NOT say:** Do not say planning was forced — say they chose with the asymmetry in front of them.

Q3 **Tell me about a time you got a finance partner to sign a risky decision.** EARN TRUST

Open (first two sentences): Finance signed each lane because the exception was written into the rule before anyone asked.

STAR (≈90 s): Situation: releasing safety stock is irreversible in practice. Task: a rule Finance could audit. Action: below 99% lead-time accuracy, no release; fat tails keep their buffer; Finance signs per lane; a stockout-and-expedite watch from day one. Result: $1.5M of working capital released, Finance-signed, no stockout in the watch window.

Land this number: 580K units / ~$1.5M (record, one-time). · **Pivot to NACF:** NACF WOC changes need a pre-signed exception clause too. · **Do NOT say:** Do not call the $0.6M recurring a result — it is modeled.

Q4 **Tell me about a process you designed that someone else now owns.** OWNERSHIP

Open (first two sentences): I designed the promise policy and the release rule as a signed planning act; planning re-applies it on its own refresh.

STAR (≈90 s): Situation: a one-off analysis decays into a new brochure. Task: leave a process, not a project. Action: drift flag per lane — p99 recomputed on refresh, the buffer decision re-opens if accuracy falls — and the watch became a standing report owned by planning. Result: the promise stays honest after I leave.

Land this number: Drift flag per lane; standing report owned by planning. · **Pivot to NACF:** Every NACF parameter I touch gets a drift flag and an owner. · **Do NOT say:** Do not say I run it today.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "Planning to the slow day, not the average."

Headline metric: p99 per lane; 99% lead-time accuracy on 200 lanes

- Cumulative p99, not mean + 2.33σ — the normal shortcut understated one skewed lane by 3.4 days.
- Winner's curse on thin lanes: size on the upper bound.
- Accuracy definition: coverage or p99-match.
- Reliable lane release 4,427; fat tail kept.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Customer Obsession; Earn Trust

Open: "Finance signed each lane because fat tails kept their buffer."

Headline metric: 580K units / ~$1.5M, zero stockouts

- Asymmetry shown lane by lane.
- Planning chose p99.
- Exception in the rule before the ask.
- Release is one-time working capital.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "I would put the interval on the page next to every p99."

Headline metric: p99 interval open-ended above at n < 330

- The record carries no window length — I do not invent one.
- S8a and S8c are the interval and validation work.
- Thin lanes: cohort pooling or a longer window.
- VERIFY-WITH-RAJESH the program's thin-lane rule.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Define 99% lead-time accuracy — what tolerance does it allow?"

The numbers: Ratio form: 1% × 5 d = 0.05 d; 1% × 30 d = 0.3 d. Per shipment vs a p99 promise: |21 − 34| ÷ 34 = 38.2% → 61.8%. Coverage form: 1 shipment in 100 beyond the promise.

Say exactly: "Written as one minus the absolute miss over the promise, one percent is a twentieth of a day on a five-day lane and a third of a day on a thirty-day lane — but against a p99 promise the average shipment misses by about thirty-eight percent, so I read ninety-nine as coverage: ninety-nine of a hundred shipments inside the promise. I would confirm which definition the record uses before I quote a tolerance."

T2 "Why p99 and not p95 — what is that buffer worth?"

The numbers: Carry per day of cover = 500 × $2.59 × 20% ÷ 365 = $0.71; p99 ⇔ a shortage-day worth > 99 × = $70; p95 ⇔ > $13.

Say exactly: "A p99 promise says a shortage day costs ninety-nine carrying days — about seventy dollars on that lane — and a line-down clears that by orders of magnitude; I showed planning the asymmetry lane by lane and they chose p99."

T3 "You released the lanes with the smallest measured σ. Isn't that a selection effect?"

The numbers: σ̂ standard error = σ ÷ √(2(n−1)) = 10.1% at n = 50; upper-bound σ 1.44; buffer 1,675 vs 1,398; 277 units held back.

Say exactly: "Yes, partly — a lane picked for a small measured spread is partly a lucky draw; on a fifty-shipment lane sizing on the upper bound holds back about two hundred seventy-seven units of the four thousand four hundred I would otherwise release. I would show that adjustment next to every release."

S8a — Elementum — actual-vs-planned lead-time distribution analysis: the brochure was wrong three ways — 3 days short on the mean, twice-too-wide on the spread, 16 days short on the tail NEW

LPs this story serves here: Dive Deep (primary — plan vs actual at the shipment grain) · Are Right, A Lot (rank, don't fit) · Insist on the Highest Standards (a "99%" plan that held 96.5% is not a 99% plan) · Earn Trust (the carrier's own scans, not my opinion)
Process family: Plan / Source seam — inbound lead-time parameter diagnosis before any buffer or policy change

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I compared plan to actual at the shipment grain on three axes — location, spread, tail — before changing a single parameter; 240 of 400 shipments ran slower than the brochure.
- **Are Right, A Lot** — I ranked the shipments instead of fitting a bell curve; the curve's "99%" plan held 96.5% — 14 of 400 shipments ran past it — and the ranked p99 held 99%.
- **Insist on the Highest Standards** — I refused to publish a p99 that carried its own error: a plan statistic had to be distribution-free and come with its sampling uncertainty.
- **Earn Trust** — I showed planners and commercial the carrier's own scans on one lane — brochure 18, mean 21, p99 34 — so the argument was the data, not my say-so.

Honesty label. I owned the question, the comparison design and the plan statistic; the data team built the per-lane distributions to my requirement. The 18 / 21 / σ 4 / p99 34 / n≈400 lane is the loop packet's example lane (secondary source — the master packet carries "18d brochure" in its crosswalk and "lane-cohort counts are not in the record"); the ERP default σ of 8 days and the 7-day manual pad are that packet's illustrative inputs. Say "illustrative inputs, real outcomes": the record is 200 lanes, 99% lead-time accuracy at the p99 promise, 580K units / ~$1.5M released. VERIFY-WITH-RAJESH that the example-lane numbers are the ones you are comfortable quoting.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The six-site manufacturer's safety stock kept growing and nobody could say why. Planning ran on carrier-brochure lead times — on my example lane, 18 days — plus a manual week of pad that every planner added. Task. Before touching a parameter I owned one question: how, exactly, was the planned lead time wrong? Action. I pulled about a year of shipment history on the lane — roughly 400 shipments — and compared plan to actual three ways. Location: the actual mean was 21 days, three days longer than the brochure, fifteen standard errors — not noise — and 240 of 400 shipments, 60%, ran slower than the number in the system. Spread: the ERP's default assumed an 8-day standard deviation; measured, it was 4 — wrong in the opposite direction, which is why planners had padded by hand. Tail: a bell curve from that mean and spread puts the 99th percentile at 30.3 days, but 14 of the 400 shipments ran longer — a "99%" plan that actually held 96.5%. Ranking the shipments puts it at 34. Result. Primary metric: shipments late against the plan on that lane, 60% → 1%, planned lead time 18 → 34 days; at program level, lead-time accuracy 99% on 200 lanes — record. The example lane is the pattern, not the headline.

20-SECOND VERSION NEW

The planned lead time was wrong in three directions at once — the mean was 3 days short, the ERP's spread was twice the real one, and the tail was 16 days out; I measured all three on the carrier's own scans and planned to the ranked p99, not a bell curve.

NACF BRIDGE — ONE SENTENCE

Every NACF inbound plan is a promised date; I would run the same plan-vs-actual on appointment and ASN dates versus receive scans, by vendor lane, before anyone resizes a buffer — location, spread, tail.

SITUATION / TASK — THE PROBLEM SPACE

S — Elementum, the same six-site manufacturer. Buffers kept growing, service was still shaky and nobody could explain why. Inventory planning ran on carrier-brochure lead times (18 days on the example lane) and every planner added an informal week of pad. The risk of doing nothing: working capital grows every quarter and every stockout triggers more padding — a wrong number plus panic.

T — Own the diagnosis: characterize how the planned lead time was wrong — bias, spread, tail — with the uncertainty on each statement, so the later policy (S8) and the formula redesign (S8b) stand on measured facts. Scope: 200 lanes by site, lane and mode; the example lane is the busy one.

Baselines on the table: planned lead time = brochure (18 d on the example lane) · actual mean 21 d · 60% of shipments slower than plan · ERP default σ_LT = 8 d (never measured) · manual pad 7 d × 500 units/day = 3,500 units on the lane · no lead-time-accuracy measurement existed.

ACTION — FIVE GATED PHASES

- **Discovery** — agreed with planning and logistics that the question was “how is the parameter wrong,” not “write a variability report”; chose the example lane because it had the history (≈8 shipments a week for a year). Exit bar: the comparison design — three tests, one lane — accepted before any number was shown.
- **Plan & Design** — requirement to the data team: promised date (PO-line schedule) vs actual receipt per shipment, keyed by lane = site × ship-from × mode, 12+ months so every season is in the sample; outputs mean, σ, p50 / p90 / p99, n and the late share against the plan in force. Exit bar: data team accepts the spec; planning agrees which plan statistic is on trial (mean, brochure + pad, normal p99, ranked p99).
- **Build & Test** — data team built the lane file; I owned the comparison: bias with its standard error, late share with a binomial interval, σ against the ERP default, then the four candidate plans scored on the same 400 shipments. Exit bar: the candidate table reproduces on a second pull; the ranked p99 reproduces the “99%” by construction and its sampling interval is printed beside it.
- **Cutover & Go-Live** — the finding went to planners as one lane, three numbers — brochure 18, mean 21, p99 34 — then the same three tests per lane for all 200; the plan statistic moved to the ranked p99 under the release rule (S8) and the formula (S8b). Exit bar: planning signs the plan statistic; the old brochure stays in the master as a labeled reference only.
- **Hypercare / Close** — drift flag per lane (p99 recomputed on refresh; review if the tail or the spread moves) and a standing plan-vs-actual page owned by planning. Exit bar: the page refreshes without me; thin lanes carry their interval on the page so nobody reads a max as a quantile.

Influence and ownership. I owned the question, the comparison and the plan statistic; the data team built the distributions; planners owned the parameters and the buffers; Finance and planning signed the policy that used it. The conflict was commercial: quoting 34 days where the brochure said 18 — answered with the number that decided it: a fast promise missed 60% of the time costs expedites and trust; the honest date costs only optics, and where speed truly mattered we fixed the lane, not the number.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **%** — shipments late against the plan on the example lane 60% (brochure) → 1% (ranked p99); program level: lead-time accuracy 99% on 200 lanes at the p99 promise (record).
- **Time** — planned lead time on the example lane 18 → 34 days: the plan now says what 99 of 100 shipments do.
- **#** — 200 lanes measured by site, lane and mode; 580K units / ~$1.5M released later under the signed rule (record, one-time — S8).
- **$** — on the example lane only: ≈$9.0K a year modeled (carry on the 3,500-unit pad + expedites on late arrivals) — an illustrative lane, never scaled by 200 and never added to S8’s ≈$0.6M or S6’s $2M.
- **Quality** — every plan statistic reported with n and an interval; the max of a thin lane is never presented as its p99.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No agreed definition of “how wrong is the planned lead time”; I defined it as three tests — location, spread, tail — and the unit of analysis (the shipment on a lane).
Scope & influence	Planners, logistics, commercial, Finance and a data team that did not report to me; the commercial pushback on 34 vs 18 resolved by the 60%-late number.
Execution	Chose a distribution-free plan statistic over a convenient formula and carried its interval; sequenced diagnosis (this story) before policy (S8) before formula (S8b).
Impact	99% lead-time accuracy on 200 lanes (record); the diagnosis is what made 580K units / ~$1.5M releasable without a stockout.
Process improvement	A standing plan-vs-actual page and drift flag per lane — the parameter can no longer decay into a new brochure.
Complexity	Skewed tails, thin lanes, seasonality inside a 12-month window, and a normal-fit shortcut that looks right and holds 96.5%.
Root cause	The root cause was the planning parameter, not the carrier: 60% of arrivals were “late” against a number the carrier never measured.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: shipment × lane (site × ship-from × mode), promised date (PO-line schedule) vs actual receipt, 12+ months; joins PO line → goods receipt → lane key; cuts by lane, mode, site, quarter. Revealed: bias +3 d (t = 15), 60% of shipments slower than plan, ERP σ 8 vs measured 4, normal-fit p99 holds 96.5%.
2 · Solution design	Options: plan to the mean · brochure + planner pad · normal-fit p99 · ranked (empirical) p99. Chosen: ranked p99 — distribution-free, defensible to the tail the business cares about; why: it is the only candidate whose coverage does not depend on an assumed shape.
3 · Math & data logic	Math: bias = 21 − 18 = +3.0 d; SE = 4/√400 = 0.20; t = 15.0 · late share 240/400 = 60% (Wilson 95% [55.1%, 64.7%]) · normal p99 = 21 + 2.33 × 4 = 30.32, 14/400 beyond → 96.5% · ranked p99 = t(⌈0.99 × 400⌉) = t(396) = 34 d · order-statistic 95% interval t(392)…t(400).
4 · Implementation	Implementation: spec → data team → lane file → comparison → planning sign-off → plan statistic moves to the ranked p99 under the S8 release rule; drift flag + refresh cadence; thin lanes print their interval. Risk: thin lanes — handled by cohort pooling or a longer window (VERIFY-WITH-RAJESH what was actually done).
5 · Probe questions	Probe questions: n and representativeness · +3 d noise vs signal · interval on a p99 from 4 observations · seasonality · thin lanes · why rank not fit — all answered with numbers below and in the Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

> **Analytics — n = 400 on one lane. Is that representative of 200 lanes?**
>
> No, and I say so. It is the busiest lane, about eight shipments a week. The master’s illustrative volume is about 10,000 shipments a year across 200 lanes — roughly 50 on an average lane — so the example shows the mechanism and the same three tests ran per lane. At n = 50 the order-statistic interval for the p99 is open-ended above: t(48) up to “unbounded” — the maximum is only a lower bound, which is why thin lanes need a longer window or cohort pooling. VERIFY-WITH-RAJESH what the program did for thin lanes.

> **Analytics — Is a three-day difference statistically meaningful or just practically meaningful?**
>
> Both. Standard error of the mean is 4 ÷ √400 = 0.20 day, so the 95% interval on the actual mean is 21 ± 0.39 — the brochure sits fifteen standard errors away. Practically, 3 days × 500 units a day is 1,500 units of exposure per cycle that the plan did not know about.

Analytics — Your p99 is the 396th of 400. What is the confidence interval?

Wide, and I print it: with n = 400 the 95% interval for the true p99 runs from the 392nd-ranked shipment (between 31 and 34 days on this lane) to the slowest one (35 or more). In-sample coverage of the ranked plan is 99% by construction; the honest sampling interval on that coverage is 97.5%–99.6% — which is why I validated on a held-out quarter (S8c) and put a drift flag on every lane.

Analytics — Why not just mean + 2.33σ?

Because lead-time tails are right-skewed and the shortcut understates them. On this lane it gives 30.32 days; 14 of 400 shipments, 3.5%, ran longer, so a “99%” plan delivered 96.5%. On the whiteboard sample — 20 shipments, mean 12.5, σ 2.64 — the normal p99 is 12.5 + 2.33 × 2.64 = 18.65 against a ranked 22: about 3.4 days short, the canon figure.

Analytics — What about seasonality in the window?

The window was twelve months or more, so every quarter is inside the sample and the p99 prices the slow season rather than ignoring it. The test I would add is the quarterly cut: the +3-day bias should carry the same sign in all four quarters before I call it a parameter error rather than one bad season. The record does not carry that cut. VERIFY-WITH-RAJESH

Shadow / closing — What would you do differently?

Lead with the thin lane, not the busy one — show the interval first. A p99 from fifty shipments is a maximum with a label on it, and the plan page should say so.

BUSINESS IMPACT MODEL — WHAT THE WRONG PLANNED LEAD TIME COST (EXAMPLE LANE, MODELED)

```
Planned lead time wrong (brochure 18 d)
+- Location error: mean 21 vs plan 18  -> 60% of shipments late vs plan (240 / 400)
+- Spread error: ERP sigma 8 vs measured 4 -> wrong the OTHER way -> planners pad by hand
+- Tail error: normal p99 30.3 holds 96.5%; ranked p99 = 34 holds 99%
+- Consequence A (stock): 7-day pad x 500 units = 3,500 units held to cover the brochure
+- Consequence B (flow): late arrivals vs the planning date -> expedites
+- Fix: plan statistic = ranked p99 (distribution-free) + interval + drift flag
```

DRIVER	BASELINE	AFTER	$/YR (LANE)	BASIS
Planned lead time	18 d (brochure)	34 d (ranked p99)	—	loop-packet example lane (secondary)
Shipments late vs plan	28 of 400 vs the 25-d planning date (7%); 240 of 400 vs the 18-d brochure	4 of 400 (1%)	—	loop-packet example lane
Carry on the manual pad	3,500 units × $2.59 = $9,065 held	pad not needed	$1,813	modeled (20% all-in carry; benchmark 15–25%)
Expedites on late arrivals	28 × 10% × $3,000	4 × 10% × $3,000	$7,200	modeled (10% share and $3K premium illustrative)
Total, this lane	—	—	**≈$9.0K**	illustrative lane — never scaled ×200, never added to S6/S8

If challenged. *“Multiply by 200.”* — “No: lanes differ by two orders of magnitude in volume; the program-level recurring value is S8’s ≈$0.6M on its own inputs, and this lane is the mechanism, not the sample.”

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: how the plan was wrong (example lane, n ≈ 400).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Sample	`n ≈ 400 (≈ 8 shipments/wk ≈ one year)`	loop-packet example lane (secondary) — a busy lane; average lane ≈ 50 (master, illustrative)	n = 400
2	Location error	`mean − brochure = 21 − 18 = +3.0 d (3 ÷ 18 = +16.7%)`	brochure 18, measured mean 21 (example lane)	+3.0 d
3	Is it noise?	`SE = σ/√n = 4/√400 = 0.20 d; t = 3.0/0.20 = 15.0; 95% CI of mean = 21 ± 1.96 × 0.20 = [20.61, 21.39]`	measured σ = 4 d	t = 15.0 — not noise
4	Late vs brochure	`240 / 400 = 60.0%; Wilson 95% [55.1%, 64.7%]`	counts of shipments slower than 18 d	60% late
5	Spread error	`ERP σ = 8 vs measured 4 → ratio 2.0, variance ratio 4.0; CI of σ ≈ 4 ± 1.96 × 4/√(2 × 399) = [3.72, 4.28]`	ERP default 8 d (illustrative); large-sample approximation	8 is far outside the interval
6	Tail error — bell curve	`21 + 2.33 × 4 = 30.32 d; 14 / 400 = 3.5% beyond → coverage 96.5%`	z = 2.33 (99%, toolkit label)	a “99%” plan that holds 96.5%
7	Tail error — vs brochure	`ranked p99 − brochure = 34 − 18 = 16 d`	ranked p99 = t(396)	16 d short on the tail
8	The manual patch	`7 d × 500 units/day = 3,500 units`	pad and demand rate illustrative	3,500 units of pad

Table 2 — the fix: four candidate plans scored on the same 400 shipments.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Plan = mean	`21 d; shipments slower: 160 of 400`	counts from the lane file (loop packet)	serves 60.0%
2	Plan = brochure + planner pad	`18 + 7 = 25 d; slower: 28 of 400`	pad illustrative	serves 93.0%
3	Plan = normal-fit p99	`21 + 2.33 × 4 = 30.32 d; slower: 14 of 400`	assumes a bell curve	serves 96.5% — not 99%
4	Plan = ranked p99	`position ⌈0.99 × 400⌉ = 396 → t(396) = 34 d; slower: 4 of 400`	distribution-free (record definition)	serves 99.0% in-sample
5	Interval on the ranked plan’s coverage	`4/400 late → Wilson [0.39%, 2.54%] late → coverage [97.5%, 99.6%]`	binomial; n = 400	in-sample 99% is not a guarantee
6	Interval on the p99 itself	`95% interval t(392)…t(400) (coverage 96.1%); n = 50 → t(48)…open; n = 100 → t(97)…open; n = 1,000 → t(983)…t(996)`	exact binomial order-statistic interval	thin lanes cannot bound a p99 from above
7	Whiteboard sample (n = 20)	`mean 12.5, σ 2.64; ranked p50 = 12, p90 = 14, p99 = 22; normal p99 = 12.5 + 2.33 × 2.64 = 18.65; gap = 22 − 18.65 = 3.35 ≈ 3.4 d`	values 9,10,11×5,12×6,13×3,14×2,15,22 (illustrative)	normal understates by ≈3.4 d (canon)

Table 3 — after: the metric, before → after.

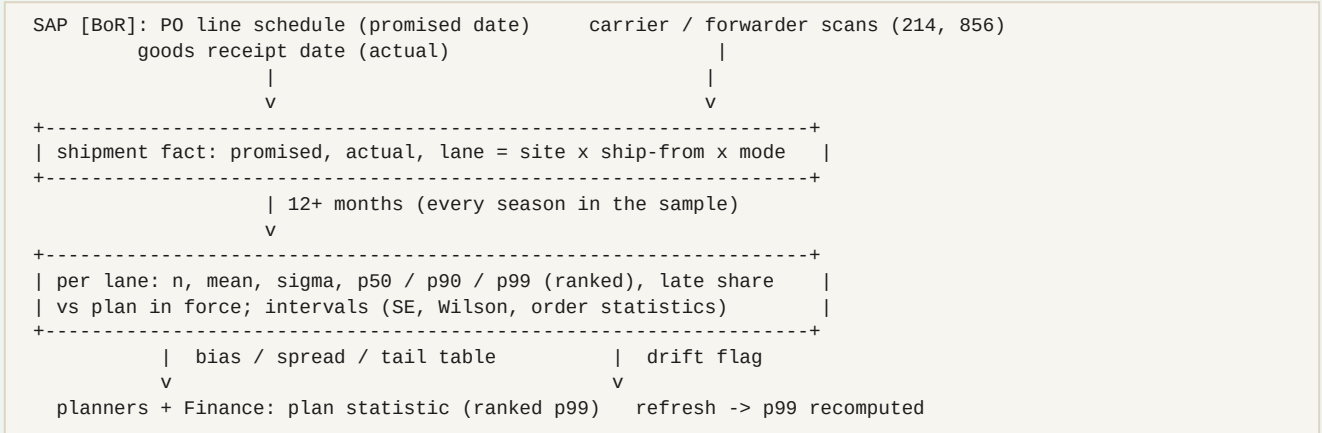
STEP	METRIC	BEFORE	AFTER	BASIS
1	Planned lead time (example lane)	18 d (brochure)	34 d (ranked p99)	example lane (secondary)
2	Shipments late against the plan	60% vs brochure (28 of 400 = 7% vs the 25-d planning date)	1% (4 of 400)	example lane
3	Lead-time cover in the plan (plan − mean)	18 − 21 = −3 d (plan below the mean)	34 − 21 = +13 d	derived; feeds S8b
4	Spread used in planning	8 d (ERP default, unmeasured)	4 d measured, per lane	illustrative default
5	Lead-time accuracy, 200 lanes	unmeasured	99% at the p99 promise	record

Table 4 — the dollars, every multiplication shown (example lane, modeled).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Pad inventory	`7 d × 500 units/day = 3,500 units; 3,500 × $2.59 = $9,065`	$2.59 = $1.5M / 580K (derived record pair)	$9,065 of capital on the pad
2	Carry on the pad	`$9,065 × 20% = $1,813/yr`	benchmark 15–25% all-in	≈$1,813/yr
3	Expedites before	`28 late × 10% × $3,000 = $8,400`	late vs the 25-d planning date; share and premium illustrative	$8,400/yr
4	Expedites after	`4 late × 10% × $3,000 = $1,200`	late vs the 34-d plan	$1,200/yr
5	Expedites avoided	`$8,400 − $1,200 = $7,200`	row 3 − row 4	$7,200/yr
6	Lane total	`$1,813 + $7,200 = $9,013`	modeled, one illustrative lane	≈$9.0K/yr — never × 200

Dollar basis labels: the lane statistics are the loop packet's example lane (secondary); every dollar is modeled on illustrative inputs; the record is the 99% / 200 lanes / 580K units / ~$1.5M quartet in S8.

SOLUTION FLOW — THE PLAN-VS-ACTUAL ANALYSIS PIPELINE



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Shipments on the lane (n) — p99 interval	n = 50 → t(48)…open	n = 400 → t(392)… t(400)	n = 1,000 → t(983)… t(996)
Carry rate on the pad	15% → $1,360	20% → $1,813	25% → $2,266
Expedite share of late arrivals	5% → $3,600 avoided	10% → $7,200	15% → $10,800
Lane total (modeled, one lane)	≈$5.0K	≈$9.0K	≈$13.1K

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

“On one busy lane the brochure said eighteen days; the actual mean was twenty-one, a three-day gap that is fifteen standard errors wide, and sixty percent of shipments ran slower than the number in the system. The system’s default spread was eight days; measured, it was four. A bell curve would have promised a ninety-ninth percentile of thirty point three, and fourteen of four hundred shipments ran past it; ranking the shipments gives thirty-four, which held ninety-nine percent. The record is ninety-nine percent lead-time accuracy on two hundred lanes. The lane is an example and the dollars around it are my model on illustrative inputs — about nine thousand a year on that one lane, and I never multiply it by two hundred.”

If challenged — the full answers

The lane is the example, not the data. “The program-level result is the record — 200 lanes, 99%, 580K units, about $1.5M. The lane shows how I got there; its counts are the loop-packet example and I say so.” The interval is wide. “Yes. With 400 shipments the p99 rests on the top four; I print the interval and validate out of sample in S8c. With 50 shipments I would not call a maximum a p99.” The dollars are small. “One lane, modeled, on purpose — I would rather show the arithmetic than a program-wide number I can’t derive.”

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you dug into the data to find the real problem.** DIVE DEEP

Open (first two sentences): Buffers kept growing and nobody could say why, so I stopped collecting opinions and pulled the shipment scans on our busiest lane. A year of history, about 400 shipments, planned date against actual receipt.

STAR (≈90 s): Situation: the plan said 18 days on that lane and every planner padded a week by hand. Task: find how the number was wrong before anyone changed a parameter. Action: I compared plan to actual three ways — the mean was 21 days, three days late and fifteen standard errors from the brochure; the ERP’s assumed spread was 8 days against a measured 4; and the tail ran to 34 days. I showed it as the carrier’s own data on one lane. Result: planners stopped debating whose buffer was right and agreed the parameter was the problem; that became the p99 promise and the 580K-unit release.

Land this number: 60% of shipments slower than the brochure (240 of 400); mean 21 vs 18 — t = 15. · **Pivot to NACF:** NACF inbound runs on promised dates — ASN ETAs and appointment slots; the same plan-vs-actual by vendor lane is the first diagnostic I would run. · **Do NOT say:** Do not say I built the distributions — the data team did, to my requirement; do not say “400 on every lane.”

Q2 **Tell me about a time the data contradicted what everyone believed.** ARE RIGHT, A LOT

Open (first two sentences): The belief was that the carrier was missing its transit times. The scans said the carrier was delivering in about 21 days and we were planning on 18.

STAR (≈90 s): Situation: service was shaky and the room's theory was carrier performance — re-contract, escalate. Task: decide whether this was carrier management or planning. Action: I did not argue; I compared actual to the planned number and to the ERP's assumed spread; the options ranked themselves — the carrier was consistent, our parameter was wrong. Result: the fix moved to planning parameters, not carrier contracts, and the lane held 99% at the ranked p99.

Land this number: Mean 21 d vs plan 18 d; p99 34 d. · **Pivot to NACF:** In NACF terms: before blaming a carrier or a vendor for dock chaos, check whether the inbound plan's lead-time parameter was ever right. · **Do NOT say:** Do not imply the carriers were perfect — say the distribution had a tail; the point is the plan ignored it.

Q3 **Tell me about a time you refused to accept a metric that looked good enough.**
INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): A bell-curve p99 looked right — 30.3 days — and it was a "99%" plan that held 96.5%. I would not ship a 99% label on a 96.5% plan.

STAR (≈90 s): Situation: the standard shortcut, mean plus 2.33 standard deviations, was what the planning system and most analysts would use. Task: choose the plan statistic. Action: I scored four candidate plans on the same 400 shipments — mean 60%, brochure plus pad 93%, normal p99 96.5%, ranked p99 99% — and printed the sampling interval on the winner so nobody mistook in-sample for guaranteed. Result: the ranked p99 became the statistic; the interval is why I also ran a held-out quarter.

Land this number: 96.5% vs 99% — 14 vs 4 of 400. · **Pivot to NACF:** For NACF placement and WOC targets, the same discipline: name the service level, then check what the distribution actually delivers. · **Do NOT say:** Do not call 99% "guaranteed"; say in-sample by construction, interval 97.5–99.6.

Q4 **Tell me about a time you persuaded a skeptical group with data.** EARN TRUST

Open (first two sentences): Commercial hated the number — 34 days where the brochure said 18. I did not defend it; I showed them what 18 had cost.

STAR (≈90 s): Situation: commercial teams quote what the system says, and a longer date reads as slower. Task: get a defensible promise adopted without an organizational fight. Action: one lane, three numbers — 18 brochure, 21 mean, 34 p99 — and the price of the status quo: a fast date missed 60% of the time costs expedites and trust, the honest date costs only optics. Where speed genuinely mattered I offered to fix the lane, not the number. Result: planning signed the plan statistic and commercial stopped quoting the brochure.

Land this number: 60% late at the quoted date vs 1% at the plan. · **Pivot to NACF:** Amazon's promise engine faces the same trade at scale; I would lead with the cost of the missed promise, not the length of the date. · **Do NOT say:** Do not say commercial was wrong to want speed — say the lever was the lane, not the label.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: “The brochure was wrong three different ways.”

Headline metric: t = 15 on the 3-day bias; 60% of shipments slower than plan

- Location: +3 d, SE 0.20 d, 15 standard errors.
- Spread: ERP σ 8 vs measured 4 — wrong the other way, hence the pad.
- Tail: normal p99 30.3 holds 96.5%; ranked p99 34 holds 99%.
- One lane, three numbers, the carrier’s own scans.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Earn Trust

Open: “Commercial wanted to keep quoting 18; I would not put an 18 on a promise that failed 60% of the time.”

Headline metric: 60% late at the brochure date vs 1% at p99

- The objection was real: 34 reads slower than 18.
- I priced the status quo — expedites and trust — instead of arguing.
- Offered the lane fix where speed mattered.
- Planning signed the statistic; the brochure became a labeled reference.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: “I led with the busiest lane because it had the data — and that was my analytical shortcut.”

Headline metric: n = 400 busy lane vs ≈ 50 on an average lane; p99 interval open-ended above at n = 50

- Honest limit: a thin lane’s max is not its p99.
- I would now show the interval first and pool thin lanes by cohort.
- The held-out quarter (S8c) and the drift flag were the backstops.
- VERIFY-WITH-RAJESH what the program actually did on thin lanes.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 “Your p99 rests on four observations. How can that be a plan?”

The numbers: n = 400 → 95% interval t(392)…t(400); coverage interval 97.5%–99.6%; n = 50 → open-ended above; held-out quarter 99.1% (loop packet; S8c).

Say exactly: “It rests on the top four, and I print the interval — ninety-seven and a half to ninety-nine point six on coverage. That is why I validated on a held-out quarter and put a drift flag on every lane; on a thin lane I would not call a maximum a ninety-ninth percentile.”

T2 “How do you know the three-day gap is not one bad quarter?”

The numbers: SE 0.20 d; t = 15.0; window ≥ 12 months; the test I would add: same sign in all four quarterly cuts.

Say exactly: “Fifteen standard errors across a twelve-month window is not noise, but one bad quarter could still dominate it — the check is the quarterly cut, same sign in all four. The record doesn’t carry that cut and I would not claim it.”

T3 “What was the counterfactual — what would the plan have delivered if you had done nothing?”

The numbers: Brochure plan late 240/400 = 60%; brochure + pad 28/400 = 7% (93% served); ranked p99 4/400 = 1%.

Say exactly: “On the same 400 shipments the brochure plan was late sixty percent of the time and the brochure-plus-pad plan seven percent; the ranked p99 plan one percent — that is the counterfactual, shipment by shipment, with the interval.”

S8b — Elementum — safety-stock formula redesign: one formula nobody could defend plus a week of pad → two honest terms; one lane released 4,835 units, one lane took 1,383 in, 580K net NEW

LPs this story serves here: Invent and Simplify (primary — two terms replace a formula and a pad) · Have Backbone; Disagree and Commit (a rule that adds stock where the tail is fat) · Frugality (less stock, better service) · Ownership (a release rule someone owns after I leave)
Process family: Plan / Source seam — inventory-policy formula and release rule on top of the measured lane distributions (S8a → S8 → S8b)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Invent and Simplify** — I replaced "$z \cdot \sqrt{(LT \cdot \sigma_d^2 + d^2 \cdot \sigma_{LT}^2)}$ plus a week of pad" with two terms a planner can read — lead-time cover to the measured p99 and demand cover over the mean lead time.
- **Have Backbone; Disagree and Commit** — The rule I wrote cuts both ways — on a lane whose measured tail is fatter than the assumption it adds stock — and I kept it two-sided when a one-way cut would have been easier to sell.
- **Frugality** — Lane A went 12,937 → 8,102 units while late arrivals fell from 7-in-100 to 1-in-100: less stock and better service from one honest number.
- **Ownership** — The release was a signed planning act with a drift flag and a standing stockout-and-expedite report planning owns — the formula does not need me to keep working.

Honesty label. I owned the formula redesign, the two worked lanes and the release rule; planners owned the parameters and Finance signed each lane. Lane A (d = 500, σ_d = 150, brochure 18 d, ERP σ_{LT} = 8 d, measured mean 21 / σ 4 / p99 34) and Lane B (d = 200, σ_d = 80, brochure 24, measured mean 30 / σ 11 / p99 58) are the loop packet's worked lanes; the roll-up split — about 150 lanes releasing an average of 4,500 units, about 50 lanes adding an average of 1,900 — is the loop packet's illustrative decomposition. The record is the net: 580K units / ~$1.5M, zero stockouts.
VERIFY-WITH-RAJESH — the master and this packet's S8 say fat-tailed lanes' buffer was "kept"; the loop packet says about 50 lanes took stock in. Say "kept, and on the worst lanes raised" only if that is what happened.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A six-site manufacturer's inventory buffers were set by a textbook formula fed an assumed lead-time swing of eight days that nobody had measured, plus a week of pad that every planner added by hand. Task. Redesign the buffer so it said what the lanes actually did, and write the rule for when capital could come out. Action. I took the formula apart: on the example lane, 97.5% of its variance came from that unmeasured eight-day swing — the formula was mostly measuring a guess. I replaced it with two terms: lead-time cover — units per day times the distance from the average lead time to the measured ninety-ninth percentile — and demand cover over the average lead time. The pad went to zero because the plan now held 99%. The rule cut both ways: release only where the measured spread was inside the assumption, and where the tail was fatter, keep or add. Result. Primary metric: safety stock on the example lane 12,937 → 8,102 units, 37% less, with late arrivals 7-in-100 → 1-in-100; the program released 580K units, about $1.5 million, with zero stockouts — record, one-time working capital.

20-SECOND VERSION NEW

I replaced a formula that was 97% an unmeasured guess, plus a hand-added pad, with two honest terms — cover to the measured p99 and demand cover — and a two-sided release rule: 580K units out net, zero stockouts.

NACF BRIDGE — ONE SENTENCE

The NACF version is the inbound booklet's INV-B: $SS = z \cdot \sigma \cdot \sqrt{LT}$ — I would show which of the two variance terms dominates before anyone moves a buffer, and I would write the release rule so it can raise a buffer as well as cut it.

SITUATION / TASK — THE PROBLEM SPACE

S — The same six-site manufacturer. Safety stock was the textbook form, $z \times \sqrt{(LT \cdot \sigma_d^2 + d^2 \cdot \sigma_{LT}^2)}$, with the brochure lead time and a σ_{LT} of 8 days that was the ERP's default, plus an informal manual pad of about a week per lane. Nobody could say which part of the buffer was protecting against demand and which against transit — and on the loud lanes the answer was always "add more."

T — Own the formula and the release rule: say what each unit of buffer is for, size it from the measured distribution (S8a), and release capital only where the measured evidence supported it — with the exception written into the rule before anyone asked.

Baselines on the table: Lane A buffer 12,937 units (25.9 days of demand) · pad 3,500 units uncounted in the ERP · 7-in-100 shipments late against the plan in force · no statement of what service level the buffer bought.

ACTION — FIVE GATED PHASES

- **Discovery** — took the ERP formula apart on one lane: the demand part contributed 405,000 to the variance, the lead-time part 16,000,000 — 97.5% of it an assumed number. Exit bar: planning and Finance agree the buffer is a parameter question, not a service-level argument.
- **Plan & Design** — two terms — lead-time cover $d \times (p99 - mean)$ using the ranked p99 from S8a, and demand cover $z \times \sigma_d \times \sqrt{(mean\ lead\ time)}$ — pad to zero; release rule: release only where the measured spread and tail are inside the assumption and lane accuracy holds; where they are not, the buffer stays or rises. Exit bar: Planning and Finance sign the rule's text, including the add-stock clause.
- **Build & Test** — the data team computed every lane; I recomputed two lanes by hand and ran the normal-formula counterfactual beside them. Exit bar: Lane A and Lane B reproduce to the unit, and the counterfactual is on the page.
- **Cutover & Go-Live** — released lane by lane under the signed rule with Finance signing each lane; stockout-and-expedite watch from day one. Exit bar: no lane released without its lane accuracy and signature. VERIFY-WITH-RAJESH wave structure and lane counts per wave are not in the record.
- **Hypercare / Close** — drift flag per lane — p99 recomputed on refresh, buffer decision re-opens if accuracy falls — and the watch report handed to planning. Exit bar: planning re-applies the rule on its own refresh without me.

Influence and ownership. I owned the formula, the rule and the two worked lanes; the data team built the distributions; Finance signed each lane; planners owned the parameters. The temptation was a blanket cut because a one-way release is the easiest thing to sell to Finance — Lane B is the page that shows why not: a blanket cut of the same size would leave about half of what the measured tail needs.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — 580K units / ~$1.5M safety stock released, net (record, one-time working capital; ≈$2.59/unit derived); ≈$0.3M/yr carry on it at 20% (modeled) — inside S8's ≈$0.6M, never added to S6's $2M.
- **#** — Lane A 12,937 → 8,102 units (−4,835); Lane B 5,238 → 6,621 (+1,383); roll-up 150 × 4,500 − 50 × 1,900 = 580,000 (loop-packet decomposition, illustrative).
- **%** — late arrivals 7-in-100 → 1-in-100 on Lane A (loop-packet example lane); lane accuracy 99% on 200 lanes (record).
- **Quality** — zero stockouts in the watch window (record); the rule is two-sided and written; the pad is gone.
- **Customer** — the production line did not stop on a released lane; the planners' pad stopped being a second, hidden policy.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No statement of what each unit of buffer protected against; I defined the two terms and the service level each one buys.
Scope & influence	Planning, Finance, the data team and six sites; Finance signed lane by lane, planning owns the report.
Execution	Hand-recomputed two lanes, ran the normal-formula counterfactual, wrote the add-stock clause into the rule.
Impact	580K units / ~$1.5M out net, zero stockouts; Lane A 37% less stock with fewer late arrivals.
Process improvement	A written, two-sided release rule and a drift flag replaced the pad as the de-facto policy.
Complexity	Skewed tails, two lanes moving in opposite directions, and a service-level definition (cycle service vs fill rate) that has to be named.
Root cause	97.5% of the old formula's variance was a default σ_LT nobody measured — the root cause was the parameter, not the planners.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: lane × (d, σ_d, brochure LT, measured LT mean / σ / ranked p99) — Lane A and B; variance decomposition of the ERP formula: 97.5% lead-time part. Revealed: a guessed σ_LT dominated the buffer.
2 · Solution design	Options: keep the ERP form with a measured σ_LT · normal-fit p99 · ranked-p99 cover + demand cover. Chosen: two terms on the ranked p99 — it does not assume the lead-time tail is a bell curve. Release rule: two-sided, signed.
3 · Math & data logic	Math: 9,437 + 3,500 pad = 12,937; after = 500 × (34 − 21) + 2.33 × 150 × √21 = 6,500 + 1,602 = 8,102; Lane B +1,383; z-table for the demand term; sensitivity to the assumed σ_LT.
4 · Implementation	Implementation: data team per lane → hand-check two lanes → signed rule → lane-by-lane release with Finance signatures → watch report and drift flag handed to planning.
5 · Probe questions	Probe questions: independence and normality of the demand term · cycle-service vs fill-rate · why z = 2.33 · blanket-cut counterfactual · what happens on a thin lane — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

> **Analytics — Your formula adds a demand term and a lead-time term. Are they independent?**
>
> The ERP form assumes demand per day and lead time are independent; I kept that for the demand term only, and it is 19.8% of the after-buffer (1,602 of 8,102). The lead-time term — 80.2% — is empirical, not a bell curve. A fifty-percent error in the demand term moves the total by about ±10%. If demand surges when transit is slow the two correlate and the true figure is higher; I would test that on the lane history before calling it closed.

> **Analytics — Why z = 2.33?**
>
> Because the signed service level was 99% and Φ(2.33) = 0.9901. The price of more is steep and nonlinear: the demand cover on Lane A is 1,131 at 95%, 1,602 at 99% and 2,124 at 99.9% — +522 units (32.6%) for the last nine-tenths of a point.

> **Analytics — Is 99% a fill rate or a cycle service level?**
>
> A cycle service level — the probability a replenishment cycle ends without a stockout, measured as lane accuracy against the p99 promise. Fill rate — the share of units shipped from stock — is a different, much higher number on the same buffer; I do not mix the two and I say which one I mean.

Analytics — Why release 4,835 units and not 8,009, which is what the textbook form with the measured σ would release?

Because the textbook form with the measured σ_LT = 4 gives 4,928 units after, i.e. a release of 8,009, and it plans to 30.3 days when 3.5% of shipments run past that. The extra 3,174 units is exactly the tail the bell curve cannot see — that is the S8a result paid for in units.

Shadow / closing — What would you do differently?

Put Lane B on the first page. The two-sided result — some lanes need more stock — was the part that earned Finance's trust, and it should have been the opening exhibit, not the second slide.

BUSINESS IMPACT MODEL — WHAT THE FORMULA REDESIGN MOVED (LANE A, LANE B, PROGRAM)

```
Safety stock = formula nobody could defend + a hand-added pad
+- Lead-time part = 97.5% of the variance, driven by an unmeasured sigma_LT = 8
+- Pad = 7 days x demand, uncounted, invisible to Finance
+- Fix: lead-time cover to ranked p99  +  demand cover over mean LT  ;  pad -> 0
+- Rule: release where measured spread/tail is inside the assumption; keep or add where it is not
+- Result: Lane A -4,835 units, Lane B +1,383 units, program 580K net, zero stockouts
```

DRIVER	BASELINE	AFTER	$ / UNITS	BASIS
Lane A buffer	12,937 units	8,102 units	−4,835 ≈ $12,523	loop-packet lane (illustrative); $2.59/unit derived
Lane B buffer	5,238 units	6,621 units	+1,383 (stock in)	loop-packet lane (illustrative)
Program net release	—	580,000 units	≈$1.5M	record (one-time)
Carry on the net release	—	—	≈$300,440/yr	modeled (20% all-in; benchmark 15–25%)

If challenged. *"Your $1.5M is a decomposition of two example lanes."* — "No: the $1.5M is the record — 580K units net, Finance-signed. The two lanes are how the rule behaves; the 150-and-50 split is the loop packet's illustration and I label it."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: why the old buffer was wrong (Lane A).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	ERP demand part	$LT_{brochure} \times \sigma_d^2 = 18 \times 150^2 = 18 \times 22{,}500 = 405{,}000$	brochure 18 d; σ_d 150 (illustrative lane)	405,000
2	ERP lead-time part	$d^2 \times \sigma_{LT}^2 = 500^2 \times 8^2 = 250{,}000 \times 64 = 16{,}000{,}000$	σ_LT = 8 d — ERP default, never measured	16,000,000
3	Combine	$\sqrt{(405{,}000 + 16{,}000{,}000)} = \sqrt{16{,}405{,}000} = 4{,}050$	sum under the root	4,050
4	Service factor	$2.33 \times 4{,}050 = 9{,}437$	z = 2.33 (99%, toolkit label)	9,437 units
5	Variance share of the guess	$16{,}000{,}000 \div 16{,}405{,}000 = 97.5\%$	derived	97.5% unmeasured
6	Manual pad	$7 \text{ days} \times 500 \text{ units/day} = 3{,}500$	pad ≈ a week (planner habit; illustrative)	3,500 units
7	Total before	$9{,}437 + 3{,}500 = 12{,}937$ (= 25.9 days of demand)	rows 4 + 6	12,937 units

Table 2 — the fix, values substituted (Lane A, the counterfactual, Lane B).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lead-time cover	`d × (p99 − mean) = 500 × (34 − 21) = 6,500`	ranked p99 = 34 (S8a); mean 21	6,500 units
2	Demand cover	`z × σ_d × √LT_mean = 2.33 × 150 × √21 = 2.33 × 150 × 4.583 = 1,602`	σ_d = 150; mean LT 21	1,602 units
3	Pad	`plan holds 99% → no manual pad needed`	S8a	0
4	Total after	`6,500 + 1,602 = 8,102`	rows 1 + 2	8,102 units
5	Release, Lane A	`12,937 − 8,102 = 4,835 (37.4%)`	before − after	4,835 units
6	Textbook-form counterfactual	`2.33 × √(21 × 150² + 500² × 4²) = 2.33 × √4,472,500 = 4,928`	normal form, measured σ_LT = 4	4,928 units — plans to 30.3 d; serves 96.5%
7	Lane B before	`2.33 × √(24 × 80² + 200² × 8²) + 7 × 200 = 2.33 × √2,713,600 + 1,400 = 3,838 + 1,400 = 5,238`	d 200, σ_d 80, brochure 24, ERP σ_LT 8	5,238 units
8	Lane B after	`200 × (58 − 30) + 2.33 × 80 × √30 = 5,600 + 1,021 = 6,621`	measured mean 30, σ 11, p99 58	6,621 units (+1,383)
9	Blanket-cut counterfactual on B	`Lane A's cut 37.4% applied to B: 5,238 × (1 − 0.3737) ≈ 3,280; ÷ 6,621 = 49.5%`	illustrative counterfactual (a blanket rule)	leaves 49.5% of the measured need — short 3,341 units ≈ 16.7 days of demand

Table 3 — after: the metrics, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Lane A buffer	12,937 units (25.9 days of demand)	8,102 units (16.2 days)	loop-packet lane
2	Lane A late arrivals	7 in 100 (brochure + pad plan)	1 in 100 (ranked p99 plan)	S8a (28/400 → 4/400)
3	Lane B buffer	5,238 units	6,621 units	loop-packet lane
4	Program net safety stock	base	−580,000 units	record
5	Stockouts in the watch window	—	0	record
6	Owner of the buffer parameter	ERP default + planner habit	signed rule + drift flag, planning	design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lane A release value	`4,835 × $2.59 = $12,523`	$2.59 = $1.5M ÷ 580K (record pair)	≈$12,523 one-time
2	Lane A carry saved	`$12,523 × 20% = $2,505`	20% all-in carry (benchmark 15–25%)	≈$2,505/yr (overlaps S8a's pad carry — do not add)
3	Roll-up, units out	`150 lanes × 4,500 = 675,000`	loop-packet decomposition (illustrative)	675,000 units
4	Roll-up, units in	`50 lanes × 1,900 = 95,000`	loop-packet decomposition (illustrative)	95,000 units
5	Net	`675,000 − 95,000 = 580,000`	ties to the record	580,000 units

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	Net value	`580,000 × $2.59 = $1,502,200`	record pair	≈$1.5M one-time (record)
7	Carry on the net	`$1,502,200 × 20% = $300,440`	modeled	≈$300,440/yr (part of S8's ≈$0.6M)

SOLUTION FLOW — FORMULA, RULE, WATCH

```
ERP: PO lines + goods receipts --> S8a lane distribution (n, mean, sigma, ranked p99)
                                    |
                                    v
per lane:  LT cover = d x (p99 - mean_LT)     demand cover = z x sigma_d x sqrt(mean_LT)
                        \                       /
                         v                     v
                    buffer = LT cover + demand cover          (pad = 0)
                                    |
       measured spread/tail inside assumption AND lane accuracy 99%?
          yes -> release (Finance signs the lane)      no -> keep or raise buffer
                                    |
            refresh: p99 recomputed; accuracy falls -> decision re-opens (drift flag)
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
ERP σ_LT assumed (Lane A release)	σ = 6 → release 2,544	σ = 8 → release 4,835	σ = 10 → release 7,142
Service level (demand cover, Lane A)	95% → 1,131	99% → 1,602	99.9% → 2,124
Measured p99 on Lane A	30 d → lead-time cover 4,500	34 d → 6,500	38 d → 8,500
Carry rate on the 580K-unit release	15% → $225,330/yr	20% → $300,440/yr	25% → $375,550/yr

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The old buffer was a textbook formula fed an eight-day lead-time swing nobody had measured — ninety-seven percent of its variance was that guess — plus a week of hand-added pad. I replaced it with two terms a planner can read: cover to the measured ninety-ninth percentile, and demand cover. On the example lane that took twelve thousand nine hundred units to about eight thousand one hundred with late arrivals falling from seven in a hundred to one; on a lane with a fatter tail it added about fourteen hundred units, because the rule cuts both ways. The program record is five hundred eighty thousand units, about one and a half million dollars, released net with zero stockouts — the lane arithmetic is the loop packet's example and I label it."

If challenged — the full answers

The split is illustrative. "The 150-and-50 decomposition is how I explain the net; the record is 580K units, about $1.5M, zero stockouts." The fat-tail lanes — kept or raised? "The rule keeps or raises where the measured tail is fatter than the assumption; whether a given lane saw stock go in is a lane-level fact I would pull from the release log rather than assert." The normal form is simpler. "And it plans to 30.3 days where 3.5% of shipments run past — I would rather defend 4,835 units than 8,009."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you simplified something complicated.** INVENT AND SIMPLIFY

Open (first two sentences): Safety stock on our lanes was a formula nobody could defend plus a week of pad nobody could count. I replaced both with two terms a planner could read.

STAR (≈90 s): Situation: the ERP buffer was z times a root of two variances, with an eight-day lead-time swing nobody had measured — 98% of the formula was that guess — and planners added a week on top. Task: say what each unit of buffer was for. Action: I split it into lead-time cover to the measured p99 and demand cover over the mean lead time, set the pad to zero, and wrote the release rule. Result: Lane A went from 12,937 to 8,102 units with late arrivals falling from seven in a hundred to one, and the program released 580K units with zero stockouts.

Land this number: 12,937 → 8,102 units; 97.5% of the old variance was one unmeasured input. · **Pivot to NACF:** INV-B is the same two-variance structure; I would show a planner which term dominates before touching a number. · **Do NOT say:** Do not say I replaced the ERP; say I changed two parameters and the rule that governs them.

Q2 **Tell me about a time you held your position when it was the harder path.** HAVE BACKBONE; DISAGREE AND COMMIT

Open (first two sentences): The easy rule to sell was a release-only rule. I wrote one that could also add stock — and defended it.

STAR (≈90 s): Situation: the obvious design was to cut buffers and report the release. Task: write a rule that would survive an audit. Action: I wrote it two-sided — release where the measured spread was inside the assumption, keep or add where it was not — and showed the alternative on Lane B: a blanket cut of Lane A's size would leave 50% of what the measured tail needs. Finance and Planning signed it with the add-stock clause. Result: 580K net, zero stockouts, and the credibility to do the next release. VERIFY-WITH-RAJESH whether a specific person asked for a one-way cut.

Land this number: Lane B +1,383 units; blanket cut leaves 49.5% of need. · **Pivot to NACF:** In placement, the same discipline: a rule that can only move one way is a target, not a policy. · **Do NOT say:** Do not name or imply a person pushed back unless it is true; say "the easy design was."

Q3 **Tell me about a time you did more with less.** FRUGALITY

Open (first two sentences): Less stock and better service — on one lane, 37% fewer units and seven late shipments in a hundred became one.

STAR (≈90 s): Situation: working capital sat in buffers that were partly folklore. Task: take capital out without moving the service line. Action: replaced the pad with a measured p99 plan and kept the buffer wherever the tail demanded it. Result: Lane A 4,835 units out (≈$12,523) with fewer late arrivals; program 580K units, about $1.5M, zero stockouts.

Land this number: 580K units / ~$1.5M net, zero stockouts. · **Pivot to NACF:** WOC targets are the NACF equivalent — fewer weeks of cover only where the measured distribution supports it. · **Do NOT say:** Do not claim recurring savings as the headline; the $1.5M is one-time working capital.

Q4 **Tell me about something you owned beyond the delivery date.** OWNERSHIP

Open (first two sentences): A release is not done when the units leave the shelf. It is done when the rule runs without you.

STAR (≈90 s): Situation: a buffer cut that nobody re-checks decays into a new brochure. Task: make the release self-correcting. Action: drift flag per lane — p99 recomputed on refresh, the buffer decision re-opens if accuracy falls — and the stockout-and-expedite watch handed to planning as a standing report. Result: planning re-applies the rule on its own refresh; the watch showed zero stockouts.

Land this number: Zero stockouts in the watch window (record). · **Pivot to NACF:** NACF inbound needs the same drift flag on every lead-time and WOC parameter. · **Do NOT say:** Do not say I ran the watch; say planning owns it.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "97.5% of the old formula's variance was a number nobody measured."

Headline metric: variance share 97.5%; 12,937 → 8,102 units

- Demand part 405,000 vs lead-time part 16,000,000.
- Two terms replace it: cover to p99 + demand cover.
- Textbook form with measured σ would release 8,009 — and serve 96.5%.
- Lane A −4,835, Lane B +1,383.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Disagree and Commit

Open: "I wrote the rule so it could add stock — that is why Finance signed it."

Headline metric: blanket cut leaves 49.5% of Lane B's measured need

- A one-way cut is the easy sell and the wrong rule.
- Lane B is the exhibit.
- Signed with the add-stock clause.
- VERIFY-WITH-RAJESH the actual stakeholder dynamics.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "The textbook form was my starting point; one skewed lane showed it understates the tail by 3.4 days."

Headline metric: normal 30.3 d vs ranked 34 d; 8,009 vs 4,835 release

- The cumulative definition came from a skewed lane.
- I would now show the demand-supply correlation test on lane history.
- Cycle service ≠ fill rate — name which one.
- VERIFY-WITH-RAJESH that this was my own starting point.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Your safety-stock formula assumes normality and independence. Why trust it?"

The numbers: Demand term = 1,602 of 8,102 = 19.8%; ±50% error → ±10% of the total; lead-time term empirical (ranked p99).

Say exactly: "Only the smaller term assumes a bell curve — about a fifth of the buffer; the lead-time term, four-fifths, is the measured ranked p99 with no distribution assumed, and a fifty-percent error in the demand term moves the total by about ten percent."

T2 "Is 99% a fill rate or a cycle service level, and what does 99.9% cost?"

The numbers: Demand cover at 99% = 1,602; at 99.9% = 2,124 (+522, +32.6%); z 2.33 → 3.09.

Say exactly: "Ninety-nine percent is a cycle service level — probability of no stockout per replenishment — not a fill rate; going to 99.9 adds about five hundred units on the demand term alone, a third more, and I would only buy it where a stockout costs more than that."

T3 "Why not just cut everything by the same percentage — it is simpler and Finance would like it?"

The numbers: Lane A cut 37.4%; applied to Lane B: 3,280 vs need 6,621 → 49.5%; short 3,341 units = 16.7 days of demand.

Say exactly: "A blanket cut of Lane A's size would leave Lane B with half the buffer its measured tail needs — about seventeen days of demand uncovered — which is a line-down waiting to happen; the two-sided rule is what let Finance sign the release."

S8c — Elementum — validation and back-test design: how I proved the p99 plan held out of sample and was not a quiet quarter — held-out quarter 99.1%, zero stockouts, and what that can and cannot show NEW

LPs this story serves here: Are Right, A Lot (primary — test my own decision out of time) · Insist on the Highest Standards (per-lane gate, no pooling) · Earn Trust (the limits of the test on the page) · Learn and Be Curious (shadow-mode the drift flags first)
Process family: Plan / Source seam — validation and monitoring design for a parameter change before capital moves

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Are Right, A Lot** — I did not trust in-sample 99%: I trained on twelve months, scored a later quarter the plan had never seen, and read the result against an interval — 99.1% [98.7%, 99.4%].
- **Insist on the Highest Standards** — The bar was per lane, 99%, before any buffer released — no pooling to hide a bad lane inside a good average.
- **Earn Trust** — I put what the test cannot show on the page — a lane has about thirteen shipments a quarter — so Finance signed knowing the limits.
- **Learn and Be Curious** — My own lesson is shadow-mode: run the drift flags against history before they go live.

Honesty label. Record: validated on a held-out quarter at 99.1% (loop packet, secondary source); zero stockouts in the watch window; Finance signed each lane; drift flag per lane. The held-out sample size (2,500 shipments = the master's illustrative ~10,000 a year ÷ 4), the old-plan late share (7%, the loop-packet example lane applied program-wide), the 150 releasing lanes and the 13-week window are illustrative inputs for the arithmetic — not the record. VERIFY-WITH-RAJESH (a) the length of the watch window, (b) which quarter was held out and whether it was a peak or a quiet quarter, (c) whether any contemporaneous control was run, (d) whether one held-out quarter or several were used.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. We had replaced brochure lead times with measured p99 plans and were about to take capital out of 150-odd lanes. A plan fitted to a year of history will look perfect on that year — that is not evidence. Task. Prove the plan would hold on shipments it had never seen, and that the result was not a quiet season. Action. I trained each lane on twelve months and scored the next quarter out of time. At program level that is about 2,500 shipments; 22 ran past the plan — 99.1% held, with a 95% interval of 98.7% to 99.4%, clearly better than the 96.5% a bell-curve plan delivers, and statistically indistinguishable from the 99% we promised. I scored the same quarter's shipments against the old brochure-plus-pad plan, so the season hits both plans equally. And I was explicit about the limit: on one lane a quarter is about thirteen shipments, so lane-level validation is the drift flag, not a quarterly test. Result. Primary metric: 99.1% on the held-out quarter against the 99% promise, then zero stockouts across the released lanes — record — with 580K units, about $1.5 million, released net.

20-SECOND VERSION NEW

I back-tested the p99 plan on a quarter it had never seen — 99.1%, interval 98.7 to 99.4 — scored the same shipments against the old plan so a quiet season could not flatter it, and kept a per-lane drift flag because a lane alone has too few shipments to validate.

NACF BRIDGE — ONE SENTENCE

For NACF placement and WOC targets I would want the same out-of-time test before any buffer comes out: train on the past, score a later quarter, and read the result as an interval, not a number.

SITUATION / TASK — THE PROBLEM SPACE

S — Elementum, six-site manufacturer. The measured p99 plans (S8a) and the formula (S8b) were ready; releasing 580K units of buffer was irreversible in practice, and the one thing a plan fitted to history will always do is look good on that history. The audience — Finance and Planning — would sign lane by lane only on evidence they could challenge.

T — Design the validation: an out-of-time test that could fail, an interval around the answer, a defense against "it was a quiet quarter," a live watch after release, and a drift flag that outlives me.

Baselines on the table: in-sample coverage 99% by construction (396 of 400 on the example lane) · no out-of-sample measurement existed · brochure-plus-pad plan late 7 in 100 on the example lane · no stockout metric tied to the release.

ACTION — FIVE GATED PHASES

- **Discovery** — named the failure mode — in-sample coverage is 99% by construction — and the second one, seasonality: a quarter that happens to be easy flatters any plan. Exit bar: Finance and Planning agree what result would make us stop.
- **Plan & Design** — train on twelve months so every season is inside the plan; hold out the next quarter; metric = arrival vs plan; score the same shipments against the old plan; pre-register the pass bar (99% promise, read as an interval) and the per-lane gate (99%, no pooling) before looking. Exit bar: bar and metric signed before the data is scored.
- **Build & Test** — scored the held-out quarter: 22 of 2,500 beyond plan, 99.1%; the paired score against the old plan on the same shipments; the power calculation. Exit bar: result reproduced by the data team; interval and power on the page.
- **Cutover & Go-Live** — release lane by lane under the signed rule; live watch — stockouts and expedites by lane — from day one; zero stockouts in the watch window. Exit bar: no lane released without its own accuracy ≥ 99%. VERIFY-WITH-RAJESH the watch window length.
- **Hypercare / Close** — drift flag per lane (p99 recomputed on refresh; decision re-opens if accuracy falls) and a standing report owned by planning; lesson recorded: shadow-mode the flags against history before they go live. Exit bar: planning re-runs the check on its own refresh.

Influence and ownership. I owned the validation design and the pass bars; the data team scored the lanes; Finance and Planning signed. The conflict was pace: releasing on the in-sample result would have been faster and indistinguishable on a slide — I held the release to the out-of-time test and wrote the per-lane gate so a good average could not hide a bad lane.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **%** — held-out quarter 99.1% against a 99% promise (loop packet; interval 98.7%–99.4% on an illustrative 2,500 shipments); lane accuracy 99% on 200 lanes (record).
- **#** — zero stockouts in the watch window (record); with 13 weeks × 150 releasing lanes = 1,950 lane-weeks that bounds the stockout rate below 0.15% per lane-week at 95% confidence (rule of three; window illustrative).
- **$** — the validation supports the 580K units / ~$1.5M release (record, one-time); it is not a separate saving. Cost of trusting the bell curve instead: ≈$79K/yr of expedites on the example inputs (modeled, a counterfactual, not a claim).
- **Quality** — a pre-registered pass bar, an interval, a paired replay, and a drift flag.
- **Customer** — no line stopped on a released lane.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No definition of “the plan held”; I defined it as arrival vs plan, out of time, as an interval.
Scope & influence	Finance, planning and the data team; Finance’s lane-by-lane signature was the acceptance test.
Execution	Pre-registered the bar, ran the out-of-time score, paired replay, power check, live watch, drift flag.
Impact	The validation is what let 580K units / ~$1.5M come out with zero stockouts.
Process improvement	A standing out-of-time check and a drift flag per lane — the validation repeats on refresh.
Complexity	Small per-lane samples, seasonality, skew, a thin-lane tail — each named and bounded.
Root cause	The risk was not the plan; it was believing an in-sample number — validation addresses the belief.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: shipments per lane, 12 months training + the next quarter held out; arrival vs plan; same shipments scored against the old plan. Revealed: 22/2,500 beyond plan, 99.1%, interval 98.7–99.4%; the old plan late ≈ 7%.
2 · Solution design	Options: in-sample fit · hold-out quarter · rolling-origin back-test · live shadow. Chosen: out-of-time hold-out plus paired replay plus live watch plus per-lane drift flag; rolling-origin is the upgrade.
3 · Math & data logic	Math: Wilson interval · z vs 96.5% = 7.13, vs 99% = 0.60 · power 99% at a true 98.0% · expected out-of-sample coverage of the ranked p99 = r/(n+1) = 98.75% ± 0.55 · rule of three on zero stockouts.
4 · Implementation	Implementation: pre-registered bars → score → paired replay → lane-by-lane signature → watch → drift flag → standing report.
5 · Probe questions	Probe questions: interval and power · seasonality · per-lane n · what zero stockouts proves · what a thin lane does — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

> **Analytics — A ranked p99 from 400 shipments, tested on new data. What coverage should you expect?**
>
> Not 99%. The r-th of n ranked observations covers, on average, r/(n+1) of the future distribution: 396/401 = 98.75% with a standard deviation of 0.55 points. So 99.1% is well inside what the method predicts — a 29% chance of a coverage that high or higher. On a thin lane the maximum of 50 shipments covers 50/51 = 98.04% ± 1.92 points and has a 7.7% chance of covering less than 95%.

> **Analytics — Is 99.1% different from 99%?**
>
> No. 22 late of 2,500; standard error under 99% is 0.20 points, z = 0.60. It is different from the bell-curve plan’s 96.5% (z = 7.13). The test is built to detect a shortfall, not a surplus: it rejects 99% only if coverage falls below 98.67%.

> **Analytics — How much power did the test have?**
>
> At 2,500 shipments: 99% power against a true coverage of 98.0%, 76% against 98.5%; the smallest shortfall it detects with 80% power is about 0.53 points (98.47% true coverage). So it is good at catching a plan that is clearly wrong and blind to a plan that is half a point short — which is why the drift flag exists.

> **Analytics — How do you know it was not seasonality?**
>
> Three defenses and one limit. The training window held every season; the metric is arrival against plan, not volume; and I scored the same held-out shipments against the old plan, so a quiet quarter flatters both plans equally — on the illustrative program numbers the old plan was late on 175 shipments and the new on 22. The limit: one held-out quarter is one draw; the upgrade is a rolling-origin back-test with a peak quarter in it. VERIFY-WITH-RAJESH which quarter was held out.

Analytics — Can you validate each lane the same way?

No. An average lane has about 50 shipments a year, roughly 13 a quarter; a lane whose true coverage is exactly 99% shows zero late shipments 87.8% of the time and one-or-more 12.2% of the time, so one late shipment in a quarter tells me nothing. Per-lane protection is the drift flag on rolling history; the quarterly test is pooled.

Shadow / closing — What would you do differently?

Shadow-mode the drift flags against history before they go live — replay a past year and count the false alarms — and start the rolling-origin back-test on day one so the peak quarter is in it.

BUSINESS IMPACT MODEL — WHAT VALIDATION PROTECTS

```
Release 580K units on a p99 plan fitted to history
+- Risk 1: in-sample 99% is by construction -> need out-of-time coverage
+- Risk 2: a quiet quarter flatters the plan -> same shipments scored vs old plan
+- Risk 3: a lane has ~13 shipments/quarter -> pooled test + per-lane drift flag
+- Risk 4: zero stockouts proves little in a short window -> rule-of-three bound
+- Payoff: Finance signs lane by lane; capital leaves; watch confirms
```

DRIVER	WITHOUT VALIDATION	WITH VALIDATION	$ / %	BASIS
Coverage evidence	in-sample 99% by construction	out-of-time 99.1% [98.7%, 99.4%]	—	loop packet (99.1%); n illustrative
Cost of the bell-curve plan	late 3.5% = 87.5 shipments / quarter	late 0.9% = 22 / quarter	≈$78,600/yr expedites avoided	modeled counterfactual (10% × $3,000)
Release supported	—	580K units / ~$1.5M	record	record (one-time)

If challenged. *"So validation saved $79K?"* — "No: that is what trusting the bell curve would have cost on illustrative inputs; the validation is the gate that made the $1.5M release signable."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: why in-sample evidence is not evidence.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	In-sample coverage	`rank ⌈0.99 × 400⌉ = 396 → 396/400 = 99.0%`	ranked p99 on the example lane (S8a)	99.0% by construction
2	Expected out-of-sample coverage	`r/(n+1) = 396/401 = 98.75%`	order-statistic (Beta(396, 5)) mean	98.75%
3	Spread of that coverage	`√[r(n+1−r)/((n+1)²(n+2))] = √[396 × 5/(401² × 402)] = 0.55 points`	Beta(396, 5) standard deviation	± 0.55 pts
4	Thin lane (n = 50)	`max covers 50/51 = 98.04% ± 1.92; P(coverage < 95%) = 0.95⁵⁰ = 7.7%`	maximum of 50 shipments as the "p99"	7.7% chance of a bad miss
5	Per-lane quarterly sample	`≈ 50/4 ≈ 13 shipments; P(0 late | 99%) = 0.99¹³ = 87.8%`	master's illustrative volume (≈50/lane/yr)	87.8% — one quarter says nothing per lane
6	The seasonality risk	`one held-out quarter = one draw; a quiet quarter flatters any plan`	design risk	needs a paired score and a peak quarter

Table 2 — the fix: the out-of-time test, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Held-out result	`2,478 of 2,500 within plan = 99.12% (22 late)`	loop packet 99.1%; n = 2,500 illustrative	99.12%

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Interval	`Wilson 95% = [98.67%, 99.42%]`	binomial, n = 2,500	[98.67%, 99.42%]
3	Standard error under 99%	`√(0.99 × 0.01 ÷ 2,500) = 0.199 pts; reject if observed < 99% − 1.645 × 0.199 = 98.67%`	one-sided 5% test	cut-off 98.67%
4	vs the promise (99%)	`z = (99.12 − 99.00) ÷ 0.199 = 0.60`	H0: coverage = 99%	not distinguishable from 99%
5	vs the bell-curve plan (96.5%)	`z = (99.12 − 96.50) ÷ √(0.965 × 0.035 ÷ 2,500) = 7.13`	H0: coverage = 96.5%	clearly better
6	Power	`P(observed < 98.67% | true 98.0%) = 99%; true 98.5% → 76%; 80%-power shortfall ≈ 0.53 pts`	normal approximation, n = 2,500	blind below ≈ 0.5 pt shortfall
7	Paired replay (same shipments)	`old plan late 175 (7%); new late 22; discordant: old-only 157, new-only 4; McNemar χ² = (157 − 4)² ÷ (157 + 4) = 145.4`	old-plan share 7% = example lane applied program-wide; the 18 / 4 split illustrative	season hits both plans equally
8	Zero stockouts	`13 wks × 150 lanes = 1,950 lane-weeks; upper bound = 3 ÷ 1,950 = 0.154% per lane-week (95%)`	rule of three; window illustrative	< 0.15% / lane-week

Table 3 — after: the metrics, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Out-of-time coverage	none (in-sample 99% only)	99.1% [98.7%, 99.4%]	loop packet; interval illustrative
2	Late vs plan, same shipments	175 of 2,500 (7%) old plan	22 of 2,500 (0.9%)	illustrative program-level
3	Stockouts after release	unmeasured	0 in the watch window	record
4	Per-lane protection	none	drift flag, rolling history	design record
5	Pass bar	not defined	99% per lane, pre-signed; no pooling	record rule

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Late at the bell-curve plan	`3.5% × 2,500 = 87.5 shipments / quarter`	96.5% coverage (S8a)	87.5 / quarter
2	Late at the ranked plan	`22 / quarter`	row above	22 / quarter
3	Excess late	`87.5 − 22 = 65.5`	difference	65.5 / quarter
4	Expedite cost	`65.5 × 10% × $3,000 = $19,650 / quarter`	10% share and $3K premium illustrative (master)	$19,650
5	Annualized (counterfactual)	`$19,650 × 4 = $78,600`	modeled — a cost avoided by not using the bell curve, not a savings claim	≈$78,600/yr
6	The release it supports	`580,000 × $2.59 = $1,502,200`	record pair	≈$1.5M one-time (record)

SOLUTION FLOW — THE VALIDATION PIPELINE

```
shipments per lane (PO schedule vs goods receipt)
       | months 1-12: fit ranked p99          months 13-15: held out, never seen
       v                                                  v
  plan statistic per lane  ------------------->  arrival vs plan, scored out of time
                                                          |  same shipments scored vs old plan (paired)
                                                          v
      pooled coverage + Wilson interval + power  -> pass bar signed in advance?
                        | yes: release lane by lane (Finance signs)
                        v
      live watch: stockouts / expedites by lane  ->  rule-of-three bound
                        v
      drift flag per lane (rolling), standing report owned by planning
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Held-out shipments	1,000 → Wilson wider (about ±0.6 pt)	2,500 → [98.7%, 99.4%]	10,000 → about ±0.2 pt
Late shipments observed (of 2,500)	12 → 99.5%	22 → 99.1%	40 → 98.4%
Watch window × releasing lanes	8 wks × 150 → bound 0.25%	13 × 150 → bound 0.154%	26 × 150 → bound 0.077%
Expedite premium on the bell-curve counterfactual	$1,500 → $39,300/yr	$3,000 → $78,600/yr	$5,000 → $131,000/yr

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

“A plan fitted to a year of history looks perfect on that year, so I held out a quarter the plan had never seen: ninety-nine point one percent held, with an interval of about ninety-eight point seven to ninety-nine point four, against ninety-six and a half for a bell-curve plan. Same shipments scored against the old plan, so a quiet season could not flatter it. After release, zero stockouts across the released lanes — which on a thirteen-week window of about two thousand lane-weeks bounds the rate under about a sixth of a percent. What the test cannot show is any single lane — a lane has about thirteen shipments a quarter — so each lane carries a drift flag. The record is five hundred eighty thousand units, about one and a half million dollars, released with zero stockouts; the interval, power and paired replay are my design arithmetic on illustrative volumes.”

If challenged — the full answers

One quarter is one draw. “Agreed — I would have preferred a rolling origin with a peak quarter, and I say so; the paired score and the live watch are the other two legs.” Zero stockouts proves little. “It bounds the rate: zero in about two thousand lane-weeks is under 0.15% per lane-week at 95% — not zero risk, a number.” The numbers are illustrative. “The 99.1% and the zero stockouts are the record; the volumes behind the intervals are my illustrative inputs and I mark them.”

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you tested your own decision and it could have failed.** ARE RIGHT, A LOT

Open (first two sentences): I was about to release 580K units on a plan fitted to history, and a plan fitted to history always looks perfect on that history. So I held a quarter out.

STAR (≈90 s): Situation: p99 plans per lane, buffer about to come out. Task: prove the plan held on shipments it had never seen. Action: trained on twelve months, scored the next quarter — 22 of 2,500 beyond plan, 99.1%, interval 98.7 to 99.4 — and scored the same shipments against the old plan so a quiet season could not flatter it. I wrote the pass bar before scoring. Result: Finance signed lane by lane; zero stockouts after release.

Land this number: 99.1% out of time vs 96.5% for a bell-curve plan. · **Pivot to NACF:** Before any NACF placement or WOC target change, the same out-of-time test on a held-back quarter. · **Do NOT say:** Do not say “proved”; say “held out of time, with an interval.”

Q2 **Tell me about a time you set a higher bar than you were asked to.** INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): The bar was 99% per lane before any buffer released — not 99% on average. Below it, no release, written into the act before anyone asked.

STAR (≈90 s): Situation: a pooled 99% can hide a bad lane inside a good average. Task: a release rule that could not be gamed by pooling. Action: per-lane accuracy bar, signed in advance; lanes under it did not release; the drift flag re-opens the decision if accuracy falls. Result: 580K units out net, zero stockouts, and no lane released on an average.

Land this number: 99% per lane across 200 lanes; 580K units / ~$1.5M. · **Pivot to NACF:** A placement model can pass on region average and fail one FC-region pair — I gate per cell. · **Do NOT say:** Do not say the bar was Finance’s; it was written into the act as my design.

Q3 **Tell me about a time you had to earn trust from a skeptical partner.** EARN TRUST

Open (first two sentences): Finance signed lane by lane because I put what the test could not show on the same page as what it could.

STAR (≈90 s): Situation: Finance was being asked to sign a release on a statistical argument. Task: make the argument checkable. Action: the interval beside the number, the power beside the interval, and the limit beside the power — an average lane has about thirteen shipments a quarter, so per-lane validation is the drift flag, not the test. Result: Finance signed each lane and the watch confirmed zero stockouts.

Land this number: 99.1% [98.7, 99.4]; per-lane n ≈ 13. · **Pivot to NACF:** Amazon’s bar-raising partners trust the analyst who names the limit first. · **Do NOT say:** Do not oversell the 2,500; it is an illustrative program-level n.

Q4 **Tell me about something you learned that changed how you work.** LEARN AND BE CURIOUS

Open (first two sentences): What I learned is to shadow-mode the drift flags against history before they go live.

STAR (≈90 s): Situation: the drift flag — rolling p99 above plan by 2 days, σ above 1.25 times the base, or four late per hundred — looked right on a whiteboard. Task: know its false-alarm rate before planners had to answer it. Action (the lesson): replay a past year and count how often the flag would have fired. Result: the flag goes live with a known alert rate; the same replay is how I would tune any threshold now.

Land this number: Rule of three: 0 in 1,950 lane-weeks → < 0.154% per lane-week. · **Pivot to NACF:** In NACF, a new inbound alert would be replayed against a prior peak before it pages anybody. · **Do NOT say:** Do not say the replay was done unless it was — say “what I would now do first.” VERIFY-WITH-RAJESH

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Are Right, A Lot

Open: "A plan fitted to a year of history looks perfect on that year."

Headline metric: 99.1% out of time, [98.7, 99.4]; expected 98.75% ± 0.55

- Hold-out quarter, never seen.
- z vs 96.5% = 7.13; vs 99% = 0.60.
- Power 99% at a true 98.0%.
- Paired replay on the same shipments.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Insist on the Highest Standards

Open: "I would not release on a pooled average."

Headline metric: 99% per lane, no pooling

- The bar was signed before scoring.
- Lanes under the bar did not release.
- The drift flag re-opens the decision.
- VERIFY-WITH-RAJESH any pressure to release early.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "My test could not tell me anything about a single lane."

Headline metric: per-lane quarterly n ≈ 13; P(0 late | 99%) = 87.8%

- One quarter is one draw — rolling origin is the upgrade.
- Zero stockouts bounds the rate; it does not prove zero.
- Shadow-mode the drift flags before go-live.
- VERIFY-WITH-RAJESH whether a peak quarter was ever held out.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "99.1% — is that different from 99%? What was your power?"

The numbers: n = 2,500, 22 late; Wilson [98.67%, 99.42%]; z vs 99% = 0.60; power 99% at 98.0%, 76% at 98.5%; blind below ≈ 0.5 pt.

Say exactly: "It is not different from 99 — the interval is 98.7 to 99.4, z is 0.6 — and it is clearly different from the bell-curve plan at 96.5. The test catches a coverage of 98 percent about 99 times in a hundred and a half-point shortfall about three in four, so I do not read it as precise; that is why the drift flag exists."

T2 "How do you know it was not just an easy quarter?"

The numbers: Same 2,500 shipments: old plan late 175 (7%), new plan 22; McNemar $\chi^2 = 145.4$; training window 12 months, all seasons.

Say exactly: "I scored the same shipments against the old plan, so an easy quarter flatters both plans equally — the old plan was late on about a hundred and seventy-five, the new on twenty-two. One quarter is still one draw, and I would add a rolling origin with a peak quarter in it."

T3 "Zero stockouts — what does that prove?"

The numbers: 13 weeks × 150 lanes = 1,950 lane-weeks; upper bound 3 ÷ 1,950 = 0.154% per lane-week (95%); per-lane quarter n ≈ 13 → P(0 late | 99%) = 87.8%.

Say exactly: "It bounds the rate, it does not prove zero: no events in about two thousand lane-weeks puts the stockout rate under a sixth of a percent per lane-week at ninety-five percent confidence. And it says nothing about any one lane — at thirteen shipments a quarter a lane at exactly ninety-nine percent shows no late shipment 88 percent of the time."

S9 — Elementum — 1,000 routes asked, 597 built first: spend Pareto, a ≥95% valid-ASN gate held twice, completeness 60% → 95%, freight spend −10% ≈ $2M in six months NEW

LPs this story serves here: Have Backbone; Disagree and Commit (primary — re-scoped an executive's 1,000-in-two-months ask) · Insist on the Highest Standards (held a cohort twice) · Dive Deep (Pareto, lane scoring, EDI survey) · Invent and Simplify (the standard was the data, not the transport)
Process family: Source / Deliver seam — inbound visibility onboarding at scale (lane scoring → standard → gate → scorecard)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Have Backbone; Disagree and Commit** — I took the 1,000-lanes-in-two-months ask two levels up with the spend curve and replaced it with 597 lanes = 80% of spend first — then committed fully.
- **Insist on the Highest Standards** — One cohort failed the ≥95% valid-ASN gate twice on a timestamp format; I held it both times — automating dirty data manufactures confident wrong answers.
- **Dive Deep** — I built the spend Pareto and lane scoring (spend × criticality × mode × volume) and the carrier EDI-capability survey before taking a scope to the executive.
- **Invent and Simplify** — The onboarding standard was the data set plus an XML/portal floor — "the standard was the data, not the transport" — and a parameterized harness replaced bespoke per-carrier maps.

Honesty label. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships.

60-SECOND PITCH

"**Situation.** The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: spend Pareto — 597 lanes = 80% of freight spend — pre-validated with Transportation and Procurement. Plan & Design: X12 850/855/856/940/945/214/997 with an XML/portal floor; cohort gate ≥95% valid ASN. Build & Test: mapping-and-test harness; scorecards per lane. Cutover: 597 live in month one; one cohort failed twice on a timestamp format — I held it. Hypercare: completeness scorecards to 95%; ~200 next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness — **60% → 95%**. $ impact: freight spend −10% ≈ $2M in six months (record); my model splits it into routing-guide compliance, mode discipline and expedites avoided, and adds ≈$0.4M/yr of modeled document-handling labor on top; 597 lanes = 80% of spend live in month one; adoption 8×."

20-SECOND VERSION NEW

A 1,000-lanes-in-two-months ask became 597 lanes carrying 80% of spend first behind a ≥95% valid-ASN gate I held twice: completeness 60% → 95%, freight spend −10% ≈ $2M in six months (record, not annualized), adoption 8×.

NACF BRIDGE — ONE SENTENCE

Inbound ASN completeness and validity are the first dependency of any NACF placement or WOC model — I would score vendors by volume Pareto and gate activation on valid-ASN rate, not on a date.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The customer's Global Head of Supply Chain wanted end-to-end inbound visibility on 1,000 global lanes in two months — scope, schedule and quality all fixed. Lanes differed by orders of magnitude in spend, volume and

criticality; carriers differed in EDI capability; completeness was 60% and high-value lanes were effectively blind.

T — Own the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor.

The goal: visibility that was actually trustworthy, at the spend that mattered first. The risk of taking the ask literally: automating dirty data manufactures confident wrong answers — 1,000 lanes at 60% completeness in two months would have shipped exactly that — and without a floor for carriers without EDI, the long tail gets priced out of the standard.

The metrics on the table, with their baselines: shipment-data completeness — 60%; high-value lane visibility — effectively blind, target 100% on the lanes carrying the spend; the activation standard — none existed, target a ≥95% valid-ASN cohort gate a lane had to clear before going live.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases:

- **Discovery** — built the spend Pareto and lane scoring model (spend × criticality × mode × volume) and the carrier EDI-capability survey; pre-validated the 597 / ~200 / remainder scope with Transportation and Procurement before taking it up. *Exit bar:* lane owners cannot contradict the cut; the executive accepts the phased plan in spend share, not lane count.
- **Plan & Design** — wrote the onboarding standard (document set + XML/portal floor — "the standard was the data, not the transport"), the ≥95% valid-ASN cohort gate, the per-lane scorecard spec (events received / expected; 997 rate; tracking-number joinability), the harness design. *Exit bar:* standard signed by Transportation; gate published to carriers before cohort 1; harness templates exist.
- **Build & Test** — held engineers to the harness (per-partner mapping templates, test scripts, DQ checklist), test files per carrier, a parallel run per lane, scorecards live. *Exit bar:* each cohort's lanes clear ≥95% valid ASN in test before activation.
- **Cutover & Go-Live (per cohort)** — activated cohorts only off the published gate; published the scorecards; recorded the hold decision when a cohort failed. *Exit bar:* cohort active only after the gate; a failing cohort held — twice, on a timestamp format.
- **Hypercare / Close** — weekly completeness scorecards; harness reused for later waves; method written up as the sourcing SOP (spend file in → phased plan out). *Exit bar:* completeness 60% → 95%; 100% visibility on high-value lanes; remaining waves behind the same gate.

The scope analysis and the standard were mine end to end — the spend Pareto and lane scoring model, the carrier EDI-capability survey, the onboarding standard, the gate, the scorecard spec and the hold decisions; engineers built the mapping templates, test scripts and harness, and Transportation held the carrier relationships and accepted the holds. The contribution: pre-validating the 597 cut with Transportation and Procurement so no lane owner could contradict it in the leadership meeting, and publishing the ≥95% gate to carriers before cohort one — a gate that only holds in private is a suggestion.

The obstacles: carriers unable to produce valid ASNs — answered with the XML/portal floor (“the standard was the data, not the transport”) and a hold rule; a cohort that failed the gate twice on a timestamp format — held both times, fixed, then activated; bespoke maps per partner — avoided by the parameterized harness. Priorities ran on spend share, not lane count — 597 lanes carrying 80% of spend first, ~200 next, the remainder after, each behind the same gate; buy-in came from the pre-validation, then the executive accepted the phased plan off the curve. The decision challenged was the 1,000-in-two-months ask itself — taken two levels up with the spend curve, deferring ~400 lanes carrying ~20% of spend, priced — and I refused activating any cohort that failed the gate.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: completeness **60%** → **95%**; 597 lanes = 80% of spend live in month one; freight −10% ≈ $2M in six months (record); adoption 8×.

- **$** — Freight spend −10% ≈ $2M in six months on the onboarded lanes (record); my three-way split (routing-guide compliance, mode discipline, expedites and detention) decomposes the record, never adds to it; ≈$0.4M/yr modeled document-handling labor on top.

- # — 597 of 1,000 lanes = 80% of freight spend live in month one; ~200 the next wave; the remainder behind the same gate (record).
- **%** — Completeness 60% → 95%; the ≥95% valid-ASN cohort gate — held twice on a cohort that failed on a timestamp format (record).
- **Time** — 597 lanes live in month one; the record $2M landed within six months (record).
- **Customer** — 100% visibility on high-value lanes; small carriers kept inside the standard by the XML/portal floor instead of being priced out (record).
- **Quality** — The gate is the product — automating dirty data manufactures confident wrong answers; adoption 8×; the method written up as the sourcing SOP (record).

Against the goal stated above: completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes = 80% of spend live in month one at ≥95% valid ASN, the remaining waves behind the same gate — with freight spend −10% ≈ $2M in six months funding the next wave. The trade was explicit and priced: ~400 lanes carrying ~20% of spend deferred so the 597 could go live at the highest data quality — and quality was never traded: a lane that slipped below its scorecard went back to “fix,” not “live.”

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	1,000 lanes in two months with scope, schedule and quality all fixed and no onboarding standard; I defined the standard and the gate.
Scope & influence	An executive two levels above my sponsor; Transportation, Procurement, carriers and engineers — none reporting to me.
Execution	597 lanes live in month one; one cohort held twice; scorecards per lane; harness parameterized for later waves.
Impact	Completeness 60% → 95%; freight spend −10% ≈ $2M in six months (record); adoption 8×.
Process improvement	The method is written up as the sourcing SOP: spend file in → phased plan out.
Complexity	Lanes differing by orders of magnitude in spend and criticality; carriers without EDI; per-partner formats.
Root cause	The root cause of blind high-value lanes was no onboarding standard and no gate — not a shortage of integrations.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: annual freight spend per lane, criticality, mode, volume; carrier EDI-capability survey; shipment-event completeness 60%. Revealed: 597 of 1,000 lanes carry 80% of spend; high-value lanes effectively blind.
2 · Solution design	Options: 1,000 lanes in two months · 597 first behind a gate · everything via EDI only. Chosen: Pareto-first, XML/portal floor, ≥95% valid-ASN cohort gate, scorecard per lane.
3 · Math & data logic	Math: S(597) = 80% (record); completeness 480/800 = 60%; implied spend base $2M ÷ 10% = $20M; document labor 90,000 × 6 min × $45/h ≈ $0.4M; gate lower bounds, grain test and control counterfactual below.
4 · Implementation	Implementation: Pareto pre-validated with Transportation and Procurement → standard and gate published to carriers → harness → cohorts activate only off the gate → weekly scorecards → SOP.
5 · Probe questions	Probe questions: is the $2M causal · what does a ≥95% gate mean with n lanes · event vs shipment grain · why 597 — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

***Challenge* — Why did you choose this story to illustrate Have Backbone; Disagree & Commit?**

Because I took the re-scope — 597 lanes first, not 1,000 — two levels up to the Global Head of Supply Chain with the spend curve and the lane owners' validation attached, then committed to delivering the remainder behind the same gate. And the gate held in public: one cohort failed twice on a timestamp format and I held it both times.

***Probing* — Why did you choose to focus on these results? What other results were important?**

Completeness 60 → 95% is the record the whole program was for, and freight −10% ≈ $2M in six months is what the executive funded the next wave on. The 597 = 80% of spend live in month one is the proof the re-scope was right.

***Probing* — You mentioned dollars — what percentage change is that year over year?**

Not YoY — the $2M is the record as measured: minus ten percent on the onboarded lanes' spend in six months, kept as stated rather than doubled into a run-rate. My three-way split of it decomposes the record, never adds to it.

***Probing* — I'm concerned about your split of the $2M — tell me more.**

The split is a judgment on how the ten points divide; to firm it I would tag every invoice line on the onboarded lanes by cause — off-guide carrier, mode change, expedite — for the six months and let the ledger set the shares.

***Challenge* — What were the lessons learned? What would you have done differently?**

Automating dirty data manufactures confident wrong answers — the gate is the product. Differently: the harness from cohort one, not cohort two.

***Challenge* — How would you implement this at Amazon?**

Vendor and carrier onboarding at scale is a Pareto plus a gate — spend share first, a published data-quality bar a partner must clear before activation, and a floor (portal/templates) so the long tail is kept inside the standard rather than priced out of it.

***Challenge* — How did these results compare to your actual goals?**

The ask was 1,000 lanes in two months; the delivery was 597 lanes = 80% of spend live in month one at ≥95% valid ASN, completeness 60 → 95%, the remaining waves behind the same gate — and the executive accepted the phased plan in spend share, not lane count.

PHASE DEEP-PROBE BANK — Q&A (13). *Answer the phase asked, then stop.*

Discovery — How did you know discovery was done?

When the spend Pareto was pre-validated by Transportation and Procurement and the scope proposal — 597 first, ~200 next, remainder after — could not be contradicted by any lane owner in the leadership meeting.

Who signed the scope?

The Global Head of Supply Chain accepted it after seeing the curve; the lane owners had already validated it. I priced the deferral in spend share, not lane count.

Plan & Design — What was in your risk register at design sign-off?

Carriers unable to produce valid ASNs, the scope being contradicted in the room, and bespoke maps per partner. Mitigations: XML/portal floor plus a hold rule, pre-validation, and a parameterized harness.

What was the entry bar for a cohort?

Standard and gate published to its carriers, mapping templates instantiated, test files in hand, a scorecard per lane. Then ≥95% valid ASN in test before activation.

How did you set the standard?

850/855/856/940/945/214/997 on X12, with XML or portal templates as the floor for carriers without EDI. The standard was the data, not the transport.

Build & Test — What was your definition of done for a lane?

Valid-ASN rate ≥95%, 997 acknowledgments flowing, tracking numbers joinable, completeness on its scorecard. Below that, the lane was fixed, not automated.

Describe your bug-bash equivalent.

The cohort test day: every carrier's file through the harness in one sitting; defects classified as format, timestamp, missing segment or join key, each with an owner. That is where the timestamp defect surfaced.

Cutover — Which cohort did you hold and how did the business take it?

One cohort failed twice on a carrier timestamp format; I held it both times. Transportation accepted because the 95% gate was published before cohort one — a gate that only holds in private is a suggestion.

What did each wave deliver?

Wave 1: 597 lanes, 80% of spend, live in month one. Wave 2: ~200 lanes on a harness that made the work mostly configuration. Wave 3: the tail, including non-EDI carriers on templates.

Hypercare — How did you know you were done? What did you watch?

Completeness 60% → 95%, 100% visibility on high-value lanes, the harness reused without me, the method adopted as the sourcing SOP. Watched weekly: per-lane completeness, the 997 rate, tracking-number joinability. A lane that slipped below its scorecard went back to "fix," not "live."

Cross-phase — Where did you trade scope for date? What did you escalate?

Deferred ~400 lanes carrying ~20% of spend so the 597 could go live at the highest data quality in month one — priced, and delivered later behind the same gate. Escalated the re-scope to the Global Head of Supply Chain — two levels up — with the spend curve and the lane owners' validation attached. Refused: activating a cohort that failed the gate.

Did you deliver the remaining 400? · Do differently?

Yes — later waves, same gate; the first 597 carried the spend and the decision. Differently: the harness from cohort one, not cohort two.

Analytics — What does “completeness 95%” count — events or shipments?

The record is shipment-data completeness 60% → 95%; my illustrative arithmetic counts four milestones per shipment (480 of 800 received = 60%). At the event grain 95% means 5% of milestones are missing; at the shipment grain, with independent misses, only 0.95^4 = 81.5% of shipments would be fully complete and the 60% start would be 0.6^4 = 13.0%. Misses cluster by carrier, so the truth sits between the union bound (80% at the 95% start) and 95%.
VERIFY-WITH-RAJESH which grain the 60% and 95% were measured at.

Analytics — A ≥95% valid-ASN gate — what does it mean with a finite test sample?

The record is the gate: the cohort must clear ≥95% valid ASN in test. The cohort sizes are not in the record and I do not quote them. As arithmetic: 38 valid of 40 is 95% observed but its one-sided 95% lower bound is 85.1%; 190 of 200 gives 91.7%; 194 of 200 gives 94.2%. To demonstrate ≥95% rather than observe it you need 196 of 200 (98%) at n = 200 (lower bound 95.5%); at n = 40 it cannot be demonstrated even at 40 of 40 (92.8%). I would add that bound to the scorecard. VERIFY-WITH-RAJESH test sizes and whether the bar was an observed or a bounded rate.

Analytics — How do you know the −10% freight spend came from visibility and not the market?

The record is −10% ≈ $2M on the onboarded lanes in six months; it carries no control. The deferred lanes are a free comparison group: about $5M of spend sits on the ~400 lanes that were not onboarded. If the deferred lanes' spend moved −2% on the market alone, the visibility effect is (−10% − (−2%)) × $20M = $1.6M; at −5% it is $1.0M. I would pull that before I defend a causal claim. VERIFY-WITH-RAJESH whether such a comparison was run.

STATISTICAL SPINE — PARETO, GRAIN, GATE AND CONTROL (EXTRA TABLES)

The record: 597 lanes = 80% of spend; completeness 60% → 95%; gate ≥95% valid ASN; freight −10% ≈ $2M in six months. The rows below are my modeled or illustrative arithmetic for a statistician's questions — labeled row by row. Cohort sizes and the number of cohorts are not in the record and are not quoted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Pareto fit through the record point	`S(k) = (k ÷ 1,000)^a with S(597) = 0.80 → a = ln 0.80 ÷ ln 0.597 = 0.4326`	one modeled curve through the one record point	a = 0.4326
2	Spend share at 300 lanes	`(0.300)^0.4326 = 59.4%`	modeled	59.4%
3	Wave 2 (~200 lanes, to 797)	`S(797) − 0.80 = 90.7% − 80.0% = 10.7%`	modeled; ~200 next wave is record	+10.7% of spend
4	Remainder (~200 lanes)	`1 − S(797) = 9.3%`	modeled; ~400 lanes ≈ 20% deferred is record	9.3% of spend
5	Marginal lane at the cut	`dS/dk = a × S ÷ k = 0.4326 × 0.80 ÷ 597 = 0.058% of spend per lane`	modeled	0.058% / lane
6	Completeness grain, independent misses	`0.60⁴ = 13.0% → 0.95⁴ = 81.5% of shipments fully complete`	four milestones per shipment (illustrative)	13% → 81.5% complete shipments
7	Completeness grain, bounds	`union bound at 95%: 1 − 4 × 0.05 = 80%; ceiling = 95%; at 60%: floor 0%, ceiling 60%`	no independence assumed	shipment-level at 95%: 80%–95%

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Gate: 38/40 valid ASN	`observed 95.0%; one-sided 95% lower bound (Clopper–Pearson) = 85.1%`	illustrative test sizes — cohort sizes are not in the record	85.1%
2	Gate: 40/40 valid ASN	`observed 100.0%; one-sided 95% lower bound (Clopper–Pearson) = 92.8%`	illustrative test sizes — cohort sizes are not in the record	92.8%
3	Gate: 95/100 valid ASN	`observed 95.0%; one-sided 95% lower bound (Clopper–Pearson) = 89.8%`	illustrative test sizes — cohort sizes are not in the record	89.8%
4	Gate: 190/200 valid ASN	`observed 95.0%; one-sided 95% lower bound (Clopper–Pearson) = 91.7%`	illustrative test sizes — cohort sizes are not in the record	91.7%
5	Gate: 194/200 valid ASN	`observed 97.0%; one-sided 95% lower bound (Clopper–Pearson) = 94.2%`	illustrative test sizes — cohort sizes are not in the record	94.2%
6	Gate: 196/200 valid ASN	`observed 98.0%; one-sided 95% lower bound (Clopper–Pearson) = 95.5%`	illustrative test sizes — cohort sizes are not in the record	95.5%
7	Gate: 490/500 valid ASN	`observed 98.0%; one-sided 95% lower bound (Clopper–Pearson) = 96.6%`	illustrative test sizes — cohort sizes are not in the record	96.6%
8	Implied spend base	`$2M ÷ 10% = $20M; total spend = $20M ÷ 0.80 = $25M; deferred = $5M`	derived from the record pair and S(597) = 80%	$5M deferred spend

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
9	Control: deferred lanes drift 0% / −2% / −5%	`(−10% − drift) × $20M = $2.0M / $1.6M / $1.0M`	illustrative difference-in-differences; the record has no control	$1.6M at −2%

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Inbound freight spend -10% ~= $2M in 6 months (record)
+- Visibility (stock): completeness 60% -> 95%, high-value lanes 100%
|    +- routing-guide compliance: off-guide carrier spend caught
|    +- mode discipline: air -> ocean/ground when ETA is trusted
|    +- expedites + detention: chase calls replaced by 214 events
+- Document handling (flow): 850/855/856/940/945/214 by X12, not by hand
+- DQ gate loop: >=95% valid ASN -> cohort activates -> scorecard per lane ->
     defects fixed as data -> harness parameterized -> cohort 3 mostly config -> adoption 8x
+- Scope loop: 597 = 80% spend first -> $ lands month one -> executive funds the next wave
```

DRIVER	BASELINE	AFTER	$	BASIS
Shipment-data completeness	60%	95%	—	record
Lanes live month one	0	597 = 80% of spend	—	record
Freight spend on onboarded lanes	~$20M over the six months (packet-derived)	−10%	≈$2M / 6 mo	record ($2M)
— routing-guide compliance	off-guide unseen	caught by 856/214	~$0.8M	modeled split of the record $2M
— mode discipline	premium modes by habit	ETA trusted	~$0.6M	modeled split
— expedite + detention avoided	blind lanes chased	events replace calls	~$0.6M	modeled split
Manual document handling	~90K docs/yr keyed	X12/XML validated	$0.4M/yr	modeled (90K × 6 min × $45/hr)

If challenged. *"Is the $2M annualized?"* — "No — it's the record as measured: minus ten percent on the onboarded lanes' spend in six months. I keep it as stated rather than doubling it into a run-rate." *"How do you know the split?"* — "I don't, to the dollar — the split is my model of where the ten percent came from, and the three slices sum to the record, never on top of it."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing
2	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	record problem	no instrument, no lever
3	Manual documents	`597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand`	illustrative volumes	~90K docs/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (record)	597 of 1,000 lanes carry 80% of spend — do them first
2	Cohort gate	`validASN rate ≥ 95% in test before activation`	published to carriers before cohort one (record)	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	“the standard was the data, not the transport” (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%

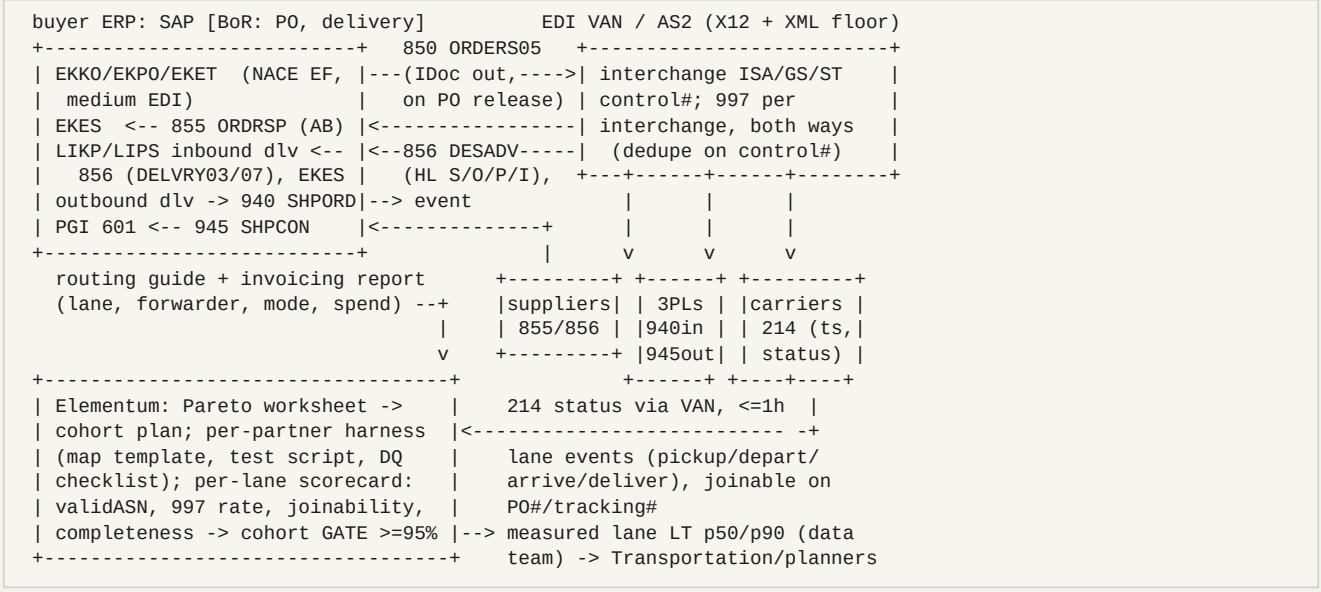
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes live month one	0	597 = 80% of spend	record
3	Freight spend on onboarded lanes	base	−10% ≈ $2M in 6 months	record
4	Document handling	~90K docs/yr keyed	X12/XML validated	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000 of spend over the six-month window`	derived from the record pair, stated as such	≈$20M (six-month base)
2	The record result	`$20M × 10% = $2M in 6 months`	record	≈$2M
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`0.8 + 0.6 + 0.6 = $2.0M`	the slices decompose the record — they never add to it	= the record $2M
7	Document labor	`90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000`	illustrative volumes (modeled add-on)	≈$0.4M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Shipments/yr on 597 lanes (illustrative)	20,000	30,000	40,000
Minutes per manual document (benchmark)	4	6	8
Loaded coordinator rate (benchmark)	$40	$45	$55
Total modeled $/yr	≈$0.2M	≈$0.4M	≈$0.9M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured number is freight spend down ten percent, about two million dollars in six months on the five hundred ninety-seven lanes that carried eighty percent of spend, with completeness sixty to ninety-five percent and adoption eight x. Inside that two million I model roughly eight hundred thousand from catching off-guide carrier spend, six hundred thousand from mode discipline once the ETA was trusted, and six hundred thousand from expedites and detention that chase calls used to cause; those three are my split, the two million is the record. On top of it I model about four hundred thousand a year of document-handling labor that X12 replaced, and that piece is entirely mine."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you pushed back on a senior leader's request.**
HAVE BACKBONE; DISAGREE AND COMMIT

Open (first two sentences): The Global Head of Supply Chain wanted 1,000 lanes of inbound visibility in two months. I told him the schedule would deliver 1,000 lanes of confident wrong data — and showed him the spend curve instead.

STAR (≈90 s): Situation: scope, schedule and quality were all fixed, completeness was 60% and high-value lanes were blind. Task: own the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. Action: I built the spend Pareto, pre-validated 597 / ~200 / remainder with Transportation and Procurement so no lane owner could contradict the cut, and took it up in spend share, not lane count. He accepted the phased plan off the curve; I committed completely. Result: 597 lanes carrying 80% of spend live in month one; ~400 lanes carrying ~20% deferred, priced, behind the same gate.

Land this number: 597 lanes = 80% of spend; ~400 deferred = ~20%. · **Pivot to NACF:** For NACF vendor onboarding I would lead with the volume Pareto and a gate, not a lane count. · **Do NOT say:** Do not name the executive's reaction beyond "accepted the phased plan"; do not quote cohort sizes.

Q2 **Tell me about a time you refused to compromise on quality under schedule pressure.**
INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): A cohort failed our ≥95% valid-ASN gate twice on a carrier timestamp format. I held it both times.

STAR (≈90 s): Situation: month one, 597 lanes going live, the executive watching. Task: protect the standard without becoming the blocker. Action: the gate was published to carriers before cohort one — a gate that only holds in private is a suggestion; the failing cohort stayed in test, we fixed the format in the mapping template, retested, and activated off the gate. Result: completeness 60% → 95%; the gate is the product — automating dirty data manufactures confident wrong answers.

Land this number: Gate ≥95% valid ASN; held twice; completeness 60% → 95%. · **Pivot to NACF:** For NACF inbound the gate is "valid ASN rate before the vendor lane feeds placement." · **Do NOT say:** Do not say how many lanes were in the failed cohort.

Q3 **Tell me about a time you used data to decide where to focus.** DIVE DEEP

Open (first two sentences): Lanes differed by orders of magnitude in spend, volume and criticality, so I scored all 1,000 before I scoped anything.

STAR (≈90 s): Situation: a flat 1,000-lane plan would spend most of the effort on the long tail. Task: decide what to do first. Action: spend × criticality × mode × volume scoring, plus a carrier EDI-capability survey, so every lane had a score and a feasibility. Result: 597 lanes = 80% of spend first, ~200 next, the rest behind the same gate — and a method written up as the sourcing SOP.

Land this number: S(597) = 80%; scoring on four factors. · **Pivot to NACF:** Placement analytics starts the same way: rank by volume × cost × criticality, then gate. · **Do NOT say:** Do not call the modeled Pareto fit the record — only the 597 = 80% point is record.

Q4 **Tell me about a time you simplified a standard so more people could meet it.** INVENT AND SIMPLIFY

Open (first two sentences): Carriers without EDI would have been priced out of the standard, so I made the standard the data, not the transport.

STAR (≈90 s): Situation: some carriers could not produce valid ASNs over EDI. Task: keep the long tail inside one standard. Action: 850/855/856/940/945/214/997 on X12 with an XML or portal floor, parameterized per-partner mapping templates and a test harness instead of bespoke maps. Result: small carriers stayed in the standard; adoption 8×; the harness was reused for later waves.

Land this number: Adoption 8×; XML/portal floor. · **Pivot to NACF:** NACF vendors without EDI need the same floor: the standard is the data. · **Do NOT say:** Do not say I built the harness; engineers did to my standard.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "597 of 1,000 lanes carried 80% of the spend."

Headline metric: S(597) = 80%; completeness 60% → 95%

- Pareto + four-factor lane scoring + EDI survey.
- Modeled wave 2 ≈ +10.7% of spend (~200 lanes).
- Shipment-grain completeness at 95% events: 80%–95%.
- Freight −10% ≈ $2M (record, not annualized).

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Disagree and Commit

Open: "I did not take the 1,000-lane ask literally."

Headline metric: 597 live in month one; ~400 deferred, priced

- Took the curve two levels up; pre-validated with owners.
- Committed fully once accepted.
- Held a failing cohort twice.
- Deferral priced: ~20% of spend, same gate.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Insist on the Highest Standards; Learn and Be Curious

Open: "A cohort failed the gate twice on a timestamp format."

Headline metric: gate ≥95% valid ASN; failed twice, held twice

- What I would do differently: a timestamp-format check in the mapping template pre-test. VERIFY-WITH-RAJESH
- Gate as an observed rate, not a bound — I would add the lower bound.
- No control on the −10% — the deferred lanes are one ($1.6M at −2% drift).
- The record is the failure and the hold; the fix is mine to describe, not to inflate.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Is the $2M causal? What is your control group?"

The numbers: Record: −10% on ≈$20M ≈ $2M, no control. Deferred lanes ≈ $5M of spend; if they drift −2%, effect = (−10% + 2%) × $20M = $1.6M; at −5%, $1.0M.

Say exactly: "The two million is the record as measured on the onboarded lanes — it has no control. The four hundred deferred lanes, about five million dollars of spend, are the natural comparison; if they drifted two percent on the market alone, the visibility effect is about one point six million, and I would pull that before defending causality."

T2 "A ≥95% valid-ASN gate — what sample size makes that meaningful?"

The numbers: 38/40: observed 95%, lower bound 85.1%; 40/40: 92.8%; 190/200: 91.7%; 194/200 (97%): 94.2%; 196/200 (98%): 95.5%.

Say exactly: "The gate was an observed rate in test — ninety-five percent valid ASN. As statistics, ninety-five observed on forty only bounds the true rate above eighty-five, and on two hundred above ninety-two; to show at least ninety-five you need ninety-eight observed on two hundred, and on forty it cannot be shown at all. I would put the bound on the scorecard next to the rate — and I will not quote cohort sizes that are not in the record."

T3 "Completeness 60 → 95% — at which grain, and what does 95% of events mean for shipments?"

The numbers: Four milestones: 0.6^4 = 13.0% → 0.95^4 = 81.5% complete shipments if independent; bounds at 95%: 80%–95%.

Say exactly: "The record is shipment-data completeness, sixty to ninety-five. If that is events, then with four milestones a shipment is fully complete eighty-one percent of the time when misses are independent, and between eighty and ninety-five when they cluster by carrier — so I report both, and I would confirm the grain before quoting a shipment-level figure."

S19 — Elementum / Walmart Marketplace — the delivery promise: one governed buffer dial instead of a flat 1.10 days — fast-promise share 7.6% → 39.6% at the held 95% floor (engagement model) NEW

LPs this story serves here: Invent and Simplify (one dial) · Customer Obsession (a promise the customer can rely on) · Dive Deep (the Φ(5.0) vs Φ(0.69) audit) · Insist on the Highest Standards (shadow gate, peak re-solve)
Process family: Deliver seam — pre-purchase promise policy for seller-fulfilled delivery (Walmart Marketplace, an Elementum client; role R-2596147 Pre-Purchase Delivery Experience)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I decomposed the 95.8% headline into two cohorts: the flat 1.10-day buffer bought Φ(5.0) — near-certainty — on the tightest cohort and Φ(0.69) ≈ 75% on the widest.
- **Invent and Simplify** — I replaced a hundred model micro-decisions with one governed dial, b = k·σ with k = 1.645, that a business owner can read every Monday.
- **Customer Obsession** — The customer blames the marketplace, not the seller, when Thursday becomes Saturday — the floor was a promise kept 95 times in 100 on every cohort.
- **Insist on the Highest Standards** — Four consecutive shadow weeks before any displayed promise, a published promotion bar and a warm rollback; and every figure said “the model” in the same breath.

Honesty label. Elementum engagement for a marketplace client — Walmart Marketplace's seller-fulfilled delivery program, where the client prints "Get it by Thursday" on inventory that sits in sellers' buildings. I led the promise-policy workstream: the decomposition, the buffer-policy alternatives, the validation gates and the governance design. The client's promise-engineering and data-science teams owned and built the promise engine and the transit models — I never configured or deployed any platform (the only platform I personally configured remains Manhattan DOM). The twelve-cohort book behind every number is the engagement's illustrative model, built so the argument could happen in numbers instead of adjectives.

60-SECOND PITCH

"**Situation.** A marketplace client promising delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and the buffer was a flat 1.10 days on every lane regardless of how variable the lane was. In the engagement model — twelve lane cohorts, about a million orders a week, transit spread from 0.22 to 1.60 days — that flat buffer printed 95.8% accuracy but put only 7.6% of orders on a fast promise, because dense metro lanes were over-insured and fat-tailed rural lanes under-insured at the same time. **Task.** Engagement lead for promise policy: own the decomposition, the alternatives, the gates and the governance recommendation; the client's data science and engineering did not report to me. **Action.** Discovery: the buffer-waste audit — the flat buffer bought Φ(5.0), near-certainty, on the tightest cohort and only Φ(0.69) ≈ 75% on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: b = 1.645σ holds 95% on every cohort by construction; the equimarginal check proved the optimum; the Q4 drill showed a frozen summer dial breaking the floor. Cutover: four shadow weeks against logged live orders before any displayed promise. Hypercare: the dial published in the weekly review header, re-solved before peak. **Result.** Primary metric: fast-promise share — **7.6% → 39.6%, +32 points at the held 95% floor**; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all on the illustrative book, and I say 'the model' in the same breath."

20-SECOND VERSION NEW

A flat 1.10-day buffer printed 95.8% accuracy and put 7.6% of orders on a fast promise; one governed dial, b = 1.645σ, kept every cohort at 95% and lifted fast-promise share to 39.6% — +32 points on the engagement's illustrative model, never a production claim.

NACF BRIDGE — ONE SENTENCE

NACF promise and appointment buffers should be risk-proportional, not flat: buffer = k × the lane's own σ, one published dial, re-solved before peak — the same logic as INV-B's z·σ·√LT.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — An Elementum engagement with Walmart Marketplace's seller-fulfilled delivery program, in the client years. The client underwrites a delivery promise on execution it does not own: the seller picks and packs, a carrier it only partially controls carries, and the customer blames the marketplace — not the seller — when Thursday becomes Saturday. The promise identity is a sum: promised date = order time + configured handling + predicted transit + buffer; the promise is kept when the handling error plus the transit error fit inside the buffer. The incumbent buffer policy was uniform — about 1.10 days on every lane — set once, owned by nobody between reviews.

T — Lead the promise-policy workstream: decompose where the incumbent buffer was wasted and where it was thin, put priced alternatives on one table, design the validation gates, and leave the client a governance instrument — not a tuning habit. Everything quantitative ran on the engagement's illustrative model: twelve lane cohorts spanning σ = 0.22 days (metro two-day) to 1.60 days (mountain rural), about a million orders a week.

Why it mattered: padding fails silently and cutting fails loudly, so every promise system drifts slow — the orders that never happened because the promise read five days file no tickets. The economics were stated up front and carried on every page: a breach costs about $12 (contact + refund risk + trust decay) against about $6.50 of contribution margin — both the model's stated assumptions, never a measured P&L. Left unfixed: the client keeps paying for insurance it doesn't need on tight lanes while underwriting known losses on loose ones, and the speed-versus-accuracy dial keeps being set implicitly by whoever tuned the last model.

The metrics on the table, with their baselines (all illustrative-model values): fast-promise share (orders promised in ≤2 days) — 7.6%; portfolio promise accuracy — 95.8% against a 95% floor; average promised day — 3.09; weekly modeled profit — $5.69M; and no governed owner of the speed-accuracy trade — the real baseline.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases, run as an engagement arc (the model, the table and the gates are the engagement record; no phase touched a production system — the client's teams built, I designed and governed):

- **Discovery** — wrote the promise identity and the accuracy factorization (accuracy ≈ P(seller ships in window) × P(carrier meets transit) × Φ(buffer/σ)); built the buffer-waste audit — days delivered early on kept promises, ranked by waste × volume — and the two-line indictment of the flat buffer: 1.10 days is 5.0 standard deviations on the tightest cohort and 0.69 on the widest. *Exit bar:* the client's own cohort table agreed in the room; "how much buffer" retired as a question, replaced by "where should it sit."
- **Plan & Design** — four policies written onto one comparison table on the same cohort set: A uniform (incumbent), B σ-proportional with one governed dial (b = k·σ), C minimum-promised-days at the floor (analytic optimum), D profit-optimal on the priced breach economics; the gates designed — shadow mode, a promotion bar, a rollback that keeps the incumbent computable. *Exit bar:* the table signed as the decision instrument; the dial (k) named as a governed, published policy instrument with an owner.
- **Build & Test** — the client's data-science team computed the table; I owned the proofs: Φ(1.645) = 0.95 holds the floor on every cohort by construction; the equimarginal condition for C (φ(b/σ)/σ equal across cohorts) verified to machine precision; the Q4 drill — σ inflated ~1.45× — showed the frozen summer dial printing 91.9%, below the floor. *Exit bar:* every policy's accuracy, speed and modeled profit reproduced independently by the client's team; the peak-regime rule written.

- **Cutover / Go-Live (design)** — shadow mode: the challenger policy computed and logged per order, never displayed, for four consecutive weeks; promotion only if would-be accuracy held the floor on logged live orders with no rise in contact signals. *Exit bar:* promotion bar published before the first shadow week; rollback = one flag back to the incumbent, kept warm.
- **Hypercare / Close** — the governance handoff: k printed in the weekly review header every Monday; tail metrics (p90 error, buffer-exceedance) on the page next to the average; the buffer-waste line standing; k re-solved per regime — published in October for peak, never frozen through it. *Exit bar:* the client's business owner — not data science — named as the dial's owner; the engagement closed on the instrument, not on a tuning.

The client's promise-engineering team owned the engine and its latency budget; their data-science team owned the transit models and σ estimates; their finance partner owned the breach price. Mine: the factorization, the waste audit, the four-policy table, the equimarginal and Q4 proofs, the shadow-mode gates and the governance design — the reframe from "a hundred ML micro-decisions" to one auditable dial a business owner reads on a Monday.

The obstacle was the room's instinct that accuracy at 95.8% meant the policy was healthy — the decomposition had to beat a green headline. The two-cohort arithmetic did it: near-certainty bought where σ was tiny, a coin-flip-plus where σ was fat. The decision I challenged: data science wanted per-lane learned buffers — more model, more value, invisible governance. I priced the difference and recommended the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The no went into the comparison table as a priced row, not a meeting minute.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: fast-promise share 7.6% → 39.6% (+32 points) at the identical 95% accuracy floor; average promised day 3.09 → 2.75; +$255K/week ≈ $13.3M/yr — every figure from the illustrative model, said with "the model" in the same breath.

- **$** — +$255K/wk vs the uniform policy ≈ $13.3M/yr (modeled on the illustrative ~1M-order book); the profit-optimal frontier (policy D: 97.8% accuracy, $6.07M/wk, ≈ +$20M/yr vs uniform) handed to the client's finance partner as a priced governance decision — shown, never claimed, never merged with the $13.3M.
- **#** — Twelve lane cohorts, ~1M orders/week, σ from 0.22 to 1.60 days; four policies on one table; one governed dial, k = 1.645 (illustrative model; the table and the dial are the engagement's design record).
- **%** — Fast-promise share +32.0 points at a held 95.0% floor; the incumbent's hidden split exposed: Φ(5.0) ≈ 99.99997% on the tightest cohort against Φ(0.69) ≈ 75.4% on the widest (model).
- **Time** — Average promised day 3.09 → 2.75 (−0.34); the tightest cohort's buffer 1.10 → 0.36 days; the widest cohort's buffer 1.10 → 2.63 days — longer on purpose, to stop underwriting known losses (model).
- **Customer** — The fast promise shown where it can be kept and withheld where it can't; the Q4 re-solve rule so peak variability never silently breaks the floor the customer experiences (engagement design record).
- **Quality** — The equimarginal check as the solver's own proof of optimality; p90 error and buffer-exceedance on the review page next to the mean, so a better average can never hide a worse tail (engagement design record).

Against the goal stated above: more fast promises without breaking the floor — +32 points at exactly 95.0%, a promise a third of a day shorter on average, and the speed-versus-accuracy trade converted from an implicit model parameter into one governed, published dial. The trade was explicit and priced: the widest cohorts got honestly slower promises so the whole book could get kept ones; nothing was bought with accuracy.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Nobody owned the speed-vs-accuracy trade; I defined it as one dial with a floor and an owner.
Scope & influence	Client promise engineering, data science, finance and seller programs — none reporting to me; the "no" to per-lane learned buffers went in as a priced row.
Execution	Four-policy table, two proofs (Φ(1.645) floor and equimarginal), Q4 drill, shadow gate and rollback designed.
Impact	+32 pts fast-promise share at a held 95% floor; +$255K/wk ≈ $13.3M/yr — all engagement model, labeled as such.
Process improvement	k published weekly and re-solved per regime — governance, not tuning.
Complexity	Twelve cohorts with σ from 0.22 to 1.60 days; heavy tails; a peak regime that breaks a frozen dial.
Root cause	A flat buffer ignores variance: it over-insures dense lanes and under-insures fat-tailed ones at once.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: twelve lane cohorts, σ 0.22–1.60 days, ~1M orders/week (the engagement's illustrative model); promise identity = order + handling + transit + buffer. Revealed: flat 1.10 d = 5.0σ on the tight cohort, 0.69σ on the wide one.
2 · Solution design	Options: A uniform · B σ-proportional with one dial · C minimum-days at the floor · D profit-optimal. Chosen: B first (auditable, one dial); D shown and never claimed.
3 · Math & data logic	Math: P(kept) = Φ(b/σ); b = kσ with Φ(k) = 0.95 → k = 1.645; tight 0.36 d, wide 2.63 d; Q4 σ × 1.45 → z = 1.13 → 87.2% per cohort, 91.9% portfolio (record-from-docs); price-of-floor, estimation-error and heavy-tail arithmetic below.
4 · Implementation	Implementation: waste audit → four-policy table → proofs → 4-week shadow → published promotion bar → k in the weekly review header; re-solved in October.
5 · Probe questions	Probe questions: σ estimation error · heavy tails · why not policy D · how the floor behaves at peak — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

> ***Challenge* — Why did you choose this story to illustrate Dive Deep?**
>
> Because the finding only exists below the headline: portfolio accuracy printed a healthy 95.8%, and the depth — per-cohort z-scores — showed the same policy was simultaneously over-insuring metros at five standard deviations and under-insuring rural lanes at 0.69. A team that stops at the portfolio number never finds thirty-two points of free speed.

> ***Probing* — Why did you choose to focus on these results? What other results were important?**
>
> The +32 points at a held floor leads because it survives any parameter change — it comes from re-allocating risk, not spending money. Also important: the honest cost (the widest cohorts' promises got longer), the Q4 regime rule, and the governance artifact — a dial the business reads weekly — which outlives any one solve.

> ***Probing* — You mentioned dollars — what percentage change is that year over year?**
>
> I don't give this one as year-over-year — every dollar here is the model's: +$255K/wk is about 4.5% of the modeled $5.69M weekly contribution, ≈ $13.3M annualized on the illustrative book. There is no production dollar in this engagement and I never promote one.

> ***Probing* — I'm concerned the whole story rests on an illustrative cohort set — tell me more.**
>
> It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers should be risk-proportional, the last buffer-day should buy the same accuracy everywhere, and a floor that disagrees with the breach economics is a governance finding.

***Challenge* — What were the lessons learned? What would you have done differently?**

The lesson: padding is not safety — a flat buffer is slow and unsafe at the same time, and only the decomposition shows it. Differently: price the breach bundle with the finance partner in discovery rather than design — the $12-versus-$6.50 conversation is what turned the floor from theology into economics, and it came one phase late.

***Challenge* — How would you implement this at Amazon?**

The same shape at the promise seam: decompose the promise identity, audit where buffer is wasted on kept promises, one governed dial per regime with the tail metrics on the page, shadow before display, re-solve before peak. At Amazon's scale a tenth of a point of buffer waste is real money — which is why the governance matters more than any single solve.

***Challenge* — How did these results compare to your actual goals?**

The goal was more fast promises without breaking the floor. The model delivered +32 points at exactly the floor and a promise a third of a day shorter — and the frontier beyond it (policy D at 97.8%) was handed to Finance as a priced decision rather than smuggled in as a default.

PHASE DEEP-PROBE BANK — Q&A (12). *Answer the phase asked, then stop.*

Discovery — How did you know discovery was done?

When the room accepted one sentence of arithmetic: the flat 1.10-day buffer is 5.0 standard deviations on the tightest cohort and 0.69 on the widest. Once that was agreed on the client's own cohort table, "how much buffer" was dead as a question and "where should it sit" replaced it.

Why a buffer-waste audit instead of starting from accuracy?

Accuracy printed 95.8% — a healthy-looking headline hiding the split. Waste — days delivered early on kept promises — is where the recoverable speed lives, and it is unarguable because it only counts promises that were kept. Ranked by waste × volume, the audit is the trim roadmap with no new models needed.

Plan & Design — Why four policies instead of recommending one?

Because the client's real decision was governance, not mathematics. A is what you run today; B is one auditable dial; C is the fastest promise the floor allows; D is what the economics want. All four on one cohort set made the trade-offs priced choices instead of expert opinion — and made B's simplicity a visible decision, not a default.

Why recommend B over the profit-optimal D?

D's optimum moves every time the breach price or margin assumption moves, and it is harder to audit. B is one dial a business owner reads on a Monday and it captures most of the value. I recommended running B and letting Finance decide whether to chase the D frontier with its own signature on the $12.

Build & Test — What exactly is the equimarginal check?

The optimality condition for policy C: the marginal accuracy bought by the last unit of buffer — $\varphi(b/\sigma)/\sigma$ — must be equal on every cohort; if a buffer-day buys more accuracy on one lane than another, moving it is free improvement. The solver's output showed the condition holding to machine precision — the model proving its own optimum.

What was the Q4 drill and what did it change?

Inflate every cohort's σ by ~1.45× — the model's peak regime — and hold the summer dial: portfolio accuracy printed 91.9%, under the floor (on a single lane taking the full inflation, the whiteboard version is $\Phi(1.645/1.45) = \Phi(1.13) \approx 87\%$ — the portfolio mix lands at 91.9%). The drill wrote the rule: k is re-solved and published before peak, never frozen through it.

Why normal errors — freight tails are skewed?

Said before anyone asked: the normal is the whiteboard version; the production recommendation was the empirical quantile per lane. The normal makes the argument legible; the quantile makes it true. Nothing in the recommendation depends on normality — only the clean Φ numbers do.

Cutover — What was the promotion bar out of shadow mode?

Four consecutive shadow weeks with would-be-promise accuracy at or above the floor on logged live orders and no rise in contact signals. A policy that can't clear its own floor in shadow has no business touching a displayed date — the same rehearse-before-commit discipline as my mock cutovers (S18).

What was the rollback?

The incumbent uniform buffer stayed computable the whole time — one flag back. A promise policy is the easiest class of change to reverse if you keep the old policy warm, and the easiest to make irreversible if you don't.

Hypercare — Why publish the dial in the review header?

Because k *is* the speed-versus-accuracy trade, and a trade the business can't see is a trade it didn't agree to. One number in the header every Monday makes the floor a standing decision instead of a buried parameter.

What keeps the policy honest after handoff?

Three standing lines on the review page: the dial, the tail metrics (p90 error and buffer-exceedance) next to the average, and the buffer-waste line. A policy whose tail and waste are public decays far more slowly than one graded on its mean.

Cross-phase — Where did you trade scope for the date? What did you refuse?

Traded: per-lane learned buffers deferred in favor of the one dial — sophistication for governance, knowingly leaving model value on the table for the first cycles. Refused: promoting on the model's average error (the tail breaks promises, so the tail went on the page), and any record dollar — the book is illustrative and I said so in every read-out.

Analytics — What does the floor cost? Price each step.

On the widest cohort (σ = 1.60 d) the buffer is 2.63 d at a 95% floor, 3.14 d at 97.5% (+0.50 d) and 3.72 d at 99% (+1.09 d); on the tightest cohort (σ = 0.22) the same steps cost 0.07 d and 0.15 d. The dial is linear in σ — so a point of floor is nearly free on dense lanes and expensive on rural ones, which is the whole argument for a risk-proportional rule. Engagement model.

Analytics — Your σ is estimated. What does that do to the floor?

With n orders in a cohort, σ̂ has relative standard error 1 ÷ √(2(n − 1)): 7.1% at n = 100, 2.2% at n = 1,000. If σ̂ is one SE low, a dial of 1.645 realises z = 1.645 ÷ (1 + SE): 1.54 at n = 100 → 93.8%; 94.6% at n = 1,000. Holding the floor within half a point at a one-SE miss needs about 584 orders per cohort per window. At about a million orders a week that is trivially met; the real exposure is regime change, which is why Q4 is the drill.

Analytics — Normal errors — what if the errors are heavy-tailed?

Production uses the empirical quantile; the whiteboard uses the normal. As a sensitivity, take Student-t errors with 5 degrees of freedom scaled to the same σ: the 95% quantile is 1.561σ, below 1.645 — the normal dial is conservative there and delivers 95.6% — but the 99% quantile is 2.607σ, above 2.326, so a normal 99% dial would deliver only 98.5%. The heavier the tail the more the empirical quantile matters at the high floors.

Analytics — With a million orders a week, is the weekly accuracy noise a problem?

No: the standard error of a 95% weekly accuracy on 1,000,000 orders is √(0.95 × 0.05 ÷ 1,000,000) = 0.022 points; on a 100,000-order cohort 0.069. A reading of 91.9% in Q4 is not noise — it is a σ regime change. At this scale the question is never “is it significant,” it is “which factor moved.”

STATISTICAL SPINE — PRICE OF THE FLOOR, Σ-ESTIMATION ERROR, HEAVY TAILS (EXTRA TABLES)

The engagement model is the record-from-docs; these tables are my sensitivity arithmetic on the same cohorts and dial — every figure engagement-model or textbook-normal, none a production measurement.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Floor 90% → k	`Φ(k) = 0.90 → k = 1.2816`	standard normal table	tight 0.28 d · wide 2.05 d

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Floor 95% → k	`Φ(k) = 0.95 → k = 1.645`	record-from-docs dial	tight 0.36 d · wide 2.63 d
3	Floor 97.5% → k	`Φ(k) = 0.975 → k = 1.96`	standard normal table	tight 0.43 d · wide 3.14 d
4	Floor 99% → k	`Φ(k) = 0.99 → k = 2.326`	standard normal table	tight 0.51 d · wide 3.72 d
5	Price of 95% → 97.5% (wide cohort)	`(1.96 − 1.645) × 1.60 = 0.50 d`	σ = 1.60 d (illustrative model)	+0.50 d of promise
6	Price of 95% → 99% (wide cohort)	`(2.326 − 1.645) × 1.60 = 1.09 d`	σ = 1.60 d (illustrative model)	+1.09 d of promise
7	Incumbent accuracy, tight / wide	`Φ(1.10 ÷ 0.22) = Φ(5.0) ≈ 100%; Φ(1.10 ÷ 1.60) = Φ(0.69) = 75.4%`	flat buffer b = 1.10 d	75.4% on the wide cohort

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	σ̂ standard error, n = 100	`1 ÷ √(2 × 99) = 7.1%`	large-sample approximation	7.1%
2	Realised accuracy at a one-SE miss, n = 100	`z = 1.645 ÷ (1 + 0.0711) = 1.536; Φ = 93.8%`	σ̂ one SE below the truth	93.8% (floor 95%)
3	Same, n = 1,000	`z = 1.645 ÷ (1 + 0.0224) = 1.609; Φ = 94.6%`	same method	94.6%
4	Orders needed for ≥ 94.5% at a one-SE miss	`z* = Φ⁻¹(0.945) = 1.5982; 1 + SE = 1.645 ÷ 1.5982; n = 1 + 1 ÷ (2 × SE²) = 584`	design arithmetic	≈ 584 orders per cohort-window
5	Heavy tails, t_5 scaled to unit variance	`95%: 2.015 ÷ √(5/3) = 1.561; 99%: 3.365 ÷ √(5/3) = 2.607`	Student-t table; sensitivity only	normal dial delivers 95.6% at 95%, 98.5% at 99%
6	Q4 regime, frozen dial	`z = 1.645 ÷ 1.45 = 1.134; Φ = 87.2% per fully-inflated cohort`	σ × 1.45 (illustrative model)	87.2% per cohort; 91.9% portfolio (record-from-docs)
7	Accuracy noise at scale	`√(0.95 × 0.05 ÷ 1,000,000) = 0.022 pts; at 100,000: 0.069 pts`	binomial	noise ≪ any real signal

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Marketplace contribution (modeled, illustrative book)
+- Conversion on fast promises (flow) -> fast share 7.6% -> 39.6%
|    fast share = SUM_cohorts volume x [handling + transit_pred + buffer <= 2d]
|    (loop) faster promise -> higher CVR -> more orders -> more seller GMV
+- Breach cost (flow, $/order) -> floor held at 95.0%
|    breach EV = (1 - Phi(b/sigma)) x $12   [contact + refund risk + trust decay]
+- Buffer waste (stock: days early on kept promises) -> harvested on tight lanes
+- Governance -> one dial k, published weekly; re-solved per regime (Q4)
+- Frontier (not claimed) -> policy D: 97.8% accuracy, +$0.38M/wk vs A
```

DRIVER	BASELINE	AFTER	$/WK	BASIS
Weekly modeled profit, policy A (uniform 1.10d)	$5.69M	—	$5.69M	illustrative model output
Weekly modeled profit, policy B (b = 1.645σ)	—	$5.94M	$5.94M	illustrative model output

DRIVER	BASELINE	AFTER	$/WK	BASIS
Fast-promise share	7.6%	39.6% (+32 pts)	inside the profit lines	illustrative model
Average promised day	3.09	2.75	inside the profit lines	illustrative model
Frontier: policy D (profit optimum)	—	97.8% accuracy, $6.07M/wk	+$0.38M vs A	illustrative; handed to Finance, never claimed

If challenged. *"Your whole result is a model."* — "Yes, and I say so in the same breath — the model exists so the client's decision could be quantitative. The structure survives any parameter set: risk-proportional buffers, equal marginal accuracy per buffer-day, a floor reconciled with the breach economics. The parameters are the client's to correct." *"Why is the $12 breach cost credible?"* — "It's the model's stated assumption — a contact, refund and return risk, and trust decay bundled — put on the table precisely so Finance could attack it. At $12 the profit optimum holds 97.8% accuracy; if Finance believes $6, the optimum falls toward the floor. Fight about the number, not the theology." *"Did any of this ship?"* — "The engagement delivered the policy table, the gates and the governance design; the client's teams owned the build. I claim the instrument and the modeled deltas, labeled — never a production P&L."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The promise identity	`EDD = t_order + handling + transit_pred + b; kept ⇔ (handling_err + transit_err) ≤ b`	per order; b = buffer in days (record-from-docs identity)	the buffer is the only policy term
2	Kept-promise probability per lane	`P(kept) = Φ(b / σ)`	ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (record-from-docs)	accuracy is a function of b and σ only
3	Incumbent buffer	`b = 1.10 days on every lane`	uniform policy A (illustrative model)	1.10 d everywhere
4	Tightest cohort, z-score	`z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997`	metro cohort σ = 0.22 d (illustrative model)	near-certainty bought — promise needlessly slow
5	Widest cohort, z-score	`z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754`	mountain-rural cohort σ = 1.60 d (illustrative model)	≈75.4% kept — a known-loss book
6	Portfolio print	`accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk`	order-weighted across 12 cohorts, ~1M orders/wk (illustrative model)	a green headline hiding both failures
7	The asymmetry	`padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)`	record-from-docs — why the system drifted slow	bias toward slow, unowned

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One governed dial	`b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%`	Φ(1.645) = 0.95 under normal errors (record-from-docs design)	k = 1.645, one number for the book

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Tightest cohort re-buffered	`b = 1.645 × 0.22 = 0.36 days`	σ = 0.22 d (illustrative)	buffer 1.10 → 0.36 d; 0.74 d of speed harvested
3	Widest cohort re-buffered	`b = 1.645 × 1.60 = 2.63 days`	σ = 1.60 d (illustrative)	buffer 1.10 → 2.63 d — slower on purpose, kept honest
4	Every cohort at the floor	`P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95 on each cohort`	by construction (design)	95.0% everywhere; no cohort subsidizes another
5	Portfolio after	`accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk`	illustrative model output, policy B	+32 pts fast share at the floor
6	Equimarginal proof (policy C)	`φ(b/σ)/σ equal across all 12 cohorts (spread < 1e-15)`	solver optimality condition (record-from-docs)	the last buffer-day buys the same accuracy everywhere
7	Peak regime rule	`σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645/1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%`	Q4 drill (illustrative model)	k re-solved and published in October, never frozen
8	Shadow gate	`4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold`	promotion bar (design)	rehearse against reality before any displayed date

Table 3 — after: the metric, before → after.

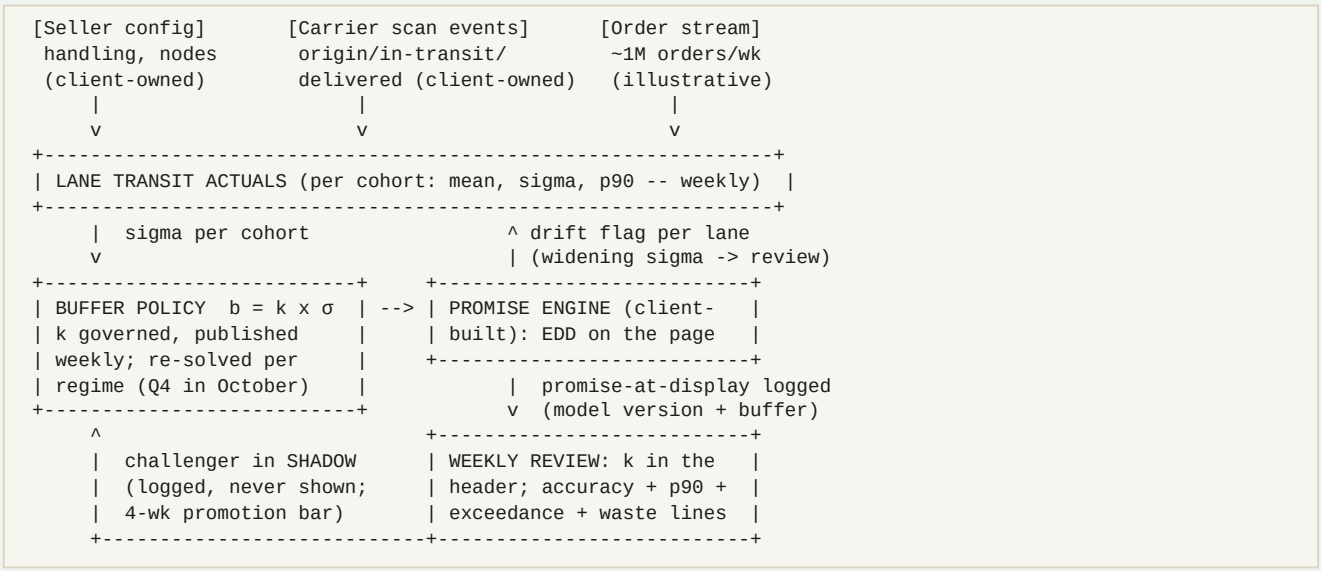
STEP	METRIC	BEFORE	AFTER	BASIS
1	Fast-promise share (≤2-day)	7.6%	39.6% (+32 pts)	illustrative model
2	Portfolio promise accuracy	95.8% (split: ~100% / 75.4%)	95.0% — the floor, held on every cohort	illustrative model
3	Average promised day	3.09	2.75 (−0.34)	illustrative model
4	Weekly modeled profit	$5.69M	$5.94M (+$255K)	illustrative model
5	Owner of the speed-accuracy trade	implicit, in the last model tune	one governed dial, published weekly, re-solved per regime	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Weekly profit, policy A	`$5,690,000 / wk, taken from the model`	illustrative model output	$5.69M/wk
2	Weekly profit, policy B	`$5,940,000 / wk, taken from the model`	illustrative model output	$5.94M/wk
3	Weekly delta	`$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)`	rows 1, 2 (illustrative)	≈ +$255K/wk
4	Annualized	`$255,000 × 52 = $13,260,000`	row 3 × 52 weeks (modeled)	≈ $13.3M/yr — "the model," said in the same breath
5	Frontier: policy D delta	`$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000`	illustrative model output; ×52 modeled	≈ $20M/yr — handed to Finance, never claimed

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	Why D holds 97.8%	`breach EV $12 > margin-per-promise-day trade at $6.50 ⇒ optimum above the 95% floor`	$12 and $6.50 are the model's stated assumptions	the floor and the economics disagree — a governance finding
7	Headline	`≈ $13.3M/yr modeled; record dollars: none`	rows 3–4	≈ $13.3M/yr, always labeled

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Breach cost bundle (model assumption)	$6	$12	$18
Profit-optimal accuracy (policy D)	falls toward the 95% floor	97.8%	above 97.8% — accuracy gets cheaper to justify
σ dispersion across cohorts (drives the free lunch)	narrow — gain shrinks	0.22–1.60 d	wider — gain grows
Fast-share gain from B vs A	≈ +10 pts	+32 pts	≈ +40 pts
Modeled $/yr vs uniform	≈ $4–5M	≈ $13.3M	≈ $20M (the D frontier)

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"On the engagement's illustrative book — twelve lane cohorts, about a million orders a week — moving from a flat 1.10-day buffer to risk-proportional buffers with one governed dial bought thirty-two points of fast-promise share at the identical ninety-five percent accuracy floor, took a third of a day off the average promise, and was worth about two hundred fifty-five thousand dollars a week, call it thirteen million a year. Every one of those numbers is the model's, and I say that in the same breath — there is no production dollar here. What I claim as mine is the instrument: the decomposition, the four-policy table, the shadow gates, and a dial the business reads every Monday."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time a headline metric hid the real problem. DIVE DEEP

Open (first two sentences): Promise accuracy printed 95.8% — green — and the room believed the policy was healthy. I split it into two cohorts.

STAR (≈90 s): Situation: a marketplace promised delivery dates on inventory it never touched, with a flat 1.10-day buffer on every lane. Task: decompose where the buffer was wasted and where it was thin. Action: z = buffer ÷ σ — 1.10 ÷ 0.22 = 5.0 on the tightest cohort (near-certainty, a needlessly slow promise) and 1.10 ÷ 1.60 = 0.69 on the widest (75.4% kept, a known-loss book). The average was hiding both failures. Result: the question changed from “how much buffer” to “where should it sit,” and the one-dial policy followed. All of this is the engagement’s illustrative model.

Land this number: Φ(5.0) vs Φ(0.69) = 75.4%; headline 95.8%. · **Pivot to NACF:** NACF appointment and ASN promise lanes: check the tightest and widest vendor lanes before trusting the average. · **Do NOT say:** Do not call any figure a production result.

Q2 Tell me about a time you simplified a complex decision. INVENT AND SIMPLIFY

Open (first two sentences): Data science wanted per-lane learned buffers. I recommended one dial — b = k·σ — and priced the difference.

STAR (≈90 s): Situation: a hundred ML micro-decisions, invisible governance. Task: leave the client an instrument, not a tuning habit. Action: four policies on one table — uniform, σ-proportional with one dial, minimum-days at the floor, profit-optimal — and the one-dial policy first because it captured most of the value and was auditable; σ improvements flow into speed automatically. The “no” to per-lane models went into the table as a priced row. Result: fast-promise share 7.6% → 39.6% at the held 95% floor on the model.

Land this number: +32 pts at a 95% floor; k = 1.645. · **Pivot to NACF:** A NACF dial a manager can read on Monday beats a model nobody can audit. · **Do NOT say:** Do not claim I built the promise engine; the client’s teams did.

Q3 Tell me about a time you protected the customer’s experience when the numbers could have been juiced. CUSTOMER OBSESSION

Open (first two sentences): Fast-promise share can be juiced by bleeding accuracy, so I pinned accuracy at 95% on every cohort and let speed float.

STAR (≈90 s): Situation: the customer blames the marketplace when Thursday becomes Saturday. Task: more fast promises without breaking the floor. Action: floor first — Φ(1.645) = 0.95 on each cohort by construction — then harvest speed: the tight cohort’s buffer 1.10 → 0.36 d, the wide cohort’s 1.10 → 2.63 d, slower on purpose. Result: +32 points of fast-promise share at exactly 95.0% on the model; no cohort subsidizes another.

Land this number: 95.0% on every cohort; wide-cohort buffer 2.63 d. · **Pivot to NACF:** The NACF promise is OTIF: protect the floor per lane, not per average. · **Do NOT say:** Do not say $13.3M a year as fact — say the model, in the same breath.

Q4 Tell me about a time you set a high bar on how a change rolled out. INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): Nothing I designed went live without four consecutive shadow weeks, a promotion bar published before week one and a rollback kept warm.

STAR (≈90 s): Situation: padding fails silently and cutting fails loudly, so every promise system drifts slow. Task: roll out a tighter promise without a loud failure. Action: shadow mode — the challenger computed and logged per order, never displayed — promoted only if would-be accuracy held the floor on logged live orders with no rise in contact signals; and the Q4 drill showed a frozen summer dial printing 91.9%, so k is re-solved and published in October. Result: a governance instrument the client’s business owner — not data science — owns.

Land this number: 4 shadow weeks; Q4 σ × 1.45 → 91.9%. · **Pivot to NACF:** Any NACF promise change shadows a peak before it displays. · **Do NOT say:** Do not say the shadow ran — it is the design of the engagement.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: “The headline was 95.8% and the policy was broken twice.”

Headline metric: Φ(5.0) vs Φ(0.69) = 75.4%

- Flat 1.10 d = 5.0σ on one cohort, 0.69σ on another.
- Price of the floor on the wide cohort: +0.50 d for 97.5%.
- σ-estimation: ≈ 584 orders per cohort-window to hold within half a point.
- All model figures.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Invent and Simplify

Open: “Data science wanted more model; I recommended one dial first.”

Headline metric: +32 pts at 95.0%; D at 97.8% shown, never claimed

- The “no” was a priced row in the table.
- D ≈ +$20M/yr rides on a $12 breach cost I cannot observe.
- B holds a floor I can defend.
- One dial, owner named.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious; Are Right, A Lot

Open: “A frozen dial broke the floor in the Q4 drill — 91.9%.”

Headline metric: Q4 σ × 1.45 → 91.9%

- Re-solve k per regime, published in October.
- Normal errors understate the 99% dial; the empirical quantile fixes it.
- No production dollar — design engagement.
- VERIFY-WITH-RAJESH any failure beyond the drill is not in the record.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 “Your σ is estimated. How many orders per cohort before the 95% floor is safe?”

The numbers: SE(σ̂) = 1 ÷ √(2(n−1)): n = 100 → 7.1% → realised 93.8%; n = 1,000 → 94.6%; ≥ 94.5% at a one-SE miss needs n ≈ 584.

Say exactly: “With a hundred orders a cohort, a σ one standard error low delivers about 94 percent against the 95 floor; to stay within half a point at a one-SE miss you need roughly 584 orders per cohort-window — trivial at a million orders a week, so the real risk is regime change, not estimation.”

T2 “Real transit errors are heavy-tailed. Does b = 1.645σ still hold 95%?”

The numbers: Student-t_5 scaled to unit variance: 95% quantile 1.561σ (normal dial delivers 95.6%); 99% quantile 2.607σ (normal dial delivers 98.5%).

Say exactly: “At the 95 percent floor the normal dial is slightly conservative on heavy tails — it delivers about 95.6 — but at 99 it would fall to about 98.5, so production uses the empirical quantile and I treat the normal rule as the whiteboard version.”

T3 “Policy D is worth $20M a year. Why didn’t you ship it?”

The numbers: B: +$255K/wk × 52 = $13.3M/yr at a 95% floor. D: ($6.07M − $5.69M) × 52 = $19.8M/yr at 97.8% — on a $12 breach cost vs $6.50 margin, both model assumptions; at $6 the optimum falls toward the 95% floor.

Say exactly: “Policy D is the priced governance decision I handed to the client’s finance partner — it rests on a twelve-dollar breach cost nobody can observe; B holds a floor I can defend and is auditable, so I recommended B first and showed D without ever claiming it.”

S20 — Elementum / Walmart Marketplace — the weekly business review for pre-purchase delivery experience: 7 outputs over 10 inputs, factor math that names the owner in minutes, and an Andon that cuts detection-to-action from ~30 days to under 24 hours (engagement design) NEW

LPs this story serves here: Dive Deep (primary — factor math, the aggregate trap) · Ownership (one owner per output) · Invent and Simplify (7 outputs, 10 inputs, one joint headline) · Bias for Action (the Andon)
Process family: Plan / Deliver seam — metric architecture and control loop for the pre-purchase promise (Walmart Marketplace; role R-2596147)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I decomposed accuracy into O1 ≈ I3 × I4 × Φ(b/σ) = 0.97 × 0.985 × 0.996 ≈ 95.2% so a red week names its factor — a regional carrier miss (97% national, 82% in the Southeast) was attributed in minutes.
- **Ownership** — Promise accuracy was owned by a committee, so by nobody between Tuesdays; every output now has one named owner and every input a roll-up and an owner.
- **Invent and Simplify** — Seven outputs over ten inputs and one joint headline, Promise Yield = fast share × accuracy; “two dashboards is zero dashboards” is a written rule.
- **Bias for Action** — The Andon — a 7-day control chart per seller-lane cell, act within 24 hours — replaces a 30-day rolling window that let a degrading cell burn customers for weeks.

Honesty label. Elementum engagement for Walmart Marketplace's pre-purchase delivery-experience organization. I designed the weekly business review — the metric architecture, the decomposition math, the thresholds and the control loop — and led the program that stood it up; the client's engineering built the pipelines and dashboards and their teams own the lines. I never configured or deployed any platform, and I never claim a production output metric moved — the engagement delivered the instrument, its sample-data proof, and the Andon design. The production-record version of this muscle is S11 (Flexport), and I keep the two separate.

60-SECOND PITCH

"**Situation.** A marketplace client whose delivery promise was owned by a committee: data science owned the model, engineering the engine, seller ops the handling, carrier programs the lanes — and the one number the customer feels, promise accuracy, was owned by nobody between Tuesdays. The scorecards ran on 30-day rolling windows, so a degrading seller or carrier did weeks of customer damage before any floor bit, and a red week was a blame negotiation. **Task.** Design and stand up the weekly business review for the pre-purchase delivery experience — outputs a customer feels, input levers a named owner can turn, and the math that connects them — plus a control loop faster than the 30-day clock. **Action.** Discovery: one glossary, one event source, audited denominators. Plan & Design: seven output metrics over ten input levers, each input with a roll-up to the output it drives and a named owner; the factorization — accuracy ≈ seller-ships-in-window × carrier-meets-transit × Φ(buffer/σ) — as the decomposition; red/green at ±5% relative week-over-week with hard reds on policy floors. Build & Test: the sample-week proof — 0.97 × 0.985 × 0.996 ≈ 95.2% — and the drill: a regional carrier miss (97% nationally, 82% in the Southeast) decomposed to its factor in minutes. Cutover: Monday data freeze, Tuesday review, reds pre-wired with cause, owner, date. Hypercare: mix-adjusted twins next to every raw output, and the Andon — a 7-day control chart per seller-lane cell, act within 24 hours. **Result.** Primary metric: red-week attribution — **a blame negotiation → minutes, by factor math**; detection-to-action 30 days → under 24 hours by design; one joint headline, Promise Yield = fast share × accuracy, baseline 41.5% on the docs' sample. The only dollar is modeled: about $580K per caught degradation episode, and I say 'modeled' in the same breath."

20-SECOND VERSION NEW

The delivery promise was owned by a committee and scored on a 30-day window; I designed the weekly review — seven outputs over ten owned inputs, the factor identity that names the red, and an Andon that acts in under 24 hours — proven on sample data; the only dollar is modeled, about $0.58M per caught episode.

NACF BRIDGE — ONE SENTENCE

The NACF version is INV-E's OTIF chain: ship-in-window × carrier-meets-transit × buffer-absorbs-residual — a weekly review where every red names a factor and an owner.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The same Elementum client engagement — Walmart Marketplace's seller-fulfilled delivery program, pre-purchase side. The promise the customer sees decomposes across three parties: the seller ships in a window, a carrier meets a transit time, a model plus a buffer absorbs the residual. The operating problem was measurement and ownership, not effort: promise accuracy, on-time delivery and late shipment lived on 30-day rolling windows that smear a step change into a slow drift; aggregate metrics hid the variance (a carrier can print 97% nationally while running 82% in the Southeast after a hub consolidation — the engagement's own worked example); three teams kept three clocks (order date, ship date, promise date); and when a week went red, the review discovered it live and negotiated blame instead of reading a decomposition.

T — Design the weekly business review for the pre-purchase delivery experience and lead the program that stood it up: the output metrics a customer feels, the input levers a named owner can turn this week, the math that connects them, the thresholds, and a control loop faster than the 30-day scorecard — then hand it to owners and leave.

Why it mattered: an output metric owned by a committee is owned by nobody between Tuesdays, and a 30-day window means a degrading cell burns customers for weeks before policy floors bite — enforcement lag was a named strategic issue in the engagement record. Left unfixed: reds without owners, action items that age into a diary, mix-shift illusions celebrated as wins (portfolio accuracy improving while every cohort degrades, because volume shifted to easy metro lanes), and the speed-versus-accuracy trade argued as theology because nobody priced a breach.

The metrics on the table, with their baselines: red-week attribution time — days of cross-team negotiation (baseline condition); detection-to-action on a degrading cell — up to ~30 days on the rolling scorecard; output ownership — committee; the joint headline — none (fast share and accuracy lived on separate lines with separate owners, tradeable against each other invisibly); the docs' sample baseline for that joint number: fast share 43.6% × accuracy 95.2% ≈ 41.5% Promise Yield.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (the metric definitions, formulas, thresholds and the Andon parameters are the engagement's design record; the sample numbers are the docs' own illustrative set — six weeks of sample data proved the mechanics before any live line):

- **Discovery** — one glossary: every metric got one definition, one owner, one grain, one date basis (order date for operating cohorts; the 30-day windows kept only for seller-facing consequences); audited the denominators (cancelled orders quietly leaving the OTD base, suppressed listings leaving penetration). *Exit bar:* three teams' three clocks reconciled to one event source; no metric without a named definition owner.
- **Plan & Design** — the architecture: seven outputs (promise accuracy ≥95%, on-time delivery ≥90% policy floor, fast-promise penetration +2 pts/qtr, late shipment ≤5%, cancellation ≤2%, WISMO contacts −10% QoQ, A/B-measured conversion lift) over ten input levers (SSS share of GMV, ML-recommended handling coverage, seller handling adherence, carrier lane SLA adherence, transit-model error and freshness, buffer calibration, inventory-feed accuracy, eligibility coverage, engine latency/cache, onboarding throughput + 30-day retention) — every input carrying a roll-up to the outputs it drives and a named owner; red/green at ±5% relative week-over-week, hard reds on the policy floors regardless of trend; one joint headline, Promise Yield = fast share × accuracy. *Exit bar:* every output decomposable into named inputs with owners; definitions frozen in writing.

- **Build & Test** — the client's engineering built the datasets and dashboard; I owned the proof on sample data: the factor identity (accuracy ≈ I3 × I4 × Φ(b/σ)) computed live — 0.97 × 0.985 × 0.996 ≈ 95.2% — then the drill: drop the Southeast carrier cell to 82% on a tenth of volume and watch the decomposition name the factor and the owner in minutes. *Exit bar:* the six-step loop (measure → variance → decompose → root-cause → act → verify) executed end-to-end on sample weeks; a red that arrives without cause, owner and date counted as a process miss.
- **Cutover / Go-Live (design)** — the operating rhythm: Monday data freeze on the Mon–Sun week, Tuesday review, pod leads pre-wire reds before the room; the action log with owner, date and visible ageing; "the meeting decides; it never discovers." *Exit bar:* two consecutive sample-cycle reviews run entirely pre-wired; the output block and the action log assigned to one named owner.
- **Hypercare / Close** — the honesty layer and the control loop: mix-adjusted twins printed next to every raw output (fixed base-quarter cohort weights; the delta *is* the mix effect) so Simpson's paradox can't celebrate volume migrating to easy lanes; and the Andon — a 7-day EWMA/CUSUM control chart per seller-lane cell, tripwire at a sustained 2× baseline breach rate with n ≥ 30 orders, action within 24 hours (widen that cell's buffer or pull its fast badge), auto-restore on recovery, designed false-trip rate under 5%. *Exit bar:* the verify step closing the loop — if an input moved and its output didn't, the roll-up hypothesis is declared wrong and re-examined, in writing.

The client's engineering built every pipeline and surface; data science owned the models behind the transit and buffer lines; seller ops and carrier programs owned their levers. Mine: the glossary and denominator audit, the output/input architecture and the roll-up map, the factorization as the decomposition instrument, the thresholds and floors, the six-step loop, the mix-adjusted twins, the Andon design and its parameters, and the handoff that left the output block owned by a person, not a committee.

The obstacle was that every team already had a dashboard and nobody wanted a new clock — overcome by making the review cheaper than the status quo: the decomposition turned the worst meeting of the week (the blame negotiation) into a five-minute read, and the pre-wire rule moved the argument out of the room. The decision I challenged: the program wanted more metrics; I held the board to seven outputs and ten inputs — a metric nobody owns or acts on came off the board, and "two dashboards is zero dashboards" went into the design note as a rule, not a preference.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: red-week attribution — from a cross-team blame negotiation to minutes, by factor math (proven on the sample-data drill); detection-to-action on a degrading cell — ~30 days → under 24 hours by design (the Andon). The only dollar is modeled: ≈$0.58M per caught episode.

- **$** — ≈$0.58M per caught degradation episode, modeled: a Southeast-type carrier miss costs ≈$192K/week in excess breaches; caught and corrected in ≤1 week instead of ~4, three weeks × $192K ≈ $576K avoided (inputs below; the $12 breach cost is the model's stated assumption). No annual figure — episode frequency is the client's data, and I won't invent it.
- **#** — 7 output metrics over 10 input levers, every input with a roll-up and a named owner; 1 joint headline (Promise Yield); 1 action log; 6 sample weeks proving the mechanics (engagement design record on the docs' illustrative set).
- **%** — Promise Yield baseline 43.6% × 95.2% ≈ 41.5% on the docs' sample, four-quarter design target 50%; red/green at ±5% relative WoW; hard reds on the floors (OTD < 90%, late shipment > 5%, cancellation > 2%); Andon false-trip design bar < 5%.
- **Time** — Red-week attribution: days → minutes (the factor math); detection-to-action: ~30 days of rolling-window lag → < 24 hours from a sustained signal (designed); Monday freeze → Tuesday review, reds pre-wired before the room.
- **Customer** — The Andon protects customers at the grain where variance lives (the seller-lane cell) instead of letting a degrading cell burn them for a month; WISMO contacts sit on the output board so promise failures are read in customer contacts, not just percentages.
- **Quality** — Mix-adjusted twins next to every raw output (the mix effect printed as its own line); denominators audited; the verify step that falsifies the lever map itself when an input moves and its output doesn't — the

mechanism that keeps the instrument honest after I leave.

Against the goal stated above: one owner of the output block instead of a committee, a decomposition that names the factor and the owner in minutes, a control loop 30× faster than the scorecard clock, and a joint headline that makes the speed-accuracy trade un-gameable at the KPI layer. The trade was explicit: seller-facing consequences stayed on the fair 30-day window — the Andon acts on cells and badges, never on scorecards — stability for sellers, speed for customers.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No agreed glossary or owner for the promise output; I defined one definition, one owner, one grain, one date basis per metric.
Scope & influence	Data science, engineering, seller ops and carrier programs — none reporting to me; every team already had a dashboard and nobody wanted a new clock.
Execution	Metric architecture, decomposition proof, red-week drill, Monday freeze / Tuesday review, Andon design and hypercare rule.
Impact	Red-week attribution days → minutes (drill); detection-to-action ~30 days → under 24 hours by design; no production metric claimed.
Process improvement	A standing review with pre-wired reds, mix-adjusted twins and an action log with owners and dates.
Complexity	Aggregates hiding cells; mix-shift illusions; rolling-window lag; a gameable pair of metrics.
Root cause	The root cause was ownership and clocks, not data: three teams on three clocks and no joint metric.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: the engagement's sample set — six weeks, seven outputs, ten inputs; fast share 43.6% × accuracy 95.2%; carrier 97% national vs 82% Southeast. Revealed: the aggregate hid the cell; the rolling window hid the lag.
2 · Solution design	Options: more metrics · fewer, owned metrics · 30-day windows · a 7-day cell chart. Chosen: 7 outputs / 10 inputs / one joint headline / Andon per seller-lane cell.
3 · Math & data logic	Math: O1 ≈ I3 × I4 × Φ(b/σ) = 0.9516; episode: I4′ = 0.9685 → O1′ = 93.6%; $192K/wk × 3 wks = $576K; Andon binomial false-trip arithmetic; mix-shift example; Promise Yield gaming test.
4 · Implementation	Implementation: glossary → architecture → sample-data proof → Monday freeze, Tuesday review → pre-wired reds → Andon with auto-restore → mix-adjusted twins.
5 · Probe questions	Probe questions: Andon false-trip rate · mix-shift · whether the joint headline can be gamed · any production claim — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

> ***Challenge* — Why did you choose this story to illustrate Insist on the Highest Standards?**
>
> Because the standards here are numbers with teeth, published in advance: hard reds on policy floors regardless of trend, a pre-wire rule that makes an unexplained red a process miss, and an Andon whose tripwire, sample floor and false-trip bar were written before the first live week. A standard that can't stop a program is decoration.

> ***Probing* — Why did you choose to focus on these results? What other results were important?**
>
> Attribution speed and detection speed lead because they are what a review is for — a WBR that can't say which lever moved is a viewing, not a review. Also important: the joint headline (Promise Yield) that stops the two factor teams trading against each other invisibly, and the mix-adjusted twins that keep a growth quarter from flattering the numbers.

***Probing* — You mentioned dollars — what percentage change is that year over year?**

None claimed — the ≈$0.58M is per caught episode, modeled from the sample-data drill, and I don't multiply it by an invented episode frequency. The year-over-year story here is mechanism speed: 30 days of enforcement lag to under 24 hours.

***Probing* — I'm concerned your baseline Promise Yield is from sample data — tell me more.**

Correct, and I say so: 43.6% × 95.2% ≈ 41.5% is the engagement docs' own illustrative six-week set, used to prove the mechanics before live lines existed. The design target — 50% in four quarters — is a target on the real book, and the first live job is re-baselining it.

***Challenge* — What were the lessons learned? What would you have done differently?**

Lesson: the decomposition is the peace treaty — attribution math is what turns a blame negotiation into a five-minute read. Differently: I'd have run the denominator audit in week one rather than mid-design; two of the "metrics" we inherited changed meaning once cancelled orders stopped quietly leaving the base.

***Challenge* — How would you implement this at Amazon?**

This is Amazon's own grammar — input levers under output metrics, reds pre-wired with an ask, COE-style mechanism fixes — so the transplant is the control loop: factor math on the promise seam, mix-adjusted twins beside raw outputs, and cell-grain Andon where a 30-day window is too slow for the customer.

***Challenge* — How did these results compare to your actual goals?**

The signed goals were one owner of the output block, decomposition in minutes, and a sub-24-hour control loop — all three delivered as the standing instrument, proven on six sample weeks, with the verify step written so the lever map itself can be falsified and corrected.

PHASE DEEP-PROBE BANK — Q&A (12). ***Answer the phase asked, then stop.***

Discovery — How did you know discovery was done?

When three teams' three clocks — order date, ship date, promise date — reconciled to one event source, and every metric on the inherited boards had exactly one definition owner or was struck. The exit memo listed the denominator findings with owners, not hopes.

What did the denominator audit find?

The classic games, running unmalevolently: cancelled orders quietly leaving the on-time base, suppressed listings leaving the penetration base. Neither was fraud — both were defaults nobody owned. The rule that fixed it: metric governance audits denominators, not just numerators.

Plan & Design — Why seven outputs and ten inputs, not more?

Because a board nobody can hold in their head is a board nobody reads. The cut rule was ruthless: an input had to be provably upstream of an output, turnable this week, and owned by a name — a metric nobody owns or acts on came off. Everything else lives in drill-down, not on the board.

Why is Promise Yield the headline instead of accuracy?

Because either factor alone is gameable: chase accuracy and you pad buffers until every promise is slow and safe; chase penetration and you shave buffers until promises are fast and broken. The product punishes both games — and the board still decomposes it one line below, because a composite that can't be decomposed is a hiding place.

Why keep the 30-day window at all?

Fairness. A seller consequence — a scorecard, a suspension — deserves a stable clock that one bad day can't trip. The design splits the clocks: 30-day for consequences, 7-day panels for operations, and the Andon acting on cells and badges in hours. Each clock does the job it's fair at.

Build & Test — Walk me through the sample-week factor proof.

Seller-ships-in-window 0.97, carrier-meets-transit 0.985, buffer absorption Φ(b/σ) 0.996 — product 95.2%, matching the sample week's printed accuracy. Then the drill: Southeast carrier adherence to 82% on 10% of volume → the carrier factor falls to 0.9685, accuracy to 93.6%, and the week-over-week factor deltas name the carrier PM in one read.

How did you test the Andon's false-trip rate?

On the sample set: replayed stable cells through the EWMA/CUSUM chart and counted trips. The n ≥ 30 floor and the sustained-signal requirement (CUSUM, not single-week thresholds) are what hold designed false trips under 5% — a noise-tripped Andon punishes small sellers and kills the mechanism's legitimacy.

Cutover — What does "pre-wired" mean mechanically?

Before Tuesday, every red line arrives with its factor decomposition, the named input that moved, the owner, and a dated action. The meeting's job is to decide on the ask — never to discover the cause. A red discovered live is logged as a process miss on the review itself.

Who could stop what?

Hard reds on policy floors stop expansion into the affected cohort automatically — eligibility growth doesn't board on a floor breach. The Andon stops a cell's fast badge without waiting for a meeting. And the output-block owner can hold any launch that would change a metric's meaning mid-flight.

Hypercare — What are mix-adjusted twins and why print both?

Every output twice: raw, and re-weighted on fixed base-quarter cohort weights. The delta between them is the mix effect, printed as its own line — so the room stops celebrating "improvements" that are volume migrating to easy metro lanes, and stops flogging teams for "degradation" that is growth into hard geographies. Re-based quarterly, by written policy.

What does the verify step do when an input moved and the output didn't?

It declares the roll-up hypothesis wrong, in writing, and re-examines the lever map — that is how the map stays honest. A WBR that never falsifies its own causal claims degrades into numerology with owners.

Cross-phase — What did you refuse?

A bigger board (metrics without owners), a faster clock for seller consequences (unfair), an annual dollar claim for the Andon (no episode-frequency basis), and running the first live weeks myself — the instrument had to be owned by the client's named owner from day one, or it would have stayed mine and died when I left.

Analytics — What is the false-trip rate of an Andon that fires on “7-day breach rate ≥ 2× baseline, n ≥ 30”?

The record carries EWMA/CUSUM parameters and a false-trip design bar under 5%; the binomial analog shows why n ≥ 30 alone cannot meet it. Baseline breach rate 5% (the 95% floor): at n = 30 a “2×” trigger is 3 breaches, which a healthy cell produces 18.8% of weeks and catches a true 2× cell only 59% of weeks; the smallest trigger under 5% at n = 30 is 5 breaches (3.3× baseline, false trips 1.6%). At n = 100, “2×” is 10 breaches: false trips 2.8%, power 55%. A 2× rule is only clean from about n = 103; below that, “sustained” — several consecutive days — has to do the work.
VERIFY-WITH-RAJESH the exact EWMA/CUSUM parameters.

Analytics — Show me a mix-shift illusion.

Two cohorts, metro and rural. Before: 97% and 88% at a 50/50 mix = 92.5%. After: each cohort degrades a point — 96% and 87% — but volume shifts to 70% metro: 93.30%. The raw number rises 0.80 points while every cohort falls; at the old mix the same week is 91.5%, down 1.0. That is the mix-adjusted twin, printed beside every raw output. Illustrative numbers.

Analytics — Can Promise Yield be gamed?

Yes, if it is the only constraint. Yield = fast share × accuracy = 0.436 × 0.952 = 41.5%. Push fast share up 2 points and let accuracy fall to 94.2% and yield rises to 43.0% — a trade the product rewards and the customer pays for. That is why the floors are hard reds regardless of trend (accuracy ≥ 95%), and why yield is a headline, not a target.

STATISTICAL SPINE — ANDON FALSE-TRIP ARITHMETIC, MIX-SHIFT, HEADLINE GAMING (EXTRA TABLE)

The record: 7 outputs over 10 inputs; the factor identity; a 7-day Andon with false-trip design bar under 5% and EWMA/CUSUM parameters; mix-adjusted twins; hard floors. The table below is my binomial and mix arithmetic on illustrative cells — it explains the design, it is not a production measurement.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Healthy-cell breach rate	$p_0 = 1 - 0.95 = 5\%$	accuracy floor 95% (record-from-docs)	$p_0 = 5\%$
2	"2× baseline" at n = 30	trigger c = 2 × 0.05 × 30 = 3 breaches; false trips P(X ≥ 3 | 30, 0.05) = 18.8%; power at p = 10%: 59%	binomial analog of the 7-day cell chart	false trips 18.8% — over the 5% bar
3	Smallest trigger under 5% at n = 30	c = 5 breaches = 3.3× baseline; false trips 1.6%; power at 10%: 18%	same method	5/30
4	"2×" at n = 100	c = 10; false trips 2.8%; power at 10%: 55%	same method	false trips 2.8% — under the bar
5	Smallest trigger under 5% at n = 300	c = 22 = 1.47× baseline; false trips 4.86%	same method	1.47× suffices
6	Orders for a clean "2×" rule	$n = 0.0475 \times (2.33 \div 0.05)^2 = 103$	$p_0(1 - p_0) = 0.0475$; 2.33σ margin to 10%	≈ 103 orders per cell-week
7	Mix-shift: before	0.5 × 97% + 0.5 × 88% = 92.5%	illustrative cohorts	92.5%
8	Mix-shift: after, raw	0.7 × 96% + 0.3 × 87% = 93.30%	both cohorts −1 pt; mix shifts to metro	93.30% (up 0.80)
9	Mix-shift: after, old mix	0.5 × 96% + 0.5 × 87% = 91.5%	mix-adjusted twin	91.5% (down 1.0)
10	Promise Yield gaming	0.436 × 0.952 = 41.5%; 0.456 × 0.942 = 43.0%	fast share +2 pts, accuracy −1 pt (illustrative)	yield up 1.4 pts while accuracy falls

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Customer promise kept + operating speed (primary outcome)
+- Attribution speed (time) -> days of negotiation -> minutes
|    O1 ≈ I3 x I4 x Phi(b/sigma): the biggest factor WoW delta names the owner
+- Detection-to-action (time) -> ~30d rolling lag -> <24h (Andon per cell)
|    (loop) late detection -> weeks of excess breaches -> WISMO + trust decay
+- Excess-breach cost per episode ($) -> modeled below
+- Gaming resistance -> Promise Yield = fast share x accuracy (one headline)
+- Honesty -> mix-adjusted twins; audited denominators; verify step
     falsifies the lever map itself
```

DRIVER	BASELINE	AFTER	$	BASIS
Red-week attribution	cross-team negotiation, days	factor math, minutes	—	engagement design record (sample-data drill)
Detection-to-action on a degrading cell	up to ~30 days	< 24 h from sustained signal	—	designed (Andon: EWMA/CUSUM, 2× baseline, n ≥ 30)
Excess breaches during one Southeast-type episode	≈16,000/wk for ~4 wks	≤1 wk	≈$0.58M avoided per episode	modeled ($12 breach assumption; illustrative book)

DRIVER	BASELINE	AFTER	$	BASIS
Joint headline	none (two gameable lines)	Promise Yield 41.5% (sample) → 50% 4-qtr target	—	record-from-docs (sample set); target is design
Annual Andon value	not claimed — episode frequency is the client's data		—	deliberately uninvented

If challenged. *"You never ran this live."* — "Correct, and I say so — the engagement delivered the instrument proven on six sample weeks; the client's owners run it. The production record for running this cadence weekly for a year is S11, with record numbers." *"The $580K rests on a $12 breach cost."* — "The $12 is the model's stated bundle — contact, refund risk, trust decay — put on the table so Finance could attack it. At $6 the episode is still ≈$290K; the mechanism argument — act in hours, not weeks — survives any price." *"Isn't a sub-24-hour Andon rash?"* — "It acts on a cell's buffer or badge — reversible in one flag, auto-restored on recovery — never on a seller's scorecard. The n ≥ 30 floor and the sustained-signal test are the guards; the 30-day window stays for consequences."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The accuracy identity	`O1 ≈ I3 × I4 × Φ(b/σ) — seller ships in window × carrier meets transit × buffer absorbs the residual`	record-from-docs factorization; three factors, three owners	every breach belongs to exactly one factor
2	Healthy sample week	`O1 = 0.97 × 0.985 × 0.996 = 0.9516 ≈ 95.2%`	I3 = 0.97, I4 = 0.985, Φ = 0.996 (docs' illustrative sample)	95.2% — matches the sample print
3	The aggregate trap	`carrier prints 97% nationally while one region runs 82%`	Southeast after a hub consolidation (record-from-docs example)	the average hides the cell
4	The rolling-window lag	`a step change entering a 30-day window moves the average ≈1/30 per day`	30-day scorecard mechanics (record-from-docs issue)	weeks before a floor bites — enforcement lag
5	The gameable pair	`fast share 43.6% and accuracy 95.2% on separate lines, separate owners`	docs' sample; either can be juiced by bleeding the other	no joint headline
6	Baseline joint number	`Promise Yield = 0.436 × 0.952 = 0.415`	rows 2, 5 (docs' sample)	≈ 41.5%

Table 2 — the fix: step by step, values substituted (the red-week drill).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The episode	`Southeast cell: carrier adherence 0.985 → 0.82 on w = 10% of volume`	82% is the docs' example; the 10% weight is illustrative	one regional carrier miss
2	New carrier factor	`I4' = 0.90 × 0.985 + 0.10 × 0.82 = 0.8865 + 0.082 = 0.9685`	volume-weighted blend (illustrative)	I4: 0.985 → 0.9685 (−1.7%)
3	New accuracy	`O1' = 0.97 × 0.9685 × 0.996 = 0.9357 ≈ 93.6%`	rows 1–2; other factors unchanged	95.2% → 93.6% — a red week
4	The attribution read	`WoW deltas: I3 0.0% · I4 −1.7% · Φ 0.0%`	computed per factor (the decomposition)	the red names I4 and the carrier PM — in minutes
5	Segmentation	`factor red isolated to one region ⇒ C-family cause, not model or seller`	lane/cohort cut of I4 (design)	cause class identified before the meeting

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	The Andon trigger	`cell 7-day breach rate sustained ≥ 2 × baseline, n ≥ 30 ⇒ act < 24 h`	EWMA/CUSUM parameters (record-from-docs design)	buffer widened or badge pulled on the cell, same day
7	Auto-restore	`control chart recovers ⇒ badge/buffer restored, no meeting needed`	design	surgical, reversible, seller-fair

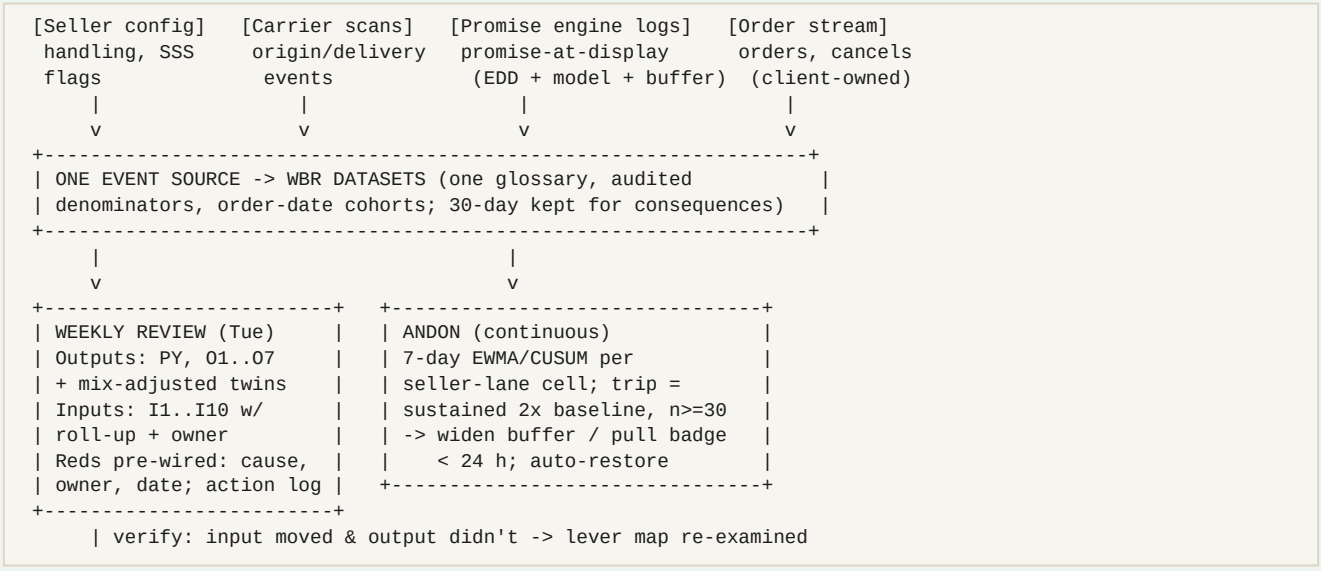
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Red-week attribution time	days (blame negotiation)	minutes (factor math)	engagement design record, proven on the drill
2	Detection-to-action, degrading cell	up to ~30 days	< 24 h from sustained signal	designed (Andon)
3	Output ownership	committee	one named owner of the output block + log	engagement design record
4	Joint headline	none	Promise Yield, 41.5% sample baseline → 50% 4-qtr target	docs' sample; design target
5	Mix honesty	raw outputs only	mix-adjusted twin + mix line per output	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Episode accuracy loss	`95.2% − 93.6% = 1.6 points`	Table 2 rows 2–3 (illustrative episode)	1.6 pts of excess breaches
2	Excess breaches per week	`1,000,000 × 1.6% = 16,000`	~1M orders/wk illustrative book	≈16,000 breaches/wk
3	Cost per breach	`$12, taken as stated`	the model's assumption: contact + refund risk + trust decay (record-from-docs)	$12 each
4	Excess cost per week	`16,000 × $12 = $192,000`	rows 2 × 3 (modeled)	≈$192K/wk
5	Detection advantage	`~4 weeks (rolling-window era) − 1 week (Andon era) = 3 weeks`	row 4 of Table 1; Andon design	3 weeks earlier action
6	Avoided per episode	`$192,000 × 3 = $576,000`	rows 4 × 5 (modeled)	≈$0.58M per caught episode
7	Annual total	`not computed — $576,000 × (episodes/yr) requires a frequency I don't have`	deliberately uninvented	per-episode only, labeled modeled

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Breach cost (model assumption)	$6	$12	$18
Episode size (regional weight × adherence drop)	5% × −10 pts	10% × −16.5 pts	15% × −20 pts
Excess cost per week	≈$50K	≈$192K	≈$430K
Weeks of detection advantage	2	3	4
Avoided per caught episode	≈$0.1M	≈$0.58M	≈$1.7M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"What the engagement delivered is an instrument: seven output metrics a customer feels over ten input levers with named owners, one joint headline — fast share times accuracy — and the factor math that attributes a red week to seller, carrier or model in minutes instead of a week of negotiation. The control loop went from a thirty-day rolling window to under twenty-four hours at the seller-lane cell. The one dollar I'll quote is modeled and I'll label it as I say it: a regional carrier degradation of the kind in the engagement record costs about a hundred ninety thousand a week in excess breaches at the model's twelve-dollar breach cost, and catching it three weeks earlier is worth roughly five hundred eighty thousand per episode. I don't annualize it — the episode frequency is the client's data, not mine."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time you found the root cause of a problem everyone was arguing about. DIVE DEEP

Open (first two sentences): Carrier on-time printed 97% nationally and the Southeast was at 82%. The review used to spend a week arguing about whose fault that was; the factor math named the cell in minutes.

STAR (≈90 s): Situation: promise accuracy was owned by a committee and a red week was a blame negotiation. Task: design a review where a red names its cause. Action: the identity accuracy ≈ seller ships in window × carrier meets transit × Φ(buffer ÷ σ) — 0.97 × 0.985 × 0.996 ≈ 95.2% — and the drill: drop the Southeast carrier cell to 82% on a tenth of volume and watch the decomposition: carrier factor 0.985 → 0.9685, accuracy 95.2% → 93.6%, the carrier factor named. Result: attribution in minutes on the sample data, with a named owner.

Land this number: 0.985 → 0.9685; 95.2% → 93.6%; attribution days → minutes. · **Pivot to NACF:** INV-E's OTIF chain is the same decomposition for NACF inbound. · **Do NOT say:** Do not say any production metric moved — the engagement delivered the instrument and its sample-data proof.

Q2 Tell me about a time you took ownership of something with no clear owner. OWNERSHIP

Open (first two sentences): The one number the customer feels — promise accuracy — was owned by nobody between Tuesdays.

STAR (≈90 s): Situation: data science owned the model, engineering the engine, seller ops the handling, carrier programs the lanes. Task: a weekly review with one owner per output and a named owner per input. Action: one glossary — one definition, owner, grain and date basis per metric — seven outputs, ten inputs with roll-ups, red/green at ±5% relative week-over-week and hard reds on policy floors, reds pre-wired with cause, owner and date before the room. Result: the review became cheaper than the status quo — a five-minute read instead of the worst meeting of the week.

Land this number: 7 outputs · 10 inputs · every one with an owner. · **Pivot to NACF:** Every NACF review metric needs a definition owner and a grain. · **Do NOT say:** Do not say the client adopted it in production.

Q3 Tell me about a time you pared something down to make it work. INVENT AND SIMPLIFY

Open (first two sentences): The program wanted more metrics. I held the board to seven outputs and ten inputs and wrote "two dashboards is zero dashboards" into the design note.

STAR (≈90 s): Situation: every team already had a dashboard and nobody wanted a new clock. Task: make the review cheaper than what it replaced. Action: a metric nobody owns or acts on comes off the board; one joint headline, Promise Yield = fast share × accuracy, 43.6% × 95.2% ≈ 41.5% on the docs' sample; mix-adjusted twins beside each raw output. Result: a board small enough to run in a meeting; four-quarter design target 50% for Promise Yield.

Land this number: 41.5% sample baseline → 50% design target. · **Pivot to NACF:** Same discipline for any placement scorecard: fewer owned metrics. · **Do NOT say:** Do not present the 50% as a result — it is a design target.

Q4 Tell me about a time you built something to act faster than the process allowed. BIAS FOR ACTION

Open (first two sentences): A 30-day rolling scorecard lets a degrading seller or carrier burn customers for weeks. I designed a control that acts in 24 hours.

STAR (≈90 s): Situation: a step change entering a 30-day window moves the average about 1/30 per day, so floors bit weeks late. Task: close the loop faster without a meeting. Action: the Andon — a 7-day control chart per seller-lane cell; sustained breach at ≥ 2× baseline with n ≥ 30 triggers a same-day action — widen the buffer or pull the badge — and auto-restores when the chart recovers. Result: detection-to-action designed from ~30 days to under 24 hours; ≈$0.58M per caught episode modeled.

Land this number: ~30 days → < 24 hours (design); ≈$0.58M per episode (modeled). · **Pivot to NACF:** NACF inbound: a per-vendor-lane control chart with an auto-restore beats a monthly scorecard. · **Do NOT say:** Do not annualize the $0.58M — episode frequency is the client's data.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "The carrier was 97% nationally and 82% in one region — the average hid the cell."

Headline metric: accuracy 95.2% → 93.6% via one factor

- O1 ≈ I3 × I4 × Φ(b/σ).
- WoW deltas by factor: I4 −1.7%.
- Mix-adjusted twin: raw +0.8 pt, adjusted −1.0 pt.
- Andon false-trip arithmetic.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Ownership

Open: "The program wanted more metrics; I held it to seven and ten."

Headline metric: 7 outputs / 10 inputs / 1 joint headline

- A metric nobody owns comes off the board.
- Hard floors regardless of trend.
- Reds pre-wired before the room.
- No production claim.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Are Right, A Lot; Learn and Be Curious

Open: "A multiplicative headline can still be gamed."

Headline metric: yield 41.5% → 43.0% while accuracy falls to 94.2%

- The joint headline needs hard floors beside it.
- n ≥ 30 alone gives 19% false trips at 2×.
- This is a design engagement — nothing in production moved.
- VERIFY-WITH-RAJESH any live Andon results are not in the record.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Your Andon fires at 2× baseline with n ≥ 30. What is the false-trip rate?"

The numbers: p_0 = 5%: n = 30, c = 3 → false trips 18.8%, power 59%; need c = 5 for < 5%; n = 100, c = 10 → 2.8%; clean 2× rule from n ≈ 103.

Say exactly: "On a single week at thirty orders a healthy cell trips about 19 percent of the time at two-times baseline and a true doubling is caught only about 59 percent — so the record's EWMA and "sustained" logic carries the false-trip bar under five percent; a clean two-times rule needs about 103 orders a cell-week."

T2 "Raw accuracy improved. Is it improving?"

The numbers: Before 0.5 × 97 + 0.5 × 88 = 92.5%; after 0.7 × 96 + 0.3 × 87 = 93.30%; at the old mix 91.5%.

Say exactly: "Not necessarily — in my example the raw number rises about 0.8 of a point while every cohort falls a point, because volume moved to the easy lanes; at the old mix it is down a full point. That is why a mix-adjusted twin prints beside every raw output."

T3 "Promise Yield is fast share × accuracy. Can a team game it?"

The numbers: 0.436 × 0.952 = 41.5%; fast share +2 pts, accuracy −1 pt → 0.456 × 0.942 = 43.0%.

Say exactly: "Yes, if it is the only constraint — two points of speed for one point of accuracy raises it from forty-one and a half to forty-three — so the accuracy floor is a hard red regardless of trend, and the yield is a headline, not a target."

S21 — Elementum / Walmart Marketplace — the stalled managed-shipping program: an incentive diagnosis, an all-or-nothing insurance contract and a shadow gate — about $3.7M a year per ten points of enrolled GMV, modeled NEW

LPs this story serves here: Dive Deep (primary — adverse selection in declared handling) · Invent and Simplify (an all-or-nothing contract) · Earn Trust (I claim the contract, never the curve) · Insist on the Highest Standards (shadow gate and graduation criteria)
Process family: Deliver seam — seller-fulfilled delivery: transit settings, handling and the promise (Walmart Marketplace; role R-2596147)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I read the stall as adverse selection, not marketing: sellers declared one-day handling to win placement and hit it about 85% of the time — the miss became the marketplace's broken promise.
- **Invent and Simplify** — I rebuilt the offer as one insurance contract — transit automation + observed handling + a participating label — with the benefit priced in the seller's own dollars.
- **Earn Trust** — I claim the contract design and wave-one mechanics only — the adoption curve after wave one belongs to the client — and I say "modeled" in the same breath as $3.7M.
- **Insist on the Highest Standards** — Shadow-run a seller's orders under managed settings before flipping; a wave that fails a published criterion does not board the next cohort.

Honesty label. Elementum engagement for Walmart Marketplace's seller-fulfilled program. The client's "Simplified Shipping Settings" (SSS) lets the marketplace manage a seller's transit settings and absorb carrier-delay accountability; adoption had stalled at roughly 38–42% of seller-fulfilled GMV. I led the adoption-program design: the incentive diagnosis, the insurance-contract offer, the wave mechanics and the measurement pair. The client's product and engineering teams built Seller Center, the enrollment flow and the observed-handling computation — I never configured or deployed any platform, and I claim the contract design and the wave-one mechanics, never the adoption curve that followed.

60-SECOND PITCH

"**Situation.** A marketplace client owning the delivery promise on inventory in sellers' buildings. Its managed-shipping program — the marketplace picks the transit settings and absorbs carrier-delay accountability — had stalled at roughly 38 to 42 percent of seller-fulfilled GMV, so the majority of promise risk still rode on manual seller configuration. And that configuration was optimistic by construction: sellers declared one-day handling to win placement and hit it about 85 percent of the time — adverse selection, not bad luck; the miss became the marketplace's broken promise. **Task.** Lead the adoption-program design: diagnose why sellers refused, rebuild the offer, design the rollout and its gates, and leave the client a measurement pair that can't be gamed. **Action.** Discovery: enrollment-funnel and refusal interviews — sellers were trading away control for a benefit that was real but unpriced and weakly communicated; an incentive-design problem, not a feature-marketing problem. Plan & Design: the offer rebuilt as an all-or-nothing insurance contract — transit automation plus observed handling plus a participating label, full bundle earns scorecard protection — with a personalized GMV-delta line in Seller Center making the benefit concrete in the seller's own dollars. Build & Test: shadow-run a seller's orders under managed settings before flipping, so day one is verified, not hoped. Cutover: waves behind published graduation criteria — a wave that fails one doesn't board the next cohort. Hypercare: the honest pair — activations per week *and* 30-day retention; enrollments without retention is churn wearing a growth costume. **Result.** Primary deliverable: the signed contract design and wave-one mechanics; the modeled economics — **about $3.7M a year per ten points of enrolled GMV** from handling-miss reduction alone — and I say 'modeled' in the same breath; the adoption curve after wave one belongs to the client, not to my record."

20-SECOND VERSION NEW

The managed-shipping program stalled at ~38–42% of GMV because sellers traded control for an unpriced benefit; I rebuilt it as an all-or-nothing insurance contract with a shadow gate and published wave criteria — about $3.7M a year per ten points of enrolled GMV, modeled; the adoption curve is the client's.

NACF BRIDGE — ONE SENTENCE

The NACF analog is vendor-managed inbound: where a vendor's declared lead time or handling is optimistic by incentive, I would move to observed performance and price the trade for the vendor.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The same Elementum client engagement — Walmart Marketplace's seller-fulfilled program, adoption side. The structural fact: the marketplace underwrites the delivery promise, but the inputs that keep it — handling time, ship nodes, carrier choices — are configuration data that thousands of independent sellers set themselves, optimistically, and update rarely. The managed-settings program existed precisely to fix this, and it had stalled at ~38–42% of seller-fulfilled GMV (record-from-docs). The refusal logic was rational: adoption asked sellers to give up control (their carrier rates, their handling declarations) for a benefit — accountability transfer — that was real but weakly priced and weakly communicated.

T — Lead the adoption-program design: diagnose the stall, rebuild the offer so adoption becomes the seller's rational choice, design the wave rollout with gates that protect both customer and seller, and define the measurement that keeps the program honest after handoff.

Why it mattered: with ~60% of GMV on manual configuration, the single largest driver of broken promises was seller-declared handling — sellers declared one day and hit it ~85% of the time (record-from-docs), and the 15% became the marketplace's broken promise at the marketplace's brand cost. Left unfixed: the promise team keeps buffering around seller optimism (taxing every customer on the lane to absorb one seller's miss), the scorecard keeps punishing sellers for carrier variance they don't control (eroding the program's legitimacy), and adoption stays stalled because the offer prices neither the benefit nor the trade.

The metrics on the table, with their baselines: managed-settings share of seller-fulfilled GMV — ~38–42%, stalled (record-from-docs); declared-handling adherence — ~85% on 1-day declarations (record-from-docs); the offer's value statement — unpriced (no seller could see what enrollment was worth in their own dollars); rollout mechanics — none designed; the measurement — enrollments counted, retention not.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (the baselines, the contract structure and the gate design are the engagement record; wave one ran as the pilot; the adoption curve beyond it is the client's, never claimed):

- **Discovery** — the refusal diagnosis: enrollment-funnel walk plus structured interviews with enrolled, refused and churned sellers (the refusals teach the most); the finding — fear of cost, loss of carrier control, distrust of the scorecard — an incentive-design problem, not a feature-marketing problem; the handling-adherence baseline written: declared 1-day, hit ~85%. *Exit bar:* the stall explained in the sellers' own words with the funnel's biggest drop-off named; the client's product owner co-signed the diagnosis.
- **Plan & Design** — the offer rebuilt as a priced insurance contract: (a) transit automation (the marketplace picks the service), (b) observed handling time (the marketplace computes the seller's real handling speed from scan history — the seller stops declaring, killing self-reported optimism at the source), (c) a participating label purchase — full bundle earns carrier-delay excusal from the seller's scorecard and the fast badge where the math clears; partial adoption earns nothing, so the hedge is all-or-nothing and self-selecting. Plus the pricing surface: a personalized line in Seller Center — "enrollment would raise your projected GMV by $X/month," computed by the client's team from the fast badge's measured conversion lift on that seller's assortment. *Exit bar:* the contract and its protection rules signed by product, seller ops and the scorecard owner; the design reference set (the dominant marketplace's equivalent bundle) documented, with the explicit decision *not* to copy its enforcement aggression while the client's own fulfillment alternative was subscale.

- **Build & Test** — the client's engineering built the enrollment flow, the observed-handling computation and the Seller Center line; I owned the verification design: shadow-run each enrolling seller's recent orders under managed settings before flipping — projected promise accuracy and badge eligibility computed on their own history, so day one is verified, not hoped. *Exit bar:* shadow results within tolerance of projection on the pilot cohort; the GMV-delta line reproducible by the client's analytics from logged lift data.
- **Cutover / Go-Live (wave one)** — wave mechanics: cohorts boarded behind published graduation criteria (shadow-verified settings, observed-handling coverage, inventory-feed accuracy at the seller's nodes, scorecard-protection rules armed); a wave that fails a criterion does not board the next cohort — publicly. *Exit bar:* wave one boarded and stable; no enrolled seller's scorecard damaged by a carrier miss the contract had excused; the refusal-reason codes flowing back into the offer design.
- **Hypercare / Close** — the measurement handoff: the honest pair on the program board — activations per week *and* 30-day retention — plus share of enrolled GMV under full-bundle protection; the kill condition written (a protection promise the scorecard fails to honor stops enrollment until fixed). *Exit bar:* the client's adoption PM owning the funnel, the pair and the gates; the engagement closed on the contract and the mechanics, with the curve explicitly left to the client's record.

The client's product team owned Seller Center and the enrollment surfaces; engineering built the observed-handling computation and the protection rules in the scorecard pipeline; seller ops ran the cohorts; their analytics computed each seller's GMV-delta line. Mine: the refusal diagnosis, the insurance-contract structure and its all-or-nothing logic, the pricing surface concept, the shadow-verification gate, the wave mechanics and graduation criteria, the measurement pair, and the recommendation on enforcement posture — carrots over sticks until the client's own fulfillment network was a credible alternative at scale.

The obstacle was the program's instinct to market the feature harder — more banners, more emails — when the stall was structural: an unpriced trade. The reframe that won the room: a seller who trusts their own ops should rationally keep control; a seller who doesn't should rationally convert — the offer's job is to make that self-selection visible in dollars, not to persuade everyone. The decision I challenged: requiring the label purchase as a hard cliff — sellers with negotiated carrier rates would read it as a tax. The recommendation: tier the protection with bundle depth rather than cliff it, and revisit when the fulfillment alternative scales. The trade went into the contract document as a priced row, not a hallway compromise.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary deliverable: the signed insurance-contract design, the shadow-verification gate, the wave-one mechanics and the measurement pair — with the modeled economics labeled: ≈$3.7M/yr per +10 points of enrolled GMV from handling-miss reduction alone. The adoption curve after wave one is the client's record, not mine.

- **$** — ≈$3.7M/yr per +10 pts of seller-fulfilled GMV enrolled, modeled: 100K orders/wk × (15% − 3%) handling-miss reduction × 50% breach-through × $12 × 52 (full chain in Table 4; the 3% and 50% are illustrative, the $12 is the model's stated assumption). No adoption "after" dollar — designed, not banked.
- **#** — One contract (three components, all-or-nothing), one pricing line per seller, one wave boarded; refusal interviews across enrolled, refused and churned cohorts; graduation criteria published before wave one (engagement design record).
- **%** — Baseline ~38–42% of seller-fulfilled GMV enrolled, stalled (record-from-docs); declared 1-day handling hit ~85% (record-from-docs); the designed measurement: % of enrolled GMV under full-bundle protection, alongside the pair.
- **Time** — Shadow verification before any seller's flip (day one verified, not hoped); wave cadence gated on criteria, not calendar; 30-day retention as the clock that keeps enrollment honest.
- **Customer** — The customer's promise stops depending on a seller's optimistic declaration: observed handling replaces declared at the source; the fast badge appears only where the math clears — the promise shown is the promise kept.
- **Quality** — The protection promise armed and audited (no enrolled seller's scorecard damaged by an excused carrier miss in wave one — the wave's exit bar); refusal reason codes flowing back into the offer; the kill condition written before launch.

Against the goal stated above: the stall diagnosed as incentive design, the offer rebuilt so the trade is explicit and priced in the seller's own dollars, wave one boarded behind published gates, and the program handed over measuring the pair that can't be gamed. The trade behind the design was itself priced: enforcement tuned down from the reference mechanism's aggression — a churned seller is a strategic loss to a marketplace still in assortment-growth mode — carrots first, the stick deferred until the fulfillment alternative is credible.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Why adoption stalled was unknown; I framed it as an incentive-design problem and tested it with funnel and refusal interviews.
Scope & influence	Client product and engineering, seller programs and sellers — none reporting to me; the contract needed all of them.
Execution	Offer redesign, shadow gate, wave criteria and the measurement pair; wave one boarded.
Impact	≈$3.7M/yr per ten points of enrolled GMV, modeled; no claim on the adoption curve.
Process improvement	Graduation gates and an honest measurement pair (activations and 30-day retention) with a kill condition.
Complexity	Adverse selection, all-or-nothing bundling, shadow replay against declared settings.
Root cause	Declared handling is a bid for placement; the stall was incentives, not awareness.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: enrollment funnel, refusal interviews, declared vs observed handling (85% hit rate), the GMV share on managed settings (~38–42%). Revealed: optimistic declaration; an unpriced benefit.
2 · Solution design	Options: more marketing · a priced all-or-nothing contract · mandatory enrollment. Chosen: insurance contract with shadow gate and published graduation criteria.
3 · Math & data logic	Math: 1M × 60% × 15% × 50% × $12 = $540,000/wk on the manual book; +10 pts → 100K × 12% × 50% × $12 × 52 = $3,744,000/yr; flip rule c = 11 misses in 200 shadow orders.
4 · Implementation	Implementation: diagnose → contract → shadow gate → waves behind criteria → activations × 30-day retention.
5 · Probe questions	Probe questions: interval on 85% · sensitivity of $3.7M · false-pass of the shadow gate · churn vs growth — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

> ***Challenge* — Why did you choose this story to illustrate Think Big?**
>
> Because the move is structural, not incremental: the program stopped marketing a settings toggle and started selling an insurance contract — accountability transfer, priced per seller, self-selecting at marketplace scale. Thinking big here meant redesigning the incentive so thousands of independent sellers each make the right call on their own.

> ***Probing* — Why did you choose to focus on these results? What other results were important?**
>
> The contract design leads because it is the thing I can honestly claim — the diagnosis, the structure, the gates, the measurement. Also important: the enforcement-posture recommendation (carrots until the alternative scales) and the kill condition — a protection promise the scorecard fails to honor stops enrollment, because one broken protection kills the program's legitimacy faster than slow adoption does.

> ***Probing* — You mentioned dollars — what percentage change is that year over year?**
>
> None claimed — the ≈$3.7M/yr is per ten points of enrolled GMV, modeled from the handling baseline, and the curve itself is the client's record. I give the unit economics so the investment argument is quantitative; I don't book the outcome.

***Probing* — I'm concerned you're taking credit for a program you didn't run to completion — tell me more.**

That's exactly why the claim is scoped: mine is the diagnosis, the contract, the gates, wave one and the measurement pair; the client's is the curve. I'd rather under-claim a design than over-claim an outcome — the same discipline as S17, where I refuse the award dollar on the carrier bid.

***Challenge* — What were the lessons learned? What would you have done differently?**

Lesson: adoption stalls are usually incentive-design problems wearing feature-marketing costumes — price the trade and the funnel moves. Differently: start the refusal interviews before touching the funnel data; the sellers' own words located the drop-off faster than the analytics did, and I ran them second.

***Challenge* — How would you implement this at Amazon?**

Amazon runs the reference version of this contract — observed handling, settings automation, protected labels, all-or-nothing accountability transfer. What I'd bring is the posture judgment: knowing when the contract's teeth fit the business (a mature fulfillment alternative, churn tolerance) and when they don't — and the measurement pair, because enrollments without retention is churn wearing a growth costume at any company.

***Challenge* — How did these results compare to your actual goals?**

The engagement's signed goals were a diagnosed stall, a rebuilt offer, a gated wave one and an honest measurement handoff — all four delivered. The goal was never "move the share N points under my name"; that outcome belongs to the owners who run the waves.

PHASE DEEP-PROBE BANK — Q&A (12). *Answer the phase asked, then stop.*

Discovery — How did you know discovery was done?

When the stall had one explanation that survived the sellers' own words, the funnel data and the product team's review: a real benefit, unpriced and weakly communicated, traded against a visible loss of control. The exit memo named the funnel's biggest drop-off and the three refusal reasons in rank order.

Why interview churned sellers, not just refusers?

Because churn is the expensive refusal — someone who tried the program and left knows exactly which promise broke. Wave-one's protection-audit exit bar came directly from a churned seller's story about a scorecard ding the program should have excused.

Plan & Design — Why all-or-nothing instead of à la carte?

Because partial adoption keeps the adverse selection: a seller who keeps declared handling keeps the optimism that breaks promises. Cede all three controls and the marketplace can underwrite you; keep any one and you keep the risk. The bundle is what makes the contract insurable — and self-selecting.

Why does observed handling matter so much?

It kills self-reported optimism at the source. Declared handling is a bid for placement (declare one day, hit it 85%); observed handling is a measurement. Moving the promise onto what sellers actually do — rather than what they say — is the single largest accuracy lever in the seller-execution family.

What was the reference mechanism and what did you refuse to copy?

The dominant marketplace's equivalent: the same three-part bundle with full accountability transfer, enforced aggressively. We copied the contract structure and the grain. We did not copy the enforcement aggression or the churn tolerance — this client was still in assortment-growth mode, its fulfillment alternative subscale, so a churned seller was a strategic loss. Tiered protection instead of a cliff, revisited when the alternative scales.

Build & Test — What exactly did the shadow run verify?

Each enrolling seller's recent orders replayed under managed settings: projected promise accuracy, badge eligibility, and the scorecard-protection rules firing where they should. Day one verified on the seller's own history, not hoped — the same calibration-before-onboarding toll as my Flexport waves (S11).

How was the GMV-delta line computed and policed?

The client's analytics computed it per seller from the fast badge's measured conversion lift applied to that seller's assortment — their model, reproducible from logged lift data. My rule on it: the line is a projection and labeled as one in Seller Center; an unlabeled promise there would be the program lying to its own sellers.

Cutover — What were the wave-one graduation criteria?

Shadow results within tolerance of projection; observed-handling coverage on the cohort; inventory-feed accuracy at the seller's ship-from nodes (a stale feed causes cancellations — the worse defect); and the protection rules armed and audited. A wave that failed a criterion did not board the next cohort — publicly.

What was the kill condition?

A protection promise the scorecard fails to honor — an enrolled seller dinged for an excused carrier miss — stops enrollment until fixed. The contract's credibility is the product; one broken protection spends it faster than any growth banks it.

Hypercare — Why activations *and* 30-day retention?

Because enrollments without retention is churn wearing a growth costume. The pair is the honest read: throughput says the funnel works; retention says the contract kept its promise. A program graded on activations alone will board sellers it should have fixed first.

What flowed back into the offer after wave one?

Refusal and churn reason codes, as data: which component sellers feared (the label requirement led), which protection they doubted, where the GMV-delta line read as marketing instead of math. Overrides with reasons are data; silent refusals are a funnel you can't fix.

Cross-phase — What did you refuse?

An adoption target under my name (the curve is the client's), a hard label-purchase cliff (a tax on sellers with negotiated rates — tiered instead), enrollment pushes ahead of the shadow gate (day one must be verified), and counting activations without retention. Every refusal is in the contract document as a priced row.

Analytics — How sure are you that handling adherence is 85%?

The 85% is a record-from-docs baseline. As statistics: a seller with 100 shipments at 85% on time has a 95% Wilson interval of 76.7% to 90.7%; with 400 shipments, 81.2% to 88.2%. That is why “observed handling” needs enough scan history per seller before it replaces a declaration.

Analytics — What does the shadow gate actually test?

Replay a seller’s recent orders under managed settings and flip only within tolerance of the projection. At a projected 3% miss rate on 200 orders, flipping when misses ≤ 10 has a 4.0% false-reject rate on a seller who truly is at 3%, and detects a seller truly at 6% with 66% probability. Illustrative gate; the record is shadow-run-before-flip.

STATISTICAL SPINE — INTERVALS, SENSITIVITY, SHADOW GATE, RETENTION (EXTRA TABLES)

The record is the engagement design and the modeled economics; the tables below add interval and sensitivity arithmetic. The adoption curve after wave one is the client’s and is not claimed.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Orders on manual configuration	`1,000,000 × (1 − 40%) = 600,000/wk`	enrolled ≈ 38–42%, 40% midpoint (record-from-docs)	600,000/wk
2	Seller-made handling misses	`600,000 × 15% = 90,000/wk`	85% hit rate (record-from-docs)	90,000/wk
3	Breach-through	`90,000 × 50% = 45,000/wk`	50% illustrative	45,000/wk
4	Cost of the manual book	`45,000 × $12 = $540,000/wk`	$12 breach bundle (model assumption)	≈$540,000/wk

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	+10 points of enrolled GMV	`1,000,000 × 10% = 100,000 orders/wk`	illustrative book	100,000/wk
2	Handling misses avoided	`100,000 × (15% − 3%) = 12,000/wk`	3% post-fix illustrative	12,000/wk
3	Breaches avoided	`12,000 × 50% = 6,000/wk`	50% illustrative	6,000/wk
4	Per week / per year	`6,000 × $12 = $72,000/wk; × 52 = $3,744,000`	modeled	≈$3.7M/yr per 10 pts
5	Sensitivity — miss reduction 6 / 12 / 15 pts	`$1.87M / $3.74M / $4.68M`	one-at-a-time	$1.87M–$4.68M
6	Sensitivity — breach-through 25% / 50% / 75%	`$1.87M / $3.74M / $5.62M`	one-at-a-time	$1.87M–$5.62M
7	Sensitivity — breach cost $6 / $12 / $18	`$1.87M / $3.74M / $5.62M`	one-at-a-time	$1.87M–$5.62M
8	Shadow gate	`n = 200, p₀ = 3%: reject if misses ≥ 11; false-reject 4.0%; power at 6% = 66%`	binomial, illustrative	c = 11
9	Churn in a growth costume	`200 enrollments × 50% retained = 100; 100 × 90% = 90`	illustrative weekly cohorts	raw enrollments say 2×; retained say 1.1×
10	Retention estimate, one wave	`40 of 50 retained = 80%; Wilson [67%, 89%]`	illustrative wave size	80% ± ≈12 pts

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Promise kept at the source + adoption that sticks (primary outcome)
+- Handling-miss rate on manual-config GMV (flow) -> declared ~85% kept
|    miss -> breach-through -> broken promise -> WISMO + trust decay
+- Enrolled GMV share (stock) -> ~38-42% stalled -> waves behind gates
|    (loop) priced offer -> rational enrollment -> observed handling
|           -> fewer misses -> badge earned -> conversion lift -> more GMV
+- Protection credibility (stock) -> kill condition; audit per wave
+- Measurement honesty -> activations/wk x 30-day retention (the pair)
+- Unit economics -> ~$3.7M/yr per +10 pts enrolled GMV (modeled)
```

DRIVER	BASELINE	AFTER (DESIGNED)	$/YR	BASIS
Enrolled share of seller-fulfilled GMV	~38–42%, stalled	waves behind gates; curve unclaimed	—	record-from-docs baseline
Handling adherence on declared 1-day	~85%	observed handling replaces declared; ~97% modeled	inside the unit economics	85%: record-from-docs; 97%: illustrative
Unit economics per +10 pts enrolled GMV	—	≈6,000 breaches/wk avoided	≈$3.7M	modeled (Table 4; $12 assumption)
Offer pricing surface	benefit abstract	personalized GMV-delta line per seller	—	engagement design record (client-computed)
Program honesty	enrollments counted	activations/wk × 30-day retention + full-bundle share	—	engagement design record

If challenged. *"Your result is a design, not an outcome."* — "Yes — and I say exactly that. Mine is the diagnosis, the contract, the gates, wave one and the measurement pair; the adoption curve is the client's record. The production-record version of me running an adoption machine end-to-end is S11: 7 to 68 behind published gates." *"The $3.7M rests on two illustrative rates."* — "It does — the 3% post-fix miss rate and the 50% breach-through are labeled

illustrative; the 85% baseline and the $12 are the engagement record's own figures. Halve both assumptions and the unit is still about a million a year per ten points — the argument survives." *"Why not just mandate enrollment?"* — "Because a mandate spends seller trust the marketplace hasn't earned yet — and a churned seller is a strategic loss in assortment-growth mode. The contract makes enrollment the rational hedge; mandates are for after the fulfillment alternative is credible."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	GMV on manual configuration	`1 − enrolled share = 1 − 40% = 60%`	enrolled ~38–42% (record-from-docs); 40% midpoint used	≈60% of seller-fulfilled GMV on declared settings
2	Orders on manual config	`1,000,000 × 60% = 600,000 / wk`	~1M orders/wk illustrative book	≈600K orders/wk
3	Handling misses	`600,000 × (1 − 85%) = 600,000 × 15% = 90,000 / wk`	declared 1-day handling hit ~85% (record-from-docs)	≈90K seller-made misses/wk
4	Breach-through	`90,000 × 50% = 45,000 / wk`	a handling miss of a day typically exhausts the buffer; 50% breach-through (illustrative)	≈45K broken promises/wk from handling alone
5	Cost of the manual book	`45,000 × $12 = $540,000 / wk`	$12 breach bundle (the model's stated assumption)	≈$540K/wk of handling-driven breach cost
6	Why it persists	`declared handling = a bid for placement; adoption = control traded for an unpriced benefit`	adverse selection + unpriced trade (record-from-docs diagnosis)	a stall no banner fixes

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The contract	`protection ⇔ (transit automation) AND (observed handling) AND (participating label)`	all-or-nothing bundle (engagement design)	full bundle → carrier-delay excusal + badge where the math clears
2	Observed handling effect	`miss rate 15% → ~3% on enrolled orders`	handling set from scan history, not declaration; 3% illustrative	−12 points of handling misses on enrolled GMV
3	Pricing the benefit	`seller line = conversion lift × seller's eligible assortment GMV = "$X/month"`	client-computed from measured badge lift (design)	the trade visible in the seller's own dollars
4	Self-selection logic	`trusts own ops ⇒ keeps control and risk; doesn't ⇒ converts`	incentive design, not persuasion	adoption as the rational hedge
5	Shadow gate	`replay seller's recent orders under managed settings; flip only within tolerance of projection`	verification design	day one verified, not hoped
6	Wave mechanics	`cohort boards ⇔ all published criteria pass; fail one ⇒ hold, publicly`	graduation gates (design; the S11 motion)	adoption earns its expansion
7	The honest pair	`program health = activations/wk AND 30-day retention AND full-bundle share`	measurement design	growth that can't be faked by churn

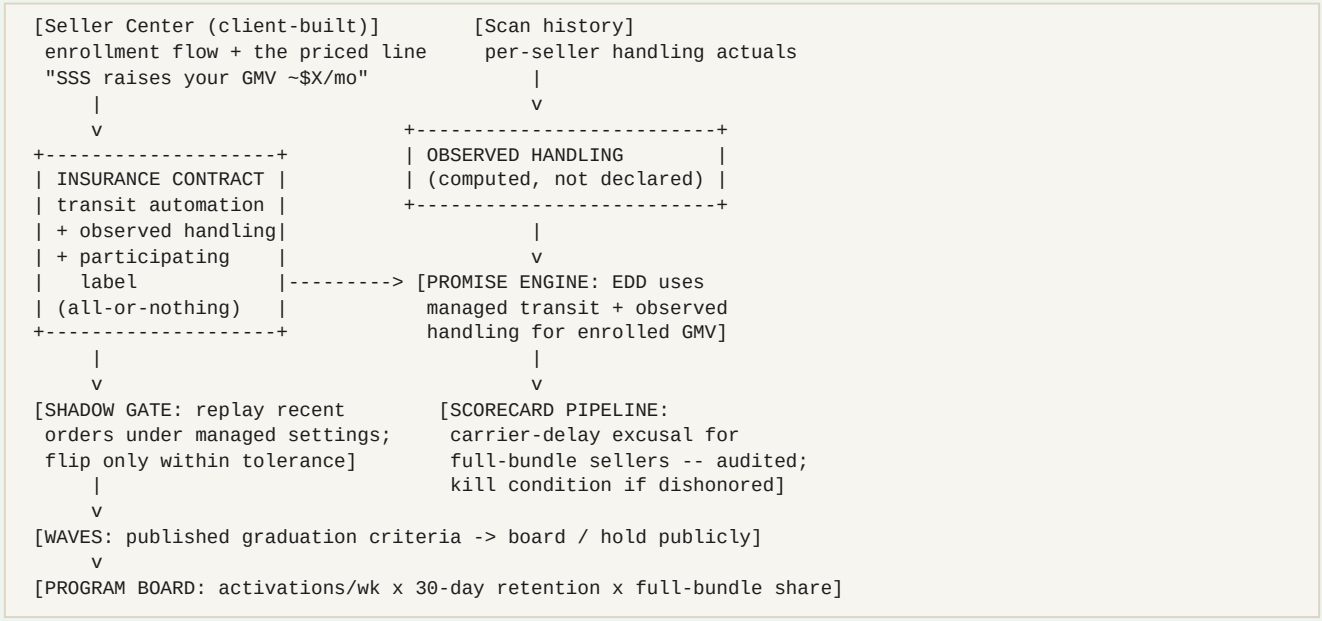
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	The offer	settings toggle, benefit abstract	priced insurance contract, benefit in seller dollars	engagement design record
2	Handling input on enrolled GMV	declared (15% miss)	observed (~3% miss, modeled)	15%: record-from-docs; 3%: illustrative
3	Rollout mechanics	none (push marketing)	shadow gate + waves behind published criteria; wave one boarded	engagement design record
4	Program measurement	enrollments	activations/wk × 30-day retention × full-bundle share; kill condition armed	engagement design record
5	Enrolled GMV share	~38–42% stalled	curve unclaimed — the client's record	record-from-docs baseline; honesty rule

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Orders in +10 pts of enrolled GMV	`1,000,000 × 10% = 100,000 / wk`	~1M orders/wk illustrative book	100K orders/wk move onto the contract
2	Handling-miss reduction	`15% − 3% = 12 points`	15% record-from-docs; 3% illustrative post-fix rate	12 pts fewer misses
3	Misses avoided	`100,000 × 12% = 12,000 / wk`	rows 1 × 2	≈12K misses/wk avoided
4	Breaches avoided	`12,000 × 50% = 6,000 / wk`	50% breach-through (illustrative)	≈6K broken promises/wk avoided
5	Weekly value	`6,000 × $12 = $72,000 / wk`	$12 breach bundle (the model's stated assumption)	≈$72K/wk
6	Annualized	`$72,000 × 52 = $3,744,000`	row 5 × 52 (modeled)	≈$3.7M/yr per +10 pts enrolled
7	Not claimed	`adoption curve × unit economics = the client's outcome`	honesty rule	unit economics only; conversion-lift upside also excluded

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Post-fix handling-miss rate (illustrative)	6%	3%	2%
Breach-through on a handling miss (illustrative)	30%	50%	70%
Breach cost (model assumption)	$6	$12	$18
Breaches avoided per +10 pts (/wk)	≈2,700	≈6,000	≈9,100
Modeled $/yr per +10 pts enrolled	≈$0.8M	≈$3.7M	≈$8.5M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The honest version first: what I delivered is the contract, not the curve. The program was stalled around forty percent of seller-fulfilled GMV because it asked sellers to trade control for an unpriced benefit — so we rebuilt it as insurance: give up transit settings, declared handling and the label, and the marketplace absorbs carrier-delay accountability and shows you what the fast badge is worth on your own assortment, in dollars. The unit economics are modeled and I label them: on the engagement record's own baseline — sellers declaring one-day handling and hitting it eighty-five percent of the time — every ten points of GMV that enrolls is worth about three point seven million dollars a year in avoided broken promises at the model's twelve-dollar breach cost. Wave one boarded behind published gates; the adoption curve after that belongs to the client's owners, and I say so."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you found the real reason a program stalled.** DIVE DEEP

Open (first two sentences): Adoption was stuck near forty percent and everyone wanted a better banner. The data said sellers were making a rational trade.

STAR (≈90 s): Situation: a managed-shipping program stalled at ~38–42% of seller-fulfilled GMV. Task: diagnose why. Action: enrollment-funnel and refusal interviews; the pattern was control traded for an unpriced, weakly communicated benefit, and declared one-day handling that sellers hit about 85% of the time — adverse selection. Result: the diagnosis moved the work from marketing to incentive design; the contract followed.

Land this number: 85% handling adherence; ~38–42% enrolled. · **Pivot to NACF:** NACF vendor ASN accuracy has the same incentive: declared dates are bids. · **Do NOT say:** Do not claim the adoption curve after wave one.

Q2 **Tell me about a time you redesigned an offer instead of pushing the existing one.** INVENT AND SIMPLIFY

Open (first two sentences): I made it one contract: give up transit settings, declared handling and take the label — and the marketplace absorbs carrier delays.

STAR (≈90 s): Situation: sellers refused a toggle. Task: a design they would choose. Action: an all-or-nothing insurance bundle — transit automation + observed handling + participating label — with a personalized GMV-delta line in Seller Center showing the benefit in the seller's own dollars. Result: signed contract design and wave-one mechanics; modeled ≈$3.7M a year per ten points of enrolled GMV.

Land this number: ≈$3.7M/yr per 10 pts (modeled). · **Pivot to NACF:** A priced, bundled commitment beats a toggle at NACF too. · **Do NOT say:** Do not call $3.7M a result.

Q3 **Tell me about a time you were precise about what was yours.** EARN TRUST

Open (first two sentences): I claim the contract design and the wave-one mechanics. The adoption curve after that belongs to the client, and I say so.

STAR (≈90 s): Situation: a program whose headline number — enrolled GMV — was moved by many hands. Task: keep my claim to what I did. Action: the honesty pair — activations per week and 30-day retention — so growth cannot be faked by churn; and in every retelling, "modeled" is in the same breath as the dollar. Result: a claim that survives a reference check.

Land this number: Wave one boarded; $3.7M modeled. · **Pivot to NACF:** In an Amazon interview, the line between "I designed" and "we shipped" is the line. · **Do NOT say:** Do not annualize or extrapolate the curve.

Q4 **Tell me about a time you held a high bar on rollout quality.** INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): Day one is verified, not hoped: we replayed a seller's orders under managed settings before flipping.

STAR (≈90 s): Situation: flipping a seller wrongly breaks a promise. Task: a gate. Action: shadow-run before flip within tolerance of the projection; waves behind published graduation criteria — fail one, hold, publicly. As a design check: at n = 200 orders and a 3% projection the gate rejects a good seller 4.0% of the time and catches a 6% seller 66% of the time. Result: wave one boarded behind the criteria.

Land this number: c = 11 misses in 200 orders (illustrative). · **Pivot to NACF:** A NACF vendor-lane activation gate on valid-ASN rate, as in S9. · **Do NOT say:** Do not quote gate numbers as the record.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: “Declared handling was a bid for placement.”

Headline metric: 85% hit rate; manual book ≈$540K/wk breach cost (model)

- 1M × 60% × 15% × 50% × $12.
- 85% interval at n = 100: [76.7%, 90.7%].
- Contract: observed handling replaces declaration.
- Sensitivity $1.9M–$5.6M on miss reduction.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Earn Trust

Open: “I would not put the adoption curve on my résumé.”

Headline metric: claim = contract + wave-one mechanics

- The curve is the client’s.
- “Modeled” in the same breath.
- Kill condition armed on the measurement pair.
- Honest pair: activations × retention.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: “The incentive diagnosis was not my first hypothesis.”

Headline metric: refusal interviews changed the frame from marketing to incentives

- The funnel alone suggested awareness.
- Interviews showed an unpriced trade.
- I would instrument seller-level retention earlier.
- VERIFY-WITH-RAJESH that this was the order of discovery.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 “Your $3.7M — which assumption dominates?”

The numbers: +10 pts: 100,000 × 12% × 50% × $12 × 52 = $3,744,000; miss reduction 6/15 pts → $1.87M/$4.68M; breach-through 25/75% → $1.87M/$5.62M; breach cost $6/$18 → $1.87M/$5.62M.

Say exactly: “It is a product of four inputs, so a halving or doubling of any one moves it proportionally — from about $1.9 million to $5.6 million on breach-through, $1.9 to $5.6 on the breach cost — and I call it modeled in the same breath; the client’s own data replaces each input.”

T2 “Eighty-five percent — what is the interval, and is 15% → 3% plausible?”

The numbers: n = 100: [76.7%, 90.7%]; n = 400: [81.2%, 88.2%]; 3% is illustrative.

Say exactly: “On a hundred shipments eighty-five percent is anywhere from about 77 to 91; the fifteen-to-three-point effect is an illustrative target — observed handling is the mechanism, and I would not quote three as measured.”

T3 “Enrollments are up — is the program working?”

The numbers: 200 enrollments × 50% retained = 100; 100 × 90% = 90; Wilson at 40/50 retained: [67%, 89%].

Say exactly: “Not necessarily — two hundred enrollments at fifty percent retention retain a hundred and a hundred enrollments at ninety retain ninety; raw enrollments say twice as good, retained activations say about the same, so the pair is activations per week and thirty-day retention, with a kill condition.”

E7 — Elementum design engagement (new-market broadband build) — equipment forecast-to-buy: a newsvendor first buy, risk-pooled central stock and a ±20% two-week re-forecast trigger, with the statistics behind each (design, not run) NEW

LPs this story serves here: Learn and Be Curious (primary — the toolkit applied, labeled) · Think Big (a repeatable launch template) · Are Right, A Lot (a decision under maximum uncertainty) · Customer Obsession (install delays caused by stockouts → 0)
Process family: Plan / Source seam — forecast-to-buy for a market launch (master E7; Google Fiber-style field-install build; finished-kit grain, no component explosion)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Learn and Be Curious** — I applied the newsvendor critical ratio and risk pooling from the MIT MicroMasters toolkit to a cold-start launch — labeled as the toolkit, not as a program I ran.
- **Think Big** — The design is a template for market N+1: build plan → adoption curve → requirement → measured lead times → first buy held central → forward-deploy as sell-through proves the curve.
- **Are Right, A Lot** — Under maximum uncertainty I priced the decision — Cu ≈ $300 vs Co ≈ $3/month → CR 0.99 — and wrote the trigger that proves the curve wrong in two weeks, not two quarters.
- **Customer Obsession** — The metric is install delays caused by stockouts → 0 in the first 90 days — a missed install is a customer waiting, priced at ≈$300 in the model.

Honesty label. Design engagement, not a run record. I have not owned a market-launch forecast end to end; the forecast number belongs to Planning — I own the equation, the measurement and the triggers (master E7). The newsvendor and pooling formulas are the MIT MicroMasters toolkit, labeled. All volumes, CVs and correlations here are illustrative. The master's E-series does not name the client; the brief ties them to Google Fiber. VERIFY-WITH-RAJESH that you are comfortable saying "Google Fiber" aloud — otherwise say "a broadband build-out client." Finished-kit grain only — no component explosion.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A broadband client launches a new market: build sites ramp by month, demand follows an adoption curve with no local history, and long-lead equipment takes weeks to months to arrive. A missed install means a re-roll and deferred revenue; extra stock means holding cost and stranding risk. Task. Design the forecast-to-buy chain so installs are never delayed by stockouts and stranded stock stays inside a band. Action. Four decisions, each with its arithmetic. First buy: a newsvendor — a missed install costs about $300, a unit held a month about $3, so the critical ratio is 0.990 and I order the 99th percentile of the first quarter's demand for the core unit; accessories with a lower miss cost sit at the 70th. Held central: one pool covers several markets' variance, √k instead of k — if the markets are uncorrelated, which I would measure. Trigger: actual take-rate versus the curve beyond ±20% for two consecutive weeks forces a new forecast version. Demand enters planning once, through the engine — no spreadsheets beside it. Result. Primary metric: install delays caused by stockouts → 0 in the first 90 days of a launch. Illustrative dollar: 2,000 installs a month × 3% reschedules × $300 ≈ $18K a month, about $216K a year per market, plus about $100K of stranding avoided — design arithmetic, not a program result.

20-SECOND VERSION NEW

For a new-market build I would buy the core unit at the 99th percentile of first-quarter demand because a missed install costs 100× a unit-month of holding, hold it central so uncertainty pools, and re-forecast when the take-rate leaves ±20% of the curve two weeks running.

NACF BRIDGE — ONE SENTENCE

NACF placement for a new FC or a new product launch has the same shape: a critical-ratio first buy, central-first, and a variance trigger — INV-C's placement model plus INV-B's service-level logic.

SITUATION / TASK — THE PROBLEM SPACE

S — Market N launches: build sites ramp by month from the project system's build plan; demand follows an adoption curve with no local history; kits and long-lead equipment have vendor lead times of weeks to months. The books: the project system for the build plan, Planning for the adoption curve, SAP MRP for supply.

T — Design the forecast-to-buy chain so installs are never delayed by stockouts and stranded stock stays inside a band — the equation, the measurement and the triggers; Planning owns the forecast number.

Baselines on the table (illustrative): 2,000 installs a month per market · 3% of installs rescheduled for stockouts · missed install ≈ $300 · holding ≈ $3/month · first buy ≈ $500K.

ACTION — FIVE GATED PHASES (DESIGN)

- **Discovery** — restated the problem as a stock-and-flow map — build plan → sites passed → adoption curve × marketing → install demand → requirement by SKU × week → MRP on measured lead times → POs → central pool → market transfers. Exit bar: the three books of record named; the primary metric agreed.
- **Plan & Design** — newsvendor first buy; central-first with risk pooling; re-forecast trigger ±20% for two weeks; invariants: demand enters planning once, lead times are measured distributions (the S8 rule), floors with hold-until dates so one market cannot steal another's launch. Exit bar: each rule has its arithmetic on one page.
- **Build & Test** — the arithmetic below — newsvendor cost curve, pooling under correlation, trigger false-alarm and detection — run on illustrative inputs; failure modes listed (a borrowed adoption curve, a brochure lead time, manual loads). Exit bar: every rule survives its own sensitivity.
- **Cutover & Go-Live (design)** — launch plan: first buy held central; forward-deploy via transfers as sell-through proves the curve; bias sign published weekly from month two. Exit bar: install delays caused by stockouts = 0 in the first 90 days.
- **Hypercare / Close (design)** — MAPE and bias per SKU × market × week; days-of-cover inside the band; the template re-run for market N+1. Exit bar: a floor breach is a governed screen with a reason code, not an email.

Influence and ownership. This is a design I owned the equation of; Planning owns the forecast, Finance the unit costs, Marketing the adoption curve. The backbone content is the invariant: demand enters planning exactly once, through the engine — manually loaded requirements beside it are blocked — and need floors stop one market starving another.

RESULTS — DESIGN OUTCOMES, ALL ILLUSTRATIVE

- **Primary metric (design target)** — install delays caused by stockouts → 0 in the first 90 days of a launch.
- **$** — 2,000 installs/month × 3% × $300 = $18,000/month ≈ $216,000/yr per market; 20% × $500K = $100,000 of stranding avoided by holding central (master, illustrative) — never a program result.
- **Statistics** — critical ratio 0.990 → z = 2.33 → first-quarter buy 9,495 units on a 6,000-unit mean; pooling saves 50% of buffer at ρ = 0 and 31% at ρ = 0.3 (k = 4).
- **Quality** — MAPE and bias per SKU × market × week from month two; bias sign before magnitude.
- **Honest limit** — not run end to end; the forecast is Planning's.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A launch with no history; I restated it as a stock-and-flow problem with a priced objective.
Scope & influence	Planning, Finance, Marketing, vendors and field operations; the design is the influence.
Execution	Newsvendor, pooling, trigger and invariants each with arithmetic and a failure-mode list.
Impact	Design arithmetic: ≈$216K/yr per market of avoided reschedules plus ≈$100K stranding avoided (illustrative).
Process improvement	A launch template for market N+1; demand enters once; floors with hold-until dates.
Complexity	Cold-start demand, correlated markets, long-lead supply, a borrowed adoption curve risk.
Root cause	Launch stockouts come from an unpriced first buy and an untested curve — not from vendor behavior.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data (design): build plan by month, adoption curve by archetype market, install-reschedule reasons joined to the supply view, vendor lead-time distributions (S8). Revealed (illustrative): the 3% baseline reschedule rate corresponds to ordering near the 79th percentile.
2 · Solution design	Options: forecast mean · p79 · p99 first buy; forward-deploy vs central-first; trigger at ±10% / ±20% / ±30%. Chosen: p99 on the core unit, p70 accessories, central-first, ±20% two weeks.
3 · Math & data logic	Math: CR = 300 ÷ 303 = 0.9901; z = 2.330; Q = 6,000 + 2.330 × 1,500 = 9,495; expected short σ·L(z) = 5.0; pooling σ√(k + k(k−1)ρ); trigger DP on consecutive exceedances.
4 · Implementation	Implementation: stock-and-flow map → rules with arithmetic → invariants → launch plan → weekly MAPE/bias → template for N+1.
5 · Probe questions	Probe questions: why p99 · what if the CV is wrong · does pooling survive correlation · trigger false alarms — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

Analytics — Show the newsvendor logic with numbers.

Critical ratio = Cu ÷ (Cu + Co) = 300 ÷ 303 = 0.9901, so order the 99.0th percentile of first-quarter demand: z = 2.330. With mean 6,000 and σ = 1,500 (CV 25%, illustrative) that is 9,495 units; expected shortfall σ·L(z) = 5.0 units, expected leftover 3,500; expected cost $12,009, versus $58,234 at the 79th percentile.

Analytics — Why does the master's 3% reschedule rate matter?

Because it pins the baseline: a 3% shortfall on a 25% CV is L(z) = 0.03 ÷ 0.25 = 0.12 → z ≈ 0.8, the 79th percentile — a first buy near the 79th percentile reproduces the 3% and the $18K a month. The design moves it to the 99th percentile: expected shortfall 0.08% of installs. Illustrative CV; the master's 3% is the anchor.

Analytics — Your CV is a guess. What if it is 35%?

Then a buy sized for 25% at the 99th percentile — 9,495 — sits at z = 1.66 on the true spread: a 4.8% chance of running short and an expected shortfall of 42 units, versus 5.0. The policy is robust in direction and fragile in tail: so the buy is re-sized as soon as the first weeks' CV is observed, and the trigger exists for the same reason.

Analytics — Why is the central pool better — and when does it stop being better?

Pooling k markets cuts the buffer from k·zσ to z·σ·√(k + k(k−1)ρ): at k = 4 and independence, 50%; at ρ = 0.3, 31.1%; at ρ = 0.6, 16.3%; at ρ = 1, nothing. New markets share a product, a campaign and an adoption archetype, so ρ is probably not zero — I would measure it on analog markets before promising the saving. VERIFY-WITH-RAJESH

Shadow / closing — What would you do differently?

Measure the cross-market correlation on analog markets before quoting a pooling saving, and archetype-review the adoption curve with Marketing first — a curve borrowed from a non-analog market is the failure mode that voids everything else.

BUSINESS IMPACT MODEL — WHAT THE FORECAST-TO-BUY DESIGN IS FOR

```
Install delays caused by stockouts -> 0 (first 90 days)
+- First buy at the critical ratio: p99 core unit, p70 accessories
+- Central pool: one buffer covers several markets (sqrt k, if uncorrelated)
+- Re-forecast trigger: take-rate vs curve beyond +/-20% for 2 weeks
+- Invariants: demand enters once; measured lead times; floors with hold-until
+- Dollars (illustrative): reschedules avoided + stranding avoided
```

DRIVER	BASELINE	AFTER	$	BASIS
Reschedules from stockouts	3% of 2,000 installs/month	→ ≈0	$18,000/month ≈ $216,000/yr per market	master illustrative
Stranding avoided	20% of a $500K first buy in-market	held central	≈$100,000	master illustrative
Expected first-quarter cost, core unit	p79: $58,234	p99: $12,009	$46,225 lower	newsvendor arithmetic (illustrative)

If challenged. *"Did you save $216K a year?"* — "No — it is design arithmetic on illustrative inputs; I did not run this launch."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the baseline in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Baseline (master)	`2,000 × 3% × $300 = $18,000/month; × 12 = $216,000/yr`	master E7 illustrative	$216,000/yr per market
2	What the 3% implies	`shortfall fraction = CV × L(z) → 0.03 = 0.25 × L(z) → L(z) = 0.12 → z ≈ 0.8 (p79)`	standard normal loss function	baseline ≈ p79 first buy
3	The 3-month demand	`μ = 2,000 × 3 = 6,000; σ = 25% × 6,000 = 1,500`	CV 25% illustrative	σ = 1,500
4	Stranding at risk	`20% × $500,000 = $100,000`	master illustrative	$100,000

Table 2 — the fix: the newsvendor first buy, every row substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Critical ratio	`Cu ÷ (Cu + Co) = 300 ÷ 303 = 0.9901 → z = Φ⁻¹ = 2.330`	Cu ≈ $300; Co ≈ $3/month (master)	order the 99.0th percentile
2	Buy at p50	`z = 0.0000: Q = 6,000 + 0.0000 × 1,500 = 6,000; short σL = 598.413; left σ(z+L) = 598.413; cost = 598.413 × $300 + 598.413 × $3 = $181,319`	normal demand, one-month holding (illustrative)	$181,319
3	Buy at p79	`z = 0.8000: Q = 6,000 + 0.8000 × 1,500 = 7,200; short σL = 180.311; left σ(z+L) = 1,380.311; cost = 180.311 × $300 + 1,380.311 × $3 = $58,234`	normal demand, one-month holding (illustrative)	$58,234
4	Buy at p90	`z = 1.2816: Q = 6,000 + 1.2816 × 1,500 = 7,922; short σL = 71.007; left σ(z+L) = 1,993.407; cost = 71.007 × $300 + 1,993.407 × $3 = $27,282`	normal demand, one-month holding (illustrative)	$27,282

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
5	Buy at p95	`z = 1.6450: Q = 6,000 + 1.6450 × 1,500 = 8,468; short σL = 31.328; left σ(z+L) = 2,498.828; cost = 31.328 × $300 + 2,498.828 × $3 = $16,895`	normal demand, one-month holding (illustrative)	$16,895
6	Buy at p99 (CR)	`z = 2.3301: Q = 6,000 + 2.3301 × 1,500 = 9,495; short σL = 5.027; left σ(z+L) = 3,500.146; cost = 5.027 × $300 + 3,500.146 × $3 = $12,009`	normal demand, one-month holding (illustrative)	$12,009
7	Buy at p99.9	`z = 3.0900: Q = 6,000 + 3.0900 × 1,500 = 10,635; short σL = 0.416; left σ(z+L) = 4,635.416; cost = 0.416 × $300 + 4,635.416 × $3 = $14,031`	normal demand, one-month holding (illustrative)	$14,031
8	Accessory at the 70th	`Cu = 0.70 ÷ 0.30 × $3 = $7.00 per missed unit`	implied by the master's p70	Cu ≈ $7

Table 3a — pooling and the trigger (substituted).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Pooling, k = 4, ρ = 0	`σ_pool = 500 × √(4 + 4·3·0) = 1,000.0; buffer = 2.33 × 1,000.0 = 2,330 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 50.0%
2	Pooling, k = 4, ρ = 0.3	`σ_pool = 500 × √(4 + 4·3·0.3) = 1,378.4; buffer = 2.33 × 1,378.4 = 3,212 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 31.1%
3	Pooling, k = 4, ρ = 0.6	`σ_pool = 500 × √(4 + 4·3·0.6) = 1,673.3; buffer = 2.33 × 1,673.3 = 3,899 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 16.3%
4	Pooling, k = 4, ρ = 1	`σ_pool = 500 × √(4 + 4·3·1) = 2,000.0; buffer = 2.33 × 2,000.0 = 4,660 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 0.0%
5	Trigger, no real shift (false alarms)	`p = 1 − Φ(2.0) = 2.28% per week per side; P(trigger within 52 weeks) = 2.55% per side → 0.051 false triggers per series-year (both sides)`	weekly noise 10% of curve (illustrative); consecutive-run recursion	0.051 / series-year
6	Trigger, true take-rate +10% vs curve	`p = P(week beyond ±20%) = 1 − Φ((0.20 − 0.10) ÷ 0.10) = 15.87% per week (one side); P(2 in a row within 4 / 12 weeks) = 6.8% / 22.0%`	weekly noise 10% of curve (illustrative); consecutive-run recursion	7% by week 4
7	Trigger, true take-rate +25% vs curve	`p = P(week beyond ±20%) = 1 − Φ((0.20 − 0.25) ÷ 0.10) = 69.15% per week (one side); P(2 in a row within 4 / 12 weeks) = 77.3% / 99.3%`	weekly noise 10% of curve (illustrative); consecutive-run recursion	77% by week 4
8	Trigger, true take-rate +40% vs curve	`p = P(week beyond ±20%) = 1 − Φ((0.20 − 0.40) ÷ 0.10) = 97.72% per week (one side); P(2 in a row within 4 / 12 weeks) = 99.8% / 100.0%`	weekly noise 10% of curve (illustrative); consecutive-run recursion	100% by week 4

Table 3 — after: the metrics, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Install delays from stockouts	3% of installs (baseline)	≈ 0.08% at the p99 buy	illustrative
2	First-quarter expected cost	$58,234 (p79)	$12,009 (p99)	newsvendor, illustrative
3	Buffer for 4 markets (independent)	4,660 units unpooled	2,330 pooled (−50%)	toolkit, illustrative
4	Re-forecast detection of a +25% shift	quarterly review	77% within 4 weeks	design arithmetic

STEP	METRIC	BEFORE	AFTER	BASIS
5	Owner of the forecast number	Planning	Planning (unchanged); I own equation, measurement, triggers	master honest line

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reschedule cost avoided	`2,000 × (3% − 0.0838%) × $300 = $17,497/month`	p79 → p99 first buy	≈ $209,967/yr per market
2	Holding cost added	`p99 leftover 3,500 × $3 = $10,500 (one month)`	newsvendor row	$10,500
3	First-quarter cost difference	`$58,234 − $12,009 = $46,225`	expected cost, p79 vs p99	$46,225 per market per quarter
4	Stranding avoided (master)	`20% × $500K = $100,000`	held central	≈ $100,000
5	Master's annual spine	`$216,000/yr + $100,000 one-time`	illustrative; never a claim	design arithmetic

SOLUTION FLOW — THE FORECAST-TO-BUY CHAIN (PROPOSED)

```
[Build plan] --milestones--> [Sites passed] --adoption curve x marketing--> [Install demand]
      --kit requirement at finished-kit grain--> [Need by SKU x week, as PIRs]
      --MRP (measured lead times, confirmed dates)--> [Planned orders -> POs]
      --ASN--> [Central 3PL pool] --STOs as sell-through proves the curve--> [Markets]

First buy: Q = mu + z*sigma, z = PhiInv(Cu/(Cu+Co)) ; central, pooled
Trigger: |take-rate / curve - 1| > 20% for 2 consecutive weeks -> new forecast version
Weekly: MAPE and bias per SKU x market x week; bias sign before magnitude
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Cost of a missed install (Cu)	$150 → CR 0.9804	$300 → CR 0.9901	$600 → CR 0.9950
Demand CV (first buy sized for 25%)	CV 15% → surplus	CV 25% → 5.0 short	CV 35% → 42 short, 4.8% risk
Cross-market correlation ρ (k = 4)	0 → 50% saved	0.3 → 31.1%	0.6 → 16.3%
Trigger: weekly noise (true shift +25%)	5% → fires fast	10% → 77% by week 4	15% → slower

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"This is a design, not a run record: I have not owned a launch forecast end to end, and the forecast number belongs to Planning. What I own is the equation, the measurement and the triggers. A missed install costs about three hundred dollars and a unit held a month about three, so the critical ratio is point nine nine and the first buy sits at the ninety-ninth percentile of the first quarter's demand; held central it pools — half the buffer for four uncorrelated markets, about thirty percent if they share a correlation of point three, which I would measure. A take-rate outside plus or minus twenty percent for two weeks forces a new forecast. The dollars — about two hundred sixteen thousand a year per market and a hundred thousand of stranding avoided — are the master's illustrative arithmetic, never a result."

If challenged — the full answers

Did you run it? “No — a design engagement; Planning owns the forecast, I own the equation, the measurement and the triggers.” Is the CV real? “Illustrative; the buy is re-sized on the first weeks’ observed spread.” Is that Google Fiber? “The master’s design engagements are Elementum client work; I would name the client only if confirmed.”

VERIFY-WITH-RAJESH

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time you learned a new method to solve a problem. LEARN AND BE CURIOUS

Open (first two sentences): I took the newsvendor model from my MIT MicroMasters coursework and applied it to a launch problem — and I label it as the toolkit.

STAR (≈90 s): Situation: a new-market build with no demand history. Task: design the first buy. Action: Cu ≈ $300 for a missed install, Co ≈ $3 a month for a unit held, critical ratio 0.990, so the 99th percentile of first-quarter demand for the core unit and the 70th for lower-cost accessories. I then stress-tested it: if the true CV is 35% rather than 25% the shortfall chance goes from about 1% to 4.8%, which is why the trigger exists. Result: a design with arithmetic, not a run record.

Land this number: CR 0.990; z 2.33; Q = 9,495 on a 6,000 mean. · **Pivot to NACF:** A new FC or product launch at NACF is the same critical-ratio problem. · **Do NOT say:** Do not say I ran the launch; do not present the CV as data.

Q2 Tell me about a time you designed for scale beyond the immediate problem. THINK BIG

Open (first two sentences): I designed the launch chain as a template for market N+1, not a one-off for market N.

STAR (≈90 s): Situation: every market launch repeats the same chain — build plan, adoption curve, requirement, supply. Task: a design that a second and third market can reuse. Action: a stock-and-flow map, invariants — demand enters planning once, lead times are measured distributions, need floors with hold-until dates — and a weekly MAPE/bias loop. Result: a template; the forecast stays with Planning.

Land this number: One chain, three books of record, five invariants. · **Pivot to NACF:** A reusable placement template for each new NACF node. · **Do NOT say:** Do not claim a deployed template.

Q3 Tell me about a decision you made with very little data. ARE RIGHT, A LOT

Open (first two sentences): With no local history, the right question was not “what is demand” but “what does being wrong cost on each side.”

STAR (≈90 s): Situation: maximum uncertainty at launch. Task: first buy. Action: price both errors, buy to the critical ratio, hold it central so uncertainty pools, and write the trigger that proves the curve wrong fast: ±20% for two weeks. The trigger catches a true 25% shift within four weeks 77% of the time and false-fires about once every 20 series-years. Result: a decision with its own error rates.

Land this number: 77% detection by week 4 at +25%; 0.051 false triggers per series-year. · **Pivot to NACF:** Placement under uncertainty: price both errors, then set the trigger. · **Do NOT say:** Do not quote the weekly noise (10%) as measured.

Q4 Tell me about a time you put the end customer first in a supply decision. CUSTOMER OBSESSION

Open (first two sentences): The metric was install delays caused by stockouts — a customer waiting for a technician.

STAR (≈90 s): Situation: a missed install means a re-roll and deferred revenue. Task: make the stockout the thing the whole chain is measured on. Action: the primary metric joined reschedule reasons to the supply view; need floors so one market could not starve another’s launch; a governed screen with a reason code instead of an email. Result: design target zero delays caused by stockouts in the first 90 days.

Land this number: Target 0 delays in 90 days; ≈$300 per missed install (model). · **Pivot to NACF:** The NACF customer is the promise date; same joined-metric discipline. · **Do NOT say:** Do not claim installs were delivered.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep; Are Right, A Lot

Open: "A 3% reschedule rate tells you the baseline was a 79th-percentile buy."

Headline metric: CR 0.990; p99 buy 9,495; cost $12,009 vs $58,234

- L(z) = 0.12 → z ≈ 0.8.
- Expected cost curve is flat near the optimum, steep to the left.
- CV 35% turns a 1% risk into 4.8%.
- All illustrative.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone

Open: "Demand enters planning exactly once — manual loads beside the engine are blocked."

Headline metric: one engine, one demand path; floors with hold-until dates

- A floor breach is a governed screen with a reason code.
- One market cannot starve another's launch.
- A borrowed adoption curve is a stop-the-line item.
- Design stance, not a run record.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "I have not owned a launch forecast end to end."

Headline metric: pooling saves 31% at ρ = 0.3, not 50%

- Measure ρ on analog markets before quoting a saving.
- The forecast belongs to Planning.
- The toolkit is labeled.
- VERIFY-WITH-RAJESH client naming.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Why the 99th percentile for the first buy — is that not wasteful?"

The numbers: CR = 300 ÷ 303 = 0.9901; z = 2.330; p99 buy 9,495: cost $12,009; p79 buy 7,200: $58,234; leftover 3,500 units × $3 = $10,500.

Say exactly: "Because a missed install costs a hundred unit-months of holding: at the ninety-ninth percentile the expected first-quarter cost is about $12 thousand against $58 thousand at the seventy-ninth, and the leftover — about thirty-five hundred units — is not waste, it is next quarter's demand."

T2 "Does the pooling benefit survive correlated markets?"

The numbers: k = 4: ρ = 0 → saves 50%; ρ = 0.3 → 31.1%; ρ = 0.6 → 16.3%; ρ = 1 → 0%.

Say exactly: "Only partly — half the buffer at zero correlation, about thirty-one percent at a correlation of point three, sixteen at point six; new markets share a product and a campaign, so I would measure it on analog markets before promising any number."

T3 "Your trigger is ±20% for two weeks. What are its false alarms and its power?"

The numbers: Weekly noise 10% (illustrative): no shift → 0.051 false triggers per series-year (both sides); true +25% → 77% within 4 weeks; +40% → 100%; +10% → 22% within 12 weeks.

Say exactly: "With ten percent weekly noise the trigger false-fires about once in 20 series-years, catches a true twenty-five percent shift within a month about 77 times in a hundred, and is nearly blind to a ten percent bias — which is why the bias sign publishes weekly beside it."

Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a Placement Analytics manager is likely to ask (rows 13–27 are new: the statistical questions the new primaries answer). Each plan points at a packet story or a named backup in the master packet; stories allocated to other interviewers are pointers only.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Walk me through how you improved forecast accuracy.**	S6 pitch — honest supply definition first, MAPE/bias at the decision grain, waves by trust; 70 → 85% on scoped class-A SKUs; planners owned the forecast, I owned its measurement.	S6
2	**Tell me about a time you used data to change an inventory policy.**	S8 pitch — brochure → measured lanes, p99 promise, 99% bar, 580K units / ~$1.5M out with zero stockouts.	S8
3	**How do you decide where / how much safety stock to hold?**	Two error signals measured at the grain; cost asymmetry sets the percentile; release only by signed rule with a watch; fat tails keep buffer.	S8 → S6
4	**What is the softest assumption in your biggest result?**	S6: the expedite count (60 × $5K). S8: the late-arrival share. Say how each would be firmed from a ledger.	S6 / S8
5	**How do you measure lead-time performance?**	Lead-time accuracy as the master defines it = 1 − |actual − promised| / promised against a per-lane p99 promise; per shipment that cannot reach 99% against a p99 promise, so say which reading (coverage, or re-measured p99 within 1%) before quoting a tolerance VERIFY-WITH-RAJESH; drift flag and refresh cadence.	S8
6	**Tell me about a time a stakeholder wanted a number that looked better.**	Planning's request to count unfirmed MRP orders as supply — refused; released planned orders only.	S6
7	**How did you handle a dependency that slipped (MPN mapping)?**	Escalated to the sponsor as a re-sequenced date with the coverage number attached; made MPN a standing metric rather than a task.	S6
8	**How would you decompose a network placement problem (e.g., inbound placement cost vs. speed)?**	Master §3.99 opening: book of record and metric → stocks and flows → broken invariant → design → sequence; then lead-time and forecast error at the lane × node grain as the two levers.	Master §3.99
9	**How do you validate a model before the business acts on it?**	Parallel run against the source (daily reconciliation to SAP; manual sample per cohort); a published gate; the old process stays alive until the match holds.	S6 (S10 pointer — Josyula's)
10	**What is your quantitative background?**	MIT Supply Chain MicroMasters + Supply Chain Network Design certificate — acquired to design what I had been implementing; the formulas are that toolkit, labeled. Never "I build models for a living."	Master §4.1 toolkit line
11	**Tell me about a time you improved a process end to end (Plan → Deliver).**	S6 as the Plan redesign (definition → gate → alert → release rule) with S8 as the Source/Deliver parameter feeding it; the two programs stay separate in dollars.	S6 + S8
12	**Another example of Dive Deep?**	Pointers only: S10 (Austria — bar raiser's) and S7 (phantom math — unallocated; no BOM-explosion content). If pressed, S7 at the level of "six sites, three waves, five months, zero auto-loaded revision overlaps."	S7 pointer

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
13	**How do you know the lead-time distribution you measured is the right one — n, window, interval?**	S8a: ~400 shipments on the busy lane; the ranked p99 with its order-statistic interval; bell-curve p99 30.3 holds 96.5%, ranked p99 34 holds 99%; thin lanes print the interval or pool by cohort VERIFY-WITH-RAJESH NEW	S8a
14	**Planned vs actual lead time — how exactly was the plan wrong?**	S8a: plan 18 d vs mean 21 d (t = 15 on the 3-day bias); ERP σ 8 vs measured 4; tail 16 days short at p99 (34 vs 18) — wrong on location, spread and tail (illustrative lane) NEW	S8a
15	**How do you set safety stock and the service level?**	S8b: two terms — cover to the p99 lead time plus demand cover; critical ratio 0.99; lane A 12,937 → 8,102 units, lane B 5,238 → 6,621 (+1,383); 580K net is the record NEW	S8b
16	**How do you pick an alert threshold — precision, recall, and the cost of a false alarm?**	S6: alert at DoC ≤ lane p90; on an illustrative lane precision 71.2%, recall 98.6%; p99 trades twelve points of precision for about one of recall; cost $12,840 vs $8,913 at $60 / $5,000 NEW	S6
17	**How do you know it was your change and not seasonality?**	S8c: held-out quarter; paired old-vs-new plan on the same shipments (22 vs 175 late of 2,500); control lanes deferred a quarter if the record has them VERIFY-WITH-RAJESH; one draw, so rolling origin is the upgrade NEW	S8c
18	**What is the uncertainty on a result you are about to act on?**	S8c: 22 late of 2,500 → 99.1%, Wilson [98.7, 99.4]; z vs 96.5% = 7.1; 0 stockouts in 1,950 lane-weeks bounds the rate below 0.154% NEW	S8c
19	**Which of 1,000 lanes do you keep, and by what criteria?**	S9: spend Pareto + four-factor lane scoring + EDI survey; 597 lanes = 80% of spend first; ~400 deferred (~20%) on the same gate NEW	S9
20	**How would you design an estimated-delivery-date / promise policy?**	S19: promise at b = k·σ with k = 1.645 for a 95% floor; accuracy pinned on every cohort, speed floats; fast-promise share 7.6% → 39.6%; re-solve k per regime NEW	S19
21	**A headline metric is green — how do you check it?**	S19: headline 95.8% hid Φ(5.0) vs Φ(0.69) = 75.4% on the wide cohort; S20: raw +0.8 pt vs mix-adjusted −1.0 pt — publish the adjusted twin beside the raw NEW	S19 / S20
22	**Design a weekly business review for a delivery experience.**	S20: 7 outputs over 10 inputs and one joint headline; factor math names the owner in minutes; hard floors beside the headline; pre-wired reds NEW	S20
23	**How do you catch a degrading seller or carrier fast without false alarms?**	S20: Andon — n ≥ 30 floor plus a sustained signal; false-trip design bar < 5%; ~30 d → < 24 h (design, nothing in production) NEW	S20
24	**How would you test an incentive or contract change before launch?**	S21: replay a seller's orders under managed settings; c = 11 misses in 200 as the pre-agreed gate (illustrative); Wilson interval on the 85% handling adherence at n = 100: [76.7%, 90.7%] NEW	S21
25	**How would you forecast and place equipment for a new market with no history?**	E7: newsvendor with critical ratio 0.99 → z 2.33 → first buy 9,495 on a 6,000 mean; pooled central stock saves ~31% at ρ = 0.3; ±20% two-week re-forecast trigger; NACF INV-C for placement (illustrative; design, not run) NEW	E7

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
26	**Which number across your stories is the softest?**	S19's ≈$13.3M/yr and S21's ≈$3.7M/yr are modeled; S9's −10% freight has no control group VERIFY-WITH-RAJESH; S8's accuracy definition VERIFY-WITH-RAJESH. Name it first, then say how I would firm it. NEW	S19 / S21 / S9
27	**How do you separate your contribution from the client's?**	S21: I claim the contract design and wave-one mechanics; the adoption curve is the client's. S19–S20 are engagement designs — nothing in production moved. NEW	S21 / S19 / S20

Cheat-numbers box — canon values with basis labels

Canon numbers for all ten primaries with their basis labels, from the master packet's §4.1 ledger; the last ten rows are new (S8a–E7).

METRIC	VALUE TO SAY	BASIS	NEVER
S6 forecast accuracy	70% → 85% on scoped class-A SKUs	record	claiming I owned the forecast
S6 safety stock	$2M released under the both-errors-improved rule	record, one-time	"$2M per year"; adding to S8
S6 recurring	≈$0.8M/yr (carry 0.2 · expedites 0.3 · planner capacity 0.3); range $0.5–1.2M	modeled	presenting as record
S6 timing / scope	shortage alert live in 3 months; eight domains; six sites; adoption 8×; planner effort −50%; shipment data 60% complete at start	record	pasting "3 months" onto the six-site ingest (S7)
S6 alert rule	days-of-cover ≤ lane lead-time p90	record mechanism	—
S8 lanes / accuracy	200 lanes; 99% lead-time accuracy at the p99 promise	record	—
S8 release	580K units / ~$1.5M, zero stockouts in the watch window	record, one-time	"$1.5M to $2M"; annualizing it
S8 recurring	≈$0.6M/yr (carry 0.3 at 20% · expedites 0.3); range $0.3–1.4M	modeled	presenting as record
S8 unit value	≈$2.59/unit	derived from the record pair	—
Site count	"six sites"	record	"six NA sites"
Toolkit	MIT Supply Chain MicroMasters + Network Design certificate; formulas are illustrations	labeled	"I build supply chain models for a living"
S7 (pointer only)	six sites, three waves, 5 months; hardest site 6 weeks vs 3-week average	record	"500+ product families"; 60→95% inside S7; any BOM-explosion walk
S8a — example lane	plan 18 d · actual mean 21 d · p99 34 d · n ≈ 400 · 240 of 400 (60%) slower than the brochure · normal p99 30.3 d holds 96.5%	illustrative (secondary-source lane)	presenting the lane as the program's data

METRIC	VALUE TO SAY	BASIS	NEVER
S8b — safety-stock lanes	lane A 12,937 → 8,102 (SS 6,500 + 1,602 vs 9,437 + 3,500 pad); lane B 5,238 → 6,621 (+1,383); 150 × 4,500 − 50 × 1,900 = 580K	illustrative decomposition; the net 580K is record	saying the program "raised" fat-tail lanes as record VERIFY-WITH-RAJESH
S8c — held-out test	22 late of 2,500 → 99.12%, Wilson [98.67, 99.42]; 0 stockouts in 1,950 lane-weeks → < 0.154%	illustrative inputs	quoting window / quarter as fact VERIFY-WITH-RAJESH
S9 — EDI onboarding	1,000 asked → 597 = 80% of spend; completeness 60% → 95%; ≥95% valid-ASN gate held twice; freight −10% ≈ $2M in six months; ~200 next wave; ~400 lanes (~20%) deferred	record	annualizing the $2M; claiming −10% was causal without a control VERIFY-WITH-RAJESH
S9 — recurring	≈$0.4M/yr documentation labor	modeled	presenting as record
S19 — delivery promise	flat 1.10 d buffer → b = k·σ, k = 1.645; fast-promise share 7.6% → 39.6% at a held 95% floor; +$255K/wk ≈ $13.3M/yr	engagement model / modeled	the 97.8% variant and its $6.07M/wk — never claimed; any production result
S19 — regime drill	Q4 σ × 1.45 → accuracy 91.9% on a frozen dial; $12 breach cost vs $6.50 margin	model / drill	a live Q4 result
S20 — pre-purchase WBR	0.97 × 0.985 × 0.996 ≈ 95.2%; Promise Yield 41.5% = 43.6% × 95.2% (four-quarter target 50%); SE carrier 82% → accuracy 93.6%; ≈$0.58M per caught episode; false-trip bar < 5%; ~30 d → < 24 h	illustrative sample / modeled / design	any live Andon result — design engagement, nothing in production
S21 — managed shipping	enrolled ~38–42% stalled; 85% handling adherence; ≈$3.7M/yr per +10 pts of enrolled GMV; claim = contract design + wave-one mechanics	modeled	owning the adoption curve; presenting $3.7M as record
E7 — new-market buy	Cu ≈ $300, Co ≈ $3 → CR 0.99 → z 2.33; ±20% two-week trigger; $216K/yr per market + $100K stranding	illustrative; design, not run	saying it ran; naming the client unless confirmed VERIFY-WITH-RAJESH

Never-say, one glance before the call.

- Never add $2M (S6) and ~$1.5M (S8) into one number; never "$1.5M to $2M."
- Never "six NA sites" — "six sites." Never paste S6's "3 months" onto the six-site ingest.
- Never claim ownership of the demand forecast or the distributions — measurement, definitions, rule and policy were mine.
- Never present the safety-stock formula as a program artifact; MRP math at finished-goods grain only — no BOM explosion.
- Never present planner capacity as cash — time redeployed, not headcount removed.
- S10 (Austria) is the bar raiser's; S11/S12, S2/S18, S1/S3 belong to Tripathy, Goforth and Lan — pointers only here.

Backup bench — use only if a primary doesn't fit

Three fallbacks for an analytics manager: two master-packet backups plus one doc-derived Flexport micro-story — S4 (Sephora — the in-transit promise rule I got wrong; the failure / Are Right, A Lot question, promise math at lane σ) and S7 (Elementum — a contract manufacturer's planning data made safe to plan on, finished-goods grain only; Learn & Be Curious by math, not meetings); the doc-derived third is the FFA Asia latency investigation from the Flexport program record (Dive Deep on production evidence; record-from-docs labels, facts identical to the same bench in Josyula's packet). S9 (the 1,000 → 597 EDI story) was promoted to a full primary above. Tell the ten primaries first.

Bench discipline. The ten primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

BACKUP — ONLY IF ASKED S4 — Sephora S/4HANA — the in-transit promise rule I got wrong, fixed through the signed policy, not a side configuration

LPs this backup can serve: Are Right, A Lot (the failure question) · Customer Obsession (backup) · Earn Trust (owning the miss) · Learn and Be Curious — the designated policy-failure story

Basis. The v1 rule was mine; the indictment came from my own program's reports. Lane σ ≈ 0.3 day and the ~0.49-day window are the record's promise math; the record carries no lane counts and no dollar for the conversion lost while I was wrong — I will not invent the loss figure.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. My first ISM ruleset barred every channel from promising units in transit; the Aging In-Transit report showed dotcom starved of fast movers one day from the building. **Task.** Correct my own policy without weakening oversell protection, through the same three signatories. **Action.** Discovery: conversion-loss evidence from the aging and replenishment reports — the entry bar to reopen the policy; I stated that the v1 rule was mine. Plan & Design: promise math — lane σ ≈ 0.3 day, a ~0.49-day window ≈ 95% coverage; Promisable = in-transit × 1[ETA ≤ window] × 1[on-time ≥ floor]; Aging In-Transit as the kill switch; both blanket options rejected in writing. Build & Test: back-test against receiving history; four regression cells added to the node × channel matrix. Cutover: released on the normal monthly cadence as a signed FSD change, lane by lane — no hotfix, no side configuration. Hypercare: kill switch armed; never fired. Lesson: both sides of a rule get a number before it ships. **Result.** Primary metric: sellable units on qualifying in-transit lanes — starved → restored inside a per-lane promise window, oversell increase = zero, kill switch never fired. ≈$0.5M/yr modeled margin recovered — the record has no dollar for the conversion lost while I was wrong, and I will not invent one.

COMPRESSED STAR

- **S —** v1 of my own ISM ruleset: no channel could promise in-transit units — safe-feeling, and starving dotcom of fast movers reliably one day from the building.
- **T —** Fix my own signed policy without weakening oversell protection; take the miss back through the same three signatories.
- **A —** Evidence pack from Aging In-Transit and Replenishment; reframed as promise math (σ ≈ 0.3 d → ~0.49-d window ≈ 95%); rule with on-time floor and kill switch; back-test against receiving history; four regression cells; released lane by lane on the monthly cadence as an FSD change.

- **R —** Starved lanes restored inside the window; oversell increase = 0; kill switch never fired; re-signed by the same three signatories; lesson institutionalized — the Locus 98% gate (S2) and the 0.5% tolerance (S18) were set from data before the event.

RESULTS LINE

Sellable units on qualifying in-transit lanes starved → restored inside a back-tested ~0.49-day window (record) · oversell increase = 0; kill switch armed, never fired (record) · four regression cells added (record) · released as a signed FSD change on the normal cadence, no hotfix (record) · ≈$0.5M/yr modeled margin recovered (inputs illustrative).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The v1 rule	`Promisable = 0 for ALL in-transit inventory`	equivalent to claiming σ → ∞ or on-time rate → 0 — infinite variability (record)	everything in transit hidden
2	One concrete lane	`Promisable = 120 × 0 = 0`	lane with InTransit = 120 units, ETA 0.4 day away (illustrative)	120 sellable units hidden
3	Across qualifying lanes	`≈300 fast-mover units/day × 0 promisable`	illustrative volume	≈300 units/day sat unsellable one day from the building
4	What starving costs (one SKU)	`100 × 60% × $20 = $1,200/day`	100 units one day out; 60% would convert; $20 margin (illustrative, not program data)	$1,200/day under-sell
5	The ignored other side	`100 × (1 − 0.97) = 3 late units; 3 × $15 = $45/day`	on-time rate 97%; $15 expedite (illustrative)	$45/day over-sell exposure — vs $1,200: the “safe” rule priced one side only

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure lane variability	`σ = 0.3 day`	trailing receiving history (record)	σ = 0.3 day
2	Choose coverage	`z at 95% = 1.645`	normal approximation on lane transit (toolkit label)	z = 1.645
3	Promise window	`w = z × σ = 1.645 × 0.3 = 0.4935`	rows 1–2	w ≈ 0.49 day
4	Coverage check	`Φ(w / σ) = Φ(0.49 / 0.3) = Φ(1.63) ≈ 0.95`	share of arrivals inside the window	≈95% covered
5	Corrected rule	`Promisable = InTransit × 1[ETA ≤ w] × 1[OTR ≥ floor]`	OTR = on-time rate over trailing receipts; floor = per-lane admission threshold, 95% here (floor illustrative)	lane by lane, not all-or-nothing
6	Lane that qualifies	`ETA 0.4 ≤ 0.49 → 1; OTR 97% ≥ 95% → 1; Promisable = 120 × 1 × 1 = 120`	the Table 1 lane, re-evaluated (ETA and OTR illustrative)	its 120 units become promisable
7	Lane that fails	`ETA 1.2 > 0.49 → 0; Promisable = 80 × 0 = 0`	a slower lane with 80 units in transit (illustrative)	stays at 0 — correctly out
8	Sample-size gate	`n = z²p(1−p)/e² = 3.84 × 0.95 × 0.05 / 0.05² = 3.84 × 0.0475 / 0.0025 ≈ 73`	OTR estimated within ±5% at 95% CI (illustrative)	a lane qualifies after ~73 receipts

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
9	Kill switch, values in	`trigger = 2 × p90 = 2 × 2.0 = 4.0 days; any unit in transit > 4.0 days → lane out`	lane p90 = 2.0 days (illustrative); armed, reviewed weekly by Inventory Control (record)	never fired (record)
10	Back-test	`rule replayed against receiving history; 4 regression cells`	before release (record)	no oversell in the back-test

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Sellable units on qualifying in-transit lanes	0	≈300 units/day restored inside the window	record mechanism; volume illustrative
2	Promise window	none — blanket exclusion	w ≈ 0.49 day at ~95% cover	record (promise math)
3	Oversell increase from the window	—	0	record
4	Kill switch	not armed	armed, never fired	record
5	Rule changes after	ad hoc	same 3 signatories, back-test + matrix regression	record — both sides priced from then on

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Would-be sales	`300 × 60% = 180 sales/day`	units restored (illustrative) × conversion (illustrative)	180 sales/day
2	Truly lost without the fix	`180 × 40% = 72 sales/day`	40% lost-sale share — the rest would have waited (illustrative)	72 lost sales/day
3	Margin recovered	`72 × $20 = $1,440/day`	$20 unit margin (illustrative)	≈$1,440/day
4	Annualized	`$1,440 × 365 = $525,600`	modeled; the mechanism is record	≈$0.5M/yr
5	Cost while v1 stood	`no dollar in the record — not claimed`	record	not claimed

TWO PROBES

Why not just hotfix it — you knew it was wrong?

Because the error was in the policy, so the fix had to be too — in the governed artifact, with regression, through the same signatures. A side configuration would have fixed the symptom and left the policy lying about what it allowed.

What did the miss cost?

The record carries no dollar for the conversion lost while v1 stood, and I will not invent one. What I can say is the mechanism: fast movers reliably one day from the building were unsellable on dotcom until the window existed; the ≈$0.5M/yr recovered is my model on illustrative inputs, labeled.

Canon numbers (basis labels): Lane σ ≈ 0.3 d; ~0.49-d window ≈ 95% coverage (record promise math) · oversell increase = 0 (record) · 4 regression cells (record) · kill switch never fired (record) · same three signatories re-signed (record) · ≈$0.5M/yr (modeled). Never a lane count; never a dollar for the loss while wrong.

BACKUP — ONLY IF ASKED S7 — Elementum — making a contract manufacturer's planning data safe to plan on: six sites, three waves, five months, zero auto-loaded revision overlaps

LPs this backup can serve: Learn and Be Curious (primary — learned the CM's data by math, not meetings) · Dive Deep · Earn Trust (external partner)

Basis. I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call; engineers built the normalization layer and quarantine to the spec; the commercial relationship belonged to the customer's procurement — my leverage was evidence. The $1.5M / 580K belong to the p99 program this backbone enabled (S8). Finished-goods grain only here; no tolerance value — the record does not carry one.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Flextronics supplied the planning data our component-demand visibility depended on, and it carried two traps documentation never confesses: pass-through assemblies that double-time demand if loaded as stocked, and overlapping revisions that rewrite history if you auto-load one. Six site systems carried six definitions of yield, lead time and order status. **Task.** I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call — with no commercial leverage; evidence was my only lever. **Action.** Discovery: sampled finished goods and reconciled the CM's planning data against its own MRP — the trap signature found by math, named in a discovery memo. Plan & Design: an explicit marker for pass-through items; an effective-date quarantine with human disposition — never auto-load; one supply equation (on-hand + production plan + released planned orders) signed by customer planning. Build & Test: mapping tables owned by named site super-users; a per-site reconciliation report as the go-live gate. Cutover: three waves — two cleanest high-volume sites, then three on parameterized maps, then the hostile legacy ERP by snapshot-diffing at six weeks against a three-week average, priced up front. Hypercare: the weekly call left standing — it had become theirs. Lesson: reconcile to the source's math, not its documentation. **Result.** Primary metric: contract-manufacturer sites with planning-grade data — 0 → 6 in three waves and five months, zero auto-loaded revision overlaps. ≈$0.3M/yr modeled — the six sites and zero overlaps are measured; the dollars are my model; the $1.5M belongs to S8.

COMPRESSED STAR

- **S —** A contract manufacturer's planning feed with two silent traps; six sites, six definitions; one hostile legacy ERP (flat files, no change-data-capture); no commercial leverage.
- **T —** Own ingest requirements, the reconciliation gate, the 2/3/1 wave plan, and chair the weekly Flextronics call.
- **A —** Found the traps by sampling finished goods against the CM's own MRP; designed the two mechanisms that made the traps impossible rather than policed (explicit marker; effective-date quarantine, human disposition); reconciliation report as the per-site gate, not a date on a slide; wave 3's six weeks priced to leadership as the plan, not a slip; made the weekly call the CM's own protection against blame for misses its feed didn't cause.
- **R —** Planning-grade sites 0 → 6 in 3 waves / 5 months; zero auto-loaded overlaps ever; phantom double-timing eliminated at source; backbone handed to the 200-lane program (S8).

RESULTS LINE

CM sites with planning-grade data 0 → 6 in three waves / five months (record) · auto-loaded revision overlaps = 0, ever (record) · hardest site 6 weeks vs the 3-week average, priced up front and absorbed in plan (record) · ≈$0.3M/yr (modeled — reconciliation labor, expedites fired at false shortages, overlap rework); the record has no dollar for this story; the $1.5M belongs to S8.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Planning-grade CM sites	`0 of 6; manual BOM-vs-MRP reconciliation at every site`	record	0 of 6
2	Manual recon burn	`6 h/wk × 6 sites = 36 h/wk`	modeled	36 h/wk of reconciliation
3	What MRP computes (finished-goods grain)	`NetReq(FG, t) = GrossReq − OnHand − SchedReceipts`	the netting identity; walk below at FG grain only (illustrative)	substituted in rows 4–5
4	Clean row	`NetReq = 500 − 150 − 200 = 150`	Gross 500; on-hand 150; receipts 200 (illustrative)	NetReq = 150; need-date = due − LT_FG
5	Distorted row — phantom FG record	`NetReq = 600 − 150 − 200 = 250; false shortage = 250 − 150 = 100`	a phantom/stale finished-goods record inflates gross requirements by 100 (illustrative)	an expedite fires on 100 units never needed
6	How often	`~5/site/mo × 6 sites × 12 mo = 360 false shortages/yr`	modeled count	360/yr
7	Timing corruption	`the same stale record pulls the need-date a lead-time early`	record mechanism	planners chase demand that was never real
8	Revision overlaps	`two revisions both partially true in [d0, d1]; ~24/yr`	auto-loading would rewrite demand for orders released under the prior revision (record rule; count modeled)	~24 history-rewrite risks/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Explicit marker	`distorted row reconciled: NetReq = 500 − 150 − 200 = 150`	the phantom record carries an explicit marker; it no longer inflates gross (record mechanism)	NetReq = 150 — matches the site's own MRP
2	Per-site gate	`count_model vs count_MRP on sampled finished goods must match`	the reconciliation report is the go-live gate, not a date on a slide (record)	a site opens on a matched report
3	Quarantine	`overlaps held for human disposition per order/effectivity`	record rule	auto-loaded overlaps = 0 — ever
4	Hostile flat-file site	`completeness = rows received / expected; a 40%-rows file would read as 60% deletions → blocked`	snapshot-diffing with a completeness flag (record mechanism)	partial files never consumed
5	Wave plan	`W1: 2 × 3 = 6 wk; W2: 3 × 3 = 9 wk; W3: 6 wk; 6 + 9 + 6 = 21 weeks`	hostile site at 6 wks vs the 3-wk average, priced up front (record)	≈5 months for six sites
6	Recon labor after	`6 → 1.5 h/wk per site — disposition queue only`	modeled	−4.5 h/wk per site

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Planning-grade CM sites	0 of 6	6 of 6, in 3 waves ≈ 5 months	record
2	Manual reconciliation	6 h/wk × 6 sites	1.5 h/wk × 6	modeled
3	Expedites on false shortages	~72/yr acted on	~0 — the marker removes the distortion	modeled
4	Auto-loaded revision overlaps	uncontrolled	0 — ever	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reconciliation labor	`(6 − 1.5) = 4.5 h/wk; × 6 sites = 27 h/wk; × 48 wk = 1,296 ≈ 1,300 h; × $60 = $77,760`	modeled	≈$0.08M/yr
2	False-shortage expedites	`360 × 20% = 72 acted on; 72 × $3,000 = $216,000`	count from Table 1 row 6; premium benchmark $2–5K (modeled)	≈$0.2M/yr
3	Overlap rework avoided	`24 × 16 h = 384 h; 384 × $60 = $23,040`	the quarantine makes the error 0 by construction (modeled)	≈$0.02M/yr
4	Total	`0.08 + 0.2 + 0.02 = $0.3M`	the record has no dollar for S7; the $1.5M belongs to S8	≈$0.3M/yr

TWO PROBES

> **How did you get an external supplier with no obligation to care?**
>
> By making the weekly call theirs: it protected Flextronics from blame for misses their feed didn't cause. Procurement held the commercial relationship; my lever was the evidence in the discovery memo, confirmed by their own engineering.

> **Why three waves instead of one?**
>
> So one surprise could not spend six sites of credibility — the two cleanest high-volume sites first, three on parameterized maps second, the hostile ERP last with its six weeks priced up front. Speed was traded, never quality: a site went live off its matched reconciliation report, not a date.

Canon numbers (basis labels): 0 → 6 sites, 3 waves, 5 months (record) · zero auto-loaded overlaps (record) · 6 weeks vs 3-week average (record) · ≈$0.3M/yr (modeled). Never "six NA sites"; never "500+ product families"; 60 → 95% belongs to S9; the $1.5M / 580K belong to S8; no tolerance value.

BACKUP — ONLY IF ASKED Doc bench — Flexport FFA — the Asia latency investigation: a 21-second P95 traced to routing, decomposed to the layer, fixed in sequence (Dive Deep)

LPs this backup can serve: Dive Deep (primary — decomposition to the layer) · Deliver Results · Learn & Be Curious · Insist on the Highest Standards (a smoke-test protocol per change)

Basis. Doc-derived micro-story from the program record (the FFA Asia latency investigation, post-MVP 2022) — every figure is RECORD-FROM-DOCS: the investigation document, written when the FF App MVP had launched to 11 partners in 10 countries and 48 of 53 import partners (90.5%) had migrated. Program-lead voice: I owned the investigation, the decomposition and the fix sequence; infrastructure teams executed the WAF and CDN changes. Not in the master packet's S-numbering; supplements canon, never overrides it. The same bench appears in Josyula's packet — facts identical.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Weeks after the FF App MVP launched to 11 partners in 10 countries — with 48 of 53 import partners already migrated — the monthly business review showed Asia page loads at a P95 of 21.19 seconds against 3.3 in the US. **Task.** Find the layer, not the anecdote, and fix it without a re-architecture in the middle of a live partner-facing launch. **Action.** I decomposed the problem before prescribing: static assets ran ~5s P95 in China against ~500ms in the US, and the app's domain had no geo-based DNS or reverse proxy — Asia traffic was riding the public internet across the Pacific; a double CDN sat on the static path. The fix was a sequence, not a project: enable the WAF, remove the double CDN (direct to S3), then a regional CDN proposal with a smoke-test protocol per change, then China tuning, then Asia tuning — each with a date and an owner. **Result.** Root cause isolated to routing — not the app, not the partners; two fixes shipped inside the investigation window and the regional plan scheduled with dates; and a standing rule imported into every later launch: the pilot's geography is a performance requirement, not a rollout detail.

RECORD-FROM-DOCS

COMPRESSED STAR

- **S —** Post-MVP, the MBR surfaced Asia P95 page load at 21.19s vs 3.3s in the US (17-day measurement window); 48 of 53 import partners (90.5%) already migrated — the app had to hold where the partners were.
- **T —** Isolate the layer causing Asia latency and sequence the fix set without destabilizing a live partner-facing launch.
- **A —** Split static from dynamic: static JS ~5s P95 in China vs ~500ms US; found no geo-based DNS / reverse proxy on the domain (public-internet routing) and a double CDN on static assets; sequenced WAF enablement → double-CDN removal (direct to S3) → regional CDN proposal → China tuning → Asia tuning, each gated by a smoke-test protocol with a named date and owner.
- **R —** Root cause isolated to routing; WAF and double-CDN fixes shipped inside the window; CDN / China / Asia work scheduled with owners; the investigation became the template for launch-geography checks.

RESULTS LINE

P95 page load 21.19s Asia vs 3.3s US (record-from-docs) · static ~5s China vs ~500ms US (record-from-docs) · 48 of 53 partners migrated = 90.5% (record-from-docs; canon's "48 of 53 within a month" — consistent) · two fixes shipped, three sequenced with dates (record-from-docs) · no post-fix latency figure exists in the record — never invent one.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Page-load P95 by region	`Asia = 21.19s; US = 3.3s; Others = 4.45s`	Datadog @view.loading_time, 04/24–05/10 window (record-from-docs)	Asia is the outlier

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	The gap factor	`21.19 ÷ 3.3 = 6.42`	row 1 (record-from-docs)	6.42× slower than the US
3	Traffic behind the number	`6.31 ÷ 2.19 = 2.88`	Asia 6.31k vs US 2.19k page views, same window (record-from-docs)	≈2.9× the US traffic riding the worst latency
4	Static-asset leg	`JS P95 = 5s China vs 500ms US; 5 ÷ 0.5 = 10`	static.flexport.com CDN behavior (record-from-docs)	10× gap on the static leg
5	Dynamic leg	`no geo-DNS / reverse proxy for trade.flexport.com → Asia GraphQL hairpins over the public internet`	infra review (record-from-docs)	routing penalty on every data call
6	Known defect on the static path	`double CDN in China — Cloudflare stacked in front of Tencent`	infra review (record-from-docs)	one avoidable hop

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Decompose before touching	`page load = static resources + GraphQL requests + JS rendering`	the metric's own definition (record-from-docs)	three legs, measured and owned separately
2	Step 1 — security gate first	`enable WAF for trade.flexport.com: China, then global`	incident-207 follow-up; done 2022/06/30 (record-from-docs)	shipped — fix 1 of 2
3	Step 2 — remove the double CDN	`Tencent CDN → direct to the S3 bucket; Cloudflare hop removed for China`	done 2022/06/24 (record-from-docs)	shipped — fix 2 of 2
4	Step 3 — static proposal, CN/Asia	`alternatives written, reviewed, signed off`	ETA 2022/07/15 (record-from-docs)	sequenced with a date
5	Step 4 — China tuning	`e.g. replicate static assets in-region`	ETA 2022/07/30; depends on step 4 (record-from-docs)	sequenced with a date
6	Step 5 — Asia tuning	`the same playbook beyond China`	ETA 2022/08/15 (record-from-docs)	sequenced with a date
7	Held in reserve	`global CDN for trade.flexport.com — built only if the metrics still demand it`	2022 Q3 or later, metrics-gated (record-from-docs)	no speculative spend
8	Verification per change	`2 user types × 2 instances = 4 smoke passes before + 4 after = 8`	login → workview → logout → re-login; internal SSO + partner users; US + China hosts (record-from-docs; the multiplication counts the checklist)	8 workflow passes around every change

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Root cause	one number — "Asia is slow"	static leg (10×) and dynamic routing isolated, each with an owner and a sequence	record-from-docs
2	Fixes	none	2 shipped (WAF; double-CDN removal); 3 sequenced with dates (07/15, 07/30, 08/15)	record-from-docs
3	Post-fix P95	21.19s	no post-fix figure exists in the record — never claimed	record-from-docs

STEP	METRIC	BEFORE	AFTER	BASIS
4	Spend discipline	—	global CDN gated on metrics, not built on speculation	record-from-docs

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Partner base exposed	`48 ÷ 53 = 0.9057 ≈ 90.5% of import partners already migrated when the gap surfaced`	migration tracker (record-from-docs)	90.5% of partners on the affected app
2	Excess wait served per window	`gap vs US = 21.19 − 3.3 = 17.89s; 6,310 × 17.89 = 112,886 partner-seconds; ÷ 3,600 ≈ 31.4 hours`	6.31k Asia page views in the 17-day window (record-from-docs); gap arithmetic illustrative	≈31.4 partner-hours of excess wait every 17 days
3	Dollar value of the wait	`no dollar in the record — not claimed`	record-from-docs	not claimed
4	Post-fix saving	`no post-fix latency figure exists → no savings chain exists`	record-from-docs	never invent one

TWO PROBES

How did you know it was routing and not the application?

By decomposition before remediation: the static/dynamic split put ~5s of the China time on assets the app server never touched, and the same assets served in ~500ms from the US — the code was constant, the path was not. The domain's missing geo-DNS / reverse proxy and the double CDN explained both halves of the gap without a single application change.

Why not just stand up a China region?

A region is a re-architecture with a compliance tail; the evidence said the bottleneck was routing and CDN topology, which fixes in weeks. The sequenced plan — WAF, double-CDN removal, regional CDN, tuning — bought most of the win at a fraction of the risk, and the smoke-test protocol meant each step was reversible.

Canon numbers (basis labels): 21.19s vs 3.3s P95 (record-from-docs) · ~5s vs ~500ms static (record-from-docs) · 48/53 = 90.5% migrated (record-from-docs) · 11 partners / 10 countries MVP (record — matches S12 canon) · fix sequence WAF → double-CDN → regional CDN → China → Asia with smoke tests (record-from-docs). Never: a post-fix latency number (none in the record); never present scheduled items as delivered.

VERIFY BEFORE YOU SAY IT NEW

Each line is a detail that is **not in the record**, or a number resting on an illustrative input; it carries VERIFY-WITH-RAJESH where it appears in a story. Tick it, or leave it out of the room.

EVERYTHING

- ☐ Which client names I am comfortable saying aloud: “Walmart Marketplace” for S19–S21, “Google Fiber” for E7 — otherwise “a marketplace client” and “a broadband build-out client.”
- ☐ That none of S9, S19, S20, S21 or E7 is also planned as a primary for another interviewer — one question, one story across the loop.
- ☐ Every “illustrative” number is said with the word “illustrative” — the example lanes are a secondary source, not the program ledger.

S6

- ☐ How p90 was originally chosen for the alert (the precision/recall table is after-the-fact design arithmetic).
- ☐ How “demonstrably improved” was operationalized for the both-errors rule (the standard-error bar is my arithmetic, not the record).
- ☐ How large the MPN-mapping delay actually was.

S8

- ☐ Which definition of “99% lead-time accuracy” the program used — ratio, coverage, or p99-match — before quoting any tolerance.
- ☐ The thin-lane rule, the lane-cohort counts and the watch-window length (none is in the record).

S8A

- ☐ That the example-lane numbers (n ≈ 400, plan 18, mean 21, p99 34) are the ones I am comfortable quoting as illustrations.
- ☐ What the program actually did on thin lanes, and whether a quarterly cut of the +3-day bias was run.

S8B

- ☐ The 150 / 50 lane split and whether fat-tail lanes were “kept” (master) or “took stock in” (loop packet).
- ☐ The actual stakeholder dynamics behind the one-way-cut argument; whether a demand–supply correlation test was run; that the textbook form was my own starting point.

S8C

- ☐ Watch-window length; which quarter was held out, and whether it was a peak or a quiet quarter.
- ☐ Whether a control (deferred lanes) or a rolling-origin back-test existed; any pressure to release early.

S9

- ☐ The grain behind completeness 60% → 95% (event or shipment); the gate test sizes and whether the bar was an observed rate or a bound.
- ☐ Whether any control comparison was run for the −10% freight, and the base period behind “≈$2M in six months.” Cohort sizes are not in the record.

S19 / S20

- ☐ Any failure or live result beyond the stated drills — both are design engagements; nothing in production moved.
- ☐ The exact EWMA / CUSUM parameters of the Andon.

S21

- ☐ The order of discovery (funnel first, interviews second) and anything claimed beyond contract design plus wave-one mechanics.

E7

- ☐ Client naming; every CV, correlation and noise input is illustrative; the design was not run — say “I designed”, not “we ran”.

> **If you cannot tick it:** say “illustrative inputs, real outcomes”, give the record (200 lanes, 99%, 580K units / ~$1.5M, zero stockouts) and how you would firm the softer input. Never invent a window, a cohort count or a dollar.

NUMBERS I MUST NOT GET WRONG NEW

The canon numbers this packet uses, each with its basis word. Say the basis in the same breath as the number.

NUMBER	WHAT IT IS	BASIS	NEVER / CAREFUL
580K units / ~$1.5M	S8 — released with zero stockouts in the watch window; ≈$2.59/unit	record, one-time; $2.59 derived	never merged with the $2M into a range; never per year
≈$0.6M / yr	S8 — carry 0.3 at 20% + expedites 0.3; range $0.3–1.4M	modeled	not a result
200 lanes · 99% · p99	S8 — promise at p99; 99% lead-time accuracy gate	record	quote a tolerance only after naming the definition
70% → 85% · $2M	S6 — class-A forecast accuracy; safety stock released under the two-signal rule	record, one-time	never add to S8; never “I owned the forecast”
≈$0.8M / yr	S6 — carry 0.2 · expedites 0.3 · planner capacity 0.3; range $0.5–1.2M	modeled	planner capacity is time, not cash
3 months · 8× · −50%	S6 — shortage alert live; adoption; planner effort	record	never paste “3 months” onto the six-site ingest
1,000 → 597 = 80%	S9 — asked vs built first; ~400 lanes (~20%) deferred; ~200 next wave	record	not “1,000 lanes live”
60% → 95% · ≥95% gate	S9 — completeness; valid-ASN gate held twice	record	grain of the 95% VERIFY-WITH-RAJESH
−10% ≈ $2M in six months	S9 — freight spend	record, not annualized	no control group for causal claims VERIFY-WITH-RAJESH
1.10 d → b = k·σ, k = 1.645	S19 — flat buffer vs governed dial; fast-promise 7.6% → 39.6% at a 95% floor	engagement model	the 97.8% variant is never claimed
+$255K/wk ≈ $13.3M/yr	S19 — modeled value of the dial	modeled	never record; $12 breach vs $6.50 margin is the sensitivity
41.5% = 43.6% × 95.2%	S20 — Promise Yield on the docs' sample; target 50%	illustrative sample / design target	nothing in production moved
~30 d → < 24 h · < 5%	S20 — Andon detection-to-action; false-trip design bar	design	no live Andon result
~38–42% · 85% · ≈$3.7M/yr	S21 — stalled enrollment; handling adherence; per +10 pts of enrolled GMV	record / modeled	claim = contract + wave one only
CR 0.99 · z 2.33 · ±20%	E7 — newsvendor first buy and the two-week trigger; $216K/yr per market + $100K stranding	illustrative; design, not run	client name only if confirmed VERIFY-WITH-RAJESH
Six sites	every program	record	say it with no regional qualifier

Basis vocabulary. *record* — measured and signed in the program · *user anchor* — a number the candidate stated · *modeled* — my arithmetic on labeled inputs · *illustrative* — example inputs chosen to show the math · *design* — specified, not run · *engagement model* — a design-engagement calculation, not a production result.

INTERVIEW-PREP PACKET 3 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS · V2 — NINE PRIMARIES

Amazon NACF Supply Chain loop — Packet 3: Rocky Goforth

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

Rocky Goforth

Senior Manager, Infrastructure — NACF

Thursday, October 30, 2026 · 10:00 am PDT · 55-minute video interview

Strategy in one line. An infrastructure owner wants to know what happens when the plan meets the floor. Open with S2 (one comingle pool instead of two warehouses, a robotics ramp behind a published 98% gate that fired at 94, two go-lives with zero customer-facing incidents) and hold S18 (dual-DC WMS migration: 0.5% tolerance published before mock one, 7% → 1.8% → 0.3%, the RF-device miss owned) for the go-live-gone-wrong / Bias for Action question. Gates, invariants, triggers and hypercare numbers are the vocabulary; vendors configured, DC Ops ran the floor. **v2:** seven new primaries (U1–U7) re-cut the same Sephora node facts — matrix, profile, reserve, constraint, runbook, AFS control, issues log — each with its own math, behavioral pack, angle cards, trap drills and NACF bridge.

LP focus — my guess, not confirmed: Insist on the Highest Standards · Bias for Action · Deliver Results or Ownership. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title and the JD in "Read me first" and should be held loosely. No interviewer biography is asserted anywhere in this packet — only title-based inference, labeled as such.

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Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

5-Minute Cram Sheet — Rocky Goforth NEW

Profile hooks (title-based inference — no biography asserted)

- **Senior Manager, Infrastructure.** Owns the physical and systems substrate: buildings, automation, WMS layer, capacity, launches, cutovers, peak readiness. His question is *what happened when the plan met the floor.*
- **His vocabulary:** dock, space, labor, throughput, node constraints, gates, rollback, freeze windows, hypercare exit — and physical truth vs system truth.
- **My moves:** open S2; name the gate that fired; put every dollar behind its basis word; end each story on one number, then one sentence to NACF inbound (doors × hours × unload rate · UPH · cube).
- **LP guess (unconfirmed):** Insist on the Highest Standards · Bias for Action · Deliver Results / Ownership — Dive Deep and Earn Trust are likely second-round.

The nine stories, in the order to reach for them

#	STORY	LPS	THE ONE NUMBER
1	**S2** · Utah DC comingle pool + robotics gate ELEVATED	Insist on the Highest Standards · Deliver · Backbone · Earn Trust	+15% processing efficiency, zero customer-facing incidents *(record)*
2	**U4** · Where the node's constraint bound NEW	Bias for Action · Frugality · Dive Deep · Earth's Best Employer	94 vs 98 → ~30% dial; ≈421 h vs ≈4,200 mis-picks *(modeled)*
3	**U5** · 343-task runbook, one irreversible instant NEW	Ownership · Deliver · Backbone · Dive Deep	343 tasks · ~815 h · 61 days; rollback never fired *(record)*
4	**S18** · WMS migration, two DCs; 0.5% tolerance ELEVATED	Ownership · Bias for Action · Learn · Insist	7.0% → 1.8% → 0.3% vs the 0.5% tolerance *(record)*
5	**U7** · 836-row log as the readiness instrument NEW	Dive Deep · Insist · Ownership · Earn Trust	441 of 836 rows Blocker/Critical; 98.0% of them closed *(record, pandas)*
6	**U1** · Node × channel × bucket-transition matrix NEW	Invent and Simplify · Dive Deep · Insist · Earn Trust	every event → exactly one bucket; Blocker + Critical open = 0 at go/no-go
7	**U3** · Merch reserve — the priced no NEW	Backbone · Customer Obsession · Earn Trust · Success & Scale	p* = 36.7% — the price of promising reserve *(modeled)*
8	**U6** · Negative AFS + the correction screen NEW	Dive Deep · Ownership · Customer Obsession · Insist	3 cause classes; negatives clean at T+1; drift 0 *(record design)*
9	**U2** · Profile both books before the design NEW	Dive Deep · Are Right · Learn · Think Big	$2.2M point → $0.74M–$2.2M range; 87% not pooling-dependent *(modeled)*

Three sentences to open with

> "At Sephora's Utah DC I turned two warehouses in one building into one comingle pool — three planning identities over one physical location — through two go-lives and a robotics ramp behind a published 98% gate: +15% processing efficiency, signed by Sephora, and zero customer-facing incidents."
>
> "The number that mattered most was 94 against 98: in week two the node's constraint moved from labor hours to pick accuracy, and a throttle DC Ops owned before the ramp bought a root-caused fix for about 421 labor hours on my model — full volume in week three."
>
> "I ran the program on one 836-row issues log, read Blocker and Critical first — 441 rows, 98% closed — and the rows where the dock, the floor, the power and the footprint were the problem each had a name and a date."

Five numbers never to get wrong

> **+15% processing efficiency · zero customer-facing incidents** — *record — Sephora-signed, DC Ops measured; the dollars are modeled*
>
> **98% gate · week 2 at 94% · ~30% throttle for a week · full volume week 3** — *record*
>
> **7.0% → 1.8% → 0.3% against the 0.5% tolerance published with Finance before mock one** — *record*
>
> **836 rows · 714 closed (85.4%) · Blocker 164 + Critical 277 = 441 (52.8%)** — *record; recomputed from the log with pandas*
>
> **343 tasks / 48,885 task-minutes ≈ 815 h · sync < 99% for 4 h · p95 > 2× for 1 h** — *record*

Never say: "400 devices failed" (say *hundreds of RF devices*) · "20+ systems" (say *about a dozen impacted systems*) · "fifteen tracks" (say *about thirty build tracks*) · any ratio of two modeled dollars · a fifty-times multiplier anywhere (the real figure in the loop is 9.7×, 7 → 68).

Question → Story Router NEW

How to use. Hear the question, find the LP, say the 15-second opener, land the number, then stop. Every question cites the real Amazon bank wording as *AIQB: LP Q#*. "Card" = the angle card under that story (data · conflict/influence/backbone · failure-learning). Story IDs: **S2** Utah DC comingle pool + robotics gate · **S18** WMS migration, two DCs; 0.5% tolerance · **U1** Node × channel × bucket-transition matrix · **U2** Profile both books before the design · **U3** Merch reserve — the priced no · **U4** Where the node's constraint bound · **U5** 343-task runbook, one irreversible instant · **U6** Negative AFS + the correction screen · **U7** 836-row log as the readiness instrument. Basis labels are in the story; *modeled* / *illustrative* are said as such.

LP	AIQB QUESTION (VERBATIM)	PRIMARY STORY → ANGLE	BACKUP	15-SECOND OPENER	ONE NUMBER
Customer Obsession	*Sometimes customers make unreasonable requests. Tell me about a time when you've had to push back or say no to a customer request.* **AIQB: Customer Obsession Q7**	**U3** · conflict card — visibility yes, promisability no	U6 (Q8 — a negative AFS is a promise we cannot keep)	"Merchandising wanted reserve stock promisable — it was in the building. I priced the ask: it pays only above about 37% release, so I said no and offered visibility instead."	**p* = 36.7% *(modeled)***
Ownership	*Describe a time when you didn't think you were going to meet a commitment you promised.* **AIQB: Ownership Q2**	**S18** · failure card — the RF-device test	U5 (Q4 — two go-lives and a runbook over one fast date)	"A device test I scheduled before training showed hundreds of RF devices failing the conversion script — and I had planned that task as low effort. I told leadership straight: training slips a week and a half, the date holds."	**+1.5 weeks; date held *(record)***
Invent and Simplify	*Give me an example of a complex problem you solved with a simple solution.* **AIQB: Invent and Simplify Q1**	**U1** · data card — one rule, one asset	S2 (one pool, three planning identities)	"Three node types, two channels, about a dozen systems — and one rule: every physical event maps to exactly one inventory bucket, tested cell by cell with numbers before and after."	**AFS 557, not 560 or 554 *(illustrative trace)***
Are Right, A Lot	*Tell me about a time when you didn't have enough data to make the right decision.* **AIQB: Are Right, A Lot Q1**	**U2** · failure card — the point became a range	U1 (the in-transit cells the floor taught us to add)	"The case said "pool them," and I did not have the demand profile. I profiled both books by node — overlap, order shape, variability — and turned a $2.2M textbook number into a $0.74M–$2.2M range with the measurement that would settle it."	**$0.74M–$2.2M *(modeled)*; 87% not pooling-dependent**
Learn and Be Curious	*Describe a time when you took on work outside of your comfort area. How did you identify what you needed to learn to be successful?* **AIQB: Learn and Be Curious Q2**	**S18** · Learn answer — month one on the floor	U2 (Q1 — the pooling math, learned in discovery)	"I had never run a WMS migration and had eight months. I wrote the charter in week one so the destination existed before my knowledge did, then spent month one on the floor until I could walk receive to ship unaided."	**month 1 on the floor; 7% → 1.8% → 0.3%**
Hire and Develop the Best	*Tell me about a time when you provided coaching for a team member. What started the coaching?* **AIQB: Hire and Develop the Best Q4**	**S15** bench (Josyula's primary — only if the question demands it)	U6 (four control-owner groups ran the screens without me)	"A QA lead feared compressed test windows — and she was right. I coached the frame and she wrote the definition of done; by cycle two she was defending the gate without me."	**critical defects per release −50% *(record)***
Insist on the Highest Standards	*Describe a time when you refused to compromise your standards around quality/customer service, etc.* **AIQB: Insist on the Highest Standards Q1**	**S2** · conflict card — the 98% gate	S18 (the 0.5% tolerance published before mock one)	"I published the standard before there was any pressure to waive it: a 98% pick-accuracy gate and a throttle dial DC Ops owned. Week two read 94, so the gate fired — bots to about 30% for a week, full volume in week three."	**98 → 94 → ~30% → full volume wk 3 *(record)***
Think Big	*Tell me about time when you were working on an initiative or goal and saw an opportunity to do something much bigger or better than the initial focus.* **AIQB: Think Big Q1**	**U2** · Think Big answer — two pools to one	S2 (the one-omni-facility ask)	"The ask was a number on pooling; the bigger prize was one physical pool with three planning identities — each channel kept its forecasting lens, and the self-transfers went to zero by construction."	**2 pools → 1; ≈260K units/yr self-transfers → 0 *(illustrative volume)***
Bias for Action	*Give me an example of a calculated risk that you have taken where speed was critical.* **AIQB: Bias for Action Q1**	**U4** · data card — the dial fires	S2 (the throttle decision)	"Week two of the robotics ramp read 94 against a published 98. The dial was built in advance and DC Ops owned it, so we throttled to about 30% that same week and root-caused the zone and slotting logic."	**94 vs 98 → ~30% → full volume wk 3 *(record)***
Frugality	*Give me an example of how you have helped save costs or eliminate waste within your role or organization.* **AIQB: Frugality Q1**	**S2** · data card — duplicate handling removed	U4 (Q4 — priced both sides of the throttle)	"Two warehouses in one building were transferring stock to themselves and paying duplicate handling for zero customer value. One comingle pool removed that cost, and DC Ops measured +15% processing efficiency."	**+15% *(record)*; ≈$1.6M/yr *(modeled)***
Earn Trust	*Tell me about a time when you had to communicate a change in direction that you anticipated people would have concerns with.* **AIQB: Earn Trust Q1**	**S2** · conflict card — Digital's concern became the design	U7 (Q6 — the worst number in the room was mine to say)	"Digital worried a virtual Retail DC would blur their forecasting lens, and they were right to. I did not argue them out of it — I made the concern the design: three planning identities over one physical location."	**3 identities, 1 location; 60+ stakeholders**
Dive Deep	*Tell me about a situation that required you to dig deep to get to the root cause.* **AIQB: Dive Deep Q2**	**U6** · data card — a negative has three causes	U7 (the log, three cuts) · U2 (the profile)	"A negative AFS is not one problem. I decomposed it to three cause classes — a duplicate event, a feed that had not caught up, an override — each with an owner, and every AFS-changing act got a screen, a role and an audit row."	**3 cause classes; drift 0 on consecutive days *(record)***
Have Backbone; Disagree and Commit	*Describe a time when you took an unpopular stance in a meeting with peers and your leader.* **AIQB: Have Backbone; Disagree and Commit Q2**	**S2** · conflict card — two go-lives, not one	U5 (Q5 — no UAT compression after the Locus RTL slipped three weeks)	"The prevailing instinct was one go-live, one date. I challenged it: one go-live puts a reserve-rule defect and a floor defect in the same room the same week — so January proved the rules and May moved the floor."	**2 go-lives, 1 defect class each**
Deliver Results	*Give me an example of a time when you were able to deliver an important project under a tight deadline.* **AIQB: Deliver Results Q1**	**U5** · Deliver answer — 61 days, peak untouched	S18 (eight months, ~3% under) · S2 (Q3, +15% signed)	"The window was 61 days, May 1 to July 1, with the Oct–Dec peak freeze behind it — so I wrote the cutover as 343 owned tasks, about 815 hours, each with a predecessor, and the two go-lives landed on their dates."	**343 tasks · ~815 h · 61 days *(record)***
Strive to Be Earth's Best Employer	*Describe a time when you implemented a change at work aimed at improving the work experience of yourself or coworkers.* **AIQB: Strive to Be Earth's Best Employer Q3**	**U4** · Employer answer — the stop condition belongs to the floor	S15 bench (QA lead elevated, not blamed — Josyula's)	"The floor was the one place a bad ramp would land. I wrote the gate and the dial before the robots arrived and DC Ops owned them — so when week two read 94, the stop condition belonged to the people who carry the consequence."	**dial owned by DC Ops before the ramp; ~30% for one week**

LP	AIQB QUESTION (VERBATIM)	PRIMARY STORY → ANGLE	BACKUP	15-SECOND OPENER	ONE NUMBER
Success and Scale Bring Broad Responsibility	*Tell me about a time when you considered everyone who was affected by your work, not just your direct customers or people within your organization.* **AIQB: Success and Scale Bring Broad Responsibility Q5**	**U3** · Success & Scale answer — a broken promise is a defect	U6 (the screens and owners that keep a promise true)	"An oversell looks like a ledger error to a planner and a broken promise to a customer. At an illustrative thirty million orders a year that is thirty thousand customers, so I treated it as a defect with an owner and a screen."	**30,000 customers/yr *(illustrative)* → zero incidents *(record)***

THE ELEVEN NON-LP PROMPTS — SAME GRID NEW

PROMPT	PRIMARY → ANGLE	BACKUP	15-SECOND OPENER	ONE NUMBER
"Tell me about a time you failed."	**S18** failure card (Apple G2)	U6 · U5 failure cards	"The RF-device test failed before training; I had planned it low-effort, I said so the same day, and I now give every single-point dependency a what-if plan."	**+1.5 wk; date held**
"Disagreed with your manager."	**S2** two go-lives (conflict card)	U5 no UAT compression	"The instinct was one go-live; I put the two-defect-class argument in front of leadership and we staged it."	**2 go-lives, 1 class each**
"Influence without authority."	**S2** · 60+ stakeholders	U3 · U6 (owners I did not manage)	"None of the systems reported to me; I owned the model, the gates and the signature chain, and made the concern the design."	**60+ stakeholders**
"Reduced cost."	**S2** · +15%, self-transfers gone	U4 priced the throttle	"One pool instead of two removed duplicate handling; the +15% is signed, the ≈$1.6M a year is my model."	**+15% record; ≈$1.6M/yr modeled**
"Dealt with ambiguity."	**U2** · profile first	S18 charter in week one	"Two books, two warehouses and no agreed model — I profiled SKU overlap, order shape and variability before anyone argued design."	**9.9%–29.3% release**
"Simplified a process."	**U1** · one rule, one asset	S2 · U7 one grammar	"Every event maps to exactly one bucket, and no story entered QA without a cell reference."	**557, not 560 / 554**
"Decision with incomplete data."	**U3** · the break-even	U2 bounded range	"I could not know the release rate, so I priced the decision at the rate where it flips."	**p* = 36.7%**
"Missed a deadline."	**U5** · Locus RTL +3 weeks	S18 training +1.5 wk	"A vendor slipped three weeks; I escalated the week it surfaced, re-slotted openly and refused to compress UAT."	**3 wk slip; UAT intact**
"Mentored someone."	**S15** bench (Josyula's)	U6 control-owner handoff	"I coached the frame, the QA lead owned the gate, and she defended it without me."	**−50% critical defects**
"Tight timeline."	**U5** · 61 days	S18 · eight months	"343 tasks, about 815 hours, every one with an owner and a predecessor."	**61 days**
"Changed your mind with data."	**U2** · $2.2M point → range	U4 constraint moved	"I had carried the textbook pooling number; the overlap and correlation tests said carry a range with a kill number."	**$0.74M–$2.2M**

Spend rule. One question, one story. If an LP is asked twice, move to the named backup, not a story allocated to another interviewer. S15 is Josyula's primary — use it only if the question demands it and keep every fact identical to his packet. Real failure chapters (Apple G1/G2/G3) are the source for every failure/learning angle; none is invented.

Read me first

The interviewer's lens (inference from title + JD — not a biography)

Inference from the title. "Infrastructure" in a fulfillment network means the physical and systems substrate the process runs on — buildings, automation and robotics, the WMS/WCS layer, capacity and launches, cutovers and peak readiness. A Senior Manager there is accountable for changes that touch the floor: go-lives that cannot drop a shipment, automation ramps that must not erode throughput, and migrations where inventory accuracy cannot wobble. He will care about gates, rollback, freeze windows, hypercare exits, and what happened when the plan met the floor.

What that means for the answers. The two Sephora distribution-center programs: S2 (two systemic warehouses in one building consolidated into one comingle pool, with a WMS upgrade and a robotics ramp on the same critical path and zero customer-facing incidents allowed) and S18 (a dual-DC legacy-WMS migration run to a 0.5% reconciliation tolerance published before the first mock, with the RF-device miss owned). Tell them as Make/Deliver process redesigns — one inventory pool instead of two, a cutover discipline instead of a date — with asset utilization (+15% processing efficiency, labor-hours, capital pooled) and reliability (zero incidents, 0.3% against 0.5%) as the results. The vendor teams configured BlueYonder and Locus; DC Ops ran the floor; I owned the model, the staging, the gates and the go/no-go.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title and the JD; if the first question lands on a different LP, the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Insist on the Highest Standards	Infrastructure owners live by gates that can stop the program: the 98% robotics gate that fired at 94 (S2) and the 0.5% tolerance published with Finance before mock one (S18) are the primary and backup for this LP in the master map.	S2 (open) · S18 (backup)
Bias for Action	When the floor breaks the plan — hundreds of RF devices failing the conversion script in a pre-training device test — he wants to see a decision taken the same day, the cost priced and the date held.	S18 (the device-test miss) · S2 (the throttle to ~30%)
Deliver Results / Ownership (one of the two)	Two go-lives, peak untouched, +15% signed by DC Ops; an eight-month migration on time and ~3% under budget with a $20M pipeline unblocked. Ownership shows in the RF miss carried with my name on the cause.	S2 (the comingle launch) · S18 (8 months, both DCs)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For Infrastructure specifically:** speak in gates, invariants and triggers — the published ramp gate and the throttle built in advance; the six cutover invariants (Σ AFS conserved; no SKU sellable in two places; negatives cleared first); rollback triggers as numbers (sync < 99% for 4 h; p95 > 2× for 1 h); hypercare exit as a number (< 2 blockers/day sustained). A gate that cannot stop the program is decoration.
- **Asset utilization is the result frame:** +15% units per labor-hour (signed), one pool instead of two (√2 pooling, modeled and labeled), self-transfers inside one roof eliminated, robotics at full volume by week three, $20M of pipeline unblocked on the date.

Loop-consistency rule. One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S6/S8 (Hoerger) · S1/S3 (Lan) · S16/S15/S10/S14 (Josyula). **v2 overlap:** U1, U3 and U6 re-cut the AFS, merch-reserve and bucket-matrix facts that sit in Lan's S1 — the numbers are identical in both packets; answer from U1/U3/U6 only when the question is about the building, and say the program is the same. S5 (the architecture council) shares the UDC program's dated chain with S2 — Lan's backup pointer, not yours; S2 and S1 share program-level artifacts (issues log, rollback triggers) — if asked, say so.. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

NACF inbound bridge — one whiteboard for nine stories NEW

Every story ends with one sentence that lands on NACF. The sentence rides on the inbound booklet's frames — CAP-A doors, CAP-B UPH, CAP-C cube, CAP-D dwell, CAP-E peak buffer, INV-D transfer break-even — on the booklet's own illustrative FC (120,000 units a day). Draw this chain first, then say which link the story's constraint lives in.

FRAME	LINK	FORMULA → SUBSTITUTION	BASIS	VALUE
A	Doors	`120,000 ÷ 6 ÷ 40 = 500 pallets; ÷ 26 = 19.2 trailers/day; ÷ 10 h = 1.92/h; ÷ 0.5 trailer per door-hour = 3.84 doors; ÷ 0.85 = 4.5`	booklet CAP-A · illustrative FC	**5 doors**
B	UPH labor	`120,000 ÷ 300 = 400 h; × 1.15 = 460 paid h; 0.7 × 300 + 0.3 × 180 = 264 blended; 120,000 ÷ 264 = 454.5 h`	booklet CAP-B · 300 UPH, 60% learning curve illustrative	**+13.6% hours**
C	Cube	`2,000,000 × 0.25 = 500,000 ft³; 1,200,000 × 0.85 = 1,020,000 effective; 500,000 ÷ 1,020,000`	booklet CAP-C · 0.25 ft³/unit illustrative	**49.0% true use**
D	Dwell	`2.0 ÷ (1 − 0.85) = 13.3 h; 2.0 ÷ (1 − 0.95) = 40.0 h; 40.0 ÷ 13.3 = 3.0×`	booklet CAP-D · qualitative queueing approximation	**85% → 95% ≈ 3× dwell**
E	Peak buffer	`156,000 ÷ 264 = 591 h; 591 − 454.5 = 136 h; ÷ 10 = 13.6`	booklet CAP-E · p90 156,000 illustrative	**≈14 flex heads**
D′	INV-D	`$3.00 ÷ $12 = 0.25 — transfer when stockout probability > 25%`	booklet INV-D · $3.00 transfer, $12 margin illustrative	**p* = 25%**

STORY → FRAME → THE QUESTION I WOULD ASK AT NACF

STORY	WHAT THE NODE TAUGHT	THE NACF SENTENCE	FRAME
S2	A published ramp gate and a dial the floor owns (94 vs 98 → ~30%).	Which link binds this week at a new FC — doors at the 85% line, UPH after the learning-curve tax, or cube at 49% effective — and who owns the stop?	A · B · C
S18	Tolerance published before mock one; a freeze sized to the longest lane; a what-if for the single-point dependency.	Publish the receipt-reconciliation tolerance before the first mock; hold 5 planned doors against 3.84 raw — about 1.2 doors of slack — so a day-one surprise does not move the date.	A · D
U1	Every event lands in exactly one bucket; zero or two is a defect.	"Stranded" inventory is an event that landed in zero states. Build the receive → stow → sellable event matrix before an FC ramps.	booklet "stranded"
U2	Profile overlap, order shape and correlation before pooling is claimed.	Do two FCs carry the same ASINs and see the same demand? Pooling pays only at low correlation; the physical move is INV-D — transfer when p > 25%.	INV-D
U3	Merch reserve is in the building and not promisable; p* = cost ÷ (gain + cost).	Promise pickable, not received. Same break-even shape as INV-D (3 ÷ 12); 10,000 reserved units at 0.25 ft³ hold 2,500 ft³ of the 1,020,000 effective.	C · D′
U4	The constraint moved from labor hours to accuracy; the throttle cost ≈421 h.	421 h is about 10.5 people for a week against CAP-B's 13.6% learning tax and CAP-E's ~14 flex heads — name the binding resource first.	B · E
U5	One irreversible instant; stop conditions written as numbers before the first task.	The first live inbound appointment is the irreversible instant; the 85% door ceiling is the slack that absorbs a re-pointed or late trailer.	A · D
U6	A negative AFS decomposes to duplicate event, feed latency, override.	Committed units above what the building can back: a double receive scan, dock-to-stock lag, a manual adjustment — zero-tolerance reconciliation of dock-received vs system-available.	D
U7	The issues log read severity-first, with the dock, power and footprint rows named.	Launch register with the dock row at the top: appointments, placards, pallet-build positions; a SKU with no footprint defaults a slot, which is wasted cube at the 85% line.	A · C

All booklet inputs are the booklet's own illustrative FC; nothing here is a Sephora figure. The people-week conversion assumes a 40-hour week (421 ÷ 40 = 10.5).

The stories — business-first, canon numbers

Nine primaries on the Sephora distribution-center work: S2 and S18 (elevated) and seven new cuts of the same node facts, U1–U7. Each: 60-second pitch → Situation/Task → Action in gated phases → Results → L6 evidence → five facets and probing → the four-table math → Twist map → behavioral pack, angle cards, trap drills, 20-second version and NACF bridge.

S2 — Sephora Utah DC — two systemic warehouses in one building → one comingle pool, coupled with a WMS upgrade and a robotics ramp behind a published gate (+15% processing efficiency, zero customer-facing incidents) ELEVATED

LPs this story can serve here: Insist on the Highest Standards (primary — the 98% gate that fired at 94) · Deliver Results (two go-lives, peak untouched) · Have Backbone (the throttle) · Strive to be Earth's Best Employer (the floor protected during the ramp) · Bias for Action (the throttle decision the week the gate failed)
Process family: Make / Deliver — DC operating model (one inventory pool, three planning identities), automation ramp governance, staged go-lives, cutover runbook, hypercare exit

Honesty label. I owned discovery through FSD for the location and inventory model, the staging and the gates; BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor and measured the +15% post-stabilization. The +15% is Sephora's signed number; every dollar is my model on an illustrative 100K-units/week DC. Program-level artifacts (dated gate chain, 836-row issues log, rollback triggers) are shared with S1/S5 — one umbrella program.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Insist on the Highest Standards. I published the robotics gate — 98% pick-accurate — before the robots arrived; when week two read 94 it fired and the bots went to ~30% for a week.

Deliver Results. Two go-lives on their dates, each carrying one defect class, peak untouched, zero customer-facing incidents and +15% processing efficiency signed by Sephora.

Have Backbone; Disagree and Commit. I challenged the single-go-live instinct and held the comingle-reserve improvements out of the release to protect zero incidents — priced as a later enhancement.

Strive to be Earth's Best Employer. The throttle and the DC-Ops-owned dial protected the floor's throughput during the ramp; the interim reporting burden was named as the price. No headcount or attrition number is claimed.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Sephora's Utah DC ran as two systemic warehouses in one building — Dotcom 801 and Retail 901 — transferring stock to itself and paying duplicate handling for zero customer value. The ask was one omni comingle facility, with a BlueYonder WMS upgrade and Locus robotics on the same critical path, an October–December peak freeze, about a dozen impacted systems, 60+ stakeholders, and zero customer-facing incidents allowed. **Task.** I owned discovery through the signed design for the location and inventory model, the staging and the acceptance gates. **Action.** I redesigned the operating model as one physical pool with three planning identities, so each channel kept its forecasting lens without a second truth; split the program into two go-lives so each carried one defect class — January proved reserve rules against live demand with the floor untouched, May moved warehouse execution; published the robotics gate at 98% pick-accurate throughput with the throttle dial built in advance and owned by DC Ops; ran three mock cutovers against the 0.5% tolerance and a 343-task runbook whose one irreversible instant was zero-then-publish, so nothing was sellable in two places. When week two of the robotics ramp hit 94 against 98, the gate fired: bots throttled to about 30% of volume for a week, zone and slotting logic root-caused, full volume week three. **Result.** Primary metric: DC processing efficiency — baseline index 100 → 115, +15%, Sephora-signed, measured by DC Ops post-stabilization; zero customer-facing incidents through both go-lives; rollback armed twice, never fired. About $1.6M a year modeled — labor at +15%, pooling carry, fewer split shipments, self-transfer handling removed — and about $2.2M of one-time safety stock released by pooling, also modeled; the +15% and the zero are the record.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The Utah DC ran as two systemic warehouses — Dotcom 801 (selling location 800) and Retail 901 — transferring stock to itself and paying duplicate handling for zero customer value. The ask: one omni/comingle facility, with a BlueYonder WMS 8.2 → 2023.1 upgrade and Locus robotics on the same critical path, about a dozen impacted systems, an Oct–Dec peak freeze, zero customer-facing incidents allowed.

The challenges sat on the critical path itself: the WMS upgrade and the robotics ramp shared it, the peak freeze bounded it, and the interim would run two live truths. The register's top three risks — two live truths during the interim, a robotics ramp with no stop condition, and a peak collision — each had a trigger, a mitigation and an owner before sign-off.

T — Discovery through FSD for the location and inventory model, the staging and the acceptance gates; 60+ stakeholders (~80 names across ~15 functions) on the steering path; none of the systems reported to me.

Baselines on the table: processing efficiency index 100 (units per labor-hour) · customer-facing incidents allowed: 0 · robotics gate 98% published in advance from the pre-ramp baseline and the incident budget · self-transfers and duplicate handling — the standing cost one pool exists to remove.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — profiled 801/901 (SKU overlap, order profiles); sponsor objectives and success criteria — one omni facility, self-transfers eliminated, zero customer-facing incidents, peak untouched; systems-impacted table. *Exit bar:* every impacted system named to an owner with a written LOE (landed 7/2); criteria signed by steering.
- **Plan & Design (5/15 → 8/14)** — council pack (three decisions + an out-of-scope fence); the 900/901 location/inventory model with three planning identities over one physical location; staged two-go-live timeline; RAID log; FSD 8/14. *Exit bar:* council sign-off 7/31; FSD 8/14; peak-freeze boundary decided; LOE-backed plan across ~30 build tracks.
- **Build & Test (Aug → Jan/Apr)** — node × channel × bucket-transition matrix as the regression asset; dated E2E days — Inbound 11/22, Dotcom 12/20, Retail 1/30 — every function in one room; Locus gate definition (98%) with the throttle spec; three mock cutovers against the 0.5% tolerance (the S18 discipline reused); the Locus RTL slipped three weeks — escalated the week it surfaced, re-slotted openly, UAT not compressed to absorb it. *Exit bar:* E2E green on three dates; Blocker + Critical open = 0 at each go/no-go; mock 3 inside tolerance.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — runbook of 343 tasks / ~815 hours, each with owner, duration and predecessor; six-step May choreography — clear negatives, transfer AFS to Uncommitted, scan-and-reserve, zero-then-publish; rollback triggers per interface (sync < 99% for 4 h; order-flow p95 > 2× budget for 1 h). *Exit bar:* six invariants held (Σ AFS conserved; no SKU sellable on 800 and 900 at once; negatives cleared first); rollback never fired.
- **Hypercare / Close (Jun → Jul)** — monthly arc of new issue rows 271 → 65 → 21 → 14 → 4 (June → October); Locus weekly gate read — week 2 at 94 vs 98 → throttle to ~30% → RCA (zone/slotting logic) → full volume week 3; hypercare exit memo; template handed to the next DC. *Exit bar:* < 2 blockers/day sustained → exit week 3 of 4; +15% measured post-stabilization by DC Ops.

Influence and ownership. Mine: the location/inventory model, the staging, the gates, the council pack, the runbook and the go/no-go recommendation. BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor, owned the robotics gate and the dial before the ramp — so when it fired, the rule fired; I did not improvise. Digital worried a virtual Retail DC would blur their forecasting lens — the three-identity design made the concern the design rather than an argument. The comingle-reserve improvements were held out of the release to protect zero incidents, priced as a later enhancement at steering.

> **Designed the process; others own it now.** I designed the location/inventory model, the gates and the throttle dial; DC Ops owned the dial before the ramp and runs the floor on it, and the hypercare template went to the next program. **Trade-off put to leadership.** The recommendation to leadership outlined the competing metrics — +15% processing efficiency and the robotics ramp vs zero customer-facing incidents and peak untouched — and made the gate (98%) and the throttle (~30%) the published answer before the robots arrived.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$1.6M/yr *modeled* (labor at +15% ≈$0.88M · pooling carry ≈$0.22M · fewer split shipments ≈$0.45M · self-transfer handling ≈$0.10M) + ≈$2.2M one-time safety-stock release *modeled* via √2 pooling; the +15% is Sephora's signed number; every dollar is my model on an illustrative 100K-units/week DC (range $1.1–2.4M/yr).
- **#** — cutover runbook 343 tasks / 48,885 task-minutes ≈ 815 hrs; issues log 836 rows / 85.4% closed; monthly new-row arc 271 → 65 → 21 → 14 → 4 (record, from the log).
- **%** — processing efficiency +15% (record, Sephora-signed); robotics gate 98%, week 2 at 94, bots throttled to ~30% of volume for a week, full volume week 3 (record).
- **Time** — council 5/15 → sign-off 7/31 → FSD 8/14; E2E 11/22 · 12/20 · 1/30; January and May go-lives; hypercare exited week 3 of 4 (record).
- **Customer** — zero customer-facing incidents through both go-lives (record) — the constraint every other number moved inside.
- **Quality** — six cutover invariants held; rollback armed for both go-lives, never fired; peak untouched (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Two systemic warehouses in one building and no agreed operating model. I owned the location/inventory model (three planning identities over one physical location), the staging and the gates, and set the success criteria with the sponsor — one omni facility, self-transfers eliminated, zero customer-facing incidents, peak untouched.
Scope & influence	BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor and owned the robotics gate and the dial; the go/no-go recommendation was mine to leadership — influence across about a dozen impacted systems and 60+ stakeholders I did not manage.
Execution trade-off + mechanism others now own	Robotics throughput vs floor stability: the 98% gate fired at 94 in week 2, bots throttled to ~30% for a week, full volume week 3. Mechanism others own now: the dial, owned by DC Ops before the ramp, and the hypercare template handed to the next program.
Impact (long-term)	Processing efficiency +15% (Sephora-signed); zero customer-facing incidents through both go-lives; ≈$1.6M/yr modeled plus ≈$2.2M one-time modeled; peak untouched.
Process improvement (standard work institutionalized)	The node × channel × bucket-transition matrix as the regression asset; a runbook of 343 tasks / ~815 hours with owner, duration and predecessor; six cutover invariants; a monthly new-row arc 271 → 4.
Complexity & consensus	Staged two go-lives (January planning / May execution) so each carried one defect class; a peak-freeze boundary; dated E2E days (11/22 · 12/20 · 1/30) with every function in one room.
Root cause at scale	The week-2 robotics miss was root-caused to zone/slotting logic and fixed before full volume — the rule fired, the cause was removed, the ramp was not pushed through.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: node × week × bot share; SKU × pool. Data: both books profiled in discovery (U2), labor hours, picks attempted/accurate, the 836-row issues log, the 343-task runbook. Cuts: accuracy by week; new rows by month (271 → 4). Revealed: the week-2 miss was zone/slotting logic, and the case rested on recurring rows more than on pooling.

2 · Solution design. Options: (A) keep two warehouses and reconcile; (B) one pool, one blended planning lens; (C) one physical pool, three planning identities (chosen); and a staging choice — one go-live vs two (chosen: two, each carrying one defect class). Why: C keeps Digital's forecasting lens without a second truth; two go-lives keep diagnosis univariate.

3 · Math & data logic. +15% → hours 240,000 / 1.15 = 208,700 (−13.0%); pooling 1 − 1/√2 = 29.3% on $7.5M → $2.2M (bounded in U2); gate T = accurate ÷ attempted ≥ 98%; throttle 100% → 30% (bounded in U4); AFS conserved (U5).

4 · Implementation. Five gated phases (3/29 → 4/30 → 5/15 → 7/31 → 8/14 → E2E 11/22 · 12/20 · 1/30 → January → May 1 → Jul 1 → hypercare); owners: me (model, staging, gates, go/no-go), BlueYonder/Locus/platform (build), DC Ops (floor, gate, dial). Controls: Blocker + Critical = 0, mock 3 ≤ 0.5%, rollback triggers.

5 · Probe questions. The six answered above in the Probing block; the zoom-ins that stand alone are U1 (matrix), U2 (profile), U3 (reserve), U4 (capacity), U5 (runbook), U6 (negatives).

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — Why 98%, and what did you actually do at 94?

98% was set from the pre-ramp baseline and the incident budget before the event, and DC Ops owned it. At 94 in week two I throttled the bots to about 30% of volume for a week, root-caused the zone and slotting logic, and ramped again — full volume by week three. A gate that cannot stop the program is decoration; mine did.

Infra — Why two go-lives instead of one big one?

So we never hunted two defect classes in the same week. January proved reserve rules against live demand while the floor was untouched; May left warehouse execution as the only remaining class. One go-live would have put a reserve defect and a floor defect in the same room.

Infra — What made you say go in May?

Counted rows, not the room: Blocker + Critical at zero, mock 3 inside the 0.5% tolerance, the 343-task runbook rehearsed, rollback armed per interface. I would have said no-go on one open Blocker; I never had to.

Infra — What were the rollback triggers and did any fire?

Sync below 99% for four consecutive hours, or order-flow p95 above twice budget for one hour — rollback per interface behind them. Armed for both go-lives; never fired. The one irreversible instant was step 4, zero-then-publish, so no SKU was ever sellable in two places.

Infra — The labor number is the biggest piece of your $1.6M. How firm?

The fifteen percent is measured and signed; the 120 FTE and the $28 loaded rate are my assumptions. Pull the DC's direct hours for the two quarters before and after and multiply by payroll — that turns the model into a record in an afternoon. If the split-shipment row didn't move, the total is about $1.1M.

> **Infra — What would you do differently?**
> Staff the interim reporting workstream on day one, not month three — for four months every report had to declare which system of record it read. And the Locus slip: escalated the week it surfaced, but I would have put the RTL on the register as a risk, not a task, from the start.

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Two warehouses in one building	`planning pools = 2 (801 retail + 800 dotcom)`	two planning identities over one roof (record)	2 pools
2	Baseline efficiency	`index = 100 units per labor-hour`	pre-UDC baseline (record)	100
3	Direct labor to run it	`120 FTE × 2,000 h/yr = 240,000 h/yr`	modeled staffing on an illustrative 100K-units/week DC	240,000 h/yr
4	Self-transfers inside one roof	`≈260,000 units/yr moved on paper between 801 and 800`	modeled	260K units/yr of non-work
5	Duplicated safety stock	`$50M × 25% = $12.5M SS; $12.5M × 60% = $7.5M`	avg inventory (illustrative) × SS share × share held in SKUs both pools carry	$7.5M duplicated poolable SS
6	Split shipments	`4,500,000 orders × 1 pt = 45,000 extra splits/yr`	~4.5M dotcom orders/yr; split rate 1 pt above the comingled state (modeled)	45,000 splits/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	One pool, three planning identities	`900/901 model → self-transfers = 0 by construction`	three planning identities over one physical location (record design)	the move is a posting, not a truck
2	Robotics gate (published first)	`T = pick-accurate completed / attempted ≥ 98%`	gate published before the ramp (record)	the dial was built in advance
3	Gate evaluated, week 2	`T = 94%; 94 < 98 → gate failed by 4 pts`	record	throttle fires
4	Throttle	`bot share 100% → 30%; manual covers (1 − 0.30) × 100,000 = 70,000 units/wk`	volume illustrative	RCA, then ramp again — full volume week 3 (record)
5	Efficiency after	`index 100 × 1.15 = 115`	+15% (record, Sephora signed, post-stabilization)	115
6	Hours for the same volume	`240,000 / 1.15 = 208,696 ≈ 208,700 h/yr`	Table 1 row 3 and the +15%	≈208,700 h/yr
7	Hours saved	`240,000 − 208,700 = 31,300; 31,300 / 240,000 = 13.0%`	arithmetic on the two rows above	31,300 h/yr = 13.0%
8	Pooling (2 equal pools → 1)	`factor = 1 − 1/√2 = 1 − 0.707 = 0.293; $7.5M × 0.293 = $2.2M`	√2 pooling (toolkit label; ceiling — bounded in U2)	≈$2.2M one-time
9	Zero-incident fence	`two staged go-lives — one defect class live at a time`	customer-facing incidents allowed: 0 (record constraint)	0 incidents through both

Table 3 — after: the metric, before → after.

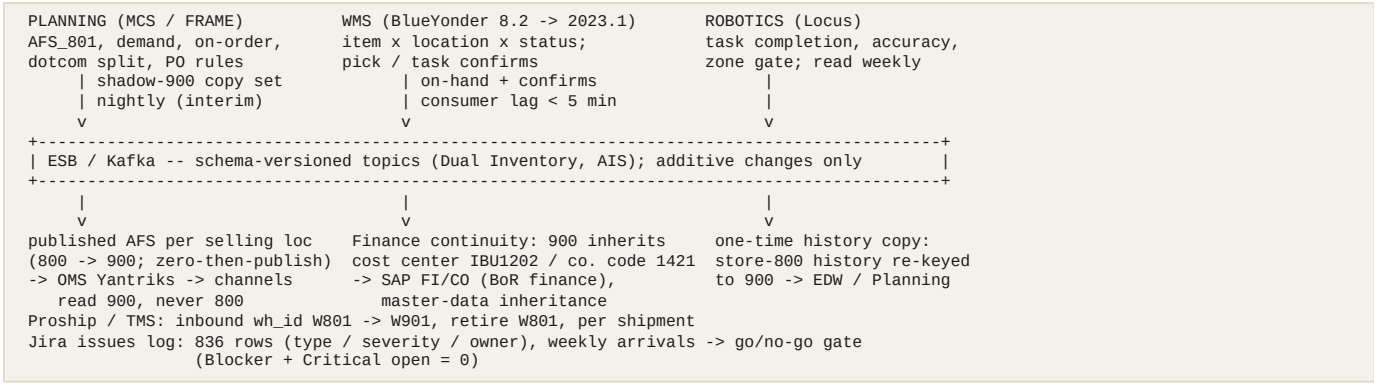
STEP	METRIC	BEFORE	AFTER	BASIS
1	Processing efficiency (units/labor-hour)	index 100	115	record (Sephora signed)
2	Direct labor hours, same volume	240,000 h/yr	≈208,700 h/yr	modeled on the record +15%
3	Self-transfers	≈260K units/yr	0	record design
4	Poolable safety stock	$7.5M	$5.3M ($7.5M − $2.2M)	modeled (√2 ceiling)
5	Split-shipment rate	baseline on ~4.5M orders	−1 pt	modeled
6	Customer-facing incidents	0 allowed	0 through both go-lives	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Labor at +15%	`31,300 h × $28/h = $876,400`	hours from Table 2 row 7; $28 loaded (modeled)	≈$0.88M/yr

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
2	Pooling release	`$7.5M × 0.293 = $2.2M`	Table 2 row 8 (one-time working capital)	≈$2.2M one-time
3	Carry on the release	`$2.2M × 10% = $220,000`	cost of capital (benchmark 8–12%)	≈$0.22M/yr
4	Split shipments	`45,000 × $10 = $450,000`	splits from Table 1 row 6; $10 per split (modeled)	≈$0.45M/yr
5	Self-transfer handling	`260,000 × $0.80 = $208,000; × 50% = $104,000`	50% — the other half already sits in the labor row (modeled)	≈$0.10M/yr
6	Total recurring	`0.88 + 0.22 + 0.45 + 0.10 = $1.65M`	rows 1, 3, 4, 5	≈$1.6M/yr
7	One-time working capital	`= row 2, stated separately`	safety-stock release	≈$2.2M

SOLUTION FLOW — THE SYSTEM & DATA MAP NEW



SENSITIVITY — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
Direct DC FTE (illustrative)	100	120	150
Loaded labor rate (benchmark)	$24/h	$28/h	$32/h
Labor $ saved	$0.62M	$0.88M	$1.25M
Cost of capital (benchmark)	8% → $0.18M	10% → $0.22M	12% → $0.26M
Split rate reduction × cost per split	0.5 pt × $8 → $0.18M	1 pt × $10 → $0.45M	1.5 pt × $12 → $0.81M
Total modeled $/yr	≈$1.1M	≈$1.6M	≈$2.4M

SPOKEN STATEMENT NEW

> "Conservatively, about one point six million a year: roughly nine hundred thousand of labor, because the same volume takes thirteen percent fewer hours at plus fifteen percent efficiency; about two hundred thousand a year of carry on the two point two million of safety stock one pool releases versus two; about four hundred fifty thousand from fewer split shipments; and about a hundred thousand from the DC no longer shipping to itself. The plus fifteen percent and the zero incidents are measured and signed by Sephora; every dollar figure is my model on an illustrative hundred-thousand-units-a-week DC, and I will tell you which input each one rests on."

IF CHALLENGED NEW

"The labor number is the biggest — how firm?"

The fifteen percent is measured; the headcount and the $28 loaded rate are my assumptions. Pull the DC's actual direct hours for the two quarters before and after and multiply by payroll — that turns the model into a record in one afternoon.

"Pooling sounds like a textbook claim."

It is, and I labeled it. √2 pooling assumes two roughly equal independent pools on the same SKUs. Measure SKU-level on-hand on the last day of 801 versus the first stable month of 901; the release is whatever the difference actually is. U2 bounds it at about $0.74M–$2.2M.

"Split shipments — evidence?"

The softest row. Parcels per dotcom order from the ship system before and after May, times the carrier's per-parcel cost. If the point didn't move, I drop the row and the total is about $1.1M.

Where this lands at Amazon. An FC launch/retrofit with automation on the critical path: publish the ramp gate and the throttle before the robots arrive, stage go-lives so each carries one defect class, hold peak sacred, and write the rollback as a number. Zero-then-publish is the same discipline as never letting one unit be promisable from two nodes at once. Asset utilization (+15% units per labor-hour; one pool), reliability (zero incidents, rollback never fired).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S2

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you held a high standard when it was costly.	Insist on the Highest Standards	The 98% gate fired at 94; a week of robot volume traded for a root-caused ramp; full volume week three.	Do not say "the robots failed" — the gate worked as designed.
Tell me about a time you had to slow something down.	Have Backbone / Bias for Action	The throttle to ~30% the week the gate failed — a decision the same week, owned by DC Ops' own rule.	Do not frame it as caution; it was a pre-built dial firing.
Tell me about launching something with zero room for error.	Deliver Results / Customer Obsession	Zero customer-facing incidents through both go-lives as the constraint; two go-lives so each carried one defect class; the six invariants.	Skip the Kafka/ESB topology unless asked.
How did you protect the floor / the people during the change?	Earth's Best Employer	The throttle protected the floor's throughput; DC Ops owned the gate and the dial; the interim's reporting burden named as the price.	Do not invent a headcount or an attrition figure; none exists in canon.
How do you plan a cutover?	Functional — launch readiness	Runbook with owner/duration/predecessor per task; one irreversible instant identified; rollback triggers as numbers; three mocks against a tolerance; hypercare exit as a number.	Never "400 devices failed" (that is S18's RF miss — say hundreds of RF devices).
What's the ROI of the consolidation?	Frugality	+15% signed; ≈$1.6M/yr modeled with the four rows and their inputs; ≈$2.2M one-time pooling, labeled textbook.	Never present the $1.6M as a signed number.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

INSIST ON THE HIGHEST STANDARDS · AIQB Q4 **Q1. "Tell me about a time when you had to make a decision between standards and delivery."**

I published the standard before there was any pressure to waive it. The robots were arriving at the Utah DC, and I wrote the gate — 98% pick-accurate completed over attempted — into the council pack, with a throttle dial and the weekly read, and DC Ops owned the gate and the dial before the ramp. A gate published before the pressure exists is most of the holding. Week two read 94. The floor owner's own rule fired and I throttled the bots to about 30% of volume for a week — a week of robot volume traded for a root-caused ramp. The zone and slotting logic was the cause; it was fixed, and week three ran at full volume. Processing efficiency ended at +15%, signed by Sephora and measured by DC Ops after stabilization. A gate that cannot stop the program is decoration; this one did.

Number to land: 98 published · 94 read · ~30% throttle · full volume week 3 · **Pivot to NACF:** At NACF I would publish the ramp gate for a new FC — the door ceiling, the UPH floor — before the first trailer, with the dial owned by the building. · **Do NOT say:** that the robots failed; that I fixed the slotting logic.

DELIVER RESULTS · AIQB Q3 **Q2. "Tell me about a time when you not only met a goal but considerably exceeded expectations."**

The constraint was zero customer-facing incidents, so I designed the program to carry one defect class at a time. About a dozen impacted systems — with a BlueYonder upgrade and Locus robotics on the same critical path and an October-to-December peak freeze. I split the work into two go-lives: January was the planning go-live — purchasing and inbound, with the reserve rules live against demand — and the floor untouched; May moved warehouse execution. Each go-live was a counted decision: Blocker plus Critical at zero, mock three inside the 0.5% tolerance, a 343-task runbook rehearsed, rollback armed. Both landed on their dates with zero customer-facing incidents, peak untouched, hypercare exited in week three of four, and +15% processing efficiency signed by Sephora. The modeled value is about $1.6 million a year; the +15% and the zero are the record.

Number to land: 2 go-lives · zero incidents · +15% signed · **Pivot to NACF:** At NACF a new FC launch has the same shape: stage it so each step carries one risk class, and decide each step on counted rows. · **Do NOT say:** that the $1.6M is signed; that I ran the floor.

EARN TRUST · AIQB Q1 **Q3. "Tell me about a time when you had to communicate a change in direction that you anticipated people would have concerns with."**

Digital worried that a virtual Retail DC would blur their forecasting lens, and they were right to worry. I did not argue them out of it; I made the concern the design: one physical location with three planning identities, so each channel keeps its lens without a second truth. Across sixty-plus stakeholders I owned none of the systems, so I wrote the success criteria with the sponsor — one omni facility, self-transfers eliminated, zero customer-facing incidents, peak untouched — and steering, meaning DC Ops, IT and Supply Chain Strategy, signed them. DC Ops owned the robotics gate and the dial before the ramp, so when it fired the floor's own rule fired. The build tracks were slotted by business criticality times effort under one issues log. Trust came from putting the concern into the design and the stop condition into the hands of the people who carry the consequence.

Number to land: 60+ stakeholders · 3 planning identities · gate and dial owned by DC Ops · **Pivot to NACF:** At NACF, placement and inbound changes cross many orgs I would not own; I would convert objections into design rows the same way. · **Do NOT say:** that Digital agreed with everything; I know their concern, not their private view.

HAVE BACKBONE; DISAGREE AND COMMIT · AIQB Q2 **Q4. "Describe a time when you took an unpopular stance in a meeting with peers and your leader."**

The prevailing instinct was one go-live: one cutover, one date, done. I challenged it. One go-live would have put a reserve-rule defect and a floor-execution defect in the same room the same week, and with sixty-plus stakeholders and zero tolerance for customer-facing incidents that is a diagnosis problem, not a schedule problem. So I proposed two go-lives so we never hunted two defect classes in one week — January for the reserve rules against live demand, May for execution — and I also held the comingle-reserve improvements out of the release, priced as a later enhancement at steering, to protect the zero. Both landed on their dates. The cost was a second cutover and the discipline of saying no to a good idea at the wrong time; the benefit was single-cause diagnosis.

Number to land: 2 go-lives, 1 defect class each; improvements held out · **Pivot to NACF:** At NACF I would stage a launch the same way and hold good ideas for the next release. · **Do NOT say:** that two go-lives were free; the second cutover has a price.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"Plus fifteen percent means 31,300 fewer hours for the same volume — 13.0% — and about 87% of the modeled benefit does not depend on pooling."

Headline metric: 240,000 → 208,700 h; ≈$0.88M labor

Index 100 → 115 is the record; hours, dollars and the pooling carry are my model on S2's illustrative DC. U2 bounds pooling at $0.74M–$2.2M; U4 prices the throttle at ≈$11.8K vs ≈$50.4K.

CONFLICT / INFLUENCE ANGLE · HAVE BACKBONE; DISAGREE AND COMMIT

"Everyone's instinct was one go-live; I argued for two."

Headline metric: 2 go-lives · 1 defect class each

One go-live would have mixed a reserve defect and a floor defect in the same week; two cost a second cutover. Held the comingle-reserve improvements out of the release; priced at steering. Digital's lens was kept by the three-identity design.

FAILURE / LEARNING ANGLE · OWNERSHIP

"For four months every report had to declare which system of record it read — I staffed that in month three, not day one."

Headline metric: interim reporting staffed month 3 of ~4 months of cost

The record's own do-differently: the interim cost was predictable. Second: the Locus RTL slipped three weeks — escalated the week it surfaced, but it should have been on the register as a risk, not a task.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

How do you know the +15% came from the consolidation and not from the robots or the WMS upgrade?

I don't split it, and Sephora's number doesn't either: +15% is whole-program processing efficiency (units per labor-hour), measured by DC Ops post-stabilization. The consolidation, the WMS upgrade and the robotics were on one critical path by design. Attribution among them would need a before/after on lanes that did not get robots.

Say: *"The +15% is the program's, signed and measured by DC Ops; I would not attribute it to one lever, and the number that would let me is a lane-level before/after."*

What was the counterfactual?

Two warehouses transferring stock to themselves (≈260,000 units a year on paper) and two safety-stock pools — the cost of not consolidating. The counterfactual dollars are modeled; the record is the efficiency and the zero incidents.

Say: *"The counterfactual is modeled; what I can prove is +15% measured by DC Ops and zero customer-facing incidents."*

What did you give up?

The comingle-reserve improvements (priced as a later enhancement), a week of robot volume (the throttle), and for four months a clean reporting picture (the interim). Each was a deliberate trade for zero incidents.

Say: *"Three things, on purpose: the reserve improvements, a week of robot volume, and four months of cleaner reporting — in exchange for zero customer-facing incidents."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. At Sephora's Utah DC I merged two systemic warehouses into one comingle pool — three planning identities over one location — through two go-lives and a robotics ramp behind a published 98% gate that fired at 94: +15% processing efficiency signed by Sephora, zero customer-facing incidents, rollback armed twice and never fired.

NACF bridge. NACF capacity chain: one node, one published ramp gate and an owned dial. A new FC's inbound ramp is CAP-A doors held at an 85% ceiling, CAP-B UPH with the learning-curve tax (+13.6% hours), CAP-C cube read at 49% effective — and the same rule that the people who own the building own the stop condition.

S18 — Sephora — legacy WMS → JDA WMS dual-DC migration: a 0.5% reconciliation tolerance published before mock one, three mocks read by cause class (7% → 1.8% → 0.3%), and the RF-device miss owned ELEVATED

LPs this story can serve here: Insist on the Highest Standards (backup — the published tolerance) · Bias for Action (the date held while training moved) · Deliver Results (backup — 8 months, both DCs, gate cleared) · Learn and Be Curious (first WMS migration; learned the floor in month one) · Ownership (the RF miss with my name on the cause)
Process family: Make / Deliver — warehouse execution system cutover across two DCs; inventory-accuracy governance; freeze, failback and hypercare discipline

Honesty label. Full lifecycle owner, accountable to sponsor and Finance; I had never run a WMS migration and learned the floor in month one. Month markers are illustrative unless stated; the RF miss is mine — I scored the task by effort when impact was what mattered. Loop-packet conflict (§1.3): that document says ~$800K budget and ~$2.5M/week; this packet's record is ~$1.8M, ~3% under, and ~$2.5M/month as my division of the $20M (modeled) — say these.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Ownership. The RF miss is mine — I scored the task by effort when impact was what mattered — and I told leadership straight, with my name on the cause.

Bias for Action. When the device test showed the conversion script failing on refreshed RF hardware, I moved training by a week and a half and held the go-live date — and put device readiness into training's entrance criteria.

Learn and Be Curious. I had never run a WMS migration; I wrote the charter in week one and walked receive-to-ship on the floor in month one until I could do it unaided.

Insist on the Highest Standards. I published the 0.5% reconciliation tolerance with Finance before mock one; mock one at 7% was the price and mock three at 0.3% was worth it.

SOURCE CONFLICT — the RF-device miss (read before you say it). Master canon (S18): ~100 of ~400 RF devices failed the conversion script *at cutover*; a rehearsed paper-pick fallback fired; ≈$45K fallback against ≈$875K of slip. Apple v11, chapter G2 (from your 2022 failure notes): the upgrade script failed on refreshed RF hardware in a device test scheduled *days before training*; manual conversion could not finish under COVID night shifts; training slipped 1.5 weeks; the notes record ~400 devices in one telling and ~100 in another — "say hundreds in the room." Precedence applied (source docs, then Apple v11, then master): every line I added to S18 says "hundreds", "the pre-training device test", "training +1.5 weeks, the date held", and none says the paper-pick $45K or the ≈19× ratio. The S18 body above has been brought to the Apple G2 wording for the RF-device miss; the master's paper-pick and ~19× lines are not carried forward. Common ground in every version: the miss was mine, scored by effort not impact, training slipped 1.5 weeks, the date held, ~3% under budget. Pick the telling you lived and use it identically in all five interviews. Listed as a source conflict on the Verify-before-you-say-it page.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Two Sephora DCs ran on a legacy WMS with no documentation, an eight-month hard deadline, a ~$1.8M budget, and a $20M ROI pipeline of four programs gated behind go-live — about $2.5M a month of slip on average (my division of the $20M, not banked). Inventory accuracy across two DCs could not wobble at cutover, and I had never run a WMS migration. **Task.** Own the full lifecycle — charter, design sign-off, the ERP ↔ WMS ↔ 3PL ↔ carrier interface inventory, UAT, cutover, hypercare — accountable to the sponsor and Finance. **Action.** I wrote the success criteria in week one — on the date, inside a published reconciliation tolerance, zero accuracy incidents, a numeric hypercare exit — and learned the floor in month one until I could walk receive-to-ship unaided. I built the interface inventory from message logs with an owner and a test cell per contract, set the mapping-table standard as data not code, and published the 0.5% tolerance with Finance before mock one so the number belonged to us before the test. Three mocks were read by cause class, not headline: 7% (units of measure, location names, status vocabularies) → 1.8% (in-transit timing) → 0.3%. Cutover ran a 72-hour in-transit freeze with the legacy system read-only for 30 days. When a device test before training showed hundreds of RF devices failing the conversion script, inventory was inside tolerance and the failure was device conversion, not data — so I held the date, let the training criterion slip 1.5 weeks, and told leadership the same day. **Result.** Primary metric: inventory reconciliation variance at cutover — 7% → 1.8% → 0.3% against the 0.5% tolerance. On time, about 3% under the ~$1.8M budget (~$54K), the $20M pipeline unblocked the next month — slip avoided, not banked. Hypercare exited week three at under two blockers a day. The RF miss is mine.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Two DCs on a legacy WMS; zero inherited documentation; a hard eight-month deadline with a $20M ROI pipeline of four programs gated behind go-live (~$2.5M/month of slip — my division, modeled); budget ~$1.8M; I had never run a WMS migration. Inventory accuracy across two DCs could not wobble at cutover.

The register at design sign-off: undocumented interface surprises, conversion mismatch above tolerance, and 3PL/carrier freeze non-compliance — each with a trigger, a data-first mitigation and an owner. The one I missed: RF device conversion sat on the log as a task, not a risk.

T — Own the full lifecycle — discovery charter, design sign-off, the ERP ↔ WMS ↔ 3PL ↔ carrier interface inventory, UAT, cutover, hypercare — accountable to the sponsor and Finance. Team: 14 core + 6 DC Ops.

Baselines on the table: conversion mismatch — gated at the 0.5% tolerance signed with Finance before mock 1 (mock 1 opened at 7%) · accuracy incidents allowed: zero · the date — eight months, hard · budget ~$1.8M · hypercare exit < 2 blockers/day.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (month 1)** — sponsor charter in week one (objectives, assumptions, success criteria, phase entry/exit criteria); interface inventory v0 from message logs; systems-impacted table with LOE per owner; the floor walked until I could follow receive → put-away → pick → ship unaided. *Exit bar:* charter signed by sponsor and Finance; every interface has a named owner.
- **Plan & Design (months 2–3)** — design sign-off pack; interface inventory v1 with a test cell per contract; conversion approach — masters → open orders → on-hand → in-transit as its own batch; mapping-table standard (data, not code); the 0.5% tolerance published with Finance before mock 1. *Exit bar:* council sign-off; tolerance signed; stories slotted by business criticality × effort.
- **Build & Test (months 4–7)** — QA demo schedule to DC Ops; UAT entrance memo (function testing done, conversion build done, test data handed off, pre-prod + perf ready) — QA could fail a build; three mock cutovers with written root causes by class: 7% → 1.8% → 0.3%; after the RF miss, the training plan and device-readiness table. *Exit bar:* mock 3 ≤ 0.5% → written go-live recommendation; training sign-off (slipped 1.5 weeks, date held).
- **Cutover & Go-Live (month 8)** — runbook; 72-hour transfer freeze (≥ the longest lane transit); conversion batches; post-load reconciliation; legacy read-only switch; the go/no-go call. the RF-device conversion — hundreds of devices caught in the pre-training device test — had already moved training 1.5 weeks while the date held; leadership was told the same day with the slip priced. *Exit bar:* post-load mismatch inside tolerance; ship confirms posting; zero accuracy incidents.
- **Hypercare / Close (4 weeks planned)** — blockers/day read daily; 30-day failback watch (legacy read-only); budget closed; exit memo with the single-point-dependency lesson. *Exit bar:* < 2 blockers/day sustained → exit week 3; ~3% under budget; pipeline programs started the next month.

Influence and ownership. Mine: the charter, the entrance/exit criteria, the interface inventory, the conversion approach, the mapping-table standard, the tolerance, the go/no-go recommendation and the fallback call; the team built, converted, tested and ran the floor; Finance co-owned the tolerance. Because the success criteria were written in week one and signed by sponsor and Finance, nobody negotiated the exit later. The RF miss is mine — I scored the task by effort when impact was what mattered — and it went upstairs the same afternoon with my name on the cause.

> **Designed the process; others own it now.** I designed the charter, the tolerance, the entrance criteria and the mapping-table standard; the team and Finance now run cutovers against them, and the standard outlived the program. **Trade-off put to leadership.** The recommendation to the sponsor outlined the competing metrics — the eight-month date and the $20M pipeline behind it vs training readiness and conversion variance — and held the date while training slipped 1.5 weeks and the 0.5% tolerance did not move.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~3% under the ~$1.8M budget (~$54K, record); the $20M ROI pipeline unblocked the next month — ~$2.5M/month of slip (my division, modeled; not banked); ≈$875K *modeled* price of a 1.5-week go-live slip (my division of the $20M; not banked); ≈$0.1M *modeled* less inventory in dispute at 0.3% vs 0.5% on an illustrative $50M on-hand.
- **#** — hundreds of RF devices failed the conversion script in the pre-training device test (a three-to-four-day estimate that proved wrong); team 14 core + 6 DC Ops across two DCs (record).
- **%** — mock variance 7% → 1.8% → 0.3% against the 0.5% tolerance published with Finance before mock 1 (record).
- **Time** — eight-month deadline held; 72-hour in-transit freeze; legacy read-only for 30 days; training slipped 1.5 weeks while the date did not move; hypercare exited week 3 of 4 (record).
- **Customer** — zero accuracy incidents at cutover; ship confirms posting; post-load mismatch inside tolerance (record exit bars).
- **Quality** — mocks read by cause class, each removing a named defect category; failback rehearsed, never used; a what-if plan now required for every single-point dependency (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	An eight-month deadline across two DCs with no charter. I wrote the sponsor charter in week one — objectives, assumptions, success criteria, phase entry/exit criteria — and set the tolerance with Finance before anyone rehearsed.
Scope & influence	14 core + 6 DC Ops across two DCs; Finance co-owned the tolerance; QA could fail a build on my UAT entrance memo; the go/no-go recommendation and the fallback call were mine to the sponsor.
Execution trade-off + mechanism others now own	Date vs readiness: training slipped 1.5 weeks while the date held; a 72-hour transfer freeze (sized to the longest lane) bought conversion integrity. Mechanisms others own now: the 0.5% tolerance, the UAT entrance memo, and the mapping-table standard (data, not code).
Impact (long-term)	Eight-month deadline held; ~3% under the ~$1.8M budget; the $20M ROI pipeline unblocked the next month — ~$2.5M/month of slip (my division, modeled; not banked); zero accuracy incidents at cutover.
Process improvement (standard work institutionalized)	Three mock cutovers read by cause class (variance 7% → 1.8% → 0.3% against 0.5%); a 30-day failback watch with legacy read-only; a numeric hypercare exit (< 2 blockers/day sustained).
Complexity & consensus	hundreds of RF devices failed the conversion script in the pre-training device test — leadership was told the same day, training slipped 1.5 weeks, the date did not move.
Root cause at scale	The single-point-dependency lesson went into the exit memo; each mock removed a named defect category rather than fixing individual records, so the fix scaled to the whole conversion.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: SKU × location × unit of measure; units in flight by lane. Data: legacy WMS and JDA conversion output, message logs for the interface inventory, three mock result sets. Cuts: variance by cause class (UoM, location names, status vocabularies, in-transit timing). Revealed: one mapping class — UoM, factor 12 on ~1% of SKUs — dominated the 7%.

2 · Solution design. Options: (A) big-bang cutover with tolerance TBD; (B) tolerance published before mock one, mocks read by cause class, freeze and failback (chosen); (C) parallel run. Chose B — the tolerance belongs to Finance and me before the test, not after.

3 · Math & data logic. $M = \Sigma|\Delta| / \Sigma$ ERP on-hand; 7.0% → 1.8% → 0.3% ≤ 0.5%; freeze F = 72 h ≥ max lane transit; slip price $2.5M / 30 = $83,333/day; 10.5 days → $875K (price of a 1.5-week go-live slip).

4 · Implementation. Five gated phases over eight months: charter week one; interface inventory with a test cell per contract; three mocks; 72-hour freeze; legacy read-only 30 days; device-readiness in training entrance criteria; hypercare exit week 3 at < 2 blockers/day.

5 · Probe questions. The six answered above in the Probing block, and the three trap drills below.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — Why publish a tolerance before you knew you could hit it?

So the number belonged to Finance and me before the test, not after. Mock one's 7% was the price; mock three's 0.3% was worth it — and the tolerance was never renegotiated.

Infra — How do you read a mock?

Never by the headline alone. Every delta got a root-cause class — units of measure, location drift, status vocabularies, then in-transit timing. If the total shrank without a named cause, the mock was not evidence. 7 → 1.8 → 0.3 each removed a category, and the fixes were mapping tables as data, not code.

Infra — Hundreds of RF devices failed the device test before training. Why not move the date?

Because inventory was inside tolerance — the failure was device conversion, not data — and the training criterion could move without moving the go-live. I held the date, let training slip 1.5 weeks, and took the miss upstairs the same afternoon with my name on the cause.

Infra — What was the freeze and why 72 hours?

A 72-hour in-transit freeze, set at or above the longest lane transit, so in-transit could be converted as its own batch with Σ legacy = Σ JDA — the class that took mock two from 1.8% to 0.3%. Legacy stayed read-only for 30 days as the failback.

Infra — The $875K avoided slip — are you banking it?

No. It is modeled arithmetic — 10.5 days at about $83K a day, from my division of the $20M over eight months — on a slip that did not happen; it is the price that made holding the date the obvious call. What I would measure is the pipeline's first program start against plan, which was the next month.

Infra — Lesson, and what rule did you leave behind?

Any single-point dependency in front of an exit criterion gets a what-if plan regardless of estimated effort. Differently: dual-path device conversion on day one, and a device model × OS × script pass/fail table before training week.

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Mock 1 variance	`M = Σ|Δ| / Σ ERP on-hand = 140,000 / 2,000,000 = 7.0%`	illustrative units on record percentages	7.0% — 14× the tolerance
2	Dominant cause — UoM	`a case booked as 12 eaches creates 12 − 1 = 11 units of error per case; factor 12 on 1% of SKUs`	record mechanism	one mapping class dominates the 7%
3	Second cause — in-transit timing	`transfers in flight at snapshot counted at neither end or both`	record mechanism	the timing class
4	The stakes	`$20M pipeline gated; about $2.5M per month of slip`	$20M record; $2.5M/month = my division, modeled	every day priced

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Tolerance signed first	`gate: M ≤ 0.5%, signed with Finance BEFORE mock 1`	never renegotiated (record)	the yardstick fixed in advance
2	UoM fix as data	`mock 2: M = 36,000 / 2,000,000 = 1.8%`	mapping tables (UoM, location, status) as data, not code (record % on illustrative units)	7.0% → 1.8%
3	Freeze + own batch	`freeze F = 72 h ≥ max lane transit; in-transit converted as its own batch with Σ legacy = Σ JDA; mock 3: M = 6,000 / 2,000,000 = 0.3%`	record % on illustrative units	1.8% → 0.3%
4	Gate evaluated	`0.3% ≤ 0.5% → go`	as a written recommendation (record)	go-live recommended
5	RF-device miss	`script failed on refreshed RF hardware (hundreds of devices); training sign-off +1.5 wk; go-live date holds`	record (Apple G2; master S18) — count, timing and fallback CONFLICT, see box	go-live on the date
6	Hypercare exit	`blockers/day < 2 sustained — the charter's own number`	record	exited week 3 of 4

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Cutover variance (three mocks)	7.0%	1.8% → 0.3% vs 0.5% tolerance	record
2	Go-live date	at risk from the RF miss	held — 8 months end to end	record
3	Budget	~$1.8M	~3% under	record
4	$20M pipeline	gated	starts on time	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Budget result	`$1,800,000 × 3% = $54,000`	record	≈$54K under budget
2	Slip price	`$2,500,000 / 30 = $83,333/day`	$20M pipeline gated (record) ÷ 8 months = $2.5M/month — my division, modeled; not banked	≈$83K/day
3	Critical-path slip avoided	`1.5 wk × 7 = 10.5 days; 10.5 × $83,333 = $875,000`	price of a 1.5-week go-live slip at $83,333/day (modeled — my division of the $20M; master S18); not banked	≈$875K — avoided, not banked
4	Fallback cost paid	`≈$45K paper-pick; $875,000 / $45,000 ≈ 19`	master-packet model only (paper-pick on a subset of devices); NOT in Apple G2 — do not say aloud	≈$45K (master only)
5	Dispute delta	`(0.5% − 0.3%) = 0.2%; × $50,000,000 = $100,000`	illustrative on-hand (modeled)	≈$0.1M one-time less in dispute

SOLUTION FLOW — THE SYSTEM & DATA MAP NEW

```
[Legacy WMS (RP)]  --extract-->  [Mapping tables as DATA: UoM | location | status]
      |                                  |
      | 72-h in-transit freeze (>= longest lane transit)
      v                                  v
[conversion batches: masters -> open orders -> on-hand -> IN-TRANSIT as its own batch]
      |                                  |
      v                                  v
[JDA WMS, two DCs] <-- interface inventory: ERP <-> WMS <-> 3PL <-> carrier (owner + test cell per contract)
      |
post-load reconciliation: M = sum|delta| / sum on-hand  vs tolerance 0.5% (signed with Finance before mock 1)
      |
go/no-go -> legacy READ-ONLY 30 days (failback) -> RF devices: conversion script failed on refreshed hardware
      |                                                  -> training sign-off +1.5 wk; go-live date held (see conflict box)
hypercare: blockers/day < 2 sustained -> exit week 3
```

SENSITIVITY — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
Average unit cost (illustrative)	$15	$25	$40
Paper-pick slowdown (master-packet model; not Apple G2)	15%	25%	40%
DC labor rate (benchmark)	$25/h	$30/h	$35/h
Inventory dispute avoided (0.5% → 0.3%)	~$60K	~$100K	~$160K
Paper-pick fallback cost (master only)	~$23K	~$45K	~$84K
Total modeled	~$40K net	~$55K net	~$75K net

SPOKEN STATEMENT NEW

> “The record numbers first: on time, about $54K under a $1.8M budget, and the $20M pipeline started the next month, where every month of slip was worth about $2.5M on average — my division of the twenty million over eight months, not banked. A slip of the 1.5 weeks the training criterion moved would, had it moved the go-live, have cost about $875K — so I held the date and moved training, and said so straight. The modeled part is smaller and I’ll label it: landing at 0.3 percent against a 0.5 tolerance meant about $100K less inventory in dispute at cutover on an illustrative $50M on-hand, and at mock one’s 7 percent that would have been $3.5M. The date, the budget, the variance and the pipeline are measured; the on-hand value is my model.”

IF CHALLENGED NEW

“Where does $50M of on-hand come from?”

It’s illustrative, two million units at $25, chosen so the percentages become dollars; the record gives me 0.3 percent, not the balance sheet. I’d firm it up with the actual ERP on-hand valuation at the freeze snapshot, which Finance already had.

“The $875K avoided slip is a counterfactual.”

Yes — record arithmetic on a slip that did not happen, so I never bank it; it is the price that made the fallback the obvious call. What I’d measure is the pipeline’s first program start against plan, which was the next month.

“What did the RF miss actually cost?”

The record is the 1.5 weeks of training sign-off — SME windows were scarce — and a go-live date that held. I have not priced a fallback and I will not invent a number; the master packet carries a modeled paper-pick figure that my 2022 notes do not, so I do not say it.

Where this lands at Amazon. Cutover discipline in its most literal form — a tolerance published with Finance before the first rehearsal, mocks read by cause class, a freeze sized to the longest lane, a tested failback, a numeric hypercare exit nobody can negotiate, and a rehearsed contingency that fired on the day. Reliability (0.3% against 0.5%, zero accuracy incidents), responsiveness (the decision the same day), cost efficiency (~3% under), the $20M pipeline on the date.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S18

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time something went wrong at go-live.	Bias for Action / Ownership	The RF miss: hundreds of devices in the pre-training device test; leadership told the same day; training slipped 1.5 weeks, the date did not.	Never "400 devices failed" or a ~100/~400 ratio; do not blame the vendor script — the scoring miss was mine.

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you set a standard others thought was too strict.	Insist on the Highest Standards	0.5% published with Finance before mock one; mock one at 7% was the price; never renegotiated.	S2's 98% gate is the primary for this LP — use it first if both land here, S18 second.
Tell me about learning something new fast.	Learn and Be Curious	Never run a WMS migration; charter in week one; the floor walked unaided by end of month one; the mocks were where I proved it.	Do not overclaim WMS configuration — the team converted and built.
Tell me about a time you delivered on a hard date.	Deliver Results	Eight months, both DCs, 0.3% vs 0.5%, ~3% under, $20M pipeline unblocked the next month.	Never bank the $875K or the $2.5M/month — slip avoided.
How do you design a freeze / failback for a systems cutover?	Functional — migration design	Freeze ≥ longest lane transit; in-transit as its own batch; legacy read-only 30 days; rollback triggers as numbers (mismatch > 0.5% for 2 days; ship-confirm posting failing > 2 h).	Keep budget at ~$1.8M and slip at ~$2.5M/month (loop-packet conflict).
What did you miss and what did you change?	Ownership	RF conversion on the log as a task, not a risk; the rule left behind: any single-point dependency in front of an exit criterion gets a what-if plan.	Do not soften it — "the miss is mine."

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

OWNERSHIP · AIQB Q2 **Q1. "Describe a time when you didn't think you were going to meet a commitment you promised."**

The miss was mine, and I said so straight. The devices team called the RF conversion three to four days and straightforward, and I planned it that way — low effort, low attention. Just to be sure, I scheduled a device test with engineering days before training, and the upgrade script failed: the refreshed RF hardware did not run the old code. Hundreds of devices, and with key people on COVID night shifts the manual conversion could not finish in time, so I told leadership straight: training sign-off, a go-live exit criterion, slips a week and a half. The root cause was my risk model — I scored the task by effort, not impact, so a single-point dependency in front of an exit criterion had no what-if plan. The date held and the migration landed on time, about 3% under budget. The rule I left behind: every cross-functional dependency gets a what-if and a mitigation whatever the effort estimate, and device readiness moved into training's entrance criteria — proven devices, not promised ones.

Number to land: training +1.5 weeks · date held · ~3% under budget · **Pivot to NACF:** At NACF a single-point dependency in front of a launch — one carrier, one door, one scanner model — gets a what-if plan regardless of size. · **Do NOT say:** an exact device count — say 'hundreds' (the 2022 notes record ~400 in one telling and ~100 in another); that the vendor script was the cause.

BIAS FOR ACTION · AIQB Q4 **Q2. "Tell me about a time when you had to gather information and respond immediately to a situation."**

I ran a check I did not have to run, and it failed. Training sign-off depended on hundreds of RF guns, printers and workstations converting, and the devices team said a few days. Just to be sure, I scheduled a device test with engineering days before training. The script failed on the refreshed hardware. The decision that mattered was fast because the options fit on one page: manual conversion on COVID night shifts could not finish before training, so something had to move, and the choice was which date. Training moved a week and a half; the go-live did not — and I told leadership that straight, with my name on the cause. Then I changed the system: device readiness went into training's entrance criteria, and the habit generalized into dress rehearsals for everything on the critical path — on UDC it became the RF-device proof-of-concept before cutover week.

Number to land: training +1.5 weeks; go-live date held · **Pivot to NACF:** At NACF a rehearsal a week before the first live trailer is how a launch absorbs a surprise on the day — I would rehearse the dependency, not just the plan. · **Do NOT say:** that I improvised; or a fallback or dollar figure the 2022 notes do not carry.

LEARN AND BE CURIOUS · AIQB Q2 **Q3. "Describe a time when you took on work outside of your comfort area. How did you identify what you needed to learn to be successful?"**

I had never run a WMS migration, and the deadline was eight months. I wrote the charter in week one — success criteria, assumptions, entry and exit criteria — so the destination existed before my knowledge did. Then I spent month one on the floor, walking receive, put-away, pick and ship until I could follow a unit through the building unaided, and reading the message logs to build the interface inventory with an owner and a test cell per contract. The mocks were where I proved the learning: each delta got a root-cause class — units of measure, location names, status vocabularies, in-transit timing — so I was learning by removing categories, not by reading documentation. By the third mock the variance was 0.3% against a 0.5% tolerance. I did not claim to configure the system; the team converted and built.

Number to land: month 1 on the floor; 7% → 1.8% → 0.3% · **Pivot to NACF:** At NACF I would walk the dock-to-stock path in my first weeks before proposing a change to it. · **Do NOT say:** that I configured JDA or converted the data myself.

INSIST ON THE HIGHEST STANDARDS · AIQB Q1 **Q4. "Describe a time when you refused to compromise your standards around quality/customer service, etc."**

I published the reconciliation tolerance — 0.5% — with Finance before the first mock. So the number belonged to Finance and to me before the test, not after. Mock one came in at 7% — fourteen times the bar — and that was the price. I did not renegotiate; I read the mock by cause class: units of measure were the dominant class — a case booked as twelve eaches is eleven units of error — then location names and status vocabularies. The fix was mapping tables as data, not code. Mock two: 1.8%. Mock three: a 72-hour in-transit freeze and in-transit converted as its own batch, 0.3%. The written go recommendation came off mock three, inside the tolerance. The tolerance was never renegotiated, and the exit bar for hypercare — under two blockers a day — was in the charter in week one, so nobody negotiated that either.

Number to land: 0.5% published first; 7.0% → 1.8% → 0.3% · **Pivot to NACF:** At NACF I would publish the reconciliation tolerance — dock-received vs system-available — before the first mock receipt. · **Do NOT say:** that 7% was a surprise to Finance; they signed the bar first.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"Seven percent, one point eight, point three — each mock removed a named category."

Headline metric: M = 140,000 / 2,000,000 = 7.0% → 0.3%

Mocks read by cause class: UoM (12 → 11 units of error per case) first, then in-transit timing as its own batch. If the total shrank without a named cause the mock was not evidence. Units are illustrative; the percentages are the record.

CONFLICT / INFLUENCE ANGLE · INSIST ON THE HIGHEST STANDARDS

"I published a tolerance with Finance before I knew I could hit it."

Headline metric: 0.5% never renegotiated

Finance co-owned the number; the sponsor signed the charter in week one; leadership was told the same day the training criterion slipped 1.5 weeks — the date held, the tolerance did not move.

FAILURE / LEARNING ANGLE · OWNERSHIP

"A three-day device task broke the training gate, and I had scored it by effort."

Headline metric: training +1.5 weeks; date held; ~3% under

Impact, not effort, is the score; a single-point dependency in front of an exit criterion gets a what-if plan. The test I ran just to be sure saved the go-live, not the training gate. Differently: a device model × OS × script pass/fail table at refresh. The rule shows in the mocks: 7.0% → 1.8% → 0.3%.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

How do you know the three mocks measured the same thing?

Same formula, same denominator (Σ ERP on-hand), same scope; the changes between mocks were named cause-class fixes. Units are illustrative; the percentages are the record.

Say: *"Same formula and denominator each time, and every change between mocks was a named cause class — if the total shrank without a named cause I did not count it."*

"A week and a half on training — that is a small failure. Give me a bigger one."

Fair. The bigger one is the Elementum UAT that crashed (Apple G1) — a lost sign-off. I keep this one because its lesson transfers best: the failure class that hits mature programs is not the big risk everyone watches, it is the three-day task nobody did a what-if on. The size of the slip is not the point; the blind spot is.

Say: *"A week and a half is small; the blind spot is not — impact decides the attention a task gets, never its effort estimate."*

What is the number that would have changed your mind on go?

Mock three above 0.5%, a failed freeze or a failback I could not describe per interface. The RF failure was a different class: it moved training, not the go.

Say: *"Mock three over half a percent — that single number was the go/no-go."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. At Sephora I took two DCs from a legacy WMS to JDA in eight months, ~3% under a ~$1.8M budget, by publishing a 0.5% reconciliation tolerance with Finance before mock one and reading three mocks by cause — 7%, 1.8%, 0.3% — and when the RF device conversion failed its pre-training test I told leadership straight — training slips a week and a half, the date holds — and made what-if plans a rule.

NACF bridge. NACF's inbound ramp is a migration of a different kind — a new FC's dock, UPH and cube all go live together — and the same discipline applies: publish the tolerance before the first mock, size the freeze to the longest lane, rehearse the fallback, and give single-point dependencies a what-if. CAP-A's 5 doors at an 85% ceiling are the slack that lets a day-one surprise be absorbed without moving the date.

U1 — The node × channel × bucket-transition matrix — one regression asset that proved every physical move on the floor lands in exactly one system bucket NEW

LPs this story serves here: Invent and Simplify (primary — one rule, one asset) · Dive Deep (primary — event traces, the 836-row log) · Insist on the Highest Standards (what "tested" means) · Earn Trust (QA demoed to DC Ops in build) · Success and Scale (backup — a new node is a row, not a project)
Process family: Make / Deliver — test and acceptance design for a comingled node; physical-event ↔ system-bucket integrity; regression asset reused through hypercare and the monthly release train

Honesty label. I owned the matrix design and its assertions as program owner of the location/inventory model; BlueYonder, ISM and platform teams built to it, QA ran it and owned the gate, DC Ops confirmed in UAT. The rule, the test families, the entry gate and the issues-log numbers are record. The cell counts (6 node-channel pairs, 14 events, 61 live cells) are **illustrative** VERIFY-WITH-RAJESH — confirm the real matrix size before saying a count aloud. The ISM/S4 bucket formula is the same program artifact as Lan's S1; here it is told from the floor-event side. Never claim configuring S/4, ISM or BlueYonder.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Invent and Simplify. About a dozen impacted systems and about thirty build tracks ran on one rule — every event maps to exactly one bucket — and one entry gate: no story enters QA without a matrix cell reference.

Dive Deep. I read one SKU, one day, event by event: an event landing in zero buckets promises 3 units that are gone (560), in two buckets hides 3 (554); the right answer is 557.

Insist on the Highest Standards. "Tested" meant every live cell asserted before/after with numbers, three dated E2E days with every function in one room, and Blocker + Critical open = 0 at each go/no-go.

Earn Trust. QA demoed each matrix family to DC Ops and planners during build, so UAT was confirmation, not discovery.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. At the Utah DC one physical pool had to serve three node types and two channels, and "available" had to move correctly on every physical event — a store ring, a pick, a pack, a ship confirm, a receipt, a transfer, an adjustment — across about a dozen impacted systems and about thirty build tracks, with zero customer-facing incidents allowed. **Task.** I owned the node × channel × bucket-transition matrix: the design and the regression asset the whole program tested against. **Action.** Discovery: a bucket inventory with exactly one owning system per bucket, and an event inventory — POS flash sale, omni pending, ship confirm, the S/4 Aggregation# posting, and SAP movement types 101, 311, 601, 701 and 702. Plan & Design: one rule — every event maps to exactly one bucket; a reversal must reference its original; a replayed event is a no-op. Build & Test: every cell carries a before/after assertion with numbers, and no story enters QA without a cell reference; three dated E2E days ran the whole flow in one room. Cutover: January and May go/no-go off counted rows — Blocker + Critical open = 0. Hypercare: ISM-vs-MB52 drift at zero SKUs on consecutive days; four in-transit cells and a de-aggregation family added from what the floor taught us. **Result.** Primary metric: customer-facing incidents — zero through both go-lives, with 836 issues logged, 441 of them Blocker or Critical (52.8%), 85.4% closed. The dollars are a what-if, labeled: break-even is about 1.8 escaped cell-classes a year.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — One physical pool, three node types (Store, Retail DC, Dotcom DC), three working definitions of "available," and a physical floor that generates events the books must follow: pick reserve, ecom pick and pack, ship confirm, receipt, transfer, adjustment. Left to the happy-path E2E, an event can land in zero buckets (oversell) or two (undersell) and nobody sees it until a customer does.

T — Own the design and the regression asset: which events exist, which bucket each one moves, what the numbers must be before and after, and the gate that stops an unproven story from entering QA. None of the systems reported to me; ~30 build tracks ran under one issues log; stories were slotted by business criticality × effort.

Baselines on the table: three working definitions of "available" · customer-facing incidents allowed: 0 · Blocker + Critical open = 0 at every go/no-go · four epic modules (reserve module, AFS engine, SNB, control screens) · per-environment workbooks Dev 310 / QA 255 / Perf 200 / Prod 221 rows · issues log 836 rows by program end.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — profiled both books; built the bucket inventory with one proposed owning system per bucket; built the event inventory — POS flash sale, omni pending, ship-to-home confirm, S/4 Aggregation# posting, and MSEG movement types 101/311/601/701/702 as events. *Exit bar:* every bucket owned by exactly one system; every event named; sessions closed 4/30.
- **Plan & Design (5/15 → 8/14)** — wrote the exactly-one-bucket rule, the reversal-references-original rule and the idempotency key into the FSD; kept Merch Reserve and Lost/Found outside S4_Avail so no double-subtract path exists; drafted the test-matrix skeleton. *Exit bar:*

FSD signed 8/14 (record) with every bucket owned by exactly one system (the master's discovery exit bar).

- **Build & Test (Aug → Jan/Apr)** — authored each cell with before/after assertions; slotted epics per module; enforced "no story enters QA without a matrix cell reference"; closed SIT contract defects before UAT; ran the dated E2E days — Inbound 11/22, Dotcom 12/20, Retail 1/30. *Exit bar:* 0 open SIT contract defects; E2E green on three dates; Blocker + Critical open = 0.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — wrote the go/no-go recommendation from counted rows; the publish switch (OMS reads published AFS, never computes); reserve-copy invariants checked per SKU. *Exit bar:* no SKU with two published AFS values; Negative AFS clean at T+1; rollback never fired.
- **Hypercare / Close (Jun → Jul)** — daily ISM-vs-MB52 drift; matrix grew from what the floor taught — four in-transit regression cells and the de-aggregation family (n lines · missing line · duplicate header) per event source. *Exit bar:* drift = 0 SKUs on consecutive days; matrix and control-owner map handed to the monthly release train.

Influence and ownership. Mine: the bucket inventory, the event inventory, the exactly-one-bucket rule, every cell's assertions, the entry gate and the go/no-go recommendation. Built by others: BlueYonder, ISM and platform teams; QA ran the cells and owned the gate; DC Ops and planners confirmed in UAT because QA had already demoed each matrix family to them during build.

> **Designed the process; others own it now.** The matrix, the entry gate and the regression families are standard work now: QA owns the gate, the control owners own the screens, and every later rule change enters as rows. **Trade-off put to leadership:** the date versus the bar — the gate was Blocker + Critical open = 0 and a countable cell list, not a feeling, so the go/no-go was arithmetic.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — no record dollar. The what-if (modeled, **illustrative**): 30,000 oversell-exposed incidents/yr ÷ 61 live cells = 491.8 per cell × $30 = ≈$14.8K/yr per escaped cell-class against ≈$26.8K one-time to author and run the matrix — break-even ≈1.8 escaped cell-classes. Not claimed as a saving.
- **#** — 836 rows in the issues log, 164 Blocker + 277 Critical = 441 (52.8%), 714 closed (85.4%) · 4 in-transit regression cells + a 3-case de-aggregation family per event source · E2E days 11/22 · 12/20 · 1/30 (record).
- **%** — Blocker + Critical open = 0 at each go/no-go; daily drift 0 SKUs on consecutive days at hypercare exit (record).
- **Time** — FSD signed 8/14 · three dated E2E days · January and May go-lives · hypercare exit week 3 of 4 (record).
- **Customer** — zero customer-facing incidents through both go-lives; no SKU with two published AFS values (record).
- **Quality** — every live cell asserted before/after; every event maps to exactly one bucket; a replayed event is a no-op (record design).

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No agreed way to prove physical events and system buckets agreed. I defined the events, the buckets and the rule, and made "tested" countable.
Scope & influence	~30 build tracks in other orgs; QA owned the gate; DC Ops and planners confirmed in UAT; the entry rule moved every track onto one asset without anyone reporting to me.
Execution trade-off + mechanism others own	Coverage versus speed: one cell reference per story, assertions with numbers. Mechanism others own now: QA's gate and the matrix families in the monthly release train.
Impact	Zero customer-facing incidents through both go-lives; 441 Blocker/Critical findings dispositioned before customers saw them.
Process improvement	Rule changes (the in-transit rule, de-aggregation) enter as new cells, not new projects; a new node or channel is a matrix row.
Complexity & consensus	Three node types, two channels, about a dozen impacted systems, three signing orgs — one FSD, one matrix.
Root cause at scale	Defects clustered at seams where two systems used different grain; the fix was a rule at the seam, not 441 individual patches.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: SKU × node × channel × bucket × event. Sources: the bucket inventory (both books), the event inventory (POS, OMS, WMS, S/4 movement types 101/311/601/701/702), the 836-row issues log (severity, owner, function). Cuts: events by bucket cardinality (0 / 1 / 2), defects by severity and by seam. Revealed: the log shows where the book is heavy — 523 of 836 rows (62.6%) sit in the Outbound, Retail and Inbound flows, where an event crosses from one system to the next — but it carries no cell tag, so the matrix's yield against it is not measured.

2 · Solution design. Options: (A) happy-path E2E only; (B) per-system unit tests; (C) an event × node-channel matrix with before/after assertions. Chose C — A and B each pass while the seam between systems is wrong. Why: the oversell bug is a seam bug, and a matrix makes coverage countable and gateable.

3 · Math & data logic. AFS = Basis − PickRes − EcomPick − EcomPack − SNB; Basis = min(S4_Avail, WMS_OnHand); cardinality rule: 0 buckets → +3 error (560), 2 buckets → −3 error (554), exactly 1 → 557; replay no-op; drift = ISM.S4_Avail − MB52 = 0; de-aggregation Σ per-line COGS = posting COGS to the cent.

4 · Implementation. Phases and gates above; owners: me (rule, cells, go/no-go), QA (gate), DC Ops/planners (UAT), control owners (screens). Controls: entry gate, E2E days, Blocker + Critical = 0, drift = 0. Risks: vocabulary drift, feed latency, override abuse (the register's top three).

5 · Probe questions. Below: why a matrix, how you know it mattered, what it missed, what 61 cells cost, and the three trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — why a matrix and not just more E2E days?

E2E proves the paths someone thought to walk; a matrix lists the cells nobody walked. We ran both — three dated E2E days — but the entry gate (no story enters QA without a cell reference) is what made coverage countable. The log tells you what broke; the matrix tells you what you never tried.

Infra — how do you know the matrix is what kept incidents at zero?

I can't isolate it and I won't claim to. The record is: zero customer-facing incidents, 441 Blocker/Critical rows dispositioned, Blocker + Critical open = 0 at each go/no-go. The number that would change my mind is a customer-facing incident traced to an event that had no cell.

Infra — what did the matrix miss?

My own first in-transit rule. The cells asserted the rule I wrote — "never promise in-transit" — and it passed every cell while starving dotcom of fast movers one day from the building. The fix added four cells and a back-test requirement: a matrix asserts the rule; only history tells you the rule is right.

Infra — what is a cell, concretely?

One event at one node-channel pair with a before state, an after state and the numbers: e.g. a store rings 3 units at 10:02 — AFS 560 → 557, SNB 40 → 43, replay at 10:04 → 557, overnight posting → S4_Avail 957, drift 0.

Infra — your 61-cell count — real?

No. Six node-channel pairs, 14 events, 84 candidate cells minus 36 structurally impossible = 48, plus 4 in-transit cells and a 3-case family on 3 event sources = 61 — all illustrative sizing. The record is the rule, the families and the log; I would confirm the actual count before quoting one.

Infra — what would you do differently?

Tag every issues-log row with its matrix cell from day one. I can say every story needed a cell reference; I cannot say what share of the 441 Blocker/Critical rows was found by a cell assertion rather than an E2E room. That is the one number that would prove the matrix's yield.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Event → bucket cardinality	unverified (0 or 2 possible)	exactly 1, asserted per live cell	±3 units per event on the worked SKU	record design
Working definitions of "available"	3	1 signed AFS	—	record
Blocker + Critical open at go/no-go	441 logged (52.8% of 836)	0 open at each go/no-go	—	record
Issues closed	836 rows	714 closed (85.4%)	—	record
Customer-facing incidents	0 allowed	0 through both go-lives	what-if ≈$14.8K/yr per escaped cell-class	record / modeled what-if

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Working definitions	`definitions = 3 (Store · Retail DC · Dotcom DC)`	record	3
2	Definition A over-promise	`channel reads raw S/4: promises 1,000; truly promisable 560; 1,000 − 560 = 440`	S1 worked SKU (illustrative); committed buckets invisible	440 units over-promised
3	Definition B over-promise	`channel reads WMS on-hand: 980 − 560 = 420`	same SKU; same blind spot	420 units over-promised
4	Units already spoken for	`Σ buckets = 200 + 120 + 60 + 40 = 420`	PickRes 200 · EcomPick 120 · EcomPack 60 · SNB 40 (illustrative)	420 units
5	Event lands in 0 buckets	`store rings 3; AFS stays 560; true 560 − 3 = 557; error = 560 − 557 = +3`	illustrative event	+3 oversell exposure
6	Event lands in 2 buckets	`AFS = 560 − 3 − 3 = 554; error = 557 − 554 = −3`	double-subtract path (illustrative)	−3 hidden units

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
7	Issues log	`164 Blocker + 277 Critical = 441; 441 / 836 = 52.8%`	record (issues log)	52.8% high-severity

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cell: supply basis	`Basis = min(1,000, 980) = 980`	asymmetric lower-of-two (record design)	980
2	Cell: AFS	`AFS = 980 − 200 − 120 − 60 − 40 = 560`	AFS = Basis − PickRes − EcomPick − EcomPack − SNB	560
3	Cell: floor above book	`Basis = min(1,000, 1,020) = 1,000; AFS = 1,000 − 420 = 580`	WMS 1,020 → OOB = −20 (illustrative)	580, not 600
4	Cell: store ring	`AFS = 560 − 3 = 557; SNB = 40 + 3 = 43`	one event → one bucket	557 / 43
5	Cell: replay	`replayed key → no-op; AFS = 557 − 0 = 557`	idempotency key (record design)	557
6	Cell: overnight posting + drift	`S4_Avail = 1,000 − 43 = 957; drift = 957 − 957 = 0`	tolerance 0 — a non-zero drift is a Sev	0
7	Cell: de-aggregation	`COGS_i = $300 × q_i / 12 → $75 + $50 + $100 + $25 + $50 = $300`	posting q = 12, C = $300; five lines (illustrative); last line absorbs rounding	$300.00 to the cent
8	Cell: late line	`SNB = 12 − 10 = 2 → aged row on the 24 h report`	one line (q = 2) late (illustrative)	2
9	Cell: in-transit lane	`Promisable = 120 × 1[0.4 ≤ 0.49] × 1[97% ≥ 95%] = 120 × 1 × 1 = 120; failing lane 80 × 0 = 0`	σ ≈ 0.3 d, window ≈ 0.49 d (record promise math); lane values illustrative	120 / 0
10	Sizing of the matrix	`6 pairs × 14 events = 84; 84 − 36 = 48; 48 + 4 + (3 × 3) = 61 live cells`	**illustrative** counts; 4 and the 3-case family are record	61

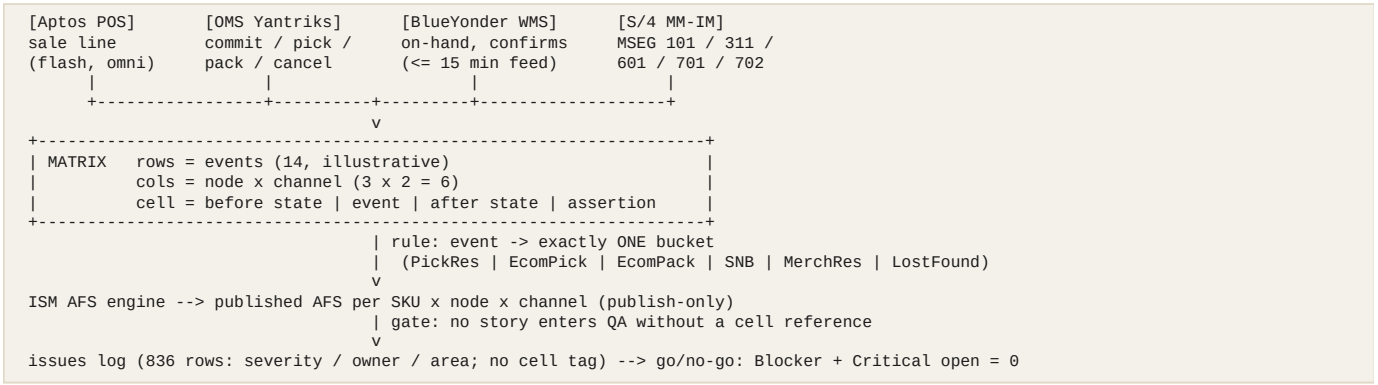
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Working definitions of “available”	3	1 signed AFS	record
2	Event → bucket cardinality	0 or 2 possible	exactly 1 on every live cell	record design
3	Blocker + Critical open at go/no-go	441 logged of 836	0 open	record
4	Issues closed	0 of 836 at kickoff	714 of 836 = 85.4%	record
5	Customer-facing incidents	0 allowed	0 through both go-lives	record
6	Drift (ISM vs MB52)	unmeasured	0 SKUs on consecutive days	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Oversell-exposed incidents	`30,000,000 × 0.10% = 30,000/yr`	S1’s illustrative orders × benchmark rate (0.05–0.25%)	30,000/yr
2	Cost per incident	`$6 + $4 + $19 + $1 = $30`	CS contact · refund · lost margin · goodwill (benchmark)	$30
3	Exposure per live cell (uniform)	`30,000 / 61 = 491.8 incidents`	assumes uniform exposure across 61 cells (illustrative)	491.8/yr
4	Dollars per escaped cell-class	`491.8 × $30 = $14,754`	modeled what-if; not claimed	≈$14.8K/yr
5	Cost to author + run the matrix	`61 × 4 h = 244 h; 244 × $110 = $26,840`	4 h per cell (illustrative); $110/h engineering rate (modeled, as S5)	≈$26.8K one-time
6	Break-even	`$26,840 / $14,754 = 1.8`	escaped cell-classes per year to pay for the matrix	≈1.8 cell-classes

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Live cells (at 4 h × $110/h)	48 → $21.1K	61 → $26.8K	80 → $35.2K
Hours per cell (at 61 cells)	2 h → $13.4K	4 h → $26.8K	6 h → $40.3K
Oversell-exposed incidents / yr	15,000 (0.05%)	30,000 (0.10%)	75,000 (0.25%)
$ per escaped cell-class (÷ 61 × $30)	$7.4K	$14.8K	$36.9K
Break-even escaped cell-classes (at $26.8K)	3.6	1.8	0.7

SPOKEN STATEMENT

> “What I can prove is the rule and the gate: every physical event maps to exactly one bucket, and no story entered QA without a cell reference. Through both go-lives there were zero customer-facing incidents, with 836 issues logged — 441 Blocker or Critical, 52.8% — and 714 closed. What I cannot prove is how many of those the matrix caught versus the E2E rooms. The dollar view is a what-if on illustrative inputs: if the matrix keeps about two cell-classes a year from escaping, it has paid for itself.”

IF CHALLENGED

“Your cell count is invented.”

The design counts are illustrative sizing and I label them that way; the rule, the three de-aggregation cases, the four in-transit cells and the log are record. I would pull the real matrix from the QA workbooks before quoting a count.

“A matrix is shelfware — how do you know it was used?”

The entry gate: a story could not enter QA without a cell reference, and QA owned the gate and could fail a build. Use is enforced by the workflow, not by goodwill.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you simplified a complex process.	Invent and Simplify	One rule (event → exactly one bucket), one asset (the matrix), one gate (cell reference to enter QA) across about a dozen impacted systems.	Do not quote 61 as the real count.
Tell me about a time you dug into the data to find a root cause.	Dive Deep	Event traces on one SKU: 0 buckets → 560, 2 buckets → 554, 1 bucket → 557; defects clustered at seams.	Never claim the matrix alone kept incidents at zero.
What does ‘tested’ mean to you?	Insist on the Highest Standards	Every live cell asserted with numbers; three dated E2E days; Blocker + Critical open = 0.	Skip the Kafka/ESB topology unless asked.
How do you test a launch at scale?	Functional — launch readiness	Coverage countable by cell; QA owns the gate; counted rows at go/no-go.	Never ‘I configured ISM’.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

DIVE DEEP · AIQB Q1 **Q1. "Tell me about a time when you were trying to understand a complex problem on your team and you had to dig into the details to figure it out."**

I stopped reading oversell reports and started reading event traces — one SKU, one day, every event, every bucket. The pattern was never "the system is wrong"; it was an event landing in zero buckets or two. Take the worked SKU: 560 available. A store rings three units. If the event hits no bucket, AFS stays 560 and we promise three units that are gone. If it hits two — sold and shipped-not-billed both decrement — AFS goes to 554 and we hide three. Same event, three possible answers, one right: 557. So I wrote one rule — every event maps to exactly one bucket, a reversal must reference its original, a replay is a no-op — and turned it into before/after assertions per cell. Then I read the issues log against it: 836 rows, 441 Blocker or Critical. The defects were not random; they clustered where two systems used different grain. The result: zero customer-facing incidents through both go-lives, and a rule the control owners could keep using after I left.

Number to land: 557 — not 560, not 554 · **Pivot to NACF:** At NACF the same trace is unload → scan → stow → sellable; I would pull one ASIN's inbound event trace and look for events that land in zero or two states. · **Do NOT say:** that I wrote the code or personally found the bugs; that the matrix alone kept incidents at zero.

INVENT AND SIMPLIFY · AIQB Q1 **Q2. "Give me an example of a complex problem you solved with a simple solution."**

About a dozen impacted systems, about thirty build tracks — and I refused to write thirty test plans. I wrote one asset: a matrix of physical events against node-and-channel cells, each with a before and after assertion, and one entry rule — no story enters QA without a cell reference. Epics were per module — reserve module, AFS engine, SNB, control screens — slotted by business criticality × effort, so every track mapped onto the same cells. That turned "is it ready?" from an argument into a count: Blocker plus Critical open equals zero, and every live cell green. When my first in-transit rule turned out wrong, the matrix absorbed the fix as four new rows, not a new project. The same logic made a new node or channel a matrix row. The result was zero customer-facing incidents through both go-lives with 85.4% of 836 logged issues closed.

Number to land: one rule · one asset · one gate (cell reference to enter QA) · **Pivot to NACF:** NACF launches many parallel tracks — a new vendor cohort, a new IXD lane; one event × node matrix gives a countable ready-to-launch gate. · **Do NOT say:** the 61-cell count as fact; that I built the tooling.

INSIST ON THE HIGHEST STANDARDS · AIQB Q2 **Q3. "Tell me about a time when you were unsatisfied with the status quo. What did you do to change it?"**

I changed what "tested" meant: not 'we ran the flow' but 'every live cell asserted, with numbers, before and after.' Each cell carried an expected state — AFS 560 to 557, SNB 40 to 43, drift zero — so a pass was a number, not an opinion. Then I put the standard in rooms, not memos: three dated end-to-end days — Inbound 11/22, Dotcom 12/20, Retail 1/30 — with every function in one room, every finding logged with a severity and an owner before anyone left. The go/no-go bar was written in advance: Blocker plus Critical open equals zero, mock three inside the 0.5% tolerance, runbook rehearsed, rollback armed. I said I would have called no-go on one open Blocker — and I never had to. The result: zero customer-facing incidents, and a gate QA, not I, defended.

Number to land: three dated E2E days; Blocker + Critical open = 0 · **Pivot to NACF:** Before any NACF launch I would publish the same bar — what the gate asserts, in numbers, and who may fail it. · **Do NOT say:** that I never faced pressure; that the bar was only mine — QA owned the gate.

EARN TRUST · AIQB Q5 **Q4. "Tell me about a time when your team's goals were out of alignment with another team you relied on in order to meet your goal."**

DC Ops did not trust a systems program to respect the floor, so I made sure nothing reached them as a surprise. During build, QA demoed each matrix family — receipts, transfers, pick and pack reserves, ship confirms — to DC Ops and the planners. By the time UAT started, their sign-off was confirmation, not discovery. QA owned the gate and could fail a build, and DC Ops saw that it could. I also kept the floor's reality inside the design: the January planning go-live (purchasing and inbound, reserve rules live against demand) left the floor untouched, and warehouse execution moved only in May. The result was go-lives where DC Ops owned the robotics gate and the dial and the floor ran on them, with zero customer-facing incidents.

Number to land: January reserve rules first; May execution second; zero incidents · **Pivot to NACF:** At NACF I would demo each inbound event family to the building before the first live trailer — the same 'confirmation, not discovery' rule. · **Do NOT say:** that I ran the floor; DC Ops did.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"Start with the log: 836 rows, 441 of them Blocker or Critical — 52.8%."

Headline metric: 441 / 836 = 52.8% high-severity; 714 closed = 85.4%

Cut the log by severity, owner and seam; the matrix made coverage countable. What it revealed: 62.6% of the rows sit in the three flows where an event crosses systems (Outbound 238, Retail 172, Inbound 113 = 523, computed from the file), which is where the matrix cells concentrate. Honest limit: rows were not tagged by cell.

CONFLICT / INFLUENCE ANGLE · EARN TRUST

"I owned a test asset for ~30 build tracks in other orgs and none of them reported to me."

Headline metric: ~30 build tracks · one entry gate

The gate (no story enters QA without a cell reference) moved every track onto one asset; QA owned it and could fail a build; DC Ops confirmed because QA had already demoed to them. Influence came from a countable rule, not rank.

FAILURE / LEARNING ANGLE · ARE RIGHT, A LOT

"My own first in-transit rule passed every cell I wrote — and it was wrong."

Headline metric: $\sigma \approx 0.3$ d → window ≈ 0.49 d ≈ 95%; 4 cells added

"Never promise in-transit" starved dotcom of fast movers one day from the building; the matrix asserted my rule, not whether the rule was right. Now: both sides of a rule get a number, back-test against history first. (Record: master S4.)

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

How do you know it was the matrix and not the E2E days or the team?

You cannot, and I would not claim it. Record: zero customer-facing incidents, 441 Blocker/Critical rows dispositioned, B+C open = 0 at go/no-go. The measurable yield is the share of the 441 first detected by a cell assertion — the log was not tagged by cell, so I cannot produce it.

Say: *"I can't isolate the matrix from the E2E days; the number that would change my mind is a customer-facing incident traced to an event with no cell."*

Why is a 61-cell matrix not over-engineering against an 836-row log?

The log is the detector; the matrix is the coverage map. At 4 h a cell, 61 cells is 244 h ≈ $26.8K (modeled); one escaped cell-class is ≈$14.8K/yr on illustrative inputs — break-even ≈1.8 cell-classes.

Say: *"The log tells me what broke; the matrix tells me what I never tried — and it costs less than two escaped cell-classes."*

What did the matrix not catch, and what changed?

The v1 in-transit rule: cells asserted my rule and it was wrong; the fix added four cells and a back-test-before-publish requirement (σ ≈ 0.3 d, window ≈ 0.49 d ≈ 95%).

Say: *"A matrix checks that the system does what the rule says; only history tells you whether the rule is right — so now both sides of a rule get a number first."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. At the Utah DC I made "tested" countable: every physical event had to land in exactly one inventory bucket, each cell asserted before and after, and no story entered QA without a cell reference — zero customer-facing incidents through both go-lives, with 441 of 836 logged issues Blocker or Critical and none reaching a customer.

NACF bridge. NACF inbound has the same seam — a trailer unloaded, a pallet received, a unit stowed, a unit sellable. The booklet's "stranded" inventory is exactly an event that landed in zero states. I would build the same event × node matrix for receive → stow → sellable so every dock event lands in exactly one inventory state before a new FC ramps.

U2 — The profile before the design — SKU overlap, order shape and demand variability per node across both books (MCS and S/4): the discovery that sized the one-pool case and bounded its $2.2M NEW

LPs this story serves here: Dive Deep (primary — both books at SKU × node grain) · Are Right, A Lot (a bounded claim with a kill number) · Learn and Be Curious (backup — the pooling math and the retail/dotcom order shapes) · Think Big (backup — two pools to one)

Process family: Plan / Source — inventory positioning and pooling analysis; the discovery profile that sized the case for one comingle pool

Honesty label. Record: Discovery (3/29 → 4/30) profiled both books — MCS and S/4 — and the 801/901 nodes for SKU overlap and order profiles per node, with node-by-node sessions across ~15 functions; the design argument started from their data. The √2 pooling release, the $50M average inventory × 25% safety stock × 60% overlap and the ≈$1.6M/yr are S2's **modeled** numbers on an illustrative 100K-units/week DC. **Not in the record:** the profile's actual outputs — measured overlap, σ ratio, correlation, order-shape cuts. Every range below is toolkit math on labeled inputs. Never say the profile "measured" a release; say it bounded one.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Dive Deep. I profiled both books at SKU × node grain before any design argument, so the debate started from their own data instead of opinions about it.

Are Right, A Lot. I refused to carry the textbook $2.2M as a point: I showed what would shrink it (unequal pools, correlation), bounded it at ≈$0.74M–$2.20M, and named the measurement that settles it.

Learn and Be Curious. I worked the variance-pooling formulas by hand on small numbers and learned the retail-vs-dotcom order shapes in the node sessions, so I could ask the right three questions of the data.

Think Big. The recommendation was structural — two pools to one, with three planning identities — and it stood even when the pooling dollars shrank, because the case never rested on them alone.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Before anyone argued design at Sephora's Utah DC, there were two inventory books — MCS and S/4 — and two systemic warehouses in one building, Dotcom 801 and Retail 901, and a business case that said "pool them." Whether pooling was worth it turned on three things nobody had put side by side: how many SKUs both pools carried, whether the two nodes' orders looked alike, and how variable and how correlated the demand was. **Task.** Own the discovery profile that the location and inventory model would stand on — about a dozen impacted systems, 60+ stakeholders, none reporting to me. **Action.** Discovery ran 3/29 to 4/30: both books profiled for SKU overlap and order profiles per node, node-by-node sessions across about fifteen functions. I turned the profile into three tests — overlap, order shape, variability — and a bounded claim: textbook √2 pooling on $7.5M of duplicated poolable safety stock is a $2.2M ceiling; unequal or correlated pools cut it to roughly $0.7M to $2.2M, and about 87% of the $1.6M a year modeled benefit does not depend on pooling at all. That fed the decision — one physical pool, three planning identities. **Result.** Primary metric: the one-time pooling claim moved from a single textbook number to a bounded range, about $0.7M to $2.2M, modeled, with discovery closed on 4/30. The +15% and the zero incidents are S2's record; the profile outputs are illustrative and I would not quote them as measured.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Two systemic warehouses in one building transferring stock to itself, two books (MCS and S/4), a case for one comingle pool resting on three unmeasured assumptions: that both pools carry the same SKUs, that their orders look alike enough to share a pick face, and that pooling their demand variability would release stock. A pooling claim built on a textbook assumption is the first number a skeptic pulls on.

T — Produce the profile the design could stand on and a claim that survives the first pull: which SKUs overlap, what each node's order looks like, how much of the $2.2M survives unequal and correlated pools — and which measurement turns the model into a record. None of the systems or analysts reported to me; ~15 functions sat in the sessions.

Baselines on the table: planning pools 2 · average inventory $50M (illustrative) × 25% safety stock × 60% held in SKUs both pools carry = $7.5M duplicated poolable stock · self-transfers ≈260,000 units/yr ≈5.0% of an illustrative 5.2M units/yr · efficiency index 100.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — pulled both books at SKU × node grain; cut for overlap (which SKUs live in both 801 and 901), order shape per node (retail store-replenishment style vs dotcom small-order style), and demand variability per SKU; ran node-by-node sessions with ~15 functions; wrote the bucket inventory. *Exit bar:* sessions closed 4/30; every bucket owned by one system; open questions carried with owner and date.

- **Plan & Design (5/15 → 8/14)** — turned the profile into the three-identity design (one physical location, three planning lenses) and into a bounded pooling claim in the sponsor pack: ceiling $2.2M, plausible floor ≈$0.74M, recurring benefit mostly independent of pooling. *Exit bar:* council sign-off 7/31 and FSD 8/14 with the range, not a point, in the case.
- **Build & Test (Aug → Jan/Apr)** — carried the profile's cuts into the matrix rows (U1) and the interim shadow monitor; the node/channel/order-shape cuts became test families. *Exit bar:* matrix families trace to a profile cut; E2E days 11/22 · 12/20 · 1/30.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — no new analysis on the critical path — the profile's job was done at design; it re-entered only as the reference for the "which book does this report read" declaration during the interim. *Exit bar:* every report declared its system of record for the interim.
- **Hypercare / Close (Jun → Jul)** — the record's own firming measurement — SKU-level on-hand on the last day of 801 against the first stable month of 901 — is the way the release becomes a record; flagged as the open measurement. *Exit bar:* pooling stays labeled "modeled" until that comparison is run.

Influence and ownership. Mine: the profile design, the three tests, the bounded claim, the design recommendation. Others': the extracts (platform/data teams), the node sessions (DC Ops, Planning, Digital, Finance), the sign-off (steering). Digital's worry that a virtual Retail DC would blur their forecasting lens was answered by the design — three planning identities over one physical location — not by an argument.

> **Designed the process; others own it now.** The profile cuts (overlap · order shape · variability) are the template the next DC's discovery reused; the three-identity design is the standing location model. **Trade-off put to leadership:** a precise-looking $2.2M versus an honest range — I took the range, because the first number a skeptic pulls on is the one resting on an assumption.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — pooling release (one-time, modeled): textbook ceiling $7.5M × 29.3% = ≈$2.2M; plausible floor ≈$0.74M at σ ratio 1/3 and ρ = 0.5. Recurring ≈$1.43M (87% of ≈$1.65M) does not depend on pooling. All modeled on S2's illustrative DC — never signed.
- **#** — 2 books profiled · 2 planning pools → 1 · 3 planning identities · self-transfers ≈260,000 units/yr (≈5.0% of illustrative volume) → 0 by construction (record design).
- **%** — release range 9.9%–29.3% of poolable stock; overlap sensitivity 40/60/80% → $1.46M / $2.20M / $2.93M (modeled).
- **Time** — sessions closed 4/30 (record); the profile preceded council 5/15 → sign-off 7/31 → FSD 8/14.
- **Customer** — zero customer-facing incidents through both go-lives (S2 record) — the constraint the one-pool design had to live inside.
- **Quality** — every pooling dollar carries its input and basis label; the claim names the measurement that would firm it.

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A case for pooling resting on assumptions nobody had measured. I defined the three tests and what each would change.
Scope & influence	~15 functions in node sessions; extracts from data teams; steering signed the criteria. I influenced by putting their numbers on one page.
Execution trade-off + mechanism others own	Point estimate versus honest range — chose the range. Mechanism others own: the three-test profile and the bounded-claim format in the sponsor pack.
Impact	A one-pool design whose recommendation survived a ≈$1.5M-smaller worst case on the one-time number.
Process improvement	Discovery ends with a profile (overlap · shape · variability) and a measurement that converts the model to a record — reused by the next DC.
Complexity & consensus	Retail and dotcom order shapes, two books and three planning lenses reconciled into one location model.
Root cause at scale	The fragile part of the case was an assumption (equal, independent pools), not a number; bounding it removed the single point of failure.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: SKU × node (801 / 901 / stores) in both books. Joins: MCS ↔ S/4 on SKU; overlap flag by node pair; demand history per SKU per node. Cuts: overlap by value; order shape per node (lines per order, units per line — illustrative until pulled); σ and correlation per overlapping SKU. Revealed (modeled): the case is most sensitive to correlation and σ ratio, and least to overlap, once the textbook is replaced by the range.

2 · Solution design. Options: (A) keep two pools and reconcile (status quo); (B) pool physically but plan as one blended identity; (C) one physical pool, three planning identities. Chose C — A keeps self-transfers; B blurs the forecasting lens Digital needed. Why: the profile showed the case did not need the safety-stock row to be big.

3 · Math & data logic. Equal independent pools: release = $1 - 1/\sqrt{n} = 1 - 1/\sqrt{2} = 29.3\%$. Unequal: $1 - \sqrt{(\sigma_1^2+\sigma_2^2)}/(\sigma_1+\sigma_2)$. Correlated: $1 - \sqrt{(\sigma_1^2+\sigma_2^2+2\rho\sigma_1\sigma_2)}/(\sigma_1+\sigma_2)$. Dollars = poolable stock × release; carry = release × 10%. Break-even sentence: the claim fails only if the recurring rows fail.

4 · Implementation. Discovery 3/29 → 4/30; design to FSD 8/14; the profile cuts carried into matrix rows; the SKU-level on-hand comparison (last day 801 vs first stable month 901) is the firming measurement. Owners: me (profile and claim), data teams (extracts), Finance (inventory

valuation).

5 · Probe questions. Below: why not trust √2, how you know the 60%, what if overlap is 20%, did the profile change the design, and the three trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — why not just trust √2?

Because it assumes two equal, independent pools on the same SKUs — three assumptions at once. I showed the release at σ ratio 1, 1/2, 1/3 and 1/4 (29.3%, 25.5%, 20.9%, 17.5%) and at correlation 0.3 and 0.5 (19.4%, 13.4%); jointly pessimistic it is 9.9%. The textbook is the ceiling, not the base.

Infra — how do you know the overlap is 60%?

I don't — it is S2's illustrative input, the share of poolable stock in SKUs both pools carry. The profile is what measures it; at 20% overlap the equal-pool release is $0.73M, at 80% $2.93M. The case never depended on getting that one right.

Infra — if the release is that uncertain, why pool at all?

Because the pooling carry is only 13% of the modeled recurring benefit. Labor at +15% (≈$0.88M, record efficiency), split shipments (≈$0.45M) and self-transfer handling (≈$0.10M) are $1.43M; self-transfers alone were ≈260,000 units a year of movement that created no value.

Infra — did the profile change the design?

It shaped it: it is why I kept three planning identities over one physical location rather than one blended lens, and why the matrix has rows per node type and order shape. I would not claim it changed the decision to pool; that case was already strong on the recurring rows.

Infra — what would you pull first at a new node?

SKU-level on-hand and sales for the two pools, joined by SKU: overlap by value, σ and correlation per SKU, then order shape — lines per order and units per line. Three pulls, one afternoon, and the claim has inputs.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Planning pools	2	1 (three planning identities)	self-transfers → 0	record design
Pooling claim (one-time)	≈$2.2M point	≈$0.74M–$2.20M range	carry ≈$0.07M–$0.22M/yr	modeled (toolkit label)
Benefit independent of pooling	not separated	≈$1.43M/yr (87%)	labor + splits + self-transfer	modeled; +15% is record
Overlap sensitivity (equal pools)	—	40/60/80% → $1.46M / $2.20M / $2.93M	—	modeled
Measurement to firm the release	none	last-day 801 vs first-month 901 SKU on-hand	turns model into record	record proposal

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Warehouses / pools	`planning pools = 2 (801 and 900/901 identities)`	record	2
2	Duplicated safety stock	`$50M × 25% = $12.5M; × 60% = $7.5M`	average inventory, SS share and overlap share are S2's illustrative inputs (modeled)	$7.5M poolable
3	Self-transfer burden	`260,000 / (100,000 × 52) = 260,000 / 5,200,000 = 5.0%`	260K units/yr modeled; 100K units/wk illustrative	5.0% of volume is non-work
4	Textbook release	`1 − 1/√2 = 1 − 0.707 = 29.3%`	equal, independent pools (toolkit label)	29.3%
5	What the textbook never saw	`σ ratio r = σ₂/σ₁ and correlation ρ unmeasured`	VERIFY — profile outputs not in the record	release could be lower

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Test 1 — overlap	`poolable = $12.5M × overlap → 40%: $5.0M · 60%: $7.5M · 80%: $10.0M`	overlap is a measured input at a new node	$5.0M / $7.5M / $10.0M
2	Test 2 — σ ratio	`f(r) = 1 − √(1+r²)/(1+r): r=1 → 29.3% · r=½ → 25.5% · r=⅓ → 20.9% · r=¼ → 17.5%`	unequal independent pools (toolkit label)	17.5%–29.3%
3	Test 3 — correlation (equal pools)	`f(ρ) = 1 − √((1+ρ)/2): ρ=0.3 → 1 − √0.65 = 19.4% · ρ=0.5 → 1 − √0.75 = 13.4%`	equal σ; correlation reduces release	13.4%–19.4%

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Jointly pessimistic	`r = ⅓, ρ = 0.5: 1 − √(1 + 1/9 + 2×0.5×⅓)/(4/3) = 1 − √1.4444/1.3333 = 1 − 1.2019/1.3333 = 9.9%`	illustrative corner of the range	9.9%
5	Dollar range	`$7.5M × 9.9% = $0.74M · $7.5M × 29.3% = $2.20M`	poolable stock × release	≈$0.74M–$2.20M
6	Firming measurement	`last-day 801 SKU on-hand vs first stable month of 901 → release = the difference`	the record's own proposal (not yet run — VERIFY)	model → record

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Planning pools	2	1	record design
2	One-time pooling claim	$2.2M point (textbook)	$0.74M–$2.20M range	modeled
3	Recurring benefit independent of pooling	not separated	$1.43M of $1.65M = 86.7%	modeled
4	Self-transfers	≈260,000 units/yr	0 by construction	record design
5	Headline total (recurring)	≈$1.65M	≈$1.50M–$1.65M	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Labor at +15%	`31,300 h × $28 = $876,400`	hours from S2 (record +15%); $28 loaded (modeled)	≈$0.88M/yr
2	Split shipments	`45,000 × $10 = $450,000`	S2 softest row (modeled)	≈$0.45M/yr
3	Self-transfer handling	`260,000 × $0.80 × 50% = $104,000`	the other half already sits in labor (modeled)	≈$0.10M/yr
4	Recurring without pooling	`876,400 + 450,000 + 104,000 = $1,430,400`	rows 1–3	≈$1.43M/yr
5	Carry on the release	`$2.2M × 10% = $220,000 · $0.74M × 10% = $73,959`	10% cost of capital (benchmark 8–12%)	≈$0.22M / ≈$0.07M
6	Headline range	`1,430,400 + 220,000 = $1,650,400 · 1,430,400 + 73,959 = $1,504,359`	rows 4–5	≈$1.65M / ≈$1.50M

SOLUTION FLOW — THE SYSTEM & DATA MAP

```
[MCS book]           [S/4 book]           [Demand history]
801 on-hand          900 / 901 on-hand    sales by SKU x node
     |                    |                    |
     +----------+----------+--------------------+
                v
JOIN on SKU -> overlap flag by value                    (Test 1: 40 / 60 / 80%)
                v
per overlapping SKU: sigma_801, sigma_901, rho          (Tests 2 and 3)
                v
release f = 1 - sqrt(s1^2 + s2^2 + 2*rho*s1*s2) / (s1 + s2)
                v
$ = poolable stock x f  ->  range 9.9% .. 29.3%         (bounded claim)
                v
order shape per node (lines/order, units/line) -> matrix rows + slotting questions
                v
DESIGN: one physical pool, three planning identities
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Overlap share (equal pools)	40% → $1.46M	60% → $2.20M	80% → $2.93M
σ ratio r (ρ = 0)	⅓ → $1.57M	½ → $1.91M	1 → $2.20M
Correlation ρ (r = 1)	0.5 → $1.00M	0.3 → $1.45M	0 → $2.20M
Cost of capital on $2.2M	8% → $0.18M	10% → $0.22M	12% → $0.26M
Headline recurring total	≈$1.50M (floor release)	≈$1.65M (ceiling release)	pooling row ≤ 13% of total

SPOKEN STATEMENT

> "The profile did two things. It told me which SKUs, which node shapes and how much variability the case rested on, and it kept me from carrying a textbook number as a fact. The pooling release is about $2.2 million at the ceiling — equal, independent pools — and about three-quarters of a million at a pessimistic corner; I show the range. What does not move is that about 87% of the modeled benefit is labor at the signed +15%, split shipments and self-transfers, none of which depend on pooling. The thing that turns the model into a record is one comparison: SKU-level on-hand the last day of 801 against the first stable month of 901."

IF CHALLENGED

"So the $2.2M in S2 is wrong?"

It is the ceiling on S2's stated inputs, and S2 labels it modeled and textbook. This story is the bounding of it — what survives if the pools are unequal or correlated — not a retraction.

"You never ran the comparison?"

I would not claim it. The comparison is the measurement the record proposes; until it is run the number stays labeled modeled. If you ask me whether pooling released that stock, my answer is: here is the range and here is the one-afternoon pull that settles it.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you used data to challenge an assumption.	Dive Deep	Both books at SKU × node grain; the equal-independent-pools assumption bounded at 9.9%–29.3%.	Never say the profile "measured" the release.
Tell me about a decision you made with incomplete data.	Are Right, A Lot	A bounded claim with a named measurement; recommendation stable across the whole range.	Do not present the 60% overlap as measured.
Tell me about learning something outside your area.	Learn and Be Curious	Variance pooling and the order shapes at retail vs dotcom nodes.	Skip the BlueYonder/Locus internals.
How do you decide whether to consolidate nodes in a network?	Functional — network design	Overlap, order shape, variability/correlation; recurring rows first, safety stock last.	Never quote $2.2M as a record.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

DIVE DEEP · AIQB Q3 **Q1. "Tell me about a problem you had to solve that required in-depth thought and analysis."**

The accepted assumption was that two pools become one and safety stock drops by 29%. That is the square-root pooling result, and it assumes the two pools are equal and independent — which retail and dotcom demand for the same SKU almost certainly are not. So in discovery I stopped quoting the textbook and asked the data three questions: which SKUs do both pools carry, how big is each pool's variability relative to the other, and how correlated are they? I put the answer as a range instead of a point. Equal and independent gives 29.3%. If one pool is a third the size of the other, 20.9%. If correlation is 0.5, 13.4%. Jointly pessimistic, 9.9%. On $7.5 million of duplicated poolable stock that is $0.74 million to $2.2 million. Then I checked whether the case still stood: about 87% of the modeled benefit — labor at the signed +15%, split shipments, self-transfers — does not depend on pooling at all. The design stood; the claim became honest.

Number to land: 9.9%–29.3% → ≈$0.74M–$2.2M; 87% independent of pooling · **Pivot to NACF:** At NACF the same three pulls decide whether two FCs' buffers belong together — same ASINs, same demand shape, correlation — before I claim any release. · **Do NOT say:** that the profile measured the release; that $2.2M is a record.

ARE RIGHT, A LOT · AIQB Q1 **Q2. "Tell me about a time when you didn't have enough data to make the right decision."**

I made the recommendation robust to the thing I did not know. I did not have measured σ or correlation for the overlapping SKUs when the sponsor pack was written. So I built the claim as a range with a kill number: the only way the one-pool recommendation fails is if the recurring rows fail, and those rest on a signed +15% efficiency, a split-shipment row I flagged as the softest, and the self-transfer move that disappears by construction. Then I named the measurement that converts model to record — SKU-level on-hand on the last day of 801 against the first stable month of 901 — and I told steering what number would change my mind: if split shipments didn't move, the total drops to about $1.1 million and I would say so. The recommendation was the same across the whole range, which is the test of a decision made with incomplete data.

Number to land: recommendation stable from $1.50M to $1.65M recurring · **Pivot to NACF:** At NACF placement decisions are made on incomplete data constantly; the habit I would bring is a range, a kill number and the one measurement that firms it. · **Do NOT say:** that I knew the overlap; that the model was validated.

LEARN AND BE CURIOUS · AIQB Q1 **Q3. "Tell me about a time when you realized you needed a deeper level of subject matter expertise to do your job well."**

I started by asking what the word "pooling" actually assumes. The business case said "pooling," and I wanted to know exactly what that word assumed. So I worked the variance-pooling toolkit — equal independent pools, unequal pools, correlation — and checked each formula by hand on small numbers until I could do it on a whiteboard. Then I learned the shapes of the two nodes' orders: retail store-replenishment style versus dotcom single-customer orders, because that is what decides whether two pools can share a pick face. I sat in the node sessions with about fifteen functions asking what a normal day looked like. By the end of discovery I could explain, in one page, why the pooling number was a ceiling and what each node's order looked like. Zero customer-facing incidents later is not mine alone, but the design started from a profile I could defend.

Number to land: three formulas hand-checked; ~15 functions in node sessions · **Pivot to NACF:** NACF placement is the same toolkit — I would arrive having learned the ASIN-level demand shapes before proposing a buffer. · **Do NOT say:** that I invented the pooling math; it is the toolkit's, and I label it.

THINK BIG · AIQB Q1 **Q4. "Tell me about time when you were working on an initiative or goal and saw an opportunity to do something much bigger or better than the initial focus."**

The ask was to merge two warehouses' reporting; the bigger structure was one physical pool. Two systemic warehouses sat in one building and moved stock to each other — about 260,000 units a year on paper, roughly 5% of volume that created no value. I proposed collapsing them into one pool with three planning identities so each channel kept its forecasting lens without a second truth. Digital worried a virtual retail DC would blur their lens; I made the concern the design — the three identities — rather than arguing. The bigger structure was riskier, so I staged it: two go-lives, each carrying one defect class. And I made sure the case did not hang on the most uncertain number: even at the pessimistic pooling corner, the recurring benefit is about $1.5 million. Big, but with the downside bounded.

Number to land: ≈260K units/yr self-transfers → 0; 2 pools → 1 · **Pivot to NACF:** At NACF a bigger structure — fewer buffers pooled upstream — needs the same discipline: bounded downside before ambition. · **Do NOT say:** that pooling alone paid for it; that the dollars are signed.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"The textbook says 29%. Here is what survives when the pools are unequal and correlated."

Headline metric: 9.9%–29.3% release; $0.74M–$2.20M

Three tests (overlap, σ ratio, correlation) turned one number into a surface; the case's dependence on pooling is ≤13%. Honest limit: the inputs are illustrative until the profile is pulled.

CONFLICT / INFLUENCE ANGLE · EARN TRUST

"Digital worried a virtual Retail DC would blur their forecasting lens — the profile gave me a way to say yes to them."

Headline metric: 3 planning identities over 1 physical location

The concern became the design: each channel kept its lens; no second truth. No one had to lose an argument. Steering signed the criteria; DC Ops and Planning confirmed the node sessions.

FAILURE / LEARNING ANGLE · ARE RIGHT, A LOT

"In S2 I say $2.2M one-time — that is the equal-pools ceiling, and I should lead with the range."

Headline metric: $2.2M point → $0.74M–$2.20M range

I label it modeled and textbook, but a point invites the first pull and the first pull finds the equal-independent assumption. Do differently: say the range first, name the kill number, and never let the ceiling stand alone.

confirm whether the sponsor pack carried the poi

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

What number would have changed your mind about pooling?

If the recurring rows failed: the split-shipment row (≈$0.45M, softest) not moving drops the total to ≈$1.1M; if the +15% were not signed the case would be different. Pooling's carry is only $0.07M–$0.22M/yr, so no outcome of the pooling math alone could flip it.

Say: *"The pooling row is at most 13% of the benefit, so the number that would change my mind is split shipments or the signed efficiency — not the safety-stock release."*

Isn't the 60% overlap just made up?

It is S2's illustrative input — said so. At 20% overlap the equal-pool release is $0.73M; at 80% it is $2.93M. The profile's job is to replace it with the measured value, join by SKU, value-weighted.

Say: *"It is an illustrative input I label as such; here is the one-afternoon pull that replaces it, and here is what the answer does to the claim across 20% to 80%."*

How do you know it was pooling and not seasonality that moved stock?

I can't — and I don't claim it. The record has +15% efficiency measured by DC Ops post-stabilization and zero incidents; the pooling release is modeled. The firming comparison is last-day 801 versus first stable month of 901 on SKU-level on-hand, seasonally matched.

Say: *"The pooling dollars are modeled; if you want it as a record, it is a seasonally-matched SKU-level on-hand comparison — that is the one measurement I would run."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. Before the design argument at Sephora's Utah DC I profiled both inventory books for SKU overlap and order shape per node and turned the textbook pooling number — $2.2M — into a bounded range of about $0.74M to $2.2M, with about 87% of the modeled benefit not depending on pooling at all; that range, not a point, went into the sponsor pack.

NACF bridge. NACF's placement and buffer questions start with the same three pulls — do two FCs carry the same ASINs, do they see the same demand shape, how correlated are they — because pooling variability upstream only pays where correlation is low, and an INV-D transfer is how the pooled unit actually moves. I would bring the range-and-kill-number habit to any buffer-positioning claim.

U3 — Merch reserve — visible, never promisable: pricing the no (and the asymmetry behind it) so one comingled pool never sells stock the floor cannot pick NEW

LPs this story serves here: Have Backbone; Disagree and Commit (primary — the priced no) · Customer Obsession (oversell is a customer defect) · Earn Trust (visibility offered, the no written as a row) · Success and Scale Bring Broad Responsibility (a customer harm, not a ledger error)
Process family: Plan / Deliver — comingled-inventory allocation at a shared node: which stock is promisable to which channel, and which is physically present but not for sale

Honesty label. Record: Merch Reserve and Lost/Found sit outside S4_Avail, so there is no double-subtract path; Merchandising wanted merch reserve promisable; I priced it (unpickable stock sold) and offered visibility without promisability; the no went into the FSD as a row, not a meeting minute; the asymmetry (oversell is a customer defect, undersell a planning variance) is the policy; comingle-reserve improvements were held out of the release to protect zero incidents and priced as a later enhancement at steering; zero customer-facing oversell incidents through both go-lives. **Illustrative:** the 20% release rate, the 10,000-unit reserve, the 50-unit worked SKU and the retail/dotcom pool split. The margin $19 and the $6 + $4 + $1 + $19 = $30 incident cost are S1's benchmark. I do not know how Merchandising defined or sized the reserve. I owned the definition and the no; Merchandising owned the reserve.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Have Backbone; Disagree and Commit. Merchandising wanted merch reserve promisable; I priced it, said no, and wrote the no into the FSD as a row so it survived the room.

Customer Obsession. I treated oversell as a customer defect and undersell as a planning variance — and accepted visible undersell to make oversell zero by construction.

Earn Trust. I gave Merchandising what they actually needed — visibility of the reserve — so the no cost them a lever, not their standing.

Success and Scale Bring Broad Responsibility. A promise on stock the floor cannot pick is a customer harm at scale; the price in dollars was the floor of the argument, not the whole of it.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. At Sephora's Utah DC one physical pool had to serve two channels, and Merchandising held some of that stock back as reserve. Merchandising wanted the reserve promisable — it was in the building, so why not sell it. **Task.** I owned the definition of "available" across Store, Retail DC and Dotcom DC, and I owned no system. **Action.** Discovery: a bucket inventory where every bucket had one owning system; merch reserve and lost/found sat outside the supply the AFS engine computes from, so there was no double-subtract path. Plan & Design: I priced the ask — stock the floor may not pick, sold — as a break-even: promise it only if more than roughly 37 of every 100 reserved units would actually be released, and on a 20% release rate that is about −$50,000 per 10,000 units, or about −$200,000 on the all-in cost. Then I offered what Merchandising actually needed: visibility without promisability. The no went into the FSD as a row, signed by Strategy, Inventory Management and Inventory Accounting. Comingle-reserve improvements were held out of the release to protect zero incidents and priced as a later enhancement. **Result.** Primary metric: customer-facing oversell incidents — recurring across three node types → zero through both go-lives, with zero units of merch reserve in published AFS. The dollars are a what-if on labeled inputs; the zero is the record.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — One physical pool, two channels, three planning identities — and stock physically in the building that was not for sale: Merchandising's reserve, lost-and-found. Merchandising wanted the reserve promisable; Finance wanted on-hand to tie to the book; dotcom wanted in-transit promisable. Most oversell bugs are vocabulary bugs, and "available" had three meanings.

T — Define, per node and per channel, which buckets are subtracted from supply and which sit outside it — and say no to the one request that would have turned unpickable stock into a promise. I owned no system; the signatories were Supply Chain Strategy, Inventory Management and FSC Inventory Accounting.

Baselines on the table: working definitions of "available" 3 · customer-facing oversell incidents recurring across three node types, target zero · worked SKU: Basis 980, buckets 420, AFS 560 · merch reserve and lost/found outside S4_Avail.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — bucket inventory with one proposed owning system per bucket; node-by-node sessions; Merchandising's need written down as a need — see the reserve — not as a mechanism. *Exit bar:* every bucket owned by exactly one system; sessions closed 4/30.
- **Plan & Design (5/15 → 8/14)** — kept merch reserve and lost/found outside S4_Avail; wrote the lower-of-two basis and the AFS equation; priced the promisable-reserve ask as a break-even; offered visibility without promisability; held the asymmetry. *Exit bar:* FSD signed 8/14;

the no is a row; every bucket owned by exactly one system.

- **Build & Test (Aug → Jan/Apr)** — matrix cells that assert the reserve never appears in published AFS (U1) and a visibility view for Merchandising; held the comingle-reserve improvements out of the release. *Exit bar:* Blocker + Critical open = 0; reserve-copy invariants checked per SKU.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — reserve rules live against demand at the January planning go-live with the floor untouched; execution moved in May; no SKU with two published AFS values. *Exit bar:* oversell incidents zero through both go-lives.
- **Hypercare / Close (Jun → Jul)** — daily ISM-vs-MB52 drift; Negative AFS and AFS Correction under control owners (U6); the improvement list handed to steering as a priced later enhancement. *Exit bar:* drift 0 on consecutive days; enhancement priced, not promised.

Influence and ownership. Mine: the bucket inventory, the break-even, the asymmetry, the offer of visibility, the row in the FSD. Merchandising owned the reserve; control owners own the screens. Digital's forecasting-lens concern was answered by the three-identity design, not by a ruling.

> **Designed the process; others own it now.** The reserve rule is now a row in the FSD and a cell family in the matrix; Merchandising sees the reserve and Inventory Control chases variance on the screens. **Trade-off put to leadership:** sell the reserve and accept unpickable-stock cancellations, or hold it back and accept visible undersell — I priced both and recommended undersell.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — no record dollar for this decision. What-if (modeled, **illustrative**): at a 20% release rate, promising 10,000 reserved units costs −$50,000 (incremental cancellation cost) to −$202,000 (S1's all-in $30 per incident) against the ≈$38,000 of expected sales; break-even release 36.7% (61.2% all-in). Not claimed as a saving.
- **#** — 3 working definitions → 1 signed AFS; 0 units of merch reserve in published AFS (record design); 3 organizations signed the row.
- **%** — break-even release rate p* = 36.7% (modeled); oversell rate behind S1's ≈$0.9M row is 0.10% of orders (benchmark 0.05–0.25%; the record is incidents → zero).
- **Time** — FSD signed 8/14; January reserve go-live; May execution go-live (record).
- **Customer** — customer-facing oversell incidents recurring → zero through two go-lives (record); no SKU with two published AFS values.
- **Quality** — reserve and lost/found outside S4_Avail — no double-subtract path; the no lives in the FSD as a signed row (record design).

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Three meanings of "available" and a request that would have promised unpickable stock; I defined which buckets are inside supply and which sit outside.
Scope & influence	Merchandising, Finance, Digital and Inventory Management each had a stake; I owned none of their systems; three orgs signed.
Execution trade-off + mechanism others own	Visible undersell versus unpickable-stock oversell — chose undersell. Mechanism others own: the FSD row and the Negative AFS / AFS Correction screens.
Impact	Oversell incidents recurring → zero through both go-lives; no SKU promised twice.
Process improvement	A priced no, written as a row; the same break-even shape (p* = cost ÷ (gain + cost)) applies to any reserve decision.
Complexity & consensus	Three node types, two channels, three planning identities, three signing orgs, one comingled pool.
Root cause at scale	Oversell is a vocabulary defect; the fix was a rule about which stock is inside supply, not 441 patches.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: SKU × node × channel × bucket. Sources: the bucket inventory, both books' on-hand, the reserve view Merchandising worked from. Cuts: stock by bucket (inside supply vs outside), by channel demand, by pool. Revealed: stock present in the building and absent from S4_Avail is physical truth the system must not promise.

2 · Solution design. Options: (A) reserve promisable; (B) reserve visible, not promisable (chosen); (C) reserve invisible. A overpromises on a release the floor cannot guarantee; C creates shadow spreadsheets. B gives Merchandising the lever they needed.

3 · Math & data logic. AFS = Basis − PickRes − EcomPick − EcomPack − SNB; MR outside S4_Avail → AFS stays 560 whether MR = 0 or 50. Break-even p* = c ÷ (g + c) = 11 ÷ 30 = 36.7%; EV per unit = p × $19 − (1 − p) × $11.

4 · Implementation. FSD row signed 8/14; matrix cells; visibility view; January/May go-lives; control owners for the screens. Risks: override abuse, vocabulary drift, feed latency (the register's top three).

5 · Probe questions. Below: how you priced it, what if the release is higher, who lost, what you deferred, and the three trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — it was in the building. Why not sell it?

Physical presence is not pickability. The reserve is stock Merchandising controls; if it is promised and not released, the customer finds out. I priced it: at a 20% release rate, 10,000 units is −$50K on the cancellation cost alone, −$202K on S1's all-in cost — and it is a customer harm beyond the dollars.

Infra — what release rate would make you say yes?

More than 36.7% on the incremental-cost view, 61.2% on the all-in view. At 40% the incremental EV is +$10K per 10,000 units — thin. I would also want a hold-until date and a release SLA — a rate alone is not a promise.

Infra — who lost?

Merchandising could not sell the reserve in the channel until it was released; they kept full visibility. The comingle-reserve improvements were held out of the release and priced as a later enhancement (master S1); the architecture-council pack had already fenced them out of scope in writing — 'the comingle reserve math stays untouched' (Apple C4).

Infra — how does one pool serve two channels without a split?

Both channels read the same Basis; every event decrements one bucket. Two-pool example: retail 300 with demand 250 (50 idle), dotcom 260 with demand 290 (30 short) — one pool of 560 meets 540 with 20 left. The cost is that the pool's AFS must be right, which is the matrix's job (U1).

Infra — what is the downside of this policy?

Visible undersell: the SKU shows less than the floor could sell. On S1's benchmark undersell costs ≈$4.75 per unit against ≈$30 for an oversell — about 6.3 to 1 — and Inventory Control chases it on the screens. I accept that trade on purpose.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Reserve promisable?	requested	no — visible only	−$50K to −$202K avoided per 10,000 units (what-if)	modeled, illustrative
Working definitions of "available"	3	1 signed AFS	—	record
Customer-facing oversell incidents	recurring across 3 node types	zero through two go-lives	S1's ≈$0.9M row is modeled	record / modeled
Stranded stock between pools	30 short while 50 idle (worked SKU)	560 vs 540 demand — 0 short	$570 margin per SKU-week (illustrative)	illustrative
Reserve physically held	2,500 ft³ per 10,000 units	unchanged — not promisable	0.25% of CAP-C's 1.02M ft³	illustrative

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Working definitions	`3 (Store · Retail DC · Dotcom DC)`	record	3
2	Worked SKU	`Basis = min(1,000, 980) = 980; AFS = 980 − 200 − 120 − 60 − 40 = 560`	S1 SKU (illustrative); MR outside S4_Avail	AFS 560
3	If the reserve were promisable	`50 reserved units → AFS = 560 + 50 = 610; over-promise exposure if unpickable = 50 units`	illustrative reserve	up to 50 units overpromised
4	Cost per oversell (all-in)	`$6 + $4 + $19 + $1 = $30`	CS contact · refund · lost margin · goodwill (benchmark)	$30
5	Incremental cost (margin never earned)	`$6 + $4 + $1 = $11; gain on a shipped sale = $19`	same inputs, margin removed from the cost side	c = $11, g = $19
6	Two pools — stranded stock	`retail 300 vs demand 250 → 50 idle; dotcom 260 vs 290 → 30 short`	illustrative split of one SKU	30 short, 50 idle
7	Physical footprint of the reserve	`10,000 × 0.25 ft³ = 2,500 ft³`	0.25 ft³/unit (booklet's illustrative); present on the floor, absent from AFS	2,500 ft³

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Break-even release (incremental)	`p* = c ÷ (g + c) = 11 ÷ (19 + 11) = 11 ÷ 30 = 36.7%`	same shape as the booklet's INV-D transfer breakeven (3 ÷ 12 = 25%)	36.7%
2	Break-even release (all-in)	`30 ÷ (19 + 30) = 30 ÷ 49 = 61.2%`	S1's $30 as the cost of a broken promise	61.2%
3	EV per promised unit at p = 20%	`0.20 × $19 − 0.80 × $11 = $3.80 − $8.80 = $-5.00`	release rate illustrative	−$5.00
4	EV per 10,000 units	`−$5.00 × 10,000 = −$50,000`	units illustrative	−$50,000
5	Same, all-in	`0.20 × $19 − 0.80 × $30 = $3.80 − $24.00 = $-20.20; × 10,000 = −$202,000`	S1 all-in cost	−$202,000
6	Option B (chosen)	`AFS = 560 whether MR = 0 or 50; MR shown in ISM, not in published AFS`	record design: MR outside S4_Avail	0 units overpromised
7	One pool, two channels	`300 + 260 = 560 supply; 250 + 290 = 540 demand; 560 − 540 = 20 left; short = 0`	same SKU, one pool	0 short

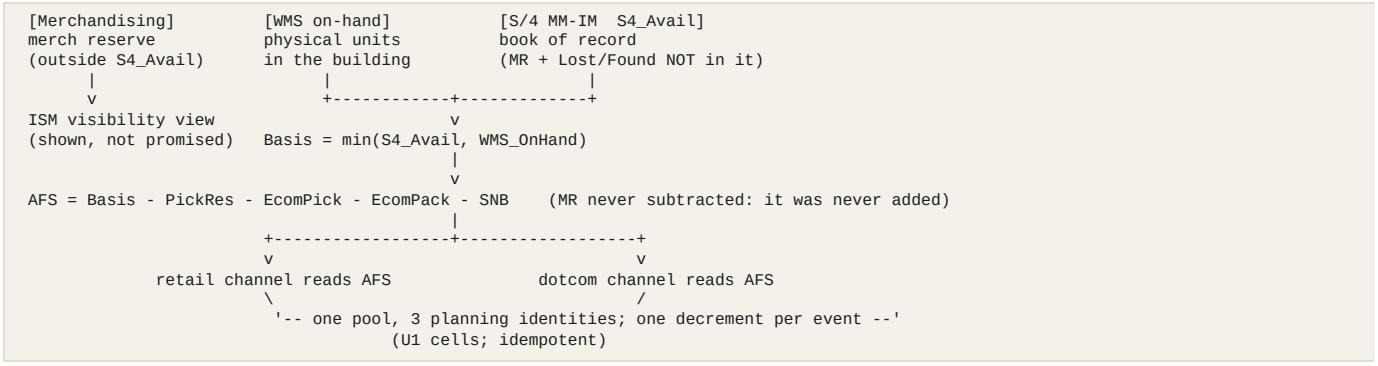
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Working definitions of "available"	3	1 signed AFS	record
2	Reserve units in published AFS	all (the ask)	0	record design
3	Oversell incidents	recurring across 3 node types	zero through two go-lives	record
4	Units short while stock idle (worked SKU)	30 short / 50 idle (two pools)	0 short / 20 left (one pool)	illustrative
5	Reserve rule	meeting minute	a signed FSD row	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Expected sales forgone (reserve not promised)	`0.20 × 10,000 × $19 = $38,000`	20% of reserve released and sold at $19 margin	$38,000 not earned
2	Cancellation cost avoided (incremental)	`0.80 × 10,000 × $11 = $88,000`	80% not released → cancelled	$88,000
3	Net, incremental view	`$88,000 − $38,000 = $50,000`	rows 1–2	$50,000 avoided
4	Net, all-in view	`0.80 × 10,000 × $30 = $240,000; − $38,000 = $202,000`	S1 all-in	$202,000 avoided
5	Stranded-stock margin (two pools)	`30 × $19 = $570 per SKU-week`	worked SKU (illustrative)	$570
6	If release were 40%	`0.40 × $19 − 0.60 × $11 = $7.60 − $6.60 = +$1.00/unit → +$10,000 per 10,000`	above p* = 36.7%	+$10,000

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Release rate p (incremental cost)	10% → −$80K	20% → −$50K	40% → +$10K

INPUT	LOW	BASE	HIGH
Cost basis at p = 20%	incremental $11 → −$50K	midpoint $20.50 → −$126K	all-in $30 → −$202K
Reserve volume at p = 20%, $11	5,000 → −$25K	10,000 → −$50K	20,000 → −$100K
Break-even release p*	36.7% (incremental)	—	61.2% (all-in)
Undersell : oversell cost	$4.75 : $30 = 1 : 6.3	—	ratio sets the policy, not one deal

SPOKEN STATEMENT

> "Merchandising asked me to make the reserve promisable. I said no, and I priced the no: promising stock that may not be released is worth it only above about thirty-seven percent release on the incremental cost, sixty-one on S1's all-in cost; on a twenty percent release rate that is about fifty thousand dollars per ten thousand units on the low view and about two hundred thousand on the high view — illustrative inputs, and I show them. What I offered was visibility without promisability, and the no went into the FSD as a signed row. The record is zero customer-facing oversell incidents through both go-lives; the dollars are what that decision would have cost."

IF CHALLENGED

> **"Isn't that just a made-up release rate?"**
> The 20% is illustrative and I tag it. What is not illustrative is the shape: the decision flips only above p* = 36.7% (61.2% all-in). The measurement I would ask Merchandising for is the historical release rate by SKU class.

> **"You priced undersell at $4.75 — where from?"**
> $19 margin × a 25% share of undersold demand that does not come back — both illustrative; it is a 1 : 6.3 asymmetry on S1's benchmark, not a measured number. If the share were 100% the ratio would still be 1.6 : 1 against promising.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you disagreed with a stakeholder and held your position.	Have Backbone; Disagree and Commit	The priced no on promisable reserve; visibility offered; the no written as a signed row.	Do not frame it as winning a fight.
Tell me about a time you protected the customer at a cost to the business.	Customer Obsession	Visible undersell accepted so oversell is zero by construction; asymmetry 1 : 6.3.	Do not claim the dollars as saved.
How do you decide whether to promise stock that is physically present?	Functional — promise design	Pickable vs present; p* = c ÷ (g + c); the promise reads published AFS, never computes.	Never 'I configured ISM'.
Tell me about a time you considered the broader impact.	Success and Scale Bring Broad Responsibility	Oversell as a customer harm at scale; the dollar price is the floor, not the whole.	Do not invent a customer count.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

HAVE BACKBONE; DISAGREE AND COMMIT · AIQB Q1 **Q1. "Tell me about a time when you strongly disagreed with your manager or peer on something you considered very important to the business."**

Merchandising wanted the merch reserve to be promisable — stock in the building, why not sell it? I disagreed, but not by arguing. I priced the ask: promising a unit that may not be released only pays if more than about 37% of reserved units actually are released; on a 20% release rate that is about minus fifty thousand dollars per ten thousand units on the cancellation cost alone and about minus two hundred thousand on S1's all-in cost. Then I offered what they actually needed: visibility — the reserve shown, not promised. The no went into the FSD as a row signed by Strategy, Inventory Management and Inventory Accounting, so it was a decision, not a meeting minute that would be re-litigated weekly. The comingle-reserve improvements were held out of the release to protect zero incidents and priced as a later enhancement at steering. I disagreed once, with a number, and committed in writing.

Number to land: p* = 36.7%; −$50K to −$202K per 10,000 units (illustrative) · **Pivot to NACF:** At NACF the same break-even decides when to promise ahead of a physical state — received-not-stowed, in transit — a number, not a feeling. · **Do NOT say:** that I won; that the dollars are saved; that the release rate was measured.

CUSTOMER OBSESSION · AIQB Q7 **Q2. "Sometimes customers make unreasonable requests. Tell me about a time when you've had to push back or say no to a customer request."**

I accepted visible undersell so that oversell would be zero by construction. The design policy was an asymmetry: an oversell is a customer defect — someone pays for something we do not have — and an undersell is a planning variance Inventory Control chases. On S1's benchmark one oversell costs about thirty dollars and an undersold unit about five, roughly a six-to-one asymmetry, so the supply basis is the lower of the two books and the reserve never enters it. That means some sellable stock shows as unavailable. I said so in the FSD and made the control owners own the screens that chase it. The result: customer-facing oversell incidents that were recurring across three node types went to zero through both go-lives. The cost of the policy is visible undersell, which I will always take over a broken promise.

Number to land: oversell incidents recurring → zero; asymmetry ≈ 1 : 6.3 · **Pivot to NACF:** At NACF a promise date is a customer commitment; I would take a slightly conservative buffer over a missed date on the same arithmetic. · **Do NOT say:** a customer count; that undersell is free.

EARN TRUST · AIQB Q3 **Q3. "Describe a time when you needed to influence a peer who had a differing opinion about a shared goal."**

The relationship was the reason I started with what Merchandising needed, not what they asked for. They asked for the reserve to be promisable; what they needed was to see it and to know when it could be released. So the first thing I said was yes: yes to visibility, in ISM, alongside AFS. Then I said no to the mechanism and showed my arithmetic — the break-even and the cancellation cost — so the no was not an opinion. I made the cost to them explicit: the reserve could not be sold in the channel until released. I put the no into the FSD as a row so nobody had to defend it in a hallway, and the comingle-reserve improvements were held out of the release and priced rather than refused. The FSD was signed by three orgs (user anchor, master S1) with the row in it. The test of trust is that the no was written down and the offer was real.

Number to land: 3 orgs signed the row · 0 open 'who owns this bucket' (both user anchor) · **Pivot to NACF:** At NACF I would open every no with the need I can say yes to and the number that makes the no checkable. · **Do NOT say:** that Merchandising agreed happily; I do not know their private view.

SUCCESS AND SCALE BRING BROAD RESPONSIBILITY · AIQB Q5 **Q4. "Tell me about a time when you considered everyone who was affected by your work, not just your direct customers or people within your organization."**

An oversell looks like a ledger error to a planner and a broken promise to a customer. At the scale of this DC — an illustrative thirty million orders a year at a benchmark 0.10% oversell rate — that is thirty thousand customers a year who paid for something we did not have. I did not let the dollar comparison be the whole argument for the no on promisable reserve. The dollar figure — fifty to two hundred thousand per ten thousand reserved units — is the floor of the case; the rest is what it does to a customer's trust in a promise. So the design treats a broken promise as a defect with an owner and a screen, not a variance to be tolerated. The record: customer-facing oversell incidents zero through two go-lives. At Amazon scale that is the difference between a rare event and a daily one.

Number to land: 30,000 incidents/yr (illustrative) → 0; zero through two go-lives · **Pivot to NACF:** At NACF a promise date on a Prime order is the same contract; I would run the same defect-not-variance logic on the promise seam. · **Do NOT say:** a real customer count; that 30,000 is measured — it is S1's illustrative orders × benchmark rate.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP	CONFLICT / INFLUENCE ANGLE · HAVE BACKBONE; DISAGREE AND COMMIT	FAILURE / LEARNING ANGLE · EARN TRUST
"I turned a stakeholder request into a break-even: promise the reserve only above 37% release." **Headline metric:** p* = 36.7% (61.2% all-in); −$50K per 10,000 units at 20% The decision is a one-line formula, p* = c ÷ (g + c), with two cost views; it flips only if the release rate is measured above the break-even. Honest limit: the 20% release rate is illustrative.	*"Merchandising wanted something I would not give them — so I gave them everything else."* **Headline metric:** visibility yes · promisability no · 3 orgs signed (user anchor) Priced no; visibility offered; the no written as an FSD row; the comingle-reserve improvements held out and priced as a later enhancement.	*"I priced the no with a release rate I never measured."* **Headline metric:** release rate 20% illustrative vs p* = 36.7% The row in the FSD carried the decision, but not the measurement that would flip it: the historical release rate by SKU class. Do differently: put the ask for that measurement in the same row.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

How do you know it was the right call and not just caution?

It is right as long as the release rate is below 36.7% (61.2% all-in). The record is zero oversell incidents; I cannot prove the counterfactual. The number that would change my mind is a measured release rate above the break-even with a hold-until date.

Say: *"The decision is right below a 37% release rate; I did not measure it, so the row I would add is that measurement — if it came back above 37%, I would reopen it."*

Isn't splitting the cost into $11 and $30 a way to get the answer you want?

No — the decision holds on both: −$50K and −$202K per 10,000 units at 20%. It flips only above p* in either view, and the two views bracket it between 36.7% and 61.2%.

Say: *"I show both cost views because the answer does not depend on which you believe — it depends on whether the release rate is above 37% or 61%."*

Who bore the cost of your no?

Merchandising: for a period the reserve was visible but not sellable in the channel until released; the comingle-reserve improvements were deferred and priced at steering.

Say: *"Merchandising bore it — visible but not sellable until release; the comingle-reserve improvements were held out of the release to protect zero incidents and priced, not promised."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. At the Utah DC I said no to making merch reserve promisable — stock in the building but not for sale — by pricing the no: it pays only above about 37% release, which on a 20% illustrative rate is −$50K per 10,000 units. I offered visibility instead, wrote the no into the FSD, and customer-facing oversell went to zero through both go-lives.

NACF bridge. NACF has the same seam on the dock: units received but not yet stowed, quarantined or in transit are in the building and not pickable. The promise should read pickable; p* = cost ÷ (gain + cost) — INV-D's 3 ÷ 12 = 25% shape — says when promising ahead of the stow pays. 10,000 reserved units at 0.25 ft³ = 2,500 ft³: CAP-C's cube in miniature.

U4 — Where the node's constraint actually bound — labor hours, then pick accuracy, then cube: one capacity envelope that priced the robotics throttle and the pooled space NEW

LPs this story serves here: Bias for Action (primary — the throttle decision the week the gate failed) · Frugality (price both sides; pay for the cheaper one) · Dive Deep (the constraint was accuracy, not robot speed) · Strive to be Earth's Best Employer (the floor protected during the ramp)
Process family: Make / Deliver — DC capacity and throughput analysis: labor-hour envelope, automation ramp, constraint identification, cube

Honesty label. Record: robotics gate 98% published before the ramp; week two at 94%; bots throttled to ~30% of volume for a week; zone/slotting logic root-caused; full volume week three; +15% processing efficiency (Sephora-signed, measured by DC Ops); DC Ops owned the gate and the dial; peak (Oct–Dec) untouched. **Modeled / illustrative (S2's DC):** 100,000 units/week, 120 FTE × 2,000 h, $28/h. **Mine to label:** $12 per mis-pick, linear hours-versus-bot-share, 10,000 attempted picks, $25 per unit and 0.25 ft³ per unit — all illustrative. I priced the trade; DC Ops ran the floor; I do not claim whose budget carried the hours.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Bias for Action. The week-2 read was 94 against a published 98; the dial, built in advance and owned by DC Ops, went to ~30% the same week — a reversible decision with a number.

Frugality. I priced both sides: ≈$11.8K of extra labor hours against ≈$50.4K of mis-pick exposure, and paid the cheaper one with existing labor — no new tooling.

Dive Deep. The robots could pick; they could not pick right — the binding constraint at week 2 was accuracy, root-caused to zone and slotting logic, not robot speed.

Strive to be Earth's Best Employer. The throttle kept the floor from absorbing the mis-picks and the rework; the interim reporting burden was named as the price. No headcount or attrition figure is claimed.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Sephora's Utah DC was ramping Locus robotics behind a published 98% pick-accuracy gate, on an illustrative 100,000-units-a-week floor running 4,615 labor hours a week. Week two read 94%. **Task.** Decide, with DC Ops, how much robot volume the floor could carry — and where the node's constraint actually sat. **Action.** Analysis first: before the ramp the binding constraint was labor hours; in week two it had moved to accuracy, not robot speed. Four points of accuracy at full bot share is 6,000 mis-picks a week against a 2,000 budget. I priced the dial: a throttle to 30% keeps bot-side errors at 1,800, inside the full-volume budget — the maximum safe share is 33.3% — at a cost of about 421 extra labor hours, roughly $11,800, against about 4,200 mis-picks avoided, roughly $50,400 on an illustrative $12 each. DC Ops owned the dial before the ramp, so the rule fired and I did not improvise. Zone and slotting logic was root-caused and fixed; full volume in week three. Pooling had freed the safety stock the node did not need to hold — about $2.2M modeled, roughly 22,000 cubic feet on illustrative unit cube. **Result.** Primary metric: the gate held — 94 against 98 became full volume in week three, with +15% processing efficiency Sephora-signed and zero customer-facing incidents. The dollars are a model; the gate, the throttle and the +15% are the record.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — One physical node, one robotics ramp, one published gate. The floor was running at baseline productivity (index 100); the plan was +15% at stabilization; the robots were the lever. The week-2 gate read 94 against 98 with the Oct–Dec peak freeze ahead. Everyone could see the gap; nobody had a number for what each response would cost.

T — Say where the constraint bound and what each dial setting cost, on one page, in the week it mattered — and make sure the rule that fired was the floor owner's, not mine.

Baselines on the table: gate 98% (record) · week 2 = 94% (record) · 240,000 h/yr = 4,615 h/wk on an illustrative 100K-units/week node · target index 115 → 4,013 h/wk · full-share exposure 6,000 mis-picks vs a 2,000 budget · peak freeze Oct–Dec.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — profiled 801/901; read the node's labor and throughput as an envelope — hours per week, units per hour, what the floor could ship; success criteria included peak untouched. *Exit bar:* criteria signed by steering; every impacted system owned.
- **Plan & Design (5/15 → 8/14)** — wrote the gate (98% pick-accurate completed ÷ attempted), the throttle dial and the weekly read in the council pack before the robots arrived; DC Ops took ownership of the gate and the dial. *Exit bar:* council 7/31; FSD 8/14; the dial owned before the ramp.
- **Build & Test (Aug → Jan/Apr)** — the Locus gate definition and throttle spec in the build; the 3-day Locus workshop; RTL slipped three weeks, escalated the week it surfaced, re-slotted openly, UAT not compressed. *Exit bar:* Locus RTL retested after the slip; E2E days

dated.

- **Cutover & Go-Live (January; May 1 → Jul 1)** — ramp read weekly; week 2 = 94 against 98 → throttle to ~30% for a week → zone/slotting RCA → week-3 re-read at the same gate. *Exit bar:* rule fired; full volume week 3.
- **Hypercare / Close (Jun → Jul)** — +15% measured post-stabilization by DC Ops; hypercare exit week 3 of 4; the dial and the gate handed to the next DC as a template. *Exit bar:* exit at < 2 blockers/day sustained.

Influence and ownership. Mine: the capacity envelope, the dial arithmetic and the go/no-go on the ramp read. DC Ops: the gate, the dial, the floor and the +15% measurement. Locus and platform teams: the build and the slotting fix.

> **Designed the process; others own it now.** DC Ops owns the dial and the weekly read; the template went to the next DC. **Trade-off put to leadership:** a week of robot volume for a root-caused ramp — recommended with the arithmetic on both sides.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — modeled on S2's illustrative DC: throttle premium ≈421 h × $28 = $11,799 against ≈4,200 mis-picks × $12 = $50,400 avoided; break-even mis-pick cost $2.81. Pooled safety stock released ≈$2.2M ≈ 22,000 ft³ (modeled). Not claimed as savings.
- # — gate 98%, week 2 at 94%, throttle ~30% for one week, full volume week 3 (record); 4,615 → 4,013 labor hours/wk at +15% (modeled on the record index).
- **%** — +15% processing efficiency (record, Sephora-signed); max safe bot share 33.3% against the ~30% dial (arithmetic).
- **Time** — one week at ~30%; week-3 re-read at the same gate; hypercare exit week 3 of 4; ~92 days of float from Jul 1 to the Oct freeze (lower bound).
- **Customer** — zero customer-facing incidents through both go-lives; peak untouched (record).
- **Quality** — the gate stayed a gate — it fired at 94 and was re-read at the same number; the cause (zone/slotting logic) was removed, not routed around.

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No agreed stop condition for the ramp. I defined the gate, the dial and the weekly read before the robots arrived.
Scope & influence	DC Ops owned the floor and the dial; Locus and platform teams the build; I owned the envelope and the recommendation.
Execution trade-off + mechanism others own	A week of robot volume for a root-caused ramp; the mechanism others own: the dial and the weekly read.
Impact	+15% processing efficiency; full volume week 3; peak untouched; zero incidents.
Process improvement	Gate → dial → RCA → re-read as standard work for the next DC.
Complexity & consensus	WMS upgrade, robotics and peak freeze on one critical path; Operations, IT and Supply Chain Strategy in steering.
Root cause at scale	The week-2 miss was zone/slotting logic — fixed at the cause before full volume.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: node × week × bot share. Data: labor hours, units, attempted and accurate picks, mis-pick cost. Cuts: accuracy by week; hours by share; exposure by share. Revealed: the constraint had moved from labor hours (pre-ramp) to accuracy (ramp) — robot speed was never the binding limit.

2 · Solution design. Options: (A) hold full volume and fix live; (B) throttle to 0 and re-test later; (C) throttle to ~30% and keep re-reading. Chose C — A's exposure is 4,000 mis-picks over budget; B costs ≈$5.1K more and loses the live volume the gate reads.

3 · Math & data logic. hours(s) = $H_0 - s(H_0 - H_1)$; premium = $(1 - s)(H_0 - H_1)$; errors = 6% × s × volume; s_max = (1 − 0.98)/(1 − 0.94) = 33.3%; z = 0.04/√(0.98 × 0.02/n).

4 · Implementation. Gate and dial published and owned by DC Ops before the ramp; weekly read; RCA; re-read; hypercare exit. Risks: ramp with no stop condition, two live truths, peak collision.

5 · Probe questions. Below: why 30, why not 0, is 94 noise, where was the constraint, and the three trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — why a 30% throttle, not 0% or 50%?

At 30% the bot-side errors are 1,800, inside the 2,000-per-week budget the gate allows at full volume; the maximum safe share is 33.3%. At 50% errors would be 3,000 — over. At 0% the premium rises to ≈$16.9K and the gate has no live bot volume to read.

Infra — is 94 versus 98 noise?

No. On an illustrative 10,000 attempted picks the standard error is 0.14 points and the gap is 28.6σ; even on 500 picks it is 6.4σ. The week-2 read was a real miss, not sampling.

> **Infra — where did the node's constraint actually bind?**
>
> Pre-ramp: labor hours — 4,615 h a week for 100K units. In the ramp: accuracy, not robot speed — the zone/slotting logic. After: labor again, at +15% — 4,013 h. I did not claim space bound; I do not know the Utah cube utilization VERIFY-WITH-RAJESH.

> **Infra — what did the throttle cost, and who paid?**
>
> About 421 extra labor hours ≈ $11,799 on the illustrative $28 — about 10.5 FTE-weeks at 40 hours. I do not claim whose budget carried them; DC Ops ran the floor.

> **Infra — did the throttle create a backlog?**
>
> Not on this arithmetic: at fixed baseline hours the node's capacity under a 30% share is ≈104,072 units a week against 100,000 — 4.1% headroom. It is a linear model; the real node's hours were not fixed.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Bot share (week 2)	100% planned	~30% (record)	errors 6,000 → 1,800	record / arithmetic
Pick accuracy (gate)	94% (week 2)	≥98% re-read week 3	full volume	record
Labor hours per week	4,615	4,013 at index 115	−602 h/wk	modeled on record +15%
Throttle week premium	—	+421 h	$11,799 vs $50,400 avoided	modeled / illustrative
Cube equivalent of pooled release	—	22,000 ft³	≈2.2% of CAP-C's 1.02M ft³	illustrative

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Baseline weekly hours	`240,000 / 52 = 4,615.4 h/wk; 100,000 / 4,615.4 = 21.67 units/h`	120 FTE × 2,000 h; 100K units/wk (S2 illustrative)	4,615 h/wk
2	Gate vs week-2 read	`98% − 94% = 4 pts`	record	4 pts short
3	Exposure at full bot share	`6% × 100,000 = 6,000 mis-picks; budget at the gate 2% × 100,000 = 2,000; excess = 4,000`	94% vs 98% on illustrative volume	4,000 over budget
4	Is it noise?	`SE(n=10,000) = √(0.98 × 0.02 / 10,000) = 0.0014; z = 0.04 / 0.0014 = 28.6`	10,000 attempts illustrative; z(n=500) = 6.4	28.6σ
5	Where the constraint bound	`pre-ramp: labor (21.67 u/h × 4,615 h = 100,000); ramp: accuracy T = 94% < 98%`	robot speed was not binding	accuracy
6	Calendar float	`Jul 1 → Oct 1 = 31 + 31 + 30 = 92 days`	peak freeze Oct–Dec (record); start date VERIFY	92 days (lower bound)

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Max safe bot share	`s_max = (1 − 0.98) / (1 − 0.94) = 0.02 / 0.06 = 33.3%`	bot-side errors ≤ the full-volume error budget	33.3%
2	Dial = ~30% (record)	`errors = 6% × 30,000 = 1,800 ≤ 2,000; avoided = 6,000 − 1,800 = 4,200`	inside the budget by 200	4,200 avoided
3	Hours at 30% share	`0.7 × 4,615.4 + 0.3 × 4,013.4 = 3,230.8 + 1,204.0 = 4,434.8`	linear in bot share (illustrative)	4,434.8 h
4	Premium vs plan	`4,434.8 − 4,013.4 = 421.4 h = 10.5% of plan hours ≈ 10.5 FTE-weeks`	40-h week (illustrative)	421.4 h
5	Capacity at fixed hours	`100,000 × 4,615.4 / 4,434.8 = 104,072 units/wk`	baseline hours held	104,072 (no backlog)
6	Upper bound premium	`0% share: 602.0 h × $28 = $16,856`	full manual week	≈$16.9K
7	Re-read	`week 3: T ≥ 98% at the same gate → full volume`	record	full volume week 3
8	Space — what pooling freed	`$2.2M / $25 = 88,000 units; × 0.25 ft³ = 22,000 ft³`	$25/unit and 0.25 ft³ illustrative	22,000 ft³

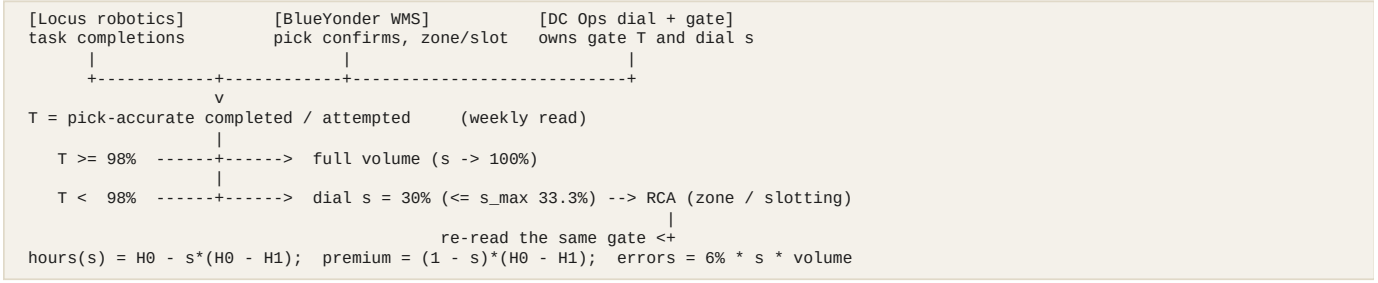
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Bot share, week 2	100% planned	~30%	record
2	Pick accuracy	94%	≥98% at week-3 re-read	record
3	Full-share mis-picks / wk	6,000	≤2,000 at the gate	illustrative volume
4	Labor hours / wk (same volume)	4,615	4,013 (index 115)	modeled on record +15%
5	Customer-facing incidents	0 allowed	0 through both go-lives	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Throttle premium	`421.4 h × $28 = $11,799`	$28 loaded (modeled)	≈$11.8K
2	Mis-picks avoided, priced	`4,200 × $12 = $50,400`	$12 per mis-pick (illustrative)	≈$50.4K
3	Net	`$50,400 − $11,799 = $38,601`	rows 1–2	≈$38.6K
4	Ratio and break-even	`$50,400 / $11,799 = 4.3×; break-even cost per mis-pick = $11,799 / 4,200 = $2.81`	decision holds above $2.81	4.3× / $2.81
5	Premium as share of the annual labor benefit	`$11,799 / $876,400 = 1.35%`	S2's ≈$0.88M/yr (modeled)	1.35%
6	Upper-bound premium	`602.0 h × $28 = $16,856`	0% share	≈$16.9K

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Mis-pick cost	$6 → $25.2K avoided	$12 → $50.4K	$20 → $84.0K
Dial setting	20% → premium $13.5K, errors 1,200	30% → $11.8K, errors 1,800	33.3% → $11.2K, errors 2,000
Attempted picks n	500 → 6.4σ	10,000 → 28.6σ	100,000 → 90.4σ
Loaded labor rate	$24 → $10.1K	$28 → $11.8K	$32 → $13.5K
Break-even mis-pick cost	—	$2.81	decision holds above this

SPOKEN STATEMENT

> "Before the robots arrived the node's binding constraint was labor hours; in week two it was accuracy — 94 against a published 98 — not robot speed. On an illustrative hundred-thousand-unit week that is about 4,000 mis-picks over the gate's budget at full share. A throttle to 30% kept the errors inside the budget, because the maximum safe share is 33%, at a cost of about 420 extra hours, roughly twelve thousand dollars, against about fifty thousand dollars of mis-picks avoided — modeled, on a twelve-dollar mis-pick I label illustrative. DC Ops owned the dial, so the rule fired; the zone and slotting logic was fixed and week three ran at full volume. The +15% and the zero incidents are the record."

IF CHALLENGED

> **"The $12 mis-pick cost is invented."**
>
> Yes, illustrative — and the decision does not depend on it: the throttle pays above $2.81 per mis-pick, so the answer holds across any plausible cost. The record is the gate, the dial and the week-3 re-read.

> **“You only fixed it because the slotting logic was wrong — the throttle was irrelevant.”**
>
> Two levers: the throttle bounded exposure while the fix was made; the fix removed the cause. I cannot separate their contributions in the record, and I would not claim to — but the dial let the re-test run on live volume.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you made a fast decision with imperfect data.	Bias for Action	Week-2 read; dial pre-built; ~30% the same week; reversible.	Do not say ‘the robots failed’.
Tell me about a time you did more with less / priced a trade-off.	Frugality	$11.8K vs $50.4K with existing labor; 4.3×; break-even $2.81.	Do not claim the dollars as savings.
How do you identify where the constraint is?	Dive Deep / Functional	Labor hours → accuracy → labor; speed was never binding.	Never claim space bound — unknown.
How did you protect the people on the floor?	Strive to be Earth’s Best Employer	Throttle kept mis-picks and rework off the floor; DC Ops owned the dial; interim reporting burden named.	Invent no headcount or attrition.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

BIAS FOR ACTION · AIQB Q1 **Q1. “Give me an example of a calculated risk that you have taken where speed was critical.”**

Week two of the robotics ramp read 94 against a published 98. The gate was published before the robots arrived, and DC Ops owned the throttle dial, so I did not need a meeting to decide whether to act; I needed to decide how far to turn the dial. I priced it that week: at full bot share four points of accuracy is about 6,000 mis-picks a week against a 2,000 budget; a throttle to 30% keeps the bot-side errors at 1,800, inside the budget — the maximum safe share is 33.3% — and costs about 420 extra labor hours. The decision was reversible: re-read the same gate in a week. I dialed to roughly 30%, root-caused the zone and slotting logic, and ran full volume in week three. Speed came from the decision being built before the pressure arrived.

Number to land: 94 vs 98 → ~30% dial → full volume week 3 · **Pivot to NACF:** At NACF the same pattern — a published ceiling, a pre-built reversible response, a weekly re-read — applies to a new FC’s inbound ramp. · **Do NOT say:** that the robots failed; that the dial was my invention — DC Ops owned it.

FRUGALITY · AIQB Q4 **Q2. “Tell me about a time when you had to make tradeoffs between quality and cost.”**

I did not buy a fix; I priced a setting on a dial that already existed. The choice in week two was between running at full volume and absorbing the mis-picks, or slowing the bots and covering the volume with the floor’s existing labor. I put both on one page: the throttle costs about 421 hours — roughly eleven thousand eight hundred dollars at an illustrative $28 — and avoids about 4,200 mis-picks, which is fifty thousand dollars at twelve dollars each, or any cost above about two dollars eighty. That is 4.3 to 1 on illustrative inputs. No new tooling, no new heads; the dial existed because DC Ops and I had built it before the ramp. And the pooled safety stock freed about 22,000 cubic feet of space the node no longer had to hold. The record is the gate holding and the +15%; the dollars are my model and I say so.

Number to land: ≈$11.8K vs ≈$50.4K (4.3×); break-even $2.81 · **Pivot to NACF:** At NACF a node’s constraint is priced the same way — flex labor against dwell, CAP-B and CAP-E — before any new capacity is bought. · **Do NOT say:** that the $12 or the $28 is measured; that this was a saving.

DIVE DEEP · AIQB Q3 **Q3. “Tell me about a problem you had to solve that required in-depth thought and analysis.”**

Everyone assumed the question was how fast the robots could go. The data said the constraint had moved. Before the ramp the node was bound by labor hours — 4,615 a week for 100,000 units. In week two the robots were picking; they were not picking accurately: 94 against 98. At a full bot share that is thousands of mis-picks, and more robot speed would have made it worse. I asked what was different about the misses, and the answer was the zone and slotting logic — the cause was in how work was assigned, not in the robots. So the fix went to the logic, and the throttle bought the time. After the fix and the week-three re-read at the same gate, the constraint went back to labor hours — now at +15%, about 4,013 a week. The mistake I avoided was optimizing the non-binding constraint.

Number to land: constraint: labor → accuracy → labor; 94 vs 98; +15% · **Pivot to NACF:** NACF’s capacity chain is the same — doors, UPH, cube, dwell — and the first job is to find which one binds this week. · **Do NOT say:** that I fixed the slotting logic myself; the platform team did.

STRIVE TO BE EARTH'S BEST EMPLOYER · AIQB Q3 **Q4. “Describe a time when you implemented a change at work aimed at improving the work experience of yourself or coworkers.”**

The floor was the one place the cost of a bad ramp would land. If the bots had stayed at full volume at 94% accuracy, every mis-pick becomes rework for someone on the floor, during a go-live. I wrote the gate and the dial before the ramp, and DC Ops owned them — so the people who knew the floor held the stop condition, not a program manager on a schedule. When the gate fired, the throttle kept the mis-picks and the rework off the floor for a week while the logic was fixed. I also named the price of the interim to leadership: for four months every report had to declare which system of record it read. I will not quote a headcount or an attrition number — there is none in the record — but the design principle is the stop condition belongs to the people who carry the consequence.

Number to land: the dial owned by DC Ops before the ramp · **Pivot to NACF:** At NACF the same principle applies to a peak ramp — the dock and the floor own the stop condition for their own safety and quality. · **Do NOT say:** a headcount, an attrition or a safety figure — none exists in the record.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"Four points of accuracy is 6,000 mis-picks a week at full share against a 2,000 budget."

Headline metric: s_max = 33.3% vs ~30% dial; 28.6σ

The throttle share has a maximum that falls out of the gate (0.02/0.06). The record's ~30% sits just inside it. The misses were not noise (28.6σ on an illustrative 10,000 attempts). Honest limit: the volume and mis-pick cost are illustrative.

CONFLICT / INFLUENCE ANGLE · HAVE BACKBONE; DISAGREE AND COMMIT

"Letting a 94% system carry full volume was the path of least resistance; the gate was published before that temptation existed."

Headline metric: 94 vs 98; dial ~30%

Holding the gate was enforcement, not escalation; the trade was visible in dollars on both sides. Record: the alternative — a 94% system carrying full volume into the combined go-live — is how an incident with a robot in the photo happens.

FAILURE / LEARNING ANGLE · OWNERSHIP

"I had the Locus RTL on the plan as a task, not as a risk."

Headline metric: 3-week RTL slip; 94 vs 98

The slip was escalated the week it surfaced and re-slotted openly without compressing UAT, but it had no trigger or owner on the register at the start. Do differently: put vendor RTL on the register as a risk from day one, and publish the zone/slotting test as a gate prerequisite.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

How do you know the throttle — and not the slotting fix — got you to 98?

I don't, and I would not claim to. The throttle bounded exposure at 1,800 errors against a 2,000 budget while the fix was made; the fix removed the cause; the week-3 re-read at the same gate is the evidence of the combination.

Say: *"I can't separate them; the throttle bought exposure control and the time to test the fix on live volume, and the week-3 re-read is the evidence of both."*

Why not 0%? Why not 50%?

At 50% errors are 3,000 against 2,000 — over budget. At 0% the premium is ≈$16.9K instead of ≈$11.8K and the gate has no bot volume to read. 30% sits inside the 33.3% maximum.

Say: *"Thirty is just inside the maximum safe share of thirty-three; fifty breaks the error budget and zero costs five thousand more with nothing to re-read."*

What is the number that would have changed your mind?

A mis-pick cost below $2.81: the throttle would not pay on the arithmetic. Or a re-read at week 3 still below 98: the dial stays, and the RCA reopens.

Say: *"Below about three dollars a mis-pick the throttle doesn't pay, and a week-three read under 98 would have kept the dial down."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. In week two of Sephora's robotics ramp the gate read 94 against a published 98; the constraint had moved from labor hours to pick accuracy, and a pre-built, DC-Ops-owned throttle to about 30% — inside the 33.3% maximum safe share — cost about 421 labor hours against roughly 4,200 mis-picks avoided, and full volume ran in week three.

NACF bridge. NACF's inbound capacity chain has the same shape: CAP-A doors held at an 85% ceiling because above it dwell goes about 3×, CAP-B's learning curve costing 13.6% more hours — the throttle week was +10.5% — and CAP-E's ~14 flex heads. The throttle's 421 hours is about ten and a half people for a week; the question for any new FC ramp is which of doors, UPH or cube binds this week, and who owns the dial.

U5 — The 343-task cutover runbook with one irreversible instant — priced rollback triggers, a three-SKU zero-then-publish trace, and what a week of slip is worth NEW

LPs this story serves here: Ownership (primary — the go/no-go and the rollback were mine to recommend) · Deliver Results (a 61-day window, peak untouched) · Have Backbone (no UAT compression; no-go on one open Blocker) · Dive Deep (the runbook arithmetic and the trace)
Process family: Deliver — cutover and go-live design: runbook, sequencing, rollback triggers, hypercare exit

Honesty label. Record: runbook 343 tasks / 48,885 task-minutes ≈ 815 hours, May 1 → Jul 1, each task with owner, duration and predecessor; six-step May choreography with step 4 (zero-then-publish) the one irreversible instant — I can name steps 1–4: clear negatives, transfer AFS to Uncommitted, scan-and-reserve, zero-then-publish **steps 5 and 6 not in the packet**; six invariants — I can name three (Σ AFS conserved; no SKU sellable on 800 and 900 at once; negatives cleared first) **other three not in the packet**; rollback triggers sync < 99% for four consecutive hours or order-flow p95 > 2× budget for one hour, consumers revert to the prior publish; armed for both go-lives, never fired; Blocker + Critical open = 0; mock 3 inside the 0.5% tolerance; monthly new-row arc 271 → 65 → 21 → 14 → 4 (June → October); hypercare exit week 3 of 4. **Modeled / illustrative:** the three-SKU trace, $110/h, 100K units/week, $30 per incident. User anchor, loop packet only (not in Apple or the master): the day-one pace in units. The weekly discovery curve is computed from the log. I built the runbook and recommendation; platform, DC Ops and vendor teams executed.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Ownership. The go/no-go recommendation, the runbook and the rollback triggers were mine; I wrote the recommendation off counted rows and I would have called no-go on one open Blocker.

Deliver Results. A 343-task, ~815-hour runbook ran inside a 61-day window with the peak freeze untouched: two go-lives, zero customer-facing incidents, hypercare exit week 3 of 4.

Have Backbone; Disagree and Commit. When the Locus RTL slipped three weeks I escalated the week it surfaced, re-slotted openly, and refused to compress UAT to absorb it.

Dive Deep. I read the cutover as arithmetic — task-hours, trigger exposure, a three-SKU trace — so every threshold had a number behind it.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Sephora's Utah DC cutover moved warehouse execution from two systemic warehouses to one comingle pool between May 1 and July 1, with zero customer-facing incidents allowed, an October–December peak freeze behind it, and about a dozen impacted systems in the path. **Task.** Own the runbook and the recommendation — what has to happen, in what order, who does it, and what makes us stop. **Action.** I built a 343-task, 48,885-minute — about 815-hour — runbook, each task with an owner, a duration and a predecessor; found the single irreversible instant, step 4, zero-then-publish, so no SKU was ever sellable on both selling locations at once; and wrote the rollback triggers as numbers per interface: sync below 99% for four consecutive hours, or order-flow p95 above twice the budget for an hour, consumers reverting to the prior publish. I wrote the go/no-go off counted rows — Blocker plus Critical at zero, mock three inside the 0.5% tolerance, runbook rehearsed — and I would have said no-go on one open Blocker. Hypercare was read as a decaying monthly arc of new issue rows, 271 → 65 → 21 → 14 → 4, and exited at under two blockers a day sustained. **Result.** Primary metric: zero customer-facing incidents through both go-lives, rollback armed twice and never fired. The runbook cost about $90,000 of effort at an illustrative $110 an hour — less than three weeks of the modeled benefit it protected.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A multi-system cutover in a fixed window: BlueYonder upgrade, comingle reserve rules, robotics ramp and about a dozen impacted systems moving the same two months, with the peak freeze behind it and no room for a customer-facing incident. The interim ran two live truths.

T — Write the runbook that makes the cutover a sequence of owned tasks, identify the instant that cannot be undone, and fix the stop conditions in numbers before anyone is tired.

Baselines on the table: 343 tasks · 48,885 task-minutes ≈ 815 h · 61 days · one irreversible instant · rollback triggers per interface · mock 3 ≤ 0.5% · Blocker + Critical open = 0 · hypercare exit at < 2 blockers/day.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — systems-impacted table with owner and written LOE per row; the interim-state risks named — two live truths, a ramp with no stop condition, a peak collision. *Exit bar:* every impacted system named to an owner (LOE landed 7/2).
- **Plan & Design (5/15 → 8/14)** — staged two-go-live timeline; peak-freeze boundary decided; the six invariants and six-step choreography drafted; rollback concept — consumers revert to the prior publish. *Exit bar:* council 7/31; FSD 8/14; freeze boundary signed.

- **Build & Test (Aug → Jan/Apr)** — three mock cutovers against the 0.5% tolerance; runbook tasks authored with owner, duration, predecessor; E2E days; the shadow monitor for the interim copy built and tested. *Exit bar:* mock 3 inside tolerance; E2E green three dates; B + C open = 0.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — executed the runbook: clear negatives, transfer AFS to Uncommitted, scan-and-reserve, zero-then-publish; inbound re-pointed W801 → W901 per shipment in Proship/TMS; triggers armed per interface. *Exit bar:* six invariants held; rollback never fired.
- **Hypercare / Close (Jun → Jul)** — daily issue read; weekly Locus gate read; control-owner handoff; hypercare exit memo; template handed to the next DC. *Exit bar:* < 2 blockers/day sustained → exit week 3 of 4.

Influence and ownership. Mine: the runbook, the invariants, the choreography, the trigger numbers and the go/no-go recommendation. Executed by: platform, BlueYonder, Locus and DC Ops teams, each owning their tasks. QA owned the gate; steering signed the criteria.

> **Designed the process; others own it now.** The runbook format (owner · duration · predecessor), the invariants and the numeric hypercare exit are the template handed to the next DC. **Trade-off put to leadership:** the date versus the bar — a counted go/no-go rather than a feeling, and a second cutover rather than two defect classes in one week.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — modeled: runbook effort 814.75 h × $110 = $89,622; one week of the modeled ≈$1.65M/yr benefit ≈ $31,731/wk → the runbook costs ≈2.8 weeks of what it protected. Not claimed as a saving.
- **#** — 343 tasks / 48,885 task-minutes ≈ 815 h; 61-day window; 836-row issues log, 85.4% closed; monthly new-row arc 271 → 65 → 21 → 14 → 4 (record, from the log).
- **%** — new-row arc −98.5% from June to October; sync trigger < 99% for 4 h; p95 trigger > 2× for 1 h (record).
- **Time** — May 1 → Jul 1; hypercare exit week 3 of 4; peak freeze Oct–Dec untouched (record).
- **Customer** — zero customer-facing incidents through both go-lives; rollback armed twice, never fired (record).
- **Quality** — six invariants held; one irreversible instant identified; Blocker + Critical open = 0 at go/no-go.

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A cutover with no agreed stop condition; I defined the invariants, the triggers and the go/no-go bar.
Scope & influence	About a dozen impacted systems, 60+ stakeholders; each task owned by someone who did not report to me.
Execution trade-off + mechanism others own	Two go-lives (a second cutover) versus one (two defect classes in the same week); the runbook format is the mechanism others own.
Impact	Zero customer-facing incidents; two go-lives landed; rollback never needed.
Process improvement	Runbook as owned tasks; irreversible instant named; triggers as numbers; hypercare exit as a number.
Complexity & consensus	Peak freeze, vendor RTL slip, interim two-truth reporting, six invariants across systems.
Root cause at scale	The cutover risk concentrated at one instant — zero-then-publish — so the design put the controls there.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: task × owner × duration × predecessor; SKU × selling location at cutover; issue × severity × day. Cuts: task-hours by day; blockers by day; negatives by SKU. Revealed: the cutover had many tasks and one irreversible instant.

2 · Solution design. Options: (A) one big-bang go-live; (B) two go-lives each carrying one defect class (chosen); (C) a parallel run. A mixes defect classes; C leaves two truths live. B costs a second cutover and buys single-cause diagnosis.

3 · Math & data logic. Σ AFS_800 + Σ AFS_900 conserved; min(AFS_800, AFS_900) = 0 at every instant; triggers: sync < 99% × 4 h → ≈24 events ≈ $714 exposure; p95 > 2× × 1 h; persistence: q^4 vs q.

4 · Implementation. Runbook of 343 tasks; six-step choreography; per-shipment inbound re-pointing; rollback per interface; daily issue read; control-owner handoff. Day-one pace planned in units — 280,000 units, 40 stores, 7 inbound appointments (user anchor, loop packet only) — and a discovery curve of 68 → 160 → 34 issues a week (record, computed from the log).

5 · Probe questions. Below: what made you say go, what did the triggers cost, what's irreversible, why two go-lives, and the three trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — what made you say go?

Counted rows: Blocker + Critical at zero, mock 3 inside 0.5%, runbook rehearsed, rollback armed per interface. I would have said no-go on one open Blocker; I never had to.

Infra — what is irreversible and how did you handle it?

Step 4, zero-then-publish: AFS_800 set to zero, then the same AFS published at 900. The reversible steps came first (clear negatives, transfer to Uncommitted, scan-and-reserve) so by step 4 the only open question was the publish.

> **Infra — what do the triggers cost if they are wrong?**
> On illustrative volume ≈595 units/h, 1% unsynced for 4 h is ≈23.8 events ≈ $714 of oversell exposure; a p95 at 120 s instead of 60 s for an hour adds a 60 s staleness window ≈ 9.9 units ≈ $298. Small against the cost of a false rollback, which is why the persistence requirement matters.

> **Infra — why two go-lives?**
> So we never hunted two defect classes in the same week. January was the planning go-live — purchasing and inbound, reserve rules live against demand, the floor untouched; May left warehouse execution as the only class.

> **Infra — what did the interim cost?**
> For four months every report had to declare which system of record it read — the price of two live truths. I would staff the interim reporting workstream on day one, not month three.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Runbook	unplanned	343 tasks · 815 h · 61 days	$89,622 effort (modeled)	record / modeled
Irreversible instants	unknown	1 (step 4)	—	record
Rollback triggers	none	2 per interface; never fired	exposure ≈$714 / ≈$297	record / illustrative
New issue rows per month	271 (June)	4 (October)	−98.5%	record
Customer-facing incidents	0 allowed	0 through both go-lives	—	record

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Runbook size	`48,885 / 60 = 814.75 h; 48,885 / 343 = 142.5 min per task`	record	≈815 h / 142.5 min avg
2	Window	`May 1 → Jul 1 = 31 + 30 = 61 days; 814.75 / 61 = 13.4 task-hours per day`	record	61 days
3	Irreversible instants	`1 — step 4, zero-then-publish`	record	1
4	Exposure of an unsynced event	`$6 + $4 + $19 + $1 = $30 per oversell`	S1 benchmark	$30
5	Interim state	`two live truths for four months`	record	4 months
6	Arc start	`271 new rows in June (the go-live month)`	record, from the log	271

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Step 1 — clear negatives	`SKU C: AFS_800 = −10 → corrected to 0 (audited); ΣAFS_800 = 560 + 120 + 0 = 680 (was 670)`	illustrative 3-SKU trace; invariant: negatives cleared first	680
2	Step 2 — transfer AFS to Uncommitted	`Σ Uncommitted = 680`	record step	680
3	Step 3 — scan-and-reserve	`reserve the scanned units; no change to Σ`	record step name; effect illustrative	680
4	Step 4 — zero-then-publish	`AFS_800 := 0 for A, B, C; then publish AFS_900 = 560, 120, 0; Σ = 0 + 680 = 680`	one irreversible instant	Σ conserved
5	Double-sellable window avoided	`publish-first would make 680 units sellable on both: 2 × 680 = 1,360 promised on 680`	why zero comes first	never both
6	False-zero window	`≤ 60 s budget; demand in 60 s = 0.165 units/s × 60 = 9.9 units`	100K units/week illustrative	≈9.9 units
7	Asymmetry applied	`undersell 9.9 × $4.75 = $47 vs oversell 9.9 × $30 = $298; ratio 6.3 : 1`	$4.75 = $19 × 25% (illustrative)	6.3 : 1
8	Trigger — sync	`1% × 595.2 u/h = 5.95 events/h × 4 h = 23.8 events × $30 = $714`	595.2 = 100,000/168 (illustrative)	≈$714

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
9	Why persistence	q = 2% healthy per-hour breach: P(4 consecutive) = 0.02^4 = 1.6×10^{-7} vs 0.02 for one hour; over 1,464 windows: 0.00023 vs 29.3 false fires	q illustrative; independence assumed	0.00023 vs 29.3

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	New issue rows per month	271 (June)	4 (271 → 65 → 21 → 14 → 4, October)	record
2	Blocker + Critical open at go/no-go	441 logged	0 open	record
3	SKU sellable on both locations at once	possible in the interim	never — min(AFS_800, AFS_900) = 0	record invariant
4	Rollbacks fired	armed twice	0	record
5	Hypercare exit	—	week 3 of 4 at < 2 blockers/day	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Runbook effort	814.75 h × $110 = $89,622	$110/h blended (modeled)	≈$89.6K
2	Weekly benefit at stake	$1.65M / 52 = $31,731	S2's modeled recurring benefit	≈$31.7K/wk
3	Runbook as weeks of benefit	$89,622 / $31,731 = 2.82 weeks	rows 1–2	≈2.8 weeks
4	Sync-trigger exposure	23.8 events × $30 = $714	illustrative volume	≈$714
5	Staleness exposure at the p95 trigger	9.9 units × $30 = $298	60 s extra window	≈$297
6	One week of slip	$1.65M / 52 = $31,731	deferred benefit (modeled)	≈$31.7K

SOLUTION FLOW — THE SYSTEM & DATA MAP

```
RUNBOOK (343 tasks, owner + duration + predecessor, May 1 -> Jul 1)
 Step 1  clear negatives      (AFS Correction, audited)        reversible
 Step 2  transfer AFS -> Uncommitted                           reversible
 Step 3  scan-and-reserve                                      reversible
 Step 4  ZERO AFS_800 -> PUBLISH AFS_900         <== the one irreversible instant
 Step 5-6 (not in the packet - VERIFY)
      |
 per-interface triggers: sync < 99% for 4 h  |  p95 > 2x budget for 1 h
      | fire -> consumers revert to the prior publish
      v
 hypercare: new rows/month 271 -> 65 -> 21 -> 14 -> 4 ; exit when < 2 blockers/day sustained
 inbound dock side: Proship/TMS re-points wh_id W801 -> W901 shipment by shipment
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Blended rate	$90 → $73.3K	$110 → $89.6K	$130 → $105.9K
Volume per hour (sync trigger exposure)	300 u/h → $360	595 u/h → $714	1,000 u/h → $1,200
Healthy per-hour breach q	1% → 0.00001 false fires	2% → 0.00023	5% → 0.0092
Weekly benefit	$1.1M/52 = $21.2K	$1.65M/52 = $31.7K	$2.4M/52 = $46.2K
Runbook as weeks of benefit	—	2.8	4.2 at the low benefit

SPOKEN STATEMENT

> "The cutover was 343 tasks and one irreversible instant. I put every control at that instant — clear the negatives, transfer to uncommitted, scan-and-reserve, then zero the old selling location and publish the new one — so no SKU was ever sellable in two places and the worst case was a sixty-second false zero, which costs about a sixth of a double-sell on S1's benchmarks. The triggers were numbers per interface, and the persistence requirement is why they would not false-fire. I wrote the go/no-go off counted rows; the record is zero customer-facing incidents and a rollback that was armed twice and never needed. The runbook cost about $90,000 of effort at an illustrative rate — under three weeks of the benefit it protected."

IF CHALLENGED

“The rollback never fired — how do you know it works?”

I don’t know it end to end under load, and I will say so. What the record has is that the triggers were armed per interface, the failback concept was consumers revert to the prior publish, and the three mocks ran against the tolerance. The right do-differently is a timed rollback drill in mock three.

“Why four hours?”

Persistence. At an illustrative 2% healthy per-hour breach, four consecutive hours is 1.6×10^{-7} per window; one hour would false-fire about 29 times over the cutover. The assumption is independence — correlated outages break it — so the p95 trigger is the fast path.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a cutover you owned.	Ownership	343 tasks; one irreversible instant; triggers as numbers; go/no-go off counted rows; zero incidents.	Do not claim I executed the tasks.
Tell me about delivering under a tight timeline.	Deliver Results	61 days; peak untouched; two go-lives; a week of slip is ≈$31.7K modeled.	Do not bank the benefit.
Tell me about a time you held a line on quality under schedule pressure.	Have Backbone	No UAT compression; no-go on one open Blocker; escalated the RTL slip the week it surfaced.	Never claim a no-go that did not happen.
How do you decide when to roll back?	Functional — cutover design	Numeric triggers per interface; persistence; revert to prior publish; a timed drill.	Never say it ‘worked in prod’.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

OWNERSHIP · AIQB Q4 **Q1. “Tell me about a time when you made a hard decision to sacrifice short term gain for something that would create long term value for the business.”**

I recommended giving up speed for certainty — two go-lives and a four-month bake — and wrote the go/no-go off counted rows rather than confidence. By the time of the May cutover Blocker plus Critical open was zero, mock three was inside the 0.5% tolerance, the 343-task runbook had been rehearsed, and rollback was armed per interface. I would have called no-go on one open Blocker, and I meant it. The runbook had one irreversible instant, step four, zero-then-publish, and I put every control at that instant: negatives cleared first, AFS transferred to uncommitted, scan-and-reserve, then zero the old selling location and publish the new. When something needed to be decided, it was decided against a number I had written before anyone was tired. The result was zero customer-facing incidents through both go-lives and rollback armed twice, never fired. The interim reporting burden — four months of every report declaring its system of record — was a cost I should have staffed earlier, and I say that too.

Number to land: 343 tasks · 1 irreversible instant · 0 incidents · rollback armed twice, never fired · **Pivot to NACF:** At NACF the owner of a new-FC launch writes the go/no-go off counted rows — doors tested, appointments scheduled, UPH validated — and the stop conditions before the first live trailer. · **Do NOT say:** that I executed the tasks; that the rollback was proven under load.

DELIVER RESULTS · AIQB Q1 **Q2. “Give me an example of a time when you were able to deliver an important project under a tight deadline.”**

The window was 61 days, May 1 to July 1, with the October to December peak freeze behind it — so slipping was not free. I wrote the runbook as 343 owned tasks, about 815 hours, each with a predecessor so the critical path was visible. A week of slip was worth about $31,700 of deferred benefit on S2’s modeled $1.65 million a year; the whole runbook cost about $90,000 of effort, under three weeks of that. So when the Locus RTL slipped three weeks I escalated the week it surfaced and re-slotted rather than letting it sit. Two go-lives landed on their dates, the monthly count of new issue rows fell 271 to 65 to 21 to 14 to 4, and hypercare exited in week three of four at under two blockers a day. The +15% processing efficiency was measured post-stabilization by DC Ops. The date held because every task had an owner and a predecessor.

Number to land: 61 days · 815 h · new rows 271 → 4 a month · exit week 3 of 4 · **Pivot to NACF:** At NACF a new-FC date lives or dies on the same arithmetic — doors, hours, appointments — written as owned tasks. · **Do NOT say:** that the $31.7K is banked; that the $90K runbook cost is a record.

HAVE BACKBONE; DISAGREE AND COMMIT · AIQB Q5 **Q3. “Tell me about a time when you pushed back against a decision that negatively impacted your team.”**

The Locus RTL slipped three weeks, and the easy way to absorb it was to compress UAT. I refused. I escalated the slip the week it surfaced — not when it hurt — re-slotted the plan openly so everyone saw the new dates, and held UAT at its full length; the robotics were retested after the slip. My argument was the bar itself: Blocker plus Critical open had to be zero at go/no-go, and compressing UAT would have moved findings from the test room to the floor. One open Blocker is a no-go; I never had to call it, because the bar was written before the pressure arrived.

Number to land: RTL slipped 3 weeks · UAT not compressed · B + C open = 0 · **Pivot to NACF:** At NACF the same line applies to a launch date — what is the bar, in numbers, that a slipped vendor does not get to lower. · **Do NOT say:** that I overruled steering; the dates were re-slotted openly, not forced.

DIVE DEEP · AIQB Q6 **Q4. "Have you ever created a metric that helped identify a need for a change in your department?"**

I did the arithmetic on the thresholds instead of accepting round numbers. The rollback trigger — sync below 99% for four consecutive hours — sounds arbitrary until you price it: on an illustrative 100,000-unit week, 1% unsynced for four hours is about 24 events, roughly $714 of oversell exposure at $30 each. The four-hour persistence is why it does not false-fire: at an illustrative 2% healthy breach, four consecutive hours is one in six million per window; a single hour would false-fire about 29 times across the cutover. And I traced three SKUs through the irreversible step to prove the invariant: the units across both selling locations sum to the same 680 before and after, and no SKU is sellable on both. The false-zero window costs about a sixth of a double-sell. Detail, not drama.

Number to land: ≈24 events ≈ $714; 0.02^4 vs 0.02; 680 conserved · **Pivot to NACF:** NACF dock and UPH plans run on the same detail — appointment slots and dwell — and I would price every threshold the same way. · **Do NOT say:** that the exposure numbers are measured; they are illustrative on 100K units/week.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"343 tasks, 48,885 minutes, 61 days — and the cutover had one irreversible instant."

Headline metric: 814.75 h · $89.6K · 2.8 weeks of benefit

The runbook arithmetic plus the trigger pricing ($714) and the persistence math show every threshold has a number behind it. Honest limit: volume, rate and q are illustrative.

CONFLICT / INFLUENCE ANGLE · HAVE BACKBONE; DISAGREE AND COMMIT

"When the vendor slipped three weeks, the easy answer was to squeeze UAT. I said no."

Headline metric: RTL +3 weeks; UAT not compressed

Escalated the week it surfaced; re-slotted openly; UAT held at full length; the no-go bar (one open Blocker) was set before the pressure.

FAILURE / LEARNING ANGLE · OWNERSHIP

"For four months every report had to say which system of record it read — and I staffed that workstream in month three."

Headline metric: interim reporting staffed month 3, not day 1

The record's own do-differently: the interim cost was predictable and I treated it as a task, not a workstream. Second honest limit: rollback armed twice, never fired — I cannot prove it under load

confirm whether a timed rollback drill ran in the m

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

How do you know the cutover worked because of the runbook and not because the team was good?

I can't separate them, and a good team is part of the design: each task had an owner, duration and predecessor. The measurable thing is the invariants held (Σ AFS conserved, no SKU sellable twice) and the monthly new-row arc 271 → 4.

Say: *"The runbook's job was to make good people's work sequenceable; the evidence is that the invariants held and the monthly new-row arc went 271 to 4 — not that the team was lucky."*

Why not one go-live and save the second cutover?

The second cutover buys single-cause diagnosis: in a one-go-live world every issue triages against two hypotheses. The cost of the second go-live is real and in the record: a deliberate four-month bake running two systems of record, with daily reconciliation (Apple C1). The decision is qualitative — triage time, not fix time, is what a 160-issue week costs, and the issues log confirms that week: 68, then 160, then 34 identified in the weeks of June 9, 16 and 23 (computed from the file).

Say: *"I paid for a second cutover to keep diagnosis univariate; that is a judgment call, and I put the price on the table at steering."*

What would have changed your mind on go?

One open Blocker; mock 3 outside 0.5%; a sync read below 99% in the rehearsal; or a rollback I couldn't describe in a sentence per interface.

Say: *"Any one of four numbers — one open Blocker, mock three over half a percent, sync under 99%, or a rollback I couldn't state per interface."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. I owned the Utah DC cutover runbook — 343 tasks, about 815 hours, a 61-day window — with one irreversible instant, zero-then-publish, rollback triggers written as numbers per interface (sync under 99% for four hours; p95 over twice budget for an hour), and a go/no-go off counted rows; zero customer-facing incidents through both go-lives and rollback never needed.

NACF bridge. NACF's new-FC launch has the same irreversible instant — the first live inbound appointment — and I would write it the same way: an owned-task runbook, a numeric go/no-go, and stop conditions before the first trailer. At Sephora the dock-side version was re-pointing each inbound shipment's warehouse ID W801 → W901 in Proship/TMS, shipment by shipment; at NACF the CAP-A 85% door ceiling is the slack that absorbs a re-pointed or late trailer.

U6 — Negative AFS and the AFS Correction screen — detecting an over-committed SKU, root-causing it to one of three seams, and giving every AFS-changing act a screen, a role and an audit row NEW

LPs this story serves here: Dive Deep (primary — decomposing a negative to its cause class) · Ownership (the control-owner map handed off) · Customer Obsession (a negative is a promise we cannot keep) · Insist on the Highest Standards (drift tolerance zero; negatives clean at T+1)
Process family: Deliver / Return — inventory-accuracy control: exception detection, root cause by cause class, control-owner governance

Honesty label. Record: AFS = Basis − PickRes − EcomPick − EcomPack − SNB; negatives cleared first at cutover; Negative AFS clean at T+1 was a go-live exit bar; daily ISM-vs-MB52 drift with tolerance 0 — a non-zero drift is a Sev, not a planning variance; control-owner map: Negative AFS → Inventory Control, Internal Audit and Finance; AFS Correction → ISM app team; the daily drift report → Inventory Control; every AFS-changing act had a screen, a role and an audit row; the register's top three risks — vocabulary drift, feed latency from WMS/OMS/POS, override abuse on AFS-changing screens — are my three cause classes. **Illustrative:** the −10 SKU, its 5/3/2 split, $30 per incident, the σ = 10 sanity check. Record-from-docs (issues-log file, Comments column): row ID 716 and row ID 818 — the only two of 836 rows that name AFS. I defined the screens, owners and cause classes; ISM app and platform teams built them; the control owners run them.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Dive Deep. A negative AFS is a symptom; I decomposed it into three cause classes — event cardinality, feed latency, override — each with a different fix and a different owner.

Ownership. Every AFS-changing act had a screen, a role and an audit row before I stepped back; the control owners ran the Negative AFS report without me.

Customer Obsession. A negative AFS is units already promised that the floor cannot back; I treated it as a customer defect with a clock, not a variance.

Insist on the Highest Standards. Drift tolerance was zero — a non-zero drift is a Sev — and the exit bar was zero SKUs of drift on consecutive days, with negatives clean at T+1.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Once Sephora's Utah DC became one comingled pool with one published AFS per node per channel, there was one new failure mode: a SKU whose committed buckets exceed what the floor can back — a negative AFS, units we have already promised and cannot ship. **Task.** Own how that is detected, root-caused and fixed, and who is allowed to change an AFS at all. **Action.** I defined the screens and the control-owner map before go-live: the Negative AFS report to Inventory Control, Internal Audit and Finance; the AFS Correction screen to the ISM app team; the daily drift report, ISM-vs-MB52, to Inventory Control, with tolerance zero — a non-zero drift is a Sev, not a planning variance. Every AFS-changing act had a screen, a role and an audit row. Root cause went by cause class — the register's top three: event cardinality, feed latency, override — so a worked SKU at minus ten decomposes into a duplicate event, a lagging WMS feed and a manual override, each with its own owner. The asymmetry behind it: oversell costs about $30 and an undersold unit about $4.75, a 6.3-to-1 ratio, so the published number sits near the 14th percentile of true availability — and the lower-of-two basis gets within a unit of that. **Result.** Primary metric: customer-facing oversell incidents zero through both go-lives; negatives clean at T+1 at cutover; drift at zero SKUs on consecutive days at hypercare exit; the control owners running the report without me.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A comingled pool publishes one AFS per node per channel and the OMS reads it, never computes it. When committed buckets exceed Basis, AFS goes negative: units promised that the floor cannot back. Before ISM, two books were reconciled by hand — about 4 FTE — and a negative surfaced as an order that could not ship.

T — Design detection, cause-class root-causing and ownership for exceptions that change AFS — so that a negative is found, explained, fixed and audited by the people who own the consequence, not by me.

Baselines on the table: ≈4 FTE reconciling two books by hand · drift unmeasured → tolerance 0 · negatives clean at T+1 as a go-live bar · the register's top-three risks (vocabulary drift, feed latency, override abuse) each with a trigger, a mitigation and an owner — mine was drift.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — bucket inventory with one owner per bucket; where in the old world negatives appeared and who chased them; sessions closed 4/30. *Exit bar:* every bucket owned by one system; open questions dated.
- **Plan & Design (5/15 → 8/14)** — wrote AFS = Basis − buckets; asymmetry (oversell customer defect / undersell planning variance); drift tolerance 0; screens and roles; the three cause classes from the risk register. *Exit bar:* FSD signed 8/14; every AFS-changing act has a

screen and a role.

- **Build & Test (Aug → Jan/Apr)** — matrix cells for negatives, replays and overrides (U1); AFS Correction audit-row assertions; SIT contract defects closed before UAT. *Exit bar:* Blocker + Critical open = 0; E2E days dated.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — step 1 of the May choreography: clear negatives first; Negative AFS clean at T+1; the screens live with control owners assigned. *Exit bar:* negatives clean at T+1; no SKU with two published AFS values.
- **Hypercare / Close (Jun → Jul)** — daily drift report; Negative AFS / AFS Correction audit review; control-owner handoff; exit into the monthly release train. *Exit bar:* drift = 0 SKUs on consecutive days; owners running the report without me.

Influence and ownership. Mine: the cause classes, the screens' definitions, the owner map, the tolerance. ISM app and platform teams built the screens; Inventory Control, Internal Audit and Finance run Negative AFS; the ISM app team runs AFS Correction.

> **Designed the process; others own it now.** Control owners and the monthly release train own the screens now. **Trade-off put to leadership:** visible undersell and a zero-tolerance drift — more investigations — against oversell as a customer defect; I accepted the investigations.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — modeled from S1: reconciliation 4 → 1 FTE = 3 × $110K = $330K/yr; baseline oversell exposure 30,000/365 = 82.2 incidents/day × $30 = $2,466/day on illustrative orders. Not claimed as savings; the record movement is incidents → zero.
- **#** — negatives clean at T+1 at cutover; drift = 0 SKUs on consecutive days; 3 cause classes; 3 control-owner orgs + the ISM app team (record).
- **%** — asymmetry 1 : 6.3; critical ratio 13.7% (modeled on S1's benchmark).
- **Time** — T+1 at cutover; hypercare exit week 3 of 4; FSD signed 8/14 (record).
- **Customer** — customer-facing oversell incidents zero through both go-lives (record).
- **Quality** — every AFS-changing act has a screen, a role and an audit row; drift tolerance 0 (record design).

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Nobody owned exceptions to the published number; I defined cause classes, owners and tolerances.
Scope & influence	Inventory Control, Internal Audit, Finance and the ISM app team — four groups that did not report to me — own the screens.
Execution trade-off + mechanism others own	Zero-tolerance drift (more investigations) against ambiguity; the mechanism others own: the screens and the audit rows.
Impact	Oversell incidents zero through both go-lives; reconciliation moved from argument to a daily report.
Process improvement	Exceptions have a class, an owner, a screen and an audit row; the same map applies to any new node or channel.
Complexity & consensus	Three orgs signed the definitions; four groups run the controls.
Root cause at scale	Negatives collapse into three seams (cardinality, latency, override); the fix is at the seam, not the SKU.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: SKU × node × channel × bucket × time. Data: ISM buckets, MB52 snapshot, the audit rows — and, for the legacy-side evidence, the 836-row WMS/Locus issues log: a keyword search of its five text columns finds only two rows that name AFS. Row ID 716 (Critical, IC): 'Dept 96 has 1800 SKUs. 400 are − AFS in MCS and 0 in BY' — 22.2% of that department's SKUs negative in the legacy book and none in the WMS. Row ID 818 (Critical): inventory holds validated moving units 'from AFS to Merch Res after about 10min' — a latency datum inside the ≤15-minute feed budget. Cuts: negatives by cause class. Revealed: the WMS defect book is nearly silent on AFS, so exceptions to the published number needed their own controls in ISM/S/4.

2 · Solution design. Options: (A) reconcile by hand (4 FTE); (B) a daily report with no owners; (C) a screen + role + audit row per AFS-changing act with a cause class (chosen). A and B leave exceptions unowned.

3 · Math & data logic. AFS = 25 − (12 + 10 + 8 + 5) = −10; critical ratio = $4.75 ÷ ($4.75 + $30) = 13.7%; z = -1.10; buffer = 1.10 × σ(10) = 11.0; lower-of-two sits 10 below the mean of the books (990 → 980); drift = 957 − 957 = 0.

4 · Implementation. Screens and owners at design; clear negatives first at cutover; T+1; daily drift at hypercare; handoff and exit memo. Risks: vocabulary drift, feed latency, override abuse.

5 · Probe questions. Below: how do you root-cause a negative, why tolerance zero, what the screen cannot see, who owns what, and the three trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — a SKU shows −10. What do you do?

Decompose by cause class. Cardinality: an event in two buckets (a duplicate SNB) — fix the event, replay is a no-op, add a matrix cell. Latency: a WMS adjustment still inside the ≤15-minute feed plus under-5-minute consumer lag — wait and re-read. Override: a manual AFS change — check role and audit row. On the illustrative split 5/3/2 the first fix takes the −10 to −5.

Infra — why drift tolerance of zero?

Because the books should agree and a difference means one of them is wrong; a non-zero drift is a Sev, not a planning variance. It also made the hypercare exit crisp: zero SKUs of drift on consecutive days. If I had to set a non-zero tolerance I would price it: each unit is up to $30.

Infra — what can the Negative AFS screen not see?

Undersell. It is a one-sided detector: a SKU showing less than the floor could sell never goes negative. The asymmetry makes that deliberate, but it means a wrong rule that starves a channel (my first in-transit rule, master S4) is invisible to it.

Infra — who owns what?

Negative AFS: Inventory Control, Internal Audit and Finance. AFS Correction: the ISM app team. The daily drift report: Inventory Control. Every AFS-changing act has a screen, a role and an audit row; override abuse was the register's third risk.

Infra — why is the lower-of-two the right buffer?

Sanity check, not a derivation: a critical ratio of 13.7% puts the published number about 1.10σ under the mean; if the two books' half-gap (10) is one σ, that is ≈11.0 units — and min(1,000, 980) sits 10 under the books' mean of 990.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Exception ownership	unowned; 4 FTE by hand	screen + role + audit row; ≈1 FTE reading the drift report	≈$0.33M/yr (S1 modeled)	modeled
Drift (ISM vs MB52)	unmeasured	0 SKUs on consecutive days	tolerance 0	record
Negatives at cutover	present	clean at T+1	—	record
Oversell incidents	recurring	zero through two go-lives	baseline ≈$2,466/day exposure (illustrative)	record / illustrative
Asymmetry	implicit	1 : 6.3 (explicit)	critical ratio 13.7%	modeled

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	A negative SKU	`Basis = min(30, 25) = 25; buckets = 12 + 10 + 8 + 5 = 35; AFS = 25 − 35 = −10`	illustrative SKU	−10
2	Exposure	`10 × $30 = $300`	S1 all-in cost per incident (benchmark)	$300
3	Cost of an oversell vs an undersell	`oversell $30; undersell = $19 × 25% = $4.75; ratio = 6.3`	$19 margin; 25% not recovered (illustrative)	6.3 : 1
4	Baseline oversell rate	`30,000,000 × 0.10% / 365 = 82.2/day × $30 = $2,466/day`	S1 illustrative orders; benchmark rate	≈$2.5K/day
5	Reconciliation burden	`≈4 FTE by hand`	record baseline	≈4 FTE
6	Where the AFS evidence sits in the log	`2 of 836 rows name AFS (0.24%); ID 716: 400 ÷ 1,800 = 22.2% of Dept 96 SKUs negative in MCS, 0 in BY`	issues-log file, Comments column (record-from-docs)	22.2%

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cause class — cardinality	`duplicate SNB event: SNB 5 → 0; AFS = 25 − (35 − 5) = −5`	illustrative 5 of the 10	−10 → −5
2	Cause class — latency	`WMS adjustment +3 arrives inside the 15-min feed: Basis = 28; AFS = 28 − 30 = −2`	illustrative 3 of the 10	−5 → −2

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
3	Cause class — override	`unauthorized 2-unit manual hold reversed via AFS Correction (role + audit row): AFS = −2 + 2 = 0`	illustrative 2 of the 10	−2 → 0
4	Decomposition check	`5 + 3 + 2 = 10 = |−10|`	illustrative split	10
5	Critical ratio	`CR = cu ÷ (cu + co) = 4.75 ÷ 34.75 = 13.7%; z = -1.10`	newsvendor (toolkit label)	13.7%
6	Buffer sanity check	`σ = (1,000 − 980) ÷ 2 = 10; buffer = 1.10 × 10 = 11.0; mean of books 990; min = 980 → 10 below`	illustrative; not a derivation	≈11.0 vs 10
7	Daily drift	`Drift = 957 − 957 = 0`	record tolerance 0	0
8	Exit bars	`negatives clean at T+1; drift = 0 SKUs on consecutive days`	record	met

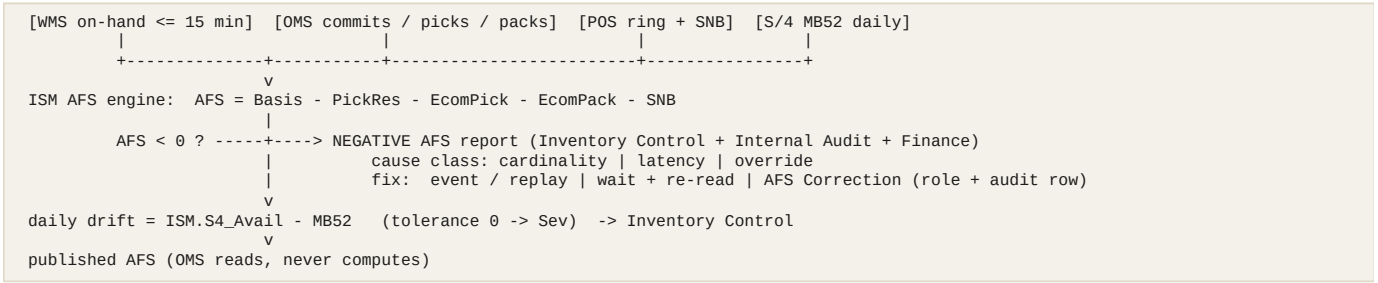
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Exception ownership	unowned / hand reconciliation	screen + role + audit row	record design
2	Drift	unmeasured	0 SKUs on consecutive days	record
3	Reconciliation FTE	≈4	≈1	modeled
4	Oversell incidents	recurring	zero through two go-lives	record
5	Negatives at cutover	present	clean at T+1	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Reconciliation labor	`(4 − 1) × $110,000 = $330,000`	S1 modeled; $110K loaded (benchmark)	≈$0.33M/yr
2	Oversell exposure per day (baseline)	`82.2 × $30 = $2,466`	S1 illustrative orders × benchmark rate	≈$2.5K/day
3	Exposure of the worked negative	`10 × $30 = $300`	illustrative SKU	$300
4	Exposure if the −10 is cured in cause-class order	`−10 → −5 → −2 → 0: remaining exposure $150 → $60 → $0`	illustrative split	$300 → $0
5	Asymmetry in dollars	`$30 vs $4.75; 30 ÷ 4.75 = 6.3`	S1 benchmark; 25% illustrative	6.3 : 1

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Undersell recovered share	50% → cu $9.50 · CR 24.1%	25% → cu $4.75 · CR 13.7%	10% → cu $1.90 · CR 6.0%
Gap between books (σ = half-gap)	10 → σ 5 → buffer 5.5	20 → σ 10 → buffer 11.0	40 → σ 20 → buffer 21.9
Oversell rate	0.05% → $1.2K/day	0.10% → $2.5K/day	0.25% → $6.2K/day
Reconciliation FTE released	2 × $95K = $0.19M	3 × $110K = $0.33M	4 × $130K = $0.52M
Cause-class split (illustrative)	1 / 1 / 8	5 / 3 / 2	8 / 1 / 1

SPOKEN STATEMENT

> “A negative AFS is units we have already promised and the floor cannot back. I did not treat it as one thing: it decomposes into three seams — an event that landed in two buckets, a feed that has not caught up, and a manual override — and each has a different fix and a different owner. Inventory Control, Internal Audit and Finance run the Negative AFS report; the ISM app team runs the AFS Correction screen; every AFS-changing act has a screen, a role and an audit row. The drift tolerance is zero because a difference means one book is wrong. The record is negatives clean at T+1, drift at zero on consecutive days at hypercare exit, and zero customer-facing oversell incidents through both go-lives; the dollar view is an asymmetry — about six to one — on benchmark inputs.”

IF CHALLENGED

“The 5/3/2 split is made up.”

It is illustrative, tagged, and I would not quote it. What is record is the three cause classes — they are the register’s top three risks — and the owner for each. The real split is a pull from the audit rows.

“How do you know the controls — not the matrix — kept incidents at zero?”

I can’t isolate them. The controls and the matrix are two halves: the matrix proves the events are mapped; the screens catch what the mapping did not. The record is the combination: zero incidents, drift at zero, negatives clean.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you found the root cause of a recurring issue.	Dive Deep	A negative decomposes into cardinality / latency / override; fix at the seam.	Never quote the 5/3/2 split as measured.
Tell me about a time you built something that others now own.	Ownership	Screens, roles, audit rows; control-owner map; they ran it without me.	Do not claim I built the screens.
Tell me about a time you put the customer ahead of an internal metric.	Customer Obsession	Visible undersell accepted so oversell is zero by construction; 1 : 6.3.	Do not claim the dollars as saved.
What does a high standard look like in inventory accuracy?	Insist on the Highest Standards	Drift tolerance 0; negatives clean at T+1; zero SKUs on consecutive days.	Do not call 0 tolerance easy.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

DIVE DEEP · AIQB Q2 **Q1. “Tell me about a situation that required you to dig deep to get to the root cause.”**

The recurring problem was a negative available-to-sell — units promised that the floor could not back — and the first instinct is to fix the SKU. I looked at what a negative actually is: committed buckets exceeding the supply basis. Take a worked SKU with a basis of 25 and 35 units committed: minus ten. I decomposed it into three cause classes — the register’s top three: an event that landed in two buckets, a feed that had not caught up, and a manual override. A duplicate event in the shipped-not-billed bucket takes the minus ten to minus five; a warehouse adjustment still inside the fifteen-minute feed takes it to minus two; an unauthorized override, removed through the AFS Correction screen with a role and an audit row, takes it to zero. Each cause class has a different fix and a different owner, so the fix is at the seam, not the SKU. That is why the matrix has a cell for each, and why the control owners had a cause class to write on every audit row.

Number to land: −10 → −5 → −2 → 0 (illustrative); 3 cause classes · **Pivot to NACF:** At NACF, a negative-available ASIN decomposes the same way — a double receive scan, dock-to-stock lag, a manual adjustment — and I would ask which seam first. · **Do NOT say:** that the 5/3/2 split is measured; that I built the screens.

OWNERSHIP · AIQB Q5 **Q2. “Describe a time when you had to transition a project you owned to a new owner.”**

I did not finish the program until the exceptions had owners who did not need me. Before go-live I mapped every AFS-changing act to a screen, a role and an audit row, and every exception report to an owner: the Negative AFS report to Inventory Control, Internal Audit and Finance; the AFS Correction screen to the ISM app team; the daily ISM-versus-MB52 drift report to Inventory Control. In hypercare the test was whether the control owners ran the report without me — part of the exit bar, along with drift at zero SKUs on consecutive days and blockers under two a day. Then the design entered the monthly release train, so a new node or channel is a matrix row and an owner row, not a project. I measure ownership by whether the process survived my leaving.

Number to land: 3 owner orgs + ISM app team; drift 0 on consecutive days · **Pivot to NACF:** At NACF I would hand a launch’s exception reports to named owners with a screen, a role and an audit row before the first live trailer. · **Do NOT say:** that I run the screens now; the control owners do.

CUSTOMER OBSESSION · AIQB Q8 **Q3. "Tell me about a time when you had to balance the needs of the customer with the needs of the business."**

The convenient metric was a clean-looking available number; the customer's interest was a promise we could keep. A negative AFS is a customer we have already told yes. So the policy is an asymmetry: an oversell is a customer defect, about thirty dollars on benchmark inputs; an undersell is a planning variance, about five. That is roughly six to one, and the asymmetry decides the design — the supply basis is the lower of the two books, so the published number sits conservatively under the truth. The price is visible undersell: some sellable stock shows as unavailable, and Inventory Control chases it. I accepted that on purpose. A sanity check supports it: a six-to-one asymmetry puts the right published number about one standard deviation below the mean, and the lower-of-two sits within about a unit of that on the worked SKU. The record is customer-facing oversell incidents going to zero through both go-lives.

Number to land: 1 : 6.3 asymmetry; critical ratio 13.7%; incidents → zero · **Pivot to NACF:** At NACF the promise date on an order is the same commitment — I would hold a conservative buffer against a missed date on the same arithmetic. · **Do NOT say:** that undersell is free; that 6.3 is measured — it rests on S1's benchmark and a 25% assumption.

INSIST ON THE HIGHEST STANDARDS · AIQB Q1 **Q4. "Describe a time when you refused to compromise your standards around quality/customer service, etc."**

I set the drift tolerance at zero, and people reasonably asked whether that was realistic. The daily check is ISM's copy of S/4 available against the MB52 snapshot, and a non-zero drift is a Sev, not a planning variance, because two books that should agree and don't mean one of them is wrong — and you do not know which one a customer will meet. The cost is more investigations — the trade I accepted. I made the standard checkable: negatives clean at T+1 at cutover, drift zero SKUs on consecutive days as the hypercare exit, and blockers under two a day. It held because the exit was a number nobody could negotiate. The thing I would say honestly is that zero tolerance is a posture, not a guarantee — it is only as good as the cause class written on each audit row.

Number to land: drift tolerance 0; exit: 0 SKUs on consecutive days · **Pivot to NACF:** At NACF I would set the same kind of zero-tolerance reconciliation on dock-received versus system-available counts, with a number for the exit. · **Do NOT say:** that zero tolerance is easy or free; that it guarantees correctness.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"A negative AFS decomposes into three seams — cardinality, latency, override."

Headline metric: −10 → −5 → −2 → 0 (illustrative)

Cause class decides the fix and the owner; a real log row shows the shape — Dept 96, 400 of 1,800 SKUs negative in the legacy book and zero in the WMS (22.2%) — and a hold feed validated at about 10 minutes against a ≤15-minute budget. Honest limit: the split is illustrative until audit rows are pulled.

CONFLICT / INFLUENCE ANGLE · OWNERSHIP

"Four groups that did not report to me had to own the exception screens before I stepped back."

Headline metric: 3 owner orgs + the ISM app team

The map was in the FSD signed by three orgs (user anchor, master S1); hypercare tested whether the control owners ran the report without me. Influence: a screen, a role and an audit row per act — not a meeting.

FAILURE / LEARNING ANGLE · INSIST ON THE HIGHEST STANDARDS

"'Clean' meant zero negatives — and zero can be reached by an override."

Headline metric: negatives clean at T+1 ≠ cause removed

The exit bar measured the symptom; override abuse was the register's third risk. Do differently: a clean requires a cause class on the audit row, not just a zero.

confirm whether cause class was recorded on au

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

How do you know the Negative AFS screen is why incidents went to zero?

I can't isolate it. The matrix proves the events are mapped; the screens catch what the mapping missed; drift at zero is the check that books agree. The record is the combination.

Say: *"I can't separate the screens from the matrix — I can tell you the three checks each caught a different class, and that the combination went to zero incidents."*

Zero-tolerance drift isn't realistic — what is the false-positive cost?

Every difference is investigated, so the cost is investigator hours; the benefit is no unexplained difference between books. A non-zero tolerance would price each unit at up to $30 of exposure; the exit bar made zero crisp.

Say: *"A zero tolerance costs investigation time, and I chose it because an unexplained difference between books is a customer promise I cannot see."*

What can your detector not see?

Undersell — a one-sided detector. A wrong rule that starves a channel never goes negative; the asymmetry makes that deliberate but unusual undersell needs a separate watch — which is what I would add.

Say: *"The Negative AFS screen is blind to undersell by design; the missing control is a watch on unusual undersell, and I would add it."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. At the Utah DC I owned exception control for the published AFS: a negative AFS decomposes into three cause classes — duplicate event, feed latency, override — each with an owner, every AFS-changing act has a screen, a role and an audit row, and drift tolerance is zero; negatives were clean at T+1, drift zero on consecutive days, and customer-facing oversell went to zero through both go-lives.

NACF bridge. NACF's equivalent is an ASIN whose committed units exceed what the building can back — a double receive scan, dock-to-stock lag (the dwell side of CAP-D) or a manual adjustment. I would stand up the same map — cause class, owner, screen, role, audit row — and a zero-tolerance reconciliation between dock-received and system-available counts before an FC ramps.

U7 — The 836-row issues log as the building's readiness instrument — severity-first reading, and the 42 rows where the dock, the floor, the power and the footprint were the problem NEW

LPs this story serves here: Dive Deep (primary — three cuts of one book: severity, flow, month) · Insist on the Highest Standards (one grammar; a visible deferral over a buried blocker) · Ownership (rows with no system owner: conveyor, aisle 45, power, guardrails) · Earn Trust (the worst number in the room was mine to say out loud)
Process family: Plan / Execute — program readiness measurement: one log, one grammar, severity-first triage, and the physical-node slice of the defect book

Honesty label. Record-from-docs, computed with pandas from *UDC_Consolidated_Issues_Log_for_BY_WMS-Locus - Copy* (sheet 1; 836 rows × 19 columns; an .xlsx container with a .CSV extension): status, priority, category, type, operation, functional area, owner and date counts are exact. Where I cut a free-text field (Operation; Functional Area) I say which labels I counted. **Limits I will state before I am asked:** the file has no close-date column, so I cannot compute time-to-close or reconstruct what was open at go/no-go from it; it is a late snapshot (last row 2025-10-01); the building slice is my own label grouping. The ACL request-row counts (Dev 310 · QA 255 · Perf 200 · Prod 221) are record from my Apple workbook (C2), not recomputed. Dollar figures in Table 4 are **modeled/illustrative** (30 min a row; $110/h blended, as S5/U5) and are not claimed as savings.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Dive Deep. I read one 836-row book three ways — severity, flow, month — and the three cuts said different things: Blockers and Criticals closed at 99.4% and 97.1% against 60.0% for Lows; Dotcom was the most severe flow; June was a spike with a 0.35 monthly decay.

Insist on the Highest Standards. One log, one grammar, no side lists; a parking-lot status meant 'decided not now' in writing with an owner — never silently deleted — and the weekly review read Blocker and Critical rows first.

Ownership. 17 of the 42 building rows (conveyor, aisle 45, Locus power, guardrails, chargers) sat with Operations, Engineering or Maintenance — not IT; they went into the same log with a name and a date.

Earn Trust. Any VP could open the log any Monday and see what the team saw; the worst number in the room was mine to say out loud, including the 271-row June spike.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The Sephora Utah DC conversion ran about thirty build tracks and sixty-plus stakeholders over about a dozen impacted systems; status by opinion would have drowned it. **Task.** Build the instrument that told us whether the building was ready — before cutover week, not during it. **Action.** One consolidated issues log with one grammar — status, priority, operation, functional area, owner on every row — read Blocker and Critical first at a weekly two-block review, with a parking-lot status for 'decided not now' and one coupled build plan beside it. **Result.** 836 rows; 714 closed — 85.4% — and 98.0% of the 441 Blockers and Criticals; the June go-live spike of 271 rows burned down 65, 21, 14, 4; and the 42 rows where the building itself was the problem — dock, appointments, conveyor, aisle 45, Locus power, footprint — were 92.9% closed with none of the open three a Blocker or Critical. Zero customer-facing incidents through both go-lives.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — About thirty build tracks, sixty-plus stakeholders, about a dozen impacted systems and a building that kept shipping. The risk I was asked to instrument was go-live readiness: defect burn-down, track completion, cutover preparedness. Issues arrived from IT, DC Ops, Engineering, Maintenance and the robotics vendor, in different vocabularies, with no shared priority scale.

T — Make a surprise show up in a Monday review, not in cutover week: one log, one build plan, one runbook — three views of the same dataset — and a way to read the book that put the dangerous rows first.

The book as it stands (computed): 836 rows · 714 closed (85.4%) · 122 not closed, of which 112 carry an open status and 10 have a blank status · Blocker 164 + Critical 277 = 441 (52.8% of the book) · bugs are only 115 rows (13.8%) — the rest are requirements, actions, questions and parked decisions: a work-triage system, not a defect list.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — agreed the grammar before the first real row: one status ladder (A open → F closed), one priority scale (0 Blocker → 4 Low), operation, functional area and owner on every row; one owner per row. *Exit bar:* grammar signed by IT and DC Ops; no side lists.
- **Plan & Design (5/15 → 8/14)** — decomposed the build plan into about thirty tracks, holiday freeze and critical path visible; physical-node work (conveyor, aisle 45 relabel, guardrails, footprint defaults) entered the log with a named owner and a date rather than living in a facilities thread. *Exit bar:* every track mapped to an E2E gate; building rows owned.
- **Build & Test (Aug → Jan/Apr)** — weekly two-block review (30 min IT, 30 min business) reading Blocker and Critical first; monthly arc watched — about 90 rows in each of February and March from the January bake plus Locus physical testing; Locus-side rows held to 86

(10.3%). *Exit bar:* every red row has a name and a date.

- **Cutover & Go-Live (January; May 1 → Jul 1)** — the log as the readiness gate: both go-lives read burn-down curves, not confidence; network ACL request workbooks per environment as the access gate (Dev 310 · QA 255 · Perf 200 · Prod 221 rows). *Exit bar:* Blocker + Critical bar from the go/no-go; access gate closed per environment.
- **Hypercare / Close (Jun → Oct)** — the 271-row June spike read as healing: 65 → 21 → 14 → 4; the open tail stayed visible and owned. *Exit bar:* decay, not plateau; every open row owned.

Influence and ownership. Mine: the grammar, the weekly read order, the parking-lot rule, the coupling of log, plan and runbook, and the readiness call off counted rows. Sephora IT/WMS (342 rows), Operations (111), Engineering/Maintenance and Locus (86) owned and closed their rows.

> **Designed the process; others own it now.** The log, the plan and the runbook are one dataset in three views, so the next program inherits an instrument, not a habit. **Trade-off put to leadership:** attention — the parking-lot status made deferral explicit so the Blocker-and-Critical half of the book got the room's time.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — modeled, not claimed: triaging 836 rows at 30 minutes each ≈ 418 h × $110 = $45,980 ≈ 1.45 weeks of the modeled ≈$31,731/week benefit — the log pays back if it prevents about a week and a half of slip.
- **#** — 836 rows; 714 closed; 441 Blocker + Critical, 432 closed; June spike 271 → 4; building slice 42 rows, 39 closed (computed from the file).
- **%** — 85.4% closed overall; 98.0% of Blocker + Critical closed; closure falls with severity — 99.4%, 97.1%, 75.4%, 69.7%, 60.0% — the read order showed in the curve.
- **Time** — fifteen months discovery through go-live; June spike decay ≈0.35 per month, half-life ≈20 days.
- **Customer** — zero customer-facing incidents through both go-lives (record).
- **Quality** — one grammar, no side lists; deferral explicit; every red row with a name and a date.

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Nobody owned 'ready'; I defined it as counted rows in one log read in a fixed order.
Scope & influence	Sixty-plus stakeholders; IT, Operations, Engineering, Maintenance and the robotics vendor all wrote to the same log — none reported to me.
Execution trade-off + mechanism others own	Attention to Blocker and Critical over a flat read of 836 rows; the mechanism others own: the log grammar and the Monday review.
Impact	Both go-lives read readiness from rows; 432 of 441 Blockers and Criticals closed.
Process improvement	Log, plan and runbook as one dataset in three views; deferral made explicit.
Complexity & consensus	Different vocabularies reduced to five fields; one scale.
Root cause at scale	The monthly shape — ramp, spike, decay — separates a healing cutover from a design problem.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: one row per issue; 19 columns. Joins: none — the discipline was one table. Cuts: severity × status (closure gradient); operation (five flows by exact label: 675 of 836 rows, 161 free-text or blank); month; owner; functional area (dock and building labels). Revealed: closure rates fall monotonically with severity; Dotcom was the most severe flow (64.5% Blocker + Critical); the June spike was 88.2% Go-Live category and decayed fast; the building slice ran more severe than the book (64.3% vs 52.8%) but z = 1.50 — directional, not proven.

2 · Solution design. Options: (A) status by opinion in decks; (B) one tool per track; (C) one log with one grammar, coupled to one build plan and one runbook (chosen). A and B make two rows describe one issue and the counts lie; C makes the count the gate.

3 · Math & data logic. Closure = closed ÷ rows by severity; Blocker + Critical share = 441 ÷ 836 = 52.8%; decay r = (4 ÷ 271)^(1/4) = 0.349, half-life = ln 2 ÷ −ln r = 0.66 months ≈ 20 days; building-slice z = (0.643 − 0.528) ÷ √(0.528 × 0.472 ÷ 42) = 1.50.

4 · Implementation. Five gated phases; the log, the build plan (~30 tracks) and the cutover runbook (343 tasks) as three views; ACL request workbooks per environment as the access gate; weekly two-block review. Risks: duplicate rows, free-text drift, priority inflation.

5 · Probe questions. Below: is 836 a defect fire, how do you know the log is honest, what does the monthly shape tell you, what is in the building slice, and the three trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — 836 issues. Isn't that a defect fire?

No — the log proves it. Bugs are 115 rows, 13.8%; the rest are requirements (159), action items, questions (23) and parked decisions. It was the program's work-triage system. What I do read as a risk signal is that 52.8% of the book is Blocker or Critical — and then the closure gradient: 99.4% of Blockers closed against 60.0% of Lows. If everything had been priority one, the curve would be flat; it is not.

Infra — what did the log say about the physical building?

Forty-two rows — 5.0% of the book — carried dock and appointment flow (23) or floor, power and footprint (19): TMS appointments with no POs attached in training (Blocker), retail pallet-build positions not available (Critical), ‘Facilities team remove half of conveyor’ (Critical), remove the west side of aisle 45 and relabel (Critical), Locus IDF temporary power until January 6 (Blocker), remove the yellow guardrail between retail and dotcom (Blocker), update large and standard footprint per SKU with large as the default (Blocker). 27 of the 42 were Blocker or Critical — 64.3% — and 39 closed; the three open were not B+C.

Infra — what does the June spike tell you?

271 rows in one month — 32.4% of the book — 239 of them Go-Live category. Then 65, 21, 14, 4: each month retained about a third of the last, r = 0.35, half-life about 20 days. A spike with fast decay is a healthy cutover; one that plateaus is a design problem. I would have wanted that picture on the wall from the first hundred rows.

Infra — how do you know the log is honest?

One row per issue, one grammar, a weekly read where the worst number was mine to say, a dedupe step in the rhythm — two rows for one issue and the counts lie — and ‘parking lot’ meaning decided-not-now in writing. The file does not carry close dates, so I will not claim a time-to-close from it.

Infra — why were Locus rows only a tenth of the book?

86 rows — 10.3% — because the robotics workstream had its own rhythm and owner, and the 98% gate with the throttle caught the zone-logic problem in week two instead of the floor. They were severe — 65 of 86 Blocker or Critical, 75.6% — and every one closed.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Readiness signal	status by opinion	836 counted rows, one grammar	decision on rows, not confidence	record
Blocker + Critical	441 identified	432 closed; 9 open	98.0% closed	record-from-docs (computed)
Go-live spike	271 rows in June	65 → 21 → 14 → 4	−98.5% by October	record-from-docs (computed)
Building slice (dock + floor)	42 rows; 27 Blocker + Critical	39 closed; 3 open, 0 B+C	92.9% closed	record-from-docs (computed)
Triage effort	unmeasured	≈418 h	≈$45,980 (modeled)	modeled

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	The book	`836 rows; dated 830; 6 undated; 2024-03-17 → 2025-10-01; 824 of the dated rows are July 2024 or later`	issues-log file, ‘Date Id’d’	836
2	Severity	`164 + 277 = 441; 441 ÷ 836 = 52.8%`	Priority column	52.8% Blocker + Critical
3	Peak month	`271 ÷ 836 = 32.4% of the book in June 2025`	Date Id’d	32.4%
4	How much of the free-text Operation field fits five flows	`238 + 172 + 113 + 90 + 62 = 675; 675 ÷ 836 = 80.7%; remaining 161`	exact labels after trim and lowercase; IC includes ‘inventory control’	80.7%
5	The building slice	`dock/appointment labels 23 + floor/space labels 19 = 42; 42 ÷ 836 = 5.0%`	Functional Area labels listed in the flow box	42 rows

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Closure gradient — Blocker	`163 ÷ 164`	closed ÷ rows	99.4%
2	Closure gradient — Critical / High	`269 ÷ 277 = 97.1%; 190 ÷ 252 = 75.4%`	closed ÷ rows	97.1% / 75.4%
3	Closure gradient — Medium / Low	`69 ÷ 99 = 69.7%; 21 ÷ 35 = 60.0%`	closed ÷ rows	69.7% / 60.0%

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Severity by flow	`Dotcom 40/62 = 64.5% · Retail 100/172 = 58.1% · Inbound 58/113 = 51.3% · Outbound 119/238 = 50.0% · IC 39/90 = 43.3%`	B+C ÷ rows per flow	Dotcom highest
5	Spike decay	`r = (4 ÷ 271)^(1/4) = 0.349; half-life = ln 2 ÷ −ln(0.349) = 0.66 mo`	geometric mean of 65/271, 21/65, 14/21, 4/14	≈20 days
6	Building slice vs the book	`27 ÷ 42 = 64.3% vs 52.8%; SE = √(0.528 × 0.472 ÷ 42) = 0.0770; z = 1.50`	one-sample proportion z (normal approximation)	z = 1.50 — directional

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Blocker + Critical open	441 identified	9 open (1 Blocker, 8 Critical)	computed from the file
2	Closed, overall	0	714 = 85.4% (open pipeline 122: 112 with an open status + 10 blank)	computed from the file
3	Monthly arc	Jun 271	Jul 65 · Aug 21 · Sep 14 · Oct 4 (−98.5%)	computed from the file
4	Building slice	42 rows, 27 B+C	39 closed (92.9%); open 3, 0 B+C	computed from the file
5	Locus-side rows	86 = 10.3% of the book	86 closed; 65 B+C = 75.6%	computed from the file

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Triage effort, the whole book	`836 × 0.5 h = 418 h; 418 × $110 = $45,980`	30 min per row all-in (illustrative); $110/h blended (modeled, as S5/U5)	≈$46.0K
2	A week of benefit	`$1.65M ÷ 52 = $31,731`	S2's modeled recurring benefit	≈$31.7K/wk
3	Break-even	`$45,980 ÷ $31,731 = 1.45 weeks`	the log pays back if it prevents ~1.45 weeks of slip	≈1.45 weeks
4	The June spike alone	`271 × 0.5 h = 135.5 h; × $110 = $14,905; ÷ 40 = 3.4 FTE-weeks`	same assumptions	≈$14.9K
5	The building slice	`42 × 0.5 h = 21 h; × $110 = $2,310`	same assumptions	≈$2.3K

SOLUTION FLOW — THE SYSTEM & DATA MAP

```
issues log (836 rows × 19 cols)  — one table, one grammar
   status A→F · priority 0→4 · operation · functional area · owner · date
      |
      +--> weekly two-block review (30 IT / 30 business): Blocker+Critical rows first
      |        every red row: a name and a date;  'parking lot' = decided-not-now, in writing
      +--> coupled build plan (~30 tracks, holiday freeze visible)  --> E2E gates 11/22 · 12/20 · 1/30
      +--> cutover runbook (343 tasks, 48,885 min)  --> 5/1 → 7/1 window
      +--> ACL request workbooks: Dev 310 · QA 255 · Perf 200 · Prod 221 = 986 rows (access gate)
building slice (Functional Area labels, exact, case-insensitive):
   dock/appointment (23): dock · dock/office · dock/receiving lanes · dock hop · shipping dock · unload ship ·
                          trouble lane · appointment · appointment creation · xdock · crossdocks
   floor/space/hardware (19): facilities · physical layout · footprint · physical · guardrail · charging · locus hardware
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Minutes per row (triage)	15 → $22,990 (0.72 wk)	30 → $45,980 (1.45 wk)	45 → $68,970 (2.17 wk)
Building-slice definition	narrow 31 rows · 64.5% · z 1.31	base 42 rows · 64.3% · z 1.50	broad 104 rows · 66.3% · z 2.78
Monthly decay r	0.50 → half-life ≈ 30 days	0.349 → ≈ 20 days (observed)	0.80 → ≈ 94 days (would read as a plateau)
Weekly benefit	$1.1M/52 = $21.2K → 2.17 wk	$1.65M/52 = $31.7K → 1.45 wk	$2.4M/52 = $46.2K → 1.00 wk

SPOKEN STATEMENT

> “I treated the program’s issues log as an instrument and read it three ways. By severity: of 836 rows, 441 were Blocker or Critical and 98% of those closed, with closure falling to 60% for Lows — so the weekly read was working on the dangerous end. By flow: Dotcom was the most severe. By month: the June go-live spike of 271 rows decayed 65, 21, 14, 4 — about a third a month — which is what a healing cutover looks like. And the part an infrastructure leader cares about: 42 rows were the building itself — dock and appointments, conveyor, aisle 45, Locus power, footprint — 64% Blocker or Critical, 39 closed, none of the three open ones severe. The statistics are directional — z of 1.5 — and the file has no close dates, so I will not claim a time-to-close. The point is that the building’s problems were rows with names and dates, not a hallway conversation.”

IF CHALLENGED

“Priority inflation — 52.8% Blocker or Critical isn’t a priority scale.”

It’s a fair flag and I’d rather say it first. Two answers: the closure gradient — 99.4%, 97.1%, 75.4%, 69.7%, 60.0% — shows the scale differentiated in practice; and the book was a go-live book, 331 rows Go-Live category, where severity runs high by construction. What I would add is a monthly check on the Blocker + Critical share so inflation shows as a trend.

“Your building slice is a label grouping you chose.”

Yes — so I ran three definitions: narrow 31 rows (64.5%, z 1.31), base 42 (64.3%, z 1.50), broad 104 (66.3%, z 2.78). The direction is the same in all three; only the broad one clears 2σ, and it is the loosest definition, so I call the result directional.

“Prove the Blocker + Critical bar was met at go/no-go from this file.”

I can’t — the file is a late snapshot with no close dates; it shows 9 B+C still open on 2025-10-01 (1 Blocker, 8 Critical), which postdates both go-lives. The go/no-go read came from the weekly review state at the time.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you used data to run a program.	Dive Deep	836 rows, three cuts; the closure gradient; the spike decay r = 0.35.	Do not call it a defect fire — bugs are 115 rows.
What did you do about things nobody owned?	Ownership	17 of 42 building rows had non-IT owners; they entered the same log with a name and a date.	Do not claim I fixed the conveyor — Operations and Maintenance did.
How do you keep a shared log honest for months?	Insist on the Highest Standards	One grammar; parking lot = decided-not-now, in writing; dedupe in the rhythm.	Do not claim a time-to-close — no close-date column.
How do you report bad news?	Earn Trust	The 271-row June spike, said out loud with the decay curve.	Do not blame IT for 342 rows.
How do you decide what to defer?	Have Backbone	Eight rows typed Parking Lot; ‘decided not now’ with an owner.	Never say a deferral was free.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

DIVE DEEP · AIQB Q4 **Q1. “Walk me through a big problem or issue in your organization that you helped to solve.”**

The problem was that a program of about thirty build tracks was going to run on opinion, and the first fix was to count. I made one consolidated log with five fields on every row and read it three ways. By severity, 441 of 836 rows were Blocker or Critical and closure fell with severity — 99.4% of Blockers, 60% of Lows. By flow, Dotcom ran most severe. By month, the shape told me the June spike was healing: 271, then 65, 21, 14, 4. What was missing was the monthly shape itself — it was a manual read well into the program — and I filled the gap by building the arc view once I saw it, which I would now have on the wall from the first hundred rows. The result was that both go-lives read readiness from rows, with zero customer-facing incidents.

Number to land: 836 rows · 441 B+C · 271 → 4 · r ≈ 0.35 · **Pivot to NACF:** NACF launch readiness is the same instrument — doors tested, appointment slots scheduled, UPH validated — as counted rows read severity-first. · **Do NOT say:** that 836 issues means 836 defects — bugs are 115.

INSIST ON THE HIGHEST STANDARDS · AIQB Q6 **Q2. “Describe the most significant continuous improvement project that you led.”**

The catalyst was that two rows describing one issue make every count lie, so the improvement was a grammar. One log, one status ladder, one priority scale, an owner on every row, and a parking-lot status that meant ‘decided not now’ in writing with an owner — never silently deleted. Dedupe was part of the weekly rhythm. The result is a book where any VP could open the file on a Monday and see what the team saw: 714 of 836 closed, 98% of the Blockers and Criticals, and the open tail of 122 visible and owned. I would rather defend a visible deferral than bury a blocker, and a log that calls everything priority one is where blockers hide.

Number to land: 85.4% closed · 98.0% of B+C · 122 open, owned · **Pivot to NACF:** At NACF I would run the same grammar on a launch register: one row per risk, an owner, a date, a severity. · **Do NOT say:** that the file proves time-to-close; it has no close dates.

OWNERSHIP · AIQB Q3 **Q3. "Give me an example of an initiative you undertook because you saw that it could benefit the whole company or your customers, but wasn't within any group's individual responsibility so nothing was being done."**

The building itself did not belong to the systems program, so I made it belong to the log. Forty-two rows were dock, floor, power or footprint: remove half of a conveyor, remove the west side of aisle 45 and relabel, get Locus IDF temporary power until January 6, remove the guardrail between retail and dotcom, set the large footprint as the SKU default. 17 of those 42 sat with Operations, Engineering or Maintenance, not IT. Putting them in the same log with a name and a date meant a Blocker about power had the same Monday review as a Blocker about a WMS interface. 39 of the 42 closed, and the three still open were not Blocker or Critical. The conveyor and the power were not mine; making them visible and dated was.

Number to land: 42 rows · 17 non-IT owners · 39 closed · 0 open B+C · **Pivot to NACF:** At NACF the dock, the floor and the power are launch dependencies too — they go on the same register as the systems. · **Do NOT say:** that I fixed the conveyor or the power; Operations and Maintenance did.

EARN TRUST · AIQB Q6 **Q4. "Tell me about a time when you uncovered a significant problem in your team."**

The June spike was the biggest number the program ever produced, and I said it first. 271 rows in one month — a third of the book — 239 of them Go-Live category. I put it in the Monday review the week it appeared with the shape that explained it: the January bake and Locus physical testing had given a ramp of about ninety a month, the combined go-live surfaced everything at once, and the decay afterwards — 65, 21, 14, 4 — was what healing looks like. A spike that decays fast is a healthy cutover; one that plateaus is a design problem. Because the number was in the room before anyone asked, nobody escalated a surprise, and the readiness call was made on rows.

Number to land: 271 rows · 88.2% Go-Live · decay ≈ 0.35/month · **Pivot to NACF:** At NACF the worst number in the WBR is the one to say first, with the curve that explains it. · **Do NOT say:** that the spike was a defect fire; it was the combined go-live surfacing.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"836 rows, and the closure rate falls with severity: 99.4%, 97.1%, 75.4%, 69.7%, 60.0%."

Headline metric: 98.0% of 441 B+C closed; r = 0.35

Three cuts of one book — severity, flow, month.

Honest limit: no close-date column, so no time-to-close; the Operation field is free text (161 rows outside five flows).

CONFLICT / INFLUENCE ANGLE · HAVE BACKBONE; DISAGREE AND COMMIT

"Well-meaning teams wanted every improvement in the release; a parking-lot status said 'decided not now' in writing."

Headline metric: 8 Parking Lot rows; Blocker + Critical first

I defended visible deferral over a buried blocker; the cost was explicit — the room's attention went to the dangerous half of the book. The disagreement was resolved by writing the deferral, with an owner, not by winning an argument.

FAILURE / LEARNING ANGLE · INSIST ON THE HIGHEST STANDARDS

"The arc — ramp, spike, decay — was a manual read well into the program."

Headline metric: shape view from month 1, not month ~6

Record do-differently: I would build the monthly shape view from the first hundred rows; the counts were always there, the picture was not. Second: 52.8% Blocker + Critical is a priority-inflation risk I should have trended monthly.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

"How do you know the log was the reason go-live went well, not the team?"

I don't, and I won't claim it. What the log gives is a testable statement: 98.0% of Blockers and Criticals closed and the June spike decayed at r ≈ 0.35. A team without the log could have done the same; what the log changed is that the readiness call was made on counted rows, and a surprise surfaced on a Monday.

Say: *"I can't separate the log from the team; I can show the readiness call came off counted rows and the spike decayed in line with a healthy cutover."*

"Your building-slice finding — is it significant?"

Directional only: 27 of 42 = 64.3% vs 52.8%, z = 1.50, about p = 0.13. Across three definitions the direction holds; only the broad one clears 2σ. The decision it informs is cheap — put physical rows in the same log with owners — and holds regardless.

Say: *"z of 1.5 — directional, not proven; I would act on it because the action costs almost nothing."*

"What number would have made you stop the go-live?"

One open Blocker at the go/no-go read, or a June-style spike that plateaued instead of decaying — a decay ratio above about 0.8 for two months. The first is the record bar; the 0.8 threshold is my own illustrative plateau test.

Say: *"A Blocker open at go/no-go, or a spike that stops decaying — I would call that a design problem."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. I ran Sephora's UDC program on one 836-row issues log with one grammar, read Blocker and Critical first: 85.4% closed, 98% of the Blockers and Criticals, the 271-row June spike decaying 65, 21, 14, 4 — and 42 rows where the dock, the floor, the power and the footprint were the problem, every one with a name and a date.

NACF bridge. NACF's launch register is the same instrument. The dock rows are CAP-A in miniature — appointments created, placards printing, pallet-build positions available before the first trailer — and the footprint rows are the CAP-C input: a SKU with no footprint or CubiScan record defaults a slot size, and a wrong default is wasted cube at the 85% effective-capacity line. I would read the register severity-first, with the dock row at the top.

Question bank — likely questions for this interviewer, with a one-line answer plan ELEVATED

Behavioral and functional questions a Senior Manager of Infrastructure is likely to ask. Each plan points at a packet story or a named backup in the master packet; stories allocated to other interviewers are pointers only. Rows 13–21 are new in v2. For the LP-by-LP version with the real Amazon wording, use the Question → Story Router at the front.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Tell me about the most complex physical / systems change you have led in a DC.**	S2 pitch — one pool, three planning identities, two go-lives, the 98% gate; end on +15% signed and zero incidents.	S2
2	**Tell me about a migration or cutover you ran.**	S18 pitch — tolerance before mock one; 7 → 1.8 → 0.3; the RF fallback; on time, ~3% under.	S18
3	**How do you decide a program is ready to go live?**	Counted rows, not confidence: Blocker + Critical = 0; mock inside tolerance; runbook rehearsed; rollback armed; a written recommendation.	S2 / S18
4	**How do you ramp automation without hurting throughput?**	Publish the gate from the pre-ramp baseline and the incident budget; build the throttle in advance; the floor owner owns the dial; RCA before re-ramp.	S2
5	**What are your rollback triggers, typically?**	Numbers per interface: sync < 99% for 4 h; p95 > 2× budget for 1 h (S2); mismatch > 0.5% for two days; ship-confirm posting failing > 2 h (S18).	S2 / S18
6	**How do you handle a vendor slipping on your critical path (Locus RTL)?**	Escalate the week it surfaces; re-slot openly; refuse to compress UAT to absorb it; retest after the slip.	S2
7	**Tell me about a time a rehearsed contingency saved you.**	Honest fit is S18's pre-training device test: hundreds of RF devices, the date held, training +1.5 weeks, the what-if rule left behind. The master's paper-pick fallback and ≈$45K are master-only — read the S18 source-conflict box before saying them.	S18
8	**How do you exit hypercare?**	A number written before go-live: < 2 blockers/day sustained; S2 exited week 3 of 4 with new rows falling 271 → 65 → 21 → 14 → 4 a month; S18 exited week 3. A calendar exit declares victory over open defects.	S2 / S18
9	**How would you approach consolidating or launching a node in the NACF network?**	Master §3.99 opening — book of record and metric → stocks and flows → the invariant broken today (a unit sellable in two places) → staged go-lives → ramp gate → hypercare number.	Master §3.99 + S2
10	**What is the utilization impact of consolidation?**	+15% units per labor-hour signed; one pool (√2 pooling, modeled); self-transfers to zero by construction; robots at full volume week three.	S2
11	**Tell me about a failure.**	S18's RF miss in full — the scoring error, the pre-training device test, the same-day escalation, training +1.5 weeks with the date held, the rule left behind.	S18
12	**How do you keep peak safe during a multi-system program?**	The Oct–Dec freeze decided as a council decision; the staged timeline around it; E2E days dated before and after; no release through peak.	S2
13	**How do you prove a program tested everything that moves inventory?** NEW	U1 — every event lands in exactly one bucket; each cell asserted before/after; no story enters QA without a cell reference.	U1
14	**How do you size a pooling / consolidation case before you design?** NEW	U2 — profile overlap, order shape, variability and correlation; carry a range with a kill number, not a point.	U2
15	**Tell me about a time you said no to a stakeholder.** NEW	U3 — priced the no at p* = 36.7%, offered visibility, wrote the no into the FSD.	U3
16	**Where did the node's constraint actually bind — labor, accuracy or space?** NEW	U4 — labor, then accuracy at 94 vs 98, then labor again; I did not claim space.	U4
17	**Walk me through your cutover runbook and what you could not undo.** NEW	U5 — 343 owned tasks, one irreversible instant (zero-then-publish), rollback as numbers, go/no-go off counted rows.	U5
18	**How do you handle a negative available-to-sell?** NEW	U6 — decompose to cause class: duplicate event, feed latency, override; screen, role and audit row for every AFS-changing act.	U6
19	**How do you read an issues log, and how do you know a building is ready?** NEW	U7 — severity-first: 441 B+C, 98.0% closed; monthly arc 271 → 4; the log is the gate, not the room.	U7
20	**Where do dock, space, power and footprint show up in a program plan?** NEW	U7 — 42 rows in the log where the physical building was the problem, each with an owner and a date.	U7
21	**How does this map to NACF inbound capacity?** NEW	Bridge page — doors × hours × unload rate (5 doors at 85%), UPH with the learning-curve tax (+13.6%), cube at 49% effective, dwell 3× from 85% to 95%.	Bridge

Cheat-numbers box — canon values with basis labels ELEVATED

Canon numbers for S2 and S18 with their basis labels, from the master packet's §4.1 ledger, then the seven new stories (U1–U7). Retail rows: 172 by exact label (Apple prints 171) — say about 170.

METRIC	VALUE TO SAY	BASIS	NEVER
S2 efficiency	+15% processing efficiency (index 100 → 115)	record, Sephora-signed, measured by DC Ops post-stabilization	presenting the dollars as signed
S2 incidents	zero customer-facing incidents through both go-lives	record	—
S2 robotics gate	98% published; week 2 at 94; ~30% throttle for a week; full volume week 3	record	"the robots failed"
S2 dollars	≈$1.6M/yr (0.88 labor · 0.22 carry · 0.45 splits · 0.10 self-transfer); ≈$2.2M one-time pooling; range $1.1–2.4M/yr	modeled on an illustrative 100K-units/week DC	—
S2 scale	about a dozen impacted systems; 60+ stakeholders (~80 names, ~15 functions); about thirty build tracks	record	"20+ systems"; "fifteen tracks"; a single stakeholder count
S2 artifacts / dates	343 tasks / ~815 hrs; 836 rows / 85.4% closed; council 5/15 → 7/31 → FSD 8/14; E2E 11/22 · 12/20 · 1/30; Jan + May go-lives	record (program-level artifacts shared with S1/S5)	—
S2 rollback triggers	sync < 99% for 4 h; order-flow p95 > 2× budget for 1 h — armed twice, never fired	record	—
S18 mocks	7% → 1.8% → 0.3% vs 0.5% tolerance published with Finance before mock 1	record	—
S18 date / budget	8 months, on time; ~$1.8M, ~3% under (~$54K)	record (this packet's canon; loop packet differs)	"~$800K"
S18 pipeline	$20M ROI pipeline unblocked the next month; ~$2.5M/month of slip	record ($20M); $2.5M/month = my division (modeled); not banked	"$2.5M/week"; banking the ~$875K
S18 RF miss	hundreds of RF devices (pre-training device test); a three-to-four-day estimate; training +1.5 weeks; date held	record (Apple G2)	"400 devices failed"; "~100 of ~400"
S18 freeze / failback / exit	72-hour in-transit freeze; legacy read-only 30 days; hypercare exit week 3 at < 2 blockers/day	record	—
S18 modeled	≈$0.1M less inventory in dispute (illustrative $50M on-hand); ≈$875K = price of a 1.5-week slip (my division, not banked)	modeled	—
U1 matrix NEW	every event → exactly one bucket (AFS 557, not 560 or 554 — worked trace); 836 rows · 441 B+C (52.8%) · 714 closed (85.4%); B+C open = 0 at go/no-go	record (log); illustrative (trace)	a cell count as fact; “zero defects”
U2 profile NEW	textbook $2.2M ceiling → $0.74M–$2.2M range; 87% of the ≈$1.65M recurring benefit not pooling-dependent; release 9.9%–29.3%	modeled	$2.2M as a point
U3 reserve NEW	p* = 36.7%; −$50K / −$202K per 10,000 units at 20% release; visibility yes, promisability no; oversell recurring → zero	modeled; illustrative inputs; zero = record	that Merchandising agreed happily
U4 constraint NEW	421 h (≈$11.8K) vs ≈4,200 mis-picks (≈$50.4K); max safe share 33.3% vs ~30% dial; 4,615 → 4,013 h/wk at +15%	modeled on an illustrative 100K-units/week DC	“space bound”

METRIC	VALUE TO SAY	BASIS	NEVER
U5 runbook NEW	343 tasks · 815 h · 61 days; ≈$31.7K/week benefit at stake; runbook ≈$89.6K; rollback never fired	record (tasks, hours, triggers); modeled ($)	banking $31.7K; “drilled under load”
U6 AFS control NEW	3 cause classes; negatives clean at T+1; drift 0 on consecutive days; asymmetry ≈ 1 : 6.3; critical ratio 13.7%	record (design, exit bar); modeled (ratios)	a measured negative count
U7 issues log NEW	98.0% of 441 B+C closed (432); 9 open (1 Blocker + 8 Critical); 42 physical rows, 39 closed; June 271 (88.2% Go-Live); r = 0.349	record (pandas); modeled fit	time-to-close; “weekly open counts”

Never-say, one glance before the call.

- Never "400 devices failed" or a ~100/~400 ratio — say hundreds of RF devices.
- Never bank the $20M pipeline, the ~$2.5M/month or the ~$875K — slip avoided, not banked.
- Never "20+ systems" or "fifteen tracks" — say about a dozen impacted systems and about thirty build tracks; never a single stakeholder count.
- Never present the ≈$1.6M/yr or the ≈$2.2M pooling as signed — the +15% and the zero are the record.
- Never claim BlueYonder or Locus configuration — vendor and platform teams built and configured; I owned the model, staging, gates and go/no-go.
- S1/S3/S5 share the UDC program's dated chain — they are Lan's; S11/S12, S6/S8 and the bar raiser's four stay out of this room.

Backup bench — use only if a primary doesn't fit

Four fallbacks for an infrastructure owner: S13 (Manhattan — Cyber Monday at 4 AM; the incident-response / Bias for Action alternate to S18, and the older pair for S2's throttle), S5 (Sephora — the architecture council without owning a system; governance of an about-a-dozen-systems change, same program family as S2 — never pair S5 with S2 for this interviewer unless asked for a second UDC example) and S15 (Sephora — the QA cadence conversion; the people / Hire & Develop question — Josyula's primary, so only if the question demands it); plus a doc-derived fourth from the Flexport program record — the Shipment-Visibility pilot roll-out and its four testing gates (launch discipline behind published graduation criteria; record-from-docs labels; same program family as Tripathy's S11, facts kept identical to her packet). Tell S2 and S18 first.

Bench discipline. The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

BACKUP — ONLY IF ASKED S13 — Manhattan — Cyber Monday, 4 AM: allocation across 3 DCs and 100 stores when the scheduler died

LPs this backup can serve: Bias for Action (primary) · Customer Obsession (primary backup — the promise ranked first) · Deliver Results

Basis. OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it; the post-mortem mechanisms were mine to drive. $8M / $6M / 99% are record; the margin figures are my model at benchmark rates. Older story — if asked for recent, pair with the UDC robotics throttle (S2): same muscle, 2024–25.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores; 4 AM Cyber Monday the VP called — backorders piling with inventory present, ~$6M of sales at risk against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction (ATP > 0, backorders rising) was the entry bar to declare an incident. Plan & Design: I ranked by promise-at-risk, not FIFO; node choice = the promise — near 0.85 days vs far 4.60. Build & Test: one bridge, every rebalance timestamped; reason-coded alerts I had already built. Cutover: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare: root cause — the allocation scheduler died silently at the daylight-saving change; the post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook. Lesson: decide reversibly, log it, find the cause later. **Result.** Primary metric: peak fill rate — 95% goal → 99% delivered. ~$8M of peak sales protected against ~$6M at risk (record); my model puts that at ≈$3.2M of gross margin at a 40% benchmark margin; structural stockouts −20% over the engagement.

COMPRESSED STAR

- **S —** 4 AM Cyber Monday: backorder lines piling with inventory present; promises aging; ~$6M at risk against a 95% fill goal; nobody had written a clearing rule for failure.
- **T —** Own the response before the cause was known, without making it worse.
- **A —** Declared the incident on the contradiction; clearing rule — promise-at-risk, not FIFO — written on the bridge log in the first hour so nobody re-argued it at noon; rebalanced across three DCs; two-hourly status; isolated the DST scheduler death through my own reason-coded alerts; blameless post-mortem → heartbeat, DST checklist, 3σ page rule, rehearsed playbook.
- **R —** Fill 99% vs the 95% goal; ~$8M protected vs ~$6M at risk (record pair); structural stockouts −20% over the engagement; the next peak ran boring.

RESULTS LINE

Peak fill 99% vs the 95% goal (record) · ~$8M of peak sales protected vs ~$6M at risk (record pair — one-time sales that didn't die, never a saving) · ≈$3.2M gross margin at a 40% benchmark (modeled) · structural stockouts −20% over the engagement (record) · detection moved from a VP phone call to one heartbeat interval.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	4 AM state	`scheduler dead; backlog = 30,000 orders; ~$6M of sales at risk vs the 95% fill goal`	record risk pair; backlog illustrative	the bridge call

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	FIFO failure	`FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age`	record rule problem	value-blind clearing
3	Naive node loads	`DC1: 22,000 / 10,000 = 2.2 days; DC3: 2,000 / 6,000 = 0.33 days`	capacities 10K / 8K / 6K orders/day (illustrative)	worst node 2.2 days behind while another idles
4	No liveness detection	`a dead scheduler was found by a VP at 4 AM, not by a page`	record problem	detection by seniority

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promise-at-risk ranking	`risk = value × urgency(promise − now); clear descending`	on the bridge log in hour 1 (record rule)	at-risk promises first, not first-in
2	Node choice = the promise	`T = handling + transit: near 0.85 d vs far 4.60 d; 4.60 − 0.85 = 3.75 d`	record pair	−3.75 days per re-pointed order
3	Load levelling	`rebalance 30,000 → 12,500 / 10,000 / 7,500: 12,500/10,000 = 1.25; 10,000/8,000 = 1.25; 7,500/6,000 = 1.25 days`	load = queue / capacity (illustrative)	worst slip 2.2 → 1.25 days
4	Liveness page (post-mortem)	`page if x < μ − 3σ = 1,200 − 3 × 150 = 1,200 − 450 = 750/hr`	μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr	caught in one interval, not by a VP

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Peak fill rate	95% goal	99% delivered	record
2	Peak sales	~$6M at risk	~$8M protected	record pair — one-time event
3	Worst-case promise slip	2.2 days	1.25 days	illustrative walk
4	Detection	a VP at 4 AM	one interval via the 3σ page	post-mortem design
5	Structural stockouts	—	−20% over the engagement	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Margin protected	`$8,000,000 × 40% = $3,200,000`	benchmark margin 35–45% (modeled)	≈$3.2M one-time
2	Orders at risk	`$6,000,000 / $150 = 40,000 orders`	AOV benchmark	40,000 orders
3	Cancels / CS avoided	`40,000 × $6.50 = $260,000`	CS contact $5–8 (benchmark; modeled)	≈$0.26M
4	Expedite premium avoided	`40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000`	re-pointed share and premium midpoint of the $5–12 benchmark (both illustrative)	≈$0.13M
5	Structural-stockout run-rate	`$400M × 4% = $16M lost sales; × 20% = $3.2M recovered; × 40% = $1,280,000`	illustrative retailer × lost-sales benchmark (2–6%) × −20% record × margin	≈$1.3M/yr modeled

TWO PROBES

> **Why act before you knew the cause?**
>
> Because waiting had an hourly price — the queue was aging while inventory sat idle — and a reversible, logged intervention did not. I checked with engineering first that manual clears could not corrupt allocation, timestamped every rebalance, and hunted the cause in parallel with alerts I had already built.

> **Isn't this a hero story?**
>
> The event is; the post-mortem is the point. Carrying the first stretch solo is one of my do-differentlys. What I would present as the result is the mechanism set — heartbeat, DST checklist, 3σ page rule, rehearsed playbook — and that the next peak ran boring.

Canon numbers (basis labels): 99% fill vs 95% goal (record) · ~$8M protected vs ~$6M at risk (record) · 3 DCs / 100 stores (record) · near 0.85 d vs far 4.60 d (record) · −20% structural stockouts (record) · ≈$3.2M margin at 40% (modeled). Never "$2MM baseline"; never present $8M as a saving.

BACKUP — ONLY IF ASKED S5 — Sephora — the architecture council without owning any system: three decisions and a fence

LPs this backup can serve: Have Backbone; Disagree & Commit (third) · Earn Trust (influence without authority) · Think Big (architecture governance). Same program family as S1/S2 — the governance story

Basis. I owned none of the systems; the design decision was the deliverable. Zero escalations and the dated gate chain are the record; the dollars are my model of what governance avoided.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The UDC location and inventory model had to clear architecture council — a challenge gate — with ~80 named participants across ~15 functions, and I owned none of those systems. **Task.** Take the design from discovery through council to sign-off and FSD without a surprise escalation. **Action.** Discovery: both inventory books profiled before any design argument; business sessions closed 4/30. Plan & Design: every impacted row pre-wired with a signed LOE by 7/2; the council asked for three separable decisions plus an out-of-scope fence (reserve math untouched; MCS decommission belongs to Global ERP). Build & Test: the "build" was the pack — architecture, staged timeline, peak-freeze boundary. Cutover: 5/15 challenge → 7/31 sign-off → 8/14 FSD. Hypercare: the fence held in writing for the whole program; a second live warehouse became a monitored shadow copy, not an argument. Lesson: a no that lives in a document survives. **Result.** Primary metric: surprise escalations on a 60+ stakeholder design — the norm on prior programs → zero; the dated gate chain held. ≈$0.5M one-time modeled (avoided rework on ~3 escalations that never happened + six weeks of benefit not deferred + a fenced second reserve engine not built).

COMPRESSED STAR

- **S —** About a dozen impacted systems, ~80 participants across ~15 functions, a council that re-litigates designs this size — and I owned none of the systems.
- **T —** Discovery (kickoff 3/29, sessions 4/30) → council 5/15 → sign-off 7/31 → FSD 8/14, without a surprise escalation.
- **A —** Profiled both books first — their numbers, not my opinion; systems-impacted table with an owner on every row; one LOE template, one deadline (7/2), one private conversation per owner before the room; three separable decisions plus the fence; the second-truth objection converted into a monitored shadow copy.
- **R —** Surprise escalations → 0 for the program's life; 100% pre-wire coverage by 7/2; gate chain held to the day with zero re-litigation; fence breaches 0; handed clean into the build (S2).

RESULTS LINE

Surprise escalations → 0 for the program's life (record) · gate chain 3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 held to the day (record) · pre-wire coverage 100% — every impacted row with a signed LOE by 7/2 (record) · both scope refusals held in writing; fence breaches 0 (record) · ≈$0.5M one-time (modeled — what did not happen).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Design size	`~80 named participants × ~15 functions; about a dozen impacted systems`	record	the blast radius
2	Norm for this size	`~3 surprise escalations per design`	industry benchmark 2–4	~3 expected
3	Price of one escalation	`960 rework hours × $110/h = $105,600`	modeled	≈$0.1M each
4	Bundle decision risk	`P(all pass) = 0.9 × 0.9 × 0.9 = 0.73`	one up/down vote on the whole design; p = 0.9 per decision (illustrative)	27% chance one objection unwinds everything

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Three separable decisions	`architecture · timeline · peak boundary → 2³ = 8 outcomes`	any partial approval preserves progress (record design)	a no on one cannot unwind all
2	Expected approvals	`E = 0.9 + 0.9 + 0.9 = 2.7 of 3`	vs the bundle's 0.73 chance of everything (illustrative p)	2.7 vs 0.73
3	Pre-wire coverage	`rows with signed LOE / impacted rows = 100% by 7/2`	one template, one deadline, one private conversation per owner (record)	100% before the room

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Gate chain held	`3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 = 77 days council→sign-off`	held to the day (record)	0 re-litigation

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Surprise escalations	~3 (the norm)	0 for the program's life	record; norm is benchmark
2	Pre-wire coverage before council	partial	100% by 7/2	record
3	Fence breaches in the log	—	0	record
4	Re-litigation after sign-off	—	0 across 77 days	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Avoided rework	`3 × 960 h = 2,880 h; 2,880 × $110 = $316,800`	~3 escalations (benchmark) × rework hours × rate (modeled)	≈$0.3M
2	Slip weeks avoided	`6 / 52 = 0.115 yr; 0.115 × $900,000 = $103,800`	six weeks of S2's ≈$0.9M/yr labor value not deferred (modeled)	≈$0.1M
3	Fenced engine not built	`2 eng × 12 wk × 40 h = 960 h; 960 × $110 = $105,600`	second reserve engine (modeled)	≈$0.1M
4	Total one-time	`0.3 + 0.1 + 0.1 = $0.5M`	all modeled — the dollars are what did not happen	≈$0.5M

TWO PROBES

> **How do you get sixty people to sign LOE you don't control?**
>
> A template, a deadline and a private conversation each — so nobody learned their cost in the room. The council was then asked for three separable decisions it could take one at a time; separability was the design.

> **What did you refuse, and did it hold?**
>
> Retuning the reserve math and folding in the MCS decommission — both written, signed and held; later scope requests were answered from the pack in writing. The legitimate concern (a second live warehouse becoming a second truth) was converted into the design as a monitored shadow copy, because converting a concern costs less than winning it.

Canon numbers (basis labels): ~80 participants / ~15 functions; about a dozen impacted systems (record) · 5/15 → 7/31 → 8/14; LOE 7/2 (record dates) · escalations 0; fence breaches 0 (record) · ≈$0.5M one-time (modeled). Never a dollar as record; never pair with S1 or S2 for the same interviewer.

BACKUP — ONLY IF ASKED S15 — Sephora — converting a 20-person Product/Engineering/QA team to a four-week cadence in six weeks — and the QA lead who now owns the gate

LPs this backup can serve: Hire and Develop the Best (primary) · Strive to be Earth's Best Employer (QA elevated, not blamed) · Insist on the Highest Standards (the gate)

Josyula's primary (Packet 5, Hire & Develop the Best) — use only if the question demands it, and keep every fact identical to that packet.

Basis. VP air cover, no line authority over anyone in the room; the QA lead authored the definition-of-done — I coached the frame, she owned the gate. The −50% and the six weeks are record; the dollars are my model on illustrative defect counts. Never invent a direct-report headcount.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The VP asked me to move a 20-person Product/Eng/QA team from a six-week hybrid cadence to four-week agile; QA feared compressed test windows — and they were right. **Task.** Convert in weeks without line authority. **Action.** Discovery: one-on-ones; an influencer per function. Plan & Design: the QA lead authored the definition-of-done — I coached the frame, she owned the gate. Build & Test: Jira benefit-ranking and review-status fields; a QA drop date with reason codes. Cutover: the four-week cadence live in six weeks. Hypercare: QA on product-business calls with pre-approved scenarios; she defended the gate without me. Lesson: development that depends on me staying is not development. **Result.** Primary metric: critical defects per release −50% after the 20-person team moved to a four-week cadence in six weeks. ≈$130K/yr modeled — the −50% and the six weeks are the record; the dollars are my model; no headcount invented.

COMPRESSED STAR

- **S —** Six-week hybrid cadence; QA (a contracting firm) and engineering (in-house) heading for a standoff; QA's fear of compressed test windows was correct arithmetic, not resistance.
- **T —** Convert to four weeks in weeks, with VP air cover and no line authority.
- **A —** Heard every real objection privately; the QA lead wrote the DoD content and entry criteria that can say no; drop date with reason codes; Jira fields; Sev-1 cause classification; rollback to six weeks named as a two-way door at kickoff and never used; facilitated the retro once and stepped back.
- **R —** Converted in six weeks; critical defects per release −50% while shipping 50% more releases (8.7 → 13/yr modeled from the cadence); the gate survived my leaving; neighbouring teams copied the ceremony from the user guide.

RESULTS LINE

Critical defects per release −50% (record) · six-week → four-week cadence, converted in six weeks (record) · 20-person team (record); releases/yr 8.7 → 13 (modeled from the cadence) · ≈$130K/yr (modeled; range $80–270K; no headcount invented) · the QA lead defending the gate by cycle 2 (record).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Cadence	`releases/yr = 52 / 6 = 8.67 ≈ 8.7`	six-week hybrid cycle (record)	8.7 releases/yr
2	Criticals	`4 per release × 8.7 = 34.8 ≈ 35/yr`	count illustrative; the record is the −50%	≈35/yr
3	The real constraint	`15 eng × 18 d = 270 eng-days → 270/3 = 90 stories; 4 QA × 18 d = 72 QA-days → 72/1 = 72 stories`	story ≈ 3 eng-days + 1 QA-day (illustrative; theory-of-constraints, toolkit label)	QA caps throughput at 72 — QA is the constraint
4	Late-land rework	`6 person-days × $800 = $4,800 per release`	re-test + hotfix after last-2-day lands (modeled)	≈$4.8K per release

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	QA-authored DoD	`entry criteria at Ready-for-QA — stories arrive ready`	the skeptic owns the gate (record)	the gate has an owner
2	Drop date	`not QA-ready by the drop date → dropped with a reason code`	silent late lands become visible re-scopes (record)	no last-2-day lands

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	Sev-1 attribution	`share caused by wrong stories vs wrong tests, classified by cause not blame`	QA put on product calls with pre-approved scenarios (record)	the trend reverser
4	Cadence converted	`releases/yr = 52 / 4 = 13; 13 / 8.7 = +50%`	converted in six weeks (record)	13 releases/yr
5	Criticals per release	`4 → 2 = −50%`	record	2 per release

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Cycle / releases per year	6 wks / 8.7	4 wks / 13 (+50%)	record
2	Critical defects	≈35/yr (4 × 8.7)	2 × 13 = 26/yr — while shipping 50% more	record −50%; counts illustrative
3	Late-land rework	≈$4.8K per release	~0 under the drop rule	modeled
4	The gate itself	mine to defend	QA lead defending it by cycle 2	record — survives my leaving

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Escaped criticals avoided	`35 − 26 = 9/yr; 9 × $10,000 = $90,000`	per-critical cost benchmark $8–15K (modeled)	≈$90K/yr
2	Late-land rework removed	`$4,800 × 8.7 releases = $41,760`	→ ~0 under the drop rule (modeled)	≈$40K/yr
3	Total	`90 + 40 = $130K`	range $80–270K; no headcount invented	≈$130K/yr

TWO PROBES

> **Who did you develop, concretely?**
>
> The QA lead — she authored the definition-of-done and its entry criteria, and by cycle two she was defending the gate on product-business calls without me. The test of development is that it survived my leaving; it did.

> **What did engineering give up?**
>
> Low-impact stories they had to drop at the QA drop date — I covered those with the business. Their objection became co-ownership through review, not a veto over the gate.

Canon numbers (basis labels): −50% critical defects per release (record) · 6 weeks to convert; 6-week → 4-week cadence (record) · 20 people (record) · 8.7 → 13 releases/yr (modeled) · ≈$130K/yr (modeled). Never a direct-report headcount.

BACKUP — ONLY IF ASKED Doc bench — Flexport — the Shipment-Visibility pilot roll-out: graduated waves behind six published criteria, four testing gates, monthly GO/NO-GO (launch discipline)

LPs this backup can serve: Insist on the Highest Standards (primary — gates that fire) · Deliver Results · Bias for Action (monthly GO/NO-GO cadence) · Customer Obsession (CSAT inside the gate)

Basis. Doc-derived micro-story from the program record (the SV pilot roll-out schedule and the SV product testing approach, 2021–22) — every figure is RECORD-FROM-DOCS. Same program family as Tripathy's S11: if she has already taken S11 deep, use this only for a launch-discipline question and keep the canon numbers identical to her packet. Program-lead voice: I owned the rollout design and the gates; data, ops and CS ran calibration, verification and UAT. Not in the master packet's S-numbering; supplements canon, never overrides it.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. A visibility product with public commitments was running a free pilot — every boarded client cost engineering, data and ops capacity before a dollar of revenue, and boarding clients before the data was trustworthy would manufacture churn at scale. **Task.** Design a roll-out that could say no — and make "no" a rule, not a judgment call. **Action.** Graduated waves: wave 1 two named clients (Ta Chen International, FAM Brands), wave 2 four of a seven-client pool, wave 3 twenty-plus, full roll-out H2 2022 — each client entering on a ~50-container calibration and graduating only on six published criteria: verification SLA 90% reviewed within 24h, data quality ≥90% and at least the incumbent Crux feed, milestone estimation ≥80%, performance to standards, ≥100 containers with key features live, CSAT ≥ the consignee average. Four testing gates stood before any launch: a 100-container live test watched end-to-end, biweekly bug bashes on the client job stories, shadow accounts with real Crux data in dedicated demo accounts, and UAT gated on data quality — 80% of shipments minor-to-no issues — not on the calendar. GO/NO-GO decisions ran monthly. **Result.** A roll-out that never had to roll back: the graduation rule, not heroics, set the pace — the launch-discipline version of S11's canon (waves held on the criteria; the held wave became the cleanest one). RECORD-FROM-DOCS

COMPRESSED STAR

- **S —** A free pilot under public commitments, with an acquired product (Crux) whose customers would transition mid-roll-out — sequencing and quality bars were the whole game.
- **T —** Own the roll-out design: wave structure, graduation criteria, testing gates and the GO/NO-GO cadence.
- **A —** Published six graduation criteria before wave 1; a ~50-container calibration per client as the entry bar; four testing gates including shadow accounts with real Crux data and a UAT data-quality bar (80% minor-to-no-issues) before any client UAT started; monthly GO/NO-GO from November; wave sizes 2 → 4-of-7 → 20+ → full H2 2022.
- **R —** A roll-out with zero rollbacks (S11 canon); every gate that fired was a rule firing, not a favor refused — the held wave became the cleanest wave.

RESULTS LINE

Waves 2 → 4-of-7 → 20+ → full H2 2022 (record-from-docs; matches S11 canon) · six graduation criteria published before wave 1 (canon) · ~50-container calibration entry bar (canon) · four testing gates with UAT gated at 80% minor-to-no-issues (record-from-docs) · monthly GO/NO-GO cadence (record-from-docs) · zero rollbacks (canon).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Client base waiting on the replacement	`79 Crux customers to migrate by end 2022`	Crux transition record (record-from-docs)	79 accounts exposed to a failed launch
2	Product state	`0 production clients; data quality unmeasured against the incumbent`	pre-pilot (record-from-docs)	unproven at any scale
3	The naive alternative	`big-bang = 79 clients × first contact with an unproven product`	the rejected option (record-from-docs)	one bad week burns every reference
4	Entry bar per client	`≥50 containers of client data required for calibration before onboarding`	graduation criterion 2a (record-from-docs)	every onboarding has a data gate

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Wave 1 cohort	`2 clients = Ta Chen + FAM Brands`	roll-out schedule (record-from-docs)	2 live, Dec 2021
2	Wave 2 cohort	`4 chosen from a named pool of 7`	roll-out schedule (record-from-docs)	4-of-7, Feb 2022

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	Wave 3 cohort	`20+ clients, Crux + non-Crux`	roll-out schedule (record-from-docs)	gradual expansion, Q2 2022
4	Full roll-out	`remaining Crux + net-new + existing 3PL clients`	H2 2022 (record-from-docs)	only after Wave 3 clears the gates
5	Calibration arithmetic — Wave 1	`2 × 50 = 100 containers before Wave 1 onboarding`	entry bar from Table 1 row 4 (record-from-docs; multiplication illustrative)	100 calibration containers
6	Calibration arithmetic — through Wave 2	`(2 + 4) × 50 = 6 × 50 = 300 containers`	same bar (record-from-docs; multiplication illustrative)	300 before any scale-up
7	Graduation gate	`6 of 6 criteria green before a wave advances — verification SLA 90%, data quality 90%+, milestone accuracy 80%+, production stability, ≥100 containers uploaded per client, zero show-stoppers`	the six published criteria (record-from-docs)	no wave skips the gate
8	Testing stack behind the gate	`4 tests sequenced: 100-container test → bug bashes every other week → shadow client accounts (Marc Fisher, Michelin) → UAT, starting only when 80% of shipments show minor-to-no issues`	testing-approach doc (record-from-docs)	UAT gated at 80%
9	Decision cadence	`2 GO/NO-GO meetings before Wave 1 — business (John, Eric) + tech (product & eng leadership)`	roll-out schedule, November (record-from-docs)	every wave is a decision, not a date

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Wave progression	plan only	W1 Dec 2021 → W2 Feb 2022 → W3 Q2 2022 → full H2 2022	record-from-docs
2	Clients live on the product	0	2 → 6 → 20+	record-from-docs
3	Rollbacks	—	0 — no wave ever reversed	canon
4	Speed control	fixed launch date	accelerate / slow down on client feedback, per the published plan	record-from-docs

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Pilot revenue, by design	`clients × $0 × ~6 months = $0 — free for the full duration of the pilot`	roll-out schedule: "no charge for the full duration of the pilot"; ~6-month estimate (record-from-docs)	$0 forgone deliberately — the gate is the product
2	Big-bang exposure avoided	`79 × 50 = 3,950 containers of one-shot onboarding risk`	client count and entry bar (record-from-docs; multiplication illustrative)	never spent — waves capped the blast radius at 2, then 6, then 20+
3	Cost of a failed Wave 1	`2 × 50 = 100 calibration containers + one feedback cycle — the maximum the plan allowed to fail at once`	record-from-docs; arithmetic illustrative	failure priced small by design
4	Dollar result claimed	`no revenue or saving dollar in the record — not claimed`	record-from-docs	not claimed

TWO PROBES

> **Why gate UAT on data quality instead of a date?**
>
> Because a client UAT against dirty data manufactures false confidence — the client signs off on a product the data then betrays. The 80% minor-to-no-issues bar meant UAT started when the product could survive a skeptical user's first week; the date followed the data, never the reverse. Shadow accounts with real Crux data let us find the defects before the client did.

> **What did the calibration actually buy?**
>
> A verified baseline per client before scale: ~50 containers through verification with the SLA clock running, so the data-quality and estimation criteria were measured on the client's own freight, not on our best lanes. It converted "ready" from a feeling into a measurement — and it made the wave-2 selection from the seven-client pool defensible to the clients not picked.

Canon numbers (basis labels): waves 2 → 4-of-7 → 20+ → full H2 2022 (record-from-docs, matches S11 canon) · six criteria incl. DQ ≥90% · ETA ≥80% · SLA 90% within 24h · ≥100 containers + features · stability · CSAT ≥ consignee average (canon; the pilot schedule phrases the SLA as 24–48h — canon wins, footnoted in Tripathy's packet) · ~50-container calibration (canon) · 100-container test · bug bashes · shadow accounts · UAT 80% bar (record-from-docs). Never: a client-revenue claim (the pilot was free); never merge with Crux-transition ARR.

Decisions not taken — the trade-off ledger NEW

An infrastructure owner asks *what else could you have done?* Each row is a fork in the Sephora node work, what I chose, the road I did not take and the number that decided it. Dollars keep their labels: record · modeled · illustrative.

FORK	TAKEN	NOT TAKEN	WHY — IN NUMBERS	STORY
Go-live shape	Two go-lives: January planning, then May execution, with a deliberate four-month interim	One big-bang go-live	One go-live puts a reserve-rule defect and a floor-execution defect in the same room in the same week; two go-lives let each carry one defect class. Cost: four months with two systems of record, and an interim reporting burden.	**S2 · U5**
Merch reserve	Visible to Merchandising, not promisable	Promisable reserve	Break-even p* = 36.7% of reserved units released; at an illustrative 20% release that is −$50K per 10,000 units (−$202K all-in). Visibility costs nothing the floor cannot pick.	**U3**
Pools	One physical pool, three planning identities	Keep Dotcom 801 and Retail 901 as two pools	About 260K units a year of self-transfers go to zero by construction (illustrative volume). The pooling dollars are a range ($0.74M–$2.2M modeled); 87% of the recurring benefit does not depend on pooling.	**U2 · S2**
Robots at 94	Throttle to about 30% for one week, DC Ops' own dial	Hold full bot volume and fix in flight	At full bot share, 94% pick accuracy is 6,000 mis-picks a week against a 2,000 budget (illustrative 100K units); the maximum safe share is 33.3%. The throttle cost about 421 extra labor hours (modeled).	**U4**
Vendor slip	Escalate the week it surfaced; re-slot openly	Compress UAT to absorb the three-week Locus RTL slip	Blocker + Critical open = 0 was the go/no-go bar; I would have said no-go on one open Blocker.	**U5 · S2**
Comingle-reserve improvements	Held out of the release; priced as a later enhancement	Ship them in the same release	Zero customer-facing incidents was the constraint; a second reserve engine is a second defect class in the same release. The out-of-scope fence was a council decision (Apple C4).	**S2**
Date vs training	Hold the go-live date; move training 1.5 weeks	Move the go-live date	A 1.5-week slip prices at about $875K on my modeled $2.5M-a-month division of the $20M pipeline — not banked. Training could absorb the delay; the pipeline could not.	**S18**
Deferral	A visible Parking Lot row with an owner	A buried or quietly dropped Blocker	A deferral you can count is a decision; one you cannot is a defect waiting. Blocker + Critical were read first in the Monday review.	**U7**

VERIFY BEFORE YOU SAY IT NEW

“Absent” = not in any source file or the master (yellow pill in the story: VERIFY-WITH-RAJESH). “Conflict” = two sources disagree; the safe wording below is the conservative one and needs no confirmation. Confirm or drop the first group tonight; use the safe wording for the rest.

#	ITEM	WHERE	TYPE	IF UNCONFIRMED, SAY
1	**Matrix size — 6 node-channel pairs, 14 events, 61 live cells**	U1	absent	Say “a matrix of node × channel × bucket cells, each asserted before and after.” Do not say a count.
2	**Whether the sponsor pack carried the $2.2M point or the range**	U2	absent	Say “I would lead with the range; the $2.2M is the equal-pools ceiling.”
3	**Utah cube utilization — did space bind?**	U4	absent	Say “I did not claim space bound; labor, then accuracy.”
4	**Steps 5 and 6 of the six-step May choreography**	U5	absent	Name steps one to four (clear negatives, transfer AFS to Uncommitted, scan-and-reserve, zero-then-publish).
5	**The other three of the six cutover invariants**	U5	absent	Name the three on the page: Σ AFS conserved; no SKU sellable on 800 and 900 at once; negatives cleared first.
6	**Whether a timed rollback drill ran in the mocks**	U5	absent	Say “armed for both go-lives, never fired” — not “drilled under load.”
7	**Whether cause class was recorded on the AFS audit rows**	U6	absent	Say “I would require a cause class on every audit row.”
8	**Close dates — the log has none**	U7	absent	Never quote time-to-close. The shape claims use created-date months and weeks only.
9	**RF-device miss: cutover day or pre-training device test; ~100 vs ~400 devices**	S18	conflict	Say “hundreds of RF devices,” “a device test before training,” “training slipped a week and a half, the date held.” Apple G2 itself says tellings differ.
10	**January go-live content: reserve rules (master) vs PO creation / inbound (Apple C1, C4)**	S2 · U1 · U3 · U5	conflict	Say “the January planning go-live — purchasing and inbound, with the reserve rules live against demand.” Never describe it as the floor go-live.
11	**Program length: “fifteen months” (Apple C1) vs “fourteen months” (Apple C2)**	S2	conflict	Say “about fifteen months from discovery to go-live.” The log itself spans 2024-03-17 to 2025-10-01.
12	**Build tracks: “15+” (Apple C1) vs “about thirty” (Apple C2)**	S2 · U1	conflict	Say “about thirty build tracks.”
13	**Retail rows: 172 by exact label (pandas) vs 171 (Apple C2 p24)**	S2 · U7	conflict	Say “about 170 retail rows.”
14	**Open pipeline: Apple’s five buckets sum to 112; 122 rows are not closed (112 explicit open + 10 with no status)**	U7	conflict	Say “122 not closed; ten of them carry no status.”
15	**Three signing orgs (Strategy, Inventory Management, Inventory Accounting) and the ISM 6/5/25 FSD date**	U3 · U6 (S1 anchors)	conflict	User anchor from master S1; Apple E1 says only “the business signed.” Say “the business owners signed the row.” The UDC FSD (8/14) is the record.
16	**Day-one pace in units (280,000 units, 40 stores, 7 inbound appointments)**	U5	conflict	Loop packet only. Say it only if you lived it; otherwise say “a daily plan in units.”

NUMBERS I MUST NOT GET WRONG NEW

Every number this packet says aloud, with its basis label and where it comes from. **Record** = Apple v11 / resume / the log itself. **Modeled** and **illustrative** are said with those words. Right-hand column = the version that gets you in trouble.

#	NUMBER — SAY IT EXACTLY LIKE THIS	BASIS	SOURCE	NEVER
SEPHORA UDC — PROGRAM RECORD				
1	+15% processing efficiency; zero customer-facing incidents through both go-lives	record	Apple C1 p21–22; resume	the dollars as signed
2	Gate 98% (99% stretch); week 2 at 94%; ~30% throttled for a week; full automation week 3	record	Apple C3 p26	“the robots failed”
3	343 tasks / 48,885 task-minutes ≈ 815 h; rollback sync < 99% for 4 h, p95 > 2× for 1 h; armed twice, never fired	record	Apple C2 p24; resume; SAP Narrative p4	“weekend” runbook
4	Kickoff 3/29/24 · sessions 4/30 · council 5/15 · LOE 7/2 · sign-off 7/31 · FSD 8/14; E2E 11/22 · 12/20 · 1/30; peak freeze Oct–Dec	record	Apple C4 p28; C1 p22	6/5/25 as this program’s FSD
5	60+ stakeholders (~80 names, ~15 functions); about a dozen impacted systems; about thirty build tracks; about fifteen months	record	Apple C1 p22, C4 p28, C2 p24	“20+ systems”; “fifteen tracks”; “fourteen months”
6	Locus retail testing slipped three weeks; Locus-side rows 86 of 836 = 10.3%	record	Apple C2 p24, C3 p26; log	—
THE 836-ROW ISSUES LOG — COMPUTED WITH PANDAS				
7	836 rows; 714 closed = 85.4%; 122 not closed (112 explicit open + 10 blank status)	record	log, column Status	“112 open”
8	Blocker 164 + Critical 277 = 441 = 52.8%; B+C closed 432 = 98.0%; open B+C 9 (1 Blocker, 8 Critical)	record	log, columns Severity × Status	“all Blockers closed”
9	Closure by severity: Blocker 99.4% · Critical 97.1% · High 75.4% · Medium 69.7% · Low 60.0%	record	log	—
10	Flows: Outbound 238 · Retail 172 · Inbound 113 · IC 90 · Dotcom 62 = 675 (80.7%)	record	log, column Operation	Retail 171 and 172 mixed
11	Monthly new rows: Feb 90 · Mar 91 · Jun 271 · Jul 65 · Aug 21 · Sep 14 · Oct 4; June 88.2% Go-Live category	record	log, column Created	“weekly open counts”
12	Weekly new rows 68 → 160 → 34; geometric decay r = (4/271)^(1/4) = 0.349, half-life ≈ 20 days	record + modeled fit	log; derived	a time-to-close
13	Owners: IT/WMS 342 · Ops 111 · Locus-side 86	record	Apple C2 p24; log	—
SEPHORA WMS MIGRATION (S18)				
14	8 months, two DCs; ~$1.8M budget, ~3% under (~$54K); $20M pipeline of four projects started the next month	record	Apple D1 p28–29; resume	“~$800K”; “$2.5M/week”
15	Mocks 7.0% → 1.8% → 0.3% against the 0.5% tolerance published with Finance before mock one	record	Apple D2 p31; resume	—
16	72-hour in-transit freeze; legacy read-only 30 days; hypercare exit < 2 blockers/day, week 3 of 4; 14 core + 6 DC Ops	record	Apple D1/D2 p29–31	—
17	Hundreds of RF devices; a three-to-four-day estimate; training +1.5 weeks; date held	record	Apple G2 p44–45	“400 devices failed”; “~100 of ~400”
18	≈$2.5M a month of slip ($20M ÷ 8); ≈$875K for a 1.5-week slip	modeled — my division, not banked	derived from Apple D1	any multiplier
THE NEW STORIES — MODELED OR ILLUSTRATIVE (SAY THE LABEL)				
19	U2: textbook $2.2M = $7.5M × (1 − 1/√2); plausible floor ≈$0.74M; 87% of the ≈$1.65M recurring does not depend on pooling	modeled	U2 tables 1–4	$2.2M as a point
20	U3: break-even p* = 36.7%; −$50K / −$202K per 10,000 units at a 20% release (illustrative)	modeled; illustrative inputs	U3 tables 1–4	that Merchandising agreed happily
21	U4: 421 h ≈ $11.8K vs ≈4,200 mis-picks ≈ $50.4K; maximum safe bot share 33.3% vs ~30% dial	modeled; illustrative DC	U4 tables 1–4	“space bound”
22	U5: ≈$31.7K a week of deferred benefit on the modeled $1.65M a year; runbook ≈$89.6K of effort ≈ 2.8 weeks	modeled	U5 tables 1–4	banking either
23	U6: −10 → −5 → −2 → 0 worked SKU (illustrative); asymmetry ≈ 1 : 6.3; critical ratio 13.7%	illustrative / modeled	U6 tables 1–4	a measured negative count
24	U1: AFS 557 — not 560, not 554 (worked trace); 61 live cells is illustrative	illustrative	U1 tables 1–4	a cell count as fact
LOOP-WIDE GUARDRAILS (OTHER INTERVIEWERS’ STORIES — DO NOT DRIFT HERE)				
25	Customers 7 → 68 = 9.7×; 170 monthly active users; CTS $0.66 → $0.03 (about 95%)	record	Apple A1 p6–7	any fifty-times multiplier; “$1.00 → $0.03”
26	Six sites — never with a regional qualifier; ~$1.5M pre-prod estimate is separate from CTS; S14 $10M won is not savings; S16 >$100K is priced, not claimed	rules	master §4.1; recon §3	a blended $1.5M–$2M range; any bill-of-materials-explosion story
27	$2.5M names three different things (S1 annual impact, S14 order value, S18 per-month slip); $1.5M names four — attach the noun every time	rule	recon §1 item 9	a bare “$2.5M”

SOURCE TRACE NEW

How to read. Precedence on conflict: (1) source documents, (2) Apple workbook v11, (3) Amazon master. **Y** = same value in the cited source. **Y, rounding** = same fact, different precision. **Y, derived** = recomputed from record inputs; the arithmetic is asserted by script. **illustrative** = labeled input, not a measured fact. **anchor only** = in the master or loop packet only, no Apple or primary-document support — say with that label. **CONFLICT → resolved** = sources disagree; the conservative wording on the page is the resolution. Math scripts behind this packet: 380 assertions pass (S2 26 · S18 16 · U1 42 · U2 27 · U3 24 · U4 40 · U5 32 · U6 33 · U7 95 · bridge 24 · back 21). Apple v11 = Apple_PMIII_Interview_Stories_v11 (page numbers are PDF pages). Master cb91 = uploaded 131-page master; the repo master's §4.1 ledger is numerically identical for every number used here.

CLAIM	STORY	SOURCE FILE	PAGE / ROW	MATCH
S2 — UTAH DC COMINGLE (MODIFIED) AND PROGRAM RECORD SHARED BY ALL NINE STORIES				
+15% processing efficiency	**S2 · U4**	Apple v11 C1	p22 "DC ops' measurement"; resume p1	**Y**
Zero customer-facing incidents, both go-lives	**all**	Apple v11 C1	p21 result 1; resume p1	**Y**
Gate 98% (99% stretch); week 2 = 94%; ~30% throttled; full automation week 3	**S2 · U4 · U5**	Apple v11 C3	p26 math box	**Y**
343 tasks / 48,885 task-minutes ≈ 815 h	**S2 · U5**	Apple v11 C2	p24 (48,885 ÷ 60 = 814.75)	**Y, rounding**
Rollback: sync < 99% × 4 h; p95 > 2× × 1 h; armed twice, never fired	**S2 · U5**	Resume; SAP Narrative	resume p1; SAP Narrative lines 101–102; Apple v11 C1/C2	**Y**
Kickoff 3/29 → 4/30 → council 5/15 → LOE 7/2 → sign-off 7/31 → FSD 8/14; peak freeze Oct–Dec	**S2 · U1–U6**	Apple v11 C4	p28 math box	**Y**
E2E days Inbound 11/22 · Dotcom 12/20 · Retail 1/30	**S2 · U1 · U5**	Apple v11 C1	p22 math box	**Y**
January vs May go-live content (reserve rules vs PO-creation / inbound)	**S2 · U1 · U3 · U5**	Apple v11 C1/C4 vs Master cb91	Apple p22, p28; master S2 p12–17	**CONFLICT → resolved**
60+ stakeholders; ~80 names; ~15 functions	**S2 · U2**	Apple v11 C1/C4	p22; p28	**Y**
About a dozen impacted systems (master "20+ touched" removed)	**S2 · U1 · U2 · U5**	Apple v11 C1/C4	p22; p26; recon S2.10	**CONFLICT → resolved**
About thirty build tracks (C1 says 15+)	**S2 · U1**	Apple v11 C2 vs C1	p24 vs p21–22; recon S2.11	**CONFLICT → resolved**
About fifteen months (C2 says fourteen; log spans 2024-03-17 → 2025-10-01)	**S2**	Apple v11 C1 vs C2	p21 vs p23–24; log Created min/max	**CONFLICT → resolved**
Locus retail testing slipped three weeks	**S2 · U5**	Apple v11 C2	p24	**Y**
≈$1.6M/yr = 0.88 + 0.22 + 0.45 + 0.10 = $1.65M; ≈$2.2M one-time = $7.5M × 0.293	**S2 · U2 · U4 · U5**	Master cb91	S2 model rows p12–17; recomputed in S2 tables (31,300 h × $28 = $876,400)	**Y, derived**
Illustrative base: 100K units/week; 120 FTE × 2,000 h; $50M × 25% × 60% = $7.5M	**S2 · U2 · U4**	Master cb91	S2 model rows; labeled illustrative	**illustrative**
THE 836-ROW ISSUES LOG — RECOMPUTED WITH PANDAS (NO CLOSE-DATE COLUMN EXISTS)				
836 rows; 714 closed = 85.4%	**S2 · U1 · U7**	UDC issues log (xlsx, 836×19; pandas)	col Status; Apple v11 C2 p24	**Y**
Blocker 164 + Critical 277 = 441 = 52.8% (52.75%)	**S2 · U1 · U7**	UDC issues log (xlsx, 836×19; pandas)	col Severity; Apple v11 C2 p24	**Y, rounding**
B+C closed 432 = 98.0%; open 9 (1 Blocker, 8 Critical)	**U7**	UDC issues log (xlsx, 836×19; pandas)	cols Severity × Status	**Y, derived**
Closure by severity 99.4 / 97.1 / 75.4 / 69.7 / 60.0%	**U7**	UDC issues log (xlsx, 836×19; pandas)	cols Severity × Status	**Y, derived**
Flows 238 · 172 · 113 · 90 · 62 = 675 (80.7%); Retail is 171 in Apple	**U7**	UDC issues log (xlsx, 836×19; pandas)	col Operation, exact-label match; Apple v11 C2 p24	**Y, rounding**
122 not closed = 112 explicit open + 10 blank status (Apple's five buckets sum to 112)	**U7**	UDC issues log (xlsx, 836×19; pandas) vs Apple v11	col Status; Apple C2 p24	**CONFLICT → resolved**
Monthly new rows Feb 90 · Mar 91 · Jun 271 · Jul 65 · Aug 21 · Sep 14 · Oct 4 (not "weekly open counts")	**S2 · U5 · U7**	UDC issues log (xlsx, 836×19; pandas)	col Created; Apple C2 p24 "monthly arc"	**Y**
June: 239 Go-Live category (88.2%); 140 B+C (51.7%); 242 closed (89.3%)	**U7**	UDC issues log (xlsx, 836×19; pandas)	cols Created × Category / Severity / Status	**Y, derived**
Weekly new rows 68 → 160 → 34	**U5 · U7**	UDC issues log (xlsx, 836×19; pandas)	col Created, ISO weeks around go-live	**Y, derived**
Owners IT/WMS 342 · Ops 111 · Locus-side 86 (10.3%)	**S2 · U7**	UDC issues log (xlsx, 836×19; pandas)	col Owner; Apple v11 C2 p24, C3 p26	**Y**
Decay r = (4/271)^¼ = 0.349; half-life ≈ 20 days	**U7**	derived	from the monthly arc; modeled fit	**Y, derived**
42 physical-building rows (dock 23 · space 19 …); 39 closed; 0 open B+C	**U7**	UDC issues log (xlsx, 836×19; pandas)	cols Area / Description; my label grouping, stated on the page	**Y, derived**
Two AFS rows (ID 716: 400 of 1,800 = 22.2%; ID 818: ~10-minute hold-feed latency)	**U6**	UDC issues log (xlsx, 836×19; pandas)	rows ID 716, ID 818	**Y**
S18 — WMS MIGRATION (MODIFIED)				
8 months, two DCs; team 14 core + 6 DC Ops	**S18**	Apple v11 D1	p28–29; resume p1	**Y**
~$1.8M budget, ~3% under (= $54,000)	**S18**	Apple v11 D1	p29	**Y, rounding**
Mocks 7.0% → 1.8% → 0.3% vs 0.5% tolerance published with Finance before mock one	**S18**	Apple v11 D2	p31; resume p1	**Y**
$20M pipeline of four projects, started the next month	**S18**	Apple v11 D1	p29	**Y**
≈$2.5M/month = $20M ÷ 8 (my division); $83,333/day; $875K = 10.5 d × $83,333	**S18**	derived from Apple v11 D1	recon S18.05; not banked	**Y, derived**
72-hour in-transit freeze; legacy read-only 30 days; hypercare exit < 2 blockers/day in week 3 of 4	**S18**	Apple v11 D1/D2	p29; p31	**Y**
RF devices: "hundreds"; three-to-four-day estimate; training +1.5 weeks; date held	**S18**	Apple v11 G2	p44–45	**Y**
Master "~100 of ~400 devices at cutover; paper-pick fallback; ≈$45K; ~19×" (removed from the spoken story)	**S18**	Master cb91 vs Apple v11	master S18 p96; Apple G2 p45 says tellings differ	**CONFLICT → resolved**
(0.5% − 0.3%) × $50M = $100K	**S18**	Master cb91	S18 model row; illustrative base	**illustrative**
U1 — NODE × CHANNEL × BUCKET-TRANSITION MATRIX (NEW)				

CLAIM	STORY	SOURCE FILE	PAGE / ROW	MATCH
Bucket inventory: one owning system per bucket; exactly-one-bucket rule; replay no-op; SAP movement types 101 · 311 · 601 · 701 · 702	**U1**	Master cb91	S1 p7 (discovery/design), p11 system map	**anchor only**
Worked trace: AFS 560 → 557 (one bucket); 554 if two; 560 if zero	**U1**	derived	U1 table 2; illustrative units	**illustrative**
6 node-channel pairs · 14 events · 61 live cells; 4 in-transit cells + 3-case de-aggregation family added	**U1**	illustrative / master	VERIFY pill on page; Recon S4.04	**illustrative**
30,000 incidents/yr ÷ 61 = 491.8 × $30 = $14.8K; 61 cells × 4 h × $110 = $26.8K; break-even ≈ 1.8 cell-classes	**U1**	derived	U1 table 4; $30 and $110/h benchmark inputs	**illustrative**
U2 — PROFILE BOTH BOOKS BEFORE THE DESIGN (NEW)				
Two books (MCS, S/4); discovery 3/29 → 4/30; node sessions across ~15 functions	**U2**	Master cb91 · Apple v11	S1 p7; Apple C4 p28	**Y**
Release f(ρ, σ) = 1 − √((1+ρ)/2) family; 29.3% ceiling; 9.9% floor; $0.74M–$2.20M; 87% = 1.43 ÷ 1.65	**U2**	derived	U2 tables 2–4 (27 script checks)	**Y, derived**
Overlap sensitivity 40/60/80% → $1.46M / $2.20M / $2.93M; ≈260K self-transfer units/yr	**U2**	derived · master S2	U2 table 2; S2 model row	**illustrative**
U3 — MERCH RESERVE, THE PRICED NO (NEW)				
Three signing orgs; zero open “who owns this bucket” rows; 0 units of merch reserve in published AFS	**U3 · U6**	Master cb91	S1 p7; Apple E1 p33 says only “the business signed”	**anchor only**
p* = c/(g+c) = 36.7%; −$50K and −$202K per 10,000 units at a 20% release (illustrative); asymmetry 1 : 6.3	**U3**	derived	U3 tables 1–4 (24 checks)	**Y, derived**
30,000,000 orders × 0.10% = 30,000 oversell incidents (benchmark 0.05–0.25%)	**U3 · U6**	Master cb91	S1 model rows; Recon S1.12	**illustrative**
U4 — WHERE THE NODE’S CONSTRAINT BOUND (NEW)				
94% vs 98%; throttle ~30%; max safe share 33.3%; 6,000 vs 2,000 mis-picks; 28.6σ	**U4**	Apple v11 C3 · derived	p26; U4 tables 1–3	**Y, derived**
4,615 → 4,013 h/wk (240,000 h ÷ 52; ÷ 1.15); +421.4 h = 10.5 people-weeks at 40 h; $11,799 at $28	**U4**	derived	U4 table 2 (40 checks); $28 loaded rate = master S2 modeled input	**Y, derived**
4,200 mis-picks avoided × $12 = $50,400; break-even $2.81; ratio 4.3	**U4**	derived	U4 table 4; $12 is an illustrative input	**illustrative**
$2.2M ÷ $25 = 88,000 units × 0.25 ft³ = 22,000 ft³; ~92 days of float Jul 1 → Oct freeze	**U4**	derived · NACF booklet	unit cube from CAP-C; calendar	**illustrative**
U5 — 343-TASK RUNBOOK (NEW)				
Window May 1 → Jul 1 = 61 days	**U5**	calendar · Apple v11 C1/C2	31 + 30 days	**Y**
Steps 1–4 of the six-step choreography; three invariants (Σ AFS conserved; never sellable on 800 and 900; negatives first)	**U5**	Master cb91	S2 p16 (cutover invariants); steps 5–6 and invariants 4–6 absent	**anchor only**
$1.65M ÷ 52 = $31,731 a week; 814.75 h × $110 = $89.6K = 2.8 weeks	**U5**	derived	U5 tables 2–4 (32 checks)	**Y, derived**
24 events ≈ $714; 0.02⁴ vs 0.02; 680 units conserved (three-SKU trace)	**U5**	derived	U5 trace; illustrative units	**illustrative**
Day-one plan 280,000 units · 40 stores · 7 inbound appointments	**U5**	Loop packet	amazon-l6-scm-loop-packet.html only; not in Apple or master	**anchor only**
U6 — NEGATIVE AFS AND THE CORRECTION SCREEN (NEW)				
Worked SKU: basis 25, 35 committed → −10; −5 → −2 → 0; ≤15-minute feed; <5-minute consumer lag	**U6**	Master cb91	S1 p11 system map; worked SKU illustrative	**illustrative**
Critical ratio 1 ÷ (1 + 6.3) = 13.7%; reconciliation 4 → 1 FTE × $110K = $330K	**U6**	derived · Master cb91	U6 tables; S1 model row	**Y, derived**
U7 — THE ISSUES LOG AS READINESS INSTRUMENT (NEW)				
836 × 0.5 h × $110 = $45,980 ≈ 1.45 weeks of $31,731; z = 1.50 (directional)	**U7**	derived	U7 tables 2–4 (95 checks); basis modeled	**Y, derived**
BRIDGE, ROUTER AND BEHAVIORAL PACKS				
120,000 units/day → 5 doors; +13.6% hours; 49.0% true utilization; dwell 13.3 h → 40 h (3.0×); ≈14 flex heads; p* = 25%	**Bridge**	NACF booklet	CAP-A · CAP-B · CAP-C · CAP-D · CAP-E · INV-D	**Y**
Real AIQB wordings: 16 LPs, router + 36 behavioral questions	**all**	AIQB Reference Guide	128 questions parsed; cited as “AIQB: LP Q#”	**Y**
9.7× (7 → 68); never a fifty-times multiplier; $0.66 → $0.03; six sites; S14 $10M, S16 >$100K, S11 ~$1.5M	**guardrails**	Apple v11 A1 · recon §3	p6–7; recon items 1–5, 84–88	**Y**

62 grouped claims; every arithmetic row above stands for the full set of recomputed steps in the story’s four tables.

INTERVIEW-PREP PACKET 4 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

Amazon NACF Supply Chain loop — Packet 4: Zhaowen Lan

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

Zhaowen Lan

Senior Manager, Supply Chain — NACF Supply Chain (peer of the hiring manager)

Thursday, October 30, 2026 · 11:00 am PDT · 55-minute video interview

Strategy in one line. A peer Senior Manager tests whether you can reach one truth across organizations that don't report to you and build a process that survives its author. Eight Sephora S/4HANA primaries, each a different interview answer from one program: open with S1 (three orgs, three meanings of "available" → one signed definition per node per channel; oversells recurring → zero) and hold S3 (Finance and Operations on one Financial On-Hand, the failure direction designed before the happy path); the six variants (S1-a data contract · S3-a de-aggregation to the cent · S1-b control screens and operating model · S1-c the priced no on merchandise reserve · S1-d the counted-rows go/no-go · S1-e the seventeen-interface study) answer the other LPs. Program-lead voice; S4 and S5 are bench stories. Start at the cram sheet and the router.

LP focus — my guess, not confirmed: Earn Trust · Ownership · Invent and Simplify or Have Backbone. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title and the JD in "Read me first" and should be held loosely. No interviewer biography is asserted anywhere in this packet — only title-based inference, labeled as such.

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

Contents NEW

Combined table of contents. Page numbers are for the printed PDF. Every NEW pill marks content added in this v2 upgrade; UPGRADED marks an existing story deepened in place; VERIFY-WITH-RAJESH marks an unsupported specific listed on the back checklist.

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5-minute cram sheet — read this, then walk in NEW

Profile hooks (inference from the title plus the inbound booklet's lens — not a biography). A peer Senior Manager of Supply Chain who shares the cadence. Listening for: how you run mechanisms other people own; the handoffs you do not want reopened; a disagreement that stays professional; what you would just fix versus bring to them; who speaks first when a shared metric goes red; and cross-functional operating judgment — scope, dependencies and what you would cut.

STORY	WHAT IT IS (ONE LINE)	LPS	THE ONE NUMBER IT LANDS
S1	One signed meaning of "available" across three node types; policy before configuration	Earn Trust · Ownership · Backbone	**oversell recurring → zero through two go-lives; ≈$2.5M = $1.0M anchor + ≈$1.5M modeled**
S3	Shipped-Not-Billed: Finance and Operations on one Financial On-Hand; failure direction first	Earn Trust · Invent and Simplify · Customer	**FOH truths 2 → 1; late line = −2, never +2; ≈$0.6M modeled**
S1-a	The data contract: ten buckets, one owner each, every event into exactly one bucket	Dive Deep · Highest Standards	**five-unit order: AFS 555 held; bad events give 550 · 550 · 565**
S3-a	De-aggregation to the cent: one posting, many lines, Σ lines = posting	Highest Standards · Dive Deep	**$33.33 + $33.33 + $33.34 = $100.00 (naive: $99.99)**
S1-b	Control screens and the operating model: every AFS-changing act has a screen, a role, an audit row	Ownership · Earn Trust · Deliver Results	**Negative AFS clean at T+1; drift = 0 SKUs on consecutive days**
S1-c	The priced no on merchandise reserve; reserve-copy invariants	Have Backbone · Customer Obsession	**560 vs 710 (150 unpickable units); break-even 30 ÷ 49 = 61%**
S1-d	Go/no-go from counted rows; two go-lives; rollback as numbers	Deliver Results · Bias for Action	**836 rows, 441 severe (52.8%); triggers: sync < 99% for 4 h, p95 > 2× for 1 h — never fired**
S1-e	Seventeen interfaces and the $42K I first refused: refuse, study, reverse	Are Right, A Lot · Frugality	**$42K funded, $25K contingent; 17 interfaces → 2 critical usages**

Three sentences to open with.

1. "Most oversell bugs are vocabulary bugs: three organizations meant three different things by 'available,' I owned no system, and I got one definition signed on 6/5/25 before any configuration."
2. "I choose the failure direction before the happy path — wrong on the safe side, visible, with a named owner."
3. "I read go/no-go from counted rows, and I say out loud which of my dollars are measured and which are my model."

Five numbers never to get wrong.

1. ≈$2.5M a year = $1.0M MCS run cost retired (user anchor) + ≈$1.5M modeled (oversell ≈$0.9M · reconciliation ≈$0.33M · change premium ≈$0.3M). Never record for the $1.5M.
2. FSD v1.1 (3/5/25) → v1.2 (5/16/25) → signed 6/5/25 by Supply Chain Strategy, Inventory Management and FSC Inventory Accounting.
3. AFS = Basis − PickRes − EcomPick − EcomPack − SNB, with Basis = min(S4_Avail, WMS_OnHand); worked SKU 980 − 420 = 560.
4. Financial On-Hand truths 2 → 1, ≈$0.6M/yr modeled, no pre-state drift dollar in the record.
5. 836 rows / 85.4% closed / Blocker 164 + Critical 277 = 52.8%; rollback triggers sync < 99% for 4 h or order-flow p95 > 2× for 1 h — armed both go-lives, never fired.

Cram sheet (2 of 2) — tell the S/4 story in 90 seconds, 5 minutes or 10 minutes

NEW

The S/4HANA program is one umbrella with eight different answers. Pick the length from the interviewer's body language and the clock, then let the probe choose the depth beat.

LENGTH	WHAT GOES IN, IN ORDER	STOP LINE	IF THEY INTERRUPT
90 seconds	S1 pitch only: situation (three meanings of "available") → task (define every bucket, sign before configuration) → one action (pre-wired every row; three signatories 6/5/25) → result (oversell recurring → zero through two go-lives; ≈$2.5M, $1M measured, $1.5M modeled).	"…the first number is measured; the other three are my model."	Answer the question, then return to the sentence you were on.
5 minutes	0:00–1:30 S1 pitch. 1:30–2:30 the one-SKU walk: 980 − 420 = 560; raw S/4 over-promises 440. 2:30–4:00 *one* depth beat chosen by the probe: Finance/controls → S3; data → S1-a; control screens → S1-b; conflict → S1-c; cutover → S1-d; testing → S1-e. 4:00–5:00 what I would do differently (the in-transit rule, S4) and the NACF bridge sentence.	"…and at NACF the same discipline is what the promise can see."	Skip the depth beat; the walk and the result are the spine.
10 minutes	0:00–1:30 S1 pitch. 1:30–3:30 the five gated phases with their exit bars (4/30 → FSD 6/5/25 → E2E → go-lives → hypercare exit). 3:30–5:00 the math: one SKU, the decrement contract (AFS 555 through five events), drift 0. 5:00–8:00 *two* depth beats: one control or finance (S3 or S1-b) and one conflict or evidence (S1-c or S1-e). 8:00–9:00 the dollar decomposition with basis words. 9:00–10:00 failure/learning (S4) and the NACF bridge.	"…which is why I'd bring the same bar to inbound."	Compress phases first, never the math or the basis words.

Which depth beat for which probe. "How did you get them to agree?" → S1 pre-wiring. "How does Finance trust it?" → S3. "How is it defined?" → S1-a. "Who owns it after you?" → S1-b. "Tell me about a no" → S1-c. "How did you decide to go?" → S1-d. "How did you scope testing?" → S1-e. "One cent?" → S3-a.

Never both ends of one program to one question. Eight answers, one umbrella — never tell two of them as if they were unrelated programs; say "same program" when you move between them. S2/S18 (UDC and WMS floor stories) are Goforth's; S14, S10, S15 and S16 are Josyula's; S11/S12 are Tripathy's; S6/S8 are Hoerger's.

Question → story router — find the story in ten seconds NEW

If the interviewer asks a behavioral question on an LP, read across: the primary story and the angle to take, the backup, the 15-second opener, and the number to land. Every one of the 16 LPs routes somewhere; two are thin and say so. Only the stories in this packet are routed — never a story allocated to another interviewer.

IF ASKED ON…	PRIMARY STORY · ANGLE	BACKUP	15-SECOND OPENER	NUMBER TO LAND
Customer Obsession	**S1-c** — priced the ask: unpickable stock sold is a cancellation	S3 — failure direction under-sell	The customer cost of selling held stock is a cancellation, so I priced it in units.	**560 vs 710 (150 units)**
Ownership	**S1-b** — every act: screen, role, audit row; owners named	S1 — took the definition nobody owned	On handoff day I wanted no AFS-changing act that only I understood.	**0 unowned buckets; drift tolerance 0**
Invent and Simplify	**S3** — one ledger, one settlement authority; the bridge has no job	S1-a — one event, one bucket	I did not improve the month-end bridge — I removed the reason for it.	**FOH 2 → 1**
Are Right, A Lot	**S1-e** — refuse, study, reverse on evidence	S4 (bench) — the in-transit rule I got wrong	My first answer was no, and then the study came back.	**$42K funded after a no**
Learn and Be Curious	**S3-a** — learned how S/4 posts aggregates before arguing	S4 — back-test before publishing	I had to learn how the book is posted before I could argue about it.	**$100.00 = 33.33 + 33.33 + 33.34**
Hire and Develop the Best	**S1-b** — develop the control owners by handing them the load-bearing artifact thin	S5 — council pre-wiring (pointer)	I measure development by whether the drift report runs without me.	**drift tolerance 0, checked daily**
Insist on the Highest Standards	**S3-a** — Σ lines = posting to the cent	S1-a (tolerance 0) · S1-d (counted rows)	A one-cent break is a failed test, not noise.	**$0.00 break in every cell**
Think Big	**S1** — book-of-record migration, ~15 functions, ~60 people	S1-d — two staged go-lives	The scale was not the system count; every promise had to keep holding while the book changed.	**2 go-lives; rollback triggers never fired**
Bias for Action	**S1-d** — reversible go: rollback as numbers	S3-a — sequenced by event source	I could move because I wrote the way back first.	**sync < 99% 4 h · p95 > 2× 1 h**
Frugality	**S1-e** — fund the study, not the fear	S9 (bench) — 597 lanes = 80% of spend	I refused to fund fear, but I did not refuse to fund evidence.	**$42K, $25K held**
Earn Trust	**S1** — three signatures; pre-wiring; objections as rows	S3 — the aging report given to Finance	Three organizations each owned a different truth; none reported to me.	**signed 6/5/25 by three orgs**
Dive Deep	**S1-a** — a five-unit order walked event by event	S3-a — the grain; S1-e count/sum/per-store	I stopped arguing about the definition and walked one order through four feeds.	**550 · 550 · 565 vs 555**
Have Backbone; Disagree and Commit	**S1-c** — the no lives in an FSD row	S3 — two refusals · S5 (fence)	Merchandising wanted held stock sellable and the room was leaning yes.	**break-even 61%**
Deliver Results	**S1-d** — dated chain held; two go-lives	S1 — MCS retired, ≈$2.5M	The chain was dated before we started, and it held through a peak freeze.	**E2E 28 then 41 days apart**
Strive to be Earth's Best Employer	**S1** — nobody learned their cost in the council room thin	S1-b — owners given real screens	Respect for people's time was the method: every row pre-wired before the room.	**zero surprise rows at the council**
Success and Scale Bring Broad Responsibility	**S1** — oversell is a customer harm, not a ledger error	S3 — under-sell on purpose · S1-c	An oversell is a customer who was told yes and then gets a cancellation.	**oversell recurring → zero**

Common non-LP prompts

IF ASKED…	ROUTE TO	NUMBER
Tell me about a time you failed / were wrong	S4 (bench): v1 rule; or S1-e: my first no	**σ 0.3 d → 0.49-d window ≈ 95% · $42K**
Disagreed with your manager	No manager dispute in the S/4 record — use S1-c (senior stakeholder) and say so no manager story	**61% break-even**

IF ASKED…	ROUTE TO	NUMBER
Influence without authority	S1: ~60 people, three signatories; S5 pointer	**signed 6/5/25**
Reduced cost	S1: $1.0M MCS run cost (user anchor); S9 bench (freight −10% ≈ $2M)	**$1.0M anchor**
Dealt with ambiguity	S1: nobody owned the definition; S1-a: ten buckets	**0 open “who owns this” rows**
Simplified a process	S3: no bridge spreadsheet; S1-a: one event, one bucket	**FOH 2 → 1**
Decision with incomplete data	S1-c: no release-reliability data, priced both sides; S1-e: threshold, not price	**p = 61% · L* = $420,000 at 10%**
Missed a deadline	No missed S/4 date in the record — do not invent; pivot to S4 (a rule I corrected) **none on record**	**28 and 41 days**
Mentored someone	Thin: S1-b handoff to control owners **no record of mentoring**	**drift tolerance 0**
Delivered under a tight timeline	S1-d: production freeze Oct–Dec, three E2E days, two go-lives	**11/22 · 12/20 · 1/30**
Changed your mind with data	S1-e: refuse → study → reverse; S4	**$42K**

Read me first

The interviewer's lens (inference from title + JD — not a biography)

Inference from the title. A second Senior Manager of Supply Chain in the same organization is the peer interviewer: someone whose team will share processes, data definitions and escalation paths with this role. Peers test whether the candidate can get to *one truth* across organizations that do not report to them, design a process that survives its author, and disagree without breaking the relationship. They also notice whether the candidate claims more than they owned — a peer has to live with the person afterwards.

What that means for the answers. Eight Sephora S/4HANA governance primaries from one program, each a different interview answer, told as process redesigns of the Plan/Deliver seam. The two originals: S1 (three organizations meant three different things by "available"; one signed definition per node per channel, policy before configuration; oversell incidents recurring → zero through two go-lives) and S3 (Finance and Operations reconciling on-hand by argument; one Financial On-Hand with a safe failure direction and a control report owned by Finance). Program-lead voice throughout — no system reported to me; SAP teams and configurators built to the FSD three orgs signed. Six variants cut from the same record answer the other LPs: S1-a the data contract and the ten buckets · S3-a de-aggregation to the cent · S1-b the control screens and operating model · S1-c the priced no on merchandise reserve · S1-d the counted-rows go/no-go across two go-lives · S1-e the seventeen-interface regression study. Their dollars are slices of S1 and S3 and are never added to them. S4 (the in-transit rule I got wrong) and S5 (the architecture council) are the same program family and appear only as backup pointers.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title and the JD; if the first question lands on a different LP, the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Earn Trust	The peer LP: one signed truth across Supply Chain Strategy, Inventory Management and FSC Inventory Accounting (S1); Finance given the real control report to own instead of a bridge spreadsheet (S3). Both are Earn Trust backups in the master map — the primary (S14) is the bar raiser's.	S1 (open) · S3 (Finance and Ops on one number) · S1-b (screens everyone can read)
Ownership	Taking the definition nobody owned without owning any system — the bucket inventory, the asymmetry, the signature chain. A peer wants to know the candidate will own the seam, not the org chart.	S1 (the definition nobody owned) · S3 (the failure direction designed before the happy path) · S1-b (who owns each screen)
Invent and Simplify / Have Backbone (one of the two)	De-aggregation with a safe failure direction is the simplification (S3); pricing Merchandising's "merch reserve promisable" ask and putting the no in the FSD as a row is the backbone (S1).	S3 (one FOH, failure direction) · S3-a (de-aggregation) · S1-c (the priced no)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.

- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For a peer specifically:** name the three organizations and what each owned (Strategy — policy; Inventory Management — execution; Inventory Accounting — the book); say what was pre-wired privately before any council; convert every objection into a document row. "A no that lives in a document survives; a no that lives in a meeting gets re-litigated weekly."
- **Say the decomposition only from this packet's model:** S1 is ≈$2.5M/yr = $1M MCS run cost retired (user anchor) + ≈$1.5M modeled (oversell ≈$0.9M · reconciliation labor ≈$0.33M · change premium ≈$0.3M). S3 is ≈$0.6M/yr, entirely modeled; the record has no pre-state drift dollar. If the room wants one cancel-rate number, keep S1's and drop S3's row.
- **Variants are slices, not additions.** S1-a to S1-e re-cut S1 and S3 facts for other LPs; any dollar in a variant is a labeled slice or an illustrative number, never added to the ≈$2.5M or ≈$0.6M. Use one story per question; if you have told S1 for Earn Trust, answer the next LP with a variant, not a re-run.

Loop-consistency rule. One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S6/S8 (Hoerger) · S2/S18 (Goforth) · S16/S15/S10/S14 (Josyula). S4 and S5 are bench stories (backup bench at the end), not primaries; S2 shares the UDC program's dated chain with S1/S5 — if asked, say it is one umbrella program. S1-e draws on the loop packet's S25 interface-regression facts (a secondary source, not in the master §4 ledger; the loop tracker also lists S25 as a Bar Raiser backup for Josyula) — if his panel spends it, drop S1-e and use S1-a or S1-d. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

The stories — business-first, canon numbers NEW

Eight Sephora S/4HANA governance primaries from one umbrella program: S1 and S3 (upgraded with LP proof lines, 20-second versions, four-table math, a behavioral pack, three angles and trap drills) and six new variants, each a complete standalone answer with its own situation, math, conflict and result. Each: LP proofs → 60-second pitch → 20-second version + NACF bridge → Situation/Task → design facets → Action in gated phases → Results → L6 evidence → Probing → four-table math → system map → sensitivity → spoken statement → follow-ups → four behavioral answers → three angles → three trap drills.

S1 — Sephora S/4HANA ↔ ISM — one signed meaning of "available" across Store, Retail DC and Dotcom DC: policy before configuration (oversell incidents recurring → zero through two go-lives) UPGRADED

LPs this story can serve here: Earn Trust (backup — three orgs signed one definition) · Ownership (backup — took the definition nobody owned) · Think Big (third — the book-of-record migration) · Success & Scale Bring Broad Responsibility (oversell is a customer harm, not a ledger error) · Have Backbone (the priced no to Merchandising)

Process family: Plan / Deliver seam — inventory availability governance (available-to-promise policy, bucket ownership, real-time decrement contract, daily reconciliation); book-of-record migration

Honesty label. Integration program lead — no system reported to me; every configurator sat in another org. I owned the definitions, the asymmetry, the decrement and reconciliation contract, the signature chain and the council ask; SAP teams and configurators built to the FSD; planners executed the business acts. The record has no pre-state oversell dollar — the ≈$1.5M is my model and I say so. Dated gate chain, issues log and rollback triggers are program-level artifacts shared with S2/S5.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Ownership — Nobody owned the definition and I owned no system — I wrote the bucket inventory with exactly one owning system per bucket and carried “vocabulary drift” at the top of my own risk register.

Earn Trust — Three organizations signed the same FSD on 6/5/25 because each owned a different truth and I pre-wired every row of the systems-impacted table before the council.

Have Backbone — Merchandising asked for merch reserve to be promisable; I priced it (unpickable stock sold) and wrote the no into the FSD as a row, not a meeting minute.

Success & Scale Bring Broad Responsibility — An oversell is a customer harm, not a ledger error — the asymmetry accepts visible undersell to make oversell zero by construction.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Sephora was moving the inventory book of record from the in-house MCS to SAP S/4HANA, with a real-time layer beside it for the digital demand S/4 cannot see in time — a web order placed, a BOPIS order packed, a POS sale rung but not yet posted. Store, Retail DC and Dotcom DC each meant something different by "available": Merchandising wanted merchandise reserve visible, Finance wanted on-hand to tie to the book, dotcom wanted in-transit promisable. Most oversell bugs are vocabulary bugs, and oversells were recurring across all three node types — a customer harm, not a ledger error. **Task.** As integration program lead for inventory supply — no system reported to me — define every bucket, its owning system, and how available-for-sale is computed and decremented in real time, and get Planning, Finance and operations to sign it before any configuration. **Action.** I built the bucket inventory with exactly one owning system per bucket; wrote the lower-of-two basis and the available-for-sale equation, the real-time decrement contract with an idempotency key per event, and a daily reconciliation of the real-time ledger to the book; pre-wired every row of the systems-impacted table with its owner's level of effort before the council so nobody learned their cost in the room, and asked the council for three decisions rather than a review. Merchandising's ask to make merch reserve promisable was priced — unpickable stock sold — and the no went into the specification as a row, not a meeting minute. Three organizations — Supply Chain Strategy, Inventory Management and FSC Inventory Accounting — signed the FSD on 6/5/25. Then a node × channel × bucket-transition regression matrix, dated end-to-end days, a January planning go-live and a May execution go-live, with the order system reading one published availability number and never computing its own. **Result.** Primary metric: customer-facing oversell incidents — recurring across three node types → zero through two go-lives; one signed availability number per node per channel; MCS retired as book of record. About $2.5M a year: $1M of MCS run cost retired, which is the measured anchor, plus about $1.5M modeled — avoided oversell, dual-book reconciliation labor, legacy change premium — and I say which is which.

20-SECOND VERSION NEW

Sephora moved the inventory book to S/4HANA and Store, Retail DC and Dotcom DC each meant something different by “available.” I owned no system, but I got one signed definition per node per channel — lower of two books minus committed buckets — before any configuration. Oversell incidents went from recurring to zero through two go-lives and MCS retired: about $2.5M a year, $1M of it measured.

NACF BRIDGE NEW

The NACF inbound version is “what can the promise see.” Units on an ASN or a dock are not available; only units that cleared receive and stow are. Same discipline: one signed definition of each state, one owner per state, and a daily check that the physical count and the system agree.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Sephora moved the book of record from MCS to S/4HANA, with ISM beside it to see digital demand S/4 cannot see in real time. Three node types, three working definitions of "available." Merchandising wanted merchandise reserve visible, Finance wanted on-hand to tie to the book, dotcom wanted in-transit promisable. Nobody owned the definition; I owned no system.

The process cost: recurring vocabulary-driven oversells — a customer harm, not a ledger error; two books reconciled by hand (≈4 FTE, modeled); the MCS run cost continuing (≈$1M/yr, user anchor); a legacy change premium on an undocumented stack. Vocabulary drift — a bucket without an owner — sat at the top of my risk register.

T — Own the integration design and data contract of inventory supply: which buckets exist, which system owns each, how available-for-sale is computed per node, and how it feeds promising, replenishment and Finance — signed by Planning, Finance and operations before configuration. Sponsor memo with the VP Supply Chain: one signed AFS per node per channel, one Financial On-Hand, zero customer-impacting incidents.

Baselines on the table: oversell incidents recurring across three node types (target zero) · availability truths: two books reconciled by hand (target one signed AFS per node per channel with daily drift as the standing check) · MCS running as book of record (target retired) · ~15 functions, ~60 people to move without authority.

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** Both books — MCS and S/4 — profiled for SKU overlap and order profiles per node type (Store, Retail DC, Dotcom DC); node-by-node sessions with ~15 functions; the live feeds that would carry the real-time layer: Aptos POS sale lines, OMS (Yantriks) commit/pick/pack/cancel events, BlueYonder WMS on-hand (≤ 15 min), FRAME channel reserves (nightly + holds), S/4 MB52 daily plus MSEG movements as events.
- **Grain and joins.** SKU × node × channel × bucket. Every event had to map to exactly one bucket; MB52 ↔ ISM copy joined daily on SKU × node (tolerance 0); WMS on-hand joined to the S/4 copy to produce OOB.
- **Cuts.** By node type, by "who says available" (Strategy, Inventory Management, Inventory Accounting), and by the bucket a unit could sit in.
- **What it revealed.** The worked SKU gives three answers to one question — raw S/4 says 1,000 (440 over-promised against the 560 that is truly promisable), raw WMS says 980 (420 over), and the committed buckets (420 units) were invisible to both. The cause was not data quality; it was that no bucket had an owner, so each reader built its own equation.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

Same SKU, two cases (illustrative units): Case A — S/4 1,000, WMS 980, committed 420. Case B — S/4 1,000, WMS 1,020, committed 420. "Exposure" is units promised above the lower of the two books after committed buckets.

OPTION	CASE A PROMISES	CASE A EXPOSURE	CASE B PROMISES	CASE B EXPOSURE	VERDICT
Raw S/4 Available (Definition A)	1,000	440	1,000	420	Rejected — committed buckets invisible
Raw WMS on-hand (Definition B)	980	420	1,020	440	Rejected — same blind spot
S/4 minus committed buckets	580	20	580	0	Rejected — over-promises when WMS is the lower book
Average of the two, minus committed	570	10	590	10	Rejected — the error always has a sign; the average hides it
min(S/4, WMS) minus committed	**560**	**0**	**580**	**0**	Chosen — oversell is a customer defect, undersell a planning variance

Other options rejected in writing: letting OMS compute its own AFS (two publishers — the one rule is that OMS reads the published number, never computes); making merch reserve promisable (priced in story S1-c); configuring to the legacy definitions first (the legacy definitions were the problem — policy before configuration).

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS, RISKS)

- **Discovery** — profiled both books (MCS and S/4) for SKU overlap and order profiles per node; node-by-node sessions with ~15 functions; the bucket inventory with one proposed owning system per bucket. *Exit bar:* every bucket owned by exactly one system; sessions closed 4/30; open questions carried with owner and date.
- **Plan & Design** — the lower-of-two basis; the AFS/FOH equations; the real-time decrement contract (event → bucket delta; idempotency key per event; a reversal must reference its original); daily ISM-vs-MB52 reconciliation with tolerance zero; the RAID log; the council pack — FSD v1.1 (3/5/25) → v1.2 (5/16/25). *Exit bar:* FSD signed 6/5/25 by Supply Chain Strategy, Inventory Management and FSC Inventory Accounting with zero open "who owns this bucket" rows.

- **Build & Test** — the node × channel × bucket-transition matrix with before/after assertions — no story enters QA without a matrix cell reference; SIT on the interface contracts before UAT; dated E2E days — Inbound 11/22, Dotcom 12/20, Retail 1/30 — business and engineering in one room. *Exit bar:* 0 open contract defects in SIT; E2E green on three dates; Blocker + Critical open = 0 at go/no-go.
- **Cutover & Go-Live** — go/no-go recommendation from counted rows; the publish-switch plan — OMS reads the published AFS, never computes; reserve-copy invariants; control-screen roles assigned; rollback triggers (sync < 99% for 4 h; order-flow p95 > 2× budget for 1 h) armed for both go-lives, never fired. *Exit bar:* January go-live (reserve rules live against demand); May go-live (execution); no SKU with two published AFS values; Negative AFS clean at T+1.
- **Hypercare / Close** — daily drift report (ISM vs MB52); Negative AFS / AFS Correction audit review; control-owner map — Negative AFS to Inventory Control, Internal Audit and Finance; AFS Correction to the ISM app team; FSD as-built note. *Exit bar:* drift = 0 SKUs on consecutive days; blockers/day under threshold; the design enters the monthly release train.

Influence and ownership. Sixty people across three orgs, none reporting to me. The method: pre-wire every row of the systems-impacted table with its owner's level of effort before the council; run working sessions to one table instead of arbitrating in meetings; convert objections into FSD rows; ask the council for three decisions rather than a review. Three signatories because each owned a different truth — Strategy the policy, Inventory Management the execution, Inventory Accounting the book; one signature would have left two orgs free to disagree later. The asymmetry held: oversell is a customer defect, undersell a planning variance Inventory Control chases through the control screens.

> **Designed the process; others own it now.** I designed the availability definition, the decrement contract and the control-owner map; Inventory Control, Internal Audit, Finance and the ISM app team now own the controls, and the drift report runs daily without me. **Trade-off put to leadership.** The recommendation to the council outlined the competing metrics — availability to sell vs oversell exposure — and resolved them as an asymmetry in the signed FSD: oversell is a customer defect, undersell a planning variance.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$2.5M/yr — $1M MCS run/support cost retired (*user anchor*) + ≈$1.5M *modeled* (avoided oversell ≈$0.9M · dual-book reconciliation labor ≈$0.33M · legacy change premium ≈$0.3M); range $1.7–5.2M; upside not in the headline: shrink ≈$0.3M, time-to-market ≈$0.2M. The record has no pre-state oversell dollar.
- **#** — oversell incidents recurring across three node types → zero through two go-lives (record); dual-book reconciliation ≈4 FTE by hand → ≈1 FTE reading the drift report (modeled).
- **%** — issues log 85.4% of 836 rows closed (program-level artifact shared with S2/S5); the oversell rate behind the ≈$0.9M row is 0.10% of orders — modeled, benchmark 0.05–0.25%; the record movement is incidents → zero.
- **Time** — FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; E2E Inbound 11/22 · Dotcom 12/20 · Retail 1/30; January planning and May execution go-lives (record dates).
- **Customer** — customer-facing oversell incidents zero through two go-lives; the asymmetry treats oversell as a customer defect and undersell as a planning variance (record).
- **Quality** — one signed AFS per node per channel; no SKU with two published AFS values; daily ISM-vs-MB52 drift = 0 SKUs on consecutive days at hypercare exit; MCS retired as book of record (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Three organizations with three meanings of "available" and no owner of the definition. I defined the lower-of-two basis, the AFS/FOH equations and the bucket inventory with one owning system per bucket, and chose the ~15 functions for node-by-node sessions.
Scope & influence	Sixty people across three orgs, none reporting to me; the council took three decisions I framed; the FSD was signed 6/5/25 by three signatories — influence up to the council and across orgs.
Execution trade-off + mechanism others now own	Oversell protection vs availability, made explicit as an asymmetry — oversell is a customer defect, undersell a planning variance. Mechanisms others own now: the daily ISM-vs-MB52 drift report and the control-owner map (Negative AFS → Inventory Control, Internal Audit and Finance; AFS Correction → the ISM app team).
Impact (long-term)	Oversell incidents recurring across three node types → zero through two go-lives; MCS retired as book of record; ≈$2.5M/yr ($1M run cost user anchor + ≈$1.5M modeled).
Process improvement (standard work institutionalized)	The node × channel × bucket-transition matrix — no story enters QA without a matrix cell reference; SIT on interface contracts before UAT; dated E2E days as entry bars.
Complexity & consensus	Every row of the systems-impacted table pre-wired with its owner's level of effort before the council; objections converted into FSD rows instead of arbitrated in meetings.
Root cause at scale	Oversell was traced to two books carrying two meanings — fixed once in a signed definition and a real-time decrement contract (idempotency key per event; a reversal references its original), not incident by incident.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Peer — You owned no system. How did you get three organizations to sign one definition?**
>
> Pre-wired every row of the systems-impacted table with its owner's level of effort before the council — nobody learned their cost in the room — ran working sessions to one table, converted objections into FSD rows, and asked the council for three decisions rather than a review. Three signatories because each owned a different truth; one signature would have left two orgs free to disagree later.

> **Peer — Why lower-of-two and not an average or the book?**
>
> Because oversell is a customer defect and undersell is a planning variance. A positive out-of-balance means the book is above the floor; promising the difference risks oversell, so the basis is the minimum, and Inventory Control chases the variance through the Negative AFS and AFS Correction screens. The asymmetry is the policy.

> **Peer — Merchandising wanted merch reserve promisable. What happened?**
>
> I priced it — unpickable stock sold — and offered visibility without promisability. The no went into the FSD as a row, not into a meeting minute. A no that lives in a document survives; one that lives in a meeting gets re-litigated weekly.

> **Peer — How do you know it was a tenth of a percent?**
>
> I don't, to the decimal. The record says incidents were recurring and went to zero; the rate is a benchmark set deliberately low — a twentieth of a typical e-commerce inventory-cancel rate. First thing I would measure: the "item unavailable after order" cancel-reason code for the twelve months before January. If the $1M MCS figure already includes enhancement labor, I drop the change-premium row and the headline is about $2.2M.

> **Peer — What did you get wrong in this program?**
>
> The first in-transit rule I wrote barred every channel from promising units in transit — safe-feeling, and it starved dotcom of fast movers a day from the building. I corrected it through the same three signatories with a per-lane promise window and a kill switch, released on the normal cadence. Lesson: both sides of a rule get a number before it ships, and back-test a ruleset against history before publishing it. (S4 — pointer; full story on request.)

> **Peer — How would this work between your team and mine here?**
> Every promise rests on one published availability number per node that downstream systems read and never recompute. Between two teams that means: one definition with one owner per bucket, the policy signed before the code, a daily drift report with tolerance zero and a named control owner — and objections turned into rows we both sign.

FACET 3 — MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Two books for one SKU	`OOB = S4_Avail − WMS_OnHand = 1,000 − 980 = 20`	S/4 Available 1,000; WMS on-hand 980 (illustrative)	+20 units — the books disagree; book above floor
2	Units already spoken for	`Σ buckets = 200 + 120 + 60 + 40 = 420`	PickRes 200 · EcomPick 120 · EcomPack 60 · SNB 40 (illustrative)	420 committed or sold-not-posted
3	Truly promisable (the yardstick)	`min(1,000, 980) − 420 = 980 − 420 = 560`	the number the fix will compute	560 units
4	Definition A — channel reads raw S/4	`1,000 − 560 = 440`	committed buckets invisible to it (record problem)	440 units over-promised
5	Definition B — channel reads WMS on-hand	`980 − 560 = 420`	same buckets invisible (record problem)	420 units over-promised
6	Oversell outcome	`30,000,000 × 0.10% = 30,000`	orders illustrative; rate benchmark 0.05–0.25%; incidents recurring across three node types (record)	≈30,000 incidents/yr — modeled
7	Reconciliation burn	`≈4 FTE reconciling two books by hand`	record baseline	≈4 FTE

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One supply basis	`Basis = min(S4_Avail, WMS_OnHand) = min(1,000, 980) = 980`	asymmetric lower-of-two (record design)	Basis = 980 units
2	Why the minimum	`Basis − S4_Avail = 980 − 1,000 = −20`	a positive out-of-balance reduces supply (record design)	the +20 is never promised
3	Available-for-sale	`AFS = 980 − 200 − 120 − 60 − 40 = 560`	AFS = Basis − PickRes − EcomPick − EcomPack − SNB; MerchRes and Lost/Found sit outside S4_Avail — no double-subtract path (record design)	AFS = 560 units
4	Floor-above-book case	`Basis = min(1,000, 1,020) = 1,000; AFS = 1,000 − 420 = 580`	WMS 1,020 → OOB = −20 (illustrative)	AFS = 580, not 600 — the 20-unit surplus stays unpromised
5	Real-time decrement	`AFS = 560 − 3 = 557; SNB = 40 + 3 = 43`	store rings 3 units at 10:02 (illustrative); each event maps to exactly one bucket	AFS 557 within the 60-second publish budget
6	Idempotency	`AFS = 557 − 0 = 557`	replayed key at 10:04 is a no-op; a reversal must reference its original (record design)	AFS stays 557
7	Overnight posting	`SNB 43 → 0; S4_Avail = 1,000 − 43 = 957`	aggregate posting lands in S/4 for 43 units (illustrative)	books move together
8	Daily drift check	`Drift = ISM.S4_Avail − MB52 = 957 − 957 = 0`	tolerance 0 — a non-zero drift is a Sev, not a planning variance (record design)	drift = 0

Table 3 — after: the metric, before → after.

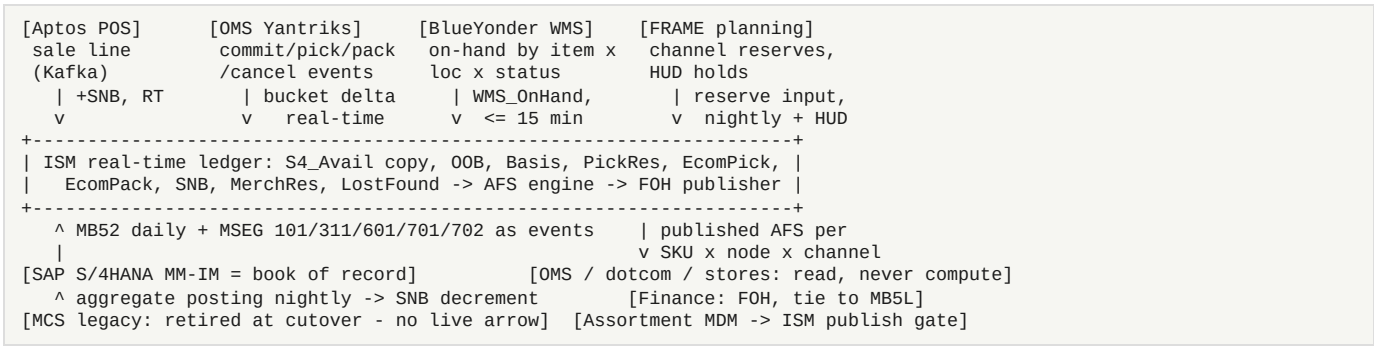
STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Working definitions of “available”	3	1 signed AFS	`3 − 1 = 2`	record
2	Worked SKU, over-promise under Definition A	440 units	0 units	`440 − 0 = 440`	record problem; units illustrative
3	Oversell incidents	recurring across three node types	zero through two go-lives	`recurring − 0 (count not in record)`	record
4	Dual-book reconciliation	≈4 FTE by hand	≈1 FTE reading the drift report	`4 − 1 = 3 FTE`	modeled on the record mechanism
5	Daily drift at hypercare exit	unmeasured	0 SKUs on consecutive days	`0 − 0 = 0`	record
6	MCS run / support cost	≈$1.0M/yr	retired	`$1.0M − $0 = $1.0M`	user anchor

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	MCS run / support retired	`≈$1.0M/yr, taken as stated`	user anchor	$1.0M/yr
2	Incidents avoided	`30,000,000 × 0.10% = 30,000`	orders illustrative; rate benchmark 0.05–0.25%; record movement: → zero	30,000 incidents/yr
3	Cost per incident	`$6 + $4 + $19 + $1 = $30`	CS contact $6 · refund $4 · lost margin ~$19 · goodwill ~$1 (benchmark)	$30 each
4	Oversell dollars	`30,000 × $30 = $900,000`	rows 2 × 3 (modeled)	≈$0.9M/yr
5	Reconciliation labor	`4 − 1 = 3; 3 × $110,000 = $330,000`	$110K loaded (benchmark)	≈$0.33M/yr
6	Tech-debt change premium	`40 × 60 h = 2,400 h; 2,400 × $125 = $300,000`	≈40 changes/yr × +60 h each on MCS × $125/h (modeled)	≈$0.3M/yr
7	Headline	`$1,000,000 + $900,000 + $330,000 + $300,000 = $2,530,000`	rows 1, 4, 5, 6	≈$2.5M/yr
8	Not in headline	`$0.3M + $0.2M = $0.5M excluded`	shrink ≈$0.3M + time-to-market ≈$0.2M (modeled upside)	excluded

Say the decomposition only from this packet's model (the loop packet says ≈$1M/yr support cost — the $1M is the shared core).

SOLUTION FLOW — THE SYSTEM & DATA MAP NEW



SENSITIVITY / IMPACT — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
Vocabulary-driven oversell rate	0.05%	0.10%	0.25%
Cost per oversell incident	$25	$30	$40
Oversell cost avoided	$0.4M	$0.9M	$3.0M
Reconciliation FTE released × loaded	2 × $95K = $0.19M	3 × $110K = $0.33M	4 × $130K = $0.52M
Change premium (changes × hrs × rate)	24 × 40 × $100 = $0.1M	40 × 60 × $125 = $0.3M	60 × 80 × $150 = $0.72M
MCS run cost (user anchor, fixed)	$1.0M	$1.0M	$1.0M
Total modeled $/yr (headline, excl. upside)	≈$1.7M	≈$2.5M	≈$5.2M

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, about two and a half million dollars a year. One million of that is measured: the MCS support cost that went away when S/4 became the book of record. About nine hundred thousand is avoided oversell — on roughly thirty million dotcom orders, a tenth of a percent going wrong at about thirty dollars each — and the record shows those incidents went to zero through both go-lives. About three hundred thousand is reconciliation labor and about three hundred thousand is the change premium we stopped paying on an undocumented stack. The first number is measured; the other three are my model, and I can show you every input.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — Walk me through the AFS on one SKU.

Basis is the lower of the two books: min(1,000, 980) = 980. Subtract what is already spoken for — pick reserve 200, e-commerce pick 120, e-commerce pack 60, shipped-not-billed 40 = 420 — so AFS is 560. A channel reading raw S/4 would have promised 1,000 and over-promised 440; reading WMS, 980 and 420. If the floor is above the book (WMS 1,020), the basis stays 1,000 and AFS is 580, not 600 — the surplus is never promised.

Numeric — What does a sale do, and what if the message is delivered twice?

A store rings 3 units: AFS 560 → 557, SNB 40 → 43, inside the 60-second publish budget. The replayed message carries the same idempotency key, so it is a no-op — AFS stays 557. A reversal must reference its original event, so a cancel can never decrement twice.

Numeric — Which of the $2.5M is measured and what is the downside case?

Only the $1.0M MCS run cost is measured (user anchor). The $1.5M is modeled: $0.9M oversell + $0.33M reconciliation + $0.3M change premium. The low case on the same model is about $1.7M (0.05% rate, $25 per incident, two FTE, 24 changes); if the $1M already includes enhancement labor I drop the premium row and the headline is about $2.2M.

Numeric — What did the gate chain look like, and where was your slack?

Kickoff 3/29, business sessions closed 4/30, council 5/15, LOE signed 7/2, sign-off 7/31, FSD 8/14; ISM FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25. E2E Inbound 11/22, Dotcom 12/20, Retail 1/30 — 28 days then 41 days apart — then January planning and May execution go-lives. The slack was the peak freeze: October to December nothing moved.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND MOMENT NEW

EARN TRUST **1. Tell me about a time you built trust with teams that don't report to you.**

Open (first two sentences): Three organizations each meant something different by "available," and I owned none of their systems. Before the council ever met, I sat with the owner of every row on the systems-impacted table and got their level of effort in writing, so nobody learned their cost in the room.

A. I ran working sessions to one table instead of arbitrating in meetings, turned every objection into a row in the FSD, and asked the council for three decisions rather than a review. I used three signatories on purpose — Strategy owned policy, Inventory Management owned execution, Inventory Accounting owned the book — because one signature would have left two orgs free to disagree later. **R.** The FSD was signed 6/5/25 by all three, and none of the three re-opened it through two go-lives.

Land the number: three signatures, 6/5/25 | **Pivot to NACF:** At NACF the same handoff is placement's recommendation versus what the FC actually receives — it gets signed at reservation, while the information is still ahead of the freight.
Do NOT say: "I aligned everyone" without naming the artifact; "I configured S/4"; S14 (the bar raiser's Earn Trust story).

OWNERSHIP **2. Tell me about something you took on that was outside your remit.**

Open (first two sentences): The riskiest line on my risk register wasn't a system — it was a bucket with no owner. I took the definition nobody owned: a bucket inventory where every bucket had exactly one owning system, and I wouldn't exit discovery until there were zero open "who owns this" rows.

A. I wrote the lower-of-two basis, the AFS equation and the real-time decrement contract before any configuration, carried "vocabulary drift" as my own top risk with a trigger and a mitigation, and left open questions each with an owner and a date at the 4/30 session close. **R.** The FSD went to signature with zero open ownership rows, and nobody had asked me to own it — the definition was the whole problem.

Land the number: zero open "who owns this bucket" rows at signature | **Pivot to NACF:** In inbound a stranded unit is the same orphan bucket — physically present, no owner, not available. I would want each stock state in the building to have exactly one owner.
Do NOT say: "It was my system," "I owned S/4," or claiming the SAP team's configuration.

THINK BIG **3. Describe the largest-scale change you have led.**

Open (first two sentences): We moved the inventory book of record from the in-house MCS to S/4HANA with a real-time layer beside it, across about fifteen functions and sixty people I had no authority over. The scale wasn't the system count — it was that every promise to every customer on three node types had to keep holding while the book underneath changed.

A. Five gated phases with dated exits — discovery to 4/30, council 5/15, LOE 7/2, sign-off 7/31 — then two go-lives so each carried one defect class: January put the reserve rules against live demand, May moved execution. Rollback was written as numbers (sync under 99% for four hours, or order-flow p95 over twice budget for an hour) and armed for both. **R.** Zero customer-facing oversell incidents through both go-lives, and MCS retired as book of record.

Land the number: two go-lives, zero customer-facing incidents; ≈$1M/yr run cost retired | **Pivot to NACF:** A new-FC launch is the same shape: stage it so each go-live carries one defect class, and write the stop condition as a number before the first unit moves.
Do NOT say: The Utah DC comingle or robotics gate — that is another interviewer's story; say "one umbrella program" if asked.

SUCCESS & SCALE BRING BROAD RESPONSIBILITY **4. Tell me about a decision where you weighed the broader impact, not just the metric.**

Open (first two sentences): Most oversell bugs are vocabulary bugs, but an oversell isn't a ledger error — it is a customer who was told yes and then gets a cancellation. So I wrote the asymmetry into the policy: oversell is a customer defect, undersell is a planning variance that Inventory Control can chase.

A. That meant lower-of-two, accepting that some sellable units stay hidden at the margin, and giving Inventory Control the Negative AFS and AFS Correction screens to chase the variance. I also priced Merchandising's ask to make merch reserve promisable — unpickable stock sold — and gave them visibility instead. **R.** Oversell incidents recurring → zero through two go-lives; the trade was visible undersell, accepted on purpose.

Land the number: oversell incidents recurring → zero through two go-lives | **Pivot to NACF:** Amazon's version is the delivery promise: a date shown honestly later is cheaper for the customer than a date broken.
Do NOT say: a measured oversell rate — 0.10% is a benchmark I set; the record is incidents → zero.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

"Before I wrote one definition, I put one SKU through every reader in the building."

440 · 420 · 560

LP: Dive Deep

One SKU, S/4 1,000, WMS 980, 420 committed. Reading raw S/4 over-promises 440; reading WMS over-promises 420; the lower-of-two minus buckets says 560.

The finding was not bad data — it was that nobody owned the buckets, so every reader built its own equation.

That table, not my opinion, is what the council saw.

CONFLICT / INFLUENCE / BACKBONE

"Merchandising wanted merch reserve promisable, and the room was leaning toward giving it to them."

3 signatures · 6/5/25

LP: Have Backbone · Earn Trust

I priced the ask — unpickable stock sold — and offered visibility without promisability.

The no went into the FSD as a row, not a meeting minute: a no that lives in a document survives; one in a meeting is re-litigated weekly.

Their visibility need was met; I did not caricature them.

FAILURE / LEARNING

"The first ruleset I authored barred every channel from promising in-transit units — it felt safe and it was wrong."

σ 0.3 d → 0.49-d window ≈ 95%

LP: Are Right, A Lot · Learn and Be Curious

The Aging In-Transit report showed dotcom starved of fast movers one day from the building.

I took it back through the same three signatories with a per-lane promise window and a kill switch — never fired.

Lesson: both sides of a rule get a number before it ships, and back-test against history first. (Full walk: S4.)

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — "How do you know it was your definition and not seasonality, or the other programs, that took oversells to zero?"

I can't isolate it statistically from the record: there is no pre-state oversell rate in the record, only "recurring across three node types → zero through two go-lives." What I defend is the mechanism: committed buckets were invisible to both raw reads (440 and 420 units on the worked SKU), and the AFS equation plus an idempotent decrement removes that path. The cross-check I would run is the "item unavailable after order" cancel-reason code for the twelve months before January, by node, against the months after; and any SKU with non-zero daily drift would be the first place I looked if incidents had persisted.

Say exactly: "I can't separate it from seasonality using the record, and I won't pretend to — I can show you the mechanism and the one query that would."

Trap — "What's the counterfactual for the $2.5M — and what would make it smaller?"

The counterfactual is the status quo: two books, MCS running. Only the $1.0M run cost is measured. The $1.5M is modeled — 30,000 incidents × $30 = $0.9M, 3 FTE × $110K = $0.33M, 40 × 60 h × $125 = $0.3M. On the model's low inputs the headline is about $1.7M; if the $1M already includes enhancement labor it is about $2.2M.

Say exactly: "One million is measured, one and a half is my model, and if the cancel data comes back at the low end the headline is one point seven."

Trap — "What number would have changed your mind about lower-of-two?"

The share of time the higher book is actually right. Promise one extra unit and you earn about $19 of margin if it exists and pay about $30 of incident cost if it does not: break-even is 30 ÷ (19 + 30) = 61%. If you net the lost margin out of the incident (pay only $11), it is 11 ÷ (19 + 11) = 37%. The books disagree by about 2% (20 of 1,000), so I would measure, from cycle-count outcomes over a quarter, how often the higher book wins the count; above roughly 61% (or 37% on the netted view) lower-of-two is too conservative and I would relax it.

Say exactly: "If the higher book wins the physical count more than about six times in ten, lower-of-two is too conservative and I would change it — and I would measure that from cycle counts, not opinion."

Where this lands at Amazon. Every fulfillment promise rests on one published availability number per node that downstream systems read and never recompute — at Amazon scale the ATP/promise seam between planning and the order pipeline. One definition, one owner, policy signed before code. Reliability (oversells to zero), cost efficiency ($1M of run cost retired), responsiveness (a real-time ledger beside the book).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S1

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you built trust across teams that didn't report to you.	Earn Trust	Three orgs, three truths, one signed FSD on 6/5/25; pre-wiring; objections as rows.	S14 (the $2.5M discrepancy) is the Earn Trust primary — it is the bar raiser's; do not use it here.
Tell me about a time you took ownership of something outside your remit.	Ownership	Nobody owned the definition of "available"; I owned no system; built the bucket inventory and the signature chain.	Never "I configured S/4" — the SAP teams configured to the signed spec.
Tell me about a decision where you had to say no to a senior stakeholder.	Have Backbone	Merch reserve promisable — priced, refused, written as an FSD row with visibility offered instead.	Do not caricature Merchandising; their visibility need was met.
Tell me about a large-scale change you led.	Think Big	The book-of-record migration — MCS retired, one real-time ledger, two go-lives, peak untouched.	Keep S2's floor story out — that is Goforth's; say "one umbrella program" if asked.
How do you define and govern a shared metric / definition between two teams?	Functional — process governance	Bucket inventory with one owner each; the equation written; signatures before configuration; daily drift with tolerance zero; control owners named.	Keep the oversell $ labeled modeled; the movement is incidents → zero.
Why does this matter beyond the ledger?	Broad Responsibility / Customer Obsession	Oversell is a customer harm, not a ledger error; the asymmetry accepts visible undersell to make oversell zero by construction.	No YoY revenue claim exists.

S3 — Sephora S/4HANA — Shipped-Not-Billed and de-aggregation: Finance and Operations on one Financial On-Hand number, with the failure direction designed before the happy path UPGRADED

LPs this story can serve here: Earn Trust (backup — Finance and Ops on one number) · Dive Deep (the grain problem — one posting per aggregate vs transaction lines) · Invent and Simplify (one ledger, one settlement authority, a safe failure direction) · Insist on the Highest Standards (Σ per-line COGS = posting COGS to the cent)

Process family: Deliver / Return-to-book seam — sale-to-posting reconciliation, financial on-hand governance, month-end close control

Honesty label. The flow, controls and ownership map were mine; S/4 and OMS engineers implemented; Inventory Accounting confirmed in UAT and took over the aging report. The record has no canonical dollar for the pre-state drift; the ≈$0.6M/yr is my model and I say so. Rode the ISM program's dated chain (FSD 6/5/25; May execution go-live).

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Earn Trust — Finance owned the book and Operations owned the count; I showed Finance the same number arriving from the physical side first, then handed them the unmatched-posting aging report to own — signed inside the FSD on 6/5/25.

Invent and Simplify — One real-time ledger, S/4 as the settlement authority, one failure direction, one control report — instead of two books and a bridge spreadsheet at every close.

Have Backbone — Two refusals held in writing on mechanism, not preference: real-time per-line posting to S/4, and a Finance-derived SNB (two numbers when the point was one).

Customer Obsession — The failure direction was chosen before the happy path: an unmatched posting under-states availability — under-sell, never over-sell — so the customer is never told yes to a unit that is gone.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. S/4 posts sales as aggregates — one posting per aggregation number with total quantity and cost — while the point-of-sale sends transaction lines. Between the sale and the posting, units were gone but the book was not decremented: a phantom-availability window in which sold units still showed as available, and Finance and Operations read different on-hand numbers and reconciled by argument, with a bridge spreadsheet at every month-end close. **Task.** Define how the real-time layer tracks that gap and reconciles it so promising, replenishment and Finance agree — under the program's zero-incident constraint, which applied to the book as much as to the customer. **Action.** I named the window and inventoried the events that move units before posting — POS flash sales, omni pending, ship-to-home confirmations — and got Finance and Operations to agree on the description of the gap before any design. The Shipped-Not-Billed bucket increments on those events and decrements only on a matched S/4 aggregate posting; de-aggregation pro-rates cost per line so the sum ties to the cent; and the failure direction was designed before the happy path — an unmatched posting keeps the bucket high, so availability under-states and Replenishment over-orders visibly instead of the customer being oversold invisibly. Finance objected at design review and wanted its own reconciliation; I showed the same number arriving from the physical side first and gave them the unmatched-posting aging report to own — a parallel spreadsheet dies when its owner owns the real report. FSC Inventory Accounting signed inside the FSD on 6/5/25; it rode the May go-live. **Result.** Primary metric: Financial On-Hand definitions — two, Finance's book versus the operations count → one shared number, availability decremented at sale or ship rather than at posting; over-sell from unmatched postings zero by construction; zero incidents. About $0.6M a year modeled — phantom-availability cancels avoided, Finance reconciliation labor, month-end close time; the record has no pre-state drift dollar and I am not claiming one.

20-SECOND VERSION NEW

S/4 posts sales as aggregates while the POS sends lines, so sold units stayed “available” until the posting and Finance and Operations reconciled on-hand by argument. I defined a Shipped-Not-Billed bucket that drops availability at the sale and clears only on a matched posting, and I picked the failure direction first: an unmatched posting makes availability low and visible, never high and invisible. Financial On-Hand went from two numbers to one; about $0.6M a year, all modeled.

NACF BRIDGE NEW

The inbound mirror is received-not-stowed. Units on an ASN, in a trailer or on a dock are the SNB of a receiving dock: not yet available. Inbound available-to-promise should count only units that cleared receive and stow, and an unmatched receipt should leave the number low and on an aging list — the same “lines must sum to the header” check I would run on ASN counts against dock counts.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — S/4 posts sales as aggregates — one posting per Aggregation# with total quantity and COGS; the Aptos POS sends transaction lines with an Order#. Between the sale and the posting, units are gone but the book is not decremented — the phantom-availability window — and Finance and Operations read different on-hand numbers and reconcile by argument.

The process cost: sold units shown as available for up to a day on the overnight batch; two books reconciled by argument; a bridge spreadsheet at every month-end close; and the gap hiding as oversell if an estimate ever decremented the book.

T — Define how ISM tracks the gap in real time and reconciles it so promising, replenishment and Finance agree, with the program's zero-incident constraint applying to the book as much as to the customer. Goal: one Financial On-Hand, AFS decremented at sale rather than posting, a safe failure direction.

Baselines on the table: Financial On-Hand truths: two (target one) · AFS decrement point: at the S/4 posting (target the sale/ship event) · over-sell from unmatched postings: a live risk (target zero by construction; aging report empty or explained at T+1).

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** S/4 aggregate postings (Aggregation#, total quantity, total COGS); Aptos POS RetailSaleTransactions lines (Order#, store, SKU, quantity); OMS omni-pending sale lines; WMS ShipmentConfirmation lines for ship-to-home.
- **Grain and joins.** One posting row covers many lines — the join runs Aggregation# ↔ POS header ↔ lines. The grain mismatch (one posting vs many lines) is the whole problem: the sum of the lines must equal the header, in quantity and in cost, or something is missing or doubled.
- **Cuts.** By event source (POS flash sale, omni pending, ship-to-home), by posting age (same day, next day, older), and by who reads which number (Finance reads the book, Operations reads the count).
- **What it revealed.** The phantom-availability window is about one day on the overnight batch: at an illustrative 12 units a day on one SKU, 12 sold units sit “available” until the posting. And the two on-hand numbers were not wrong — Finance’s was right about the book, Operations’ was right about the floor; the defect was that nobody owned the gap between them.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

One SKU, 12 units sold today (illustrative). The posting arrives for 10 units; one line (q = 2) is late. “Exposure” is the AFS error versus truth, in units.

OPTION	DAY-OF-SALE EXPOSURE	LATE-LINE (Q = 2) ERROR	VERDICT
Status quo — decrement at the S/4 posting	+12 (all sold units still “available”)	+2 until the next aggregate	Rejected — this is the problem

OPTION	DAY-OF-SALE EXPOSURE	LATE-LINE (Q = 2) ERROR	VERDICT
Real-time per-line posting to S/4	0	0	Rejected — throughput and posting cadence are why S/4 posts aggregates
Estimate-clear — release SNB at batch close	0	+2, invisible (reads as oversell later)	Rejected — the error hides inside AFS
Finance-derived SNB	0	depends on whose number	Rejected — two numbers; the point was one
event-fed SNB, cleared only on a matched posting	**0**	**−2, visible: aged row at 24 h**	Chosen — under-sell is a planning variance, over-sell is a customer defect

Why the chosen option is right even though it is "wrong" by −2: the error has a sign, the sign points at under-sell, and it lands on a report with a named owner.

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS, RISKS)

- **Discovery** — problem statement (the phantom-availability window); event inventory (POS flash sale, omni pending, ship-to-home confirm, S/4 Aggregation# posting); who-reads-which-number map. *Exit bar:* Finance and Ops agree on the description of the gap and which events move units before posting; Internal Audit named as reviewer.
- **Plan & Design** — SNB bucket definition (increment on events; decrement only on matched posting); join spec Aggregation# ↔ POS header ↔ lines; per-line COGS pro-rata de-aggregation with the last line absorbing rounding; the failure-direction statement (unmatched ⇒ SNB stays high); control-owner map. *Exit bar:* signed inside the FSD 6/5/25; Finance accepts one FOH; cost ownership confirmed S/4-owned, ISM visibility-only.
- **Build & Test** — de-aggregation test family (n lines; missing line; duplicate header); the SNB-never-negative assertion; the unmatched-posting aging report; sequenced by event source — POS flash sales, then omni pending, then ship-to-home — each added to SIT only when its increments and matched decrements reconciled. *Exit bar:* every posting matches or ages visibly; Σ per-line COGS = posting COGS to the cent in every cell; UAT = confirmation by Inventory Accounting.
- **Cutover & Go-Live** — riding the May go-live: SNB opening balance reconciled to unposted sales at the snapshot; SNB publish back to S/4 enabled; rollback cheap because S/4 stayed settlement authority. *Exit bar:* aging report empty or explained at T+1; no negative SNB; Finance reads FOH from ISM.
- **Hypercare / Close** — daily aging review with Inventory Accounting; control-owner handoff. *Exit bar:* aging report owned by Finance without me; month-end FOH ties to MB5L through the ISM number, not a bridge spreadsheet (mechanism canonical; days-to-tie not in record).

Influence and ownership. Mine: the SNB definition, the join spec, the de-aggregation rule, the failure-direction statement, the control-owner map and the test families; S/4 and OMS engineers implemented; Inventory Accounting confirmed in UAT and took over the aging report. Two refusals: real-time per-line posting to S/4 (throughput and posting cadence are why S/4 posts aggregates — ISM is the real-time ledger, S/4 stays settlement authority) and a Finance-derived SNB (that is two numbers; the whole point was one).

> **Designed the process; others own it now.** I designed the SNB definition, the join spec and the de-aggregation rule; Inventory Accounting owns the aging report and the daily review, and S/4 remained settlement authority. **Trade-off put to leadership.** The recommendation to Finance and Operations outlined the competing metrics — availability accuracy at the sale event vs the integrity of the financial posting — and chose the failure direction before the happy path: under-sell, never over-sell.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$0.6M/yr *modeled* (phantom-availability cancels avoided ≈$0.45M · Finance reconciliation labor ≈$0.14M · month-end close time ≈$0.05M; range $0.25–1.2M); the record has no pre-state drift dollar and I am not claiming one.

- # — Financial On-Hand truths 2 → 1 (record, FSD signed 6/5/25); over-sell from unmatched postings = 0 by construction.
- **%** — design KPI: match rate ≥ 99.5% within 24 h; the modeled phantom-cancel rate is ~0.10 pt of orders → ~0 (modeled, labeled).
- **Time** — AFS decrement point moved from the S/4 posting to the sale/ship event (record); rode the May execution go-live; aging report empty or explained at T+1 (record exit bar).
- **Customer** — the failure direction protects the customer: an unmatched posting under-states availability — under-sell, never over-sell (record mechanism).
- **Quality** — Σ per-line COGS = posting COGS to the cent; SNB never negative; month-end FOH ties to MB5L through the ISM number, not a bridge spreadsheet (record mechanism; days-to-tie not in record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A phantom-availability window between the sale event and the S/4 aggregation posting, with Finance and Operations reading different numbers and nobody owning the gap. I wrote the problem statement, the event inventory and the who-reads-which-number map.
Scope & influence	Finance (Inventory Accounting), Operations, S/4 and OMS engineers; rode the May go-live; Inventory Accounting confirmed in UAT and took the aging report — influence across a Finance organization I did not sit in.
Execution trade-off + mechanism others now own	Real-time per-line posting to S/4 (refused — S/4 stays settlement authority) vs an SNB bucket in ISM; the failure direction chosen before the happy path — under-sell, never over-sell. Mechanism others own now: the unmatched-posting aging report, owned by Finance without me.
Impact (long-term)	Financial On-Hand truths 2 → 1 (FSD signed 6/5/25); month-end FOH ties to MB5L through the ISM number, not a bridge spreadsheet; ≈$0.6M/yr modeled.
Process improvement (standard work institutionalized)	The de-aggregation test family (n lines; missing line; duplicate header); the SNB-never-negative assertion; the match-rate design KPI (≥ 99.5% within 24 h); a daily aging review handed to Inventory Accounting.
Complexity & consensus	Two refusals held in writing; sequencing by event source (POS flash sales, then omni pending, then ship-to-home) so each source carried its own defect class.
Root cause at scale	The structural cause — the AFS decrement point sitting at the S/4 posting instead of the sale/ship event — moved once, for every channel, rather than reconciled by hand each month.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Peer — Finance wanted their own reconciliation. How did you turn that around?**
>
> I showed the same number arriving from the physical side first and gave them the real report to own — the unmatched-posting aging report, reviewed daily with Inventory Accounting. A parallel spreadsheet dies when its owner owns the real report. They signed inside the FSD on 6/5/25.

> **Peer — Why decrement only on a matched posting? Isn't that just moving cost to Replenishment?**
>
> Yes, and I priced it that way on purpose: an unmatched posting under-states availability and shows up as an aged row Finance clears the same day, so the over-order is small and visible. Decrementing on an estimate would hide the gap as oversell. The measure is the aging report's p95 in hours and the match rate inside 24 hours; above 99.5% the Replenishment side rounds to zero.

> **Peer — Your cancel row looks borrowed from S1. Double-counted?**
>
> Different window: S1's rate is the vocabulary problem across node types; this is the posting-lag window inside one node. If the room wants one number, I keep S1's and drop this row — the mechanism still stands on the record.

> **Peer — Why not post every line to S/4?**
>
> Throughput and posting cadence — S/4 posts aggregates for a reason. ISM is the real-time ledger; S/4 stays the settlement authority. The price is a matching cycle to maintain; the benefit is one book.

> **Peer — 32 hours a week of Finance time — source?**
>
> Benchmark, not record — four people at eight hours a week is what Inventory Accounting described, not what a timesheet says. I would count manual journal entries and bridge lines per month before and after; that is in the posting log and it is not opinion.

> **Peer — Lesson?**
>
> Design the failure direction before the happy path. Differently: Internal Audit in discovery, not review — the day Inventory Accounting asked me why a posting had aged before I noticed it, the control had an owner.

FACET 3 — MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Financial On-Hand truths	`truths = 2 (Finance's number vs Operations' number)`	reconciled by argument (record)	2
2	Phantom window	`W = t_post − t_sale ≈ 1 day`	AFS decremented at posting, not at sale (record design problem)	sold units "available" up to a day
3	Exposure per SKU	`12 units/day × 1 day = 12`	illustrative sales rate (matches the posting example below)	12 phantom units per SKU-day
4	Cancels it causes	`30,000,000 × 0.10% = 30,000`	orders illustrative; rate modeled	≈30,000 phantom cancels/yr
5	Bridge labor	`32 h/wk × 52 = 1,664`	bridge spreadsheet between the two books (modeled)	1,664 h/yr
6	Month-end bridging	`6 × 8 h × 12 = 576`	+1 day of close spent bridging (modeled)	576 h/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	SNB ledger rule	`SNB += each sale/ship event; SNB −= matched posting only`	POS / OMS / WMS events; decrement needs the Aggregation# match (record design)	walked on 12 units below
2	Walk — sales land	`SNB = 0 + 12 = 12`	12 units sold through the day (illustrative); AFS falls at each sale	AFS honest all day
3	Window closed	`exposure = 12 units/day × 0 days = 0`	decrement moved from t_post to t_sale	phantom window = 0
4	De-aggregation	`COGS_i = $300 × q_i / 12 → $75.00 + $50.00 + $100.00 + $25.00 + $50.00 = $300.00`	posting q_P = 12, C_P = $300.00; five lines q = 3, 2, 4, 1, 2 (illustrative); last line absorbs rounding	Σ = $300.00 to the cent
5	Matched posting	`SNB = 12 − 12 = 0`	12 posted with an Aggregation# match	books tie
6	Late line	`SNB = 12 − 10 = 2`	one line (q = 2) late, 10 matched (illustrative)	aged row visible at 24 h
7	Failure direction	`AFS_reported − AFS_true = −2`	unmatched ⇒ SNB high ⇒ AFS_reported ≤ AFS_true (record design)	under-sell, never over-sell

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
8	Match-rate KPI	`matched ÷ received ≥ 99.5% within 24 h`	design target (record)	≥ 99.5%

Table 3 — after: the metric, before → after.

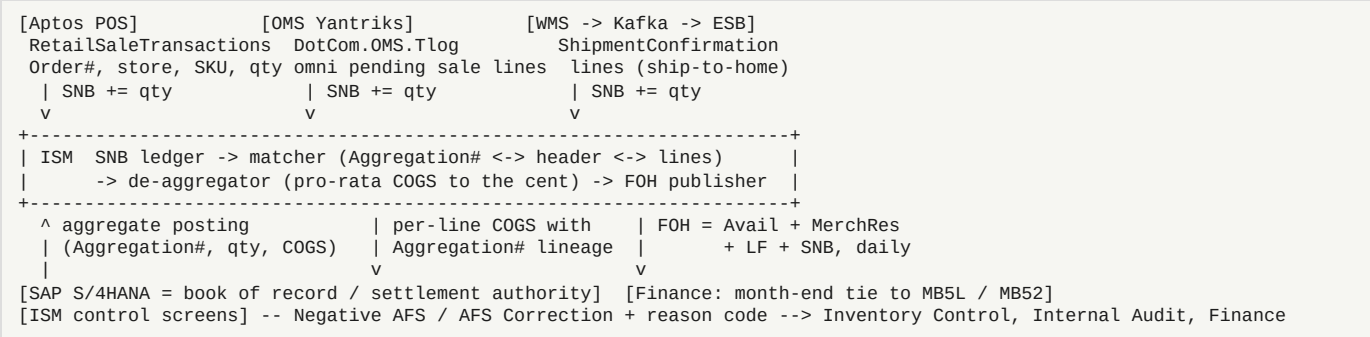
STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Financial On-Hand truths	2	1 — signed inside the FSD 6/5/25	`2 − 1 = 1`	record
2	AFS decrement point	at S/4 posting	at sale / ship	`moved`	record
3	Phantom-window exposure per SKU-day	12 units	0 units	`12 − 0 = 12`	record design; units illustrative
4	Phantom-availability cancels	~0.10 pt of orders	~0	`0.10 − 0 = 0.10 pt`	modeled
5	Finance bridge hours	32 h/wk	~0	`32 − 0 = 32 h/wk`	modeled
6	Month-end close	+1 day bridge	ties to MB5L	`1 − 0 = 1 day`	record design; days-to-tie not in record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Phantom cancels avoided	`30,000 × $15 = $450,000`	$15 all-in per cancel, range $10–20 (modeled); 30,000 from Table 1	≈$0.45M/yr
2	Reconciliation labor	`1,664 × $85 = $141,440`	hours from Table 1; $85/h benchmark (modeled)	≈$0.14M/yr
3	Month-end close	`576 × $85 = $48,960`	hours from Table 1 (modeled)	≈$0.05M/yr
4	Total	`$450,000 + $141,440 + $48,960 = $640,400`	all modeled; the record has no pre-state drift dollar	≈$0.6M/yr (range $0.25–1.2M)

Finance reads FOH from ISM; S/4 remains settlement authority; the aging report with a named Finance owner is the real product.

SOLUTION FLOW — THE SYSTEM & DATA MAP NEW



SENSITIVITY / IMPACT — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
Phantom-cancel rate avoided	0.05 pt	0.10 pt	0.15 pt
Cost per cancel (benchmark)	$10	$15	$20
Cancel $ avoided	$0.15M	$0.45M	$0.90M

INPUT	LOW	BASE	HIGH
Bridge hours per week × loaded rate	20 × $75 = $0.08M	32 × $85 = $0.14M	48 × $95 = $0.24M
Close days saved × rate	0.5 day → $0.02M	1 day → $0.05M	2 days → $0.11M
Total modeled $/yr	≈$0.25M	≈$0.6M	≈$1.2M

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, about six hundred thousand a year: roughly four hundred fifty thousand of cancels that stop happening because availability drops at the sale instead of at the posting, about a hundred forty thousand of Finance hours that were going into a bridge spreadsheet, and about fifty thousand of month-end close time. What is measured is that Finance and Operations went from two on-hand numbers to one, decremented at sale or ship, with the failure direction pointing at under-sell. Every dollar is my model on an illustrative thirty-million-order scale; the record has no pre-state drift dollar and I am not claiming one.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — Walk me through twelve units and a late line.

Twelve units sold through the day, so SNB goes 0 → 12 and AFS falls at each sale — the window is 12 × 0 = 0. The posting arrives for 10 matched units: SNB = 12 − 10 = 2, and those two sit on the aging report at 24 hours. Because S/4 already decremented the posting, AFS under-states by 2 — under-sell, never over-sell.

Numeric — Prove the de-aggregation ties.

Posting: 12 units, $300.00. Cost per unit is $300 ÷ 12 = $25.00, so lines of 3, 2, 4, 1 and 2 units carry $75, $50, $100, $25 and $50 — and 75 + 50 + 100 + 25 + 50 = $300.00. When the cost per unit does not divide evenly the last line absorbs the cent, so the sum is exact by construction.

Numeric — Which of the $0.6M would you defend and which would you drop?

Defend the mechanism, price the rest: $0.45M cancels + $0.14M labor + $0.05M close = $0.64M, all modeled. The cancel row is three quarters of it and is the weakest; if the room wants one cancel number I keep S1's and drop this row, leaving $0.14M + $0.05M = $0.19M — and the mechanism still stands on the record.

Numeric — What is the match rate and why 99.5%?

Matched postings ÷ received within 24 hours ≥ 99.5% is a design KPI, not a measured result. At an illustrative 2,000 postings a day, 0.5% is 10 unmatched rows a day on the aging report — a list one analyst can clear the same day; at 5% it would be 100 rows and the report stops being a control.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND MOMENT NEW

EARN TRUST **1. Tell me about a time you got two teams with competing numbers to agree.**

Open (first two sentences): Finance and Operations read different on-hand numbers and reconciled by argument at every close. I did not ask Finance to trust mine — I showed them the same number arriving from the physical side first.

A. Finance wanted its own reconciliation. I put both numbers on one table, showed the gap was a window — sold units not yet posted — not an error, and then gave them the unmatched-posting aging report to own, with Inventory Accounting reviewing it daily. A parallel spreadsheet dies when its owner owns the real report. **R.** Financial On-Hand went from two numbers to one, signed inside the FSD on 6/5/25.

Land the number: two numbers → one, signed 6/5/25 | **Pivot to NACF:** The same move at an inbound dock: let the receiving team own the received-not-stowed aging list so the planning number and the dock number are one.
Do NOT say: "Finance signed because I was persuasive"; a claimed days-to-tie figure — the record has none.

INVENT AND SIMPLIFY **2. Tell me about a time you simplified something complex.**

Open (first two sentences): The month-end bridge was a spreadsheet between two books that disagreed for a mechanical reason. Instead of improving the spreadsheet, I removed the reason.

A. Four moving parts became one rule: SNB goes up on every sale or ship event and goes down only on a matched posting; S/4 stays settlement authority; the failure direction is under-sell; one aging report. The simplification is that the bridge no longer has a job. **R.** One Financial On-Hand number; month-end FOH ties to MB5L through the ISM number rather than a bridge; about $0.6M a year modeled.

Land the number: one rule, one report, one number | **Pivot to NACF:** For inbound: replace the three-way reconciliation of ASN, dock count and system on-hand with one ledger of received-not-stowed.
Do NOT say: "I built the matcher" — engineers implemented the design.

HAVE BACKBONE; DISAGREE AND COMMIT **3. Tell me about a time you refused something a stakeholder preferred.**

Open (first two sentences): Two proposals were on the table that would each have shipped faster: post every line to S/4 in real time, and let Finance derive the SNB itself. I refused both, on mechanism.

A. Per-line posting fights the throughput and cadence S/4 posts aggregates for — ISM is the real-time ledger and S/4 stays settlement authority. A Finance-derived SNB is two numbers, and the whole point was one. Both refusals went into the FSD as rows with the reason, and each person who proposed one got a replacement that served their need — the real-time view, and a report they owned. **R.** No rework of posting cadence; FSD signed 6/5/25.

Land the number: two refusals, both in the FSD as rows | **Pivot to NACF:** In placement, "just send everything to the nearest FC" is the same shape: fast, and it moves cost somewhere nobody is watching.
Do NOT say: Naming the people who proposed them; implying S/4 is slow.

CUSTOMER OBSESSION **4. Tell me about a time you designed for the failure case to protect the customer.**

Open (first two sentences): Before I designed the happy path I wrote down which way the system should be wrong. An unmatched posting could make availability too high — the customer is told yes to a unit that is gone — or too low — a unit sits unsold for a day.

A. I picked too low, on purpose: SNB stays high until the posting matches, so AFS under-states and Replenishment may over-order visibly. On the worked SKU the late line of 2 units is a −2, not a +2. I priced both sides: an over-stated unit costs about $30 as an incident, an under-stated one about $7.60 of expected lost margin (40% lost-sale share × $19). **R.** Over-sell from unmatched postings is zero by construction.

Land the number: −2 visible, never +2 invisible | **Pivot to NACF:** For inbound: if a receipt is uncertain, show the lower available-to-promise; the cost of a broken date is higher than a late-shown one.
Do NOT say: "Zero oversell" as a measured rate; the 40% and $19 are benchmark inputs, say so.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

"One posting of twelve units, five lines, three hundred dollars — and they have to sum to the cent."

$300.00 = 75 + 50 + 100 + 25 + 50

LP: Dive Deep

The grain mismatch — one posting, many lines — was the whole problem.

Cost per unit $300 ÷ 12 = $25.00; each line is q × $25; the last line absorbs rounding.

The test family: n lines, a missing line, a duplicate header — each has one expected SNB.

CONFLICT / INFLUENCE / BACKBONE

"Finance wanted to keep its own reconciliation, and I understood why."

2 numbers → 1

LP: Earn Trust · Have Backbone

I showed the same number arriving from the physical side first.

Then I refused a Finance-derived SNB — two numbers — and gave them the aging report to own.

A parallel spreadsheet dies when its owner owns the real report.

FAILURE / LEARNING

"The thing I would do differently is when I brought Internal Audit in."

T+1 aging report — owner named late

LP: Learn and Be Curious

I treated Internal Audit as a reviewer at the end; the day Inventory Accounting asked why a posting had aged before I noticed, the control had an owner — but it should have had one in discovery.

No days-to-tie figure exists in the record; I do not claim one.

The cancel row is the weakest dollar; I keep it labeled modeled.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — "Your biggest dollar row borrows S1's rate. Strip it — what is left?"

Different window — S1 is vocabulary across node types, this is posting lag inside one node — but if the room wants one number I keep S1's and drop this row: $141,440 + $48,960 = $190,400, about $0.19M, entirely modeled, and the mechanism still stands on the record.

Say exactly: "If you want one cancel number I keep S1's; what stays here is about nineteen cents on the dollar of that — a hundred ninety thousand — and the mechanism."

Trap — "Why is under-sell the right failure direction? What number would change your mind?"

Compare unit costs: an over-stated unit that turns into an incident costs about $30 (S1's benchmark build); an under-stated unit costs expected lost margin, 40% lost-sale share × $19 = $7.60. Under-sell is right whenever the chance that an over-stated unit becomes an incident exceeds 7.60 ÷ 30 ≈ 25%. If cycle data showed fewer than one in four over-stated units ever became incidents, I would loosen the rule.

Say exactly: "Under-sell wins as long as more than about one in four over-stated units would turn into a customer incident — and I would measure that, not assume it."

Trap — "Halve the S/4 team's capacity. What do you cut, and what can you never cut?"

I cut by event source, last first: POS flash sales stay, omni pending next, ship-to-home last — each source was already sequenced to enter SIT only when its increments and matched decrements reconciled, so a cut source simply stays on the old overnight path. I never cut the matched-only clear, the SNB-never-negative assertion or the aging report with a named owner — without them the failure direction is gone.

Say exactly: "I would cut the sources, never the controls: ship-to-home waits; matched-only clear and the aging report do not."

Where this lands at Amazon. Every fulfillment event moves units before Finance posts them — the same window at a vastly larger grain. The transferable design: one real-time ledger, the financial system as settlement authority, a safe failure direction chosen before the happy path, and a control report with a named owner instead of a bridge spreadsheet. Reliability (over-sell zero by construction), cost efficiency (bridge labor and close time), one number for two organizations.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S3

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you got Finance and Operations to agree.	Earn Trust	FOH 2 → 1; the aging report given to Finance to own; signed inside the FSD on 6/5/25.	Never claim a pre-state drift dollar — the ≈$0.6M is modeled.
Tell me about a problem that needed you to understand the data at its grain.	Dive Deep	One posting per aggregate vs transaction lines; the Aggregation# ↔ header ↔ lines join; COGS pro-rated to the cent.	Skip the table names unless asked.
Tell me about simplifying something complex.	Invent and Simplify	One ledger, one settlement authority, one failure direction, one control report — instead of two books and a bridge.	Do not say "I built the matcher" — engineers implemented the design.
How do you design controls for a process?	Functional — control design	Choose the failure direction first; a never-negative assertion; a KPI with a denominator (match rate ≥ 99.5% in 24 h); an aging report with a named owner; rollback that is cheap because the authority never moved.	Keep S14's standing reconciliation report out — bar raiser's.
What did you refuse and why?	Have Backbone	Per-line posting to S/4; a Finance-derived SNB — both refused on mechanism, not preference.	—
What would you do differently?	Learn and Be Curious	Internal Audit in discovery, not review.	No days-to-tie figure exists — do not invent one.

S1-a — Sephora S/4HANA ↔ ISM — the inventory supply data contract: ten buckets, one owning system each, and every event into exactly one bucket (the double-count path closed on paper before any code) NEW

LPs this story carries (with proof lines below): Dive Deep (primary — three failure classes found at event grain) · Insist on the Highest Standards (drift tolerance zero) · Invent and Simplify (one event, one bucket) · Ownership (zero open "who owns this bucket" rows at signature)
Process family: Plan / Deliver seam — data contract design: buckets, systems of record, event-to-bucket mapping, publish budgets, daily reconciliation

Honesty label. I owned the contract — the bucket inventory, the owner column, the event-to-bucket map, the idempotency and reversal rules, the publish and lag budgets and the drift check — not the build. The ISM app team and SAP configurators implemented to the signed FSD. The unit walk below is illustrative; the equation, rules and budgets are record design. Two ownership cells are tagged VERIFY-WITH-RAJESH because the master names the systems that feed each bucket but not every bucket-to-system pairing.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Dive Deep — I stopped arguing about "available" and walked a five-unit order event by event until I found three distinct ways one bad event moves AFS by ±5 — double-bucket, replay, double reversal.

Insist on the Highest Standards — The daily drift check has tolerance zero — a non-zero ISM-vs-MB52 row is a Sev, not a planning variance — and I refused to write a "reasonable tolerance" into the FSD.

Invent and Simplify — The rule that replaced a page of exceptions: every event maps to exactly one bucket, and a move between buckets is one atomic event, never two.

Ownership — The contract has an owner column; I did not exit design until it had zero open "who owns this bucket" rows, and I carried vocabulary drift as my own top risk.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. ISM sits beside S/4 as a real-time ledger, so for every SKU at every node a unit can sit in one of ten buckets — S/4 available, WMS on-hand, the out-of-balance between them, the basis, three committed buckets, shipped-not-billed, merchandise reserve and lost-and-found. If two systems can each move the same unit, or one system can move it twice, available-for-sale is wrong in a way no one can trace. **Task.** Write the data contract: which buckets exist, which system owns each, how every event moves units, and how fast each number must reach the order system — signed before configuration. **Action.** I wrote a bucket inventory with one owner per bucket; an event-to-bucket map where every event touches exactly one bucket; an idempotency key per event and a rule that a reversal must reference its original; a 60-second publish budget, a WMS feed of at most 15 minutes and consumer lag under 5 minutes; and a daily ISM-versus-MB52 comparison with tolerance zero. I proved it on a five-unit order walked event by event, because that walk is where the three failure classes showed up. **Result.** The FSD was signed 6/5/25 with zero open "who owns this bucket" rows; no SKU carries two published AFS values; drift was zero SKUs on consecutive days at hypercare exit.

20-SECOND VERSION NEW

ISM tracks ten inventory buckets beside S/4. I wrote the contract that gives every bucket one owning system and forces every event into exactly one bucket, with an idempotency key so a replay is a no-op and a reversal that must name its original. On a five-unit order the contract keeps AFS at 555 where three bad events would have moved it ±5; the FSD was signed 6/5/25 with zero unowned buckets.

NACF BRIDGE NEW

Inbound needs the same contract: a unit on an ASN, at the dock door, in receiving, in stow and on a rack is five states with one owner each, and a pallet moves between them as one atomic event. Inbound available-to-promise is then opening on-hand plus units that cleared receive and stow — and a daily count-versus-system check with a stated tolerance.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Three readers of the same SKU — Store, Retail DC and Dotcom DC — each computed "available" from a different subset of what is really ten buckets, and nothing said which system was allowed to move a unit between them. Events came from the POS (Aptos), the order system (OMS, Yantriks), the WMS (BlueYonder) and S/4 itself (MB52 daily plus movement events).

T — Own the data contract of inventory supply: buckets, owners, event rules, budgets and the standing reconciliation, signed by Supply Chain Strategy, Inventory Management and FSC Inventory Accounting before any configuration.

Baselines on the table: bucket owners — undefined (target 1 each, zero open rows) · events able to touch more than one bucket — unknown (target 0) · replay and reversal behavior — undefined (target idempotent, reversal references its original) · publish latency — unbudgeted (target 60 s) · WMS feed ≤ 15 min · consumer lag < 5 min · daily drift — unmeasured (target 0 SKUs).

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** Every feed that moves a unit: Aptos POS sale lines, OMS commit/pick/pack/cancel events, BlueYonder on-hand by item × location × status (≤ 15 min), FRAME channel reserves and HUD holds, and S/4 (MB52 daily plus MSEG movements as events — receipts, transfers, deliveries, count differences).
- **Grain.** SKU × node × bucket × event. One order of five units was traced across all four feeds, one event at a time, with the bucket balance after each.
- **Cuts.** By feed, by bucket, and by failure class: what happens when an event touches two buckets, arrives twice, or is reversed twice.
- **What it revealed.** Three distinct ways a single five-unit order moves AFS by five units off truth: a pick recorded as a new pack without releasing the pick (−5), a commit delivered twice with no key (−5), and a cancel applied twice (+5 — an oversell in waiting). None needs bad data; each only needs an unwritten rule.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

Same SKU, state AFS = 560 (Basis 980 − 200 − 120 − 60 − 40; illustrative). A 5-unit web order commits, picks, packs and ships; then a bad event arrives. "Error" is AFS minus truth (555).

OPTION	WHAT IT DOES	AFS ERROR	VERDICT
Rich events — one event updates several buckets	pick event writes EcomPack +5 and forgets EcomPick −5	−5	Rejected — a half-applied move is a double count
At-least-once delivery, dedupe downstream	commit arrives twice; the consumer sees two orders	−5	Rejected — dedupe belongs at the event, not in each reader
Reversals as free-standing events	a cancel applied twice returns 10 units for a 5-unit order	+5	Rejected — a reversal must reference its original
Each consumer computes its own AFS from raw buckets	OMS and the dotcom site each run the equation	two published values	Rejected — two publishers; OMS reads, never computes

OPTION	WHAT IT DOES	AFS ERROR	VERDICT
15-minute snapshot polling instead of events	sales appear when the snapshot lands	up to a 15-min window	Rejected — misses the real-time purpose
one event, one bucket delta; key per event; reversal names its original; one publisher	atomic move between buckets, replay is a no-op	**0**	Chosen — the error cannot be written

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS, RISKS)

- **Discovery** — the bucket inventory — ten rows with one proposed owning system each — and the event inventory by feed. *Exit bar:* every bucket owned by exactly one system; sessions closed 4/30; open questions carried with owner and date.
- **Plan & Design** — the equation and the contract: lower-of-two basis, AFS = Basis − PickRes − EcomPick − EcomPack − SNB; event → one bucket; idempotency key per event; reversal references original; 60-second publish budget; WMS feed ≤ 15 min; consumer lag < 5 min; drift tolerance zero; FSD v1.1 (3/5/25) → v1.2 (5/16/25). *Exit bar:* signed 6/5/25 by three orgs with zero open “who owns this bucket” rows.
- **Build & Test** — the node × channel × bucket-transition matrix with before/after assertions — the five-unit order is the template cell; test families for replay, duplicate, out-of-order and double reversal; SIT on the interface contracts before UAT. *Exit bar:* 0 open contract defects in SIT; every matrix cell has an owner.
- **Cutover & Go-Live** — the publish-switch plan — OMS reads the published AFS and never computes; no SKU with two published values; control-screen roles assigned. *Exit bar:* January and May go-lives; Negative AFS clean at T+1.
- **Hypercare / Close** — daily ISM-vs-MB52 drift to Inventory Control; consumer-lag and publish-latency monitors against budget. *Exit bar:* drift = 0 SKUs on consecutive days; blockers/day under threshold; exited into the monthly release train.

Influence and ownership. Mine: the bucket inventory, the owner column, the event map, the idempotency and reversal rules, the budgets, the drift definition and the matrix template cell. Built by: the ISM application team and SAP configurators; the POS, OMS and WMS teams exposed the events. The hard part was not the rules — it was getting three orgs to accept that a row without an owner blocks signature.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (BASIS-LABELED)

- **#** — Buckets with an owning system: ten of ten at signature; zero open “who owns this bucket” rows (record). Events able to touch more than one bucket: 0 by contract (record design).
- **%** — Budgets: publish ≤ 60 s; WMS feed ≤ 15 min; consumer lag < 5 min; drift tolerance 0 — all record design. No SKU with two published AFS values.
- **Time** — FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; E2E Inbound 11/22 · Dotcom 12/20 · Retail 1/30; January and May go-lives (record dates).
- **Quality** — Daily ISM-vs-MB52 drift = 0 SKUs on consecutive days at hypercare exit (record). The three failure classes closed: double-bucket −5, replay −5, double-reversal +5 each reduce to 0 on the five-unit walk.
- **$** — No new dollars. The contract’s value is a slice of S1’s modeled oversell row — illustrative 25% × $900,000 = $225,000 VERIFY-WITH-RAJESH — and is never added to S1’s ≈$2.5M.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Three readers, ten buckets, no owner for the definition of a move. I defined the buckets, owners and the event map from first principles by walking one order.
Scope & influence	Four feeds (POS, OMS, WMS, S/4) and three signatory orgs; none reported to me; the contract was signed 6/5/25.
Execution trade-off	Atomic one-bucket events cost more events and a stricter producer contract; I accepted that to make double-counting unwritable. Under-sell accepted over over-sell.
Impact	No SKU with two published AFS values; drift zero on consecutive days; ≈$0.225M illustrative slice of S1's oversell row, never added.
Process improvement	The five-unit order is the template cell of the transition matrix; no story enters QA without a matrix cell reference.
Complexity	Real-time events, a ≤ 15-minute floor feed and a nightly aggregate posting reconciled under one definition.
Root cause	The cause was three missing rules (one bucket per event, a key per event, reversals name originals), fixed once in the contract rather than per incident.

PROBING — FACET 3: MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Starting state	`AFS = 980 − 200 − 120 − 60 − 40 = 560`	S4_Avail 1,000; WMS 980; PickRes 200, EcomPick 120, EcomPack 60, SNB 40 (illustrative; Basis = min(1,000, 980))	560 units
2	Truth after a 5-unit web order commits	`AFS = 980 − 200 − 125 − 60 − 40 = 555`	EcomPick 120 + 5 = 125 (illustrative)	555 units
3	Failure class A — one event writes two places	`AFS = 980 − 200 − 125 − 65 − 40 = 550`	pick recorded as EcomPack +5 without EcomPick −5: both 125 and 65 carry the 5 units	550; error 550 − 555 = −5
4	Failure class B — commit replayed, no key	`AFS = 980 − 200 − 130 − 60 − 40 = 550`	EcomPick 125 + 5 = 130	550; error −5
5	Failure class C — cancel applied twice	`AFS = 980 − 200 − 115 − 60 − 40 = 565`	EcomPick 125 − 5 − 5 = 115 for a 5-unit order	565; error 565 − 555 = +5 (oversell in waiting)
6	Floor staleness at the 15-minute WMS feed	`2 units/h × (15 ÷ 60) h = 0.5`	floor-only changes (damage, lost, relocation) — illustrative 2 units/h; sales do not wait for the feed	≈0.5 unit worst case

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Rule 1 — one event, one bucket	`commit: EcomPick 120 → 125; AFS = 980 − 200 − 125 − 60 − 40 = 555`	event maps to exactly one bucket delta (record design)	AFS 555
2	Rule 1 — a move is one atomic event	`pick: EcomPick 125 − 5 = 120 and EcomPack 60 + 5 = 65 together; AFS = 980 − 200 − 120 − 65 − 40 = 555`	both legs commit or neither does	AFS stays 555
3	Ship confirm	`EcomPack 65 − 5 = 60 and SNB 40 + 5 = 45; AFS = 980 − 200 − 120 − 60 − 45 = 555`	event → SNB; WMS and S/4 books have not moved yet	AFS stays 555

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Overnight posting and floor feed	`S4_Avail 1,000 − 5 = 995; WMS 980 − 5 = 975; SNB 45 − 5 = 40; AFS = min(995, 975) − 200 − 120 − 60 − 40 = 555`	both books and SNB move together at the posting (record design)	AFS stays 555
5	Rule 2 — replay with the same key	`AFS = 555 − 0 = 555`	second delivery of the commit key is a no-op (record design)	AFS 555
6	Rule 3 — reversal names its original	`cancel of the 5-unit order: EcomPick 125 − 5 = 120; a second reversal finds the original already reversed → rejected; AFS = 980 − 200 − 120 − 60 − 40 = 560`	a reversal must reference its original (record design)	AFS 560 = starting state
7	Latency budget	`60 s + 15 min = 60 + 900 = 960 s`	publish budget plus WMS feed — worst case from a floor-only change to a published AFS	960 s = 16 min
8	Daily drift	`drift = ISM.S4_Avail − MB52 = 995 − 995 = 0 for every SKU × node row`	tolerance 0 (record design); a non-zero row is a Sev	0 SKUs

Table 3 — after: the metric, before → after.

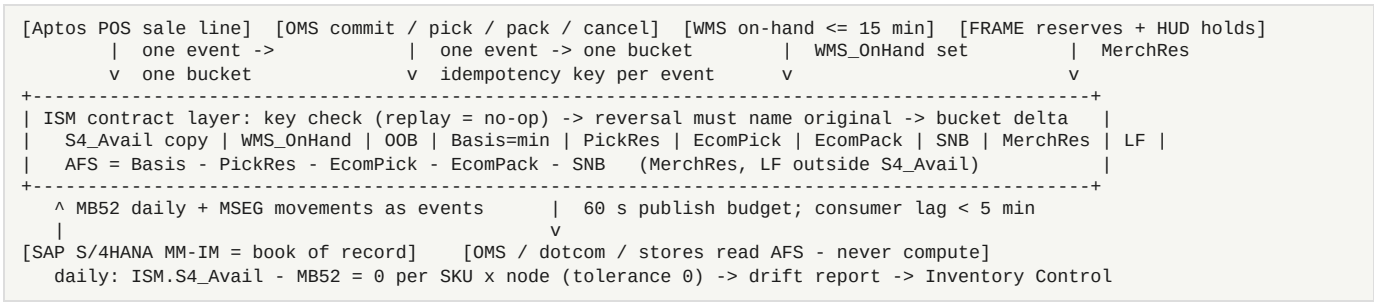
STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Owners per bucket	undefined	1 owning system each; 0 open rows	`10 − 10 = 0 unowned`	record
2	Events able to touch > 1 bucket	unknown	0 by contract	`0`	record design
3	Replay of the same key	AFS 550 (−5)	AFS 555 (0)	`5 units`	record design; units illustrative
4	Reversal applied twice	AFS 565 (+5)	rejected; AFS exact	`5 units`	record design; units illustrative
5	Two published AFS values per SKU	possible	0 SKUs	`0`	record
6	Daily drift	unmeasured	0 SKUs on consecutive days	`0`	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	S1 oversell row (parent)	`30,000,000 × 0.10% = 30,000; 30,000 × $30 = $900,000`	orders illustrative; rate benchmark 0.05–0.25%; $30 = $6 + $4 + $19 + $1 (modeled)	$900,000 (S1's row)
2	Event-level share (illustrative)	`$900,000 × 25% = $225,000`	25% attributable to event-level double counting VERIFY-WITH-RAJESH; illustrative	≈$0.23M — a slice of S1, not additive
3	Price of one class-C event	`1 incident × $30 = $30`	one +5 over-state becomes at most one incident per order (benchmark)	$30 per event
4	Check	`$225,000 ÷ $30 = 7,500`	implied incidents in the slice	7,500 of 30,000 incidents

Walk values are illustrative units; the equation, the three rules, the budgets and the tolerance are record design. The dollar slice is illustrative, flagged for verification, and sits inside S1's ≈$2.5M — never on top of it.

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY / IMPACT — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Share of oversell from event-level double counting	10% → $0.09M	25% → $0.23M	40% → $0.36M
Parent oversell row (S1)	$0.4M	$0.9M	$3.0M
WMS feed interval	5 min	15 min	30 min
Worst-case floor-only exposure at 2 units/h	0.17 unit	0.5 unit	1.0 unit
Publish budget + feed (seconds)	60 + 300 = 360	60 + 900 = 960	60 + 1,800 = 1,860

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, there is no new dollar here, and I will not manufacture one: this contract is the mechanism inside S1's two and a half million, and my illustrative slice is about a quarter of the oversell row, about two hundred twenty-five thousand, which I would drop before defending. What I can defend is on the record: ten buckets, ten owners, zero open rows when three organizations signed on 6/5/25, and drift of zero SKUs on consecutive days at hypercare exit.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — Walk me through the five-unit order.

Start AFS 560. The order commits: EcomPick 120 → 125, AFS 555. Pick moves 5 from EcomPick to EcomPack in one event, 555. Ship moves 5 from EcomPack to SNB, 555. The overnight posting moves S/4, the WMS feed and SNB together by 5 — min(995, 975) − 200 − 120 − 60 − 40 = 555. AFS never moves except at the commit.

Numeric — What breaks if one rule is missing?

Without "one event, one bucket" the pick leaves 5 units in two buckets: AFS 550, −5. Without the key, a replayed commit does the same, −5. Without "a reversal names its original", a cancel applied twice gives back 10 for a 5-unit order: 565, +5 — that one is an oversell waiting to happen.

Numeric — How stale can the number be?

Events are real time inside a 60-second publish budget. The floor feed is up to 15 minutes, so the worst case for a change that only the floor sees — a damage or a relocation — is 60 + 900 = 960 seconds, about 16 minutes; at an illustrative 2 units an hour that is half a unit. Sales do not wait for the feed because they decrement the committed buckets immediately.

Numeric — Why tolerance zero? Isn't that unrealistic?

Because both books are event-fed, a non-zero drift means a missing or duplicated event — a defect with a cause — not noise. At an illustrative 5,000 SKUs × 10 nodes = 50,000 rows VERIFY-WITH-RAJESH, "within 1%" would tolerate 500 rows of unexplained difference; zero forces each into the report, where Inventory Control names the cause.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND

MOMENT NEW

DIVE DEEP **1. Tell me about a time you dug into the details and found something others missed.**

Open (first two sentences): Everyone was arguing about the definition of “available.” I stopped, took one five-unit order, and walked it event by event through four feeds, writing the bucket balance after each one.

A. The walk showed three separate ways the same order could leave AFS five units off — a pick written to two buckets, a commit replayed without a key, a cancel applied twice. None needed bad data, only a rule nobody had written. I turned each into a rule and the walk into the template cell of the test matrix. **R.** The contract was signed 6/5/25 with zero open ownership rows.

Land the number: three failure classes, ±5 units each | **Pivot to NACF:** The same walk on an inbound pallet — ASN, door, receive, stow — shows where available-to-promise could be double counted.
Do NOT say: Claiming a measured defect rate — the walk is illustrative; the rules are record design.

INSIST ON THE HIGHEST STANDARDS **2. Tell me about a standard you held when others wanted to relax it.**

Open (first two sentences): In design review someone asked for a drift tolerance — “within 1% is fine.” I held at zero.

A. Both books are fed by events, so a non-zero difference means a missing or duplicated event. I showed that 1% of an illustrative 50,000 SKU × node rows is 500 unexplained rows a day, and that zero turns each into a named row for Inventory Control. I did write a rule for the exceptions: the row is a Sev, with a reason code, not a quiet variance. **R.** Drift = 0 SKUs on consecutive days at hypercare exit.

Land the number: tolerance 0 vs 500 rows | **Pivot to NACF:** At inbound: a receive count compared with the ASN should have a stated tolerance and a named owner — not “close enough.”
Do NOT say: Claiming the 50,000 rows is real VERIFY-WITH-RAJESH.

INVENT AND SIMPLIFY **3. Tell me about a time you replaced a complicated process with a simple rule.**

Open (first two sentences): The first draft of the event handling had a page of exceptions. I replaced it with one sentence: every event touches exactly one bucket.

A. A move between buckets became one atomic event with two legs that commit together, so “half-applied” could not exist. The exceptions page collapsed into the idempotency key and the reversal-names-original rule. **R.** On the five-unit walk AFS holds at 555 through every event.

Land the number: one page of exceptions → one sentence | **Pivot to NACF:** In placement, the same simplification is “each unit has exactly one planned destination.”
Do NOT say: “I invented a new data model” — it is the SAP/OMS events under one rule.

OWNERSHIP **4. Tell me about a time you took on something no one owned.**

Open (first two sentences): The contract needed an owner column, and the awkward finding was that for some buckets nobody could say who was allowed to move them.

A. I did not exit design until the column had no blanks, and I carried “a bucket with no owner” as my own risk. Where two systems both claimed a bucket I wrote the tie-break into the FSD with the signatories. **R.** Zero open rows at signature; after handoff Inventory Control owns the drift report. Two pairings are flagged for verification VERIFY-WITH-RAJESH.

Land the number: 0 unowned buckets at signature | **Pivot to NACF:** Stranded inbound stock is the same orphan bucket.
Do NOT say: Claiming I owned any of the systems.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

"I traced one five-unit order through four feeds and wrote the balance after every event."

560 → 555 → 555

LP: Dive Deep

Truth after the commit is 555; three bad events give 550, 550 and 565.

None needed bad data — each needed a missing rule.

The walk became the template cell for the matrix.

CONFLICT / INFLUENCE / BACKBONE

"The room asked for "within one percent" on drift and I said no."

tolerance 0

LP: Insist on the Highest Standards · Have Backbone

Event-fed books drifting means a missing or duplicate event.

I priced the alternative: an illustrative 500 unexplained rows a day.

The compromise was a reason-coded Sev row, not a tolerance.

FAILURE / LEARNING

"The contract held; the first promising rule I put on top of it — no channel may promise units in transit — was wrong."

120 hidden units → 120 promisable

LP: Are Right, A Lot

A contract that says what a bucket is does not say whether a rule on top of it is right.

The in-transit rule (S4) starved fast movers; I corrected it through the same signatories with a window and a kill switch.

Lesson: back-test any rule against history before publishing; the data contract is the right place to write the kill switch.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — "Your 25% slice — where does it come from?"

It does not come from the record and I say so: it is an illustrative share I would drop before defending. What I can defend is the mechanism: three failure classes at event grain. To get a real share I would take a quarter of reason-coded incident rows and count how many trace to a replay, a double-bucket or a double reversal.

Say exactly: "The twenty-five percent is a placeholder, flagged for verification; the real number is a count of coded incidents by failure class."

Trap — "What is the dependency risk in this contract?"

Four producers have to honour it — POS, OMS, WMS, S/4 — and I own none. Dependency order: the key and reversal rules need the OMS and POS teams first because they emit the events; the 15-minute feed is the WMS team's; the drift check needs S/4's MB52. The one I would cut if a team slipped is the 60-second budget on non-sale events — never the key or the reversal rule.

Say exactly: "I would relax a latency budget before I relaxed idempotency or the reversal rule — one costs minutes, the other costs trust."

Trap — "How would you know the contract is working without waiting for a customer incident?"

Three leading checks: the drift report (target 0 rows), the count of events rejected as duplicates (non-zero is healthy; a spike means a producer is retrying), and the count of reversals rejected for a missing original (should be zero; any is a producer bug).

Say exactly: "A healthy contract rejects duplicates quietly and rejects zero orphan reversals."

Where this lands at Amazon / NACF. Amazon's promise seam is the same contract at a vastly larger grain: every event moves a unit between states exactly once, and the order system reads one published number. The NACF version is inbound states from ASN to stow with one owner each.

S3-a — Sephora S/4HANA — de-aggregation to the cent: one posting, many lines, and the rule that makes the sum of the parts equal the whole (the grain problem behind Financial On-Hand) NEW

LPs this story carries (with proof lines below): Insist on the Highest Standards (primary — Σ per-line COGS = posting COGS to the cent) · Dive Deep (the grain: one posting vs transaction lines) · Learn and Be Curious (learning how S/4 posts) · Bias for Action (sequenced by event source)
Process family: Deliver / Return-to-book seam — cost allocation and matching at the aggregate-versus-line grain; Finance control design

Honesty label. The de-aggregation rule, the join spec, the test families and the aging-report spec were mine; S/4 and OMS engineers implemented; Inventory Accounting confirmed in UAT and owns the aging report. Cost ownership stayed with S/4 (WAC$ S/4-owned, ISM visibility-only). The $100.00, $175.00 and 2,000-postings-a-day figures are illustrative; the rounding mode and the line-order key are design choices I flag VERIFY-WITH-RAJESH because the record says "the last line absorbs rounding" without naming either.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Insist on the Highest Standards — Σ per-line COGS must equal the posting COGS to the cent in every test cell — a one-cent break is a failed row, not a rounding variance.

Dive Deep — The problem lived one level below the dashboards: one S/4 posting per Aggregation# against many POS lines per Order#, so the join and the allocation had to be written at line grain.

Learn and Be Curious — I learned how S/4 posts sales — aggregates with total quantity and COGS — well enough to explain to Finance why per-line posting was never the answer.

Bias for Action — I sequenced the build by event source — POS flash sales first, then omni pending, then ship-to-home — each added to SIT only when its increments and matched decrements reconciled.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. S/4 posts sales as aggregates — one posting per aggregation number with a total quantity and a total cost — while the POS sends transaction lines. ISM needs cost and quantity per line to decrement the Shipped-Not-Billed bucket and to give Finance a number it can trust, so every posting has to be spread back onto its lines. **Task.** Define the de-aggregation exactly: how a posting's quantity and cost are matched to lines, how the cent is handled, and what happens when a line is missing, late or duplicated. **Action.** I wrote the join Aggregation# to POS header to lines; the pro-rata cost rule — each line gets posting COGS × its quantity ÷ the posting quantity, rounded, with the last line taking the difference so the sum is exact; a SNB-never-negative assertion; and three test families — n lines, a missing line, a duplicate header — each with one expected answer. Finance owns the aging report for postings that do not match. **Result.** Every posting either matches to the cent or ages visibly on a report with an owner; the unmatched rate is a design KPI (match ≥ 99.5% within 24 hours); Financial On-Hand moved from two truths to one on the signed FSD of 6/5/25.

20-SECOND VERSION NEW

S/4 posts one aggregate per aggregation number; the POS sends many lines. I wrote the rule that spreads a posting's cost back onto its lines — quantity-weighted, rounded, the last line absorbing the cent — so the lines always sum to the posting to the cent, and any posting that cannot match ages on a report Finance owns. It is the engine behind Financial On-Hand moving from two numbers to one.

NACF BRIDGE NEW

Inbound has the same grain problem: an ASN header with many lines versus a dock count by pallet. "Lines sum to the header" is the check that would have caught the hidden 98.0% on one class in a 99.0% overall accuracy number — an aggregate hides what the lines show. I would run the same spread-and-tie on received quantities by ASN line.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — One S/4 posting per Aggregation# carries a total quantity and a total COGS; the Aptos POS holds the lines with an Order#. Between sale and posting the book is behind; once the posting lands, ISM has to retire exactly the units and dollars that posting covers — no more, no less — against lines that may arrive late, twice or not at all.

T — Define the de-aggregation and matching rule, the exception handling and the control report so that Σ line cost = posting cost to the cent and SNB can never go negative — inside the ISM FSD signed 6/5/25.

Baselines on the table: Σ per-line COGS vs posting COGS — untested at the grain (target = $0.00 difference in every cell) · SNB negative values — a possible silent path (target never negative) · unmatched postings — invisible (target aged row at 24 h, owner named) · match rate — unmeasured (target ≥ 99.5% within 24 h).

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** S/4 aggregate postings (Aggregation#, quantity, COGS); Aptos RetailSaleTransactions headers and lines (Order#, store, SKU, qty); OMS and WMS shipment lines; S/4 cost stays S/4-owned (WAC$ read-only to ISM).
- **Grain and joins.** Posting ↔ header ↔ lines is one-to-many-to-many; the quantity must sum exactly, the cost must sum to the cent. Cut by lines per posting and by whether the unit cost divides evenly into cents.
- **What it revealed.** A break is not an exception, it is guaranteed: $100.00 over three equal units is $33.333… a unit, and no way of rounding each line independently makes the three lines sum to $100.00 — they sum to $99.99. Left alone, every such posting leaves a one-cent hole that Finance would chase by hand.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

Posting of 3 units, $100.00; three lines of 1 unit (illustrative). “Break” is Σ lines − posting.

OPTION	LINES (ROUNDED TO THE CENT)	Σ LINES	BREAK	VERDICT
Round each line independently	$33.33 + $33.33 + $33.33	$99.99	−$0.01	Rejected — a break on every non-divisible posting
Bank the cents in a rounding account	$33.33 × 3 and $0.01 to a GL residual	$100.00 with a residual	$0.00 + a $0.01 residual	Rejected — creates a balance Finance must explain
Largest-remainder allocation	equal remainders, so an order key is needed anyway	$100.00	$0.00	Rejected — more logic for the same answer
last line absorbs the difference	$33.33 + $33.33 + $33.34	**$100.00**	**$0.00**	Chosen — deterministic if the line order is written down VERIFY-WITH-RAJESH

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS, RISKS)

- **Discovery** — the event inventory (POS flash sale, omni pending, ship-to-home confirm, S/4 Aggregation# posting), the who-reads-which-number map, the lines-per-posting profile and the rounding-class profile. *Exit bar:* Finance and Operations agree on the description of the gap; Internal Audit named as reviewer.
- **Plan & Design** — join spec Aggregation# ↔ header ↔ lines; the pro-rata cost rule with the last line absorbing; rounding mode and line order written into the spec VERIFY-WITH-RAJESH; SNB-never-negative assertion; unmatched-posting aging report spec; WAC$ S/4-owned, ISM visibility-only. *Exit bar:* signed inside the FSD 6/5/25; Finance accepts one FOH.
- **Build & Test** — three test families — n lines, a missing line, a duplicate header — with expected SNB and cost for each; sequenced by event source, each added to SIT only when its increments and matched decrements reconciled. *Exit bar:* every posting matches or ages visibly; Σ per-line COGS = posting COGS to the cent in every cell; UAT = confirmation by Inventory Accounting.
- **Cutover & Go-Live** — rode the May go-live: SNB opening balance reconciled to unposted sales at the snapshot; SNB publish back to S/4 enabled. *Exit bar:* aging report empty or explained at T+1; no negative SNB.

- **Hypercare / Close** — daily aging review with Inventory Accounting; handoff of the report. *Exit bar:* aging report owned by Finance without me; match ≥ 99.5% within 24 h as the design KPI.

Influence and ownership. Mine: the join spec, the allocation rule, the exception cases, the test families and the aging-report spec. Built by: S/4 and OMS engineers. Confirmed and owned after: Inventory Accounting. Decision I held: cost stays S/4-owned — ISM carries it for visibility, so there is never a second cost number.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (BASIS-LABELED)

- **Quality** — Σ per-line COGS = posting COGS to the cent in every test cell; SNB never negative; every unmatched posting visible on the aging report (record mechanism).
- **#** — Financial On-Hand truths 2 → 1 (record, FSD signed 6/5/25); three test families, each with one expected SNB and cost (record design).
- **%** — Design KPI: match ≥ 99.5% within 24 h (design target). At an illustrative 2,000 postings a day, 0.5% = 10 aged rows a day — a list one analyst clears VERIFY-WITH-RAJESH.
- **Time** — SNB opening balance reconciled at the May go-live snapshot; aging report empty or explained at T+1 (record exit bar).
- **$** — No new dollars. Illustrative slice of S3's modeled Finance bridge-labor row: 1,664 h × 30% × $85 = $42,432 VERIFY-WITH-RAJESH — inside S3's ≈$0.6M, never added to it.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Nobody had written what "spread a posting onto its lines" means when the cents do not divide. I wrote it, including the cases.
Scope & influence	Finance, Inventory Accounting, S/4 and OMS engineers, Internal Audit as reviewer; Inventory Accounting took the report.
Execution trade-off	Last-line-absorbs over a rounding account: one line carries a cent's distortion in exchange for no residual balance anywhere.
Impact	The grain problem behind Financial On-Hand truths 2 → 1; every posting matches or ages visibly.
Process improvement	Test families as standard work: n lines, missing line, duplicate header — each with one expected answer.
Complexity	Three event sources with different latencies, one posting grain; sequenced by source.
Root cause	The cause was grain, not data quality; fixed in the allocation rule once, not in a monthly bridge.

PROBING — FACET 3: MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Naive rounding	`3 × $33.33 = $99.99`	posting $100.00 over three equal units; each line rounded alone (illustrative)	$99.99; break −$0.01
2	Unit cost does not divide	`$100.00 ÷ 3 = $33.3333`	posting COGS ÷ posting quantity	$33.3333 a unit
3	A second case	`$175.00 ÷ 8 = $21.875`	posting 8 units, $175.00 (illustrative)	$21.875 a unit
4	Naive rounding, second case	`$65.63 + $65.63 + $43.75 = $175.01`	lines of 3, 3, 2 units rounded independently, half-up (illustrative)	$175.01; break +$0.01
5	Missing line	`unmatched units = 8 − 6 = 2`	posting 8 units; lines present 3 + 3 = 6 (illustrative)	2 units no line explains
6	Duplicate header	`Σ q = (3 + 3 + 2) × 2 = 16 vs posting 8`	the same Order# arrives twice (illustrative)	16 ≠ 8 — doubled

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Rule	`COGS_i = round(C_P × q_i ÷ q_P, 2); last line = C_P − Σ prior lines`	record design; rounding mode and line order flagged VERIFY-WITH-RAJESH	defined for every case
2	Case 1, lines 1–2	`round($100.00 × 1 ÷ 3, 2) = $33.33`	each of two lines	$33.33 + $33.33 = $66.66
3	Case 1, last line	`$100.00 − $66.66 = $33.34`	the last line absorbs the cent	$33.34
4	Case 1, tie	`$33.33 + $33.33 + $33.34 = $100.00`	Σ lines = posting	$100.00 exact
5	Case 2, lines 1–2	`round($175.00 × 3 ÷ 8, 2) = round($65.625, 2) = $65.63`	half-up (illustrative mode); two lines	$65.63 + $65.63 = $131.26
6	Case 2, last line	`$175.00 − $131.26 = $43.74`	unrounded would be $43.75; the −$0.01 is absorbed here	$43.74
7	Case 2, tie	`$65.63 + $65.63 + $43.74 = $175.00`	Σ lines = posting	$175.00 exact
8	Missing line — guard	`attempted clear 8 against SNB 6: 6 − 8 = −2 → blocked`	SNB-never-negative assertion (record design)	clear 6; 2 to the aging report
9	Duplicate header — guard	`Σ q = 16 > 8 → hold the second set`	quantity may not exceed the posting (record design)	8 matched; 8 held as duplicate
10	SNB opening at cutover	`SNB = 1,200 − 1,150 = 50`	unposted sales at the snapshot: lines received 1,200, matched postings 1,150 (illustrative)	opening SNB 50

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Case 1 break	−$0.01 ($99.99)	$0.00 ($100.00)	`$0.00 − (−$0.01) = $0.01`	record design; units illustrative
2	Case 2 break	+$0.01 ($175.01)	$0.00 ($175.00)	`$0.01`	record design; units illustrative
3	Residual GL balance from rounding	a rounding account	none	`$0.01 per break → $0`	record design
4	Missing-line unmatched units	invisible	2 units on the aging report	`2`	record design
5	Duplicate set	double count	8 held	`8`	record design
6	Match rate	unmeasured	≥ 99.5% in 24 h	`design KPI`	design target

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Aged rows a day at KPI	`2,000 × 0.5% = 10`	illustrative 2,000 postings a day VERIFY-WITH-RAJESH	10 rows
2	At 5% unmatched	`2,000 × 5% = 100`	the report stops being a control at this volume	100 rows
3	S3 reconciliation labor (parent)	`32 h/wk × 52 = 1,664; 1,664 × $85 = $141,440`	modeled (S3)	$141,440
4	Slice from line-level breaks	`1,664 × 30% = 499.2 h; 499.2 × $85 = $42,432`	30% illustrative VERIFY-WITH-RAJESH; slice, not additive	≈$0.04M

Rounding is half-up here (illustrative); the point is that any mode, with the last line absorbing the difference, ties exactly. Case 2 shows a different shape: the naive sum is +$0.01, the chosen rule is exact.

SOLUTION FLOW — THE SYSTEM & DATA MAP

```
[Aptos POS lines]        [OMS omni lines]       [WMS ShipmentConfirmation]
 Order#, SKU, qty ----+---- sale / ship events ----+---- (all SNB += qty) ---->  SNB ledger
                                                                                    |
[S/4 aggregate posting: Aggregation#, q_P, C_P] --> matcher: Aggregation# <-> header <-> lines
                                                                |
     +------------------------------+-----------------+-----------------------+
     |                              |                                         |
 q matched = q_P ?             q matched < q_P                          q matched > q_P
 spread cost pro-rata          clear matched only                       hold duplicate set
 last line absorbs cent        SNB never < 0 (assert)                   aging report
 SNB -= matched                unmatched -> aging report                (owner: Inventory Accounting)
     |
 per-line COGS with Aggregation# lineage -> FOH publisher -> Finance: month-end tie to MB5L / MB52
```

SENSITIVITY / IMPACT — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Postings a day	1,000	2,000	5,000
Unmatched at 0.5%	5 rows	10 rows	25 rows
Unmatched at 5%	50 rows	100 rows	250 rows
Share of bridge hours from line-level breaks	15% → $0.02M	30% → $0.04M	50% → $0.07M
Parent labor row (S3)	$0.08M	$0.14M	$0.24M

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, I am not adding a dollar to the story I already told: this is the engine inside S3's six hundred thousand, and the slice I can put on it is about forty thousand — illustrative, and I would drop it before defending it. What is on the record is the control: every posting matches to the cent or ages on a report Finance owns, SNB can never go negative, and Financial On-Hand moved from two numbers to one.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — Why can't each line just be rounded?

Because $100.00 over three units is $33.3333 a unit; rounding each line alone gives $33.33 three times, $99.99, a one-cent hole on every non-divisible posting. With the last line absorbing it, $33.33 + $33.33 + $33.34 = $100.00.

Numeric — Show me a case where naive rounding goes the other way.

Posting 8 units, $175.00: $21.875 a unit, lines of 3, 3 and 2 units. Naive half-up gives $65.63 + $65.63 + $43.75 = $175.01 — a plus one-cent break. The rule gives the last line $175.00 − $131.26 = $43.74, so it sums to $175.00 exactly.

Numeric — A posting says 8 units but only 6 lines are in. What happens?

SNB would go 6 − 8 = −2 if I let it clear, so the assertion blocks it: 6 units clear, 2 go to the aging report as an unmatched posting, and Finance sees them the same day. A negative SNB would quietly add supply to AFS — an invisible over-sell.

Numeric — How many exceptions a day can your team actually clear?

At an illustrative 2,000 postings a day VERIFY-WITH-RAJESH, a 99.5% match rate is 10 aged rows a day; at 95% it is 100. The KPI is chosen so the report stays a list a person can clear — a control only works while it is small enough to read.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND MOMENT NEW

INSIST ON THE HIGHEST STANDARDS **1. Tell me about a time you insisted on a standard that others thought was too strict.**

Open (first two sentences): Someone said a one-cent difference on a posting is rounding noise. I said it is a failed test.

A. I showed that a one-cent break is guaranteed on every non-divisible posting, so “noise” is a hole on every one of them, and that Finance would chase each by hand. I wrote the last-line-absorbs rule and made Σ line cost = posting cost to the cent the pass condition for every cell in every test family. **R.** Every cell tied; nothing accumulates in a rounding account.

Land the number: $0.00 break in every cell | **Pivot to NACF:** On inbound receipts: the standard is lines sum to the ASN header exactly, or the row ages.
Do NOT say: Quantifying the cent as a dollar claim.

DIVE DEEP **2. Tell me about a time the answer was hidden below the aggregate.**

Open (first two sentences): The Finance number and the Operations number were both right; the problem was one grain down.

A. I laid one posting next to its lines and showed that the join is one-to-many and the quantity and cost must both sum. The three failure shapes — a missing line, a duplicate header, a non-divisible cost — each have one expected result. **R.** The three test families became standard work.

Land the number: one posting, three failure shapes | **Pivot to NACF:** Same pattern as an overall inbound accuracy that hides the class that is off.
Do NOT say: Table names unless asked.

LEARN AND BE CURIOUS **3. Tell me about something you had to learn quickly to do the job.**

Open (first two sentences): I was designing a bucket that depended on how S/4 posts sales, and I had to learn it before I could argue about it.

A. I sat with the S/4 and Finance teams and learned why S/4 posts aggregates — throughput and posting cadence — and what the posting carries: an Aggregation#, a quantity, a cost. That is why I refused per-line real-time posting and designed the match on the ISM side. **R.** The design held S/4 as the settlement authority; I never claim to have configured it.

Land the number: S/4 stays the settlement authority | **Pivot to NACF:** The same: learn the receiving system’s posting cadence before promising a number off it.
Do NOT say: “I configured S/4.”

BIAS FOR ACTION **4. Tell me about a time you had to deliver in sequence under a hard date.**

Open (first two sentences): There were three event sources and one date, so I did not try to land them together.

A. I sequenced by source — POS flash sales first, then omni pending, then ship-to-home — and each entered SIT only after its increments and matched decrements reconciled. A late source slipped alone, not the date. **R.** Rode the May go-live with SNB opening balance reconciled at the snapshot.

Land the number: three sources, one date, sequenced | **Pivot to NACF:** Inbound launches the same way: one door class or one lane group at a time, each reconciled before the next.
Do NOT say: Dates beyond the May go-live.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

"I put one posting next to its three lines and asked what the cents add up to."

$99.99 vs $100.00

LP: Dive Deep

$100.00 ÷ 3 = $33.3333; independent rounding gives $99.99.

The break is guaranteed on every non-divisible posting.

Last-line-absorbs gives $33.33 + $33.33 + $33.34 = $100.00.

CONFLICT / INFLUENCE / BACKBONE

"The easy answer to a penny is a rounding account, and I did not want to create one."

$0.00 residual

LP: Insist on the Highest Standards · Have Backbone

A rounding account creates a balance someone must explain every month.

I wanted the sum right on the line, not corrected downstream.

Who proposed the account, if anyone, is VERIFY-WITH-RAJESH; I do not claim a specific disagreement.

FAILURE / LEARNING

"What I would write on day one: which rounding mode, and which line is last."

rounding mode · line order

LP: Are Right, A Lot

The rule says "the last line absorbs"; it needs a stated mode and a stated order or two runs could differ by a cent.

The record does not carry a miss here — VERIFY-WITH-RAJESH whether one happened.

Differently: write the ordering key and mode into the spec and the test cell on day one.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — "Is the cent worth a rule? What would change your mind?"

The rule costs one line of spec and one test cell. What would change my mind is if breaks were rare — they are not: a cost that divides evenly is rare; $33.33-type splits happen on every posting whose unit cost is not a whole cent. If Finance showed me a rounding account they already reconcile for free, I would revisit it.

Say exactly: "It costs one line of spec to avoid a cent on every non-divisible posting; if Finance already had a free rounding account I would revisit it."

Trap — "What is the dependency, and what do you cut if S/4's team has no capacity?"

The dependency chain: the Aggregation# on the posting (S/4), the Order# join (POS), and the shipment lines (WMS). I cut ship-to-home last-in, first-out — it rides the old overnight path longer. I never cut the SNB-never-negative assertion or the aging report.

Say exactly: "I would cut a source, never the assertion or the report."

Trap — "A posting arrives twice. What happens to Financial On-Hand?"

Nothing, by two guards: the posting's Aggregation# is idempotent — a second delivery is a no-op — and quantity matched can never exceed the posting quantity. On the walk with 8 units, a doubled set is 16 > 8, so 8 clear and 8 hold as a duplicate on the aging report.

Say exactly: "A double delivery is a no-op, and a doubled set of lines holds on the aging report — it cannot move Financial On-Hand."

Where this lands at Amazon / NACF. At Amazon scale the lines-to-header tie is the same control wherever aggregates meet transactions: ASN lines to the header, dock pallets to receipts. It is how an aggregate accuracy number stops hiding the class that is off.

S1-b — Sephora S/4HANA ↔ ISM — control screens and the operating model: Negative AFS, AFS Correction and the drift report, so every change to available-for-sale has a screen, a role and an audit row before I stepped back NEW

LPs this story carries (with proof lines below): Ownership (primary — a mechanism others own after I leave) · Insist on the Highest Standards (an audit row for every AFS-changing act) · Earn Trust (Inventory Control, Internal Audit and Finance as named owners) · Deliver Results (hypercare exit on counted rows)
Process family: Plan / Deliver seam — control design and operating model: exception screens, role map, reason codes, hypercare handoff

Honesty label. I designed the control-owner map, the screens' purpose and exit bars, the reason-code requirement and the hypercare handoff; the ISM application team built the screens, and Inventory Control, Internal Audit, Finance and the ISM app team run them. The SKU example and the daily row counts are illustrative; the roles, the T+1 bar and the zero drift tolerance are record design. Whether the screens enforce a separate approver for a correction is VERIFY-WITH-RAJESH — the record says "a screen, a role and an audit row," not maker-checker.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Ownership — Every AFS-changing act got a screen, a role and an audit row before I handed off — the drift report runs daily without me and the owners are named in the FSD.

Insist on the Highest Standards — A negative AFS is never clamped to zero silently: it appears on a screen, with a reason code, and the exit bar is "clean at T+1."

Earn Trust — I gave Internal Audit and Finance seats in the control design rather than a review at the end — they own the screens that would catch my own design's errors.

Deliver Results — Hypercare exited on counted rows — drift = 0 SKUs on consecutive days and blockers per day under threshold — not on a calendar.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. ISM computes available-for-sale from ten buckets in real time. That is only as good as what happens on the days the number is wrong — a negative AFS because commitments exceed the lower of the two books, a stale reservation that no longer corresponds to an order, a drift between ISM and S/4. If anyone can change an AFS by hand, or nobody owns the exception, the real-time ledger becomes the next spreadsheet. **Task.** Design the control layer and the operating model so the system stays trustworthy after the program team leaves. **Action.** I defined three controls — a Negative AFS screen, an AFS Correction screen with a reason code and an audit row, and a daily ISM-versus-S/4 drift report with tolerance zero — and a control-owner map: Negative AFS to Inventory Control, Internal Audit and Finance; AFS Correction to the ISM application team; the drift report to Inventory Control. I set the exit bars before go-live: Negative AFS clean at T+1; drift zero SKUs on consecutive days. **Result.** At hypercare exit every AFS-changing act had a screen, a role and an audit row; the drift report runs without me; and the design entered the monthly release train. The dual-book reconciliation that took about four people by hand became about one reading a report — my model of the record mechanism.

20-SECOND VERSION NEW

A real-time availability number is only trustworthy if the days it is wrong have an owner. I designed three controls — a negative-AFS screen, a reason-coded AFS correction and a daily drift report at tolerance zero — and put a named owner on each in the signed spec. When I handed off, every change to available-for-sale had a screen, a role and an audit row; reconciliation went from about four people by hand to about one reading a report (modeled).

NACF BRIDGE NEW

Inbound needs the same operating model: a negative or implausible available-to-promise (more promised than received-and-stowed) must land on a screen with an owner, and any manual change to inbound availability needs a reason code and an audit row. A daily "system versus count" drift report for each building closes the loop.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A real-time availability ledger beside the book of record creates a new class of exception: a negative AFS, a stale reservation, a drift between the ledger and S/4 — and a new temptation: a manual fix nobody can trace. Before the program, reconciling the two books took about four people by hand (modeled).

T — Design controls and a role map so every AFS-changing act is visible and owned, with exit bars that hold up in hypercare, and hand the mechanism to the people who would run it.

Baselines on the table: AFS-changing acts with a screen, a role and an audit row — none defined (target 100%) · Negative AFS — unmonitored (target clean at T+1) · ISM-vs-MB52 drift — unmeasured (target 0 SKUs on consecutive days) · reconciliation ≈4 FTE by hand (target ≈1 FTE reading a report, modeled).

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** The bucket ledger (SKU × node × bucket), the published AFS stream, S/4 MB52 snapshots and MSEG movements, and the cause classes of exceptions seen in testing: committed above supply, stale reservation, feed lag, count adjustment.
- **Grain.** SKU × node, per publish cycle; one row on the screen per SKU × node where AFS < 0 or where drift ≠ 0.
- **What it revealed.** On a SKU with S/4 500, WMS 460 and 480 committed, AFS is −20 on the lower-of-two basis but +20 if a reader trusts the S/4 book. The negative is the screen working, not a bug — and clamping it to zero would hide 20 units of over-commitment.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

The question is who may change AFS and how. Same SKU, AFS −20 (illustrative).

OPTION	WHO CAN CHANGE AFS	TRACEABILITY	VERDICT
Anyone with order-system access adjusts by hand	many	none	Rejected — the next spreadsheet
App team fixes by direct database change on request	one team, off-screen	ticket only	Rejected — no reason code or audit row at the act
Clamp negatives to zero and move on	nobody	hides the cause	Rejected — hides 20 units of over-commitment
No manual change ever	nobody	n/a	Rejected — a count error needs a legitimate path
one AFS Correction screen: reason code, audit row, a named owning role	ISM app team	**every act: who, what, why, when**	Chosen — with Negative AFS to Inventory Control, Internal Audit and Finance

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS, RISKS)

- **Discovery** — inventory every way AFS can change and every exception class; name the readers — Inventory Control, Internal Audit, Finance, the ISM app team. *Exit bar:* every AFS-changing act mapped to a screen or removed.
- **Plan & Design** — the control-owner map — Negative AFS → Inventory Control, Internal Audit and Finance; AFS Correction → ISM app team; drift report → Inventory Control; reason-code requirement on every correction; exit bars (Negative AFS clean at T+1; drift 0). *Exit bar:* in the FSD signed 6/5/25.
- **Build & Test** — screen specs and the matrix cells that exercise a negative, a correction and a drift row; UAT = Inventory Control confirming it can work the list. *Exit bar:* 0 open contract defects in SIT; owners trained on a rehearsal list VERIFY-WITH-RAJESH.
- **Cutover & Go-Live** — control-screen roles assigned before January and May go-lives; rollback triggers armed. *Exit bar:* Negative AFS clean at T+1 after each go-live.
- **Hypercare / Close** — daily drift report; Negative AFS / AFS Correction audit review; hypercare exit memo and FSD as-built note. *Exit bar:* drift = 0 SKUs on consecutive days; blockers/day under threshold; the design enters the monthly release train.

Influence and ownership. Mine: the map of AFS-changing acts, the owner map, the reason-code rule, the exit bars and the handoff memo. Built by: the ISM application team. Run by: Inventory Control (negative AFS and drift), Internal Audit (sampling), Finance (value review) and the ISM app team (corrections). I deliberately did not make myself an owner of any screen.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (BASIS-LABELED)

- **Quality** — Every AFS-changing act has a screen, a role and an audit row (record); Negative AFS clean at T+1 (record exit bar); drift = 0 SKUs on consecutive days at hypercare exit (record).
- **#** — Three controls, four owning roles; reconciliation ≈4 FTE by hand → ≈1 FTE reading the drift report (modeled on the record mechanism).
- **Time** — Controls assigned before the January (planning) and May (execution) go-lives; exit into the monthly release train after hypercare (record).
- **$** — No new dollars. The dollar behind this story is S1's reconciliation row: 3 FTE × $110,000 = $330,000 (modeled), inside S1's ≈$2.5M — never added to it.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No one had listed the ways AFS could change. I wrote the list and removed the ones that had no owner.
Scope & influence	Four owning roles across Inventory Control, Internal Audit, Finance and the ISM team; signed 6/5/25.
Execution trade-off	A screen with a reason code slows a correction by a minute; I accepted that for an audit row on every act.
Impact	The reconciliation burden shifted from argument to a report (≈4 → ≈1 FTE, modeled).
Process improvement	Exit bars on counted rows: T+1 negative clean; drift zero on consecutive days.
Complexity	Two go-lives, each with the controls assigned before it, each with a rollback trigger armed.
Root cause	The cause of "shadow spreadsheets" was an unowned exception; fixed by naming owners, not by policing.

PROBING — FACET 3: MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Basis, lower of two	`Basis = min(500, 460) = 460`	S4_Avail 500; WMS_OnHand 460 (illustrative)	460
2	Out-of-balance	`OOB = 500 − 460 = 40`	book above floor	+40
3	Committed	`200 + 150 + 80 + 50 = 480`	PickRes 200, EcomPick 150, EcomPack 80, SNB 50 (illustrative)	480
4	AFS on the lower-of-two	`AFS = 460 − 480 = −20`	Basis − committed	−20 — a negative
5	AFS if a reader trusts S/4	`AFS = 500 − 480 = 20`	raw S/4 read (the old definition A)	+20 — masks −20
6	Exposure if clamped to zero and ignored	`20 units × $30 = $600`	$30 incident cost (S1 benchmark build)	$600 per SKU-day

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Negative AFS row	`AFS = −20 < 0 → row on the Negative AFS screen at the next publish`	60-second publish budget (record design); reason-code classes: committed above supply · stale reservation · feed lag · count	1 row, owner Inventory Control
2	Resolve by AFS Correction	`PickRes = 200 − 25 = 175`	25 stale reservations for orders already cancelled (illustrative); reason “stale reservation”	PickRes 175
3	AFS after correction	`AFS = 460 − 175 − 150 − 80 − 50 = 5`	same Basis 460	+5 — clean
4	Audit row	`SKU, node, bucket PickRes, before 200, after 175, Δ −25, reason, user, time`	every AFS-changing act (record: screen, role, audit row)	1 audit row
5	Drift after the correction	`drift = ISM.S4_Avail − MB52 = 500 − 500 = 0`	the correction moves a committed bucket, not a book	0
6	T+1 exit check	`open negative rows at T+1 = 0`	exit bar: Negative AFS clean at T+1 (record)	0 rows
7	Workload check — negatives	`40 × 6 ÷ 60 = 4.0 h`	illustrative 40 rows a day, 6 min each VERIFY-WITH-RAJESH	4.0 h/day
8	Workload check — drift	`10 × 15 ÷ 60 = 2.5 h`	illustrative 10 rows a day, 15 min each VERIFY-WITH-RAJESH	2.5 h/day

Table 3 — after: the metric, before → after.

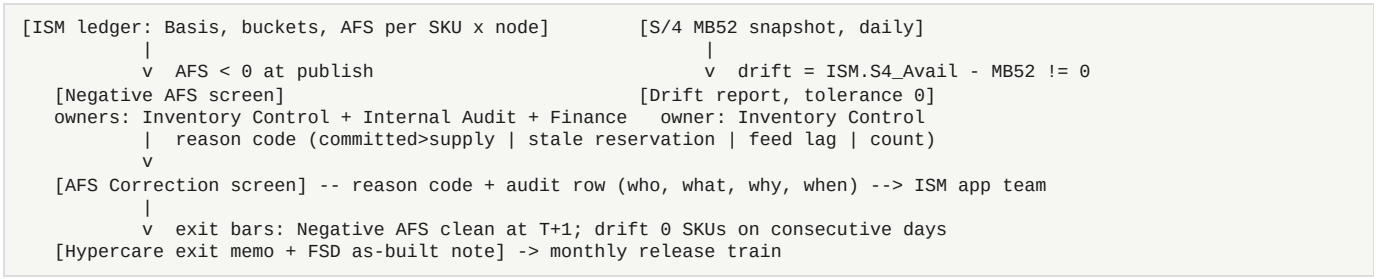
STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	AFS-changing acts with screen, role and audit row	none defined	100%	`all`	record
2	Negative AFS at T+1	unmonitored	0 rows open	`0`	record exit bar
3	Daily drift	unmeasured	0 SKUs on consecutive days	`0`	record
4	Reconciliation people	≈4 FTE	≈1 FTE	`4 − 1 = 3 FTE`	modeled on the record mechanism
5	Daily workload on one reader	—	6.5 h of a 7.2 h day	`7.2 − 6.5 = 0.7 h slack`	illustrative

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Workload sum	`4.0 + 2.5 = 6.5`	rows above (illustrative)	6.5 h/day
2	One person's productive day	`1,800 h ÷ 250 days = 7.2`	1,800 productive h a year (illustrative)	7.2 h/day
3	Fits one FTE?	`7.2 − 6.5 = 0.7`	slack > 0	yes, with 0.7 h slack
4	Reconciliation labor released	`4 − 1 = 3; 3 × $110,000 = $330,000`	S1 modeled row; $110K loaded (benchmark)	≈$0.33M/yr (inside S1)

The lower-of-two shows −20 where the S/4 book shows +20 — that gap is the whole reason the screen exists. The workload check tests whether "about one FTE" is plausible; it does not replace a time study.

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY / IMPACT — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Negative rows a day × minutes each	20 × 6 = 2.0 h	40 × 6 = 4.0 h	80 × 6 = 8.0 h
Drift rows a day × minutes each	5 × 15 = 1.25 h	10 × 15 = 2.5 h	20 × 15 = 5.0 h
Total daily workload	3.25 h	6.5 h	13.0 h
FTE needed at 7.2 h a day	0.45	0.90	1.81
Reconciliation $ if 3 FTE released × loaded	2 × $95K = $0.19M	3 × $110K = $0.33M	4 × $130K = $0.52M

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, there is no new dollar here; the dollar is S1's reconciliation row, about three hundred thirty thousand, modeled. What I can say flat is the record: Negative AFS clean at T+1, drift at zero SKUs on consecutive days, and every AFS-changing act with a screen, a role and an audit row when I stepped back. And I can show you the check that "about one person" is plausible — six and a half hours of a seven-point-two hour day, on illustrative volumes.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — Show me a negative AFS and how it clears.

S/4 500, WMS 460, committed 480: Basis = min(500, 460) = 460, AFS = 460 − 480 = −20. A reader on raw S/4 would see +20. The screen shows the row; Inventory Control finds 25 stale reservations, a correction releases them — PickRes 200 → 175 — and AFS = 460 − 175 − 150 − 80 − 50 = +5, with an audit row.

Numeric — Why not clamp negatives to zero?

Because clamping hides the cause: −20 means 20 units are over-committed, and at about $30 an incident that is $600 on one SKU-day that nobody would see. A negative is the lower-of-two doing its job.

Numeric — Is “about one FTE” real?

Plausibility check, not a time study: at an illustrative 40 negative rows at 6 minutes and 10 drift rows at 15 minutes, 4.0 + 2.5 = 6.5 hours against a 7.2-hour productive day. At double the volume it needs 1.8 people. I would run a two-week time study before committing the headcount.

Numeric — What proves the controls work?

Three counted rows: open negative rows at T+1 = 0, drift SKUs = 0 on consecutive days, and AFS-changing acts without an audit row = 0. Anything else is a feeling.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND MOMENT NEW

OWNERSHIP **1. Tell me about a time you built something that outlasted your involvement.**

Open (first two sentences): On the day I handed off, I wanted to be able to leave without a single AFS-changing act that only I understood.

A. I listed every way AFS could change, gave each a screen, a role and an audit row, and put the owners in the signed spec — Inventory Control for negatives and drift, Internal Audit and Finance as reviewers, the ISM app team for corrections. I did not make myself an owner of any of them. **R.** The drift report runs daily without me and the design entered the monthly release train.

Land the number: every act: screen, role, audit row | **Pivot to NACF:** At NACF I would design inbound controls so the building runs them: one screen for implausible available-to-promise, one for drift versus the dock count.
Do NOT say: “I ran the screens.”

EARN TRUST **2. Tell me about a time you built trust with a control function like audit or finance.**

Open (first two sentences): Internal Audit and Finance are usually reviewers at the end; I put them in the design at the start.

A. The Negative AFS screen has Inventory Control, Internal Audit and Finance as owners, so the people who would test the design own the thing that catches its errors. A correction carries a reason code and an audit row. **R.** No control owner asked for a parallel report.

Land the number: three control owners on one screen | **Pivot to NACF:** At NACF: the finance and ops partners both read the inbound exception list.
Do NOT say: Claiming audit findings or opinions — none in the record.

INSIST ON THE HIGHEST STANDARDS **3. Tell me about a time you wouldn’t accept a quick fix.**

Open (first two sentences): The quick fix to a negative AFS was to clamp it to zero. I would not accept it.

A. A negative is a lower-of-two result; clamping hides over-commitment. I required the row on a screen with a reason code and an owner, and made “clean at T+1” the exit bar. **R.** Negative AFS clean at T+1 after each go-live.

Land the number: −20 not 0 | **Pivot to NACF:** In inbound: never clamp a negative promise; show it.
Do NOT say: Counting incidents prevented — none measured.

DELIVER RESULTS **4. Tell me about a time you delivered a result on a hard exit criterion.**

Open (first two sentences): Hypercare exit was defined before go-live as numbers, not a date.

A. Exit on drift = 0 SKUs on consecutive days and blockers per day under threshold; each go-live had its controls assigned and rollback triggers armed — sync under 99% for four hours, or order-flow p95 above twice budget for one hour; neither fired. **R.** Exited into the monthly release train.

Land the number: drift 0 on consecutive days | **Pivot to NACF:** The same: an inbound launch exits on a door-to-stock time and a count tie, not a week.
Do NOT say: A hypercare duration — not in the record.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

“Same SKU, two readers: one sees plus twenty, the other minus twenty.”

+20 vs −20

LP: Dive Deep

S/4 500, WMS 460, committed 480: lower-of-two −20; raw S/4 +20.

Clamping to zero would hide 20 units; $600 at $30 an incident.

The screen is the lower-of-two doing its job.

CONFLICT / INFLUENCE / BACKBONE

“Everyone wanted me to be the owner of the screens, and I said no.”

4 owning roles

LP: Ownership · Have Backbone

A control owned by the author does not survive the author.

I named Inventory Control, Internal Audit, Finance and the ISM app team in the spec.

Handoff on counted rows: T+1 clean, drift 0.

FAILURE / LEARNING

“The honest gap: the record says “a screen, a role and an audit row” and does not say who approves a correction.”

maker-checker?

LP: Insist on the Highest Standards

If one person can propose and approve a correction, the audit row is a record, not a control.

Whether the screens separated the two is VERIFY-WITH-RAJESH.

What I would do now: require a second role for corrections above a unit threshold.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — “How do you know the controls caused the result and not the other programs?”

I cannot isolate it statistically and I would not claim to. What I defend is the mechanism: every AFS-changing act traceable, negatives visible, drift measured. The counterfactual is the status quo — a ledger where the same exceptions would be fixed by hand and traced by nobody.

Say exactly: “I can’t separate it from the rest of the program, and I won’t pretend to; I can show what each control would have caught.”

Trap — “What would you cut if Internal Audit had no time?”

I would cut their seat on the Negative AFS screen to a monthly sample of corrections, not their review of the reason-code list. I would never cut the audit row on a correction, the T+1 bar, or the drift report.

Say exactly: “Audit sampling can be thinner; the audit row, T+1 and drift cannot.”

Trap — “What is the dependency risk — who must act for this to work?”

Inventory Control must work the lists daily; the ISM team must keep corrections available; Finance must read the value impact. If Inventory Control’s capacity is the constraint — 6.5 of 7.2 hours on one person at base volume — the first thing I cut is non-exception drift cases, never negatives.

Say exactly: “The binding dependency is one team’s daily capacity; I protect negatives first.”

Where this lands at Amazon / NACF. Amazon’s equivalent is the exception review on promise data: a negative or implausible available-to-promise on a screen with an owner, a reason-coded override with an audit trail, and a daily system-versus-physical drift check.

S1-c — Sephora S/4HANA ↔ ISM — the priced no: Merchandising wanted merchandise reserve promisable; I priced it, wrote the no into the FSD as a row, and made the reserve copy provably equal to the plan (reserve-copy invariants and the old-versus-new AFS reconciliation) NEW

LPs this story carries (with proof lines below): Have Backbone, Disagree and Commit (primary) · Customer Obsession (unpickable stock sold is a broken promise) · Are Right, A Lot (priced both sides; named what would change my mind) · Earn Trust (visibility offered; Merchandising not caricatured)
Process family: Plan / Deliver seam — reserve policy and cutover: merchandise reserve, channel reserves, publish switch, old-versus-new reconciliation

Honesty label. The ask, the pricing, the no-as-a-row and the reserve-copy invariants were mine; FRAME planning, the ISM app team and OMS configurators built to the signed FSD. The record says the no was priced as "unpickable stock sold" and written as an FSD row and that merch reserve got visibility; it does not carry the units, the sell-through or the number of reserve holds below, which are illustrative and flagged VERIFY-WITH-RAJESH. The four invariants are my statement of the "reserve-copy invariants" the record names; their exact wording is VERIFY-WITH-RAJESH.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Have Backbone, Disagree and Commit — I said no to Merchandising's promisable-reserve ask in the room and in writing — priced, with the alternative on the table — and I was ready to commit if the evidence changed.

Customer Obsession — Held reserve is stock nobody can pick; selling it is a cancellation waiting to happen, and the customer is the one who finds out.

Are Right, A Lot — I priced both sides — the margin earned if the hold releases in time, the incident if it does not — and wrote the break-even that would flip my answer.

Earn Trust — Merchandising's underlying need was visibility of the reserve; I met it with its own column in FOH and the screens, so the no did not leave them blind.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Merchandising holds stock back — launches, allocations, channel reserves — and that stock sits in the building but cannot be picked until a hold is released. Merchandising wanted it counted as sellable, so a product could take orders before the hold lifted. In the council the room was leaning toward yes. **Task.** Decide, and make the reserve data trustworthy enough that either answer was safe at cutover. **Action.** I priced the ask: every promisable unit that is unpickable becomes an oversell incident if the hold does not release in time. I proposed instead that merch reserve is a visible bucket — its own column in the Financial On-Hand and on the control screens — but outside available-for-sale; the no went into the specification as a row with the pricing, not a meeting minute. For the January planning go-live I wrote the reserve-copy invariants — the reserve in ISM equals the reserve in the plan at every nightly cut; reserve never sits inside the S/4 available figure; it never enters AFS as supply; there is one published AFS and the order system reads it — and a reconciliation of old versus new AFS by SKU, node and channel where every difference has a reason code. **Result.** Merchandising kept full visibility; unpickable stock was never sold; January and May go-lives with oversell incidents recurring → zero; the no survived because it lived in the FSD.

20-SECOND VERSION NEW

Merchandising wanted held-back stock to count as sellable. I priced it — held stock is unpickable, so every unit promised is an oversell incident if the hold doesn't lift — wrote the no into the FSD as a row, and gave them what they actually needed: reserve visible in its own column, outside available-for-sale. I then wrote four invariants so the reserve copy provably matched the plan at the January go-live.

NACF BRIDGE NEW

At NACF the equivalent is inbound stock that is physically present but not promisable — held for QA, cross-dock allocation or a launch. It belongs in a visible column, outside inbound available-to-promise, until the hold lifts; promising it is promising what the floor cannot yet release.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Merchandise reserve — stock held back for launches, allocations and channel plans — is in the building but unpickable. It sits outside S/4 available and is fed from the planning system's channel reserves nightly plus intraday holds. Merchandising asked for it to be promisable; the council leaned that way.

T — Decide the policy with evidence, and make the reserve data safe to go live on: the copy of reserves in ISM had to equal the plan, old and new AFS had to reconcile, and the order system had to read one number.

Baselines on the table: promisable reserve — requested (target: visible, not promisable) · reserve copy vs plan — untested (target 0 differences at each nightly cut) · old-vs-new AFS differences without a reason code — unknown (target 0 at go/no-go) · SKUs with two published AFS values — possible (target 0).

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** The channel-reserve and hold feed from planning (nightly plus intraday holds); the reserve column in the ISM ledger; S/4 available (which excludes merch reserve); order history for the demand that would see a promisable reserve.
- **Grain.** SKU × node × channel × bucket. The reconciliation compared AFS under the legacy definition and under the new one for every SKU × node × channel row.
- **What it revealed.** Counting reserve as sellable adds exactly the held units to AFS (illustratively 560 → 710 on the worked SKU), which is the size of the over-promise if the hold is not lifted before demand arrives. The size of that error, not the principle, carried the decision.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

Worked SKU: Basis 980, committed 420, AFS 560; merch reserve 150 held for a launch (illustrative). If demand takes the whole promisable amount before release, "over-promise" is the units sold that cannot be picked.

OPTION	AFS SHOWN	UNITS SOLD ABOVE PICKABLE	VERDICT
Promisable reserve (the ask)	560 + 150 = 710	up to 150	Rejected — unpickable stock sold
Promisable if the hold releases within a window	560 + 150 while the window is open	up to 150 if release slips	Rejected — the release is Merchandising's decision, not a supply fact
Invisible reserve (folded into S/4 or hidden)	560	0	Rejected — Merchandising blind; Finance cannot read FOH
Reserve inside the S/4 figure and subtracted again	560 − 150 = 410	0 but under-states by 150	Rejected — a double subtract
own bucket: visible in FOH and screens, outside AFS	**560**	**0**	Chosen — visibility without promisability

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS,

RISKS)

- **Discovery** — who sets reserves (Merchandising through planning), how often (nightly plus intraday holds), where they sit relative to S/4 available (outside). *Exit bar:* the reserve data path mapped to one owner per step.
- **Plan & Design** — the pricing of the ask; the policy (visible, not promisable) written as an FSD row; reserve-copy invariants; the old-versus-new AFS reconciliation by SKU × node × channel; council ask framed as three decisions. *Exit bar:* FSD signed 6/5/25 with the no as a row.
- **Build & Test** — node × channel × bucket-transition matrix cells for reserve set, hold, release and expire; the invariant checks as assertions. *Exit bar:* every reserve transition has a matrix cell; E2E green 11/22 · 12/20 · 1/30.
- **Cutover & Go-Live** — January planning go-live — the reserve rules live against demand; the publish-switch plan: OMS reads ISM AFS, never computes; go/no-go read from counted rows of the old-versus-new reconciliation. *Exit bar:* no SKU with two published AFS values; every difference has a reason code.
- **Hypercare / Close** — daily reserve-copy equality check; visibility reports to Merchandising; drift report. *Exit bar:* reserve-copy differences 0 on consecutive days.

Influence and ownership. Mine: the pricing, the policy row, the four invariants, the reconciliation design and the council framing. Built by: the planning-system and ISM teams. Decision-maker for any future change: the three FSD signatories, through the regression matrix and the release train.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (BASIS-LABELED)

- **Customer** — Unpickable stock was never sold; oversell incidents recurring → zero through two go-lives (record).
- # — One policy row in the FSD; four invariants; zero SKUs with two published AFS values; reserve visible as its own column (record design / record).
- **Time** — FSD signed 6/5/25; January planning go-live with reserve rules live against demand; May execution go-live (record).
- **Quality** — Old-versus-new AFS difference rows unexplained at go/no-go: 0 (record exit bar); reserve copy equals the plan at the nightly cut.
- **$** — No new dollars. Illustrative per-hold price of the no: 150 × 60% × $30 = $2,700 VERIFY-WITH-RAJESH — a worked example, not a savings claim, never added to S1's ≈$2.5M.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Whether reserve counts as available was undefined; I defined it with a price.
Scope & influence	Merchandising, Planning, Finance, the SAP and ISM teams; the council leaned toward yes and the FSD said no.
Execution trade-off	Visibility without promisability: some potential early sales are forgone to make oversell zero by construction.
Impact	Oversell recurring → zero through two go-lives; MCS retired.
Process improvement	Reserve transitions in the matrix; invariants as assertions; old-versus-new reconciliation as the go/no-go evidence.
Complexity	Nightly feed plus intraday holds, an outside-S/4 bucket, three node types and channel reserves.
Root cause	The root cause was a definition that mixed "in the building" with "pickable"; fixed once.

PROBING — FACET 3: MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	AFS without reserve	`AFS = 980 − 200 − 120 − 60 − 40 = 560`	Basis min(1,000, 980) = 980; committed 420 (illustrative, S1 worked SKU)	560

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	AFS if reserve were promisable	`560 + 150 = 710`	merch reserve 150 held for a launch (illustrative) VERIFY-WITH-RAJESH	710
3	Over-promise if the hold slips	`710 − 560 = 150`	units shown that cannot be picked	150 units
4	Incidents if 60% sell before release	`150 × 60% = 90`	60% sell-through before release (illustrative) VERIFY-WITH-RAJESH	90 incidents
5	Cost of those incidents	`90 × $30 = $2,700`	$30 = $6 + $4 + $19 + $1 (S1 benchmark build)	$2,700 per hold
6	Double-subtract trap	`560 − 150 = 410`	reserve already outside S4_Avail, subtracted again (record design says no double-subtract path)	under-states by 150

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Policy	`MerchRes sits outside S4_Avail; AFS = Basis − PickRes − EcomPick − EcomPack − SNB`	record design	reserve never enters AFS
2	Invariant 1 — copy equals plan	`Σ reserve (ISM) − Σ reserve (plan) = 0 per SKU × node at each nightly cut`	copy equality, tolerance 0 VERIFY-WITH-RAJESH	0 differences
3	Invariant 2 — never inside S/4 available	`AFS = 980 − 420 = 560, not 560 − 150`	no double-subtract	560
4	Invariant 3 — never promisable	`AFS(with reserve held) − AFS(reserve = 0) = 0`	changing MerchRes does not move AFS	0
5	Invariant 4 — one publisher	`published AFS values per SKU = 1`	OMS reads ISM AFS, never computes	1
6	Release	`reserve 150 → 0; stock moves to the sellable pool and AFS rises by the released units after the S/4 side posts`	visible in the FOH column until then	visible, then promisable
7	Old-vs-new reconciliation	`20,000 − 18,600 = 1,400 rows differ; explained 1,400; unexplained = 1,400 − 1,400 = 0`	illustrative counted rows VERIFY-WITH-RAJESH; go/no-go bar: 0 unexplained	0 unexplained → go
8	Break-even that would flip the no	`p × $19 = (1 − p) × $30 → p = 30 ÷ 49 = 0.612`	$19 margin if released in time; $30 incident if not (benchmarks)	p = 61%

Table 3 — after: the metric, before → after.

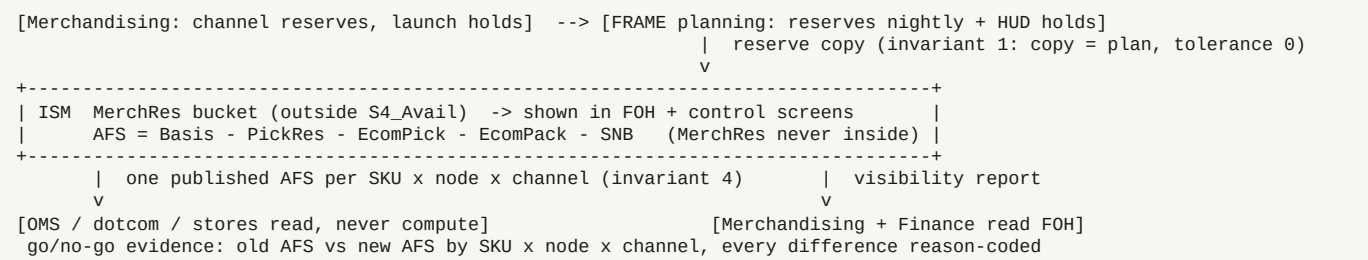
STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Reserve promisable	requested	visible, not promisable	`policy`	record
2	Over-promise exposure on the worked hold	150 units	0 units	`150 − 0 = 150`	illustrative
3	Unexplained old-vs-new differences at go/no-go	unknown	0	`0`	record exit bar; counts illustrative
4	SKUs with two published AFS values	possible	0	`0`	record
5	Oversell incidents	recurring	zero through two go-lives	`recurring − 0`	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Per-hold price of the no	`150 × 60% = 90; 90 × $30 = $2,700`	illustrative units, sell-through and benchmark incident cost	$2,700 per hold
2	If 200 holds a year	`200 × $2,700 = $540,000`	200 holds a year VERIFY-WITH-RAJESH — a bound on exposure, not a savings claim	$0.54M exposure bound
3	Margin forgone by not promising (if all release in time)	`150 × $19 = $2,850`	$19 margin per unit (benchmark)	$2,850 per hold
4	Break-even reliability	`p × $19 = (1 − p) × $30 → p = 30 ÷ 49 = 0.612`	$19 margin if released in time; $30 incident if not (benchmarks); promising wins only above this reliability	p = 61% — unmeasured, so no

Merch reserve counts and sell-through are illustrative; the break-even 30 ÷ (19 + 30) = 61% reuses S1's benchmark inputs. The $0.54M is an exposure bound for a hypothetical number of holds, not a result.

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY / IMPACT — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Sell-through before release	30%	60%	90%
Incidents on a 150-unit hold	45	90	135
Cost per hold at $30	$1,350	$2,700	$4,050
Break-even release reliability	30 ÷ (19 + 30) = 61%	same	same
Holds a year (exposure bound)	100 → $0.27M	200 → $0.54M	400 → $1.08M

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, there is no new dollar in this story and I will not manufacture one. What I will say is the decision rule: promising a held unit pays about nineteen dollars if the hold releases in time and costs about thirty if it does not, so it only wins if release is reliable more than sixty-one percent of the time — and nobody measured that, so the answer was no, in writing, with visibility instead. The record is that oversell incidents went from recurring to zero through two go-lives.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — What did the ask do to AFS?

On the worked SKU AFS is 560. Counting a 150-unit reserve as sellable shows 710 — an over-promise of 150 if the hold slips. Subtracting it again when it is already outside S/4 would show 410 and under-state by 150; neither is right, which is why reserve is its own bucket outside AFS.

Numeric — What number would have changed your mind?

Release reliability. Promising earns about $19 a unit if the hold lifts in time and costs about $30 an incident if it does not: p × 19 = (1 − p) × 30, so p = 30 ÷ 49 = 61%. If Merchandising could show holds lifting on time more than about six times in ten, promisable would pay; nobody had that number.

Numeric — How did you know the reserve copy matched the plan?

Invariant one: Σ reserve in ISM minus Σ reserve in the plan is zero per SKU and node at every nightly cut, tolerance zero. At go/no-go I read the old-versus-new AFS reconciliation as counted rows — on illustrative numbers, 20,000 rows, 1,400 different, every one with a reason code, zero unexplained.

Numeric — Are 200 holds a year real?

No — it is a bound for the question "how big could the exposure be." At 60% sell-through and $30 an incident a 150-unit hold exposes $2,700; the record has no hold count and I do not claim one.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND MOMENT NEW

HAVE BACKBONE; DISAGREE AND COMMIT **1. Tell me about a time you disagreed with a senior stakeholder.**

Open (first two sentences): Merchandising wanted held-back stock to be sellable, and the room was leaning toward giving it to them.

A. I priced the ask instead of arguing it: unpickable stock sold is a cancellation. I offered visibility — reserve in its own column in the Financial On-Hand and on the control screens — and put the no into the FSD as a row with the pricing, so it could not be re-litigated every week. I told them the number that would change my mind. **R.** Merchandising's need was met, the FSD was signed 6/5/25, and oversell went from recurring to zero through two go-lives.

Land the number: the no lives in an FSD row | **Pivot to NACF:** At NACF: when planning wants held inbound stock promised, the same price-and-visibility answer.
Do NOT say: Caricaturing Merchandising; "they didn't understand."

CUSTOMER OBSESSION **2. Tell me about a time you protected the customer against an internal request.**

Open (first two sentences): The customer cost of selling held stock is a cancellation email, which is why I kept coming back to the unit nobody can pick.

A. I turned "reserve counts" into units: 560 shown with the reserve out, 710 with it in; 150 units the customer can be told yes to and cannot receive. **R.** Unpickable stock was never sold.

Land the number: 560 vs 710 | **Pivot to NACF:** The delivery date a customer is shown must be one the building can keep.
Do NOT say: Claiming a CSAT or NPS change.

ARE RIGHT, A LOT **3. Tell me about a decision you made with incomplete data.**

Open (first two sentences): I did not have the number that settles whether reserve should be promisable — release reliability — so I priced both sides.

A. $19 of margin if the hold lifts in time, $30 of incident if not: the break-even is 61%. Nobody had the reliability data, so the default went to the side with the customer cost; I wrote the break-even into the row so the decision could be reopened with evidence. **R.** The decision held through both go-lives.

Land the number: 61% break-even | **Pivot to NACF:** Make the number that would change your mind part of the decision.
Do NOT say: Presenting 61% as a measured release rate.

EARN TRUST **4. Tell me about a time you said no and kept the relationship.**

Open (first two sentences): Merchandising's need was real — they needed to see their reserve — so the no had to come with a yes.

A. The reserve got its own column in FOH and on the screens and a visibility report; Merchandising owns the holds, I owned the policy. **R.** The no did not become a standing dispute.

Land the number: visible, not promisable | **Pivot to NACF:** At NACF: the planner's hold is visible in the same view as the inbound plan.
Do NOT say: Naming individuals.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

"Same SKU, two answers: 560 or 710."

560 vs 710

LP: Dive Deep

AFS = 980 − 420 = 560 with the reserve out; 710 with a 150-unit hold in.

The difference is the over-promise if the hold slips.

Break-even release reliability 30 ÷ (19 + 30) = 61%.

CONFLICT / INFLUENCE / BACKBONE

"The room was leaning toward yes."

1 FSD row

LP: Have Backbone

I priced the ask and offered visibility.

The no went into the FSD as a row, not a minute.

I named the number that would reopen it.

FAILURE / LEARNING

"The no I wrote had no review trigger."

p = 61%

LP: Are Right, A Lot

The record does not say the row carried a review condition — VERIFY-WITH-RAJESH.

What I would do differently: write the break-even and a review date into the row itself.

A no with no "what would change my mind" is just a veto.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — "You said no to your own business partner. What did Merchandising lose?"

The ability to take orders before a hold released — a potential early-sales margin of up to 150 × $19 = $2,850 on the worked hold if everything released in time. I did not hide it; I compared it with a 90-incident exposure of $2,700 at 60% sell-through and said the decision flips if reliability exceeds 61%.

Say exactly: "They lose early orders on held stock — about two thousand eight hundred fifty dollars on the worked hold if the release is on time; the decision flips above sixty-one percent reliability."

Trap — "What would you cut if cutover capacity halved?"

The invariants stay; what I cut is breadth of the old-versus-new reconciliation — start with the highest-volume nodes and channels, and defer long-tail rows to the day-after check. I would never cut invariant 1 (copy equals plan) or the one-publisher rule.

Say exactly: "I would thin the reconciliation scope, never the reserve-copy check or the single publisher."

Trap — "How do you know the reserve copy was right at go-live?"

Counted rows, not confidence: Σ reserve difference at the nightly cut is zero per SKU and node, and every old-versus-new difference row has a reason code. The go/no-go is a recommendation read from those counts, taken by the forum.

Say exactly: "Two counted rows: reserve-copy difference zero and unexplained differences zero — read, not argued."

Where this lands at Amazon / NACF. Amazon's version is the buy-box or detail-page promise on stock that is held, quarantined or pre-allocated: it should be visible to planners and excluded from the promise until released.

S1-d — Sephora S/4HANA ↔ ISM — go/no-go from counted rows: two go-lives, an 836-row issues log read by severity, and two rollback triggers armed in numbers and never fired NEW

LPs this story carries (with proof lines below): Deliver Results (primary — dated gates held through two go-lives) · Bias for Action (reversible decisions: rollback written as numbers before go-live) · Insist on the Highest Standards (a phase graduates on counted rows, never on confidence) · Think Big (book-of-record migration across ~15 functions)

Process family: Plan / Deliver seam — release and cutover governance: go/no-go design, rollback triggers, reconciliation evidence, staged go-live

Honesty label. I wrote the go/no-go recommendation from counted rows, the readiness checklist, the rollback triggers and the publish-switch plan; the forum took the decision and the SAP and ISM teams executed the cutover. The dated gate chain, the 836-row issues log and the rollback triggers are program-level artifacts shared with S2 and S5 — one umbrella program, so if two of these stories go to one loop I say so. Row counts below are record (836, 85.4%, 164, 277) or derived from them; anything else is illustrative and flagged VERIFY-WITH-RAJESH.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Deliver Results — FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; E2E Inbound 11/22 · Dotcom 12/20 · Retail 1/30; January planning and May execution go-lives — the dated chain held and oversell stayed at zero.

Bias for Action — Rollback was written as numbers — sync under 99% for four hours, or order-flow p95 above twice budget for one hour — and armed for both go-lives, so deciding to go was reversible.

Insist on the Highest Standards — A phase graduated on counted rows, never confidence: Blocker + Critical open = 0 at go/no-go, with an overall-percent-closed figure explicitly not accepted as a substitute.

Think Big — A book-of-record migration across about fifteen functions and sixty people, staged as two go-lives so each carried one defect class.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Moving the inventory book of record from MCS to S/4 with a real-time layer beside it meant every promise on three node types had to keep holding through the switch; the program log held 836 issues, and more than half — 441 — were Blocker or Critical at some point. **Task.** Own the go/no-go recommendation and the cutover design for the inventory supply integration, with no authority over the teams that would execute. **Action.** I wrote the go/no-go as a checklist read from counted rows: Blocker plus Critical open equal to zero; reconciliation clean for a stated run of days; monitors live; rollback rehearsed; control owners named; comms drafted. I split the cutover into two go-lives — January for planning, with the reserve rules live against demand, May for execution — so each carried one defect class. And I wrote rollback as two numbers before anyone asked: ISM-to-S/4 sync under 99% for four hours, or order-flow p95 above twice its budget for one hour. Both go-lives armed both triggers; neither fired. **Result.** The dated chain held — E2E Inbound 11/22, Dotcom 12/20, Retail 1/30 — oversell incidents recurring across three node types went to zero through both go-lives, and MCS was retired as book of record.

20-SECOND VERSION NEW

I turned an 836-row issues log into a go/no-go: Blocker plus Critical open must be zero, reconciliation clean for a stated number of days, rollback rehearsed. I split cutover into two go-lives and wrote rollback as numbers — sync under 99% for four hours, or order-flow p95 over twice budget for an hour — armed both times, never fired; oversell stayed at zero.

NACF BRIDGE NEW

A new-FC or lane launch needs exactly this: a go/no-go from counted rows (door-to-stock time inside budget, dock-count-to-system tie clean N days) and a rollback written as a number before the first trailer arrives — such as redirecting inbound to the prior building if stow lag stays above twice its budget for an hour.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The cutover touched every promise on Store, Retail DC and Dotcom DC. The log held 836 issues; Blocker 164 plus Critical 277 is 441, or 52.8% of the log, so “percent closed” alone could not be trusted to describe readiness. Production changes were frozen October through December.

T — Write the go/no-go recommendation, the publish-switch plan and the rollback triggers for the inventory supply integration, and carry the program through two go-lives with zero customer-impacting incidents.

Baselines on the table: issues log 836 rows, 85.4% closed · Blocker + Critical 441 rows (52.8%) · oversell incidents recurring (target zero) · MCS as book of record (target retired) · E2E chain dated 11/22, 12/20, 1/30 (28 then 41 days apart).

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** The issues log by severity and status; E2E results by matrix cell; reconciliation results by day; monitor coverage; rollback rehearsal evidence; control-owner assignments.
- **Grain.** One row per issue, cut by severity and status; one row per matrix cell for test evidence.
- **What it revealed.** “85.4% closed” is true and uninformative: 836 × 85.4% ≈ 714 closed and 122 open — but if those 122 were all severe, 122 ÷ 441 = 27.7% of the severe rows would still be open at the same overall figure. The overall percentage hides the severity mix, the same way an aggregate accuracy hides the class that is off. So the gate had to be a count at the severity grain.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

Four rejected ways to decide “go,” and the one I chose, tested on the same log (836 rows, 441 severe, 122 open — illustrative worst case: all 122 severe).

OPTION	RULE	WHAT IT WOULD HAVE SAID	VERDICT
Calendar	go on the date	go — regardless of 122 open	Rejected — the date is a plan, not evidence
Executive confidence	go when leaders feel ready	depends on the room	Rejected — confidence is not a count
Overall % closed ≥ 85%	go at 85.4%	go — even if all 122 open rows were severe	Rejected — hides the severity mix
Single reconciliation snapshot	go if drift = 0 today	go — one clean day proves little	Rejected — N days running, not one
severity-grain counted rows plus the readiness checklist	Blocker + Critical open = 0; reconciliation clean N days; monitors live; rollback rehearsed; owners named; comms drafted	**no-go until each row is a count**	Chosen — a recommendation read from the checklist; the forum decides

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS, RISKS)

- **Discovery** — the program's dependency map and the list of defect classes by node and bucket; sessions closed 4/30. *Exit bar:* every bucket owned; open questions with owner and date.
- **Plan & Design** — FSD signed 6/5/25; the go/no-go checklist and thresholds written before testing; rollback triggers as numbers; the two-go-live split (January planning with reserve rules live, May execution). *Exit bar:* thresholds signed by the same three orgs.
- **Build & Test** — E2E Inbound 11/22, Dotcom 12/20, Retail 1/30 — 28 then 41 days apart — each green before the next; the matrix assertions. *Exit bar:* Blocker + Critical open = 0 at go/no-go; 0 open contract defects in SIT.
- **Cutover & Go-Live** — go/no-go recommendation from counted rows; the publish-switch plan — OMS reads ISM AFS and never computes; reserve-copy invariants; control-screen roles assigned; both triggers armed. *Exit bar:* January go-live and May go-live; no SKU with two published AFS values; Negative AFS clean at T+1.
- **Hypercare / Close** — daily drift; blockers per day against a threshold; triggers monitored; hypercare exit memo and FSD as-built note. *Exit bar:* drift = 0 SKUs on consecutive days; exit into the monthly release train.

Influence and ownership. Mine: the checklist, the thresholds, the recommendation, the trigger numbers and the publish-switch plan. Decision-makers: the go/no-go forum. Executors: SAP and ISM teams, WMS and OMS owners. The recommendation was read from counts, so the forum's decision could be made on evidence rather than on my confidence.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (BASIS-LABELED)

- **Customer** — Customer-facing oversell incidents recurring across three node types → zero through two go-lives (record).
- **Time** — FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; E2E 11/22 · 12/20 · 1/30; January planning go-live; May execution go-live (record).
- **#** — Rollback triggers armed at both go-lives, never fired (record); Blocker + Critical open = 0 at go/no-go (record exit bar); 836-row log, 85.4% closed (record, shared with S2/S5).
- **Quality** — No SKU with two published AFS values; drift = 0 SKUs on consecutive days at hypercare exit; MCS retired as book of record (record).
- **$** — No new dollars. The cost of a one-month slip in retiring MCS is the user anchor over twelve months: $1.0M ÷ 12 = $83,333 a month — modeled, inside S1's anchor, and VERIFY-WITH-RAJESH on when MCS run cost actually stops.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	What "ready" means was not defined; I defined it as a checklist of counts.
Scope & influence	About fifteen functions and sixty people; a forum took the decision on my recommendation.
Execution trade-off	Two go-lives (more cutover work) for isolating defect classes; a higher bar than overall percent closed.
Impact	Oversell recurring → zero; MCS retired.
Process improvement	The checklist and the "graduate on counted rows" rule were standard work for later releases.
Complexity	A production freeze October–December, dated E2E days, and three node types.
Root cause	The risk was deciding on an aggregate; fixed by deciding at severity grain.

PROBING — FACET 3: MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Issues log	`836 rows; 836 × 0.854 = 714`	record: 836 rows, 85.4% closed	≈714 closed; 836 − 714 = 122 open

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Severe rows	`164 + 277 = 441; 441 ÷ 836 = 0.527`	record: Blocker 164, Critical 277	441 rows = 52.8% of the log
3	Non-severe rows	`836 − 441 = 395`	derived	395 rows
4	What 85.4% could hide	`122 ÷ 441 = 0.277`	worst case: all 122 open rows severe (illustrative worst case)	27.7% of severe rows open at 85.4% overall
5	E2E spacing	`Inbound 11/22 → Dotcom 12/20 = 28 days; Dotcom 12/20 → Retail 1/30 = 41 days`	record dates	28 and 41 days
6	Rollback — sync trigger	`99% of 12,000 events/h = 11,880; 12,000 − 11,880 = 120`	illustrative 12,000 events an hour VERIFY-WITH-RAJESH	trigger at more than 120 unapplied per hour for 4 h

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Gate 1 — severity	`Blocker + Critical open = 441 − 441 = 0`	all severe rows closed at go/no-go (record exit bar)	0 severe open
2	Gate 1 — implication	`open rows are non-severe: 395 − (714 − 441) = 122`	derived; snapshot of the final log VERIFY-WITH-RAJESH	122 open, all below Critical
3	Gate 2 — reconciliation	`drift rows = 0 on N consecutive days`	N stated before testing VERIFY-WITH-RAJESH	0 rows × N days
4	Gate 3 — checklist	`monitors live + rollback rehearsed + owners named + comms drafted = 4 of 4`	launch readiness checklist (record design)	4 of 4
5	Trigger A — sync	`trigger if sync < 99% for 4 h; 4 × 120 = 480 unapplied`	at the illustrative 12,000 events an hour	480 unapplied events
6	Trigger B — order flow	`trigger if p95 > 2 × budget for 1 h; 2 × 60 = 120 s`	budget = the 60-second publish budget (my reading) VERIFY-WITH-RAJESH	p95 > 120 s for 1 h
7	Two go-lives	`January: reserve rules live against demand; May: execution`	one defect class each	2 go-lives
8	Result check	`triggers fired = 0 of 2 per go-live`	armed both, never fired (record)	0 firings

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Go/no-go basis	confidence / % closed	counted rows by severity	`changed`	record design
2	Blocker + Critical open at go/no-go	441 rows in the log	0	`441 − 441 = 0`	record
3	Rollback triggers	none written	2 armed	`2`	record
4	Go-lives	one big switch	2 staged	`2 − 1 = 1`	record
5	Oversell incidents	recurring	zero through two go-lives	`recurring − 0`	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Monthly MCS run cost	`$1,000,000 ÷ 12 = $83,333`	user anchor ÷ 12	≈$0.08M a month

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Cost of a one-month slip	`1 × $83,333 = $83,333`	assumes run cost stops at cutover VERIFY-WITH-RAJESH	≈$0.08M
3	Cost of a three-month slip	`3 × $83,333 = $250,000`	same assumption	≈$0.25M
4	Check against the anchor	`$83,333 × 12 = $999,996`	rounding	≈$1.0M/yr

Only the log counts (836, 85.4%, 164, 277, 52.8%) and dates are record; 122 open and the worst-case 27.7% are derived arithmetic; the event volume and N days are illustrative. The slip cost is the $1M user anchor spread over twelve months — not a result.

SOLUTION FLOW — THE SYSTEM & DATA MAP

```
[Issues log 836 rows] -> by severity: B 164 + C 277 = 441 (52.8%) -> go/no-go gate 1: B + C open = 0
[E2E 11/22 Inbound -> 12/20 Dotcom -> 1/30 Retail]  -> gate 2: reconciliation clean N days
[Checklist: monitors live | rollback rehearsed | owners named | comms drafted]  -> gate 3
                                    |
                      recommendation (mine) --> forum decision
                                    |
     Jan go-live (planning, reserve rules live)  ->  May go-live (execution)
                                    |
armed during each: sync < 99% for 4 h  OR  order-flow p95 > 2x budget for 1 h  -> rollback (never fired)
                                    |
hypercare exit: drift = 0 SKUs on consecutive days; blockers/day under threshold -> release train
```

SENSITIVITY / IMPACT — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Open severe rows hidden at 85.4%	0 → 0%	60 → 13.6%	122 → 27.7%
Unapplied-events trigger per hour (at 12,000/h)	5,000/h → 50	12,000/h → 120	30,000/h → 300
Unapplied events across the 4-hour window	200	480	1,200
MCS slip (months) × $83,333	1 → $0.08M	3 → $0.25M	6 → $0.5M

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, I do not claim a dollar for the cutover; the record is outcomes — two go-lives, oversell incidents going from recurring to zero, both rollback triggers armed and never fired. The only number I will put on timing is the anchor: a million dollars a year of MCS run cost is about eighty-three thousand a month, and I would say that is modeled from the anchor and depends on when the cost actually stops.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — 85.4% of 836 is closed. Why is that not good enough?

836 × 85.4% ≈ 714 closed and 122 open. If all 122 were severe, 122 of the 441 severe rows — 27.7% — would be open at the same overall figure. So the rule is a count at severity grain: Blocker plus Critical open equals zero. At that point the 122 open rows are all below Critical — an arithmetic fact I would verify against the final log.

Numeric — How do you write a rollback trigger?

As a threshold, a duration and an owner before go-live: sync under 99% for four hours, or order-flow p95 above twice budget for one hour. At an illustrative 12,000 events an hour, 99% is 11,880, so more than 120 unapplied an hour — 480 across four hours — pulls the trigger. Both armed at both go-lives, never fired.

Numeric — Why two go-lives, not one?

So each carries one defect class. January put the reserve rules against live demand, planning-side; May moved execution. If something broke in January it was a rules problem, not a floor problem. The E2E days were spaced 28 and 41 days apart so each cycle had time to close before the next.

Numeric — What did a delay cost?

I do not claim a cost of delay; I can bound it: the $1M MCS anchor is about $83,333 a month, so a three-month slip is about $250,000 if the run cost stops at cutover — which I would verify.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND MOMENT NEW

DELIVER RESULTS **1. Tell me about a time you delivered a complex result on a fixed date.**

Open (first two sentences): The chain was dated before we started: FSD signed 6/5/25, three E2E days, two go-lives — and an October to December production freeze that removed any slack.

A. I wrote the exit bar for every phase as a count and recommended go or no-go from the checklist. I split cutover so each go-live carried one defect class and kept both triggers armed. When something slipped it slipped inside a gate, not over one. **R.** Both go-lives landed; oversell recurring → zero; MCS retired.

Land the number: zero incidents through two go-lives | **Pivot to NACF:** Inbound launches need the same chain: dated test days, a freeze, a staged opening.
Do NOT say: The Utah DC floor story or the robotics gate — Goforth's.

BIAS FOR ACTION **2. Tell me about a time you moved fast on a risky decision.**

Open (first two sentences): The decision to go was risky; what made it fast was that it was reversible.

A. I wrote the rollback before go-live as two numbers — sync under 99% for four hours, or order-flow p95 above twice budget for an hour — so the decision to go did not require certainty, only a way back. **R.** Armed at both go-lives; never fired.

Land the number: two triggers, never fired | **Pivot to NACF:** The same: launch a lane with a written redirect trigger.
Do NOT say: Saying the triggers "would have worked" — they were never tested in anger.

INSIST ON THE HIGHEST STANDARDS **3. Tell me about a time you wouldn't sign off on a number others accepted.**

Open (first two sentences): 85.4% closed sounded good, and I would not accept it.

A. I showed the same overall figure with all 122 open rows severe is 27.7% of the severe rows still open, and made Blocker plus Critical open equal zero the gate. **R.** Go/no-go read from counts; the forum decided.

Land the number: 27.7% hidden | **Pivot to NACF:** On inbound: 99.0% overall accuracy hiding 98.0% on one class.
Do NOT say: Presenting the 27.7% as what the log showed — it is a worst case.

THINK BIG **4. Tell me about the largest change you have led.**

Open (first two sentences): We moved the inventory book of record while every promise on three node types had to keep holding.

A. I broke it into five gated phases and two go-lives, with each dated gate written before the work and the whole thing frozen through peak. About fifteen functions, sixty people, none reporting to me. **R.** Zero customer-facing incidents through both go-lives.

Land the number: ~15 functions · ~60 people · 2 go-lives | **Pivot to NACF:** The scale at NACF is the number of buildings; the discipline is the same.
Do NOT say: The 80 names and a dozen core systems — UDC program figures; say them only if asked about the umbrella program.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

"The log said 85.4% closed; I asked how many of the open ones were severe."

27.7%

LP: Dive Deep

836 × 0.854 ≈ 714 closed; 122 open.

Worst case all 122 are severe: 122 ÷ 441 = 27.7%.

The gate is a count at severity grain, not a percent.

CONFLICT / INFLUENCE / BACKBONE

"The date was the most powerful number in the room."

441 → 0

LP: Have Backbone · Insist on the Highest Standards

I made the checklist a recommendation, not a decision.

The forum could still decide to go; it could not do so without reading the counts.

Whether any forum disagreed with my recommendation is VERIFY-WITH-RAJESH.

FAILURE / LEARNING

"The triggers never fired — which means I never learned whether the thresholds were right."

99% · 4 h · 2× · 1 h

LP: Are Right, A Lot

Armed and never fired is a good outcome and an unverified calibration.

If I did it again I would add a rehearsal that injects a breach to prove the trigger fires.

Whether a rehearsal was run is VERIFY-WITH-RAJESH.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — "You said go/no-go was your recommendation. What if the forum overruled you?"

Then the checklist rows are on the page: Blocker plus Critical open, reconciliation days, monitors, rollback, owners, comms. A forum can accept a risk on a row; it cannot do so unseen. The recommendation is read from the checklist; the decision is theirs.

Say exactly: "The forum owns the decision; I own making every row a count they can see."

Trap — "What do you cut if the date is fixed and the log is not clean?"

I cut scope before I cut the gate: defer a node type or a channel to the second go-live — January and May were already a staged split — and never relax Blocker plus Critical equals zero, rollback armed, or owners named.

Say exactly: "Move scope between go-lives; never relax the severity gate or the rollback."

Trap — "What is the dependency that could have broken this?"

The WMS feed and the order system had to be ready to the same contract; the S/4 side owned MB52 and posting. The single point I watched was the sync rate to S/4, because it was one of the two triggers: below 99% for four hours means ISM and the book are diverging, and that is a rollback.

Say exactly: "The dependency I watched most was the sync to S/4, because it was a rollback trigger."

Where this lands at Amazon / NACF. A staged opening is how a new building ramps: dated test days, a go/no-go from counted rows, and a rollback written as a number — the FC launch playbook in one page.

S1-e — Sephora S/4HANA migration — seventeen interfaces, two global dashboards and the $42K I first refused: refuse the unscoped ask, fund the study, reverse on evidence, and test with count-and-sum arithmetic NEW

LPs this story carries (with proof lines below): Are Right, A Lot (primary — changed position on evidence, in public) · Frugality (fund the study, not the fear) · Earn Trust (the public reversal bought credibility) · Dive Deep (three cheap checks, each owning one failure mode)
Process family: Plan / Deliver seam — cross-program arbitration and regression-test design: scope, budget, non-regression testing before UAT

Honesty label. I owned the arbitration — scope, budget and a testing strategy both programs would sign — not the global reporting layer. Facts here — 17 interfaces, two critical usages, $42K funded, $25K contingent, count-and-sum checks, six core files plus the supply-chain family — are from the loop packet's S25, a secondary source that is not in the master packet's canon ledger: VERIFY-WITH-RAJESH to confirm the figures before saying them. LOOP-CONSISTENCY FLAG: the loop packet lists S25 as a Bar Raiser backup; use it here only if it is not being spent with the Bar Raiser. I never price the avoided regression.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

Are Right, A Lot — "Potentially impacted" is not a scope, so I refused — then funded a study, and when it mapped real risk onto the two critical usages I reversed my own refusal on the record.

Frugality — I refused to fund fear — an unscoped ask — and funded exactly what the study proved: $42K for the files feeding two critical usages, with $25K held contingent and never pre-spent.

Earn Trust — The public reversal bought more credibility with the global program than the original refusal had cost; the first no shipped with a funded study so being wrong was cheap.

Dive Deep — Count, sum and per-store matching — three checks, each naming the failure it catches — instead of clicking through screens.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. During the North America S/4 migration, the global reporting program — the layer behind the dashboard worldwide leadership reads and the portal the global brands use — said our migration potentially touched seventeen of its interfaces and asked for a non-regression testing budget. **Task.** I owned the arbitration: scope, budget and a testing strategy both programs would sign. **Action.** I said no: "potentially impacted" is not a scope. But I funded a joint study to turn it into a list. It came back: most of the seventeen did not matter, but the files feeding two critical global usages did. I reversed myself on the record and funded the $42K non-regression option, scoped to those files, and held the second option — $25K to adapt mappings — strictly contingent on a mapping actually changing. The tests were arithmetic: for each file, a delta extract from the S/4 environment checked on count, sum and per-store match against legacy, and sequenced before business UAT. **Result.** $42K spent against a risk my first answer would have left naked; the two global-critical usages crossed the migration behind named gates; migration dates held; and the contingency was not pre-spent. I do not put a dollar on the regression that did not happen.

20-SECOND VERSION NEW

The global reporting program said the S/4 migration "potentially" touched seventeen of its interfaces and wanted test budget. I refused the unscoped ask but funded a study, which found two critical usages at risk; I reversed myself publicly, funded $42K scoped to those files, held $25K contingent, and tested with count, sum and per-store arithmetic before UAT.

NACF BRIDGE NEW

Any NACF change that touches a shared feed — a new inbound system, a re-slotted receiving flow — has downstream readers like planners' dashboards. Same move: do not fund fear; fund a study that converts "potentially impacted" to a list, then test the list with counts, sums and per-node matching so aggregate matches cannot hide a shifted mapping.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The NORAM S/4 migration potentially affected 17 interfaces feeding the global reporting layer, with two critical usages on top — the worldwide executive dashboard and the global brand portal. The reporting program asked for regression-test budget; the migration program — mine — had initially refused.

T — Own the arbitration: the scope, the budget and a testing strategy both programs would sign, without delaying the migration's hard dates.

Baselines on the table: candidate interfaces — 17 ("potentially impacted") · critical usages — 2 · regression-test budget — unscoped ask · options on the table — $42K non-regression (Option 1), $25K mapping-change adaptation (Option 2) · test method — undefined.

FACET 1 — DATA ANALYSIS — WHAT I PROFILED, AT WHAT GRAIN, WHAT IT REVEALED NEW

- **Data.** The interface inventory (17 candidates), the mapping from interfaces to the two critical usages, the file formats each usage consumes, and delta extracts from the S/4 environment (US and Canada data) against legacy.
- **Grain.** Interface → file → usage; within a file, store or storage-location rows. Checks ran at the store level, not the total.
- **What it revealed.** Most of the 17 did not feed anything critical; the union of the files behind the two critical usages — six core files plus the supply-chain file family — was the real scope. A regression almost never invents plausible data: it drops rows, doubles rows through a bad join, or shifts a mapping.

FACET 2 — SOLUTION DESIGN — OPTIONS I PUT ON THE TABLE, AND WHY ONE WON NEW

Budget options as presented to the two programs; the avoided regression is deliberately not priced.

OPTION	COST	COVERS	VERDICT
Fund the unscoped ask (all 17)	unscoped	"potentially" everything	Rejected — fear is not a line item
Fund nothing	$0	nothing	Rejected — leaves the two critical usages naked
Fund a joint study first	small VERIFY-WITH-RAJESH	converts "potentially" to a list	Chosen as step one
Option 1 — non-regression tests on the critical files	$42K	files behind the two critical usages	**funded on the study's evidence**
Option 2 — adapt mappings if one changes	$25K	only if a mapping actually changes	**as contingent — not pre-spent**

ACTION — FACET 4: IMPLEMENTATION IN FIVE GATED PHASES (PHASES, OWNERS, CONTROLS, RISKS)

- **Discovery** — the unscoped ask and my first answer — no; the joint study converting 17 "potentially impacted" interfaces to a named list. *Exit bar:* a file list tied to the two critical usages.

- **Plan & Design** — the reversal, on the record; Option 1 funded at $42K scoped to the MVP file list; Option 2 held at $25K contingent; test design — delta extract per interface, count-and-sum invariants per store or storage location. *Exit bar:* both programs sign the testing strategy.
- **Build & Test** — delta extract from S/4 (US and Canada) per interface; three checks per file; non-regression tests sequenced ahead of business UAT and both ahead of the migration's hard dates. *Exit bar:* every file passes count, sum and per-store checks.
- **Cutover & Go-Live** — the two critical usages cross the migration behind named gates. *Exit bar:* gates green; contingency untouched unless a mapping changed.
- **Hypercare / Close** — the contingency stays contingent; lessons carried into the testing strategy. *Exit bar:* no regression on the two critical usages.

Influence and ownership. Mine: the refusal, the study's framing, the reversal, the two budget options, the test design and the sequencing. Built and run by: the migration team and the global reporting program's owners. The global program decided what its critical usages were; I decided what I would pay for.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (BASIS-LABELED)

- **$** — $42K spent against a risk my first answer would have left naked (record); the $25K contingency never pre-spent (record); the avoided regression is deliberately unpriced.
- **Quality** — The two global-critical usages crossed the migration behind named arithmetic gates instead of hope (record).
- **Time** — Migration dates held with the global layer protected (record); tests sequenced ahead of business UAT.
- **Team** — The public reversal bought more credibility with the global program than the original refusal had cost (record).
- **Whether the $25K was ever spent** — not in the record VERIFY-WITH-RAJESH

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	"Potentially impacted" — I converted it to a named list by funding a study.
Scope & influence	Two programs, one budget, a global reporting layer I did not own.
Execution trade-off	$42K and calendar spent on tests that, if all goes perfectly, find nothing — against the program's own budget discipline.
Impact	Two critical global usages protected; dates held.
Process improvement	Count, sum, per-store checks as a reusable non-regression pattern.
Complexity	Seventeen interfaces, two dashboards, a hard migration date and two separate programs.
Root cause	The root cause of the conflict was an unscoped ask; fixed by a list.

PROBING — FACET 3: MATH & DATA LOGIC, FOUR TABLES, EVERY PLUG-IN SHOWN

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Candidate interfaces	`17`	record	17
2	Budget on the table	`$42,000 + $25,000 = $67,000`	Option 1 + Option 2 (record)	$67,000 maximum
3	Contingent share	`$25,000 ÷ $67,000 = 0.373`	Option 2 ÷ both	37%
4	Funded share at the start	`$42,000 ÷ $67,000 = 0.627`	Option 1 ÷ both	63%
5	Failure A — rows dropped	`5,000 − 4,950 = 50`	legacy 5,000 rows; S/4 delta 4,950 (illustrative)	50 rows dropped
6	Failure B — rows doubled by a bad join	`$4,000,000 ÷ $2,000,000 = 2.0`	S/4 sum $4,000,000 vs legacy $2,000,000 (illustrative)	ratio 2.0 — doubled

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
7	Failure C — a shift that cancels	`1,200 + 800 = 2,000 and 1,000 + 1,000 = 2,000`	stores A and B, legacy vs S/4 (illustrative)	totals match; stores do not

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Check 1 — count	`legacy rows − S/4 rows = 5,000 − 4,950 = 50`	catches drops (record design); pass = 0	FAIL on A
2	Check 2 — sum	`S/4 sum − legacy sum = $4,000,000 − $2,000,000 = $2,000,000`	catches doubles; pass = 0	FAIL on B
3	Check 3 — per store (A; B)	`1,000 − 1,200 = −200; 1,000 − 800 = 200`	catches shifts that cancel in the total; pass = 0 per store	FAIL on C, though 2,000 − 2,000 = 0
4	Pass case	`5,000 − 5,000 = 0; $2,000,000 − $2,000,000 = $0; 1,200 − 1,200 = 0; 800 − 800 = 0`	all three zero	file passes
5	Sequencing	`non-regression tests before UAT; both before the migration's hard dates`	record	tests first
6	Contingency	`Option 2 spent only if a mapping changes: expected mapping changes = 0`	the strategy assumed none (record)	$0 pre-spent
7	Break-even question (not a price)	`p × L* = $42,000 → L* = $42,000 ÷ 10% = $420,000`	at p = 10% chance of a regression (illustrative); L* is the loss above which funding pays — never a claim of the regression's value	L* = $420,000 at p = 10%

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Interfaces in test scope	17 "potentially impacted"	files behind 2 critical usages (6 core + supply-chain family)	`scope cut`	record
2	Budget committed	unscoped	$42,000	`scoped`	record
3	Budget held contingent	$0 defined	$25,000 not pre-spent	`$25,000`	record
4	Checks per file	none defined	3 (count, sum, per-store)	3	record design
5	Critical usages protected	2 at risk	2 behind named gates	2	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Committed	`$42,000 spent`	record	$42K
2	Contingent, not spent	`$67,000 − $42,000 = $25,000`	contingency held	$25K
3	Threshold at p = 5%	`$42,000 ÷ 5% = $840,000`	illustrative probability	$840,000
4	Threshold at p = 25%	`$42,000 ÷ 25% = $168,000`	illustrative probability	$168,000

Row counts and dollar sums in the failure cases are illustrative. The break-even thresholds are a way to read the decision — at what loss $42K pays at a given probability — not a valuation of the regression, which I deliberately never price.

SOLUTION FLOW — THE SYSTEM & DATA MAP

```
[Global reporting program: 17 candidate interfaces]  --"potentially impacted"-->  ask: test budget
        |  my first answer: NO (unscoped)  +  fund a joint study
        v
[Study: interface -> file -> usage]  --> 2 critical usages: executive dashboard, global brand portal
        |  list = 6 core files + supply-chain file family
        v
[Option 1: $42K non-regression tests, scoped to the list]   [Option 2: $25K mapping adaptations - contingent]
        |
        v  per file: delta extract from S/4 (US + Canada) vs legacy
   count (drops) -> sum (doubles) -> per-store match (shifts that cancel)  -> gate
        |  tests ahead of business UAT; both ahead of the migration's hard dates
```

SENSITIVITY / IMPACT — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Probability of a regression without tests	5%	10%	25%
Break-even loss L* = $42,000 ÷ p	$840,000	$420,000	$168,000
Contingent share of total budget	25 ÷ 67 = 37%	37%	37%
Share of the 17 interfaces needing tests	not in record	not in record	not in record VERIFY-WITH-RAJESH

SPOKEN TOTAL-IMPACT STATEMENT — SAY IT LIKE THIS

Conservatively, there is no dollar to claim here and I would not take one: forty-two thousand was spent, twenty-five thousand was held and not pre-spent, and the value of the executive dashboard not breaking is deliberately unpriced. What is on the record is the sequence — refuse the unscoped ask, fund the study, reverse on evidence — and that the two critical usages crossed behind arithmetic gates.

FACET 5 — HARDEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Numeric — Why $42K, and why not the whole seventeen?

Because the study showed that most of the seventeen did not feed anything critical. $42K bought count, sum and per-store tests for the files behind the two critical usages — six core files plus the supply-chain family. The $25K for mapping changes stayed contingent because the strategy assumed no mapping would change.

Numeric — Show me a regression that totals hide.

Stores A and B: legacy 1,200 and 800 visits, S/4 1,000 and 1,000. Both totals are 2,000, so count and sum pass in aggregate — but store A is off by −200 and B by +200. Per-store matching is the check that sees a mapping shift cancelling in the total.

Numeric — How do you read $42K against a risk you never priced?

As a threshold, not a price: at a 10% chance of a silent regression, $42K pays for itself against any loss above $420,000; at 5% above $840,000; at 25% above $168,000. I never put a dollar on the dashboard breaking — I ask whether anyone believes the loss is smaller than that.

Numeric — Were you simply wrong the first time?

The first no was right on the information available: an unscoped ask with no interface list. What made it safe is that the no shipped with a funded study, so being wrong was cheap and fast to find out.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A DIFFERENT 90-SECOND MOMENT NEW

ARE RIGHT, A LOT **1. Tell me about a time you changed your mind based on data.**

Open (first two sentences): I refused a budget request, then reversed myself in public when the study came back.

A. I said no to an unscoped "potentially impacted" ask but funded a small joint study to turn it into a list. It showed real risk on two critical usages, so I funded the $42K option scoped to those files and told both programs I had changed my position and why. **R.** The usages crossed behind named gates and the public reversal bought credibility.

Land the number: $42K funded after a no | **Pivot to NACF:** At NACF: refuse the unscoped "this might affect inbound" and fund the study that names the interfaces.
Do NOT say: Claiming the dashboard would have broken — unknown and unpriced.

FRUGALITY **2. Tell me about a time you got more by spending less.**

Open (first two sentences): I refused to fund fear, but I did not refuse to fund evidence.

A. A small study replaced an open-ended budget; Option 2, $25K, was written as contingent and never pre-spent. I scoped tests to the files the critical usages consume. **R.** $42K spent against a risk, $25K held back.

Land the number: $42K, not $67K | **Pivot to NACF:** Inbound: spend on the lanes that matter, hold the rest.
Do NOT say: Calling it a savings — it is a refused open-ended ask.

EARN TRUST **3. Tell me about a time you admitted you were wrong.**

Open (first two sentences): The global program had a legitimate concern and I had said no.

A. I reversed on the record as soon as the evidence arrived, and said so to both programs rather than quietly funding. The first no had shipped with a study, so it never read as a stonewall. **R.** The reversal bought more credibility than the refusal had cost.

Land the number: public reversal | **Pivot to NACF:** Same posture with any planning partner: name the change of mind.
Do NOT say: Saying I was right all along.

DIVE DEEP **4. Tell me about a time you designed a test that caught what totals miss.**

Open (first two sentences): The tests had to find a regression that looks plausible, so I started from how migrations actually fail.

A. They drop rows, double rows through a bad join, or shift a mapping. Count catches drops, sum catches doubles, per-store catches shifts that cancel in the total — three cheap checks, each owning one failure mode. **R.** Delta extracts per interface passed or failed on arithmetic.

Land the number: 3 checks, 3 failure modes | **Pivot to NACF:** Inbound: the same per-door and per-lane matching, not the building total.
Do NOT say: Illustrative row counts as real.

SAME STORY, 3 ANGLES — DIFFERENT OPENER, HEADLINE, LP NEW

DATA / ANALYTICAL

“The totals matched and two stores were wrong by two hundred each.”

1,200 / 800 vs 1,000 / 1,000

LP: Dive Deep

Count 5,000 vs 4,950 catches 50 dropped rows; sum ratio 2.0 catches a double.

Per-store matching catches the shift: −200 and +200.

Three checks, each owning one failure mode.

CONFLICT / INFLUENCE / BACKBONE

“I told another program’s lead no.”

17 → 2

LP: Have Backbone · Frugality

“Potentially impacted” is not a scope.

The no came with a funded study.

I did not fund the unscoped ask.

FAILURE / LEARNING

“Was I wrong the first time?”

$42K

LP: Are Right, A Lot

The first no was right on the information available, and shipped with a study so being wrong was cheap.

The cost was calendar and some credibility; the public reversal repaid it.

Differently: ask for the interface list on day one of the program, not on the first request.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH THE EXACT SENTENCE NEW

Trap — “You said no to a legitimate risk. How is that not a failure?”

The no was to the unscoped version; the same answer funded a study. A wrong no costs a regression on a global-critical asset; a wrong yes costs $42K — so I ran the loop that made the no cheap: refuse, study, reverse on evidence. I would run it again.

Say exactly: “The no shipped with a funded study, so being wrong was cheap and fast.”

Trap — “What did you cut from the testing scope, and what could you never cut?”

I cut from the paid scope the interfaces the study cleared — they did not feed the critical usages — and held the mapping-adaptation option as contingent. I never cut the per-store match, because it is the check that catches the shift that cancels in the total.

Say exactly: “I cut the interfaces the study cleared; I never cut the per-store check.”

Trap — “How do you know the tests protected anything?”

I do not and I do not claim to — I deliberately do not price the avoided regression. What I can say is the arithmetic gates were in place and passed, and that I know exactly which three failure modes they would have caught.

Say exactly: “I can’t prove a regression that didn’t happen; I can show the three failures the gates would have caught.”

Where this lands at Amazon / NACF. Whenever a change touches a shared feed — at Amazon scale, an inbound system change under planners’ dashboards — the move is the same: refuse the unscoped ask, fund the list, test with counts, sums and per-node matches.

Peer operating drills — scope, dependencies, what I would cut, what I would never cut NEW

A peer tests cross-functional operating judgment: not whether the design was clever, but how I scope, who I depend on, what I would cut when capacity or time halves, and what I refuse to cut. Cut orders below are my judgment from the record's gates and exit bars; they are not claims of something that happened.

STORY	CUT FIRST	NEVER CUT	BINDING DEPENDENCY
S1	Long-tail node × channel cells to the second go-lives	Bucket owners; lower-of-two; signed FSD row for each no	Three signatory orgs; the council's three decisions
S3	Ship-to-home source — rides the overnight path longer	Matched-only clear; SNB never negative; the aging report with a named owner	S/4 Aggregation# on the posting; Inventory Accounting
S1-a	The 60-second budget on non-sale events	Idempotency key; reversal names its original; one bucket per event	POS, OMS, WMS and S/4 all honouring one contract
S3-a	Non-POS sources first (omni, ship-to-home)	Σ lines = posting to the cent; the duplicate-header guard	Order# join keys from the POS
S1-b	Internal Audit's read of every correction → a monthly sample (the audit row is still written on every one)	The audit row; T+1 clean; the drift report	Inventory Control's daily capacity (6.5 of 7.2 hours)
S1-c	Breadth of old-vs-new reconciliation (start with high-volume nodes)	Reserve-copy equality; one publisher; the no as an FSD row	Planning's reserve feed (nightly plus holds)
S1-d	Scope between go-lives (a node type or channel)	Blocker + Critical open = 0; armed rollback; owners named	The sync to S/4 (a rollback trigger)
S1-e	Interfaces the study cleared from the paid scope	The per-store match; the contingent option held	The global program's own owners deciding what is critical

EIGHT DRILLS, WITH THE EXACT SENTENCE NEW

Trap — "You have half the people and the date does not move. What do you cut?"

Scope before gates, sources before controls, breadth before depth. Cut: long-tail node × channel cells to the second go-live, the ship-to-home source, non-sale latency budgets, and the breadth of the old-versus-new reconciliation. Never cut: the bucket owners, matched-only clear, idempotency, the audit row on a correction, Blocker + Critical open = 0, or an armed rollback.

Say exactly: "I cut scope and sources, never the controls: owners, idempotency, the audit row, the severity gate and the rollback stay."

Trap — "Which dependency worried you most, and what did you do about it?"

The ISM-to-S/4 sync, because it was a rollback trigger — sync below 99% for four hours — and because the S/4 side owned MB52 and the posting. I made it visible instead of trusting it: a daily drift report at tolerance zero, an owner per open question with a date, and dated E2E days 28 and 41 days apart so each cycle could close before the next.

Say exactly: "The sync to S/4 — so I made it a rollback trigger and a daily drift report, not an assumption."

Trap — "Everyone wants their thing in. How do you scope?"

By consumers, not candidates. S1-e: seventeen candidate interfaces became the files behind two critical usages. S1: epics slotted by business criticality × effort, and no story enters QA without a matrix cell reference. Scope is a list with an owner for each line, and the council is asked for three decisions, not a review.

Say exactly: "Scope is a list of what is consumed, with an owner per line — not a list of what might be touched."

Trap — “What would you stop doing in your first ninety days at NACF?”

I would stop myself from building a new dashboard before one number has one owner. First: the book of record and the primary metric; then a stocks-and-flows map of inbound; then the invariant already broken — typically a unit in two stocks or a number with two owners. This is a plan, not a record.

Say exactly: “I would not add a dashboard before one inbound number has one owner and one daily drift check.”

Trap — “What do you want me to just fix, and what do you want me to bring to you?”

Fix what is inside the envelope and reversible today: a stale reservation corrected on the screen with a reason code, a vendor or door re-slot. Bring policy: a rule change, a bucket definition, a threshold — those ride the release train and are signed by the same signatories, never a side configuration.

Say exactly: “Fix reversible acts with an audit row; bring anything that changes a definition or a threshold.”

Trap — “A metric we both touch goes red. Who speaks first?”

The owner of the input speaks first, with the decomposition already written. In ISM that is the owning control role — Inventory Control for drift and negatives — and a red drift row arrives with a reason code and an owner before the review, not in it.

Say exactly: “The owner of the input speaks first, with the cause and the date already written.”

Trap — “Two teams’ priorities conflict and neither reports to you. How do you resolve it?”

Price both sides and write the number that would flip the answer. S1-c: promising held stock earns about $19 a unit if it releases in time and costs about $30 an incident if not; break-even 61%. S1-e: $42K against a regression I never price, read as a threshold at a chosen probability. Then the decision lives in a document.

Say exactly: “I put a price on both sides and write down the number that would change my mind.”

Trap — “How would you know a cut was a mistake?”

Each cut has a tripwire written in advance — the way the rollback triggers were: a threshold, a duration and an owner. For example, if the ship-to-home source stays on the overnight path and its unmatched rows exceed the aging-report capacity — about 10 a day at the 0.5% KPI on an illustrative 2,000 postings — I pull it back in.

Say exactly: “Every cut ships with a tripwire: a threshold, a duration and an owner.”

NACF bridge — AFS on the inbound side: what the promise can see NEW

Use this in the last minute of any S/4 answer, or when asked how the work transfers. Every inbound number below is from the inbound booklet's worked week (illustrative there: one vendor, three SKUs, 12,000 units on six trailers); the S/4 side is record design from S1.

S/4 / ISM CONCEPT	INBOUND ANALOG	INBOUND BOOKLET ANCHOR	LINE TO SAY
AFS = Basis − committed buckets	Inbound AFS = opening on-hand + units that clear receive and stow	F: planned 1,000 + 6,000 = 7,000; actual 1,000 + 5,880 = 6,880	The promise sees only what cleared stow.
Shipped-Not-Billed (sold, not yet posted)	Received-not-stowed (on the dock or in receiving)	Availability starts at stow, not at the dock	A unit on the dock is the SNB of a building.
Lower-of-two basis	Promise on the lower of ASN and dock count	ASN 6,000 vs counted 5,880: promise 5,880	Wrong on the safe side, visible.
Merch reserve: visible, not promisable	Stranded / held stock: present, not sellable	"Stranded — physically present, not sellable… inbound's silent inventory"	A stow without a sellable record is not availability.
Drift report, tolerance zero	Dock count vs system receipt, daily, per building	Accuracy 99.0% overall hides 98.0% on F	An aggregate hides the line.
Aggregate posting vs lines (S3-a)	ASN header vs ASN lines	0.02 × 6,000 = 120 units short on F alone	Lines must sum to the header or the row ages.
Negative AFS screen (S1-b)	Promised more than received-and-stowed	Cover 7.00 days → 6.76 days	Never clamp a negative promise; show it.

WORKED ON THE BOOKLET'S FAST MOVER — FOUR SMALL TABLES NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Planned inbound AFS	`1,000 + 6,000 = 7,000`	opening on-hand 1,000; PO 6,000 (booklet, illustrative)	7,000 units
2	Dock count short 2%	`0.02 × 6,000 = 120`	2% short on F only	120 units
3	Received	`6,000 − 120 = 5,880`	step 2	5,880 units
4	Aggregate accuracy	`(5,880 + 4,000 + 2,000) ÷ 12,000 = 0.990`	M and S at plan (booklet)	99.0% overall
5	Line accuracy on F	`5,880 ÷ 6,000 = 0.980`	the aggregate hides this	98.0% on F

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promise on the lower of two	`min(6,000, 5,880) = 5,880`	ASN vs dock count — the lower-of-two asymmetry	promise 5,880
2	Inbound AFS after stow	`1,000 + 5,880 = 6,880`	only units that cleared receive and stow	6,880 units
3	Gap to plan	`7,000 − 6,880 = 120`	the shortage equals the whole miss	120 units
4	Cover at the close	`1,000 + 5,880 − 3,500 = 3,380; 3,380 ÷ 500 = 6.76`	week sales 3,500; demand 500 a day	6.76 days

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
1	Inbound AFS vs plan	7,000	6,880	`7,000 − 6,880 = 120`	booklet, illustrative
2	Cover (days)	7.00	6.76	`7.00 − 6.76 = 0.24`	booklet, illustrative

STEP	METRIC	BEFORE	AFTER	Δ / CHECK	BASIS
3	Accuracy	99.0% overall	98.0% on F	`99.0 − 98.0 = 1.0 pt`	booklet, illustrative

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Shortage as demand	`120 ÷ 500 = 0.24`	days of F demand	0.24 days
2	Shortage as a share of the week	`120 ÷ 3,500 = 0.0343`	F weekly demand 3,500	3.43%
3	Pass-through check	`0.24 × 500 = 120`	the days lost equal the units short	120 units

THE BRIDGE IN 20 SECONDS

“At Sephora the promise could see only what the ledger said was available: lower of two books, minus what was already committed. Inbound is the same discipline one step earlier — the promise sees opening on-hand plus the units that cleared receive and stow, on the lower of the ASN and the dock count, and a unit on the dock is just a shipped-not-billed unit of the building. One owner per state, and a daily check that the count and the system agree.”

Question bank — likely questions for this interviewer, with a one-line answer plan

UPGRADED

Twenty-four questions (the original twelve plus twelve new). Each plan points at a story in this packet or a bench story; stories allocated to other interviewers are pointers only.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Tell me about a time you aligned multiple organizations on one definition.**	S1 pitch — three truths, one signed FSD 6/5/25; pre-wiring; objections as rows; oversells recurring → zero.	S1
2	**Tell me about getting Finance to trust an operational number.**	S3 pitch — one FOH; failure direction; the aging report given to Finance to own.	S3
3	**How do you influence without authority?**	Pre-wire every row with its owner's LOE; working sessions to one table; three decisions not a review; a no that lives in a document. (S5 pointer.)	S1 (S5 pointer)
4	**Tell me about a time you were wrong.**	S4 (bench) — the in-transit rule; or S1-e — my first no, reversed on evidence. Keep S4 to the pointer unless the room asks for the full walk.	S4 / S1-e
5	**How would our two teams share a process / a metric here?**	One definition, one owner per bucket (S1-a); drift report tolerance zero with a named owner (S1-b); rule changes ride a release train.	S1-a / S1-b
6	**Tell me about designing for failure.**	S3 — failure direction before the happy path; unmatched ⇒ under-sell; aging report at T+1; late line −2, never +2.	S3
7	**How do you handle a stakeholder who wants a capability that sells more?**	S1-c — merch reserve promisable: priced (560 vs 710), visibility offered, no written as a row; break-even 61%.	S1-c
8	**How do you decompose a cross-functional process problem?**	Master §3.99 — book of record and metric first; every bucket a unit can sit in; flows and their documents; the broken invariant; minimal design; sequence.	Master §3.99
9	**What is the business value of the book-of-record migration?**	≈$2.5M/yr: $1M measured MCS run cost (anchor) + ≈$1.5M modeled with its three rows; incidents to zero is the record movement.	S1
10	**How do you keep a design from being re-litigated after sign-off?**	Three signatories; the no as an FSD row (S1-c); changes ride the release train as FSD changes (S4 pointer).	S1 / S1-c
11	**Tell me about a release / change-management discipline you set up.**	Rule changes ride the monthly cadence; the matrix is the regression gate; no story enters QA without a cell reference; go/no-go from counted rows (S1-d).	S1 / S1-d
12	**What would a peer say is hard about working with you?**	I convert verbal agreements into rows people have to sign — slower in the week, faster over the program. S1 pre-wiring as the example.	S1
13	**Walk me through available-for-sale on one SKU — and how an event moves it.** NEW	S1-a — 980 − 420 = 560; commit → 555; pick, pack, ship move it between buckets; the posting moves both books together.	S1-a

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
14	**How do you reconcile an aggregate to its lines? What about one cent?** NEW	S3-a — $100.00 over three units: last line absorbs → $33.33 + $33.33 + $33.34; missing line and duplicate-header guards.	S3-a
15	**Who owns the exceptions after you leave?** NEW	S1-b — Negative AFS to Inventory Control, Internal Audit and Finance; AFS Correction to the ISM app team; drift report to Inventory Control.	S1-b
16	**Tell me about a no you wrote down.** NEW	S1-c — the FSD row, the pricing, the number that would flip it.	S1-c
17	**How do you decide go or no-go?** NEW	S1-d — a checklist read from counted rows: Blocker + Critical open = 0; reconciliation clean N days; rollback armed; 836 rows, 441 severe.	S1-d
18	**A neighboring team asks for a test budget. How do you scope it?** NEW	S1-e — refuse the unscoped ask, fund a study, reverse on evidence; $42K; count, sum, per-store.	S1-e
19	**Half the capacity, same date: what do you cut?** NEW	Peer operating drills — cut scope and sources, never the controls; each cut has a tripwire.	Drills
20	**Which dependency worried you, and what did you do?** NEW	The sync to S/4 (rollback trigger); daily drift; dated E2E days; owner and date on every open question.	Drills / S1-d
21	**How does this transfer to NACF inbound?** NEW	NACF bridge — AFS = opening + units that cleared receive and stow; lower of ASN and dock count; one owner per state.	Bridge
22	**Tell me about a time you missed a deadline.** NEW	Honest: there is no missed S/4 date in the record — do not invent one; offer S4 (a rule I corrected) and say why **none on record**.	S4 pointer
23	**Tell me about developing someone.** NEW	Thin: S1-b — control owners handed the screens; measure by whether the drift report runs without me **no mentoring record**.	S1-b
24	**Tell me about a time you changed your mind.** NEW	S1-e — refuse, study, reverse in public; S4 — back-test before publishing.	S1-e / S4

Cheat-numbers box — canon values with basis labels UPGRADED

Canon numbers for the eight S/4 primaries and the bench stories, with basis labels, from the master packet's §4.1 ledger. The S1-a to S1-e rows are re-cuts of S1 and S3 facts; none adds a new dollar.

METRIC	VALUE TO SAY	BASIS	NEVER
S1 primary metric	customer-facing oversell incidents recurring across three node types → zero through two go-lives	record	an oversell count or rate as record
S1 headline $	≈$2.5M/yr = $1M MCS run cost retired + ≈$1.5M modeled (oversell ≈$0.9M · recon ≈$0.33M · change premium ≈$0.3M); range $1.7–5.2M	user anchor + modeled	presenting the $1.5M as record; adding upside into the headline
S1 signatures / dates	FSD v1.1 (3/5/25) → v1.2 (5/16/25) → signed 6/5/25 by Supply Chain Strategy, Inventory Management, FSC Inventory Accounting	record	one signatory
S1 gate chain	kickoff 3/29 → council 5/15 → LOE 7/2 → sign-off 7/31 → FSD 8/14; E2E 11/22 · 12/20 · 1/30; January planning + May execution go-lives; 836 rows / 85.4% closed	record (shared program artifacts with S2/S5)	—
S1 design	lower-of-two basis; AFS = Basis − PickRes − EcomPick − EcomPack − SNB; idempotency key per event; daily ISM-vs-MB52 drift tolerance 0; OMS reads, never computes	record design	"I configured S/4" / "I wrote ABAP"
S1 reconciliation FTE	≈4 FTE by hand → ≈1 FTE reading the drift report	modeled	stating it as record
S3 primary metric	Financial On-Hand truths 2 → 1; AFS decremented at sale/ship; over-sell from unmatched postings = 0 by construction	record (FSD 6/5/25)	—
S3 dollars	≈$0.6M/yr (cancels 0.45 · recon labor 0.14 · close 0.05); range $0.25–1.2M	modeled — no pre-state drift dollar in record	any record drift dollar; double-counting S1's cancel rate
S3 design KPI	match rate ≥ 99.5% within 24 h; aging report empty or explained at T+1; Σ per-line COGS = posting COGS to the cent	record design	a days-to-tie figure
S1-a contract NEW	ten buckets (incl. derived OOB and Basis), one owner each; one event → one bucket; 60 s publish; WMS feed ≤ 15 min; consumer lag < 5 min; drift tolerance 0	record design (walk units illustrative)	a measured defect rate; the 25% slice as fact
S3-a rounding NEW	$100.00 over 3 units: $33.33 + $33.33 + $33.34; $175.00 over 8 units: $65.63 + $65.63 + $43.74	record rule; amounts illustrative	claiming rounding mode / line order are in the record
S1-b controls NEW	Negative AFS → Inventory Control, Internal Audit, Finance; AFS Correction → ISM app team; drift → Inventory Control; Negative AFS clean at T+1	record	a maker-checker claim; workload numbers as fact

METRIC	VALUE TO SAY	BASIS	NEVER
S1-c reserve NEW	merch reserve outside S4_Avail; visible, not promisable; 560 vs 710 on the worked SKU; break-even 30 ÷ (19 + 30) = 61%	record design; units and benchmarks illustrative	a measured release rate; a hold count
S1-d cutover NEW	836 rows; 85.4% closed; Blocker 164 + Critical 277 = 441 = 52.8%; triggers sync < 99% 4 h, p95 > 2× 1 h; E2E 28 then 41 days apart	record (derived arithmetic labeled)	a fired trigger; a cost of delay as record
S1-e interfaces NEW	17 interfaces; 2 critical usages; $42K funded; $25K contingent, not pre-spent; count, sum, per-store	record (loop packet S25 — not in master canon)	a price on the avoided regression
S4 (bench)	starved lanes → restored; oversell +0; kill switch never fired; ≈$0.5M/yr modeled margin	record / modeled	a dollar for the conversion lost while I was wrong
S5 (bench)	surprise escalations → 0; ~80 names across ~15 functions; 5/15 → 7/31 → 8/14; ≈$0.5M one-time modeled	record / modeled	—
SAP tenure	S/4HANA integration program lead 3+ years (Apr 2023 – present); client-SAP data integration at Elementum 2017–19; 15 years full-lifecycle SCM systems	record	“10 years of SAP configuration”; “SAP functional owner”

Never-say, one glance before the call.

- Never “I configured S/4,” “I deploy SAP,” or “I wrote ABAP” — integration program lead; SAP teams and configurators built to the signed FSD. Manhattan DOM is the only platform personally configured in canon.
- Never present the ≈$1.5M (S1) or the ≈$0.6M (S3) as record; the $1M MCS figure is the user anchor and the only measured dollar here. Never add the variants’ slices to S1 or S3.
- Never give the S1 and S3 cancel rates as two results in one room — keep S1’s, drop S3’s row ($0.19M left).
- Never invent a days-to-tie figure, a direct-report headcount, a pre-state oversell dollar, or a price for the avoided regression (S1-e).
- Never inflate a multiple: customers 7 → 68 is 9.7×. Say “six sites,” never a regional qualifier on it. Finished-goods grain only — no component or bill-of-materials stories. Never turn a single dollar figure into a range you cannot source.
- S4 and S5 are bench stories; S14 is the bar raiser’s; S2/S18 are Goforth’s — “one umbrella program” if asked, no floor story here.

Backup bench — use only if a primary doesn't fit

Four fallbacks for a peer Senior Manager — S4 and S5 are the question bank's pointers, here in full bench form: S4 (Sephora — the in-transit promise rule I got wrong, fixed through the same three signatories; the failure / Are Right, A Lot question), S5 (Sephora — the architecture council without owning a system; Backbone and influence without authority — same program family as S1, so use only when the question is about governance rather than the definition) and S9 (Elementum — the 597-lane Pareto two levels up; Backbone / Think Big outside the Sephora program); plus a doc-derived fourth from the Flexport program record — the FFA Audit Logs PRD (controls and process integrity for a partner-facing platform; record-from-docs labels; same platform as Tripathy's S12, facts kept identical to her packet). Tell S1 and S3 first.

Bench discipline. The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

BACKUP — ONLY IF ASKED S4 — Sephora S/4HANA — the in-transit promise rule I got wrong, fixed through the signed policy, not a side configuration

LPs this backup can serve: Are Right, A Lot (the failure question) · Customer Obsession (backup) · Earn Trust (owning the miss) · Learn and Be Curious — the designated policy-failure story

Basis. The v1 rule was mine; the indictment came from my own program's reports. Lane σ ≈ 0.3 day and the ~0.49-day window are the record's promise math; the record carries no lane counts and no dollar for the conversion lost while I was wrong — I will not invent the loss figure.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. My first ISM ruleset barred every channel from promising units in transit; the Aging In-Transit report showed dotcom starved of fast movers one day from the building. **Task.** Correct my own policy without weakening oversell protection, through the same three signatories. **Action.** Discovery: conversion-loss evidence from the aging and replenishment reports — the entry bar to reopen the policy; I stated that the v1 rule was mine. Plan & Design: promise math — lane σ ≈ 0.3 day, a ~0.49-day window ≈ 95% coverage; Promisable = in-transit × 1[ETA ≤ window] × 1[on-time ≥ floor]; Aging In-Transit as the kill switch; both blanket options rejected in writing. Build & Test: back-test against receiving history; four regression cells added to the node × channel matrix. Cutover: released on the normal monthly cadence as a signed FSD change, lane by lane — no hotfix, no side configuration. Hypercare: kill switch armed; never fired. Lesson: both sides of a rule get a number before it ships. **Result.** Primary metric: sellable units on qualifying in-transit lanes — starved → restored inside a per-lane promise window, oversell increase = zero, kill switch never fired. ≈$0.5M/yr modeled margin recovered — the record has no dollar for the conversion lost while I was wrong, and I will not invent one.

COMPRESSED STAR

- **S —** v1 of my own ISM ruleset: no channel could promise in-transit units — safe-feeling, and starving dotcom of fast movers reliably one day from the building.
- **T —** Fix my own signed policy without weakening oversell protection; take the miss back through the same three signatories.
- **A —** Evidence pack from Aging In-Transit and Replenishment; reframed as promise math (σ ≈ 0.3 d → ~0.49-d window ≈ 95%); rule with on-time floor and kill switch; back-test against receiving history; four regression cells;

released lane by lane on the monthly cadence as an FSD change.

- **R —** Starved lanes restored inside the window; oversell increase = 0; kill switch never fired; re-signed by the same three signatories; lesson institutionalized — the Locus 98% gate (S2) and the 0.5% tolerance (S18) were set from data before the event.

RESULTS LINE

Sellable units on qualifying in-transit lanes starved → restored inside a back-tested ~0.49-day window (record) · oversell increase = 0; kill switch armed, never fired (record) · four regression cells added (record) · released as a signed FSD change on the normal cadence, no hotfix (record) · ≈$0.5M/yr modeled margin recovered (inputs illustrative).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The v1 rule	`Promisable = 0 for ALL in-transit inventory`	equivalent to claiming σ → ∞ or on-time rate → 0 — infinite variability (record)	everything in transit hidden
2	One concrete lane	`Promisable = 120 × 0 = 0`	lane with InTransit = 120 units, ETA 0.4 day away (illustrative)	120 sellable units hidden
3	Across qualifying lanes	`≈300 fast-mover units/day × 0 promisable`	illustrative volume	≈300 units/day sat unsellable one day from the building
4	What starving costs (one SKU)	`100 × 60% × $20 = $1,200/day`	100 units one day out; 60% would convert; $20 margin (illustrative, not program data)	$1,200/day under-sell
5	The ignored other side	`100 × (1 − 0.97) = 3 late units; 3 × $15 = $45/day`	on-time rate 97%; $15 expedite (illustrative)	$45/day over-sell exposure — vs $1,200: the “safe” rule priced one side only

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure lane variability	`σ = 0.3 day`	trailing receiving history (record)	σ = 0.3 day
2	Choose coverage	`z at 95% = 1.645`	normal approximation on lane transit (toolkit label)	z = 1.645
3	Promise window	`w = z × σ = 1.645 × 0.3 = 0.4935`	rows 1–2	w ≈ 0.49 day
4	Coverage check	`Φ(w / σ) = Φ(0.49 / 0.3) = Φ(1.63) ≈ 0.95`	share of arrivals inside the window	≈95% covered
5	Corrected rule	`Promisable = InTransit × 1[ETA ≤ w] × 1[OTR ≥ floor]`	OTR = on-time rate over trailing receipts; floor = per-lane admission threshold, 95% here (floor illustrative)	lane by lane, not all-or-nothing
6	Lane that qualifies	`ETA 0.4 ≤ 0.49 → 1; OTR 97% ≥ 95% → 1; Promisable = 120 × 1 × 1 = 120`	the Table 1 lane, re-evaluated (ETA and OTR illustrative)	its 120 units become promisable
7	Lane that fails	`ETA 1.2 > 0.49 → 0; Promisable = 80 × 0 = 0`	a slower lane with 80 units in transit (illustrative)	stays at 0 — correctly out
8	Sample-size gate	`n = z²p(1−p)/e² = 3.84 × 0.95 × 0.05 / 0.05² = 3.84 × 0.0475 / 0.0025 ≈ 73`	OTR estimated within ±5% at 95% CI (illustrative)	a lane qualifies after ~73 receipts

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
9	Kill switch, values in	`trigger = 2 × p90 = 2 × 2.0 = 4.0 days; any unit in transit > 4.0 days → lane out`	lane p90 = 2.0 days (illustrative); armed, reviewed weekly by Inventory Control (record)	never fired (record)
10	Back-test	`rule replayed against receiving history; 4 regression cells`	before release (record)	no oversell in the back-test

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Sellable units on qualifying in-transit lanes	0	≈300 units/day restored inside the window	record mechanism; volume illustrative
2	Promise window	none — blanket exclusion	w ≈ 0.49 day at ~95% cover	record (promise math)
3	Oversell increase from the window	—	0	record
4	Kill switch	not armed	armed, never fired	record
5	Rule changes after	ad hoc	same 3 signatories, back-test + matrix regression	record — both sides priced from then on

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Would-be sales	`300 × 60% = 180 sales/day`	units restored (illustrative) × conversion (illustrative)	180 sales/day
2	Truly lost without the fix	`180 × 40% = 72 sales/day`	40% lost-sale share — the rest would have waited (illustrative)	72 lost sales/day
3	Margin recovered	`72 × $20 = $1,440/day`	$20 unit margin (illustrative)	≈$1,440/day
4	Annualized	`$1,440 × 365 = $525,600`	modeled; the mechanism is record	≈$0.5M/yr
5	Cost while v1 stood	`no dollar in the record — not claimed`	record	not claimed

TWO PROBES

> **Why not just hotfix it — you knew it was wrong?**
>
> Because the error was in the policy, so the fix had to be too — in the governed artifact, with regression, through the same signatures. A side configuration would have fixed the symptom and left the policy lying about what it allowed.

> **What did the miss cost?**
>
> The record carries no dollar for the conversion lost while v1 stood, and I will not invent one. What I can say is the mechanism: fast movers reliably one day from the building were unsellable on dotcom until the window existed; the ≈$0.5M/yr recovered is my model on illustrative inputs, labeled.

Canon numbers (basis labels): Lane σ ≈ 0.3 d; ~0.49-d window ≈ 95% coverage (record promise math) · oversell increase = 0 (record) · 4 regression cells (record) · kill switch never fired (record) · same three signatories re-signed (record) · ≈$0.5M/yr (modeled). Never a lane count; never a dollar for the loss while wrong.

BACKUP — ONLY IF ASKED S5 — Sephora — the architecture council without owning any system: three decisions and a fence

LPs this backup can serve: Have Backbone; Disagree & Commit (third) · Earn Trust (influence without authority) · Think Big (architecture governance). Same program family as S1/S2 — the governance story

Basis. I owned none of the systems; the design decision was the deliverable. Zero escalations and the dated gate chain are the record; the dollars are my model of what governance avoided.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The UDC location and inventory model had to clear architecture council — a challenge gate — with ~80 named participants across ~15 functions, and I owned none of those systems. **Task.** Take the design from discovery through council to sign-off and FSD without a surprise escalation. **Action.** Discovery: both inventory books profiled before any design argument; business sessions closed 4/30. Plan & Design: every impacted row pre-wired with a signed LOE by 7/2; the council asked for three separable decisions plus an out-of-scope fence (reserve math untouched; MCS decommission belongs to Global ERP). Build & Test: the "build" was the pack — architecture, staged timeline, peak-freeze boundary. Cutover: 5/15 challenge → 7/31 sign-off → 8/14 FSD. Hypercare: the fence held in writing for the whole program; a second live warehouse became a monitored shadow copy, not an argument. Lesson: a no that lives in a document survives. **Result.** Primary metric: surprise escalations on a 60+ stakeholder design — the norm on prior programs → zero; the dated gate chain held. ≈$0.5M one-time modeled (avoided rework on ~3 escalations that never happened + six weeks of benefit not deferred + a fenced second reserve engine not built).

COMPRESSED STAR

- **S —** A dozen core systems (20+ touched), ~80 participants across ~15 functions, a council that re-litigates designs this size — and I owned none of the systems.
- **T —** Discovery (kickoff 3/29, sessions 4/30) → council 5/15 → sign-off 7/31 → FSD 8/14, without a surprise escalation.
- **A —** Profiled both books first — their numbers, not my opinion; systems-impacted table with an owner on every row; one LOE template, one deadline (7/2), one private conversation per owner before the room; three separable decisions plus the fence; the second-truth objection converted into a monitored shadow copy.
- **R —** Surprise escalations → 0 for the program's life; 100% pre-wire coverage by 7/2; gate chain held to the day with zero re-litigation; fence breaches 0; handed clean into the build (S2).

RESULTS LINE

Surprise escalations → 0 for the program's life (record) · gate chain 3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 held to the day (record) · pre-wire coverage 100% — every impacted row with a signed LOE by 7/2 (record) · both scope refusals held in writing; fence breaches 0 (record) · ≈$0.5M one-time (modeled — what did not happen).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Design size	`~80 named participants × ~15 functions; 20+ systems touched`	record	the blast radius
2	Norm for this size	`~3 surprise escalations per design`	industry benchmark 2–4	~3 expected
3	Price of one escalation	`960 rework hours × $110/h = $105,600`	modeled	≈$0.1M each

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Bundle decision risk	`P(all pass) = 0.9 × 0.9 × 0.9 = 0.73`	one up/down vote on the whole design; p = 0.9 per decision (illustrative)	27% chance one objection unwinds everything

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Three separable decisions	`architecture · timeline · peak boundary → 2³ = 8 outcomes`	any partial approval preserves progress (record design)	a no on one cannot unwind all
2	Expected approvals	`E = 0.9 + 0.9 + 0.9 = 2.7 of 3`	vs the bundle's 0.73 chance of everything (illustrative p)	2.7 vs 0.73
3	Pre-wire coverage	`rows with signed LOE / impacted rows = 100% by 7/2`	one template, one deadline, one private conversation per owner (record)	100% before the room
4	Gate chain held	`3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 = 77 days council→sign-off`	held to the day (record)	0 re-litigation

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Surprise escalations	~3 (the norm)	0 for the program's life	record; norm is benchmark
2	Pre-wire coverage before council	partial	100% by 7/2	record
3	Fence breaches in the log	—	0	record
4	Re-litigation after sign-off	—	0 across 77 days	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Avoided rework	`3 × 960 h = 2,880 h; 2,880 × $110 = $316,800`	~3 escalations (benchmark) × rework hours × rate (modeled)	≈$0.3M
2	Slip weeks avoided	`6 / 52 = 0.115 yr; 0.115 × $900,000 = $103,800`	six weeks of S2's ≈$0.9M/yr labor value not deferred (modeled)	≈$0.1M
3	Fenced engine not built	`2 eng × 12 wk × 40 h = 960 h; 960 × $110 = $105,600`	second reserve engine (modeled)	≈$0.1M
4	Total one-time	`0.3 + 0.1 + 0.1 = $0.5M`	all modeled — the dollars are what did not happen	≈$0.5M

TWO PROBES

> **How do you get sixty people to sign LOE you don't control?**
>
> A template, a deadline and a private conversation each — so nobody learned their cost in the room. The council was then asked for three separable decisions it could take one at a time; separability was the design.

> **What did you refuse, and did it hold?**
>
> Retuning the reserve math and folding in the MCS decommission — both written, signed and held; later scope requests were answered from the pack in writing. The legitimate concern (a second live warehouse becoming a second truth) was converted into the design as a monitored shadow copy, because converting a concern costs less than winning it.

Canon numbers (basis labels): ~80 participants / ~15 functions; a dozen core systems, 20+ touched (record) · 5/15 → 7/31 → 8/14; LOE 7/2 (record dates) · escalations 0; fence breaches 0 (record) · ≈$0.5M one-time (modeled). Never a dollar as record; never pair with S1 or S2 for the same interviewer.

BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

LPs this backup can serve: Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

Basis. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	record problem	no instrument, no lever
3	Manual documents	`597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand`	illustrative volumes	~90K docs/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (record)	597 of 1,000 lanes carry 80% of spend — do them first
2	Cohort gate	`validASN rate ≥ 95% in test before activation`	published to carriers before cohort one (record)	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	“the standard was the data, not the transport” (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes live month one	0	597 = 80% of spend	record
3	Freight spend on onboarded lanes	base	−10% ≈ $2M in 6 months	record
4	Document handling	~90K docs/yr keyed	X12/XML validated	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000/yr`	derived from the record pair, stated as such	≈$20M/yr
2	The record result	`$20M × 10% = $2M in 6 months`	record	≈$2M
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`0.8 + 0.6 + 0.6 = $2.0M`	the slices decompose the record — they never add to it	= the record $2M
7	Document labor	`90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000`	illustrative volumes (modeled add-on)	≈$0.4M/yr

TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

Canon numbers (basis labels): 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

BACKUP — ONLY IF ASKED Doc bench — Flexport — the FFA Audit Logs PRD: an immutable who/what/when trail across ~100 partner orgs, nine categories at a 100%-captured bar (controls & process integrity)

LPs this backup can serve: Insist on the Highest Standards (primary — 100% capture as the written bar) · Ownership (compliance as a launch feature) · Dive Deep (the event taxonomy to the grain) · Earn Trust (sign-off over assumption)

Basis. Doc-derived micro-story from the program record (the FFA Audit Logs PRD, 2022) — every figure is RECORD-FROM-DOCS. Same platform as Tripathy's S12: use only for a controls / process-integrity question and keep S12's canon numbers identical to her packet. Program-lead voice: I owned the PRD, the event taxonomy and the sign-off chain; engineering built the pipeline and dashboards. Not in the master packet's S-numbering; supplements canon, never overrides it.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The Freight Forwarding App serves external partner organizations — roughly a hundred of them, making thousands of changes a day to schedules, routes, documents and permissions. When a partner disputes "who changed this?", the answer has to come from an immutable trail — and the security organization's compliance bar ("important events") was deliberately vague. **Task.** Define the audit-log product: what gets captured, at what standard, phased how — so compliance is a launch feature, not an incident response. **Action.** I wrote the PRD: an immutable who/what/when record; nine event categories — logins/logouts, work-item changes, route updates, messages, document uploads/downloads, schedule updates, exception changes, assignment-rule changes, permission changes — each held to a written 100%-captured bar; priorities P0/P1/P2 per category; delivery phased Alpha (a login/logout dashboard on the log platform, Q4 2022) → Beta → MVP (Q1 2023); and the vague-bar risk mitigated structurally — PM buy-in plus CISO sign-off written into the plan, the "trusted date" for audits recorded in a governed document, and the eight dependent FFA teams mapped before build. **Result.** The capture standard was defined, prioritized and signed before build: 100% of nine categories, a phased path that did not pretend everything could be captured on day one, and zero ambiguity about who approved the scope. RECORD-FROM-DOCS

COMPRESSED STAR

- **S —** A partner-facing platform with ~100 partner orgs and thousands of changes a day; no immutable trail; a compliance bar stated vaguely enough to move.
- **T —** Own the audit-logs PRD: taxonomy, capture standard, phasing, and the sign-off chain that makes the bar stick.
- **A —** Nine event categories at a written 100%-captured bar with P0/P1/P2 priorities; phased Alpha → Beta → MVP so capture breadth grew honestly; CISO sign-off and PM buy-in as the mitigation for the vague "important events" definition; the "trusted date" for audits recorded in a governed document; eight dependent teams mapped.
- **R —** Capture standard defined and signed before build (record-from-docs). The record carries no post-launch compliance dollar, and none is claimed.

RESULTS LINE

Nine event categories at a 100%-captured bar (record-from-docs) · ~100 partner orgs · thousands of changes/day (record-from-docs — platform-wide; never merge with canon's 53-partner import migration set) · phased Alpha (Q4 2022) → Beta → MVP (Q1 2023) with P0/P1/P2 (record-from-docs) · CISO sign-off as the written mitigation (record-from-docs) · no post-launch dollar in the record — never invent one.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Audit coverage	`0 of 9 event categories captured — no dashboard, no tool, no store`	PRD problem statement (record-from-docs)	"who changed this data?" unanswerable
2	Organizations affected	`≈100 partner organizations on the app`	PRD hypotheses (record-from-docs)	every org's changes invisible
3	Change volume projected	`thousands of changes/day once all partners and internal operators are live`	PRD hypotheses (record-from-docs)	untracked writes at scale
4	Explicit non-goals	`reads/views, IP addresses, automated cascades — out of scope this iteration`	PRD scope section (record-from-docs)	scope held to writes by humans

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Capture bar	`9 of 9 categories at 100% capture, each with a sample query`	PRD metrics section (record-from-docs)	login/logout · work-item edits · route updates · messages · document up/downloads · schedule updates · exception changes · assignment-rule changes · permission changes
2	Event schema	`8 parameters per event = user FID + date + time + action type + object + field + previous value + current value`	PRD Epic 1 (record-from-docs)	every change fully described
3	Immutability	`deletion of captured events = 0 — by construction`	PRD acceptance criteria (record-from-docs)	the log cannot be rewritten
4	Phase arithmetic	`Alpha (Q4 2022): 1 category live — login/logout on a Sumo Logic dashboard → Beta (Q1 2023): 9 of 9 tracked with full parameters → MVP (Q1 2023): every product area publishes links + sample queries`	PRD launch phases (record-from-docs)	proof → coverage → discoverability
5	Trusted-start mechanism	`each team registers a target date in Confluence; history is trusted only from the latest start date`	PRD risk table (record-from-docs)	honest about partial history
6	Volume sizing	`2,000 changes/day × 365 = 730,000 events/yr; × 8 parameters = 5,840,000 stored values/yr`	"thousands/day" from Table 1 row 3 (record-from-docs); 2,000 as the low-end reading and the arithmetic are illustrative	≈0.73M immutable events/yr

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Categories captured at 100%	0 of 9	9 of 9 — the stated success metric	record-from-docs
2	Alpha exit	—	Sumo Logic dashboard with login/logout metrics	record-from-docs
3	MVP exit	—	all product areas linked, with sample queries	record-from-docs
4	Security sign-off	vague CISO requirements	CISO representatives sign the "important events" definition before development	record-from-docs

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Investigation deflection	`2 h × $60/h = $120 per "who changed this" hunt; × 100 hunts/yr = $12,000/yr`	every input illustrative — no rate or count exists in the record	≈$12K/yr, illustrative only
2	Compliance exposure	`no dollar in the record — not claimed`	record-from-docs	not claimed
3	Post-launch dollar	`none in the record — never invent one`	record-from-docs	not claimed

TWO PROBES

Why 100% capture as the bar — isn't that over-engineering?

Because an audit trail with holes is worse than none — it lends false confidence to a dispute answer. The 100% bar is per category, phased: Alpha proved the pipeline on the highest-volume, simplest events (logins), then breadth grew by priority. The bar was written down precisely so "good enough" could not drift downward under delivery pressure.

What did the vague "important events" definition actually risk?

Scope that moves after build — the classic compliance trap where the auditor's definition surfaces at the audit, not at the PRD. The mitigation was structural, not heroic: PM buy-in and CISO sign-off on the taxonomy up front, so the definition of "important" was the signed document, not someone's memory. When the bar can move, pin it to a signature.

Canon numbers (basis labels): nine categories at 100% capture (record-from-docs) · ~100 partner orgs · thousands of changes/day (record-from-docs) · Alpha Q4 2022 → MVP Q1 2023 (record-from-docs) · CISO sign-off + "trusted date" in a governed document (record-from-docs). Never: merge ~100 partner orgs with S12's 53-partner migration set; never claim a prevented-breach dollar (none in the record).

VERIFY BEFORE YOU SAY IT — every tagged item NEW

Every VERIFY-WITH-RAJESH tag in the new stories, in one place. For each: confirm it from your own memory or the source, or drop it. Anything left unconfirmed should be said as “illustrative” or not said.

WHERE	ITEM TO CONFIRM	IF YOU CANNOT CONFIRM
S1-a	Ownership pairings for PickRes, MerchRes and LostFound — the master names the feeds (OMS, WMS, planning) but not every bucket-to-system pairing; “ten buckets” is my count of the named fields including derived OOB and Basis.	Confirm or say “ten fields.”
S1-a	The 25% event-level share of S1’s oversell row ($225,000) and 5,000 SKUs × 10 nodes = 50,000 rows.	Drop both unless you have counts.
S3-a	Rounding mode (half-up) and line-order key for “the last line absorbs”; 2,000 postings a day; the 30% share of bridge hours ($42,432); who proposed a rounding account (none claimed).	Say “the rule is last line absorbs”; keep the rest illustrative.
S1-b	Whether AFS Correction needs a second approver (maker-checker); the rehearsal list for control owners; 40 negative rows × 6 min and 10 drift rows × 15 min; 1,800 productive hours.	Say “screen, role, audit row” only.
S1-c	The 150-unit hold, 60% sell-through, 200 holds a year; the exact wording and tolerance of the four reserve-copy invariants; the 20,000 / 1,400 old-vs-new row counts; whether the FSD row carries a review trigger.	Keep as worked example, labeled illustrative.
S1-d	The 12,000 events an hour; the number of clean days N; “sync” = events applied ÷ received (my reading); the order-flow budget = the 60-second publish budget (my reading); the snapshot at which 85.4% was true; whether MCS run cost stops at cutover; whether any forum disagreed; whether a breach rehearsal was run.	Say the two trigger thresholds only; derive nothing else.
S1-e	All S25 figures (17, two usages, $42K, $25K, six core files plus the supply-chain family) come from the loop packet, not the master canon; the study’s cost; whether the $25K was ever spent; how many of the 17 were in scope. LOOP-CONSISTENCY: the loop packet lists S25 as a Bar Raiser backup.	Confirm figures; do not use if spent with the Bar Raiser.
Router	Hire and Develop the Best and Strive to be Earth’s Best Employer are thin; there is no manager-disagreement, missed-deadline or mentoring story in the S/4 record.	Keep answers short and honest; pivot to the nearest real story.
Drills	The cut orders and tripwires are my judgment from the record’s gates, not events.	Say “I would,” never “I did.”
Bridge	Inbound numbers (7,000, 6,880, 99.0%, 98.0%, 6.76) are the booklet’s illustrative week, not Amazon data.	Always say “in a worked example.”

NUMBERS I MUST NOT GET WRONG — with basis labels NEW

The canon and worked numbers used in this packet. Read the basis column aloud in your head before each one.

NUMBER	WHAT IT IS	BASIS	STORY
≈$2.5M/yr	$1.0M MCS run cost retired + ≈$1.5M modeled	user anchor + modeled	S1
range $1.7–5.2M	low / high of the same model; upside shrink ≈$0.3M + time-to-market ≈$0.2M excluded	modeled	S1
FSD 3/5/25 → 5/16/25 → signed 6/5/25	three signatory orgs	record	S1
E2E 11/22 · 12/20 · 1/30	28 then 41 days apart; January planning and May execution go-lives	record	S1, S1-d
836 rows / 85.4% / 52.8%	Blocker 164 + Critical 277 = 441 of 836	record (program-level)	S1-d
sync < 99% for 4 h · p95 > 2× for 1 h	armed both go-lives, never fired	record	S1, S1-d
FOH truths 2 → 1	AFS decrement at sale/ship; match ≥ 99.5% in 24 h	record / design KPI	S3
≈$0.6M/yr	cancels $0.45M + labor $0.14M + close $0.05M; range $0.25–1.2M	modeled	S3
560 = 980 − 420	worked SKU; raw S/4 over-promises 440, raw WMS 420	illustrative units	S1, S1-a
555 · 550 · 550 · 565	five-unit order and three bad events	illustrative units	S1-a
$100.00 = 33.33 + 33.33 + 33.34	last line absorbs the cent	record rule; amounts illustrative	S3-a
−20 vs +20	lower-of-two vs raw S/4 on S/4 500, WMS 460, committed 480	illustrative units	S1-b
61% = 30 ÷ 49	break-even release reliability for promising held stock	benchmark inputs	S1-c, S1 trap
$42K · $25K · 17 · 2	funded · contingent · interfaces · critical usages	record (loop packet)	S1-e
6.5 of 7.2 hours	one reader’s daily workload on illustrative volumes	illustrative	S1-b

INTERVIEW-PREP PACKET 5 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS · V2

Amazon NACF Supply Chain loop — Packet 5: Venkata Josyula

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

Venkata Josyula

Principal Product Manager, FBA — treat as BAR RAISER

Thursday, October 30, 2026 · 12:00 pm PDT · 55-minute video interview

> **Strategy in one line.** Treat as the Bar Raiser: independent of the hiring team, holding a veto, probing LP depth to the fifth why and consistency across every story and every interviewer's notes. Six primary stories across four employers — S14 (Manhattan: the $2.5M order-value discrepancy, transparency over a quiet fix), S16 (Flexport: four systems → one pipeline; the security-review miss owned upward), S10 (Elementum: Austria — the data said measurement), S15 (Sephora: the gate handed to the skeptic), S13 (Manhattan: Cyber Monday, 4 AM) and S9 (Elementum: a thousand lanes asked, 597 first) — so any of the sixteen LPs has a story; plus the three real failure arcs (G1–G3). Answer the LP asked, then stop; keep facts identical when the LP switches; every dollar with its basis label.

LP focus — my guess, not confirmed: Earn Trust · Ownership or Hire & Develop the Best · Are Right, A Lot or Broad Responsibility. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title and the JD in "Read me first" and should be held loosely. No interviewer biography is asserted anywhere in this packet — only title-based inference, labeled as such.

Bar-raiser lens: LP depth + I-versus-we + data honesty + consistency. Start with the 5-minute cram sheet, then the Question → Story Router (all 16 LPs, real AIQB wording) and the Bar-Raiser Self-Audit; the Trap Drills and the probes grid carry the classic bar-raiser follow-ups; the failure arcs carry "tell me about a time it failed".

Canon: amazon/amazon-l6-scm-master-packet.html (195 pp; the 131-page cb91 copy agrees on every headline number used), Apple story workbook v11 and the source documents (precedence: documents, v11, master). The SOURCE TRACE at the back lists every quantitative claim with file and page. Every number below is canon; every dollar carries its basis label (record · user anchor · modeled · priced).

Contents NEW

Every NEW or changed section carries a pill: NEW added in v2 · ELEVATED an existing story deepened with LP proof lines, the five facets, an impact model, a behavioral pack, three angles, trap drills and a 20-second bridge · UPDATED existing section extended. The original 46-page content is kept.

Front

5-minute cram sheet NEW 3

Question → Story Router — all 16 LPs (AIQB wording) + common prompts NEW 4

Bar-Raiser Self-Audit — strongest evidence line per LP NEW 6

Read me first — lens, LP focus, answer strategy 7

LP-coverage grid (six stories) and consistency rules UPDATED 9

The six primaries

Story index and order of preference UPDATED 10

S14 — Manhattan — the $2.5M order-value discrepancy (Earn Trust) ELEVATED 11

S16 — Flexport — four systems → one pipeline; the security-review miss (Ownership) ELEVATED 18

S10 — Elementum — Austria: the data said measurement (Are Right) ELEVATED 25

S15 — Sephora — the gate handed to the skeptic (Hire & Develop) ELEVATED 32

S13 — Manhattan — Cyber Monday, 4 AM (Bias for Action) NEW 38

S9 — Elementum — a thousand lanes asked, 597 first (Backbone / Think Big) NEW 44

Bar-raiser back matter

Top-impact re-cuts — S11, S12, S18, S2 NEW 50

Classic bar-raiser probes × six stories NEW 52

Failure arcs — G1, G3, G2 (and M1) NEW 53

Question bank — likely questions, one-line plans UPDATED 54

Backup bench and design cases — S19, S20 (design cases), RAPID Crux, FFA Asia latency UPDATED 56

NUMBERS I MUST NOT GET WRONG — canon values and basis labels UPDATED 64

VERIFY BEFORE YOU SAY IT — confirm list, source conflicts, my inputs NEW 66

SOURCE TRACE — claim, file, page, recon status; arithmetic register NEW 67

5-MINUTE CRAM SHEET NEW

Profile hooks (inference from title and slot — not a biography). Venkata Josyula — Principal Product Manager, FBA. Bar Raiser: independent of the hiring team, holds the bar at "better than half the people at this level", holds a veto. Thursday, October 30, 2026 · 12:00 pm PDT · 55 minutes. He probes LP depth to the fifth why, asks *what did YOU do versus the team*, wants the number that proves it, wants the failure and what changed, and compares notes with the other four interviewers.

THE SIX STORIES, IN ORDER OF PREFERENCE — AND THE ONE NUMBER EACH LANDS

#	STORY	COMPANY	LPS IT CARRIES	LAND THIS NUMBER
1	**S14** — the $2.5M order-value discrepancy I would not fix quietly	Manhattan	Earn Trust · Backbone · Customer Obsession	~200 orders / $2.5M of order value → 0 (user anchor); $10M won — never savings
2	**S16** — four systems → one pipeline; the security-review miss	Flexport	Ownership · I&S · Learn · Frugality	Accuracy 60% → 90% (record); 2 weeks lost, no review skipped; >$100K priced (BRD $103,680), never saved
3	**S10** — the data overturned the explanation; Austria → 25 sites	Elementum	Are Right · Dive Deep · Earn Trust · Broad Resp.	DQ ~55% true → ~99% within a quarter; Austria ≈$500K/yr → ~$8M across 25 locations (record)
4	**S15** — the gate handed to the skeptic	Sephora	Hire & Develop · Earth's Best Employer · Highest Standards	Critical defects −50% (record); six- → four-week cadence
5	**S13** — Cyber Monday, 4 AM NEW	Manhattan	Bias for Action · Customer Obsession · Dive Deep · Deliver Results	99% fill vs 95% goal; ~$6M on the line → ~$8M of the day's sales protected (record)
6	**S9** — 1,000 lanes asked, 597 first NEW	Elementum	Backbone · Think Big · Highest Standards · Frugality	Completeness 60% → 95% (record); 597 lanes first = 80% of spend and adoption 8× (user anchors)
F	**G1 / G3 / G2** — the real failure arcs NEW	Elementum / Flexport / Sephora	Are Right · Earn Trust · Ownership	G1: 5 users × 10 scenarios passed, 2 days from sign-off · G3: 2 weeks · G2: 1.5 weeks

THREE SENTENCES TO OPEN WITH

> **1.** "For fifteen years I have run supply-chain programs the same way: find the number that is broken, prove it with data I pull myself, and gate the work until that number closes."
>
> **2.** "I will be specific about what I did versus what the team did, and I will tell you for each number whether it is on the record, my own tracking, modeled or priced."
>
> **3.** "The part of every story I care most about is what I would do differently — so please ask me that on any of them."

THE FIVE NUMBERS NEVER TO GET WRONG

1. **7 → 68 customers = 9.7×** (Flexport, S11) — never "50×", never "10×".
2. **~200 orders / $2.5M of order value → 0** (S14, user anchor) and the **$10M won — never counted as savings**; the ≈$1.5M is illustrative expected loss avoided.
3. **>$100K a year priced, never saved** (S16) — the BRD priced it at **$103,680** on thirty hours a week of WBR work at an assumed ninety dollars an hour; my hours/decision-risk split is a model; the miss was **2 weeks**.
4. **~55% true data quality → ~99% within a quarter**, **Austria ≈$500K/yr → ~$8M across 25 locations** (S10) — never a month count I cannot source, never "$1M", never "400+ lanes".
5. **99% fill vs a 95% goal; about $6M on the line when the page fired, about $8M of the day's sales protected** (S13, record) — the $8M means "protected" only — and "**six sites**", never "six NA sites".

THE SIX BAR-RAISER PROBES — ANSWERS LIVE IN THE TRAP DRILLS OF EVERY STORY

(1) What was *your* contribution — I vs we · (2) What would you do differently · (3) What is the data that proves it · (4) Tell me about a time it failed · (5) Why was it the right long-term decision, not the short-term one · (6) Who disagreed with you, and were they right. One-page matrix: "Classic bar-raiser probes × six stories", back of the packet.

> **Route, do not improvise.** If the question does not map in two seconds: Router → LP row → opener → number → stop. If he asks for another example on a spent LP: backup column. If he asks for a failure: G1 first, never a disguised win.

QUESTION → STORY ROUTER — all 16 Leadership Principles NEW

One question, one story. Find the LP, read the AIQB question number, say the opener, land the number, stop. If the primary was already used earlier in the loop (or with this interviewer), take the backup — the S-codes are canon story codes; **S9, S10, S13–S16** are developed in full in this packet, **G1–G3** are the failure arcs, and pointers (S11, S12, S18) carry master-canon numbers only. Question wording is abbreviated from the Amazon Interview Question Bank (AIQB).

LP	IF THE QUESTION IS… (AIQB: THIS LP, Q#)	PRIMARY STORY · ANGLE	BACKUP IF SPENT	15-SECOND OPENER	LAND THIS NUMBER
Customer Obsession	AIQB Q1 "difficult interaction with a customer" · Q7 "push back or say no" · Q8 "balance the customer with the business"	**S14** — the customer told first; they set the patch priority (Q1)	S13 — promise-at-risk ranked first; the price named (Q8)	*"Two days before an executive review, my customer's order and warehouse systems disagreed on about two hundred orders — $2.5 million of order value — and I decided they would hear it from me first."*	**~200 orders / $2.5M of order value → 0 (user anchor)**
Ownership	AIQB Q1 "outside your area" · Q2 "wouldn't meet a commitment" · Q3 "initiative nobody owned" · Q4 "short-term sacrifice for long-term value"	**S16** — nobody owned the Monday number; I wrote myself out of it (Q3); the two-week miss owned the same day (Q2)	S14 — the OMS–WMS seam (Q1) · S9 — the long tail held behind the same gate (Q4)	*"A weekly review of a visibility program ran on a hand-stitched sheet only I could fix. Nobody owned the number, so I did — and then I wrote myself out of it."*	**Accuracy 60% → 90% (record); no longer the single point of failure**
Invent and Simplify	AIQB Q1 "complex problem, simple solution" · Q3 "simpler for customers" · Q5 "usual approach wouldn't address"	**S16** — four systems → one pipeline; seven definitions frozen; parity gate	S9 — "the standard was the data, not the transport" (XML/portal floor)	*"I replaced a spreadsheet stitched from four systems with one pipeline — and the simplification was seven frozen metric definitions, not the tool."*	**7 metrics; parity within 1% (record)**
Are Right, A Lot	AIQB Q1 "not enough data" · Q4 "bad decision" · Q5 "error in judgment" · Q6 "your idea was not the best"	**S10** — I measured the room's explanation instead of arguing it (Q6, Q1)	S15 — went to QA first (Q6) · Q4/Q5 failures → G1, G3 (failure page)	*"Everyone believed the carriers were late. I pulled three years of shipments in SQL and found our own data quality was about 55%, not the 40% we reported."*	**40% reported → ~55% true (record) → ~99% within a quarter**
Learn and Be Curious	AIQB Q1 "deeper subject-matter expertise" · Q2 "outside your comfort area" · Q3 "didn't know what to do next"	**S16** — learned enough of the data stack to write the BRD infra built from (Q1, Q2)	S18 pointer — never ran a WMS migration; learned the floor in month one (Q3)	*"I had never written requirements for a data pipeline. I learned the stack well enough to write the BRD the infra team built to — and I was explicit about what I did not build."*	**60% → 90% (S16); mocks 7% → 1.8% → 0.3% (S18)**
Hire and Develop the Best	AIQB Q1 "helped a team member develop their career" · Q4 "provided coaching"	**S15** — the QA lead authored and defended the gate	S10 — regional super-users ran the sites after two shadowed reviews	*"I gave the quality gate of a twenty-person team to its most skeptical member — the QA lead — and measured her development by whether the gate survived my leaving."*	**Critical defects −50% (record); six- → four-week cadence**
Insist on the Highest Standards	AIQB Q1 "refused to compromise" · Q4 "standards vs delivery" · Q6 "continuous improvement project you led"	**S9** — a cohort held twice at ≥95% valid ASN (Q1, Q4)	S15 — the gate that can say no (Q6) · S14 — standing reconciliation	*"One cohort failed my ninety-five percent valid-ASN gate twice on a timestamp format with a date waiting. I held it both times, because the gate was published before cohort one."*	**≥95% valid-ASN gate (record: resume); held twice (user anchor)**
Think Big	AIQB Q1 "opportunity much bigger" · Q2 "changed a function's direction" · Q4 "drove adoption for your vision"	**S9** — a one-time ask turned into a standard, a gate and a scorecard (Q4)	S12 pointer — flipped the 18-month roadmap · S10 — Austria → 25 locations	*"I turned a thousand-lane ask into a standard, a gate and a scorecard — the 597 lanes carrying eighty percent of spend went first, and the method became the sourcing SOP."*	**Completeness 60% → 95% (record); 597 lanes first = 80% of spend and adoption 8× (user anchors)**
Bias for Action	AIQB Q1 "calculated risk, speed critical" · Q2 "tight deadlines, not all options" · Q4 "respond immediately"	**S13** — declared the incident at 4 AM before the cause was known (Q1)	S14 — two days; a census in a morning (Q2)	*"At four a.m. on Cyber Monday the allocation queue was aging with inventory idle and I did not know why. I acted before the cause was known — reversibly, logged, and checked with engineering."*	**99% fill vs 95% goal (record)**

QUESTION → STORY ROUTER (continued) NEW

LP	IF THE QUESTION IS… (AIQB: THIS LP, Q#)	PRIMARY STORY · ANGLE	BACKUP IF SPENT	15-SECOND OPENER	LAND THIS NUMBER
Frugality	AIQB Q1 "save costs / eliminate waste" · Q5 "creative solution without resources" · Q6 "requested funding"	**S10** — executed savings only; Finance co-owned the method first (Q1)	S9 — one harness, later cohorts mostly configuration (Q5) · S16 — priced >$100K (BRD $103,680), never saved (Q6)	*"I signed the savings method with Finance before I claimed a dollar — contracted guide rate minus paid rate, per shipment, executed shipments only."*	**Austria ≈$500K/yr → ~$8M across 25 locations (record) · S16 >$100K priced, not saved**
Earn Trust	AIQB Q1 "communicate a change people would have concerns with" · Q4 "not able to meet a commitment" · Q6 "uncovered a significant problem"	**S14** — refused the quiet fix; the cost put on my own calendar (Q1, Q6)	S16 — the miss owned upward the same day (Q4)	*"I was asked to fix it quietly. I said a silent correction is a loan against trust at a terrible interest rate — and offered to run the war room myself."*	**$2.5M of order value (user anchor); ≈$1.5M expected corruption loss avoided (illustrative; not a saving)**
Dive Deep	AIQB Q2 "dig deep to the root cause" · Q3 "in-depth analysis" · Q6 "a metric you created"	**S10** — every blank classified by defect class and burned down weekly (Q2, Q3)	S13 — the contradiction: inventory present, backorders rising · S14 — the match-rate metric (Q6)	*"I classified every blank in three years of shipments — missing rate, wrong-grain lane, one carrier in five spellings, brochure transit times — and burned each class down weekly."*	**~55% → ~99% within a quarter (record)**
Have Backbone; Disagree and Commit	AIQB Q1 "strongly disagreed with your manager or peer" · Q2 "unpopular stance" · Q3 "committed despite disagreeing"	**S9** — 597 first vs all 1,000 at once, taken two levels up; then I committed to the rest (Q2, Q3)	S14 — the quiet fix, privately, with a census (Q1)	*"An executive asked for a thousand lanes in two months on sixty-percent-complete data. I said that was the wrong target and took the spend curve two levels up."*	**597 of 1,000 lanes first = 80% of spend (user anchor)**
Deliver Results	AIQB Q1 "tight deadline" · Q2 "unanticipated obstacles" · Q3 "exceeded expectations"	**S13** — 99% against a 95% goal on the worst night of the year (Q2, Q3)	S9 — 597 lanes first, the rest behind the same gate; completeness 60% → 95% (Q1)	*"The fill-rate goal for Cyber Monday was 95%. We delivered 99%. About six million was on the line when the page fired; about eight million of the day's sales were protected."*	**99% vs 95% goal · ~$6M on the line → ~$8M protected (record)**
Strive to be Earth's Best Employer	AIQB Q3 "change to improve coworkers' work experience" · Q6 "supported or empowered someone"	**S15** — QA from blamed afterthought to owner of the standard; Sev-1s by cause, not blame (Q3, Q6)	S16 — wrote myself out as the single point of failure	*"On that team QA was the blamed afterthought. I made her the owner of the standard, classified every Sev-1 by cause rather than blame, and made overload visible so people got help."*	**Critical defects −50% (record); the drop-date rule**
Success and Scale Bring Broad Responsibility	AIQB Q8 "left something better than you found it" · Q5 "everyone affected" · Q2 "changed a decision for unintended consequences"	**S9** — the XML/portal floor kept small carriers inside a strict standard (Q5)	S10 — one site's fix became 25 sites' instrument; a no-hit pages a human (Q8)	*"A strict data standard prices out small carriers. I made the standard the data, not the transport — an XML or portal floor — so the long tail stayed inside it."*	**≥95% valid-ASN gate with a portal floor; Austria → 25 locations (record)**

COMMON NON-LP PROMPTS → STORY NEW

IF THEY ASK…	PRIMARY · ANGLE	BACKUP	NUMBER
"Tell me about a time you failed."	**G1** — the Elementum UAT that crashed (the criteria I never wrote)	G3 (S16 — the security review; 2 weeks) · G2 (Sephora RF-device script; 1.5 weeks)	**5 users × 10 scenarios passed; sign-off 2 days out**
"Disagreed with your manager."	**S14** — the quiet fix; census in hand; the war room on my calendar	S12 pointer — the roadmap flip, privately first	**break-even P* 0.43% (illustrative)**
"Influence without authority."	**S15** — no line authority; the gate handed to the skeptic	S9 — two levels above my sponsor	**critical defects −50% (record)**
"Reduced cost."	**S10** — executed-only savings, Finance-co-owned	S12 pointer (~$5M avoided, said once) · S16 priced, never saved	**Austria ≈$500K/yr → ~$8M across 25 locations (record)**
"Dealt with ambiguity."	**S13** — no cause, a VP on the phone, a rule on the log	S16 — nobody owned the number	**99% vs 95%**
"Simplified a process."	**S16** — four systems → one pipeline	S9 — standard = the data, not the transport	**7 metrics; parity ≤1%**
"Decision with incomplete data."	**S13** — triage on promise-at-risk before the cause	S14 — sample census in one morning	**~$6M on the line → ~$8M protected**
"Missed a deadline."	**G2** — training slipped 1.5 weeks; the program did not	S16 — two weeks lost, no Monday review skipped	**mocks 7% → 1.8% → 0.3%**
"Mentored someone."	**S15** — the QA lead	S10 — super-users; the rules file	**gate survived my leaving**
"Delivered under a tight timeline."	**S14** — two days to the review	S13 — hours · S9 — 597 lanes first	**~200 orders / $2.5M of order value → 0 (user anchor)**
"Changed your mind with data."	**S10** — carriers were not late; we could not measure	S15 — go to QA first; pull the real baseline	**40% → ~55% true**

BAR RAISER SELF-AUDIT — the strongest evidence line for each of the 16 LPs NEW

Read this the night before. For each LP: the one line I would stake the answer on, the number that carries it, and the softest part he is most likely to press — so I name it before he does. Every number is canon with its basis label.

LP	STORY	STRONGEST EVIDENCE LINE	NUMBER (BASIS)	SOFT SPOT TO NAME FIRST
Customer Obsession	S14	Told the customer first; they set the patch priority; before/after evidence on every order.	**~200 orders / $2.5M of order value → 0 (user anchor)**	No satisfaction score exists; the evidence is order-level and is my own program tracking — not in the workbook. The $10M is won, never saved.
Ownership	S16	Owned a number nobody owned, wrote myself out as its single point of failure, and owned the two-week miss the same day.	**Accuracy 60% → 90%; the two-week miss absorbed, no review skipped (record)**	I did not build the pipeline — infra did; "no longer a single point of failure" is the outcome, and there is no count after "one".
Invent and Simplify	S16	Four systems and a hand-stitched sheet became one pipeline; seven definitions frozen; parity gate.	**7 metrics; parity within 1% (record)**	The ticket cross-reference is a bridge; Support owned the real key.
Are Right, A Lot	S10	Measured the room's explanation instead of arguing it: true data quality ~55%, not 40%.	**40% → ~55% true → ~99% within a quarter (record)**	I claimed measurement became enforceable — not that carriers improved.
Learn and Be Curious	S16	Learned enough of the data stack to write the BRD infra built from.	**Accuracy 60% → 90% (record)**	Learning, not building: the code was infra's.
Hire and Develop the Best	S15	The QA lead authored and defended the gate — by cycle two and after I left.	**Critical defects −50% (record); six- → four-week cadence**	No direct-report headcount; I coached by influence. The +50% releases is cadence arithmetic and the $0.13M is modeled.
Insist on the Highest Standards	S9	Held a failing cohort twice at ≥95% valid ASN, with the date waiting.	**≥95% valid-ASN gate (record: resume); held twice (user anchor)**	Cohort sizes and counts are not in the record — never quoted; the twice-held cohort is my program tracking.
Think Big	S9	A one-time 1,000-lane ask became a standard, a gate and a scorecard; the method became the sourcing SOP.	**Completeness 60% → 95% (record); adoption 8× (user anchor)**	The 8× is my own tracking, not in the workbook; the ≈$2M of freight is as measured in six months — never annualized, never added to another $2M.
Bias for Action	S13	Declared the incident at 4 AM before the cause was known; reversible, timestamped, checked with engineering.	**99% fill vs 95% goal (record)**	The counterfactual is not provable; engineering fixed the scheduler.
Frugality	S10	Finance co-owned the savings method before a dollar was claimed; executed shipments only.	**~$8M across 25 locations, executed-only (record)**	S16's >$100K is priced (BRD $103,680), never claimed; ≈$0.5M analyst capacity is modeled.
Earn Trust	S14	Refused the quiet fix and put the cost of the honest path on my own calendar.	**$2.5M of order value (user anchor); P* 0.43% (illustrative)**	≈$1.5M expected corruption loss is avoided risk — not a saving; the $2.5M is order value, not savings.
Dive Deep	S10	Classified every blank by defect class and burned each down weekly by reason code.	**~55% → ~99% within a quarter (record)**	Per-class counts are not in the record; the exact month count (two vs three) is not asserted.
Have Backbone; Disagree and Commit	S9	Took the 597-first phasing for a thousand-lane ask two levels up with the spend curve, then committed to delivering the rest.	**597 of 1,000 lanes first = 80% of spend (user anchor)**	He accepted the phasing; I never say he was wrong. A thousand lanes were asked; 597 went first — never "400+ lanes".
Deliver Results	S13	99% fill against a 95% goal on Cyber Monday; promises kept in the right order.	**~$6M on the line → ~$8M of the day's sales protected (record)**	The $8M is only ever "protected" — never "at risk" (that is the ~$6M) and never a saving.
Strive to be Earth's Best Employer	S15	Sev-1s classified by cause, not blame; the drop-date rule made overload visible so people got help.	**Critical defects −50% (record)**	No HR or comp-cycle experience claimed; the drop-date rule is a mechanism, not a number.
Success and Scale Bring Broad Responsibility	S9	The XML/portal floor kept small carriers inside a strict standard; S10's no-hit pages a human.	**≥95% valid-ASN gate (record: resume); Austria → 25 locations (record)**	Small-carrier counts are not in the record.

Audit totals. 16 of 16 LPs carry a Y-story with a labeled number: 13 have a record number (Apple v11, the resume or a source document); 3 — Customer Obsession and Earn Trust (S14) and Have Backbone (S9) — rest on user anchors, which I say as "my own tracking". S9 carries four LPs (the heaviest), S10 and S16 three each, S13, S14 and S15 two each — if an LP is asked twice in the loop, or two LPs on the same story, switch to the backup in the Router and keep the number identical.

Read me first

The interviewer's lens (inference from title + JD — not a biography)

Inference from the title and the slot. A Principal Product Manager from FBA — outside the hiring team, the last interview of the loop — is the profile of a Bar Raiser: an interviewer trained to be independent of the hiring manager, to hold the bar at "better than half the people at this level," and to hold a veto. Bar raisers are less interested in the supply-chain content than in *how the candidate thinks and whether the stories are consistent*: they drill one story to the fifth "why," they switch LPs mid-story to see whether the candidate re-tells the same facts honestly from a different angle, and they compare numbers across stories and across interviewers' notes. A PM from FBA additionally brings a seller/customer lens — who was the customer, what did they experience, what did you give up for them.

What that means for the answers. Six primaries across four employers, so the bar raiser can pick any of the sixteen LPs and still see range and the same voice across fifteen years: S14 (Manhattan — the $2.5M order-value discrepancy, transparency over a quiet fix), S16 (Flexport — four systems into one pipeline, the security-review miss owned upward the same day), S10 (Elementum — Austria: "the noise said carriers; the data said measurement," one site's fix rolled to 25), S15 (Sephora — a 20-person team moved to a four-week cadence by handing the gate to the skeptic), S13 (Manhattan — Cyber Monday at 4 AM: a reversible, logged call before the cause was known) and S9 (Elementum — a thousand lanes asked, 597 first, taken two levels up). Each is told business-first and ends on a number with a basis label; the failure arcs G1–G3 are prepared on their own page, because bar raisers probe for lessons. The Router and the Self-Audit at the front say which story answers which LP. The LP-coverage grid and the consistency-rules box below exist because this interviewer will test exactly those two things.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title and the JD; if the first question lands on a different LP, the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Earn Trust (the flagship)	Bar raisers almost always carry a trust/integrity LP, and they want the expensive version — the customer told first, the cost priced onto the candidate. S14 is the master map's Earn Trust primary; S16's same-day upward ownership of the miss is the recent echo.	S14 (open) · S16 (owned the miss upward)
Ownership / Hire & Develop the Best (one of the two)	Bar raisers take LPs the HM did not assign; Ownership (S16 — writing a single-point-of-failure system out of my own hands; owning the scoping miss) and Hire & Develop (S15 — the QA lead who now owns the gate) are both primaries in the master map and both unassigned elsewhere in this loop.	S16 (Ownership) · S15 (Hire & Develop / Earth's Best Employer)
Are Right, A Lot / Success & Scale Bring Broad Responsibility (one of the two)	Judgment under a confident room, and what you do once you are right: S10 is primary for both in the master map — the data overturning the explanation everyone held, then one site's method becoming 25 sites' instrument with Finance's signature on the method before the number.	S10 (both) · backup: S16 (priced the status quo before build)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For the bar raiser specifically:** answer the LP asked, then stop — bar raisers fill their worksheet from probes, and over-answering gives them inconsistencies to find. When they switch LPs mid-story, keep the facts identical and change only the emphasis (the twist maps below are built for that). Volunteer the failure beat before they ask for it (S16's two weeks; S14's "what if he had said no").
- **Consistency is the bar:** every number must match what the other four interviewers heard. The consistency-rules box carries the never-say items that would show up as a discrepancy in the debrief: 9.7× not 50×; "six sites" not "six NA sites"; every dollar with its basis label; the $10M won, never saved; the >$100K priced, never claimed.
- **Customer lens (FBA):** name the customer in each story — the VP steering on a wrong number (S16), the QA lead and the release's end users (S15), the site and the carriers who could not dispute the scorecard (S10), the retail customer told first (S14); the shoppers whose promises were ranked first (S13); the executive who needed visibility and the small carriers who had to stay inside the standard (S9).

> **Loop-consistency rule.** One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets: S11/S12 (Tripathy) · S6/S8 (Hoerger) · S2/S18 (Goforth) · S1/S3 (Lan) — in this packet S11, S12, S18 and S2 appear only as re-cut pointers (one facet, master numbers, on request), never as the full telling. S16 is the same program family as S11 — if the bar raiser has read notes from the HM's interview, say so: "the review the HM

heard about is the one this pipeline replaced.". If pushed for "another example" on an LP, use the backup column of the Router or the backup bench; a re-cut pointer is the last resort, and then only the facet on its card.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

Bar-raiser instruments — LP coverage and consistency rules

LP-coverage grid — which Leadership Principles each of the six stories can serve UPDATED

Y = can open on this LP (the story is primary, backup or third for it in the master map, or the beat is explicit in the record) · **P** = a secondary angle, use only if the Y stories for that LP are spent · blank = do not stretch the story to it. **All sixteen LPs are covered by at least one Y across the six stories**; the Think Big gap of the four-story version is closed by S9 (a one-time thousand-lane ask turned into a standard, a gate and a scorecard). The Router at the front picks one primary and one backup per LP from this grid.

STORY	CO	OWN	I&S	ARL	L&C	H&D	IHS	TB	BFA	FRU	ET	DD	HB	DR	EBE	BR
S14 Manhattan — $2.5M	Y	Y							Y		Y		Y			P
S16 Flexport — pipeline		Y	Y		Y		P			Y	P	P		P		
S10 Elementum — Austria	P	P		Y			P			Y	Y	Y	P	P		Y
S15 Sephora — cadence			P			Y	Y				P		P	P	Y	
S13 Manhattan — Cyber Monday NEW	Y	P		P			P		Y			Y	P	Y		
S9 Elementum — 597 lanes NEW		P	P				Y	Y		Y		P	Y	P		Y

CO Customer Obsession · OWN Ownership · I&S Invent and Simplify · ARL Are Right, A Lot · L&C Learn and Be Curious · H&D Hire and Develop the Best · IHS Insist on the Highest Standards · TB Think Big · BfA Bias for Action · FRU Frugality · ET Earn Trust · DD Dive Deep · HB Have Backbone; Disagree and Commit · DR Deliver Results · EBE Strive to be Earth's Best Employer · BR Success and Scale Bring Broad Responsibility

Consistency-rules box — the data a bar raiser cross-checks across stories and interviewers

Say it the same way in every room.

- **9.7×, never 50× and never 10×.** Flexport customers 7 → 68 = 9.7× (S11, the HM's story). If the bar raiser quotes a multiple back, correct it gently to 9.7×.
- **"Six sites," never "six NA sites."** The Elementum manufacturer is six sites; the CM-ingest program (S7) is six sites, three waves, five months — no region prefix, no "500+ product families."
- **Every dollar carries its basis label** — record · user anchor · modeled · priced. In this packet: $2.5M of order value, S14 (user anchor) · $10M initiative, S14 (user anchor — won, never saved) · >$100K/yr (priced, never claimed) · $500K/yr Austria and ~$8M across 25 sites (record roll-up) · ≈$0.13M/yr (S15, modeled) · ≈$0.1M/yr (S14, modeled) · ≈$0.5M/yr Austria analyst capacity (modeled).
- **Three $2.5M figures exist in my canon and they are different things — attach the noun the first time I say each:** S14's $2.5M is the order value behind the OMS↔WMS discrepancy at Manhattan (user anchor); S1's ≈$2.5M/yr is the annual impact of Sephora's book-of-record work (user anchor + modeled, Lan's story); S18's ≈$2.5M a month is my division of the $20M pipeline over eight months (modeled, Goforth's story). Never let them blur.
- **Four ~$1.5M figures exist and only one is in this packet:** S14's ≈$1.5M is the *expected corruption loss avoided* — risk math, not money (modeled). The other three are other interviewers' and each carries its noun: S1's ≈$1.5M modeled (Lan); S8's ~$1.5M of inventory carrying cost released (Apple — state the carry rate, 20% in S8 versus 10% in S2/S6; Hoerger); S11's ~$1.5M pre-production figure (user anchor; Tripathy). Never "$1.5M to $2M."
- **System accuracy 60% → 90% is one number:** it is the S16 number (the review's data accuracy after the pipeline). S11 carries the same figure for the same program family, so I tell it in one story only (S16), say "system accuracy", and never call it ETA accuracy.
- **Failure stories are consistent:** S16 — the two-week security-review miss, owned upward the same day; S18 (Goforth) — the RF miss; S4 (Lan's pointer) — the rule I got wrong. The lesson sentences are fixed: "security, privacy and data-access reviews are program dependencies with lead times"; "any single-point dependency in front of an exit criterion gets a what-if plan"; "both sides of a rule get a number before it ships."
- **Voice:** program lead who owned definitions, rules, gates and signatures; engineers, SIs and vendor teams built; the only platform personally configured in canon is Manhattan DOM (S13/S14 era). Never "we" for an action that was mine; name the team's part once.
- **S13 and S9 numbers, said one way:** the ~$8M has one meaning — about eight million of the day's sales *protected* (record pair with the ~$6M on the line when the page fired) — never "at risk", never "GMV", never a saving. S9: a thousand lanes were asked; 597 first = 80% of spend (user anchor), the remainder behind the same gate; the ≈$2M of freight cost (user anchor) is as measured in six months, never annualized; never "400+ lanes".
- **S10 timing:** ~99% data quality *within a quarter* — true under Apple's two-month bar and the ~3-month canon; never "in two months", never "400+ lanes".
- **The three real failures have fixed lessons:** G1 (Elementum UAT) — "test at the customer's shape"; G3 (Flexport security review) — "planned, not discovered"; G2 (Sephora RF devices) — "effort is not impact". RF devices: say "hundreds", never "400 failed".
- **No invented numbers:** no direct-report headcount (S15), no realized dollars on the $10M initiative (S14), no claimed savings on the >$100K (S16), no Austria "$1M" or "99% in two months" (S10). "I don't have a canonical figure and won't invent one; what I can defend is…"

The stories — six primaries across four employers, business-first, canon numbers UPDATED

Order of preference if the bar raiser lets me choose: **S14** (Earn Trust) → **S16** (Ownership, with the failure beat) → **S10** (Are Right / Dive Deep / Frugality) → **S15** (Hire & Develop) → **S13** (Bias for Action) → **S9** (Backbone / Think Big). Each story: 60-second pitch → LP proof lines → Situation/Task → Action in gated phases → Results → L6 evidence → the five facets (data analysis · solution design · math & data logic · implementation · probe questions) → Probing → impact model and the four-table math → Twist map → Behavioral pack (four AIQB questions, four separate 90-second answers) → three angles → three Trap Drills → 20-second version and NACF bridge.

#	STORY	EMPLOYER	LPS IT CARRIES	HEADLINE (BASIS)	STATUS
1	**S14** — the OMS↔WMS discrepancy (~200 orders, $2.5M of order value)	Manhattan	Earn Trust · Backbone · Customer Obsession · Bias for Action	~200 orders / $2.5M of order value → 0 before the review (user anchor); $10M won — never savings	ELEVATED
2	**S16** — four systems → one pipeline; the security-review miss	Flexport	Ownership · Invent & Simplify · Learn · Frugality	Accuracy 60% → 90% (record); no longer a single point of failure; >$100K priced (BRD $103,680), never saved; 2 weeks lost	ELEVATED
3	**S10** — Austria: the data said measurement	Elementum	Are Right · Dive Deep · Earn Trust · Frugality · Broad Resp.	DQ ~55% true → ~99% within a quarter; $500K at Austria → ~$8M across 25 European locations (record)	ELEVATED
4	**S15** — the gate handed to the skeptic	Sephora	Hire & Develop · Earth's Best Employer · Highest Standards	Critical defects per release −50% (record: resume); six- → four-week cadence in six weeks	ELEVATED
5	**S13** — Cyber Monday, 4 AM	Manhattan	Bias for Action · Customer Obsession · Dive Deep · Deliver Results	Fill 99% vs 95% goal; ~$6M on the line → ~$8M of the day's sales protected (record pair)	NEW *promoted from bench*
6	**S9** — a thousand lanes asked, 597 first	Elementum	Backbone · Think Big · Highest Standards · Frugality · Broad Resp.	Completeness 60% → 95% (record); 597 lanes = 80% of spend, adoption 8×, ≈$2M of freight cost as measured (user anchors)	NEW *promoted from bench*

S14 — Manhattan Associates — the OMS↔WMS discrepancy (~200 orders, $2.5M of order value) my manager wanted to fix quietly: transparency as the reconciliation control, two days before an executive review ELEVATED

LPs this story can serve here: Earn Trust (primary — the flagship integrity story) · Have Backbone, Disagree & Commit (backup — refusing the quiet fix) · Ownership (the seam between OMS and WMS) · Bias for Action (third — two days to the review) · Customer Obsession (the customer told first and given the priority)
Process family: Deliver / Return-to-book — order-to-fulfillment reconciliation control; incident handling under a customer-relationship deadline

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF EACH NEW

Earn Trust	I briefed the customer's IT and finance leads on root cause, plan and timeline before the executive review — not in it — with a $10M initiative riding on the room, and attached the cost of that choice to my own calendar.
Have Backbone; Disagree & Commit	I told my manager privately, with a census in hand, that the quiet cross-table script was the riskier path; he accepted hesitantly; had he said no, I would have put the risk assessment in writing, recorded the decision with its owner and named the tripwire.
Customer Obsession	The customer set the priority: their own prioritized order list was patched first, per line, with before/after evidence for every order.
Bias for Action	Census in a morning, options memo by afternoon, customer pack and verification method on day two, patch before the review — and the permanent fix scheduled, not rushed.

Honesty label. The census, the risk argument, the customer pack and the war room were mine; the consultants traced the handshake, wrote and verified the per-line patch, and built the permanent fix. The $10M initiative is never counted as savings. **Basis:** the ~200 orders, the $2.5M of order value and the $10M initiative are a **user anchor** — my own record of this engagement; they are not in the Apple story workbook, the resume or a source document — so I say "my own tracking", never "the dashboard". Manhattan is the era in which I personally configured Manhattan DOM — the only platform configuration claim in canon.

SAFE-SAY BOX NEW

The most attackable number: ~200 orders / $2.5M of order value, and the $10M. Say: "About two hundred orders and two-point-five million dollars of order value, made consistent in the open. That is order value, not a saving — and the ten-million initiative was won at the review, which is a fact about the relationship, not a saving."

If he asks "what is your source for that?" "That is my own tracking of the engagement — the order list and the before-and-after evidence per order — not a company dashboard or a workbook figure. The number I would stand behind is: about two hundred orders to zero before the review, with evidence on every order."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Two days before an executive review at a Manhattan Associates retail customer, the order-management and warehouse-management systems did not reconcile — about two hundred orders, $2.5 million of order value — and a $10 million initiative rode on that review. My manager and the room favored a quiet back-end correction: a cross-table script, written in under two days, and nobody tells the customer. **Task.** I was the consultant who would run whichever path was chosen; I owned the recommendation and then the war room. **Action.** Facts before path: I built the discrepancy census by cause on a sample of the two hundred orders — which handshake step, which order states — and named the fear of losing the initiative as legitimate. Then I disagreed on risk, not principle: a script across related tables written in two days cannot be verified; an IT audit would turn a data issue into a reputation issue on a $10M relationship; the expected corruption loss from the silent path was about $1.5M on illustrative odds — risk, not money. I offered to run the war room personally so the cost of my position landed on me. He accepted, hesitantly. The customer got root cause, plan and timeline before the review — no apology-first — prioritized their own orders, asked for a per-line, reversible patch on that set with before-and-after evidence per order, and gave time for a clean fix to the handshake. **Result.** Primary metric: the OMS–WMS discrepancy — about 200 orders and $2.5M → zero on the customer's prioritized set before the review, then zero through the permanent fix; a standing reason-coded reconciliation report read before every executive review from then on. The $10M initiative was won — a fact about the relationship that I never count as savings. Lesson: silent corrections are loans against trust at a terrible interest rate.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Two days before an executive review, OMS and WMS didn't reconcile — ~200 orders / $2.5M — and a $10M initiative rode on the review. My manager and the room favored a quiet back-end correction.

The risks of the quiet path: a rushed cross-table script corrupting data (my own illustration: expected corruption loss ≈$1.5M, from 0.3 × 2 × $2.5M of order value), a near-certain audit finding turning a data issue into a reputation issue on a $10M relationship, and the customer surprised at the review. The customer in this story is the retailer's executives and the IT and finance teams who would otherwise find out from an audit.

T — I was the consultant who would run whichever path was chosen; I owned the recommendation and then the war room. Goal: the discrepancy to zero and a review the relationship could survive, without a silent correction.

Baselines on the table: OMS↔WMS discrepancy ~200 orders / $2.5M (target zero with before/after evidence per order) · new discrepancies unmonitored (target zero on a standing daily report) · two days to the review.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (day 1, morning)** — discrepancy census by cause (handshake step × order state) on a sample of the ~200; exposure statement (review in two days; $10M initiative); both options named honestly. *Exit bar:* the room agrees the facts before the path; the fear of losing the initiative named as legitimate.
- **Plan & Design (day 1, afternoon)** — options memo on risk, not principle: a two-day cross-table script is error-prone; an IT audit turns a data issue into a reputation issue; transparency = root cause + action plan + I run the war room. *Exit bar:* manager accepts, hesitantly; my labor attached to my position.
- **Build & Test (day 2)** — customer communication pack (root cause, plan, timeline — no apology-first); war-room plan (bridge, log, cadence); verification method for any patch (per line, reversible, before/after recorded). *Exit bar:* customer briefed before the review; war room staffed.
- **Cutover (review + patch)** — took the customer's prioritized order list; ran the bulk patch per line with verification; scheduled the clean root-cause fix. *Exit bar:* patch verified; discrepancy to zero on prioritized orders; review held with the facts on the table.
- **Hypercare / Close** — the permanent OMS–WMS handshake fix; the reason-coded reconciliation report as a standing control read before every executive review. *Exit bar:* $10M initiative won; handshake reconciles; new discrepancies = 0 on the daily report.

Influence and ownership. Mine: the census, the risk argument, the customer pack and the war room; the consultants traced the handshake, wrote and verified the per-line patch, and built the permanent fix. The argument was about risk, not virtue, and my labor was attached to my position. Had my manager said no, I would have documented the risk assessment in writing and asked for the decision recorded with its owner — disagree and commit, with the tripwire (any audit query on those orders) named; I would not have run the silent script without that record.

> **Designed the process; others own it now.** I designed the customer pack, the war room and the verification method; the standing reason-coded reconciliation is read before every executive review by the account team, not by me. **Trade-off put to leadership.** The recommendation to my manager outlined the competing metrics — two days of speed with a cross-table script vs the risk of an error-prone patch and a reputation issue — and attached my own labor to the transparent path.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~200 orders / $2.5M discrepancy → zero (user anchor); the $10M initiative won — a fact about the relationship, never counted as savings; ≈$0.1M/yr *modeled* from the standing reconciliation (range $55–160K); the ≈$1.5M expected corruption from the silent script is *avoided risk*, not a saving.
- **#** — ~200 orders patched per line, reversible, with before/after evidence per order, in the customer's priority order (user anchor).
- **Time** — two days to the executive review; the customer briefed before the review, not in it; the clean root-cause fix scheduled after (user anchor).
- **Customer** — the customer prioritized their own orders, asked for the bulk patch on that list, and gave time for a clean fix — the review held with the facts already on the table (user anchor).
- **Quality** — per-line reversibility — no bulk script across related tables; the standing reason-coded OMS↔WMS reconciliation read before every executive review; new discrepancies = 0 on the daily report (user anchor).
- **%** — no percentage claim exists in my notes for this story — say the counts and the control; do not invent a rate.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Two days to an executive review, a $10M initiative at stake, and a manager's preferred quiet fix. I ran the discrepancy census by cause (handshake step × order state) and named both options honestly before choosing the path.
Scope & influence	My manager persuaded on risk, not virtue; the customer's executives briefed before the review; the consultants traced the handshake and built the patch and the permanent fix — influence one level up and outward to the customer.
Execution trade-off + mechanism others now own	A two-day cross-table script (speed) vs a per-line reversible patch plus transparency (risk) — the recommendation was framed as risk, and my labor was attached to my position. Mechanism others own now: the reason-coded OMS↔WMS reconciliation read before every executive review.
Impact (long-term)	~200 orders / $2.5M discrepancy → zero; the $10M initiative won (a fact about the relationship, never counted as savings); new discrepancies = 0 on the daily report.
Process improvement (standard work institutionalized)	The verification method for any patch (per line, reversible, before/after recorded); the war-room plan (bridge, log, cadence); the standing reconciliation as a control, not a project.
Complexity & consensus	The fear of losing the initiative named in the room; the customer prioritized their own orders and gave time for a clean fix — the review held with the facts already on the table.
Root cause at scale	The census by handshake step × order state led to a permanent OMS–WMS handshake fix, not just the patched orders — the trending cause removed.

THE FIVE FACETS — DATA ANALYSIS · SOLUTION DESIGN · MATH & DATA LOGIC · IMPLEMENTATION · PROBE QUESTIONS NEW

1 · DATA ANALYSIS

Data. The ~200 orders where OMS status did not equal the mapped WMS status — $2.5M of order value, $12,500 average per order (derived). **Grain:** order line × handshake step. **Cuts:** (a) which handshake step diverged, (b) the order state at divergence — on a *sample* of the ~200, because two days did not allow classifying all of them before choosing a path. **What it revealed:** the break clustered at the OMS–WMS handshake rather than scattering — a systemic, fixable-at-source defect, and the very tables a quiet script would have rewritten. The standing version of the cut is four reason classes (status mapping · timing · failed message · manual edit) that must sum to the delta. I do not have per-class counts in my notes and will not quote any.

2 · SOLUTION DESIGN

Options considered: (A) a quiet cross-table script written in under two days, customer not told; (B) transparency — root cause, plan and timeline to the customer first, a per-line reversible patch on the customer's own priority list, the clean handshake fix scheduled after the review; (C) an apology-first note with no plan (rejected — gives the customer nothing to decide). **Chosen: B.** Why: it was verifiable line by line, reversible, survived an audit, and put the priority decision with the customer. The argument was on risk, not principle.

3 · MATH & DATA LOGIC

Formulas. Average discrepancy = $2.5M / 200. Quiet-path expected loss = P(error) × blast radius = 0.3 × (2 × $2.5M) — both illustrative. Break-even P* = cost of the honest path / blast radius. Match rate = orders with OMS status = map(WMS status) / orders in flight. Classes: Σ reason classes = Δ; a row with no class is an investigation, not a bucket. Exposure at detection = unreconciled value built over ~60 days (illustrative) → about one day's worth under a daily report. All four tables below substitute every input.

4 · IMPLEMENTATION

Phases and owners. Day 1 AM census (me) → day 1 PM options memo to my manager (me) → day 2 customer pack, war-room plan and patch-verification method (me; consultants staffed) → review + per-line patch on the customer's list (consultants wrote and verified the patch; I ran the war room) → permanent handshake fix (consultants) → standing reconciliation read before every executive review (account team). **Controls:** per-line reversibility; evidence rows = set size; customer priority order; tripwire = any audit query on those orders. **Risks:** corruption, audit finding, customer surprised at the review.

5 · PROBE QUESTIONS

Hardest follow-ups with numeric answers are in the Probing section (the $10M, "what if he said no", "why not apologize and fix", "$0.1M is small") and in the three Trap Drills at the end of this story: the counterfactual and break-even P* = 0.43%; my contribution versus the consultants'; who disagreed and what they got right.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

BR — You're claiming the $10M?

No. The initiative was won in the review the facts survived; I claim the control and the posture, not the contract value. The only recurring dollar I model is ≈$0.1M a year from the standing reconciliation, labeled; the ≈$1.5M is my illustration of expected corruption loss from the silent script — avoided risk, not money earned.

BR — Your manager wanted the quiet fix. What exactly did you say, and what if he had said no?

Privately first, with the census. I disagreed on risk, not principle: an unverifiable script plus an audit exposure versus a customer told the truth with a plan — and I offered to run the war room so the cost landed on me. He accepted, hesitantly. Had he said no, I would have documented the risk assessment in writing and asked for the decision recorded with its owner, with the tripwire named; I would not have run the silent script without that record.

BR — Why not just apologize and fix it?

No apology-first. The customer pack was root cause, action plan and timeline — to their IT and finance before the executive review, not in it. They responded by prioritizing their own orders and giving us time for a clean fix; an apology without a plan would have given them nothing to decide.

BR — What control outlived the war room?

The reason-coded OMS↔WMS reconciliation as a standing check, read before every executive review rather than two days before one — status mapping, timing, failed message, manual edit; a row with no class is an investigation, not a bucket. The handshake fix removed the cause; the report made sure the next one surfaced early.

BR — A hundred thousand a year is small for a flagship story.

Yes, and I would rather say small and true. The story is about the control and the choice, not the run-rate. The hardest input is how many discrepancies arise per month; I assumed the two hundred built over about sixty days. I would read the daily match report's reason-code counts for a quarter and price exactly what surfaced.

BR — Anything more recent where you disagreed upward?

Yes — re-sequencing my manager's roadmap at Flexport on cost-to-serve evidence (the hiring manager heard that one), and at Sephora the rule I had written myself and had to correct through the same three signatories. Same posture: disagree on evidence, carry the consequence.

BUSINESS IMPACT MODEL — KPI TREE AND DRIVERS NEW

```
Customer trust that survives an audit -> $10M initiative won (user anchor; never counted as savings)
+- OMS<->WMS discrepancy ~200 orders / $2.5M -> 0 (user anchor)
|    +- recovery: per-line reversible patch, before/after evidence, customer-prioritized set first
|    +- prevention: standing reason-coded reconciliation (daily); exposure at detection ~60 d -> ~1 d
+- Risk not taken: silent multi-table script = P(error) x blast radius (+ audit finding)
+- Trust loop (+): tell first -> customer sets priority -> gives time -> clean fix -> initiative won -> latitude next time
```

DRIVER	BASELINE	AFTER	$	BASIS
OMS↔WMS discrepancy	~200 orders / $2.5M	0, evidence per order	—	user anchor
$10M initiative	at risk	won	—	user anchor — never counted as savings
Standing reconciliation	war rooms; unmonitored	daily report	≈$0.1M/yr	modeled (range $55–160K)
Silent script not run	—	—	≈$1.5M expected corruption	illustrative avoided risk — not a saving

MATH — THE FOUR TABLES (EVERY INPUT SUBSTITUTED) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The discrepancy	`$2,500,000 / 200 = $12,500`	user anchor pair ($2.5M of order value, ~200 orders); average derived	≈$12.5K per order
2	The clock and the stakes	`2 days to the executive review; $10M initiative riding on it`	user anchor	two days
3	Exposure build-up	`$2,500,000 / 60 = $41,667`	~$2.5M built over ~60 days — the 60 days is illustrative	≈$41.7K per day unreconciled
4	The tempting path	`one quiet script across related tables, written in <2 days; no verification, customer not told`	my argument (user anchor)	fast, silent, unverifiable

Table 2 — the fix: values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Census by cause	`Δ grouped by (handshake step × order state) on a sample of the ~200 — before proposing any path`	my method (user anchor)	facts before path
2	Quiet path priced — blast radius	`2 × $2,500,000 = $5,000,000`	2× blast radius illustrative	$5.0M
3	Quiet path priced — expected loss	`0.3 × $5,000,000 = $1,500,000`	P(error) = 0.3 illustrative; plus a near-certain audit finding	≈$1.5M avoided risk — not a saving
4	Honest path priced — my cost	`6 × 24 × $150 = $21,600`	one war room: 6 people × 24 h (3 days × 8 h) × $150/h — illustrative	≈$21.6K one-time
5	Break-even P*	`$21,600 / $5,000,000 = 0.43%`	honest-path cost / blast radius	quiet path wins only if P(error) < 0.43% — and an audit would still find it

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	Patch walk	`customer-prioritized set first; per line; reversible; evidence rows = set size`	user anchor	0 on the set before the review
7	Standing control	`match rate = orders with OMS status = map(WMS status) / orders in flight; Σ 4 reason classes = Δ`	my design	a row with no class is an investigation

Table 3 — after: the metric, before → after.

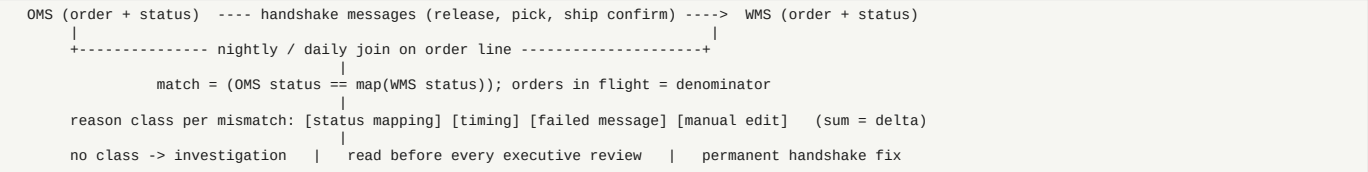
STEP	METRIC	BEFORE	AFTER	BASIS
1	OMS↔WMS discrepancy	~200 orders / $2.5M	0 on the prioritized set before the review; 0 through the clean fix	user anchor
2	New discrepancies	unmonitored	0 on the daily report	my design
3	Exposure at detection	~$2.5M over ~60 days	~1 day's worth ≈ $41,667	my design; per-day value derived
4	Customer	would learn at the review or from an audit	briefed first; set the priority; gave time for a clean fix	user anchor
5	$10M initiative	at risk	won in the review the facts survived	user anchor — never counted as savings

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Expected corruption avoided	`2 × $2,500,000 × 0.3 = $1,500,000`	illustrative probabilities	≈$1.5M avoided risk — not money earned
2	Manual fixes avoided	`100 × 12 × 80% × $30 = $28,800`	100 discrepancies/month (illustrative) × 12 × 80% needing a hand-fix × $30 per fix (benchmark)	≈$29K/yr modeled
3	Stuck cash carried	`100 × $12,500 × 12 × 7 / 365 × 10% = $28,767`	$12.5K avg × 100/month × 12 = $15.0M a year passing through mismatch; average 7 days stuck; 10% cost of capital — all illustrative	≈$29K/yr modeled
4	War rooms avoided	`2 × $21,600 = $43,200`	2 repeat war rooms a year × $21.6K (illustrative)	≈$43K/yr modeled
5	Recurring value of the control	`$28,800 + $28,767 + $43,200 = $100,767`	sum of the three modeled lines; canon range $55–160K	≈$0.1M/yr modeled
6	$10M initiative	`won — excluded from any savings column by rule`	user anchor — a fact about the relationship	never claimed

Inputs not in the master (80% hand-fix share, 7 days stuck, $150/h, 6-person war room) are my explicit assumptions chosen to reproduce the canon's ≈$29K / ≈$30K / ≈$43K split and the ≈$0.1M total; replace them with the daily match report's actual reason-code counts after a quarter.

SOLUTION FLOW — THE SYSTEM AND DATA MAP NEW



SENSITIVITY — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
P(script error) — illustrative	0.1	0.3	0.5
Expected corruption on a $5.0M blast radius	0.1 × $5,000,000 = $500,000	0.3 × $5,000,000 = $1,500,000	0.5 × $5,000,000 = $2,500,000
Discrepancies arising per month (illustrative)	50	100	150
Cost per manual fix (benchmark)	$20	$30	$40
Total modeled recurring (canon range)	≈$55K	≈$0.10M	≈$0.16M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS NEW

> "Two and a half million of order value made consistent in the open, with before-and-after evidence on every order, and the ten-million initiative won — both are my own tracking, and neither is a saving I claim. What the standing reconciliation is worth going forward is small and honest: about a hundred thousand a year in my model — roughly thirty in manual fixes, thirty in cash not stuck behind mismatched orders, forty in war rooms we do not run. The bigger number is the one we did not lose: the silent script carried an illustrative expected one and a half million of corruption plus an audit finding on a ten-million relationship; that is risk math, not money in the bank."

IF CHALLENGED — THE FULL ANSWERS NEW

A hundred thousand a year is small for a flagship story.

Yes, and I would rather say small and true. The story is the control and the choice, not the run-rate. The hardest input is how many discrepancies arise per month; I assumed 100. I would read the daily match report's reason-code counts for a quarter and price exactly what surfaced.

> **The 0.3 error probability is made up.**
> It is illustrative and labeled. At 0.1 the expected loss is $0.5M, at 0.5 it is $2.5M; the decision does not turn on it, because break-even is 0.43% — the quiet path only wins if a two-day cross-table script has under a half-percent chance of going wrong, and the audit finding is separate.

> **Thirty dollars a fix and ten percent cost of capital are assumptions.**
> Both are benchmarks and I ranged them: $20–40 a fix and 8–12% move the recurring value from about $55K to $160K. I would take the minutes from the war-room log, which timestamped every per-line patch, and the cost of capital from the customer's finance team.

Where this lands at Amazon. The purest Earn Trust pattern — tell the customer first, price the honest path's cost onto yourself, and leave a standing control so the courage is rarely needed again. The transferable control is not the courage; it is the reason-coded reconciliation read before every review. Reliability (a seam that reconciles daily), responsiveness (the customer decided the priority), trust that survives an audit.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S14

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you had to deliver bad news to a customer.	Earn Trust / Customer Obsession	Root cause, plan, timeline before the review; the customer set the priority; no apology-first.	Never count the $10M as impact.
Tell me about a time you disagreed with your manager.	Have Backbone, Disagree & Commit	On risk, not principle; war room offered; the "what if he had said no" answer ready — document, record the owner, name the tripwire.	Do not make the manager the villain — his fear was legitimate and named as such.
Tell me about acting fast under a deadline.	Bias for Action	Two days: census by cause in a morning, options memo in an afternoon, customer pack on day two, per-line patch before the review.	Do not imply the permanent fix was done in two days — it was scheduled after.
Tell me about owning a problem across system boundaries.	Ownership	The seam between OMS and WMS had no owner; the census, the patch method and the standing report gave it one.	Consultants built the patch and the fix — say so.
How do you design a reconciliation control?	Functional — control design	Reason-coded classes that sum to the delta; a row with no class is an investigation; read before every review, not two days before one.	No invented percentage — my notes have none here.
Why did this matter beyond the one customer?	Broad Responsibility	A silent fix would have taught the practice that quiet scripts are acceptable; the standing control and the posture became the practice's norm.	Do not claim practice-wide dollars — none exist.

BEHAVIORAL PACK — FOUR LIKELY QUESTIONS, FOUR SEPARATE 90-SECOND STAR ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY) NEW

Q1 **AIQB Customer Obsession Q1 — "Describe a difficult interaction you had with a customer. How did you deal with it? What was the outcome? How would you handle it differently?"** CUSTOMER OBSESSION · EARN TRUST

> **Opener (first two sentences):** "Two days before an executive review, my customer's order and warehouse systems disagreed on about two hundred orders — $2.5 million. I decided they would hear it from me, with a plan, before the review rather than from an audit after it."

S — A retail customer on Manhattan's order and warehouse systems; about 200 orders and $2.5M did not reconcile; a $10M initiative rode on the review in two days.

T — Own the customer conversation and the recovery, whichever path my manager chose.

A — I built a pack for their IT and finance leads: root cause at the handshake step, an action plan, a timeline — no apology-first, because an apology with no plan gives them nothing to decide. I asked them to set the priority and they chose their own orders. We patched that list per line, reversibly, with before/after evidence for every order, and scheduled the clean fix after the review.

R — Zero discrepancy on their prioritized set before the review; the review held with the facts already on the table; they gave us time for the clean fix; the $10M initiative was won — which I never count as savings.

LAND THIS NUMBER
~200 orders / $2.5M → 0 before the review.

PIVOT TO NACF
At NACF the same move is telling a seller or an FC GM about a receive shortage before the weekly review — root cause, plan, timeline — instead of letting the gap surface in an escalation.

DO NOT SAY
That the customer "thanked" or "loved" it (not in my notes); the $10M as impact; "apology-first".

Q2 **AIQB Have Backbone; Disagree and Commit Q1 — "Tell me about a time when you strongly disagreed with your manager or peer on something you considered very important to the business. What was it and how did you handle it? Knowing what you know now, would you do anything differently?"** HAVE BACKBONE; DISAGREE & COMMIT

> **Opener (first two sentences):** "My manager wanted a quiet back-end script to fix those two hundred orders before the review, and I told him — privately, with a census in hand — that I thought it was the riskier path. I disagreed on risk, not principle, and I put my own time behind it."

S — The room favored a script across related tables written in under two days with the customer not told; the fear — losing the $10M initiative — was legitimate and I said so.

T — Change the decision or commit to it; I was the consultant who would run whichever path won.

A — Facts first: the census by handshake step and order state. Then an options memo: an unverifiable script, an IT audit turning a data issue into a reputation issue, an expected corruption of about $1.5M on illustrative odds. I offered to run the war room so the cost of my position landed on me. He accepted, hesitantly. Had he said no, I would have written the risk assessment down, asked for the decision to be recorded with its owner, named the tripwire — any audit query on those orders — and not run the silent script without that record.

R — The honest path ran; the discrepancy went to zero on the prioritized set; the initiative was won. The relationship lesson: private first, evidence first, my labor attached.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
≈$1.5M expected corruption avoided — illustrative P = 0.3, avoided risk, not a saving.	At NACF I would disagree the same way with a plan owner: in private, with the arithmetic, with my own hours attached — and commit with a named tripwire if overruled.	That he was wrong or foolish; that he "came around" gladly; any claim the $1.5M was realized.

Q3 **AIQB Bias for Action Q2 — "Tell me about a time when you worked against tight deadlines and didn't have time to consider all options before making a decision. How much time did you have? What approach did you take? What did you learn?"** BIAS FOR ACTION

> **Opener (first two sentences):** "I had two days, about two hundred discrepant orders and no root cause. I did not wait to classify them all — I sampled by handshake step and order state in one morning, and that sample decided the path."

S — Two days to an executive review; cause unknown; every hour of delay narrowed the options.

T — Get enough facts to choose a path within hours — and keep every move reversible.

A — Morning: sample census. Afternoon: options memo to my manager. Day two: customer pack, war-room plan, and the verification method for any patch — per line, reversible, before/after recorded. I did not pretend two days was enough for the permanent fix; that was scheduled after the review.

R — The customer was briefed before the review; the prioritized set was patched with evidence; the clean fix followed. Speed bought options, not shortcuts.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Two days; ~200 orders.	At NACF peak you get hours, not days: sample the exceptions by cause, act reversibly, schedule the permanent fix.	"I fixed the root cause in two days"; any percentage — my notes have none for this story.

Q4 **AIQB Dive Deep Q6 — "Have you ever created a metric that helped identify a need for a change in your department? What was the metric? Why did you create it? How did this and other information influence change? What was the outcome of the change?"**
DIVE DEEP · HIGHEST STANDARDS

> **Opener (first two sentences):** "After the war room I did not want the next discrepancy found two days before a review. So I turned the one-off census into a standing, reason-coded reconciliation that is read before every executive review."

S — The ~200-order gap had built unseen, over roughly 60 days on illustrative timing — nothing compared OMS to WMS on a schedule.

T — Make early detection the default instead of a hero moment.

A — Match rate = orders where OMS status equals the mapped WMS status over orders in flight. Four reason classes — status mapping, timing, failed message, manual edit — that must sum to the delta; a row with no class is an investigation, not a bucket. The permanent handshake fix removed the cause; the report made the next one surface early.

R — New discrepancies at zero on the daily report; exposure at detection from about $2.5M built over ~60 days to about one day's worth; ≈$0.1M a year modeled, range $55–160K.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Exposure at detection ~60 days → ~1 day.	At NACF: an ASN-versus-receipt match rate by reason code — the booklet's 10,000 × 0.92 × 0.88 = 8,096-unit example leaves a 1,904-unit gap that must sum to named classes.	A percentage improvement (none exists); calling the $1.5M a saving.

SAME STORY, THREE ANGLES — DIFFERENT OPENER, HEADLINE METRIC AND LP EACH NEW

(i) Data / analytical
DIVE DEEP

> "Before I proposed anything, I wanted to know what the $2.5 million was made of."

Headline metric: $12.5K average × ~200 orders; census by handshake step × order state

- Sampled the ~200 instead of waiting to classify all of them.
- The break clustered at the handshake — systemic, not random.
- The tables a quiet script would rewrite were the tables the handshake wrote.
- Standing version: four classes that sum to the delta.
- No per-class counts in my notes — I say so.

(ii) Conflict / influence / backbone
HAVE BACKBONE; DISAGREE & COMMIT

> "My manager and I disagreed, and he was right about two things."

Headline metric: ≈$1.5M expected corruption (illustrative P 0.3) vs break-even P* 0.43%

- Right about the clock and about the fear of losing the initiative.
- Wrong, in my judgment, that speed justified an unverifiable script.
- Private first; census in hand; risk, not virtue.
- Attached my labor: I ran the war room.
- If no: written risk, decision owner, tripwire.

(iii) Failure / learning
OWNERSHIP · LEARN AND BE CURIOUS

> "The part I would do differently is that it took a near-miss for anyone to read a reconciliation."

Headline metric: Exposure at detection ~60 days → ~1 day

- Master lesson: silent corrections are loans against trust at a terrible interest rate.
- ~$41.7K a day went unreconciled with no standing check.
- I built the control after the incident, not before it.
- Do differently: propose the daily reconciliation the moment the handshake is in scope.
- VERIFY-WITH-RAJESH confirm you had scope to propose it earlier; if not, say "I did not have scope — I would ask in week one."

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS NEW

T1 "It worked. But how do you know it was the right decision — what is the counterfactual?"

Numeric answer. Quiet path: E[loss] = 0.3 × (2 × $2.5M) = $1.5M on illustrative odds, plus a near-certain audit finding. Break-even P* = $21,600 / $5.0M = 0.43%: the quiet path is cheaper on expected loss only if a two-day cross-table script has under a half-percent chance of going wrong. A good outcome does not prove a good decision; verifiability did.

Say exactly: "Nobody ran the silent script, so I cannot prove the counterfactual. I can show the decision rule: for the quiet path to win on expected loss, the chance of a two-day script across related tables going wrong would have to be under half a percent — and an audit would still find it."

T2 "What was YOUR specific contribution versus the consultants'?"

Numeric answer. Mine: the sample census, the risk argument, the customer pack, the war room, the verification method. Theirs: tracing the handshake, writing and verifying the per-line patch, building the permanent fix. The standing report is read by the account team, not by me.

Say exactly: "I owned the census, the recommendation, the customer pack and the war room. The consultants traced the handshake, wrote and verified the per-line patch and built the permanent fix. I did not write the patch."

T3 "Who disagreed with you, were they right — and what number would have changed your mind?"

Numeric answer. My manager. Right: the two-day clock was real and the fear of losing a $10M initiative was legitimate. The number that would have changed my mind: evidence the script could be verified line by line before the review — at which point it is the per-line patch I ran, with the customer in the loop. P(error) under 0.43% with no audit exposure would also flip the arithmetic; neither was plausible.

Say exactly: "He was right about the clock and about the fear. Where I disagreed was that speed justified an unverifiable script. If someone had shown me a way to verify it line by line before the review, I would have changed my mind — and that is, in effect, the patch we ran."

20-SECOND VERSION AND NACF BRIDGE NEW

20-SECOND VERSION NEW

"Two days before an executive review, a customer's order and warehouse systems disagreed on about two hundred orders — $2.5 million — and the room wanted to fix it quietly. I disagreed on risk, told the customer first with a plan, ran the war room myself, and left a reason-coded reconciliation behind. The discrepancy went to zero before the review."

NACF BRIDGE NEW

"At NACF the same seam is plan versus receipt. The booklet's week-44 example: 10,000 units ordered × 0.92 fill × 0.88 on-time = 8,096 units, 81.0% of plan — a 1,904-unit gap. I would run it as reason-coded classes that sum to the gap, read before the weekly review, and tell the seller or the FC first."

S16 — Flexport — four systems into one Snowflake pipeline: writing a bus-factor-one operating metric out of my own hands, and the security-review miss I owned upward the day it surfaced ELEVATED

LPs this story can serve here: Ownership (primary — a single point of failure → pipeline; owning the miss) · Invent and Simplify (backup — four systems → one) · Frugality (the >$100K priced baseline) · Learn and Be Curious (learned the stack to write the BRD) · Earn Trust (zero review deliveries skipped while paying for the miss)
Process family: Deliver — operating-metric production for a weekly business review; data-integrity and ownership of a decision instrument

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF EACH NEW

Ownership	I took a bus-factor-one system that I had built by hand and wrote it out of my own hands; and I owned the two-week security-review miss upward the same day it surfaced, with zero Monday reviews skipped while I paid for it.
Invent and Simplify	Four systems and a hand-stitched sheet became one pipeline; seven metric definitions froze into code; a cross-reference table shipped now while the real key was pushed upstream.
Frugality	I priced doing nothing in the BRD at more than $100K a year — the BRD's own arithmetic is $103,680 on thirty hours a week of WBR work at an assumed $90 an hour, and my split into hours and decision risk is a model — so the infra team could prioritize the build themselves. Priced, never claimed as savings.
Learn and Be Curious	I learned enough of the data stack to write the first business requirements document for a program's operating metrics that the infra team could build from.

Honesty label. I wrote the BRD and owned the cutover; the infra team owned the build. The >$100K/yr is a priced baseline: the BRD priced it at $103,680 on thirty hours a week of WBR work at an assumed ninety dollars an hour; my split of it (≈$30K hours + ≈$70K decision risk) is a model — priced, not claimed as savings. **Basis:** the BRD arithmetic is record-from-docs; 60% → 90%, parity within 1%, the 4% catch and the two-week miss are record (Apple v11 A3/G3); any count after "bus factor one" is a user anchor, so I say "no longer a single point of failure". The two-week security-review miss is mine, owned upward the same day. Same program family as S11 (the HM's story) — never both to one interviewer.

SAFE-SAY BOX NEW

The most attackable number: >$100K a year. Say: "More than a hundred thousand dollars a year — the BRD priced it at $103,680 on thirty hours a week of WBR work at an assumed ninety dollars an hour; my split into hours and decision risk is a model. Priced, not a claimed saving."

If he asks "what is your source for that?" "That is the BRD's own arithmetic on page one — three people, five hours, two reviews, at an assumed rate — not a Finance figure. The number I would stand behind is system accuracy 60% to 90%, with parity within one percent for consecutive weeks."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The Monday business review that steered Flexport's visibility rollout ran on four systems that did not talk, stitched together in spreadsheets by one person — me. I was the single point of failure; data accuracy 60%; 1.5–2 hours per snapshot; support tickets with no container ID so they could not be joined to anything. A VP was steering a program on a number one person could break and nobody could check. **Task.** Nobody had written requirements for a program's operating metrics; I wrote the business requirements and owned the cutover; the infrastructure team owned the build. **Action.** I documented the as-is and priced doing nothing in the BRD at more than $100K a year — the BRD priced it at $103,680 on thirty hours a week of WBR work at an assumed ninety dollars an hour; my split into my hours and the decision risk of a VP steering on a wrong number is a model — so that the infra team could prioritize the build themselves. The BRD froze the seven metric definitions, specified the feed, the failure alerts and a cross-reference table for the ticket join while the real key was pushed upstream, and set the acceptance gate: parallel run within 1% per metric for consecutive weeks, every mismatch explained by cause. Mid-build a data-access security review surfaced that I had not planned; it cost two weeks. I owned it upward the same day as a scoping miss, re-sequenced the build, and kept the review alive on the manual pull — zero Mondays skipped. The deck moved onto the pipeline when parity held; the manual pull stayed as rollback a week beyond. **Result.** Primary metric: review data accuracy 60% → 90% once four systems became one pipeline; from a single point of failure to a pipeline; the first automated catch was a 4% ticket-sync drop that had been invisible in the manual era. The >$100K a year is a priced baseline — the BRD's arithmetic, not a Finance figure — and not a claimed saving. Lesson: every kickoff now asks whose approval the data needs and what its lead time is.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The Monday review ran on four systems that didn't talk, stitched in Google Sheets by one person — me, the single point of failure: accuracy 60%, 1.5–2 hours per snapshot, tickets carried no container ID. A VP was steering on a number one person could break.

The cost of leaving it: a wrong number steering the rollout, and silent data loss nobody would catch — the 4% ticket-sync drop was invisible in the manual era. The customer in this story is the VP making rollout and funding decisions on the number.

T — Nobody had written requirements for a program's operating metrics. I wrote the BRD and owned the cutover; the infra team owned the build. Goal: one pipeline producing the same seven numbers, accurate, and owned by more than one human.

Baselines on the table: WBR accuracy 60% · single point of failure (me) · snapshot effort 1.5–2 hours per snapshot · ticket-to-container joinability none · no alarm on silent data loss.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — documented the as-is; named the join gap (tickets without container ID); priced the status quo at >$100K/yr in the BRD (BRD arithmetic $103,680 on thirty hours a week at an assumed $90/h; my split, with VP decision risk, is a model). *Exit bar:* infra prioritizes it themselves off the price; scope = the seven metrics, nothing new.
- **Plan & Design** — the BRD — seven frozen definitions, daily/hourly feed, failure alerts, cross-reference table then a real key upstream; acceptance = parallel run within 1% for consecutive weeks; the RAID log — missing the data-access security review. *Exit bar:* BRD accepted by infra; acceptance gate published.
- **Build & Test** — extract/load specs per system; parity checks; the parallel-run procedure; when the security review surfaced mid-build, re-sequenced and owned the miss upward the same day. *Exit bar:* review closed (2 weeks); parallel run within 1% for consecutive weeks; mismatches explained (mostly late-arriving tickets on the manual side).

- **Cutover & Go-Live** — the cutover memo; the Monday deck moved onto the pipeline; the manual pull kept as rollback. *Exit bar:* first Monday off the pipeline, same seven numbers; zero review deliveries skipped through the build.
- **Hypercare / Close** — watched the failure alerts; triaged the first automated catch (a 4% ticket-sync drop); definitions into code; institutionalized the kickoff question. *Exit bar:* accuracy 60% → 90%; the pipeline steers the review without me.

Influence and ownership. The build was the infra team's — the ETL → S3 → SQS → Snowflake → Looker pipeline and the alerts; mine were the as-is, the pricing, the BRD's frozen definitions, the acceptance gate, the parallel-run procedure and the cutover memo. The lever was pricing the status quo in the BRD — thirty hours a week at an assumed $90 an hour is $103,680, and my own split adds the decision risk of a wrong number, not just hours, as a model — so they could prioritize it themselves. The category error was mine: I treated the security review as an engineering concern rather than a program dependency with a lead time.

> **Designed the process; others own it now.** I designed the BRD, the acceptance gate and the cutover memo; the infra team owns the pipeline and its alerts, and the definitions live in code rather than in a person. **Trade-off put to leadership.** The recommendation to the infra lead and the VP outlined the competing metrics — build speed vs the data-access security review and decision risk on the Monday numbers — and re-sequenced the build rather than route around the review.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — >$100K/yr priced baseline (BRD: $103,680 at an assumed $90/h — record-from-docs) — priced, *not* claimed as savings; my model splits it ≈$30K hours + ≈$70K decision risk (≈$50K of which the accuracy gain retires), both modeled; bus-factor exposure ≈$15K/yr modeled.
- **#** — four systems → one pipeline; no longer a single point of failure; zero review deliveries skipped through the build (record).
- **%** — review data accuracy 60% → 90%; parity within 1% for consecutive weeks; first automated catch a 4% ticket-sync drop (record).
- **Time** — the security-review miss cost two weeks — owned upward the same day, build re-sequenced, review closed (record).
- **Customer** — the VP steered on the same seven numbers the first Monday off the pipeline; the manual pull stayed as rollback until parity had held a week beyond (record).
- **Quality** — seven frozen definitions moved into code (no drift); every parallel-run mismatch explained by cause (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A Monday review fed by a manual pull across four systems with a bus factor of one, owned by nobody. I documented the as-is, named the join gap (tickets without container ID) and priced the status quo at >$100K/yr in the BRD ($103,680 on its arithmetic; my split is a model).
Scope & influence	The infra team — not mine — prioritized the build off the price; the VP steered the review off the pipeline from the first Monday; my seat was the BRD, the acceptance gate and the cutover memo.
Execution trade-off + mechanism others now own	Build speed vs the data-access security review: when it surfaced mid-build I re-sequenced and owned the miss upward the same day (two weeks). Mechanisms others own now: the pipeline and its failure alerts (infra), the seven frozen definitions in code.
Impact (long-term)	Four systems → one pipeline; no longer a single point of failure; review data accuracy 60% → 90%; the pipeline steers the review without me.
Process improvement (standard work institutionalized)	Acceptance = parallel run within 1% for consecutive weeks; the parallel-run procedure and cutover memo; the manual pull kept as rollback until parity held a week beyond; the kickoff question institutionalized (which reviews belong in the RAID log).
Complexity & consensus	Zero review deliveries skipped through the build; the security review treated as a dependency with a lead time rather than argued around.
Root cause at scale	The join gap fixed as a cross-reference table then a real key upstream; the first automated catch — a 4% ticket-sync drop — triaged by cause, not patched by hand.

THE FIVE FACETS — DATA ANALYSIS · SOLUTION DESIGN · MATH & DATA LOGIC · IMPLEMENTATION · PROBE QUESTIONS NEW

1 · DATA ANALYSIS

Data RECORD-FROM-DOCS: BRD TSR-852. As-is: Zendesk ticket data plus a manual pull of containers from Crux into Google Sheets, matched by container ID by hand — 1.5–2 hours for a week-over-week snapshot; tickets carried no container ID, so they could not be joined to anything. **The seven metrics:** coverage (automated), coverage (with human inputs), tickets generated, tickets per active container, 4PL % within SLA, 3PL % within SLA, and the 4PL-vs-3PL delta. **Cuts that mattered:** parallel-run gap per metric per week; every mismatch classified by cause. **What it revealed:** most parity mismatches were the *manual* side being wrong (late-arriving tickets), and — once automated — a 4% ticket-sync drop that had been invisible in the manual era.

2 · SOLUTION DESIGN

Options and trade-offs RECORD: APPLE V11 A3: I ran *no bake-off* — Snowflake was the house warehouse and conformance with the existing stack was the point; my decision was narrower: get ticket and container data into that stack with the definitions intact. The trade was scope: the BRD froze the seven existing metrics and named new metrics out of scope — a narrower, fundable pipeline this quarter over a perfect one someday — and cutover waited for consecutive matching weeks, not for my confidence. **Chosen: one pipeline** — ETL → S3 → SQS → Snowflake → Looker — built by the infra team to my BRD. BRD requirements: daily-or-hourly feed (BR-3), an alert when the feed fails (BR-4), Looker views over SQL so the seven metrics need no manual reconciliation, seven definitions frozen, a cross-reference table for the ticket join while the real key is pushed upstream to Support. Acceptance gate: parallel run within 1% per metric for consecutive weeks.

3 · MATH & DATA LOGIC

Formulas. Parity gap = |pipeline − manual| / manual ≤ 1% per metric. Silence alert: 1 − sync rate > 1% → page. Priced baseline = hours term + decision-risk term, where hours = h/wk × 52 × $/h and decision risk = 52 weeks × P(wrong steer) × cost of a wrong steer. Break-even P for the >$100K = (100,000 − hours term) / (52 × cost per wrong steer). Retired share of risk = (wrong-rate before − wrong-rate after) / wrong-rate before. Substituted below.

4 · IMPLEMENTATION

Phases/owners. Discovery (as-is, price) → Plan & Design (BRD, RAID log, acceptance gate) → Build & Test (infra built; I wrote parity checks and the parallel-run procedure; security review surfaced) → Cutover (memo; deck moved; manual pull kept as rollback a week beyond parity) → Hypercare (alerts, first catch triaged). **Controls:** parity gate, rollback, definitions in code, failure alerts. **Risk not on the register:** the data-access security review — now item one at every kickoff.

5 · PROBE QUESTIONS

See Probing (fifth-why on the review, "are you claiming $100K", soft decision-risk dollar) and the three Trap Drills below: what proves the >$100K (and the break-even P), my contribution vs infra's, and why two weeks was the right long-term call.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

BR — Why is this Ownership and not just automation?

Both halves are ownership: taking the bus-factor-one system I had built and writing it out of my own hands, and owning the security-review miss upward the same day it surfaced — with zero review deliveries skipped while I paid for it.

BR — You are claiming $100K of savings?

No — pricing, not claiming. The number existed so the infra team could prioritize the build, and it did: the BRD priced it at $103,680 on thirty hours a week of WBR work at an assumed ninety dollars an hour; my split into hours and decision risk is a model. The claimed record is the accuracy move and the first automated catch; the split of the price is my model and I will not call any of it realized savings.

BR — Why didn't you know about the security review? Fifth why.

Because I treated it as an engineering concern rather than a program dependency with a lead time — a category error, not a scheduling slip. The fix is institutional: every kickoff now asks whose approval the data needs and what its lead time is. I would trade the two weeks again; I would plan them now.

BR — Who was the customer and what did they experience?

The VP steering the rollout. What they experienced was continuity — the same seven numbers on the first Monday off the pipeline, zero skipped Mondays — and then a capability they never had: an alert instead of silence when 4% of tickets stopped syncing.

BR — The decision-risk dollar is soft.

Yes — one wrong number in ten becoming a wrong steer worth a squad-week; range $12K to $117K. I would firm it by tagging every review action item for a quarter and counting how many were reversed when a number was corrected.

BR — How does this connect to the review the hiring manager heard about?

It is the same program: the review in S11 ran by hand for over a year; this pipeline is what retired the manual pull, with the definitions in code. Different story, same facts.

BUSINESS IMPACT MODEL — KPI TREE AND DRIVERS NEW

```
VP steers the program on a number that is right and survives me
+- Status quo priced >$100K/yr (BRD: $103,680 on 30 h/wk at an assumed $90/h — priced, not claimed)
|    +- hours term (my pull) ~$30K  +  decision-risk term ~$70K
+- Review data accuracy 60% -> 90% (record): 7 definitions frozen in code; join rate -> ~100%; parity gate <=1%
+- Bus factor one -> no longer a single point of failure (risk stock retired)   +- Silent data loss -> alerts (first catch: 4% ticket-sync drop)
(loop held) security review unplanned -> 2-wk slip -> manual pull keeps the review alive -> 0 skipped Mondays -> trust held
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Review data accuracy	60%	90%	—	record
Status quo priced	>$100K/yr	retired by the build	—	record-from-docs: BRD $103,680 at an assumed $90/h — priced, not savings
— hours inside the price	~4 h/wk	~0	≈$30K	modeled
— decision risk inside the price	wrong-steer exposure	≈75% retired by the accuracy gain	≈$70K → ≈$50K retired	modeled
Single point of failure	one person (bus factor one)	no longer one, plus alerts	≈$15K	modeled; the count after is a user anchor

MATH — THE FOUR TABLES (EVERY INPUT SUBSTITUTED) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Accuracy / fragility	`accuracy 60%; bus factor 1; 5 × 40% = 2 weeks in 5 wrong`	record (60%; single point of failure — Apple A3, WBR doc p2); weeks-in-5 derived from 1 − 60%	one person, wrong 2 weeks in 5
2	Hours term	`4 × 52 = 208 h; 208 × $145 = $30,160`	4 h/wk and ≈$145/h are my own model inputs (illustrative) — not the BRD's 1.5–2 h per snapshot	≈$30K/yr
3	Decision-risk term	`52 × 10% = 5.2 wrong-steer weeks; 5.2 × $13,500 = $70,200`	P(wrong steer) 10% and ≈ a squad-week = $13.5K — both illustrative; range $12–117K	≈$70K/yr
4	Priced baseline	`$30,160 + $70,200 = $100,360`	my model (hours + decision risk) — a second lens on the same >$100K; priced, NOT claimed as savings	>$100K/yr priced
5	The BRD's own business case — the primary derivation of the >$100K (see SOURCE TRACE)	`3 × 5 × 2 = 30 h/wk; 30 × $90 × 48 = $129,600 → 6 × $90 × 48 = $25,920; $129,600 − $25,920 = $103,680`	record-from-docs: BRD TSR-852 p1–2 — 3 resources × 5 h × 2 WBRs = 30 h/wk at an ASSUMED $90/h, 48 weeks, cut to 1 h/wk each… the doc prints $10,880/month; the arithmetic is $10,800	$103,680 is the number to say; my model's $100,360 is a second lens — call the split a model
6	Join problem	`tickets carried no container ID → joins impossible; Ticket ID ↔ Crux container number needed`	record (BRD problem statement)	ticket metrics unlinkable

Table 2 — the fix: values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Parity gate	`|pipeline − manual| / manual ≤ 1% per frozen metric, consecutive weeks`	BRD acceptance gate (record)	cutover unarguable

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Pass example	`(71.2 − 70.9) / 71.2 = 0.42%`	coverage: manual 71.2% vs pipeline 70.9% (illustrative)	0.42% ≤ 1% → pass
3	Investigate example	`(461 − 456) / 456 = 1.1%`	tickets: manual 456 vs pipeline 461 (illustrative)	1.1% > 1% → investigate: 5 late tickets, the manual side was wrong
4	Silence alert	`1 − 96% = 4% > 1% threshold → alert`	threshold illustrative; the 4% catch is record	silence never mistaken for health
5	Join fix	`documented cross-reference table now; a real key pushed upstream to Support`	record	join rate → ~100%
6	Definitions into code	`seven frozen metrics with owners`	record	no drift; no longer a single point of failure

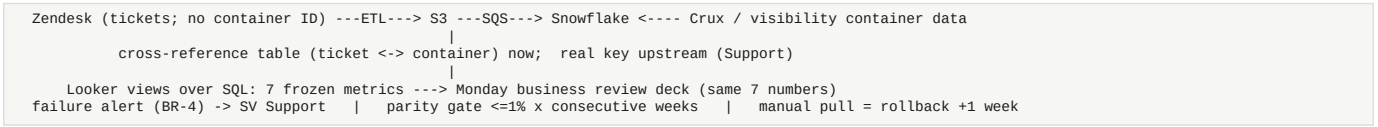
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Review data accuracy	60%	90%	record
2	Single point of failure	one person (bus factor one)	no longer one	record (qualitative; any count is a user anchor)
3	Manual hours (my pull)	~4 h/wk	~0	modeled
4	Silent data loss	invisible	alerted; first catch a 4% ticket-sync drop	record
5	Security-review miss	not on the register	2 weeks lost; owned upward the same day; re-sequenced; 0 Mondays skipped	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Hours term	`4 × 52 × $145 = $30,160`	modeled	≈$30K/yr
2	Decision-risk term	`52 × 10% × $13,500 = $70,200`	illustrative P and cost	≈$70K/yr
3	Priced baseline	`$30,160 + $70,200 = $100,360`	priced — NOT savings	>$100K/yr
4	Share of risk retired	`(40% − 10%) / 40% = 75%`	wrong rate 40% (1 − 60%) → 10% (1 − 90%)	75% retired
5	Risk retired	`$70,200 × 75% = $52,650`	modeled	≈$50K/yr
6	Bus-factor exposure	`20% × $75,000 = $15,000`	20% key-person attrition (benchmark) × $75K rebuild + blind weeks (illustrative)	≈$15K/yr
7	Modeled total inside the price	`$27,000 + $47,000 + $17,000 = $91,000`	my model's sensitivity base (hours $27K + risk $47K + bus factor $17K); the $30K + $50K + $15K split above sums to $95K — if asked about the split, say "about ninety thousand, modeled"; the priced baseline is "about a hundred thousand"	≈$90K/yr modeled
8	Break-even P for >$100K	`($100,000 − $30,160) / (52 × $13,500) = 9.9%`	the risk term must carry at least $69,840	P(wrong steer) ≈ 10%: below it the priced baseline falls under $100K

SOLUTION FLOW — THE SYSTEM AND DATA MAP NEW



SENSITIVITY — WHAT MOVES THE PRICED BASELINE NEW

INPUT	LOW	BASE	HIGH
P(wrong steer) — illustrative	5%	10%	15%
Decision-risk term = 52 × P × $13,500	52 × 5% × $13,500 = $35,100	52 × 10% × $13,500 = $70,200	52 × 15% × $13,500 = $105,300
Priced baseline = $30,160 + term	$30,160 + $35,100 = $65,260	$30,160 + $70,200 = $100,360	$30,160 + $105,300 = $135,460

The honest reading: the >$100K needs roughly one wrong steer in ten Mondays; at 5% the price is about $65K. That is why it is priced, never claimed — and why the realized test is Monday restatements after cutover.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS NEW

> "The number is a price, not a saving: the manual status quo was priced in the BRD at more than a hundred thousand dollars a year — thirty hours a week of review work at an assumed ninety dollars an hour is $103,680 — and that is what let the infra team prioritize the build. Inside that my model puts about thirty thousand on my hours and about seventy on decision risk — a VP steering on a number that was wrong two weeks in five. Taking accuracy from sixty to ninety retires about fifty of that, and no longer being a single point of failure is worth about fifteen. The claimed record is the accuracy move, the one-percent parity, and the four-percent drop we caught; the splits are modeled, and I will not call any of it realized savings."

IF CHALLENGED — THE FULL ANSWERS NEW

You never claimed the $100K, so why is it here?

Because it is the number in the BRD that let the infra team prioritize the build; it is a priced baseline — the BRD's thirty hours a week at an assumed ninety dollars an hour — not a saving. What I would measure now is hours actually retired and whether any Monday number was restated after cutover — that is the honest realized version.

The decision-risk number is the soft part.

Yes. It rests on one wrong steer in ten Mondays, worth a squad-week; the break-even is about 10%, and at 5% the price is about $65K. I would firm it by tagging every review action item for a quarter and counting how many were reversed when a number was corrected.

Bus factor is not a dollar.

Agreed, so I priced it as expected loss: 20% attrition times the rebuild plus blind weeks. The real test is whether the seven numbers publish when I am on vacation — and after cutover they did.

Where this lands at Amazon. The Ownership pattern: price the status quo in the owning team's own rates so they prioritize it themselves, keep the old path alive until the new one has earned parity, and treat security, privacy and data-access reviews as dependencies with lead times. Reliability (a number that survives its author), responsiveness (alerts instead of silence), cost efficiency (priced, honestly).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S16

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you took ownership of a problem that wasn't yours to fix.	Ownership	Nobody had written requirements for operating metrics; I wrote the BRD and wrote myself out of the system I had built.	Do not claim the build — infra built; I owned BRD and cutover.
Tell me about a mistake and what you learned.	Ownership / Earn Trust	The security review: two weeks, owned upward the same day, zero Mondays skipped; the kickoff question that outlived it.	Do not minimize it as "a dependency slipped" — it was my scoping miss.
Tell me about simplifying something.	Invent and Simplify	Four systems → one pipeline; seven definitions into code; a cross-reference table now, the real key upstream.	Skip the ETL stack names unless asked.
Tell me about a time you made a business case.	Frugality	The status quo priced at >$100K/yr in the owner's own rates — including decision risk; it got prioritized.	Never "saved $100K."
How do you know when a new system can replace the old one?	Highest Standards	Parity within 1% per metric for consecutive weeks; mismatches explained by cause; old path alive a week beyond.	—
Tell me about learning something outside your expertise.	Learn and Be Curious	Learned enough of the data stack to write a BRD infra could build from; the first requirements ever written for a program's metrics.	Do not overclaim engineering depth.

BEHAVIORAL PACK — FOUR LIKELY QUESTIONS, FOUR SEPARATE 90-SECOND STAR ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY) NEW

Q1 AIQB Are Right, A Lot Q5 — "We don't always make the right judgment all the time. Tell me about a time when you made an error in judgment. What was the impact? What did you learn? How have you applied what you learned?" ARE RIGHT, A LOT · OWNERSHIP

> **Opener (first two sentences):** "Mid-build, a data-access security review surfaced that I had not planned for. It cost two weeks, it was my scoping miss, and I told my leadership the same day."

S — I was moving the Monday business review of a visibility program onto one pipeline; my BRD risk register had the join gap, parity failure and bus factor one — not the security review.

T — Keep every Monday review alive while absorbing the slip, and fix the cause rather than the symptom.

A — I called it a scoping miss, not "a dependency slipped"; re-sequenced the build; kept the review running on the manual pull; closed the review. Fifth why: I had treated it as an engineering concern instead of a program dependency with a lead time. The fix is institutional — every kickoff now asks whose approval the data needs and what its lead time is.

R — Two weeks lost; zero Monday reviews skipped; parity run completed; cutover accepted. I would take the two weeks again — I would just plan them.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Two weeks lost; zero reviews skipped.	Any new lane, door, shift or system change at an FC has approvals with lead times; I put them on the plan on day one, not when they surface.	"A dependency slipped"; blaming infra or security; claiming the review was optional.

Q2 **AIQB Ownership Q3 — "Give me an example of an initiative you undertook because you saw that it could benefit the whole company or your customers, but wasn't within any group's individual responsibility so nothing was being done."** OWNERSHIP

> **Opener (first two sentences):** "Nobody owned the numbers that steered a program's weekly review except me — and I was the single point of failure. So I wrote requirements nobody had asked for and wrote myself out of the system."

S — The Monday review ran on four systems that did not talk, stitched in Google Sheets by one person — me: bus factor one, accuracy 60%, 1.5–2 hours per snapshot.

T — Nobody had written requirements for a program's operating metrics. I wrote the BRD and owned the cutover; the infra team owned the build.

A — Documented the as-is, froze seven metric definitions, specified the feed, the failure alerts and a ticket cross-reference, and set the acceptance gate: parallel run within 1% per metric for consecutive weeks. I kept the manual pull as rollback a week beyond parity.

R — Accuracy 60% → 90%; no longer a single point of failure; the first automated catch was a 4% ticket-sync drop invisible in the manual era; the pipeline steers the review without me.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Accuracy 60% → 90%; no longer a single point of failure.	The weekly inbound capacity review at NACF should survive any one analyst: frozen definitions, a parity run before any feed replaces a manual one.	"I built the pipeline" — infra built it; I owned the BRD, gate and cutover. Do not conflate with S11's ETA accuracy 60 → 90%.

Q3 **AIQB Frugality Q6 — "Give an example of a time you requested additional funding/budget to complete a project. Why was it needed? Did you try to figure out another approach? Did you get the additional resources? Why or why not?"** FRUGALITY

> **Opener (first two sentences):** "I could not get a build prioritized by saying the spreadsheet was painful, so I priced doing nothing in the BRD — and I was careful to call it a price, not a saving."

S — The infra team had its own roadmap and I had no authority over it; my manual pull was a bus-factor-one risk to a VP's decisions.

T — Make the status quo legible in dollars, including the risk of a wrong decision, not just my hours.

A — The BRD's own arithmetic was $103,680 — thirty hours a week at an assumed $90 an hour. My own split, a model: hours term ≈ $30K (4 h/wk × 52 × ≈$145); decision-risk term ≈ $70K (one wrong steer in ten Mondays at about a squad-week). I told them the risk term is the soft one — the break-even is about 1 in 10 Mondays — and scoped the build to the seven metrics, nothing new.

R — The infra team prioritized the build off the priced BRD. The price is more than $100K a year — priced, not saved; the realized test is Monday restatements after cutover.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
>$100K/yr — priced, not claimed.	At NACF I would price a labor or placement proposal with Finance's rates, name each rate as assumed or Finance's own, and say which input is soft before they find it.	"Saved $100K"; calling the decision-risk dollar realized.

Q4 **AIQB Invent and Simplify Q1 — "Give me an example of a complex problem you solved with a simple solution. What made the problem complex? How do you know your solution addressed the problem?"** INVENT AND SIMPLIFY · HIGHEST STANDARDS

> **Opener (first two sentences):** "I would not move the deck onto the pipeline until it matched the manual pull within one percent per metric for consecutive weeks — and until every mismatch had a cause."

S — The VP was steering a rollout on seven numbers; a wrong number on cutover day would have cost more than the delay.

T — Make the cutover unarguable.

A — Parity gate |pipeline − manual| / manual ≤ 1% per metric; every mismatch classified — most turned out to be late-arriving tickets on the manual side; failure alerts live before cutover; manual pull kept as rollback a week beyond parity.

R — First Monday off the pipeline the VP saw the same seven numbers; the first automated catch was the 4% ticket-sync drop.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Parity within 1% per metric for consecutive weeks.	Same standard for retiring a manual placement or labor plan: a tolerance published before the parallel run, rollback kept alive until parity holds.	A specific number of consecutive weeks (the record says "consecutive weeks").

SAME STORY, THREE ANGLES — DIFFERENT OPENER, HEADLINE METRIC AND LP EACH NEW

(i) Data / analytical

DIVE DEEP

> "The first thing I did was list what the Monday number was actually made of."

Headline metric: Seven metrics; accuracy 60%; 1.5–2 h per snapshot; a 4% drop caught

- BRD as-is: tickets with no container ID; a hand join in Sheets.
- Parallel run: mismatches classified by cause; mostly late tickets on the manual side.
- Silence alert: 1 − 96% = 4% > 1% threshold.
- The manual era could not see a 4% loss at all.

(ii) Conflict / influence / backbone

EARN TRUST · OWNERSHIP

> "I had no authority over the infra roadmap, and a review that could not be skipped — so I priced my problem in the BRD and re-sequenced rather than route around the security review."

Headline metric: >$100K priced; two weeks, zero skipped Mondays

- Priced in the BRD (priced, not saved); scoped to seven metrics only.
- Security review: re-sequence, do not argue around a data-access control.
- Told leadership the same day; the VP never saw a skipped Monday.
- Honest tension: the cross-reference table is a bridge, not the real key — Support owned the key.

(iii) Failure / learning

LEARN AND BE CURIOUS

> "I missed a dependency that cost two weeks — and the miss was in how I thought, not in my calendar."

Headline metric: Two weeks lost; 0 reviews skipped

- Real failure arc G3 (Apple v11): the BRD priced the build and never the access review; the parallel run kept the Monday review alive.
- Category error: security review treated as engineering, not a program dependency with a lead time.
- Fix: every kickoff asks whose approval the data needs and its lead time.
- Softest number I should have firmed: the decision-risk dollar — tag a quarter of action items.
- I would trade the two weeks again; I would plan them now.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS NEW

T1 "What is the data that proves the $100K — and what number would change your mind?"

Numeric answer. Hours term 4 × 52 × $145 = $30,160; risk term 52 × 10% × $13,500 = $70,200; priced $30,160 + $70,200 = $100,360. Break-even P(wrong steer) ≈ 10%: at 5% the price is ≈ $65K. The BRD's own labor case is 30 × $90 × 48 − 6 × $90 × 48 = $103,680 (30 → 6 h/wk at an assumed $90, 48 weeks) — record-from-docs, and the number to say; my split above is a second lens, a model (both exceed $100K; say "over a hundred thousand"). Nothing here is realized savings.

Say exactly: "It is a price, not a saving. The part that would change my mind is how often a wrong number becomes a wrong steer: below about one Monday in ten, the price falls under a hundred thousand. The realized test is how many Monday numbers were restated after cutover."

T2 "What was YOUR specific contribution versus the infra team's?"

Numeric answer. Mine: the as-is, the pricing, the BRD with seven frozen definitions, the acceptance gate, the parallel-run procedure, the cutover memo, the first-catch triage. Infra's: the ETL → S3 → SQS → Snowflake → Looker pipeline and its alerts. Support owned the real ticket key.

Say exactly: "I wrote the requirements, the gate and the cutover memo and ran the parallel run. The infra team built the pipeline and the alerts. I did not build it, and I will not say 'we built it'."

T3 "Why was taking two weeks the right long-term decision — versus shipping around the review?"

Numeric answer. Two weeks lost; the manual pull absorbed the gap at about 2 × 4 = 8 extra hours of my time — against putting customer ticket data into an unreviewed pipeline (BR-1 in the BRD: customer data must be protected). The pipeline was to outlive me; a control bypassed once is a control bypassed.

Say exactly: "The short-term cost was two weeks and about eight hours of manual pulling. The long-term cost of routing around a data-access review is a pipeline nobody can trust — which defeats the point of replacing my spreadsheet."

20-SECOND VERSION AND NACF BRIDGE NEW

20-SECOND VERSION NEW

"The weekly review of a visibility program ran on a hand-stitched spreadsheet only I could fix. I priced it, wrote the requirements, froze seven definitions and moved it onto one pipeline behind a one-percent parity gate. Accuracy went from 60 to 90 percent — and I missed a security review that cost two weeks, which I owned upward the same day."

NACF BRIDGE NEW

"NACF's capacity review — dock doors, UPH labor, cubic fill, weeks of cover — is only as honest as its feeds. A receive-scan feed that silently drops 4% makes the 85% door ceiling read wrong. I would freeze the definitions, parallel-run any new feed against the manual plan, and alert on silence."

S10 — Elementum — Austria: "the noise said carriers; the data said measurement" — routing-guide data quality ~55% → ~99% within a quarter, $500K/yr on a Finance-co-owned method, then ~$8M across 25 European locations ELEVATED

LPs this story can serve here: Are Right, A Lot (primary — the data overturned the room's explanation) · Success & Scale Bring Broad Responsibility (primary — one site's fix rolled to 25 sites) · Dive Deep (backup — measured the measurement) · Earn Trust (the Finance-co-owned method before any dollar) · Customer Obsession (a scorecard the carriers could not dispute)
Process family: Deliver — transportation performance measurement (on-time delivery), routing-guide governance, freight-savings control

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF EACH NEW

Are Right, A Lot	I pulled three years of shipment history in SQL myself and measured the room's explanation instead of arguing it: true data quality was ~55%, not the 40% proxy, and the defects were ours, not the carriers'.
Dive Deep	I classified every blank — missing rate, wrong-grain lane, one carrier under five spellings, brochure transit times posing as promises — and burned the classes down weekly by reason code to ~99% within a quarter.
Earn Trust	Finance co-owned the savings method before I claimed a dollar — contracted guide rate minus paid rate, executed shipments only — so the carriers never disputed the scorecard or the dollars.
Success & Scale Bring Broad Responsibility	One site's fix became 25 sites' instrument at a rules file plus one DQ cycle each; a no-hit pages a human instead of silently defaulting a carrier.

Honesty label. I owned the analysis and the rules design and pulled the data myself; the data platform team built the engine; Finance co-owned the savings method before any dollar was claimed. Site-wave counts and dates beyond "Austria first, then 25 sites" are not in the record. Guardrails: $8M, never "$1M"; say "within a quarter" — true under Apple's two-month bar and the ~3-month canon — never "in two months" and never "400+ lanes"; the eight-to-ten hours is the before and "a small fraction" the after (the 1–2 hrs/wk is a user anchor). **Basis:** 40% reported → ~55% true, ≈97%, $500K at Austria, the ~$8M across 25 European locations and 8–10 hrs/wk are record (Apple v11 F2); the ~$5M freight base and the 10% behind it, the ~8% Incoterm share and the 1–2 hrs/wk after are my own anchors.

SAFE-SAY BOX NEW

The most attackable number: how long to ~99% — and the $8M. Say: "From about fifty-five percent true data quality to about ninety-nine within a quarter. At Austria that was worth about five hundred thousand a year; across twenty-five European locations, about eight million of validated freight savings, executed shipments only."

If he asks "what is your source for that?" "The eight million is the per-shipment ledger — contracted guide rate minus paid rate, executed shipments only — on a method Finance co-owned before my first claim. The five-million freight base and the ten percent behind the Austria figure are my own anchor, and the exact month count differs between my notes. The number I would stand behind is the ledger."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Transportation at the customer's Austria site could not measure on-time delivery. Reports said data quality was 40%; the team spent 8–10 hours a week reconciling promised against actual dates by hand; the freight base was about $5M a year (my own anchor) with no enforceable on-time standard — and everyone, including my own sponsor, believed the carriers were late. **Task.** I owned the analysis and the rules design; the goal was an on-time number the carriers could not dispute, and savings claimed only on a method Finance co-owned first. **Action.** I pulled three years of shipment history in SQL myself — the room already had an explanation, and a request queue would have been slower and arguable. True data quality was about 55%, not 40%, and the defects were ours: missing rates, wrong-grain lanes, one carrier under five spellings, brochure transit times posing as promises. The problem was measurement, not carriers. I specified the promised-delivery-date as a routing-guide rules engine — lane key by ship-from, ship-to, mode, volume band, Incoterm and SLA, with Incoterm needed on a share of lanes — about 8% in my own anchor — and a no-hit paging a human rather than defaulting a carrier; defects worked as classes with reason-code alerts and a weekly burn-down; and the savings method signed with Finance before the number existed — contracted guide rate minus paid rate, per shipment, executed only. The scorecard was released only when automated on-time matched a manual sample per lane cohort; the sponsor was unhappy with the delay for weeks and right to want speed; the carriers never disputed the result. Then scale: a rules file plus one data-quality cycle per site, regional super-users after two shadowed reviews. **Result.** Primary metric: routing-guide data quality — about 55% true, 40% reported → about 99% within a quarter. About $500K a year at Austria on the Finance-co-owned, executed-only method, then about $8M across 25 European locations — both record; manual reconciliation from eight-to-ten hours a week to a small fraction; about $0.5M a year of analyst capacity modeled on top.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Transportation at the customer's Austria site could not measure on-time delivery. Reports said data quality was 40%; the team spent 8–10 hours a week reconciling promised vs actual dates by hand; the freight base was ≈$5M/yr (user anchor) with no enforceable OTD. Everyone — the site, the carriers' account teams, my own sponsor — believed the carriers were late.

The cost of acting on the room's explanation: a scorecard on 55% data the carriers would dispute and win, the manual reconciliation continuing at every site, and any savings claim disputed after the fact. The customers: the site that needed an enforceable standard, and the carriers who deserved to be measured against a real promise.

T — Own the analysis and the rules design. Goal: an OTD the carriers could not dispute, and savings claimed only on a method Finance co-owned first. The real challenge: the data had to overturn the room's explanation.

Baselines on the table: routing-guide data quality 40% reported, ~55% true once measured · manual reconciliation 8–10 hours a week per site · enforceable OTD: none, on a ≈$5M/yr freight base (user anchor).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — pulled three years of shipment history in SQL myself; defect census (missing rates, wrong-grain lanes, five spellings of one carrier, brochure transit times); true DQ ~55% vs 40% reported; the problem reframed as measurement, not carriers. *Exit bar:* sponsor accepts the reframe on the evidence; defect classes named and counted.
- **Plan & Design** — rules-engine spec (lookup by ship-from/to, mode, volume, Incoterm, SLA; a no-hit pages a human, never a default carrier); reason-code alert design; the savings method signed with Finance before any dollar was claimed (contracted guide rate − paid rate, per shipment, executed only); the scorecard

DQ gate (automated OTD = manual sample per lane cohort). *Exit bar:* spec signed by transportation; Finance method signed; scorecard gate published.

- **Build & Test** — held engineering to the spec; weekly defect burn-down by reason code; manual-sample comparison per lane cohort (≈97% agreement). *Exit bar:* DQ trending to ~99%; automated OTD matches the manual sample per cohort.
- **Cutover & Go-Live (Austria)** — released the carrier scorecard only after the gate passed; retired the manual reconciliation (8–10 hrs/wk → a small fraction); stood up the savings ledger per shipment. *Exit bar:* a scorecard the carriers could not dispute; $500K/yr on the Finance-co-owned method.
- **Hypercare / Close → scale** — super-user model per region; one-page runbook (add a lane · read a no-hit · when to escalate); two shadowed rule reviews per site; rolled out site by site. *Exit bar:* 25 EU sites ≈ $8M; ~99% DQ; adoption cost per site = a rules file + one DQ cycle, in days.

Influence and ownership. I pulled the data myself because the room already had an explanation; the data platform team built the engine; regional super-users ran the rules files after two shadowed reviews each. Mine: the SQL pull, the defect census, the rules-engine spec, the reason-code alert design, the Finance-co-owned method and the scorecard gate. The sponsor was unhappy with the delay for weeks — answered with the defect classes trending down weekly and the manual-sample gate; he was right to want speed, and the carriers never disputed the result. Refused: a scorecard on 55% data, a silent default carrier on a no-hit, and any savings number without a Finance-co-owned method.

> **Designed the process; others own it now.** I designed the rules-engine spec, the reason-code alerts and the savings method; regional super-users run the rules files and the DQ cycle at 25 sites without me. **Trade-off put to leadership.** The recommendation to the sponsor and Finance outlined the competing metrics — speed to a carrier conversation vs a baseline the carriers could not dispute — and held the scorecard until DQ reached ~99%, with savings counted executed-only.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — $500K/yr at Austria on the Finance-co-owned, executed-only method (record; the 10%-of-a-~$5M-base arithmetic is my user anchor) — then ~$8M across 25 EU sites (record roll-up; ~$320K/site average, derived); plus ≈$0.5M/yr *modeled* analyst capacity (range $0.2–0.7M).
- **#** — 25 EU sites on the same rules file and one DQ cycle each; three years of shipment history pulled in SQL for the baseline (record).
- **%** — routing-guide data quality ~55% true (40% reported) → ~99% within a quarter (record); ≈97% agreement automated vs manual sample.
- **Time** — ~99% DQ within a quarter; adoption cost per site = a rules file + one DQ cycle, in days (record).
- **Customer** — a carrier scorecard the carriers could not dispute — automated OTD matched the manual sample per lane cohort before release (record).
- **Quality** — manual reconciliation 8–10 hrs/wk → a small fraction per site (the after is a user anchor); the savings method signed with Finance before any dollar; negotiated-but-not-executed never counted (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	The room already had an explanation — "the carriers." I pulled three years of shipment history in SQL myself, built the defect census and reframed the problem as measurement; the sponsor had not asked for that.
Scope & influence	The data platform team built the engine; regional super-users ran the rules files after two shadowed reviews each; Finance co-owned the savings method before any dollar; the sponsor, unhappy with the delay for weeks, was answered with the defect census.
Execution trade-off + mechanism others now own	Speed (a carrier beating now) vs a baseline that could not be disputed: DQ to ~99% before any scorecard was released. Mechanisms others own now: a rules file + one DQ cycle per site, the super-user model, the one-page runbook and the Finance-co-owned, executed-only savings method.
Impact (long-term)	$500K/yr at Austria (record) → ~$8M across 25 EU sites (record roll-up); manual reconciliation 8–10 hrs/wk → a small fraction per site.
Process improvement (standard work institutionalized)	Reason-code alerts; weekly defect burn-down by reason code; a no-hit pages a human, never a default carrier; adoption cost per site measured in days.
Complexity & consensus	A carrier scorecard the carriers could not dispute — automated OTD matched the manual sample per lane cohort (≈97% agreement) before release.
Root cause at scale	Five spellings of one carrier, wrong-grain lanes and brochure transit times were the trending cause — fixed in the rules engine and the DQ cycle once, then rolled to 25 sites.

THE FIVE FACETS — DATA ANALYSIS · SOLUTION DESIGN · MATH & DATA LOGIC · IMPLEMENTATION · PROBE QUESTIONS NEW

1 · DATA ANALYSIS

Data. Three years of shipment history pulled in SQL, at shipment × leg grain from the customer's SAP delivery and transport tables (actual goods-issue date, planned begin/end, forwarder), joined to (a) the routing guide on the lane key — ship-from, ship-to, mode, volume band, Incoterm, SLA; (b) actual delivery timestamps from carrier proof-of-delivery, 214 events and portals; (c) a carrier-name map. **Cuts:** by defect class and by lane cohort. **What it revealed:** reported data quality 40% was a field-populated proxy; true quality — valid promised ∧ valid actual ∧ resolvable lane — was ~55%, so about 45% of shipments could not be validly scored; four defect classes (missing rates, wrong-grain lanes, one carrier under five spellings, brochure transit times posing as promises) were ours. Manual reconciliation ran 8–10 hours a week per site.

2 · SOLUTION DESIGN

Options: (1) take the room's explanation — beat the carriers with a scorecard now (rejected: a scorecard on ~55% data would be disputed, and the carriers would win); (2) fix measurement first — a promised-date rules engine, then a scorecard (chosen). **Design:** promised = ship date + transit(lane key); Incoterm in the key (needed on ~8% of lanes — a user anchor — because it changes who owns the clock); a no-hit pages a human, never a default carrier; defects worked as classes with reason-code alerts and a weekly burn-down; scorecard released only when automated OTD matches a manual sample per lane cohort (≈97% agreement); savings = contracted guide rate − paid rate per executed shipment, method co-owned with Finance first.

3 · MATH & DATA LOGIC

Formulas. True DQ = shipments with valid promised ∧ valid actual ∧ resolvable lane ÷ all shipments. OTD = deliveries with actual ≤ promised ÷ valid promises. Savings = Σ (contracted guide rate − paid rate) over executed shipments. Austria = base × 10%. Roll-up average = total ÷ sites; implied base = total ÷ 10%. Analyst capacity = hours saved per site per week × sites × weeks × rate. Substituted below.

4 · IMPLEMENTATION

Phases: Discovery (SQL pull, defect census) → Plan & Design (rules-engine spec, reason-code alerts, Finance-co-owned method, scorecard gate) → Build & Test (data platform team built to my spec; weekly burn-down; manual-sample comparison) → Cutover at Austria (scorecard released after the gate; manual reconciliation retired 8–

10 h/wk → a small fraction) → Hypercare/scale (regional super-users, one-page runbook, two shadowed rule reviews per site). **Controls:** manual reconciliation kept alive until the match held (free rollback); executed-only ledger. **Risks:** Incoterm owning the clock; a silent default carrier; savings disputed after the fact.

5 · PROBE QUESTIONS

Probing carries "how did you know they were wrong", "is the $8M real", "defend the delay", "what is Broad Responsibility here". Trap Drills: carriers-improving vs measurement, the $8M and my contribution, and why delaying was right and who was right about speed.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

BR — Everyone believed the carriers were late. How did you know they were wrong?

I didn't — I measured. Three years of shipment history in SQL showed true data quality was about 55%, not the 40% proxy, and the defects were ours: missing rates, wrong-grain lanes, five spellings of one carrier, brochure transit times posing as promises. The data overturned the explanation; I did not argue with it.

BR — Is the $8M real? Fifth why.

It is the Finance-co-owned, executed-only method applied at each site, not a projection: contracted guide rate minus paid rate, per shipment, only on shipments that moved at the guide rate. Negotiated-but-not-executed never counted. The $500K Austria pilot is the method's proof; the $8M is the same ledger across 25 sites; what firms it is the per-shipment ledger Finance read. The ten percent is my own anchor, not the proof — Austria at $500K sits above the ~$320K/site simple average.

BR — Your sponsor wanted speed and you delayed weeks. Defend that.

I traded date for evidence, never scope: a scorecard on 55% data would have been disputed and the carriers would have won. I showed him the defect classes trending down weekly and the gate the scorecard had to pass. He was right to want speed; the carriers never disputed the result.

BR — What is Broad Responsibility here?

One site's fix became 25 sites' instrument — and the responsibility was in the method: the savings rule was signed before the number so no site could claim what it had not executed, and a no-hit paged a human rather than silently defaulting a carrier.

BR — Who was the customer?

The site, which needed an enforceable standard; and the carriers, who deserved to be measured against a real promise — the scorecard's legitimacy with them is why the savings held.

BR — What would you do differently?

Start with the industry benchmark for lane transit times instead of proving the brochure wrong lane by lane — same answer, faster. And sign the method before the number — which held, and which I would do again.

BUSINESS IMPACT MODEL — KPI TREE AND DRIVERS NEW

```
Enforceable OTD + executed freight savings: $500K Austria -> ~$8M (record)
+- Measurement integrity (stock): true DQ ~55% (40% reported) -> ~99%
|    +- promised date = ship + transit(lane_key), Incoterm in key (~8% of lanes, user anchor); no-hit pages a human
+- Savings ledger (flow): (guide rate - paid rate) x executed shipments;  Austria ~$500K/yr (the ~$5M x 10% is a user anchor);  25 sites ~$8M
+- Manual reconciliation: 8-10 hrs/wk -> a small fraction per site (1-2 is a user anchor; modeled $)
+- Defect loop (+): alerts by reason code -> weekly burn-down -> DQ rises -> fewer no-hits -> scorecard trusted
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Routing-guide data quality	~55% true (40% reported)	~99% within a quarter	—	record
Austria savings (guide − paid, executed)	$0	10% of ~$5M (user anchor)	$500K	record ($500K); 10% × $5M is a user anchor
Network roll-up, 25 EU sites	$0	~$8M	~$8M	record
Manual reconciliation, 25 sites	8–10 h/wk each	1–2 h/wk (user anchor)	≈$0.5M	modeled

MATH — THE FOUR TABLES (EVERY INPUT SUBSTITUTED) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reported data quality	`40% — a field-populated proxy`	record	the number leadership saw
2	True data quality	`valid promised ∧ valid actual ∧ resolvable lane ÷ all shipments ≈ 55%`	three years of shipment history pulled in SQL (record)	~55% true
3	Shipments that could not be validly scored	`100% − 55% = 45%`	derived from the definition	≈45% of shipments
4	Defect classes	`missing rate · wrong-grain lane · carrier name in five spellings · brochure transit posing as a promise`	record	4 named classes
5	Manual reconciliation	`(8 + 10) / 2 = 9 h/wk per site (range midpoint) × 9 × 25 = 225 h/wk across 25 sites`	8–10 h/wk per site (record); same burden at every site is illustrative	≈225 h/wk
6	Freight base without an enforceable OTD	`≈$5M a year at Austria`	record	no basis for a carrier conversation

Table 2 — the fix: values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promised-date rule	`promised = ship_date + transit(lane_key); lane_key = (from, to, mode, volume band, Incoterm, SLA)`	Incoterm needed on ~8% of lanes (user anchor); a no-hit pages a human, never a default carrier (record: design)	a promise per shipment, from the guide

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	OTD definition	`OTD = deliveries with actual ≤ promised ÷ valid promises only`	brochure promises excluded (record)	enforceable
3	Release gate	`automated vs manual sample per lane cohort ≈ 97% agreement`	record	scorecard released only after the gate
4	Burn-down	`defect classes worked weekly → ~99% within a quarter`	record	~99% DQ
5	Savings method	`saving = Σ (contracted guide rate − paid rate) per executed shipment`	Finance-co-owned before any claim; negotiated-not-executed never counted (record)	results follow in Table 4

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Routing-guide data quality	~55% true (40% reported)	~99% within a quarter	record
2	OTD	unenforceable	enforceable; ≈97% agreement vs manual sample	record
3	Manual reconciliation per site	8–10 h/wk	1–2 h/wk (user anchor; 83% at the midpoints)	before: record; after: user anchor; midpoint cut derived
4	Austria savings	$0	$500K/yr, executed-only	record
5	Network roll-up	$0	~$8M across 25 EU sites	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Austria	`$5,000,000 × 10% = $500,000`	~$5M freight base × 10% (user anchor; the $500K result is record — Finance-co-owned, executed-only)	$500K/yr
2	Network average per site	`$8,000,000 / 25 = $320,000`	~$8M record roll-up ÷ 25 sites — average derived, stated as such	~$320K/site
3	Implied freight base	`$8,000,000 / 10% = $80,000,000`	at my 10% anchor; derived — not volunteered	~$80M implied base
4	Austria vs average	`$500,000 / $320,000 = 1.56 × the average`	above the simple average — no claim about why	Austria above average
5	Analyst capacity returned	`(9 − 1.5) × 25 × 50 × $55 = $515,625`	7.5 h/wk saved × 25 sites × 50 weeks × $55/h — modeled add-on; assumes all 25 sites carried Austria's burden	≈$0.5M/yr modeled

SOLUTION FLOW — THE SYSTEM AND DATA MAP NEW

```
customer SAP (shipments: actual GI, planned begin/end, forwarder) --nightly, 3 yrs history--> shipment fact (SQL)
routing guide (rules per site; versioned) --rules file by regional super-user--> rules engine: promised = ship + transit(lane_key)
carrier POD / 214 / portals --actual delivery ts (<=24h join)--> OTD score  |  carrier_name_map (5 spellings -> 1 id)
no-hit -> pages a human   |   defect classifier (reason codes) -> weekly burn-down   |   scorecard gate: auto = manual sample
contracted rates (guide) + executed invoices (paid) -> savings ledger: (guide - paid) x executed;  Finance-co-owned method
```

SENSITIVITY — LOW / BASE / HIGH (ANALYST CAPACITY, THE SOFTEST LINE) NEW

INPUT	LOW	BASE	HIGH
Hours saved per site per week	6	7.5	8.5
Loaded analyst rate	$50	$55	$65
Sites with the same burden (illustrative)	15	25	25
Capacity returned $/yr	6 × 15 × 50 × $50 = $225,000	7.5 × 25 × 50 × $55 = $515,625	8.5 × 25 × 50 × $65 = $690,625
If realized savings ran at 5% instead of 10% on the implied base	—	$80,000,000 × 5% = $4,000,000	—

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS NEW

> "The measured numbers are about five hundred thousand a year at Austria, on a method Finance co-owned before I claimed a dollar — and about eight million when the same rules file and one data-quality cycle went to twenty-five European locations. Eight million over twenty-five sites is about three hundred twenty thousand a site, which implies roughly eighty million of freight at my own ten-percent anchor — a derived figure, not a measured one; Austria sits above that simple average. On top I model about half a million a year of analyst capacity — that piece is entirely my model."

IF CHALLENGED — THE FULL ANSWERS NEW

Is the eight million really measured?

It is the Finance-co-owned, executed-only method applied at each site, not a projection. The ten percent is my own anchor and I do not lean on it; what firms the eight million is the per-shipment ledger Finance read. If realized savings had run at five percent on the implied base, the number would be about four million — so the ledger, not the rate, is the proof.

The half million of labor is the softest.

It assumes all twenty-five sites carried Austria's 8–10 hours a week and got to a small fraction, which I cannot prove; a one-week time log at three sites before their rules file went live would replace the assumption with data. Range $0.2–0.7M.

> **Why not count negotiated rate cuts too?**
> Because negotiated-but-not-executed is a promise, not a saving. The method counts only shipments that moved at the guide rate — and that discipline is why the carriers never disputed the scorecard or the dollars.

Where this lands at Amazon. The cleanest "used data to overturn the room's explanation" story — and the model for how a savings claim should be built: method signed with Finance before the number, executed-only, then scaled site by site with the adoption cost measured in days. Reliability (an enforceable promise), cost efficiency ($500K → ~$8M record), asset utilization (analyst hours returned, modeled).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S10

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you were right and others were wrong.	Are Right, A Lot	The measurement reframe on three years of data; the fix was a rules engine, not a carrier beating.	Do not gloat — the sponsor was right to want speed.
Tell me about a time your work had impact beyond your team / site.	Broad Responsibility	Austria's method → 25 European locations ≈ $8M; adoption in days; super-users after two shadowed reviews.	Never "$1M"; never claim site-wave counts or dates beyond "Austria first."
Tell me about a time you dug into data others had accepted.	Dive Deep	Pulled the SQL myself; the defect census; true DQ ~55% vs 40% reported.	S8 (measured lanes) is Hoerger's — do not use it here.
How do you make a savings claim credible?	Earn Trust / Frugality	Finance co-owns the method before the number; executed-only; negotiated never counted; the ledger per shipment.	—
Tell me about a time you delayed a launch.	Highest Standards / Backbone	The scorecard held until automated OTD matched the manual sample per cohort; sponsor unhappy for weeks; carriers never disputed it.	Never "99% in two months" — say "within a quarter."
Who was the customer and what changed for them?	Customer Obsession	The site got an enforceable standard; the carriers got a fair promise; the team got 8–10 hours a week back.	The eight-to-ten hours is the before; the after is "a small fraction" (1–2 is a user anchor).

BEHAVIORAL PACK — FOUR LIKELY QUESTIONS, FOUR SEPARATE 90-SECOND STAR ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY) NEW

Q1 **AIQB Dive Deep Q2 — "Tell me about a situation that required you to dig deep to get to the root cause. How did you know you were focusing on the right things? What was the outcome? Would you have done anything differently?"** DIVE DEEP · ARE RIGHT, A LOT

> **Opener (first two sentences):** "Everyone — including my own sponsor — believed the carriers were late. I pulled three years of shipment history myself, because the data could not be argued with and a request queue could be."

S — Transportation at the customer's Austria site could not measure on-time delivery; reports said data quality was 40%; the team reconciled promised against actual dates by hand, 8–10 hours a week, on about a $5M freight base (my anchor).

T — Own the analysis and rules design; produce an OTD the carriers could not dispute.

A — SQL on three years of shipments; a defect census. True quality was about 55%, not 40%, and the defects were ours: missing rates, wrong-grain lanes, one carrier under five spellings, brochure transit times posing as promises. I reframed the problem as measurement, not carriers — and showed the sponsor the classes, not an opinion.

R — Data quality ~55% true → ~99% within a quarter; a scorecard the carriers never disputed; the foundation for $500K a year at Austria.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
40% reported vs ~55% true.	At NACF, lead-time and receipt data drive placement and safety stock; I would measure the measurement before blaming a vendor or a carrier.	"I knew the carriers were fine" — I did not know; I measured. Do not gloat — the sponsor was right to want speed.

Q2 **AIQB Insist on the Highest Standards Q1 — "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?"** INSIST ON THE HIGHEST STANDARDS · HAVE BACKBONE

> **Opener (first two sentences):** "My sponsor wanted the carrier scorecard out. I held it for weeks until automated on-time matched a manual sample for every lane cohort — and he was right to want speed."

S — A scorecard released on 55% data would be disputed — and the carriers would have won.

T — Trade date for evidence, never scope.

A — I showed him the defect classes trending down weekly and the gate the scorecard had to pass — automated OTD agreeing with a manual sample at about 97% per cohort — and kept manual reconciliation running meanwhile. Refused: a scorecard on 55% data, a silent default carrier on a no-hit, any savings number without a Finance-co-owned method.

R — Scorecard released after the gate; carriers never disputed it; manual reconciliation retired — from eight-to-ten hours a week to a small fraction.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
≈97% agreement per cohort; DQ ~99% within a quarter.	At NACF I would publish the bar (an 85% door ceiling, a 95% data gate) before the first release so that holding it is not a personal judgment call.	"99% in two months" (say "within a quarter"); blaming the sponsor.

Q3 **AIQB Frugality Q1 — "Give me an example of how you have helped save costs or eliminate waste within your role or organization. What was the situation? What was the impact?"** FRUGALITY · EARN TRUST

> **Opener (first two sentences):** "I got Finance to sign the savings method before I had a number to claim: contracted guide rate minus paid rate, per shipment, executed shipments only."

S — Freight savings claims are routinely disputed after the fact; the carriers would test every dollar.

T — Make the claim unarguable before it existed.

A — The method was written and signed first; the ledger was per shipment; negotiated-but-not-executed never counted; the scorecard gate meant the OTD underneath was also undisputed.

R — $500K a year at Austria — then ~$8M across 25 European locations; both record roll-ups (the 10% of a ~$5M base is my own anchor); the carriers never disputed it.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
$500K at Austria; ~$8M across 25 European locations.	At NACF I would sign the savings definition with Finance before a placement or transfer program reports a dollar.	"$1M routing guide"; counting negotiated rates.

Q4 **AIQB Success and Scale Bring Broad Responsibility Q8 — "Tell me about a time you left something better than how you found it. What was it and how did you change or impact it? How did your actions influence things for the better?"** BROAD RESPONSIBILITY · HIRE AND DEVELOP

> **Opener (first two sentences):** "One site's fix only matters if twenty-four other sites can adopt it without me, so I packaged it as a rules file, one data-quality cycle and a one-page runbook."

S — Austria worked; the sponsor wanted the method across Europe.

T — Make adoption cost days, not projects — and keep scale from inflating the claim.

A — Regional super-users ran each site's rules file after two shadowed reviews; the runbook was one page — add a lane, read a no-hit, when to escalate. A no-hit pages a human, so scale could not hide a silent default carrier.

R — 25 EU sites at roughly $320K a site on average (a simple average, derived); ~$8M record roll-up; the method outlives me.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
~$8M; 25 sites; adoption in days.	The same packaging — rules file, shadowed reviews, one-page runbook — is how I would roll a placement rule across FCs.	Site-wave counts or dates beyond "Austria first, then 25 sites".

SAME STORY, THREE ANGLES — DIFFERENT OPENER, HEADLINE METRIC AND LP EACH NEW

(i) Data / analytical

DIVE DEEP

> "I measured the measurement."

Headline metric: 40% reported vs ~55% true → ~99%

- Three years in SQL; shipment × leg grain.
- True DQ = valid promised ∧ valid actual ∧ resolvable lane.
- Four defect classes burned down by reason code.
- ≈45% of shipments could not have been validly scored.

(ii) Conflict / influence / backbone

INSIST ON THE HIGHEST STANDARDS

> "My sponsor was unhappy with the delay for weeks — and he was right to want speed."

Headline metric: ≈97% agreement per cohort before release

- I traded date for evidence, not scope.
- The defect classes trending weekly were my answer, not an argument.
- Manual reconciliation stayed alive — a free rollback.
- The carriers never disputed the result.

(iii) Failure / learning

LEARN AND BE CURIOUS · FRUGALITY

> "I proved the brochure wrong lane by lane, when an industry benchmark would have got me there faster."

Headline metric: ~99% within a quarter; $8M record

- Apple v11 F2: build the audit ledger's format during the Austria pilot, not alongside the first claims.
- Do differently: start with the industry benchmark for lane transit times — same answer, faster.
- The 10% is my own anchor; the per-shipment ledger is the proof of the $8M, not the rate.
- The ≈$0.5M analyst capacity assumes 25 sites carried Austria's burden — a time log at three sites would firm it.
- Signing the method before the number held — I would do that again.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS NEW

T1 **"How do you know the improvement was your measurement fix and not the carriers getting better — or seasonality?"**

Numeric answer. I did not claim an OTD improvement. The record claims: data quality ~55% → ~99%, an enforceable OTD (≈97% agreement with a manual sample), and executed savings on a per-shipment ledger (guide rate − paid rate). Counterfactual for savings is the contracted guide rate itself, so carrier behavior and seasonality cannot inflate it.

> **Say exactly:** "I did not claim carriers got better or worse — I claimed the measurement became enforceable. The savings are contracted guide rate minus paid rate on shipments that actually moved at the guide rate, which is why neither seasonality nor carrier behavior can inflate the ledger."

T2 **"What was YOUR specific contribution to the $8M — versus the platform team, Finance and the site super-users?"**

Numeric answer. Mine: the SQL pull and defect census, the rules-engine spec, the reason-code alert design, the Finance-co-owned method, the scorecard gate, the runbook. Platform team: built the engine. Finance: signed the method and read the ledger. Super-users: ran each site's rules file after two shadowed reviews. If realized savings had been 5% not 10% on the implied $80M base, the roll-up would be $4M — the ledger is the proof.

> **Say exactly:** "I owned the analysis, the rules design, the method and the gate; the data platform team built the engine, Finance co-owned the method and read the ledger, and regional super-users ran the sites. The eight million is the sum of those per-shipment ledgers, not my projection."

T3 **"Why was delaying the scorecard right long-term vs short-term — and who disagreed and were they right?"**

Numeric answer. My sponsor disagreed on timing; right that speed matters. A scorecard on ~55% data would leave 100% − 55% = 45% of shipments unscored or wrongly scored — disputable, and the carriers would have won. Cost: weeks. Benefit: carriers never disputed the result, and $500K became ~$8M on the same method.

> **Say exactly:** "He was right to want speed. I traded weeks for a baseline the carriers could not dispute — on that data, nearly half the shipments could not have been validly scored. The proof is that no carrier ever disputed the scorecard."

20-SECOND VERSION AND NACF BRIDGE NEW

20-SECOND VERSION NEW

"Everyone believed the carriers were late. I pulled three years of shipments in SQL, showed our data quality was about 55%, not 40, and that the defects were ours. We fixed the rules engine, took quality to about 99% within a quarter, signed the savings method with Finance first — and Austria's $500K became about $8M across 25 European locations."

NACF BRIDGE NEW

"Inbound lead-time truth sets safety stock. The booklet's INV-B: SS = z × σ_d × √LT = 1.645 × 120 × √7 ≈ 522 units; at a 5-day lead time, 441 — 81 units less. I would measure the lead-time distribution from receipts before trusting any quoted transit time."

S15 — Sephora — converting a 20-person Product/Engineering/QA team to a four-week cadence in six weeks, by handing the gate to the skeptic (critical defects per release −50%) ELEVATED

LPs this story can serve here: Hire and Develop the Best (primary — the QA lead who now owns the gate) · Strive to be Earth's Best Employer (primary — QA elevated, not blamed) · Insist on the Highest Standards (the definition-of-done that can say no) · Have Backbone (naming QA's fear as correct)
Process family: Make — product delivery operating model: release cadence, definition-of-done, quality gate ownership

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF EACH NEW

Hire and Develop the Best	The most skeptical person on the team — the QA lead — authored the definition-of-done; I coached the frame, she owned the gate, and she was defending it to engineering without me by cycle two and after I left.
Strive to be Earth's Best Employer	QA moved from blamed afterthought to owner of the standard: Sev-1s were classified by cause, not blame, and the drop date made overload visible so people got help.
Insist on the Highest Standards	The definition-of-done had entry criteria that could say no, evaluated at Ready-for-QA — and critical defects per release fell 50% while releases rose 50%.
Have Backbone; Disagree & Commit	I told the room QA's fear was correct arithmetic — which is what made the QA lead willing to write the gate — and named rollback to six weeks as a two-way door at kickoff.

Honesty label. VP air cover, no line authority over anyone in the room; the QA lead authored the definition-of-done — I coached the frame, she owned the gate. The −50% (resume) and the six-week conversion (resume, Apple D3) are record; "50% more releases" is arithmetic from the cadence (52 ÷ 6 → 52 ÷ 4), not a measured count; the dollars are my model on illustrative defect counts. Never invent a direct-report headcount.

SAFE-SAY BOX NEW

The most attackable number: "fifty percent more releases" and the dollars. Say: "Critical defects per release fell fifty percent while we converted from a six-week cadence to a four-week one in six weeks. The release count per year is arithmetic from the cadence — fifty-two over six versus fifty-two over four — not a measured count, and the hundred and thirty thousand is my model."

If he asks "what is your source for that?" "The minus-fifty percent and the six-week conversion are on my resume and in my story record; the dollars are my own model on an illustrative defect base. The number I would stand behind is minus fifty percent on critical defects per release."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Newly joined at Sephora, the VP of Supply Chain asked me to move a 20-person Product, Engineering and QA team from a six-week hybrid cadence to four-week agile. QA — a contracting firm — and in-house engineering were heading for a standoff: QA feared compressed test windows, and the arithmetic said they were right. The business cost of getting it wrong was a faster cadence shipping more escaped critical defects — the one outcome the VP could not afford. **Task.** Convert the cadence in weeks, not quarters, without line authority and without the standoff becoming the story; success was the four-week cadence with no rise in escaped criticals and the QA window protected by rule. **Action.** One-on-ones with every person first, an influencer named per function, and the problem stated honestly — QA's fear is correct. Then I handed the load-bearing artifact to the most skeptical person: the QA lead authored the definition-of-done with entry criteria that could say no; I coached the frame, she owned the content, engineering shaped it through review rather than veto. Jira got benefit-ranking and review-status fields; late heavy stories got a drop date with a reason code instead of a silent slide into the last two days; QA sat on the product-business calls with pre-approved scenarios so Sev-1s were classified by cause, not blamed. I facilitated the retrospective once and stepped back from the review line. **Result.** Primary metric: critical defects per release — baseline → −50%, on a cadence that went from six weeks to four (about 50% more releases a year — arithmetic from the cadence, not a measured count); the 20-person team on a four-week cadence in six weeks; the gate defended by its owner to engineering without me by cycle two, and it survived my leaving. About $130K a year modeled — escaped-critical cost avoided and late-land rework removed — and none of it is a record figure; no headcount was invented to get there.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Newly joined, the VP of Supply Chain asked me to move a 20-person Product/Engineering/QA team from a six-week hybrid cadence to four-week agile. QA (a contracting firm) and engineering (in-house) were heading for a standoff: QA feared compressed test windows — and they were right.

The risks: the standoff becoming the story, dev consuming the shrunken test window, Sev-1s mis-attributed to QA — and a faster cadence shipping more escapes, the exact failure the VP could not afford. The customers: the release's end users in supply chain operations, and the QA team who would otherwise be blamed for the arithmetic.

T — Convert the cadence in weeks, not quarters, without the standoff becoming the story. VP air cover, no line authority over anyone in the room. Success criteria written with the VP: four-week cadence, no rise in escaped criticals, QA window protected by rule.

Baselines on the table: release cadence six-week hybrid (≈8.7 releases/yr) · escaped critical defects per release — the baseline the −50% is measured against (count illustrative; the −50% is record) · the QA test window being compressed by arithmetic.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (weeks 1–2)** — one-on-ones with every person (objections, motivations); an influencer per function; the problem statement — QA's fear is correct — and the success criteria. *Exit bar:* every function's real objection heard privately; the QA lead agreed to author the gate.
- **Plan & Design (weeks 2–3)** — ceremony calendar for a four-week cycle; the definition-of-done frame (what a gate needs to survive pressure — the QA lead wrote the content); Jira field spec (benefit ranking, review status); user-guide outline. *Exit bar:* DoD with entry criteria written by QA and reviewed with engineering; fields live.
- **Build & Test (weeks 3–5)** — the user guide (grooming → completion); live dashboard; the QA drop date with reason codes; QA seated on product-business calls with pre-approved scenarios. *Exit bar:* cycle 1 completed on the new cadence; drop-date rule in force; Sev-1 cause classification running.
- **Cutover (week 6)** — facilitated the retro once; data read-out to the VP; handed the gate to the QA lead to defend herself; rollback (reverting to six weeks) named as a two-way door at kickoff and never used. *Exit bar:* converted in 6 weeks; critical defects −50% under the drop-date rule.

- **Hypercare / Close** — stepped back from the review line; neighbouring teams copied the ceremony from the user guide. *Exit bar:* the gate survived my leaving; Sev-1 trend from story-accuracy defects reversed.

Influence and ownership. Mine: the one-on-ones, the ceremony calendar, the DoD frame, the Jira field spec, the drop-date rule and the retro — facilitated once. Hers: the DoD content and the gate, defended by her. Engineering's objection became co-ownership through review, not veto, with me covering the low-impact stories they had to drop. The assumption I challenged was that QA's fear was resistance — it was correct arithmetic, and saying so is what made the QA lead willing to author the gate.

> **Designed the process; others own it now.** I designed the ceremony calendar, the DoD frame and the drop-date rule; the QA lead owns the gate and defends it, and neighbouring teams copied the ceremony from the user guide. **Trade-off put to leadership.** The read-out to the VP outlined the competing metrics — release frequency (8.7 → 13 per year, modeled) vs escaped critical defects — and showed both moving the right way under a gate that could say no.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$130K/yr *modeled* (escaped-critical cost avoided ≈$90K + late-land rework removed ≈$40K; range $80–270K); none of it is a record figure, and no headcount was invented to get there.
- **#** — 20-person Product/Eng/QA team; releases per year 8.7 → 13 (*modeled* from the cadence: 52 ÷ cycle weeks).
- **%** — critical defects per release −50% (record: resume) — on a cadence that went from six to four weeks (8.7 → 13 releases a year is cadence arithmetic, not a measured count).
- **Time** — six-week hybrid → four-week agile, converted in six weeks (record); the QA lead defending the gate by cycle 2.
- **Customer** — QA moved from blamed afterthought to owner of the standard, with the data protecting them; end users got more frequent releases with fewer escaped criticals (record mechanism).
- **Quality** — a QA-authored definition-of-done with entry criteria that can say no; the drop date with reason codes; Sev-1 cause classification — and the gate survived my leaving (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A VP ask to change a 20-person team's cadence, with QA and engineering heading for a standoff and no line authority over anyone in the room. I chose an influencer per function and set the success criteria.
Scope & influence	20-person Product/Engineering/QA team with VP air cover; the QA lead authored and defended the gate; neighbouring teams copied the ceremony from the user guide — influence by frame, not by rank.
Execution trade-off + mechanism others now own	Release frequency (8.7 → 13/yr modeled) vs test-window quality: a QA-authored definition-of-done with entry criteria that can say no, and a drop date with reason codes. Mechanism others own now: the gate, owned and defended by the QA lead.
Impact (long-term)	Critical defects per release −50% on a six-to-four-week cadence (releases a year 8.7 → 13, modeled from the cadence); the gate survived my leaving; the Sev-1 trend from story-accuracy defects reversed.
Process improvement (standard work institutionalized)	A ceremony calendar for the four-week cycle; Jira benefit-ranking and review-status fields; Sev-1 cause classification; a user guide (grooming → completion) that other teams reused.
Complexity & consensus	QA (a contracting firm) vs engineering (in-house): engineering's objection became co-ownership through review, not a veto; rollback to six weeks named as a two-way door at kickoff and never used.
Root cause at scale	QA's fear was correct arithmetic about compressed windows — addressed by a mechanism (DoD + drop date), not a heroics week, so the defect reduction held across cycles.

THE FIVE FACETS — DATA ANALYSIS · SOLUTION DESIGN · MATH & DATA LOGIC · IMPLEMENTATION · PROBE QUESTIONS NEW

1 · DATA ANALYSIS

Data. The release calendar (six-week hybrid cycle → 8.7 releases a year); the Sev-1 / critical-defect log by release and by cause (story-accuracy vs code); the order in which stories reached QA within a cycle (late heavy lands in the last two days); and a QA-versus-engineering capacity comparison. **Grain:** story × cycle-day; defect × cause × release. **Cuts:** arrival day, defect cause, release. **What it revealed:** QA's fear was correct arithmetic — on an illustrative back-of-envelope, 15 engineers × 18 days ÷ 3 days a story = 90 stories against 4 QA × 18 days ÷ 1 day a story = 72, so QA capped throughput — and the reason codes on late stories turned a silent pattern into a visible one. Counts are illustrative; the record is the −50%.

2 · SOLUTION DESIGN

Rejected: converting the cadence with no gate (the standoff becomes the story and escapes rise); a heroics week (does not hold across cycles). **Chosen:** a mechanism — a QA-authored definition-of-done with entry criteria evaluated at Ready-for-QA; a drop date with reason codes so late heavy stories get re-scoped visibly instead of sliding silently into the last two days; QA seated on the product-business calls with pre-approved scenarios; Sev-1s classified by cause; rollback to six weeks named as a two-way door at kickoff. Why the skeptic writes it: a gate she owns survives pressure; one I write dies when I leave.

3 · MATH & DATA LOGIC

Formulas. Releases/yr = 52 ÷ cycle weeks. Criticals/yr = criticals per release × releases/yr. Constraint = stories one function can clear in the window = people × days ÷ days per story. Avoided cost = (criticals before − criticals after) × cost per critical. Rework = person-days × $/day × releases. Count per year, not per release — halving per release at 13 releases still nets about nine fewer a year. Substituted below.

4 · IMPLEMENTATION

Phases (weeks): Discovery 1–2 (one-on-ones, influencer per function, success criteria with the VP) → Plan & Design 2–3 (ceremony calendar, DoD frame, Jira fields) → Build & Test 3–5 (user guide, dashboard, drop-date rule, QA on product calls) → Cutover week 6 (retro facilitated once; gate handed to the QA lead) → Hypercare (stepped back; neighbouring teams copied the ceremony). **Owners:** me — calendar, frame, fields, rule, retro; QA lead — DoD content and the gate. **Controls:** drop-date reason codes; Sev-1 cause classification. **Risk:** standoff, dev consuming the test window, Sev-1s mis-attributed to QA.

5 · PROBE QUESTIONS

Probing carries "why Hire & Develop with no reports", "faster cadence means more escapes — why did yours fall", "where does four criticals come from", fifth-why on the skeptic. Trap Drills: seasonality/other-causes, my contribution vs the QA lead's, and what would change my mind on the $130K.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

BR — Why is this Hire and Develop? You had no reports.

Because the development mechanism is the story, not the org chart: the most skeptical person authored the load-bearing artifact; I coached the frame, she owned the gate, and she was defending it to engineering without me by cycle two. Development that depends on me staying is not development. I have managed onshore/offshore delivery teams earlier in my career; I have not run a large HR org with comp cycles and won't claim that muscle.

BR — Faster cadence usually means more escapes. Why did yours fall?

Because the gate got stronger as the window got shorter: entry criteria authored by the person with the most to lose, and a drop date that converted silent late lands into visible, reason-coded re-scopes. The −50% is the record; the defect counts behind the dollars are illustrative and labeled.

BR — Where does four criticals per release come from?

It is illustrative, not record — the record gives me the fifty percent drop, not the base count. I would pull the Sev-1 log for the last four six-week releases and use the actual mean.

BR — What did engineering give up, and how did you handle it?

Low-impact stories dropped under the drop-date rule — dev's date protected, QA's window protected, and I covered some of the drops myself. Their objection to the DoD became co-ownership through review, not veto.

BR — Fifth why: why hand the artifact to the skeptic?

A gate the skeptic owns survives pressure; one I write dies when I leave. And naming her fear as correct arithmetic is what made her willing to write it — the standoff was never about resistance.

BR — What would you do differently?

Go to QA first, and put her name on the review line from week one.

BUSINESS IMPACT MODEL — KPI TREE AND DRIVERS NEW

```
Release quality at a faster cadence (escaped criticals/yr, $ avoided)
+- Releases/yr (flow): 52 / cycle weeks: 6w -> 4w = 8.7 -> 13 (+50%)   [modeled from the cadence]
+- Critical defects per release (record -50%)  <- DoD entry criteria (QA-authored) + drop date + QA on product calls
+- Release-end rework (late-land re-test + hotfix) -> ~0 under the drop rule
(loop broken) Sev-1 blamed on QA -> standoff -> gate weakens -> more Sev-1s   [broken by: the skeptic owns the gate]
(loop built)  gate says no -> dev drops low-impact story -> window held -> fewer criticals -> gate trusted
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Cadence / releases per year	6 wks / 8.7	4 wks / 13	—	record cadence; releases modeled (52 ÷ weeks)
Critical defects per release	4 (illustrative)	2	—	record −50%; count illustrative
Escaped-critical cost	≈$350K	≈$260K	≈$90K avoided	modeled ($10K per critical; benchmark $8–15K)
Late-land rework	≈$4.8K per release	≈$0	≈$40K	modeled (6 person-days × $800 × 8.7)
Total modeled	—	—	≈$130K	range $80–270K; no headcount invented

MATH — THE FOUR TABLES (EVERY INPUT SUBSTITUTED) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Cadence	`52 / 6 = 8.7 releases per year`	six-week hybrid cycle (record)	8.7 releases/yr
2	Criticals per year	`4 × 8.7 = 34.8 ≈ 35`	count illustrative; the record is the −50%	≈35/yr
3	The real constraint — engineering	`15 × 18 / 3 = 90 stories`	15 engineers × 18 days ÷ 3 days per story (illustrative; theory-of-constraints, toolkit label)	90 stories
4	The real constraint — QA	`4 × 18 / 1 = 72 stories`	4 QA × 18 days ÷ 1 day per story (illustrative)	72 stories — QA caps throughput; QA was right
5	Late-land rework	`6 × $800 = $4,800 per release`	re-test + hotfix after last-two-day lands (modeled)	≈$4.8K per release
6	Cost of escaped criticals	`35 × $10,000 = $350,000`	$10K per critical (benchmark $8–15K; modeled)	≈$350K/yr

Table 2 — the fix: values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Cadence after	`52 / 4 = 13 releases per year`	four-week cycle (record)	13 releases/yr (+50% — cadence arithmetic)
2	QA-authored DoD	`entry criteria evaluated at Ready-for-QA — stories arrive ready`	the skeptic owns the gate (record)	the gate has an owner
3	Drop date	`not QA-ready by the drop date → dropped with a reason code`	silent late lands become visible re-scopes (record)	no last-two-day lands
4	Criticals per release after	`4 × 50% = 2`	−50% record applied to the illustrative base of 4	2 per release
5	Criticals per year after	`2 × 13 = 26`	illustrative count	26/yr

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	Fewer per year	`35 − 26 = 9`	counts illustrative	9 fewer a year on the cadence arithmetic (modeled)

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Cadence	six weeks (8.7/yr)	four weeks (13/yr)	record cadence; releases modeled
2	Critical defects per release	baseline	−50%	record
3	Conversion	standoff risk	converted in six weeks	record
4	Gate ownership	none	the QA lead defended it by cycle two and after I left	record
5	Sev-1s from story-accuracy defects	rising, mis-attributed to QA	trend reversed; classified by cause	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Cost after	`26 × $10,000 = $260,000`	$10K per critical (modeled)	≈$260K
2	Escaped-critical cost avoided	`$350,000 − $260,000 = $90,000; or 9 × $10,000 = $90,000`	modeled; 9 fewer × $10K	≈$90K/yr
3	Late-land rework removed	`$4,800 × 8.7 = $41,760`	modeled; ~0 under the drop rule	≈$40K/yr (rounded from $41.8K)
4	Total	`$90,000 + $40,000 = $130,000`	modeled; range $80–270K; no headcount invented	≈$130K/yr
5	Record vs modeled	`record: −50%, four-week cadence, six weeks; modeled: every dollar and the base count`	—	say it in that order

SOLUTION FLOW — THE OPERATING-MODEL MAP NEW

```
grooming -> story ready? --(DoD entry criteria, QA-authored)--> Ready-for-QA -> test window -> release (every 4 weeks)
                |  no                                                                              ^
                +-> drop date reached? -> dropped with reason code -> re-scope (visible, not a silent slide)
Sev-1 -> cause classification (story-accuracy | code | environment) -> QA on product-business calls, pre-approved scenarios
rollback: revert to six weeks (two-way door named at kickoff, never used)
```

SENSITIVITY — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
Cost per critical (benchmark)	$8K	$10K	$15K
Avoided = 9 × cost per critical	9 × $8,000 = $72,000	9 × $10,000 = $90,000	9 × $15,000 = $135,000
Total modeled (canon range)	$80K	≈$130K	$270K

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS NEW

> "The record is a four-week cadence in six weeks and critical defects per release down fifty percent while we shipped fifty percent more releases — and the gate surviving my leaving. The dollars are my model: about ninety thousand a year of escaped-critical cost on an illustrative base of four per release at ten thousand each, and about forty thousand of late-land rework at benchmark rates — about a hundred and thirty thousand, none of it a record figure, and I did not invent a headcount to get there."

IF CHALLENGED — THE FULL ANSWERS NEW

Four criticals a release is made up.

It is illustrative, not record — the record gives me the fifty percent, not the base. I would pull the Sev-1 log for the last four six-week releases and use the actual mean; at a base of 2 the avoided cost roughly halves, and the −50% stays the headline.

How do you know the fifty percent was the gate and not luck?

I cannot isolate it statistically. The evidence is the mechanism: reason codes moved late lands from silent to visible, the story-accuracy Sev-1 trend reversed, and the same definitions applied before and after. I would compare the post-conversion cycles to the prior four on that one definition.

Why does the 72-versus-90 arithmetic matter?

Because it is what turned QA's fear from resistance into a constraint everyone could see. It is illustrative, a theory-of-constraints back-of-envelope — the point is the order of magnitude, and that QA was right.

Where this lands at Amazon. Develop people by giving them the load-bearing artifact and the frame, not by writing it for them — and measure development by whether the gate survives your leaving. The Earth's Best Employer angle: QA moved from blamed afterthought to owner of the standard, with the data protecting them. Reliability (fewer escaped criticals), responsiveness (a four-week instead of a six-week cadence), quality owned at the source.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S15

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you developed someone.	Hire and Develop the Best	The QA lead: frame coached, content hers, gate defended by her by cycle 2 and after I left.	Never invent a direct-report headcount.

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about making a workplace better for a team.	Earth's Best Employer	QA from blamed to owner; Sev-1s classified by cause, not blame; the drop date made overload visible and got people help.	Do not claim engagement or attrition numbers — none exist.
Tell me about a time you raised the bar on quality.	Highest Standards	A definition-of-done that can say no, evaluated at Ready-for-QA; criticals −50% per release on a four-week cadence.	Keep the per-release base count labeled illustrative.
Tell me about resolving a conflict between two teams.	Earn Trust / Backbone	One-on-ones first; QA's fear named as correct arithmetic; engineering's veto turned into review.	Do not make engineering the villain.
How do you change a process without authority?	Functional — change management	Hear every real objection privately; hand the artifact to the skeptic; name the rollback as a two-way door; step back on evidence.	—
Tell me about a change that stuck after you left.	Deliver Results / Ownership	Neighbouring teams copied the ceremony; the gate survived my leaving.	No dollars as record — ≈$130K is modeled.

BEHAVIORAL PACK — FOUR LIKELY QUESTIONS, FOUR SEPARATE 90-SECOND STAR ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY) NEW

Q1 AIQB Hire and Develop the Best Q1 — "Tell me about a time when you helped one of your team members develop their career. How did you help that team member? What was the result?" HIRE AND DEVELOP THE BEST

> **Opener (first two sentences):** "The most skeptical person on a twenty-person team wrote the quality gate for a faster release cycle. I coached the frame; she owned the content — and she was defending it to engineering without me by the second cycle."

S — A VP asked me to move a 20-person Product, Engineering and QA team from a six-week to a four-week cadence. QA — a contracting firm — feared compressed test windows; engineering is in-house; I had no line authority.

T — Convert in weeks, not quarters, without the standoff becoming the story.

A — One-on-ones with every person; then I asked the QA lead to author the definition-of-done. I coached what a gate needs to survive a release-day argument — entry criteria that can say no — and she wrote them. Engineering shaped it through review, not veto.

R — Converted in six weeks; critical defects per release −50%; she defended the gate by cycle two and it survived my leaving. I measure development by whether the work survives me.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Gate defended by her by cycle two; survived my leaving.	At NACF I would give an FC ops lead the receive-quality or dock-appointment bar to author and defend, and measure it the same way.	A direct-report headcount; HR or comp-cycle experience (I have managed onshore/offshore delivery teams; not a large HR org).

Q2 AIQB Strive to be Earth's Best Employer Q3 — "Describe a time when you implemented a change at work aimed at improving the work experience of yourself or coworkers. What was the goal? Why did you feel this change was necessary? What was the outcome?" STRIVE TO BE EARTH'S BEST EMPLOYER

> **Opener (first two sentences):** "QA was being blamed for escaped defects that were really story-accuracy problems, so I changed how Sev-1s were classified — by cause, not by who touched the release last."

S — Sev-1s were landing on QA by default; the shortened window made it worse; QA felt like an afterthought.

T — Make the standard — and the pressure — visible and fair.

A — Cause classification on every Sev-1; QA seated on the product-business calls with pre-approved scenarios; a drop date with reason codes so overload showed up as data and got people help instead of blame; I covered some low-impact drops myself.

R — Critical defects per release −50% on the four-week cadence; the Sev-1 trend from story-accuracy defects reversed; QA moved from blamed afterthought to owner of the standard, with the data protecting them.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Criticals −50% per release; cadence six weeks → four.	At NACF the floor feels the pressure first at peak; I would make overload visible with reason codes (UPH, dwell) rather than let it read as underperformance.	Engagement or attrition numbers — none exist.

Q3 AIQB Are Right, A Lot Q6 — "Tell me about a time when you discovered that your idea was not the best course of action. What was your idea? Why wasn't your idea the best course of action? How did you find out? What did you learn from the experience?" ARE RIGHT, A LOT

> **Opener (first two sentences):** "I started by assuming QA's pushback was resistance. The arithmetic said otherwise: with the cycle shortened, QA could clear about 72 stories to engineering's 90 — QA was the constraint, and they were right."

S — QA and engineering heading for a standoff over a shortened test window.

T — Decide whether this was a change-management problem or a capacity problem.

A — I did the back-of-envelope: 15 engineers × 18 days ÷ 3 days a story = 90; 4 QA × 18 days ÷ 1 day a story = 72. I said so in the room — QA's fear is correct — and that is what made the QA lead willing to write the gate. Numbers are illustrative; the order of magnitude is the point.

R — The argument moved from "QA is slow" to "the window needs a gate and a drop date"; criticals −50% per release on a four-week cadence.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
72 vs 90 stories — illustrative.	Whichever constraint caps inbound — doors, labor, cube — I name it with arithmetic before arguing about people.	Present the 72/90 as record; claim a direct-report headcount.

Q4 **AIQB Insist on the Highest Standards Q6 — "Describe the most significant continuous improvement project that you led. What was the catalyst to this change and how did you go about it? What was the outcome?"** INSIST ON THE HIGHEST STANDARDS

> **Opener (first two sentences):** "I made the gate able to say no — and then made sure the people who would be told no had shaped it."

S — A faster cadence usually means more escapes, and the VP could not afford that.

T — Four-week cadence with no rise in escaped criticals and the QA window protected by rule.

A — Entry criteria evaluated at Ready-for-QA; a drop date that converted silent late lands into reason-coded re-scopes; engineering shaped the criteria through review and I covered the low-impact stories they had to drop.

R — Critical defects per release −50% on the new cadence (8.7 → 13 releases a year, modeled from the cadence — not a measured count); rollback to six weeks never used.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Criticals −50% per release on a four-week cadence.	At NACF the equivalent is a dock-appointment or receive-quality gate with a published ceiling (the booklet's 85% door utilization) that can say no at peak.	The per-release base count as record; ≈$130K as record.

SAME STORY, THREE ANGLES — DIFFERENT OPENER, HEADLINE METRIC AND LP EACH NEW

(i) Data / analytical

DIVE DEEP

> "A four-week cadence sounds like an agile problem. It was a capacity-constraint problem."

Headline metric: 72 vs 90 stories (illustrative); 35 → 26 criticals/yr

- QA capped throughput on the arithmetic.
- Reason codes made late heavy lands visible.
- Count per year, not per release: 9 fewer a year.
- Base count illustrative; the −50% is record.

(ii) Conflict / influence / backbone

HAVE BACKBONE · EARN TRUST

> "QA and engineering were heading for a standoff and I had no authority over either."

Headline metric: Gate defended by her by cycle two

- One-on-ones first; every real objection heard privately.
- I told the room QA's fear was correct.
- Engineering's objection became review, not veto; I covered low-impact drops.
- Hard feedback went sideways through the issues-log rows — some was genuine overload, and the evidence got people help.

(iii) Failure / learning

LEARN AND BE CURIOUS

> "If I did it again I would go to QA first and put her name on the review line from week one."

Headline metric: Six weeks to convert; base count not pulled up front

- Real failure arc at Sephora is G2 (the RF-device script: training slipped 1.5 weeks) — on the failure page.
- I did not pull the real Sev-1 baseline first — the dollars rest on an illustrative 4 per release.
- Do differently: pull the last four releases' Sev-1 log in week one.
- I covered drops myself — fine once, not a mechanism.
- No HR or comp-cycle experience; I will not claim it.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS NEW

T1 **"How do you know it was your change and not seasonality or a lighter release mix?"**

Numeric answer. I cannot isolate it statistically. Record: −50% critical defects per release (resume); the cadence went from six weeks to four. Mechanism evidence: reason-coded late lands became visible; story-accuracy Sev-1s reversed. On the illustrative base, 4 × 8.7 = 34.8 → 2 × 13 = 26 criticals a year. I would compare the four prior six-week releases to the post-conversion cycles on one Sev-1 definition.

> **Say exactly:** "I cannot rule out every other cause statistically. What I can show is the mechanism — late heavy stories turned from silent slides into reason-coded re-scopes, and the Sev-1s from story-accuracy defects reversed. The fifty percent is the record; the base count under the dollars is illustrative."

T2 **"What was YOUR contribution versus the QA lead's?"**

Numeric answer. Mine: one-on-ones, the ceremony calendar, the DoD frame, the Jira field spec, the drop-date rule, facilitating the retro once, the VP read-out. Hers: the DoD content and the gate, defended by her. Engineering: shaped it through review. VP: air cover — I had no line authority.

> **Say exactly:** "I owned the frame, the calendar, the fields and the drop-date rule. She wrote the definition-of-done and owned the gate. I coached; I did not write it — which is exactly why it survived my leaving."

T3 **"Is the $130K real — and what would change your mind?"**

Numeric answer. Record: none. Modeled: 9 × $10,000 = $90,000 + $4,800 × 8.7 = $41,760 (rounded to $40K) ≈ $130K. At a base of 2 criticals a release, fewer per year = 2 × 8.7 − 1 × 13 = 4.4 ≈ 4.4 → ≈ $44K avoided. The −50% stays the headline either way.

> **Say exactly:** "No — none of the dollars is a record figure. It is about a hundred and thirty thousand on an illustrative base of four criticals a release; at a base of two the escaped-critical piece roughly halves. What I defend is the minus fifty percent, and I would pull the real Sev-1 log to firm the rest."

20-SECOND VERSION AND NACF BRIDGE NEW

20-SECOND VERSION NEW

"I moved a 20-person Product, Engineering and QA team from six-week to four-week releases in six weeks by handing the quality gate to the most skeptical person — the QA lead. Critical defects per release fell 50% as we converted to a four-week cadence, and the gate survived my leaving."

NACF BRIDGE NEW

"Same logic as dock doors and labor: a faster cadence only holds if the constraint owns a gate that can say no. The booklet's door model — 85% utilization ceiling, and at 95% dwell goes from 13.3 to 40 hours — is that gate; I would make the ops lead the author and the defender of it."

S13 — Manhattan Associates — Cyber Monday, 4 AM: allocation across 3 DCs and 100 stores when the scheduler died — a rule written on the bridge log in hour one, 99% fill against a 95% goal NEW

LPs this story can serve here: Bias for Action (primary — a reversible, logged call before the cause was known) · Customer Obsession (the promise ranked first, not the queue) · Dive Deep (the noise said capacity; the trace said a dead scheduler) · Deliver Results (99% vs 95%) · Insist on the Highest Standards (silence became the alarm) — the oldest era in my set; for a recent twin, pair it with the UDC robotics throttle (S2: a dial and a gate written before the crisis).

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF EACH NEW

Bias for Action	At 4 AM, with the cause unknown, I declared the incident on the contradiction (inventory available, backorders rising) and chose a reversible, timestamped intervention over waiting — because waiting had an hourly price.
Customer Obsession	I ranked the clear by promise-at-risk instead of first-in-first-out, so the promises whose miss cost customers most were kept first — and I named the price: some cheap early orders waited.
Dive Deep	The noise said "capacity — add staff"; my own reason-coded alerts and the trace said the allocation scheduler had stopped at the daylight-saving change, with inventory sufficient and nothing draining the queue.
Insist on the Highest Standards	The post-mortem did not stop at the clear: a scheduler heartbeat so silence is the alarm, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook — and the next peak ran boring.

Honesty label. I was the OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it, and the client's teams executed the clears. The triage call, the clearing rule, the rebalance plan, the cadence and the post-mortem mechanisms were mine to drive. $8M, $6M, 99% and 95% are record (Apple v11 M1); every margin and cost figure below is my model at benchmark rates, labeled; the 0.85-day / 4.60-day node-distance pair is a user anchor. The $8M has one meaning — about eight million of the day's sales protected: the final sales on orders inside the failure window, not the whole day's revenue — one-time, never a saving, never "at risk", never "the day's GMV".

SAFE-SAY BOX NEW

The most attackable number: $6M versus $8M — one meaning each. Say: "About six million dollars was on the line when the page fired; about eight million dollars of the day's sales were protected — fill held at ninety-nine percent against a ninety-five percent goal." The $8M means protected, only: never "at risk", never a saving.

If he asks "what is your source for that?" "The six million is my sizing at 4 AM from the backorder lines; the eight million is the final sales figure for the orders inside the failure window, not the whole day's revenue. The number I would stand behind is ninety-nine percent fill against the ninety-five percent goal."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. I was the Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores for a strategic retailer. At 4 AM on Cyber Monday the VP called: backorder lines piling up with inventory present, ~$6M of sales on the line against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction — ATP above zero while backorders rose — was my bar to declare an incident. Plan & Design: rank by promise-at-risk, not FIFO; pick the node by the promise — near 0.85 days versus far 4.60 (my own anchor). Build & Test: one bridge, every rebalance timestamped, reason-coded alerts I had already built. Cutover: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare: root cause — the allocation scheduler died silently at the daylight-saving change; the post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook. **Result.** Peak fill 99% against the 95% goal; ~$8M of the day's sales protected against ~$6M that was on the line when the page fired; structural stockouts −20% over the engagement. Lesson: decide reversibly, log it, find the cause later.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — OMS consultant on order allocation, promising and the OMS–WMS handshake for a strategic retailer — 3 DCs, 100 stores — on call 6 AM–9 PM for the holiday promotion. 4 AM Cyber Monday, the VP of Supply Chain on the phone: backorder lines piling up with plenty of inventory; promises aging; ~$6M of sales on the line against a 95% fill-rate goal.

Every hour of waiting cost promises — the queue aged while inventory sat idle — and a wrong intervention could corrupt allocation. Peak was the one window the retailer could not absorb a miss, and nobody had written a clearing rule for failure.

T — I owned the response: clear the right orders first and hold the promise through the peak, acting before the root cause was known, without making it worse.

Baselines on the table: fill rate — 95% goal, ~$6M on the line at 4 AM · time to detection — my backorder-variance page and the VP's call came together; nothing alarmed on the scheduler's silence itself · structural stockouts — the standing run-rate the engagement later cut.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (04:00–05:00)** — triage read: symptom (allocation stalled), exposure (~$6M vs the 95% goal), what I did not know (root cause); decided to act before the cause. *Exit bar:* bridge open; exposure sized; the question framed as "which orders first", not "what broke".
- **Plan & Design (first hour)** — the clearing rule (rank by promise-at-risk, not FIFO), the rebalance plan across three DCs (near ~0.85 d vs far ~4.60 d), two-hourly comms on the bridge log. *Exit bar:* rule on the log; the VP agrees the ranking.
- **Build & Test (morning)** — manual clears in promise-at-risk order; rebalance so no node drowns while others idle; read my reason-coded alerts for the cause. *Exit bar:* queue draining; alerts point at the scheduler.
- **Cutover (root-cause fix)** — handed the isolated DST scheduler death to engineering with the evidence; verified automated allocation resumed; stopped manual clears; timestamped the handoff. *Exit bar:* scheduler restored; fill trending to 99%.
- **Hypercare / Close (post-mortem)** — blameless, off the log timeline; mechanisms driven with owners — heartbeat, DST checklist, 3σ page rule, rehearsed rebalance playbook, surge-scaling rebuilt. *Exit bar:* live before the next peak.

Influence and ownership. Mine: the triage read, the exposure sizing, the ranking rule, the rebalance plan, the two-hourly cadence, the isolation of the cause through my reason-coded alerts, and the post-mortem mechanisms. Engineering fixed the scheduler once I isolated it; the client's teams executed the clears. The obstacles: the VP wanting the fastest clear (one sentence on fastest versus best, then the rule written in the log); acting without corrupting allocation (checked with engineering

first, every rebalance timestamped); and carrying the first stretch solo, which is one of my do-differentlys. The decisions I challenged: FIFO, and "wait for the root cause".

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~$8M of the day's sales protected against ~$6M that was on the line when the page fired (record pair — one-time sales that did not die; never a saving); ≈$3.2M gross margin on that at a 40% benchmark (modeled); structural-stockout margin ≈$1.3M/yr on an illustrative $400M retailer (modeled).
- **#** — 3 DCs and 100 stores; the backorder queue drained in promise-at-risk order with every rebalance timestamped on one log (record).
- **%** — Peak fill 99% against the 95% goal (record); structural stockouts −20% over the engagement (record).
- **Time** — Node choice = the promise: near ~0.85 days vs far ~4.60 (user anchor); status every two hours until 9 PM; a silent scheduler went from no alarm of its own to one heartbeat interval.
- **Customer / quality** — Promises whose miss cost most were kept first; blameless post-mortem; heartbeat, DST checklist, 3σ page rule and rehearsed playbook shipped — the next peak ran boring (record). The trade: some early, cheap orders waited.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	4 AM, no root cause, a VP on the phone: I declared the incident on a contradiction (ATP above zero, backorders rising) and acted reversibly while hunting the cause in parallel.
Scope / influence	Three DCs, 100 stores, two-hourly leadership status until 9 PM, a clearing rule the VP agreed in the first hour; engineering and client teams executed under my log.
Execution	One bridge, one log, one decision list; every rebalance timestamped; the handoff to automated allocation verified and logged.
Impact	99% fill against a 95% goal; ~$8M of the day's sales protected against ~$6M on the line (record pair, not savings).
Process improvement	Silence became the alarm: heartbeat, DST checklist, 3σ page rule, rehearsed rebalance playbook, surge-scaling rebuilt.
Complexity	Promise-at-risk ranking versus FIFO across nodes with different distances (0.85 d vs 4.60 d) and capacities.
Root cause	Noise: capacity. Cause: the allocation scheduler stopped at the daylight-saving change; the engine was not wrong — it was not running.

THE FIVE FACETS — DATA ANALYSIS · SOLUTION DESIGN · MATH & DATA LOGIC · IMPLEMENTATION · PROBE QUESTIONS NEW

1 · DATA ANALYSIS

Data. Backorder lines against inventory-on-hand (ATP above zero while backorders rose — the contradiction that triggered the incident), node queue ÷ capacity, promise times, and the reason-coded alert stream I had built earlier. **Grain:** order line × node × promise time. **What it revealed:** not a capacity shortage — a stalled queue; alerts clustered on the allocation scheduler at the clock change. The ~$6M on the line was sized from the backorder lines against the 95% fill goal and quoted in every status.

2 · SOLUTION DESIGN

Options: (A) FIFO clear — the default, processes cheap orders while at-risk promises age; (B) add staff to the backlog — treats a stalled scheduler as a capacity problem; (C) wait for the root cause before touching allocation — each hour has a price; (D) promise-at-risk ranking with reversible, timestamped manual clears and a node choice by promise. **Chosen: D.** The rule went on the bridge log in hour one so nobody re-argued it at noon.

3 · MATH & DATA LOGIC

Rules. Risk = value × urgency(promise − now), clear descending. Node time T = handling + transit: near ≈0.85 d vs far ≈4.60 d, a 3.75-day gap per re-pointed order. Load = queue ÷ capacity; rebalance to equalize. Liveness page when allocations per hour fall below μ − 3σ (a dead scheduler reads 0). Orders at risk = $ at risk ÷ average order value. All substituted in the four tables below.

4 · IMPLEMENTATION

Phases and owners. 04:00 incident declared (me) → first hour rule + rebalance plan (me, VP agreed) → morning clears and alert-based isolation (client teams execute, me directing) → scheduler fixed (engineering, on my evidence) → 9 PM status close → post-mortem mechanisms with owners (me driving). **Controls:** one log, timestamps, engineering check before each class of manual clear. **Risks:** corrupting allocation; leadership noise; solo fatigue — the one I under-managed.

5 · PROBE QUESTIONS

Hardest follow-ups with numeric answers are in the Probing section and the three Trap Drills at the end of this story: the counterfactual for the $8M, my contribution versus engineering's, and the VP I disagreed with on fastest versus best.

PROBING — CHALLENGE Q&A TUNED TO A BAR RAISER

BA — Why act before you knew the cause?

Because waiting had an hourly price — the queue was aging while inventory sat idle — and a reversible, logged intervention did not. I checked with engineering first that manual clears could not corrupt allocation, timestamped every rebalance and hunted the cause in parallel with alerts I had already built.

CO — Why promise-at-risk and not FIFO?

FIFO processes cheap orders while at-risk promises age. Promise-at-risk clears the orders whose miss costs most first — the ranking is the policy. The cost is explicit: some early, cheap orders waited.

DD — How did you find the root cause?

Through my own reason-coded alerts: the pattern pointed at the allocation scheduler, which had died silently at the daylight-saving change. Engineering fixed it once I isolated it.

BR — Isn't this a hero story?

The event is; the post-mortem is the point. Carrying the first stretch solo is one of my do-differentlys. What I present as the result is the mechanism set and that the next peak ran boring.

BUSINESS IMPACT MODEL — KPI TREE AND DRIVERS NEW

```
Peak sales protected ~$8M vs ~$6M on the line (record pair; one-time, never a saving)
+- Peak fill rate 95% goal -> 99% delivered (record)
|    +- backorder queue (stock) drained by promise-at-risk, not FIFO
|    +- node choice = the promise: near 0.85 d vs far 4.60 d (user anchor)  |  load_n = queue_n / capacity_n -> re-point -> loads level
|    +- scheduler liveness: heartbeat + DST checklist + 3-sigma page -> detect a silent scheduler in 1 interval, on its own alarm
+- Gross margin on protected sales (= $8M x margin, modeled)
+- Structural stockouts -20% over the engagement (record) -> margin run-rate (modeled)
```

DRIVER	BASELINE	AFTER	$	BASIS
Peak fill rate	95% goal	99%	—	record
Peak sales protected	~$6M on the line	~$8M protected	one-time event	record pair
Gross margin protected	—	≈$3.2M	one-time	modeled ($8M × 40%; benchmark 35–45%)
Cancel / CS contacts avoided	≈40K × $6.50	0	≈$0.26M	modeled ($6M ÷ $150 AOV; CS $5–8)
Expedite premium avoided	—	near-node sourcing	≈$0.13M	modeled (benchmark $5–12/order)
Structural stockouts	—	−20%	≈$1.3M/yr	modeled margin run-rate on an illustrative $400M retailer

MATH — THE FOUR TABLES (EVERY INPUT SUBSTITUTED) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	4 AM state	`scheduler dead; backlog 22,000 + 6,000 + 2,000 = 30,000 orders; ~$6M on the line vs the 95% fill goal`	record risk pair; backlog and node split illustrative (DC2 = 30,000 − 22,000 − 2,000)	the bridge call
2	FIFO failure	`FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age`	record rule problem	value-blind clearing
3	Naive node loads — DC1	`22,000 / 10,000 = 2.20 days`	capacities 10,000 / 8,000 / 6,000 orders per day (illustrative)	worst node 2.2 days behind
4	Naive node loads — DC2 and DC3	`6,000 / 8,000 = 0.75 d; 2,000 / 6,000 = 0.33 d`	same illustrative capacities	DC3 idles at 0.33 d
5	Detection	`a dead scheduler raised no alarm of its own — the page was on backorder variance and the VP called the same hour`	record problem	silence unalarmed

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promise-at-risk ranking	`risk = value × urgency(promise − now); clear descending`	on the bridge log in hour one (record rule)	at-risk promises first, not first-in
2	Node choice = the promise	`4.60 − 0.85 = 3.75 days`	near ≈0.85 d vs far ≈4.60 d (user anchor pair)	−3.75 days per re-pointed order
3	Load levelling — DC1	`12,500 / 10,000 = 1.25 days`	rebalance 30,000 → 12,500 / 10,000 / 7,500 (illustrative)	was 2.20
4	Load levelling — DC2 and DC3	`10,000 / 8,000 = 1.25 d; 7,500 / 6,000 = 1.25 d`	same illustrative capacities	every node at 1.25 d — worst slip 2.2 → 1.25 days
5	Liveness page (post-mortem)	`page if allocations/hr < μ − 3σ: 1,200 − 3 × 150 = 750`	μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr	caught in one interval, on its own alarm

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Peak fill rate	95% goal	99% delivered	record
2	Peak sales	~$6M on the line	~$8M protected	record pair — one-time event
3	Worst-case promise slip	2.2 days	1.25 days	illustrative walk
4	Detection	variance page + a VP call at 4 AM; no alarm on silence	one interval via the 3σ page / heartbeat	post-mortem design
5	Structural stockouts	—	−20% over the engagement	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Margin protected	`$8,000,000 × 40% = $3,200,000`	benchmark margin 35–45% (modeled)	≈$3.2M one-time
2	Orders at risk	`$6,000,000 / $150 = 40,000 orders`	AOV $150 benchmark	40,000 orders
3	Cancels / CS avoided	`40,000 × $6.50 = $260,000`	CS contact $5–8 (benchmark; modeled)	≈$0.26M
4	Expedite premium avoided	`40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000`	re-pointed share and $8 premium (midpoint-ish of the $5–12 benchmark) — both illustrative	≈$0.13M
5	Structural-stockout run-rate	`$400,000,000 × 4% = $16,000,000 lost sales; × 20% = $3,200,000 recovered; × 40% = $1,280,000`	illustrative $400M retailer × lost-sales benchmark (2–6%) × −20% record × 40% margin	≈$1.3M/yr modeled

SOLUTION FLOW — THE SYSTEM AND DATA MAP NEW

```
order surge --> OMS allocation scheduler --(reserves inventory per order)--> node queues (DC1 / DC2 / DC3) --> WMS pick/pack/ship
                    |                                                                |
        heartbeat: allocations/hr < mu - 3 sigma  -> PAGE a named human (post-mortem)        queue_n / capacity_n = load_n
                    |                                                                |
 reason-coded alerts on backorder variance          clear order = promise-at-risk (value x urgency); re-point by node time T
                    |                                                                |
 bridge log: every decision timestamped   ---->   two-hourly status (orders at risk / saved / time to clear)   ---->   post-mortem
```

SENSITIVITY — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
Benchmark margin on protected sales	35%	40%	45%
Margin protected on $8M	$8,000,000 × 35% = $2,800,000	$8,000,000 × 40% = $3,200,000	$8,000,000 × 45% = $3,600,000
CS cost per contact (benchmark)	$5.00	$6.50	$8.00
Cancels / CS avoided on 40,000 orders	40,000 × $5 = $200,000	40,000 × $6.50 = $260,000	40,000 × $8 = $320,000
Stockout run-rate inputs: lost-sales share × effect × margin	2% × 15% × 35%	4% × 20% × 40%	6% × 25% × 45%
Structural-stockout margin run-rate ($400M, illustrative)	$400,000,000 × 2% × 15% × 35% = $420,000	$400,000,000 × 4% × 20% × 40% = $1,280,000	$400,000,000 × 6% × 25% × 45% = $2,700,000

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS NEW

> "On Cyber Monday the allocation scheduler died silently at the daylight-saving change. We held 99% fill against a 95% goal and protected about eight million dollars of the day's sales against the six million I sized as on the line at 4 AM — one-time sales that did not die, not a saving. My model puts that at roughly three million in margin at a benchmark forty percent. The result I would put weight on is the mechanism: a heartbeat so silence is the alarm, a daylight-saving checklist, a three-sigma page rule and a rehearsed rebalance playbook — and the next peak ran boring."

IF CHALLENGED — THE FULL ANSWERS NEW

The $8M — is that not just the day's revenue?

No, and I would not claim that. About $6M was the exposure I sized at 4 AM from the backorder lines against the 95% goal; about $8M is the peak-sales figure that held through the event at 99% fill. It is revenue protected on orders inside the failure window — I would rather claim the defensible protected number than the whole day's sales. It is one-time and never a saving.

Manual rebalancing is not a scalable answer.

Agreed, and I did not pretend otherwise. The night's work was deliberately throwaway so the promises were not; the permanent fixes — heartbeat, checklist, 3σ page, rehearsed playbook, surge-scaling rebuilt — waited for the post-mortem where they belonged. Right unscalable thing at 4 AM, boring scalable thing after.

The stockout run-rate is soft.

It is the biggest and softest number, and I label it so: a 4% lost-sales benchmark on an illustrative $400M retailer at −20% and a 40% margin. At the low and high ends it runs about $0.42M to $2.7M. I would measure it from the retailer's own out-of-stock lines times unit margin, before and after the playbook.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S13

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you made a decision with incomplete information.	Are Right / Bias for Action	Declared the incident on the contradiction; ranked by promise-at-risk; reversible and timestamped; cause hunted in parallel.	Never imply the cause was known at 4 AM.
Tell me about a time you worked against a tight deadline / under pressure.	Deliver Results	99% vs 95%; status every two hours until 9 PM; one log, one decision list.	Never call $8M a saving.
Tell me about a time you failed or would do differently.	Are Right A Lot / Ownership	DST checklist should have existed; paged on-call engineering sooner rather than carrying the first stretch solo.	Never say the scheduler failure was engineering's fault.
Tell me about a time you balanced the customer with the business.	Customer Obsession	Promise-at-risk first; price named: cheap early orders waited.	Never quote a satisfaction score — none in the record.

BEHAVIORAL PACK — FOUR LIKELY AIQB QUESTIONS, FOUR SEPARATE 90-SECOND STAR ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY) NEW

Q1 AIQB Bias for Action Q1 — "Give me an example of a calculated risk that you have taken where speed was critical… What steps did you take to mitigate the risk? …Knowing what you know now, would you have done anything differently?" BIAS FOR ACTION

> **Opener (first two sentences):** "At four in the morning on Cyber Monday the allocation queue was aging with inventory sitting idle, and I did not know why. I decided to act before the cause was known — but only in ways that were reversible, logged and checked with engineering first."

S — 3 DCs, 100 stores, ~$6M of sales on the line against a 95% fill goal; the VP on the phone; peak was the one window the retailer could not absorb a miss.

T — Own the response; clear the right orders first without making allocation worse.

A — Declared the incident on the contradiction (ATP above zero, backorders rising). Wrote the clearing rule on the bridge log in the first hour — promise-at-risk, not FIFO — and got the VP to agree it. Mitigations for the risk: engineering confirmed manual clears could not corrupt allocation; every rebalance timestamped; a rollback was the rehearsed manual mode I did not yet have, so I kept changes small and one-way-reversible.

R — Fill 99% against 95%; ~$8M of the day's sales protected against ~$6M on the line; the next peak boring. Differently: the DST checklist should have existed, and I would page on-call engineering sooner.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
99% fill vs 95% goal.	At NACF peak the equivalent is a receive-scan or dock-schedule stall: act on a reversible, logged rule while the cause is hunted — a silent stall costs dwell by the hour.	That the cause was known; $8M as a saving; "I fixed the scheduler" (engineering did).

Q2 AIQB Customer Obsession Q8 — "Tell me about a time when you had to balance the needs of the customer with the needs of the business." CUSTOMER OBSESSION

> **Opener (first two sentences):** "The default way to clear a backlog is first-in-first-out, and it would have cleared cheap orders while the promises that mattered most kept aging. I ranked by promise-at-risk and said out loud what that cost."

S — A backlog across three DCs with different distances to the customer — near ≈0.85 days, far ≈4.60 — and a VP who wanted the fastest clear.

T — Protect the customer promise, not the queue statistic.

A — One sentence on the difference between fastest and best, then the rule written in the log so nobody re-argued it at noon. Re-pointed orders to nodes that kept the promise; rebalanced so no DC drowned while another idled; two-hourly status of orders at risk, orders saved, time to clear.

R — 99% fill; promises whose miss cost most were kept first. The trade I named: some early, cheap orders waited.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
≈0.85 d vs ≈4.60 d — a 3.75-day gap per re-pointed order (user anchor).	At NACF the same trade is placement: a unit sent to the node that keeps the promise instead of the node that clears the dock fastest.	Any customer-satisfaction score or compliment as evidence (the VP's appreciation email is real, but it is color, not a metric).

Q3 AIQB Dive Deep Q2 — "Tell me about a situation that required you to dig deep to get to the root cause. How did you know you were focusing on the right things?" DIVE DEEP

> **Opener (first two sentences):** "The noise said capacity — throw staff at the backlog. I didn't believe it, because inventory was sitting there. I went to the reason-coded alerts I had already built and followed them to a scheduler that had stopped."

S — Backorders rising with sufficient inventory — a contradiction capacity could not explain.

T — Find the cause without stopping the clears.

A — Read the alert stream by reason code while the clears ran; the pattern pointed to the allocation scheduler; handed engineering the evidence; they confirmed it had stopped at the daylight-saving change. I verified automated allocation resumed before stopping manual clears.

R — Cause isolated while the queue drained; the post-mortem made silence itself an alarm — heartbeat plus a 3σ page: μ 1,200 − 3 × 150 = 750 allocations/hr (illustrative).

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Contradiction: ATP above zero while backorders rose.	At NACF: a dock that is "full" but not receiving is the same contradiction — check the scan stream by reason code before adding labor.	That I fixed the scheduler; any per-class alert counts (not in the record).

Q4 AIQB Insist on the Highest Standards Q2 — "Tell me about a time when you were unsatisfied with the status quo. What did you do to change it?" INSIST ON THE HIGHEST STANDARDS

> **Opener (first two sentences):** "The status quo was that a dead scheduler had no alarm of its own — my page fired on backorder variance, and the VP called at four in the morning. I wasn't willing to leave it that way — so the post-mortem made silence the alarm."

S — Silence had no alarm: my backorder-variance page fired and the VP called the same hour.

T — Make the next failure detectable in one interval and the manual mode rehearsed.

A — Blameless post-mortem off the log timeline. Mechanisms with owners: a scheduler heartbeat, a daylight-saving checklist, a 3σ page rule so anomalies wake a human, a written and rehearsed rebalance playbook, surge-scaling rebuilt.

R — The next peak ran boring — the heartbeat paged nothing because nothing died; structural stockouts −20% over the engagement.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Silence: no alarm → one heartbeat interval.	At NACF: a heartbeat on receive scans and appointment-scheduling jobs, with a named human on the page.	"Zero incidents since" — only "the next peak ran boring"; a percentage on detection time (none in the record).

SAME STORY, THREE ANGLES — DIFFERENT OPENER, HEADLINE METRIC AND LP EACH NEW

(i) Data / analytical

DIVE DEEP

> "Inventory was sitting there and backorders were rising — that contradiction was the whole diagnosis."

Headline metric: μ − 3σ = 1,200 − 450 = 750 allocations/hr page line (illustrative)

- Contradiction first: ATP above zero, backorders rising.
- Reason-coded alerts — built before the incident — pointed at the scheduler.
- Node time: near ≈0.85 d vs far ≈4.60 d.
- Load = queue ÷ capacity; rebalance to 1.25 d each (illustrative).
- No per-class alert counts in the record — I say so.

(ii) Conflict / influence / backbone

HAVE BACKBONE; DISAGREE & COMMIT

> "The VP wanted the fastest clear. I said fastest and best are different — and wrote the rule in the log."

Headline metric: 3.75-day gap per re-pointed order (4.60 − 0.85; user anchor)

- One sentence, then the rule on the log in hour one.
- He was right that speed mattered; wrong that FIFO was fast for the promises.
- The VP agreed the ranking within the first hour.
- Status every two hours — same numbers — until 9 PM.
- Price named: cheap early orders waited.

(iii) Failure / learning

OWNERSHIP · ARE RIGHT, A LOT

> "The failure in that night was mine: the checklist did not exist and I carried the first stretch alone."

Headline metric: Silence: no alarm → one heartbeat interval

- A daylight-saving checklist should have existed before that Monday.
- I should have paged on-call engineering sooner instead of carrying the first stretch solo.
- The log was right; the escalation was slower than it should have been.
- Fix: heartbeat, DST checklist, 3σ page, rehearsed playbook.
- Honest scale: the real failure arcs are on the failure page (G1–G3); this is the smaller one.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS NEW

T1 "What is the data that proves your action — not luck, not the day's sales?"

Numeric answer. Hard evidence: fill 99% against a 95% goal; ~$6M sized as on the line at 4 AM, ~$8M of the day's sales protected (record pair). Soft: the counterfactual — what would have happened without my ranking — is not in the record, and I will not invent one. The decision rule I can show: re-pointing a near node saves 4.60 − 0.85 = 3.75 days per order, and levelling loads cut the worst illustrative slip from 22,000/10,000 = 2.2 days to 1.25.

Say exactly: "The record is 99% fill against a 95% goal, with about eight million of the day's sales protected against six million that was on the line. I cannot prove the counterfactual; what I can show is the rule — promise-at-risk ranking and a 3.75-day gap per re-pointed order — and that the next peak ran boring."

T2 "What was YOUR specific contribution — the engineers fixed the scheduler, the client's teams ran the clears?"

Numeric answer. Mine: the triage read and incident declaration, exposure sizing, the clearing rule, the rebalance plan, the cadence, the alert-based isolation, and driving the post-mortem mechanisms. Theirs: engineering fixed the scheduler once isolated; client teams executed clears.

Say exactly: "I ran the incident: I declared it, sized it, wrote the clearing rule, planned the rebalance, kept the log and the two-hourly status, and isolated the cause through my own alerts. Engineering fixed the scheduler and the client's teams executed the clears."

T3 "Why was that the right long-term decision, not just the right 4 AM decision — and what would you do differently?"

Numeric answer. The 4 AM work was deliberately unscalable; the long-term decision was to make silence the alarm and rehearse the manual mode. Differently: the DST checklist should have existed before that Monday, and I would have paged on-call engineering sooner instead of carrying the first stretch solo.

Say exactly: "At 4 AM I traded elegance for survival on purpose. The long-term decision was the heartbeat, the checklist, the 3σ page and a rehearsed playbook. What I would change: the checklist should have existed before that Monday, and I would page engineering sooner rather than carrying the first stretch alone."

20-SECOND VERSION AND NACF BRIDGE NEW

20-SECOND VERSION NEW

"On Cyber Monday at 4 AM the allocation queue was aging with inventory idle. I declared the incident, ranked the clear by promise-at-risk instead of first-in-first-out, rebalanced across three DCs, and isolated a dead scheduler through my own alerts. Fill held at 99% against a 95% goal, about eight million protected, and the next peak ran boring."

NACF BRIDGE NEW

"The NACF twin is a silent stall in receiving: dwell grows fast as utilization climbs — the booklet's 13.3 hours at 85% door utilization becomes 40 hours at 95%. I would put a heartbeat on receive scans and a 3σ page on throughput, with a rehearsed manual mode and a named human."

S9 — Elementum — a thousand lanes asked, 597 first: a spend Pareto taken two levels up, and a ≥95% valid-ASN gate held twice NEW

LPs this story can serve here: Have Backbone; Disagree & Commit (primary — 597 vs 1,000, taken two levels up, then committed) · Think Big (the staged thousand-lane vision; adoption 8× — a user anchor; the method became the sourcing SOP) · Insist on the Highest Standards (the gate held twice) · Frugality (a harness that made later cohorts mostly configuration) · Deliver Results · Broad Responsibility (the portal floor kept small carriers inside the standard).

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF EACH NEW

Have Backbone; Disagree & Commit	I took the re-scope — a thousand lanes asked, 597 first — two levels up to the Global Head of Supply Chain with the spend curve and the lane owners' validation attached, then committed to delivering the rest behind the same gate.
Think Big	I turned a one-time 1,000-lane ask into a staged vision with a standard, a gate and a scorecard; the harness made later cohorts mostly configuration, adoption reached 8× (user anchor), and the method was written up as the sourcing SOP.
Insist on the Highest Standards	One cohort failed the ≥95% valid-ASN gate twice on a timestamp format (my own tracking); I held it both times — a gate that only holds in private is a suggestion.
Frugality	Instead of bespoke maps per partner I held engineers to one parameterized harness (templates, test scripts, DQ checklist), so later cohorts were mostly configuration — scope cut by spend share, not by cutting quality.

Honesty label. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships and accepted the holds. **Cohort sizes and the number of cohorts are not in my notes — I never quote them. Basis.** A thousand lanes across twenty-five European locations, phased by value, and completeness ~60% → 95% are record (Apple v11 F1, resume), as is the ≥95% valid-ASN cohort gate (resume). **User anchors** — traced only to my earlier story notes, not to Apple or a source document: 597 lanes = 80% of spend, freight −10% ≈ $2M of freight cost in six months (never annualized), adoption 8×, and the gate held twice on a timestamp format. I say "a thousand lanes were asked; 597 first" and "the remainder behind the same gate" — never "400+ lanes", never a count of what went live. My three-way split of the $2M of freight cost and the ≈$0.4M/yr document labor are modeled.

SAFE-SAY BOX NEW

The most attackable number: 597 lanes, 80% of spend, ≈$2M. Say: "A thousand lanes were asked; I phased by value, and 597 lanes carrying about eighty percent of freight spend went first — the rest waited behind the same gate. Completeness went from sixty to ninety-five percent."

If he asks "what is your source for that?" "The thousand-lane program across twenty-five European locations and the sixty-to-ninety-five completeness are in my story record and resume; the 597, the eighty percent, the two million of freight cost and the eight-fold adoption are my own program tracking. The number I would stand behind is completeness 60% to 95%."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Elementum's customer — the Global Head of Supply Chain — wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and the high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard — X12 850/855/856/940/945/214/997 with an XML/portal floor — and a cohort gate of ≥95% valid ASN, published to carriers before cohort one. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: the 597 lanes went first; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; the remainder behind the same gate. **Result.** Completeness 60% → 95%; a thousand lanes asked, 597 carrying 80% of spend first (user anchor); freight spend −10% ≈ $2M of freight cost in six months (user anchor, not annualized); adoption 8× (user anchor). Lesson: automating dirty data manufactures confident wrong answers — the gate is the product.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The customer's Global Head of Supply Chain wanted end-to-end inbound visibility on 1,000 lanes in two months — scope, schedule and quality all fixed. Lanes differed by orders of magnitude in spend, volume and criticality; carriers differed in EDI capability; completeness was 60% and high-value lanes were effectively blind.

Taking the ask literally would have shipped confident wrong answers — 1,000 lanes at 60% completeness in two months — and, without a floor for carriers lacking EDI, would have priced the long tail out of the standard.

T — Own the scope analysis and the onboarding standard, taking the re-scope two levels up with the evidence attached — visibility that was trustworthy, at the spend that mattered first.

Baselines on the table: completeness 60% (target ≥95%) · high-value lane visibility — effectively blind (target 100%) · activation standard — none existed (target a ≥95% valid-ASN gate every lane clears before it goes live).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — spend Pareto and lane-scoring model (spend × criticality × mode × volume) from the routing guide and the invoicing report; carrier EDI-capability survey; the 597-first / remainder scope pre-validated with Transportation and Procurement. *Exit bar:* no lane owner can contradict the cut; the executive accepts the plan in spend share, not lane count.
- **Plan & Design** — onboarding standard (document set + XML/portal floor), the ≥95% valid-ASN cohort gate, the per-lane scorecard spec (events received ÷ expected; 997 rate; tracking-number joinability), the harness design. *Exit bar:* standard signed by Transportation; gate published to carriers before cohort one.
- **Build & Test** — held engineers to the harness — per-partner mapping templates, test scripts, DQ checklist; test files per carrier; parallel run per lane. *Exit bar:* each cohort's lanes clear ≥95% valid ASN in test before activation.
- **Cutover (per cohort)** — activated cohorts only off the published gate; recorded the hold decision when a cohort failed — twice, on a timestamp format. *Exit bar:* cohort active only after the gate.

- **Hypercare / Close** — weekly completeness scorecards; harness reused for later waves; method written up as the sourcing SOP (spend file in → phased plan out). *Exit bar:* completeness 95%; high-value lanes 100%; remaining waves behind the same gate.

Influence and ownership. The scope analysis and the standard were mine end to end — the Pareto and scoring model, the carrier survey, the onboarding standard, the gate, the scorecard spec and the hold decisions. Engineers built the mapping templates, test scripts and harness; Transportation held the carrier relationships and accepted the holds. The obstacles: carriers unable to produce valid ASNs (answered with the XML/portal floor and a hold rule); a cohort that failed twice on a timestamp format; bespoke maps per partner (avoided by the parameterized harness). The decision I challenged was the 1,000-in-two-months ask itself; the one I refused was activating any cohort that failed the gate.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — Freight spend −10% ≈ $2M of freight cost in six months on the onboarded lanes (user anchor — kept as measured, never annualized). My split — ~$0.8M routing-guide compliance, ~$0.6M mode discipline, ~$0.6M expedites and detention — decomposes the $2M of freight cost and never adds to it (modeled). ≈$0.4M/yr document-handling labor on top (modeled add-on).
- **#** — a thousand lanes asked; 597 lanes = 80% of freight spend went first; the remainder behind the same gate (user anchor for the 597 and the 80%).
- **%** — Completeness 60% → 95% (record: Apple F1, resume); the ≥95% valid-ASN gate held twice on one cohort and adoption 8× (user anchor).
- **Time** — the 597 lanes went first; the $2M of freight cost landed within six months (user anchor).
- **Customer / quality** — 100% visibility on high-value lanes; small carriers kept inside the standard by the XML/portal floor; method written up as the sourcing SOP. The priced trade: the remainder, about 20% of spend, waited behind the same gate — quality never traded.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A literal ask (1,000 lanes, two months) with 60% completeness and unknown carrier EDI capability; I turned it into a scored, phased plan with a data-quality bar.
Scope / influence	Re-scoped an executive's own ask two levels above my sponsor, with the lane owners already on side; Transportation, Procurement, carriers and engineers worked to my standard.
Execution	Gate published before cohort one; harness, scorecards and hold decisions recorded; the 597 lanes first.
Impact	Completeness 60 → 95%; −10% ≈ $2M of freight cost in six months (user anchor, not annualized); adoption 8× (user anchor).
Process improvement	Pareto + published gate + portal floor became the sourcing SOP: spend file in, phased plan out.
Complexity	Seven X12 document types across carriers with different capability, an XML/portal floor, and a timestamp-format defect class that only surfaced at the gate.
Root cause	The noise: "we need all lanes". The cause: completeness 60% — automating it would manufacture confident wrong answers.

THE FIVE FACETS — DATA ANALYSIS · SOLUTION DESIGN · MATH & DATA LOGIC · IMPLEMENTATION · PROBE QUESTIONS NEW

1 · DATA ANALYSIS

Data. The routing guide and the invoicing report — all 1,000 lanes ranked by annual freight spend and scored by spend × criticality × mode × volume; a carrier EDI-capability survey; per-lane event completeness (events received ÷ expected), the 997 acknowledgment rate and tracking-number joinability. **Grain:** lane × shipment event. **What it revealed:** the spend curve was steep — 597 lanes carried 80% of freight spend, so the first wave could carry the business case; the remainder carried the other ~20% of spend.

2 · SOLUTION DESIGN

Options: (A) all 1,000 lanes in two months as asked — slow and dirty; (B) phase by region — spreads the proof too thin; (C) phase by spend share behind a published ≥95% valid-ASN gate, with an XML/portal floor for carriers without EDI. **Chosen: C** — the platform's credibility was the scarce resource, so the first cohorts had to carry the business case and clear the bar. The standard defined the data, not the transport.

3 · MATH & DATA LOGIC

Rules. S(k) = cumulative spend share; S(597) = 80%. Completeness = events received ÷ events expected (480 / 800 = 60%; 760 / 800 = 95%, illustrative volumes). Gate: valid-ASN rate ≥ 95% in test before activation. Documents per year = lanes × shipments × documents (597 × 50 × 3 ≈ 90,000, illustrative). Labor = docs × minutes ÷ 60 × rate. The −10% on the $2M implies a ≈$20M addressable base — derived from the user-anchor pair, never a record itself.

4 · IMPLEMENTATION

Phases and owners. Discovery and Pareto (me, validated with Transportation and Procurement) → standard and gate (me; signed by Transportation) → harness and test files (engineers, to my standard) → per-cohort activation off the gate (me deciding holds; Transportation accepting) → weekly scorecards and the SOP write-up (me). **Controls:** gate published before cohort one; scorecard per lane; hold rule. **Risks:** carriers unable to produce valid ASNs; scope contradicted in the room; bespoke maps per partner.

5 · PROBE QUESTIONS

Hardest follow-ups with numeric answers are in the Probing section and the three Trap Drills at the end of this story: the data that proves 597 was the right cut, who disagreed and what they got right, and the long-term versus short-term price of putting the remainder behind the gate.

PROBING — CHALLENGE Q&A TUNED TO A BAR RAISER

> **BR — You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached: 597 lanes carried 80% of the spend, so I argued the phased plan in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **HS — Why hold a cohort twice? Was the date more important?**
>
> The gate was published to carriers before cohort one, so Transportation accepted the hold. A cohort that fails ≥95% valid ASN on a timestamp format would put wrong dates into a visibility product — the first read by the customer would be bad data. Held both times; fixed; then activated.

> **TB — How did you drive adoption?**
>
> The standard defined the data, not the transport — an XML or portal floor kept small carriers inside it — and a scorecard per lane made progress visible; the harness made later cohorts mostly configuration; the method was written up as the sourcing SOP. Adoption reached 8× (user anchor).

> **DR — Is the $2M annualized?**
>
> No — it is my own tracking as measured: minus ten percent on the onboarded lanes' freight spend in six months, about two million dollars of freight cost, kept as stated. My three-way split decomposes that number; it never adds to it.

BUSINESS IMPACT MODEL — KPI TREE AND DRIVERS NEW

```
Inbound freight spend -10% ~= $2M of freight cost in 6 months (user anchor; not annualized)
+- Visibility (stock): completeness 60% -> 95%; high-value lanes 100%
|    +- routing-guide compliance: off-guide carrier spend caught (modeled ~$0.8M of the measured $2M, user anchor)
|    +- mode discipline: premium modes -> ocean/ground when ETA is trusted (modeled ~$0.6M)
|    +- expedites + detention: events replace chase calls (modeled ~$0.6M)
+- Document handling (flow): 850/855/856/940/945/214 by X12 or portal, not by hand (modeled ~$0.4M/yr)
+- DQ gate loop: >=95% valid ASN -> cohort activates -> scorecard per lane -> defects fixed as data -> harness parameterized -> adoption 8x (user anchor)
+- Scope loop: 597 = 80% of spend first -> value lands first -> executive funds the next wave
```

DRIVER	BASELINE	AFTER	$	BASIS
Shipment-data completeness	60%	95%	—	record
Lanes first	0	597 = 80% of spend	—	user anchor
Freight spend on onboarded lanes	≈$20M/yr (derived)	−10%	≈$2M in 6 months	user anchor ($2M of freight cost); base derived
— routing-guide compliance / mode discipline / expedites + detention	—	—	~$0.8M / ~$0.6M / ~$0.6M	modeled split — decomposes the measured $2M (user anchor)
Manual document handling	~90K docs/yr keyed	X12/XML validated	≈$0.4M/yr	modeled (90K × 6 min × $45/h)

MATH — THE FOUR TABLES (EVERY INPUT SUBSTITUTED) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing
2	Scope asked vs cut	`1,000 − 597 = 403 lanes in the remainder (I say "the remainder", not the count); 100% − 80% = 20% of spend`	597 lanes = 80% of spend (user anchor); phasing priced in spend share, not lane count	the remainder ≈ 20% of spend
3	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	my framing	no instrument, no lever
4	Manual documents	`597 × 50 × 3 = 89,550 ≈ 90,000 docs/yr keyed by hand`	597 lanes × ~50 shipments × 3 docs (illustrative volumes)	~90K docs/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (user anchor)	597 of 1,000 lanes carry 80% of spend — do them first
2	Cohort gate	`valid-ASN rate ≥ 95% in test before activation`	the ≥95% gate is record (resume); publication before cohort one and the twice-held hold are user anchors	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	"the standard was the data, not the transport" (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%
5	Scorecard per lane	`events received ÷ expected · 997 rate · tracking-number joinability`	my spec (Apple F1 scorecard: completeness · tender acceptance · OTD)	below the line → back to "fix", not "live"

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes first	0	597 = 80% of spend	user anchor
3	Freight spend on onboarded lanes	base	−10% ≈ $2M of freight cost in 6 months	user anchor
4	Adoption	1×	8×	user anchor
5	Document handling	~90K docs/yr keyed	X12/XML validated	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000`	derived from the user-anchor pair; stated as derived	≈$20M/yr
2	The measured result	`$20,000,000 × 10% = $2,000,000`	user anchor — in six months, not annualized	≈$2M

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`$800,000 + $600,000 + $600,000 = $2,000,000`	the slices decompose the measured $2M (user anchor) — they never add to it	= the measured $2M
7	Document labor	`90,000 × 6 / 60 = 9,000 h; 9,000 × $45 = $405,000`	90,000 docs × 6 min ÷ 60 = 9,000 h; × $45/h (illustrative volumes; modeled add-on)	≈$0.4M/yr

SOLUTION FLOW — THE SYSTEM AND DATA MAP NEW

```
buyer ERP (PO, delivery) --850--> EDI VAN / AS2 (X12 + XML/portal floor) <--855 ack-- suppliers
  3PL: 940 in / 945 out · carriers: 214 status · 997 per interchange, both ways
                          |
routing guide + invoicing report (lane, forwarder, mode, spend) --> Pareto worksheet
  --> cohort plan: 597 first | remainder behind the same gate
                          |
per-partner harness (map template + test script + DQ checklist)
  --> per-lane scorecard: validASN, 997 rate, joinability, completeness
                          |
cohort GATE: validASN >= 95% in test  --pass--> activate   --fail--> HOLD (back to fix)
  measured lane events --> Transportation / planners
```

SENSITIVITY — LOW / BASE / HIGH NEW

INPUT	LOW	BASE	HIGH
Shipments/yr on the 597 lanes (illustrative)	20,000	30,000	40,000
Documents/yr (× 3 per shipment)	20,000 × 3 = 60,000	30,000 × 3 = 90,000	40,000 × 3 = 120,000
Minutes per manual document / loaded rate (benchmark)	4 min / $40	6 min / $45	8 min / $55
Document labor per year (modeled)	60,000 × 4 / 60 × $40 = $160,000	90,000 × 6 / 60 × $45 = $405,000	120,000 × 8 / 60 × $55 = $880,000
If the measured cut had been −5% / −10% / −15%, the same $2M implies a base of	$2,000,000 / 5% = $40,000,000	$2,000,000 / 10% = $20,000,000	$2,000,000 / 15% = $13,333,333

The last row is the honesty check on the derived base: the user-anchor pair constrains the base but does not fix it — I never present the ≈$20M as a record.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS NEW

> "The number I track is freight spend down ten percent — about two million dollars of freight cost in six months, my own program tracking — on the five hundred ninety-seven lanes that carried eighty percent of spend, with completeness sixty to ninety-five percent and adoption eight-fold, also my own tracking. Inside that two million I model roughly eight hundred thousand from catching off-guide carrier spend, six hundred thousand from mode discipline once the ETA was trusted, and six hundred thousand from expedites and detention that chase calls used to cause — that split is mine; the two million is my tracking, and I do not annualize it. On top I model about four hundred thousand a year of document-handling labor that X12 replaced."

IF CHALLENGED — THE FULL ANSWERS NEW

> **The split of the two million is the softest part.**
>
> Yes — it is my judgment on how the ten points divide. To firm it I would tag every invoice line on the onboarded lanes by cause — off-guide carrier, mode change, expedite — for the six months and let the ledger set the shares. The two million itself is my own tracking as measured.

> **Ninety thousand documents is a guess.**
>
> Fifty shipments a lane and three documents each are illustrative; the ISA and GS control-number counts on the VAN for the first full quarter would give the exact document volume in an afternoon. At 60,000 to 120,000 documents the labor runs about $160K to $880K.

> **Labor saved is not cash.**
>
> Correct — nine thousand hours is coordinator capacity moved from keying to exceptions; the number Finance can see is headcount held flat while lanes grew. That is why I label it modeled and keep it separate from the measured $2M (user anchor).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S9

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you disagreed with a senior leader / pushed back on a goal.	Have Backbone	597 vs 1,000 taken two levels up with the spend curve and lane owners' validation; then committed to the remainder behind the same gate.	Never say he was wrong or foolish; never quote cohort sizes.
Tell me about a time you refused to compromise quality for a date.	Highest Standards	Gate published before cohort one; cohort held twice on a timestamp format; put the remainder (~20% of spend) behind the same gate, priced.	Never call the $2M a run-rate.
Tell me about a time you thought bigger than the initial ask.	Think Big	Staged vision: 597 first → the remainder behind the same gate; harness; adoption 8× (user anchor); SOP.	Never invent a number of cohorts.

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you got more done with limited resources.	Frugality	Parameterized harness — later cohorts mostly configuration; portal floor instead of custom EDI for every carrier.	Never present the ≈$0.4M labor as cash.

BEHAVIORAL PACK — FOUR LIKELY AIQB QUESTIONS, FOUR SEPARATE 90-SECOND STAR ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY) NEW

Q1 AIQB Have Backbone; Disagree and Commit Q2 — "Describe a time when you took an unpopular stance in a meeting with peers and your leader. What was it? Why did you feel strongly about it? What did you do? What was the outcome?" HAVE BACKBONE; DISAGREE & COMMIT

> **Opener (first two sentences):** "I was asked to put a thousand lanes live in two months on data that was sixty percent complete. I said that was the wrong target — and I took that to an executive two levels above my sponsor with the spend curve in my hand."

S — The Global Head of Supply Chain wanted 1,000 lanes visible in two months; completeness was 60%; lanes differed by orders of magnitude in spend.

T — Own the scope analysis; change the target without losing the executive.

A — Built the spend Pareto and scoring model, and pre-validated the cut with Transportation and Procurement so no lane owner could contradict it in the room. Argued in spend share: 597 lanes carry 80% of spend — do them first, behind a published ≥95% valid-ASN gate; the remainder behind the same gate. Then I committed: the remainder behind the same gate.

R — He accepted the phasing off the curve; the 597 went first; completeness 60% → 95%; the remainder followed behind the same gate.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
597 of 1,000 lanes = 80% of spend.	At NACF the unpopular stance is "ship the top vendors that carry most of the inbound volume first, behind a data-quality bar" — Pareto, then gate.	That he was wrong; the number of cohorts or their sizes; the −10% as proof the phasing was right.

Q2 AIQB Think Big Q4 — "Tell me about a time when you drove adoption for your vision/ideas. How did you know your vision/idea was adopted by others? How did you drive adoption for your vision/ideas? How did you track adoption?" THINK BIG

> **Opener (first two sentences):** "The vision was that a carrier's data had to clear a published bar before it counted — and that every carrier, even without EDI, had a path to clear it. Adoption went eight-fold, and I tracked it with a scorecard per lane."

S — No activation standard existed; carriers had very different EDI capability; the long tail risked being priced out.

T — Make the standard adoptable by everyone, not just the big carriers.

A — The standard defined the data, not the transport: X12 850/855/856/940/945/214/997 with an XML or portal floor. Published the gate to carriers before cohort one. A per-lane scorecard — events received over expected, 997 rate, tracking-number joinability — showed adoption weekly. The harness made later cohorts mostly configuration.

R — Adoption 8× (user anchor); completeness 95%; the method was written up as the sourcing SOP — spend file in, phased plan out.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Adoption 8× (user anchor).	At NACF: a vendor ASN standard with a floor for small vendors, tracked as one scorecard per vendor.	Cohort counts; "global rollout" claims beyond my notes; the $2M as proof of adoption.

Q3 AIQB Insist on the Highest Standards Q4 — "Tell me about a time when you had to make a decision between standards and delivery. What tradeoffs did you have to make? What was the outcome?" INSIST ON THE HIGHEST STANDARDS

> **Opener (first two sentences):** "One cohort failed my ninety-five percent valid-ASN gate twice on a timestamp format, with a go-live date waiting. I held it both times, and I could do that because the gate had been published to carriers before cohort one."

S — A cohort of carriers kept sending timestamps in a format that broke the valid-ASN check; leadership wanted lanes live.

T — Protect the data quality the whole program was for, without breaking the date for everyone else.

A — Classified defects by type (format, timestamp, missing segment, join key) on the cohort test day; recorded the hold decision with an owner; Transportation accepted the hold because the gate was public; fixed; retested; activated only after it cleared. The trade I priced earlier: the remainder, about 20% of spend, waited behind the same gate — never a lane live below its scorecard.

R — Completeness 60% → 95%; no cohort activated below the gate; "a lane that slipped below its scorecard went back to fix, not live".

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Gate ≥95% valid ASN, held twice.	At NACF: do not accept an ASN feed into dock planning below the bar — the door plan (19.2 trailers/day → 5 doors) assumes the ASN is right.	The number of cohorts; "never missed a date" (not in my notes).

Q4 AIQB Frugality Q5 — "Tell me about a time when you generated a creative solution to a problem or project without requiring additional resources. What was the problem? What was the solution and how did you come up with it? What was the outcome?" FRUGALITY

> **Opener (first two sentences):** "The expensive way to onboard a thousand lanes is a custom map per partner. I held the engineers to one parameterized harness instead, so that by the third cohort the work was mostly configuration."

S — Hundreds of carriers, seven X12 document types, a fixed two-month ask and a fixed team.

T — Get more lanes live per engineer-week without lowering the bar.

A — Per-partner mapping templates, test scripts and a DQ checklist as one harness; one test day per cohort with defects classified; a portal floor instead of custom EDI work for small carriers. Cut scope by spend share rather than cutting quality.

R — Later cohorts were mostly configuration; the 597 went first; ≈$0.4M/yr of manual document keying modeled as replaced — a model, not cash.

LAND THIS NUMBER	PIVOT TO NACF	DO NOT SAY
Harness: cohort one bespoke → later cohorts mostly configuration.	At NACF: reuse one vendor-onboarding harness across vendors and FCs rather than bespoke feeds.	Any headcount or budget figure (none in my notes); calling the labor a saving.

SAME STORY, THREE ANGLES — DIFFERENT OPENER, HEADLINE METRIC AND LP EACH NEW

(i) Data / analytical

DIVE DEEP

> "I ranked all thousand lanes by spend before anyone agreed what 'first' meant."

Headline metric: S(597) = 80% of spend; completeness 480/800 → 760/800

- Routing guide + invoicing report; spend × criticality × mode × volume.
- The curve was steep: 597 lanes = 80% of spend.
- Completeness = events received ÷ expected.
- Defects classed on the cohort test day: format, timestamp, missing segment, join key.
- No cohort counts or sizes in my notes.

(ii) Conflict / influence / backbone

HAVE BACKBONE; DISAGREE & COMMIT

> "The executive asked for a thousand lanes in two months. I thought that would produce a thousand confident wrong answers."

Headline metric: a thousand asked; 597 first = 80% of spend; the remainder behind the same gate

- Right about wanting visibility everywhere.
- Wrong, in my judgment, that all 1,000 could clear a quality bar in two months.
- Pre-validated with lane owners; argued in spend share.
- Committed to delivering the remainder behind the same gate.
- Refused: activating a cohort that failed the gate.

(iii) Failure / learning

LEARN AND BE CURIOUS · OWNERSHIP

> "What I would do differently is start the harness at cohort one and measure tender acceptance from the first wave."

Headline metric: Acceptance instrumented from cohort three → would catch a carrier-data problem two cohorts sooner

- Completeness was measured from day one; tender acceptance only from cohort three (Apple v11 F1).
- Earlier signal would have caught a carrier-data problem two cohorts sooner.
- The harness: from cohort one, not cohort two (master S9).
- A cohort failing twice is the gate working — and an earlier harness would have shortened it.
- The real failure arcs are on the failure page (G1–G3).

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS NEW

T1 "What is the data that proves 597 was the right cut — and that the −10% came from onboarding, not rate changes or seasonality?"

Numeric answer. Hard (my own tracking): the Pareto — 597 of 1,000 lanes = 80% of spend; completeness 60 → 95%; the gate held twice. Soft: the −10% ≈ $2M of freight cost is my own tracking as measured on the onboarded lanes in six months; it does not isolate onboarding from rate or season. My split ($0.8M + $0.6M + $0.6M = $2.0M) is a model; the ledger tagged by cause would firm it. The implied base ($2M / 10% = $20M) is derived; at −5% the same $2M would imply $40M — so the pair constrains the base but does not fix it.

Say exactly: "The measured result is freight down ten percent, about two million in six months, on the onboarded lanes. It does not by itself separate onboarding from rates or season; the way to separate it is to tag every invoice line by cause. The split I show is my model of the ten points; the two million is my own tracking."

T2 "Who disagreed with you, and were they right?"

Numeric answer. The Global Head of Supply Chain, on 1,000 in two months. Right: he needed visibility across the network and the lane count was his own commitment. Wrong, in my judgment: 1,000 lanes at 60% completeness in two months would have produced confident wrong answers and priced out the long tail. Evidence that would have changed my mind: carriers clearing ≥95% valid ASN in test across all lanes — which the cohort test day showed they did not on one timestamp format, twice.

Say exactly: "He was right about the need and about the lane count being his own commitment. Where I disagreed was that every lane could clear a quality bar in two months. I took the spend curve to him, committed to the rest behind the same gate — and delivered it."

T3 "Why was putting the remainder behind the gate the right long-term decision and not just the safe short-term one — and what would you do differently?"

Numeric answer. Short-term cost: the remainder, about 20% of spend, waited. Long-term: the gate is the product — the standard, the harness and the SOP outlived the program; the 597 carried the spend and the decision. Differently: the harness from cohort one, not cohort two; tender acceptance on the scorecard from cohort one, not cohort three.

Say exactly: "I phased by value: 597 lanes carrying about eighty percent of spend went first, and the remainder waited behind the same gate so the first cohorts could clear at the highest data quality. What I would change: I would start the harness at cohort one, and put tender acceptance on the scorecard from the first wave rather than the third."

20-SECOND VERSION AND NACF BRIDGE NEW

20-SECOND VERSION NEW

"An executive asked for a thousand lanes visible in two months on data that was sixty percent complete. I took a spend Pareto two levels up — 597 lanes carried eighty percent of spend — set a ninety-five percent valid-ASN gate, held a cohort twice, and completeness went sixty to ninety-five percent."

NACF BRIDGE NEW

"At NACF the same lever is inbound vendor data. The booklet's door plan — 19.2 trailers a day over 10 hours at 0.5 unload rate and 85% utilization gives 4.5, so five doors — assumes the ASN is right. I would phase vendors by inbound volume and gate ASN quality before it feeds dock planning."

TOP-IMPACT RE-CUTS — the biggest story from each of the other employers, cut for a bar raiser NEW

These are the highest-impact records in the master packet at Flexport (S11, S12) and Sephora (S18, S2). Each is another interviewer's primary, so the rule is: **use a re-cut only when the bar raiser asks for it, tell only the facet below, never the full telling, and keep every number identical to the master canon.** Each card is cut the way a bar raiser probes: what was mine versus the team's, the number that proves it, the long-term-versus-short-term call, who disagreed, and what I would change.

S11 Flexport — the week I would not print: a flattering 99.4% discarded, and a wave held in public

EARN TRUST · DIVE DEEP · HIGHEST STANDARDS · DELIVER RESULTS

WHY THIS ONE	Data honesty under pressure. Leadership said the program was not showing value (in my account it would drop from #3 to #14 of 16 initiatives — user anchor, so I say "not showing value" and skip the ranks unless asked) and I had been reporting activity, not evidence. The facet that is only mine here: the review I built would not print a number that could not survive the next Monday.
15-SECOND OPENER	"Leadership told me the program was not showing value, and they were right — I had been reporting activity, not evidence. So I built a review that threw away the best-looking week it ever had."
I VS THE TEAM	**Mine:** seven frozen metric definitions; the review format (copied from the LSA review, not invented); the six graduation criteria published to Sales before wave 1; running the review by hand every Monday; the call to discard the 99.4% week; holding wave 3 in public. **Theirs:** nine teams / ~100 people built ingestion, monitors and kill switches; infra later built the pipeline that retired my hand-run (that is S16).
THE NUMBER	7 → 68 customers: 68 / 7 = 9.7× · cost-to-serve $0.66 → $0.03 per container: ($0.66 − $0.03) / $0.66 = 95.45% — say "about ninety-five percent" · zero rollbacks · both public targets (68 customers, 170 monthly users) two months early (record: Apple A1/A2 and the resume). The direct CTS dollars are only ≈$50K/yr, modeled on an illustrative container count — say that out loud. The ~$1.5M pre-production figure (S11, my estimate of a month and a half versus six) is a separate *user anchor*, never merged with CTS; for the pre-production test prefer the documented 500 containers in 20 minutes at 99.6% connectivity and 86% of containers trackable in the dry run.
LONG VS SHORT TERM	"Three touches is not a week": printing 99.4% would have taught the VP a number that could not survive the next Monday. Wave 3 was held in public on an ETA-accuracy miss from one bad source rather than take a bad first impression at scale — and the held customers became the cleanest wave.
WHO DISAGREED	Sales wanted the whole pool boarded at once. Right: the public targets were real and the pool was waiting. Wrong on pace — and because the six criteria were published before wave 1, the hold was the rule firing, not my opinion.
DO DIFFERENTLY	Start the review at kickoff, not after the critique (master).
DO NOT SAY	"10×", "50×", "$1.00 → $0.03", "95.5%", "85% → ~98%", "70% baseline"; any automation percentage or review-hours figure (the packets differ — keep to "by hand every Monday"). S11 is the hiring manager's story: facet only.

S12 Flexport — the plan with my manager's name on it: operator-first vs partner-first

HAVE BACKBONE · THINK BIG · FRUGALITY · EARN TRUST

WHY THIS ONE	Disagreeing with the owner of a committed, socialized 18-month plan, with no authority over the roadmap — and winning by making it his decision. Answers "who disagreed, and were they right?" with a concession.
15-SECOND OPENER	"The roadmap was operator-first, eighteen months, already socialized — and my manager owned it. I had no authority over it, so I brought him the model before anyone else saw it."
I VS THE TEAM	**Mine:** the time study × loaded rates (baselines 339.5 min per shipment; 2,200 operator interactions per departed shipment); the −25% vs −50% comparison; the 8-week rebuild/reuse rule; the written reversal trigger; showing my manager privately first. **Theirs:** engineering produced the re-estimates with confidence buffers; the Bellevue onsite ratified the flip. The decomposition behind the ~$5M is my model — the record does not itemize it.
THE NUMBER	about $5M of avoided cost, said once (record: resume "$5M cost avoidance in 6 months"; avoidance, never an annual reduction, never with "15% lower overhead") · first value in month 6 instead of month 18 (18 − 6 = 12 months earlier) · engineering effort to first value −66% · CTS −50% vs −25% at the work-item grain · 48 of 53 import partners migrated within a month · the import-side app went live April 2022 with eleven partners in ten countries.
LONG VS SHORT TERM	Operators lived in the old screens two more quarters — priced and published, not hidden; at the end of the 18-month path they would still have spent 20% of their time there. New scope was estimated and swapped, never absorbed (burn-up 55.8% → 83.2%).
WHO DISAGREED	My manager. He argued two points and I adjusted one; the onsite ratified the flip with his name on it. The record does not say which point I conceded — so say the risk-register version: operators feeling abandoned, answered by the priced two-quarter schedule. VERIFY-WITH-RAJESH *(confirm which point you conceded)*
DO DIFFERENTLY	The master lists a lesson, not a regret: bring the model, name the owner, write the reversal trigger. Say the cost I published — two more quarters of old screens — as a priced trade, not an apology.
DO NOT SAY	"$5M savings"; "avoidance and annual reduction" as two results; "MVP" for the April 2022 import-side app (the export MVP is Q1 2023); that this was a Sephora roadmap (the loop packet tells it that way; the master record is Flexport FF App). Tripathy-adjacent: facet only.

TOP-IMPACT RE-CUTS (continued) NEW

S18 Sephora — the tolerance I published before I could meet it: RP → JDA, two DCs, three mocks

HIGHEST STANDARDS · LEARN AND BE CURIOUS · DELIVER RESULTS · OWNERSHIP

WHY THIS ONE	A standard set before the test and never renegotiated — and a story of not knowing: I had never run a WMS migration, with no inherited documentation and an eight-month deadline.
15-SECOND OPENER	"I had never run a WMS migration. Two DCs, eight months, no documentation. In week one I wrote the success criteria, and the first number I published was the tolerance Finance and I would hold the first mock to."
I VS THE TEAM	**Mine:** the charter; entrance and exit criteria; the interface inventory; the conversion approach; the mapping-table standard (data, not code); the 0.5% tolerance; the go/no-go recommendation; the fallback call. **Theirs:** 14 core + 6 DC Ops built, converted, tested and ran the floor; Finance co-owned the tolerance.
THE NUMBER	Mock variance 7% → 1.8% → 0.3% against the 0.5% tolerance (record) · on time (eight months) · ~3% under the ~$1.8M budget: $1,800,000 × 3% = $54,000 (record) · a $20M pipeline of four projects was waiting on it — about $2.5M a month on average, my division of the $20M over the eight months (modeled), a slip avoided and not banked · zero accuracy incidents.
LONG VS SHORT TERM	Publishing the tolerance before mock 1 meant 7% was the price, not an argument. When hundreds of RF devices failed the conversion script I did not roll back — inventory was inside tolerance and the failure was device conversion, not data — I paid ≈$45K of modeled paper-pick cost to hold the date.
WHO DISAGREED	The record names the challenge, not a person: "why not roll back?" The answer was a rehearsed fallback, and leadership heard it the same afternoon with my name on the cause. I do not invent a disagreer.
DO DIFFERENTLY	Dual-path device conversion on day one, and a device model × OS × script pass/fail table before training week (master). The miss itself is G2 on the failure page.
DO NOT SAY	"400 devices failed" (say hundreds of devices — see the failure page); "$800K budget"; "$2.5M a week"; any multiplier on the fallback; the avoided slip as a saving. Goforth's story: facet only.

S2 Sephora — the gate that could stop the program: robotics held at 94 against a published 98

HIGHEST STANDARDS · HAVE BACKBONE · DELIVER RESULTS · EARTH'S BEST EMPLOYER

WHY THIS ONE	The standard cost something: a published gate that fired. Answers "when did your bar actually stop a launch?"
15-SECOND OPENER	"I published a ninety-eight percent robotics gate before the ramp. In week two we hit ninety-four — and the gate did what a gate is for: it stopped the ramp."
I VS THE TEAM	**Mine:** discovery through the functional design for the location and inventory model; the staging (two go-lives so each carried one defect class); the gates; the throttle built in advance. **Theirs:** BlueYonder, Locus and platform teams built and configured; DC Ops ran the floor. Sixty-plus stakeholders (the discovery deck listed about eighty names across fifteen functions); about a dozen impacted systems.
THE NUMBER	+15% DC processing efficiency (record — DC Ops' own measurement) · zero customer-facing incidents through both go-lives · gate 98% published; week two 94; bots throttled to ~30% of volume; full volume in week three. The ≈$1.6M/yr (the model sums to $1.65M; I say about $1.6M) is my model on an illustrative 100K-units-a-week DC — never record.
LONG VS SHORT TERM	A week at reduced robot volume to protect a zero-incident constraint; one comingle pool replaced two systemic warehouses paying duplicate handling for zero customer value.
WHO DISAGREED	The record names no person. The argument was with a number I had published: 98 was set from the pre-ramp baseline and the incident budget before the event, so nobody had to win a debate at 94.
DO DIFFERENTLY	Staff the interim reporting workstream on day one, not month three — for four months every report had to declare which system of record it read (master).
DO NOT SAY	That the +15% is a dollar figure; the $1.6M or $2.2M as record; "a dozen core, 20+ touched" (20+ is unverified); a named opponent. Goforth's story: facet only.

CLASSIC BAR-RAISER PROBES × SIX STORIES NEW

The six questions a bar raiser asks of every story, answered once per story in one grid. Each cell is the answer in spoken order; the full numeric answers sit in each story's Trap Drills. If a cell says "go to G1", go — a disguised win in place of a failure is the one answer that loses the loop.

BAR-RAISER PROBE	S14 MANHATTAN $2.5M ORDER VALUE	S16 FLEXPORT PIPELINE	S10 ELEMENTUM AUSTRIA	S15 SEPHORA CADENCE	S13 MANHATTAN CYBER MONDAY	S9 ELEMENTUM 597 LANES
1 · What was YOUR contribution — I vs we?	Census, risk argument, customer pack, war room, verification method. Consultants traced the handshake, wrote and verified the patch, built the fix. *I did not write the patch.*	As-is, pricing, the BRD with seven frozen definitions, acceptance gate, parallel run, cutover memo. Infra built the pipeline and alerts. *I did not build it.*	SQL census, rules spec, reason-code alerts, Finance-co-owned method, scorecard gate. Platform team built the engine; Finance read the ledger; super-users ran sites.	Frame, calendar, Jira fields, drop-date rule, VP read-out. *The QA lead wrote the definition of done and owned the gate.* I had no line authority.	Declared and sized the incident, wrote the clearing rule, the rebalance plan, the cadence, isolated the cause by my alerts. Engineering fixed the scheduler; clients ran the clears.	Spend Pareto, carrier survey, onboarding standard, gate, scorecard spec, hold decisions. Engineers built maps and harness; Transportation held carrier relationships.
2 · What would you do differently?	Propose the standing daily reconciliation the moment the handshake was in scope — not after the near-miss. I built the control after, not before.	Put the access review in the BRD and price it next to the build. Kickoff question: "whose approval does this data need, and how long does it take?"	Build the audit-ledger format during the Austria pilot, not alongside the first claims; start from an industry benchmark for lane transit times.	Go to QA first; put her name on the review line from week one; pull the real Sev-1 log in week one instead of an illustrative base.	A daylight-saving checklist before that Monday; page on-call engineering sooner instead of carrying the first stretch solo.	Start the harness at cohort one and put tender acceptance on the scorecard from cohort one — not cohort three.
3 · What is the data that proves it?	~200 orders / $2.5M of order value → 0 (user anchor), before/after evidence per order, evidence rows = set size. No percentage exists — I do not invent one.	Parity within 1% per frozen metric for consecutive weeks; accuracy 60% → 90%; the first alert caught a 4% sync drop.	Data quality ~55% true (40% reported) → ~99% within a quarter; ≈97% agreement with a manual sample; ~$8M across 25 locations = sum of per-shipment ledgers (guide rate − paid rate), executed shipments only.	Critical defects per release −50% (record, resume). Releases 8.7 → 13 a year is cadence arithmetic (52 ÷ 6, 52 ÷ 4), not a measured count. The dollars are modeled on an illustrative base — never the headline.	Fill 99% against a 95% goal; about $6M on the line when the page fired, about $8M of the day's sales protected (record). The counterfactual is not provable and I say so.	Completeness 60% → 95% (record); 597 lanes = 80% of spend (user anchor); freight −10% ≈ $2M in six months as measured (user anchor) — not isolated from rate or season.
4 · Tell me about a time it failed	The patch did not fail. The near-miss: ~60 days of drift (illustrative timing) because nothing compared OMS to WMS on a schedule. A real failure: G1.	The two-week security-review miss (G3): scoping, owned upward the same day; the parallel run kept Monday alive — zero Mondays skipped.	No outright failure in the outcome; soft miss: audit-ledger format built late (Apple F2). For a real failure go to G1 — do not stretch S10.	Outcome did not fail; soft miss: dollars rest on an illustrative base because I did not pull the real Sev-1 log first. Real failures: G2, G1.	The smaller failure: no checklist, first stretch solo, and a dead scheduler had no alarm of its own — my variance page fired and the VP called the same hour. The real arcs are G1–G3 on the failure page.	Tender acceptance instrumented only from cohort three; a cohort failed the gate twice — the gate working, not the plan failing.
5 · Why the right LONG-term decision, not the short-term one?	A silent fix is a loan against trust. Honest path ≈ $21.6K of my time vs quiet path expected loss ≈ $1.5M (illustrative); break-even P* 0.43%.	Two weeks of schedule vs an unreviewed customer-data pipeline that was meant to outlive me; the parallel run kept every Monday alive.	Weeks of delay bought a baseline carriers could not dispute: about $500K a year at Austria became ~$8M across 25 locations on the same method.	She owns the gate, so it survived my leaving: critical defects per release −50% (record, resume).	Made silence the alarm: heartbeat, DST checklist, 3σ page, rehearsed manual mode — the next peak ran boring.	A thousand lanes were asked; 597 first (80% of spend, user anchor) so the first cohorts cleared at the highest quality, and the rest waited behind the same gate; the standard, harness and SOP outlived the program.
6 · Who disagreed — and were they right?	My manager. Right on the clock and the fear of losing the $10M initiative; wrong that speed justified an unverifiable script.	No opponent on the pipeline. The tension was the review: shipping around it was faster — I declined and re-sequenced.	My sponsor, on weeks of delay. Right that speed matters; the carriers never disputed the scorecard.	Engineering objected to the gate; QA feared a standoff. QA's fear was correct; engineering's objection became review, not veto.	The VP wanted the fastest clear. Right that speed mattered; wrong that first-in-first-out was fast for the promises.	The Global Head of Supply Chain, on a thousand lanes in two months. Right on the need; I committed to delivering the rest behind the same gate.

FAILURE ARCS — G1, G3, G2 (the three real ones) and the smaller M1 NEW

From the user's Apple story workbook v11, Part 6 (written from the Elementum, Flexport and Sephora source documents). Never open with these; own them completely when asked — AIQB: Are Right, A Lot Q4 ("a bad decision"), Q5 ("an error in judgment"), Earn Trust Q4 ("not able to meet a commitment"). Order of use: **G1 first** (customer-facing, the largest), G3 (recent, Flexport), G2 (a missed date). Each has the same spine: the miss · the first wrong move · the root cause that was mine · what changed permanently · the number that proves it stuck.

G1 — Elementum · the UAT that crashed: the performance criteria I never wrote

ARE RIGHT Q5 · EARN TRUST · HIGHEST STANDARDS

THE MISS	Two days from UAT sign-off, five key users had completed all ten end-to-end scenarios (5 × 10 = 50 user-scenario passes, derived). Then the sponsor opened access to all their users; concurrent logins crashed a system my criteria never load-tested. The sign-off date passed and user confidence had to be rebuilt.
FIRST WRONG MOVE	I re-ran the scenarios, they passed, and I sent screenshots saying it worked — defending instead of digging. That reflex is the actual failure.
ROOT CAUSE (MINE)	My criteria covered explicit needs and ignored the implicit one, performance. The sponsor did something I had not planned for — an implicit requirement I failed to surface, not their fault.
WHAT CHANGED	Numeric entrance and exit criteria on every critical scenario — uptime, page load, concurrent load — run at the customer's shape, not ours. The first hour of any escalation now goes to reproducing the customer's shape, not proving our innocence.
PROOF IT STUCK	The Flexport launch gate carried a performance proof — 500 containers in 20 minutes at 99.6% connectivity — before any customer saw the product; the JDA UAT entrance criteria (S18) are the same lesson institutionalized.
I VS WE	Mine: the criteria (the miss), the root cause taken to the sponsor with my name on it, the rewritten UAT contract. Team: simulated the concurrency and shipped the fix round the clock.
SAY EXACTLY	"Two days before UAT sign-off five users had passed all ten scenarios. Then the sponsor opened access to everyone, concurrent logins crashed the system, and the date passed. My first move was wrong — I sent screenshots that said it worked. The root cause was mine: my criteria tested what users do and never how many at once. Every gate I have run since carries a load number — the Flexport launch proof was 500 containers in 20 minutes at 99.6% connectivity."
SOURCE NOTE	The loop packet tells G1 as "dozens of users, slipped days"; the Apple v11 record says all users and a passed sign-off date. Use Apple v11 (the user's vetted record).
NACF BRIDGE	A launch-readiness review carries a peak-shape line: the dock plan is tested at peak arrivals, not an average day — the booklet's 19.2 trailers a day needs 4.5 doors at 85% utilization, so five.

G3 — Flexport · the security review I discovered instead of planned

ARE RIGHT Q5 · EARN TRUST Q4 · DIVE DEEP

THE MISS	The data-access review for customer-content ticket data was under-planned by two weeks — discovered mid-build, never scoped in the BRD, which priced the build and never the review.
FIRST WRONG MOVE	Treating the access review as a formality instead of a workstream with a lead time.
ABSORBED BY	The parallel run: the manual path kept the Monday review alive — zero skipped review weeks. By design (a correctness gate) and by luck (it doubled as a schedule buffer); now I plan the buffer.
WHAT CHANGED	A standing line on every data program: whose approval does this data need, and what is its lead time — priced in the BRD next to the build cost.
PROOF IT STUCK	Cutover on the ≤1% parity gate; system accuracy 60% → 90%; no review week skipped. The miss cost margin, not delivery.
I VS WE	Mine: the scoping miss, owned upward the same day; the re-sequence; closing the review (cross-reference table documented, data classes named, who can read what). Infra built the pipeline.
SAY EXACTLY	"I planned the build and never the data-access review. It surfaced mid-build and cost two weeks. I told leadership the same day, kept the Monday review alive on the manual path, and closed the review properly. Now every kickoff has a line: whose approval does this data need, and how long does it take."

G2 — Sephora · the RF-device script: the low-effort task that broke the training gate

ARE RIGHT Q4 · EARN TRUST · DELIVER RESULTS

THE MISS	On the RP → JDA upgrade, device conversion was estimated at three to four days and planned as low effort. A device test days before training showed the upgrade script failed on the refreshed RF hardware. With key people on COVID night shifts, manual conversion could not finish in time: training sign-off slipped 1.5 weeks. The go-live date held.
ROOT CAUSE (MINE)	I scored the task by effort, not impact: a single-point dependency in front of a go-live exit criterion had no what-if plan. I accepted an effort estimate as a risk assessment.
WHAT CHANGED	Every cross-functional dependency gets a what-if and a mitigation plan regardless of effort; device readiness moved into training's entrance criteria; the rehearsal habit became the mock cutovers.
PROOF IT STUCK	Mocks 7% → 1.8% → 0.3% against the 0.5% tolerance; the migration landed on time, ~3% under the ~$1.8M budget ($1,800,000 × 3% = $54,000). The rehearsed paper-pick fallback (≈$45K modeled) held the date.
SAY EXACTLY	"I planned a three-day device task as low effort. The script failed on new RF hardware, manual conversion could not finish before training, and training slipped a week and a half — the date held. The root cause was mine: I scored the risk by effort, not by impact. Since then every dependency in front of an exit criterion has a what-if plan."
SOURCE NOTE	Device count: one source document records ~400 devices, another ~100; the master says ~100 of ~400 failed. Say "hundreds of devices" and never "400 failed".

M1 — Manhattan · Cyber Monday: the smaller one (told inside S13)

OWNERSHIP · ARE RIGHT Q1

No daylight-saving checklist existed before that Monday, and I carried the first stretch of the incident alone instead of paging on-call engineering sooner; a dead scheduler had no alarm of its own — my backorder-variance page fired and the VP called the same hour. Fixed with a heartbeat, a DST checklist, a 3σ page and a rehearsed manual mode. Never offered as *the* failure — it is the smaller arc after G1, G3 and G2.

If he pushes: "those are soft failures." G1 is not soft — a lost sign-off, a crashed system, a customer's confidence spent. The set is deliberately shaped: one customer-facing failure (G1), one scoping failure (G3), one planning failure (G2) — the three ways programs actually break. What they share is the pattern: the root cause was mine, the fix was structural, and there is a number in a later story proving the lesson held.

Question bank — likely questions for this interviewer, with a one-line answer plan UPDATED

Bar-raiser questions: LP-depth probes, mid-story LP switches, softest-number challenges and the consistency check. Each plan points at a packet story; stories allocated to other interviewers are pointers only. Rows 13–28 are new: AIQB-worded behavioral prompts for S13, S9 and the failure arcs, and the I-versus-we probes.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Tell me about a time you earned trust the hard way.**	S14 pitch — tell the customer first; cost priced onto me; the standing control; the $10M initiative won (user anchor), never claimed as savings.	S14
2	**Tell me about a mistake you made and what changed because of it.**	S16 — the security review; two weeks; owned upward the same day; zero Mondays skipped; the kickoff question.	S16
3	**Tell me about developing someone.**	S15 — the QA lead who authored and defended the gate; development measured by whether the gate survived my leaving.	S15
4	**Tell me about a time the data disagreed with everyone.**	S10 — three years in SQL; true DQ ~55%; measurement, not carriers; Finance-co-owned method before the number.	S10
5	**Switch LP mid-story: "OK — now tell me that same story as Frugality / Customer Obsession / Bias for Action."**	Keep the facts identical; change emphasis per the twist map (e.g., S16 as Frugality = the >$100K priced; S14 as Bias for Action = the two days; S10 as Customer Obsession = the carriers could not dispute).	Twist maps
6	**Which of your numbers would you least want me to check?**	Name the softest input in the story just told and how to firm it (S16 decision risk; S15 criticals base count; S10 the $0.5M analyst capacity; S14 discrepancies per month) — never defend a soft number as record.	All four
7	**What did you give up for the customer?**	S14 — my time in a war room and the risk of being wrong in front of them; S15 — the stories I covered so dev could drop them; S10 — weeks of schedule for evidence.	S14 / S15 / S10
8	**What is your biggest weakness as a leader?**	I convert verbal agreements into documents people must sign — slower in the week; and I under-planned a dependency with a lead time once (S16) and now over-ask that question at every kickoff.	S16
9	**Tell me about a time you had to scale something.**	S10 — Austria → 25 sites at a rules file plus one DQ cycle each; the super-user model; the method signed before the number so scale could not inflate the claim.	S10
10	**How do you decide what not to build?**	S16 — ship with the cross-reference table, push the real key upstream; S15 — the drop-date rule; S14 — no bulk script across related tables.	S16 / S15 / S14
11	**Why should we hire you over someone who has run an Amazon network region?**	Fifteen years designing the supply chain — definitions, lead times, buffers, one truth — and keeping it honest in a weekly review; I have not run an Amazon region and won't pretend otherwise; the pattern transfers, the scale is the reason I am here.	Master TMAY close
12	**Is there anything you've told the other interviewers that you'd state differently to me?**	No — and name the two numbers most likely to be cross-checked: 7 → 68 is 9.7×, and the three different $2.5M figures (Manhattan order value; Sephora annual impact; Sephora per-month slip, modeled) are different things.	Consistency box
13	**"Tell me about a time you made a bad decision." (AIQB: Are Right Q4)**	G2 — scored a risk by effort not impact; training slipped 1.5 weeks while the date held; what-if on every dependency since.	G2
14	**"Tell me about an error in judgment." (AIQB: Are Right Q5)**	G1 — the UAT that crashed; first move defended instead of dug; criteria rewritten; 500 containers / 20 min / 99.6% proof.	G1
15	**"A time you were not able to meet a commitment." (AIQB: Earn Trust Q4)**	S16 / G3 — two weeks on the security review, owned upward the same day; zero Mondays skipped.	S16 / G3
16	**"An unpopular stance in a meeting with peers and your leader." (AIQB: Have Backbone Q2)**	S9 — a thousand lanes were asked; 597 first, taken two levels up with the spend curve; 597 lanes = 80% of spend (user anchor).	S9
17	**"You committed to a decision you disagreed with." (AIQB: Have Backbone Q3)**	S9 — after the phasing was accepted I committed to delivering the remaining lanes behind the same gate; S14 — manager accepted hesitantly, I ran the war room.	S9 / S14
18	**"Tight deadlines, not all options considered." (AIQB: Bias for Action Q2)**	S14 — two days: census in a morning, options memo, customer pack, per-line reversible patch. S13 for hours rather than days.	S14 / S13
19	**"A calculated risk where speed was critical." (AIQB: Bias for Action Q1)**	S13 — 4 AM, declared the incident before the cause was known; reversible, timestamped, checked with engineering; 99% vs 95%.	S13
20	**"Significant unanticipated obstacles." (AIQB: Deliver Results Q2)**	S13 — a dead scheduler on the biggest night; or G2 — new RF hardware that ran no old code.	S13 / G2
21	**"A decision between standards and delivery." (AIQB: Highest Standards Q4)**	S9 — a cohort held twice at ≥95% valid ASN with a date waiting; S15 — the gate that can say no.	S9 / S15
22	**"An opportunity much bigger than the initial focus." (AIQB: Think Big Q1)**	S9 — a one-time thousand-lane ask became a standard, a gate and a scorecard; the method became the sourcing SOP; adoption 8× (user anchor).	S9
23	**"How have you helped save costs or eliminate waste?" (AIQB: Frugality Q1)**	S10 — executed savings only, on a method Finance co-owned before the number: about $500K a year at Austria → ~$8M across 25 locations. S16 is priced, never saved.	S10
24	**"You said 'we' — what exactly did YOU do?"**	The Mine / Theirs line of the story just told (Trap T2 in each story; the I-vs-we row of the probes grid). One sentence each, then the specific artifact I wrote.	Probes grid

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
25	**"Which story would you least want me to dig into?"**	Name it first: S15's dollars (modeled on an illustrative base; the −50% is the record) or S13's counterfactual (not provable) — then say how to firm each.	S15 / S13
26	**"What would a former manager say you got wrong?"**	I defended before I dug (G1); I under-planned a dependency with a lead time (G3); I scored a task by effort (G2) — each with the permanent rule. Never a humble-brag.	G1 / G3 / G2
27	**"Tell me about the last time data changed your mind."**	S10 — carriers were not late; we could not measure: reported 40%, true ~55%. S9 — freight −10% is as measured (user anchor), not proof of cause.	S10 / S9
28	**"Biggest impact you have had at any employer."**	Pointer only, master numbers: S11 — 7 → 68 customers = 9.7×, CTS $0.66 → $0.03; S12 — about $5M avoided, said once; S2 — +15% processing efficiency (DC Ops' own measurement). Re-cut cards on the re-cut page.	S11 / S12 / S2

Backup bench and design cases — use only if a primary doesn't fit UPDATED

Four fallbacks for the bar raiser, none of them another interviewer's primary: **S19** and **S20** are **design cases — not lived results** (the Walmart Marketplace delivery-promise material is my own interview-prep case work on illustrative data; every number is "in the model", and the lived record for the theme is S8 / Apple workbook F4) — plus two doc-derived Flexport benches from the program record: the RAPID Crux customer transition (Earn Trust / Customer Obsession: 79 customers, transition-or-cancel decision rules) and the FFA Asia latency investigation (Dive Deep; facts identical to Hoerger's bench) — both record-from-docs, both from the S11/S12 program family, so keep the shared numbers identical to Tripathy's packet. S13 (Cyber Monday) and S9 (597 lanes first) were promoted from this bench to full primaries in v2. Tell S14, S16, S10, S15, S13 and S9 first; a bar raiser compares notes, so a bench story must carry exactly the master's numbers.

Bench discipline. The six primaries are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

DESIGN CASE — NOT A LIVED RESULT S19 — Design case: a marketplace delivery-promise buffer as one governed dial — in the model, +32 points of fast-promise share at the same 95% floor (all model; no delivered dollar)

LPs this design case can serve: Dive Deep (alternate when S8/S10 are spent) · Invent and Simplify (alternate) · Frugality (speed bought with zero new spend) — design-flavored probes; always as "I designed / in the model / here is how I would", never as a result I delivered

Basis — a design case, not a lived result. This is a case I worked through in my own interview preparation for a marketplace delivery-promise role, on an illustrative twelve-cohort model (war book and solver model; sample data). I did not deliver these results for any employer, and I never configured or deployed any platform. Every quantity is "in the model" — say it in the same breath; there is no production dollar and none is ever promoted. The lived record for promise-at-the-tail is S8 (Elementum; Apple workbook F4): 200 lanes, 99% lead-time accuracy, 580,000 units and about $1.5M of inventory carrying cost released — state the carry rate when you say it. Say that one if he wants a result. Labels (metro two-day, mountain rural) are the model's illustrative cohort names.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation (the case). A marketplace printing delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and in the case the buffer is a flat 1.10 days on every lane regardless of variability. In the model — twelve illustrative lane cohorts, about a million orders a week, transit σ from 0.22 to 1.60 days — that flat buffer prints 95.8% accuracy but puts only 7.6% of orders on a fast promise: dense metro lanes over-insured and fat-tailed rural lanes under-insured at the same time. **Task (as I designed it).** Own the decomposition, the alternatives, the validation gates and the governance recommendation. **Action (what I designed).** Discovery: the buffer-waste audit — the flat buffer buys Φ(5.0), near-certainty, on the tightest cohort and only Φ(0.69) ≈ 75% on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: b = 1.645σ holds 95% on every cohort by construction; the equimarginal check proves the optimum; the Q4 drill shows a frozen summer dial breaking the floor. Gates I would set: four shadow weeks against logged live orders before any displayed promise; the dial published in the weekly review header and re-solved before peak. **Result (in the model).** Fast-promise share 7.6% → 39.6%, +32 points at the held 95% floor; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all in the model, on an illustrative book. It is a design, not a delivered result, and I say so in the same breath.

COMPRESSED STAR

- **S —** A flat 1.10-day buffer on every lane: 95.8% headline accuracy hiding over-insurance on tight lanes and under-insurance on loose ones; fast-promise share 7.6% (in the model).
- **T —** Design case: decompose the waste, put priced alternatives on one table, design the gates, leave a governance instrument — not a tuning habit.
- **A —** Promise identity and accuracy factorization; buffer-waste audit; four policies on one table; proofs (Φ(1.645) = 0.95 per cohort, equimarginal check, Q4 drill at ~1.45× σ); shadow mode for four weeks (designed); the dial k owned by a business owner and printed every Monday.
- **R —** In the model: fast-promise share 7.6 → 39.6% (+32 pts) at exactly the 95% floor; average promised day 3.09 → 2.75; +$255K/wk ≈ $13.3M/yr — modeled, labeled; the widest cohorts get honestly slower promises (1.10 → 2.63 days) on purpose.

RESULTS LINE

In the model: fast-promise share 7.6% → 39.6% (+32.0 points) at a held 95.0% floor · average promised day 3.09 → 2.75 (model) · +$255K/wk ≈ $13.3M/yr vs uniform (modeled on the illustrative ~1M-order book) · the profit-optimal frontier (97.8% accuracy, ≈ +$20M/yr) handed to Finance as a priced decision — shown, never claimed, never merged with the $13.3M.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The promise identity	`EDD = t_order + handling + transit_pred + b;` `kept ⇔ (handling_err + transit_err) ≤ b`	per order; b = buffer in days (design, from my prep docs identity)	the buffer is the only policy term
2	Kept-promise probability per lane	`P(kept) = Φ(b / σ)`	ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (design, from my prep docs)	accuracy is a function of b and σ only
3	Incumbent buffer	`b = 1.10 days on every lane`	uniform policy A (illustrative model)	1.10 d everywhere
4	Tightest cohort, z-score	`z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997`	metro cohort σ = 0.22 d (illustrative model)	near-certainty bought — promise needlessly slow
5	Widest cohort, z-score	`z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754`	mountain-rural cohort σ = 1.60 d (illustrative model)	≈75.4% kept — a known-loss book

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	Portfolio print	`accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk`	order-weighted across 12 cohorts, ~1M orders/wk (illustrative model)	a green headline hiding both failures
7	The asymmetry	`padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)`	design, from my prep docs — why the system drifted slow	bias toward slow, unowned

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One governed dial	`b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%`	Φ(1.645) = 0.95 under normal errors (design, from my prep docs design)	k = 1.645, one number for the book
2	Tightest cohort re-buffered	`b = 1.645 × 0.22 = 0.36 days`	σ = 0.22 d (illustrative)	buffer 1.10 → 0.36 d; 0.74 d of speed harvested
3	Widest cohort re-buffered	`b = 1.645 × 1.60 = 2.63 days`	σ = 1.60 d (illustrative)	buffer 1.10 → 2.63 d — slower on purpose, kept honest
4	Every cohort at the floor	`P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95 on each cohort`	by construction (design)	95.0% everywhere; no cohort subsidizes another
5	Portfolio after	`accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk`	illustrative model output, policy B	+32 pts fast share at the floor
6	Equimarginal proof (policy C)	`φ(b/σ)/σ equal across all 12 cohorts (spread < 1e-15)`	solver optimality condition (design, from my prep docs)	the last buffer-day buys the same accuracy everywhere
7	Peak regime rule	`σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645/1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%`	Q4 drill (illustrative model)	k re-solved and published in October, never frozen
8	Shadow gate	`4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold`	promotion bar (design)	rehearse against reality before any displayed date

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Fast-promise share (≤2-day)	7.6%	39.6% (+32 pts)	illustrative model
2	Portfolio promise accuracy	95.8% (split: ~100% / 75.4%)	95.0% — the floor, held on every cohort	illustrative model
3	Average promised day	3.09	2.75 (−0.34)	illustrative model
4	Weekly modeled profit	$5.69M	$5.94M (+$255K)	illustrative model
5	Owner of the speed-accuracy trade	implicit, in the last model tune	one governed dial, published weekly, re-solved per regime	my design

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Weekly profit, policy A	`$5,690,000 / wk, taken from the model`	illustrative model output	$5.69M/wk
2	Weekly profit, policy B	`$5,940,000 / wk, taken from the model`	illustrative model output	$5.94M/wk
3	Weekly delta	`$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)`	rows 1, 2 (illustrative)	≈ +$255K/wk
4	Annualized	`$255,000 × 52 = $13,260,000`	row 3 × 52 weeks (modeled)	≈ $13.3M/yr — "the model," said in the same breath
5	Frontier: policy D delta	`$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000`	illustrative model output; ×52 modeled	≈ $20M/yr — handed to Finance, never claimed
6	Why D holds 97.8%	`breach EV $12 > margin-per-promise-day trade at $6.50 ⇒ optimum above the 95% floor`	$12 and $6.50 are the model's stated assumptions	the floor and the economics disagree — a governance finding
7	Headline	`≈ $13.3M/yr modeled; record dollars: none`	rows 3–4	≈ $13.3M/yr, always labeled

TWO PROBES

> **The whole story rests on an illustrative cohort set.**
>
> It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers risk-proportional, the last buffer-day buying the same accuracy everywhere, and a floor that disagrees with the breach economics treated as a governance finding.

> **Why one dial instead of per-lane learned buffers?**
>
> In the design I would put per-lane learned buffers on the same table as a priced row — more model, more value, invisible governance — and recommend the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The difference sits in the comparison table as a priced row, not a meeting minute.

Design-case numbers (all in the model; basis labels): 7.6% → 39.6% fast-promise share (+32 pts, model) · 95.8% → 95.0% accuracy held at the floor (model) · 3.09 → 2.75 promised days (model) · +$255K/wk ≈ $13.3M/yr (modeled) · 12 cohorts, ~1M orders/wk, σ 0.22–1.60 days, k = 1.645 (illustrative model) · breach ≈$12 vs margin ≈$6.50 (model assumptions). Never a production dollar; never a delivered result.

DESIGN CASE — NOT A LIVED RESULT S20 — Design case: the pre-purchase WBR for a marketplace delivery promise — input levers under output metrics, the five-minute red-week decomposition, and the Andon that would replace a 30-day lag (design; no delivered result)

LPs this design case can serve: Insist on the Highest Standards (alternate — hard reds on policy floors) · Earn Trust (alternate — mix-adjusted twins, honest denominators) · Dive Deep (alternate — the factor math) — cadence/mechanism probes; always as a design

Basis — a design case, not a lived result. I designed the weekly business review on paper — metric architecture, decomposition math, thresholds and control loop — as part of my own interview preparation for a marketplace delivery-promise role, on illustrative sample data (six sample weeks). I did not stand it up for any employer, never configured or deployed any platform, and never claim a production output metric moved. The production-record WBR story is S11 (Flexport) — keep the two separate. The lived record for promise-at-the-tail is S8 (Elementum; Apple workbook F4): 200 lanes, 99% lead-time accuracy, 580,000 units and about $1.5M of inventory carrying cost released — state the carry rate when you say it. Say that one if he wants a result.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation (the case). A marketplace whose delivery promise is owned by a committee: data science owns the model, engineering the engine, seller ops the handling, carrier programs the lanes — and promise accuracy, the one number the customer feels, is owned by nobody between Tuesdays. Scorecards run on 30-day rolling windows, so a degrading seller or carrier does weeks of customer damage before any floor bites, and a red week becomes a blame negotiation. **Task (as I designed it).** Design the weekly business review — outputs a customer feels, input levers a named owner can turn, and the math that connects them — plus a control loop faster than the 30-day clock. **Action (what I designed).** Discovery: one glossary, one event source, audited denominators. Plan & Design: seven output metrics over ten input levers, each input with a roll-up and a named owner; the factorization — accuracy ≈ seller-ships-in-window × carrier-meets-transit × Φ(buffer/σ) — as the decomposition; red/green at ±5% relative week-over-week with hard reds on policy floors. Proof: the sample week — 0.97 × 0.985 × 0.996 ≈ 95.2% — and the drill: a regional carrier miss (97% nationally, 82% in the Southeast) decomposed to its factor in minutes. Cadence: Monday data freeze, Tuesday review, reds pre-wired with cause, owner, date. Control: mix-adjusted twins next to every raw output, and the Andon — a 7-day control chart per seller-lane cell, act within 24 hours. **Result (in the model).** Red-week attribution from days of negotiation to minutes by factor math, proven on the sample-data drill; detection-to-action from ~30 days to under 24 hours by design. A design case — nothing here is a delivered production result.

COMPRESSED STAR

- **S —** Output metric owned by a committee; 30-day rolling windows smearing step changes into drift; three teams keeping three clocks; a red week discovered live and negotiated.
- **T —** Design the WBR for the pre-purchase delivery experience — metric architecture, decomposition math, thresholds and control loop — as a design case, with owners named so it can be handed over.
- **A —** One glossary and denominator audit; 7 outputs over 10 input levers with roll-ups and named owners; the factor identity as the decomposition; ±5% WoW red/green with hard reds on floors; Monday freeze → Tuesday review, reds pre-wired; mix-adjusted twins; a cell-grain Andon (7-day EWMA/CUSUM, 2× baseline breach, n ≥ 30, act in 24 h, false-trip bar < 5%).
- **R —** In the model: attribution days → minutes; detection-to-action ~30 days → < 24 h (designed); Promise Yield 43.6% × 95.2% ≈ 41.5% baseline on the sample set, four-quarter design target 50%; ≈$0.58M per caught episode (modeled).

RESULTS LINE

Red-week attribution: a cross-team negotiation → minutes by factor math (proven on the sample-data drill) · detection-to-action ~30 days → under 24 hours by design (the Andon) · 7 outputs / 10 inputs / 1 joint headline / 6 sample weeks (my design) · ≈$0.58M per caught degradation episode (modeled; no annual figure — episode frequency is unknown and I won't invent it).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The accuracy identity	`O1 ≈ I3 × I4 × Φ(b/σ) — seller ships in window × carrier meets transit × buffer absorbs the residual`	design, from my prep docs factorization; three factors, three owners	every breach belongs to exactly one factor
2	Healthy sample week	`O1 = 0.97 × 0.985 × 0.996 = 0.9516 ≈ 95.2%`	I3 = 0.97, I4 = 0.985, Φ = 0.996 (docs' illustrative sample)	95.2% — matches the sample print
3	The aggregate trap	`carrier prints 97% nationally while one region runs 82%`	Southeast after a hub consolidation (design, from my prep docs example)	the average hides the cell
4	The rolling-window lag	`a step change entering a 30-day window moves the average ≈1/30 per day`	30-day scorecard mechanics (design, from my prep docs issue)	weeks before a floor bites — enforcement lag
5	The gameable pair	`fast share 43.6% and accuracy 95.2% on separate lines, separate owners`	docs' sample; either can be juiced by bleeding the other	no joint headline
6	Baseline joint number	`Promise Yield = 0.436 × 0.952 = 0.415`	rows 2, 5 (docs' sample)	≈ 41.5%

Table 2 — the fix: step by step, values substituted (the red-week drill).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The episode	`Southeast cell: carrier adherence 0.985 → 0.82 on w = 10% of volume`	82% is the docs' example; the 10% weight is illustrative	one regional carrier miss
2	New carrier factor	`I4' = 0.90 × 0.985 + 0.10 × 0.82 = 0.8865 + 0.082 = 0.9685`	volume-weighted blend (illustrative)	I4: 0.985 → 0.9685 (−1.7%)

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	New accuracy	`O1' = 0.97 × 0.9685 × 0.996 = 0.9357 ≈ 93.6%`	rows 1–2; other factors unchanged	95.2% → 93.6% — a red week
4	The attribution read	`WoW deltas: I3 0.0% · I4 −1.7% · Φ 0.0%`	computed per factor (the decomposition)	the red names I4 and the carrier PM — in minutes
5	Segmentation	`factor red isolated to one region ⇒ C-family cause, not model or seller`	lane/cohort cut of I4 (design)	cause class identified before the meeting
6	The Andon trigger	`cell 7-day breach rate sustained ≥ 2 × baseline, n ≥ 30 ⇒ act < 24 h`	EWMA/CUSUM parameters (design, from my prep docs design)	buffer widened or badge pulled on the cell, same day
7	Auto-restore	`control chart recovers ⇒ badge/buffer restored, no meeting needed`	design	surgical, reversible, seller-fair

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Red-week attribution time	days (blame negotiation)	minutes (factor math)	my design, proven on the sample-data drill
2	Detection-to-action, degrading cell	up to ~30 days	< 24 h from sustained signal	designed (Andon)
3	Output ownership	committee	one named owner of the output block + log	my design
4	Joint headline	none	Promise Yield, 41.5% sample baseline → 50% 4-qtr target	docs' sample; design target
5	Mix honesty	raw outputs only	mix-adjusted twin + mix line per output	my design

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Episode accuracy loss	`95.2% − 93.6% = 1.6 points`	Table 2 rows 2–3 (illustrative episode)	1.6 pts of excess breaches
2	Excess breaches per week	`1,000,000 × 1.6% = 16,000`	~1M orders/wk illustrative book	≈16,000 breaches/wk
3	Cost per breach	`$12, taken as stated`	the model's assumption: contact + refund risk + trust decay (design, from my prep docs)	$12 each
4	Excess cost per week	`16,000 × $12 = $192,000`	rows 2 × 3 (modeled)	≈$192K/wk
5	Detection advantage	`~4 weeks (rolling-window era) − 1 week (Andon era) = 3 weeks`	row 4 of Table 1; Andon design	3 weeks earlier action
6	Avoided per episode	`$192,000 × 3 = $576,000`	rows 4 × 5 (modeled)	≈$0.58M per caught episode
7	Annual total	`not computed — $576,000 × (episodes/yr) requires a frequency I don't have`	deliberately uninvented	per-episode only, labeled modeled

TWO PROBES

> **Your baseline Promise Yield is sample data.**
>
> Correct, and I say so: 43.6% × 95.2% ≈ 41.5% is the case docs' own illustrative six-week set, used to prove the mechanics before any live line exists. The 50% target is a design target for a real book, and the first live job would be re-baselining it.

> **How is this different from the Flexport review (S11)?**
>
> S11 is the production record — a review I ran by hand every Monday for over a year (automation 40 → 90%, accuracy 60 → 90%). This design case is where I carried the review's mathematics — the factor decomposition and the control loop — furthest, as a design. I keep the two separate and never borrow a number across them.

Design-case numbers (all in the model; basis labels): 7 outputs / 10 input levers / 1 joint headline (my design) · 0.97 × 0.985 × 0.996 ≈ 95.2% (sample-week proof) · 97% national vs 82% Southeast (the case's worked example) · 30 days → < 24 h (designed) · Promise Yield 41.5% baseline, 50% target (docs' sample) · ≈$0.58M per caught episode (modeled: ≈$192K/wk × 3 weeks). Never a production output claim.

BACKUP — ONLY IF ASKED **Doc bench — Flexport — the RAPID Crux customer transition: 79 customers, $185,892 ARR, 294,581 containers — transition where it fits, cancel honestly where it doesn't (Earn Trust / Customer Obsession)**

LPs this backup can serve: Earn Trust (primary — honest cancels) · Customer Obsession (per-customer calls for the largest accounts) · Have Backbone, Disagree & Commit (internal pushback priced, not overridden) · Deliver Results (every customer decided by end 2022)

Basis. Doc-derived micro-story from the program record (RAPID — the Crux customer transition, 2022) — every figure is RECORD-FROM-DOCS. Crux was the acquired visibility product; its customers moved to Shipment Visibility (Tripathy's S11 program family) — if the room connects them, keep her packet's canon numbers identical. Program-lead voice: I owned the transition design and the decision rules; account owners ran the per-customer calls. Not in the master packet's S-numbering; supplements canon, never overrides it.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The acquired visibility product, Crux, still carried 79 customers — $185,892 ARR and 294,581 containers a year — and the decision was taken to move them onto Shipment Visibility by end 2022. The catch: SV did not fit every segment; some customers were better served by an honest cancel than a force-fit. **Task.** Decide every customer — transition or cancel — without burning the revenue that fit or the trust of those that didn't. **Action.** Segment-level decision rules instead of 79 one-offs: truck brokers to a white-label partner product or cancel; forwarders, customs brokers and 3PLs to the SV API or cancel; the two competitor SCM-SaaS accounts canceled on published dates (August 1 and October 24, 2022). The two largest accounts by volume got per-customer calls — one moving a national home-improvement retailer's freight, the other a 600-truck carrier relationship. Internal pushback — migration keep-the-lights-on cost, feature parity, the timing of consignee acquisition — was answered with priced options in the decision document, not overridden by title. **Result.** Every one of the 79 customers had a decided, dated path — transition where SV fit, cancel honestly where it did not — with the segment table and the decision rules as the record. RECORD-FROM-DOCS

COMPRESSED STAR

- **S —** 79 acquired-product customers ($185,892 ARR; 294,581 containers) across eight segments, on a product being retired by end 2022; the replacement did not fit every segment.
- **T —** Own the transition design: decision rules per segment, dates per account, and the handling of internal dissent.
- **A —** A segment table with a transition-or-cancel rule per segment; a white-label partner product for truck brokers; the SV API for forwarders/brokers/3PLs; competitor SaaS accounts canceled on published dates; per-customer calls for the two largest-volume accounts; pushback on KTLO cost, feature parity and consignee timing priced into the decision document.
- **R —** 79 of 79 customers decided and dated (record-from-docs). The record carries no preserved-ARR figure, and none is claimed.

RESULTS LINE

79 customers · $185,892 ARR · 294,581 containers (record-from-docs) · eight segments with a rule each (record-from-docs) · two competitor SaaS accounts canceled on published dates (record-from-docs) · two largest accounts handled per-customer (record-from-docs) · no preserved-ARR or churn dollar in the record — never invent one.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Customers on the retiring product	`79 across the printed segment table`	RAPID doc, customer-by-type table (record-from-docs)	79 accounts to decide
2	Revenue carried	`$15,491 MRR × 12 = $185,892 ARR`	printed totals (record-from-docs)	$185,892 ARR
3	Volume carried	`294,581 containers tracked (2021)`	printed total (record-from-docs)	294,581 containers/yr
4	The fit problem	`SV fits shippers and consignees; carriers, brokers, forwarders, 3PLs and competitor SaaS are not consignees`	RAPID opportunity section (record-from-docs)	one product cannot absorb all 79

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Segment count check	`2 + 14 + 39 + 9 + 3 + 5 + 1 + 4 + 2 = 79`	GKA · Enterprise · SMB · Motor Carrier · Truck Broker · Freight Forwarder · Customs Broker · 3PL · SCM SaaS rows (record-from-docs)	rows reconcile to the printed total
2	Container check	`17,061 + 17,983 + 4,153 + 12,359 + 221,044 + 11,871 + 432 + 4,218 + 5,460 = 294,581`	segment rows (record-from-docs)	reconciles exactly
3	Concentration	`221,044 ÷ 294,581 = 0.7504 ≈ 75% of volume on 3 truck-broker accounts`	segment rows (record-from-docs)	the decision's center of gravity
4	Rule — green segments	`2 + 14 + 39 = 55 customers → transition to Shipment Visibility`	RAPID opportunity section (record-from-docs)	55 transitioned
5	Rule — carriers & brokers	`9 + 3 = 12 → white-label through Transmission, else cancel at wind-down`	RAPID decision section (record-from-docs)	12 decided
6	Rule — forwarders, customs brokers, 3PLs	`5 + 1 + 4 = 10 → SV API, else cancel at wind-down`	RAPID decision section (record-from-docs)	10 decided

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
7	Rule — competitor SaaS	`2 → cancel: Shippabo 2022/08/01; Shipwell 2022/10/24`	RAPID appendix (record-from-docs)	2 canceled on published dates
8	Closure check	`55 + 12 + 10 + 2 = 79`	rows 4–7 (record-from-docs)	every customer under a rule — eight segments carry a designed forward rule; the ninth printed row (SCM SaaS) was immediate cancel, already executed on the dates in row 7
9	Two largest accounts	`per-customer handling, not a segment rule — AV Logistics and ITS Logistics, the volume leaders`	RAPID sections 3 and 7 (record-from-docs)	Transmission partnership modeled on ITS's existing usage

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Customers decided	0 of 79	79 of 79, each with a dated path	record-from-docs
2	Paths	one undifferentiated book	55 → SV · 12 → Transmission-or-cancel · 10 → SV-API-or-cancel · 2 → canceled	record-from-docs
3	Largest accounts	segment averages	per-customer calls for the two biggest-volume accounts	record-from-docs
4	Internal dissent	—	KTLO cost, feature parity and consignee-timing objections priced inside the decision document — answered, not overridden	record-from-docs

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	ARR identity check	`$15,491 × 12 = $185,892`	printed MRR and ARR totals (record-from-docs)	totals internally consistent
2	ARR on the committed SV path	`$16,500 + $22,560 + $30,000 = $69,060`	GKA + Enterprise + SMB rows (record-from-docs)	$69,060 ARR transitioning to SV
3	ARR under transition-or-cancel decision	`$17,760 + $139,512 + $14,820 + $600 + $2,400 + $10,800 = $185,892`	the six non-green rows (record-from-docs)	equals the printed ARR total exactly
4	Concentration of the decision	`$139,512 ÷ $185,892 = 0.7505 ≈ 75% of decided ARR in 3 truck-broker accounts`	segment rows (record-from-docs)	why the two largest got per-customer calls
5	Average account	`$185,892 ÷ 79 = $2,353 ARR; 294,581 ÷ 79 = 3,729 containers`	printed totals (record-from-docs)	the typical account was small — rules, not 79 one-offs
6	Preserved-ARR claim	`no preserved-ARR or churn dollar in the record — not claimed`	record-from-docs	never invent one

TWO PROBES

How do you justify canceling a paying customer?

By pricing the alternative honestly: force-fitting a segment the product doesn't serve converts this year's revenue into next year's churn plus support cost, and it burns the reference. A clean cancel with notice and a path — a white-label partner product, or the API if they fit it — kept the relationship honest; two of the 79 were competitor SaaS accounts where canceling was the only truthful option.

How did you handle the internal pushback?

As inputs to price, not objections to defeat: migration KTLO cost, feature parity and consignee-acquisition timing each got an option with a cost in the decision document, and the decider chose with the prices on the table. Pushback that is priced becomes design input; pushback that is overridden becomes a later surprise.

Canon numbers (basis labels): 79 customers · $185,892 ARR · 294,581 containers (record-from-docs) · segment table incl. Enterprise 14 · SMB 39 · Motor Carrier 9 (record-from-docs) · two SaaS cancels dated Aug 1 / Oct 24, 2022 (record-from-docs) · transition deadline end 2022 (record-from-docs). Never: a saved-ARR percentage; never merge with S11's 7 → 68 customer canon.

BACKUP — ONLY IF ASKED **Doc bench — Flexport FFA — the Asia latency investigation: a 21-second P95 traced to routing, decomposed to the layer, fixed in sequence (Dive Deep)**

LPs this backup can serve: Dive Deep (primary — decomposition to the layer) · Deliver Results · Ownership (the standing rule imported into later launches) · Insist on the Highest Standards (a smoke-test protocol per change)

Basis. Doc-derived micro-story from the program record (the FFA Asia latency investigation, post-MVP 2022) — every figure is RECORD-FROM-DOCS: the investigation document, written when the FF App MVP had launched to 11 partners in 10 countries and 48 of 53 import partners (90.5%) had migrated. Program-lead voice: I owned the investigation, the decomposition and the fix sequence; infrastructure teams executed the WAF and CDN changes. Not in the master packet's S-numbering; supplements canon, never overrides it. The same bench appears in Hoerger's packet — facts identical.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Weeks after the FF App MVP launched to 11 partners in 10 countries — with 48 of 53 import partners already migrated — the monthly business review showed Asia page loads at a P95 of 21.19 seconds against 3.3 in the US. **Task.** Find the layer, not the anecdote, and fix it without a re-architecture in the middle of a live partner-facing launch. **Action.** I decomposed the problem before prescribing: static assets ran ~5s P95 in China against ~500ms in the US, and the app's domain had no geo-based DNS or reverse proxy — Asia traffic was riding the public internet across the Pacific; a double CDN sat on the static path. The fix was a sequence, not a project: enable the WAF, remove the double CDN (direct to S3), then a regional CDN proposal with a smoke-test protocol per change, then China tuning, then Asia tuning — each with a date and an owner. **Result.** Root cause isolated to routing — not the app, not the partners; two fixes shipped inside the investigation window and the regional plan scheduled with dates; and a standing rule imported into every later launch: the pilot's geography is a performance requirement, not a rollout detail. RECORD-FROM-DOCS

COMPRESSED STAR

- **S —** Post-MVP, the MBR surfaced Asia P95 page load at 21.19s vs 3.3s in the US (17-day measurement window); 48 of 53 import partners (90.5%) already migrated — the app had to hold where the partners were.
- **T —** Isolate the layer causing Asia latency and sequence the fix set without destabilizing a live partner-facing launch.
- **A —** Split static from dynamic: static JS ~5s P95 in China vs ~500ms US; found no geo-based DNS / reverse proxy on the domain (public-internet routing) and a double CDN on static assets; sequenced WAF enablement → double-CDN removal (direct to S3) → regional CDN proposal → China tuning → Asia tuning, each gated by a smoke-test protocol with a named date and owner.
- **R —** Root cause isolated to routing; WAF and double-CDN fixes shipped inside the window; CDN / China / Asia work scheduled with owners; the investigation became the template for launch-geography checks.

RESULTS LINE

P95 page load 21.19s Asia vs 3.3s US (record-from-docs) · static ~5s China vs ~500ms US (record-from-docs) · 48 of 53 partners migrated = 90.5% (record-from-docs; canon's "48 of 53 within a month" — consistent) · two fixes shipped, three sequenced with dates (record-from-docs) · no post-fix latency figure exists in the record — never invent one.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Page-load P95 by region	`Asia = 21.19s; US = 3.3s; Others = 4.45s`	Datadog @view.loading_time, 04/24–05/10 window (record-from-docs)	Asia is the outlier
2	The gap factor	`21.19 ÷ 3.3 = 6.42`	row 1 (record-from-docs)	6.42× slower than the US
3	Traffic behind the number	`6.31 ÷ 2.19 = 2.88`	Asia 6.31k vs US 2.19k page views, same window (record-from-docs)	≈2.9× the US traffic riding the worst latency
4	Static-asset leg	`JS P95 = 5s China vs 500ms US; 5 ÷ 0.5 = 10`	static.flexport.com CDN behavior (record-from-docs)	10× gap on the static leg
5	Dynamic leg	`no geo-DNS / reverse proxy for trade.flexport.com → Asia GraphQL hairpins over the public internet`	infra review (record-from-docs)	routing penalty on every data call
6	Known defect on the static path	`double CDN in China — Cloudflare stacked in front of Tencent`	infra review (record-from-docs)	one avoidable hop

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Decompose before touching	`page load = static resources + GraphQL requests + JS rendering`	the metric's own definition (record-from-docs)	three legs, measured and owned separately
2	Step 1 — security gate first	`enable WAF for trade.flexport.com: China, then global`	incident-207 follow-up; done 2022/06/30 (record-from-docs)	shipped — fix 1 of 2
3	Step 2 — remove the double CDN	`Tencent CDN → direct to the S3 bucket; Cloudflare hop removed for China`	done 2022/06/24 (record-from-docs)	shipped — fix 2 of 2
4	Step 3 — static proposal, CN/Asia	`alternatives written, reviewed, signed off`	ETA 2022/07/15 (record-from-docs)	sequenced with a date
5	Step 4 — China tuning	`e.g. replicate static assets in-region`	ETA 2022/07/30; depends on step 4 (record-from-docs)	sequenced with a date
6	Step 5 — Asia tuning	`the same playbook beyond China`	ETA 2022/08/15 (record-from-docs)	sequenced with a date

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
7	Held in reserve	`global CDN for trade.flexport.com — built only if the metrics still demand it`	2022 Q3 or later, metrics-gated (record-from-docs)	no speculative spend
8	Verification per change	`2 user types × 2 instances = 4 smoke passes before + 4 after = 8`	login → workview → logout → re-login; internal SSO + partner users; US + China hosts (record-from-docs; the multiplication counts the checklist)	8 workflow passes around every change

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Root cause	one number — "Asia is slow"	static leg (10×) and dynamic routing isolated, each with an owner and a sequence	record-from-docs
2	Fixes	none	2 shipped (WAF; double-CDN removal); 3 sequenced with dates (07/15, 07/30, 08/15)	record-from-docs
3	Post-fix P95	21.19s	no post-fix figure exists in the record — never claimed	record-from-docs
4	Spend discipline	—	global CDN gated on metrics, not built on speculation	record-from-docs

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Partner base exposed	`48 ÷ 53 = 0.9057 ≈ 90.5% of import partners already migrated when the gap surfaced`	migration tracker (record-from-docs)	90.5% of partners on the affected app
2	Excess wait served per window	`gap vs US = 21.19 − 3.3 = 17.89s; 6,310 × 17.89 = 112,886 partner-seconds; ÷ 3,600 ≈ 31.4 hours`	6.31k Asia page views in the 17-day window (record-from-docs); gap arithmetic illustrative	≈31.4 partner-hours of excess wait every 17 days
3	Dollar value of the wait	`no dollar in the record — not claimed`	record-from-docs	not claimed
4	Post-fix saving	`no post-fix latency figure exists → no savings chain exists`	record-from-docs	never invent one

TWO PROBES

> **How did you know it was routing and not the application?**
>
> By decomposition before remediation: the static/dynamic split put ~5s of the China time on assets the app server never touched, and the same assets served in ~500ms from the US — the code was constant, the path was not. The domain's missing geo-DNS / reverse proxy and the double CDN explained both halves of the gap without a single application change.

> **Why not just stand up a China region?**
>
> A region is a re-architecture with a compliance tail; the evidence said the bottleneck was routing and CDN topology, which fixes in weeks. The sequenced plan — WAF, double-CDN removal, regional CDN, tuning — bought most of the win at a fraction of the risk, and the smoke-test protocol meant each step was reversible.

Canon numbers (basis labels): 21.19s vs 3.3s P95 (record-from-docs) · ~5s vs ~500ms static (record-from-docs) · 48/53 = 90.5% migrated (record-from-docs; the source prints 90.5%, exact 90.57% — keep the lower, printed figure) · 11 partners / 10 countries MVP (record — matches S12 canon) · fix sequence WAF → double-CDN → regional CDN → China → Asia with smoke tests (record-from-docs). Never: a post-fix latency number (none in the record); never present scheduled items as delivered.

NUMBERS I MUST NOT GET WRONG — canon values with basis labels UPDATED

Canon numbers for S14, S16, S10, S15, S13 and S9, the failure arcs and the re-cut pointers, with their basis labels (record = Apple v11 or a source document; user anchor = my own earlier packet only), plus the cross-loop figures a bar raiser is most likely to cross-check — from the master packet's §4.1 ledger.

METRIC	VALUE TO SAY	BASIS	NEVER
S14 discrepancy	~200 orders / $2.5M of order value → 0 on the prioritized set before the review; per-line reversible patch with before/after evidence	user anchor	a percentage — none exists; "$2.5M" without its noun; "savings"
S14 initiative	$10M initiative won	user anchor — never counted as savings	any realized dollar on it
S14 modeled	≈$0.1M/yr standing reconciliation (range $55–160K); ≈$1.5M expected corruption loss from the silent script	modeled / avoided risk	calling the $1.5M expected loss a saving
S16 accuracy / single point of failure	WBR system accuracy 60% → 90%; parity within 1% for consecutive weeks; first alert caught a ~4% ticket-sync drop; no longer a single point of failure	record (Apple A3, WBR doc)	"bus factor ≥2" as a count; calling it ETA accuracy; telling the 60 → 90% in two stories
S16 dollars	>$100K a year — the BRD priced it at $103,680 (30 → 6 hrs/wk of WBR work at an assumed $90/hr); my split ≈$30K hours + ≈$70K decision risk is a model	priced (BRD) / modeled	"saved $100K"; "infra's own loaded rates"; stacking "1.5–2 → ~4 hrs/wk" on it
S16 miss	data-access security review — 2 weeks lost; owned upward the same day; zero WBR deliveries skipped	record (Apple A3, G3)	"a dependency slipped"
S15 team / cadence	20-person Product/Eng/QA team; six-week hybrid → four-week agile in six weeks; releases 8.7 → 13 a year (52 ÷ 6, 52 ÷ 4 — from the cadence, not a measured count)	record (resume) / modeled (releases)	a direct-report headcount; "+50% releases" as a measured result
S15 quality / dollars	critical defects per release −50%; ≈$0.13M/yr (≈$90K + ≈$40K; range $80–270K)	record (resume) / modeled	the four-per-release base count as record
S10 data quality	~55% true (40% reported) → ~99% within a quarter; ≈97% agreement with a manual sample; manual reconciliation 8–10 hrs/wk → a small fraction	record (Apple F2)	"99% in two months" or any month count; "1–2 hrs/wk" as the after
S10 dollars	about $500K a year at Austria → ~$8M across 25 European locations, executed shipments only, per-shipment ledger (~$320K/site derived); the $5M freight base / 10% behind the Austria figure; ≈$0.5M/yr analyst capacity	record (Apple F2) / user anchor (base, 10%) / modeled	"$1M routing guide"; "400+ lanes"; counting negotiated-not-executed
S10 design	lane key (from, to, mode, volume band, Incoterm, SLA); Incoterm on ~8% of lanes; no-hit pages a human	user anchor (the 8%)	site-wave counts or dates beyond "Austria first"
S13 fill / sales	Fill 99% vs 95% goal; ~$6M on the line when the page fired; ~$8M of the day's sales protected; 3 DCs and 100 stores	record (Apple M1, resume)	the "day's GMV"; "$8M at risk"; "saved $8M"; the $8M as a saving
S13 modeled	≈$3.2M gross margin on the $8M at a 40% benchmark; ≈$1.3M/yr structural-stockout margin on an illustrative $400M retailer	modeled / illustrative	either as record
S13 node time	Near node ~0.85 days vs far ~4.60 days — a 3.75-day gap per re-pointed order	user anchor	a counterfactual
S9 lanes / gate	a thousand lanes asked; 597 first = 80% of spend; the remainder behind the same gate; completeness 60% → 95%; ≥95% valid-ASN gate; a cohort held twice; adoption 8×	record (completeness, gate: resume) / user anchor (597, 80%, held twice, 8×)	"400+ lanes"; "~200 next / ~400 deferred"; cohort sizes; a lane geography
S9 dollars	Freight −10% ≈ $2M in six months on the onboarded lanes (as measured); my split $0.8M + $0.6M + $0.6M; ≈$0.4M/yr document labor	user anchor / modeled	annualizing the $2M; the $20M implied base; confusing it with S6's $2M safety stock
G1 UAT	5 users passed 10 scenarios; sign-off 2 days away; concurrent logins crashed it; proof: 500 containers / 20 min / 99.6%	record (Apple v11)	"dozens of users"; "slipped a few days"
G2 RF devices	Training slipped 1.5 weeks, date held; mocks 7% → 1.8% → 0.3% vs 0.5%; about three percent under a ~$1.8M budget, roughly $54K; "hundreds" of RF devices	record (Apple G2, D1, D2)	"400 devices failed" or "~100 of ~400"; "$800K budget"
G3 security review	2 weeks; zero skipped Mondays; parity ≤1%; system accuracy 60% → 90%	record	"a dependency slipped"
Re-cut S11	7 → 68 customers = 9.7×; CTS $0.66 → $0.03, about ninety-five percent; a flattering 99.4% week discarded; the ~$1.5M pre-prod figure is separate	record / user anchor (the $1.5M)	"10×", "50×", "$1.00 → $0.03", "95.5%", automation %

METRIC	VALUE TO SAY	BASIS	NEVER
Re-cut S12	about $5M avoided, said once; effort to first value −66%; value month 6 vs 18; 48 of 53 import partners in a month; the import-side app, April 2022	record (resume, Apple B1–B2)	annual reduction; "15% lower overhead"; "savings"; "MVP" for April 2022
Re-cut S18 / S2	S18: 7% → 1.8% → 0.3% vs 0.5%, on time, about three percent under ~$1.8M; about $2.5M a month is my division of the $20M (modeled). S2: +15% processing efficiency (DC Ops' own measurement); gate 98%, week two 94, throttle ~30%; about a dozen impacted systems	record / modeled ($2.5M a month)	a multiplier on the fallback; "a dozen core, 20+ touched"; the $1.6M or $2.2M as record
Which $2.5M / $1.5M	$2.5M = S14 order value (user anchor) · S1 annual impact (Lan's story; user anchor) · S18 per month, my division (modeled). $1.5M = S1 modeled · S8 carrying cost released (Apple) · S11 pre-production (user anchor) · S14 expected corruption loss (modeled)	attach the noun on first mention	a bare "$2.5M" or "$1.5M"
Cross-loop multiples	7 → 68 = 9.7× (S11, HM's story); 170 monthly users and 32,000 shipments are targets	record	"10×" / "50×"; 32,000 as achieved
Cross-loop site count	"six sites" (Elementum)	record	"six NA sites"
Platform configuration	Manhattan DOM — the only platform personally configured	record	"I configured S/4" / BlueYonder / Locus / Snowflake
NACF bridges	19.2 trailers/day → 4.5 → 5 doors; dwell 13.3 h at 85% vs 40 h at 95%; 8,096 units = 81.0%, 1,904 gap; safety stock 522, 441 at LT = 5	booklet worked examples (illustrative inputs)	presenting them as Amazon data

Never-say, one glance before the call.

- Never count the $10M initiative as savings; never claim the >$100K as savings; never call S14's ≈$1.5M expected loss a saving.
- Never "10×" or "50×" (9.7×); never "six NA sites"; never "$1.5M to $2M"; never "$1M routing guide"; never "99% in two months"; never "400+ lanes".
- Never invent a direct-report headcount, a days-to-tie figure, a percentage for S14, or site-wave counts for S10.
- Never a bare "$2.5M" or "$1.5M" — three different $2.5Ms (S14 order value, S1 annual impact, S18 per-month slip) and four different $1.5Ms (S1 modeled, S8 carrying cost released, S11 pre-production, S14 expected loss); attach the noun on first mention.
- S19–S21 are design cases, never lived results — "in the model", with every number prefixed; the lived record is Apple F4.
- Never "we" for an action that was mine; name the team's part once — consultants (S14), infra (S16), data platform team (S10), the QA lead (S15).
- S11/S12, S6/S8, S2/S18, S1/S3 are the other four interviewers' — pointers only; if the bar raiser has read their notes, connect (S16 ↔ S11; S14 ↔ S12 "more recent upward disagreement") without re-telling.

VERIFY BEFORE YOU SAY IT NEW

The addendum rule: **VERIFY-WITH-RAJESH is a last resort** — only a fact that is in none of the source documents, the Apple workbook v11 or either master. Everything else below is a source conflict already resolved by precedence (source documents, then Apple v11, then master) and the conservative figure.

A · FACTS TO CONFIRM — ABSENT FROM EVERY SOURCE (VERIFY-WITH-RAJESH)

#	WHERE	WHAT TO CONFIRM	DEFAULT IF YOU DO NOT
V1	S14 — failure / learning card	Whether you had the standing to propose the daily reconciliation before the incident (the card says "propose it the moment the handshake is in scope").	No source records it. **If unconfirmed:** say only "I built the control after the incident, not before it."
V2	S12 re-cut — who disagreed	Which of your manager's two points you adjusted. The master says "he argued two points, I adjusted one" and does not name them.	No source names them. **If unconfirmed:** use the risk-register line (operators feeling abandoned → the priced two-quarter schedule) and do not attribute it to him.

B · SOURCE CONFLICTS — RESOLVED, KNOW THEM BEFORE THE CALL

ITEM	STORY	RESOLUTION
three $2.5Ms, four $1.5Ms (S14.05)	S14	rule 9: never a bare "$2.5M" or "$1.5M"; attach the noun on first mention — S14 order value, S1 annual impact, S18 per-month slip; S1 modeled, S8 carrying cost released, S11 pre-production, S14 expected loss.
>$100K/yr (S16.01)	S16	priced, not a saving. Safe wording: ">$100K a year — the BRD priced it at $103,680 on thirty hours a week of WBR work at an assumed ninety dollars an hour; my split into hours and decision risk is a model." Never "infra's own loaded rates".
$10,880 / month (S16.02)	S16	BRD prints $10,880 but $129,600 / 12 = $10,800. If a monthly figure is quoted, say $10,800.
~99% within a quarter (S10.02)	S10	Apple v11: "inside the two-month bar"; master: three months. Safe wording: "within a quarter" — true under both. Never "in two months", never a month count.
"25 EU sites / 400+ lanes" (S10.08)	S10	never say 400+; Apple F1/F2 = a thousand lanes, 25 locations (see S9).
~$8M of the day's sales protected (S13.04)	S13	Apple uses $8M as both "at risk" and "protected". Safe wording: "about six million on the line when the page fired; about eight million of the day's sales protected." Never "at risk", never GMV, never a saving.
a thousand lanes asked; 597 first (S9.01)	S9	Apple F1 + resume: a thousand lanes onboarded across 25 European locations, goal hit; master: 597 first, ~200 next, ~400 deferred. Safe wording: "a thousand-lane program; I phased by value — 597 lanes carrying 80% of spend went first, the remainder behind the same gate."
1,000 global lanes → 25 European locations (S9.09)	S9	say "a thousand lanes across twenty-five European locations" when geography is asked.
"400+ lanes" (S10 phrasing) (S9.10)	S9	pick one — Apple: 1,000 lanes / 25 locations. Never "400+".
all users (not "dozens") (own check)	G1	Loop packet: "dozens of users, slipped days". Apple v11: all users, sign-off date passed. Apple v11 used (the vetted record).
"hundreds" of RF devices (S18.09)	G2	sources disagree (~400 vs ~100). Say "hundreds of RF devices" and "a three-to-four-day estimate"; never pair ~100 and ~400 as a ratio; never "400 failed".
170 monthly users; 32,000 shipments (S11.02 / S11.03)	S11	"170 monthly active users — the public target, landed"; 32,000 is a target, never achieved; never "400 to 3X".
about $5M avoided (S12.03)	S12	Apple carries both "annual cost reduction + 15% lower overhead" and "avoidance"; the resume says "$5M cost avoidance in 6 months". Say "about five million of avoided cost" once; never "annual"; never "15% lower overhead".
import-side app live April 2022 (S12.11)	S12	Apple contradicts itself (MVP Q1 2023 in the math box). Say "the import-side app went live April 2022 with eleven partners in ten countries".
about a dozen impacted systems (S2.10)	S2	say "about a dozen impacted systems"; "20+ touched" is unverified. Also say "about thirty build tracks" and "about fifteen months", never "fifteen tracks" or "fourteen months" (S2.11, S2.16).
S19–S21 Walmart-client numbers (S19.01 / S20.01 / S21.01)	bench	design cases built from my own interview prep (war book, WBR deep dive; illustrative data) — never lived results; every number is prefixed "in the model". The lived record for that theme is Apple F4.
S11 automation % and review hours	S11 (re-cut)	The master says automation 40% → 90% and review hours 1.5–2 → ~4; the loop packet says 60% → 97% and 30 → 6. Three different manual-effort measures exist (BRD As-Is 1.5–2 h per snapshot; BRD-priced 30 h/wk; Apple "about four hours a week at its worst"). Resolved (S11.15): none is stacked or quoted in S11; say "by hand every Monday"; the BRD figures appear only inside S16, labeled.
S12 told as Sephora	S12 (re-cut)	The loop packet tells the roadmap flip as a Sephora story; the master record (and the sources) are Flexport FF App. Resolved: Flexport.
S18 budget and slip	S18 (re-cut)	The loop packet says ~$800K and ~$2.5M a week; the master says ~$1.8M and ~$2.5M a month ($54K is 3% of $1.8M; the $2.5M a month is my division of the $20M over eight months, modeled). Resolved: ~$1.8M, per month, said as "about".
cb91 vs repo master	all	The uploaded 131-page master (cb91) and the repo 195-page master agree on every headline number used here (S2, S9–S16, S18). cb91 has no S19–S21; those benches rest on the repo master only.

C · INPUTS I CHOSE — ILLUSTRATIVE OR BENCHMARK, LABELED AS SUCH IN EVERY TABLE

STORY	INPUTS THAT ARE MINE, NOT THE RECORD
S14	~60 days of drift; P(error) 0.3; blast radius 2× the $2.5M of order value; war room 6 people × 24 h × $150/h; 100 discrepancies a month × 80% hand-fixed × $30; $12.5K × 100 × 12 passing through mismatch, 7 days stuck, 10% cost of capital
S16	≈$145/h and 4 h/wk; P(wrong steer) 10%; ≈ a squad-week = $13,500 per wrong steer; 20% key-person attrition × $75K rebuild; example parity values
S10	7.5 h/wk saved × 25 sites × 50 weeks × $55/h; the same 8–10 h/wk burden at every site
S15	Base of 4 criticals a release; $10K per critical (benchmark $8–15K); 15 engineers × 18 days ÷ 3 days per story; 4 QA × 18 days ÷ 1 day per story; 6 person-days × $800
S13	Capacities 10,000 / 8,000 / 6,000 orders a day; backlog 30,000 split 22,000 / 6,000 / 2,000; AOV $150; margin 35–45%; CS contact $5–8; μ 1,200 and σ 150 allocations an hour; the $400M retailer
S9	200 shipments × 4 milestones = 800 expected, 480 received; 597 × ~50 shipments × 3 documents; 6 minutes a document at $45/h; the 60K / 90K / 120K document sensitivity

One rule for all of section C. Say the input and its label in the same breath — "on an illustrative base of four a release", "at a benchmark ten thousand dollars a critical". If he asks where it came from: "It is my assumption; here is how I would replace it with the record."

SOURCE TRACE — every quantitative claim, with file and page NEW

Precedence: (1) source documents, (2) Apple story workbook v11, (3) Amazon master canon (repo 195 pp and the uploaded cb91 131 pp, both treated as canon). Never the larger figure when sources differ; unresolved items use the conservative figure and the recon's safe wording. Status vocabulary is the numbers-reconciliation (NUMBERS-RECON.md): **MATCH** = Apple v11 or a source document carries the figure; **ROUNDING** = same figure, rounded; **CONFLICT** = sources differ, safe wording in the last column; **UNSUPPORTED-anchor** = only my earlier packet (GFiber v8 / master) carries it — said as a user anchor; **UNSUPPORTED-modeled** = derived arithmetic — said as modeled or illustrative. Pages are PDF page numbers of the named file.

Status count. Recon registry (all 21 stories): **202 claims** — MATCH 85 · ROUNDING 8 · CONFLICT 32 · UNSUPPORTED-anchor 32 · UNSUPPORTED-modeled 45. This packet's trace: **94 rows** — MATCH 49 · ROUNDING 3 · CONFLICT 16 · UNSUPPORTED-anchor 14 · UNSUPPORTED-modeled 12. Basis I say aloud: MATCH → *record* (or *record-from-docs* where a source document carries it) · ROUNDING → record, said rounded · CONFLICT → the recon's safe wording, in the column on the right · UNSUPPORTED-anchor → *user anchor* (my own earlier packet only) · UNSUPPORTED-modeled → *modeled / illustrative*. Recon ID = the row in NUMBERS-RECON.md; "own check" = verified against the cited pages here, not in the recon registry.

NUMBER	STORY	CLAIM	SOURCE FILE · PAGE	RECON STATUS	RECON ID	BASIS I SAY · SAFE WORDING
~200 orders / $2.5M of order value	S14	OMS↔WMS discrepancy before the review	master (repo, 195 pp) p185 master cb91 (131 pp) p77	UNSUPPORTED-anchor	S14.01	user anchor — master / GFiber v8 only; no Apple chapter, no source doc. Say "about two hundred orders and two-point-five million of order value"; never as savings.
$10M initiative	S14	initiative riding on the review — won, never savings	master (repo, 195 pp) p185 master cb91 (131 pp) p77	UNSUPPORTED-anchor	S14.02	user anchor. Say "the ten-million initiative was won at the review — a fact about the relationship, not a saving."
two days	S14	clock to the executive review	master (repo, 195 pp) p185 master cb91 (131 pp) p77	UNSUPPORTED-anchor	own check	user anchor — master only (not in the recon registry; no Apple or source-doc text).
≈$1.5M	S14	expected corruption loss from a silent script (0.3 × 2 × $2.5M of order value)	master (repo, 195 pp) p185 master cb91 (131 pp) p79	UNSUPPORTED-modeled	S14.03	modeled / illustrative — "avoided risk, not money"; never a saving.
≈$0.1M/yr ($55–160K)	S14	standing reconciliation	master (repo, 195 pp) p186 master cb91 (131 pp) p79	UNSUPPORTED-modeled	S14.04	modeled (labor $29K + carry $30K + war room $43K = $102K).
three $2.5Ms, four $1.5Ms	S14	dollar nouns that collide across stories	— (see note)	CONFLICT	S14.05	rule 9: never a bare "$2.5M" or "$1.5M"; attach the noun on first mention — S14 order value, S1 annual impact, S18 per-month slip; S1 modeled, S8 carrying cost released, S11 pre-production, S14 expected loss.
60% → 90%	S16	WBR system accuracy	master (repo, 195 pp) p77 master cb91 (131 pp) p86 Apple v11 p11	MATCH	S16.04	record; say "system accuracy" and tell it in S16 only (S11.09).
1.5–2 hours	S16	manual snapshot effort (As-Is, per snapshot)	master (repo, 195 pp) p77 master cb91 (131 pp) p86 BRD_-_SV_Zendesk_Data_in_Snowflake p2	MATCH	S16.09	record-from-docs (BRD As-Is). A different measure from the BRD's 30 hrs/wk — never stacked with it.
seven metrics	S16	frozen definitions	BRD_-_SV_Zendesk_Data_in_Snowflake p3 master (repo, 195 pp) p77 master cb91 (131 pp) p86	MATCH	S16.08	record-from-docs (BRD lists metrics 1–7).
within 1%	S16	parity gate manual vs pipeline	Apple v11 p10 master (repo, 195 pp) p77 master cb91 (131 pp) p86	MATCH	S16.05	record.
~4% ticket-sync drop	S16	first automated catch	master (repo, 195 pp) p77 master cb91 (131 pp) p86	MATCH	S16.06	record — "about four percent".
>$100K/yr	S16	priced status quo — three derivations exist	BRD_-_SV_Zendesk_Data_in_Snowflake p1 master (repo, 195 pp) p77 master cb91 (131 pp) p86	CONFLICT	S16.01	priced, not a saving. Safe wording: ">$100K a year — the BRD priced it at $103,680 on thirty hours a week of WBR work at an assumed ninety dollars an hour; my split into hours and decision risk is a model." Never "infra's own loaded rates".
≈$30K + ≈$70K	S16	split of the >$100K (hours + decision risk)	master (repo, 195 pp) p78 master cb91 (131 pp) p88	UNSUPPORTED-modeled	S16.11	modeled — the ledger's own total row (≈$90K) differs from 30 + 70; say "about a hundred thousand" and "my model" for any split.
2 weeks	S16	security-review miss	Apple v11 p11 master (repo, 195 pp) p77 master cb91 (131 pp) p86	MATCH	S16.07	record.
3 × 5 h × 2 WBRs at $90/h	S16	BRD labor case inputs (rate is the BRD's assumption)	BRD_-_SV_Zendesk_Data_in_Snowflake p1	MATCH	S16.01	record-from-docs; the $90/h is an assumed rate in the BRD, not a measured one.
$129,600/yr	S16	BRD labor cost per year (as-is)	BRD_-_SV_Zendesk_Data_in_Snowflake p2	MATCH	S16.01	record-from-docs.
$25,920/yr after	S16	BRD labor after (1 h/wk)	BRD_-_SV_Zendesk_Data_in_Snowflake p2	MATCH	S16.01	record-from-docs.
$103,680	S16	BRD saving case: 30 → 6 hrs/wk	BRD_-_SV_Zendesk_Data_in_Snowflake p2	MATCH	S16.01	priced (BRD arithmetic recomputed: $129,600 − $25,920); the number to say for ">$100K"; priced, never claimed as savings.
$10,880 / month	S16	BRD monthly cost typo	BRD_-_SV_Zendesk_Data_in_Snowflake p2	CONFLICT	S16.02	BRD prints $10,880 but $129,600 / 12 = $10,800. If a monthly figure is quoted, say $10,800.
bus factor 1 → ≥2	S16	single point of failure removed	master (repo, 195 pp) p77 master cb91 (131 pp) p86	UNSUPPORTED-anchor	S16.10	user anchor for any count after "one" — Apple and the WBR doc say only "single point of failure"; say "no longer a single point of failure".
no bake-off; scope trade; 4 systems	S16	options and trade-offs paragraph	Apple v11 p11 Apple v11 p10	MATCH	S16.03	record (Apple v11 A3; WBR doc "4 systems").
40% reported → ~55% true	S10	data quality at the measured grain	master (repo, 195 pp) p114 master cb91 (131 pp) p56 Apple v11 p35	MATCH	S10.01	record (Apple F2).
~99% within a quarter	S10	time to data quality 99%	Apple v11 p36 master (repo, 195 pp) p114 master cb91 (131 pp) p56	CONFLICT	S10.02	Apple v11: "inside the two-month bar"; master: three months. Safe wording: "within a quarter" — true under both. Never "in two months", never a month count.
8–10 hrs/week manual reconciliation	S10	before state	Apple v11 p35	MATCH	S10.03	record (Apple F2).
"1–2 hrs/wk" after	S10	after state	master (repo, 195 pp) p114 master cb91 (131 pp) p56	UNSUPPORTED-anchor	S10.04	user anchor — say "from eight-to-ten hours to a small fraction".
$500K/yr Austria	S10	Austria pilot, executed-only	master (repo, 195 pp) p114 master cb91 (131 pp) p56 Apple v11 p36	MATCH	S10.05	record (Apple F2); method co-owned with Finance before the first claim.
≈97% agreement	S10	auto vs manual sample	master (repo, 195 pp) p118 master cb91 (131 pp) p59	MATCH	S10.05	record (Apple F2).

NUMBER	STORY	CLAIM	SOURCE FILE · PAGE	RECON STATUS	RECON ID	BASIS I SAY · SAFE WORDING
$5M base × 10%	S10	arithmetic behind the Austria $500K	master (repo, 195 pp) p114 master cb91 (131 pp) p56	UNSUPPORTED-anchor	S10.06	user anchor — say "about five hundred thousand a year at Austria"; the base and the 10% are mine.
~$8M across 25 locations	S10	roll-up, executed shipments only	master (repo, 195 pp) p114 master cb91 (131 pp) p56 Apple v11 p35	MATCH	S10.07	record (Apple F2). "About eight million, executed-only, auditable per shipment." Never "$1M"; the resume's "$27M+" is a different aggregate.
25 locations	S10	sites in the roll-out	Apple v11 p35 master (repo, 195 pp) p114 master cb91 (131 pp) p56	MATCH	S10.07	record.
"25 EU sites / 400+ lanes"	S10	lane count	master (repo, 195 pp) p114 master cb91 (131 pp) p56	CONFLICT	S10.08	never say 400+; Apple F1/F2 = a thousand lanes, 25 locations (see S9).
~8% of lanes need Incoterm	S10	lane-key design	master (repo, 195 pp) p114 master cb91 (131 pp) p56	UNSUPPORTED-anchor	S10.09	user anchor.
≈$0.5M/yr analyst capacity	S10	time saved across sites (7.5 h/wk × 25 × 50 × $55)	master (repo, 195 pp) p114 master cb91 (131 pp) p56	UNSUPPORTED-modeled	S10.10	modeled; range $0.2–0.7M.
20-person team	S15	Product/Eng/QA team converted	master (repo, 195 pp) p56 master cb91 (131 pp) p82 Apple v11 p3	MATCH	S15.01	record (Apple D3, resume).
4-week cadence in 6 weeks	S15	conversion	master (repo, 195 pp) p56 master cb91 (131 pp) p82	MATCH	S15.02	record (Apple D3, resume).
critical defects per release −50%	S15	primary metric	master (repo, 195 pp) p56 master cb91 (131 pp) p82	MATCH	S15.03	record via the resume; Apple D3 omits it. Said without "+50% releases".
releases 8.7 → 13 a year	S15	cadence arithmetic (52 ÷ 6, 52 ÷ 4)	master (repo, 195 pp) p57 master cb91 (131 pp) p84	UNSUPPORTED-modeled	S15.04	modeled from the cadence — "not a measured count"; "50% more releases" is the same arithmetic.
≈$0.13M/yr	S15	dollars (escaped criticals ≈$90K + late-land rework $41,760)	master (repo, 195 pp) p56 master cb91 (131 pp) p82	UNSUPPORTED-modeled	S15.05	modeled on illustrative defect counts; no headcount.
99% fill vs 95% goal	S13	Cyber Monday fill rate	master (repo, 195 pp) p179 master cb91 (131 pp) p72 Apple v11 p40	MATCH	S13.02	record (Apple M1, resume).
~$6M on the line	S13	exposure sized at 4 AM	Apple v11 p40 master (repo, 195 pp) p179 master cb91 (131 pp) p72	MATCH	S13.03	record (Apple M1) — "on the line when the page fired".
~$8M of the day's sales protected	S13	peak sales protected — one meaning	master (repo, 195 pp) p179 master cb91 (131 pp) p72 Apple v11 p40	CONFLICT	S13.04	Apple uses $8M as both "at risk" and "protected". Safe wording: "about six million on the line when the page fired; about eight million of the day's sales protected." Never "at risk", never GMV, never a saving.
3 DCs / 100 stores	S13	scale	Apple v11 p40 master (repo, 195 pp) p179 master cb91 (131 pp) p72	MATCH	S13.01	record (Apple M1, resume).
4 AM / every two hours until 9 PM	S13	incident cadence	Apple v11 p40 master (repo, 195 pp) p179 master cb91 (131 pp) p72	MATCH	own check	record (Apple M1).
near ~0.85 d vs far ~4.60 d	S13	node time	master (repo, 195 pp) p179 master cb91 (131 pp) p72	UNSUPPORTED-anchor	S13.08	user anchor — GFiber v8 only; low-use.
structural stockouts −20%	S13	engagement result	master (repo, 195 pp) p179 master cb91 (131 pp) p72	MATCH	S13.05	record via the resume (Apple M1 omits it).
3σ page rule	S13	post-mortem mechanism	Apple v11 p41 master (repo, 195 pp) p179 master cb91 (131 pp) p72	MATCH	own check	record (Apple M1).
$3.2M margin; $400M-retailer cases	S13	modeled dollars	master (repo, 195 pp) p179 master cb91 (131 pp) p72	UNSUPPORTED-modeled	S13.07	modeled — "$8M / $6M / 99% are the record; the rest is my model".
a thousand lanes asked; 597 first	S9	the ask and the phasing	Apple v11 p34 master (repo, 195 pp) p99	CONFLICT	S9.01	Apple F1 + resume: a thousand lanes onboarded across 25 European locations, goal hit; master: 597 first, ~200 next, ~400 deferred. Safe wording: "a thousand-lane program; I phased by value — 597 lanes carrying 80% of spend went first, the remainder behind the same gate."
597 lanes = 80% of spend	S9	Pareto cut	master (repo, 195 pp) p99 master cb91 (131 pp) p51	UNSUPPORTED-anchor	S9.02	user anchor (GFiber v8 only; the spine of the story).
~200 next / ~400 deferred ≈ 20%	S9	later waves — not said	master (repo, 195 pp) p100 master cb91 (131 pp) p52	UNSUPPORTED-anchor	S9.03	user anchor — say "the remainder behind the same gate"; never "400+ lanes".
60% → 95%	S9	completeness	master (repo, 195 pp) p99 master cb91 (131 pp) p51 Apple v11 p34	MATCH	S9.04	record (Apple F1, resume) — only in this story.
≥95% valid-ASN gate	S9	cohort gate	master (repo, 195 pp) p99 master cb91 (131 pp) p51 Apple v11 p34	MATCH	S9.05	record (resume).
cohort held twice	S9	timestamp-format failures	master (repo, 195 pp) p99 master cb91 (131 pp) p51	UNSUPPORTED-anchor	S9.08	user anchor — the held-gate beat; "my program tracking".
freight −10% ≈ $2M in six months	S9	measured result on the onboarded lanes	master (repo, 195 pp) p99 master cb91 (131 pp) p51	UNSUPPORTED-anchor	S9.06	user anchor, as measured; never annualized; not S6's $2M safety stock.
adoption 8×	S9	carrier adoption	master (repo, 195 pp) p99 master cb91 (131 pp) p51	UNSUPPORTED-anchor	S9.07	user anchor — no Apple support.
1,000 global lanes → 25 European locations	S9	geography of the program	Apple v11 p34	CONFLICT	S9.09	say "a thousand lanes across twenty-five European locations" when geography is asked.
"400+ lanes" (S10 phrasing)	S9	S9 vs S10 lane counts	master (repo, 195 pp) p114 master cb91 (131 pp) p56	CONFLICT	S9.10	pick one — Apple: 1,000 lanes / 25 locations. Never "400+".
$0.8M + $0.6M + $0.6M = $2.0M	S9	decomposition of the $2M; ≈$0.4M/yr document labor	master (repo, 195 pp) p102 master cb91 (131 pp) p53	UNSUPPORTED-modeled	S9.11	modeled — "my split decomposes the record, never adds"; the $20M implied base is not volunteered (S9.12).
5 users × 10 scenarios; 2 days from sign-off	G1	UAT before the crash	Apple v11 p43 Apple v11 p44	MATCH	own check	record (Apple v11 G1).
500 containers / 20 min / 99.6%	G1	performance proof that stuck	Apple v11 p43 Technology_Program_Management_Status_Week_of_November_19th_2021 p10 master (repo, 195 pp) p67	MATCH	S11.18	record (Apple A2, TPM status 11/19/21 p10).
all users (not "dozens")	G1	who the sponsor opened access to	Apple v11 p43	CONFLICT	own check	Loop packet: "dozens of users, slipped days". Apple v11: all users, sign-off date passed. Apple v11 used (the vetted record).
1.5 weeks	G2	training sign-off slip	Apple v11 p44 master (repo, 195 pp) p26 master cb91 (131 pp) p96	MATCH	S18.10	record.
3–4 days estimate	G2	device task estimate	Apple v11 p44	MATCH	S18.09	record (one source doc says 1–2 days).

NUMBER	STORY	CLAIM	SOURCE FILE · PAGE	RECON STATUS	RECON ID	BASIS I SAY · SAFE WORDING
"hundreds" of RF devices	G2	RF device count (~100 of ~400 in the master)	Apple v11 p45 master (repo, 195 pp) p26 master cb91 (131 pp) p96	CONFLICT	S18.09	sources disagree (~400 vs ~100). Say "hundreds of RF devices" and "a three-to-four-day estimate"; never pair ~100 and ~400 as a ratio; never "400 failed".
7% → 1.8% → 0.3% vs 0.5%	G2 / S18	mock cutover variance vs tolerance	Apple v11 p44 master (repo, 195 pp) p26 master cb91 (131 pp) p96	MATCH	S18.03	record (Apple D2, resume); tolerance published with Finance before mock 1.
~3% under ~$1.8M	G2 / S18	migration budget	master (repo, 195 pp) p26 master cb91 (131 pp) p96 Apple v11 p44	ROUNDING	S18.02	say "about three percent under a $1.8M budget — roughly $54K". The loop packet's ~$800K / ~$2.5M a week is the conflict (S18.11); the master and Apple agree on $1.8M.
≈$45K paper-pick fallback	S18	modeled fallback cost	master (repo, 195 pp) p27 master cb91 (131 pp) p98	UNSUPPORTED-modeled	S18.12	modeled — "forty-five thousand dollars of fallback to protect the date"; never a multiplier ("~20×", "~55×").
~$2.5M a month of slip	S18	division of the $20M pipeline over eight months	master (repo, 195 pp) p26 master cb91 (131 pp) p96	UNSUPPORTED-modeled	S18.05	modeled — "about $2.5M a month on average — my division of the $20M over the eight months; not banked". The loop packet's "$2.5M a week" is wrong.
2 weeks (security review)	G3	under-planned	Apple v11 p45	MATCH	S16.07	record.
zero skipped review weeks	G3	parallel run absorbed the miss	Apple v11 p46	MATCH	own check	record (Apple G3).
7 → 68 customers = 9.7×	S11	multiple	master (repo, 195 pp) p63 master cb91 (131 pp) p62	ROUNDING	S11.01	record; "seven to sixty-eight — nine-point-seven times". The WBR doc's "MAC to 50X" is never spoken.
170 monthly users; 32,000 shipments	S11	public targets	master (repo, 195 pp) p63 master cb91 (131 pp) p62	CONFLICT	S11.02 / S11.03	"170 monthly active users — the public target, landed"; 32,000 is a target, never achieved; never "400 to 3X".
$0.66 → $0.03	S11	cost-to-serve per container	master (repo, 195 pp) p63 master cb91 (131 pp) p62	MATCH	S11.05	record (Apple A1/A2; WBR doc "66 cents to 3 cents").
about 95% (not 95.5%)	S11	CTS reduction: 1 − 0.03 / 0.66 = 95.45%	Apple v11 p7	ROUNDING	S11.06	say "about ninety-five percent"; never "95.5%" and never "$1.00 → $0.03" (S11.07: "about a dollar in the hand-carried pilot weeks" only if asked).
99.4% week discarded	S11	honesty example	master (repo, 195 pp) p63 master cb91 (131 pp) p62	MATCH	S11.16	record (Apple A2).
nine teams / ~100 people	S11	program scope	master (repo, 195 pp) p63 master cb91 (131 pp) p62	MATCH	S11.23	record-from-docs (Path to 4PL).
~$1.5M pre-production; 1.5 vs 6 months	S11	pre-production estimate — separate from CTS	master (repo, 195 pp) p63 master cb91 (131 pp) p62	UNSUPPORTED-anchor	S11.22	user anchor — "my estimate of a month and a half versus six"; prefer 500 containers / 20 min / 99.6% and 86% tracked.
≈$50K/yr direct CTS	S11	modeled	master (repo, 195 pp) p64 master cb91 (131 pp) p64	UNSUPPORTED-modeled	S11.25	modeled on illustrative container counts.
about $5M avoided	S12	cost avoidance, said once	master (repo, 195 pp) p70 master cb91 (131 pp) p67	CONFLICT	S12.03	Apple carries both "annual cost reduction + 15% lower overhead" and "avoidance"; the resume says "$5M cost avoidance in 6 months". Say "about five million of avoided cost" once; never "annual"; never "15% lower overhead".
import-side app live April 2022	S12	launch timing	Apple v11 p16	CONFLICT	S12.11	Apple contradicts itself (MVP Q1 2023 in the math box). Say "the import-side app went live April 2022 with eleven partners in ten countries".
−66% effort; month 6 vs 18	S12	time-to-value	master (repo, 195 pp) p71 master cb91 (131 pp) p67	MATCH	S12.01 / S12.02	record (Apple B2, resume).
339.5 min; 2,200 interactions	S12	baselines	master (repo, 195 pp) p70 master cb91 (131 pp) p67	MATCH	S12.06	record (Apple B2).
48 of 53 import partners	S12	migrated within a month	master (repo, 195 pp) p70 master cb91 (131 pp) p67	MATCH	S12.12	record — always "import partners".
burn-up 55.8% → 83.2%	S12	two quarters	master (repo, 195 pp) p70 master cb91 (131 pp) p67	MATCH	S12.10	record (html master only).
+15% processing efficiency	S2	DC Ops' own measurement	master (repo, 195 pp) p18 master cb91 (131 pp) p12	MATCH	S2.02	record — "DC Ops' own measurement".
gate 98%; week two 94; ~30% throttle	S2	robotics gate	master (repo, 195 pp) p18 master cb91 (131 pp) p12	MATCH	S2.07	record (Apple C3).
about a dozen impacted systems	S2	scope	master (repo, 195 pp) p18 master cb91 (131 pp) p12	CONFLICT	S2.10	say "about a dozen impacted systems"; "20+ touched" is unverified. Also say "about thirty build tracks" and "about fifteen months", never "fifteen tracks" or "fourteen months" (S2.11, S2.16).
≈$1.6M/yr modeled	S2	sum of the model rows is $1.65M	master (repo, 195 pp) p18 master cb91 (131 pp) p12	UNSUPPORTED-modeled	S2.17	modeled on an illustrative 100K-units-a-week DC; $1.65M printed as "about $1.6M" (the conservative reading).
0.5% tolerance	S18	published with Finance before mock 1	master (repo, 195 pp) p26 master cb91 (131 pp) p96	MATCH	S18.03	record.
$20M pipeline of four projects	S18	waiting on the migration	master (repo, 195 pp) p26 master cb91 (131 pp) p96	MATCH	S18.04	record (Apple D1, resume).
S19–S21 Walmart-client numbers	bench	delivery-promise buffers, WBR, SSS adoption	— (see note)	CONFLICT	S19.01 / S20.01 / S21.01	design cases built from my own interview prep (war book, WBR deep dive; illustrative data) — never lived results; every number is prefixed "in the model". The lived record for that theme is Apple F4.
19.2 trailers/day → 4.5 → 5 doors	bridge	dock plan	NACF booklet p7	MATCH	own check	booklet worked example (illustrative inputs).
13.3 h at 85% vs 40 h at 95%	bridge	dwell vs utilization	NACF booklet p8 NACF booklet p8	MATCH	own check	booklet worked example.
8,096 units; 81.0%; 1,904 gap	bridge	plan vs receipt	NACF booklet p13 NACF booklet p13	MATCH	own check	booklet worked example.

SOURCE TRACE (continued) — the arithmetic register NEW

Every computed figure in the six primaries, the re-cuts and the failure arcs, printed with its operands substituted. The build script recomputes each one and asserts the printed text matches. Inputs not labeled otherwise in the story's own table are record; inputs the story labels illustrative or benchmark are listed on the VERIFY page, section C.

STORY	FIGURE	PRINTED ARITHMETIC (EVERY OPERAND SUBSTITUTED; RECOMPUTED BY THE BUILD SCRIPT — INPUTS ARE RECORD UNLESS THE STORY'S OWN TABLE LABELS THEM USER ANCHOR, ILLUSTRATIVE OR MODELED)
S14	avg per order	$2,500,000 / 200 = $12,500
S14	per day	$2,500,000 / 60 = $41,667
S14	blast	2 × $2,500,000 = $5,000,000
S14	E[loss]	0.3 × $5,000,000 = $1,500,000
S14	war room	6 × 24 × $150 = $21,600
S14	P*	$21,600 / $5,000,000 = 0.43%
S14	one day	$41,667
S14	avoided	2 × $2,500,000 × 0.3 = $1,500,000
S14	manual	100 × 12 × 80% × $30 = $28,800
S14	carry	100 × $12,500 × 12 × 7 / 365 × 10% = $28,767
S14	war rooms	2 × $21,600 = $43,200
S14	total	$28,800 + $28,767 + $43,200 = $100,767
S14	lo	0.1 × $5,000,000 = $500,000
S14	base	0.3 × $5,000,000 = $1,500,000
S14	hi	0.5 × $5,000,000 = $2,500,000
S16	wrong weeks	5 × 40% = 2
S16	hours	4 × 52 = 208
S16	hours $	208 × $145 = $30,160
S16	wrong steers	52 × 10% = 5.2
S16	risk $	5.2 × $13,500 = $70,200
S16	priced	$30,160 + $70,200 = $100,360
S16	brd h/wk	3 × 5 × 2 = 30
S16	brd before	30 × $90 × 48 = $129,600
S16	brd after	6 × $90 × 48 = $25,920
S16	brd saving	$129,600 − $25,920 = $103,680
S16	pass	(71.2 − 70.9) / 71.2 = 0.42%
S16	inv	(461 − 456) / 456 = 1.1%
S16	silence	1 − 96% = 4%
S16	hours	4 × 52 × $145 = $30,160
S16	risk	52 × 10% × $13,500 = $70,200
S16	priced2	$30,160 + $70,200 = $100,360
S16	share	(40% − 10%) / 40% = 75%
S16	retired	$70,200 × 75% = $52,650
S16	bus	20% × $75,000 = $15,000
S16	canon total	$27,000 + $47,000 + $17,000 = $91,000
S16	be P	($100,000 − $30,160) / (52 × $13,500) = 9.9%
S16	s lo	52 × 5% × $13,500 = $35,100
S16	s base	52 × 10% × $13,500 = $70,200
S16	s hi	52 × 15% × $13,500 = $105,300
S16	p lo	$30,160 + $35,100 = $65,260
S16	p base	$30,160 + $70,200 = $100,360
S16	p hi	$30,160 + $105,300 = $135,460
S16	t1a	4 × 52 × $145 = $30,160
S16	t1b	52 × 10% × $13,500 = $70,200
S16	t1c	$30,160 + $70,200 = $100,360
S16	t1d	30 × $90 × 48 − 6 × $90 × 48 = $103,680
S16	t3	2 × 4 = 8
S10	unscorable	100% − 55% = 45%
S10	mid h/wk	(8 + 10) / 2 = 9
S10	total h/wk	9 × 25 = 225
S10	recon cut	83%
S10	austria	$5,000,000 × 10% = $500,000
S10	avg site	$8,000,000 / 25 = $320,000
S10	implied base	$8,000,000 / 10% = $80,000,000
S10	austria ratio	$500,000 / $320,000 = 1.56
S10	analyst	(9 − 1.5) × 25 × 50 × $55 = $515,625
S10	sens lo	6 × 15 × 50 × $50 = $225,000
S10	sens base	7.5 × 25 × 50 × $55 = $515,625
S10	sens hi	8.5 × 25 × 50 × $65 = $690,625
S10	at 5%	$80,000,000 × 5% = $4,000,000
S10	trap 45	100% − 55% = 45%
S15	rel before	52 / 6 = 8.7
S15	crit before	4 × 8.7 = 34.8
S15	eng stories	15 × 18 / 3 = 90
S15	qa stories	4 × 18 / 1 = 72
S15	rework per rel	6 × $800 = $4,800
S15	crit $ before	35 × $10,000 = $350,000

STORY	FIGURE	PRINTED ARITHMETIC (EVERY OPERAND SUBSTITUTED; RECOMPUTED BY THE BUILD SCRIPT — INPUTS ARE RECORD UNLESS THE STORY'S OWN TABLE LABELS THEM USER ANCHOR, ILLUSTRATIVE OR MODELED)
S15	rel after	52 / 4 = 13
S15	crit/rel after	4 × 50% = 2
S15	crit after	2 × 13 = 26
S15	fewer	35 − 26 = 9
S15	cost after	26 × $10,000 = $260,000
S15	avoided	$350,000 − $260,000 = $90,000
S15	avoided9	9 × $10,000 = $90,000
S15	rework yr	$4,800 × 8.7 = $41,760
S15	total	$90,000 + $40,000 = $130,000
S15	sens lo	9 × $8,000 = $72,000
S15	sens base	9 × $10,000 = $90,000
S15	sens hi	9 × $15,000 = $135,000
S15	trap 130a	9 × $10,000 = $90,000
S15	trap 130b	$4,800 × 8.7 = $41,760
S15	trap base2	2 × 8.7 − 1 × 13 = 4.4
S13	backlog	22,000 + 6,000 + 2,000 = 30,000
S13	dc1 naive	22,000 / 10,000 = 2.20
S13	dc2 naive	6,000 / 8,000 = 0.75
S13	dc3 naive	2,000 / 6,000 = 0.33
S13	node gain	4.60 − 0.85 = 3.75
S13	lv1	12,500 / 10,000 = 1.25
S13	lv2	10,000 / 8,000 = 1.25
S13	lv3	7,500 / 6,000 = 1.25
S13	3 sigma	1,200 − 3 × 150 = 750
S13	margin	$8,000,000 × 40% = $3,200,000
S13	orders	$6,000,000 / $150 = 40,000
S13	cs	40,000 × $6.50 = $260,000
S13	repointed	40,000 × 40% = 16,000
S13	exp	16,000 × $8 = $128,000
S13	lost sales	$400,000,000 × 4% = $16,000,000
S13	recovered	$3,200,000
S13	sso margin	$1,280,000
S13	m lo	$8,000,000 × 35% = $2,800,000
S13	m base	$8,000,000 × 40% = $3,200,000
S13	m hi	$8,000,000 × 45% = $3,600,000
S13	cs lo	40,000 × $5 = $200,000
S13	cs base	40,000 × $6.50 = $260,000
S13	cs hi	40,000 × $8 = $320,000
S13	sso lo	$400,000,000 × 2% × 15% × 35% = $420,000
S13	sso base	$400,000,000 × 4% × 20% × 40% = $1,280,000
S13	sso hi	$400,000,000 × 6% × 25% × 45% = $2,700,000
S9	completeness before	480 / 800 = 60%
S9	remainder lanes	1,000 − 597 = 403
S9	remainder spend	100% − 80% = 20%
S9	docs	597 × 50 × 3 = 89,550
S9	completeness after	760 / 800 = 95%
S9	base	$2,000,000 / 10% = $20,000,000
S9	record	$20,000,000 × 10% = $2,000,000
S9	split	$800,000 + $600,000 + $600,000 = $2,000,000
S9	hours	90,000 × 6 / 60 = 9,000
S9	labor	9,000 × $45 = $405,000
S9	docs lo	20,000 × 3 = 60,000
S9	docs base	30,000 × 3 = 90,000
S9	docs hi	40,000 × 3 = 120,000
S9	lab lo	60,000 × 4 / 60 × $40 = $160,000
S9	lab base	90,000 × 6 / 60 × $45 = $405,000
S9	lab hi	120,000 × 8 / 60 × $55 = $880,000
S9	alt base	$2,000,000 / 5% = $40,000,000
S9	alt base 10	$2,000,000 / 10% = $20,000,000
S9	alt base 15	$2,000,000 / 15% = $13,333,333
RC	S11 multiple	68 / 7 = 9.7
RC	S11 CTS	($0.66 − $0.03) / $0.66 = 95.45%
RC	S12 months earlier	18 − 6 = 12
RC	S18 under budget	$1,800,000 × 3% = $54,000
G	G1 user-scenarios	5 × 10 = 50
G	G2 under budget	$1,800,000 × 3% = $54,000

INTERVIEW-PREP BOOKLET · NACF INBOUND FLUENCY · BUSINESS-FIRST, SYSTEMS AS ENABLERS

NACF Inbound — how capacity and inventory are managed

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6) · interview loop October 29–30, 2026

The panel this booklet serves

1 · Snigdha Tripathy — Sr Manager, Supply Chain — NACF · **HIRING MANAGER** · ex-EU IXD / placement leader ($80MM savings 2021; built the IXD team ground-up, +$50MM 2020; 50+ FC launches Amazon.in; Kaizen/Lean trainer; 11+ yrs Amazon)

2 · Ryan Hoerger — Manager, Placement / Transfer Analytics · ex-Sr BIE · Chicago Booth MBA, Siebel Scholar · Nike analytics · statistical rigor

3 · Rocky Goforth — Sr Manager, Infrastructure

4 · Zhaowen Lan — Sr Manager, Supply Chain — peer

5 · Venkata Josyula — Principal PM, FBA · bar raiser

> **Strategy in one line.** Sound like someone who has run inbound — not someone who memorized a glossary. Every mechanism below is spoken as a business process with a constraint, a metric and a lever; every number is worked end-to-end so the arithmetic can be done out loud in the room. Inbound in one breath: *the right inventory, reserved into the right node, received and stowed inside capacity, made available where the demand is — with the plans traveling ahead of the freight and the variance closing the loop.*

Honesty labels, on every mechanism: **[well-known public Amazon practice]** — published, talkable-in-public Amazon mechanisms (AWD, IXD, appointment scheduling, random stow, IPI components, WBR) · **[industry standard]** — textbook supply chain (UPH labor planning, z·σ·√LT safety stock, S&OP, queueing approximations) · **[illustrative]** — numbers invented to make the math legible. No confidential Amazon internals are asserted anywhere; no number is claimed as Amazon's own.

How to use. §1–§4 are the fluency: read the diagrams until you can redraw them, then recompute every green math block until the substitutions feel like your own. §5 maps your master-packet stories onto inbound concepts. §6 is per-interviewer: ten questions each of the five will likely ask, with a one-to-two sentence answer pointer, then three questions to ask them. §7 is the math appendix — every formula symbolic → substituted → computed, for the night before. §8 is one week at one FC — the same 12,000 units walked from the forecast through the close, for the whiteboard.

Canon: amazon/amazon-l6-scm-master-packet.html (stories S2, S6, S8, S9, S11, S12, S13, S18, S19 with their basis labels) · panel packets 1–5 (per-interviewer lenses). Sibling booklet file; the five panel packets are untouched by this document.

The 60-second answer — "what is NACF inbound?"

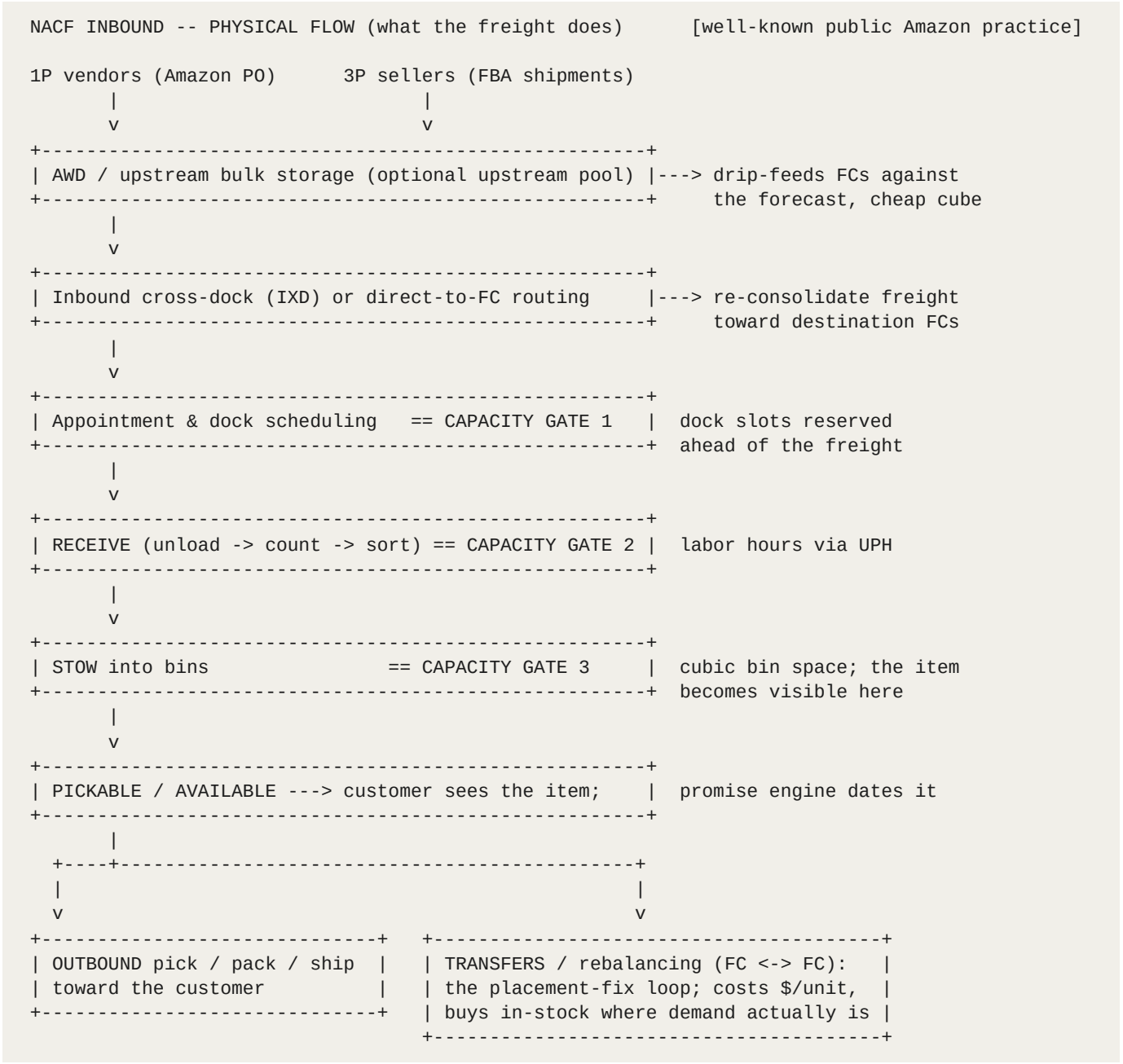
> **SAY THIS VERSION OUT LOUD, THEN STOP**
>
> "Inbound is everything from the vendor or seller's dock to the moment an item is pickable in front of a customer promise — and it is managed as two coupled problems. **Capacity**: three physical currencies — dock-appointment slots, labor hours, and cubic storage — planned from the demand forecast through a receive plan and a UPH-based labor plan, with peak buffers sized off the forecast distribution, not the average. **Inventory**: where the units land and how long they last — placement by distance-to-customer against cost, weeks of cover by SKU, safety stock positioned where the variability actually is, and a transfer loop that rebalances what placement got wrong. The information flow travels ahead of the freight: forecast → inbound plan → placement decision → capacity reservation, and the actuals come back through the WBR as variance. My job in a seat like this is to keep that loop honest — measured lead times, not brochure ones; buffers released on evidence; capacity added against the constraint that actually binds."

Everything after this page unpacks that paragraph. The discipline that makes it credible: every mechanism carries its label, and every quantity in the worked examples can be recomputed from the row above it — do the arithmetic out loud in the interview and the fluency reads as experience.

> **The one mental model to keep on top.** Inbound is a *reservation system* feeding a *positioning system*: you reserve capacity (slots, hours, cube) against a plan, and you position inventory (placement, cover, buffers) against demand. Most inbound failures are one system lying to the other — freight arrives that capacity never reserved, or inventory sits where demand isn't. The metrics in §2–§3 exist to catch exactly those two lies early.

1 · The inbound machine, end-to-end

The physical flow first — what the freight does. Each node is labeled with the constraint that binds it; the three capacity gates are the whole of §2. **[well-known public Amazon practice]** for the node names (AWD, IXD, appointment scheduling, random-stow-style bin visibility); the diagram itself is **[illustrative]**.



NODE	WHAT IT IS (BUSINESS LANGUAGE)	WHAT CONSTRAINS IT	THE METRIC THAT WATCHES IT	LABEL
Vendor / seller ship	1P purchase orders and 3P FBA shipments; the plan's first promise is the ship date and the ASN (advance shipment notice).	Vendor lead time and OTIF (on-time-in-full); ASN data quality.	OTIF %, fill rate, ASN accuracy.	**[industry standard]**

NODE	WHAT IT IS (BUSINESS LANGUAGE)	WHAT CONSTRAINS IT	THE METRIC THAT WATCHES IT	LABEL
AWD / upstream	Low-cost bulk storage upstream of the FC network; holds inventory cheap and replenishes the network as the forecast firms.	Upstream cube; replenishment lead time into FCs.	Upstream weeks of cover; replenishment dwell.	**[well-known public Amazon practice]**
IXD / direct FC	Inbound cross-dock sorts and re-consolidates inbound freight toward destination FCs; some freight books direct.	Cross-dock throughput; linehaul schedule.	IXD dwell time; trailer utilization.	**[well-known public Amazon practice]**
Appointment / dock	Carriers book delivery appointments; the dock door is the first scarce, schedulable resource in the building.	Dock doors × receiving hours × unload rate.	Dock utilization %, trailer dwell, missed-appointment count.	**[well-known public Amazon practice]**
Receive	Unload, count, inspect, sort to stowable state; where physical freight becomes system inventory.	Labor hours (units per hour, UPH) and conveyor/sort rates.	Receive UPH, backlog in trailers/cartons, receive accuracy.	**[industry standard]**
Stow	Items into bins; the item becomes findable and sellable — availability starts here, not at the dock.	Cubic bin space and stow labor.	Stow UPH, bin utilization %, time-to-available.	**[well-known public Amazon practice]**
Pickable / available	Inventory the promise engine can sell against; the boundary between inbound and outbound.	Nothing inbound — this is the finish line.	In-stock rate; time from dock to available.	**[industry standard]**
Transfers / rebalancing	FC-to-FC moves that fix placement error after the fact; the network's insurance policy, priced per unit.	Transfer cost vs stockout cost (the breakeven in §3).	Transfer units/wk, transfer cost/unit, avoided stockouts.	**[well-known public Amazon practice]**

The information flow — plans travel ahead of the freight

Every physical move above is preceded by an information move; when the two drift apart, the WBR is where it gets caught. **[industry standard]** structure; Amazon vocabulary (WBR, promise engine) **[well-known public Amazon practice]**.

```
INFORMATION FLOW -- plans travel ahead of the freight     [industry standard]

demand forecast (SKU x region x week)
       |
       v
inbound plan: what to buy / book, and when it must land  <---> vendor & seller
       |                                                       lead times, OTIF
       v
placement decision: WHICH node receives it --
  distance-to-customer vs cost vs node capacity      <--- the hiring manager's
       |                                                  home turf (EU IXD placement)
       v
capacity reservation: dock slots + labor plan (UPH -> hours) + cube
       |
       v
receive schedule -> stow plan -> availability signal
       |
       v
promise engine / search in-stock ---> the customer sees a delivery date
       |
       v
actuals vs plan --> variance --> next WBR input (the loop that keeps it honest)
```

> **How to sound like you've run it.** Name the seam, not the node. "The dock" is trivia; "the seam where the inbound plan meets the appointment calendar — and what happens when a vendor ships three days early into a full dock" is experience. Each worked example in §2–§3 is one of those seams, with the arithmetic attached.

2 · Capacity management — the three currencies and the planning loop

Inbound capacity is three currencies that convert into each other only through the plan. **[industry standard]** framework; all numbers **[illustrative]**.

CAPACITY TYPE	UNIT OF ACCOUNT	WHAT DRIVES CONSUMPTION	HOW YOU PLAN IT	WHEN IT BINDS
Dock / appointment slots	Trailer-unloads per day = doors × hours × unload rate	Truckload arrivals; vendor/carrier appointment discipline; trailer mix (floor-loaded vs palletized).	Receive plan → trailers/day → doors at an unload rate, capped at ~85% utilization (worked in CAP-A, CAP-D).	First, at peak — it is the least elastic currency (you cannot stack trailers at a door).
Labor hours	Hours = units ÷ UPH, grossed up for indirect work and learning curves	Units received/stowed; associate proficiency; process (conveyable vs hand-stack).	UPH standards → direct hours → paid hours → headcount by shift (CAP-B).	Second — elastic but slow: hiring and the learning curve lag the plan by weeks.
Cubic storage	Cubic feet = on-hand units × cube per unit	Inventory target (days of cover), unit cube, bin geometry / honeycomb loss.	Inventory plan → cube needed vs effective bin capacity (CAP-C).	Last, then suddenly — it looks fine at 49% and is a wall at 90%; the least flexible currency mid-peak.

The planning loop — forecast to floor

```
CAPACITY PLANNING LOOP                                          [industry standard]

demand forecast (units/day, with a distribution -- p50 ... p90)
       |
       v
receive plan: units/day -> cartons -> pallets -> trailers/day (CAP-A)
       |
       +---------------> dock reservation: trailers/hour vs doors at ~85% ceiling
       |
       v
labor plan: units / UPH = direct hours -> +indirect % -> /shift-hours = heads (CAP-B)
       |
       v
cube check: on-hand units x cube/unit vs effective bin capacity (CAP-C)
       |
       v
peak buffer: fixed capacity to ~p50-p70, flex (temp labor, OT, appointment
smoothing) for the tail (CAP-E) -- then WBR reads actuals vs all of it
```

The metrics that watch capacity

METRIC	DEFINITION / FORMULA	RED FLAG	LABEL
Dock utilization %	`utilization = scheduled unloads ÷ (doors × hours × unload rate)`	Sustained > ~85% — dwell goes nonlinear (CAP-D).	**[industry standard]**
Dwell time	`dwell = trailer arrival → door departure (hours)`	Rising dwell with flat volume = variability, not volume, is the problem.	**[industry standard]**
Receive backlog	`backlog_days = unreceived units ÷ receive capacity (units/day)`	> 0.5 day carried into the next shift; it compounds.	**[industry standard]**

METRIC	DEFINITION / FORMULA	RED FLAG	LABEL
Forecast variance	`variance = (actual − plan) ÷ plan, by week and by node`	Same-sign variance three weeks running = a bias, not noise; fix the plan, not the floor.	**[industry standard]**

CAP-A · REQUIRED RECEIVE CAPACITY — UNITS/DAY → TRAILERS → DOCK DOORS

One illustrative FC receiving 120,000 units/day. Every row recomputes from the rows above; say the chain out loud and you sound like the floor.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Inbound volume	`V = 120,000 units/day`	illustrative receive plan	120,000 units/day
2	Cartons per day	`120,000 ÷ 6 = 20,000`	6 units/carton (illustrative)	20,000 cartons/day
3	Pallets per day	`20,000 ÷ 40 = 500`	40 cartons/pallet (illustrative)	500 pallets/day
4	Truckloads per day	`500 ÷ 26 = 19.2`	26 pallets per 53-ft trailer [industry standard]	19.2 trailers/day
5	Trailer arrival rate	`19.2 ÷ 10 = 1.92`	10 receiving hours/day (illustrative)	1.92 trailers/hour
6	Raw doors required	`1.92 ÷ 0.5 = 3.84`	unload rate 0.5 trailer/door-hour = 2 h/trailer (illustrative)	3.84 doors
7	Doors at the 85% ceiling	`3.84 ÷ 0.85 = 4.5 → round up`	utilization ceiling ~85% [industry standard]	**5 dock doors planned**

The sentence to say: "Twelve trailers an hour isn't the plan — the plan is five doors, because at 85% utilization the doors stop being fungible and dwell starts compounding."

CAP-B · LABOR HOURS FROM UPH — WITH THE LEARNING CURVE PRICED IN

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Direct receive hours	`120,000 ÷ 300 = 400`	mature-associate 300 UPH (illustrative)	400 direct hours/day
2	Paid hours (indirect gross-up)	`400 × 1.15 = 460`	15% indirect allowance (illustrative)	460 paid hours/day
3	Headcount	`460 ÷ 10 = 46`	10-hour shifts (illustrative)	46 associates/day
4	Learning curve, week 1	`300 × 0.60 = 180 UPH`	new associate at 60% of standard (illustrative)	180 UPH effective
5	Blended UPH at 30% new	`0.7×300 + 0.3×180 = 210 + 54 = 264`	Q4 ramp: 30% of staff in week 1–2 (illustrative)	264 UPH blended
6	Direct hours at blended UPH	`120,000 ÷ 264 = 454.5; 454.5 − 400 = 54.5; 54.5 ÷ 400 = 13.6%`	same volume as step 1	454.5 h = **+54.5 h (+13.6%)**

The sentence to say: "The hiring plan isn't the headcount math — it's the learning-curve math. Thirty percent new hires isn't a 30% problem, it's a 13.6% tax on every received unit for two weeks, and the peak plan has to pay it before the first trailer arrives."

CAP-C · CUBIC CAPACITY — UNITS → CUBE → % OF THE BUILDING

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	On-hand target	`I = 2,000,000 units`	illustrative network-position target for this FC	2.0M units
2	Cube required	`2,000,000 × 0.25 = 500,000`	0.25 ft³/unit average (illustrative)	500,000 ft³
3	Nominal utilization	`500,000 ÷ 1,200,000 = 41.7%`	1.2M ft³ usable bin capacity (illustrative)	41.7% of nominal
4	Effective capacity (honeycomb)	`1,200,000 × 0.85 = 1,020,000`	15% partial-bin loss [industry standard]	1,020,000 ft³ effective
5	True utilization	`500,000 ÷ 1,020,000 = 49.0%`	against effective, not nominal	**49.0% — plan against this**
6	Days of cover check	`2,000,000 ÷ 120,000 = 16.7 days`	same 120K units/day flow as CAP-A	16.7 days
7	Surge consumption	`3 × 120,000 × 0.25 = 90,000 ft³; 90,000 ÷ 1,020,000 = 8.8%`	a 3-day inbound surge ahead of forecast	8.8% of effective cube in 3 days

The sentence to say: "Cube is the currency that lies — 42% nominal is 49% effective, and a three-day vendor early-ship surge eats nine points of the building before the WBR sees it."

CAP-D · WHY DWELL EXPLODES ABOVE ~85% DOCK UTILIZATION

Queueing approximation: with variability in arrivals and unload times, expected wait scales with ρ/(1−ρ) and total dwell with service ÷ (1−ρ), where ρ = utilization **[industry standard — Kingman-style approximation, qualitative use]**.

STEP	UTILIZATION P	QUEUE FACTOR P/(1−P)	DWELL AT 2.0 H SERVICE: 2 ÷ (1−P)	READING
1	70%	`0.70 ÷ 0.30 = 2.33`	`2.0 ÷ 0.30 = 6.7 h`	comfortable
2	85%	`0.85 ÷ 0.15 = 5.67`	`2.0 ÷ 0.15 = 13.3 h`	the knee — the ceiling in CAP-A
3	95%	`0.95 ÷ 0.05 = 19.0`	`2.0 ÷ 0.05 = 40.0 h`	a 2-hour unload becomes a 40-hour trailer stay
4	85% → 95%	`19.0 ÷ 5.67 = 3.4×`	`40.0 ÷ 13.3 = 3.0×`	**+10 points of utilization ≈ 3× the dwell**

The sentence to say: "Above about 85% the building stops failing linearly — the last ten points of utilization triple your dwell. So the plan holds dock slack the way inventory holds safety stock: you don't apologize for either until the week you needed it."

CAP-E · PEAK BUFFER SIZING — P50 VS P90

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Peak-week distribution	`p50 = 120,000; p90 = 156,000; 156,000 − 120,000 = 36,000`	illustrative; p90/p50 = 1.30	+36,000 units/day at p90
2	Build to p50 — one p90 day	`36,000 ÷ 120,000 = 0.30 day`	shortfall carried as backlog	0.30 day of backlog
3	Build to p50 — three p90 days	`3 × 36,000 = 108,000; ÷ 120,000 = 0.9`	backlog compounds across the shift handoffs	~1 full day of backlog
4	Build to p90 — a p50 day	`36,000 ÷ 156,000 = 23%`	idle capacity on the median day	23% idle
5	Labor to flex the tail	`156,000 ÷ 264 = 591 h; 591 − 454.5 = 136 h; 136 ÷ 10 = 13.6 ≈ 14`	blended 264 UPH from CAP-B; 10-h shifts	**≈14 flex associates**

The sentence to say: "Fixed capacity to the median, flex the tail. Building the building to p90 means paying 23% idle on the average day; building to p50 means a three-day p90 run is a full day of backlog — so the answer is temp labor, overtime and appointment smoothing, sized at about fourteen heads on this volume."

3 · Inventory management, inbound lens — where the units land and how long they last

Capacity decides how fast freight moves; inventory decisions decide whether the right units are in the right building. This is the hiring manager's home turf — placement was her EU IXD charter — so the placement math (INV-C) gets said slowly, with the constraint and the shadow price named.

Placement strategy — distance-to-customer vs cost

The mechanism. Amazon positions inventory close to forecasted demand so that the outbound leg — the expensive, promise-bound leg — is short **[well-known public Amazon practice]**. Inbound placement is the upstream half of that decision: which node receives the units, given regional demand, node capacity and inbound cost. The trade is never "cheapest inbound" — it is distance-weighted delivered cost under a capacity constraint, with the customer promise as the boundary condition. Speed and cost are one table, not two arguments.

Network safety-stock positioning. Because pooling variability upstream beats duplicating buffers everywhere **[industry standard]**, the cheap place to hold uncertainty is upstream (AWD-style bulk) and the fast place is forward (FC pickface); the worked question is how much buffer each layer carries (INV-B) and when a transfer beats a stockout (INV-D).

Weeks of cover, health, and the FBA seller view

CONCEPT	DEFINITION	OPERATOR'S ACTION RULE	LABEL
Weeks of cover (WOC)	WOC = on-hand ÷ weekly forecast	< 2 → replenish / expedite · 2–4 → healthy · 4–6 → watch · > 6 → excess action (worked in INV-A).	**[industry standard]**
Excess / aged	Units above the cover ceiling, and units aging against storage-fee cliffs.	Price, transfer, or remove — in that order of preference.	**[industry standard]**
Stranded	Physically present, not sellable (listing, hold, or data break) — inbound's silent inventory.	Fix the listing/hold at the source; a stow without a sellable record is not availability.	**[well-known public Amazon practice]**
In-stock rate	Share of demand-facing selection available to promise.	The north-star inventory output; every inbound metric is an input to it.	**[industry standard]**
IPI (FBA seller view)	Inventory Performance Index components: excess inventory %, sell-through, stranded inventory, in-stock rate.	Sellers manage to the same four levers; restock limits make network capacity legible to sellers **[well-known public Amazon practice]**.	**[well-known public Amazon practice]**
Transfers / rebalancing	FC↔FC moves that correct placement error.	Run them against the breakeven (INV-D), never as a reflex.	**[well-known public Amazon practice]**

INV-A · WEEKS OF COVER — THREE SKUS, THREE ACTIONS

STEP	SKU	FORMULA → SUBSTITUTION	WOC	ACTION (RULE FROM THE TABLE ABOVE)
1	A	`4,200 ÷ 600/wk = 7.0`	7.0	> 6 → excess action: to reach 4.0 WOC move 4,200 − 4.0×600 = 4,200 − 2,400 = **1,800 units** (transfer feeds INV-D)
2	B	`1,350 ÷ 450/wk = 3.0`	3.0	2–4 → healthy; no action, watch variance
3	C	`500 ÷ 500/wk = 1.0`	1.0	< 2 → expedite: stockout in 500 ÷ 7 = 71.43/day; 500 ÷ 71.43 = **7.0 days** absent receipt; pull the inbound appointment forward or re-source

The sentence to say: "WOC turns a warehouse full of units into three sentences: A is seven weeks of stock and I'm moving 1,800 units; B is fine; C is a stockout next Friday unless the appointment moves."

INV-B · SAFETY STOCK — Z·Σ·√LT, SUBSTITUTED, AND THE TWO LEVERS

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Demand and variability	`d = 500/day; σ_d = 120/day`	illustrative SKU at one FC	—
2	Demand over lead time	`500 × 7 = 3,500`	lead time LT = 7 days (illustrative)	3,500 units cycle exposure
3	σ over lead time	`σ_LT = 120 × √7 = 120 × 2.6458 = 317.5`	independent daily demand [industry standard]	317.5 units
4	Safety stock at 95%	`SS = 1.645 × 317.5 = 522; 522 ÷ 500 = 1.044 ≈ 1.04 days`	z(95%) = 1.645 [industry standard]	**522 units = 522 ÷ 500 = 1.04 days**
5	Service lever: 99%	`SS = 2.326 × 317.5 = 739; 739 − 522 = +217; 217 ÷ 522 = 41.6%`	z(99%) = 2.326	+217 units = **+41.6% buffer for +4 pts**
6	Lead-time lever: 7 → 5 days	`SS = 1.645 × 120 × √5 = 1.645 × 268.3 = 441; 522 − 441 = −81; 81 ÷ 522 = 15.5%`	√5 = 2.2361; 120 × 2.2361 = 268.3	−81 units = **−15.5% buffer, free**

The sentence to say: "Service points are bought at an accelerating price — 95 to 99 costs forty percent more buffer — but lead time is the free lever: two days off the lane releases fifteen percent of the buffer with the service level untouched. That's why measured lead times, not brochure ones, are an inventory program" (this is exactly the S8 story — §5).

INV-C · PLACEMENT MINI-MODEL — 2 FCS × 3 DEMAND REGIONS, CAPACITY-CONSTRAINED

Distance-weighted delivered cost per unit **[illustrative]**. Demand: R1 = 2,000/wk, R2 = 3,000/wk, R3 = 1,000/wk (6,000 total). FC-West capacity 2,500 units/wk; FC-East 4,000.

COST $/UNIT	R1 (2,000)	R2 (3,000)	R3 (1,000)
FC-East	$1.20	$2.40	$3.60
FC-West	$3.40	$2.10	$1.30

STEP	DECISION	FORMULA → SUBSTITUTION	WEEKLY COST
1	Unconstrained optimum: R1→East, R2→West, R3→West	2,000×1.20 + 3,000×2.10 + 1,000×1.30 = 2,400 + 6,300 + 1,300	$10,000
2	West capacity binds at 2,500: give West its cheapest regions first — R3 (1,000), then R2 partial (1,500)	1,000×1.30 + 1,500×2.10 = 1,300 + 3,150	West serves 2,500
3	East takes R1 + residual R2 (1,500); load check 2,000 + 1,500 = 3,500 ≤ 4,000	2,000×1.20 + 1,500×2.40 = 2,400 + 3,600	East serves 3,500 ✓
4	Constrained total	2,400 + 1,300 = 3,700; 3,700 + 3,150 = 6,850; 6,850 + 3,600 = 10,450	**$10,450**
5	Capacity premium	10,450 − 10,000 = 450; 450 ÷ 10,000 = 4.5%	**$450/wk (+4.5%)**
6	Shadow price check	(2.40 − 2.10) × 1,500 = 0.30 × 1,500 = 450	each added West slot is worth $0.30 on R2 — the model agrees with itself

The sentence to say: "Placement is a constrained assignment, not a map. Here the constraint costs four and a half points, and the shadow price — thirty cents a unit — is the number I'd take to the capacity team to argue for one more West slot, and to the finance review to explain the premium."

INV-D · TRANSFER BREAKEVEN — WHEN REBALANCING BEATS A STOCKOUT

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Transfer cost per unit	$1.10 handling + $1.90 freight = $3.00	illustrative FC↔FC economics	$3.00/unit
2	Expected stockout cost per unit	$12 × 0.40 = $4.80	lost margin $12/unit; P(stockout before inbound lands) = 40% (illustrative)	$4.80/unit
3	Decision rule	3.00 < 4.80 → transfer	transfer when cost < expected loss	**transfer wins by $1.80/unit**
4	Breakeven probability	p* = 3.00 ÷ 12 = 0.25	transfer when stockout probability > 25%	p* = 25%
5	At scale (SKU A's 1,800 units)	spend 1,800 × 3.00 = 5,400; expected loss 1,800 × 4.80 = 8,640; net 8,640 − 5,400 = 3,240	the 1,800 units from INV-A	**+$3,240 net — and the premium is priced, not argued**

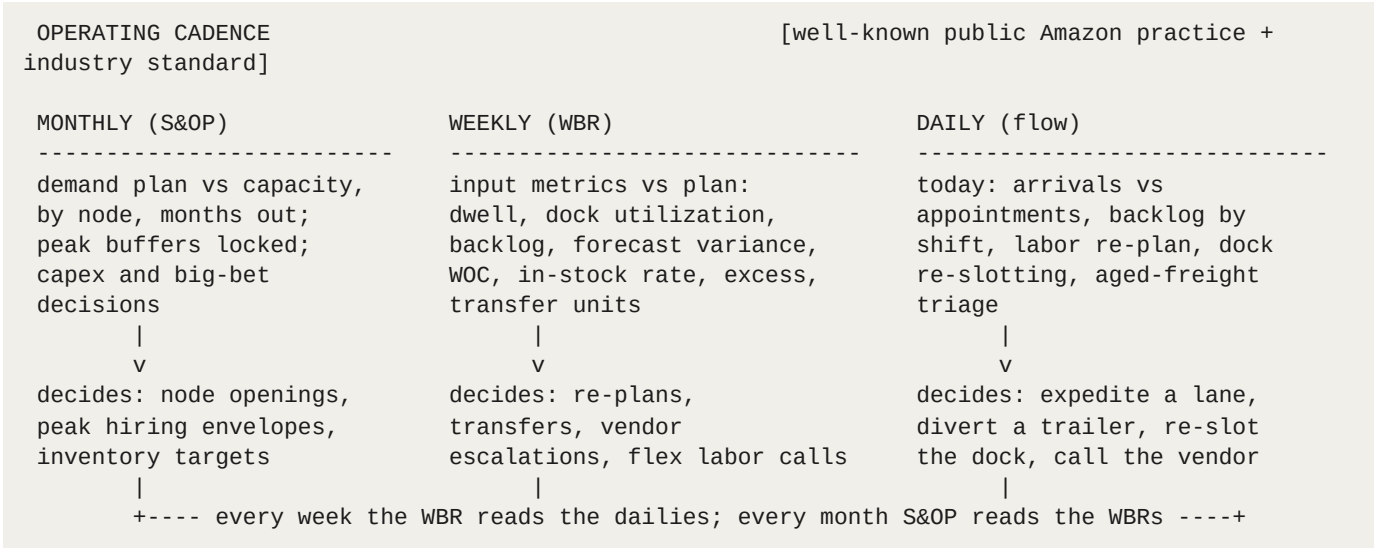
The sentence to say: "Transfers are insurance with a deductible. At a twelve-dollar margin the breakeven is a twenty-five percent stockout probability — below that I let it ride, above it I move the freight, and either way the decision is arithmetic, not adrenaline."

INV-E · VENDOR OTIF / FILL-RATE → THE IN-STOCK CHAIN

STEP	LINK	FORMULA → SUBSTITUTION	VALUE
1	Planned PO volume	`10,000 ÷ 7 = 1,428.6 ≈ 1,429/day`	10,000 units (illustrative)
2	× fill rate	`10,000 × 0.92 = 9,200`	9,200 shipped (800 short)
3	× on-time	`9,200 × 0.88 = 8,096`	8,096 OTIF units = **81.0% of plan**
4	Weekly deficit	`10,000 − 8,096 = 1,904`	1,904 units (19.0% of plan)
5	Safety-stock absorption	`522 ÷ 1,429 = 0.37 days; residual 1,904 − 522 = 1,382`	the INV-B buffer covers about a third of a day — it was sized for demand noise, not vendor failure
6	In-stock proxy for the week	`10,000 − 1,382 = 8,618; 8,618 ÷ 10,000 = 86.2%; 100 − 86.2 = 13.8 ≈ 14 pts`	**~86% in-stock absent recovery — a 14-point hit from one vendor's 92/88 week**

The sentence to say: "OTIF isn't a vendor scorecard, it's the first link of the in-stock chain: fill times on-time is eighty-one percent of plan, and a cycle buffer sized for demand noise absorbs a third of a day of it. So vendor management is inventory management — measured lead times, escalation thresholds, and a recovery path that starts at the ASN, not at the stockout."

4 · Operating cadence — one drumbeat, three tempos



Peak (Q4) planning rhythm [industry standard]

WHEN	WHAT GETS DECIDED	THE INBOUND-SPECIFIC WORK
Q2–early Q3	Network peak plan: capacity envelopes by node, hiring envelopes, inventory targets.	p50/p90 forecast distribution locked (CAP-E); cube plan vs effective capacity (CAP-C); vendor appointment and freight-plan commitments.
Late Q3	Readiness: flex labor pipelines, overflow/storage contingency, freeze windows.	Learning-curve staffing math paid early (CAP-B step 5–6); appointment smoothing rules published to vendors/carriers.
Q4 in-peak	Execution: daily flow cadence dominates; WBR shifts to peak inputs.	Dwell and backlog watched hourly (CAP-D); transfers only against the breakeven (INV-D); escalations on same-sign forecast variance.
January	Debrief: what the plan got wrong, written down while it's fresh.	Variance autopsy by node; buffers released on evidence, not relief (the S6/S8 release rule, §5).

Escalation norms. Escalate with the arithmetic attached: the metric, its frozen definition, the variance, the root cause so far, and the decision you need [industry standard]. Same-sign variance three periods running is a plan problem and goes up as a re-plan request, not a floor problem and goes down as pressure. A miss you report with a recovery path is a WBR input; a miss discovered by someone else is a trust problem — the S11 discipline (§5).

5 · Tie-to-your-stories — the master packet, spoken in inbound

Each story maps to one inbound concept; the third column is the sentence that makes the mapping out loud. Canon numbers carry their basis labels; stories stay allocated to their panel owners — here they are *vocabulary*, not re-told STARs.

STORY (CANON)	INBOUND CONCEPT	THE SENTENCE THAT TIES THEM
S8 — Elementum measured lanes (200 lanes; promise at p99; 99% lead-time accuracy; 580K units / ~$1.5M buffer released, record, one-time)	Inbound lead-time reliability	"Every inbound plan and every safety-stock formula is only as honest as its lead times — I replaced brochure lead times with measured distributions and released buffer on evidence; INV-B's free lever is that story."
S6 — Elementum supply/demand view (forecast accuracy 70% → 85%; $2M safety stock released under the both-errors rule, record, one-time)	Network safety-stock positioning	"Buffer gets released where forecast error AND lead-time error both improved — a signed release rule, not a victory lap; that's how I'd run NACF's inventory health reviews."
S2 — Sephora UDC comingle (two warehouses → one pool; +15% efficiency, signed; robotics behind a published 98% gate)	Capacity pooling	"Pooling variance beats adding capacity: one pool removed duplicate handling before anyone asked for a bigger building — the same logic as holding cube slack upstream instead of over-building the FC."
S13 — Manhattan Cyber Monday (3 DCs / 100 stores; 99% vs 95% fill; ~$8M protected vs ~$6M at risk, record pair)	Peak capacity under failure	"Peak is when the CAP-D nonlinearity stops being theory — I rebalanced by promise-at-risk at 4 AM, and the post-mortem shipped the heartbeat and the playbook so the next peak ran boring."
S18 — Sephora dual-DC WMS migration (0.5% tolerance published with Finance; mocks 7% → 1.8% → 0.3%; rehearsed fallback for ~100 of ~400 RF devices)	Launch / cutover readiness	"New capacity or a new process goes live behind a published tolerance and a rehearsed fallback — exactly how I'd bring a new IXD lane or a peak process into the network."
S19 — Elementum promise dial (k = 1.645σ one governed dial; fast-promise share 7.6% → 39.6% at a held 95% floor — all model, labeled)	Promise from inventory position	"The date the customer sees is priced from inventory position and lane variability — placement (INV-C) sets the position, the dial prices the promise."

STORY (CANON)	INBOUND CONCEPT	THE SENTENCE THAT TIES THEM
S11 / S12 — Flexport visibility & roadmap (boarding behind published gates; CTS $0.66 → $0.03 across 7 → 68 customers; month-18 plan → value in month 6)	Upstream visibility into inbound	"Inbound starts before the truck: ASN-grade visibility and graduation gates upstream are what let a dock plan survive contact with real vendors — and re-sequencing on cost-to-serve is how I'd order the roadmap to get there."
S9 — Elementum lane onboarding (597 lanes = 80% of spend live in month one; ≥95% valid-ASN gate held twice; completeness 60% → 95%)	ASN / inbound data quality	"Bad ASNs manufacture surprise at the dock — a gate on data quality upstream is capacity protection downstream."

> **Discipline in the room.** These are one-sentence bridges, not re-tellings — the full STARs belong to their allocated interviewers per the panel packets. Never mix the loop packet's conflicting numbers into these bridges; never present any modeled dollar as record.

6 · Per-interviewer questions — what they'll ask, and what to ask them

For each interviewer: ten likely NACF-inbound questions with a one-to-two sentence answer pointer (the booklet section carries the depth), then three questions to ask them. The first six in each list are the questions to expect from that seat's lens; the last four press the part of the job that person actually owns — placement, the distribution, the building, the peer handoff, or the seller.

Snigdha Tripathy — Sr Manager, Supply Chain · HIRING MANAGER

> **Lens.** Placement and IXD are her home turf (EU IXD leader; built the team ground-up; 50+ FC launches). She judges fit for the actual work: network process design, value made legible, and judgment under unit economics. Talk placement slowly and precisely.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
1	**Walk me through how you'd decide where inbound inventory lands across our network.**	Walk INV-C slowly: a cost matrix, the capacity that binds, and a shadow price that reproduces the premium — $450 either way in that example. §8 stage 6 is the same move on one week, where splitting 2,350 units saves $5,640 and the shadow-price check prints that same $5,640; then bridge to S8, because those lane costs are only honest if the lead times were measured.
2	**How do you know the network has enough inbound capacity for Q4?**	Say CAP-A, CAP-B, and CAP-E as one chain: units to trailers to doors held at the 85% ceiling, UPH to heads, fixed capacity near the median and flex on the tail. §8 runs it on 12,000 units — the dense equivalent is 1.92 trailers, the tender is 6, and a 7th arrival on a 6-slot block is what actually breaks the day.
3	**Tell me about a time you built a process or capability from nothing.**	Use S11: a boarding process stood up behind gates that were published before wave 1, with a weekly input/output review. That is the same shape as a ground-up IXD build, mechanism first and volume second, and S18 is the launch version of it if she pulls on the fifty-plus FC openings.
4	**A strategic vendor misses OTIF three weeks running into peak. What do you do?**	Price the miss before the conversation. INV-E turns 92% fill times 88% on-time into 8,096 units, 81% of plan, and about a 14-point in-stock hit a 522-unit cycle buffer was never sized to absorb — §8's 120-unit shortage is the small version of that pass-through — and the escalation carries the recovery path and a re-plan, the way S8 replaced brochure lead times with measured ones.
5	**How do you run your weekly business review — inputs, outputs, decisions?**	Use §4's three tempos and S11's mechanics: a short list of frozen inputs, each signed by its owner and tagged to the output it moves, reds already decomposed with a date. Three weeks of same-sign variance comes up as a re-plan request, which is how §8's 1,650-unit slow-mover over-order would be handled the week it repeats.
6	**How do you trade speed against cost in placement without losing the customer promise?**	Put the promise in as the boundary condition, then show one priced table. INV-C's constrained premium was 4.5% with a $0.30 shadow price on the displaced unit, INV-D transfers only when cost sits below margin times probability (breakeven 25% on the $3 and $12 example), and S19 is the promise-side version: one governed dial, with the option you decline kept in the table as a priced row.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
7	**Where would you run a Kaizen on the dock-to-available path, and what stays frozen through peak?**	Start on the wait that adds no units — §8 stage 2, where 6 doors × 2.0 h × 0.5 rate = 6 slots and the 7th trailer dwells 4.0 h. Through peak, CAP-E keeps the flex on labor, overtime, and appointment smoothing, and the placement objective and the safety-stock z stay frozen until the January debrief writes down what the plan got wrong.
8	**IXD dwell and an FC dock are both red the same week. Which do you pull, and how do you make the cost of waiting visible?**	Pull the pool whose units are already inside a customer promise, and price the hours on the other one. §8's marginal trailer is 4.0 h of dwell, and under a FIFO yard that delay lands on an expected 1,714 units — that slide is the escalation, the same habit as making an IXD saving legible in dollars rather than as a red cell.
9	**How do you ramp inbound at a new FC — doors, the labor curve, and the first units you place there?**	Doors from CAP-A off the launch forecast, labor from CAP-B with the learning curve paid before trailer one (30% of the crew at 60% UPH blends to 264 and costs 13.6% more hours), and the first inventory as an INV-C assignment into a building CAP-C still shows as empty. S18 is the gate around the ramp: a published tolerance, mocks, and a fallback already rehearsed.
10	**Infrastructure wants the dock held near 70% and Finance wants the cube earning. How do you settle it?**	Put both constraints on one page with a price. CAP-D is why dock slack is insurance — from 85% to 95%, dwell goes from 13.3 h to 40 h, about 3× — CAP-C is why cube is read on effective capacity, 49.0% in that example against 41.7% nominal, and the settlement is INV-C's shadow price, what the next slot is worth per unit, taken to both teams as the same number.

Questions to ask them

- "Which inbound outcome does this seat own end-to-end in the first year — placement, the capacity envelope, or inventory health — and which two or three WBR inputs would tell you, six months in, that the hire was right?"
- "Where does NACF inbound hurt most going into this peak — dock slots, the labor curve, cube, or placement accuracy — and what would you want moved first?"
- "You built EU IXD and placement from the ground up. What did that teach you about the order of people, process, and mechanisms that I should copy walking in?"

Ryan Hoerger — Manager, Placement / Transfer Analytics

> **Lens.** Ex-Sr BIE, Booth MBA, Siebel Scholar, Nike analytics — statistical rigor is the love language. Show the distribution behind the number, name your assumptions, and let the math be audited.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
1	**How would you model optimal inventory placement across FCs?**	Start from the cost matrix, let capacity bind, then check that the shadow price reproduces the premium. INV-C does it in six steps and the $450 appears twice, which is the audit; §8 stage 6 is the two-node case, where the $5,640 of savings also equals transfer cost avoided minus the $0.60 of extra inbound, so the model agrees with itself before anyone is asked to believe it.
2	**When is a transfer worth doing?**	Transfer when its cost is below margin times the probability of stocking out before the next inbound lands. On INV-D's economics that is $3.00 against $12 × 0.40 = $4.80, and the breakeven probability is 3.00 ÷ 12 = 25%; §8 applies the same rule once the trailers are already at ONT9-style, and the fast mover fails it while the medium and slow movers pass.
3	**How do you size and position safety stock across a network?**	Size it with INV-B, SS = z × σ × √LT, and name the lever you pulled. Moving z from 1.645 to 2.326 added 217 units (+41.6%), while cutting lead time from 7 days to 5 released 81 units (−15.5%) with the service level held; pool the rest upstream and release it only under S6's both-errors rule, because §8's fast mover is a buy with the z term left out — cover lands on the 7-day lead time and a 120-unit shortage passes straight through.
4	**How do you measure forecast quality, and what do you do with the variance?**	Use the §2 definition, (actual − plan) ÷ plan, by week and by node. Three weeks of the same sign is bias, so the action is a re-plan; S6 is the program behind that sentence, forecast accuracy from 70% to 85%, with buffer released only where both error terms had actually moved.
5	**Do you size buffers to p50 or p90?**	Fix the building near the center of the distribution and flex the tail. CAP-E is the arithmetic: a p50 plan carries 0.30 day of backlog on one p90 day and about a full day across three, a p90 building sits 23% idle on the median day, and the flex on these numbers is 36,000 ÷ 264 ÷ 10 ≈ 14 heads.
6	**How do you keep a model from going stale or getting gamed?**	Put a drift flag and a refresh cadence on the distributions, which is what S8 did for lanes, and freeze the metric definitions under the owner's signature the way S11 did. A truncated week stays a data miss: show the hole in the denominator before you show a ratio that has been cleaned up.
7	**The placement residual looks biased on slow movers. What do you inspect before you change the model?**	Separate bias from noise on the velocity slice before you touch the objective. §8's slow mover is the picture: sell-through 350 ÷ 3,400 = 10.3%, 8.71 weeks of cover against a 4.0 target, an over-order of 1,650 units that a WOC residual would have flagged — and only then do you decide whether the feature, the capacity constraint, or the loss is the thing to change.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
8	**How would you test, out of sample, that a transfer policy beats its breakeven?**	Freeze the rule on a prior window — here p* = 3.00 ÷ 12 = 25% — and score the stockouts you actually saw against the transfers the rule would have fired. INV-D step 5 is the in-sample version, +$3,240 on 1,800 units at an assumed 40% probability; the out-of-sample question is whether that net survives when the probability is estimated from history rather than typed in.
9	**Two planners hand you two different sigmas for the same SKU. What has to be true before safety stock moves?**	Same series, same lead-time window, and a stated assumption about independent days, because INV-B's √LT is exactly that assumption. §8's sigmas — 80, 40, and 15 units a day — stay labeled inputs until the window and the autocorrelation are in the room; resizing safety stock off an unstated σ is how the fast mover's 348-unit buffer quietly disappears from the buy.
10	**The capacity team wants average dwell as the WBR headline. What do you put next to it, and when do you refuse the formula?**	Put the marginal trailer next to the mean, because CAP-D's queue factor moves from 5.67 at 85% to 19 at 95% while an average can still look calm. §8 reports both from the same yard — expected dwell (6 × 2.0 + 4.0) ÷ 7 = 2.29 h, and 4.0 h on the trailer that waits — and once ρ = 7/6 the Kingman form is undefined, so the residual count replaces it.

Questions to ask them

- "When operators override the placement or transfer recommendation, do you score the override against the model's counterfactual — and who is usually right?"
- "What is the refresh cadence and the drift test on the demand and lead-time distributions the placement model actually sees?"
- "Which input do you trust least today — σ, the stockout probability, or the transfer cost per unit — and what observation would change it?"

Rocky Goforth — Sr Manager, Infrastructure

> **Lens.** Infrastructure owns the physical envelope — buildings, docks, automation, peak readiness. Speak capacity as physics: constraints, nonlinearities, launch discipline.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
1	**What actually constrains an FC's inbound throughput — and which one binds first at peak?**	Three currencies, in a fixed order: dock slots first, because a door takes one trailer; labor second, because hiring lags the plan by weeks; cube last, and then as a wall. CAP-A and CAP-C are the two ends of that, and §8 is a week where the Wednesday block fails at 7 trailers on 6 slots while the mod is still at 25.7% effective before the receipt even lands.
2	**Why does dwell blow up when docks run hot?**	The wait scales like ρ ÷ (1 − ρ), and CAP-D runs it: at 85% the factor is 5.67 and a 2.0 h unload dwells about 13.3 h; at 95% the factor is 19 and dwell is 40 h, roughly 3× for ten more points of utilization. §8 is the region past that formula — ρ = 7/6 makes (1 − ρ) negative — so the residual trailer replaces the approximation.
3	**How should we be planning infrastructure for Q4?**	Follow §4's peak rhythm with CAP-E underneath: lock the p50/p90 distribution and the effective-cube plan in Q2 and early Q3, pay the learning-curve hours in late Q3, and in peak watch dwell and backlog on the daily tempo. S13 is the in-peak posture I want Infrastructure beside me on — rebalance by promise at risk, then leave the heartbeat and the playbook so the next peak is dull.
4	**Tell me about bringing automation into a live building.**	S2. Robotics came in behind a published 98% gate and a throttle that existed before go-live; week two printed 94%, the fleet was held near 30% of volume, the cause was found, and full volume was back in week three. The gate is what made the install an infrastructure change with a stop condition.
5	**How do you launch new capacity or process without wobbling the network?**	S18: a 0.5% tolerance published with Finance, three mocks that moved discrepancies from 7% to 1.8% to 0.3%, and a fallback already rehearsed when about 100 of about 400 RF devices failed. A new door bank or a new inbound lane gets those same three things — a published tolerance, mocks against it, and a fallback someone has already run.
6	**When capacity and cost fight, how do you decide?**	Price the constraint and take that price to both sides. INV-C turned one more West slot into $0.30 a unit and $450 a week, because (2.40 − 2.10) × 1,500 = 450 and that matched the premium; §8 turns "send 2,350 units direct to the sister node" into $5,640 versus transferring them after all six trailers have arrived here, and CAP-E is the building-sized version of the same choice, idle percent versus backlog days, already computed.
7	**Six doors, and a carrier puts seven trailers on one block. What do you change — the building, the appointment grid, or the carrier?**	Change the grid and the tender. §8's full receiving day is 6 × 10 × 0.5 = 30 trailer-slots and this vendor's six trailers are 6 ÷ 150 = 4% of the week, so the 2.0 h block is what failed — 6 × 2.0 × 0.5 = 6 slots and the seventh trailer dwells 4.0 h — and the operating rule is that ASN trailers take those six doors while the unscheduled yard-drop holds the 4.0 h.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
8	**Cube reads fine, then one bulky receipt puts a mod near 80%. How early do you want that signal, and in what unit?**	On the ASN, in cubic feet against effective capacity, before the door is reserved. CAP-C is the warning that nominal utilization understates the building — 41.7% nominal was 49.0% effective — and §8's slow mover is 16.8% of the units and 43.5% of the receipt cube, taking the mod from 25.7% to 79.9% effective in one drop, with another 860 units of that SKU (688 ft³) touching the 90% line.
9	**Where should Supply Chain stop asking Infrastructure for peak space and change the flow instead?**	When the cube is a placement error the flow can still decline. §8's 1,650 excess slow-mover units are 1,650 × 0.80 = 1,320 ft³, 19.4% of that mod's effective capacity, so moving them is a flow decision, while a new rack is the ask when CAP-A says the doors are the currency that binds — bring the shadow price of the space and the WOC of the units in the same sentence as the square footage.
10	**We staffed mature UPH, the crew is full of new hires, and the unload rate misses. Who owns that, and why was it a surprise?**	The rate standard belongs to the building, the share of new heads belongs to the plan, and the miss belongs on the readiness page either way. CAP-B prices it ahead of the trailer: 0.7 × 300 + 0.3 × 180 = 264 blended UPH, and 120,000 ÷ 264 = 454.5 h against 400 h, a 13.6% tax for the first two weeks — that line sits next to the door plan at the late-Q3 review.

Questions to ask them

- "Which physical constraint is actually binding NACF inbound right now — a door bank, a sort or conveyor rate, or a mod that is fuller on effective cube than the nominal number says — and how do you want that called at the WBR?"
- "How far ahead are door counts and appointment blocks locked, and what is the path on a morning when a carrier's tender overruns the block the way seven trailers on six slots does?"
- "What should Supply Chain put on the ASN or the capacity reservation — cube by SKU, trailer count, or the learning-curve headcount — that Infrastructure still receives too late?"

Zhaowen Lan — Sr Manager, Supply Chain · peer

> **Lens.** A peer who shares the cadence. He is listening for how you run mechanisms other people own — the WBR, the handoffs, the disagreement that stays professional.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
1	**How do you run a WBR that people actually trust?**	Keep the list short and the definitions frozen. S11 ran five to seven input metrics, each signed by the person who owns it, each tagged to the output it is supposed to move, and a red arrived already decomposed with a date on the fix — the trust is in that pre-read.
2	**An input metric owned by another team is red every week. What do you do?**	Decompose it with the owner before the meeting and bring the dated fix. When the definition itself is wrong, correct it in the room: S11 discarded a 99.4% week rather than defend a denominator that had been cut short, and the number stays the owner's while the escalation carries the variance and the date.
3	**How do you manage the S&OP-to-WBR-to-daily handoff?**	§4 assigns each tempo a decision: S&OP locks the envelope, the WBR reads variance against that envelope, and the day re-slots the dock and calls the vendor. §8's Wednesday seven-on-six is a daily decision; the 1,650-unit over-order on the slow mover, once it is the same sign across weeks, is the re-plan the two of us would take back to S&OP.
4	**Tell me about a disagreement with a partner team and how it resolved.**	S12. The numbers went to the partner privately first, the reversal trigger was written before the plan moved, and the plan's owner ratified the change in the room. The disagreement stayed on the sequence and the cost, and both were still visible after the meeting.
5	**How do you keep inventory healthy across the network — excess vs in-stock?**	Apply INV-A per SKU and keep in-stock as the output. Under two weeks you expedite, two to four you watch, over six you move units; §8's close is that policy on one receiving week — fast mover 0.966 WOC, medium 3.00, slow mover 8.71 — and a stranded unit is not available just because it is inside the building.
6	**What would your first 90 days look like in this seat?**	Book of record and the primary metric first, then a stocks-and-flows map of inbound, then the invariant that is already broken, and one input/output review actually running by week six. I would rather walk into your WBR at week six with §4's short list alive than with a ninety-day plan that has not yet met a red metric.
7	**I call a SKU excess and you call it strategic cover. How do we leave the room with one number?**	Compute the band and the safety stock, and let the adjective catch up to the arithmetic. On §8's slow mover, safety stock is 65 units and the close is 3,050, so the word "strategic" has to explain 2,985 units above the safety stock, 1,650 of them past a 4.0-week target; if there is a real spike, it goes into the forecast and the WOC moves with it.
8	**Walk me through the handoff you don't want reopened: placement's recommendation versus the trailers the FC actually receives.**	That handoff gets signed at reservation, while the information is still ahead of the freight. §8 stage 6 values the split at $5,640 before the six trailers are tendered entirely to ONT9-style; once they are in the yard the inbound cost is sunk and the only open decision is the INV-D test on what to transfer, so Wednesday is an unload of a node that was already chosen.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
9	**A metric we both touch goes red. Who speaks first, and what would you hear as me stepping on your team?**	The owner of the input speaks first, with the decomposition already written — that is the S11 rule I want between peers. A new number you have not seen, or a redefinition of dwell or WOC in the meeting, is me stepping on your team; three weeks of same-sign variance is the escalation §4 already gives both of us.
10	**What do you want me to just fix, and what do you want me to bring to you?**	Fix what sits inside the envelope and is reversible today: a door re-slot, a vendor call, a 120-unit discrepancy claim. Bring the plan-level breaks — three weeks of same-sign forecast variance, a SKU that will cross its lead time with no safety stock the way §8's fast mover does at 6.76 days against 7.00, or a transfer that spends past the breakeven — and bring them before the WBR does.

Questions to ask them

- "Where are the seams between your team and this seat — especially who decides where a unit is placed, versus what this building actually receives?"
- "Which shared input is the last to turn green at your WBR, and what do you want a peer to do about it before they put it on your agenda?"
- "When you and a peer have split on excess versus in-stock, what made the call feel fair once the meeting was over?"

Venkata Josyula — Principal PM, FBA · bar raiser

Lens. Principal PM for FBA plus bar-raiser duties: the seller is a customer, LP answers must be clean STARs with honest numbers, and every mechanism should end at a customer outcome.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
1	**How does inbound look from the seller's side of FBA?**	From the seller's dock it is still a promise: ship, appointment, check-in, available. IPI's four components — excess, sell-through, stranded, in-stock — are the §3 health math made visible to the seller, and restock limits are network capacity translated into a constraint they can see; §8's three SKUs are that scoreboard with the arithmetic showing, sell-through 50.9%, 25.0%, and 10.3%, one SKU you replenish and one you stop sending.
2	**Why Amazon — and why this role?**	The three reasons in the master packet land on the same seam. Elementum was the forecast made honest, Flexport was the weekly review with frozen inputs, Sephora was one version of the truth across buildings — and this seat is where the forecast becomes a reservation, the review closes the variance, and two nodes have to share one set of numbers.
3	**Tell me about a time you failed.**	S18's RF miss: I had scored the risk by how small the task looked, and the right scale was impact — about 100 of about 400 devices failed at cutover. The rehearsed fallback fired, so the customer outcome held, and the lesson I kept is that a single point of failure gets a what-if plan on the page before the weekend, whatever its size on the punch list.
4	**Seller experience and network cost pull apart — how do you trade them?**	Price the trade, publish the price, and keep the rejected option in the table. INV-D is that page for a unit already in the network — $3.00 against a $12 margin, breakeven probability 25% — S19 is the same discipline on the promise, one dial, with the decline written as a priced row, and a seller's in-stock and our transfer bill belong on that page together.
5	**How would you improve FBA inbound?**	Start upstream of the door, where a surprise is still cheap. S9 and S11 are ASN quality and visibility before the truck, the appointment and the check-in-to-available clock are the seller-visible half of the §1 flow, AWD is the cheap cube that keeps a slow bulky unit out of an FC pick mod, and §8's slow mover — 0.80 ft³ and 10.3% sell-through — is the unit that pool is for.
6	**Tell me about a time you chose long-term over short-term pressure.**	S12. A committed 18-month roadmap was re-sequenced so the value landed in month 6, at about 66% less effort. The reversal trigger was written down before the plan swung, and the cost of delaying the operators was priced and published, so the long-term choice was a document the room could audit.
7	**A restock limit just cut a seller whose selection was in stock. Was that capacity working, or a defect?**	Check which currency was scarce, and on which SKU. If the FC was already at a CAP-D dwell and the limit stopped a trailer the block could not take, the limit was doing the job of the 85% ceiling; if the cut SKU looks like §8's fast mover — 50.9% sell-through, cover inside one lead time — while a 10.3% sell-through unit still flowed, the limit used an average and constrained the healthy unit, so I would want velocity and cube in the rule, the split stage 5 already shows inside one receipt.

#	LIKELY QUESTION THEY WILL ASK	HOW TO ANSWER
8	**Tell me about a time the data changed your mind.**	S6. I was ready to release safety stock on a better forecast alone, and the rule we wrote required forecast error and lead-time error both to have moved, so the buffer stayed where only one error had improved. INV-C's shadow-price check is the same habit on a placement number: the premium and the shadow price have to print the same dollar, $450 in that example and $5,640 in §8, before I will defend it.
9	**When is holding a seller's inventory upstream, AWD-style, the wrong answer for that seller?**	When the unit is fast, the demand is local, and the replenishment lead time into the FC consumes the promise. INV-B puts lead time under the square root, so days there are the expensive term, and §8's fast mover burned a 1,000-unit opening in 2.00 days and then sat at 6.76 days against a 7-day bridge with no safety stock under it — the upstream pool is the right home for the slow bulky unit (0.80 ft^3, 8.71 weeks of cover) and the wrong home for the unit a customer will miss this week.
10	**What does raising the bar look like on an inbound mechanism — a higher target, or something else?**	A frozen definition, a figure the room can recompute from the row above it, and a recommendation you will walk back when the arithmetic flips. S11's discarded 99.4% week, INV-D's 25% breakeven, and §8's choice to keep the fast mover in place once a transfer would only relocate a $6,000 stockout and add $1,500 of cost — that is the bar I would want a candidate to clear in the room.

Questions to ask them

- "Where does FBA inbound lose seller trust most sharply right now — the appointment, the check-in-to-available clock, or restock limits — and which of those can this Supply Chain seat actually move?"
- "How has AWD changed the conversation with sellers about who holds the slow, bulky units, and where does that conversation still stall?"
- "In the first month, what does a Supply Chain Manager do that makes a Principal PM's mechanism easier to run — and what do they do that you end up routing around?"

7 · Math appendix — every formula, symbolic → substituted → computed

The night-before page. Each row: the formula in symbols, the substitution from the booklet's worked example, the computed result, and where it lives. All inputs **[illustrative]** unless labeled otherwise; recompute each row from the row above it.

#	FORMULA (SYMBOLIC)	SUBSTITUTION	COMPUTED	SECTION
A1	`trailers/day = V ÷ u_c ÷ c_p ÷ p_t`	`120,000 ÷ 6 ÷ 40 ÷ 26`	`= 19.2 trailers/day`	CAP-A
A2	`doors = (trailers/day ÷ hours) ÷ unload_rate ÷ ρ_max`	`(19.2 ÷ 10) ÷ 0.5 ÷ 0.85`	`= 4.5 → 5 doors`	CAP-A
A3	`direct hours = V ÷ UPH; paid = ×(1+indirect)`	`120,000 ÷ 300 = 400; ×1.15`	`= 460 paid h; ÷10 = 46 heads`	CAP-B
A4	`blended UPH = (1−n)·UPH + n·UPH·LC`	`0.7×300 + 0.3×300×0.6 = 264`	`120,000 ÷ 264 = 454.5 h (+13.6%)`	CAP-B
A5	`util_true = I·cube ÷ (C × 0.85)`	`2.0M×0.25 ÷ (1.2M×0.85)`	`= 500,000 ÷ 1,020,000 = 49.0%`	CAP-C
A6	`queue factor = ρ ÷ (1−ρ); dwell ≈ service ÷ (1−ρ)`	`ρ=0.95: 19.0; 2.0 ÷ 0.05`	`= 40 h dwell (vs 13.3 h at 85%)`	CAP-D
A7	`flex heads = (p90 − p50) ÷ UPH_blend ÷ shift`	`36,000 ÷ 264 ÷ 10`	`= 13.6 ≈ 14 flex associates`	CAP-E
B1	`WOC = on-hand ÷ weekly forecast`	`A: 4,200÷600; B: 1,350÷450; C: 500÷500`	`= 7.0 / 3.0 / 1.0 weeks`	INV-A
B2	`excess units = on-hand − WOC_target × forecast`	`4,200 − 4.0×600`	`= 1,800 units to move`	INV-A
B3	`SS = z × σ_d × √LT`	`1.645 × 120 × √7 = 1.645 × 317.5`	`= 522 units = 1.04 days`	INV-B
B4	`SS(99) − SS(95) = (z99 − z95)·σ_LT`	`(2.326 − 1.645) × 317.5`	`= +217 units (+41.6%)`	INV-B
B5	`SS(LT=5) = z × σ_d × √5`	`1.645 × 120 × 2.2361`	`= 441 units (−81; −15.5%)`	INV-B
B6	`cost = Σ d_r × c_{r,f} s.t. node capacity`	`2,000×1.20 + 1,000×1.30 + 1,500×2.10 + 1,500×2.40`	`= $10,450/wk (vs $10,000 unconstrained)`	INV-C
B7	`shadow price = (c_far − c_near) × units_displaced`	`(2.40 − 2.10) × 1,500`	`= $450/wk = the capacity premium`	INV-C
B8	`transfer if cost_t < margin × P(stockout); p* = cost_t ÷ margin`	`3.00 < 12 × 0.40 = 4.80; p* = 3.00 ÷ 12`	`= transfer; p* = 25%; net +$3,240 on 1,800 units`	INV-D
B9	`OTIF_units = PO × fill × on_time`	`10,000 × 0.92 × 0.88`	`= 8,096 units = 81.0% of plan`	INV-E
B10	`in-stock proxy = 1 − (deficit − SS) ÷ PO`	`1 − (1,904 − 522) ÷ 10,000`	`= 86.2% weekly in-stock proxy`	INV-E

Recompute discipline: A1→A2 share 19.2; A3→A4 share the 120,000 and 300 UPH; B3→B4→B5 share σ_d = 120 and z values; B2's 1,800 feeds B8's scale row; B9's deficit feeds B10. If any row doesn't reproduce from its own substitution, the row is wrong — say the formula, not the memory.

§8 · One week at one FC — the complete worked example

One building, one week, one vendor, three SKUs, and the same figures carried from the forecast through the close. Every stage below is symbolic formula, then the substitution, then the result — the same shape as the CAP and INV blocks. If a later row quotes a number you cannot find above it, the row is wrong. All inputs are **[illustrative]**; the rates, the z, and the transfer economics are the ones already used in §2–§3 so the arithmetic is familiar out loud. ONT9-style and PHX7-style are labels for the whiteboard, not claims about any real building.

> **Assumptions to say before the first number.** Demand is flat across seven days. The Wednesday receipt is stowed the same day, so a 2.0 h unload slips the promise and does not lose Wednesday's sales. PHX7-style opening on-hand for this vendor is zero. In the stress block, service is deterministic and all seven trailers are in the yard at block open. √7 = 2.6458 and z = 1.645, as in INV-B. Transfer cost is $3.00/unit and lost margin is $12/unit, as in INV-D. Placement is decided on the planned PO; the 2% shortage is discovered at receive, so health at the close uses actuals.

The scenario card — week 44, reused everywhere below

CONSTANT	VALUE	WHERE IT IS REUSED
Building, week	ONT9-style FC, week 44	Every stage. Illustrative; not a real site's data.
The shipment	One vendor, 3 SKUs, 12,000 units, 6 trailers	Stages 1–7. 12,000 ÷ 6 = 2,000 units/trailer.
Doors, day, block	6 doors; 10 h receiving day; stress block is 2.0 h	Stage 2. The block is one service deep.
Unload rate	0.5 trailer/door-hour, so service = 1 ÷ 0.5 = 2.0 h	Stage 2. Same rate as CAP-A.
Pack factors	6 units/carton, 40 cartons/pallet, 26 pallets per dense trailer	Stage 1. Same factors as CAP-A.
Labor standards	Receive 300 UPH, stow 240 UPH, indirect 15%, shift 10 h	Stages 3 and 5. Receive UPH matches CAP-B.
The mod	4,000 bins × 2.0 ft³ = 8,000 ft³ nominal; × 0.85 = 6,800 ft³ effective	Stage 5. The 15% honeycomb matches CAP-C.
Service and lead time	z = 1.645; LT = 7 days; √7 = 2.6458	Stage 7. Same constants as INV-B.
Money	Inbound ONT9-style $0.80/unit; PHX7-style $1.40/unit; transfer $3.00; margin $12	Stage 6. Transfer and margin match INV-D.
Sister node	PHX7-style demand: F 500, M 700, S 1,050 per week; opening 0	Stage 6 only. This vendor's first units there.

SKU	ROLE	UNITS ON THE PO	WEEKLY FORECAST	PER DAY	OPENING ON-HAND	Σ / DAY	FT³ / UNIT
F	Fast	6,000	3,500	500	1,000	80	0.15
M	Medium	4,000	1,400	200	1,600	40	0.30
S	Slow	2,000	350	50	1,400	15	0.80
Total		12,000	5,250		4,000		

Buy targets that produced the PO, proved in stage 1: F back to 1.0 week of cover, M to 3.0 weeks, S would have needed 350 units to stay at 4.0 weeks and was ordered to a 2,000-unit multiple instead. The 1,650-unit gap on S is the number stage 6 and stage 7 both spend.

Stage 1 · Inbound forecast → units per week → trailer equivalents

1A · THE BUY THAT EQUALS 12,000 — RECEIPT = TARGET ON-HAND − OPENING + FORECAST SALES

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Opening cover, F	`1,000 ÷ 500 = 2.00 days; 1,000 ÷ 3,500 = 0.286 wk`	scenario card	2.00 days — gone by Tuesday close
2	Opening cover, M	`1,600 ÷ 1,400 = 1.143 wk`	scenario card	1.14 weeks
3	Opening cover, S	`1,400 ÷ 350 = 4.00 wk`	scenario card	4.00 weeks — already the healthy center
4	F receipt	`target OH = 1.0 × 3,500 = 3,500; 3,500 − 1,000 + 3,500 = 6,000`	target 1.0 WOC; sales = forecast	**6,000 units**
5	M receipt	`target OH = 3.0 × 1,400 = 4,200; 4,200 − 1,600 + 1,400 = 4,000`	target 3.0 WOC	**4,000 units**
6	S receipt the band wants	`target OH = 4.0 × 350 = 1,400; 1,400 − 1,400 + 350 = 350`	target 4.0 WOC, the INV-A center	350 units
7	S over-order	`2,000 − 350 = 1,650`	vendor order multiple is 2,000	**1,650 units — remembered at stages 6 and 7**
8	Week inbound	`6,000 + 4,000 + 2,000 = 12,000`	F + M + the multiple on S	**12,000 units**

F's target close equals one lead time of demand and contains no safety stock. That choice is quiet here and loud in stage 4.

1B · UNITS → CARTONS → PALLETS → TRAILERS — THE TENDER IS NOT THE DENSE EQUIVALENT

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Cartons	`12,000 ÷ 6 = 2,000`	6 units/carton, as CAP-A	2,000 cartons
2	Pallets	`2,000 ÷ 40 = 50`	40 cartons/pallet, as CAP-A	50 pallets
3	Dense trailer-equivalents	`50 ÷ 26 = 1.923`	26 pallets per 53-ft trailer	**1.92 trailers**
4	What the vendor tendered	`6 trailers; 12,000 ÷ 6 = 2,000 units each`	the ASN	6 trailers
5	How full each trailer is	`50 ÷ 6 = 8.333 pallets; 8.333 ÷ 26 = 0.321`	50 pallets across 6 trailers	**32.1% of a dense trailer**
6	Door currency vs cube currency	`6 ÷ 1.923 = 3.12`	tendered ÷ dense	**3.12× — plan doors off 6, cube off 50 pallets**

The sentence to say: "Twelve thousand is a cover equation plus an order multiple — six thousand, four thousand, two thousand — and the two thousand includes sixteen hundred fifty the band did not ask for. Dense, that is one-point-nine trailers; the tender is six, a third full, so the dock feels six arrivals and the building feels fifty pallets."

Stage 2 · Dock-door capacity — 7 trailers against 6 slots, and the dwell that forms

The week was reserved as six trailers spread across five days. Wednesday morning the yard has all six ASN trailers (the 12,000 units) plus one unscheduled yard-drop that is not on this PO. The block they have fallen into is sized at six door-slots. Carried in from stage 1: 6 trailers, 2,000 units each, service rate 0.5.

2A · DOES THE BLOCK HAVE THE SLOTS? — DOORS × HOURS × UNLOAD RATE

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Slots in the block	`S = 6 × 2.0 × 0.5 = 6.0`	doors × hours × unload rate	**6 door-slots**
2	Arrivals	`6 ASN + 1 unscheduled = 7`	the week's 6 trailers, bunched, plus a yard-drop with none of the 12,000	**7 trailers**
3	Utilization	`ρ = 7 ÷ 6 = 1.1667`	arrivals ÷ slots	**116.7%**
4	The 85% ceiling	`0.85 × 6 = 5.1`	same ceiling as CAP-A	even 6 scheduled trailers were over the ceiling
5	Why Kingman stops here	`1 − ρ = 1 − 7/6 = −0.167`	dwell ≈ service ÷ (1 − ρ) needs ρ below 1	the form is undefined — count the residual instead
6	The building's week, for scale	`6 × 10 × 0.5 = 30/day; 30 × 5 = 150; 6 ÷ 150 = 0.040`	full receiving day, five days	30 slots/day, 150/week; this vendor is **4.0% of the week**

The week had the doors. The two-hour block did not. That is a bunching failure, and it is the one to say out loud before anyone proposes a seventh door.

2B · EXPECTED DWELL ONCE P IS ABOVE 1 — DETERMINISTIC, ALL SEVEN PRESENT AT BLOCK OPEN

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Service time	`1 ÷ 0.5 = 2.0 h`	the unload rate on the card	2.0 h on a free door
2	Who starts at once	`min(7, 6) = 6 trailers start; 7 − 6 = 1 waits`	6 doors, simultaneous arrival	1 trailer in queue
3	Dwell of a trailer that starts	`wait 0 + service 2.0 = 2.0`	door free at t = 0	2.0 h
4	Dwell of the marginal trailer	`wait = 2.0; dwell = 2.0 + 2.0 = 4.0`	first door frees at the end of one service	**4.0 h**
5	Expected dwell	`(6 × 2.0 + 1 × 4.0) ÷ 7 = 16 ÷ 7 = 2.286`	FIFO: each of the 7 equally likely to be last	**2.29 h**
6	Expected wait	`(6 × 0 + 1 × 2.0) ÷ 7 = 2 ÷ 7 = 0.286`	same weights	0.29 h
7	Spill into the next block	`7 × 2.0 − 6 × 2.0 = 14 − 12 = 2.0 door-hours`	work demanded minus work the block can do	**1 trailer, and 1 of the next block's 6 slots**
8	FIFO hit on this PO	`6 × (1/7) × 2,000 = 12,000 ÷ 7 = 1,714 units`	P(a given ASN trailer waits) = 1/7; 2,000 units on it	**1,714 units delayed 2.0 h**
9	ASN-priority alternative	`delayed PO units = 0; unscheduled dwell = 4.0 h`	the 6 ASN trailers take the 6 doors	the 12,000 stay on a 2.0 h clock; the yard-drop holds the 4.0 h

The sentence to say: "Seven over six is past the formula — one minus rho is negative — so the number is the residual trailer. Six start and dwell two hours; the seventh dwells four; the average is sixteen over seven, two-point-two-nine hours. Give the ASN the doors and the twelve thousand stay on a two-hour clock. Leave it FIFO and about seventeen hundred units slip two hours, and the next block opens already down one slot."

Stage 3 · Labor plan — UPH to hours to heads on the shift

Labor is reserved on the plan, 12,000 units, before the shortage in stage 4 is known. The plan spread those units across five days; the yard in stage 2 put them on one shift. Both rosters are below, because the average is what gets staffed and the bunch is what shows up. Carried in: 12,000 units, 300 UPH, 15% indirect, 10 h shifts.

3 · RECEIVE LABOR — THE BUNCH IS THE SHIFT THAT HAS TO BE ROSTERED

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Direct hours, the plan	`12,000 ÷ 300 = 40.0`	mature receive UPH, as CAP-B	40.0 direct hours
2	Paid hours	`40.0 × 1.15 = 46.0`	15% indirect	46.0 paid hours
3	Heads if the week arrives on one shift	`46.0 ÷ 10 = 4.6 → round up to 5`	10 h shift; a partial head is not a person you can roster	**5 receivers**
4	Slack on that crew	`5 × 10 − 46.0 = 4.0; 4.0 ÷ 50 = 0.080`	paid hours the roster actually contains	4.0 h slack, 8.0% idle
5	The spread the average suggests	`12,000 ÷ 5 = 2,400/day; 2,400 ÷ 300 = 8.0; 8.0 × 1.15 = 9.2; 9.2 ÷ 10 = 0.92`	five even days	0.92 → **1 receiver looks like enough**
6	What 1 receiver does with the bunch	`40.0 ÷ 1 = 40.0 direct hours of work on a 10 h shift`	stage 2 put the week on one morning	about 4 shifts of backlog from one rostering choice

The sentence to say: "Forty direct hours, times one-fifteen, is forty-six paid hours, divided by a ten-hour shift is four-point-six, so five people. The daily average says one person and point-nine of a head. Staff the bunch. The one hundred twenty units we find short in the next stage free four-tenths of an hour — you do not re-roster a shift for that."

Stage 4 · Receive accuracy — a 2% shortage on the fast mover, and what it does to AFS

AFS here means units the promise can see: opening on-hand, which was already sellable, plus units that clear receive and will clear stow. The shortage never reaches stow, so it never becomes AFS. Carried in: PO of 6,000 / 4,000 / 2,000, opening 1,000 / 1,600 / 1,400, daily demand 500 / 200 / 50.

4A · THE COUNT — 2% OF F, AND WHY 99% OVERALL IS THE WRONG HEADLINE

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Shortage	`0.02 × 6,000 = 120`	2% short on F only; M and S count clean	**120 units, all SKU-F**
2	Received, F	`6,000 − 120 = 5,880`	step 1	5,880 units
3	Received, the PO	`5,880 + 4,000 + 2,000 = 11,880`	M and S at plan	11,880 units
4	Accuracy	`11,880 ÷ 12,000 = 0.990; 5,880 ÷ 6,000 = 0.980`	actual ÷ PO	**99.0% overall, 98.0% on F, 100% on M and S**
5	F AFS at the moment of receipt	`planned 1,000 + 6,000 = 7,000; actual 1,000 + 5,880 = 6,880; 7,000 − 6,880 = 120`	opening was already AFS	**AFS lower by 120**
6	Those 120 as demand	`120 ÷ 500 = 0.24 days; 120 ÷ 3,500 = 0.0343`	F demand from the card	0.24 days, 3.43% of the week

4B · DOES THE SHORTAGE MISS THIS WEEK'S PROMISE, OR THE BRIDGE TO THE NEXT RECEIPT?

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Mon–Tue sales, F	`2 × 500 = 1,000`	opening cover was 2.00 days (stage 1)	opening is consumed exactly; AFS hits zero at Tuesday close
2	Wed–Sun sales, F	`5 × 500 = 2,500; 5,880 − 2,500 = 3,380 left`	assumption: stow finishes Wednesday, so the day is slipped by 2.0 h and not lost	2,500 sold; the receipt covers them
3	Week sales, all three	`F 3,500; M 1,400; S 350`	each week's forecast, fully served under the assumption	this week's in-stock holds
4	Close, F, if the count had been clean	`1,000 + 6,000 − 3,500 = 3,500; 3,500 ÷ 500 = 7.00 days`	the stage-1 buy, which targeted 1.0 WOC and no safety stock	7.00 days — exactly the lead time
5	Close, F, actual	`1,000 + 5,880 − 3,500 = 3,380; 3,380 ÷ 500 = 6.76 days`	the 120 removed	**3,380 units, 6.76 days**
6	The pass-through	`7.00 − 6.76 = 0.24 days; 0.24 × 500 = 120`	steps 4 and 5	**the shortage equals the entire lead-time miss**
7	Close, M	`1,600 + 4,000 − 1,400 = 4,200`	M counted clean	4,200 units — used again in stages 5–7
8	Close, S	`1,400 + 2,000 − 350 = 3,050`	S counted clean	3,050 units — used again in stages 5–7

The sentence to say: "Two percent of the fast mover is one hundred twenty units. The building's accuracy reads ninety-nine percent, and that average is hiding the only SKU that was bought to land exactly on its lead time. This week's sales still clear. The one hundred twenty comes off the bridge: seven-point-zero-zero days becomes six-point-seven-six, and there was no safety stock underneath it."

Stage 5 · Stow and cubic fill — the mod before, at the peak, and at the close

This vendor's three SKUs live in one illustrative pick mod, so the fill move is visible. A whole FC would swallow 12,000 units; the mod is the grain a stow lead actually manages. Carried in: cube per unit from the card, received units from stage 4 (5,880 / 4,000 / 2,000), closes 3,380 / 4,200 / 3,050.

5A · BIN CAPACITY, THEN FILL BEFORE AND AFTER THE RECEIPT

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Nominal capacity	`4,000 × 2.0 = 8,000 ft³`	bins × cube per bin	8,000 ft³
2	Effective capacity	`8,000 × 0.85 = 6,800; per bin 2.0 × 0.85 = 1.70`	15% honeycomb, as CAP-C	**6,800 ft³ effective**
3	Opening cube	`1,000×0.15 + 1,600×0.30 + 1,400×0.80 = 150 + 480 + 1,120 = 1,750`	opening units × cube	1,750 ft³
4	Opening fill	`1,750 ÷ 8,000 = 0.21875 = 21.9% nominal; 1,750 ÷ 6,800 = 0.2574 = 25.7% effective`	against both denominators	**25.7% effective — plan against this**
5	Receipt cube, actual	`5,880×0.15 + 4,000×0.30 + 2,000×0.80 = 882 + 1,200 + 1,600 = 3,682`	stage-4 received units	3,682 ft³
6	Who ate the cube	`units: 2,000 ÷ 11,880 = 0.168 = 16.8%; cube: 1,600 ÷ 3,682 = 0.435 = 43.5%`	S versus the receipt	**S is 16.8% of units and 43.5% of cube**
7	Peak, before the week's sales	`1,750 + 3,682 = 5,432; 5,432 ÷ 6,800 = 0.7988 = 79.9%; 5,432 ÷ 8,000 = 0.679 = 67.9% nominal`	opening + receipt	**79.9% effective (67.9% if you trust nominal)**
8	Headroom to the 90% wall	`0.90 × 6,800 = 6,120; 6,120 − 5,432 = 688; 688 ÷ 0.80 = 860`	CAP-C's "wall at 90%"	**688 ft³, or 860 more units of S**
9	Bins the receipt touches	`3,682 ÷ 1.70 = 2,165.9 → 2,166; 2,166 ÷ 4,000 = 0.5415 = 54.2%`	receipt cube ÷ effective cube per bin	2,166 bins, 54.2% of the mod

5B · STOW LABOR, AND THE CUBE THAT IS STILL THERE AT THE CLOSE

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Stow hours on the plan	`12,000 ÷ 240 = 50.0 direct; 50.0 × 1.15 = 57.5 paid`	stow 240 UPH; same 15% indirect	57.5 paid hours
2	Stow heads on the bunched shift	`57.5 ÷ 10 = 5.75 → round up to 6; slack = 6 × 10 − 57.5 = 2.5 h`	same rounding as the receivers	**6 stowers, 2.5 h slack**
3	What the shortage saves	`120 ÷ 240 = 0.5 direct hour; 11,880 ÷ 240 = 49.5`	the 120 units from stage 4	half an hour — hold the roster of 6
4	Close cube	`3,380×0.15 + 4,200×0.30 + 3,050×0.80 = 507 + 1,260 + 2,440 = 4,207`	closes from stage 4	4,207 ft³
5	Close fill	`4,207 ÷ 6,800 = 0.6187 = 61.9%; S share 2,440 ÷ 4,207 = 0.580 = 58.0%`	effective denominator	**61.9% effective; S is 58% of what is left**

The sentence to say: "The mod opens at twenty-five-point-seven percent effective, not the twenty-two percent nominal. The receipt takes it to seventy-nine-point-nine, and the slow mover is sixteen-point-eight percent of the units and forty-three-point-five percent of the cube. Eight hundred sixty more of those units would put the mod on the ninety-percent wall. Stow is six people on the shift that the freight actually arrived."

Stage 6 · Placement — two nodes, put the units where the demand is, pick the cheaper plan

This decision belongs with the information flow, before the tender. It is worked here, after the freight is in the building, so both numbers can be said: what the split was worth on Tuesday, and what is still worth doing once the six trailers are in the ONT9-style yard. Planned units, from stage 1: F 6,000, M 4,000, S 2,000. PHX7-style weekly demand, from the card: F 500, M 700, S 1,050, opening zero.

COST TO HAVE THE UNIT THERE, $/UNIT	SERVE ONT9-STYLE DEMAND	SERVE PHX7-STYLE DEMAND
Inbound to ONT9-style, then transfer if needed	`$0.80`	`$0.80 + $3.00 = $3.80`

COST TO HAVE THE UNIT THERE, $/UNIT	SERVE ONT9-STYLE DEMAND	SERVE PHX7-STYLE DEMAND
Inbound to PHX7-style, then transfer if needed	`$1.40 + $3.00 = $4.40`	`$1.40`

6A · WHERE EACH SKU GOES — DEMAND FIRST, LEFTOVER ONLY WHERE COVER CAN TAKE IT

STEP	DECISION	FORMULA → SUBSTITUTION	WEEKLY RESULT
1	F stays entirely at ONT9-style. Even the whole 6,000 leaves that node at one week.	`close = 1,000 + 6,000 − 3,500 = 3,500; WOC = 3,500 ÷ 3,500 = 1.00`	ONT 6,000 / PHX 0. PHX's 500 is a separate buy, not a slice of this PO.
2	M: give PHX its week, keep the rest at ONT. ONT's band survives the cut.	`ONT keeps 4,000 − 700 = 3,300`	ONT 3,300 / PHX 700
3	S: ONT keeps the 350 the 4.0 WOC band asked for in stage 1. The over-order goes where the demand is.	`PHX gets 2,000 − 350 = 1,650`	ONT 350 / PHX 1,650 — the same 1,650
4	Totals	`ONT 6,000 + 3,300 + 350 = 9,650; PHX 0 + 700 + 1,650 = 2,350; 9,650 + 2,350 = 12,000`	**9,650 / 2,350**
5	Trailer-equivalents at this vendor's 2,000 units per trailer	`9,650 ÷ 2,000 = 4.825; 2,350 ÷ 2,000 = 1.175`	about 5 trailers and about 1, not 6 and 0

6B · COST OF THE SPLIT VERSUS COST OF SIX TRAILERS HERE, THEN A TRANSFER TO REPAIR IT

STEP	PLAN	FORMULA → SUBSTITUTION	COST
1	Split, direct inbound, no transfer	`9,650 × 0.80 = 7,720; 2,350 × 1.40 = 3,290; 7,720 + 3,290 = 11,010`	**$11,010**
2	All 12,000 inbound to ONT9-style	`12,000 × 0.80 = 9,600`	$9,600 sunk once the trailers are here
3	Then transfer the 2,350 the split would have sent direct	`2,350 × 3.00 = 7,050; 9,600 + 7,050 = 16,650`	$16,650
4	Premium for deciding late	`16,650 − 11,010 = 5,640`	**$5,640**
5	Shadow-price check — the model has to agree with itself	`extra inbound = 2,350 × (1.40 − 0.80) = 2,350 × 0.60 = 1,410; transfer avoided = 7,050; 7,050 − 1,410 = 5,640`	$5,640 again
6	The same check by SKU	`M: 700 × (3.00 − 0.60) = 700 × 2.40 = 1,680; S: 1,650 × 2.40 = 3,960; 1,680 + 3,960 = 5,640`	F contributes $0 — it was not split

Optimum on Tuesday: **split, 9,650 here and 2,350 direct to PHX7-style.** The booking that was actually made — six trailers, all to ONT9-style — leaves $5,640 on the table. After the yard, the $9,600 inbound is sunk, and the open question is the INV-D test on each SKU, not a wish to rerun Tuesday.

6C · AFTER THE TRAILERS ARE HERE — TRANSFER ONLY WHAT PASSES THE BREAKEVEN, ON ACTUAL CLOSES

STEP	SKU	FORMULA → SUBSTITUTION	DECISION
1	F, the test	`transfer 500 × $3.00 = $1,500; PHX loss avoided = 500 × $12 = $6,000; ONT then shorts the same 500 × $12 = $6,000`	net +$1,500 worse. **Keep F. Expedite, do not transfer.**
2	M, the test	`cost 700 × 3.00 = 2,100; PHX loss if you do not = 700 × 12 = 8,400; 2,100 < 8,400`	**Transfer 700. Wins by $6,300.**
3	M, ONT after the move	`(4,200 − 700) ÷ 1,400 = 3,500 ÷ 1,400 = 2.50 WOC`	still inside the 2–4 band (actual close from stage 4)
4	S, three quantities you might move	`to the 6.0 ceiling: 3,050 − 6×350 = 3,050 − 2,100 = 950; to cover PHX's week: 1,050; to the 4.0 center: 3,050 − 4×350 = 3,050 − 1,400 = 1,650`	950 leaves PHX short 100; 1,050 leaves ONT at 5.71 WOC, still in the watch band
5	S, the quantity chosen	`1,650 × 3.00 = 4,950; PHX week loss avoided ≥ 1,050 × 12 = 12,600; ONT left at 1,400 ÷ 350 = 4.00 WOC`	**Transfer 1,650. It is the stage-1 over-order, and it is the INV-A move down to 4.0 WOC.**
6	Cash still to spend, forward-looking	`2,100 + 4,950 = 7,050 on M and S; $0 on F`	$7,050. The $5,640 from Tuesday is no longer available to save; this $7,050 is the repair.

The sentence to say: "Tuesday's optimum was ninety-six fifty here and twenty-three fifty direct to the sister node, eleven thousand ten dollars against sixteen thousand six fifty if I haul them later — a fifty-six forty gap, and the shadow price prints the same fifty-six forty, so I trust it. The trailers came here anyway. Forward from the yard I keep the fast mover, because moving it spends fifteen hundred dollars to relocate a stockout, and I transfer seven hundred of the medium and the sixteen fifty of the slow. That sixteen fifty is the same over-order the buy created in stage one."

Stage 7 · Inventory health at the close — who is thin, who is fine, who gets moved

Closes are the stage-4 actuals: F 3,380, M 4,200, S 3,050. Sales this week equaled forecast, so sell-through uses units sold ÷ (opening + received). Safety stock uses the card: z = 1.645, √7 = 2.6458, σ of 80 / 40 / 15. The action rule is INV-A's: under 2 weeks expedite, 2–4 watch, over 6 move.

7A · WOC, SAFETY STOCK, AND SELL-THROUGH — THREE SKUS, THREE ANSWERS

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	WOC, F	`3,380 ÷ 3,500 = 0.9657 ≈ 0.966`	close ÷ weekly forecast	**0.966 weeks — under 2, expedite**
2	SS, F	`σ_LT = 80 × 2.6458 = 211.664; SS = 1.645 × 211.664 = 348.19 ≈ 348; 348 ÷ 500 = 0.696 days`	z × σ × √LT	348 units, 0.70 days
3	F against the bridge	`LT units = 7 × 500 = 3,500; LT+SS = 3,500 + 348 = 3,848; gap = 3,848 − 3,380 = 468; and 120 + 348 = 468`	the shortage plus the safety stock the buy never held	**short 120 vs the lead time, short 468 vs lead time plus safety stock**
4	Sell-through, F	`3,500 ÷ (1,000 + 5,880) = 3,500 ÷ 6,880 = 0.5087 = 50.9%`	sold ÷ (open + received)	50.9%
5	WOC, M	`4,200 ÷ 1,400 = 3.00`	stage-4 close	**3.00 weeks — healthy**
6	SS, M	`σ_LT = 40 × 2.6458 = 105.832; SS = 1.645 × 105.832 = 174.09 ≈ 174; cover days = 4,200 ÷ 200 = 21.0; LT+SS days = 7 + 174 ÷ 200 = 7.87; 21.0 ÷ 7.87 = 2.67×`	same z and √LT	174 units of safety stock; position is 2.67× the floor
7	Sell-through, M	`1,400 ÷ (1,600 + 4,000) = 1,400 ÷ 5,600 = 0.250 = 25.0%`	sold ÷ (open + received)	25.0%
8	WOC, S	`3,050 ÷ 350 = 8.714 ≈ 8.71; days = 3,050 ÷ 50 = 61.0`	stage-4 close	**8.71 weeks, 61 days — over 6, move it**
9	SS, S	`σ_LT = 15 × 2.6458 = 39.687; SS = 1.645 × 39.687 = 65.29 ≈ 65; 3,050 − 65 = 2,985 above the safety-stock floor`	same z and √LT	65 units. "Strategic cover" would have to explain the other 2,985.
10	Sell-through, S	`350 ÷ (1,400 + 2,000) = 350 ÷ 3,400 = 0.1029 = 10.3%`	sold ÷ (open + received)	10.3% — the excess showing up as a rate

7B · THE CORRECTIVE ACTION — ONE PER SKU, EACH ONE A NUMBER ALREADY ON THIS PAGE

STEP	SKU	FORMULA → SUBSTITUTION	ACTION
1	F — at risk of stockout	`expedite 468 = 120 shortage + 348 safety stock; 468 ÷ 500 = 0.94 days of demand; 3,380 + 468 = 3,848 = the LT+SS target`	**Pull the next ASN forward by 468 units. Do not fund it by transferring F out — stage 6c already priced that as a loss.**
2	M — healthy	`WOC 3.00 is the center of the 2–4 band. After the stage-6 transfer of 700: 2.50 WOC, still inside the band.`	**No inventory action. The 700-unit move is a placement repair, and the band still holds.**
3	S — overstocked	`move 1,650; 3,050 − 1,650 = 1,400; 1,400 ÷ 350 = 4.00 WOC; cost 1,650 × 3.00 = 4,950`	**Transfer 1,650 to PHX7-style. Same 1,650 as the stage-1 over-order.**
4	Cube the S move gives back	`1,650 × 0.80 = 1,320 ft³; 1,320 ÷ 6,800 = 0.194 = 19.4% of the mod`	19.4 points of effective fill, freed without a rack

The sentence to say: "Three SKUs, three sentences. Fast mover, zero-point-nine-seven weeks, six-point-seven-six days against a seven-day lead time — expedite four hundred sixty-eight, which is the one hundred twenty we never received plus the three hundred forty-eight of safety stock we never bought, and keep every unit of it here. Medium, three-point-zero-zero weeks, healthy, and it can still spare seven hundred for the sister node and land at two-and-a-half. Slow mover, eight-point-seven weeks and ten percent sell-through — transfer the sixteen fifty, four thousand nine hundred fifty dollars, back to four weeks, and that sixteen fifty is the order multiple from the first page of the week."

The one-page close — stage, metric, value, what breaks

STAGE	METRIC	VALUE	WHAT BREAKS IF IT IS MISMANAGED
1 · Forecast	Units and trailer-equivalents	12,000 = 6,000 + 4,000 + 2,000. 50 pallets = 1.92 dense trailers. Tendered 6, at 32.1% full.	Ordering S to the 2,000-unit multiple overstocks by 1,650. Buying F to 1.0 WOC with no safety stock leaves a 7-day bridge and nothing under it.
2 · Dock	Slots, ρ, dwell	6 × 2.0 × 0.5 = 6 slots against 7 arrivals. ρ = 116.7%. Marginal dwell 4.0 h. E[dwell] = 2.29 h. The week was 4.0% full.	FIFO delays an expected 1,714 units by 2.0 h and spills one slot into the next block. Quoting Kingman past ρ = 1 hides the residual trailer. A seventh door was never the constraint.
3 · Labor	Hours and heads	12,000 ÷ 300 = 40 h × 1.15 = 46 h ÷ 10 = 4.6 → 5 receivers on the bunched shift. The spread averages 0.92 → 1.	Rostering the average leaves ~40 direct hours with one person, and the backlog carries into Thursday. The 120-unit shortage frees 0.4 h — hold the crew.
4 · Accuracy	Shortage and AFS	0.02 × 6,000 = 120 of F. Received 11,880 (99.0% overall, 98.0% on F). F AFS 6,880 vs 7,000. Cover 6.76 days vs 7.00.	A 99% headline hides the only SKU sitting on its lead time. This week's sales still clear; the 120 is the entire miss versus the next receipt.
5 · Stow / cube	Fill and stow heads	Effective mod 6,800 ft³. Fill 25.7% → 79.9% peak → 61.9% at close. Nominal at the peak would have said 67.9%. S is 43.5% of receipt cube. 6 stowers.	Another 860 units of S, 688 ft³, hits the 90% wall. Reading nominal fill understates the peak by 12 points. The slow mover is the cube problem.
6 · Placement	Split vs all six trailers here	Optimum 9,650 / 2,350. Cost $11,010 vs $16,650. Savings $5,640, matched by the shadow-price check. After arrival: keep F, transfer M 700 and S 1,650, spend $7,050.	Six trailers into ONT9-style leave PHX's medium and slow demand unfilled and leave 8.71 weeks of S in this mod. Deciding after the yard turns a $1,410 inbound premium into a $7,050 transfer.
7 · Health	WOC and the action	F 0.966 WOC, expedite 468. M 3.00 WOC, hold (2.50 after the 700). S 8.71 WOC, transfer 1,650 for $4,950, back to 4.00, freeing 1,320 ft³.	F stocks out inside the next lead time. S sits until the storage cost and the wasted cube. M is the control: inside the band, and still able to fund the sister node.

SAY THIS VERSION OUT LOUD — ABOUT SIXTY SECONDS, THEN STOP

"Week 44, an ONT9-style building: one vendor, three SKUs, twelve thousand units on six trailers. Cover targets rebuild that — fast mover to one week, medium to three — except the slow mover arrives as a two-thousand-unit multiple against a three-fifty need, sixteen fifty too many. Dense, one-point-nine trailers; tendered, six, so the dock feels six arrivals. Wednesday, seven trailers hit a six-slot block: the queueing formula does not apply, the seventh dwells four hours, the average is two-point-two-nine. Staff the bunch, five receivers and six stowers. Receive is short one hundred twenty of the fast mover — the whole gap from a seven-day lead time to six-point-seven-six days, because safety stock was never bought. The mod goes from twenty-five-point-seven percent full to seventy-nine-point-nine. Splitting twenty-three fifty units to the sister node was worth five thousand six hundred forty dollars. At the close: expedite four sixty-eight of the fast mover and keep it, leave the medium at three weeks, transfer the same sixteen fifty of the slow mover. Same twelve thousand, three actions."